MIDLANDS STATE UNIVERSITY

GENERAL INFORMATION AND REGULATIONS

6

- Agriculture, Environment and Natural Resources Management
- Built Environment, Art and Design
- Business Sciences
- Engineering and Geosciences
- Medicine and Health Sciences
- Science and Technology

2024 – 2028

VOLUME 2

GENERAL INFORMATION AND REGULATIONS

6

Agriculture, Environment and Natural Resources Management
Built Environment, Art and Design
Business Sciences
Engineering and Geosciences
Medicine and Health Sciences
Science and Technology

2024 – 2028

2024 Midlands State University

The Yearbook is as far as possible, accurate and up to date at the time of going to the printers. However, it should be noted that not all programmes of study or modules described herein would necessarily be on offer each year.

Compiled by:
Midlands State University
P Bag 9055
Gweru Zimbabwe

CONTENTS

Addresses and Telephone Numbers 1

Principal Officers of the University……..2

University Council 3

Midlands State University Act ……6

Staff Qualifications 39

Faculties and Teaching Departments ……..85

Historical Note 88

Faculty Governance Ordinance No. 4 92

Departmental Governance Ordinance No. 5 96

Academic Staff Grading, Tenure and Promotions Ordinance No. 3 .....102.

Administrative Staff, Appointments, Grades, Promotion, Grading and Tenure Ordinance (Grades 4-8) Ordinance 6 118

Junior Non-Teaching Staff Ordinance (Grade 9-14) 122

Rules of Student Conduct Discipline Ordinance No. 2..... 124.

Innovation and Industrialisation Ordinance of 2023 134

General Academic Regulations For Undergraduate Certificates and Diplomas..... 142

General Academic Regulations For Undergraduate Degrees 146

General Academic Regulations For Postgraduate Diplomas And Taught Masters Degrees..... 168

Generalacademic Regulations For Master Of Philosophy Degrees And Doctor Of Philosophy Degrees 177

FACULTY OF AGRICULTURE, ENVIRONMENT AND NATURAL RESOURCES MANAGEMENT ……..189

Department of Agricultural Economics and Development..... 192.

Department of Agronomy and Horticulture..... 241.

Department of Animal and Wildlife Sciences ..... 306.

Department of Land and Water Resources Management ……. 352

FACULTY OF BUILT ENVIRONMENT, ART AND DESIGN ……..391

Department of Surveying and Geomatics ..... 394.

FACULTY OF BUSINESS SCIENCES ……405

Department of Accounting Sciences ……411
Department of Economic Sciences ……..459
Department of Information and Marketing Sciences ……..494
Department of Management Sciences ……568
Department of Supply Chain, Insurance and Risk Sciences …… 595

Department of Hospitality, Tourism and Leisure Sciences ……..615
Graduate School of Business Leadership ……..641
FACULTY OF ENGINEERING AND GEOSCIENCES ……..662

Faculty Regulations for Undergraduate Degree Programmes ..... 665.

Department of Fuels and Energy Engineering ……666
Department of Mechanical Engineering…….687.

Department of Metallurgical and Materials Engineering ..... 703.

Department of Mining Engineering..... 720.

Department of Geosciences ..... 737.

FACULTY OF MEDICINE AND HEALTH SCIENCES ……763

FACULTY OF SCIENCE AND TECHNOLOGY……..776

Faculty Regulations for Undergraduate and Postgraduate Degrees 779

Department of Applied Biosciences and Biotechnology..... 784

Department of Chemical Sciences..... 822

Department of Computer Science..... 922

Department of Applied Mathematics and Statistics..... 994

Department of Applied Physics and Telecommunications..... 1043

Department of Food Science and Nutrition 1124

LIBRARY DEPARTMENT 1154

POSTAL ADDRESS	MIDLANDS STATE UNIVERSITY PRIVATE BAG 9055 SENGA ROAD GWERU, ZIMBABWE
PHYSICAL ADDRESS	MIDLANDS STATE UNIVERSITY SENGA ROAD GWERU, ZIMBABWE
TELEPHONE	
Main Lines	263 (054) 2260331, 2260416, 2260417, 2260431, 2260450, 2260476, 2260496, 2260525, 2260541,2260554, 2260568, 2260575, 2260592, 2260596, 2260632
Marketing and Communications Office	263-54-260359
Student Affairs	263-54-260422
Student Health Services	263-54-260752
Batanai Complex	263-54-260641, 260210, 260251, 260283
Faculty of Business Sciences	263-54-260432
Hellenic Hall	263-54-227213

BUSINESS SCIENCES AND LAW

Graduate School of Business Leadership	263-54-223523
Faculty of Law	263-54-223205

FAX NUMBERS

Main Campus	263-54-260233
Faculty of Business Sciences	263-54-260442
Batanai Complex	263-54-260708
Graduate School of Business Leadership	263-54-223795
Faculty of Law	263-54-223795

UNIVERSITY BUSINESS HOURS

Monday – Thursday: 0800 hours – 1245 hours - 1400 hours – 1645 hours Friday: 0800 hours –

1245 hours – 1400 hours – 1615 hours

PRINCIPAL OFFICERS OF THE UNIVERSITY CHANCELLOR

His Excellency Cde Dr Emmerson D Mnangagwa
LLB (University of London), Hon.LLD (MSU, Zim), Hon.LLD (UZ), Hon.DPIR (GZU), Hon.LLD (UNZA), Hon.DDSS (ZNDU), Hon.DEST (CUT), Hon.DTech. (HIT), DPG (BUSE), Hon.DEd (NUST)

VICE-CHANCELLOR

V N Muzvidziwa — BSc (UZ), MSc (UZ), PhD (Waikato, New Zealand)

PRO-VICE-CHANCELLORS

M Maphosa — Academic Affairs and Administration
BSc (UZ), MSc (Makerere), PhD (Makerere

G C Mugumbate — Research, Innovation and Industrialisation
BSc (UZ), GCE (UZ), MSc (UZ) PhD (University of Cape Town)

G Manyatera — Infrastructure and Campus Development
LLB Hons. (UZ), LLM (Turin, Italy), MSc. CLEF (Turin, Italy), LLD (UP)

REGISTRAR

T Zishiri — GCEDE (UZ), Dip IR (UZ), Dip in training Mgt (IPMZ), Dip in Pers Mgt (IPMZ), Dip. Sal. Admin (Stallion Consultancy), BA (UZ), MBA (NUST)

BURSAR

S Ncube — ND, HND Accounting (Byo Poly), BComm Accounting (Oxford Brookes), MCom Accounting (MSU), ACCA (UK), RPA (PAAB)

LIBRARIAN

N P Chibanda — BA (ZOU), MSc (NUST), HND Lib & Info (Hre Poly), ND Lib & Info (Byo Poly)

SENIOR P R O C T O R

E Mandipa, LLB Hons (UZ), LLM (UP)

UNIVERSITY COUNCIL

SECTION 10 OF THE MIDLANDS STATE UNIVERSITY ACT [CHAPTER 25:21] AS AMENDED

1 THE CHANCELLOR AND THE VICE-CHANCELLOR (AND THE PRO-VICE CHANCELLORS WHO MAY ATTEND THE COUNCIL MEETING AT THE INVITATION OF THE VICE CHANCELLOR BUT SHALL NOT HAVE VOTING PRIVILEGES), WHO SHALL BE EX-OFFICIO MEMBERS; AND

a) Chancellor, Cde E. D. Mnangagwa

LLB (University of London), Hon. LLD (MSU, Zim.), Hon. LLD (UZ), Hon. DPIR (GZU), Hon. LLD (UNZA), Hon. DDSS (ZNDU), Hon. DEST (CUT), Hon. DTech. (HIT), DPG (BUSE), Hon. DEd. (NUST)

b) V. N. Muzvidziwa Vice Chancellor
BSC. Hons. (UZ) MSc. (UZ), PhD (Waikato, New Zealand)

The Pro-Vice Chancellors who may attend the Council meeting at the invitation of the Vice Chancellor but shall not have voting privileges

c) M. Maphosa, Pro-Vice Chancellor, Academic Affairs and Administration,
BSc. Hons. (UZ) MSc. (Makerere) PhD. (Makerere)

d) G. Mugumbate , Pro – Vice Chancellor, Research, Innovation and Industrialisation,
BSc. (UZ) GCE. (UZ) MSC. (UZ) PhD (University of Cape Town)

c) G. Manyatera, Pro-Vice Chancellor, Infrastructure and Campus Development, LLB Hons. (UZ), LLM (Turin, Italy), MSc. CLEF (Turin, Italy), LLD (UP)

2 THE PRESIDENT OF THE STUDENTS' REPRESENTATIVE COUNCIL, WHO SHALL BE AN EX-OFFICIO MEMBER

D. Simonje SRC President
Bachelor of Laws Hons. student

3 PERSONS APPOINTED BY THE MINISTER, INCLUDING A MINISTRY REPRESENTATIVE

a) M. E. Mwamuka Chairperson of Council,
Cert.in Medical Ultrasound (The Burwin Institute of Medical Ultrasound,Canada); Dip. in Adult Ed. (UZ); Further Ed. Teaching Certificate (City and Guilds of London Institute); The Society of Radiographers Diploma (Mpilo Hospital)

b) C. Chimbetete, Vice Chairperson of Council
PhD in Public Health (Swiss Society of Public Health); PhD in Global Health (University of Genève); Master in Public Health (UZ); Bachelor of Medicine and Surgery (UZ), Dip. in HIV Mgt. (College of Medicine South Africa)

c) K. C. Maregere
PhD Public Mgt & Governance (UJ), MBA (NUST), Bachelor of Adult Ed. (UZ), Dip. in Training Mgt. (IPMZ), Diploma in Adult Education, University of Zimbabwe

d) S Makunde, MBA (ZOU), Bachelor of Industrial Eng. (NUST)

e) T. S. Makamure, MBA, BSc. Hons Soc., Dip in Pers. Mgt. (IPMZ)

f) P. Mtetwa, LLBS Hon. (UZ)
g) Z. Mguni-Makwavarara, DPhil (UCT), MA in African Lang. & Lit. (UZ), B.A. Gen. (UZ), GCE (UZ)
h) P. S. Whata, MBL (UNISA), BAcc. Hons (UZ), CA (ICAZ), CA (South Africa), ACMA (ICAZ)

i) S. T. Kambarami, BSc. Civil Eng. (University of Kentucky, USA)
j) M. Mukewa, BAcc. Sci. Hons. (UNISA), CIA, CA.

k) C. Matowanyika-Nemaunga, BSc (Hons) Postgrad. Cert. in Architecture (University of East London), BA Hons. Architecture RIBA (University of Greenwich, London)

l) M. Mutema, MBA (MSU), MSc Dvt. Studies (Royal Agricultural College, Cirencester, England); CIS, B. Comm Hons in Acc. (NUST); Certified Forensic Auditor

m) W. Dzuda, BAcc. Hons (UZ), CA (Zim.), RPA

Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development Representative

n) B Murungweni, Master of Pub.Sector Mgt. (AU); BSci.Psy. (ZOU), Dip.in Training Mgt. (IPMZ), Dip. in Pers. Mgt. (IPMZ)

4 PERSONS WHO ARE MEMBERS OF THE ACADEMIC STAFF APPOINTED BY THE SENATE, OTHER THAN THE VICE-CHANCELLOR AND ANY PRO-VICE-CHANCELLOR

a) C. Hungwe, BSc Hons. (UZ), MSc. (UZ), DPhil (UNISA)

b) T. M. Mashingaidze, BA Gen. (UZ), BA Special Hons (UZ), PhD. (University of Minnesota)

5 TWO PERSONS ELECTED BY THE WORKERS' COMMITTEE OF THE UNIVERSITY

a) V. Hove, MSULA President, BSc. Bio & Biochem (UZ), MSc. Food Sci. & Nutrition (MSU)

b) MSUNTSA President - vacant

Registrar and Council Secretary

T. Zishiri
MBA (NUST), BA (UZ), Grad. CE, Dip. Ed. Foundations (UZ), Dip. IR (UZ), Dip. in Training Mgt. (IPMZ), Dip in Pers. Mgt. (IPMZ), Dip. Sal. Admin. (Stallone Consultancy)

Chapter 25:21

MIDLANDS STATE UNIVERSITY ACT

Acts 4/1999, 22/2001, 14/2002, 4/2022

ARRANGEMENT OF SECTIONS

Sections

1. Short title.
2. Interpretation.
3. Establishment of University.
4. Objects and powers of the University.
5. Membership of the University.
6. Prohibition against discrimination in membership of University.
7. Chancellor.
8. Vice-Chancellor.
9. Pro-Vice-Chancellors.
10. Council.
11. Functions of Council.
12. Disqualifications for appointment as Council member.
13. Vacation of office by Council member.
14. Chairman and vice-chairman of Council.
15. Executive Committee of Council.
16. Senate.
17. Functions of Senate.
18. Academic Board.
19. Registrar.
20. Bursar.
21. Librarian.
22. Convocation.
23. Terms and conditions of service of staff.
24. Appointment and grading of staff.
25. Promotion of staff.
26. Staff Disciplinary Committee.

27. Student Disciplinary Committee.

28. Finance Committee.

28A. Service of process.

29. Statutes.

30. Regulations.

31. Validity of decisions of Council, Senate, Convocation and boards and committees.

SCHEDULE: Statutes of the University.

ACT

To establish the Midlands State University and to provide for matters connected therewith or incidental thereto.

[amended by General Laws Amendment (No.2) Act 2002]

[Date of Commencement: 28 May, 1999.]

ENACTED by the President and the Parliament of Zimbabwe.

1 Short title

This Act may be cited as the Midlands State University Act [*Chapter 25:21*].

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

2 Interpretation

In this Act—

"academic staff" means all persons employed, whether full-time or part-time, by the University as;

(*a*) professors, lecturers of any class or persons engaged in research; or

(*b*) holders of posts declared by the Senate to be academic posts;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

"Academic Appointments Board" means the Board established in terms of subsection (1) of section *twenty four*;

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

"administrative staff" means all persons employed by the University who are categorised in terms of the Statutes as members of the administrative staff;

"Bursar", means the person holding office as Bursar of the University in terms of section *twenty*;

"chairman of department" means a person appointed in terms of the Statutes to be chairman of a teaching department or head of an institute or centre controlled by the University;

"Chairman of the Council" means the person elected to be chairman of the Council in terms of section *fourteen;*

"Chancellor" means the President in his capacity as chancellor of the University in terms of section *seven*;

"Council" means the University Council established in terms of section *ten*;

"employees" means all persons employed by the University who are categorised in terms of the Statutes as employees;

"faculty" means a faculty of the University established in terms of the Statutes;

"Faculty Board" means a faculty board established in terms of the Statutes;

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

"institute" means an institute established in terms of the Statutes;

"Librarian" means the person holding office as Librarian of the University in terms of section *twenty-one*;

"Minister" means the Minister of Higher Education or any other Minister to whom the President may, from time to time, assign the administration of this Act;

[The Minister of Higher Education, Science and Technology Development was assigned by SI 221/2018 w.e.f.19th October,2018.]

"non-academic staff" means all persons employed by the University who are not members of the academic staff;

"Pro-Vice-Chancellor" means a person holding office as Pro-Vice-Chancellor in terms of section *nine*;

"professor" means a professor of the University;

"Registrar" means the person holding office as Registrar of the University in terms of section *nineteen*;

"regulations" means regulations made by the Senate under section *thirty*;

"Senate" means the Senate established in terms of section *sixteen*;

"senior", in relation to the staff of the University, means the Registrar, the Bursar and such other members of staff as the Council may determine from time to time;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

"Statutes" means the Statutes of the University set out in the Schedule as amended from time to time or replaced in terms of section *twenty-nine*;

"Students' Union" means any association of students recognized by the Council as the Students' Union;

"technical staff" means all persons employed by the University who are categorised in terms of the Statutes as members of the technical staff;

"University" means the Midlands State University constituted in terms of this Act;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

"Vice-Chancellor" means the person holding the office of Vice-Chancellor in terms of section *eight*.

"workers" means all persons employed by the University who are categorised in terms of the Statutes as workers.

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

3 Establishment of University

(1) There is hereby constituted a university to be known as the Midlands State University.

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(2) The University shall be a body corporate with perpetual succession and shall be capable of suing and being sued in its corporate name and, subject to this Act, of performing all acts that bodies corporate may by law perform

4 Objects and powers of the University

(1) The objects of the University are the advancement of knowledge, the diffusion and extension of arts, science and learning, the preservation, dissemination and enhancement of knowledge that is relevant for the development of the people of Zimbabwe through teaching and learning, research, community service, innovation and industrialisation and, so far as is consistent with those objects, the nurturing of the intellectual, aesthetic, social and moral growth of the students at the University.

[Subsection (1) substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(2) For the achievement of its objects, the University shall, subject to this Act, have the following powers—

(*a*) to provide for research and courses of instruction, whether on a full-time or part-time basis by correspondence or extra-murally, and to take such other steps as may appear necessary and desirable for the advancement and dissemination of knowledge;

(*b*) to hold examinations and to confer degrees, including honorary degrees, diplomas, certificates and other awards, upon persons who have followed a course or courses of study approved by the Senate and additionally, or alternatively, have satisfied such other requirements as may be determined by the Senate;

(*c*) to provide courses not leading to degrees, diplomas or certificates, including training for persons wishing to enter the University;

(*d*) to provide opportunities for staff and students and such other persons as the University may approve to engage in productive activity in any field in which the University may from time to time be engaged;

(*e*) to promote research into economic, political, social, cultural, scientific and other matters generally and with particular reference to the interests of Zimbabwe;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*f*) to institute professorships, lectureships, research fellowships, staff development fellowships and other posts and offices and to make appointments thereto;

(*g*) to institute and award fellowships, scholarships, bursaries, prize medals, exhibitions, and other distinctions, awards and forms of assistance consistent with its objects;

(*h*) to erect, equip and maintain laboratories, offices, halls of residence, lecture halls, libraries, museums and other buildings and structures required for the promotion of its objects;

(*i*) to enter into such contracts and to establish such trusts and to appoint such staff as the University may require;

(*j*) to regulate and provide for the residence of officers, staff, students and employees;

(*k*) to provide and maintain recreational facilities for officers, staff, students and employees;

(*l*) to demand and receive such fees as may from time to time be prescribed by or in terms of the Statutes;

(*m*) to establish pension, superannuation or provident or other credit fund schemes for the benefit of its staff or any section thereof and to enter into arrangements with the Government or any organization or person for the operation of such schemes;

(*n*) to acquire any property, movable or immovable, and to take, accept and hold any property which may become vested in it by way of purchase, exchange, grant, donation, lease, testamentary disposition or otherwise;

(*o*) to sell, mortgage, let on hire, exchange, donate or otherwise dispose of any property held by it;

(*p*) to invest in land or securities such funds as may be vested in it for the purpose of endowment, whether for general or specific purposes, or such other funds as may not be immediately required for current expenditure;

(*q*) to borrow money for any purpose which the Council thinks fit;

(*r*) to lend money in the form of short-term loans to its staff on terms and conditions approved by the Council;

(*s*) to do all such acts and things, whether or not incidental to the powers specified in this subsection and whether inside or outside Zimbabwe, as may be requisite in order to further its objects or any of them.

5 Membership of the University

The University shall consist of—

(*a*) a Chancellor; and

(*b*) a Vice-Chancellor; and

(*c*) 1 or more Pro-Vice-Chancellors; and

(*d*) members of the Council; and

(*e*) members of the Senate; and

(*f*) members of staff; and

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*g*) registered students; and

(*h*) the Convocation.

6 Prohibition against discrimination in membership of University

(1) No test of religious or political belief, race, ethnic origin, nationality or gender shall be imposed upon or required of any person in order to entitle him to be admitted as a member of staff or student of the University or to hold any office therein or privilege thereof.

(2) Nothing in subsection (1) shall be construed as preventing the University from giving preference to citizens or residents of Zimbabwe when making appointments or promotions or when admitting students.

7 Chancellor

(1) The President of Zimbabwe shall be Chancellor of the University.

(2) The Chancellor shall be the head of the University.

(3) The Chancellor shall have the right—

(*a*) to preside over any assembly or meeting held by or under the authority of the University; and

(*b*) upon the recommendation of the Council and the Senate, to confer degrees, diplomas, certificates and other awards and distinctions of the University and to withdraw or restore such awards.

8 Vice-Chancellor

(1) The Vice-Chancellor shall be appointed by the Chancellor after consultation with the Minister and the Council and shall hold office for such period as is provided in his contract of employment.

(2) Subject to the general control of the Council, the Vice-Chancellor shall be the chief academic, administrative and disciplinary officer of the University, with general responsibility for maintaining and promoting the efficiency, effectiveness and good order of the University.

(3) Subject to subsections (4) and (5), the Vice-Chancellor may—

(*a*) suspend from duty any member of staff of the University;

(*b*) subject to section *six*, prohibit the admission of a student or any person to the University;

(*c*) prohibit any student or group of students or person or group of persons from entering or remaining on such part or parts of the University campus as he may specify;

(*d*) expel or suspend, indefinitely or for such period as he may specify, any student or group of students;

(*e*) dissolve or suspend, indefinitely or for such period as he may specify, the Students' Union or any of its committees or organs, or prohibit or suspend, indefinitely or for such period as he may specify, any activity or function of the Students' Union or any of its committees or organs;

(*f*) impose any other penalty or give any other order in respect of—

(i) a member of staff, which is recommended by the Staff Disciplinary Committee in terms of subsection (6) of section *twenty-six*;

(ii) a student, which is recommended by the Student Disciplinary Committee in terms of subsection (6) of section *twenty-seven*.

(4) The Vice-Chancellor shall not expel a student for misconduct unless the student has been found guilty of misconduct by the Student Disciplinary Committee in terms of section *twenty-seven*.

(5) Any action taken by the Vice-Chancellor in terms of subsection (3) shall be subject to ratification by the Council.

9 Pro-Vice-Chancellors

(1) 1 or more Pro-Vice-Chancellors may be appointed by the Council with the approval of the Minister and in accordance with the Statutes.

(2) A Pro-Vice-Chancellor shall assist the Vice-Chancellor in the performance of his functions and, in addition, shall have such functions as may be specified in the Statutes.

(3) The Vice-Chancellor may delegate to a Pro-Vice-Chancellor, either absolutely or subject to conditions, any of his functions in terms of this Act and may at any time amend or withdraw any such delegation:

Provided that the delegation of a function in terms of this subsection shall not prevent the Vice-Chancellor from himself exercising that function.

10 Council

[Substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(1) Subject to this Act and any general directions as to policy given by the Minister, the government and executive authority of the University shall be vested in the Council which shall consist of not less than ten (10) and not more than twenty (20) members including—

(a) the Chancellor and the Vice-Chancellor (and the Pro-Vice Chancellors who may attend the Council meeting at the invitation of the Vice Chancellor but shall not have voting privileges), who shall be *ex-officio* members; and

(b) 10 persons appointed by the Minister including a Ministry representative; and

(c) 2 persons who are members of the academic staff appointed by the Senate, other than the Vice-Chancellor and any Pro-Vice-Chancellor; and

(d) the President of the Students' Representative Council, who shall be an *ex-officio* member; and

(e) 2 persons elected by the workers' committee of the University.

(2) In appointing members to the Council, the Minister or any appointing authority under this section shall—

(a) ensure that both genders are equally represented and for the avoidance of doubt that women constitute half the membership of the Council; and

(b) ensure that there is fair regional representation;

(c) appoint members of relevant and appropriate professions.

11 Functions of Council

(1) Subject to this Act and the Statutes, the Council shall—

(*a*) appoint, with the approval of the Minister, the Pro-Vice Chancellors and the Registrar; and

(*b*) appoint the Bursar, the Librarian, the academic staff, the administrative staff and other employees of the University:

Provided that the Council may delegate its duties under this paragraph to such committee as may be prescribed in the Statutes;

(*c*) on the recommendation of the Senate, institute professorships, associate professorships and other academic offices, and abolish or hold in abeyance any such offices;

(*d*) receive and, if the Council considers it proper to do so, give effect to reports and recommendations from the Senate on those matters upon which the Senate is authorized or required by this Act or the Statutes to make reports and recommendations;

(*e*) cause to be prepared annually a statement of the income and expenditure of the University during the previous financial year, and of the assets and liabilities of the University on the last day of that previous financial year;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*f*) submit statements of income and expenditure referred to in paragraph (*e*) to audit by an auditor appointed by the Council, and shall publish such statements and the auditor's report thereon;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*g*) cause to be prepared annually estimates of income and expenditure for the following financial year;

(*h*) cause to be prepared and made available to the public a report on the activities of the University during each year;

(*i*) to prepare and refer to the Senate any proposed amendment, repeal or replacement of the Statutes.

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(2) Without limitation on any other powers conferred on the Council by this Act, the Council shall have the following powers—

(*a*) to receive recommendations from the Senate for the conferment, withdrawal or restoration of degrees, including honorary degrees, and diplomas, certificates and other awards and distinctions of the University and, if approved, to submit them to the Chancellor;

(*a*1) to amend, repeal or replace the Statutes in terms of subsection (2) of section *twenty-nine*;

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*b*) to administer the property of the University and to control its affairs and functions;

(*c*) to exercise on behalf of the University such of the powers set out in subsection (2) of section *four* as are not exercisable in terms of this Act by any other authority;

(*c*1) to delegate its functions to the Executive Committee or to any other committee or officer of the University:

Provided that the delegation or any such function shall not prevent the Council from exercising that function;

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*d*) to do such other acts as it considers to be necessary for the proper administration of the University and the achievement of its objects.

12 Disqualifications for appointment as a Council member

The Minister shall not appoint a person as a member of Council and no person shall be qualified to hold office as a member of Council who—

(*a*) is not a citizen of Zimbabwe or permanently resident in Zimbabwe; or

(*b*) has in terms of any law in force in any country—

(i) been adjudged or otherwise declared insolvent or bankrupt and has not been rehabilitated or discharged; or

(ii) made an assignment to, or arrangement or composition with, his creditors which has not been rescinded or set aside; or

(*c*) has, within the period of 5 years immediately preceding the date of his proposed appointment, been sentenced in any country to a term of imprisonment of or exceeding 6 months, whether or not any portion has been suspended, imposed without the option of a fine and has not received a free pardon.

13 Vacation of office by Council member

A member shall vacate office and his office shall become vacant—

(*a*) after giving the Minister such period of notice of his intention to resign as may be fixed in his conditions of appointment or, if no such period has been fixed, after the expiry of 1 month or of such other period of notice as he and the Minister may agree; or

(*b*) on the date he begins to serve a sentence of imprisonment, the term which is not less than 6 months, whether or not any portion has been suspended, imposed without the option of a fine in any country; or

(*c*) if he becomes disqualified in terms of paragraph (*a*) or (*b*) of section *twelve* to hold office as member; or

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*d*) if he is absent without the permission of the Council from three consecutive meetings of the Council.

14 Chairman and vice-chairman of Council

(1) The Council shall elect a chairman and a vice-chairman from amongst its members to hold office for such period and subject to such terms and conditions as are prescribed in the Statutes.

(2) At all meetings of the Council at which the Chancellor is not present the chairman of the Council or, in his absence, the vice-chairman shall preside.

(3) If at any meeting of the Council the Chancellor and the chairman and vice-chairman of the Council are all absent, the members of the Council who are present shall elect a person from amongst their number to preside at the meeting.

15 Executive Committee of Council

(1) There shall be a principal committee of the Council to be known as the Executive Committee.

(2) The Executive Committee shall consist of—

(*a*) the chairman and vice-chairman of the Council; and

(*b*) the Vice-Chancellor and every Pro-Vice-Chancellor; and

(*c*) 10 members of the Council, of whom—

(i) 4 shall be appointed by the Minister; and

(ii) 3 shall be appointed by the Senate; and

(iii) 3 shall be appointed by the Council.

(3) The chairman and vice-chairman of the Council shall be the chairman and vice-chairman, respectively, of the Executive Committee.

(4) The Executive Committee shall exercise such of the functions of the Council as the Council may delegate to it.

(5) Any delegation of functions by the Council in terms of subsection (4) may be made absolutely or subject to conditions and may be amended or withdrawn at any time.

(6) A delegation of any function by the Council in terms of subsection (4) shall not prevent the Council from itself exercising that function.

16 Senate

(1) Subject to this Act, the academic authority of the University shall be vested in the Senate, which shall consist of—

(*a*) the Vice-Chancellor, the Pro-Vice-Chancellors, deans, professors, deans, the chairmen of departments and the Librarian, who shall be *ex officio* members; and

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*b*) 1 member of the permanent academic staff from each faculty, who shall be elected annually by such staff; and

(*c*) 6 students elected annually by the Students' Union:

Provided that such students shall not be entitled to attend deliberations of the Senate on matters which are considered by the chairman of the Senate to be confidential.

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(2) If the permanent academic staff or the Students' Union fails or declines to elect a person in terms of paragraph (*b*) or (*c*) of subsection (1), the Council may appoint a suitably qualified person to fill the vacancy.

17 Functions of Senate

The Senate shall have the following functions—

(*a*) to promote the advancement of knowledge through research;

(*b*) to formulate and carry out the academic policy of the University;

(*c*) to regulate the programmes, subjects and courses of study and the examinations held by the University;

(*d*) to regulate the admission of students to the University and to the courses of instruction held by the University;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*e*) to recommend to the Chancellor, through the Council, the conferment of degrees, including honorary degrees, diplomas, certificates and other awards and distinctions of the University and the withdrawal and restoration of such awards;

(*f*) to fix, subject to consultation with any sponsors and subject to the approval of the Council, the times, modes and conditions of competitions for fellowships, scholarships and prizes;

(*g*) to appoint examiners for examinations conducted by the University;

(*h*) to cause to be prepared estimates of expenditure required to carry out the academic work of the University and to submit them to the Council;

(*i*) subject to the approval and direction of the Council, to formulate, modify and revise the organization of faculties, departments, institutions, centres and units;

(*j*) to recommend to the Council the institution, abolition or holding in abeyance of professorial chairs and other academic offices;

(*k*) without derogation from the powers of the Council, to propose changes to the Statutes;

(*l*) to make any regulations it is authorized to make by or in terms of this Act;

(*m*) to appoint committees, which may include persons who are not members of the Senate, to exercise any of the functions of the Senate, other than the power to make regulations;

(*n*) to make periodic reports on its activities to the Council and to report on any matter referred to it by the Council;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*o*) to do such other acts as the Council may authorize or direct it to do.

18 Academic Board

(1) There shall be a principal committee of the Senate to be known as the Academic Board.

(2) The Academic Board shall consist of—

(*a*) the Vice-Chancellor, who shall be chairman; and

(*b*) every Pro-Vice-Chancellor; and

(*c*) 4 full professors of the University, elected annually by the Senate; and

(*d*) all the deans of faculties; and

(*e*) the Librarian; and

(*f*) 2 members of the Senate, elected annually by the persons referred to in paragraph (*a*) and (*b*) of subsection (1) of section *sixteen*.

(3) The Academic Board shall exercise such of the functions of the Senate as the Senate, with the approval of the Council, may delegate to it.

(4) Any delegation of functions by the Senate in terms of subsection (3) may be made absolutely or subject to conditions and may be amended or withdrawn at any time.

(5) A delegation of any function by the Senate in terms of subsection (3) shall not prevent the Senate from itself exercising that function.

19 Registrar

(1) There shall be a Registrar of the University who shall be appointed by the Council with the approval of the Minister in the manner provided in the Statutes.

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(2) Subject to the directions of the Council, the Registrar shall be responsible for the general administration of the University, and shall perform such other functions as may be specified in the Statutes.

(3) The Registrar, or a member of his staff whom he may authorize to act for him, shall be secretary of the Council and the Senate and their principal committees.

20 Bursar

(1) The Bursar shall be appointed by the Council with the approval of the Minister.

[Subsection (1) substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(2) Subject to the directions of the Council, the Bursar shall act as the accountant of the University and shall be responsible for the safeguarding of its funds and, in accordance with the general directions of the Vice-Chancellor, for authorizing its investments and expenditure.

(3) The Bursar shall perform such additional functions as may be specified in the Statutes.

21 Librarian

(1) The Librarian shall be appointed by the Council with the approval of the Minister.

[Subsection (1) substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(2) Subject to the directions of the Senate and the Vice-Chancellor, the Librarian shall be responsible for the administration and safeguarding of the libraries of the University.

22 Convocation

(1) The Convocation of the University shall consist of all persons whose names appear on the Convocation Roll maintained by the Registrar.

(2) The Vice-Chancellor and all Pro-Vice-Chancellors, deans, professors, lecturers, chairmen of departments, the Registrar, the Librarian and the Bursar shall be *ex officio* members of the Convocation.

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(3) All graduates of the University who signify in writing addressed to the Registrar that they desire to be members of the Convocation and who inform the Registrar of their address shall be entitled to have their names entered on the Convocation Roll.

(4) Subject to section *seven*, the Vice-Chancellor or such other person as he may appoint shall be chairman of meetings of the Convocation.

(5) Meetings of the Convocation shall be held at such times and places as the Council may direct.

(6) The Convocation may deal with any matter relating to the University which the Council, in accordance with the Statutes, may refer to the Convocation.

23 Terms and conditions of service of staff

The terms and conditions of service for each category of staff employed by the University, including the Vice-Chancellor and the Pro-Vice-Chancellors, shall be determined by the Council in terms of this Act and, subject to the Labour Act [*Chapter 28:01*], such terms and conditions shall provide that—

(*a*) any person so employed shall be entitled to resign from his employment on giving such notice in writing to the Registrar as may be fixed in such terms and conditions;

(*b*) any person so employed shall, subject to any exception that may be provided for in special cases, retire from his office at such time or in such circumstances as may be fixed in such terms and conditions;

(*c*) no summary termination of the employment of any person so employed shall take place except for good cause and, in the event of any such termination, the person concerned shall have a right of appeal to the Council, whose decision shall be final.

24 Appointment and grading of staff

(1) Every appointment to the academic staff shall be made by an Academic Appointments Board consisting of—

(*a*) the Vice-Chancellor or his nominee, who shall be chairman; and

(*b*) 2 members of the Council who are not members of staff of the University; and

(*c*) the chairman of the department to which the appointment is made; and

(*d*) 1 other member, approved by the Vice-Chancellor, of the department to which the appointment is made; and

(*e*) the dean of the faculty to which the appointment is made; and

(*f*) the chairman of a department approved by the Council as being related to the department to which the appointment is made.

(2) Subject to this Act and the Statutes, the Council shall appoint boards of selection for the purpose of appointing members of staff other than the Vice-Chancellor, Pro-Vice-Chancellors, the Registrar, the Bursar, the Librarian and academic staff.

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(3) The Council shall appoint a Grading Committee for the purpose of determining all matters relating to the grades and points of entry upon salary scales by members of staff of the University.

25 Promotion of staff

(1) Every promotion of a person to a post or grade within the academic staff shall be effected by an Academic Staff Promotions Committee consisting of—

(*a*) the Vice-Chancellor or his nominee, who shall be chairman; and

(*b*) every Pro-Vice-Chancellor; and

(*c*) 4 members of the Council who are not members of staff of the University, appointed by the Council; and

(*d*) all deans of faculties; and

(*e*) at least 3 full professors of the University appointed by the Senate once every 3 years.

(2) Every promotion of a person to a post or grade within the non-academic staff shall be effected by a Non-Academic Staff Promotions Committee appointed by the Council and consisting of—

(*a*) a Pro-Vice-Chancellor, who shall be chairman; and

(*b*) the Registrar; and

(*c*) the Bursar; and

(*d*) the Librarian; and

(*e*) 2 members of the Council who are not members of staff of the University; and

(*f*) 1 representative of each category of non-academic staff prescribed in the Statutes; and

(*g*) 1 representative of the faculty where the person to be promoted works.

26 Staff Disciplinary Committee

[Substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(1) There shall be a staff disciplinary committee which shall consist of the following members appointed by the Vice-Chancellor—

(a) a Pro-Vice-Chancellor, who shall be the chairperson; and

(b) a senior member of the academic or administrative staff; and

(c) a member of the Council; and

(d) a member of the academic or administrative staff of similar status to the person charged;

(2) 2 members of the staff disciplinary committee shall form a quorum.

(3) All matters to be decided at any meeting of the staff disciplinary committee shall be decided by a simple majority and, in the event of an equality of votes, the chairman or person presiding shall have a casting vote in addition to his or her deliberative vote.

(4) The functions of the staff disciplinary committee shall be to investigate any breach of a statute, regulation or ordinance or other misconduct on the part of any member of the academic or administrative and general staff and, subject to subsection (6), to recommend to the Vice-Chancellor the punishment to be imposed on or order to be made in respect of the member if it finds him or her guilty of such misconduct.

(5) A person charged with misconduct referred to in subsection (4) shall have a right of audience before the staff disciplinary committee.

(6) Where the staff disciplinary committee has found a person guilty of misconduct referred to in subsection (4), the Committee shall recommend to the Vice-Chancellor any one or more of the following—

(a) that the person's employment be terminated;

(b) that the person shall pay to the University a fine not exceeding level 5;

(c) that the person be demoted;

(d) that the person be censured or reprimanded;

(e) such other penalty or order as may be provided for by or in terms of the Statutes.

27 Student Disciplinary Committee

(1) There shall be a Student Disciplinary Committee which shall consist of the following members appointed by the Vice-Chancellor¾

(*a*) a Senior Proctor, who shall be chairman; and

(*b*) 4 members of the academic staff one of whom shall belong to the same department or faculty to which the student charged belongs; and

(*c*) 2 students nominated by the Students' Union, one of whom shall belong to the same department or faculty to which the student charged belongs; and

(d) a person registered in terms of the Legal Practitioners Act [*Chapter 27:07*].

[Para (d) substituted by section 7 of Act 4/2022 w.e.f. 2nd September, 2022.]

(2) 5 members of the Student Disciplinary Committee shall form a quorum.

(3) All matters to be decided at any meeting of the Student Disciplinary Committee shall be decided by a simple majority and, in the event of an equality of votes, the chairman or person presiding shall have a casting vote in addition to his deliberative vote.

(4) The functions of the Student Disciplinary Committee shall be to investigate any breach of a Statute, regulation or ordinance or other misconduct on the part of any student and, subject to subsection (6), to recommend to the Vice-Chancellor the punishment to be imposed on the student if it finds him guilty of such misconduct.

(5) A student charged with misconduct referred to in subsection (4) shall have a right of audience and to be legally represented before the Student Disciplinary Committee.

(6) Where the Student Disciplinary Committee has found a student guilty of misconduct referred to in subsection (4), the Committee shall recommend to the Vice-Chancellor the imposition upon the student of any one or more of the following punishments.

(*a*) expulsion or suspension from the University;

(*b*) the withdrawal of any academic or University privilege, benefit, right or facility other than to follow courses of instruction and present himself for examination;

(*c*) the imposition of a fine not exceeding level 3, which fine may be deducted from any allowances payable to the student and shall be paid to the University;

[Amended by Act 22 of 2001;
substituted by Part VI, section 7 of Act 4/2022 w.e.f. 2nd September, 2022.]

(d) a censure or reprimand;

(*e*) such other penalty as may be provided for by or in terms of the Statutes.

28 Finance Committee

(1) Subject to this Act, there shall be a Finance Committee of the Council consisting of—

(*a*) …

(*b*) …

[Paras (a) and (b) repealed by Act 4/2022 w.e.f. 2nd September, 2022.]

(*c*) every Pro-Vice-Chancellor; and

(*d*) 3 members of the Council who are not members of the University staff and one of them shall be the Chairperson of the Finance Committee;

[Para (d) amended by Act 4/2022 w.e.f. 2nd September, 2022.]

and

(*e*) 2 persons appointed by the Senate from among its members; and

(*f*) the Bursar, who shall be the secretary of the Finance Committee; and

(*g*) the Registrar; and

(*h*) a representative of the deans of faculties appointed by the Council.

(2) The functions of the Finance Committee shall be as provided in the Statutes.

28A Service of process

Any writ or other process, notice or document may be served on the University, Council or Senate by serving the same on the Registrar.

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

29 Statutes

(1) Subject to this Act, the University shall be administered in accordance with the Statutes.

(2) With the approval of the Minister the Council may, by statutory instrument, amend, repeal or replace the Statutes set out in the Schedule in order to prescribe all matters which, in terms of this Act, are required or permitted to be prescribed in Statutes or which, in the opinion of the Council, are necessary or convenient to be so prescribed for carrying out or giving effect to this Act or for the proper administration of the University.

(3) Subject to the Act, Statutes made in terms of subsection (2) may provide for—

(*a*) the appointment, conditions of service and functions of the Vice-Chancellor, the Pro-Vice-Chancellors, the Registrar, the Bursar, the Librarian and all members of staff and the categorisation of such members of staff;

(b) ….

[repealed by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*c*) the election or appointment of persons to the Council and the Senate and to committees of the Council and the Senate;

(*d*) the terms of office of members of the Council and its committees, other than *ex officio* members thereof;

(*e*) the terms of office and conditions of service of the chairman and vice-chairman of the Council;

(*f*) the convening of meetings of the Council, the Senate and the Convocation and of committees of the Council and the Senate, the quorum at such meetings and the procedure to be adopted thereat;

(*g*) the terms of office of members of the Academic Appointments Board and the Academic Staff Promotions Committee, boards of selection, the Grading Committee, the Non-Academic Staff Promotions Committee and the Finance Committee, the quorum at such meetings and procedure to be adopted thereat;

(*h*) the functions of the Academic Appointments Board, boards of selection, the Grading Committee, the Academic Staff Promotions Committee, the Non-Academic Staff Promotions Committee, the Finance Committee and the Convocation;

[amended by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

(*i*) the persons who may enter into contracts and sign documents on behalf of the University, and the procedure to be followed in relation to transactions entered into by or on behalf of the University;

(*j*) the establishment and organization of faculties, departments, centres and institutes;

(*k*) the functions of deans of faculties, chairmen of departments and heads of centres and institutes;

(*l*) the holding of congregations of the University for the purpose of conferring degrees, diplomas, certificates and other honours and awards;

(*m*) the seal of the University and its use and custody;

(*n*) the academic year of the University;

(*o*) the discipline of members of staff and students and the procedures to be adopted in respect thereof;

(*p*) fees and charges to be paid for anything done by any person in terms of this Act;

(*q*) empowering any person specified in the Statutes to make ordinances or rules in respect of any matter referred to in paragraphs (*a*) to (*p*).

30 Regulations

(1) Subject to this Act and the Statutes and with the approval of the Council, the Senate may make regulations prescribing any matter which, in the opinion of the Senate, is appropriate to be prescribed for the better carrying out of the Senate's functions.

(2) Regulations may provide for—

(*a*) teaching within the University, whether generally or in relation to specific subjects;

(*b*) progress reports to be submitted by members of the academic staff;

(*c*) schemes of study and research and the conditions subject to which such schemes may be embarked upon or continued;

(*d*) the use of libraries of the University;

(*e*) the conduct and supervision of examinations;

(*f*) the award of degrees, diplomas, certificates and other academic honours and awards, other than honorary degrees, honours or awards.

(3) The Senate may at any time amend or repeal any regulations.

(4) Regulations need not be published in the *Gazette*, but the Senate shall publish them in such manner as the Senate considers will best make them known to the persons to whom they apply.

31 Validity of decisions of Council, Senate, Convocation and boards and committees

No decision or act of the Council, the Senate, the Convocation or any board or committee established by or in terms of this Act shall be invalid solely on the ground that—

(*a*) the Council, Senate, Convocation, board or committee, as the case may be, consisted of fewer than the number of members for which provision is made by or in terms of this Act; or

(*b*) a disqualified person acted as a member of the Council, Senate, Convocation, board or committee, as the case may be;

if the duly qualified members who took the decision or did or authorized the act constituted a quorum of the membership of the Council, Senate, Convocation, board or committee, as the case may be.

SCHEDULE

(SECTIONS 2 AND 29)

STATUTES OF THE UNIVERSITY

ARRANGEMENT OF STATUTES

Statute

1. Appointment of Pro-Vice-Chancellors.

2. Length of appointment of members of Council.

3. Casual vacancies in Council.

4. Meetings and quorum of Council.

5. Resolutions by circulation among members of Council.

5A. Chairman and Vice-Chairman of Council.

6. Meetings and quorum of Senate.

7. Convocation.

8. Faculties.

9. Deans.

10. Functions of deans.

11. Evaluation of performance of dean.

12. Termination of appointment of dean.

13. Composition of faculty boards.

14. Functions of faculty boards.

15. Meetings of faculty boards.

16. Teaching departments.

17. Institutes and centres.

18. Appointment of Registrar, Bursar and Librarian.

19. Procedure generally.

20. Financial procedures.

21. Auditors.

22. University seal.

23. Ordinances.

24. Arrangements with other universities, affiliated bodies, etc.

Appointment of Pro-Vice-Chancellors

1.

(1) For the purpose of considering appointments to the office of Pro-Vice-Chancellor, there shall be a joint committee of the Council and the Senate consisting of—

(*a*) the chairman of the Council, who shall be the chairman of the joint committee; and

(*b*) the vice-chairman of the Council; and

(*c*) the Vice-Chancellor; and

(*d*) 2 persons appointed by the Council from among its members who are not members of the Senate; and

(*e*) 3 persons appointed by the Senate from among its members.

(2) The joint committee constituted by this Statute shall consider each applicant for appointment to a vacancy in the office of Pro-Vice-Chancellor and shall make recommendations thereon to the Council, and the Council, after considering the joint committee's recommendations, shall make the appointment concerned with the approval of the Minister.

(3) A Pro-Vice-Chancellor shall hold office for 3 years from the date of his appointment and, subject to subsection (2) and the approval of the Minister, shall be eligible for re-appointment.

Length of appointment of members of Council

2.

(1) Members of the Council, other than *ex-officio* members, shall hold office for 4years, and shall be eligible for re-appointment or re-election for one further term, as the case may be.

[Substituted by Act 4/2022 w.e.f. 2nd September, 2022.]

(2) Any member of the Council, other than an *ex officio* member, may resign his membership at any time by notice in writing addressed to the Minister.

[Amended by Act 4/2022 w.e.f. 2nd September, 2022.]

Casual vacancies in Council

3.

Any casual vacancy occurring among the appointed or elected members of the Council shall be filled as soon as possible by the person, persons or body which appointed or elected the member whose place has become vacant, and the person so appointed or elected shall hold office for the remainder of the period for which the person whose place he fills was appointed or elected.

Meetings and quorum of Council

4.

(1) The Council shall meet at least 3 times a year.

(2) The quorum of the Council shall be one-half of the members holding office at the time of the meeting.

Resolutions by circulation among members of Council

5.

A resolution proposed by the Executive Committee of the Council, other than for the purpose of making a Statute, which is on its authority circulated by the Registrar to all members of the Council and which receives the written agreement of not less than two-thirds of such members shall, upon receipt of such agreement by the Registrar, have the same force and effect as a resolution passed at a meeting of the Council.

Chairman and Vice-Chairman of Council

5A.

(1) The Council shall elect from among its members a Chairman and a Vice-Chairman, each to hold office for a period not exceeding three years, and each of whom shall be eligible for re-election.

(2) If both the Chairman and the Vice-Chairman are absent from a meeting of the Council the members present shall appoint from among themselves a chairman for the purpose of that meeting.

(3) If both the Chairman and the Vice-Chairman are at the same time unable to carry out their duties the Council may elect one of its members to perform such duties during such time.

[inserted by the General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

Meetings and quorum of Senate

6.

(1) The Senate shall meet at least three times a year.

(2) The quorum of the Senate shall be one-half of the members holding office at the time of the meeting.

(3) The Vice-Chancellor or, in his absence, a Pro-Vice-Chancellor, shall be chairman of the Senate.

Convocation

7.

There shall be no quorum for meetings of the Convocation, the proceedings of which shall be regulated by the chairman subject to any general or special direction of the Council.

Faculties

8.

The University shall include such faculties as may from time to time be established by the Council.

Deans

9.

(1) There shall be a dean of each faculty, who shall be appointed by a selection board appointed by the Council and chaired by the Vice-Chancellor or a Pro-Vice-Chancellor.

(2) A dean shall hold office for 3 years and, on the expiry of his term of office, shall be eligible for re-appointment, subject to statute 12.

Functions of deans

10.

(1) A dean shall be responsible to the Vice-Chancellor for—

(*a*) the character and quality of his faculty's academic programmes and teaching; and

(*b*) the proper direction, control and management of the staff, property, equipment and finances of his faculty; and

(*c*) contributing to the evolution and maintenance of an environment conducive to learning at the University.

(2) Without limiting subsection (1), a dean's functions include—

(*a*) the planning and control of his faculty's budget; and

(*b*) implementing the policies and objectives of the University and his faculty; and

(*c*) developing and maintaining relations with alumni of his faculty and with other organizations and persons, with a view to enriching the academic programmes offered by his faculty; and

(*d*) soliciting funds and other assistance to support programmes and projects of his faculty; and

(*e*) co-operating with other faculties, agencies and entities to further the programmes and projects of his faculty; and

(*f*) performing such other functions as may be prescribed by Ordinance.

Evaluation of performance of dean

11.

The performance of every dean shall be evaluated annually by a committee appointed by the Vice-Chancellor and consisting of senior academic staff and administrative staff.

Termination of appointment of dean

12.

(1) A dean may resign his office as such by giving the Vice-Chancellor one month's written notice or such longer or shorter notice as he and the Vice-Chancellor may agree.

(2) Subject to the approval of the Council, the Vice-Chancellor may suspend or terminate the appointment of a person as dean—

(*a*) on the recommendation of a committee referred to in section 11; or

(*b*) for other good cause.

(3) A Dean whose appointment has been suspended or terminated in terms of subsection (2) may appeal to the Council within fourteen days of being notified of the suspension or termination, and on any such appeal the Council may confirm, vary or rescind the suspension or termination, as the case may be, or give such other direction in the matter as it thinks appropriate.

Composition of faculty boards

13.

For each faculty there shall be a faculty board which shall consist of—

(*a*) all the full-time academic staff of the faculty; and

(*b*) such persons as may be assigned to the faculty board by the Senate; and

(*c*) two students elected annually by the students in the faculty.

Functions of faculty boards

14.

A faculty board shall have the following functions—

(*a*) to regulate, subject to the approval of the Senate, the teaching and study of the subjects assigned to the faculty;

(*b*) to report to the Senate on any matter specifically relating to the work of the faculty;

(*c*) to deal with any matter referred or delegated to it by the Senate;

(*d*) to appoint committees, which may include a minority of persons who are not members of the faculty board, to carry out any of the duties or exercise any of the powers of the board.

Meetings of faculty boards

15.

(1) The dean of each faculty shall call regular meetings of the faculty board at which matters relating to the policy of the faculty and appointments shall be discussed and recommendations adopted for submission to the appropriate authorities on appointments.

(2) The Vice-Chancellor and every Pro-Vice-Chancellor shall be entitled to attend any meeting of a faculty board or any committee thereof.

Teaching Departments

16.

(1) The teaching departments and their allocation to faculties shall be prescribed by ordinance.

(2) A chairman of department shall be appointed by the Vice-Chancellor, on behalf of the Council, from among, the full-time members of the academic staff of the department, after consultation with each member of the academic staff in the department and the dean of the faculty to which the department is allocated.

(3) A chairman of department shall hold office for a period of 3 years, or such other period as may be determined by ordinance, and shall be eligible for re-appointment.

(4) A department may be allocated to two or more faculties.

Institutes and centres

17.

The Council, after consultation with the Senate, may establish institutes or centres of learning within or outside the University and shall appoint the head of any such institute or centre and give directions as to its studies and research and administration.

Appointment of Registrar, Bursar and Librarian

18.

(1) For the purpose of considering appointments to the offices of Registrar, Bursar and Librarian, there shall be a joint committee of the Council and the Senate consisting of—

(*a*) the chairman of the Council, who shall be the chairman of the joint committee; and

(*b*) the vice-chairman of the Council; and

(*c*) the Vice-Chancellor; and

(*d*) every Pro-Vice-Chancellor; and

(*e*) two persons appointed by the Council from among its members who are not members of the Senate; and

(*f*) three persons appointed by the Senate from among its members.

(2) The joint committee constituted by this Statute shall consider each applicant to a vacancy in the office of Registrar, Bursar or Librarian and shall make recommendations thereon to the Council.

Procedure generally

19.

(1) Except as otherwise specifically provided by the Act or, these Statutes, in the absence of the chairman or vice-chairman at a meeting of a board or committee the members present shall elect from those present a chairman to preside over that meeting.

(2) Except as otherwise specifically provided by the Act or these Statutes, the quorum at any meeting of a board or committee shall be as fixed by the person or authority that appointed the board or committee.

(3) Except as otherwise specifically provided by the Act or these Statutes, each board or committee shall determine and may make rules for the time, place and procedure of its meetings.

(4) The minutes of a meeting of a board or committee shall be laid on the table at the next following meeting of the body that appointed it.

(5) At a meeting of board or committee, in the event of an equality of votes on any matter, the person presiding shall have a casting vote in addition to his deliberative vote.

(6) Subsections (3), (4) and (5) shall apply, *mutatis mutandis*, to the Council and the Senate, save that minutes of the Council shall be sent to the Chancellor and the Minister and a report of each meeting of the Council shall be laid on the table at a meeting of the Council.

Financial procedures

20.

(1) The Council shall fix the financial year of the University.

(2) The Finance Committee shall submit to the Council, before the beginning of the financial year, draft estimates of income and expenditure, and such estimates, amended as the Council thinks fit, shall be approved by the Council before the beginning of the financial year.

(3) The Council may revise the estimates during the course of the financial year and give directions for the manner in which amendments of expenditure estimates may be made, which directions may make provision for delegating the powers of revision so long as such delegation does not extend to altering the total estimated expenditure.

(4) As soon as practicable after the end of the financial year, a balance sheet and income and expenditure account with supporting schedules shall be submitted to the auditors.

(5) The audited accounts, with any comments thereon made by the auditors, shall be submitted to the Council.

Auditors

21.

(1) Subject to subsections (2) and (3), the Council shall, before the beginning of each financial year, appoint as auditors persons who are registered as public auditors in terms of the Public Accountants and Auditors Act [*Chapter 27:12*].

(2) No person shall be appointed as an auditor in terms of subsection (1) if he, or any of his partners or employees, holds any other office in the University.

(3) If no appointment of new auditors is made before the beginning of any financial year, the auditors in office shall continue in office.

(4) An auditor appointed in terms of subsection (1) shall be entitled at all reasonable times to require any officer, employee or agent of the University—

(*a*) to produce all accounts and other records relating to the financial affairs of the University as may be in the custody of such officer, employee or agent; and

(*b*) to provide such information or explanation as, in the opinion of the auditor, is necessary for the purposes of the audit.

University seal

22.

(1) There shall be a seal of the University, of such design as may be approved by the Council.

(2) The seal of the University shall be kept in the custody of the Registrar and, subject to the directions of the Council, shall be affixed to—

(*a*) certificates, degrees and diplomas conferred by the University; and

(*b*) any document attested by the signature of the Vice-Chancellor and the Registrar.

Ordinances

23.

(1) The Council may, with the approval of the Minister, make ordinances providing for any matter referred to in paragraphs (*a*) to (*p*) of subsection (3) of section 29 of the Act.

(2) The Registrar shall publish any ordinance made in terms of subsection (1) in such manner as the Council may direct, being a manner which the Council considers will best make the ordinance known to the persons to whom it applies.

Arrangements with other universities affiliated bodies, etc

24.

(1) The Council may make arrangements with any other university whereby students of the University may be registered as students of such other university and so enabled to study for, enter the examinations of and be afforded the degrees of such other university.

(2) The Council may affiliate to the University any other institution or any branch or department of any other institution, and may¾

(*a*) recognise selected members of the staff of any affiliated institution, branch or department as teachers of the University; and

(*b*) admit the members of any affiliated institution, branch or department to any of the privileges of the University; and

(*c*) accept attendance at courses of study in any affiliated institution, branch or department in place of attendance at courses of study at the University;

on such terms and conditions and subject to such rules as the Council may determine from time to time.

[amended by General Laws Amendment (No.2) Act 2002 promulgated on the 24th January, 2003 - with retrospective effect, in terms of clause 47 - from the 4th February, 2002

STAFF QUALIFICATIONS

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Executive Dean
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L Mawuru, BA Hons (GZU), MA (GZU)

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N Mupaso, BSc (UZ), BSc Special Honours (LSU), MSc (UZ), PDTE (MSU)
S Matindike, BCom (MSU), BSc Special Honours (LSU), MCom (MSU), PhD (MSU),
N Mhaka, BCom (MSU), MCom (MSU)
M Wiri, Diploma in Horticulture (Harare Polytechnic), BSc (BUSE), MSc in Food security (BUSE), MSc in Agricultural and Applied Economics (UZ)
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T H Madzaramba, MSc (Lundi, Sweden, BTech (CUT)
A Gumbo, MSc (UZ), MSc (BUSE) BSc (MSU)

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C Munhande, BA Hons (UZ), MA (UZ),

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G T Ncube, A (UZ), MA (UZ), PhD

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P Mukwara, CA (UZ), BA (UZ), MPhil (UZ)
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V Mkhwananzi BA (MSU)

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DEPARTMENT OF ARCHITECTURE

Lecturers

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C Mutsambiwa (Dipl.-Eng.) Arch (Bauhaus University, Germany); MACZ. MIAZ. MNCAQS
T J Mutasa BArch Hons (NUST), Master of Architecture (NUST). MIAZ

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B Mavhima BSc (Hons) (UZ), MSc (UZ), MSc (UZ)
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FACULTY OF BUSINESS SCIENCES
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L Makusha, MCom (MSU), BCom (MSU), PGCE (UNISA)
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B Mweku, BCom (MSU), PDTE (MSU), MCom (MSU)
TRE Manhiwa, BCom (MSU), PDTE (MSU), MCom (MSU)
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B Mabenge, BSc (CUT), MPhil (CUT)
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NPVA Washaya, BBA (Solusi), MSc (Ehwa)

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R Mandishekwa, BCom (MSU), MCom (MSU), DPhil (MSU)
P Makoni, BCom (MSU), MCom (MSU)
A Mautsa, BCom (MSU), MCom (MSU)
A R Chiweshe, BSc (MSU), MCom MSU
N V Zivanayi, BSc (MSU), MCom (MSU)
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DEPARTMENT OF INFORMATION AND MARKETING SCIENCES
Chairperson of Department
CT Chipfumbu, BSc (MSU) PGDE (MSU), MCom (MSU) PhD (NMMU)

Lecturers
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B Munyoro, ND (HEXCO), PDTE (MSU), BTech (HIT), MSc (NUST)
D R Hungwe, PDTE (MSU), BCom (MSU), MSc (NUST)
H Hogo, PDTE (MSU), BCom (NUST), MSc (NUST)
S Marambi, PDTE (MSU), BSc (MSU), MSc (MSU)
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M Zhou, BSc (MSU), PDTE (MSU), MSc (MSU), PhD (UNISA)
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M Muduva, Dip (ALI), CHTE (HIT), BTech (HIT), MSc (CUT), MLC (BUSE)
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T Kwarayi, BCom (MSU), BSc (LSU), MCom (MSU)
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DEPARTMENT OF MANAGEMENT SCIENCES
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B G Mashiri, Dip. Ed (UZ), BSc (CUT), MSc (CUT), DPhil (CUT)
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M W Muchemwa), HND P&S (HEXCO), HND Mkt (HEXCO), BSc (MSU), MSc (CUT
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DEPARTMENT OF TOURISM, HOSPITALITY AND LEISURE SCIENCES
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T Makwara, BCom (MSU), MCom (MSU), PDTE (MSU)
R Magwaza, BCom (MSU), MCom (MSU) PDTE (MSU)
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N Y Manungo, BCom (MSU), MCom (MSU)
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S Hungwe, LLM (Wits), LLB (Wits)
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I Hore, LLM (MSU), LLB (UZ)
C Mashoko, LLM (MSU) LLB (UZ), BL (Hons) (UZ)
N Ndlovu, LLM (MSU), LLB (UZ), PGDME (Stellenbosch)
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DEPARTMENT OF PRIVATE LAW

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A Takawira, LLM (Stellenbosch), LLB (UZ)

E Mandipa, LLM (UP), LLB (UZ)
T G Kasuso, LLD (UNISA), LLM (UNISA), LLB (UZ)
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S Muchemwa, BEd (Solusi University), MEd (Solusi University), PhD (University of Eastern Africa, Baraton)
W Magwa, PGD (UZ), Grad CE (UZ), BA (UZ), Med (UZ), PhD (UNISA)

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E Hungwe, Dip-Ed (UZ); BED (MSU); MED (UZ), PhD (Free State University, SA)
A Rwodzi, PhD (UKZN), B.A (UNISA), B.A Hons (UZ), M.A (UZ), Cert.Ed. (UZ)
S Njini, Dip Ed, (UZ), BSc (UZ), M.Sc. (UZ), B. Com Acc (MSU); M.Ed. (MSU)
E Munanga, B.Sc (UZ), Grad.CE (UZ), FETD (Hexco), M.Ed. (ZOU)
S Shumba, Grad CE (UZ), B.A. General (UZ), B.A. Special Hons (UZ), MEd (ZOU), MSc (UZ), PhD (UNISA)
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E Manyumwa, CE (UZ), BEd (UZ), MEd (UZ), PhD (NMMU, SA)
L Makonye, Dip Ed (UZ); BEd (MSU); Med (GZU); PhD (UKZN, SA.).
C Muchandiona, Dip Ed (UZ), BECE (MSU), MECDP (MSU)
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J Maera, MECDP (MSU), BEd ECE (MSU), Dip Ed (UZ)
E R Machiridza, Dip in ECD (MMTC), BEd Hons ECD (GZU), Masters in ECD (MSU).
S Mnkandla, Dip Ed (UZ), Bed (MSU), MEd (MSU)
C Zishiri, Dip Ed (UZ), Dip in DSS (UZ), B.Ed (GZU), M.Ed (GZU), Msc (BUSE), PhD (WUA
N Manzunzu, Dip Ed (UZ); BEd- (MSU), MEd (GZU), PhD- (Witwatersrand, S.A)
L Bema, Dip Ed (UZ), BECE(MSU), MECDP(MSU)
O Musweweshiri, Dip Ed (UZ), BECE(MSU), MECDP(MSU)
S Silempa, Dip Ed (UZ), Dip in Special Edu (UZ), Bed Management (Technikon, SA), Masters in Special Needs Edu (GZU)

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W Samkange, CE (UZ); Bed (UZ); Med (ZOU); PhD (Fort Hare, S.A)

Senior lecturers
L Museva, Dip in Adult Education (UZ), Bachelor of Adult Education (UZ), PGDTE (MSU), MAE (UZ) PhD (Durban University of Technology, SA)
W Dzimiri, CE (UZ), B.Ed (UZ), Med (UZ), PhD (Georgia State Univ., USA) Grad CE (Georgia State Univ. USA)

Lecturers
W Mazani, Dip Ed (UZ), BEd (UZ), MEd (Zou), Dip HR (IPMZ), PhD (UNISA)
G Bhebhe, Dip Ed (UZ), BEd EAPPS (ZOU), MEd (ZOU), PhD (UFH)
S Musingarambwi, CE (UZ), BA (UNISA), MPhil (NMMU), PhD (NMMU)
R Guvhu, Dip Edu (UZ); BEd (ZOU), MEd (UZ), PhD (UFS, S.A)
J Katsande, Dip eD – UZ, BEd (UZ) MSc (UZ)
W Muchabaiwa, Dip Edu (UZ), BEd (GZU); MEd (UZ); M.Sc. (UZ), DPhil (UNISA)
A Chafa, Cert (IPMZ), Dip Ed (UZ), BEd (MSU), Med (MSU)
S Gama, Dip. Ed (UZ), HND (HEXCO), B. Ed (ZOU), BTech Ed (Tecknikon Pretoria), MEd (ZOU), DPhil (ZOU)

DEPARTMENT OF SCIENCE TECHNOLOGY AND DESIGN EDUCATION

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T Usai, CE (UZ), BSc (UZ), MSc (UZ), PhD (MSU)
L Chitanana, BEd (UZ), PGDE (UZ), Med (MASU), PhD (UKZN)

Lecturers
C Musarurwa, BA Hons (UNISA), BEd (UZ), MSc Ed (UZ), MPhil (NMMU), PhD (University of Botswana)
M Mapope, PGDTE (GZU), BSc Special Hons (LSU), B.Tech Hons(CUT), MSc (MSU)
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J Kaifa, Dip Ed (UZ), BSc Ed (BUSE), MSc Ed (BUSE)
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F Jamba, Dip ED (UZ), BSc (MSU), MSc (MSU)
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M Chari, Dip Ed(UZ), BEd (NUST), MSc (NUST)
P Ziwada, Dip Ed (UZ), HND (HEXCO), Hons (HIT), MSc (NUST)

B Chibonga, Dip Ed(UZ), Bachelor of Science Education (BUSE), MSc in Physics Edu (UZ)

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Madzvamuse S, NC (HEXCO), ND (HEXCO), ND- Tec-Voc -Education (HEXCO), Bed (Solusi University

FACULTY OF ENGINEERING AND GEOSCIENCES

Acting Dean
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DEPARTMENT OF FUELS & ENERGY ENGINEERING

Acting Chairperson of Department
Vacant
Senior Lecturers
P Zvawanda , BTech (CUT), MSc (Gubkin – Russia), PhD (UKZN-South Africa)
P Masukume , LD (Enrique-Cuba), PGD (UZ), MSc (UZ), PhD (UZ)
L Gunda, BEng Electronics (NUST), MEng Electronics (Stellenbosch, RSA), FETC (Masvingo Poly)

Lecturers
J Chigerwe, NC, ND (Hre Poly), HND (Hre Poly), BTech (NUST), MSc (UZ) I Sundure, BEng (NUST), MSc (UZ)T Tafireyi, MSc (NUST); BSc Computer Science (NUST) Zim
A Mbonje, BEng (UST – Algeria), MSc (UST – Algeria)
H Samambgwa, BSc (NUST), MSc (NUST)
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Technicians

DEPARTMENT OF GEOSCIENCES

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A Mamuse, PhD (Applied Geology) (Curtin Australia); MSc (Applied Geology) (Curtin Australia); Postgraduate Diploma (Applied Geology); BSc Honours (Geology); Diploma (Gemology); BSc General;
T Dzvinamurungu, PHD in Geology (UJ, SA); MSc in Geology (SA); BSc in (Chemistry & Geology, UZ).

Lecturers
S Ncube, BSc Geology (UNZA, Zambia) Msc in Mineralogy (Hull, UK)
E Chinwada, Msc in Exploration Geophysics; MSc in Eng. WREM; BSc Physics Special Honours ; BSc General (Physics & Mathematics);
R Musinga, N.Diploma Mining Geology (ZSM, Zim), BTech Honours in Geology (TUT, SA) MSc in Economic Geology (Wits, SA)
A Mutakaya, PhD (Limpopo, SA) MSc Mathematical Modelling (NUST, Zim); BSc Honours in Mathematics(GZU, Zim) ; Grad DE (GZU, Zim); BSc (General) Mathematics and Geography (GZU, Zim);
P Chifoto, MSc in Physics (UKZN, SA); Bsc Honours in Applied Physics (MSU, Zim)
H Dewah, Bsc Honours Applied Physics;(UZ, Zim) MSc in Geophysics (NUST, Zim)
T Senderayi, MSc LGIS (UZ), BSC Gen (UZ), PostGradDip GIS (UZ), PDT (MSU)
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T Gwavuya, BSc Hons Applied Geology (MSU, Zim)
A Gumbo, BSc Hons Applied Geology (MSU, Zim)
T Paraziva, BSc Hons Applied Geology (MSU, Zim)

DEPARTMENT OF MECHANICAL ENGINEERING

Acting Chairperson of Department
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E Madzudzo, MSc (UZ), BEng Hons (NUST), Post Graduate Diploma in Higher Education (NUST), Zim D Ndiyamba, MEng (NUST), BTech (NUST), Diploma IPMZ
V Tambaoga, MSc (University of Oran-Mohamed Boudiaf), BSc (University of Oran-Mohamed boudiaf)
S Mateta, MTech (HIT), HND (HEXCO), ND (HEXCO)

DEPARTMENT OF MINING ENGINEERING

Acting Chairperson of Department
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P Mpesi, BEng. Hons Mining (MSU), MSc (UZ)
J Dohwe MSc in Mechatronics and Artificial Intelligence (UZ), BEng (Hons) Mechatronics Engineering (CUT)

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J Sauti, BEng Hons Mining Engineering (MSU, Zim)

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Chairperson of Department
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P Mutimutema, BSc (Hons) (UZ), MEng (Stellenbosch, SA)
T Matupire,BSc (Oran, ALG), MSc (Oran , ALG) S Gumbochuma, BSc (Hons) (UZ), MSc (Eng) (Witwatersrand, SA)

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Assistant Registrar

Research Innovation and Industry Linkage Officer
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C Nyoni, BA Hons (MSU), MADS (MSU)

DEPARTMENT OF ANATOMY

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DEPARTMENT OF ANAESTHESIA AND CRITICAL MEDICINE

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P Chuma, MBChB (UZ), MMed (UZ)

DEPARTMENT OF BIOCHEMISTRY

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Lecturers
T E Mudhluli, PDT (ZOU), BSc (UZ), MSc (UZ),
A Vengesai, BSc (UZ), MPhil (UZ) PhD (UZ)

DEPARTMENT OF CHEMICAL PATHOLOGY

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DEPARTMENT OF CLINICAL PHARMACOLOGY

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DEPARTMENT OF COMMUNITY MEDICINE

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DEPARTMENT OF HAEMATOLOGY

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DEPARTMENT OF IMMUNOLOY

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DEPARTMENT INTERNAL MEDICINE

Chairperson of Department
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DEPARTMENT OF MEDICAL MICROBIOLOGY

Chairperson of Department
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DEPARTMENT OF OBSTETRICS AND GYNECOLOGY

Chairperson of Department
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DEPARTMENT OF PAEDIATRICS

Chairperson of Department

T Nyamukapa, Dip /IPCC, (Sydney), MBChB (UZ), MMed (UZ)

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DEPARTMENT OF PHARMACY

Chairperson of Department
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DEPARTMENT OF PHYSIOLOGY

Chairperson of Department/Senior Lecturer
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DEPARTMENT OF PSYCHIATRY

Chairperson of Department
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Lecturers
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I Magura, BSc (UZ), MSc (UZ)

DEPARTMENT REHABILITATION

Chairperson of Department
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Lecturers
Vacant

DEPARTMENT OF SURGERY

Chairperson of Department
B Paketh, MBChB (UZ), FCS Ortho, COSECSA

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FACULTY OF SCIENCE AND TECHNOLOGY

Executive Dean
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DEPARTMENT OF APPLIED BIOSCIENCES AND BIOTECHNOLOGY

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DEPARTMENT OF CHEMICAL SCIENCES

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L Gedi, MSc (UZ), HND (Masvingo Poly), ND (Masvingo Poly) NC (Masvingo Poly)
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DEPARTMENT OF COMPUTER SCIENCE

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DEPARTMENT OF FOOD SCIENCE AND NUTRITION

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DEPARTMENT OF APPLIED PHYSICS AND TELECOMMUNICATIONS

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FACULTY OF SOCIAL SCIENCES

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L S Hombarume, BSc Hons (MSU), MMSc (MSU), PGDTE (MSU
L S Linjesa, BSc Hons (WUA), MSc (UZ)
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DEPARTMENT OF COMMUNITY STUDIES

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T R Bvirindi, BSc (UZ), MSc (UZ), PhD (Free State-SA)
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G Madzudzo, BSc Hons (UZ), MSc (UZ), MSc (Akershus-Norway)
T Choto, BSc Hons (UZ), MSc (UZ)
P Gandah, BSc Hons (UZ), MSc (UZ)
M Tshuma, BSc Hons (UZ), MA (UZ)
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DEPARTMENT OF GEOGRAPHY, ENVIRONMENTAL SUSTAINABILITY AND RESILIENCE BUILDING

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DEPARTMENT OF GOVERNANCE AND PUBLIC MANAGEMENT

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DEPARTMENT OF HUMAN RESOURCE MANAGEMENT

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DEPARTMENT OF MUSIC BUSINESS, MUSICOLOGY AND TECHNOLOGY

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P Magwati, Dip (UZ), Dip (MAM) BEd (GZU), MTED (UZ) DPhil (GZU)
W Chimbudzi, PGDTE (MSU), BSc Hons (MSU), MPhil (MSU)
T Shoko, BEd (UZ), MTED (UZ)

I Machafa, Dip (UZ), BEd (UZ), BA Hons (UKZN), MA (UKZN)
P D Bajilla, BSc Hons (MSU), MPhil (MSU)
A Mutavati, PGDTE (MSU), BSc Hons (MSU), MSc (MSU)
A Moyo, BSc Hons (MSU), MSc (MSU)
K F Moyo, BSc Hons, (MSU), MSc (MSU)

Instructor
E Chikomo, Dip (UZ), PGDTE (MSU), BSc Hons (MSU), B. Tech (Pretoria) MPhil (UCT)

Technicians
T Ngwerume BSc (MSU)
W Chikondo BSc (MSU)

DEPARTMENT OF PEACE AND SECURITY STUDIES

Chairperson of Department
K C Ngwaru, BA (UZ), MSc (AU)

Senior Lecturers
D Goredema, BA (UZ), MA (MSU), DPhil (MSU)

Lecturers
J Muwanzi, BSc Hons (WUA), MSc (WUA), MSc (MSU), DPhil (UKZN)
N Moyo-Nyoni, BA Hons (TCZ), BA Special Hons (LSU), MCom (UKZN), DPhil (DUT, SA)
F Ngwaru, BSc Hons (WUA), MSc (WUA), MSc (NUST), MSc (UZ) MED (UZ), DPhil (UDSM, Tanzania)
F J Chidhume, BA (MSU), MSc (MSU)
T B Matsvimbo, BSc Hons (MSU), MSc (MSU), MA MSU
M Changa, BSc Hons (GZU), BSc Special Hons (GZU), MSc (GZU)
S Njeru, BSc (UZ), MSc (AU), PhD (AU)
E Ndawana, BA Hons (UZ), MSc (ZOU), DPhil (UJ, SA)
A Zevure, BA Hons (UZ), MSc (ZOU)
D Mutanda, B A Hons (UZ), M Phil (UZ), DPhil (UJ, SA)
N Makwanise, BA (UZ) MSc (AU) DPhil (DUT, SA)
Chivombo, BSc Hons (GZU), MSc (MSU)
E Manyonda, BA (Solusi University), MA (AU)

SCHOOL OF SOCIAL WORK

Executive Director
N G Muridzo, Dip SWK (UZ), BSc (UZ), MSc (UZ), DPhil (Wits)

Lecturers
T Simango, Cert. in Leadership (USA), BSc SWK (UZ), BSc M&E (LSU), MSc (MSU)
A A Matanga, BSc Hons (UZ), MSc (MSU)
S Charamba, BSc Hons (UZ), MSc (MSU)
D Kanyere, Dip SWK (UZ), BSc Hons (UZ), MSc (UZ), Dip Resource Mobilisation (IFRD), NC in ECD (St Gabriel's College)
W Kurevakwesu, BSc Hons (BUSE), MSc (UZ)
L Nyamaka, Cert. in M&E (BUSE), BSc Hons (BUSE), MSc (MSU)
R Nhau, Executive Cert. in M&E (UZ), BSc Hons (BUSE), MSc (MSU)
A Tome, BSc Hons (UZ), MSc (UZ)
S T Jacob, BSc Hons (UZ), MSc (MSU)
F Takavarasha, BSc Hons (UZ), MSc (UZ)
N Risinamhodzi, BSc Hons (BUSE), MSc (MSU), Advanced Dip. Family therapy (Connect ZIST)
T Simango, BSc Hons (UZ), MSc (MSU), BSc Special Hons M&E(LSU), MSc M&E (LSU)
F Maushe, Dip SWK (UZ), BSc (UZ), MSc (UZ), PDTE (BUSE)
T Tendengu, BSc Hons (BUSE), MSc (UZ)
E Matope, BSc Hons (MSU), MSc (MSU)
M Razawu, BSc Hons (MSU), MSc (MSU)

T Sukulawo, BSc Hons (MSU), MSc (MSU)
A Tshugulu, Cert. in Project Mgmt (BUSE), BSc Hons (MSU), MSc (MSU)
V Kahonde, Dip SWK (WUA) BSc Hons (WUA), MSc (MSU)
O Mushapaidze, BSc Hons (BUSE), MSc (MSU)

Staff Development Fellows
D Mahamba, BSc (MSU)

CENTRE FOR PUBLIC POLICY AND DEVOLUTION

Coordinator
V Chakunda, PGDTE (MSU), BSc (MSU), Mcom (MSU), DPhil (Fort Hare-SA)

Research Fellows
S Mugari, BSc (ZOU), MPhil HIV/AIDS (Stellenbosch), PhD (UP), PGDip HIV/AIDS (Stellenbosch, Cert (GTC)
V Makanza, B Admin (UZ), MA (Bradford UK), PGDTE (MSU), Cert. (London, UK), Cert.nt (Bradford, UK)
M Chindoga, BA (Walter Sisulu University – SA), Hons (Fort Hare – SA), MSc (Fort Hare – SA), PhD (Fort Hare – SA)

Research Officers
O Mujana, BA (MSU), MSc (UZ)

INSTITUTES

Gender Institute

Executive Director

E Chauraya, CE (UZ), BED (UZ), MED (UZ), DED (UNISA)

Deputy Director and Senior Research Fellow

N Matope, PGDBA Gen (UZ), MES (UZ), PhD (Nelson Mandela)

Research Fellows

S Mhembwe, DIP. ED (UZ), BA (UZ), MAD (MSU, PhD (UNISA)

D Mawere, CE (UZ), BED (UZ), MED (UZ), PhD (UNISA)

D Nhengu, BA Hons (UZ), BSc MA (AAU), PhD (AU)

M.K. Chanhuwa, BSc (RU) LLB (RU), MCOM TAX (RU)

N Maruzani, BA GEN (UZ), GRAD CE (UZ), MED (MSU, MSC (GZU)

T T. Mugodzwa, CE (UZ), BED (UZ), MES (UZ)

A Maposa, Dip (GZU), Dip (IPMZ, BSc (MSU), MSc (MSU)

Midlands State University National Language Institute

Executive Director

B.Taringa, B.ED (UZ), Dip (GTC), M.ED (UZ), DPhil (UZ)

Deputy Director

Vacant

Research Fellows

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T. Khoza, M.ED (Solusi University), BA (ZOU) ,Diploma (UZ),DipEd(UZ), MA (University of Venda), PhD (University of Venda)

S Dlodlo, PGTE (MSU), BA Hons (MSU), MA (MSU), PhD (UNISA)

T F Muringa, PGDTE, (MSU), BA (MSU), MA (MSU), PhD (MSU)

Language Researchers

N. Nhemachena, Dip (IPMZ), Dip (HTC), B.ED (MSU), BSC Hons (ZOU), M.ED (MSU)

A.Mudenda,), PGDE (MSU), BA (UZ), M.ED (MSU)

L. Viriri, PGDE (MSU), BA (UZ), M.ED (GZU),

A.Nombembe, DipEd (HTC), B.ED (MSU), MA (MSU)

H. Ndhlovu Ncube, Dip (UCE), B.Ed. (MASU), MA (MSU), Med (GZU),

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A.M. Phiri, BED (GZU), MED (GZU),

B. Maruzani, (Morgenster Teacher's College), English Elementary Braille Cert (Pretoria), Cert of Proficiency in reading & writing Braille (Murewa Resource Centre), Dip ED, BED (GZU), MED (GZU)

S. W. Rwakonda, Cert (UZ), Cert (UCE), Bed (UZ), MEd (GZU),

M. Dube, BA Hons (LSU), Dip (UZ), BA (UZ), MA (GZU)

S. Mapangisana, BA Hons (UZ), MA (UZ)

L. Matsilele, PGDE (GZU) BA (GZU), M.ED (GZU),

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R. Marandure, Dip (MTC), Certificate (UNESCO/ UNICEF), Cert (SASLA), Advanced Cert (SASLA), BSc (ZOU), MEd (GZU),

F. Chikwature, DipEd (UCE), Dip (MTC), BED (GZU), Med (GZU)

T. Maphosa, DipEd (MTC), B. ED (GZU), Med (GZU),

Z. Chinyowa, Cert (UZ), BA Hons (UZ), MA (UZ)

Language Research Assistant

L. Njabulo, DipEd (Gwanda Zintec), BEd (GZU),

L. Ngomacheza, Bed (GZU)

Tugwi Mukosi Multi-Disciplinary Research Institute

Executive Director

Prof Jephias Matunhu, CE (MTC), B Ed (UZ), M Ed (UZ), MBA (ZOU), PhD (Fort Hare)

Associate Research Professor

J Mapuva, BED (UZ), BA UNISA, MPhil (UWC, SA), MPA (UWC, SA), PHD (UWC)

Research Fellows

F Muromo, CE(UZ), BEDEAPPS (UZ), Cert in M & E(UZ), PGDPPM (UZ), MSc RUP (UZ), MPhil (UCT), PhD (UKZN)

M Zvavahera, BSc Hons (MSU), MSc Zoology (Walter Sisulu University), PhD (Fort Hare)

VICE-CHANCELLOR'S OFFICE

VICE-CHANCELLOR

V N Muzvidziwa, BSc (UZ), MSc (UZ), PhD (New Zealand)

Assistant to the Vice Chancellor
H T Ngoshi, BA (UZ), MA (UZ), D. Lit (Pretoria).

Business Development Director
A I Jongwe, Dip (IPMZ), BSc (UZ), MBA (UZ)

Acting Professional Assistant
N. Bhowa BA Hons (MSU) MSc (GZU)

PRO-VICE-CHANCELLORS

Pro-Vice Chancellor - Research, Innovation and Industrialisation
G C Mugumbate, GCE (UZ), BSc (UZ), MSc (UZ), PhD (UCT)

Research Fellow
T Ngwenya, BSc (MSU), MPhil (MSU)

Research Assistant
P Muguti, BSc (Lupane)

RESEARCH AND INNOVATION DIVISION

Acting Executive Director

G Mehlana, BSc Hons (MSU), PDTE (MSU), PhD (UCT)

Deputy Director

Vacant

Finance Officer
N Phiri, CIS (CGI), BCom (MSU), MCom (MSU)

SHE Officer

Vacant

Research Manager

W Kusena, PDTE (MSU), PSEA (UNICEF), BSc Hons (MSU), MSC (MSU), PHD (UKZN),

Innovation Manager

Vacant

Business Development Manager

Vacant

Grants and Consultancy Manager

Vacant

Postgraduate Studies Manager

U Saidi, PGDTE (MSU), BA Hons. (MSU), MA (MSU), DPhil

PRO-VICE CHANCELLOR ACADEMIC AFFAIRS AND ADMINISTRATION
M Maphosa, BSc (UZ), MSc (Makerere), PhD (Makerere)

Research Fellow
O S Mapako, BSc (UZ), MSc (Lupane)
Research Assistant
T F Manyadza, BSc (MSU)

PRO-VICE CHANCELLOR'S OFFICE-INFRASTRUCTURE AND CAMPUS DEVELOPMENT
G Manyatera, LLD (UP) MSc, IUC (Turin), LLM (Turin), LLB (UZ)

Research Fellow
Vacant

INTERNATIONAL RELATIONS AND ALUMNI OFFICE
Executive Director

L Chikoko, PDTE (MSU), PSEA (UNICEF) BSc Hons (UZ), MSC (UZ), PHD (NMU)

BURSAR'S DEPARTMENT

Bursar
S Ncube, ND, HND (Byo Poly), BCom (UK), MCom (MSU), ACCA (UK), RPA (PAAB).

Deputy Bursar Planning and Control
M Paraffin, Diploma (CU), Advanced Diploma (CIMA), BCom (NUST), MBA (MSU)

Deputy Bursar Accounting and Finance

N Makaza, BCom (MSU), MCom (GZU), CGIAZ.

Principal Assistant Bursar Accounting and Finance
V Nderecha, BCom (MSU), MCom (MSU)

Principal Assistant Bursar Planning
Vacant

Assistant Bursar Non-Faculties
C Bhebhe, Certificate (UZ), BBA (Azalia), MBA (MSU)

Assistant Bursar Faculties
W Manganda, BCom (GZU)

Assistant Bursar Management Accounts
Vacant

A/Assistant Bursar Projects
S Mahlangu, BCom (MSU), MCom (GZU)

Assistant Bursar Stores
T Mutandwa, BCom (MSU), MSc (GZU)

Assistant Bursar Payables
A Chirume, BCom (MSU)

Assistant Bursar Assets
A Dzawanda, BCom (MSU)

Assistant Bursar Salaries
N Makonya, BCom (MSU), MCom (MSU)
Assistant Bursar Agro Business
T Makaza, BCom (MSU)

Assistant Bursar Cashbooks
S Muridzo, BCom (MSU)

REGISTRY DEPARTMENT

REGISTRAR

Mr T Zishiri, GCEDE (UZ), Dip IR (UZ), Dip in training Mgt (IPMZ)Dip in Pers Mgt (IPMZ), Dip. Sal. Admin (Stallion Consultancy), BA (UZ), MBA (NUST)

HUMAN RESOURCE SECTION

Deputy Registrar
S Masuku, Dip PR (LCCI), BSc (MSU), MSc (MSU)

Principal Assistant Registrar
T Nyamande, Dip (IPMZ), BSc (UZ), MSc (GZU)

A/Principal Assistant Registrar
A Munhande, Cert PM (IPMZ), Dip (UZ), BSc (MSU), MSc (MSU)

Senior Human Resource Officer
O Chikandwa, Dip (IPTA), Dip PM (IPMZ), BSc (RCU)

ACADEMIC AFFAIRS SECTION

Deputy Registrar
L Mangwanya, BSc (UZ), MSc (UZ), DPhil (UZ)

EXAMINATIONS OFFICE

Director
L Shava, BCom (ZOU) MSc (GZU)

Principal Assistant Registrar Exams
R Takayendisa, Dip (UZ), BED (MSU), MSc (GZU)

Assistant Registrar
R W Muwani, GCE (UZ), BA (UZ), MSc (WUA)

CENTRAL RECORDS AND ARCHIVING
Manager
E Mudondo, ND (HEXCO), BSc (ZOU), MSc (NUST)

WORK-RELATED LEARNING
Work Related Learning Coordinator
T Matsiya, BA Hons (MSU), BSc Special Honours (LSU), MA Development Studies (MSU)

A/ Principal Assistant Registrar Admissions
E Bwerinofa, BSc (MSU), MA (MSU)

Assistant Registrar Admissions
L Mashava, Diploma (UZ), Diploma (IPMZ), BTED (CUT), MSc (CUT)

C Masukume, HND (Gweru Poly), BSc (ZOU), MSc (ZOU)

SECRETARIAT SERVICES

Director Secretariat
D Nemaramba, BA (MSU), MA (MSU), PHD (RHODES UNIVERSITY)

A/Assistant Registrar
A Sakala, ND (Harare Poly), BSc (ZOU)

LEGAL SERVICES DEPARTMENT

Legal Officer

S R Mafa, (Prosecutors & Magistrates Cert.) Dip (HTC), LLB Hons (MSU), LLM, **(**MSU**),**

Assistant Legal Officer, Contracting & Compliance

G Tapanya, LLB (Walter Sisulu University), LLM (University of KwaZulu-Natal)

Assistant Legal Officer - Litigation

T W Govha, LLB (UZ)

PERFORMANCE CONTRACTING AND COMPLIANCE UNIT

Director

Vacant

Manager

J Gono, BSc (MSU), MSc (GZU)

MEAL Officer

D Mapuranga, Certificate (UZ), Diploma (Masvingo General Hospital), BSc (CUT), BSc (LSU), MSc (LSU), MSc (LSU)

N H Chiheve, BSc (MSU)

CENTRAL SERVICES, AMENITIES AND MAINTENANCE
Director

K Magadzire, BTech (UZ), MCom (UZ)

Deputy Director Central Services & Amenities
Vacant

Deputy Director Maintenance
Vacant

Property Management Officer
J Mvindo, BSc (NUST)

Senior Artisan Supervisor
S Chatizwa, NC (Gweru Polytechnic), Class 1 (Manpower)

RISK AND LOSS CONTROL DEPARTMENT,
Director
G Mvududu, BCom (MSU), MCom (MSU), Cert, (IRM, UK), COBIT 5 Cert (APMG), Cert (IIZ)

Risk Manager
Vacant

IT Risk / Data Protection Manager
Vacant

Loss Control Manager
S P Chademana, BCom (MSU), MCom (MSU)

MARKETING AND COMMUNICATIONS DEPARTMENT
Director
M D Mawere, BAcc Hons (UZ), BBA (IMM), MBA (MSU)

Communications Manager
J K Bhowa, PGD (MSU), BA (MSU), MSc (MSU)

Marketing Manager
L Nyamuranga Rubatika, Certificate (MAZ), PGD (IMM), BSc (MSU), MSc (MSU)

EVENTS MANAGEMENT
C Mugwamba, Diploma (ICCL), Diploma (IPMZ), BSc Honours (ZOU), MSc (MSU)

Events Coordinator
M R Nyoni, Diploma (LCCI), BCom (GZU), MCom (GZU)

LIBRARY

Librarian
N P Chibanda, ND (Byo Poly), HND (Hre Poly), BA (ZOU), MSc (NUST)

Deputy Librarian- Acquisitions, Resource Description, Materials and Metadata Services
F Masunungure, ND (Hre Poly), HND (Hre Poly), BA (ZOU), MSc (NUST)

Deputy Librarian - Scholarly Communications, Research Support, Innovations and Digital Services
E Makeke, NC (Hre Poly), ND (Hre Poly), BSc (ZOU), MSc (NUST)

Deputy Librarian- Teaching and Learning Services
Vacant

Library System Manager
Vacant

Sub Librarians
E Poswo, NC (Harare Poly), ND (Harare Poly), HND (Harare Poly), BA (ZOU), MSc (NUST)

K Tokwe, ND (Harare Poly), BSc (ZOU), MSc (NUST)

S Mazhindu, BSc (NUST), MSc (NUST)

Assistant Librarian
K Mafa, NC (Bulawayo Poly), ND (Bulawayo Poly), BA (ZOU)

Assistant Librarian
R Masimbiti, NC (Harare Poly), ND (Harare Poly), HND (Harare Poly), BSc (ZOU), PGD (NUST)

Assistant Librarian
E Matsa, NC (Gweru Poly), ND (Gweru Poly), BSc (ZOU)

Assistant Librarian
C Chuma, NC (Bulawayo Poly), ND (Bulawayo Poly), HND (Bulawayo Poly), BCom (ZOU), MSc (NUST)

INFORMATION TECHNOLOGY SERVICES

Director
L Dandira, Bsc Hons (MSU)

E-Learning Manager
T Mukute**,** Bsc Hons, (MSU), Msc Hons (CUT)

User Support Manager
M Mandinika, Bsc Hons (MSU), MSc Hons (MSU)

Network Manager
R Muchekeni, HND (HEXCO) BSc (MSU),MSc (NESU)

Business Solutions Manager
T Marongere, BSc (MSU), MSc (CUT)

DIVISION OF STUDENT AFFAIRS

Executive Dean
F C Chakauya, LLB Hons (Wits), LLM (UCT)

CATERING SERVICES DEPARTMENT

Catering Manager
M Chipato, BCom (MSU), MCom (MSU)

Assistant Catering Manager

B Choga, Certificate (Speciss College), Diploma (Speciss College), PGDE (ZOU)
BCom (MSU), MCom (CUT)

CHAPLAINCY DEPARTMENT
Acting Senior Ecumenical Chaplain
R Chinodakufa, BA Hons (GZU), MA (GZU)

COUNSELLING
Counselling Coordinator
E Mabika, BSc (ZOU), MSC (GZU)

HEALTH SERVICES
Director
P Toungana, MBChB (UZ)

Administrative Matron
M Muchikichi, Diploma in General Nursing (PGH), Diploma in Theatre Nursing (PGH)
BSc (ZOU)

SPORTS AND RECREATION DEPARTMENT
Sports and Recreation Coordinator
O Nayoto, BSc (BUSE)

Residence Department
Residence Coordinator
P Ncube, Diploma (DBC), BCom (MSU)

PROJECTS

Director
J Mapetere, NC in Plumbing (Further Ed Exam Board Zim), Cert. in Metal Work, Cert. in Site Supervision (Crown Agents Manpower Dev Services)

Deputy Director
T R Chibwana, Bachelor of Quantity Surveying Honours Degree (NUST)
Quantity Surveyor
Vacant

Architect
Vacant

PROCUREMENT MANAGEMENT UNIT

Procurement Director
F F Mazwi, HND (HEXCO), MBA (ZOU)

Procurement Manager (Tenders and Compliance)
E Mugabe, National Diploma (Gweru Poly), BSc (CUT), MSc (CUT)

Procurement Manager (Systems and Enterprises)
S Ntini, ND (HEXCO), BCom (MSU), MSc (BUSE), CIPS

INTERNAL AUDIT
Chief Internal Auditor

T M Machiwana, Dip (SAAA), HND (SAAA), BCom Honours (GZU), MCom (GZU)

Senior Internal Auditor
N L Nyanyiwa , BCom (GZU), MCom (GZU)

Senior Information and Technology Auditor
S Kapayapundo, HND (Bulawayo Poly), BBA (Solusi University)

Internal Auditor
J T Mbofana, BCom (MSU)

Internal Auditor
E Saki, National Diploma (Gweru Poly), HND (Gweru Poly), BCom (MSU)

Internal Auditor
R Mudzengi, BBA (Helderberg College)

HARARE CAMPUS
Campus Director
T M Mashingaidze, BA Gen (UZ), MA (UZ), PhD (Univ of Min, USA)

REGISTRY

Deputy Registrar (Academic Affairs and Admin)

GT Gwatidzo, PGD (HEXCO), BSc (UZ), MBA (NUST)
Assistant Registrar
P Chifamba, Diploma (UZ), BSc (BUSE), MSc (ZOU)

BURSAR
Principal Assistant Bursar
T Gumbo, Certificate (SAAA), BCom (MSU), MCom (MSU)

LIBRARY
Sub Librarian
C Munhu, Diploma (HTC), BSc (ZOU), MSc (NUST)

ZVISHAVANE CAMPUS
Campus Director
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REGISTRY
Deputy Registrar (Academic Affairs and Admin)
M Dandadzi, BSc (UZ), MSc (CUT)

STUDENT AFFAIRS
HEALTH SERVICES
Deputy Director
B Gudu, MBChB (UZ)

Principal Assistant Bursar
T Mvura, BCom (MSU), MCom (MSU)

Sub Librarian
P Chigamani, BSc (NUST), MSc (NUST)

Counselling Department
I Munyira, BSc (ZOU), MSc (ZOU)

MSU RADIO STATION
Managing Director
Vacant

Director of Station
Vacant

Station Manager
Vacant

KWEKWE CAMPUS
Campus Director
G Tarugarira, Dip.Ed. Tech (UZ), Dip. in Training Management (IPMZ),HND (IPMZ) BA Hons (UZ), Grad CE (UZ); MA (UZ), PhD (MSU) Zimbabwe)

REGISTRY
Deputy Registrar (Academic Affairs and Admin)
N Shava CE, BEd (UZ), MBA (MSU)

BULAWAYO LEARNING CENTER
Coordinator
G Bhebhe, Diploma (Hillside Teachers College), BED (UZ), MED (ZOU), Phd (Fort Hare)

Assistant Registrar
T Dlodlo, Dip (NUST), BSc (UZ), MSc (UZ)

MSU ENTERPRISES
Executive Director
U Guyo, Certificate (InnoVenton (SA), PGD (MSU), BSC (MSU), MA (UZ), PHD (MMU)

She Officer
Vacant

Deputy Bursar Enterprise
D Zvidzayi, HND (Byo Poly) BCom (NUST), MBA (MSU), Intermediate Diploma CIS.

Assistant Bursar Enterprise
R Marivadze, BCom (MSU), MCom (MSU)

Press and Publication
Production Manager
B Ncube, Diploma in Printing and Production (England)

Clothing and Textiles
Production Manager
Vacant

Production Supervisor
S Masuku, Diploma (MCD), Diploma (MSU), BSC (MSU)

Marketing and Sales Manager
G Madhora, Diploma (IMM), Diploma (ZIM), BCom (ZOU), MCom (MSU)

FOOD AND BEVERAGES
Production Manager

Vacant

Production Supervisor
E Madzokere, BSc (CUT), MSc (UZ)

CHEMICAL MANUFACTURING UNIT

Production Manager
V Chitsa, Post Grad Dip (MSU), BSC (MSU), MA (MSU)

PHARMACETICUALS UNIT
Production Manager
Vacant

Production Supervisor
Vacant

QUALITY ASSURANCE
Quality Control Manager
Vacant

Quality Controller
Vacant

AGRO-ENTERPRISE

Agrobusiness Director
J Matarutse, BSc (MSU), MSc (MSU)

Assistant Farms Coordinator Gwenhoro

E Mushavi, Certificate (Mupfure College), Diploma (Mlezu College)

Assistant Farms Coordinator Pittscottie

T T Giniyo, Diploma (MSU)

Assistant Farms Coordinator Kwekwe

Vacant

RETAIL SHOP
Retail Manager
B Mautsa, BCom (MSU), MSc (MSU),

Assistant Retail Manager
G Deure, Dip (CDS), BCom (MSU), MCom (MSU)

Accountant
M Mutangi, BCom ((MSU), MCom (MSU)

MSU NATIONAL PATHOLOGY, RESEARCH AND DIAGNOSTIC CENTRE

Director
L Mabaya, PGD (MSU), BSc (UZ), MSc (UZ), DPhil (UZ)

Radiologist
W Chirambasukwa, MBChB (UZ), FCR (RCSI)

Histopathologist
S Bopoto, PGD (OLC), MBChB (UZ), MMEd (UZ), MBA (Nottingham Business School)

Manager – Diagnostic Services Laboratory
L Nhari, BSc (UZ)

Manager – Diagnostic and Innovation Research Laboratory
E Nyagumbo, PGD (CUT), BSc (NUST), MSc (CUT)

Senior Radiographer
A Chiremba, BSc Honours (UZ), MSc (NUST)

Chief Scientist
Vacant

Marketing Officer
C Karanda, BCom (MSU)

Accountant
T Mbwembwe, Bcom (GZU)

Quality Officer
Vacant

Safety Officer
Vacant

COAL TAR PROJECT
Production Manager
Vacant
Process Chemist
Vacant

Production Engineer
Vacant

FACULTIES AND TEACHING DEPARTMENTS

FACULTY OF AGRICULTURE ENVIRONMENT AND NATURAL RESOURCES MANAGEMENT
Department of Agricultural Economics and Development
Department of Agronomy and Horticulture
Department of Land and Water Resources Management
Department of Animal and Wildlife Management

FACULTY OF ARTS AND HUMANITIES
Department of Archaeology, Cultural Heritage and Museum Studies
Department of Media, Communication, Film and Theatre Art Studies
Department of Languages, Literature and Cultural Studies
Department of History, Heritage and International Studies
Department of Development Studies
Department of Religious Studies and Ethics
Communication Skills Centre

FACULTY OF BUILT ENVIRONMENT, ART AND DESIGN
Department of Architecture

Department of Civil Engineering
Department of Estate Management
Department of Quantity Surveying
Department of Regional and Urban Planning
Department of Surveying and Geomatics

- **FACULTY OF BUSINESS SCIENCES**
  Department of Accounting Sciences
  Department of Management Sciences
  Department of Economic Sciences
  Centre for Entrepreneurial Studies and Innovation
  Department of Supply Chain, Insurance and Risk Sciences
  Department of Information and Marketing Sciences Management
  Department of Tourism, Hospitality and Leisure Sciences Graduate School of Business Leadership

- **FACULTY OF EDUCATION**
  Department of Humanities, Business Development and Arts Education
  Department of Educational Foundations, Primary Education and Pedagogy
  Department of Educational Policy Studies and Leadership
  Department of Science Technology and Design Education

- **FACULTY OF LAW**
  Department of Private Law Department of Procedural
  Law Department of Public Law

- **FACULTY OF MEDICINE**
  Department of Anatomy
  Department of Anesthesia and Critical Medicine
  Department of Biochemistry
  Department of Chemical Pathology
  Department of Clinical Pharmacology
  Department of Community Medicine
  Department of Haematology
  Department of Histopathology
  Department of Immunology
  Department of Internal Medicine
  Department of Medical Microbiology
  Department of Pediatrics
  Department of Pharmacy
  Department of Physiology
  Department of Psychiatry
  Department of Rehabilitation
  Department of Surgery

FACULTY OF ENGINEERING AND GEOSCIENCES

Department of Fuels and Energy Engineering
Department of Geosciences
Department of Mechanical Engineering
Department of Mining Engineering
Department of Metallurgical and Materials Engineering

- **FACULTY OF SCIENCE AND TECHNOLOGY**
  Department of Applied Biosciences and Biotechnology
  Department of Chemical Sciences
  Department of Computer Science

Department of Food Science and Nutrition
Department of Applied Mathematics and Statistics
Department of Applied Physics and Telecommunications

- **FACULTY OF SOCIAL SCIENCES**

Department of Applied Psychology
Department of Community Studies
Department of Geography, Environmental Sustainability and Resilience Building
Department of Governance and Public Management
Department of Human Resource Management
Department of Music Business, Musicology and Technology
Department of Peace and Security Studies
School of Social Work
Centre for Public Policy and Devolution

HISTORICAL NOTE

Devolution

The idea of a University in the Midlands dates back to the foundation of the National University of Science and Technology when Gweru, which was identified as a possible site for a second university campus, lost its bid to Bulawayo and the Province of Matabeleland. Two other opportunities to host institutions of higher learning (the Open University and the Catholic University) were also missed by the Midlands Province when the two universities went to Harare instead. It was in the midst of such disappointments that two initiatives gradually converged to give birth to what has since become the Midlands State University. The then President of the Republic of Zimbabwe, His Excellency, Cde. Robert Gabriel Mugabe, on the nudging of the provincial political leadership of the Midlands, acceded to the idea of a national university being built in the Midlands and announced that at two provincial rallies held at Gokwe and Mataga growth points. The President's declarations coincided with the Ministry of Higher Education and Technology's policy of devolution, which was aimed at expanding access to higher education by converting teachers' and technical colleges into degree-granting institutions. These colleges would offer Bachelor of Education and Bachelor of Technology degree programmes.

In line with the devolution policy, the then Minister of Higher Education and Technology, the Hon, Dr S.I Mudenge appointed in October 1995 the Committee on the Devolution of Bachelor of Education and Bachelor of Technology Degree Programmes, which was chaired by Professor C.J Chetsanga to investigate the feasibility of upgrading teachers' and technical colleges into degree granting places. The Committee recommended that Gweru Teachers College spearhead the process, followed by Masvingo and Chinhoyi Teachers' Colleges and that Gweru starts by offering Bachelor of Education degrees. The next Minister of Higher Education, Hon. Dr I.M.C Chombo set up in June 1996 another committee chaired by Prof. H.A.M Dzinotyiwei to advise on the details of getting degree programmes going at Gweru Teachers College. The Dzinotyiwei Committee recommended that the college should offer BA-Education, BSc – Education and B. Com- Education degree programmes under some special relationship with the University of Zimbabwe for four years before becoming an independent and fully-fledged university. A third committee of professionals chaired by Dr R.J Zvobgo, the Principal of Gweru Teachers College made the final recommendations on the initial staffing needs and degree programmes. These recommendations laid the basis on which the first-degree students were enrolled for Bachelor of Commerce with Education and Bachelor of Science with Education degrees offered by the University of Zimbabwe through the Faculty of Education's Department of Teacher Education.

Establishment of the University

In the meantime, although the devolution policy inaugurated an irrevocable process of bringing university education to the Midlands, there was a strong feeling, especially in the Province, that what was being done did not quite amount to the President's promise of a fully-fledged state university in the province. Responding to these feelings, but without losing sight of constraints imposed on the Government by declining national funds, the Minister of Higher Education and Technology transformed the devolution project at Gweru into Zimbabwe's third state university by means of the State University in the Midlands Act of April 1999.

The new University, whose name was later changed to the Midlands State University, was to be initially housed at the Gweru Teachers College premises. The mandate of the institution was contained in its broad objectives which are the advancement of knowledge, the diffusion and extension of arts, science and learning, preservation, dissemination and enhancement of knowledge that is relevant for the development of the people of Zimbabwe through teaching and research and, so far, as is consistent with the objectives, the nurturing of the intellectual, aesthetic, social and moral growth of the students at the University.

Inaugural Council

The University Act provided for governance made up of the Chancellor, who is the President of the Republic of Zimbabwe, a Council, Vice Chancellor, Pro Vice Chancellors and Senate.

To immediately manage the implementation of the Act, the Government, through the Ministry of Higher Education and Technology, appointed an inaugural Council with the following 21 members drawn from all walks of life and professions:

Mr C F Dube	Dube Manikai and Hwacha Legal Practitioners, as Chairperson
Mr W S Muringani	Zimbabwe National Water Authority
Mr R A Mabhena	Ministry of Primary and Secondary Education
Mr W Chidhakwa	Managing Director of the Export Processing Zone
Mr G G Nhemachena	Town Clerk of the City of Gweru
Mr W P Channon	Makwiro Platinum
Dr R F Wutaunashe	Harare Medical Practitioner
Mrs A S Kamba	Former Director of the National Archives of Zimbabwe and member of the Public Service Commission
Mr C Shumba	District Administrator, Chinhoyi
Dr M S D Mutopo	Production Director of the National Breweries and Chairperson of the Liquor Undertaking Board
Dr V Hungwe	Director of Lands
Prof N M Bhebe	MSU Vice-Chancellor
Prof R J Zvobgo	MSU Pro-Vice-Chancellor
Mr C S Nziramasanga	National Railways of Zimbabwe

Founding Vice-Chancellor and Pro-Vice-Chancellor

In consultation with the then Minister of Higher Education and Technology, Dr I. Chombo, and with the approval of the then President and Chancellor of the University, His Excellency, Cde R.G Mugabe, the inaugural Council appointed a distinguished academic and historian, Professor Ngwabi Bhebe, as the first Vice-Chancellor with effect from December 1999 and a renowned educationist, Professor Rungano Zvobgo, as Pro-Vice- Chancellor in January 2000.

Campus Sites

The University was initially granted use of facilities of the former Gweru Teachers College, Batanai Complex in Senga and the Senga Institute of Manpower Planning and Development. The site originally offered by the City of Gweru had to be abandoned in deference to the Zimbabwe National Army, who are using it as a training ground. The University was later granted 164 hectares of State land by the Vungu Rural District Council. This land stretches westwards from the former Gweru Teachers College. In addition, the University later acquired 188.6ha of land from TelOne, where more facilities were put up to aid in teaching and learning. As of the year 2024, this campus is home to the Graduate School of Business Leadership, Faculty of Law, Faculty of Education, Faculty of Business Sciences, Midlands State University National Pathology Research and Diagnostic Centre, Midlands State University National Language Institute and Innovation Hub under the Research and Innovation Directorate.

As part of its growth strategy, the University adopted a multicampus approach. To date, the University has a presence in Gweru, Zvishavane, Harare, Kwekwe and Bulawayo.

In fulfilment of the ten-faculty vision, the University, further, established the following faculties:

1. Faculty of Medicine and Health Sciences
2. Faculty of Engineering and Geosciences
3. Faculty of Built Environment, Art and Design

The University also embarked on a rebranding exercise for all its academic programmes to align them to the provisions of the heritage-based Education 5.0 philosophy. The rebranding exercise has resulted in changes to some faculty names as well as the introduction of new academic programmes.

The University's ten faculties and three research institutes are distributed as follows:

- The Gweru Main Campus hosts five faculties, namely; Social Sciences, Science and Technology, Medicine and Health Sciences, and Agriculture, Environment and Natural Resources Management. In addition, the Gweru campus also accommodates the following units:
  - MSU National Language Institute
  - MSU National Pathology Research and Diagnostic Centre
  - Gender Institute
  - Centre for Public Policy and Devolution
- The Graduate School of Business Leadership Learning Centre currently houses the faculties of Business Sciences, Education and Law,
- Zvishavane Campus situated at the old Shabanie Mine Office Complex accommodates the faculties of Engineering and Geosciences, & Arts and Humanities. The Campus also hosts a research institute namely, the Tugwi-Mukosi Multi-Disciplinary Research Institute which was established in 2018 and whose objective is to generate and disseminate continuous valid information on the social, economic, political, industrial and environmental impact of the dam on society.
- Harare Campus situated at number 1, Kwame Nkrumah Avenue is home to the Harare Weekend School that offers degree programmes in the Faculty of Business Sciences and postgraduate programmes in the faculties of Arts and Humanities as well as Social Sciences.
- Kwekwe Campus, which is scheduled to host the Faculty of Law, the Faculty of Built Environment, Art and Design as well as some projects under the Faculty of Agriculture, Environment and Natural Resources Management.
- Batanai Learning Centre in the Senga suburb hosts the departments of Food Science and Nutrition, Banking and Finance and Tourism and Hospitality Management,
- Bulawayo Learning Centre, which offers programmes in the Faculty of Education.
- The School of Social Work located at the corner of Chinhoyi and Grant Street offers both undergraduate and postgraduate programmes in Social Work.

Academic Programmes and Vision of the University

The first task for the Council and its Vice-Chancellor designates was to come up with proposals for academic programmes, which in turn would provide a basis for drawing up a university budget, staff recruitment and student admissions. It was also necessary to have a provisional working vision and mission statement for the new institution, which would be refined and concretized in due course in consultation with all the institutional and national stakeholders.

Such a vision envisaged a ten-faculty university whose principal mission was to train human resources innovatively and to generate, through original and applied research, scientific knowledge, which would substantially contribute to the social and economic development of Zimbabwe. The vision also has in sight a modern university whose delivery methods aim at breaking down traditional boundaries between vocational training and academic teaching using semesterised modular systems to allow the student as the major client to participate in the formulation and management of his/ her own programme; and above all at exposing the student to work-related learning to apply theoretical principles and knowledge to real work situations.

Against this background, Council adopted on the 25th of November 1999, 'proposed Academic Programmes' of eight faculties with the proviso that the Vice-Chancellor investigates the feasibility of adding the Faculty of Law. The faculties which were provisionally approved with their cognate departments were:

Faculty of Architecture, Art and Design

Department of Architecture

Department of Design and Technology
Department of Fashion and Textiles
Department of Visual and Performing Arts

Faculty of Commerce and Information Systems
Department of Accounting
Department of Business Studies
Department of Insurance
Department of Tourism Studies
Department of Economics

Faculty of Education
Department of Adult Education
Department of Curriculum and Arts Education
Department of Educational Administration Department of Educational Foundations

Faculty of Engineering
Department of Chemical and Production Engineering
Department of Environmental Engineering
Department of Industrial/Management Engineering
Department of Solar and Renewable Energy Engineering
Department of Structural Engineering
Department of Surveying

Faculty of Social Sciences
Department of Human Resource Management
Department of Social Counselling and Social Work
Department of Media and Society
Department of Nursing and Health Care
Department of Sport, Culture and Leisure

Faculty of Natural Resources Management
Department of Agro-Business Department of Agro-Meteorology
Department of Horticulture Department of Forestry
Department of Game and Wildlife Management
Department of Water Resources, Arid and Semi-Arid Agriculture

Faculty of Sciences
Department of Biology
Department of Chemistry
Department of Mathematics
Department of Physics
Department of Computer Science and Information Systems

Faculty of Arts
Department of African Language Studies
Department of Economic and Social History
Department of English
Department of Geography

The same Council meeting deliberated on what programmes were feasible to start within the year 2000 taking into account the following factors:

- the short time available for planning and recruitment of staff before the opening of the University;
- the shortage of funds and the facilities available at the Gweru Teachers College

The Council finally settled for those programmes that would require the least investment in teaching equipment and specialized buildings; that could be easily taught at Batanai and Gweru Teachers College and which would complement and strengthen the teaching of the programmes initiated by the Government at Gweru Teachers College in association with the University of Zimbabwe. Basically, the decision meant that the majority of the programmes would be the Gweru Teachers College-UZ ongoing ones with a few new ones added to give a start to the University. The decision too involved some minimal reorganisation of existing academic structures primarily in order to separate pedagogical professional training from content disciplines. This helped to clear the way for proper semester/modular planning and teaching and ensured that both the teachers' training and the content disciplines would get equal exposure to practical/attachment teaching.

The upshot of the decision were the following Faculties and their Departments which constituted the initial offering of the new University:

Faculty of Arts and Social Sciences
Department of African Languages Department of English
Department of Economic and Social History
Department of Geography and Environmental Studies
Department of Human Resource Management
Department of Media and Society Studies

Faculty of Commerce
Department of Accounting
Department of Business Studies
Department of Economics
Department of Tourism and Hospitality Management

Faculty of Education
Department of Applied Education
Department of Educational Foundations and Administration

Faculty of Science
Department of Biology
Department of Chemistry
Department of Computer Science and Information Systems
Department of Home Economics
Department of Mathematics Department of Physics
Department of Surveying and Geomatics

To obviate wastage of expensive infrastructure, valuable equipment and teaching consumables through disuse and neglect, following the phasing out of diploma programmes at Gweru Teachers College, the University had to bring forward the introduction of the **Faculty of Natural Resources Management and Agriculture** with the following departments:

1. Department of Wildlife and Livestock Management
2. Department of Land and Water Resources Management
3. Department of Crop Production and Horticulture

Opening of the University
The University started its first semester on 27 March 2000 with the following enrolment of students and teaching staff:

Under the University of Zimbabwe intake	**198**
Under Midlands State University	**400**
Teaching Staff	**70**

As of February 2024, the student enrolment stands at 27 000 which is underpinned by a staff complement of 890 academic staff. Since its inception, the University has produced over 80 000 graduates.

Strategic Plan
The University drew up its foundational long-term strategic plan. The exercise provided an opportunity to bring together the University community and all the national stakeholders. The Strategic Plan provides a guide and a basis for the development of the institution in terms of the programmes that must be taught; the physical plant, infrastructure and equipment that must be provided; projections of staff and student numbers by programmes; and work out a long-term business plan for the University.

Building on the foundational long-term strategic plan, the University introduced and implemented several other strategic initiatives to further its goals.

Midlands State University Enterprises
In keeping with the Education 5.0 philosophy, the University has made significant efforts to generate third-stream income through industrialisation which has resulted in the establishment of the **MSU Enterprises Division**, with interests in agriculture, chemical manufacturing, clothing and textile, and press and publications.

Furthermore, the University established a vibrant **Midlands State University National Pathology Research and Diagnostic Centre** which was fully funded by the Government of Zimbabwe through the Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development. The Centre offers pathology consultancy and high technology-driven clinical services and diagnostic imaging services through innovation, biomedical research and development.

The University also established a **National Language Institute**. The Centre offers language services that include editing and translation of documents, and the provision of Sign Language and Braille facilities. The Centre was fully funded by the Government of Zimbabwe through the Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development.

In 2021, the University was licenced for **MSU Radio** and the station was allocated two frequencies, one in Zvishavane at 90.3 FM and another in Gweru at 101.7 FM. MSU Radio manned shifts started on February 18, 2023. At the moment the station is on air every day of the week from 0600hrs to 2100hrs broadcasting programmes such as; Breakfast Show, Mid-Morning Show, Lunchtime Listening, Drive Time Show and The After Drive. The year 2024 saw the launch of the **Our Destiny (OD) Stores**. Established as a retail shop, OD Stores pursue the Education 5.0 blueprint's agenda to locally produce and distribute goods & services for the nation.

True to our motto, "Our Hands, Our Minds, Our Destiny", we rely on ourselves to build a university which is truly instrumental to the development of Zimbabwe and beyond.

FACULTY GOVERNANCE ORDINANCE NO. 4

1. ***Executive Deans***

i. There shall be an Executive Dean of each Faculty, who shall be appointed by a selection board appointed by the Council and chaired by the Vice-Chancellor or a Pro-Vice-Chancellor.
ii. An Executive Dean shall hold office for three years and, on the expiry of his term of office, shall be eligible for re-appointment.

2. ***Functions of Executive Deans***

i. An Executive Dean shall be responsible to the Vice-Chancellor for:
    a. the character and quality of his faculty's academic programmes and teaching;
    b. the proper direction, control and management of the staff, property, equipment and finances of his faculty; and
    c. contributing to the evolution and maintenance of an environment conducive to learning at the University.
ii. Without limiting subsection (i), an Executive Dean's functions include:
    a. implementing the policies and objectives of the University and his faculty;
    b. ensuring regular meetings of the Faculty Planning Committee to oversee, plan and direct the

academic and human resource development of the whole Faculty;

c. planning and control of the faculty budget;
d. developing and maintaining relations with alumni of his faculty and with other organizations and persons, with a view to enriching the academic programmes offered by his faculty;
e. establishing links and partnerships with industry, commerce, professional establishment and informal sectors in order to support student internships, sandwich modules and work-related learning
f. undertaking fundraising to support teaching and research in the faculty;
g. co-operating with other faculties, agencies and entities to further the programmes and projects of his faculty;
h. maintaining an overview of the faculty's quality assurance;
i. ensuring that the faculty office provides accurate registered students statistics to departments and lecturers, so as to ensure that only registered students attend lectures;
j. ensuring that student registers are constantly updated and well maintained so that the faculty only provides service to registered students;
k. ensuring that departments are adequately provided for in the University teaching timetable;
l. ensuring that the faculty has adequate teaching space to accommodate all its classes;
m. ensuring that all registered students in the faculty have properly kept and updated records of all their academic profiles;

n. in liaison with Chairpersons ensure that teaching and other duties are adequately catered for before academic personnel proceed on leave, whether ordinary, contact or sabbatical leave and;
o. performing such other functions as may be prescribed by the Ordinance
p. establishing links and partnerships with industry,
commerce, professional establishment and informal sectors in order to support student internships, sandwich modules and work-related learning
q. undertaking fundraising to support teaching and research in the Faculty;

3. ***Evaluation of Performance of Executive Deans***

The performance of every Executive Dean shall be evaluated by a committee appointed by the Vice-Chancellor and consisting of senior academic staff and administrators.

4. **Termination of Appointment of Executive Deans**

i. An Executive Dean may resign his office as such by giving the Vice-Chancellor one month's notice or such longer or shorter notice as he and the Vice-Chancellor may agree.
ii. Subject to the approval of the Council, the Vice-Chancellor may suspend or terminate the appointment of a person as an Executive Dean: -
    a. on the recommendation of the committee referred to in section11; or;
    b. for other good cause.
    c. an Executive Dean whose appointment as such has been suspended or terminated in terms of subsection (2) may appeal to the council within fourteen days of being notified of the suspension or termination and on any such appeal the Council may confirm, vary or rescind the suspension or termination, as the case may be, or give such other direction in the matter as it thinks appropriate.

5. ***Composition of Faculty Boards***

For each Faculty there shall be a Faculty Board which shall consist of:
a. the Executive Dean;
b. all the full-time academic staff of the Faculty; such persons as may be assigned to the Faculty Board by the Senate; and
c. two students elected annually by the students in the Faculty.

6. ***Functions of Faculty Boards***

A Faculty Board shall have the following functions:
a. to regulate, subject to the approval of the Senate, the teaching and study of the subjects assigned to the faculty;
b. to report to the Senate on any matter specifically relating to the work of the faculty;

c. to deal with any matter referred or delegated to it by the Senate; and
d. to appoint committees, which may include a minority of persons who are not members of the Faculty Board, to carry out any of the duties or exercise any of the powers of the Board.

7. ***Meetings of Faculty Boards***

i. The Executive Dean of each faculty shall call for at least three (3) meetings of the Faculty Board at which matters relating to the policy of the faculty and appointments shall be discussed and recommendations adopted for submission to the appropriate authorities on appointments.
ii. The Vice-Chancellor and every Pro-Vice-Chancellor shall be entitled to attend any meeting of a Faculty Board or committee thereof.

THE DEPARTMENTAL GOVERNANCE ORDINANCE NO. 5

1. The Departmental Board

1.1 There shall be a Departmental Board for each Department which shall consist of:

1.1.1 the Chairperson of the Department;

1.1.2 all full-time members of the academic staff of the Department;

1.1.3 honorary and part-time lecturers in the Department or their representatives, as determined by the Departmental Board;

1.1.4 where relevant, at least one representative of the technical staff in the Department or, where technical staff are assigned to the Faculty and not to Departments and the Board considers it helpful to have representatives of such staff, at least one representative of the technical staff in the Faculty, elected annually by such staff, provided that the representative(s) so elected shall not be entitled to attend for deliberations on matters considered by the Chairperson to be confidential;

1.1.5 two students in the Department elected annually by the students from amongst themselves, provided that the students so elected shall not be entitled to attend for deliberations on matters considered by the Chairperson to be confidential;

1.2 The Vice-Chancellor, the Pro-Vice-Chancellor(s) and the appropriate Executive Dean and Deputy Dean shall be entitled to attend Departmental Board meetings in an ex-officio capacity.

1.3 From time to time, the Chairperson of the Department, after consultation with members of the Departmental Board, may invite other persons to attend Departmental Board meetings.

1.4 Each Departmental Board shall meet at least twice every semester and shall maintain a proper system of agendas and minutes for such meetings.

1.5 Subject to Section 3 of this Ordinance, each Departmental Board shall regulate its own procedures, including the establishment of a quorum.

2. Duties and Powers of the Departmental Board

Subject to the provisions of the University Statutes, the authority of the Senate, provisions of this Ordinance and such limitations as the Faculty of which the Department forms a part may impose, the Departmental Board;

2.1 shall arrange for, conduct and control the teaching and instruction of students within the Department and the setting and marking of examination papers in accordance with regulations approved by Senate, the general academic policy agreed by the Faculty Board and the approved administrative procedures of the University.

2.2 shall make recommendations to the Faculty Board for the establishment of new courses and the amendment of existing regulations and syllabi relating to studies within the Department.

2.3 may delegate functions and responsibilities to individuals or groups of individuals within the Department;

2.4 shall exercise such powers as may be conferred upon it by the Faculty Board, the Senate or the Vice-Chancellor;

2.5 may provide consultancy services on matters concerning the subject assigned to the Department within the limitations of its capabilities and subject to the general University policy on consultancy services;

2.6 shall monitor the implementation of the University's conditions relating to the undertaking of private remunerative work in respect of the members of the Department;

2.7 may make recommendations in respect of estimates of expenditure in the Departmental Budget Committee;

2.8 shall suggest a preliminary short-list of candidates for appointment to academic and technical posts within the Department, taking into account the need to maintain strict confidentiality in handling applications, and shall forward such short-lists to the relevant Board of Selection for consideration;

2.9 shall formulate general guidelines on pure and applied research and suggest means of funding research programmes in the Department.

2.10 shall submit recommendations to the Senate in respect of the appointment of External Examiners and any other consultants in the Department.

3. Chairpersons of Departments

3.1 There shall be a chairperson of each department appointed by the Vice-Chancellor, on behalf of the University Council, from among the full-time members of the Academic staff of the Department.

3.1.1 The Chairperson of the Department shall, as part of management participate in the formulation, implementation and evaluation of the academic policies of the University and shall promote academic excellence in teaching, research and University service.

3.1.2 The Departmental Chairperson shall also provide academic leadership to the Department by planning, directing and coordinating the formulation and implementation of the academic plans and programme of the Department.

3.1.3 The Departmental Chairperson shall have held the rank of Senior Lecturer or above in a reputable University. In the absence of such an individual, the Chairperson shall be at least tenured and should have amassed substantial experience in teaching, active research, University service and should reveal a strong commitment to the development and attainment of the Departmental goals.

3.2 ***Before appointing a Departmental Chairperson***

3.2.1 the Executive Dean of the Faculty shall identify suitable candidates guided by the stated criterion.

3.2.2 the Executive Dean of the Faculty together with the Pro-Vice-Chancellor (Academic) shall consult all full-time members of the academic staff in the Department on the nominations made by the Dean.

3.2.3 the Pro-Vice-Chancellor (Research and Academic Affairs) together with the Executive Dean of the Faculty shall make a recommendation, in concurrence with all the full-time members of the Academic staff in the Department to the Vice Chancellor.

3.2.4 The Vice-Chancellor shall appoint the Departmental Chairperson.

3.3 The Chairperson of a Department shall hold office as such for a period of up to three years and shall be eligible for re-appointment for a further term of three years provided that he shall not hold office for a continuous period exceeding six years.
On completion of the second term of office, a Chairperson of the Department shall not be eligible for further appointment to the position of Chairperson of Department until a three-year period has elapsed.

3.4 After consultation with the Pro-Vice-Chancellor, the Dean of the Faculty and the Chairperson concerned, the Vice-Chancellor may terminate the appointment of a Chairperson of the Department as such by giving him/her two months' notice in writing.

3.5 The Chairperson of a Department may resign his appointment as such by giving the Vice-Chancellor two months' notice in writing.

3.6 Where the Chairperson of a Department is unable, either because of his absence from the University or for any other reason, to carry out his functions as Chairperson, the Vice-Chancellor may, subject to the provisions of section
3.2 of this Ordinance appoint an Acting Chairperson of the Department for such period and under such conditions as he may determine, provided that the period of appointment does not exceed the balance of the period of office of the substantive Chairperson.

3.7 A Chairperson of a Department shall be paid a responsibility allowance at a rate of the difference between the lecturer's top notch and the top-notch of the Associate Professor grade, provided that no allowance shall be payable where the Chairperson is absent or unable to perform his functions for a period of 30 days or more.

3.8 An Acting Chairperson of the Department who is appointed Acting Chairperson for a period in excess of 30 days shall be paid a responsibility allowance at a rate of the difference between the Lecturer's top notch and the top-notch of the Associate Professor grade.

4. Duties and Powers of the Chairperson of Department

In addition to his teaching, research and other duties and responsibilities, the Chairperson of Department shall: -

4.1 serve as executive officer of the Department in the implementation of Departmental policy, as determined by the Departmental Board and other University authorities, and be responsible for the day- to-day administration of the Department;

4.2 serve as Chairperson of Departmental Board meetings.

4.3 ensure that constant departmental meetings are held to guide lecturers on the academic and Human Resource affairs in the department;

4.4 represent the Department on the appropriate University Committees and other bodies as required;

4.5 make recommendations in respect of estimates of expenditure in the Department Budget Committee;

4.6 make recommendations to the Faculty Board for the introduction of new modules as well as amendment of all modules, existing regulations and syllabi relating to studies within the Department;

4.7 ensure that the Department provides accurate statistics of registered students to lecturers so as to ensure that only registered students attend lectures;

4.8 ensure that lecturers are adequately provided for in the University teaching timetable;

4.9 ensure that lecturers are utilizing the venues allocated to them on the time table;

4.10 setting and marking of examination papers in accordance with regulations approved by Senate, the general academic policy agreed by the Faculty Board and the approved administrative procedures of the University;

4.11 ensure that lecturers turn up for lectures and tutorials on time, give a fair number of assignments per semester assess fairly and comment meaningfully on student assignments and maintain clear and accurate records of student assignment assessments;

4.12 make recommendations with respect to probation, advancement and promotion of academic staff within the Department, provided that :-

4.12.1 in so doing, the Chairperson shall consult all full-time academic members of the Department;

4.12.2 if the Chairperson is himself/herself a candidate for promotion, the appropriate recommendations shall be made by the Executive Dean of the relevant Faculty after consultation with all full-time academic members of the Department;

4.13 ensure that teaching and other duties are adequately catered for before academic personnel proceed on leave whether ordinary, contact or sabbatical leave;

4.14 shall monitor the implementation of the University conditions relating to the undertaking of private remunerative work in respect of the members of the department and;

delegate functions and responsibilities to individuals or groups of individuals within a department.

ACADEMIC STAFF APPOINTMENTS, GRADING, TENURE AND PROMOTIONS ORDINANCE 3 OF 2000 AS AMENDED

ARRANGEMENT OF SECTIONS

Preamble

1. Interpretation
2. Title and Date of Commencement
3. Objectives
4. Structure of Academic Staff Grades
5. Grading and Initial Appointment
6. Grading and Salary Structure
7. Promotion

8. Tenure

PREAMBLE

In terms of section 29 (3) (h) of the Midlands State University Act [Chapter 25:21] as amended, the Council of Midlands State University, with the approval of the Minister of Higher and Tertiary Education, Innovation, Science and Technology Development, or such other Minister to whom the President may from time to time assign the administration of the Midlands State University Act on the recommendation of the University Senate, makes the following Ordinance:

1. INTERPRETATION

In this Ordinance, **'academic staff''** means all persons employed by the University as Professors/Research Professors, Associate Professors/Associate Research Professors, Senior Lecturers/Senior Research Fellows, Lecturers/Research Fellows, Chief Technicians, Senior Technicians, Researchers, Research Assistants, Technicians and Technical Assistants in teaching and/or research departments and institutes, who are contractually required to carry out teaching, examination and administrative duties and to conduct research and innovation.

2. TITLE AND DATE OF COMMENCEMENT8

This Ordinance is cited as the Academic Staff Appointments, Grading, Tenure and Promotions Ordinance Number 3 of 2000 as amended and shall take effect from 1 February 2024.

3. **OBJECTIVES**

In making this Ordinance, Council seeks to:

3.1 Promote Academic Staff whose performance demonstrates particular merit in teaching, consultancy, student supervision, research, innovation, enterprise development,

scholarship and creative activity, administration, service and leadership in the University and professional practice including service to the community.

3.1.1 Research and Scholarship imply basic, and applied multi-disciplinary research.
3.1.2 Innovation implies the invention, creation and implementation of new processes, products, services and methods of delivery which result in significant improvements in outcomes, efficiency, effectiveness or quality.
3.1.3 Enterprise development implies an act of investing time and capital to establish, expand or improve business.
3.1.4 Creative work entails generation of content attuned to innovation and industrialisation.
3.1.5 Leadership entails more than administrative work; it can entail getting people to understand and believe in the vision and work to achieve set goals.
3.1.6 Service to the Community entails staff voluntary involvement in activities of different dimensions that assist the community

3.2 Provide an objective, fair and systematic method of assessment in order to ensure a consistent application of standards for the appointments and promotions of staff.

4. STRUCTURE OF ACADEMIC STAFF GRADES

a. The following structure of grades for Academic Staff shall apply within the University:

Title	Grade
Professor/Research Professor	3a2
Associate Professor/Associate Research Professor	4a
Senior Lecturer/Senior Research Fellow	5a
Lecturer/Research Fellow/ Chief Technician	6a
Researcher/ Teaching Assistant /Senior Technician	7a
Staff Development Fellow/ Research Assistant/Technician	8a
Technical Assistant/ Language Assistant	9a

b. The salary scale applicable to each grade is subject to approval by the University Council.

c. The University Council or its authorised committees shall have sole discretion and in so doing, may designate that posts be established at any of the grades listed in section 4.1 above, but normally each of the other academic posts on Department or Faculty establishment may be filled up to professorial level.

5. GRADING AND INITIAL APPOINTMENT OF ACADEMIC STAFF

a. On initial appointment to the University's Academic Staff, a successful candidate shall be graded according to his or her qualifications, experience, published research, proven innovations and industrialisation.

b. **Qualifications**

i. The basic qualification for appointment to the University's academic staff at lecturer/research fellow grade is normally a PhD degree but at least a credit in a Master's degree may be considered in the critical shortage areas as approved by Senate. This is subject to review from time to time.
ii. The basic qualification for appointment at Teaching Assistants/Research Assistants/Staff Development Fellow grade is a good Honours Degree in the relevant field. Senate may review this from time to time.
iii. The basic qualification for appointment at Technician grade is a professional diploma and a degree in the relevant field.
iv. The basic qualification for an appointment at Technical Assistant grade is a professional diploma.

c. Experience

i. In granting recognition to relevant postgraduate experience, the University shall:
5.3.3.1 Make no distinction between professional and non-professional experience.
5.3.3.2 Recognise in full, the time spent as a full-time member of the academic (teaching and /or research) staff of a reputable university/institute.
5.3.3.3 In all disciplines, any postgraduate experience in the appropriate discipline shall be recognized as relevant.

5.4 Research, Innovation and Industrialization Output

5.4.1 The University recognises published research in journals, including books and book chapters, filed/registered Intellectual Property Rights (IPR), innovation, creative work and enterprise development for the purposes of determining the grade on initial appointment.
5.4.2 The Appointment Board shall take cognisance of the quality of the Research, Innovation, and Industrialization and decide whether it meets the conditions set by ZIMCHE.
5.4.3 The Appointment Board shall ensure that the assessment and weighting of published research and filed/registered IPR, creative work, innovation and enterprise development are consistent with standards implemented by the Promotions Committee (Refer to Annexure 1).

6. GRADING AND SALARY STRUCTURE

a. There shall be only one Academic (Teaching and Research) Grading and Salary Structure, and the same salary scale shall apply to all posts in all Faculties and Institutes within the University.

7. PROMOTION

a. For one to be promoted, he/she has to be a permanent/full-time member of staff or have a long-term contract of two or more years that has been renewed more than once.

b. Normally, the basic qualification for promotion is a PhD degree from Lecturer/Research Fellow grades up to Professor/Research Professor grades; BSc/BA and professional diploma for Technical Assistant, and Master's degree or a professional diploma from Researcher Technicians upwards.

c. **Conditions for promotion**

i. Annual performance reviews shall be considered as part of the promotion requirements.
ii. No member of staff at the level of lecturer /Research Fellow/ Researcher / Chief Technician /Senior Technician/Technician shall be promoted while on probation.
iii. No member of staff shall be promoted while on full-time study leave.

iv. **Criteria for Promotion**

In assessing the suitability of members of the academic staff for promotion to a higher grade, the University shall take into account the following broad criteria.

1. **Professor/Research Professor/ Associate Professor/Associate Research Professor/Senior Lecturer/Senior Research Fellow**
   a. Teaching or, for research staff, contribution to knowledge through research, consultancy, advisory work, part-time teaching, etc.
   b. Research;
   c. Innovation and;
   d. Industrialization
   e. Service to the University and Community ;

7.3.4.2 Chief Technician/ Senior Technician

7.3.4.2.1 Continuous assessment work on practicals.
7.3.4.2.2 Technical support for research, laboratory projects and field work
7.3.4.2.3 Research and mobilization of funds;
7.3.4.2.4 Innovations and
7.3.4.2.5 industrialization
7.3.4.2.6 Service to the University and Community.

7.4 Assessment of Teaching

In making an assessment of a candidate's teaching, the University regards the following general areas as being of central importance:

7.4.1 For lecturer grades - teaching methods which incorporate innovation and enterprise development/industrialization;

7.4.1.1 Module content which incorporates innovation, and industrialization;
7.4.1.2 The general performance of students in the module taught by the candidate for promotion;

7.4.1.3 The quality of the candidate's supervision of undergraduate and postgraduate students; the orientation of students' projects towards innovation and enterprise development/industrialisation; and
7.4.2.4 The development of new and effective techniques of instruction, including the production of teaching materials.

7.4.2 For Technician grades

7.4.2.1 Work related to facilitation of continuous assessment components in learning and teaching;
7.4.2.2 Continuous assessment work on practicals;
7.4.2.3 Laboratory work such as maintenance of laboratory equipment;
7.4.2.4 Field work, such as the planning and carrying out of field experiments

7.4.3 Researcher

7.4.3.1 Work related to consultancy and research output.
7.4.3.2 Laboratory work, such as maintenance of laboratory equipment; and
7.4.3.3 Field work such as the planning and carrying out of field work.

7.5 Assessment of Research, Innovation, Industrialization and Community Service

7.5.1 Assessment of Research

Research is a necessary component in academic life. Those who successfully conduct research geared at solving the problems of the world deserve credit. No one can be accepted as an academic without involvement in research.

In assessing a candidate's research, scholarship and creative work (hereinafter referred to as "research"), the University recognizes that research relates to origination of new knowledge that advances and/or promotes a discipline and/or its relevance to scholarship and society. It has several dimensions, and therefore assessment of the candidate's research shall include any of the following dimensions:

8

7.5.1.1 The accumulation of data that confirms an existing theory;

7.5.1.2 The application of existing theory to facts specific to given contexts;

7.5.1.3The generation of new theory and its empirical testing;

7.5.1.4 The generation of new methodologies for dealing with problems in the discipline or in practice;

7.5.1.5 Demonstrable mentorship of students and other Faculty in research as evidenced by authorship patterns in multi-authored scholarly publications;

7.5.1.6 Normally, the University would take into account for promotion purposes only research which has been published, been accepted for publication or is in a written-up form in which it can be assessed;

7.5.1.7 In addition to assessing a candidate's research from the point of view of its quality, the University would also make an assessment in terms of the quantity of research output attributable to the candidate, sustained rate of productivity, the individual member's contribution to the research, relevance and impact;

7.5.1.8 Evidence of students supported by one's research funds;

7.5.1.9 Evidence of resource mobilisation for improving University asset base.

7.5.2 Assessment of Innovation

In assessing a candidate's innovation, the University recognise the intricate relationship between research and innovation. It has several dimensions, and therefore, assessment of the candidate's innovation accompanied by evidence, shall include any of the following dimensions:

7.5.2.1 Coming up with new discoveries, inventions or artworks;

7.5.2.2 Coming up with new methods, new approaches, new techniques, and new products and/or services;

7.5.2.3 Creating an original design or invention that qualifies for intellectual property protection (prototypes, designs, practical solutions, etc);

7.5.2.4 Translation of research ideas or products and intellectual property into commercial value (goods and services);

7.5.2.5 Number of Intellectual Property generated and registered (rate of productivity);

7.5.2.6 Range of traceable products, technologies and services in Industry, Commerce and Social Services;

7.5.2.7 Improve the efficiency of existing processes and/or resource use.

7.5.3 Assessment of Industrialisation

In assessing Industrialisation, the University recognises the candidate's contribution to university-linked industrialisation processes taking cognisance of the linkages among research, innovation and industrialisation. It has several dimensions, and therefore, assessment of

the candidate's industrialisation, accompanied by evidence, shall include any of the following dimensions:

7.5.3.1 Distinct role in IP commercialisation processes;

7.5.3.2 Functional systems or models developed that have been adopted by Industry and Commerce;

7.5.3.3 Adopted procedures and processes for industrialisation;

7.5.3.4 Technology transfer for the delivery of goods and services;

7.5.3.5 Income realised by the University from the utilisation of Intellectual

Capital

7.5.3.5.1 Number of joint research contracts with innovative companies

7.5.3.5.2 Number of innovation-driven companies created by the University

7.5.3.5.3 Number of licence agreements

7.5.3.5.4 Number of patents in use by industry

7.5.3.6 Evidence of new service industry, software, new artifacts and artistic creative products that have been developed and deployed.

7.5.4 Assessment of Community Service

7.5.4.1 In assessing a candidate's community service, the University is conscious of the fact that every member of the academic staff should be a good University citizen, performing his/her duties conscientiously and well, attending meetings and participating in committees to which he/she is assigned. The University regards such activities as adequate University service.

7.5.4.2 If a candidate has taken a leadership role in University service, such as serving as Dean, Chairperson of a Department, Chairperson of a Committee or in organizing a Faculty or Department's contribution to such activities as Vacation School or Open Day, or organizing vacation research for students, or is involved in students counselling and does these activities outstandingly well, the University would regard such services as better than adequate.

7.5.4.3 If a candidate has taken a leadership role, such as undertaking research to solve a national problem or is Chairperson of a Parastatal Board, the University shall regard that as adequate

7.5.4.4 If a candidate has been involved in community services, projects and programmes with accolades attributable to the University, it shall be regarded as adequate community service with appropriate scoring.

7.6 Criteria for Promotion to Specific Grades (Refer to Annexure 1 and 2)

7.6.4 Criteria for Promotion to Technician

The criteria for promotion to Technician are as follows:

7.7.1.1 Acquisition of a relevant first degree would normally qualify for promotion to the Technician grade.

7.6.5 Criteria for Promotion to Senior Technician

The criteria for promotion to Senior Technician are as follows:

7.7.2.1 For Senior Technician- An assessment at the level of **satisfactory** in work related to facilitation of continuous assessment work in teaching, laboratory and field research work, innovation leading to industrialization and Service to the University and Community;

7.7.2.3 For Senior Technician - An assessment at the level of **outstanding** in at least one of the following: work related to facilitation of continuous assessment work in learning and teaching, research, innovation, creative work, enterprise development and service to the University and Community;

7.6.6 Criteria for Promotion to Chief Technician

The criteria for promotion to Chief Technician are as follows:

7.7.3.1 An assessment at the level of **satisfactory** in work related to facilitation of continuous assessment in teaching, laboratory and field research work, innovation, creative work leading to industrialization and Service to the University and Community;

7.7.3.2 An assessment at the level of **satisfactory** research shall normally mean at least three (3) refereed research publications.

7.7.3.3 An assessment at the level of **outstanding** in at least one of the following: work related to facilitation of continuous assessment in teaching, research, innovation for industrialization and service to the University and Community;

7.7.3.4 Outstanding research shall normally mean at least five (5) refereed research publications and at least one filed IPR or one (1) enterprise development or assessed equivalents.

7.6.7 Criteria for promotion to the Senior Lecturer/Senior Research Fellow Grade

The criteria for promotion to the Senior Lecturer/Senior Research Fellow Grade are as follows:

7.7.4.1 An assessment at the level of **satisfactory** in teaching and/or research, consultancy, innovation, creative work, enterprise development and Service to the University and Community;

7.7.4.2 Satisfactory teaching shall normally mean performing well in at least half of the areas listed in section 7.4.1;

7.7.4.3 An assessment at the level of **outstanding** in at least one of the following: teaching, research, consultancy, innovation**,** creative work, enterprise development and service to the University and Community;

7.7.4.4 For Senior Lecturer, outstanding research, **innovation, creative work and enterprise development** shall normally mean at least twelve (12) refereed research publications and at least one filed IPR or one (1) enterprise development or assessed equivalents and satisfactory research shall normally mean at least twelve (12) refereed research publications.

7.7.4.5 Outstanding teaching shall normally mean performing well in all the areas listed in section 7.4;

7.7.4.6 For Senior Research Fellow, satisfactory research shall normally mean twenty two (22) refereed publications and at least one (1) filed IPR or one enterprise development or assessed equivalents;

7.7.4.7 The criteria for promotion to the Senior Research Fellow Grade shall be as for promotion to the Senior Lecturer grade except that in place of teaching, contributing to knowledge in the candidate's discipline through research and consultancy, innovation, creative work, enterprise development, and advisory work, shall be considered where such contributions are verifiable through documentary evidence.

7.7.5 Criteria for promotion to the Associate Professor/Associate Research Professor Grade

7.7.5.1 The criteria for promotion to the Associate Professor/Associate Research Professor grade are as follows:

7.7.5.1.1 For Associate Professor, at least twenty five (25) refereed research publications and at least one filed IPR or one (1) enterprise development or assessed equivalents which merit international recognition in the candidate's own field as testified by external assessors, examiners or reviewers of the candidate's work;

7.7.5.1.2 Satisfactory teaching;

7.7.5.1.3 For Associate Research Professor, at least thirty-five (35) refereed research publications and at least one filed IPR or one (1) enterprise development or assessed equivalents which merit international recognition in the candidate's own field as testified by external assessors, examiners or reviewers of the candidate's work; and

7.7.5.1.4 For both Associate Professor and Associate Research Professor Satisfactory Service to the University and Community.

7.7.6 Criteria for promotion to the Professorial/Research Professorial Grade

7.7.6.1 The criteria for promotion to the Professorial/Research Professorial Grade are as follows:

7.7.6.1.1 For Professorial Grade at least forty (40) refereed research publications and at least one filed IPR or one (1) enterprise development or assessed equivalents which merits international recognition in the candidate's own field as testified by external assessors, examiners or reviewers of the candidate's work, together with general quality of research which is of a level higher than that expected for promotion to the Associate Professor/Associate Research Professor Grade (Refer to Annexure 1).

7.7.6.1.2 For Research Professorial Grade, at least fifty (50) refereed research publications and at least two filed IPR or two (2) enterprise development or assessed equivalents which merit international recognition in the candidate's own field as testified by external assessors, examiners or reviewers of the candidate's work, together with general quality of research which is of a level higher than that expected for promotion to the Associate Professor/Associate Research Professor Grade (Refer to Annexure 1).

7.7.6.1.3 An international reputation for scholarship in the candidate's work; in some fields, academic books and substantial publications in scholarly journals;
7.7.6.1.4 Generally, cumulative and recognized international excellence;
7.7.6.1.5 Satisfactory teaching; and
7.7.6.1.6 Satisfactory service to the University and Community.

7.8 For the sake of clarity, and to emphasize the difference expected in the research record of candidates for promotion to Senior Lecturer/Senior Research Fellow; Associate Professor/Associate Research Professor and Professor/Research Professor, the following shall apply:

7.8.1 For promotion to Senior Lecturer/Senior Research Fellow, the quality of the research needs not be a level meriting the candidate's international recognition;

7.8.2 Whereas, for promotion to Associate Professor/Associate Research Professor the candidate's research must be adjudged to be a quality meriting international recognition but needs not have the element of sustained intellectual excellence expected of the research of a candidate for the professorship grade;

7.8.3 An Associate Professor/Associate Research Professor shall be required to demonstrate the sustained record required for promotion to a Professor/Research Professor in the period after his/her promotion or appointment as Associate Professor/Associate Research Professor.

7.9 Promotion Procedures for Academic Staff

7.9.1 Annually, in the early part of each calendar year, the Chairperson/Head of a Department shall place in the file of each academic member of that Department a report concerning the member's teaching, research, innovation, creative work, enterprise development and service to the University and Community; in the previous year. This report shall be available to the member of staff concerned, the Faculty/Institute and the appropriate members of the University administration.

7.9.2 The annual report filed in terms of section 7.10 shall form the basis of a Chairperson/Head of Department's report to the promotions or appropriate Academic Appointment Board for the purposes of applications for promotion and consideration of the granting of tenure (see section 8 of the Ordinance). The annual report shall have an additional section for innovation, consultancy, creative work and enterprise development.

7.9.3 A member of staff, with a PhD for Lecturer/Research Fellow and MSc/MA/MED for Technician/Researcher; BSc/BA/BED for Technical Assistant grades, wishing to apply for promotion should do so in respect to the annual invitation to apply for promotion issued by the University Administration. In the first instance, such applications shall be submitted to the Chairperson/Head of the member's Department who will submit them to the Promotion Committee through the Executive Dean of the Faculty/Executive Director of an Institute, a dossier containing the following documents:

7.9.3.1 The candidate's complete, up-to-date curriculum vitae;
7.9.3.2 The candidate's application for promotion;
7.9.3.3 The candidate's Research, Innovations and Industrialisation ;
7.9.3.4 Copies of each of the annual reports on the member since initial appointment or since the date of last promotion whichever is the most recent;
7.9.3.5 A summary of the Departmental Board's view on the candidate's application;
7.9.3.6 A report containing the Head of Department/Chairperson's recommendation on the candidate's application, which shall first have been made available to the candidate for comment, but in all other respects should be regarded as confidential to the Promotions Committee; and
7.9.3.7 Any comments on the Head of Department/Chairperson's reports which the candidate may wish to submit to the Promotions Committee.

7.9.4 In circumstances where the candidate expressly wishes the Departmental Board to consider any difference of opinion between him/herself and the Departmental Chairperson/Head of Department, he/she may request the Departmental Board to do so in which event the outcome of the Departmental Board's deliberations would be included in the dossier submitted through the Executive Director/Dean to the Promotions Committee.

7.9.5 The dossier for promotions in Associate Professor/Associate Research Professor/Professor/Research Professor grades shall also include a list, in order of preference, of assessors to whom the Promotions Committee may refer. This list shall be prepared by the Departmental Board and shall be as follows:

7.9.5.1 Associate Professor/Associate Research Professor or Professor/Research Professor: Six (6) assessors, at least four of whom shall be external assessors.

7.9.5.2 In the case where a candidate relies on Teaching or Service to the University and Community in claiming an assessment at the level of outstanding: three (3) assessors internal to the University but external to the Department shall be appointed to consider the candidate's credentials.

7.9.6 The Executive Dean of each Faculty/Executive Director of each Institute shall forward to the Promotions Committee the dossier received from the Head/Chairperson of Department on each candidate, together with a report and recommendation by him/herself as Executive Dean/Director. The Executive Dean/Director's report shall be confidential, except if he/she disagrees with the Head of Department/Chairperson's report, the Executive Dean/Director is required to make his report available to the Head of Department/Chairperson and the candidate, in which event the candidate's comments on the Executive Dean/Director report be included in the final dossier submitted to the Promotions Committee.

7.9.7 The Promotions Committee shall have final authority in all promotion matters. If a "prima facie" case for promotion is judged to exist then the Committee shall seek the opinion of external assessors, in which event the assessors' reports shall be confidential to the Promotions Committee.

7.9.8 In all instances where:

7.9.8.1 The Head/Chairperson of Department is a candidate for promotion, the Executive Director/Dean of the Faculty/Institute shall undertake the responsibilities of the Head/Chairperson of Department as laid down in this Ordinance;

7.9.8.2 The Executive Dean/Director of a Faculty/Institute is a candidate for promotion, the dossier shall be submitted by the Departmental Head/Chairperson direct to the Promotions Committee;

7.9.8.3 After one has been promoted, they have to wait for at least three years before they can be promoted to the next grade;

7.9.8.4 Normally members of staff with PhDs are eligible to apply for promotion;

7.9.8.5 A full Professor should have evidence of having supervised to completion at least five (5) students whilst it should be at least two (2) PhD students to completion for Associate Professorship.

7.9.8.6 No member of staff at the level of lecturer /Research Fellow/ Researcher / Chief Technician /Senior Technician/Technician shall be promoted while on probation.

7.9.8.7 A Lecturer/Research Fellow shall not be eligible for promotion to the level of Professor/Research Professor; however, he/she can be promoted to be Associate Professor/Associate Research Professor if all other conditions are satisfied. Consequently, a Senior Lecturer/Senior Research Fellow is eligible for promotion to Professor/Research Professor if all other conditions are satisfied.

7.10 Composition of the Annual Performance Report

7.10.1 The Annual Performance Report placed in each academic member of staff's personal file each year by the Head/Chairperson of the Department shall include an assessment of the member's teaching, research, innovations and industrialization as stipulated in items 7.4 and 7.5. The annual report shall have an additional section for innovation, consultancy, creative work and enterprise development.

7.10.2 **Report on Teaching/Consultancy**
In compiling the Section of the Annual Performance Report on a member's teaching the Head/Chairperson of the Department shall include an evaluation arrived at after implementing at least the following means of assessment:

7.10.2.1 **Peer Evaluation of Teaching/Consultancy**

The Departmental Board shall establish a panel of at least two academic members of the Department, with the responsibility of attending lectures/observing consultancy work of their colleagues on different occasions

and without prior notice to the person giving the lecture/doing consultancy work. The panel's report shall be considered by the Departmental Board and, after constructive, open discussion with the member of staff concerned, shall become a component in the assessment of each member of staff's teaching/consultancy. The evaluation of teaching/consultancy shall also focus on innovation, creative work and enterprise development.

7.10.2.2 **Evaluation by Students**

Students will be provided with an opportunity to complete an evaluation form in respect of each course. In the first instance, evaluation forms completed by students shall be submitted to the Chairperson of Department who shall discuss the views expressed with the member of staff concerned and the Departmental Board before including any formal comment in the Annual Performance Report.

7.10.2.3 **Assessment by External Examiners**

Any comments by External Examiners on a member's teaching shall be included in the Annual Performance Report.

7.10.2.4 **Seminars**

All Departments are required to encourage members of staff to hold seminars on relevant subjects of their choice to which staff and students should be invited to attend. Assessment of presentations in seminars and public lectures may become part of the evaluation of the members of staff's application for promotion.

7.10.2.5 **Report on Research, Innovation, Creative Work and Enterprise Development**

The section of the Annual Performance Report on a member's research shall comprise:
A report prepared by the member of staff, on his/her peer reviewed publications, innovation, creative work, filing or registration of IPR; enterprise development; conference papers, conference talks and the status of his/her work in progress during that year.

7.10.2.6 **Report on Service to the University and Community**

The section of the Annual Performance Report on a member's University service shall comprise:

7.10.2.6.1 A report by the Chairperson/ Head of Department/Unit on the member's University service during the year in question;
7.10.2.6.2 Any written comments by the member's Chairperson/Head of Department/Unit report made in terms of Section 7.10.1 above.

7.11 Criteria for granting promotions

7.11.1 For Senior Lecturer/Senior Research Fellow grade, four (4) hard copies and electronic version of files shall be submitted to the Promotions Committee. If the Department, represented by the Executive Director/Faculty Executive Dean, has a *prima facie* case supporting the promotion, the Promotions Committee shall either grant or decline the promotion based on content of the application and whether standard procedure was followed in submitting the application.

7.11.2 The dossier for promotions in Lecturer/Research Fellow grades shall also include a list, in order of preference, of assessors to whom the Promotions Committee may refer. This list shall be prepared by the Departmental Board and shall be as follows:

7.11.2.1. Associate Professor/Associate Research Professor or Professor/Research Professor: Six (6) assessors, at least four of whom shall be external assessors.

7.11.2.2. The criteria for granting promotions to Associate Professor/Associate Research Professor and Professor/Research Professor grades shall be based on recommendations from the six (6) assessors.

7.11.2.3. Six (6) files in hard and a soft copy format/electronic version, with an orderly arrangement of documents stated in section 7.9.3, shall be submitted to the Promotions Committee.

7.11.3 The Promotions Committee, which has sole discretion on the outcome of the application at this stage, shall make a decision to:

7.11.3.1 decline the application if (i) it does not meet the requirements stipulated for the grade or (ii) the file is in an unacceptable arrangement of documents referred to in section 7.9.3 or (iii) some vital documents are missing from the file or (iv) procedures stated in section 7.9 were not followed;

7.11.3.2 send the six (files) to assessors if the application meets the standards for the grade;

7.11.4 If a decision to send the applicant's six (6) hard or soft copy files to assessors is arrived at by the Promotions Committee, at least four (4) out of the six (6) hard or soft copy files shall be submitted to external assessors who should be at least Associate Professors for the Associate Professorship/Associate Research Professorship grade applications and Full Professors for Professorship/Research professorship grade applications.

7.11.5 The remainder after sending at least four (4) of the six (6) submitted hard or soft copy files shall be sent to internal assessors who should be at least Associate Professors for the Associate Professorship/Associate Research Professorship grade applications and Full Professors for Professorship/Research Professorship grade applications.

7.11.6 The Promotions Committee may grant promotion to Associate Professorship/Associate Research Professorship and Professorship/Research Professorship grades if at least four (4) of the six (6) assessors return a positive recommendation.

7.11.7 The Promotions Committee shall not grant promotion to Associate Professorship/Associate Research Professorship and Professorship/Research

Professorship grades if less than four (4) of the six (6) assessors return a positive recommendation.

7.11.8 A decision on the application shall remain pending for not more than three (3) years after the first sitting of the Promotions Committee.

8. TENURE

8.1 Only citizens and permanent residents of Zimbabwe, in accordance with Section 6 (2) of the University Act shall be appointed on permanent/full-time terms of service, which permit the granting of tenure in terms of this Ordinance, should the University so determine.

8.2 Persons who are not citizens or residents of Zimbabwe shall be appointed only on fixed-term contracts. In the event that they have completed one contract term successfully, such members shall be entitled to being considered for tenure, and therefore, further provisions of this section of the Ordinance do apply to such staff.

8.3 All academic staff appointed by the University shall be required to serve a probationary period before being considered for tenure provided that Appointment Boards, may, where it appears that the candidate so merits it, decide that a candidate be appointed at the level of Associate Professor/Associate Research Professor or Professor/Research Professor with immediate tenure.

8.4 The duration of the probationary period shall be:-

8.4.1 In the case of staff who, before the appointment, had tenure at another reputable University or institution of higher education, tenure maybe transferrable for those academic staff appointed from similar or recognizable institutions.

8.4.2 In the case of staff who had not previously had tenure: three (3) years.

8.4.3 Technicians: one (1) year

8.5 The criteria for the granting of tenure on completion of the requisite probationary period shall be: -

8.5.1 Satisfactory Teaching, Community Engagement, Research, Innovation and Industrialisation or for Research staff, contributing to knowledge as stipulated in item 7.4;

8.5.2 Satisfactory Research, that is, five (5) publications in refereed journals for Lecturer/Research fellow grade;

8.5.3 For Researchers and Technicians, satisfactory research has five (5) publications;

8.5.4 Satisfactory University Service.

8.6 The procedures for determining whether or not to grant tenure shall be as stipulated above for the consideration of applications for promotion and, therefore, the provisions of sections 7.4; 7.5; 7.6; 7.7; and 7.8 shall apply except that:

8.6.1 the initiation of the process shall be made by the University Administration at a time suitable in advance of the scheduled date of a member's probationary period of service.

8.7 If a member of staff is not granted tenure after the completion of his/her probationary period, the University may either:

8.7.1 Extend the probationary period by up to two further years with permission for the member of staff to apply for tenure before the expiry of that time; or terminate the member of staff's employment with the University.

8.7.2 For Researchers, extend the probationary period by up to two (2) further years with permission for the member of staff to apply for tenure before the expiry of that time; or terminate the member of staff's employment with the University.

8.7.3 For Technicians, extend the probationary period by one (1) year; or terminate the member of staff's employment with the University.

8.8 If at the end of the probationary period a member of staff is granted tenure, then the appointment shall be without time limit up to the age fixed by the University for retirement except that:

8.8.1 A member may resign his/her appointment by giving not less than six months' notice in writing.

8.8.2 For Technicians/Researchers a member may resign his/her appointment by giving not less than three months' notice in writing.

8.8.3 If a member fails to serve the required notice period, he/she shall be liable to pay the University an amount equivalent to his/her gross salary for the period not served.

8.8.4 A member may not resign while he/she is on Sabbatical Leave or on Contact Visit.

8.8.5 A member who resigns while he/she is on Sabbatical Leave or on Contact Visit shall be required to repay the University Sabbatical airfares received, salary received whilst on Sabbatical or Contact Visit, University contributions to member's pension scheme and six months' salary in lieu of notice.

8.8.6 Any period preceding or spent on such leave or visit shall not be counted as a period or portion of a period of notice.

8.8.7 The University Council may terminate the appointment for "good cause" by giving the member of staff not less than six months' notice or paying the member's salary in lieu thereof. This should be done in accordance with the University Code of Conduct.

8.9 Before terminating an appointment in terms of section 8.8.7 above, the Council: -

8.9.1. Shall inform the member in writing of the matters alleged against him/her and give the member the opportunity of replying in writing to those charges;

8.9.2 May, and if so requested by the member of staff, before considering such dismissal, refer the case to the Staff Disciplinary Committee established in terms of section 26 of the University Act.

8.10 Good Cause for the termination of an appointment in terms of section 8.8.7 means:

8.10.1 Conviction for any offence which the Council considers to be such as to render the person concerned unfit for the execution of the duties of his/her office,

8.10.2 Any physical or mental incapacity which the Council considers to be such as to render the person concerned unfit to hold his/her office,

8.10.3 Conduct of an immoral, scandalous or disgraceful nature which the Council considers to be such as to render the person concerned unfit to continue to hold his/her office,

8.10.4 Conduct which the Council considers to be such as to constitute failure or inability of the person concerned to perform the duties of his/her office or to comply with the conditions of tenure of his/her office.

ADMINISTRATIVE STAFF: APPOINTMENTS, GRADES, PROMOTION, GRADING AND TENURE ORDINANCE (GRADES 4 – 8), ORDINANCE 6

ARRANGEMENT OF SECTIONS

1. Interpretation
2. Title and date of commencement
3. Objectives
4. Structure of Administrative Staff (Grades 4 – 8)
5. Grading and Initial Appointment
6. Promotion
7. Tenure

PREAMBLE

In terms of section 29 (3) (h) of the University Act No.4 of 1999, the Council of the Midlands State University, with the approval of the Minister of Higher and Tertiary Education, Innovation, Science and Technology Development or such other Minister to whom the President may from time to time assign the administration of the Midlands State University Act, on the recommendation of the University Senate, makes the following Ordinance, which shall be administered in conjunction with other labour legislations and in-house policies and procedures.

1. INTERPRETATION

(i) In this Ordinance – **"administrative staff"** refers to all persons employed by the University in the grades 4 to 8 ranges and are not members of the teaching staff.

(ii) **'Promotion'** refers to upward movement of an employee from one job to another higher one or job category, with increase in salary, status and responsibilities.

(iii) **'Internal Applicant'** refers to a person who at the time of recruitment for the vacancy is employed by the University on permanent conditions with tenure having being duly granted by the responsible committee.

(iv) **"Substantive appointment"** means an appointment made under the provisions of these rules after due selection by prescribed methods of recruitment and includes an appointment made on probation followed by confirmation on completion of the period of probation.

(v) **University Service** refers to the period served in full time employment of the University

2. This Ordinance is cited as the Administrative Staff Ordinance (Grades 4-8) Number 6 of 2024 and shall take effect from 1 February 2024

3. OBJECTIVES

In making this ordinance, Council seeks to: -

3.1 Attract and retain administrative staff whose performance demonstrates particular merit in administrative duties (and related duties), service and leadership in the University and professional practice including service to the community;

3.2 provide an objective, fair systematic method of assessment in order to ensure a consistent application of standards for the appointment and promotion of staff.

4. STRUCTURE OF ADMINISTRATIVE STAFF GRADES (4 – 8)

4.1 The following minimum entry qualifications shall apply for this category of staff within the University:

Grade	Minimum Entry Qualifications

4	Masters
5	Masters - Administrative Bachelors-Technical
6	Masters
7	Bachelor's degree
8	Bachelor's degree-Administrative HND-Technical

4.2 The salary scale applicable to each grade is subject to approval by the University Council.

4.3 Minimum entry experience shall be determined by job specific requirements.

4.4 The University Council, or its authorized Committees shall have sole discretion in the University, and in so doing, may designate that posts be established at any of the grades listed in section 4.1.

4.5 **Categories of Grades**

Grade	Designations
4	Deputy Registrar; Deputy Bursar; Deputy Librarian; Chief Internal Auditor; Director-Marketing and Communications; ITS Director; Director of Projects, Procurement Director, University Engineer, Director-Health Services, Director-Performance Contracting, Director-University Secretariat; Legal Officer; Director-Central Services, Amenities and Maintenance
5	Deputy Director; Central Records and Archives Manager; Examinations Manager; Risk Officer; Quantity Surveyor; Systems Manager; Network Manager; User Support Manager; Business Solutions Manager; e-Learning Manager; Projects Manager; Architect; Production Manager; Food and Beverages Manager; Pharmaceuticals Manager; Enterprise Sales and Marketing Manager
6	Examination Officer; Principal Assistant Registrar; WRL Coordinator; Sub Librarian; Catering Services Manager; Procurement Manager; Senior Ecumenical Chaplain; Sports & Recreation Coordinator; Farms Coordinator; Senior Internal Auditor; Chief Business Analyst; ICT Projects Officer; Chief Software Developer; Chief ICT Security Officer, Webmaster; Marketing/Communications Manager; Sports Coordinator; Campus Coordinator; Counselling Coordinator; International Students and Alumni Officer; Residences Coordinator; Administrative Matron; Principal Assistant Bursar; Senior Internal Auditor; Assistant Legal Officer;

7	Assistant/Senior Assistant Registrar; Assistant/Senior Assistant Bursar; Internal Auditor; Industrial Liaison Officer; Computer Liaison Officer; Network/Systems/Software/Hardware Engineer; Systems Analyst; Server Administrator; Helpdesk Supervisor; Business Analyst; Ethical Hacker; Software Quality Assurance Officer; Senior Assistant Librarian; Assistant Catering Services Manager; Property Management Officer, University Counselor; Nurse- in- Charge; Production Supervisor; Sales and Marketing Officer; Senior HR Officer;
8	Administrative Assistant/Senior Administrative Assistant; Communications Officer; Head Chef; Personal Assistant; Graphic Designer; Media Officer; Communications Officer; Events Coordinator; Transport & Logistics Officer; Clothing Supervisor; Artisan Supervisor; Printer; Central Services Campus Supervisor; Deputy Chaplain; Deputy Sports Coordinator; Full-Time Warden; Head Chefs; Environmental Health Technician; Internal Audit Assistant; Procurement Specialist; HR Officer; Industrial Relations Officer; Welfare Officer; HR Systems Coordinator

5. GRADING AND APPOINTMENT

5.1 On initial appointment to the University Administrative staff category, a successful candidate shall be graded according to his or her qualifications and experience.

5.2 There shall be only one Administrative Staff Grading and Salary Structure, and the same salary scale shall apply to all posts in all relevant sections within the University.

5.3 **Qualifications**

5.3.1 The basic qualification for appointment to the University's Administrative staff is normally a first degree but a Master's degree and/or Professional Practicing Certificates may be required in areas as approved by the University. This is subject to review from time to time.

5.4 **Experience**

5.4.1 In granting recognition to relevant postgraduate experience, the University shall:

5.4.1.1 Make no distinction to relevance between professional and non-professional experience.

5.4.1.2 Recognise in full, the time spent as a full-time member of staff of a reputable university or tertiary organisation.

6 PROMOTION

6.1 Promotion is not an entitlement but shall depend on the availability of a vacancy.

6.2 The normal recruitment and selection procedure would apply when filling a higher-level vacant position. In this instance, the normal Staff Appointment Board would be convened to interview candidates for any vacant position.

6.3 The qualifying candidate shall be appointed accordingly.

7.TENURE

7.1 On joining the University, each employee shall enter into a three (3) month probationary period. During this period, either party may terminate the contract by providing a two (2) week notice period.

7.2 The criteria for the granting of tenure on completion of the requisite probationary period shall be: -

7.2.1 Satisfactory performance of duties as assigned by the respective head of department;

7.2.2 Satisfactory University Service;

7.2.3 Outstanding performance in the areas of innovation, creative work and enterprise development.

7.3 The appropriate final authority shall be the Non-Teaching Staff Promotions and Tenure Board.

JUNIOR NON-TEACHING STAFF ORDINANCE (GRADES 9 – 14)

ARRANGEMENT OF SECTIONS

1. Interpretation
2. Title and date of commencement
3. Objectives
4. Structure of junior non-teaching staff grades
5. Appointment Procedures
6. Tenure

PREAMBLE

In terms of section 29 (3) (h) of the University Act No.4 of 1999, the Council of the Midlands State University, with the approval of the Minister of Higher and Tertiary Education, Innovation, Science and Technology Development or such other Minister to whom the President may from time to time assign the administration of the Midlands State University Act on the recommendation of the University Senate, makes the following Ordinance. This Ordinance shall be read in conjunction with other labour legislations and in-house policies and procedures:

1. INTERPRETATION

(i) In this Ordinance – **'junior non-teaching staff'** refers to all persons employed by the University in the grade 9 to 14 ranges and are not members of the teaching staff excluding those on a probation period.

(ii) **'Promotion'** refers to upward movement of an employee from one job to another higher one or job category, with increase in salary, status and responsibilities.

(iii) **'Internal Applicant'** refers to a person who at the time of recruitment for the vacancy is employed by the University on permanent conditions with tenure having being duly granted by the responsible committee.

(iv) '**External Applicant'** refers to a person whom at the time of recruitment for a vacancy is not employed by the University or otherwise employed by the University on fixed-term conditions of service.

(v) **"Substantive appointment"** means an appointment made under the provisions of these rules after due selection by prescribed methods of recruitment and includes an appointment made on probation followed by confirmation on completion of the period of probation.

2. This Ordinance is cited as the Junior Non-Teaching Staff Ordinance Number 7 of 2024 and shall take effect from 1 February 2024

3. OBJECTIVES

In making this ordinance, Council seeks to:-

a. Hire the most-qualified candidate to fill an available position in a cost effective and timely manner.
b. Provide an objective, fair systematic method of assessment in order to ensure a consistent application of standards for the appointment and promotion of staff.

4. STRUCTURE OF JUNIOR NON-TEACHING STAFF GRADES (9 – 14)

a. The following structure of grades shall apply for junior non-teaching staff within the University:

LISTING OF NON TECHNICAL POSTS

Grade	Post
9	Supervisor –landscaping/ Cleaning, Chief Accounting Assistant, Chief Library Assistant, Records Supervisor, Internal Audit Assistant, Assistant IT Auditor, Procurement Expeditor, Chief Data

	Capture Clerk, Student Developer, Procurement Officer, Registry Supervisor, Cook,
10	Accounting Assistant, Binder, Senior/ Chief Secretary, Driver, Senior Library Assistant I, Security Officer, Senior Data Capture Clerk, Driver, Supervisor, Drug Stores Controller, Senior Waiter, Records Clerk
11	Stores Clerk, Cook, Senior Waiter, Senior Library Assistant II, Secretary, Receptionist/ Switchboard Operator, Ambulance Technician, Health Service Clerk, Accounts Clerk, Cashier
12	Security Sergeant, Data Capture Clerk, Records Clerk, Supervisor-Agriculture Practice, Waiter, Blockman, Janitor, Gym Attendant, HR Clerk, Sergeant
13	Security Corporal, Library Technical Assistant, Nurse Aide, Marketing Clerk, Office Orderly, Data Capture Clerk –Bursar's, Sewage Pond Attendant, Charge hand, Dispensary Assistant, Lifesaver
14	Artisan Assistant, Welder, Fuel Attendant, Messenger, House Maid, Office Orderly, Security Guard

LISTING OF TECHNICAL POSTS

Grade	Post
9	Electrician, Plumber, Mechanic, Bricklayer, Technician, Nurse in Charge, Telephone Technician
10	Braillist,
11	Ambulance Technician
12	Special Needs Assistant
13	Reprographic Technician, Disability Technician Assistant, Lab Assistant

5. GRADING AND APPOINTMENT

a. On initial appointment to the University's Junior Non-Teaching Staff compliment, a successful candidate shall be graded according to their qualifications and experience.
b. There shall be only one Junior Non-Teaching Staff Grading and Salary Structure, and the same salary scale shall apply to all posts in all relevant sections within the University.

7 PROMOTION

7.1 Promotion is not an entitlement but shall depend on the availability of a vacancy.
7.2 The normal recruitment and selection procedure shall apply when filling a vacant position. In this instance, the normal Appointments Board would be convened to interview candidates for any vacant position.
7.3 The qualifying candidate shall be appointed accordingly.

8 TENURE

8.1 On joining the University, each employee shall enter into a three (3) month probationary period. During this period, either party may terminate the contract by providing a two (2) week notice period.
8.2 The criteria for the granting of tenure on completion of the requisite probationary period shall be:-
8.2.1 Satisfactory performance of duties as assigned by the respective head of department.
8.2.2 Satisfactory University Service.
8.3 The appropriate final authority shall be the Non-Teaching Staff Promotions and Tenure Board, Junior Non - Teaching Staff.

THE RULES OF STUDENT CONDUCT AND DISCIPLINE ORDINANCE NO. 2, 2000

ORDINANCE NO. 2

In terms of Section 29 (iii) (a) of the Midlands State University Act (Chapter 25:21 of 1999), the Council of the State University in the Midlands with the approval of the Minister of Higher Education, makes the following Ordinance:-

1. The Ordinance may be cited as "The Rules of Student Conduct and Discipline Ordinance, 2000, Ordinance No. 2".
2. It constitutes the first set of Rules of Student Conduct and Discipline for the Midlands State University.
3. The Rules of Student Conduct and Discipline set out in the Schedule to this Ordinance shall come into effect from the 27th of March 2000.
4. In terms of Section 27 (4) of the University Act, the Student Disciplinary Committee hereby delegates to the Officials referred to in the Rules of Student Conduct and Discipline the power of investigating and exercising disciplinary authority in respect of misconduct by any student to the extent and in the manner set out in the Rules of Student Conduct and Discipline.
5. In terms of Section 27 (6) of the University Act, the Student Disciplinary Committee may recommend to the Vice-Chancellor the imposition of one or more of the following punishments:-
   - 5.1 expulsion or suspension from the University.
   - 5.2 the withdrawal of any academic or University privilege, benefit, right of facility other than to follow courses of instruction and present himself for examination.
   - 5.3 the imposition of a fine which shall be determined by the University from time to time, which fine may be deducted from any allowance payable to the student and shall be paid to the University
   - 5.4 a censure or rimand.
   - 5.5 such other penalty as may be provided for by or in terms of the statutes

- **SCHEDULES**

1. INTERPRETATION

The University Officers charged with the administration of these rules will at all times seek to implement the letter and spirit of the University Act and will in particular have regard to the following principles:-

- 1.1 the University is a society in which high standards of communal life must be established and maintained for the benefit of both present and future members of the University.
- 1.2 a high level of personal integrity and a development sense of responsibility towards others are as important to the University as outstanding scholastic achievement.
- 1.3 a proper concern for the reputation of the University and what it ought to stand for makes it incumbent upon its members to live decent and ordered lives.
- 1.4 individual or collective action by members of the University which constitutes a breach of these rules may require to be punished notwithstanding that the motive or goal of such action was commendable in the belief of such members.

2. UNDERTAKING AT REGISTRATION

When registering as a member of the University a student shall be given a copy of these rules and shall sign a statement in which he/she acknowledges that he/she has been furnished with the rules, and he/she undertakes to conduct himself/herself while a student of the University in accordance therewith and with any amendments duly made thereto.

3. STUDENT CONDUCT

3.1 No student of the University shall:

3.1.1 use the University premises contrary to University Regulations, Residence, Faculty or Departmental rules or do any act reasonably likely to cause such misuse.

3.1.2 damage or deface any property of the University or do any act reasonably likely to cause damage or defacement thereto.

3.1.3 disrupt teaching, study, research or administrative work, or prevent any member of the University or its staff from carrying on his/her study or work, or do any act reasonably likely to cause disruption or prevention and

3.1.4 engage in any conduct whether on or off the campus which is or is reasonably likely to be harmful to the interests of the University, members of the University staff or students.

3.2 The following would be regarded by the University as stances of breaches of the rule contained in 3.1 above:

3.2.1 displaying violence by word or act towards any member of the University, whether academic or administrative staff or student, or a guest of the University, or any visitor to the University or in any way intimidating or obstructing the free movement of such a member, guest or visitor.

3.2.2 disrupting or seeking to disrupt any proper function of the University whether it be an official function, Council meeting, Senate meeting, Faculty or Committee meeting, lecture, teaching session, the function of any University Society or day-today administrative activity.

3.2.3 Seeking to prevent a speaker invited by any section of this University Community from lawfully expressing his/her views.

3.3 Students are informed that:-

3.3.1 if a group of students form a common intention to commit certain acts and to assist each other in their commission, and in due course a breach of these rules is committed by one or some of the group, then each member of the group may be held to have committed that breach of the rules, who foresaw that the breach would occur, or who must, in the view of the Student Disciplinary Committee, have foreseen that it would occur. A member of such a group can avoid this happening to him/her by taking clear and unequivocal steps, before such a breach is committed, to show that he/she dissociates himself from the acts of the group with whom he/she has so far been associating.

3.3.2 Where a number of students have committed a breach of these rules and only one or more of these students can be identified, the University will not hesitate to take disciplinary action against those students who can be identified and against whom there is sufficient evidence to warrant investigation.

3.3.3 Where a student commits an act which is both an offence according to the laws of the country and one which after investigation appears to be a breach of the disciplinary rules of the University, the University may punish such a student notwithstanding that he/she is prosecuted and/or punished by the courts of the country.

3.4 A student shall obey any rules made from time to time by the Vice-Chancellor and shall further obey all instructions given by the Vice-Chancellor, the Proctors, and all those persons whom the Vice-Chancellor has charged to assist him in the maintenance of discipline, and in this regard:-

3.4.1 any Executive Dean, Deputy Dean, Chairperson of Department, Librarian, Deputy Librarian or member of the senior administrative staff may order any member of a gathering of students which is committing, or whose activities are likely to lead to a breach of Rule 3 above, to disperse, and may further order any such member to furnish his/her full name or accompany the member of staff for an interview with the Vice-Chancellor or a Proctor, or give both such orders. For the purpose of this section, Senior Administrative Staff shall include the Registrar, the Deputy Registrar (Academic), the Bursar, the Senior Assistant Registrar, Director of Works and Estates, Executive Dean of Students, Deputy Dean of Students, Assistant to the Vice-Chancellor, and Wardens of Halls of Residence.

3.4.2 if a student misbehaves in a lecture or teaching session or interferes with the conduct of a lecture or teaching session, the member of staff conducting such lecture or teaching or teaching session may order the student to leave or to cease interference.

3.4.3 Failing to comply with any order as above constitutes a serious offence.

4. THE POWERS OF UNIVERSITY PROCTORS

4.1 The Vice-Chancellor shall from time to time appoint as many Proctors as he deems necessary from among the academic staff who are not members of the Student Disciplinary Panel or Wardens. One Proctor shall be known as the Senior Proctor with responsibility for organizing and supervising the work of the proctors and shall be a member of the presentation of cases before the Student Disciplinary Committee.

4.2 A Proctor shall be charged with ensuring the proper observance of these rules by students on or off the University site and to this end shall, in addition to his power under Rule 3.4 have the following powers:-

4.2.1 to receive and investigate reports of students misconduct.

4.2.2 to summon any student to appear before him/her either to answer a charge or complaint against him/her or to answer questions with regard to any matter under investigation by him/her.

4.2.3 to proceed in the absence of a student who in the opinion of the Proctor has been duly summoned but has failed to appear;

4.2.4 to recommend to the Registrar that a student be summoned to appear before the Student Disciplinary Committee to answer a charge or complaint against him/her or to answer questions with regard to any matter under investigation by the Student Disciplinary Committee or Proctors.

4.2.5 to reprimand a student.

4.2.6 to impose a fine on a student which shall be determined by the University from time to time.

4.2.7 to order a student to pay to the University the amount of any financial loss caused to the University by such student.

4.2.8 to withdraw an existing student privilege, other than residence, for a period not exceeding one (1) academic term.

4.3 when a resident of a Hall of Residence makes a report to a Proctor of alleged misconduct, the Proctor shall communicate such a report to the Warden of the Hall of Residence concerned before taking action against the resident.

5. POWERS OF WARDENS

5.1 A Warden of a hall of Residence shall have the power to investigate any breach of these rules by a student of his/her Hall of Residence committed within any Hall of Residence and to make any of the following orders in respect of such student adjusted by him to have committed a breach of these rules:-

5.1.1 to reprimand a student.

5.1.2 to withdraw an existing resident student privilege, eg exeat.

5.1.3 to impose a fine which shall be determined by the University from time to time.

5.1.4 to order a student to pay the University the amount of any financial loss caused to the University. Such an order may be made on the person or persons who caused the loss or where identity cannot be established, on members of the residence in which the loss was sustained.

5.2 Where a Warden wishes to impose penalties 5.1.3 or 5.1.4 above, he/ she shall first furnish the Senior Proctor with a full report concerning the alleged offence and the proposed penalty. On receipt of this report the Senior Proctor may, either:-

5.2.1 confirm the proposed penalty and order of the Warden and in the event of variation, order the warden to execute such varied order or

5.2.2 institute a further investigation of the matter before making such order at the conclusion as he/she deems fit or

5.2.3 recommend to the Registrar that a Student Disciplinary Committee be convened to examine the case.

5.3 A Warden of a Hall of Residence and the Warden of Non –Resident students shall have the power to report any matter of student conduct to the Proctor or through the Registrar to the Student Disciplinary Committee.

5.4 Where a warden has imposed any of the penalties set out in Section 5.1.1 to 5.1.4 above on a student, he/she shall submit a report to the Senior Proctor, Registrar and Executive Dean of Students.

6. RULES FOR HALLS OF RESIDENCE

6.1 ***General***

A resident shall obey all rules made and instructions given by the Warden, Deputy and Sub Warden of the residence and shall refrain from conduct which:

6.1.1 may bring discredit upon his/her hall of residence or

6.1.2 is prejudicial to the welfare of other residents of the hall.

6.2 ***Powers of Hall Committee Members***

Members of a Hall Committee shall have the power, to investigate, and if necessary reprimand residents for any infringements of the rules contained in this section, and report such investigation or reprimand to the Warden.

6.3 ***Damage to Hall Property***

A resident shall be liable to compensate the University in full for any damage caused by him/her to University property. Damage caused to a study – bedroom shall be presumed to have been caused by the resident to whom such a room has been allocated unless the contrary is proved.

6.4 ***Fire***

A resident having knowledge of the outbreak of fire in, or adjacent to Hall premises shall as soon as possible:

6.4.1 inform the Warden, Deputy or Sub-Warden

6.4.2 summon the Municipal Fire-Brigade

6.4.3 inform the Director of Works and Estates.

6.5 ***Vacation Residence***

6.5.1 a resident occupy a study-bedroom during University vacation with the prior written authority of the Executive Dean of Students on the recommendation of the Executive Dean of the appropriate Faculty and the Warden.

6.5.2 applications for vacation residence must be submitted through the prescribed channels and on the prescribed form.

6.5.3 a resident granted leave to reside in a Hall of residence during vacation who no longer wishes to avail himself/herself of this privilege shall furnish the Executive Dean of Students with at least 3 days' written notice of such a fact. Omission to do so will render a resident liable to monetary penalty.

6.6 ***Exeats***

A resident may not absent himself/herself from a hall overnight, say on Saturday nights without previously informing and obtaining the permission of the Warden. To be absent from a Hall for two or more consecutive nights, he/she requires the leave of the Executive Dean of his/her Faculty in addition to that of the Warden. A resident of a Hall may be required to be in the Hall every night by such time as may be laid down in Regulations unless prior permission has been granted by the Warden or Sub-Warden to return to the hall at a late hour.

6.7 ***Visitors***

6.7.1 Resident students' parent may visit them in their rooms from 1000 to 2230 hours.

6.7.2 Male students may be entertained in female halls and female students may be entertained in male halls between the following hours:

Monday to Friday	1200 to 2230 hours
Saturday	1200 to 0000 hours
Sunday	1030 to 2230 hours

6.7.3 Outside the prescribed visiting hours, all parts to the halls except the Common Rooms and entrance foyers are out of bounds to members of the opposite sex.

6.7.4 Special arrangements may be made by application to the Warden of the Hall concerned.

6.7.5 These provisions apply to all students – undergraduate and Post-Graduate.

6.7.6 No visitor or non-resident student may make unauthorized use of accommodation in hall of residence. Students introducing visitors or non- resident students to the Hall may be held responsible by the Warden for the conduct of such visitors, and non-resident students making unauthorized use of the Hall facilities shall be guilty of misconduct.

6.8 ***Withdrawal from Residence***

If a student should leave the University or withdraw from residence before the end of the session for which he/she has been admitted, fees already paid by him/her are not returnable and the balance or all fees for that session not yet paid become immediately payable, except that a student who gives proper notice before the end of a semester that he/she wishes to vacate residence for the remainder of the session may be refunded the balance of residence fees in respect of the remaining whole terms of the session.

6.9 ***Loss of Valuables***

A resident shall report as soon as possible to the Warden the loss of any article from hall premises in circumstances raising any suspicion of theft.

6.10 ***Relationship of Residents and Staff***

A resident shall not require a member of the organization providing cleaning services to perform a service outside the scope of his/her normal employment duties.

6.11 ***Illness***

A resident who is confined to bed shall, so far as he/she is able, ensure that his/her illness is reported to the University Student Health Services.

7. RULES FOR THE USE OF VEHICLES

7.1 A student wishing to keep or use a motor vehicle including a motorcycle, motor scooter or motorized bicycle within the boundaries of the University site shall notify the Registrar in writing on the form prescribed.

7.2 Save for the prior written permission of the Registrar, a student shall not, within university grounds:

7.2.1 Park a vehicle in a parking place marked "for staff and visitors."
7.2.2 park a vehicle in any place at which parking by any person has been prohibited.
7.2.3 bring a vehicle within any university building.
7.2.4 ride or drive a vehicle on any part other than roads, tracks or parking places.
7.2.5 leave a vehicle in an unusable condition for a period longer than is reasonably required to effect necessary repairs.

7.3 Whenever a vehicle registered with the university is driven, ridden or parked in contravention of the rules set out in section 7.2 it shall be presumed that it was so driven, ridden or parked by the person in whose name the vehicle has been registered with the University unless the contrary is proved.

7.4 ***Penalties***

7.4.1 The wardens, proctors and such other persons so authorized by the Vice Chancellor shall have power to investigate breaches of the Rules contained in this section and to impose a fine which shall be determined by the University from time to time.
7.4.2 In the case of a third or subsequent offence, the name of the offender, with particulars of his/her previous offences under this section, shall be reported to the proctors who shall exercise appropriate authority in terms of rule 4

8. RULES OF PROCEDURE IN DISCIPLINARY PROCEEDINGS BEFORE THE STUDENT DISCIPLINARY COMMITTEE.

8.1 The Chairperson of the Committee shall regulate proceedings notwithstanding, best fitted to do substantial justice and at all times in accordance with the principles of natural justice. More particularly a student charged with the breach of the Rules of Student Conduct and Discipline shall at any investigation thereof before the Committee and with no derogation of his/her rights in terms of section 27(v) of the University Act:-

8.1.1 be furnished with a full and fair opportunity to meet such allegations if he/she so desires;
8.1.2 be permitted to present any relevant facts or call any witness capable of giving testimony relevant to the investigation;
8.1.3 be permitted to put questions to witnesses save those which are irrelevant, frivolous or vexatious;
8.1.4 be permitted to be present at times save when the committee is deliberating upon its decisions of the matter;
8.1.5 be advised as fully and clearly as possible of the Committee's decisions or recommendations.

8.2 The Proctors and Wardens shall conduct any proceedings before the Student Disciplinary Committee in accordance with Rule 8.1 save the provisions of section 27(vii) of the University Act will not be applicable.

8.3 In the event of the Legal Proctor conducting an investigation before the Student Disciplinary Committee it shall further be his/her duty;

8.3.1 to elicit all evidence brought to his/her attention which is relevant to the investigation and admissible, whether favourable to or prejudicial to the student whose conduct is the subject thereof;

8.3.2 to ensure the conduct of the investigation is consistent with the principles of natural justice;

8.3.3 if so required by the Committee, to advise the Committee as to the issues which they have to decide and as to any point of law or procedure so as to ensure the conduct of the investigation is consistent with the principles of natural justice.

8.3.4 to be absent at all times from the deliberations of the Committee upon its final judgment.

8.4 A notice to a student summoning him/her to appear before the Committee for investigation of an alleged breach of the University Rules of Student Conduct and Discipline shall be contained in a letter addressed to him/her and advising him/her of:

8.4.1 the place at which he/she is to attend;

8.4.2 the date and time at which he/she is to attend, provided that such date shall be not less than five (5) days after the date which such notice is received;

8.4.3 the rule which he/she is alleged to have contravened and full particulars of his/her alleged contravention.

8.4.4 his/her right to be accompanied and represented before the committee by a legal practitioner.

8.4.5 his/her right to call witnesses to attend and give any relevant testimony on his/her behalf before the Committee.

8.4.6 the right to furnish to the Proctors in advance of the investigation any information which he/she deems relevant to the investigation and which he/she wishes to have given due consideration.

8.5 A member of the Committee, who has acquired, other than in the course of his/her University life, knowledge of evidence in an investigation of misconduct to be held before the Committee, shall not participate in such investigation.

8.6 The Committee shall only find a student to have committed a breach of the Rules of Student Conduct and Discipline when it is satisfied beyond reasonable doubt that the student has committed such breach.

8.7 In the event of the Committee finding a student to have committed a breach of the Rules, either on the student's own admission or at the conclusion of an investigation, it shall, before determining the punishment it should impose or the terms of its recommendation to the Vice-Chancellor, permit such a student a full opportunity to make a statement or produce evidence which he/she wishes to be taken into consideration in mitigation of his/her punishment.

8.8 The Chairperson of the Committee shall keep full note of any proceedings before it, but this should not be a verbatim record.

8.9 As soon as possible, after the conclusion of a Student Disciplinary Committee hearing, the Chairperson shall convey a full report to the Registrar who shall inform the student concerned of the Committee's judgment.

8.10 The Chairperson of the Committee shall keep full note of any proceedings before it, but this should not be a verbatim record.

8.11 As soon as possible, after the conclusion of a Student Disciplinary Committee hearing, the Chairperson shall convey a full report to the Registrar who shall inform the student concerned of the Committee's judgment.

INNOVATION AND INDUSTRIALISATION ORDINANCE NUMBER 8 OF 2023

ARRANGEMENT OF SECTIONS

1. Preamble
2. Title
3. Interpretation
4. Application
5. Objectives
6. Structure of the Staff
7. Promotion
8. Probation/Tenure
9. Administration of Innovations
10. Transfer of Technology
11. Students in the Innovation Hub

1. PREAMBLE

In terms of section 29 (3) (h) of the University Act No.4 of 1999, the Council of the Midlands State University, with the approval of the Minister of Higher and Tertiary Education, Innovation, Science and Technology Development, or such other Minister to whom the President may from time to time assign the administration of the Midlands State University Act on the recommendation of the University Senate, makes the following Ordinance:

2. TITLE

This ordinance may be cited as the Innovation and Industrialisation Ordinance Number 8 of 2023.

3. INTERPRETATION

In this Ordinance:

3.1. **Commercialisation** shall refer to the transfer of intellectual property to bring or to create new products or services to the market; create awareness, establish distribution channels, and promote the trial and purchase of the goods and services.

3.2. **Community innovators** shall refer to innovators who are outside the university and are from the community.

3.3. **Contracted research** shall refer to legally bound research between the University and external partners. This research is conducted under agreed terms and conditions such as *inter alia* funding and ownership of research outputs.

3.4. **Enterprise shall refer to any profitable commercial venture.**

3.5. **Heritage-Based Education 5.0** shall refer to an education system that includes teaching, research, community service, innovation, and industrialisation with a heritage-based philosophy that emphasises industrialisation through innovation and knowledge of Zimbabwe's natural resources.

3.6. **Industrialisation** shall refer to the setting up of industries and processes that shall translate innovations and research outputs into goods and services.

3.7. **Innovation** shall refer to using relevant knowledge and resources to convert ideas into new products, processes, or practices or improve the existing product(s), process, or practice.

3.8. **Innovator** shall refer to an individual, company, or group of individuals who create new methods, ideas, or products.

3.9. **Intellectual Property (IP)** shall refer to mental creations such as inventions, literary and artistic works, designs and commercial symbols, names, and images. Patents, trademarks, trade secrets, utility models, industrial designs, and copyright laws can all protect intellectual property in various forms.

3.10. **Invention** means the act of putting ideas or objects together in a unique and novel way to create something that did not exist before.

3.11. **Spin-out** refers to any venture that springs out of knowledge capital created by any of the University's internal stakeholders. It can be a start-up, spin-off, or joint venture.

3.11.1. **Joint Venture** shall mean a commercial arrangement between two or more participants who agree to do business together wherein the parties create an entity by contributing equity and share the revenues, expenses, and control of the enterprise in proportion to their capital contribution, e.g., Special Purpose Vehicle (SPV)

3.11.2. Special Purpose Vehicle shall refer to a company, which comprises the university and other shareholders, registered with a specific mandate to undertake a specific project of commercial interest.

3.11.3. **Spin-off** company shall mean a business created by the University whose technology is owned by the University, financed by the University, and managed by the University.

3.11.4. **Start-up** company shall mean a business in the first stages of operations.

3.12. **Technology** shall refer to the application of scientific knowledge for practical purposes.

3.13. **Technology transfer (TT)** shall be defined as the transfer of new technologies, inventions/creations, and discoveries from university/research institutions to parties capable of commercialization for the framework.

3.14. **University** shall mean Midlands State University.

3.15. **University Company** shall mean a company wholly owned by the University.

4. APPLICATION

This ordinance applies to all innovators including University students and staff members, Faculties, Institutes, Centres as well as the community.

5. OBJECTIVES

In making this Ordinance, Council seeks to:

5.1 Promote a culture of research, innovation, and industrialisation within the University and beyond by supporting innovative initiatives by University staff members, students, and the community.

5.2 Provide for the commercialisation of university intellectual property into commercial industrial ventures or enterprises.

5.3 Enable the conversion of innovations and inventions from university and community research and development outputs into goods and services.

5.4 Foster collaboration with the community, industry, and commerce thus enabling technology development and transfer.

5.5 Promote the creation and sustainable growth of Start-ups, Spin-offs, Spin-off companies, and Special Purpose Vehicles.

5.6 Protect intellectual property by providing adequate legal protection against unauthorized use.

5.7 Create a framework for promoting and protecting innovators.

6. STRUCTURE OF RESEARCH AND INNOVATION AND ENTERPRISE DIVISIONS

6.1. The University shall establish a Research and Innovation Division and an Enterprise Division to serve as integrating factors and conduits for releasing University innovation outputs that lead to commercialisation and industrialisation through the following:

a) recruit and assist innovators in their innovation process and in starting new businesses;
b) engage in the development of products, processes, and services; c) facilitate technology transfer and promote enterprises;
c) create a conducive environment for inventions and innovations;
d) ensure innovators are promoted and facilitated;
e) stimulate and manage the flow of knowledge and technology across innovators and markets; and
f) facilitate application for intellectual property rights
g) facilitate the acquisition of relevant technology

6.2. The Research and Innovation Division shall have a structure approved by the University Council and shall be headed by an Executive Director.

6.3. The Enterprise Division shall have a structure approved by the University Council and shall be headed by an Executive Director.

6.4. The salary scale applies to each grade and is subject to approval by the University Council.

6.5. The University Council or its authorised committees shall have sole discretion and in so doing, may designate those posts to be established.

7. GRADING AND INITIAL APPOINTMENT OF INNOVATION AND ENTERPRISE STAFF

On initial appointment, a successful candidate shall be graded according to his or her qualifications and experience.

8. PROMOTION OF INNOVATION STAFF

8.1. Promotion will be guided by existing relevant ordinances.

8.2. The normal recruitment and selection procedure would apply when filling a higher-level vacant position. In this instance, the normal Staff Appointment Board would be convened to interview candidates for any vacant position.

8.3. The qualifying candidate shall be appointed accordingly.

8.4. The staff appointed to Innovation posts shall be promoted following the relevant Ordinance.

9. TENURE/PROBATION

9.1. On joining the University, each employee shall enter into a three (3) months probationary period. During this period, either party may terminate the contract by providing a two (2) week notice period.

9.2. The criteria for the granting of tenure on completion of the required probationary period shall be: -

9.2.1. Satisfactory performance of duties as assigned by the respective head of the department.

9.2.2. Satisfactory University Service.

10. UNIVERSITY INNOVATION HUB AND INDUSTRIAL AND SCIENCE PARKS

10.1. The university shall establish Industrial and Science Parks for nurturing innovative initiatives from the University Innovation Hub. The Innovation Hub shall:

a) assist Start-up companies to develop and launch new business models including developing business and marketing plans;
b) provide business and technical services, laboratory facilities, advisory, network, and linkages;
c) encourage entrepreneurship, innovation, and incubation of ideas;
d) assist the innovators in creating a sustainable, scalable, and profitable business model;
e) offer seminars or workshops that are relevant to business development; and
f) assist in commercialising new technologies.

10.2. The University Industrial and Science Parks: The University shall establish one or more Industrial and Science Parks for purposes of commercialisation and production of goods and services coming from innovators using the following criteria:

a) production of prioritised goods and services arising from prototypes, industrial designs, or other IP rights generated from the University Innovation Hub or related University research and innovation processes;
b) Start-up;
c) Spin-off; and
d) Special Purpose Vehicle.
e)

11. INNOVATION STREAMS INTO THE INNOVATION HUB

11.1. *Faculty Staff/Student/Graduate Innovators from the University's Faculties, Institutes, and Centres*: This stream shall focus on innovations generated by university staff and students.

11.2. *Industry-based Innovators*: Through the University Innovation Hub, industrialists, professionals, business practitioners, or entrepreneurs can turn their intellectual or artistic creations, designs, inventions, discoveries, or other forms of IP into monetised innovations for commercialisation and industrial growth.

11.3. *Community-based Innovators*: This stream draws on innovations generated by individuals engaged in diverse economic activities but not affiliated with the University.

11.4. *University Commissioned Innovations*: This stream is derived from research commissioned by the University to provide solutions and generate innovations to address priority national development issues.

12. ADMINISTRATION OF INNOVATIONS, INVENTIONS, AND CREATIONS

12.1. The Research and Innovation Division and its staff shall be responsible for recruiting, selecting, adopting, and managing Innovations in the Innovation hub.

12.2. The Technical Committee shall evaluate MSU's research products/services/technologies/IP/IPR before commercialisation. They shall preside over disputed areas in the selection, protection, and commercialization of innovations and recommend the course of action to the Executive Director, Research and Innovation.

13. THE INCUBATION PROCESS

13.1. The Incubation process shall be as follows:

13.1.1. Predisclosure of innovation/invention/creation

13.1.2. Full disclosure of innovation/invention/creation

13.1.3. Assessment of innovation/invention/creation

13.1.4. Signing of Incubation agreement and non-disclosure agreements.

13.1.5. IP protection, filing, and enforcement.

13.1.6. Product or technology development to prototyping demonstrable in a real environment.

13.1.7. The signing of Business partnership agreement(s) and registration of the company(ies). The agreement shall specify the revenue-sharing model.

13.1.8. Business and commercialization strategy development and funding to Market

14. OBLIGATIONS OF THE UNIVERSITY

The University shall:

14.1. Form an Innovation Committee with representatives from all faculties, institutes, and centres.

14.2. Encourage the creation, registration, and management of intellectual property rights.

14.3. Encourage the creation and/or upkeep of functional research and innovation infrastructure and equipment, such as laboratories, workshops, field sites, and industrial facilities.

14.4. Create and promote research, innovation, and industrialisation platforms to advance new knowledge, intellectual creations, innovations, scientific discoveries, technological inventions, arts, and scholarship. The platforms must

include, among other things:

14.4.1. Conducting a "Research and Innovation Expose" as the institution's prime annual showcase;

14.4.2. Promoting the conduct of research, innovation, and industrialisation seminars, dialogues, workshops, exhibitions, and conferences organised at the local, national, and international levels by Departments, Faculties, Institutes, and Centres following the University vision and dictates;

14.4.3. Recognizing achievements and excellence in research and innovation among innovators; and

14.4.4. Promoting skills and professional development training in relevant areas of research, innovation, and IP commercialisation among innovators;

14.4.5. Promoting a conducive environment for promoting productivity and sustainability of the University's research-innovation-industrialization ecosystem.

14.5. The University shall primarily support the development and productivity of the University Research-Innovation-Industrialisation Ecosystem;

14.6. Resident innovators shall be incentivised according to their agreement and University policies.

15. OBLIGATIONS OF THE INNOVATOR

The Innovator shall:

15.1. be guided by the Research and Innovation Division in establishing clear periodic performance targets for outputs and outcomes in the execution of innovation projects;

15.2. develop innovations to produce IP, products, goods, and services over a reasonably defined period;

15.3. individually or jointly work with others and University partners in industry, commerce, and with international research and innovation partners to cause the development, registration, and commercialisation of IPs;

15.4. use funds for the intended purpose;

15.5. provide the necessary acquittal for the funds received;

15.6. ensure confidentiality;

15.7. conform to ethics and regulatory requirements governing the project;

15.8. avail their prototypes or any such outputs of innovation to any dissemination platforms as may be requested by the University and

15.9. report progress according to the terms and conditions of the contract.

15.10. report to the Innovation Hub as per the contract agreement of resident innovators.

15.11. governed by the provisions of the Midlands State University Act (No. 4 of 1999 (Chapter 25:21) and any statutes, ordinances, regulations, or rules made under the authority of the Act, thus for resident innovators

15.12. sign non-disclosure and contract agreements when they join the Innovation Hub.

16. MANAGEMENT OF INNOVATION OUTPUTS AND IPRs

16.1. The University innovation outputs and IPRs shall be governed following the University IP Policy, Spin-out Policy, and Commercialisation Policy.

16.2. The Principal Investigator (PI) or lead innovator shall be regarded as the originator of the original idea and strategic leader of the innovation process. There shall be only one PI to an Innovation Project or Programme

16.3. The Co-Principal Investigator (Co-PI) shall be regarded as the co-originator and next-biggest contributor to the original idea, providing strategic leadership support to the research/innovation process. There may be more than one Co-PI to a project or programme.

16.4. The PI, Co-PI, and any other persons who have contributed significantly to the innovation shall be regarded collectively as innovators.

16.5. A Start-up company, where the Innovator(s) shall get at least 70% of the net income and the University gets up to 30% of the net income in the first 5 (five) years, and then innovators shall get 90% and the University 10% in the subsequent years.

16.6. A Spin-off company, where the University shall be entitled to 70% and Innovators, not more than 30%.

16.7. A Joint-venture with external parties or collaborative research, where the shareholding shall be stipulated according to the equity brought in by the university and other parties; and specified in a joint-venture agreement (thus sharing agreement shall be case specific).

16.8. In all cases, the University share shall be allocated to the relevant stakeholders.

16.9. The following guidelines shall apply to the transfer of IP

16.9.1. Creators may not assign, or license rights in, applicable intellectual property to third parties without the written consent of the University

16.9.2. All assignments must be in writing and must conform to the requirements of the Ordinance

17. GOVERNANCE AND SHARE OWNERSHIP STRUCTURE OF COMPANIES

17.1. The Governance and share ownership structure of companies shall be as follows:

17.1.1. Category A: Start-up Companies

a) Directorship: The Principal Innovator and Co-Innovators shall be among the principal directors or appoint their representatives on the Start-up Company Board, while the University shall appoint a Director(s) to the Board to represent its interests.

b) Shareholding: The Principal Innovator and Co-Innovators shall have a majority shareholding in any Start-up, with the University holding no more than 30% equity in each Start-up regardless of the investment including technical and expert support.

c) The Start-up company shares shall not be sold without the written consent of the University.

17.1.2. Category B: Spin-off Companies

a) Directorship: Directors shall be appointed by the University Council on the recommendation of the Vice Chancellor for all Spin-off companies.

17.1.3. Category C: Special Purpose Vehicle Companies

(a) The University may use Special Purpose Vehicle(s) (SPV) primarily for purposes of undertaking joint ventures and/or managing specific strategic projects.

(b) Directorship: Directors that represent University interests in the SPV shall be appointed by the University Council upon the recommendation from the Vice Chancellor and the shareholders of the Special Purpose Vehicle.

(c) Shareholding: The level of shareholding by the University in an SPV shall be proportional to the level of involvement and/or investment by the institution in the joint venture or project in question.

GENERAL ACADEMIC REGULATIONS FOR UNDERGRADUATE CERTIFICATES AND DIPLOMAS OF THE MIDLANDS STATE UNIVERSITY

1. PREAMBLE

1.1 Senate shall be the final authority for the interpretation of these Regulations.

1.2 Senate has the power to exempt any student from any of the regulations.

1.3 A student who has started a programme following one set of regulations will not be affected by regulations subsequently adopted unless agreed to in writing by the student.

1.4 Every student, by the act of registering with a Faculty, agrees to abide by all the rules, regulations and policies of the University, the Faculty, and the relevant Department.

1.5 In these Regulations the following terms shall be used as described:-

Programme: a plan of study lasting a period of time, which leads to the award of a diploma or degree of the University.

Level: a defined portion of a programme.

Subject: a field of study offered by a Department.

Module: a separately examinable component within a subject.

Project: a defined practical assignment, which is separately examinable.

Work Related Learning Experience: a prescribed period of hands-on experience in a relevant work setting.

Continuous Assessment: prescribed assignments to be completed within a given period and forming a part module.

2. GENERAL PROVISIONS

2.1 The Diplomas offered by the University are carried out primarily through taught modules with continuous assessment and examinations offered at the end of a semester.

2.2 Studies may be pursued on a full-time or part-time basis as prescribed by Faculties in their Regulations.

3. FACULTY REGULATIONS

3.1 There shall be Faculty Regulations for Diploma studies, which shall be read in conjunction with the General Academic Regulations. Specific restrictions on the choice of subjects, modules and additional requirements may be specified in the Departmental Regulations.

FACULTY OF EDUCATION

Diploma in Adult Education (UDAE)

4. FACULTY OF SOCIAL SCIENCES

- Certificate in Social Work
- Diploma in Social Work

5. ENTRY REQUIREMENTS

5.1 ***Normal Entry***
Holders of at least a certificate or at least five 'O' Level Passes including English.

5.2 ***Mature Entry***
The Faculty or Department may use their discretion on maturity entry.

6. WORK RELATED LEARNING

A prescribed period of hands-on practical experience in a relevant industrial, commercial / professional or community setting as prescribed by the Faculty.

7. DURATION OF CERTIFICATES AND DIPLOMAS

The minimum duration of the diploma shall be as prescribed in Faculty regulations.

8. INTERRUPTION OF STUDY

8.1 A student who breaks his or her studies for more than two consecutive semesters without the approval of the Senate shall normally be deemed to have lost any accumulated credits. Such a student may be allowed to re-apply for admission to the University

8.2 A student who wishes to interrupt his or her study programme shall apply in advance to the Faculty through the Head of Department and Deans of the appropriate Faculty, stating reasons why he/she wants to interrupt his/her study programme. Permission must duly be granted before he/she leaves the University. The decision of the Senate shall be communicated to the applicant by the Deputy Registrar Academic.

9. ELIGIBILITY FOR EXAMINATIONS

9.1 A student shall attend all such lectures, tutorials, seminars and practical work and undertake all other activities and assignments as approved by the University.

9.2 Each department shall, with the approval of Senate, determine the requirements for the modules they offer. A student who does not fulfil the requirements for any module shall not be allowed to take the examination for that module.

10. MARKING SCHEME AND CLASSIFICATION

Diplomas shall be awarded the following categories; distinction, merit and pass
The following marking scheme shall be used for the modules and programmes:

80% and above	Distinction
60%-79%	Merit
50%-59%	Pass
Below 50%	Fail

11. FAILURE TO SATISFY EXAMINERS

11.1 A candidate who fails to satisfy the examiners may be required by senate to write 'supplementary examinations' or to 'repeat' or to 'withdraw.'

11.2 If a student is repeating a module(s) he or she shall only be credited with the marks obtained during the 'repeat' examination.

11.3 "Withdraw" means that the student shall withdraw from the university.

11.4 For the prescribed project candidates shall be informed in advance of the deadline for submission of such project.

11.5 "Carry over" means carryover modules from previous level to the Subsequent level.

11.6 “Retake” means redo module(s) failed in previous semester within the same level.

12. PLAGIARISM

12.1 Students must be guided by the University’s Anti-plagiarism and Responsible use of Artificial Intelligence (AI) Policy

12.2 Plagiarism is an academic offence and the University does not tolerate it whatsoever.

12.3 Plagiarism is the unacknowledged use of another person’s material or ideas. As such, plagiarism is an academic offence in the sense that theft is in ordinary daily life.

12.4 Recommendations on the severity of the penalty shall be determined by theappropriate Departmental Board or Panel of Examiners. Cases of plagiarism shallbe handled in the following manner: -

12.5 ***Minor Cases of Plagiarism***

- **First Offence:** in the case of plagiarism being discovered in a piece ofwork such as an essay or laboratory report or dissertation the student shall get a Chairperson’s warning but shall be given an opportunity tore-do and re-submit an acceptable piece of work after one week and shall be awarded a maximum mark of 50%.
- **Second Offence:** The student shall get a Dean’s warning and shall beawarded a mark of zero.
- **Third Offence:** Senate shall take disciplinary measures such as suspension or expulsion of the student who will have been awarded amark of zero.

12.6 ***Major Cases of Plagiarism***

- In the case of plagiarism being discovered in a project at end of the semester, that candidate shall be denied the opportunity to resubmit the project.

  15.5.1.1 The new project shall be submitted in the second semester after the discovery of the offence.

  15.5.1.2 The new project shall be awarded a maximum mark of50%.

- In the case of plagiarism being discovered in a project for the second time and after re-submission, a mark of zero shall be awarded and recorded, and Senate shall take disciplinary action either to suspend orexpel the student.

GENERAL ACADEMIC REGULATIONS FOR UNDERGRADUATE DEGREES

1. PREAMBLE

1.1 Senate shall be the final authority for the interpretation of these regulations.

1.2 The Senate reserves the right to alter, amend, replace or cancel any of the academic regulations.

1.3 No student who has started a programme following one set of regulations shall be set at a disadvantage by a regulation or regulations subsequently adopted.

1.4 There shall be Special Regulations for each Faculty which shall be subject to approval by the Senate and which shall include provision for admission to Programmes, Subjects and Modules within the Faculty and schemes of examinations for these Programmes.

1.5 No student may register for modules/programmes with another University without the approval of the Senate.

1.6 The General Academic Regulations shall take precedence over the Special Faculty Regulations.

1.7 Detailed syllabuses for modules in the subject will not form part of the General Academic Regulations or Special Faculty Regulations but will be submittedto the appropriate Faculty Boards for approval.

In these regulations the following shall be used as defined: - **"Programme"** a plan of study, lasting over a specific period, which leads to a degree, diploma or certificate; **"Level"** a prescribed period of study normally made up of two semesters at the end ofwhich decisions determining issues of progression are made. **"Semester"** a prescribedperiod of study normally made up of 15 weeks made up of, but not limited to, 13 weeks of teaching and two weeks of examinations.

"Module" for purposes of teaching and examinations a subject may be divided into one or more components called modules.

"Project" a defined practical/ research assignment, which is separately examinable. **"Work Related Learning"** a prescribed period of practical experience in a relevant industrial/commercial/professional or community setting. **"Continuous Assessment"** prescribed assignments to be completed within a specific period and forming part ofa module.

"Carry Over" means carryover modules from previous semester level to the subsequent level.

"Retake" means redo module(s) failed in a previous semester within the same level.

"Pending and Review" applies specifically to Level 3 students who are carrying modules

"Discontinue" means that the candidate must discontinue from the programme in which he/she has failed. Such students will be free to apply for admission/transfer into a different programme and his/her application will be considered through the normal admission procedures.

"Withdraw" means the student must withdraw from the University. Once withdrawn the student may not apply for admission until a period of four successivesemesters has elapsed.

"Aegrotat" a degree offered on concessionary terms after one has failed to sit for examinations due to illness or other unavoidable intervention of a 'force majeure' nature.

"Supplement" means rewrite module(s) failed with marks in the range of 40-49% **"Deferments"** means suspension of studies for a period of at least one semester. **"Presumed withdrawn"** a student is presumed to have withdrawn from his/her studieswhen after completing all registration formalities, the student without explanationfails to submit or partially submits continuous assessment or work-related learningreports and does not write scheduled end of semester examinations.

"Credit" An educational award which in accumulation with other educational awardsleads to the award of a degree/diploma certificate. One credit shall consist of 10 notional hours.

1.8 The Registrar shall maintain a schedule of programmes and modules and their codesfor use in computerized student records.

2. PROGRAMMES

2.1 The University may offer programmes for undergraduate Bachelor's Degrees at General and Honours levels.

2.2 GENERAL DEGREES

A 'General' degree requires the study of a degree of modules over four to six semesters.The programme shall include work related learning, which shall normally be done during the third level or as prescribed by the Special Faculty Regulations.

Students may select modules from a number of options in accordance with provisions in the Special Faculty Regulations. Normally one or two modules will be identified as 'major modules' to be studied throughout the programme, with up to five modules being studied as 'minors', each for two, three, or four semesters.

2.3 HONOURS DEGREES

2.3.1 The structure of the Honours Degree Programmes shall be prescribed in the Special Faculty Regulations. These structures may vary with the particular requirements of different faculties and departments, but all Honours Programmesshall normally contain the following elements:

2.3.1.1 one or more 'major modules' shall be studied over at least six orten semesters. Honours degrees shall normally include a period of work-related learning during the third level. Modules in the Honours degrees shall be studied intensively and progressively (i.e. studies inthe fifth to eighth semester assume prior knowledge of the modules studied from the first to the fourth semester). They shall be taught and examined at the level requiring a greater depth and breath of knowledge and understanding than expected of modules in 'general' degrees programmes.

2.3.1.2 a candidate for the Honours degree shall normally be required to complete a project or dissertation within his/ her programme of studyand this work may be based on the tasks and research performed during work Related learning.

2.3.1.3 the combination of modules within the Honours Programme shall be prescribed so as to focus on specific topics and to disallow a wide choice of disparate options.

2.3.1.4 one or more subsidiary modules may be studied at the 'General' levelwithin the Honours Programme but assessment in these modules willnot contribute to the final classification of the degree awarded or willbe assigned a light weighting in the overall calculations of the final marks. Normally 60% of the modules taken in the Honours degree will be in the major subject(s).

2.3.1.5 the following are in the General and Honours Degree Programmes offered by the University

2.4 SPECIAL HONOURS DEGREES

A 'Special Honours' degree requires the study of a degree over two semesters.The programme shall normally be done by candidates who are holders of a recognized undergraduate general/honours degree or as prescribed by the Special Faculty Regulations.

3. ENTRY REQUIREMENTS

3.1 Normal Entry

For normal entry candidates should:

3.1.1 have satisfied the general requirements as prescribed below; and

3.1.2 have satisfied the special requirements for entry into the particular programme chosen; and

3.1.3 have passed English Language at 'O' level with a C or better for all programmes

3.1.4 have passed Mathematics at 'O' level with a C or better in those programmes in which Mathematics is a prerequisite as prescribed by Faculty regulations.

3.1.5 Candidates from academic jurisdictions whose qualifications are deemed to be not equivalent to entry requirements set in the General Regulations will be required to take a foundation course.

GENERAL REQUIREMENTS

Passes in at least 5 subjects at Ordinary Level and at least two subjects atAdvanced Level or their equivalents.

The following are acceptable to the University:

Ordinary Level Passes or Equivalent

Ordinary Level of the Zimbabwe School Examinations Council

Ordinary Level of the Associated Examinations Board's General Certificate of Education Ordinary Level of the Cambridge School Certificate

Ordinary Level of the University of London's General Certificate of Education

Advanced Level Pass or Equivalent

Advanced Level of the Zimbabwe School Examinations Council.

Advanced Level of the Associated Examinations Board's General Certificate of Education;Advanced Level of the Cambridge Higher School Certificate;

Advanced Level of the University of London's General Certificate of Education;

3.1.6 General Subject Provisions

Subjects must have been chosen from the approved list below and restrictions against combinations overlapping must have been observed.

3.1.7 Approved Subjects for Admission Purposes

Subjects approved by the Zimbabwe School Examinations Council and/or theAssociated Examinations Board; and/or the Cambridge Local Examinations Syndicate and/or the London Certificate of Education: -

Level	
OA	Accounting
O	Accounts
OA	Accounts, Principles of
OA	Ancient History
A	Ancient History and Literature
A	Animal Science
OA	Applied Mathematics
O	Applied Statistics
OA	Art
OA	Arts Crafts
OA	Bible Knowledge

OA	Biology
O	Bookkeeping
OA	Botany
OA	Building Technology
OA	Business Management
OA	Business Studies
OA	Business Enterprise Skills
OA	Chemistry
OA	Commerce
OA	Commercial Studies
OA	Computer Studies
A	Computing Science
OA	Design Technology
OA	Divinity
O	Drama and Theatre Arts
A	Economics and Political Studies
O	Economic and Political Affairs
A	Economic and Social History
A	Economic Geography
OA	Economic History
O	Economic Principles
OA	Economics
A	Electronic Systems
O	Electricity and Electronics
O	Elementary Physiology
O	Elements of Sociology
OA	Engineering Drawing
OA	Engineering Science
O	English Language
OA	English Literature
O	Environmental Biology
OA	Environmental Studies
OA	Family and Religious Studies
OA	Fashion and Fabrics
OA	Food and Technology Design
OA	Food and Nutrition
OA	French
O	French Literature
O	French Studies
O	General Mathematics
O	General Papers (Use of English)
OA	General Principles of English LawO General Science
OA	Geography
OA	Geology
A	Government and Politics/Political Studies
O	Government, Economics and Commerce
OA	Health Science
OA	History
OA	History, Ancient
O	History and Appreciation of Music
O	Home Economics
OA	Horticulture
OA	Human Biology
OA	Indigenous Language
OA	Literature in Indigenous Language
OA	Law
OA	Mathematics

A	Mathematics, Applied
O	*Metal Work
O	*Metal Engineering
A	*Metal Technology
OA	Music
OA	Ndebele
OA	Physical Science
OA	Physics
OA	Physics with Chemistry
OA	Political Studies
OA	Portuguese
O	Principles of Economics
OA	Psychology
OA	Religious Studies
O	Rural Biology
OA	Shona
A	Software Engineering
A	Sport Management
A	Sport Technology
OA	Social Science
OA	Sociology
OA	Statistics
O	*Survey
OA	Technical Drawing
OA	Technical Graphics
OA	Theatre Arts
OA	Textile Technology
OA	Technical Graphics
O	*Woodwork
OA	Zoology

*Not more than one subject indicated above by an asterisk may be recognized for the purpose of satisfying Ordinary Level requirements.
Senate on the recommendation of the Registrar may accept other subjects and other Examining Boards.

3.1.8 Restrictions against the combination of overlapping subjects.

In the selection of subjects for the purpose of satisfying the general requirement subjects listed under column A in the table below cannot be counted with corresponding subjects listed under column B.

COLUMN A	**COLUMN B**
Accounting	Accounts, Principles of Accounts, Bookkeeping
Art	History of Art
Biology	Rural Biology, Botany, Zoology, General Science
Chemistry	Physical Science, Physics with Chemistry, General Science
Economic Geography	Geography, Environmental Studies
Economics	Economic Principles, Commerce Elementary Physiology, Human Biology
Elements of Sociology	Sociology
Engineering Drawing	Technical Drawing
Environmental Studies	Geography
General Mathematics	Mathematics
General Science	Physics, Physical; Physics with Chemistry
Geography	Economic Geography
Government and Political Studies	Government and Politics
Health Science	Human Biology
Mathematics	Pure and Applied Mathematics, Pure Mathematics; Applied Mathematics
Physical Science	Physics with Chemistry, Chemistry,

	General Science, Physics
Physics	Physics with Chemistry, Physical Science, General Science
Pure and Applied	Pure Mathematics, Applied Mathematics,
Social Science	Sociology, Zoology, Human Biology

3.1.9 Faculty Requirements

For admission to a particular programme of study and/or for subject/modules within theprogramme there may be specific restrictions on the choice of subjects in the generalrequirements and/or additional requirements concerning entry. Such additional requirements shall be prescribed in the Faculty Regulations.

3.2 *Special Entry*

The following persons may apply for Special Entry and for permission to proceed to a firstdegree with exemption from the whole or part of the normal entry requirements: -

3.2.1 a person who has obtained a degree of this or another university or degree awarding institution.

3.2.2 a person who has obtained from a University or an Institution of similar status, academic qualifications (other than degrees) acceptable to Senate. A person who has obtained an appropriate number of subjects at an approved examination and has subsequently passed an intermediate or equivalent examination at a University acceptable to Senate;

3.2.3 students who qualify under this regulation for Special Entry may apply to the Senate and shall be granted exemptions as prescribed in Faculty regulations. Permission may be given by the Senate in accordance with Zimbabwe National Qualifications Framework for candidates to complete the programme for a Bachelor's degree in less than the normal required period provided that no student shall be allowed direct entry to the final level of any programme.

3.2.4 students who apply for admission under this regulation may be required to attendinterviews and/or special tests at the University to determine their suitability for admission to Bachelor's degree studies.

3.3 *Mature Entry*

Persons who are at least 25 years of age for males and 23 years for females on the first dayof the academic year in which admission is sought and who are not eligible for entry underthe Normal or Special Entry Regulations may apply for Mature Entry provided that:

3.3.1 applicants must have passed at least five approved 'O' Level subjects including English Language and Mathematics (where required) or equivalents, and must have demonstratedpotential suitability for University studies by virtue of their attainments and/or relevant work experience.

3.3.2 normally, applicants should have completed their full-time school or college education at least five years before the start of the academic year in which admission is sought.

3.3.3 applicants who wish to be considered under the Mature Entry provisions may berequired to attend interviews and/or special tests at the University designed to assess their command of the English Language, numeracy and reasoning ability and general suitability for admission to the Bachelor's degree studies.

3.3.4 applicants who have previously attended Mature Entry tests and/or interviews without success will not be considered for admissions under this form of entry unless in the intervening period they have acquired additional qualifications and/or experience.

3.4 Submission of Applications

3.4.1 Applications must be submitted either in hard copy or soft copy to Admissions, Registration and Student Statistics Office or online through the university online admissions platform.

3.4.2 The closing dates for receipt of application forms for Normal Entry shallbe advertised by the Registrar's Office each year. Late applicants may be considered upon payment of the prescribed late application fee.

3.4.3 The closing date for Special Entry and Mature Entry applications shall also be advertised by the Registrar's Office.

3.5 General Provisions

3.5.1 Every student must satisfy the University that he/she has adequate command of the English Language.

New students are required to register and pass a Communication Skills module, upon registration for a Bachelor's degree studies.

3.5.2 Students admitted under the Special Entry provision may be exempted from this requirement.

3.5.3 A student shall not register simultaneously for more than one programme at this University or any other institution without the permission of the Senate.

3.5.4 Registration will take place in accordance with the arrangements prescribed eachyear through the Registrar's Office.

3.5.5 A student's registration shall not be confirmed until he/she has fulfilled the requirements for payment of fees.

3.5.6 Normally, no student shall be admitted to any programme or any module more than two weeks after its commencement. Any exception to this regulation must have the written endorsement of the Chairperson of the Department and the Dean of the Faculty concerned and will be subject to approval through the Registrar's Office.

3.5.7 Students who enter or return to the University late shall not be entitled to extra tuition.

3.5.8 Such students shall be liable to pay late registration fees, unless the Registrar has given permission for such waiver of late registration.

3.5.9 A student registered for a module is expected to attend all classes prescribed for suchmodule. Where tutorials, seminars, fieldwork, vacation work and practical sessions areprescribed a student is required to attend and complete any assignment set.

3.5.10 If a student is unable to attend classes for health reasons for longer than 72 hours, he/she must notify the appropriate Faculty Officer as soon as possibleand submit certification in support thereof by a medical practitioner registeredin accordance with the Medical, Dental and Allied Health Professions Act. For absence on grounds other than health reasons, prior permission from the Dean on the recommendation of the Chairperson of the Department concerned shall benecessary.

3.5.11 After taking due consideration of the academic progress of the student, the Senatemay require or allow a student originally registered for one programme or subjectto register for another programme or subject on the completion of either the FirstSemester or Second Semester of the programme for which he/ she is registered.

3.5.12 Normally, no programme shall commence with fewer than ten students.

3.6 Refund of tuition fees

3.6.1 A student who leaves the University before the end of the session for which he/she has been admitted shall need to give a proper notice in writing to the Registrar and may upon written application to the Bursar, obtain a refund on tuition fees.

3.6.2 Application fees are non-refundable.

3.7 Deferment of Studies

3.7.1 A student who wishes to defer his/her studies for whatever reason should:

3.7.1.1 complete deferment of studies forms in triplicate.

3.7.1.2 submit completed forms to the Chairperson of Department and Dean of Faculty for approval.

3.7.1.3 submit deferment forms to the Bursar's Department (StudentAccounts Section) for noting.

3.7.1.4 submit signed deferment forms in triplicate to the Admissions,Registration and Statistics office.

3.8 The student should make sure that the deferment process is completed before leaving the University. Upon return to the University, resumption of studies forms shouldbe completed and submitted to the Chairperson,

Dean, and Bursar's Department for approval.

3.9 Students who fail to complete this process shall be deemed to have attended lectures and their student accounts will be debited with fees for the respective semester.

3.10 No retrospective authorization will be allowed.

3.11 Fees will be charged pro rata for the taught period on deferment.

3.12 Results for all students who deferred during their semester should indicate that the student deferred.

4. STRUCTURE OF PROGRAMMES

The duration of Bachelor's Degree Programmes shall be prescribed by Faculty Regulations. Each programme shall be divided into levels as follows:

4.1 a level shall include two teaching semesters, each broken down into two periods of study consisting of not less than 15 weeks. At the end of the second semester of a level, decisions determining issues of progression shall be made.

4.2 the possible combination of modules on offer within a specific semester shall be in accordance with the Faculty Regulations and shall be subject to approval by the Chairperson(s) of the Department(s) and the Deans(s) concerned.

5. GRADING AND DEGREE CLASSIFICATION

5.1 All Bachelor's degrees shall be classified in the following classes: First Class,Upper Second Class, Lower Second Class and Pass.

5.2 The following Grading and Degree Classification System shall be adopted for all modules and programmes:

75% and above	1	(First Class)
65%- 74%	2.1	(Upper Second Class)
60%-64%	2.2	(Lower Second Class)
50%-59%	3	(Pass)
49% and below	Fail	(Fail)

6. ASSESSMENT OF CANDIDATES FOR BACHELOR'S DEGREE

6.1 Normally, evaluation shall be based on continuous assessment as well as Universityexaminations. Unless otherwise approved by the Senate, continuous assessment will contribute between 25%and 50% of the overall assessment.

6.2 Each department shall determine which items of continuous assessment andpractical work shall be included in the continuous assessment and shall define the relevant weighting to each item of continuous assessment or practical work. Eachdepartment shall maintain written records of each student's performance in these elements of continuous assessment. Items incorporated in continuous assessment may include assignments, tests, essays, fieldwork, laboratory work and projects orany other as shall be determined by the department concerned.

6.3 Students shall normally take University examinations at the end of each period of study as prescribed in the Faculty and/or Departmental Regulations.

6.4 External Examiners shall be appointed to moderate all University examinations (except supplementary/special examinations).

6.5 All matters relating to the conduct of University examinations shall be the responsibility of the Registrar.

6.6 To be admitted to any University examinations, a candidate must: -

6.6.1 be registered as a student of the University in accordance with the GeneralAcademic Regulations;

6.6.2 have satisfactorily completed approved modules of study at the University.'Satisfactory Completion' of modules shall require submission of written work, attendance of all lectures, seminars, tutorials, work related learning placement, and research and innovation activities as stated in the Faculty and Departmental Regulations.

6.7 Exclusion from a University Examination shall require the authority of the Senate.

6.8 The Examiners may require any candidate to attend an oral examination and/or write a special examination.

6.9 Candidates studying Midlands State University degree programmes shall be required to have attained a minimum of 120 number of credits for undergraduates and 90 credits for Masters acquired during their period of study in order to graduate subject to faculty specifications. These are subject to Faculty and Departmental specification.

6.10 Candidates who would have attained credits from other reputable institutions shall be allowed to transfer their credits to Midlands State University. It is the prerogative of Senate to determine if such credits would be added to students' academic record.

7. INTERRUPTION OF STUDY

7.1 A student who discontinues studies for more than two consecutive semesters without the approval of the Senate shall normally be deemed to have lost any accumulated credits. Such a student may be allowed to re-apply for admission to the University.

7.2 A student who wishes to interrupt his or her study programme shall apply in advance to the Faculty through the Head of Department and Dean of the appropriate Faculty, stating reasons why he/she wants to interrupt his/her study programme. Permission must duly be granted before he/she leaves the University. The decision of the Senate shall be communicated to the applicant by the Deputy Registrar Academic Affairs.

8. ELIGIBILITY OF EXAMINATIONS

8.1 A student shall attend all such lectures, tutorials, seminars and practical work and undertake all other activities and assignments as approved by the University.

8.2 Each department shall, with the approval of Senate, determine the requirements for the modules they offer.

8.3 A student who does not fulfil the requirements for any module shall not be allowed to take the examination for that module.

9. DETERMINATION OF CANDIDATES' RESULTS

12.7 The Senate, on the recommendation of the Faculty Boards of Examiners, shall determine results.

12.8 Departmental Panels of Examiners shall comprise all full-time lecturing staff in that Department, the External Examiner(s) and, where appropriate, as determined by the Departmental Panel, part-time and/ or teaching assistants for the module/ subject concerned.

12.9 Faculty Boards of Examiners shall comprise the Dean and Deputy Dean of the Faculty, the Chairperson of each Department and the External Examiner(s) for the Department plus one other member of the Department nominated by the Departmental Panel from each Department.

12.10 The Departmental Panel of Examiners shall:

- agree, for each candidate on marks in terms of percentages, for continuous assessment, for the University examinations and overall marks (combining the continuous assessment and University examinations marks) in modules and, where required, in terms of the Faculty Regulations, in subjects.
- recommend to the Faculty Board of Examiners whether a candidate should pass or fail the relevant module(s) taken.

- Departmental Panel of examiners will meet at the end of the examination session to ratify the marks obtained by candidates.
- Departmental Panel of Examiners will meet at the end of the secondsemester of a level to ratify the marks obtained by candidates and to makerecommendations to Faculty Boards on issues of progression by students to the next level. In the case of repeat students, the panel of examiners mayhowever make a recommendation to withdraw a student at this stage;
- Where module prizes are available for award, make recommendations to the Faculty Board at the end of the second semester of a level for the awardof these prizes.

12.11 The Faculty Board of Examiners shall: -

- consider the recommendations of the Panels of Examiners and recommendto Senate an overall result for each candidate and any other conditions it may deem appropriate,
- meet at the end of end of a defined examination period each study period/ semester of a level to ratify the marks obtained by candidates.
- meet at the end of the second semester of a level to ratify the marks obtained by candidates and to make recommendations to Senate on issues of progression by students to the next level. In the case of repeat level students, the Faculty Board may have however, make a recommendation towithdraw or discontinue a student at this stage;
- make recommendations to the Senate at the end of the second semesterof a final level with regard to the award of prizes, which may be available for candidates within the programme.

12.12 In determining results, all Departmental Panels of Examiners and Faculty Boards of Examiners shall have regard to all special requirements as prescribed in the Faculty Regulations. Such regulations may require candidates to satisfy the examiners in thecontinuous assessment and University examinations separately or in aggregate.

13. PROVISIONS FOR PASSING AND PROCEEDING IN A PROGRAMME

13.1 Students' progress automatically from semester one to semester two at each level with the exception of special cases which would have been approved by Senate e.g.,cases of students applying for deferment of studies and cases of students failing four modules in the first semester of the year.

13.2 Each Scheme of Examinations shall indicate which modules must be passed before acandidate may be allowed to proceed to a subsequent level of the programme (or portion thereof).

13.3 Normally a student shall not be allowed to proceed in a subject without having passed the previous formal examination(s) in that subject and having satisfied the entire prerequisites for proceeding in that subject as may be specified in the FacultyRegulations.

13.4 A student who passes sufficiently well in one level in aggregate may be permitted to proceed to a subsequent level carrying a module or modules subject to the provisionsof the Faculty Regulations.

14. FAILURE TO SATISFY THE EXAMINERS

14.1 A candidate who fails to satisfy the Examiners in terms of these General AcademicRegulations and Faculty Regulations may be required by Senate to: -

a. retake if Faculty regulations prescribe such examinations.

b. proceed to the next level of the programme carrying one or more modules from a preceding level.

c. repeat

d. discontinue

e. pending and review

f. withdraw and reapply after two years

14.2 ***Special Examinations***

- Senate may allow a candidate to write a special examination whereextenuating circumstances exist.
- Students affected by extenuating circumstances may apply to their Faculties within one week from the missed examination for their claims tobe considered.
- Faculties would consider students claims and make recommendations to the Academic Board.
- A student may claim that extenuating circumstances have contributed materially or significantly to:

  a. poor performance in any formally assessed work.

  b. absence from examinations or other assessment event.

  c. failure in examinations or other assessed work.

- Valid extenuating circumstances shall normally be:

  - circumstances which are unexpected, significantly disruptive andbeyond a student's control and which may have affected his/her academic performance.
  - an illness or serious accident at the time of an assessment or in theperiod leading up to a formal assessment.
  - severe emotional or mental stress at the time of an assessment orimmediately before an assessment or e.g. through bereavement, social, matrimonial family problems, experience of assault, robbery or other traumatic event, eviction/homeless in unavoidable circumstances, unavoidable involvement in legal proceedings.
  - other factors totally outside the student's control, e.g. for part timestudents, unforeseen and essential work commitment, unforeseendecisions taken by the company or practice which prevent them from completing their assignment (in both these circumstances, a letter from the employer must be supplied)

- Evidence in support of claims must be in the form of an originaldocument (not a photocopy) written and supplied by an appropriate third party giving details of the circumstances with dates and if possiblestating how students' assessment has been affected. Third party should be a person who knows the student in a professional capacity and can give a first-hand account of the circumstances.
- Acceptable evidence of extenuating circumstances shall include:

  - a medical certificate issued at the time of the illness, specifying the nature of the illness and the dates affected and confirming thatthis is the doctor's own diagnosis. It is not sufficient for the doctorto write a letter stating that the student saw him/her and claimed to be suffering from stress etc.
  - a letter from the University Student Counselling Services
  - a letter from solicitor; summons to attend court; an eviction notice; a report from a police officer;
  - death certificate (that is, of a close relative)
  - a letter from a transport official confirming serious unforeseen disruption to transport or a letter from a member of the University staff confirming absence of other group members due to the samedisruption;
  - a report from the invigilator (if the student falls ill during an examination) outlining the

extenuating circumstances together with medical evidence if appropriate.

- Students affected by extenuating circumstances shall be allowed to sit for aspecial examination during the period as scheduled by the Examinations Department/ when supplementary examinationsare being written.
- Special examinations shall be assessed in the same way as regular/ sessional examinations.

14.3 ***Supplementary Examinations***

- Senate may allow a candidate (second level second semester or final level second semester students only) to write a supplementary examination to enable him/her to pass a module of a programme in terms of the Faculty Regulations.
- Normally supplementary examinations shall be written at the end of the programme by a student who fails with a supplementable mark during the last semester of the programme in line with the Faculty Regulations.
- To be eligible for supplementary examination in a module a student must obtain a supplementable mark of at least 40% in the overall assessment.
- Where supplementary examinations have been allowed the examiners willnot incorporate continuous assessment in the determination of the result forthe supplementary examination.
- Supplementary examinations shall be marked as "Pass" or "Fail" and the overall maximum mark awarded in a supplementary examination shall be 50%.
- If a student is required to write a supplementary examination this fact and the results obtained in the supplementary examination in terms of "Pass" or "Fail" should be recorded on the official transcript of his/her academic results.
- Where a dissertation or project is prescribed in any programme, candidatesshall be informed in advance of the deadline of submission of such dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normallywithin three months of the original submission deadline. Unless otherwisedetermined by Senate, the maximum mark allowed for such work should be50%.

14.4 ***Retake***

See definition of terms (Subsection 1.7)

14.5 ***Carry Over***

- The number of modules carried from the previous level to the subsequent

  level may be limited by the Faculty Regulations.
- For all levels other than the Work-Related Learning Level, the maximum number permissible under this provision shall not exceed 25% of the numberof modules normally scheduled in a particular level of a programme.
- For the Work-Related Learning level, students may only carry modules

  from the previous level of the programme.
- Normally where a prerequisite is involved a candidate will be required to

  pass it before taking higher modules.
- Students carrying over modules will be re-examined at the next regularly scheduled examinations, normally taken at the next semester examinations.
- Students may be required by the Senate to undertake continuous assessmentwith the carry over modules. This assessment will then be taken into account in the usual way in determining the overall assessment.

14.6 ***Repeat***

- A candidate, who is not proceeding to the subsequent level of the programme but has passed at least 50% of the modules in that level of theprogramme, may be allowed to repeat failed modules.
- 'Repeat' failed modules will be allowed in cases where a candidate has passed 50% of the modules in that level of the programme but cannot proceed to the next level. In such cases the student will be allowed to retain credits for the passed modules.
- Repeat a level means repeat all modules of the level. This will apply to candidates who will have failed more than 50% of the modules taken in that level. In this case candidates will be required to repeat all modules both passed and failed modules in that level.

14.7 ***Discontinue***

- A candidate who passes less than 25% of the modules in the level of a

  programme will be required to discontinue.

14.8 ***Withdraw***

A candidate who:

a. has failed the same level of the programme twice, or

b. has failed two different programmes will be required to withdraw.

15. WORK-RELATED LEARNING

15.1 Programmes at the University shall normally include at least ten months of supervised Work-Related Learning during the third level. In exceptional cases, periods of Work-Related Learning maybe prescribed by the Faculty Regulations.

15.2 Assessment of the Work-Related Learning will be carried out in accordance with the following regulations: -

- To obtain overall pass, a student must pass the Academic Supervisors Assessment Report the Employer's Assessment Report and the Work-Related Learning Report (where Applicable).
- A student who fails Academic Supervisors Assessment Report and the

  Employers Assessment Report will be required to repeat the level.
- A student who fails the Work-Related Learning Report (where applicable)but has passed the Academic Supervisors Assessment Report may be allowed to re-submit the Work-Related Learning Report within two (2) months from the date of publication of results, unless otherwise determined by Senate. The maximum mark allowable for such referred work is 50%.
- The Academic Supervisors Assessment Report mark and the EmployersAssessment Report shall be determined by the Departmental Panel of Examiners from the marks awarded by the Work-Related Learning Supervisors and the Academic Supervisors.
- Two copies of the Work-Related Learning Report (where applicable) ina format approved by the faculty must be submitted to the Departmentwithin three weeks before the end of the level.
- A student who fails to meet the date for submission of the Work-RelatedLearning Report (where applicable) will normally be considered to have failed the report.
- The General Regulations on repeat, discontinue and withdraw shall apply to Work-Related Learning.

15.3 A student who fails the Work-Related Learning level shall not proceed to the next level of the programme.

16. APPEALS

16.1 Any candidate who, having failed to satisfy the Examiners, is required to withdrawfrom the University or to discontinue from a programme, has a right to appeal against the decision.

16.2 A committee shall be set up by Senate to consider such an appeal.

16.3 Any candidate who wishes to lodge an appeal against withdrawal or discontinue must do so in writing to the Registrar within twenty-one (21) days after the publication of the Examination results.

16.4 On appeal, the candidate must state clearly the grounds of the appeal.

Where a candidate is appealing on medical grounds, a medical practitionerregistered in terms of the Health and Allied Professions Act must substantiate themedical grounds in writing. Any other evidence, which the candidate wishes to submit in support of his/her case, must be lodged with the written appeal.

16.5 The Registrar will refer all punctually submitted appeals submitted in time to theTermination of Studies Appeals Committee.

16.6 The Appeals Committee will consider as legitimate grounds for appeal new evidence of mitigating circumstances (except mere lack of diligence or other fault on the part of the student), which was not previously available to the Examiners. Extenuating circumstances of a 'force majeure' nature which explainand are directly relevant to the student's academic performance and which he/ she could not reasonably have been expected to have foreseen or avoided will beconsidered.

16.7 The Committee will be empowered to hear an appellant orally and seek such

information and evidence, as it may consider pertinent.

16.8 No right to oral hearing is conferred upon appellants and the University will not reimburse any expenses incurred by an appellant in making a personal appearancebefore the Committee.

16.9 The Committee shall make recommendations on each case, as it deems appropriate.

Its recommendations shall be submitted to Senate for approval, or to the Academic Board or the Vice Chancellor on behalf of Senate for consideration.

16.10 All re-mark requests will not be entertained unless directed by Senate. However,a student who has been given a **withdraw** or a **discontinue** decision may appealin writing to the Registrar within twenty-one days after the publication of the examination results.

17. AEGROTAT PROVISIONS

17.1 If a candidate, having completed a substantial component of his/her programme,is prevented by serious illness or other sufficiently substantiated cause, from completing the prescribed requirements for that level of the programme, he/she may be deemed by Senate to have satisfied the Examiners for that level upon the recommendation of the Board of Examiners concerned and upon such other conditions as Senate may decide, provided that: -

- the candidate will not normally be exempted from presenting a thesisor a dissertation where such is prescribed; and
- the award of the Aegrotat Degree shall be without classification.

17.2 Where a student qualifies for an Aegrotat Degree, he/she may opt instead to writespecial examination in order that an overall grade may be determined and formallycredited to the student. Application for such an option must be submitted to the Registrar in writing not later than four weeks before the scheduled examinations.

17.3 Senate may require any candidate, irrespective of his/her programme or Faculty whose examination performance has been adversely affected by sufficiently substantiated circumstances of 'force majeure' nature to write a special examinationat a future date, normally not later than three (3) months after the date of the last examinations missed. In such circumstances, unless otherwise stipulated by the Senate, the mark

obtained in the special examination will be counted in the overallassessment for the purpose of degree classification.

17.4 Exceptionally, Senate on the recommendations of the Academic Board of Examiners may permit such special examinations for candidates in respect of the performance in a supplementary examination. In such circumstances, the mark obtained in the specialsupplementary examination must be treated in accordance with Regulation 11.3.5.

17.5 A candidate who wishes to be considered for an aegrotat degree must apply in writing together with written substantiation for his/her case, to the Registrar within ten days of the end of the University examinations for the programme concerned. Acertificate from a medical practitioner registered in terms of the Health and Allied Professions Act must support appeals that are submitted on medical grounds.

17.6 A candidate who is awarded an aegrotat degree may not re-enter the examination for the same degree, but shall be eligible to apply to proceed to an appropriate higher degree

18. PLAGIARISM

18.1 Students must be guided by the University's Anti-plagiarism and Responsible use of Artificial Intelligence (AI) Policy

18.2 Plagiarism is an academic offence and the University does not tolerate it whatsoever.

18.3 Plagiarism is the unacknowledged use of another person's material or ideas. As such, plagiarism is an academic offence in the sense that theft is in ordinary daily life.

18.4 Recommendations on the severity of the penalty shall be determined by theappropriate Departmental Board or Panel of Examiners. Cases of plagiarism shallbe handled in the following manner: -

18.5 ***Minor Cases of Plagiarism***

- **First Offence:** in the case of plagiarism being discovered in a piece ofwork such as an essay or laboratory report or dissertation the student shall get a Chairperson's warning but shall be given an opportunity tore-do and re-submit an acceptable piece of work after one week and shall be awarded a maximum mark of 50%.
- **Second Offence:** The student shall get a Dean's warning and shall beawarded a mark of zero.
- **Third Offence:** Senate shall take disciplinary measures such as suspension or expulsion of the student who will have been awarded amark of zero.

18.6 ***Major Cases of Plagiarism***

- In the case of plagiarism being discovered in a project at end of the semester, that candidate shall be denied the opportunity to resubmit the project.

  15.5.1.3 The new project shall be submitted in the second semester after the discovery of the offence.

  15.5.1.4 The new project shall be awarded a maximum mark of50%.

- In the case of plagiarism being discovered in a project for the second time and after re-submission, a mark of zero shall be awarded and recorded, and Senate shall take disciplinary action either to suspend orexpel the student.

19. MISCONDUCT AT EXAMINATIONS

Subject to Ordinance 2, (the Rules of Student Discipline Ordinance) any candidate found using unauthorized materials, including electronic gadgets or attempting to obtain information from other candidates or their papers, or otherwise guilty of misconduct during the examination may be disqualified not only in the examination, but in the whole examination, and further disciplinary action may be taken by the University.

20. PUBLICATION OF RESULTS

20.1 The Registrar shall be responsible for the publication of the results of the University examinations as approved by Senate.

20.2 Queries on examination results shall be entertained within fourteen days from the date of publication of results.

21. ACADEMIC TRANSCRIPT

On leaving the University each student may obtain, on application to the Registrar, onecopy of formal transcript of his/her complete academic record at the University.

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22. AWARD OF DEGREE

The award of degree and diplomas shall be subject to the approval by University Council.Candidate completing the requirements for such awards will be entitled to receive formal certificates of the University, bearing the University seal and signed by the Vice- Chancellor and the Registrar confirming the award.

23. DURATION OF STUDY PROGRAMMES

The following shall be the duration for the completion of the undergraduate diplomasand degrees:

PROGRAMME PERMITTED	NORMAL PERIOD	MAXIMUM PERIOD (INCLUDING REGISTRATION)
Full time - 2 Years Diploma	2 Years	4 Years
Block Release/ Visiting Diploma	3 Years	6 Years
Full time - 2 Years Degree	2 Years	4 Years
Full time - 4 Years Degree	4 Years	8 Years
Full time - 5 Years Degree	5 Years	10 Years
Block Release/ Visiting Degree	3 Years	6 Years

GENERAL ACADEMIC REGULATIONS FOR POSTGRADUATE DIPLOMAS AND TAUGHT MASTER'S DEGREES

1. PREAMBLE

1.1 Senate shall be the final authority for the interpretation of these Regulations.

1.2 Senate reserves the right to alter, amend, repeal, suspend or replace any of these Regulations.

1.3 Senate has the power to exempt any student from any of the Regulations.

1.4 A student who has started a programme of study following one set of regulations shall not be affected by regulations subsequently adopted unless agreed to in writing by the student.

1.5 There shall be Academic Regulations for each Faculty which shall be subject to approval by the Senate and which shall include provision for admission to programmes.

1.6 The General Academic Regulations shall take precedence over the Faculty Regulations.

1.7 Every student, by the act of registering, agrees to abide by all the rules, regulations and policies of the University and of the relevant Faculty and Department.

1.8 Students shall be held liable for contravening any of these Regulations.

1.9 In these Regulations the following terms shall be used as described: -

"Programme"

A plan of study lasting over a period of time, which leads to the award of a diploma or degree of the University.

"Level"

A prescribed period of study.

"Semester"

A prescribed period of study normally made up of 15 weeks including 15 weeks of teaching including two weeks of examinations.

"Subject"

A field of study offered by the Department

"Module"

A separately examinable component within a subject.

"Project/ Research"

A defined practical assignment, which is separately examinable.

"Work-Related Learning"

A prescribed period of hands-on practical experience in a relevant industrial, commercial/professional or community setting.

"Continuous Assessment"

Prescribed assignment to be completed within a given period and forming a part of a module.

"Carry Over"

Means carry over modules from previous level to the subsequent level.

"Retake"

Means redo module(s) failed in a previous semester within the same level.

"Aegrotat"

A degree offered on concessionary terms after one has failed to sit for examinations due to illness or other unavoidable interventions of a 'force majeure' nature.

"Deferment"

Means suspension of studies for a period of at least one semester.

"Presumed withdrawn"

A student is presumed to have withdrawn from his/her studies when after completing all registration formalities, the student without explanation fails to submit or partially submits continuous assessment or work-related learning reports and does not write scheduled end of semester examinations.

"Credit"

An educational award which in accumulation with other educational awards leads to the award of a degree and diploma. One credit shall consist of 10 notional hours.

2. GENERAL PROVISIONS

2.1 Studies are pursued on a full-time basis.

2.2 The degrees may be offered in all Faculties and Departments. Fields of study or research are determined by Senate on the recommendation of the Department and Faculty concerned.

2.3 A students shall not register simultaneously for more than one programme at this University or any other institution without the permission of the Senate.

3. FACULTY REGULATIONS

There shall be Faculty and Departmental Regulations, which shall be read in conjunction with the General Postgraduate Diplomas and Taught Masters Regulations. Specific restrictions on the choice of modules and additional requirements may be prescribed in the Faculty Regulations.

4. APPLICATION

4.1 Application forms for the various programmes shall be obtainable from:

Admissions, Registration and Student Statistics Office

Midlands State University

P Bag 9055

Gweru

Zimbabwe

Harare Campus

Number 1 Kwame Nkrumah Avenue

Harare Zimbabwe

Zvishavane Campus

C/O Main Campus

Alternatively, forms can be downloaded from the University Website: www. msu.ac.zw

4.2 The completed application forms should be returned to the Admissions, Registration and Student Statistics Office.

4.3 Applications shall be considered in the first instance by the Chairperson of Department in consultation with the teaching staff of the Department in which the candidate wishes to study. The considerations shall be sent to the Faculty Dean for approval.

5. GENERAL REQUIREMENTS FOR ADMISSION

5.1 *Post-Graduate Diplomas*

To be admitted to a programme leading to the award of a Postgraduate Diploma, a candidate must normally have obtained a pass in the first degree in an appropriate field of study at Midlands State University or any other approved University.

5.2 *Taught Masters*

5.2.1 To be admitted to a programme leading to the award of a taught Master's degree, a candidate must normally have obtained a good pass in the first degree or experience in an appropriate field of study at Midlands State University or any other approved University.

5.3 The Chairperson of a department shall submit in writing to the Faculty Higher Degrees Committee a statement on the suitability or otherwise of each applicant.

6. STRUCTURE OF SEMESTER

Shall be as prescribed in the Faculty Regulations

7. DURATION OF STUDY PROGRAMMES

7.1 The following shall normally be the duration for the completion of the post-graduate diploma and taught Masters degrees:

PROGRAMME	MINIMUM PERIOD	MAXIMUM PERMITTED DURATION (Including Registration)

POSTGRADUATE DIPLOMA		
Full-Time	4 Semesters	8 Semesters
MASTERS PROGRAMMES		
Full-Time	5 Semesters	9 Semesters

8. PROGRAMME OF STUDY

8.1 Each student is required to pursue a prescribed programme of study under the direction of a supervisor, who shall be a member of the academic staff of the University in the appropriate discipline. Normally, one or more supervisors, either from within or from outside the University, will also be appointed. If the major part of the student's research programme is being conducted at an institution outside the University, a co- supervisor at the institution shall, where possible, be appointed.

8.2 The supervisors shall report on each student's progress to the appropriate Departmental Board and/or to the Faculty Higher Degrees Committee.

8.3 Students must maintain regular contact with the supervisor and may be required to attend scheduled continuous assessment sessions for certain periods at the University as directed by their approved supervisor(s).

8.4 No break in the normal continuity of study shall be permitted, except by permission of the Senate on the recommendation of the Faculty Higher Degrees Committee. If a student who is doing continuous assessment as component of the degree is unable to attend scheduled sessions for health reasons or other reasons for longer than 72hours, he/she must notify the appropriate Department through his/her supervisor as soon as possible and where the absence is due to health reasons, submit documentary evidence in support thereof by medical practitioner registered in accordance with the Health and Allied Professions Act.

9. INTERRUPTION OF STUDY PROGRAMME

A student registered for a post-graduate programme may not break his/her programme of study for more than two consecutive semesters without the approval of Senate on the recommendations of the Faculty Higher Degrees Committee.

10. PROGRAMME STRUCTURE FOR POSTGRADUATE DIPLOMAS AND TAUGHT MASTERS DEGREES

The programme structure for the Post-Graduate Diplomas and taught Master's Degrees shall comprise:

10.1 Core Modules

10.2 Prescribed Electives (to be defined by Department)

10.3 Project/Dissertation and supervision

11. PROJECT/DISSERTATION TOPIC AND SUPERVISION

11.1 The Chairperson of Department and members of his/her Graduate Studies Committee shall, in respect of each student, approve a dissertation or project area and propose a Supervisory Committee and /or Supervisor for the candidate.

12. GRADING SYSTEM

12.1 Student performance in a module shall be graded as follows:

12.1.1 DIPLOMA/DEGREE CLASSIFICATION

Range of Marks %	**Interpretation**
80-100	Distinction
70-79	Merit
60-69	Credit
50-59	Pass
Below 50	Fail

13. ELIGIBILITY FOR EXAMINATIONS

13.1 A student shall attend all such lectures, tutorials, seminars and practical work and undertake all other activities and assignments as approved by the University.

13.2 Each department shall, with the approval of Senate, determine the requirements for the modules they offer. A student who does not fulfil the requirements for any module shall not be allowed to take the examination for that module.

14. MODULE CREDIT REQUIREMENTS FOR POSTGRADUATE STUDIES

The minimum credits that the registered student is required to earn to graduate shall vary and are prescribed according to the programme regulations.

15. PROGRESSION

15.1 If a candidate fails all modules at the end of the first semester, he or she shall repeat failed modules.

15.2 If a candidate obtains the minimum number of credits required but fails in not more than two core modules, he or she can proceed to the second level, but must pass all such core modules before obtaining the degree. To proceed from one level to another, a student should at least pass 75% of the modules.

16. DEFERMENT OF EXAMINATION

16.1 *On grounds of ill-heath:*

a student, who has satisfied all the requirements but is unable to take the main (end of semester) examinations on grounds of ill health shall, on application to the Registrar, and on provision of a Medical Certificate issued or endorsed by the University Medical Officer, be allowed to defer the semester examinations, and shall be allowed to take them

at the next offering. Subsequent application for deferment, on grounds of ill health, shall be subject to a Medical Certificate issued by a properly constituted Medical Board.

16.2 *On grounds other than ill-health*:

in cases of deferment on grounds of other than ill-health, the appropriate Dean shall invite the applicant for an interview and advise the University as appropriate. It shall be the student's responsibility to satisfy the University beyond reasonable doubt why he/she wishes to defer the examinations.

16.3 In all cases of deferment of examinations, the applicant(s) shall obtain written responses from the Registrar before leaving the University.

17. SCHEME OF EXAMINATION

17.1 The examination shall consist of written papers and such practical work/project/ dissertation as may be prescribed by departmental regulations.

71.1.1 A candidate may also be examined orally on the substance of his/her dissertation.

18. SUBMISSION OF DISSERTATION

18.1 A candidate shall submit a proposal or project of his/her dissertation for approval to the appropriate department not less than six months before the completion of the dissertation. After the proposal of the dissertation has been approved, it may not be changed except with the permission of the Senate on the recommendation of the Department and the Faculty Higher Degrees Committee.

18.2 After completing his/her programme of study, the candidate shall submit to the Faculty Board through the Chairperson of Department a dissertation which shall comply with the following:

18.2.1 Each copy of dissertation shall be signed by the candidate and the Supervisors/Supervisory Committee.

18.2.2 The dissertation or thesis shall normally be written in English and the presentation must be satisfactory.

18.2.3 A dissertation submitted for a Degree shall consist of the candidate's own account of his/her research. There shall be a declaration to the effect that the dissertation is the candidate's own work produced from research undertaken under supervision. It may describe work done in conjunction with the candidate's Supervisor provided that the candidate states clearly his/her share in the investigation and that his/her statement is certified by the Supervisor

18.2.4 A dissertation / project submitted for Taught Master's Degree and Postgraduate Diploma must normally range between the following:

Taught Master's Degree	15 000 - 20 000 words
Post-Graduate Diploma	10 000 - 15 000 words

19. PLAGIARISM

19.1 Students must be guided by the University's Anti-plagiarism and Responsible use of Artificial Intelligence (AI) Policy

19.2 Plagiarism is an academic offence and the University does not tolerate it whatsoever.

19.3 Plagiarism is the unacknowledged use of another person's material or ideas. As such, plagiarism is an academic offence in the sense that theft is in ordinary daily life.

19.4 Recommendations on the severity of the penalty shall be determined by theappropriate Departmental Board or Panel of Examiners. Cases of plagiarism shallbe handled in the following manner: -

19.5 ***Minor Cases of Plagiarism***

19.5.1 **First Offence:** in the case of plagiarism being discovered in a piece ofwork such as an essay or laboratory report or dissertation the student shall get a Chairperson's warning but shall be given an opportunity tore-do and re-submit an acceptable piece of work after one week and shall be awarded a maximum mark of 50%.

19.5.2 **Second Offence:** The student shall get a Dean's warning and shall beawarded a mark of zero.

19.5.3 **Third Offence:** Senate shall take disciplinary measures such as suspension or expulsion of the student who will have been awarded amark of zero.

19.6 ***Major Cases of Plagiarism***

19.6.1 In the case of plagiarism being discovered in a project at end of the semester, that candidate shall be denied the opportunity to resubmit the project.

19.6.1.1 The new project shall be submitted in the second semester after the discovery of the offence.

19.6.1.2 The new project shall be awarded a maximum mark of50%.

19.6.2 In the case of plagiarism being discovered in a project for the second time and after re-submission, a mark of zero shall be awarded and recorded, and Senate shall take disciplinary action either to suspend orexpel the student.

20. MISCONDUCT AT EXAMINATIONS

20.1 Subject to Ordinance 2, (the Rules of Student Discipline Ordinance) any candidate found using unauthorized materials, including electronic gadgets or attempting to obtain information from other candidates or their papers, or otherwise guilty of misconduct during the examination may be disqualified not only in the examination, but in the whole examination, and further disciplinary action may be taken by the University.

NB: Items 19 and 20 are additions which originally were not there.

21. EXAMINERS AND DETERMINATION OF RESULTS

21.1 *WRITTEN MODULES*

Two Examiners of whom one shall be a moderator shall be appointed by Senate on the recommendation of the Faculty Higher Degrees Committee to examine written modules in respect of the taught Master's Degree programmes and Post Graduate Diploma programmes.

1.2 APPEALS

1.2.1 All re-mark requests will not be entertained unless directed by Senate. However, a student who has been given a withdraw or a discontinue decision may appeal in writing to the Registrar within twenty-one days after the publication of the examination results.

1.2.2 An application entered on a candidate's behalf by a person other than the aggrieved candidate himself/herself shall not be entertained.

1.2.3 No action shall be taken on an application which is submitted outside the time stipulated in 21.2.1. Review shall not proceed unless the review fee is fully paid.

1.2.4 If it emerges a complaint for review is frivolous or ill motivated, the Faculty Board may prescribe further sanctions, which may include barring the complainant from the University examination for a stated period or an indefinite period.

22. PUBLICATION OF RESULTS

22.1 Results of examinations taken at the end of each semester shall normally be published by the Registrar before the commencement of the following semester.

22.2 A result statement indicating a student's performance in the examination shall be made available to the student via his/her e-learning account.

23. AWARD OF DEGREE

23.1 A Post-Graduate Diploma or Taught Master's degree shall be awarded to a candidate who has been properly admitted to the University, has followed the approved programme of study over the period, and has fulfilled both the University and Faculty/Departmental requirements.

24. TRANSCRIPT OF ACADEMIC RECORD

At the end of a student's programme, the University shall issue to the particular student a complete transcript of his/ her academic record.

25. CANCELLATION OF AWARD

Notwithstanding previous confirmation of an award of a taught Master's Degree and Post Graduate Diploma, Senate may at any time cancel an award, even with retrospective effect, if it becomes known that:

25.1 a candidate had entered the University with false qualifications, or

25.2 a candidate had impersonated someone else, or

25.3 a candidate had been guilty of an examination malpractice.

25.4 that there are other reasons that would have led to the withholding of confirmation of the award in the first place.

In any such event, the decision of Senate shall be published and the candidate notified. Such cancellation and the reasons for it shall be entered on the candidate's transcript.

GENERAL ACADEMIC REGULATIONS FOR MASTER OF PHILOSOPHY AND DOCTOR OF PHILOSOPHY DEGREES

1. PREAMBLE

1.1 The Senate shall be the final authority for the interpretation of these Regulations.
1.2 The Senate reserves the right to alter, amend, repeal, suspend or replace any of these Regulations.
1.3 A candidate who has begun a programme of study following one set of Regulations shall not be affected by a new set of Regulations.
1.4 The General Academic Regulations shall take precedence over Faculty Regulations.
1.5 Every candidate, by the act of registering, agrees to abide by all the rules, regulations and policies of the University and of the relevant Faculty and Department.
1.6 Candidates shall be held liable for contravening any of these regulations.
1.7 In these Regulations the following terms shall be used as described:

"Concept Note"
A one (1) to five (5) paged document that outlines an area and nature of study.

"Deferment"
Suspension of studies for a period of at least one semester, that is, taking time off a study period of up to two consecutive semesters.

"Forced Termination of Studies"
A candidate's studies may be terminated on academic or non-academic grounds such as when they fail to meet the requirements set out in the General Academic Regulations for Master of Philosophy and Doctor of Philosophy Degrees.

"Level"
A prescribed phase of study.

"Presumed Withdrawn"
A candidate is presumed to have withdrawn from their studies, when after the candidate has completed all registration formalities and without explanation, fails to submit progress report.

"Programme"
A plan of study lasting over a period of time, which leads to the awarding of a degree of the University.

"Progress Report"
A document that outlines work done in a particular semester.

"Registration"
Registration shall mean providing respective documents as may be required, and paying semester fees as set out in the fees structure on or before the beginning of every new semester, and obtaining confirmation of registration.

"Semester"
A prescribed period of study shall normally made up of six (6) months.

"Subject"
A field of study offered by a department.

"**Supervision**"

Supervision shall mean mentoring of the supervisee who shall be expected to work independently but under expert guidance and oversight.

"Thesis"
A defined practical assignment, which is separately examinable.

"Voluntary Termination of Studies"

This is a voluntary decision by the student to terminate their studies at the University.

2. GENERAL PROVISIONS

2.1 Departments, Faculties, Schools and Institutes are responsible for ensuring that all students admitted into an MPhil/PhD/DPhil programme have met the admissions requirements; that the research is reasonable, and capable of yielding significant results; and that candidates have fulfilled all requirements for the MPhil/PhD/DPhil degree prior to being recommended to the Senate for the aforesaid award.

2.2 The objective of the Masters degree by research (MPhil) and doctoral study (PhD/DPhil) is to guide students to do original research. Thesis produced should be original and suitable for at least two publications in an accredited peer-reviewed indexed journal as a contribution towards knowledge and insight of the discipline.

2.3 The degrees shall be offered in all Departments within all Faculties; Schools, and Institutes. Fields of study or research shall be determined by Senate.

2.4 The maximum timeframe expected for a decision on provisional and full registration from the date of receipt of the application shall normally be 21 working days respectively.

3. REGULATIONS

3.1 There shall be Departmental, Faculty, School or Institute Regulations, to be read in conjunction with the General Academic Regulations for Master of Philosophy and PhD/Doctor of Philosophy Degrees.

3.2 Area of study/specialisation and any other additional requirements may be prescribed and/or suggested the by Department, Faculty, School or Institute Regulations.

4. POSTGRADUATE DEGREES

The University shall offer Master of Philosophy (MPhil) and PhD/Doctor of Philosophy (PhD/DPhil) either be by Traditional/Research thesis, or Thesis by publication.

5. GENERAL REQUIREMENTS FOR ADMISSION

Application for Master of Philosophy and PhD/Doctor of Philosophy shall be open throughout the University's calendar year.

5.1 Master of Philosophy Degree

5.1.1 Applicants must have obtained an appropriate honours degree (ZNQF Level 8), or equivalent.

5.2 PhD/Doctor of Philosophy

5.2.1 A candidate must be in possession of a pass in a relevant Master's or MPhil degree (ZNQF Level 9) or its equivalence, in a field offered by the University.

5.3 Application Requirements and Procedure

5.3.1 Application for MPhil or PhD/DPhil shall be made online on https://ww5.msu.ac.zw on the MSU-Postgraduate Studies Portal.

5.3.2 A one-to-four paged concept note, certified copies of academic, and professional qualifications shall be required.

5.3.3 A motivational statement, or curriculum vitae or resume, and or any such additional documents, and or artefacts, recordings or publications **may be** requested.

5.3.4 Proficiency in English language, or any such, as may be deemed relevant to be used for training shall be verified.

6. REGISTRATION

6.1 Provisional Registration

Students shall register online at the beginning of each semester on the MSU-Postgraduate Studies Portal.

6.1.1 **The candidate shall be advised of the need to register, and will only be regarded as a student after successfully completing registration procedures. The purpose of provisional registration shall be to:**

i. allow the candidate time to develop an acceptable supervised proposal for the thesis; and
ii. allow the candidate to demonstrate suitability for the degree by meeting the requirements established by the University.

6.1.2 The candidate shall be given one semester (six months) to complete and defend (at a faculty organised seminar series) the full proposal. The semester will not be counted as part of the training period. Extension of period rests with Senate.

6.2 Full Registration

At the end of the provisional registration period, a candidate shall be required to apply for full registration provided they have successfully presented their final supervised proposal at a scheduled Faculty, School or Institute Seminar Series. If the candidate fails to meet the requirements for full registration within the stipulated time, the Faculty Higher Degrees Committees, the School, Institute or Departmental Board may recommend period of extension for approval by Senate, or termination of studies.

6.2.1 MPhil students must complete their studies within 24 months (2 years), and PhD/DPhil four years (48 months) after registering for full registration, continuation, thereafter is subject to approval by Senate.

6.3 Progress Reporting

There shall be Candidate and Supervisor(s) Progress Reporting submitted to the university online vial the MSU-Postgraduate Studies Portal.

7 ROLES OF THE RESEARCH AND INNOVATION DIVISION, DEPARTMENTAL BOARDS, FACULTY/SCHOOL/INSTITUTE HIGHER DEGREES COMMITTEES AND DEANS COMMITTEE ON ACADEMIC AFFAIRS

7.1 Research and Innovation Division

7.1.1 The Research and Innovation Division shall render administrative and operational support to Departments, Faculties, Schools and Institutes with respect to MPhil and PhD/DPhil studies in the University.

7.2 Boards and Higher Degrees Committees

7.2.1 In the case of a Provisional Registration, the Departmental Board, School, or Institute shall reflect on the **concept note**, guided by, *inter alia*, whether it is:

a. valuable and has an impact to humanity, the discipline, innovation, industry and commerce, and academia in general;
b. researchable, and or may lead to the development, promotion or creation of new knowledge, goods and or services, and or
c. whether the department and, or the university has the capacity in terms of the required infrastructure, supervision (even as externally hired), funding or whether there is potential for self and or external funding, and;

7.2.1.1 Suggest and recommend supervisor(s).
7.2.1.2 Recommend to the Faculty Higher Degrees Committee granting of a provisional or Full Registration.

7.2.2 Faculty, School or Institute, shall grant or reject provisional registration thereafter recommend to DCAA to approach Senate for ratification.

7.2.3 In the case of a Full Registration:

7.2.3.1 the Departmental Board, School, or Institute shall reflect on **the supervised proposal**, guided by whether the proposed study is:

a. valuable and has an impact to humanity, the discipline, innovation, industry and commerce, and academia in general;
b. researchable, and or may lead to the development, promotion or creation of new knowledge, goods and or services, and or
c. Whether the department and, or the university has the capacity in terms of the required infrastructure, supervision (even as externally hired), funding or whether there is potential for self and or external funding.
d. Recommend to the Faculty Higher Degrees Committee for the granting of Full Registration.

7.2.3.2 Satisfy itself that the Departmental Board fulfilled Section 7.2.1 and grant, or reject provisional registration as recommended by the Departmental Board; report the decision to the Deans Committee on Academic Affairs (DCAA) for Senate decision.

7.2.4 In the case of a full registration, the Faculty Higher Degrees Committee shall satisfy itself that **the supervised proposal**:

a. Addresses the research, innovation and industrialisation needs of the university and country, humanity and or,
b. Potentially leads to the development, promotion or creation of new knowledge, goods and or services, and or
c. Whether the department and, or the university has the capacity in terms of the required infrastructure, supervision (even as externally hired), funding or whether there is potential for self and or external funding.

7.2.5 Recommend to DCAA for the granting of a full registration or termination of studies.

7.3 The Deans Committee on Academic Affairs (DCAA)

7.3.1 Recommend to Senate for Provisional, Full Registration, Termination or Extension of Studies decisions.

8. UPGRADE FROM MPHIL TO PhD/DPHIL

A candidate registered for the MPhil degree may apply for upgrading to PhD/DPhil, after completion of not less than twelve (12) months' studies if their supervisor(s) so recommend(s), upgrade registration to PhD/DPhil. The application shall be accompanied by a written recommendation from the Faculty Higher Degrees Committee, School or Institute, submitted to the Research and Innovation Division. DCAA shall make a determination and recommend to Senate for approval. The duration of further study, if any, which shall be required to pursue for the PhD/DPhil shall be prescribed by Senate.

9. DOWNGRADE FROM PhD/DPHIL TO MPHIL

A candidate registered for the PhD/DPhil, who wishes to downgrade to MPhil may apply for permission to downgrade to MPhil. The following shall be required by DCAA for recommendation to Senate:

i. Supervisor(s), Faculty Higher Degrees Committee, School or Institute written recommendation.

10. SUPERVISION OF MPHIL OR PhD/DPHIL

10.1 Supervision Requirements

An MPhil or PhD/DPhil candidate shall conduct research under the guidance of a supervisor(s) as approved by Senate. All members of the supervisory team shall be suitably qualified scholars or may be experts/practitioners who shall report on each candidate's progress at the end of each semester. Normally, a supervisor shall be permitted to supervise a maximum of five (5) MPhil candidates and four (4) PhD/DPhil.

10.2 Supervisor qualities and qualifications

10.2.1 The MPhil or PhD/DPhil supervisor must be a doctoral graduate with proven academic accomplishments, and or industrial experience in research; and must have a solid or respectful publication record in peer reviewed journals or assessed equivalents.
10.2.2 Preferably, be in the professorial grade or have credentials that qualify them to be in that grade.
10.2.3 The MPhil or PhD/DPhil supervisor must be knowledgeable and possess deep, current content knowledge, and stays ahead of the supervisee for effective mentoring; demonstrates awareness of research, innovation and industrialisation needs and preferably, have experience in building

collaborative relationships with a range of colleagues (internal or external) within own and adjacent disciplines and or research areas, research groups, and with stakeholders and other users of research.

10.2.4 Normally, two supervisors shall be assigned to each project, a main supervisor plus a co-supervisor. At least one of them must be an academic at the Midlands State University. In cases where the University has no capacity to supervise, all supervisors may be drawn from any national, regional, international university or research institute or institution.

10.2.5 Senate reserves the right to add or remove supervisor(s) from the recommended supervisory team.

10.3 Duties, Expectations of Supervisor and Supervisee

10.3.1 The collective duties of the Supervisor shall be to:

10.3.1.1 Initiate the supervisee into sound research culture and practices acceptable in the field chosen for study; assist the supervisee progress semesterly and complete of set targets.

10.3.1.2 Guide supervisee on topic selection and refinement; proposal and thesis development; mentor candidate in paper presentations and article publications; lead on experiments, and compilation of the final thesis report in readiness for examination.

10.3.1.3 Assist in the search and application of research grants, and where applicable, facilitate filing for the protection of intellectual property (IP), through patents, copyrights, or industrial designs.

10.3.1.4 The supervisor and supervisee shall thrive to identify or develop innovations, or create new products, or knowledge for dissemination to the society.

10.3.1.5 The medium language shall normally be English (UK). Where other languages, other than English, shall be used, translations into English (UK) shall be required at the expense of the candidate.

10.4 **Supervision Allowances**

10.4.1 Supervisors shall normally be paid an allowance at the end of every semester as shall be determined by the University, provided the candidate being supervised is registered and has been supervised for the semester upon submission of progress reports.

10.4.2 The supervisor(s) shall be paid the allowance for the semester in which the final thesis is submitted for examination.

10.5 **Supervisee.**

10.5.1 A supervisee or candidate shall maintain regular contact with the university and their supervisor(s).

10.5.2 No break in the normal continuity of study shall be permitted, except by permission of the Senate.

10.5.3 An MPhil or PhD/DPhil candidate **may** be engaged in teaching at the University, provided that the total demand made by teaching duties does not exceed eight hours of work per week.

11. Deferment of Studies

11.1 Deferment of studies shall be permitted. Normally, the deferment shall not be more than two consecutive semesters. The candidate must have the deferment process finalised before leaving the University as deferment shall not normally be done in retrospect.

12.1 Any extensions to the deferment period shall be approved by Senate.

12.2 At the expiry of the deferment, the candidate shall be required to regularise resumption of their studies.

12. Termination of Studies and Supervision

12.1 Termination of studies and or supervision shall be permissible.

12.1.1 Termination shall be 'Voluntary' wherein a candidate or appointed supervisor shall intentionally express desire to terminate their studies or supervisor contract any time of the year by notifying, in writing or otherwise, the university through the Research and Innovation Division.

12.1.2 A candidate who wishes to voluntarily terminate their studies shall complete a 'Termination of Studies Form,' and submit a completed and signed form to the Research and Innovation Division together with a supporting letter detailing the reasons for intending to terminate their studies.

12.1.3 The Research and Innovation Division, together with Department and Faculty, School, or Institute may interview, even separately, the applicant after which studies shall be terminated. DCAA shall approach Senate for ratification.

12.1.4 Termination shall be 'Forced Termination' wherein the candidate shall be obliged to cease undertaking studies due to:

12.1.4.1 Lapse in the maximum permissible period required to have completed the programme, or;
12.1.4.2 Failure to fulfil full registration requirements within the stipulated time, or due to;
12.1.4.3 Poor academic performance or gross failure to progress.
12.1.4.4 Failure to re-register after deferment of studies period has lapsed and no communication granted by the University.
12.1.4.5 Gross violation or failure to abide by binding and other University regulations, policy or procedure or;
12.1.4.6 Where the candidate has provided false information at the time of application or at any point thereafter.

12.2 A recommendation to terminate studies shall be made by the Faculty Higher Degrees Committee, School or Institute to the DCAA.

12.3 Termination of supervision

12.3.1 Supervisor(s) may voluntarily terminate or have their supervision contract terminated. A Supervisor may voluntarily express desire to terminate supervision contract with the University in writing.

12.3.2 The Supervisor's contract may be terminated should there be gross evidence of lack of supervision or violations of any standing University rules and regulations or as a result of failure to undertake the expected responsibilities of a supervisor.

13. Thesis Submission for Examination

13.1 Only a supervised thesis shall be eligible for examination. Candidates shall be examined by a board of examiners appointed by Senate.

13.2 MPhil candidates must have published at least one and PhD/DPhil at least two publications in reputable refereed journals emanating from their masters or doctoral research.

13.3 The candidate shall present their work during a scheduled seminar series, have the thesis language-edited (proof of which must be an authentic editorial certificate or letter); thesis be subjected to an anti-plagiarism check with similarity index (SI) of less than 15% before the thesis is submitted for examination.

14. Examinations

14.1 There shall be examinations of the thesis produced at the end of the candidate's period of study.

14.2 The Faculty Higher Degrees Committee, on recommendation of the departmental board; School or Institute shall recommend names of at least six (6) distinguished scholars, two of whom must be from Midlands State University, two from outside Midlands State University but in Zimbabwe, and two from an international institution.

14.3 Three examiners from the recommended six (6) as in **Section 15.2**, one from Midlands State University, and two from outside the University shall be appointed to examine the submitted final thesis, and produce examiners' reports within 30-days from the date of appointment.

14.4 Upon receipt of the three examiners' reports, an Internal Examination Committee chaired by the Executive Director (or designate) Research and Innovation Division shall make a determination on whether or not the candidate proceeds for an oral examination/*viva voce*.

14.5 If the thesis submitted for a degree is judged to be inadequate, the Internal Examination Committee may recommend to the Academic Board that the candidate be permitted to re-submit for examination in a revised form within a stipulated period.

14.5.1 If granted by the Academic Board, the candidate shall register for the stipulated period.

14.5.2 The candidate shall be deemed to have failed the degree if they do not re-submit the revised thesis within the stipulated period, and **NO** candidate shall re-submit thesis for examination more than once. A candidate shall proceed to appear for oral examination or *viva voce* if at least two of the examiners passes the thesis in accordance to the set guidelines:

14.5.2.1 The candidate shall be given one (1) month notice in writing of the pending examination.

14.5.2.2 The oral examination committee shall comprise of the Executive Director Research and Innovation Division (Chairing) or designate, Postgraduate Studies Manager, at least two (2) examiners; at least one (1) supervisor; the Executive Dean of the Faculty or Executive Director of the Institute; the Chairperson of the Department or Coordinator; one (1) Senior Academic member from the

Department; one (1) Senior Academic member nominated by the Executive Dean of the Faculty or Executive Director of the School or Institute, the Deputy Registrar (Academic Affairs) and Research and Innovation Division technical and support staff nominated by the Executive Director.

14.6 Examination Verdict

14.7 The final decision shall be endorsed by the Academic Board on the recommendation of the Examination Committee based on the thesis and oral examination (*viva voce*).

14.8 Four categories shall be used as follows;

14.8.1 The final examination decision to be recommended to the Academic Board for MPhil and PhD/DPhil candidates shall fall within one of the following categories:

14.8.1.1 Pass.

14.8.1.2 Pass subject to minor corrections to the satisfaction of the candidates' supervisor(s) and the department.

14.8.1.3 Pass subject to major corrections to the satisfaction of the candidates' supervisor(s), department, faculty, school or institute.

14.8.1.4 Fail.

14.9 In the event of a candidate failing, the candidate may submit a duly motivated appeal in writing to the Executive Director Research and Innovation Division within fourteen working days from the date of notification.

14.9.1 If the appeal is granted, an independent examiner shall be appointed who shall act as an arbiter. The verdict of the arbiter shall be final and binding.

15. Deferment of Oral Examination

15.1.1 The oral examination may be deferred and candidate shall be allowed to take them at the next offering on the grounds of ill-health subject to a Medical Certificate issued by a properly constituted Medical Board. If on the grounds other than ill-health, the Executive Director Research and Innovation Division shall invite the applicant for an interview, thereafter and the University appropriately.

15.1.2 It shall be the candidate's responsibility to satisfy the University beyond reasonable doubt on why they wish to defer the oral examination (*viva voce*).

16. Publication of results and thesis archiving

16.1.1 Thesis Examination Results shall be published by the Registrar upon satisfactory fulfilment of all the stipulated requirements for the degree.

16.1.2 A result pass list indicating the candidate's performance in the examination shall be made available to the candidate.

16.1.3 After the final thesis has been approved, print copies shall be bound in standard form Art Vellum or cloth; overcast; edges uncut; lettered boldly up back in gold (0.5-1.25

cm) degree; date and name; before the award of the degree is published by the Registrar.

16.1.4 Two bound copies plus a soft copy in PDF must be presented to the University with the approval of the Departmental Chairperson.

16.1.5 Print and soft copies of the thesis that have been accepted for the award of the degree shall be distributed for archiving by the University.

17. Award of Degree

17.1 An MPhil or PhD/DPhil degree appropriately designated shall be awarded to a candidate who was properly admitted to the University, has followed the approved programme of study over the stipulated period, and has fulfilled both the University, Faculty and Department, School or Institute requirements.

17.2 The result of an examination held for the award of an MPhil or PhD/DPhil degree shall be presented to Senate for confirmation. Thereafter, the degree shall be conferred under the seal of the University upon each successful candidate.

18. Cancellation of Award

18.1.1 Notwithstanding confirmation of an award of a degree, Senate may at any time cancel an award, even with retrospective effect, if it becomes known that:

18.1.1.1 a candidate had entered the University with false qualifications, or

18.1.1.2 a candidate had impersonated someone else, or

18.1.1.3 a candidate had been guilty of an examination malpractice, or

18.1.1.4 that there are other reasons that would have led to the withholding of confirmation of the award in the first place.

18.1.1.5 In any such event, the decision of Senate shall be published on the University Notice Board and the candidate notified.

18.2 At the end of a candidate's programme, the University shall issue to the particular candidate a complete transcript of their academic record.

19. DOCTOR OF PHILOSOPHY DEGREE (*HONORIS CAUSA*)

19.1 Honorary degrees may be conferred upon those who have achieved distinction through outstanding intellectual or creative achievements, or through outstanding leadership in education, business, public service, or any other discipline.

19.2 Recommendations for such awards shall be initiated by a faculty, through the Departmental and Faculty Boards

19.3 The Honorary Degrees and Distinctions Committee shall meet to consider the submission from the Faculty and make recommendations to Senate and Council.

FACULTY OF AGRICULTURE, ENVIRONMENT AND NATURAL RESOURCES MANAGEMENT
REGULATIONS FOR THE UNDERGRADUATE AND POSTGRADUATE DEGREE PROGRAMMES

1. INTRODUCTION

1.1. These regulations shall be read in conjunction with the General Academic Regulations for the Undergraduate Degrees herein after referred to as General Regulations.

1.2. The Faculty of Agriculture, Environment and Natural Resources Management offers the following undergraduate degree programmes:

1.2.1. Bachelor of Science Honours degree in Agricultural Economics and Development (HAGED)

1.2.2. Bachelor of Science Honours degree in Agricultural Value Chain and Marketing (HAVCM)

1.2.3. Bachelor of Science Honours degree in Applied Agricultural Sciences and Business (HAASB)

1.2.4. Bachelor of Science Honours degree in Information Technology for Agricultural Systems

1.2.5. Bachelor of Science Honours degree in Agronomy (HAGRO)

1.2.6. Bachelor of Science Honours degree in Crop Production and Horticulture (HCRH)

1.2.7. Bachelor of Science Honours degree in Horticulture (HHORT)

1.2.8. Bachelor of Science Honours degree in Crop Protection (HCPR)

1.2.9. Bachelor of Science Honours degree in Agronomy Majoring in Crop Improvement and Seed Systems (HACPR)

1.2.10. Bachelor of Science Honours degree in Animal and Wildlife Science (HAWS)

1.2.11. Bachelor of Science Honours degree in Animal Products Processing Technology (HAPPT)

1.2.12. Bachelor of Science Honours degree in Dairy Sciences (HDS)

1.2.13. Bachelor of Science Honours degree in Wildlife Ecology and Management (HWEM)

1.2.14. Bachelor of Science Honours degree in Poultry Science (HPS)

1.2.15. Bachelor of Science Honours degree in Biosystems Engineering (HBE)

1.2.16. Bachelor of Science Honours degree in Environmental Science and Technology (HEST)

1.2.17. Bachelor of Science Honours degree in Water and Sanitation Technology (HWST)

1.2.18. Bachelor of Science Honours degree in Land and Water Resources (HLWRM)

1.2.19. Bachelor of Science Honours degree in Agricultural Water Management (HAWM)

1.3. The Faculty of Agriculture, Environment and Natural Resources Management offers the following postgraduate degree programmes:

1.3.1. Master of Science degree in Applied Agricultural Economics (MAAE)

1.3.2. Master of Science degree in Agricultural Informatics and Data Analytics (MAIDA)

1.3.3. Master of Science degree in Crop Science (MCS)

1.3.4. Master of Science degree in Crop Protection (MCPR)

1.3.5. Master of Science degree in Horticulture Sciences (MHORT)

1.3.6. Master of Science degree in Animal Breeding and Biotechnology (MABB)

1.3.7. Master of Science degree in Land and Water Resources Management (MLWRM)

2. ENTRY REQUIREMENTS

2.1. ***NORMAL ENTRY***

For normal entry to an undergraduate degree programme in the Faculty of Natural Resources Management and Agriculture, a prospective student must satisfy the entry regulations specified in the General Regulations and in so doing, must also meet the following Faculty requirements:
Must have passed O' Level Mathematics and English Language with C or better and obtained A' Level passes in at least two of the following subjects or their recognised equivalents:
Biology/Agriculture, Chemistry, Physics, Mathematics, Geography.
Each Department may prescribe specific combinations of these A' Level subjects.

2.2. ***SPECIAL ENTRY***

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the respective programme.

2.3. ***MATURE ENTRY***

Refer to Section 3.3 of General Regulations.

3. DEFINITION OF TERMS

3.1. ***FIELD TRIP***

A field trip is an excursion outside the University Campus under the supervision of academic and/or technical staff of the university. It may last for part of a day or be conducted over a longer period. The field trip can form part of a module or if it is of a sufficient duration, form a module in its own right.

3.2. For the definitions of programme, semester, module, level, Work-Related Learning and credit, refer to General Regulations.

4. GENERAL PROVISIONS

4.1.1. The BSc. Agriculture, Environment and Natural Resources Management programme is a four-year, full - time programme.

4.1.2. In Level One (1) and Level Two (2) of the degree programme, students will take **CORE** Agricultural Practice (AGPR 131, AGPR 132, AGPR234 and AGR235) modules. The practicals are normally conducted on Fridays. Students absent from these practicals three times in a semester without the written

permission of the Executive Dean of the Faculty will automatically fail the module.

4.1.3. In their third level, students go on Work-Related Learning for a period of at least ten (10) months and during this period they should undertake a Research Project.

4.1.4. For each semester, all pre-requisites that apply to a module shall be listed in the special regulations for the degree programme of which the module forms a part. A student shall not register for a module unless he/she has passed all pre-requisites for that module.

4.1.5. A student may include in his/her programme, modules offered by other Departments.

4.1.6. A student's selection of modules will be subject to the approval of the Dean of the Faculty and Chairpersons of all respective Departments. The selection will also be subject to:

4.1.7. A student enrolling for modules worth at least 72 notional credit hours in each semester except for the Work-Related Learning level.

4.1.8. Registration for modules to be taken in a semester is completed within the first two weeks of the semester.

5. ASSESSMENT

5.1. The assessment of a module shall be based on a written examination and continuous assessment (assignments, tests and practicals) submitted during the semester according to a schedule laid down by the appropriate Departmental Board. The examination shall normally contribute 60% of the final mark while continuous assessment accounts for 40%.

5.2. Practicals and tutorials are compulsory and failure to attend practicals may result in a student being barred from sitting the final examinations. A student may not be allowed to sit an examination if he/she has no coursework mark.

6. PROVISIONS FOR PROGRESSION

6.1.1. Students progress automatically from semester one (1) to semester two (2) at each level with the exception of special cases where the Senate directs otherwise.

6.1.2. To proceed to the next level the student must have accrued at least 108 notional credit hours. However, a student may be allowed to proceed carrying failed modules in order to enable him or her to accrue the required 144 notional credit hours per level.

6.1.3. Students carrying over modules will be re-examined at the next regularly scheduled examination normally taken at the next semester examinations.

7. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of General Regulations.

8. REMARKING OF EXAMINATIONS

Refer to Section 11.1 of General Regulations.

9. GRADING AND DEGREE CLASSIFICATION

9.1. To be awarded the degree, a student must pass all core modules at all levels and must have accrued at least 552 notional credit hours.

9.2. The classification of all modules and degree programmes shall be as stated in Section 5.2 of the General Regulations.

10. DEGREE WEIGHTING

For purposes of undergraduate degree weighting the results of Levels 1, 2, 3 and 4, the weighting shall be as follows:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

For degree weighting at the postgraduate level, refer to Departmental regulations.

DEPARTMENT OF AGRICULTURAL ECONOMICS AND DEVELOPMENT

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN AGRICULTURAL ECONOMICS AND DEVELOPMENT (HAGED)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To address the demand for Agricultural Economics techniques and development analysis in the areas of sustainable food production, natural resource management and overall economic growth.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have at least five 'O' Level passes including English Language, Science and at least a B in Mathematics and any two 'A' Level passes in the following subjects Agriculture, Mathematics, Economics, Management of Business, Sociology, Accounts, Geography and Biology.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation

3. PROGRAMME CHARACTERISTICS

Areas of study: Micro/Macro/Environment Economics for Agriculture, Agricultural Development, Econometrics

Specialist Focus: Agricultural Value Chains Data, Policy analysis, Value Chain Development

Orientation: Oriented towards innovation and application of theory to solve real-life problems
Distinctive Features: Knowledge of the role of agricultural entrepreneurship and marketing in agriculture, agribusiness and agro-industrial development

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Private Sector – Bankers, commodity brokers, Agribusiness Finance Managers, Applied Economists, Consultants, Marketing Agents/Sales Representatives, Extensionists, education, real estate, and public relations.
Public Sector – Researchers (Applied Economists, Marketing Agents,) Extensionists, Lecturers, natural resource and environmental management, Education and Public Relations
Self-Employment – Farmers, Consultants
Further Studies: Masters and Doctoral studies in Agricultural Economics and Development, Economics, Development, Policy

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), and Final examination (60%).

6. INTENDED LEARNING OUTCOMES

- Students to be able to analyse trade and other macroeconomic policies
- Ability to analyze domestic and international trends in agribusiness value chains and markets
- Good understanding of relationships and roles amongst stakeholders in agroindustry value chains: input suppliers, producers, traders, processors, manufacturers and consumers
- Identify problems in agricultural production and development
- To develop new practices methods and technologies that enhance production efficiency
- Students to master econometric software
- Create an understanding and application of entrepreneur skills for employment creation.
- Students should have the ability to apply effective communication and extension

7. GENERAL PROVISIONS

7.1. The BSc in Agricultural Economics and Development is a four-year full-time programme.
7.2. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.
7.4. Students can include in their programme of study, modules offered by other departments.
7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAMME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

12.1 The degree programme shall have at least 32 taught modules spread over the six teaching semesters.

Level 1 Semester 1

Code	Module Title	Prerequisite	Credits
*AGRO135	Plant Biology and Physiology		12
*AGPR131	Agricultural Practice I		12
*LWR135	Principles of Soil Sciences		12
*CS131	Basic Communication Skills		12
*HCS135	Introduction to Information Technology		12
*AGED131	Introduction to Agricultural Economics		12
*AGED133	Mathematics for Agricultural Economics		12
*AGED139	Introduction to Agricultural Planning & Development		12

Level 1 Semester 2

Code	Module Title	Prerequisite	Credits
*AGED132	Micro Economics for Agriculture 1	AGED131	12
*AGED134	Macro-Economics for Agricultural Economics 1	AGED131	12
*AGED135	Introduction to Rural Sociology		12
*AGED138	Statistics for Agricultural Economics		12
*AGPR132	Agricultural Practice II	AGPR 131	12
*HAWS122	Principles of Animal Production		12
*AGRO133	Principles of Crop Production		12

Level 2 Semester 1

Code	Module Title	Prerequisite	Credits
*AEGS231	Gender Studies		12
*AGED232	Financial Management		12
*AGED233	Microeconomics for Agricultural Economics 11		12
*AGED234	Macroeconomics for Agricultural Economics 11		12
*AGED231	NGOs and Agricultural Development		12

Level 2 Semester 2

Code	Module Title	Prerequisite	Credits
*AGED235	Economics of Environmental Management		12

*AGED236	Research Methods for Agricultural Economics		12
*AGED237	Agribusiness Management		12
*AGED238	Econometrics 1		12
*AGED239	Project Management		12
*AGED240	Farm Business Management		12
AGRO234	Value addition in Agriculture	(Optional)	12
LWR246	Climate Change and Adaptation	(Optional)	12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report		30

Level 3 Semester 2: Work-Related Learning

*AGPR 331	Work-Related Learning Report	All level 1.1 modules	45
*AGPR 332	Academic Supervisor's Assessment Report	All level 1.1 modules	30
*AGPR 333	Employer's Assessment Report	All level 1.1 modules	45

Level 4 Semester 1

Code	Module Title	Prerequisites	Credits
*AGED431	Agricultural Price Analysis and Forecasting		12
*AGED432	Agricultural Policy Analysis		12
*AGED435	International Trade and Policy		12
*AGED438	Agricultural Production Economics	AGED233	12

Options: (Choose 1)

AGED437	Banking and Capital Markets	AGED232	12
AGED440	Econometrics 11	AGED238	12

Level 4 Semester 2

Code	Module Title	Prerequisites	Credits
*AGED433	Entrepreneurship in Agriculture		12
*AGED436	Insurance and Risk Management in Agriculture		12
*AGED439	Agricultural Extension		12
*AGED434	Disaster and Relief Management		12
*AGED414	Dissertation	AGED236	12

14. MODULE SYNOPSIS

AGPR 131 Agricultural Practice 1

Farm orientation, Poultry production/ Feed mixing (University farm), Soil sampling and crop fertilization, Plant population and yield estimation, Irrigation scheduling, Blood samples and smears, Administering veterinary drugs, Cattle identification and detention, Surveying in agriculture, Introduction to vegetable production, Farm week practicals (all semesters), Crops planting and fertilizing, Pig production(examined in semester 1.1), Dairying university farm (examined in semester 1.1)

AGPR 132 Agricultural Practice II

Pig production at University farm, Dairying of University farm, Control of internal parasites in livestock, Use and calibration and spraying equipment, Ox harness and use of ox-drawn ploughs, Tractor handling and ploughing/ contour construction, Soil coding for land

classification, Vegetation coding in rangeland management, Farm budgets and Agric financing, Castration and dehorning (large and small stock), Game farming- visit a commercial undertaking, Field visits: i) research station/ research trials ii) Large scale/ small scale farming

AGED 131 Introduction to Agricultural Economics
Economic systems and organization, theory of supply and demand, market equilibrium and nature of markets. Micro – economics. Macro – Economics. The consumption, investment and savings functions. The international economy. Monetary and fiscal policies. Employment and growth policies. The role of the agricultural industry in the national economy.

AGED 132 Micro economics for Agriculture 1
Economics of agricultural production- theoretical foundations; Physical and economic relations, The factor- product relationship, the factor- factor relationship, the product- product relationship. The product supply and input demand; Product supply, Demand for inputs. Production Economics; Efficiency of resource use, Technical, Allocative and Economic efficiency, Risk and uncertainty. Economics of market demand

AGED 133 Mathematics for Agricultural Economics
The module is designed to give students a basic understanding of mathematical tools and techniques required in economics, policy formulation and business decision-making. The module seeks to show the relevance of mathematical tools/concepts such as integration linear algebra, differentiation and optimization theory in the field of economics. Trigonometry, probability, functions, graphs, solving equations, matrix algebra, multivariate calculus, differentiation and integration

AGED 134 Macro economics for Agriculture 1
Definition of macro-economics. The module will look into the branch of economics at the national level through variables such as macro-economic issues, inflation, the gross national product, employment, national accounting, demand/supply, national economics equilibrium, monetary policy, bank rate, foreign currency reserves and balance of payments, foreign currency exchange market, food policy, agricultural policies, and production theories and agriculture at the elementary level.

CS131 Basic Communication Skills
Refer to the Department of Communication Skills

HCS135 Introduction to Information Technology
Refer to the Department of Computer Science

AGRO135 Plant Biology & Physiology
Refer to the Department of Livestock and Wildlife Management.

LWR135 Principles of Soil Science
Refer to the Department of Land and Water Resources Management.

AGED 135 Introduction to Rural Sociology
Examining Rural social systems and structures and how they impact positively or negatively on development. Examining social change theories and how they relate to the development discourse

AGED139 Introduction to Agricultural Planning & Development

Theoretical approaches used to study organization and management, Examines the structure and behaviour as well as the main tasks of management. Looking in detail and questioning the applicability of management theory. Examining the application of management theory in relation to specific areas of activity. Current theory issues in reformation and improving management of Agricultural Development.

HAWS 122 Principles of Animal Production
Refer to the Department of Animal and Wildlife Sciences.

AGRO103 Principles of Crop Production
Refer to the Department of Agronomy.

AGED 232 Financial Management

Sources of agricultural finance in Zimbabwe. Interest rates. Working capital. Capital budgeting and appraisal techniques. Case studies of capitalization schemes. Cashflow budgeting. Basic interpretation of financial accounts; profitability analysis and ratios; equity ratios; Working capital ratios. Contracting as a source of financing. Impact of national programmes in agricultural finance and development. Taxation.

AGED 233 M1CRO – Economics for Agriculture 11
Basic demand relationships, Elasticity of demand, Homogeneity condition. Developments in demand theory; New theories of demand, Equilibrium and Exchange, Definition of equilibrium, Partial vs General equilibrium, Interference with equilibrium. Analysis of agricultural markets, Competitive markets and Pareto optimality, Policy intervention in markets. Welfare Economics and Economics of trade. Food and Agricultural policy.

AGED 234 Macro economics for Agriculture 11
Looks at economic behaviour and policies related to consumption, savings, investment, balance of trade, equilibrium in the goods and money market, domestic and international debt, and their implication in a more rigorous manner

AGED 235 Economics of Environmental Management
Environment and the micro-economy; Markets, Market failure and property rights, Government property rights, Economic instruments for environmental management, Attaching money values to environmental goods. Environment and the macro-economy; the green economy, Macroeconomic policy and the environment, Environment and the international economy, Environmental regulation, comparative advantage and international competitiveness, trans boundary externalities and environmental policy, trade liberalization and environmental impacts, Economics of pollution control, theory of pollution economics, the practical use of economic instruments in pollution control, Natural resource economics Management of renewal resources, market and institutional failure in the natural resource sector

ENT205 Entrepreneurship
Refer to the Department of Entrepreneurship.

LWR246 Climate Change and Adaptation
Refer to the Department of Land and Water Resources Management.

AGED 236 Research Methods

Research proposal formulation, problem definition, methods of data collection, Questionnaire formulation, and interviewing techniques, sampling procedures for surveys, organisation and implementation of surveys, survey quality control, analyses and interpretation, and results presentation. Software; SPSS, GENSTAT, SAS etc

AGED 237 Agribusiness Management
The module seeks to appropriately recall all values from other modules and contextualize them to advantage Agricultural Business Management through: Forms of business and analysis of business environment. Value chain in agricultural operations. Management principles. Cooperate governance. Operations research. Agribusiness viability and analysis.

AGED 238 Farm Business Management
Background: Functions of management-Forms of business organization in commercial agriculture. Farm records and Information systems management.
Labour management farm planning and budgeting Gross Margin Analysis. Whole farm budget and farm planning. Partial budget. Break-even budget. Cash flow budget. Sources of agricultural finance. Agricultural Support Schemes with special reference to.Government support scheme. Contract Production schemes Management of working capital. –Investment appraisal-Stock valuations and depreciation Farm trading accounts, Profit and loss account, Farm balance sheet, its analysis and interpretation. Variances analysis, Audit of farm operation and budgetary control. Business ethics.

AGED 238 Econometrics 1
The module gives empirical content to economic theory. Uses mathematics and statistical tools to analyze economic behaviour i.e. estimate relationships, infer the implications of the estimates, test hypotheses about economic behaviour and predict economic behaviour. The tools include regression models, specification testing, general linear models, problems of single equation models and simultaneous equation models. The module content focuses on econometric techniques that provide students with the understanding needed in modern economics.

AEGS231 Gender Studies
The module empowers the students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, Gender Issues in Zimbabwe, Redressing Gender Imbalances, Empowerment and Strategies for creating a gender-responsive environment. Every student has to pass the module to graduate.

AGED 239 Project Management
This module is about the modes and mechanisms through which development assistance is channelled, via investment in developing countries, for the promotion of agricultural and rural development. It is thus primarily about the rationale, context, and methods of planning, appraising and evaluating development projects and programmes. Projects and programmes are widely used when attempting to allocate limited resources for specific development purposes as effectively as possible, and a core part of the module is on methods for appraising the financial and economic efficiency of rural and agricultural development projects. These methods of appraisal are informed by economic theories and, in particular, those of applied welfare economics. While the module emphasises financial and economic efficiency, other important issues in assessing project design and impact are presented. Planning and management techniques for the project cycle are covered; including project identification and

logical framework analysis. Approaches for social and environmental appraisal of projects are also reviewed.

AGED 231 NGOS and Agricultural Development

Types of non – profit organisations in agricultural development. Registration, and management structures of non – profit organizations. Funding and financial management, scope, corporate governance and ethics in NGOs. Civil society theory. Participation theory. NGOs and agricultural development. NGOs and humanitarian relief. farming systems, stakeholder consultations in agricultural development, and links with governmental and other developmental agencies.

AGED 431 Agricultural Price Analysis and Forecasting

This course is aimed at developing an understanding of the principles of price theory and how these principles can be applied to problems in agriculture starting from the supply and demand of agricultural products. The role of price in agricultural production is examined at the onset. Specific topics include price determination and discovery, movement of prices over time and modelling of cycles and seasonality, using index numbers to correct for inflation, detailed analysis of agricultural commodities price trends, empirical estimation of demand and supply, demand shift and income effect, supply response and long-term competitiveness. Students are also introduced to the modelling of market structures, market outlook and different methods of forecasting prices, and hedging and its effectiveness for agricultural commodities.

AGED 432 Policy Analysis

This course offers a multi-disciplinary approach to understanding agricultural and food policy in Zimbabwe and the world. It examines the agricultural policy process in terms of rationale, content, and consequences. It thereby helps students develop a better understanding of policy-making entities at various levels of government. It also provides a broad understanding of how policy actions in agriculture impact not only farmers' incomes, but also the well-being of consumers, the economic viability of rural communities, and the quality of our nation's environmental resources. Looks at economic behaviour and policies related to consumption, savings, investment, the balance of trade, equilibrium in the goods and money market, domestic and international debt, and their implication in a more rigorous manner.

AGED 433 Entrepreneurship in Agriculture

Role of entrepreneurship in agricultural development and the economy. Theories of entrepreneurship. Processes of establishing a business. Sources of funding. Feasibility studies. Planning production and marketing of enterprise. Entrepreneurship and public policy. Characteristics, nature and problems of small agricultural ventures. Advantages and disadvantages of big versus small. Management techniques. Linkages.

AGED 434 Disaster and Relief Management

Introduction to disaster management. Assessment of disasters and emergency complex situations. Impact of disasters on communities and agriculture. Needs analysis during disasters. Disaster management and recovery plans. Ethics in disaster management. Long-term planning in disaster management. Case studies of disasters affecting agriculture and communities.

AGED 435 International Trade and Policy

International trade familiarizes students with theories and policies that guide international trade. This course is usually taken in one of the first semesters of a degree program. Students learn why international trade is important for Zimbabwe and other nations, and they examine how it affects production, profit and the economy.

AGED 436 Insurance and Risk Management in Agriculture

Concepts of risk management in agriculture. Theories and definitions of risk management. The measurement of risk. And the stochastic basis of risk. The Beta. Risk adjustment methods and techniques used in dealing with risk, risk financing techniques. Hedging against risk, the futures options and contracts. Insurance in agriculture; crop, animal, farm vehicles. Policy wordings, underwriting and claims.

AGED 437 Banking and Capital Markets

The course deals with the description of the activities of banks in agriculture, financial intermediaries, central banks, regulatory agencies, and monetary theory. Money market institutions. Capital market institutions. It shores up knowledge in bank management and capital markets as they relate to agriculture.

AGED 438 Agricultural Production Economics

Resource allocation, production selection, and scale of operation of agricultural firms including risk and uncertainty associated with agricultural production. Analysis of agricultural production economic models using econometrics, mathematical programming and simulation. Introduction to the systems science perspective. Techniques for Analysis of Risk, environmental Value, technological change and technology adoption.

AGED 439 Agricultural Extension

BACKGROUND: history of agricultural extension in Zimbabwe- Definition of extension. Extension ethics- communication skills: Mass media: Kinescis: art of listening. Conducting successful meetings. Extension methods and training programme planing: projects, programmes and Policy in Agricultural Extension- interpretation of Agricultural policy and planning programmes. Farmer participation in extension programme planning, monitoring and evaluation. Research – Extension- Linkages. Agricultural extension change management and support- what is change management- Donor Organisations in agricultural development and extension. Privatization of extension services- is this an option for the future? Female Extension Agents and Gender in Agricultural development and extension- Agrarian reform-challenges for extension.

AGED 440 Econometrics 11

This module in econometrics is designed to provide students will the tools required to evaluate and carry out empirical research. It focuses on the violation of classical liner regression model assumptions. Issues such as heteroscedasticity, autocorrelation, model specification and multicollinearity are discussed. Modelling with dummy variables, simultaneous equations, time series and panel data econometrics will be studied.

AGED 414 Dissertation

This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board,

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN AGRICULTURAL VALUE CHAIN AND MARKETING (HAVCM)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load:		504
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To address the demand of Agricultural Economics techniques and development analysis in the areas of sustainable food production, natural resource management and overall economic growth.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have at least five 'O' Level passes including English Language, Science and at least a B in Mathematics and any two 'A' Level passes in the following subjects Agriculture, Mathematics, Economics, Management of Business, Sociology, Accounts, Geography and Biology.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Micro/Macro/Environment Economics for Agriculture, Agricultural Development, Econometrics

Specialist Focus: Agricultural Value Chains Data, Policy analysis, Value Chain Development

Orientation Oriented towards innovation and application of theory to solve real-life problems

Distinctive Features: Knowledge of the role of agricultural entrepreneurship and marketing in agriculture, agribusiness and agro-industrial development

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Private Sector – Bankers, commodity brokers, Agribusiness Finance Managers, Applied Economists, Consultants, Marketing Agents/Sales Representatives, Extensionists, education, real estate, and public relations.

Public Sector – Researchers (Applied Economists, Marketing Agents,) Extensionists, Lecturers, natural resource and environmental management, Education and Public Relations

Self-Employment – Farmers, Consultants

Further Studies: Masters and Doctoral studies in Agricultural Economics and Development, Economics, Development, Policy

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), and Final examination (60%).

6. INTENDED LEARNING OUTCOMES

- Students to be able to analyse trade and other macroeconomic policies
- Ability to analyze domestic and international trends in agribusiness value chains and markets
- Good understanding of relationships and roles amongst stakeholders in agroindustry value chains: input suppliers, producers, traders, processors, manufacturers and consumers
- Identify problems in agricultural production and development
- To develop new practices methods and technologies that enhance production efficiency
- Students to master econometric software
- Create an understanding and application of entrepreneur skills for employment creation.
- Students should have the ability to apply effective communication and extension skills.

7. GENERAL PROVISIONS

7.1. The BSc in Agricultural Value Chain and Marketing is a four-year full-time programme.
7.2. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.
7.4. Students can include in their programme of study, modules offered by other departments.
7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAMME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

The degree programme shall have at least 32 taught modules spread over the six teaching semesters.

Level 1 Semester 1

Code	Module Title	Prerequisite	Credits
AWS101	Agricultural Biology		12
AGPR131	Practices in Farm Entrepreneurial skills 1		6
CS131	Basic Communication Skills		12
HSC115	Introduction to Information Technology		12
AGED131	Introduction to Agricultural Economics		12
AGED133	Mathematics for Agricultural Economics		12
AVCM101	Introduction to Agricultural Value Chain		12
AGED132	Micro Economics for Agriculture 1	AGED131	12
AGED134	Macro-Economics for Agricultural Economics 1	AGED131	12
AGED135	Introduction to Rural Sociology		12
AGPR132	Practices in Farm Entrepreneurial Skills 2	AGPR131	6
AVCM 102	Agricultural Value Chains Development		12
AGRO133	Principles of Crop Production		12

Level 2

AVCM 201	Communication and Innovation in Agriculture		12
AGED233	Microeconomics for Agricultural Economics 11		12
AGED234	Macroeconomics for Agricultural Economics 11		12
AEGS231	Gender Studies		12
AGED235	Farm Business Management		12
AVCN 203	ICT in Agriculture Value Chains		12
AVCM209	Econometrics and Socio-economic data analysis		12
AVCM 206	Research and Innovation Skills		12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report		30

Level 3 Semester 2

AGPR301	Work-Related Learning Report		45
AGPR302	Academic Supervisor's Assessment Report		30
AGPR303	Employer's Assessment Report		45

Level 4

AGED432	Agricultural Policy Analysis		12
AVCM401	Agricultural Trade		12
AGED438	Agricultural Production Economics	AGED233	12
AGED431	Agricultural Price Analysis and Forecasting		12
AGED407	Banking and Capital Markets	AGED232	12
AGED440	Econometrics 11	AGED239	12
AGED412	Rural Development		12

Level 4 Semester 2

AVCM 432	Agriculture Value Chains Financing		12
AGED436	Insurance and Risk Management in Agriculture		12
AGED439	Agricultural Extension		12
AGED414	Dissertation	AGED236	24

Electives	**(choose 1)**	
AGED433	Entrepreneurship in Agriculture	12
AGED411	Agricultural Tourism	12

14. MODULE SYNOPSIS

AGPR131 practices in farm entrepreneurial skills 1

This module gives first-year students in the field of Agriculture, a broad understanding of the fundamental entrepreneurial skills essential in starting a farming business, screening of opportunities, developing a business concept and translating ideas into business opportunities. The module will also cover certain animal production enterprises such as ruminant production systems (beef, dairy, small ruminants); non-ruminant production systems (broilers, layers, piggery and rabbitry) and intensive wildlife management enterprises. A field trip to an animal farm shall take place during each semester.

AGPR132 Practices in Farm Entrepreneurial Skills 2

In addition to entrepreneurial skills learnt in AGPR102, students shall learn to finance a farming business, do cash flow budgets and apply for a successful bank loan. Production enterprises shall include the production of maize, wheat, leafy vegetables, onions, tomatoes, field beans, tobacco, cotton, soyabean, citrus, mango and banana. Students shall also be exposed to the production of major flowers e.g. carnations and roses. Students shall be exposed to aspects of marketing, human resources, pricing, government policies and taxation. Students shall visit a crop production farm during each semester.

AGED 131 Introduction to Agricultural Economics

Economic systems and organization, theory of supply and demand, market equilibrium and nature of markets. Micro – economics. Macro – Economics. The consumption, investment and savings functions. The international economy. Monetary and fiscal policies. Employment and growth policies. The role of the agricultural industry in the national economy.

AGED 132 Micro Economics for Agriculture 1

Economics of agricultural production- theoretical foundations; Physical and economic relations, The factor- product relationship, the factor- factor relationship, the product- product relationship. The product supply and input demand; Product supply, Demand for inputs. Production Economics; Efficiency of resource use, Technical, Allocative and Economic efficiency, Risk and uncertainty. Economics of market demand

AGED 133 Mathematics for Agricultural Economics

The module is designed to give students a basic understanding of mathematical tools and techniques required in economics, policy formulation and business decision-making. The module seeks to show the relevance of mathematical tools/concepts such as integration linear algebra, differentiation and optimization theory in the field of economics. Trigonometry, probability, functions, graphs, solving equations, matrix algebra, multivariate calculus, differentiation and integration

AGED 134 Macro Economics for Agriculture 1

Definition of macro-economics. The module will look into the branch of economics at the national level through variables such as macro-economic issues, inflation, the gross national product, employment, national accounting, demand/supply, national economics equilibrium, monetary policy, bank rate, foreign currency reserves and balance of payments, foreign currency exchange market, food policy, agricultural policies, and production theories and agriculture at the elementary level.

CS131 Basic Communication Skills
Refer to the Department of Communication Skills

HCS135 Introduction to Information Technology
Refer to the Department of Computer Science

AWS101 Agricultural biology
Refer to the Department of Livestock and Wildlife Management.

AGED 135 Introduction to Rural Sociology
Examining rural social systems and structures and how they impact positively or negatively on development. Examining social change theories and how they relate to the development discourse

AVCM102 Introduction to Agricultural Value Chain Development
Theoretical approaches used to study organization and management, Examines the structure and behaviour as well as the main tasks of management. Looking in detail and questioning the applicability of management theory. Examining the application of management theory in relation to specific areas of activity. Current theory issues in reformation and improving management of Agricultural Development.

AVCM 101 Introduction to Agricultural Value Chains
This course focuses on the role of agriculture in economic development and how to develop appropriate interventions to improve the role of the sector at different stages of economic growth. Topics to be covered include the role of agriculture in economic development - product contribution, market contribution, factor contribution and foreign exchange contribution; pre-developmental theories of Thomas Malthus and David Ricardo; agricultural development models; agriculture in developing and developed countries and planning for agricultural development. Institutional constraints, economic efficiency, structural change and technology adoption in developing countries. Farming systems, agricultural development policies, sector planning and implications for long-term economic growth and development. Poverty alleviation and agriculture in the international economic context.

HAWS 122 Principles of Animal Production
Levels of organization – molecular, cellular, tissue, organ etc.
Students to develop skills and knowledge of the principles of efficient production including, feeding practices, breeds, management, housing and handling facilities, marketing, diseases and sanitation under tropical conditions for beef, dairy, small ruminants, pigs and poultry. Students to be introduced to the basic principles of animal production and be able to identify problems in livestock production in the Tropics.

AGRO133 Principles of Crop Production
Refer to the Department of Agronomy.

AGED 201 Agri-Business Law
The module will cover the following aspects as a matter of principles of law but more so as they apply to the current challenges in the agricultural sector; Sources of law, basics of law of sale defining and applying the legal terms of thing, price, ownership and risk as well as sellers obligations and buyers remedies. Contract law, definition, essentials and application to agricultural funding, operations and marketing. Modes of access to land (land tenure). The law of agency, challenging of agency. Employment contracts. Collective bargaining. National Social Security Authority (NSSA) Act.

AVCM 203 ICT in Agricultural Value Chain
ICTs are increasingly being used in the agricultural sector. In this regard, this module focuses on ICTs in Agriculture, the role of ICTs in agriculture (productivity, food and nutrition security, access markets, and employment), ICT potential for agriculture value chain and agribusiness development, a framework for engaging youth in agriculture, the role of social media in agricultural development as platforms for creating awareness, information exchanges and technology adoption.

AGED 233 Micro – economics for Agriculture 11
Basic demand relationships, Elasticity of demand, Homogeneity condition. Developments in demand theory; New theories of demand, Equilibrium and Exchange, Definition of equilibrium, Partial vs General equilibrium, Interference with equilibrium. Analysis of agricultural markets, Competitive markets and Pareto optimality, Policy intervention in markets. Welfare Economics and Economics of trade. Food and Agricultural policy.

AGED 234 Macroeconomics for Agriculture 11
Looks at economic behaviour and policies related to consumption, savings, investment, balance of trade, equilibrium in the goods and money market, domestic and international debt, and their implication in a more rigorous manner

AGED 235 Economics of Environmental Management
Environment and the micro-economy; Markets, Market failure and property rights, Government property rights, Economic instruments for environmental management, Attaching money values to environmental goods. Environment and the macro-economy; the green economy, Macroeconomic policy and the environment, Environment and the international economy, Environmental regulation, comparative advantage and international competitiveness, trans boundary externalities and environmental policy, trade liberalization and environmental impacts, Economics of pollution control, theory of pollution economics, the practical use of economic instruments in pollution control, Natural resource economics Management of renewal resources, market and institutional failure in the natural resource sector

AVCM 236 Research and Innovation Methods
Research proposal formulation, problem definition, methods of data collection, Questionnaire formulation, and interviewing techniques, sampling procedures for surveys, organisation and implementation of surveys, survey quality control, analyses and interpretation, and results presentation. Software; SPSS, GENSTAT, SAS etc.

AGED 237 Agribusiness Management
The module seeks to appropriately recall all values from other modules and contextualize them to advantage Agricultural Business Management through: Forms of business and analysis of business environment. Value chain in agricultural operations. Management principles. Cooperate governance. Operations research. Agribusiness viability and analysis.

AGED 238 Farm Business management
Background: Functions of management-Forms of business organization in commercial agriculture. Farm records and Information systems management.
Labour management farm planning and budgeting, Gross Margin Analysis. Whole farm budget and farm planning. Partial budget. Break-even budget. Cash flow budget. Sources of agricultural finance. Agricultural Support Schemes with special reference to. Government support scheme. Contract Production schemes Management of working capital. Investment

appraisal, Stock valuations and depreciation, Farm trading accounts, Profit and loss account, Farm balance sheet, its analysis and interpretation. Variances analysis, Audit of farm operation and budgetary control. Business ethics.

AVCM 209 Econometrics and Data Analysis
The module gives empirical content to economic theory. Uses mathematics and statistical tools to analyse economic behaviour i.e. estimate relationships, infer the implications of the estimates, test hypotheses about economic behaviour and predict economic behaviour. The tools include regression models, specification testing, general linear models, problems of single equation models and simultaneous equation models. The module content focuses on econometric techniques that provide students with the understanding needed in modern economics.

AEGS231 Introduction to Gender Studies
The module empowers the students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, Gender Issues in Zimbabwe, Redressing Gender Imbalances, Empowerment and Strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

AGED 239 Project Management
This module is about the modes and mechanisms through which development assistance is channelled, via investment in developing countries, for the promotion of agricultural and rural development. It is thus primarily about the rationale, context, and methods of planning, appraising and evaluating development projects and programmes. Projects and programmes are widely used when attempting to allocate limited resources for specific development purposes as effectively as possible, and a core part of the module is on methods for appraising the financial and economic efficiency of rural and agricultural development projects. These methods of appraisal are informed by economic theories and, in particular, those of applied welfare economics. While the module emphasises financial and economic efficiency, other important issues in assessing project design and impact are presented. Planning and management techniques for the project cycle are covered; including project identification and logical framework analysis. Approaches for social and environmental appraisal of projects are also reviewed.

AGED 231 NGOS and Agricultural Development
Types of non – profit organisations in agricultural development. Registration, and management structures of non – profit organizations. Funding and financial management, scope, corporate governance and ethics in NGOs. Civil society theory. Participation theory. NGOs and agricultural development. NGOs and humanitarian relief. farming systems, stakeholder consultations in agricultural development, and links with governmental and other developmental agencies.

AVCM 402 Agriculture Marketing and Value Chain Financing
In addition to exploring marketing principles, this module also addresses unique marketing concepts related to Agricultural and Food Marketing. The module explores the now ubiquitous concept of value chains in agricultural supply. It delves into agricultural value chain development, value addition along input, and commodity value chains, vertical and lateral coordination issues of chain development, supplier competence, forms of governance, strategies to reduce costs of governance, power asymmetries, concentration and the increasing importance of standards.

AGED 432 Policy Analysis
This course offers a multi-disciplinary approach to understanding agricultural and food policy in Zimbabwe and the world. It examines the agricultural policy process in terms of rationale, content, and consequences. It thereby helps students develop a better understanding of policy-making entities at various levels of government. It also provides a broad understanding of how policy actions in agriculture impact not only farmers' incomes, but also the well-being of consumers, the economic viability of rural communities, and the quality of our nation's environmental resources. Looks at economic behaviour and policies related to consumption, savings, investment, the balance of trade, equilibrium in the goods and money market, domestic and international debt, and their implication in a more rigorous manner.

AGED 433 Entrepreneurship in Agriculture
Role of entrepreneurship in agricultural development and the economy. Theories of entrepreneurship. Processes of establishing a business. Sources of funding. Feasibility studies. Planning production and marketing of enterprise. Entrepreneurship and public policy. Characteristics, nature and problems of small agricultural ventures. Advantages and disadvantages of big versus small. Management techniques. Linkages.

AGED 434 Disaster and Relief Management
Introduction to disaster management. Assessment of disasters and emergency complex situations. Impact of disasters on communities and agriculture. Needs analysis during disasters. Disaster management and recovery plans. Ethics in disaster management. Long-term planning in disaster management. Case studies of disasters affecting agriculture and communities.

AVCM 402 Agricultural Trade and Policy
International trade familiarizes students with theories and policies that guide international trade. This course is usually taken in one of the first semesters of a degree program. Students learn why international trade is important for Zimbabwe and other nations, and they examine how it affects production, profit and the economy. The theories of international trade (absolute advantage, comparative advantage), gains from trade and balance of payments; public finance, taxes, public expenditure and public debt, international trade agreements including EU-ACP, and the role of WTO will be covered.

AGED 436 Insurance and Risk Management in Agriculture
Concepts of risk management in agriculture. Theories and definitions of risk management. The measurement of risk. And the stochastic basis of risk. The Beta. Risk adjustment methods and techniques used in dealing with risk, risk financing techniques. Hedging against risk, the futures options and contracts. Insurance in agriculture; crop, animal, farm vehicles. Policy wordings, underwriting and claims.

AGED 437 Banking and Capital Markets
The course deals with the description of the activities of banks in agriculture, financial intermediaries, central banks, regulatory agencies, and monetary theory. Money market institutions. Capital market institutions. It shores up knowledge in bank management and capital markets as they relate to agriculture.

AGED 438 Agricultural Production Economics
Resource allocation, production selection, and scale of operation of agricultural firms including risk and uncertainty associated with agricultural production. Analysis of agricultural production economic models using econometrics, mathematical programming and simulation. Introduction to the systems science perspective. Techniques for Analysis of Risk, environmental Value, technological change and technology adoption.

AGED 439 Agricultural Extension
History of agricultural extension in Zimbabwe- Definition of extension. Extension ethics-communication skills: Mass media: Kinescis: art of listening. Conducting successful meetings. Extension methods and training programme planning: projects, programmes and Policy in Agricultural Extension- interpretation of Agricultural policy and planning programmes. Farmer participation in extension programme planning, monitoring and evaluation. Research – Extension- Linkages. Agricultural extension change management and support- what is change management- Donor Organisations in agricultural development and extension. Privatization of extension services- is this an option for the future? Female Extension Agents and Gender in Agricultural development and extension- Agrarian reform-challenges for extension.

AGED 440 Econometrics 11
This module in econometrics is designed to provide students will the tools required to evaluate and carry out empirical research. It focuses on the violation of classical liner regression model assumptions. Issues such as heteroscedasticity, autocorrelation, model specification and multicollinearity are discussed. Modelling with dummy variables, simultaneous equations, time series and panel data econometrics will be studied.

AGED 411 Agricultural Tourism
Origins and means of food production and the importance of sustainable practices. Definitions, reasons for agricultural tourism, and its importance at the local and national levels. Examples of agricultural tourism in Zimbabwe and worldwide; Agro – recreation and leisure businesses in farms, botanical gardens, city gardens, golf courses, ranches, safaris, agro and forest Estates, agro – processing plants, hunting clubs, boating, polocross, horse racing. the viability and preservation of farms and ranches, natural resources conservation, economic and enterprise development, entrepreneurial education (including business planning, risk management, food safety, marketing, customer relations, product development, diversification and value-added), policy analysis, regulatory policies and education of such to landowners, and public education

AGED412 Rural Development
Leading issues in rural development planning are discussed in detail. The course focuses on rural poverty alleviation, the role of agriculture and agro-industries in the improvement of the quality of rural lives, and social and economic development of rural areas, evaluation of alternative theories of rural development, and how to address hunger and rural poverty through innovations, infrastructural development, community participation, agro-processing and value addition. Attention is paid to the management of development institutions, participatory and integrated planning, infrastructure and physical resources, sectoral and spatial planning, management of local authorities, gender and demographics in development administration, decentralized planning and planning laws, and processes and planning techniques for rural and regional development.

AGED 414 Dissertation
This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board,

AGED431 Agricultural Price Analysis and Forecasting
This course is aimed at developing an understanding of the principles of price theory and how these principles can be applied to problems in agriculture starting from the supply and demand of agricultural products. The role of price in agricultural production is examined at the onset.

Specific topics include price determination and discovery, movement of prices over time and modelling of cycles and seasonality, using index numbers to correct for inflation, detailed analysis of agricultural commodities price trends, empirical estimation of demand and supply, demand shift and income effect, supply response and long-term competitiveness. Students are also introduced to the modelling of market structures, market outlook and different methods of forecasting prices, and hedging and its effectiveness for agricultural commodities.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED AGRICULTURAL SCIENCES AND BUSINESS (HAGSB)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To develop a BSc. Hons. Applied Agricultural Sciences and Business-related degree programme relevant to Zimbabwe and the developing regions through contributing towards sustainable food and fibre systems and business processes

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have at least five 'O' Level passes including English Language, Science and at least a B in Mathematics and any two 'A' Level passes in the following subjects Agriculture, Mathematics, Economics, Management of Business, Sociology, Accounts, Geography and Biology.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Micro/Macro/ Economics for Agriculture, Statistics, Econometrics, Data base management, Mathematics for Programming, Content management, Project management,
Specialist Focus: Agriculture Policy and Development
Orientation Oriented towards innovation and application of theory to solve real-life problems
Distinctive Features: Knowledge development in Applied Agricultural research, ICT, statistical and econometric skills.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Private Sector – Land-use Planners, eCommerce System Designers, Data Base Designers, Supply Chain Managers, Commodity Brokers, Agribusiness Finance Managers, Consultants, Marketing Agents/Sales Representatives, Extensionists, education, real estate, and public relations.
Public Sector – Researchers (Applied Economists, Marketing Agents,) Extensionists, Lecturers, natural resource and environmental management, Education and Public Relations
Self-Employment – Farmers, Consultants
Further Studies: Masters and Doctoral studies in Precision Agriculture, Information Technology, Agricultural Economics, Economics, Development, Policy

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), and Final examinations (60%).

6. INTENDED LEARNING OUTCOMES

- The programme is designed to train students in the fields of both agricultural sciences and IT
- Students to identify and define different theories, current thoughts and practices in informatics
- Acquaint students with the principles of management of information systems and services
- Equip students with the necessary knowledge, techniques and skills to respond intelligently with ICT solutions to business, socio-economic and management problems
- Students to be able to apply the acquired knowledge and skills in the ICT field
- To develop new practices methods and technologies that enhance production efficiency
- Students to master econometric and statistical software
- Ability to carry out national and international research, outreach programs and teaching in various aspects of ICT as it applies to agribusiness and agricultural economics
- Create an understanding and application of entrepreneur skills for employment creation.
- Students should have the ability to apply effective communication and extension skills.

7. GENERAL PROVISIONS

7.1. The BSc in Agricultural Sciences and Business management degree is a four-year full-time programme.

7.2. In Level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAMME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 11 of Faculty Regulations.

13. PROGRAMME STRUCTURE

The degree programme shall have at least 36 taught modules spread over the six teaching semesters.

Level 1 Semester 1

Code	Module Title	Credits
	Prerequisite	
*AWS 101	Agricultural Biology	12
*AGPR 131	Agricultural Practice I	12
*LWR 108	Agricultural Chemistry	12
*CS131	Basic Communication Skills	12
*HCS135	Introduction to Information Technology	12
*AGED131	Introduction to Agricultural Economics	12
*AGED133	Mathematics for Agricultural Economics	12

Level 1 Semester 2

*AGED132	Micro Economics for Agriculture 1	12
*AGED134	Macro-Economics Economics 1	12
*HAWS122	Principles of Animal Production	12
*AGRO133	Principles of Crop Production	12
*AGPR 132	Agricultural Practice 2	12
AGSB103	Agricultural Production Technology & Intellectual Property Rights	12
AGSB108	Agribusiness management 1	12

Level 2 Semester 1

*AGED210	Agri-Business Law	12
*AGED232	Financial Management	12
*AGED235	Economics of Environmental Management	12
AGSB212	Business Statistics	12
Options	*Choose 2*	
AGSB201	Sustainable Food Production	12
AGSB202	Food Safety and Standards	12
AGSB203	Economics of pests and disease management	12
AGSB204	Food and Water Security Theories and Strategies	12

Level 2 Semester 2

AGSB205	Research Methods and experimental designs	12
AGSB206	Farm and Agribusiness management	12
AGSB207	Innovation Enterprise management	12
AGSB208	Imports and Export processes	12

Options	***Choose 2***	
LWR246	Climate Change and Adaptation	12
AGBS209	Environmental Impact Assessment for Agricultural Development	12
AGBS210	Natural Resources Economics	12
AGBS211	Farm Accounting and Taxation	12
Level 3 Semester 1: Work-Related Learning		
*AGPR330	Work-Related Preliminary Report	30
Level 3 Semester 1 and 2: Work-Related Learning		
AGPR 331	Work-Related Learning Report All level 1.1 modules	45
AGPR 332	Academic Supervisor's Assessment Report	30
AGPR 332	Employer's Assessment Report	45
Level 4 Semester 1		
*AGED431	Agricultural Price Analysis and Forecasting	12
*AGED433	Entrepreneurship in Agriculture	12
*AGED436	Insurance and Risk Management in Agriculture	12
Options	*Choose 2*	
*WAP417	Sustainable Management of Natural Resources	12
AGSB401	Precision Agriculture	12
AGSB402	**Post-harvest Technologies and Value Addition**	**12**
AGSB403	Agri-cosmetology	12
Level 4 Semester 2		
AGSB404	Welfare Economics and Public Policy	12
AGSB405	Linear Programming and Decision Analysis	12
AGSB406	Agri-Informatics	12
AGED414	Dissertation	24
Options	***Choose 1***	
AGSB407	Consultancy in Agriculture	12
AGSB408	Agricultural Production Technology and Intellectual Property Rights	12
AGSB409	Pharmaceutical Agriculture	12
AGSB410	Contracts, Derivatives, Futures Markets	12

14. MODULE SYNOPSIS

AGPR131 Practices in Farm Entrepreneurial Skills 1
This module gives first-year students in the field of Agriculture, a broad understanding of the fundamental entrepreneurial skills essential in starting a farming business. The module will also cover certain animal production enterprises such as ruminant production systems (beef, dairy, small ruminants); non-ruminant production systems (broilers, layers, piggery and rabbitry) and intensive wildlife management enterprises. A field trip shall take place.

AGPR132 Practices in Farm Entrepreneurial Skills 2
In addition to entrepreneurial skills learnt in AGPR102, students shall learn to finance a farming business, do cash flow budgets and apply for a successful bank loan. Production enterprises shall include the production of maize, wheat, leafy vegetables, onions, etc.

Students shall also be exposed to the production of major flowers e.g. carnations and roses. Students shall be exposed to aspects of marketing, human resources, pricing, government policies and taxation. Students shall visit a crop production farm during each semester.

AGED 131 Introduction to Agricultural Economics
This module gives students knowledge of economic systems and organisation theory. The module will cover the following topics; economic systems and organization, theory of supply and demand, market equilibrium and nature of markets. In macroeconomics topics to be covered include the consumption, investment and savings functions, international economy and the role of the agricultural industry in the national economy.

AGED 132 Micro Economics for Agriculture 1
This module gives students knowledge on the economics of agricultural production. The main areas to be covered include physical and economic relations, The factor- product relationship, the factor-factor relationship, the product- product relationship, the product supply and input demand; efficiency of resource use, technical, allocative and economic efficiency.

AGED 133 Mathematics for Agricultural Economics
The module is designed to give students a basic understanding of mathematical tools and techniques required in economics, policy formulation and business decision-making. The module seeks to show the relevance of mathematical tools/concepts such as integration linear algebra, differentiation and optimization theory in the field of economics. Trigonometry, probability, functions, graphs, solving equations, matrix algebra, multivariate calculus, differentiation and integration

AGED 134 Macro Economics for Agriculture 1
The module will look into the branch of economics at the national level through variables such as macro-economic issues, inflation, the gross national product, employment, national accounting, demand/supply, national economics equilibrium, monetary policy, bank rate, foreign currency reserves and balan ce of payments, foreign currency exchange market, food policy, agricultural policies, and production theories and agriculture at the elementary level.

CS131 Basic Communication Skills
Refer to the Department of Communication Skills

HCS115 Introduction to Information Technology
Refer to the Department of Computer Science

AWS101 Agricultural Biology
Refer to the Department of Livestock and Wildlife Management.

LWR108 Agricultural Chemistry
Refer to the Department of Land and Water Resources Management.

AGED 105 Introduction to Rural Sociology
The modules will equip students with the skill of examining rural social systems, and structures and how they impact positively or negatively on development and examining social change theories and how they relate to the development discourse.

HAWS 122 Principles of Animal Production
Refer to the Department of Animal and Wildlife Sciences.

AGRO133 Principles of Crop Production

Refer to the Department of Agronomy.

AGSB108 Agribusiness Management 1
The module will introduce concepts and theories to agribusiness management. Features of the management function, Pillars of the management function, planning-organizing-leading-controlling, Economic, Social, and Environmental Performance, introduction to management theory (Scientific theory by Frederick W. Taylor, Administrative theory by Henri Fayol, Bureaucratic theory by Max Weber, Human relations theory by Elton Mayo, X&Y theory by Douglas McGregor.

AGSB408 Agricultural Production Technology and Intellectual Property Rights
Technological innovations are a key to driving productivity. The module will explore the economics of precision agriculture, hydroponics, vertical agriculture and the use of ICT. In addition, indigenous technological innovations for productivity will also be looked at. An understanding of how trademarks, patents, copyrights, trade secrets and the legal instruments surrounding their use will enable students to be ready for the international market.

AGSB212 Business Statistics
Describing, summarizing, graphically presenting, measuring and analyzing statistical data, probability distributions, variance applications and control charts. Exploratory and descriptive statistics, probability, probability distributions, sampling and sampling distributions, statistical inference and estimation, hypothesis testing, non-parametric tests, correlation regression, introduction to multivariate analysis

AGSB201 Sustainable Food Production
With ever-increasing populations and declining natural resources coupled with climate change, the issue of sustainable food production as seen increased attention. The module will explore the concepts and principles of sustainability and the frameworks applicable. The module will look into methods of production that use processes and systems that are: non-polluting, conserve non-renewable energy and natural resources, are economically efficient, are safe for workers, communities and consumers, and do not compromise the needs of future generations. Constraints and challenges to sustainability are fully looked at including the current debate on developed and developing nations.

AGSB202 Food Safety and Standards
Food quality has no compromise due to its effect on the health of both humans and animals. In addition, the use of pesticides and chemicals poses a threat to humans and the environment. Principles of Hazard Analysis Critical Control Point. The module will look into the management of toxins along the supply chain and the various microbiological analyses performed. Principles of Safety Health and Environment ate taught. Processes of quality control, Consumer council of Zimbabwe, Iso certification (ISO 22000, organic certification, rainforest certification, Euro certification.

AGSB203 Economics of Pests and Disease Management
Pests and diseases cause economic losses. This module will begin by looking at the evolution of pests and diseases and their management. It looks at both scientific ways and indigenous pest and disease management systems. Biological control, Cultural control, Regulatory or legislative control, Chemical control, Mechanical control, Integrated Pest Management. Insecticides, Pesticides, Fungicides. Classification of pesticides, Fungicides: Protective, Systemic, Organic-Inorganic, Pesticide Toxicity, Decision-making process of pest control

AGSB204 Food and Water Security Theories and Strategies

The module explores the critical changes in thinking about security through an examination of the main alternative concepts of security that have emerged in recent years. Major theories to be discussed in the module include food entitlement theory, food availability decline theory and endowment theory, among others. The module looks at food and water strategies like irrigation, intercropping, water harvesting, contract farming and conservation.

AGSB205 Research Methods and Experimental Designs

Research proposal formulation, problem definition, methods of data collection, Questionnaire formulation, and interviewing techniques, sampling procedures for surveys, organization and implementation of surveys, survey quality control, analyses and interpretation, and results presentation. Software; SPSS, GENSTAT, SAS etc. Introduction to ANOVA, Simple Randomized Design, Randomized Complete Block Design, Mean separation, LSD, DMRT, Turkeys Test, Dunnet's test, Correlation and Regression Analysis

AGSB206 Farm and Agribusiness Management

The module seeks to appropriately recall all values from other modules and contextualize them to advantage Agricultural Business Management through: Forms of business and analysis of business environment. Value chain in agricultural operations. Management principles. Cooperate governance. Operations research. Agribusiness viability and analysis. Farm records and Information systems management. -Labour management farm planning and budgeting: -Gross Margin Analysis. Whole farm budget and farm planning. Partial budget. Break-even budget. Cash flow budget. Sources of agricultural finance. Agricultural Support Schemes with special reference to. Government support scheme-. Contract Production Schemes Management of working capital. Investment appraisal-Stock valuations and depreciation. Farm trading accounts-Profit and loss account, Farm balance sheet, its analysis and interpretation. Variances analysis-Audit of farm operation and budgetary control. Business ethics.

AGBS211 Farm Accounting and Taxation

This module is intended to provide the basic accounting principles from books of original entry to the balance sheet as accounting ratio. The module will also cover stock store control and purchases and sales control for accounting purposes. An understanding of Zimbabwe's tax system will be reviewed. Farm taxation methodology including livestock taxation as well as capital gains taxation and procedures. Tax planning, evasion, tax avoidance and farm auditing principles will be taught in this module. Students will also be exposed to farm accounting software

AGED 201 Agri-Business Law

The module covers the following aspects as a matter of principles of law but more so as they apply to the current challenges in the agricultural sector; sources of law, contract law, operations and marketing, modes of access to land, the law of agency, challenging of agency and employment contracts.

AGED 232 Financial Management

The modules will equip students with the skill of examining of managing financial resources in an agricultural entity. The main areas to be covered include sources of agricultural finance in Zimbabwe, interest rates, capital budgeting and appraisal techniques, Cash flow budgeting and contracting as a source of financing.

AGED 235 Economics of Environmental Management

The modules will equip students with an understanding of the environment and the micro-economy, and macroeconomic policy and the environment. Other topics to be covered include

market failure, property rights, Government property rights, Economic instruments for environmental management, and economics of pollution control, natural resource economics management of renewal resources, market and institutional failure in the natural resource sector.

ENT205 Entrepreneurship
Refer to the Department of Entrepreneurship.

LWR246 Climate Change and Adaptation
Refer to the Department of Land and Water Resources Management.

GS201 Introduction to Gender Studies
The module empowers the students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, Gender Issues in Zimbabwe, Redressing Gender Imbalances, Empowerment and Strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

AGED211 Environmental Impact Assessment for Agricultural Development
The module deals with the concept of sustainability and its application to farming, offering suggestions for improving sustainability in various aspects of farming. Sustainable agriculture is a discipline that addresses current issues such as climate change, increasing food and fuel prices, poor-nation starvation, water pollution, soil erosion, fertility loss, pest control, and biodiversity depletion.

Qualitative Analysis
At the end of this module, students must be able to: (1) Differentiate among data preparation, management, analysis, and interpretation, (2) Differentiate analyses treating text as an object of analysis v. as an index of experience (3) Differentiate between case-oriented and variable-oriented analyses (4) Describe generic technical approaches to analysis (e.g., coding, counting, quoting, visually displaying) (5) Describe the defining features of qualitative content/thematic analysis, grounded theorizing, and narrative/discourse analysis and (6)Describe issues surrounding secondary analysis of qualitative data.

AGED217 Natural Resource Economics
This course aims to familiarise students with the differing categories of natural resources and an explanation and critique of the economic theory associated with their optimal use. The general context is that of the 'sustainable development /carrying capacity' debate and the potential constraints that natural resource availability may pose to the processes of economic development. The course will have a multidisciplinary flavour and some basic knowledge of the physical and biological sciences is helpful. The history of concern over 'limits to growth' will be discussed alongside explanations for the failure of earlier forecast 'doomsday' predictions to materialise. The issue of climate change will also be incorporated since environmental ‘assimilative capacity’ is now recognised as a pressing resource constraint issue and with it policies aimed at ‘decarbonising’ economic activity. Application of Economics in Non-Renewable Natural resource Management Issues, Non-Renewable resource Mineral extraction, Hotelling and Ricardian Scarcity, Rents, Market power in Non-Renewable, Resource Industries, Application of Economics in Renewable Natural resource Management Issues. Elements of profitability in Natural Resources; revenue, cost, taxes and subsidies. The concept of time in natural Resources; interest payment and discounting criteria of profitability in Natural resource investment; risks and uncertainties. Cost-benefit analysis.

Application of economic principles to problems in Natural Resources, such as multiple uses of forest lands, including wildlife, recreation, watershed, timber production and consumption.

AGSB207 Innovation Creativity and Enterprise Management
The module will look at principles of innovation and creativity for agricultural production. It will explore the theoretical framework of innovation and the drivers and inhibitors of innovation in agriculture.

AGSB208 Imports and Export Processes
Increased global and regional trade requires experts who understand the processes to unlock value in food and agricultural commodities markets. The module will look at export and import procedures, licensing, accreditation and certification, and the regulatory framework for both exports and imports.

AGSB410 Contracts, Derivatives, Futures Markets
Market volatility is a big challenge to agricultural producers. An in-depth analysis of the various models of contract farming will be done. Additionally, the understanding of derivative markets and futures contracts is important for the development and growth of agriculture and agribusiness. The use of forwards, futures, options, and swaps commodities and underlying assets such as stocks, bonds, interest rates, and currencies will be examined.

AGSB401 Precision Agriculture
The application and understanding of satellite technology, site-specific crop management, precision agriculture in extension services and its limitations and advantages are essential to productivity and profitability. Application of computers, GPS, GIS, remote sensing, and application control concepts will be taught.

AGSB402 Post Harvest Technologies and Value Addition
A great amount of agricultural produce is wasted when demand is less than supply. The module will look at techniques applied to agricultural commodities after harvest, preservation, conservation, quality control/enhancement, processing, packaging, storage, distribution, and marketing. The principles and benefits of value addition will be explored.

AGSB405 Linear Programming and Decision Analysis
The module deals with the practical use of linear programming and decision analysis techniques. The module develops an understanding of the theory behind such techniques and discusses their practical and theoretical limitations and their most useful attributes.

AGSB406 Agri-Informatics
A wide range of agri-informatics is the ultimate goal of this module. Information processing and management, creation of databases, environmental data, data acquisition, data analysis, environmental modelling, smart agriculture, software engineering, automated systems, smart phone applications geospatial technology and database development and management.

AGSB407 Consultancy in Agriculture
Experts in agriculture are increasingly sort after for consultancy purposes. This module will look at the following. This module will look at the following: competency analysis, soliciting for consultancy, responding to a call, negotiation skills, budgeting, log frame, timeframe,

report generation, teambuilding, conflict management, ethics and other areas of professional consulting.

AGSB103 Agricultural Production Technology and Intellectual Property Rights
Technological innovations are a key to driving productivity. The module will explore the economics of precision agriculture, hydroponics, vertical agriculture and the use of ICT. In addition indigenous technological innovations for productivity will also be looked at. An understanding of how trademarks, patents, copyrights, trade secrets and the legal instruments surrounding their use will enable students to be ready for the international market.

AGSB403 Agri-Cosmetology
Plants and animals have generated a billion-dollar industry in the cosmetic industry. There is growth potential in the commercial cultivation and processing of plants and animals for domestic regional and international markets. The module will look at plant-animal interactions. It begins with organic and inorganic chemistry. Principles dermatology and basic human skin and hair properties. Plant chemistry is also explored for infusion into cosmetic industry. Oils and fragrance extraction and fiber production are also taught in this module.

AGSB409 Pharmaceutical Agriculture
Herbs and plants have been used for medicinal purposes since time immemorial. This module will look at incorporating IKS of pharmaceutical plants and the economics of their production and distribution. The module will explore the production of pharmaceutical plants/ medicinal herbs and analysis of APIs and therapeutic proteins. Value addition of pharmaceutical plants and animal products.

AGED 401 Agriculture and Food Marketing
The module addresses unique marketing concepts related to agricultural and food marketing. Main topics to be taught are key roles of marketing in agriculture, price discovery; the food price dilemma, market structures, marketing principles, agricultural marketing management; agricultural marketing extension. Further the module looks ate a market-oriented approach to post harvest management.

AGED 432 Policy Analysis
This course offers a multi-disciplinary approach to understanding agricultural and food policy in Zimbabwe and the world. It examines the agricultural policy process in terms of rationale, content, and consequences. It thereby helps students develop a better understanding of policy-making entities at various levels of government. It also provides a broad understanding of how policy actions in agriculture impact not only farmers' incomes, but also the well-being of consumers.

AGED 433 Entrepreneurship in Agriculture
The modules will equip students with the skill of understanding the role of entrepreneurship in agricultural development and the economy. Topics to be covered include theories of entrepreneurship, processes of establishing a business, sources of funding, feasibility studies, planning production and marketing of enterprise and public policy.

AGED 435 International Trade and Policy
The modules will equip students with theories and policies that guide international trade. This course allows students to learn why international trade is important for Zimbabwe and other nations, and they examine how it affects production, profit and the economy.

AGED 436 Insurance and Risk Management in Agriculture

The module looks at concepts of risk management in agriculture. The main areas that are covered include theories and definitions of risk management, the measurement of risk, the stochastic basis of risk, the Beta Risk adjustment methods and techniques used in dealing with risk, hedging against risk, and the future options and contracts.

AGED 437 Banking and Capital Markets
The course deals with the description of the activities of banks in agriculture, financial intermediaries, central banks, regulatory agencies, and monetary theory. Money market institutions. Capital market institutions. It shores up knowledge in bank management and capital markets as they relate to agriculture.

AGED 444 Dissertation
This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board,

AGED416 Institutional Economics for Development
The module incorporates theory of institutions into economics. It builds on, modifies and extends the neoclassical theory. The module combines theoretical and empirical research examining the role of institutions in furthering or preventing economic growth. The issues covered include transaction costs, political economy, property rights, hierarchy and organization and public choice.

AGED 421 Agricultural Trade and Law
This course introduces the legal framework, policies and jurisprudence of the World Trade Organization (WTO). It is designed to give students an overview of the regulation of trade in goods, services and intellectual property that is necessary for international legal practice, as well as provide the policy analysis that will equip students to better understand the trade dimensions of international legal disputes. Students will also analyze laws pertaining to monopolies and trade protection and discuss theories that can help them become successful in international trade. The course introduces the main structural characteristics of the WTO, covers procedural issues in the WTO dispute settlement mechanism, addresses the traditional free trade doctrines: most-favoured nation treatment, national treatment, and transparency, discusses fair trade doctrines: anti-dumping measures, countervailing duties, safeguard measures, and protection of intellectual property, covers trade-related public policy issues, namely, exceptions to the free trade doctrines based on GATT Article XX. Finally, it looks into developing countries and agricultural trade under the Agreement on Agriculture.

AGED422 Welfare Economics and Public Policy
Principles of welfare economics applied to policy issues in agriculture and natural resources. Review of measures of household welfare, willingness to pay, and notions of Pareto optimality, aggregate welfare and market failure. Practical methods of comparative statics analysis of the effect of public policies on consumer and firm behaviour, and on market equilibrium. Theory of externalities and welfare implications of market versus non-market allocation of public goods examined. Applications include evaluation of such policies as taxes, price supports, quotas, pollution controls, environmental damage liability, and intellectual property rights.

AGED423 Linear Programming and Decision Analysis
The module deals with the practical use of linear programming and decision analysis techniques. The module develops an understanding of the theory behind such techniques and discusses their practical and theoretical limitations and their most useful attributes.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN INFORMATION TECHNOLOGY FOR AGRICULTURAL SYSTEMS (HITAS)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	504
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To address the demand of Information Technology techniques and analysis for Agricultural Systems in the areas of sustainable food production, natural resource management and overall economic growth.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have at least five 'O' Level passes including English Language, Science and at least a B in Mathematics and any two 'A' Level passes in the following subjects Agriculture, Mathematics, Economics, Management of Business, Sociology, Accounts, Geography and Biology.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Micro/Macro/ Economics for Agriculture, Statistics, Econometrics, Data base management, Mathematics for Programming, Content management, Project management,

Specialist Focus: Economic Data analysis, ICT, Agricultural Economics

Orientation Oriented towards innovation and application of theory to solve real life problems

Distinctive Features: Knowledge development in Applied Agricultural research, ICT, statistical and econometric skills.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Private Sector – Land-use Planners, eCommerce System Designers, Data Base Designers, Supply Chain Managers, Commodity Brokers, Agribusiness Finance Managers, Consultants, Marketing Agents/Sales Representatives, Extensionists, education, real estate, and public relations.

Public Sector – Researchers (Applied Economists, Marketing Agents,) Extensionists, Lecturers, natural resource and environmental management, Education and Public Relations

Self-Employment – Farmers, Consultants

Further Studies: Masters and Doctoral studies in Precision Agriculture, Information Technology, Agricultural Economics, Economics, Development, Policy

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), and Final examination (60%).

6. INTENDED LEARNING OUTCOMES

- The programme is designed to train students in the fields of both agricultural sciences and IT
- Students to identify and define different theories, current thoughts and practices in informatics
- Acquaint students with the principles of management of information systems and services
- Equip students with the necessary knowledge, techniques and skills to respond intelligently with ICT solutions to business, socio-economic and management problems
- Students to be able to apply the acquired knowledge and skills in the ICT field
- To develop new practices methods and technologies that enhance production efficiency
- Students to master econometric and statistical software
- Ability to carry out national and international research, outreach programs and teaching in various aspects of ICT as it applies to agribusiness and agricultural economics
- Create an understanding and application of entrepreneur skills for employment creation.
- Students should have the ability to apply effective communication and extension skills.

7. GENERAL PROVISIONS

7.1. The Bachelor of Science Honours Degree in Information Technology for Agricultural Systems is a four-year full-time programme.

7.2. In Level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAMME ASSESSMENT

Refer To Section 6 Of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

- The degree programme shall have at least 32 taught modules spread over the six teaching semesters.

Level 1 Semester 1

Code	Module Title	Prerequisite	Credits
AWS101	Agricultural Biology		12
AGPR131	Practices in Farm Entrepreneurial skills 1		6
CS131	Basic Communication Skills		12
HSC135	Introduction to Information Technology		12
AGED131	Introduction to Agricultural Economics		12
AGED133	Mathematics for Agricultural Economics		12
AGED139	Introduction to Agricultural Planning & Development		12

Level 1 Semester 2

AGED132	Micro Economics for Agriculture	AGED131	12
AGED134	Macro-Economics for Agriculture	AGED131	12
AGED105	Introduction to Rural Sociology		12
AGPR132	Practices in Farm Entrepreneurial Skills 2	AGPR131	6
HAWS122	Principles of Animal Production		12
AGRO133	Principles of Crop Production		12

Level 2 Semester 1

HITAS201	Introduction to Biostatistics	12
HITAS202	Mathematics for Programming	12
HITAS203	Introduction to Content Management Systems	12
AGED232	Financial Management	12
AEGS201	Gender Studies	12

Electives **Choose 1**

AGED201	Agri-Business Law	12
AGED231	NGOs and Agricultural Development	12

Level 2 Semester 2

Code	Module Title	Prerequisite	Credits
HITAS 204	Programming for Internet and Mobile Devices		12
HITAS205	Data Base Systems		12
HITAS206	Research Methods and Innovation Skills		12
HITAS 207	Econometrics and Socio-economic Data analysis		12
AGED239	Project Management		12

Electives **Choose 1**

AGED237	Agribusiness Management	12
AGED238	Farm Business Management	12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

*AGPR331	Work-Related Learning Report	45
*AGPR332	Employer's Assessment Report	30
*AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

HITAS401	Business Information Systems Design and ICT security		12
HITAS402	Web Designing		12
AGED432	Agricultural Policy Analysis		12
AGED405	International Trade		12
AGED431	Agricultural Price Analysis and Forecasting		12
Electives	**Choose 1**		
AGED437	Banking and Capital Markets	AGED202	12
AGED440	Econometrics 11	AGED209	12
AGED412	Rural Development		12

Level 4 Semester 2

HITAS403	Geographical Information System		12
HITAS404	Operating Systems		12
AGED413	Agriculture Value Chain		12
HITAS 414	Dissertation	HITAS 205	24
Electives	**(Choose 1)**		
AGED433	Entrepreneurship in Agriculture		12
AGED439	Agricultural Extension		12
AGED436	Insurance and Risk Management in Agriculture		12

14. Module Synopsis

AGPR131 Practices in Farm Entrepreneurial Skills 1
This module gives first-year students in the field of Agriculture, a broad understanding of the fundamental entrepreneurial skills essential in starting a farming business. The module will also cover certain animal production enterprises such as ruminant production systems (beef, dairy, small ruminants); non-ruminant production systems (broilers, layers, piggery and rabbitry) and intensive wildlife management enterprises. A field trip shall take place.

AGPR132 Practices in Farm Entrepreneurial Skills 2
In addition to entrepreneurial skills learnt in AGPR102, students shall learn to finance a farming business, do cash flow budgets and apply for a successful bank loan. Production enterprises shall include the production of maize, wheat, leafy vegetables, onions, etc. Students shall also be exposed to the production of major flowers e.g. carnations and roses. Students shall be exposed to aspects of marketing, human resources, pricing, government policies and taxation. Students shall visit a crop production farm during each semester.

AGED 131 Introduction to Agricultural Economics
This module gives students knowledge of economic systems and organisation theory. The module will cover the following topics; economic systems and organization, theory of supply and demand, market equilibrium and nature of markets. In macroeconomics topics to be covered include the consumption, investment and savings functions, international economy and the role of the agricultural industry in the national economy.

AGED 132 Micro Economics for Agriculture
This module gives students knowledge on the economics of agricultural production. The main areas to be covered include physical and economic relations, The factor- product relationship, the factor-factor relationship, the product- product relationship, the product supply and input demand; efficiency of resource use, technical, allocative and economic efficiency.

AGED 133 Mathematics for Agricultural Economics
The module is designed to give students a basic understanding of mathematical tools and techniques required in economics, policy formulation and business decision-making. The module seeks to show the relevance of mathematical tools/concepts such as integration linear algebra, differentiation and optimization theory in the field of economics. Trigonometry, probability, functions, graphs, solving equations, matrix algebra, multivariate calculus, differentiation and integration

AGED 134 Macro Economics for Agriculture
The module will look into the branch of economics at the national level through variables such as macro-economic issues, inflation, the gross national product, employment, national accounting, demand/supply, national economics equilibrium, monetary policy, bank rate, foreign currency reserves and balance of payments, foreign currency exchange market, food policy, agricultural policies, and production theories and agriculture at the elementary level.

CS131 Basic Communication Skills
Refer to the Department of Communication Skills

HCS135 Introduction to Information Technology
Refer to the Department of Computer Science

AWS101 Agricultural Biology
Refer to the Department of Livestock and Wildlife Management.

AGED 135 Introduction to Rural Sociology
The modules will equip students with the skill of examining rural social systems, and structures and how they impact positively or negatively on development and examining social change theories and how they relate to the development discourse

AGED139 Agricultural Development and Planning
This course focuses on the role of agriculture in economic development and how to develop appropriate interventions to improve the role of the sector at different stages of economic growth. Topics to be covered include the role of agriculture in economic development - product contribution, market contribution, factor contribution and foreign exchange contribution; agricultural development models and agriculture in developing and developed countries.

HAWS 122 Principles of Animal Production
Refer to the Department of Animal and Wildlife Sciences.

AGRO133 Principles of Crop Production
Refer to the Department of Agronomy.

AGED 201 Agri-Business Law
The module covers the following aspects as a matter of principles of law but more so as they apply to the current challenges in the agricultural sector; sources of law, contract law,

operations and marketing, modes of access to land, the law of agency, challenging of agency and employment contracts.

AGED 232 Financial Management
The modules will equip students with the skill of examining of managing financial resources in an agricultural entity. The main areas to be covered include sources of agricultural finance in Zimbabwe, interest rates, capital budgeting and appraisal techniques, Cash flow budgeting and contracting as a source of financing.

HITAS 201 Introduction to Biostatistics
The overall aim of the module is to provide students with an understanding of the nature and structure of data and data collection, and a good working knowledge of basic descriptive statistics and inference.

HITAS 202 Mathematics for Programming
The module covers pure mathematical tools used in economics. It covers number systems, set algebra, functions, equilibrium analysis, matrix algebra, and differentiation with a focus on maxima, minima and applications, unconstrained and constrained optimization, and various applications to the field of agriculture.

HITAS 203 Introduction to Content Management Systems
The module introduces the concept of a Content Management System (CMS) and why it is widely used in the market. Components of CMS, Features of CMS, Advantages and disadvantages of CMS.

HITAS 204 Programming for Internet and Mobile Devices
This module introduces Android-based operating systems frameworks, architecture, design and engineering issues, and methodologies for developing applications for smartphones, tablet computers, and other mobile devices. It covers memory and power management, user interface design for mobile devices and unique user interactions using multi-touch technologies, object-oriented design using model-view-controller paradigm, event handling, animation, audio and video, multi-threading, networking, data storage, and performance considerations.

HITAS 205 Data Base Systems
This module consists of two main components namely data structure, algorithms, and data base systems. The Data structure and algorithms focuses on logical structures of data as well as the design, implementation and analysis of algorithms operating on these structures and their application to solving practical problems. The module covers basic and essential topics in data structures, including array based lists, linked lists, skip lists, hash tables, recursion, binary trees, scapegoat trees, red–black trees, heaps, sorting algorithms, graphs, and binary tree. Data base systems introduces the basic concept of data management and guides towards the concepts of its application. It includes representing information with the relational database model, manipulating data with an interactive query language (SQL) and database programming, database development including internet applications, and database security, integrity and privacy issues.

HITAS 206 Research and Innovation Skills
The module is designed to shape students to have an understanding of the importance of multi-disciplinary and interdisciplinary partnerships in research and innovation, to appreciate the different tools, software packages, methods and approaches in data analysis and to be able to draw conclusions. The module will capacitate students with skills to implement context-specific 'hands-on' research activities under targeted conditions (e.g. archival, field,

laboratory, industrial, civil and commercial conditions). Students will learn to structure and compose various types of professional documents relevant to their professions such as case study reports, community project reports, grant proposals, letters, memos, curriculum vitae, feature articles, minutes, survey research reports and software manuals. Students will be moulded to be able to; articulate a research problem diagnosis and translate it into a good or service, identify and distinguish an innovation, explain intellectual property (IP) registration processes and identify opportunities for marketing IPs and package and present information for a wider audience using different approaches and technologies. Research proposal formulation, problem definition, methods of data collection, Questionnaire formulation, and interviewing techniques, sampling procedures for surveys, organisation and implementation of surveys, survey quality control, analyses and interpretation, results presentation. Software; SPSS, GENSTAT, SAS etc

HITAS 207 Econometrics and Socioeconomic Data Analysis
The module integrates economic theory, statistics and mathematics to analyze economic phenomena. The econometrics component covers the methodology of econometric research, correlation, Ordinary Least Squares, properties of Least Square Estimators and Multiple Regression. The socioeconomic data analysis part introduces students to SPSS, R, and STATA as software for practical data analysis for case studies.

AGED 237 Agribusiness Management
The module seeks to appropriately recall all values from other modules and contextualize them to advantage. Agribusiness Management shall cover specific topics that include: Forms of business and analysis of business environment, Value chain in agricultural operations, Management principles, cooperate governance, Operations research and Agribusiness viability and analysis.

AGED 238 Farm Business Management
The modules will equip students with the skill of managing a farm business. It covers the following areas functions of management, forms of business organization in commercial agriculture, farm records and Information systems management, labour management farm planning and budgeting, gross Margin Analysis, partial budget, sources of agricultural finance, Government support schemes, contract production schemes, and management of working capital. Further, the course will cover farm accounting.

AEGS231 Gender Studies
The module empowers the students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, and Gender Issues in Zimbabwe, Redressing Gender Imbalances, Empowerment and Strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

AGED 239 Project Management
This module is about the modes and mechanisms through which development assistance is channelled, via investment in developing countries, for the promotion of agricultural and rural development. It is thus primarily about the rationale, context, and methods of planning, appraising and evaluating development projects and programmes. While the module emphasises financial and economic efficiency, other important issues in assessing project design and impact are presented.

AGED 231 NGOs and Agricultural Development
The modules will have knowledge of types of non – profit organisations in agricultural development, their registration, management structures of non – profit organizations, funding and financial management, scope, corporate governance and ethics in NGOs. The module also looks at civil society theory, farming systems, stakeholder consultations in agricultural development, and links with governmental and other developmental agencies.

HITAS 401 Business Information Systems Design and ICT Security
This module consists of two main components: Business Systems Analysis and Design: This component examines the analysis and design of business systems. It introduces the systems development life cycle (SDLC), the fundamental four-phase model (planning, analysis, design, and implementation) common to all development projects. This includes business value creation; business systems requirements analysis and specification; logical and physical design of business systems. It overviews object-oriented systems analysis and design and describes the Unified Process and its extensions. Multimedia systems and applications: This is an introductory component in digital media. Students will be introduced to the principles and current technologies of multimedia systems. Issues in effectively representing, processing, and retrieving multimedia data such as sound and music, graphics, images and video will be addressed. The practical component of the course is organized around learning about and using various software for manipulating digital sound, digital images, and digital video.

HITAS 402 Web Designing
The module introduces students to web concepts, and understand that the internet supports a range of services such as the World Wide Web (WWW), file transfer, email, and IM. Understand the relationship between the browser and the webserver. Html coding- learning how to create a web page, the basics of CSS- learning how to style and format web pages. Primary protocols, URL, hyperlink, web hosting etc. Web Publishing, Legal issues and general terms of use.

HITAS 403 Geographical Information System
The purpose of this module is to acquaint agricultural and environmental science students with fundamental principles and applications of Geographic Information Systems. The module provides the students with theoretical principles and techniques for handling and analyzing spatial data in the following application areas: natural resources management, agriculture, land management, water resources management; transport network systems; solid and liquid waste management; wildlife management, etc. The module is also designed to guide students to become thinkers regarding the use of spatial data/information to solve agricultural and environmental management challenges for informed decisions. After their successful completion of the module, the students should be able to use GIS technology in natural resources management, agriculture, land management, water resources management; transport network systems; solid and liquid waste management; and wildlife management, to provide goods and services at local, national, regional and global scales. For example, the students should be able to map Zimbabwe's key natural resources.

HITAS 404 Operating Systems
Operating systems familiarize students with resource management and hardware abstraction, the different operating system types, hardware addresses and types of addresses.

AGED 432 Policy Analysis
This course offers a multi-disciplinary approach to understanding agricultural and food policy in Zimbabwe and the world. It examines the agricultural policy process in terms of rationale,

content, and consequences. It thereby helps students develop a better understanding of policy-making entities at various levels of government. It also provides a broad understanding of how policy actions in agriculture impact not only farmers' incomes but also the well-being of consumers.

AGED 433 Entrepreneurship in Agriculture
The modules will equip students with the skill of understanding the role of entrepreneurship in agricultural development and the economy. Topics to be covered include theories of entrepreneurship, processes of establishing a business, sources of funding, feasibility studies, planning production and marketing of enterprise and public policy.

AGED 435 International Trade and Policy
The modules will equip students with theories and policies that guide international trade. This course allows students to learn why international trade is important for Zimbabwe and other nations, and they examine how it affects production, profit and the economy.

AGED 436 Insurance and Risk Management in Agriculture
The module looks at concepts of risk management in agriculture. The main areas that are covered include theories and definitions of risk management, the measurement of risk, the stochastic basis of risk, the Beta Risk adjustment methods and techniques used in dealing with risk, hedging against risk, and the future options and contracts.

AGED 437 Banking and Capital Markets
The course deals with the description of the activities of banks in agriculture, financial intermediaries, central banks, regulatory agencies, and monetary theory. Money market institutions. Capital market institutions. It shores up knowledge in bank management and capital markets as they relate to agriculture.

AGED 439 Agricultural Extension
The modules will equip students with the knowledge of the history of agricultural extension in Zimbabwe, mass media, kinesics, extension methods and training programme planning: projects, programmes and policy in agricultural extension, farmer participation in extension programme planning, monitoring and evaluation. The course further looks at donor organisations in agricultural development and extension, female extension agents and gender in agricultural development and extension.

AGED 440 Econometrics 11
This module in econometrics is designed to provide students with tools required to evaluate and carry out empirical research. It focuses on violation of classical liner regression model assumptions. Issues such as heteroscedasticity, autocorrelation, model specification and multicollinearity are discussed. Modelling with dummy variables, simultaneous equation, time series and panel data econometrics will be studied.

AGED412 Rural Development
The module focuses on rural poverty alleviation, the role of agriculture and agro-industries in the improvement of the quality of rural lives, and social and economic development of rural areas, evaluation of alternative theories of rural development, and how to address hunger and rural poverty through innovations, infrastructural development, community participation, agro-processing and value addition. Attention is paid to the management of development institutions, participatory and integrated planning, infrastructure and physical resources.

HITAS 414 Dissertation
This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board,

AGED431 Agricultural Price Analysis and Forecasting
This course is aimed at developing an understanding of the principles of price theory and how these principles can be applied to problems in agriculture starting from the supply and demand of agricultural products. The role of price in agricultural production is examined at the onset. Specific topics include detailed analysis, modelling and forecasting of agricultural commodities prices. Students are also introduced to the modelling of market structures, market outlook and hedging and its effectiveness.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN APPLIED AGRICULTURAL ECONOMICS (MAAE)

Duration	:	2 years
Credit Load	:	270
Minimum Credit Load	:	270
Maximum Credit Load	:	330
Total MBKS Credit Load	:	270
Level		9

1. PURPOSE OF THE PROGRAMME

To address the demand of agricultural economist, food security, poverty reduction and overall economic growth.

2. ENTRY REQUIREMENTS

To be eligible in the programme, a candidate must normally be a holder of at least a lower second class (2.2) first degree in Agriculture, Agricultural Economics; Economics; or any other related programme from a recognized University.

3. PROGRAMME CHARACTERISTICS

Areas of study: Agriculture Economics
Specialist Focus: Agriculture Economics
Orientation Oriented towards innovation and application of theory to solve real agricultural challenges
Distinctive Features: Knowledge development in Applied Agricultural Economics

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Private Sector – Agribusiness intelligence analysts, Bankers, commodity brokers, Agribusiness Finance Managers, Consultants, Marketing Agents/Sales Representatives, Extension Officers
Public Sector – Researchers, Lecturers
Self-Employment – Farmers, Consultants
Further Studies: Doctoral studies in Agricultural Economics, Environmental Economics, Economics

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/field visits, dissertation
Assessment Methods: Written and oral examinations, tests, quiz, group presentations, dissertation. Continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- To equip graduates with specialized training so that they can better position their own businesses, or the organizations in which they work, for growth in a global marketspace.
- Enable students to conceptualize, formulate and implement solutions to organizational challenges using new and existing technologies.
- To develop new practices methods and technologies that enhance production efficiency.
- Create e-Policies that address the ethical use of information and devices in the agribusiness environment.

7. GENERAL PROVISIONS

7.1. The programme shall normally be studied over a minimum of four semesters on a block release basis.

7.2. This programme is a taught Master's programme with modular work in the first two semesters and research and dissertation writing in the third and fourth semesters.

7.3. Continuous assessment consists of lectures, presentations, seminars and tests.

7.4. Lectures and taught modules shall normally be at the Midlands State University

7.5. The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months

7.6. Each student has to identify at least two supervisors who shall be approved by the Department of Agricultural Economics and Development

7.7. There shall be oral presentations for the Project Proposal and Research Findings. Weighting of these, together with the final written document, shall be as follows:

a) Dissertation Proposal 20%
b) Presentation of Research Findings 20%
c) Final Written Document 60%

7.8. Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

7.9. Candidates shall be expected to submit at least three executive bound copies of the dissertation.

7.10. The format of the dissertation shall be provided for by the Department of Agricultural Economics and Development.

8. PROGRAMME ASSESSMENT

- Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations.
- Formal examinations shall be held at the end of each semester.
- Continuous assessment shall account for 40 % of the overall assessment, while the examination shall account for 60 % of the overall assessment.
- The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 32 credits in the first level in order to proceed to the second level (dissertation).

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

- The degree programme shall have at least 9 taught modules spread over the two teaching semesters.

Level 1 Semester 1

Code	Description	Credits
*MAAE 701	Advanced Microeconomic Theory for Agriculture	18
*MAAE 702	Advance Macroeconomics for Agriculture	18
*MAAE 703	Mathematics for Agricultural Economists	18
*MAAE 704	Statistics for Agricultural Economists	18

Level 1 Semester 2

*MAAE 705	Econometrics	18
*MAAE 706	Research Methods	18
*MAAE 707	Production Economics	18
*MAAE 711	Agricultural Trade and Policy	18

Options (Choose Two)

MAAE 708	Environmental and Natural Resources	18
MAAE 710	Agricultural Marketing and Price Analysis	18
MAAE 712	Agribusiness Management	18
MAAE 713	Agricultural Finance and Risk Management	18
MAAE 714	Economics of Climate Change	18

Level 2 Semesters 1 and 2

*MAAE 715	Dissertation	90

14. Module Synopsis

MAAE 701 Advance Microeconomic Theory for Agriculture

The module provides a review of basic microeconomic concepts, consumer theory: axioms of rationality, utility maximization, income and substitution effects, mathematical derivation using Slutsky's equation, theory of Revealed Preference, theory of the firm, risk and uncertainty in agriculture, mathematical and calculus approaches to risk analysis, multi-period consumption, consumption efficiency and gains from exchange through markets.

MAAE 702 Advance Macroeconomics for Agriculture

The module focuses on the microeconomic foundations in dynamic macroeconomics, neoclassical growth model, overlapping generations model, real business cycle theory, unemployment, new Keynesians models, monetary policy trade-offs, and their applications using local national datasets.

MAAE 703 Mathematics for Agricultural Economics

The module provides a review of common mathematical notations, number systems, set algebra, functions, equilibrium analysis, matrix algebra, maxima, minima and applications, unconstrained and constrained optimization, and applications to the field of agriculture.

MAAE 704 Statistics for Agricultural Economists

In the module, focus is on the review of basic statistics, parametric inference, maximum likelihood estimation, the method of moments, Hypothesis testing, Bayesian statistics, Principal component analysis, Generalized Linear Models, Forecasting and time series analysis, and use of statistical software (STATA, SPSS, R) for practical data analysis.

MAAE 705 Econometrics
This module provides a review of linear regression models, non-linear models, bootstrapping and Monte Carlo simulation, binary choice models, GMM and M estimators, and applications using statistical software STATA, SPSS and R.

MAAE 706 Research Methods
This module introduces postgraduate students to the basic ideas about conduction research. Students will learn methods for reading technical papers, selecting research topics, devising research questions, planning research, project management and ethics. The evaluation is based on the assignments throughout the course and a final project report and presentation. The module provides students with knowledge, skills and practical experience to develop and manage their research. Conceptualizing and implementing novel research can be an overwhelming task. Students will be made aware of what is involved in getting a project started from conceptualizing the idea, designing the research question, designing the experiments to best answer the questions, ethical research standards and project management.

MAAE 707 Production Economics
The module covers the theory of production economics with emphasis on applications to agriculture and natural resources. Topics include the derivation, estimation and use of production, cost, profit, revenue, demand and supply functions. Discusses the concepts of efficiency and productivity. Production response over time and under risk. Agricultural production economic theory under static and dynamic situations. Analysis of allocation of factors of production, production efficiency, demand for factors of production and supply of agricultural products, costs of products and farm growth.

MAAE 708 Environmental and Natural Resource
This course studies the application of economic analysis to the management of the environmental and natural resources. This module considers the role of economic theory in understanding and solving environmental and resource problems and discuss empirical examinations of the theory. Policy implications will be addressed with particular attention paid to analyzing practical applications of the theories in different social, economic and political environments.

MAAE 709 Agricultural Value Chains Development and Financing
The module will cover the following topics; Why is there interest in value chain finance in agriculture?, Understanding agricultural value chain finance, Value chain business models, Agricultural value chain finance instruments, Innovations, strategy and design recommendations for programmes dealing with agricultural value chains and agricultural value chain finance.

MAAE 710 Agricultural Marketing
The module addresses the following topics pertaining to agriculture; The role of prices; demand analysis (theory), focusing on review of consumer behavior, utility maximization, elasticity, substitution & income effects of price changes, Engel curves. Demand analysis (empirical consideration), demand for a typical farm product, formulating demand equations, demand for agricultural inputs. Supply analysis, supply relation in agriculture, estimating supply functions (direct & indirect approaches). Estimation problems, model specification, using cross-section and time-series data, interpreting estimated parameters, statistical tests, and forecasts.

MAAE 711 Agricultural Trade and Policy

The module covers the following issues; an understanding of why nations trade and the role of supply and demand factors in determining trade; a basis for evaluating international competitiveness and comparative advantage; an appreciation for how agricultural trade is related to growth and development; an ability to evaluate the welfare implications of policies affecting production, consumption, and trade; an understanding of the implications of protectionism, free trade, managed trade, regional trade blocs, and multilateral trade liberalization, and the role for international trade institutions.

MAAE 712 Agribusiness Management

This module introduces students to Micro-economic Concepts and their Application to Business; Business Behaviour in Different Market Structures; Production and Costs; The Firm and Its Environment; Demand Estimation: Regression Analysis; Decision Making Under Uncertainty; Product Pricing; Types of agribusiness ownership, such as Sole Ownership, Partnership, Corporate Structure, etc. Managing the agricultural business, including strategic management, comparative advantage and competitive advantage, value-chain analysis. Business analysis and control – financial Statements, profit and loss statement, financial ratio analysis, and performance and activity analysis.

MAAE 713 Agricultural Finance and Risk Management

MAAE 714 Economics of Climate Change

The module discusses the basic climate change science and the core economic concepts, models and analyses related to emission mitigation and the contemporary discussions and negotiations on a national and international level. A particular focus will be given to recent advances in the research on the valuation of costs and benefits over time under uncertainty.

MAAE 715 Dissertation

This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN AGRICULTURAL INFORMATICS AND DATA ANALYTICS (MAIDA)

Duration	:	2 years
Credit Load	:	270
Minimum Credit Load	:	270
Maximum Credit Load	:	330
Total MBKS Credit Load	:	270
Level		9

1. PURPOSE OF THE PROGRAMME

To address the demand of Information Technologies, Data processing and Analytical techniques in Agriculture for sustainable food and fiber production, natural resources management and overall economic growth.

2. ENTRY REQUIREMENTS

To be eligible in the programme, a candidate must normally be a holder of at least a lower second class (2.2) first degree in Agriculture, Agricultural Economics; Economics; Mathematics; Statistics; Information Technology or any other related programme from a recognized University.

3. PROGRAMME CHARACTERISTICS

Areas of study: Agriculture, Information Technology, Data Management and Analysis
Specialist Focus: Agricultural Informatics, Data Analytics
Orientation Oriented towards innovation and application of theory to solve real life challenges
Distinctive Features: Knowledge development in Applied Information Technology, Agricultural research, Data processing and analytical skills.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Private Sector – Agribusiness intelligence analysts, Data scientists, Bankers, commodity brokers, Agribusiness Finance Managers, Consultants, Marketing Agents/Sales Representatives, Extension Officers
Public Sector – Researchers, Lecturers
Self-Employment – Farmers, Consultants
Further Studies: Doctoral studies in Data Analytics, Agricultural Informatics

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), and Final examinations (60%).

6. INTENDED LEARNING OUTCOMES

- To equip graduates with specialized training so that they can better position their own businesses, or the organizations in which they work, for growth in a global marketspace.
- Enable students to conceptualize, formulate and implement solutions to organizational challenges using new and existing technologies.
- To develop new practices methods and technologies that enhance production efficiency.
- Create e-Policies that address the ethical use of information and devices in the agribusiness environment.

7. GENERAL PROVISIONS

7.1. The programme shall normally be studied over a minimum of four semesters on a block release basis.

7.2. This programme is a taught Masters programme with modular work in the first two semesters and research and dissertation writing in the third and fourth semesters.

7.3. Continuous assessment consists of lectures, presentations, seminars and tests.

7.4. Lectures and taught modules shall normally be at the Midlands State University

7.5. The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.

7.6. Each student has to identify at least two supervisors who shall be approved by the Department of Agricultural Economics and Development

7.7. There shall be oral presentations for the Project Proposal and Research Findings. Weighting of these, together with the final written document, shall be as follows:

- Dissertation Proposal 15%
- Presentation of Research Findings 15%
- Final Written Document 70%
-

7.8. Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

7.9. Candidates shall be expected to submit at least three executive bound copies of the dissertation

7.10. The format of the dissertation shall be provided for by the Department of Agricultural Economics and Development.

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations.

8.2. Formal examinations shall be held at the end of each semester.

8.3. Continuous assessment shall account for 40 % of the overall assessment, while the examination shall account for 60 % of the overall assessment.

8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 32 credits in the first level in order to proceed to the second level (dissertation).

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

13.1. The degree programme shall have at least 32 taught modules spread over the six teaching semesters.

Level 1 Semester 1

*MAIDA 701	Emerging Issues in e-Agriculture	18
*MAIDA 702	Data Science: Analytics and Visualizations	18
*MAIDA 703	Qualitative and Quantitative Techniques in Agriculture Analysis	18
*MAIDA 704	Geographical Information Systems	18

Level 1 Semester 2

*MAIDA 705	Research Methods	18
*MAIDA 706	Socio-economic Data Base Systems Design	18
*MAIDA 707	Big Data Analytics in Agriculture	18
*MAIDA 708	Agricultural Value Chains Development and Financing	18

OPTIONS (CHOOSE TWO)

MAIDA 709	Agricultural Policy Informatics	18
MAIDA 710	Agricultural Trade and Policy	18
MAIDA 711	Information Security, Ethics and Legal Aspects	18
MAIDA 712	Sustainable development and Smart Agriculture	18
MAIDA 713	Cyber Analytics	18

Level 2 Semester 1 & 2

*MAIDA 715	Dissertation	90

14. MODULE SYNOPSIS

MAIDA701 Emerging Issues in e-Agriculture

The module discusses contemporary issues in e-Agriculture. More emphasis will be placed on; Technology for changing times, Mobile telephony, from mobile phones to smartphones, Mobile financial services, Use of ICTs in agriculture, e-Agriculture strategies in ICT policies, Online learning, Growing use of big/open/real-time data collection and analysis, Digital dilemma and Appropriate technologies.

MAIDA702 Data Science: Analytics and Visualizations

The module covers; Supervised learning and applications. Multicol9linearity, ridge regression, the LASSO and the elastic net. Parametric and nonparametric logistic regression and nonlinear regression. Survival regression. Regression extensions: Random forests MARS and Conjoint analysis. Neural networks.

MAIDA703 Qualitative and Quantitative Techniques in Agriculture Analysis

A wide range of quantitative and qualitative techniques are applied to the analysis of management problems. This module will provide students with the skills to apply a wide range of quantitative and qualitative techniques to a variety of management problems in the various areas of agriculture. A critical feature of the module is the use of managerial oriented cases to focus students on the application of quantitative and qualitative techniques to management problems. Particular emphasis will be placed on computer-based applications of quantitative and qualitative techniques.

MAIDA704 Geographical Information Systems
This module seeks to equip students with the theoretical understanding and practical skills in geo-informatics that underpin natural resources and environmental management. The module comprises basic theoretical and practical concepts in Geographic Information Systems (GIS) and remote sensing (RS). The GIS component presents the concepts and techniques for collecting or creating, processing, storing and analyzing spatial data; while the RS component addresses the theory and techniques requisite for handling optical satellite images. Topics covered include the nature of GIS, spatial data capture and storage, spatial data models, spatial analysis, satellite remote sensing and image analysis.

MAIDA705 Research Methods
This module introduces postgraduate students to the basic ideas about conduction research. Students will learn methods for reading technical papers, selecting research topics, devising research questions, planning research, project management and ethics. The evaluation is based on the assignments throughout the course and a final project report and presentation.

MAIDA706 Socio-economic Data Base Systems Design
The module covers the following topics; Information in Social Context, Socio-economic data, Aggregate and disaggregate data, Cross-sectional and longitudinal data, Sources of socio-economic data, Database design, determining data to be stored, Principle of orthogonal design, determining data relationships, Concept mapping, logically structuring data, Entity-Relationship model, Database normalization, Conceptual schema and Physical design of the database.

MAIDA707 Big Data Analytics in Agriculture
This module examines the theoretical and practical techniques used in the analysis of large data sets. It covers data collection, data pre-processing, predictive, prescriptive and descriptive analytics as well as social network analytics. The module will facilitate discussion of case studies that involve big data analytics including fraud detection, web analytics and recommender systems. Due to the data and technology-intensive nature of the material, this module will be delivered primarily through face-to-face instruction and hands-on lab sessions.

MAIDA708 Agricultural Value Chains Development and Financing
The module will cover the following topics; value chain finance in agriculture, understanding agricultural value chain finance, Value chain business models, Agricultural value chain finance instruments, Innovations, strategy and design recommendations for programmes dealing with agricultural value chains and agricultural value chain finance.

MAIDA709 Agricultural Policy Informatics
The module addresses the following topics pertaining to agriculture; Public policy theory, domestic and international policy, introduction to policy and practice, international policies and organizations, urban politics, Information and Communication Technology policy, legal and regulatory policy frameworks.

MAIDA710 Agricultural Trade and Policy
The module covers the following issues; an understanding of why nations trade and the role of supply and demand factors in determining trade; a basis for evaluating international competitiveness and comparative advantage; an appreciation for how agricultural trade is related to growth and development; an ability to evaluate the welfare implications of policies affecting production, consumption, and trade; an understanding of the implications of protectionism, free trade, managed trade, regional trade blocs, and multilateral trade liberalization, and the role for international trade institutions.

MAIDA711 Information Security, Ethics and Legal Aspects
This module introduces students to theoretical and practical aspects of information security, ethics and legal aspects as it relates to data and information in organizations. The module aims to provide students with the skills and knowledge necessary for the planning and implementation of policies and procedures for the security of an organization's information assets.

MAIDA712 Sustainable Development and Smart Agriculture
The module gives an overview of climate-smart agriculture as an approach to address the interlinked challenges of achieving sustainability, increasing food security and responding to climate change. The module describes an overall framework for building resilience and increasing efficiency in various agricultural production systems. Issues to be addressed to implement climate-smart agriculture and make progress towards efficient and resilient agriculture production systems and food systems.

MAIDA713 Cyber Analytics
The module will address the following aspects; Reviewing, from a statistical perspective, the cyber-infrastructure ecosystem including distributed computing, and multi node and distributed file eco systems, such as Amazon Web Services. Structured and unstructured data sources, including social media data and image data. Setting up of large data structures for analysis. Algorithms and techniques for computing statistics and statistical models on distributed data. Software to be used include, Hadoop, Map reduce, SAS, SAS Data loader for Hadoop.

MAIDA715 Dissertation
This is a scientific report of between 15 000 and 20 000 words, based on supervised research by the student. The dissertation should be presented to a panel of the Departmental Board.

DEPARTMENT OF AGRONOMY AND HORTICULTURE

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN AGRONOMY (HAGRO)

Duration	:	4 years
Actual Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	540
Total MBKS Credit Load	:	384
ZNQF Level	:	8

1. PURPOSE OF PROGRAM

1.1. The programme aims at contributing towards sustainable food and cash crop production, economic growth and overall food security.

1.2. The Department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical knowledge and practical skills in field crop production.
b. Production of competitive and innovative graduates for various career opportunities in private and public sectors as well as self-employment.
c. Departmental participation in agricultural development through research, innovation, industrialization and outreach programmes.
d. Collaborating with relevant stakeholders.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have:

- At least two 'A' Level passes in Biology/Crop science/Horticulture and any other science subject.
- At least five 'O' Level passes including English Language, Science and Mathematics.

SPECIAL ENTRY

Relevant Diploma based on Departmental Board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Plant/crop biology & ecology, production practices/technologies; agricultural statistics and research methods; climate change and adaptation; molecular biology and biotechnology; plant breeding; post-harvest technology and crop protection, pathology nematology and weed management, horticultural crops and field crops.

Specialist Focus: Sustainable crop production, plant breeding, crop protection and biotechnology technologies and value addition.

Orientation: Research, innovation and application of theory to solve practical agronomic-related problems

Distinctive Features: Knowledge, innovation and skills development

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability • Private Sector – Agronomists, Plant Breeders, Crop Protection officers, Technical and Sales Representatives, Farm Managers, Consultants.

• Public Sector – Researchers (Agronomists, Plant Breeders, Crop Protectionists) Extensionists, Lecturers.

• Self-Employment – Farmers, Consultants

Further Studies Master's Degree level in the various fields of Agronomy (crop production, crop protection, biotechnology and breeding, crop processing etc).

They can also proceed to do PhD and DPhil studies in these areas.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory and field practicals, group work, industrial and farm/field visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quizzes, group and individual presentations, work-related learning reports, final year dissertations, continuous assessment (30-50%), and Final examinations (50-70%).

6. LEARNING OUTCOMES

On completion of this programme a graduate will be able to:

- Understand and apply key scientific principles and theories underpinning Agronomy.
- Apply practical skills in sustainable crop production, scientific research and knowledge transfer.
- Develop credible solutions using contemporary technologies in climate–smart agriculture, molecular biology and biotechnology.
- Work cooperatively in groups, displaying tolerance to diversity.
- Research, innovate and provide solutions in Agronomy.
- Create enterprises in the agricultural sector.

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.2. Students can include in their programme of study, modules offered by other departments.

7.3. A student shall be enrolled for at least four modules at any time during the semester

7.4. The module combination should be feasible in terms of the timetable.

7.5. A student may change their registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAMME ASSESSMENT

Coursework: Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30-50% of the final mark.

Written Examinations: A three-hour final examination, which shall contribute to 50-70% of the final mark.

Practical Examination: The duration of practical exams, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. Grading And Degree Classification

Refer to Section 9 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 10 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, students are expected to acquaint themselves with introductory modules of the degree programme. Choice of modules should be done with guidance of Departmental staff.

13.2. The degree programme shall have at least 36 taught modules spread over the 6 teaching semesters.

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
*AGRO135	Plant Biology		12
*LWR135	Principles of Soil Science		12
*AGED131	Introduction to Agricultural Economics		12
*HAWS132	Principles of Biochemistry		12
*HORT131	Principles of Microbiology		12
*HCS135	Introduction to Information Technology		12
*AGPR131	Agricultural Practice 1		6
*CS131	Basic Communication Skills		12

Level 1 Semester 2

*AGRO132	Introduction to Genetics	(HAWS132)	12
*HORT133	Introduction to Horticulture		12
*HORT132	Introduction to Entomology and Parasitology		12
*LWR137	Soil fertility and plant mineral nutrition	(LWR135)	12
*AGRO136	Introduction to Statistics		12
*AGPR132	Agricultural Practice 2	(AGPR131)	6

Level 2 Semester 1

*AGRO231	Crop Ecology and Physiology	(AGRO135)	12
*AGRO233	Principles of Crop Production	(AGRO135)	12
*HORT231	Biology and Management of Plant Pathogens		12
*AGRO239	Statistical Methods and Experimental Design	(AGRO136)	12
*AEGS231	Gender Studies		12
*ENT131	Entrepreneurship 1		12

Optional ***(Choose one module from those on offer in the current semester)***

AGRO248	GIS and Land Use Planning		12
AGRO232	Principles of Plant Breeding	(AGRO132)	12
LWR233	Principles of Irrigation		12

AGRO236	Agroforestry	(AGRO135)	12
AGPR234	Value Addition in Agriculture		12

Level 2 Semester 2

*AGRO246	Research Methods and Scientific Writing		12
*HORT235	Biology and Management of Plant Insect Pests	(HORT132)	12
*AGRO234	Weed Science and Management	(AGRO135)	12
*AGED240	Farm Business and Management	(AGED131)	12
*HAWS238	Natural Resource Management		12

Optional (*Choose two from modules on offer in the current semester)*

AGRO238	Seed Science and Production	(AGRO135/AGRO132)	12
AGRO255	Plant Breeding Practice	(AGRO 232)	12
HORT232	Olericulture	(HORT133)	12
LWR236	Irrigation Systems Design		12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

*AGPR331	Work-Related Learning Report	45
*AGPR332	Employer's Assessment Report	30
*AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

Code	**Module Description**	**Prerequisite**	**Credits**
*AGRO450	Annual Food Crop Production	(AGRO233)	12
*AGRO447	Molecular Biology and Plant Biotechnology	(AGRO135)	12
*AGRO446	Global Environmental Issues and Climate change		12
*AGED439	Agricultural Extension		12

Optional (*Choose two from modules on offer in the current semester)*

AGRO422	Plant Breeding Methods	(AGRO232)	12
AGED401	Agriculture and Food Marketing	(AGED131, AGED240)	12
HORT 445	Floriculture		12
HORT446	Pesticide Science		12
AGRO434	Crop processing and Storage of Grain Crops	(AGRO233)	12
LWR435	Project Planning and Feasibility Studies		12
HORT442	Landscape Horticulture		12
AGRO433	Basic Agricultural Meteorology		12

Level 4 Semester 2		**Prerequisites**	**Credits**
*AGRO451	Industrial and Cash Crop Production	(AGRO233)	12
*AGHORT480	Dissertation		24
*LWR240	Soil and Water Conservation		12

Optional (*Choose two from modules on offer in the current semester)*

AGRO432	Applied Plant Breeding	(AGRO232)	12
AGRO412	Advanced Research Methods in Natural Sciences		12
HORT435	Post Harvest Physiology and Technology		12
AGRO424	Applied Molecular Biology and Plant Biotechnology		12

** Core modules (Obligatory)*

14.0 Module Synopses

AGRO135 Plant Biology
The module covers the following basic understanding of plant physiological concepts; cell biology; plant growth and development; seed, germination and dormancy; plant growth regulators; plant mineral nutrition; plant water relations; photoperiodism and vernalization; photosynthesis at cellular level (light and dark reactions, C3, C4, CAM photosynthetic systems, photo-respiration).

AGRO132 Introduction to Genetics
The module covers Introduction to genetics, Cellular division and chromosomes, Mendelian principles of inheritance, Gene interaction, chromosome theory of heredity, Sex determination and sex linkage, Nucleic acids, genetic code and protein synthesis, Gene and chromosomal mutations, Maternal effects and Cytoplasmic heredity.

AGRO136 Introduction to Statistics
Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion. Sampling techniques; Data presentations and summarization techniques. statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA. Regression and correlation.

HCS135 Introduction to Information Technology
This module is aimed at assisting students to achieve their full academic potential by equipping them with the necessary communication and computer skills essential for their degree program.

AGPR131 Agricultural Practice 1
The module provides hands-on learning to students, to compliment theoretical modules on agricultural production. Farm (University farm) orientation; Poultry, Dairy & Pig production, Cattle identification and detention, Soil sampling, Crop planting and fertilizing, Plant population and yield estimation, Introduction to vegetable production, Irrigation scheduling, Blood samples and smears, administering veterinary drugs, surveying in agriculture, Farm week practicals (all semesters).

AGPR132 Agricultural Practice II
This module is part II of the Agricultural practice module. Pig production and Dairying of University cattle, Control of internal parasites in livestock; Use & calibration of spraying equipment; Ox harness & ox-drawn ploughing; Tractor handling & ploughing/ contour construction; Soil coding; Vegetation coding in rangeland management; Farm budgets and Agric financing; Castration and dehorning, Game farming - farm visit; Field visits (research station/Large scale/small scale farming areas)

AGRO231 Crop Ecology and Physiology
The module aims to cover the following aspects; growth analysis (Liebig's law, parameters of size, efficiency and duration of the photosynthetic system, Infrared Gas Analyser); canopy and architecture (total radiation, the light spectrum, photosynthetically active radiation (PAR); canopy photosynthesis and respiration; plant water relations; temperature and plant growth; partitioning of assimilates; modelling.

AGRO232 Principles of Plant Breeding
The module covers probability analysis and chi-square tests in inheritance studies, Gene control models in eukaryotes, introduction to population genetics, Quantitative Inheritance, and Components of heritability in plant breeding. Linkage and recombination in plant breeding, gene mapping in diploids, Reproductive systems and pollination control crop plants, Polyploidy in Plant Breeding, Interspecific hybridization, Tissue Culture in Plant Breeding, principles of Mutation Breeding, Principles of recombinant DNA Technology in plant breeding.

AGRO236 Agro Forestry
In this module, students learn to create resilient agricultural systems that are more productive and sustainable. Basic concepts in Agroforestry, classification of Agroforestry systems, Plant aspects of Agroforestry, soil aspects in Agroforestry, component interactions in Agroforestry, propagation of tree species, factors influencing the adoption of Agroforestry systems, design of Agroforestry systems

HORT235 Biology and Management of Plant Pathogens
This module covers issues to deal with the structure, physiology, reproduction and classification of bacteria and other procaryotes, fungi, viruses, parasitic higher plants and their role as plant pathogens; Concepts of Disease epidemiology, Disease Cycle; Mechanisms of infection; Plant defence mechanisms against pathogens; Physiological or Abiotic Diseases to plants; Plant Disease management - cultural, chemical, preventative within the concept of integrated disease management. Lifecycles and management of key crop diseases

AGRO248 GIS and Land Use Planning
The course deals with both theoretical and practical principles of geographical information systems and remote sensing as they relate to assessment of agricultural and environmental resources. The course will equip students with knowledge and skills of assessing, evaluating, analysing and mapping distribution and changes in natural resources and the environment using ground surveys, GPS, existing maps and satellite images.

AGRO234 Weed Science and Management
The module covers evolution of weeds, adaptations of weeds as pests of disturbed areas, seed production, dormancy, germination, seed bank, weed interference, allelopathy and methods to study allelopathy, vegetative propagation mechanisms, weed dispersal. Weed management tactics:

manual, cultural, legislative, biological weed control, chemical. classification of herbicides, herbicide selectivity, soil applied herbicides, foliar applied herbicides, herbicide application.

AGRO233 Principles of Crop Production
The module gives an overview of plant interactions with their environment and practices/techniques of nurturing a crop. Classification of crops; Agro-ecological factors affecting crop growth & yield; Land use classification and evaluation; Practices on crop establishment and protection, Soil fertility (manure and fertilizers) Cropping systems (e.g., Intercropping; agroforestry); Harvesting and post-harvest techniques; Irrigation principles plus scheduling; Organic farming principles.

AGRO239 Statistical Methods and Experimental Design
Introduction to experimental designs, data analysis and interpretation. Analysis of variance for one factor, two factor experiments and multi-factor experiments; Multiple comparison of means; Partitioning of sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS, R) for all designs: Regression and Correlation.

AGRO246 Research Methods and Scientific Writing
The module equips students with research and scientific writing competencies. Planning-formulation of research questions and design, measurement tools and sampling. Referencing data sources. Research ethics and plagiarism. Research methods. Development of research proposals, experimental protocols and concept notes. Data handling, analysis & interpretation. Report / thesis writing, presentations and dissemination. Students produce and submit a term paper on a selected topic on a relevant research topic.

AGRO238 Seed Science and Production
The module covers seed science, production and associated legislation. Introduction to the physiology of development, ripening, and dormancy of crop seeds. Seed multiplication. Methods of harvesting, drying, processing, storage and marketing of seed. Tests for purity, germination capacity, and vigour, moisture contents, uniformity and health. Maintaining varietal purity. Procedures for certification, release and registration of cultivars. Seed laws and regulations.

AGRO255 Plant Breeding Practice
This module covers key issues that are required in developing the techniques of plant breeding. The focus will be given on induction of variation, emasculation techniques, mating systems and pollination techniques, proper field designs, selection techniques. The student will also learn how to manage a breeding program by using computer software.On completion of this module the student should be able to manage a breeding program using appropriate computer software.

HORT235 Biology and Management of Plant Insect Pests
This module covers important pests of crops, biology and ecology, life cycles, population ecology, population monitoring and economic decision making, monitoring and pest risk assessments, pest management strategies- cultural, physical, biological, IPM, surveillance, quarantine, botanicals, pheromones, insect growth regulators, plant resistance and genetic modifications of plants or insects.

AGPR330, AGPR331, AGPR332 and AGPR333 -Work-Related Learning
Work-Related Learning (WRL) for at least 8 months of the third year is a prerequisite for registration for fourth year modules. The purpose of WRL is to make students relate theory to practice in real work situations, facilitate students to acquire world of work experience and skills, facilitate career choices by students and help students to identify entrepreneurial and business opportunities, among others. WRL is assessed as two components viz. continuous assessment and WRL report. The continuous assessment component consists of at least two assessment reports by academic staff that visit the students during WRL plus a workplace supervisor's report. The Work-Related Preliminary (WRP) and WRL reports are comprehensive and critical analysis of the activities, processes and

outputs of the organization to which the student was attached and the drawing of lessons learnt from the WRL experiences. The student is required to identify a specific problem in systems, processes and/or in the organization to which they are seconded, analyse the problem and solutions. Credit will be given to reports that show critical thinking and comprehensive analysis of the problem identified during WRL.

AGRO434 Crop Processing and Storage of Grain Crops
The module covers the following aspects; Introduction to post-harvest technology of grain crops. Post-harvest losses. Grain properties and characteristics. Harvesting and drying of grain crops. Crop storage (causes of in –store losses and their control; types and management of storage structures). Threshing and shelling of grain crops. Primary processing of agricultural products (grains and oilseeds); value addition of grain crops.

AGHORT480 Dissertation
The module prepares students in areas of proposal writing, conducting of research, data interpretation and publication. They get an opportunity to engage in an independent study, with limited supervision and guidance. This is achieved through a program that exposes students to: research proposal writing, data collection & organization, data analysis & statistical competence, data interpretation & presentation, quality scientific writing, oral presentations.

AGRO446 Global Environmental Issues and Climate Change
The module outlines global environmental issues and ocean-atmosphere phenomena of importance in agriculture. Climate change; The role of water bodies and ice caps in regulating the global climatic systems; El-nino and La-nina effect; The ozone layer; Pollution and contamination of the environment; pollution from agro-chemicals; WHO limits on concentration of pesticides and other pollutants in potable water and food; Deforestation; erosion; desertification; measures to combat degradation; Sustainable development paradigms.

AGRO447 Molecular Biology and Plant Biotechnology
This module presents an overview of the techniques and underlying theory of molecular biology, plant tissue culture and genetic engineering, research and commercial applications, in the area of plant biotechnology. It exposes students to issues and challenges encountered in the area of molecular biology and plant biotechnology.

AGRO450 Annual Food Crop Production
The module covers classification, adaptation and agronomy of selected annual field crops grown for food. Classification, importance, growth and development of selected groups of annual food crops. Regional and national potential; limitations to production; Agronomy: cultivar selection, land preparation, planting, weed, pest and disease management, irrigation, harvesting and post-harvest handling of cereal grain, legume grain; plantain; root and tuber crops.

AGRO451 Industrial and Cash Crop Production
The module highlights the economic importance, agronomy and marketing of cash and industrial crops; Regional potential and limitations to production; Economic importance; Agronomy of cash and industrial crops; cultivar selection, soil requirements, tillage, planting, fertilizer application, weeding, crop protection, irrigation, harvesting, post-harvest processing and marketing cash and industrial beverage, sugar, fibre, oilseeds, drug, nut, silvicultural and citrus crops.

HORT446 Pesticide Science
This module covers the role of pesticides, pesticide classification, modes of activity, formulation, laws and regulations governing use of pesticides in Zimbabwe, pesticide registration in Zimbabwe, equipment used to apply pesticides, calibration, pesticide hazards to humans, wildlife, and the environment, management of pesticide resistance and pesticides in integrated pest management programmes.

AGRO422 Plant Breeding Methods
The module covers breeding methods for self-pollinated crops and breeding methods for cross pollinated crops; Population improvement methods; Types of cultivars: pure lines, multi-lines, various types of hybrids, synthetics, composites, and open pollinated varieties.

AGRO432 Applied Plant Breeding
In this module the student acquires knowledge on applications in plant breeding. The student studies breeding for insect and disease resistance, temperature and moisture stress, quality and adaptation in a specific crop. On completion of the module the student should be able to successfully initiate a breeding program for the crop studied.

AGRO424 Applied Molecular Biology and Plant Biotechnology
The aim of this module is to expose the student to molecular markers used in plant breeding and crop protection, and the application of these techniques in breeding programs and crop protection. After completing this module, the student should be able to apply these techniques in plant breeding and crop protection.

HAWS132 Principles of Biochemistry
Refer to the Department of Animal and Wildlife Science.

AGED131 Introduction to Agricultural Economics
Refer to the Department of Agricultural Economics and Development

CS131 Basic Communication Skills
Refer to the Department of Communication Skills

HCS135 Introduction to Information Technology
Refer to the Department of Computer Science

HORT131 Principles of Microbiology
Refer to Horticulture degree programme

LWR135 Principles of Soil Science
Refer to the Department of Land and Water Resources Management

HORT132 Introduction to Entomology and Parasitology
Refer to Horticulture degree programme

LWR137 Soil Fertility and Plant Mineral Nutrition
Refer to the Land and Water Resources Management Department

ENT131 Entrepreneurship 1
Refer to the Entrepreneurship Department

AGED240 Farm Business and Management
Refer to the Department of Agricultural Economics and Development

HORT232 Olericulture
Refer to the Horticulture Degree programme

LWR236 Irrigation Systems Design

Refer to land and Water Resources Management Department

AEGS231 Gender Studies
Refer to the Gender Studies Department

HAWS238 Natural Resource Management
Refer to the Animal and Wildlife Sciences Department

AGED401 Agriculture and Food Marketing
Refer to the Department of Agricultural Economics and Development

HORT445 Floriculture
Refer to Horticulture Degree Programme

LWR405 Project Planning and Feasibility Studies
Refer to the Land and Water Resources Management Department

LWR240 Soil and Water Conservation
Refer to the Land and Water Resources Management Department

HORT435 Post Harvest Physiology and Technology
Refer to Horticulture Degree Programme

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN CROP PRODUCTION AND HORTICULTURE (HCRH)

Duration	:	4 years
Actual Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	540
Total MBKS Credit Load	:	384
ZNQF Level	:	8

1. PURPOSE OF PROGRAM

1.1. The programme aims at contributing towards sustainable food and cash crop production, economic growth and overall food security

1.2. The Department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical knowledge and practical skills in field and horticultural crop production.
b. Production of competitive and innovative graduates for various career opportunities in private and public sectors as well as self-employment.
c. Departmental participation in agricultural development through research, innovation, industrialization and outreach programmes.
d. Collaborating with relevant stakeholders.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have:

- At least two 'A' Level passes in Biology/Crop Science/Horticulture and any science subject.
- At least five 'O' Level passes including English Language, Science and Mathematics.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Plant/crop biology & ecology, production practices/technologies; agricultural statistics and research methods; climate change and adaptation; molecular biology and biotechnology; plant breeding; post-harvest technology and crop protection.

Specialist Focus: Agronomy, horticultural therapy, climate-smart agriculture, organic farming, urban agriculture, hydroponics aquaponics precision agriculture, greenhouse management.

Orientation: Research, innovation, and application of theory to solve practical agronomic and horticulture-related problems.

Distinctive Features: Knowledge and skills development is a key feature of the programme.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability • Private Sector – Agronomists, Horticulturists, Plant propagators, Nursey managers Plant Breeders, Crop Protection officers, Technical and Sales Representatives, Farm Managers, and Consultants.

• Public Sector – Researchers (Agronomists, Horticulturists, Plant propagators Plant Breeders, Crop Protectionists) Extensionists, Lecturers.

• Entrepreneurship – Farmers, Nursery Practitioners, Consultants.

Further Studies Master's Degree level in the various fields of Agronomy and Horticulture (crop production, crop protection, biotechnology and breeding, crop processing etc).
They can also proceed to do PhD and DPhil studies in these areas.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (30-50 %), and Final examinations (50-70%).

6. LEARNING OUTCOMES

On completion of this programme a graduate will be able to:

- Apply key scientific principles and theories underpinning agronomy and horticulture.
- Apply practical skills in sustainable crop field production, vegetable flower, fruit and nut scientific research and knowledge transfer.
- Develop credible solutions using contemporary technologies in climate-smart agriculture, molecular biology and biotechnology.
- Work cooperatively in groups, displaying tolerance to diversity.
- Research, innovate and provide solutions in agronomy and horticulture
- Create enterprises in the agricultural sector.

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.2. Students can include in their programme of study, modules offered by other departments.
7.3. A student shall be enrolled for at least four modules at any time during the semester
7.4. The module combination should be feasible in terms of the timetable.
7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAMME ASSESSMENT

Coursework: Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30-50% of the final mark.
Written Examinations: A three-hour final examination, which shall contribute to 50-70% of the final mark.
Practical Examination: The duration of practical exams, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 10 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1 In the first year, students are expected to acquaint themselves with introductory modules of the degree programme. Choice of modules should be done with guidance of Departmental staff.

13.2 The degree programme shall have at least 36 taught modules spread over the 6 teaching semesters.

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
*AGRO135	Plant Biology		12
*LWR135	Principles of Soil Science		12
*AGED131	Introduction to Agricultural Economics		12
*HAWS132	Principles of Biochemistry		12
*HCS135	Introduction to Information Technology		12
*AGPR131	Agricultural Practice 1		6
*HORT131	Principles of Microbiology		12
*CSC131	Basic Communication Skills		12

Level 1 Semester 2

*AGRO132	Introduction to Genetics	(HAWS132)	12
*HORT132	Introduction to Entomology and Parasitology		12
*AGRO136	Introduction to Statistics		12
*AGPR132	Agricultural Practice 2	(AGPR131)	6
*HORT133	Introduction to Horticulture		12
*LWR137	Soil Fertility and Plant Mineral Nutrition		12

Level 2 Semester 1

*AGRO231	Crop Ecology and Physiology	(AGRO135)	12
*AGRO239	Statistical Methods and Experimental Design	(AGRO136)	12
*AGRO233	Principles of Crop Production	(AGRO 135)	12
*HORT231	Biology and Management of Plant Pathogens		12
*AEGS231	Gender Studies		12
*ENT131	Entrepreneurship 1		12

Optional (*Choose one module from modules on offer in the current semester)*

AGRO248	GIS and Land Use Planning		12
AGRO232	Principles of Plant Breeding	(AGRO132)	12
LWR233	Principles of Irrigation		12
AGRO236	Agroforestry	(AGRO135)	12
HORT233	Plant Propagation & Nursery Management	(HORT 133)	12
AGPR234	Value Addition in Agriculture		12

Level 2 Semester 2

*AGRO234	Weed Science and Management	(AGRO135)	12
*AGRO246	Research Methods and Scientific Writing		12
*HORT235	Biology and Management of Plant Insect Pests	(HORT132)	12

*HORT232	Olericulture		12
*HAWS238	Natural Resource Management		12
* AGED240	Farm Business Management	(AGED131)	12

Optional ***(Choose one module from modules on offer in the current semester)***

AGRO238	Seed Science & Technology	(AGRO135/AGRO132)	12
LWR236	Irrigation Systems Design		12
AGRO255	Plant Breeding Practice		12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report		30

Level 3 Semester 2: Work-Related Learning

*AGPR331	Work-Related Learning Report		45
*AGPR332	Employer's Assessment Report		30
*AGPR333	Academic Supervisor's Report		45

Level 4 Semester 1

*AGRO450	Annual Food Crop Production	(AGRO233)	12
*AGRO447	Molecular Biology and Plant Biotechnology	(AGRO135)	12
*AGRO446	Global Environmental Issues and Climate Change		12
*HORT444	Plantation Horticulture	(HORT133)	12
*AGED439	Agricultural Extension		12

Optional ***(Choose one module from modules on offer in the current semester)***

HORT443	Herbal Horticulture	(HORT133)	12
HORT434	Mushroom Production	(HORT231)	12
HORT437	Organic Farming		12
HORT442	Land Scape Horticulture	(HORT133)	12
HORT446	Pesticide Science		12
HORT445	Floriculture	(HORT133)	12
AGRO434	Crop Processing and Storage of Grain Crops		12

Level 4 Semester 2

*AGHORT480	Dissertation		24
*HORT435	Postharvest Physiology and Technology		12
*LWR240	Soil & Water Conservation		12
*AGRO451	Industrial and Cash Crop Production	(AGRO233)	12

Optional ***(Choose one module from modules on offer in the current semester)***

HORT418	Processing of Spices Herbal Plants and Essential oils		12
HORT442	Landscape Horticulture	(HORT133)	12
AGRO424	Applied Molecular Biology and Plant Biotechnology	(AGRO447)	12
AGRO442	Advanced Research Methods in Natural Sciences		12

**** Core modules (Obligatory)***

14. MODULE SYNOPSES

AGPR131 Agricultural Practice 1

The module provides hands-on learning to students, to complement theoretical modules on agricultural production. Farm (University farm) orientation; Poultry, Dairy & Pig production, Cattle identification and detention, Soil sampling, Crops planting and fertilizing, Plant population and yield estimation, Introduction to vegetable production, Irrigation scheduling, Blood samples and smears, administering veterinary drugs, surveying in agriculture, Farm week practical's (all semesters).

AGPR132 Agricultural Practice II

This module is part II of the Agricultural practice module. Pig production and Dairying of University cattle, Control of internal parasites in livestock; Use & calibration of spraying equipment; Ox harness & ox-drawn ploughing; Tractor handling & ploughing/ contour construction; Soil coding; Vegetation coding in rangeland management; Farm budgets and Agric financing; Castration and dehorning, Game farming - farm visit; Field visits (research station/Large scale/small scale farming areas)

AGRO135 Plant Biology

The module covers the following basic understanding of plant physiological concepts; cell biology; plant growth and development; seed, germination and dormancy; plant growth regulators; plant mineral nutrition; plant water relations; photoperiodism and vernalization; photosynthesis at cellular level (light and dark reactions, C3, C4, CAM photosynthetic systems, photo-respiration).

AGRO132 Introduction to Genetics

The module covers cell division, nucleic acids, central dogma of life, and protein synthesis. Mendelian principles: segregation and independent assortment. Environmental effects and gene expression. Gene interaction and lethality; sex determination and sex linkage. Gene mutation and induced genetic change (biotechnology). Gene structure and regulation. Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources

AGRO136 Introduction to Statistics

Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion. Sampling techniques; Data presentations and summarization techniques. statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA. Regression and correlation.

HCS135 Introduction to Information Technology

This module is aimed at assisting students to achieve their full academic potential by equipping them with the necessary communication and computer skills essential for their degree program.

AGRO231 Crop Ecology and Physiology
The module aims to cover the following aspects; growth analysis (Liebig's law, parameters of size, efficiency and duration of the photosynthetic system, Infrared Gas Analyser); canopy and architecture (total radiation, the light spectrum, photosynthetically active radiation (PAR); canopy photosynthesis and respiration; plant water relations; temperature and plant growth; partitioning of assimilates; modelling.

AGRO232 Principles of Plant Breeding
The modules cover basic statistics applied in plant breeding, Gene control models in eukaryotes, introduction to population genetics, quantitative genetics in plant breeding, Linkage and recombination in plant breeding, gene mapping in diploids, Reproductive systems and pollination control in crop plants, Polyploidy in Plant Breeding, Interspecific hybridisation, Tissue Culture in Plant Breeding, principles of Mutation Breeding, Principles of recombinant DNA Technology in plant breeding.

AGRO234 Weed Science and Management
The module covers evolution of weeds, adaptations of weeds as pests of disturbed areas, seed production, dormancy, germination, seed bank, weed interference, allelopathy and methods to study allelopathy, vegetative propagation mechanisms, weed dispersal. Weed management tactics: manual, cultural, legislative, biological weed control, chemical. classification of herbicides, herbicide selectivity, soil-applied herbicides, foliar applied herbicides, herbicide application.

AGRO233 Principles of Crop Production
The module gives an overview of plant interactions with their environment and practices/techniques of nurturing a crop. Classification of crops; Agro-ecological factors affecting crop growth & yield; Land use classification and evaluation; Practices on crop establishment and protection, Soil fertility (manure and fertilizers) Cropping systems (e.g., Intercropping; agroforestry); Harvesting and post-harvest techniques; Irrigation principles plus scheduling; Organic farming principles

AGRO239 Statistical Methods and Experimental Design
Introduction to experimental designs, data analysis and interpretation. Analysis of variance for one factor, two-factor experiments and multi-factor experiments; Multiple comparisons of means; Partitioning of the sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS, R) for all designs: Regression and Correlation.

AGRO246 Research Methods and Scientific Writing
The module equips students with research and scientific writing competencies. Planning-formulation of research questions and design, measurement tools and sampling. Referencing data sources. Research ethics and plagiarism. Research methods. Development of research proposals, experimental protocols and concept notes. Data handling, analysis & interpretation. Report / thesis writing, presentations and dissemination. Students produce and submit a term paper on a selected topic on a relevant research topic.

AGRO238 Seed Science and Production
The module covers seed science, production and associated legislation. Introduction to the physiology of development, ripening, and dormancy of crop seeds. Seed multiplication. Methods of harvesting, drying, processing, storage and marketing of seed. Tests for purity, germination capacity, and vigour, moisture contents, uniformity and health. Maintaining varietal purity. Procedures for certification, release and registration of cultivars. Seed laws and regulations.

AGRO423 Applied Plant Breeding
The module covers goals of plant breeding, Plant Introduction and Domestication, Genetic Consequences of Hybridisation, Introduction to theory of selection, breeding methods with self-pollinated crops; Breeding methods with cross pollinated plants; Principles of marker assisted selection in plant breeding, Seed production and marketing; Future of plant breeding

AGRO446 Global Environmental Issues and Climate Change
The module outlines global environmental issues and ocean-atmosphere phenomena of importance in agriculture. Climate change; The role of water bodies and ice caps in regulating the global climatic systems; El-nino and La-nina effect; The ozone layer; Pollution and contamination of the environment; pollution from agro-chemicals; WHO limits on the concentration of pesticides and other pollutants in potable water and food; Deforestation; erosion; desertification; measures to combat degradation; Sustainable development paradigms.

AGRO424 Applied Molecular Biology and Plant Biotechnology
The modules covers plant tissue culture systems and methods (includes cell culture systems, micro-propagation, somatic embryogenesis and organogenesis); application of biotechnology in the conservation of crop genetic resources (in vitro and cryopreservation); understanding of traditional and molecular assisted plant breeding; molecular markers, genomics, gene transfer systems, including case studies and regulatory and ethical issues; cell & molecular understandings of plant processes important in horticulture, agriculture and forestry.

AGRO450 Annual Food Crop Production
The module covers classification, adaptation and agronomy of selected annual field crops grown for food. Classification, importance, growth and development of selected groups of annual food crops. Regional and national potential; limitations to production; Agronomy: cultivar selection, land preparation, planting, weed, pest and disease management, irrigation, harvesting and post-harvest handling of cereal grain, legume grain; plantain; root and tuber crops.

AGRO451 Industrial and Cash Crop Production
The module highlights the economic importance, agronomy and marketing of cash and industrial crops; Regional potential and limitations to production; Economic importance; Agronomy of cash and industrial crops; cultivar selection, soil requirements, tillage, planting, fertilizer application, weeding, crop protection, irrigation, harvesting, post-harvest processing and marketing cash and industrial beverage, sugar, fibre, oilseeds, drug, nut, silvicultural and citrus.

HORT131 Principles of Microbiology
The module provides an introduction to the study of microorganisms relevant to the areas of agriculture, food and industry. Introduction to the range of microorganisms; Bacteria and fungi, Viruses and viroids, Protozoa. Microorganisms as agents of

diseases; Beneficial aspects of microbes; Food spoilage; Food and industrial microbiology.

HORT133 Introduction to Horticulture
The module covers all basics of Horticulture. Definition, characteristics and importance of horticulture; Branches of horticulture, classification of horticultural crops; Management of environmental factors influencing horticultural plants; growing structures, Pruning and training of horticultural crops; Innovations in the horticultural industry - hydroponics, aeroponics, aquaponics, Harvesting, postharvest handling and storage.

HORT132 Introduction to Entomology and Parasitology
This module covers beneficial and injurious insects, classification, adaptation characteristics, insect anatomy and physiology, nervous system, digestive system, locomotion, insect reproduction, and metamorphosis. Mites, ticks and nematodes-morphology, classification, reproduction and economic importance.

HORT231 Biology and Management of Plant Pathogens
This module covers issues to deal with the structure, physiology, reproduction and classification of bacteria and other procaryotes, fungi, viruses, parasitic higher plants and their role as plant pathogens; Concepts of Disease epidemiology, Disease Cycle; Mechanisms of infection; Plant defence mechanisms against pathogens; Physiological or Abiotic Diseases to plants; Plant Disease management - cultural, chemical, preventative within the concept of integrated disease management. Lifecycles and management of key crop disease

HORT232 Olericulture
This module seeks to teach students the science and art of vegetable production. Classification of vegetables and their economic importance; Production and management of the following groups: Solanaceous crops, Cucurbit crops, Cole crops, Legume crops, Leafy Vegetables, Alliums, Root crops, Traditional Vegetables, Export Vegetables, Herbs and Spices

HORT233 Plant Propagation and Nursery Management
This module introduces students to the main concepts of propagating plants and managing nurseries. The regeneration of plants from vegetative and reproductive tissue and organs. Methods, and techniques for greenhouse, nursery, field and orchards. Nursery layouts for efficient production, handling and distribution, programming production, nursery practices, field and container growing of woody ornamentals, mechanization, quality control, and record keeping.

HORT235 Biology and Management of Plant-Insect Pests
This module covers important pests of crops, biology and ecology, life cycles, population ecology, population monitoring and economic decision making, monitoring and pest risk assessments, pest management strategies- cultural, physical, biological, IPM, surveillance, quarantine, botanicals, pheromones, insect growth regulators, plant resistance and genetic modifications of plants or insects.

HORT445 Floriculture
The module covers important aspects of cut flower and ornamental flower production. Factors affecting flower and ornamental production. Ecological requirements of

flower and ornamental production. Greenhouse flower and ornamental production. Botany and physiology of cut flowers (Rose, Carnations, Proteas and Chrysanthemums), Flower cooling and temperature requirements, and flower preservation.

HORT434 Mushroom Production
This module covers techniques and practices in the production of mushrooms. Mushroom biology, spawn production, preparation of media and tissue cultures, cultivation techniques, types of substrates, mycorrhiza and edible fungi, preparing and spawning straw substrate, oyster mushroom cultivation.

HORT435 Postharvest Physiology and Technology
The module introduces the students to basic postharvest handling of fruits, vegetables and cereals at domestic, small scale and commercial levels. Storage physiology and biochemistry of fruits, vegetables and ornamental crops. Influence of pre-harvest grading. storage, packing, transportation and marketing. Effects of storage temperature CO_2, O_2, ethylene on ripening.

HORT437 Organic Farming
The module focuses on organic production systems, soil and plant nutrition, nutrient cycling. Organic pest and disease control, managing and saving weeds. Livestock integration in organic farming. Marketing of organic products. Research and development.

HORT444 Plantation Horticulture
The module provides an overview of the production of fruit & nut trees. Plant growth and development in relation to tree fruit production; emphasis on canopy development and pruning, flowering and fruit set and development and dormancy. Overview of the Zimbabwe fruit and nut industry. Fruits include both indigenous and exotics.

HORT442 Landscape Horticulture
The module aims to address aesthetic components of Horticulture, Definition of Landscape horticulture, sustainable landscapes, effective landscaping, benefits of landscaping (aesthetic, environmental, social, economic, health), Elements of design, Principles of design, the design process, lawn establishment, turf management, Ground covers, Edible landscaping, xeriscaping, Landscape function software.

HORT413 Herbal Horticulture
The module addresses agronomic aspects of herb and spice production and their commercial utilization. Herbal botany, Biology and physiology of herbal plants and spices. Compounds of medicinal plants. Propagation methods of medicinal plants. Intensive herb and spices production. Research trends in herbal and spice production. Marketing. Plants include Aloe vera, Russian comfrey, Chinese mint, Balbernella, Crotolaria spp, Moringa olifera and many others.

AGPR330, AGPR331, AGPR332 & AGPR333: Work-Related Learning
Work-Related Learning (WRL) for at least 8 months of the third year is a prerequisite for registration for fourth year modules. The purpose of WRL is to make students relate theory to practice in real work situations, facilitate students to acquire world of work experience and skills, facilitate career choices by students and help students to identify entrepreneurial and business opportunities, among others. WRL is assessed as two components viz. continuous assessment and WRL report. The continuous assessment component consists of at least two assessment reports by academic staff that visit the students during WRL plus

a workplace supervisor's report. The Work-Related Preliminary (WRP) and WRL reports are comprehensive and critical analyses of the activities, processes and outputs of the organization to which the student was attached and the drawing of lessons learnt from the WRL experiences. The student is required to identify a specific problem in systems, processes and/or in the organization to which they are seconded, and analyse the problem and solutions. Credit will be given to reports that show critical thinking and comprehensive analysis of the problem identified during WRL.

AGHORT480 Dissertation

The module prepares students in areas of proposal writing, conducting of research, data interpretation and publication. They get an opportunity to engage in an independent study, with limited supervision and guidance. This is achieved through a program that exposes students to: research proposal writing, data collection & organization, data analyses & statistical competence, data interpretation & presentation, quality scientific writing, oral presentations.

AGED131 Introduction to Agricultural Economics

Refer to Department Agricultural Economics and Development

HAWS132 Principles of Biochemistry

Refer to the Department of Animal and Wildlife sciences

LWR135 Principles of Soil Science

Refer to the Department of Land and Water Resources management

LWR240 Soil and Water Conservation

Refer to the Department of Land and Water Resources management

AEGS231 Gender Studies

Refer to Gender studies Department

AGRO248 GIS and Land Use Planning

Refer to the Agronomy programme

LWR233 Principles of Irrigation

Refer to the Department of Land and Water Resources

ENT131 Entrepreneurship 1

Refer to the Department of Entrepreneurship.

AGED439 Agricultural Extension

Refer to Department Agricultural Economics and Development

HORT448 Processing of Spices Herbal Plants and Essential oils

Refer to the HHORT programme

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN HORTICULTURE (HHORT)

Duration	:	4 years
Actual Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	540
Total MBKS Credit Load	:	384
ZNQF Level	:	8

1. PURPOSE OF PROGRAM

1.1. The programme aims at contributing towards increased export horticultural production thus improving export earnings, gross domestic product (GDP) growth and nutrition security as well as the overall well-being of the population.

1.2. The Department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical and practical knowledge in both open-field and protected-culture horticultural crop production.
b. Production of competitive and innovative graduates for various career opportunities in private and public sectors as well as self-employment.
c. Departmental participation in agricultural development through research, innovation, industrialization and outreach programmes.
d. Collaborating with relevant stakeholders.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have:

- At least two 'A' Level passes in Biology/Crop Science/Horticulture and any other science subject.
- At least five 'O' Level passes including English Language, Science and Mathematics.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Plant biology, production techniques and technologies; agricultural statistics and research methods; climate change and adaptation; molecular biology and biotechnology; plant breeding; post-harvest technology and crop protection.

Specialist Focus:

Vegetable, fruit, flower, and mushroom production as well as advanced horticultural technologies such as greenhouses, and hydroponics among others.

Research, innovation and application of theory to solve practical horticulture-related challenges

Orientation Research, innovation and application of theory to solve practical agronomic-related problems

Distinctive Features: A special focus will be directed towards developing complete graduates with requisite theoretical knowledge, practical skills and problem-solving abilities.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability • Private Sector –
Horticulturists, Greenhouse Managers, Florists, Pomologists, Vegetable Specialists, Technical and Sales Representatives, Estate Managers, Consultancy and Advisory services.

• Public Sector –
– Researchers, Plant Quarantine Services, Crop Inspectors, Lecturers
• Self-Employment – Farmers, Contract-growers for local Supermarkets as well as exporters, Market-Linkages for Horticultural produce for both local and export markets.
Further Studies - Master's Degree level in the various fields of Horticulture (Horticultural Sciences, crop protection, biotechnology, breeding).
Further opportunities also exist for advanced studies (PhD or DPhil) in these areas of specialization.

5. PROGRAMME DELIVERY

Teaching and Learning Methods:
Lectures, tutorials, presentations, practicals (field and laboratory), group work, industrial and farm/field visits, work-related learning, dissertations
Assessment Methods: Examinations, tests, quizzes, presentations, work-related learning reports, final year dissertations, continuous assessment (30-50%), and Final examination (50-70%).

6. LEARNING OUTCOMES

On completion of this programme a graduate will be able to:

- Apply key scientific principles and theories underpinning Horticulture.
- Apply practical skills in Horticultural crop production and research.
- Research, innovate and provide solutions in Horticulture.
- Create enterprises in the agricultural sector.

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.2. Students can include in their programme of study, modules offered by other departments.
7.3. A student shall be enrolled for at least four modules at any time during the semester.
7.4. The module combination should be feasible in terms of the timetable.
7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAMME ASSESSMENT

Coursework: Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30-50% of the final mark.
Written Examinations: A three-hour final examination, which shall contribute to 50-70% of the final mark.
Practical Examination: The duration of practical exams, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 9 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, students are expected to acquaint themselves with introductory modules of the degree programme. The choice of modules should be done with the guidance of Departmental staff.

13.2. The degree programme shall have at least 36 taught modules spread over the 6 teaching semesters.

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
*AGED131	Introduction to Agricultural Economics		12
*AGPR131	Agricultural Practice 1		6
*AGRO135	Plant Biology		12
*HAWS132	Principles of Biochemistry		12
*HCS135	Introduction to Information Technology		12
*LWR135	Principles of Soil Sciences		12
*HORT131	Principles of Microbiology		12
*CS131	Basic Communication Skills		12

Level 1 Semester 2

*AGPR132	Agricultural Practice 2	(AGPR131)	6
*AGRO132	Introduction to Genetics	(AWS135)	12
*AGRO136	Introduction to Statistics		12
*HORT133	Introduction to Horticulture	(AGRO135)	12
*HORT132	Introduction to Entomology and Parasitology		12
*LWR137	Soil Fertility and Plant Mineral Nutrition		12

Level 2 Semester 1

*AGRO231	Crop Ecology and Physiology	(AGRO135)	12
*ENT131	Entrepreneurship 1		12
*AGRO239	Statistical Methods and Experimental Designs		12
*AEGS231	Gender Studies		12
*HORT231	Biology and Management of Plant Pathogens	(HORT131)	12
*HORT233	Plant Propagation & Nursery Management	(HORT133)	12

Optional ***(Choose two modules from modules on offer in the current semester)***

AGRO232	Principles of Plant breeding	(AGRO 132)	12
AGRO233	Principles of Crop Production	(AGRO 135)	12
LWR233	Principles of Irrigation		12
AGPR234	Value Addition in Agriculture		12

Level 2 Semester 2

*HORT232	Olericulture	(HORT133)	12
*HORT235	Biology and Management of Plant Pests	(HORT132)	12
*AGRO246	Research Methods and Scientific Writing	(AGRO239)	12
*AGRO234	Weed Science and Management	(AGRO135)	12
*AGED240	Farm Business and Management	(AGED131)	12
*HAWS 238	Natural Resource Management		12

Optional ***(Choose one module from modules on offer in the current semester)***

HORT213	Spice and Herbal Plant Production	12
HORT212	Viticulture and Oenology	12
LWR236	Irrigation Systems Design	12
AGRO255	Plant Breeding Practice	12

Level 3 Semester 1: Work-Related Learning

*AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

*AGPR331	Work-Related Learning Report	45
*AGPR332	Employer's Assessment Report	30
*AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

*HORT 444	Plantation Horticulture	12
*HORT445	Floriculture	12
*AGRO447	Molecular Biology and Plant Biotechnology	12
*HORT437	Organic Farming	12

Optional ***(Choose two modules from modules on offer in the current semester)***

AGRO450	Annual Food Crop Production	12
HORT416	Pesticide Science	12
AGED 401	Agriculture and Food Marketing	12
AGRO446	Global Environmental Issues and Climate Change	12
AGED439	Agricultural Extension	12

Level 4 Semester 2		**Prerequisites**	**Credits**
*AGHORT480	Dissertation		24
*AGRO424	Applied Molecular Biology and Plant Biotechnology		12
*HORT435	Post Harvest Physiology and Technology		12
*HORT442	Landscape Horticulture		12

Optional ***(Choose one module from modules on offer in the current semester)***

LWR240	Soil and Water Conservation	12
HORT448	Processing of Spices, Herbal Plants and Essential oils	12
AGRO451	Industrial and Cash Crop Production	12

14. MODULE SYNOPSES

HORT131 Principles of Microbiology

The module provides an introduction to the study of microorganisms relevant to the areas of agriculture, food and industry. Introduction to the range of microorganisms; Bacteria and

fungi, Viruses and viroids, Protozoa. Microorganisms as agents of diseases; Beneficial aspects of microbes; Food spoilage; Food and industrial microbiology.

HORT132 Introduction to Entomology and Parasitology
This module covers beneficial and injurious insects, classification, adaptation characteristics, insect anatomy and physiology, nervous system, digestive system, locomotion, insect reproduction, metamorphosis. Mites, ticks and nematodes-morphology, classification, reproduction and economic importance.

HORT133 Introduction to Horticulture
The module covers all basics of Horticulture. Definition, characteristics and importance of horticulture; Branches of horticulture, classification of horticultural crops; Management of environmental factors influencing horticultural plants; growing structures, Pruning and training of horticultural crops; Innovations in the horticultural industry - hydroponics, aeroponics, aquaponics, Harvesting, postharvest handling and storage.

HORT232 Olericulture
This module seeks to teach students the science and art of vegetable production. Classification of vegetables and their economic importance; Production and management of the following groups: Solanaceuos crops, Cucurbit crops, Cole crops, Legume crops, Leafy Vegetables, Alliums, Root crops, Traditional Vegetables, Export Vegetables, Herbs and Spices.

HORT233 Plant Propagation and Nursery Management
This module introduces students to the main concepts of propagating plants and managing nurseries. The regeneration of plants from vegetative and reproductive tissue and organs. Methods, and techniques for greenhouse, nursery, field and orchards. Nursery layouts for efficient production, handling and distribution, programming production, nursery practices, field and container growing of woody ornamentals, mechanization, quality control, record keeping.

HORT231 Biology and Management of Plant Pathogens
This module covers issues to deal with the structure, physiology, reproduction and classification of bacteria and other procaryotes, fungi, viruses, parasitic higher plants and their role as plant pathogens; Concepts of Disease epidemiology, Disease Cycle; Mechanisms of infection; Plant defence mechanisms against pathogens; Physiological or Abiotic Diseases to plants; Plant Disease management - cultural, chemical, preventative within the concept of integrated disease management. Lifecycles and management of key crop diseases

HORT235 Biology and Management of Insect Pests
This module covers important pests of crops, biology and ecology, life cycles, population ecology, population monitoring and economic decision making, monitoring and pest risk assessments, pest management strategies- cultural, physical, biological, IPM, surveillance, quarantine, botanicals, pheromones, insect growth regulators, plant resistance and genetic modifications of plants or insects.

HORT212 Viticulture and Oenology
This module covers the agronomic and commercial aspects of grape cultivation and its relevance in wine making. Botany and culture of grapes, vineyard management, Training and pruning, Diseases and pests of grapes, Commercial production of grapes, Post-harvest handling of grapes, Wine production process (yeast selection, fermentation, filling and filtration). Organic viticulture, Research in grape management.

HORT213 Spice and Herbal Plant Production
Students are taught the agronomic aspects of herb and spice production and their commercial utilization. Biology & physiology of herbs & spices; Intensive production of selected spices (e.g., poppy, cardamom, cinnamon, ginger, pepper and sesame and herbs (used as fresh or dried aromatic leaves) such as rosemary, and thyme. Production of medicinal plants such as Lemon grass, Cannabis Aloe vera, Moringa olifera and others. Research trends in herbal and spice production. Marketing.

HORT445 Floriculture
The module covers important aspects of cut flower and ornamental flower production. Factors affecting flower and ornamental production. Ecological requirements of flower and ornamental production. Greenhouse flower and ornamental production. Botany and physiology of cut flowers (Rose, Carnations, Proteas and Chrysanthemums), Flower cooling and temperature requirements, flower preservation.

HORT435 Post Harvest Physiology and Technology
The module introduces the students to basic postharvest handling of fruits, vegetables and cereals at domestic, small scale and commercial levels. Areas covered include storage physiology and biochemistry of fruits, vegetables and ornamental crops. Influence of pre-harvest grading. storage, packing, transportation and marketing. Effects of storage temperature CO_2, O_2, ethylene on ripening.

HORT437 Organic Farming
The module focuses on organic production systems, soil and plant nutrition, nutrient cycling. Organic pest and disease control, managing and saving weeds. Livestock integration in organic farming. Marketing of organic products. Research and development.

HORT442 Landscape Horticulture
The module aims to address aesthetic components of Horticulture. Definition of Landscape horticulture, sustainable landscapes, effective landscaping, benefits of landscaping (aesthetic, environmental, social, economic, health), Elements of design, Principles of design, the design process, lawn establishment, turf management, Ground covers, Edible landscaping, xeriscaping, Landscape function software.

HORT448 Processing of Spices, Herbs and Essential oils
The module introduces students to post-harvest handling and processing of spices and herbs. Storage physiology, biochemistry, drying, processing, grading, storage, packing, transportation and marketing of herbs and spices. Compounds and oils in spices, herbal and medicinal plants. Aromatics and their trade.

AGHORT480: Dissertation
The dissertation provides an opportunity for the students to undertake and contribute to a piece of original research work, in an area related to crop science. The students are required to design an experiment (or investigation, survey or other means) to test a hypothesis or proposition, to plan and execute the research work, to evaluate the outcomes and draw valid conclusions. The research work is carried out individually, under guided supervision.

AGRO246 Research Methods and Scientific writing
The module equips students with research and scientific writing competencies. Planning-formulation of research questions and design, measurement tools and sampling. Referencing data sources. Research ethics and plagiarism. Research methods. Development of research

proposals, experimental protocols and concept notes. Data handling, analysis & interpretation. Report / thesis writing, presentations and dissemination. Students produce and submit a term paper on a selected topic on a relevant research topic.

AGPR330, AGPR331, AGPR332 & AGPR333: Work-Related Learning
Work-Related Learning (WRL) for at least 8 months of the third year is a prerequisite for registration for fourth year modules. The purpose of WRL is to make students relate theory to practice in real work situations, facilitate students to acquire world of work experience and skills, facilitate career choices by students and help students to identify entrepreneurial and business opportunities, among others. WRL is assessed as two components viz. continuous assessment and WRL report. The continuous assessment component consists of at least two assessment reports by academic staff that visit the students during WRL plus a workplace supervisor's report. The Work-Related Preliminary (WRP) and WRL reports are comprehensive and critical analysis of the activities, processes and outputs of the organization to which the student was attached and the drawing of lessons learnt from the WRL experiences. The student is required to identify a specific problem in systems, processes and/or in the organization to which they are seconded, analyse the problem and solutions. Credit will be given to reports that show critical thinking and comprehensive analysis of the problem identified during WRL.

AGED131 Introduction to Agricultural Economics
Refer to the Department of Agricultural Economics and Development.

LWR135 Principles of Soil Science
Refer to Land and Water Resources Management Department.

HAWS132 Principles of Biochemistry
Refer to Animal and Wildlife Sciences Department

ENT131 Entrepreneurship 1
Refer to Entrepreneurship Department Regulations.

AGED439 Agricultural Extension
Refer to Agricultural Economics and Development Regulations.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN CROP PROTECTION (HCPR)

Duration	:	4 years
Actual Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	540
Total MBKS Credit Load	:	384
ZNQF Level	:	8

2. PURPOSE OF PROGRAM

2.1. The programme aims at contributing towards export horticultural production thus improving export earnings, gross domestic product (GDP) growth and nutrition security as well as overall well-being of the population.

2.2. The Department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical knowledge in both and open field and protected horticultural crop production.
b. Production of competitive and innovative graduates for various career opportunities in private and public sectors as well as self-employment.
c. Departmental participation in agricultural development through research, innovation, industrialization and outreach programmes.
d. Collaborating with relevant stakeholders.

2. ENTRY REQUIREMENTS

NORMAL ENTRY

Prospective students should have:

- At least two 'A' Level passes in Biology/Crop Science/Horticulture plus any other science subject.
- At least five 'O' Level passes including English Language, Science and Mathematics.

SPECIAL ENTRY

Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Plant pathogen Biology, Ecology and management, plant insect pest Biology, Ecology and Management, Weed Biology, Pesticide Science, Molecular Biology and Biotechnology, Microbiology, Plant breeding, Climate Change and adaptation; plant biology and ecology; Production practices and technologies, agricultural statistics and Research methods and Post-harvest technology.

Specialist Focus: Sustainable crop protection practices and technologies, research, Innovation and Application of Theory to solve Practical Crop Protection related problems.

Distinctive Features: Knowledge and skills development, integrated Pest Management (IPM) Resistance management, plant health diagnosis, Precision Agriculture.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability • Private Sector – Crop protectionists, Plant breeders, Agronomists Technical and Sales Representatives, farm Mangers, Consultancy.
• Public Sector – Researchers, (Crop Protectionists, Plant breeders, Agronomists, Extensionists, Lecturers.
• Entrepreneurship – Farmers, Consultants.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, Tutorials, laboratory and practicals, group work, industrial and farm/field visits, work-related learning, dissertations,
Assessment Methods: Written and Oral examinations, Tests, Quizzes, group and individual presentations, Work-Related Learning Reports, Final year dissertations, Continuous assessment (30-50%), and Final examination (50-70%) depending on the module.

6. LEARNING OUTCOMES

On completion of this programme a graduate will be able to:

- Understand and apply key scientific principles and theories underpinning Agronomy.
- Apply practical skills in sustainable crop production, scientific research and knowledge transfer.
- Develop credible solutions using contemporary technologies in climate–smart agriculture, molecular biology and biotechnology.
- Work cooperatively in groups, displaying tolerance to diversity.
- Research, innovate and provide solutions in Agronomy.
- Create enterprises in the agricultural sector.

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.2. Students can include in their programme of study, modules offered by other departments.
7.3. A student shall be enrolled for at least four modules at any time during the semester
7.4. The module combination should be feasible in terms of the timetable.
7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAMME ASSESSMENT

Coursework: Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30-50% of the final mark.

Written Examinations: A three-hour final examination, which shall contribute to 50-70% of the final mark.

Practical Examination: The duration of practical exams, where they are deemed. necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 10 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, students are expected to acquaint themselves with introductory modules of the degree programme. Choice of modules should be done with guidance of Departmental staff.

13.2. The degree programme shall have at least 36 taught modules spread over the 6 teaching semesters.

Minimum Body of Knowledge And Skills Modules

MODULE	CREDITS
Level I	
*AGRO135 Plant Biology	12
*HAWS132 Principles of Biochemistry	12
*LWR135 Principles of Soil Science	12
*AGRO136 Introduction to Statistics	12
*AGRO132 Principles of Genetics	12
*HORT131 Principles of Microbiology	12
*HORT132 Introduction to Entomology and Parasitology	12
*AGED131 Introduction to Agricultural Economics	12
*CS131 Basic Communication Skills	12
*HCS135 Introduction to Information Technology	12
*LWR137 Soil Fertility and Plant Mineral Nutrition	12
*AGPR131 Agricultural Practice 1	6
*AGPR132 Agricultural Practice 2	6
Level II	
*HORT133 Introduction to Horticulture	12
*AGRO233 Principles of Crop Production	12
*AGRO239 Statistical Methods and Experimental Design	12
*AGRO231 Crop Ecology and Physiology	12
*HORT231 Plant Pathogen Biology and Ecology	12
*HORT235 Insect Pest Biology and Ecology	12
*AGRO234 Weed Biology and Ecology	12
*AEGS231 Gender Studies	12
*ENT131 Entrepreneurship 1	12
*HAWS238 Natural Resources Management	12
*AGRO246 Research Methods and Scientific Writing	12
AGRO248 GIS and Land Use Planning	12
AGRO232 Principles of Plant Breeding	12
LWR233 Principles of Irrigation	12
AGRO236 Agro Forestry	12
AGPR234 Value Addition in Agriculture	12
AGRO238 Seed Science and Production	12
AGRO455 Plant Breeding Practice	12
HORT232 Olericulture	12
LWR236 Irrigation Systems Design	12

Level III

*AGPR330 Work-Related Learning Preliminary Report	30
*AGPR331 Work-Related Learning Report	45
*AGPR332 Employer's Assessment Report	30
*AGPR333 Academic Supervisor's Report	45

Level IV

*AGHORT480 Dissertation	24
*AGRO450 Annual Food Crop Production	12
*AGRO451 Industrial and Cash Crop Production	12
*AGRO447 Molecular Biology and Plant Biotechnology	12
*AGHORT435 Plant Nematode Biology, Ecology and Management	12
*HORT446 Pesticide Science	12
*AGRO436 Weed Management	12
*AGHORT436 Insect Pest Management	12
*AGHORT437 Plant Pathogen Management	12
*AGED439 Agricultural Extension	12
*AGRO446 Global Environmental Issues and Climate Change	12
AGRO422 Plant Breeding Methods	12
AGED401 Agriculture and Food Marketing	12
HORT445 Floriculture	12
AGRO434 Crop Processing and Storage of Grain Crops	12
LWR435 Project Planning and Feasibility Studies	12
HORT442 Landscape Horticulture	12
AGRO433 Basic Agricultural Meteorology	12
AGRO424 Applied Molecular Biology and Plant Biotechnology	12
AGRO432 Applied Plant Breeding	12
AGRO412 Advanced Research Methods in Natural Sciences	12
HORT435 Post Harvest Physiology and Technology	12

Core modules (obligatory)

14. MODULE SYNOPSES

AGRO135 Plant Biology

The plant cell and organelles; Plant water relations; Photosynthesis – the ecological significance of photosynthesis in primary production of organic molecules, the Light Reaction, The Calvin Cycle, C3/C4 photosynthesis, CAM photosynthesis. Photoperiodism and vernalization. Fruit and seed formation: double fertilization and formation of seed tissues, Seed structure, chemical composition of seed, apomixis. Germination: What is germination? Phases in germination: seed hydration, genetic and metabolic events during germination, role of plant growth regulators in germination. Seed dormancy**:** Mechanisms of dormancy: Innate, induced and enforced dormancy, Significance of dormancy in attuning cycles of plant growth to seasonal changes in environment. Plant growth and development: meristems, cell expansion, cell differentiation, acid growth theory, tropisms, cell differentiation, apical dominance, bud dormancy; Plant growth regulators – synthesis and transport and physiological effects of auxins, gibberellins, cytokines, abscisic acid and ethylene on plant growth and development; Use of artificial plant growth regulators in Agriculture.

HORT131 Principles of Microbiology

Characteristics of eukaryotic and prokaryotic cells and a range of micro-organisms viz. Bacteria, Fungi, Viruses, Viroid and Protozoa; their classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens),

use of micro-organisms in industrial microbiology (cheese, beer and wine making) and in silage making; Immunology; techniques of isolating and identifying micro-organisms; Microscopy.

AGRO136 Introduction to Statistics
Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean. Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation, and summarization techniques; tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HORT132 Introduction to Entomology and Parasitology
Position of insects in the world: reproductive capacity, beneficial and injurious insects. Insect external anatomy: the exoskeleton, ecydysis. Internal anatomy and physiology: The nervous system-sight, sound and vibration detection, chemical communication (pheromones), the digestive system, locomotion, insect reproduction, metamorphosis; Insect classification, life cycles, population ecology, population monitoring and economic decision making; Mites, ticks and spiders; Nematodes-morphology, classification, reproduction and economic importance.

HORT133 Introduction to Horticulture
Definition, characteristics and importance of horticulture; branches of horticulture (pomology, olericulture, landscape horticulture, viticulture, ornamental horticulture); methods of propagating horticultural crops, Environmental factors affecting horticultural production at different phonological stages; Types and uses of controlled environments in horticultural production (manipulation of the plant environment); Pruning and training of horticultural crops; Harvesting, post-harvest handling and storage.

AGRO132 Principles of Genetics
Cell division: mitosis and meiosis; nucleic acids, replication and protein synthesis. Mendelian principles: segregation and independent assortment. Environmental effects and gene expression. Gene interaction and lethality; sex determination and sex linkage. Gene mutation and induced genetic change (biotechnology). Gene structure and regulation. Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources.

AGRO231 Crop Ecology and Physiology
Measurement of Crop Growth – Growth analysis, parameters of size, efficiency and duration of the photosynthetic system, advantages and limitations of growth analysis; Gas analysis, Infra-red gas analyser, calibration for use in the absolute and differential modes, different ways of setting up the INRGA for measuring CO_2 exchange, advantages and limitations of gas analysis, canopy analysers; The radiant environment and the crop; total radiation, the light spectrum, photosynthetically active radiation (PAR), factors that affect radiation receipts on the earth's surface, radiation interception by plant leaves and canopies, extinction coefficient (k) of canopies; Efficiency, size and duration of the photosynthetic

system and factors that affect them. Management of factors that affect the efficiency, size and duration of the photosynthetic system (choice of crops and varieties, planting date, planting populations and patterns, fertilizer/manure application, irrigation and pest and disease control) to maximize economic yield. Growth stages of important crops in Zimbabwe (maize, wheat, soybean and cotton) and effect of stress on growth and dry matter partitioning to economic yield. Plant adaptation to water and temperature stress.

AGRO239 Statistical Methods and Experimental Design
Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two-factor experiments and multi-factor experiments; Multiple comparisons of means; Partitioning of the sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for the goodness of fit and independence; Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS) for all designs: Regression and Correlation.

AGRO234 Weed Biology and Ecology
Definition, characteristics and adaptation; Classification and identification of common weeds; Reproduction and dispersal mechanisms; Weed-crop interactions;
Evolution of weeds: the r and k strategies, ruderals and agrestals. Adaptations of weeds as pests of disturbed areas: relative growth rate and leaf area partitioning ratio, seed production, dormancy, periodicity of weed germination, formation of seed banks. Weed interference: weed crop competition, types of competition, methods of studying weed crop competition, weedy and weed-free periods to avert yield loss; Allelopathy and methods to study allelopathy. Vegetative propagation: mechanisms, advantages and disadvantages. Weed dispersal. Biology of problem weeds in Zimbabwe (terrestrial and aquatic); Parasitic weeds of the tropics; Poisonous plants of Zimbabwe.

HORT231 Plant Pathogen Biology and Ecology
Structure, physiology, reproduction and classification of bacteria, fungi, viruses and other pathogens; Concepts of Disease; Disease epidemiology, Stages in the Development of Disease (Disease Cycle); Mechanisms of infection by pathogens; Plant defence mechanisms against pathogens; Fungi as Plant Pathogens; Bacteria and other procaryotes as Plant Pathogens; Viruses as Plant Pathogens; Parasitic Higher Plants as Plant Pathogens; Physiological or Abiotic Diseases; Plant

HORT235 Insect Pest Biology and Ecology
Biology and ecology of major crop pests of important crops (food, storage cash and horticultural crops). Life cycles, damage caused by insect pests to crops and their products insect pest population dynamics, r and k strategies, insect pest voltines, agro ecosystems, trivial and transboundary movement.

AGRO233 Principles of Crop Production
Definition and origin, domestication and classification of cultivated plants; factors that affect crop productivity; crop water requirements & efficiency, water-logging and effects on crop growth. Zimbabwe's agro=ecological zones, rainfall type, pattern, length and quality, crops or crop varieties selection. Land capability and suitability. Crop establishment, methods of planting and planting time. Plant competition Cropping systems and patterns; Biological nitrogen fixation, crop yield function; quality of crop products; Tillage methods; Principles of Irrigation, types of irrigations, drainage, scheduling; importance of plant breeding in crop production; Grain Storage; Agricultural Pollution.

AGPR330, AGPR331, AGPR332 & AGPR333: Work-Related Learning

Work-Related Learning (WRL) for at least 8 months of the third year is a prerequisite for registration for fourth-year modules. The purpose of WRL is to make students relate theory to practice in real work situations, facilitate students to acquire a world of work experience and skills, facilitate career choices by students and help students to identify entrepreneurial and business opportunities, among others. WRL is assessed as two components viz. continuous assessment and WRL report. The continuous assessment component consists of at least two assessment reports by academic staff that visit the students during WRL plus a workplace supervisor's report. The Work-Related Preliminary (WRP) and WRL reports are comprehensive and critical analyses of the activities, processes and outputs of the organization to which the student was attached and the drawing of lessons learnt from the WRL experiences. The student is required to identify a specific problem in systems, processes and/or in the organization to which they are seconded, and analyse the problem and solutions. Credit will be given to reports that show critical thinking and comprehensive analysis of the problem identified during WRL.

AGHORT480 Research Project

The rationale behind offering this module is to prepare students in areas of quality proposal and dissertation scientific writing, conducting of research, data collection, analysis, interpretation, verbal results presentation and publication. The primary objective of this module is to allow students to engage in independent study, with limited supervision and guidance. The dissertation shall focus on one of the following areas weed, insect pest, pathology and nematology issues,

AGRO447 Molecular Biology and Plant Biotechnology

This module presents an overview of the techniques and underlying theory of molecular biology, plant tissue culture and genetic engineering, understanding of traditional and molecular assisted plant breeding; molecular markers, genomics, gene transfer systems, including case studies and regulatory and ethical issues; cell & molecular understandings of plant processes important in horticulture, agriculture and forestry research and commercial applications, and issues/challenges in the area of plant biotechnology. It introduces students to the principles, practices and application of plant tissue culture and transformation in science, agriculture and industry. It gives students hands-on experience and training in plant tissue culture and genetic engineering techniques. It exposes students to issues and challenges encountered in the area of molecular biology and plant biotechnology. It acquaints students with experimental design and analysis of plant biotechnology experiments.

AGHORT435 Plant Nematode Biology, Ecology and Management

Important plant parasitic nematodes of key agronomic and horticultural crops: their biology and ecology, surveillance, quarantine systems and management. Plant parasitic classification, life cycles, population dynamics, monitoring and economic decision making, risk assessments. Management- cultural, physical, biological, chemical, use of botanicals, pheromones, growth regulators plant resistance and genetic modifications of plants. All essential aspects of Integrated Nematode Management (INM) will be covered.

AGRO450 Annual Food Crop Production

The module covers classification, adaptation and agronomy of selected annual field crops grown for food. Classification, importance, growth and development of selected groups of annual food crops. Regional and national potential; limitations to production; Agronomy: cultivar selection, land preparation, planting, weed, pest and disease management, irrigation, harvesting and post-harvest handling of cereal grain, legume grain; plantain; root and tuber crops.

AGRO451 Industrial and Cash Crop Production
The module highlights the economic importance, agronomy and marketing of cash and industrial crops; Regional potential and limitations to production; Economic importance; Agronomy of cash and industrial crops; cultivar selection, soil requirements, tillage, planting, fertilizer application, weeding, crop protection, irrigation, harvesting, post-harvest processing and marketing cash and industrial beverage, sugar, fibre, oilseeds, drug, nut, silvicultural and citrus.

AGRO436 Weed Management
Weed management tactics: manual –hoe weeding, slashing, biological, integrated, cultural, physical, mechanical, chemical/herbicide and breeding weed management, legislative (preventative) weed management tactics, weed control, case studies of aquatic weed control. Chemical weed management: history of herbicides, classification of herbicides according to chemical structure, penetration and movement of herbicides in plants, metabolism of herbicides and herbicide selectivity, soil applied herbicides, foliar applied herbicides, herbicide application and how to develop effective weed management programs.

AGHORT437 Plant Pathogen Management
Disease management options - cultural, chemical, preventative, biological, breeding/host plant tolerance, regulatory, surveillance, quarantine, population monitoring, botanicals, pheromones, insect growth regulators plant resistance and genetic modifications, regulatory and integrated disease management. Spray application technology; Integrated Disease Management (IDM).

AGHORT436 Insect Pest Management
Management of key insect pests of agronomic pests and horticultural crops, management methods such as cultural, chemical, preventative, physical, IPM, biological, regulatory, surveillance, quarantine, population monitoring, botanicals, pheromones, insect growth regulators plant resistance and genetic modifications. Economic decision making. Pest surveillance, monitoring and pest risk assessments. All essential aspects of integrated insect pest management (IPM) will be covered.

HORT446 Pesticide Science
Introduction to Pesticide Science. Principles and concepts of pest and disease management d uring cultivation of crop plants.
Chemicals for crop protection, a historical review. Chemistry and classification of the active ingredients of pesticides. Physicochemical and
biological properties, biochemistry and toxicology of the main groups of pesticides, includin g insecticides, herbicides, fungicides, bactericides,
nematicides and other plant protection products. Plant uptake, translocation and metabolism of pesticides. Biochemical modes of action and mechanisms of selective toxicity.
Economics of agrochemical use. Agrochemical and the environment; movement in plants; pesticide resistance causes, mechanisms and management; implications for management strategies; regulations and registration process, application technologies and equipment: nozzle types, calibration of sprayers. Safe and effective use of agrochemicals.

AGED131 Introduction to Agricultural Economics
Refer to the Department of Agricultural Economics and Development.

LWR135 Principles of Soil Science
Refer to Land and Water Resources Management Department.

HAWS132 Principles of Biochemistry

Refer to the Animal and Wildlife Sciences Department

ENT131 Entrepreneurship 1
Refer to Entrepreneurship Department Regulations.

AGED439 Agricultural Extension
Refer to Agricultural Economics and Development Regulations.

AEGS231 Gender Studies
Refer to Agricultural Economics and Development Regulations.
CS131 Basic Communication Skills
Refer to Communication Skills Regulations.
HCS135 Introduction to Information Technology
Refer to Computer Science Regulations.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN AGRONOMY MAJORING IN CROP IMPROVEMENT AND SEED PRODUCTION (HACPR)

Duration	:	4years
Actual Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	540
Total MBKS Credit Load	:	384
ZNQF Level	:	8

1. PURPOSE OF PROGRAM

1.1. The programme aims at contributing towards sustainable food and cash crop production, economic growth and overall food security

1.2. The Department will accomplish its mission by:

e. Developing curricula that provide appropriate theoretical knowledge and practical skills in field crop production.
f. Production of competitive and innovative graduates for various career opportunities in private and public sectors as well as self-employment.
g. Departmental participation in agricultural development through research, innovation, industrialization and outreach programmes.
h. Collaborating with relevant stakeholders.

2. ENTRY REQUIREMENTS

NORMAL ENTRY
Prospective students should have:

- At least two ‘A’ Level passes in Biology and any other science subject.
- At least five ‘O’ Level passes including English Language, Science and Mathematics.

SPECIAL ENTRY
Relevant Diploma based on departmental board evaluation.

3. PROGRAMME CHARACTERISTICS

Areas of study: Sustainable crop production, crop improvement, using modern techniques; seed biology, health, production and distribution, seed systems.

Specialist Focus: Crop improvement for adaptation to biotic and abiotic stress, crop improvement for nutritional quality, seed production and distribution systems.
Orientation: Research, innovation and application of theory to solve practical crop improvement as well as seed production and distribution related problems.
Distinctive Features: Knowledge, innovation and skills development.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability • Private Sector – Agronomists, Plant Breeders, Crop Protection officers, Technical and Sales Representatives, Farm Managers, Consultants.
• Public Sector – Researchers (Agronomists, Plant Breeders, Crop Protectionists) Extensionists, Lecturers.
• Self-Employment – Farmers, Consultants.
Further Studies Master's Degree level in the various fields of Agronomy (crop production, crop protection, biotechnology and breeding, crop processing etc).
They can also proceed to do PhD and DPhil studies in these areas.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, work-related learning reports, final year dissertations, continuous assessment (30-50%), and Final examination (50-70%).

6. LEARNING OUTCOMES

On completion of this programme a graduate will be able to:

- Apply key scientific principles and theories underpinning Agronomy.
- Apply practical skills in sustainable crop production, scientific research and knowledge transfer.
- Develop credible solutions using contemporary technologies in climate-smart agriculture, molecular biology and biotechnology.
- Work cooperatively in groups, displaying tolerance to diversity.
- Research, innovate and provide solutions in Agronomy with a bias on crop improvement aspects
- Create enterprises in the crop production processing sector.

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.2. Students can include in their programme of study, modules offered by other departments.
7.3. A student shall be enrolled for at least four modules at any time during the semester.
7.4. The module combination should be feasible in terms of the timetable.
7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAMME ASSESSMENT

Coursework: Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30-50% of the final mark.

Written Examinations: A three-hour final examination, which shall contribute to 50-70% of the final mark.

Practical Examination: The duration of practical exams, where they are deemed. necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 10 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, students are expected to acquaint themselves with the introductory modules of the degree programme. The choice of modules should be done with the guidance of Departmental staff.

13.2. The degree programme shall have at least 36 taught modules spread over the 6 teaching semesters.

MODULE	CREDITS
Level I	
*AGRO135 Plant Biology	12
*HAWS132 Principles of Biochemistry	12
*LWR135 Principles of Soil Science	12
*AGRO136 Introduction to Statistics	12
*AGRO132 Principles of Genetics	12
*HORT131 Principles of Microbiology	12
*HORT132 Introduction to Entomology and Parasitology	12
*AGED131 Introduction to Agricultural Economics	12
*CS131 Basic Communication Skills	12
*HCS135 Introduction to Information Technology	12
*AGHORT133 Seed Biology	12
*HORT133 Introduction to Horticulture	12
*AGPR131 Agricultural Practice 1	6
*AGPR132 Agricultural Practice 2	6
LWR137 Soil Fertility and Plant Mineral Nutrition	12
Level II	
*AGRO233 Principles of Crop Production	12
*AGRO239 Statistical Methods and Experimental Design	12
*AGRO231 Crop Ecology and Physiology	12
*AEGS231 Gender Studies	12
*AGHORT239 Seed Production, Handling and Processing	12

*AGRO234 Weed Science and Management	12
*AGED240 Farm Business Management	12
*ENT131 Entrepreneurship 1	12
*AGHORT240 Varietal Evaluation, Release and Seed Certification	12
*AGRO246 Research Methods and Scientific Writing	12
*AGRO232 Principles of Plant Breeding	12
HORT232 Olericulture	12
AGHORT238 Fundamentals in Seed Health and Quarantine	12
HORT231 Biology and Management of Plant Pathogens	12
AGHORT241 Seed Testing and Storage	12
HORT235 Biology and Management of Plant-Insect Pests	12

Level III

*AGPR330 Work-Related Learning Preliminary Report	30
*AGPR331 Work-Related Learning Report	45
*AGPR332 Employer's Assessment Report	30
*AGPR333 Academic Supervisor's Report	45

Level IV

*AGHORT480 Dissertation	24
*AGRO450 Annual Food Crop Production	12
*AGRO451 Industrial and Cash Crop Production	12
*AGRO447 Molecular Biology and Plant Biotechnology	12
*AGHORT438 Plant Genetic Resources Conservation	12
*HAWS238 Natural Resources Management	12
*AGRO446 Global Environmental Issues and Climate Change	12
*AGRO422 Plant Breeding Methods	12
*AGRO424 Applied Molecular Biology and Plant Biotechnology	12
*AGRO432 Applied Plant Breeding	12
AGED439 Agricultural Extension	12
AGRO433 Basic Agricultural Meteorology	12
AGHORT439 Seed Technologies	12
AGHORT440 Quantitative Genetics	12

Core modules (obligatory)

14.0 Module Synopses

AGRO135 Plant Biology

The module covers the following; cell biology; plant growth and development; seed, dormancy and germination, plant growth regulators, Plant water relations; photoperiodism and vernalization; Photosynthesis – the ecological significance of photosynthesis in primary production of organic molecules, the Light Reaction, The Calvin Cycle, C3/C4 photosynthesis, CAM photosynthesis.

HAWS132 Principles of Biochemistry

The module will provide students with fundamental knowledge of biochemistry and specific knowledge of compounds and biochemical pathways that occur in plants. The student will be able to describe the structure and chemical and physical properties of amino acids, sugars, proteins, carbohydrates, and lipids. Explain the relationship between structure and function of macromolecules including proteins, enzymes, carbohydrates, and lipids. Perform calculations related to enzyme kinetics, protein-ligand binding, energetics of solute transport, and oxidation-reduction reactions. Outline pathways of intermediary metabolism, C3 and C4 photosynthesis, photorespiration, photophosphorylation, cell wall biosynthesis, nitrate and sulfate assimilation, and plant secondary metabolism. Explain Michaelis-Menten enzyme

kinetics. Explain the principles of liquid chromatography and electrophoresis and apply various techniques to the purification and characterization of proteins. Explain how light energy is captured and converted to chemical forms of energy. Explain the chemi-osmotic theory including its structural and thermodynamic requirements. Design experiments using techniques of genetics and biochemistry to address questions about the structure and function of biomolecules, metabolic pathways, and cellular processes.

HORT131 Principles of Microbiology
Characteristics of eukaryotic and prokaryotic cells and a range of micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; their classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens), use of micro-organisms in industrial microbiology (cheese, beer and wine making) and in silage making; Immunology; techniques of isolating and identifying micro-organisms; Microscopy.

AGRO136 Introduction to Statistics
Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean. Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HCS135 Introduction to Information Technology
This module is aimed at assisting students to achieve full academic potential through equipping them with the necessary communication and computer skills essential for their degree program.

AGPR131 Agricultural Practice 1
The module provides hands-on learning to students, to compliment theoretical modules on agricultural production. Farm (University farm) orientation; Poultry, Dairy & Pig production, Cattle identification and detention, Soil sampling, Crops planting and fertilizing, Plant population and yield estimation, Introduction to vegetable production, Irrigation scheduling, Blood samples and smears, administering veterinary drugs, surveying in agriculture, Farm week practicals (all semesters).

AGPR132 Agricultural Practice II
This module is part II of the Agricultural practice module. Pig production and Dairying of University cattle, Control of internal parasites in livestock; Use & calibration of spraying equipment; Ox harness & ox-drawn ploughing; Tractor handling & ploughing/ contour construction; Soil coding; Vegetation coding in rangeland management; Farm budgets and Agric financing; Castration and dehorning, Game farming - farm visit; Field visits (research station/Large scale/small scale farming areas)

HORT132 Introduction to Entomology and Parasitology
Position of insects in the world: reproductive capacity, beneficial and injurious insects. Insect external anatomy: the exoskeleton, ecydysis. Internal anatomy and physiology: The nervous system-sight, sound and vibration detection, chemical communication (pheromones), the digestive system, locomotion, insect reproduction, metamorphosis;

Insect classification, life cycles, population ecology, population monitoring and economic decision making; Mites, ticks and spiders; Nematodes-morphology, classification, reproduction and economic importance.

HORT133 Introduction to Horticulture
Definition, characteristics and importance of horticulture; branches of horticulture (pomology, olericulture, landscape horticulture, viticulture, ornamental horticulture); methods of propagating horticultural crops, Environmental factors affecting horticultural production at different phonological stages; Types and uses of controlled environments in horticultural production (manipulation of the plant environment); Pruning and training of horticultural crops; Harvesting, post-harvest handling and storage.

AGRO132 Principles of Genetics
Cell division: mitosis and meiosis; nucleic acids, replication and protein synthesis. Mendelian principles: segregation and independent assortment. Environmental effects and gene expression. Gene interaction and lethality; sex determination and sex linkage. Gene mutation and induced genetic change (biotechnology). Gene structure and regulation. Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources.

LWR135 Principles of Soil Science
Soil formation: Parent material characteristics; Influence of factors of soil formation. Physical properties of soils: soil separates and soil textural classes, inter-particle forces, flocculation and dispersion, and soil structure. Mineralogy of soils: particle mineralogy and its effects on physical and chemical properties (1:1, 2:1 and 2:2 phyllosilicates, Fe oxides), crystalline and amorphous clay colloids, origin and magnitude of permanent and pH-dependent charge, soil water relations; soil air and temperature; soil organic matter. Soil chemical environment: ion adsorption and exchange, ion exchange capacities, base cations, exchangeable aluminium, hydroxy aluminium. Soil microbiology and biochemistry: Plant mineral nutrition; cycles; movement, and uptake by plants; soil pH; Soil classification, land capability classification; soil management; management of saline and sodic soil.

AGED131 Introduction to Agricultural Economics
Economic systems and organisation, theory of supply and demand, market equilibrium and nature of markets. Micro – economics. Macro – economics. The consumption, investment and savings functions. The international economy. Monetary and fiscal policies. Employment and growth policies. The role of the agricultural industry in the national economy.

AGHORT133 Seed Biology
Flowering in angiosperms and gymnosperms; Plant meristems; Floral induction and initiation in Plants; Pollination and Fertilisation; Apomixes; Seed formation and development; somatic embryogenesis, artificial seed; Seed maturation, chemical composition of seeds, seed structure, seed germination and the physiological factors affecting germination; seed dormancy. Seed Agronomy.

AGRO231 Crop Ecology and Physiology
Measurement of Crop Growth – Growth analysis, parameters of size, efficiency and duration of the photosynthetic system, advantages and limitations of growth analysis; Gas analysis, Infra-red gas analyser, calibration for use in the absolute and differential

modes, different ways of setting up the INRGA for measuring CO_2 exchange, advantages and limitations of gas analysis, canopy analysers; The radiant environment and the crop; total radiation, the light spectrum, photosynthetically active radiation (PAR), factors that affect radiation receipts on the earth's surface, radiation interception by plant leaves and canopies, extinction coefficient (k) of canopies; Efficiency, size and duration of the photosynthetic system and factors that affect them. Management of factors that affect the efficiency, size and duration of the photosynthetic system (choice of crops and varieties, planting date, planting populations and patterns, fertilizer/manure application, irrigation and pest and disease control) to maximize economic yield. Growth stages of important crops in Zimbabwe (maize, wheat, soybean and cotton) and effect of stress on growth and dry matter partitioning to economic yield. Plant adaptation to water and temperature stress.

AGRO239 Statistical Methods and Experimental Design
Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two factor experiments and multi-factor experiments; Multiple comparison of means; Partitioning of sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for goodness of fit and independence; Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS) for all designs: Regression and Correlation.

HORT231 Biology and Management of Plant Pathogens
This module covers issues to deal with the structure, physiology, reproduction and classification of bacteria and other procaryotes, fungi, viruses, parasitic higher plants and their role as plant pathogens; Concepts of Disease epidemiology, Disease Cycle; Mechanisms of infection; Plant defence mechanisms against pathogens; Physiological or Abiotic Diseases to plants; Plant Disease management - cultural, chemical, preventative within the concept of integrated disease management. Lifecycles and management of key crop diseases

AGRO234 Weed Science and Management
The module covers the evolution of weeds, adaptations of weeds as pests of disturbed areas, seed production, dormancy, germination, seed bank, weed interference, allelopathy and methods to study allelopathy, vegetative propagation mechanisms, and weed dispersal. Weed management tactics: manual, cultural, legislative, biological weed control, chemical. classification of herbicides, herbicide selectivity, soil-applied herbicides, foliar-applied herbicides, herbicide application.

HORT235 Biology and Management of Plant-Insect Pests
This module covers important pests of crops, biology and ecology, life cycles, population ecology, population monitoring and economic decision making, monitoring and pest risk assessments, pest management strategies- cultural, physical, biological, IPM, surveillance, quarantine, botanicals, pheromones, insect growth regulators, plant resistance and genetic modifications of plants or insects.

LWR137 Soil Fertility and Mineral Nutrition
Factors controlling growth of plants; Elements involved in plant nutrition and their sources; Soil organic matter and plant nutrient availability; Factors controlling various nutrient availabilities in soils; Soil plant relations affecting nutrient up-take and growth; Nutrient use efficiency; Types of fertilizers (mineral versus organic) and their use;

Compost manure making; Biological Nitrogen Fixation; Agro-forestry; Soil fertility management in crop production; Conditions and processes of plant nutrient uptake, distribution and assimilation; Nutrient requirements of various crops; Design of fertilizer trials; Survey of fertilizer usage; Research findings on important soils and crops

AGRO233 Principles of Crop Production

Definition and origin, domestication and classification of cultivated plants; factors that affect crop productivity; crop water requirements & efficiency, water-logging and effects on crop growth. Zimbabwe's agro=ecological zones, rainfall type, pattern, length and quality, crops or crop varieties selection. Land capability and suitability. Crop establishment, methods of planting and planting time. Plant competition Cropping systems and patterns; Biological nitrogen fixation, crop yield function; quality of crop products; Tillage methods; Principles of Irrigation, types of irrigations, drainage, scheduling; importance of plant breeding in crop production; Grain Storage; Agricultural Pollution.

AGHORT240 Varietal Evaluation, Release and Seed Certification

Importance of varietal evaluation; Relationship of varietal evaluation with selection. Analysis of multi-location variety data; the role of genotype x environment interaction in selection. Varietal release regulations and processes.

Brief history, Objectives. Certification authority. Manpower requirements. Elements of Sound Seed Certification programme. Minimum certification standards. Field inspection. Pre and Post-harvest control (varietal purity, seed borne disease). Seed quality tests in the laboratory. Summary of certification scheme. Certification of blends, phytosanitary certification.

AGHORT239 Seed Production, Processing and Handling

The seed industry and its developmental stages. Seed production systems. Essential requisites of a seed Programme. Agronomic and climatic aspects of Seed reproduction. Determinants of seeds quality. Seed multiplication. Production of seed for the farmer, contract seed production. Harvesting and drying. Seed processing certification. Seed legislation, seed marketing, Field Inspection, test for cultivar authenticity. Transportation, Seed processing Principles, Pre-cleaning and conditioning. Basic cleaning Dimensional sizing. Specific gravity separation. Surface texture separation. Air separators. Electronic separators. Miscellaneous cleaning equipment. Commercial seed treatments. Seed handling. Accessories. Design and layout of the processing plant.

AGHORT241 Seed Testing and Storage

Types of seeds. Seed sampling, physical purity, moisture, germination and seed vigour analysis. Seed moisture relationships and storage. Life span of seeds. Seed deterioration. Possible causes of seed deterioration. Seed storage, purposes and precepts. Storage pests; storage pathogens. Types of storage. Conditioned storage. Seed packaging.

AGHORT238 Fundamentals of Seed Health and Quarantine

Concept of seed-borne mycoflora. Seed pests. Plant diseases caused by seed-borne pathogens. Seed diseases of major tropical crops. Transmission of seed disease agents. Mode of infection by seed-borne inoculum. Micro-climate and seed disease development. General morphology and taxonomy of seed-borne fungi, bacteria, nematode and viruses. Principles and methods of seed health testing. Modern

techniques for detection of seed-borne inoculum. Ecological aspects of seed health testing. Survey of seed diseases, seed-borne diseases and causal agents with reference to taxonomic distinctions. Seed health, mycotoxins and human health. Control of field and storage pests and diseases of seeds. Integrated pest management approach to seed disease control. Seed health and quarantine programmes.

AGPR330, AGPR331, AGPR332 & AGPR333: Work-Related Learning

Work-Related Learning (WRL) for at least 8 months of the third year is a prerequisite for registration for fourth-year modules. The purpose of WRL is to make students relate theory to practice in real work situations, facilitate students to acquire world of work experience and skills, facilitate career choices by students and help students to identify entrepreneurial and business opportunities, among others. WRL is assessed as two components viz. continuous assessment and WRL report. The continuous assessment component consists of at least two assessment reports by academic staff that visit the students during WRL plus a workplace supervisor's report. The Work-Related Preliminary (WRP) and WRL reports are comprehensive and critical analysis of the activities, processes and outputs of the organization to which the student was attached and the drawing of lessons learnt from the WRL experiences. The student is required to identify a specific problem in systems, processes and/or in the organization to which they are seconded, analyse the problem and solutions. Credit will be given to reports that show critical thinking and comprehensive analysis of the problem identified during WRL.

AGHORT480 Dissertation

The rationale behind offering this module is to prepare students in areas of quality proposal and dissertation scientific writing, conducting of research, data collection, analysis, interpretation, verbal results presentation and publication. The primary objective of this module is to give students an opportunity to engage in an independent study, with limited supervision and guidance. The dissertation shall focus on one of the following areas weed, insect pest, pathology and nematology issues,

AGRO447 Molecular Biology and Plant Biotechnology

This module presents an overview of the techniques and underlying theory of molecular biology, plant tissue culture and genetic engineering, understanding of traditional and molecular assisted plant breeding; molecular markers, genomics, gene transfer systems, including case studies and regulatory and ethical issues; cell & molecular understandings of plant processes important in horticulture, agriculture and forestryresearch and commercial applications, and issues/challenges in the area of plant biotechnology. It introduces students to the principles, practices and application of plant tissue culture and transformation in science, agriculture and industry. It gives students hands-on experience and training in plant tissue culture and genetic engineering techniques. It exposes students to issues and challenges encountered in the area of molecular biology and plant biotechnology. It acquaints students with experimental design and analysis of plant biotechnology experiments.

AGRO450 Annual Food Crop Production

The module covers classification, adaptation and agronomy of selected annual field crops grown for food. Classification, importance, growth and development of selected groups of annual food crops. Regional and national potential; limitations to production; Agronomy: cultivar selection, land preparation, planting, weed, pest and disease management, irrigation, harvesting and post-harvest handling of cereal grain, legume grain; plantain; root and tuber crops.

AGRO451 Industrial and Cash Crop Production

The module highlights the economic importance, agronomy and marketing of cash and industrial crops; Regional potential and limitations to production; Economic importance; Agronomy of cash and industrial crops; cultivar selection, soil requirements, tillage, planting, fertilizer application, weeding, crop protection, irrigation, harvesting, post-harvest processing and marketing cash and industrial beverage, sugar, fibre, oilseeds, drug, nut, silvicultural and citrus.

AGRO433 Basic Agricultural Meteorology

Scope of agricultural meteorology; Aspects of physical climatology important to Agriculture; Heat balance of the soil- soil temperature; Water and the hydrological cycle in Agriculture; Microclimate (influencing factors and effects on agricultural production); Meteorological instrumentation (familiarisation with the operation and interpretation of data from various meteorological equipment including rain gauge, maximum and minimum thermometer, wet and dry bulb thermometer, Class A evaporation pan and cup-anemometer). Weather hazards affecting crop production; Applied / operational agro meteorology (case studies).

AGED439 Agricultural Extension

Overview of agricultural extension: what is extension? Historical evolution of extension and modern extension services; Context of agricultural extension: operational environment, institutional context; Agricultural knowledge system components: technology generation, technology transfer, technology utilization and policy; Case studies of agricultural knowledge systems; Approaches to extension: General clientele approaches- Ministry Based Extension, Training and Visit, Integrated Rural Development (IRD) approaches, Farmer Field School Approach, Animation Rurale, Commodity Based Extension, Extension as a Commercial Service, Client Based and Client Controlled Extension; Adoption theory (Lionberger, 1968): awareness, interest, evaluation, trial, adoption; Developing and delivering extension programs: extension programme planning, centralized (blue print) planning, participatory planning, combination of centralized and participatory planning: Needs assessment; Participatory Rural Appraisal; Monitoring and evaluation of extension programs; Strengthening farmer organizations; Management and structure of extension organizations. Use of IT systems in agricultural extension.

AGHORT439 Seed Technologies

Seed quality enhancement and improvement through controlled priming, pelleting, coating, methods of breaking various forms of dormancy and other novel technologies. Seed mixing, grading and seed dressing. Methods of improving longevity in artificial seed. Effects of various genetic i.e. seed covering layers, plant growth regulators, enzymes and environmental factors affecting germination i.e. light, temperature and water will be studied.

AGHORT438 Plant Genetic Resources Conservation

Reasons for plant genetic resources conservation; Assessment of Genetic Diversity in Plants, Biochemical, molecular, and cytogenetic methods of plant genetic resources characterization and evaluation; utilization and classification of plant genetic resources. Plant Genetic Resources Conservation Methods and Management Strategies in plant genetic resources conservation and genebank management. Plant Genetic Resources Documentation and Information Management. Management of passport, characterization, evaluation and inventory information in genebanks. SDDE

AGRO424 Applied Molecular Biology and Plant Biotechnology

The aim of this module is to expose the student to molecular markers used in plant breeding and crop protection, and application of these techniques in breeding programs and crop protection. RFLP (restriction fragment length polymorphism), PCR (polymerase chain reaction), RAPD (random amplified polymorphic DNA), AFLP (amplified fragment length polymorphism), SSR (single sequence repeats), SNPs (Single nucleotide polymorphism) proteomics. After completing this module, the student should be able to apply these techniques in plant breeding and crop protection.

AGRO432 Applied Plant Breeding

In this module the student acquires knowledge on applications in plant breeding. The student studies breeding for insect and disease resistance, temperature and moisture stress, quality and adaptation in a specific crop. On completion of the module, the student should be able to successfully initiate a breeding program for the crop studied.

AGRO422 Plant Breeding Methods

The module will cover the goals of plant breeding, and breeding methods for self-pollinated crops: pure line selection, mass selection, pedigree method, bulk method, backcross method, breeding methods for cross-pollinated crops: backcross method, mass selection and pedigree method. Population improvement methods: simple recurrent selection, recurrent selection for general combining ability, recurrent selection for specific combining ability, and reciprocal recurrent selection. Types of cultivars: pure lines, multi-lines, various types of hybrids, synthetics, composites, and open-pollinated varieties.

AGHORT440 Quantitative Genetics in Plant Breeding

Students will learn mating systems, mating designs, genotype X environmental interaction, Inbreeding depression and heterosis, genetic variance, heritability, biometrical techniques in plant breeding, assessment of variability, aids to selection; quantitative trait loci analysis in plant breeding, choice of parents and breeding procedures, varietal adaptation.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN CROP SCIENCE (MCS)

Duration	:	2.5 years
Actual Credit Load	:	384
Minimum Credit Load	:	360
Maximum Credit Load	:	450
Total MBKS Credit Load	:	288
Level		9

1. PURPOSE OF THE PROGRAMME

This programme aims to give graduates in various areas of Crop Science and related fields an opportunity to broaden their knowledge in Crop Science. Emphasis will be placed on the scientific aspects of Crop Production, Crop Improvement and Crop Protection that are relevant to agriculture.

On completion of this programme, graduates will be able to:

a. Develop an understanding of the disciplines related to crop production, protection or improvement; make them familiar with chemical, biological, and physical principles that relate to crop plants and their environment. Develop their skills to apply the principles related to crops, soils, and environment to solve crop production, improvement and protection problems.
   Develop their leadership, communication, and technical skills in crop sciences.
b. Manage agricultural enterprises and develop entrepreneurship skills to create employment and improve the economy of the country.

2. ENTRY REQUIREMENTS

To be eligible in the programme, a candidate must normally be a holder of at least a good first degree in Agronomy, Crop Science or Horticulture or related fields from a recognised University.

3. PROGRAMME CHARACTERISTICS

Areas of study: Crop Science
Specialist Focus: Crop Production; Crop Improvement
Orientation Oriented towards innovation and application of theory to solve real food and nutrition related challenges.
Distinctive Features: Knowledge and skills development in crop production and improvement.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:After completing the programme, graduates should be able to:

- Teach at universities and colleges.
- Work as agronomists, crop protectionists, plant breeders and horticulturalists.
- Work as researchers in public and private institutions.
- Work as consultants.
- Create their own business ventures.

- Work in NGOs,
- National Agricultural Research Systems (NARS) and International, Researchers, Consultants

Further Studies: Doctoral studies in Agronomy, Plant Breeding and other related programmes.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory and field practicals, group work, industrial and farm/field visits, dissertation

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, dissertation. Continuous assessment (30-50%), Final exam (50-70%)

6. INTENDED LEARNING OUTCOMES

- To equip graduates with specialized training so that they can better position their own businesses, or the organizations in which they work, for growth in a global marketspace.
- Enable students to conceptualize, formulate and implement solutions to organizational challenges using new and existing technologies.
- To develop new practices methods and technologies that enhances production efficiency.
- Create e-Policies that address ethical use of information and devices in the agribusiness environment.

7. GENERAL PROVISIONS

- Refer to Section 8 of General Regulations.
- The programme shall normally be studied over a minimum of four semesters on a block release basis.
- This programme is a taught Masters programme with modular work in the first two semesters and research and dissertation writing in the third, fourth and fifth semesters.
- Continuous assessment consists of lectures, presentations, seminars and tests.
- Lectures and taught modules shall normally be at the Midlands State University
- The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.
- Each student must identify at least two supervisors who shall be approved by the Department of Agronomy and Horticulture
- There shall be oral presentations for the Project Proposal and Research Findings. The weighting of these, together with the final written document, shall be as follows:
    - Dissertation Proposal 15%
    - Presentation of Research Findings 15%
    - Final Written Document 70%

Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

Candidates shall be expected to submit at least three executive-bound copies of the dissertation.

The format of the dissertation shall be provided for by the Department of Agronomy and Horticulture

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations, laboratory and or field practicals

8.2. Formal examinations shall be held at the end of each semester.

8.3. Continuous assessment shall account for 30-50% of the overall assessment, while the examination shall account for 50-70 % of the overall assessment.

8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 120 notional credit hours in the first level in order to proceed to the second level.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

The taught courses/modules (Level 1 Semester 1 & 2) will contribute 50 % and the dissertation (semester3, 4 & 5) will contribute 50%.

13. PROGRAMME STRUCTURE

The following structure would allow candidates to major in Crop Production or Crop Improvement.

Level 1 Semester 1
CROP PRODUCTION OPTION

Code	Module Description	Credits
MCS702	Advanced Crop Physiology	24
MCS703	Advanced Crop Protection	24
MCS708	Agroclimatology and Environmental Issues	24

Electives (Choose one)

MCS722	Integrated Agricultural Systems & Rural Development	24
MCS723	Crop Simulation Modelling	24

Level 1 Semester 2
Core Modules:

MCS701	Advanced Experimental Design and Analysis	24
MCS705	Advanced Soil Fertility and Plant Nutrition	24
MCS706	Advanced Soil Plant Water Relations	24

ELECTIVES: Choose any one from

MCS704	Entrepreneurship and Agribusiness Management	24
MCS707	Advanced Horticultural Technologies	24
MCS709	Advanced Seed Biology and Technology	24
MCS724	Practical Applications of Climate Smart Agriculture	24

CROP IMPROVEMENT OPTION
Core Modules:

Level 1

MCS702	Advanced Crop Physiology	24
MCS709	Advanced Seed Biology and Technology	24
MCS711	Quantitative Genetics in plant breeding	24
MCS701	Advanced Experimental Design and Analysis	24
MCS712	Population Genetics in plant breeding	24
MCS 714	Plant Biotechnology	24
MCS715	Plant Genetic Resources Conservation	24
MCS721	Plant Breeding for Stress & Nutritional Quality	24

Level 2 Semester 1 and 2; Semester 3 level 1

MCS 870	Dissertation	192

14. Module Synopsis

MCS 701 Advanced Experimental Designs and Analysis
Methods of research and experimentation in Crop Science and Production. Data Collection and Organisation. Experimental designs and data analysis in crop production. Regression models and analysis. Intensive practicals on use of statistical software for data processing. Principle component analysis. Survey Methodology and Scientific Communication.

MCS 702 Advanced Crop Physiology
Effects of environmental factors (temperature, photoperiod, rainfall, etc.) on crop growth and development. Adaptation of crop plants to stress factors. Analysis of growth and development. Maximisation of crop yield through optimisation of photosynthetic potential: C3, C4and CAM plants. Photosynthate partitioning in relation to yield, modification of yield potential by chemical and cultural means. Phytohormones and plant, growth regulators in crop production. Physiological role of mineral nutrition. Crop geometry and competition. Photosynthesis, phototropism and photo-morphogenesis; and their relationship with photo-synthetically active radiation (PAR). Artificial lighting and light quality for crop production.

MCS 703 Advanced Crop Protection
Cultural, physical, chemical, biological, host-plant resistance, genetic and biotechnological methods of pest, disease, nematode & weed control. Environmental aspects of chemical control of pests; Economics pest control, Pest scouting; Integrated Pest Management. Applications of molecular methods in crop protection.

MCS704 Entrepreneurship and Agribusiness Management
Role of entrepreneurship in agricultural development and the economy. Theories of entrepreneurship. Processes of establishing a business. Sources of funding. Feasibility studies. Planning production and marketing of enterprise. Entrepreneurship and public policy. Characteristics, nature and problems of small agricultural ventures. Advantages and disadvantages of big versus small enterprises management techniques. The agri-business finance function; financial statements. Financial analysis; financial control; cash budgeting; capital budgeting; credit management; financial planning and management; short–term financing; long term financing; capital markets. Mergers and acquisitions.

MCS705 Advanced Soil Fertility & Plant Nutrition
General concepts of soil fertility; Factors affecting soil fertility; soil reactions. Essential plant nutrients: forms, functions, deficiency symptoms, transformations; Techniques of evaluating soil fertility: deficiency symptoms, yield responses, plant tissue analyses; Approaches to managing and maintaining soil fertility: cropping systems, organic fertilizers, organic matter

decomposition and theories of humus formation, organic fertilizer sources, soil amendments, fertilizers-types, manufacturing, formulation, methods of application and management; Fertilizer recommendations for crops; Biological management of soil fertility and rhizosphere effect. Fertilizer pollution,

MCS706 Advanced Soil Plant Water Relations
Concepts of water potential and movement in soil – plant – atmosphere continuum; water and nutrient uptake by plant roots; water budgets; effects and measurement of water in soil and plants, research techniques in studies on soil plant water relations and their applications. Irrigation planning and design; management of irrigation water; control and management of salts in irrigation; agronomic and socio-economic factors in irrigation management; evaluating irrigation systems; use of industrial and urban effluent for irrigation; selected irrigation problems with special reference to Zimbabwe.

MCS707 Advanced Horticultural Technologies
Greenhouse and other types of structures for protected agriculture. Production under protected structures permanently or for crop season extension. Greenhouses designs and their operation: permanent or retractable structures. Plastic and net-houses. Hydroponics production. Plasticulture: low tunnels, walk-in tunnels and plastic mulches. Hydroponic media and hydroponic production systems. Nutrient film and other techniques. Plant culture in hydroponics. Fertigation techniques in protected agriculture: pH, EC, macronutrient and micronutrient optimization. Postharvest handling of fresh produce, and their processing. Pollinating agents in greenhouses. Transplant production in protective structures. Bio-control of pests in greenhouses.

MCS708 Agro-Climatology and Environmental Issues
Definition and concepts of agro-climatology. Economic significance and importance of weather; Introduction to agro-meteorology: greenhouse effect, energy budgets; Atmospheric moisture and precipitation; weather teleconnections, Global climatic change and variability and agriculture, vulnerability, adaptation and mitigation. International conventions on climate Weather patterns over Zimbabwe. Extreme weather events and management. Use of weather data for irrigation scheduling. Weather effects on pest and disease incidence. Crop weather modelling. Environmental impact assessment, agricultural pollution.

MCS709 Advanced Seed Biology and Technology
Principles of seed production, seed processing, quality control and storage. Seed treatment (protection) and seed physiological enhancement. Seed development, maturation, dormancy and germination. Seed anatomy and germination physiology. Seed composition. Environmental and endogenous factors affecting dormancy and germination. Primary and secondary dormancy. The harvesting processes, seed cleaning, conditioning, testing, seed certification and quality control. Seed vigour test methods. Viability test methods. Seed health testing and seed borne diseases. Factors affecting seed demand and demand forecasting, marketing and distribution. Formal and informal seed production systems, seed industry development, varietal registration and plant breeder's rights. Seed Regulations and the Zimbabwe Seed Certification Notice, Site selection and isolation standards, Inspections.

MCS710 Advanced Plant Breeding and Genetics
Goals of plant breeding. Plant breeding organizations in Zimbabwe. International breeding organisations Patterns of evolution in crop plants. Hybridization, mutation in crop improvement. Ideotype concept in crop improvement. Breeding for quality. Breeding for abiotic and biotic stresses. Varietal development, Varietal adaptation and model, application of Biotechnology to conventional breeding, Future of plant breeding

MCS711 Quantitative Genetics in Plant Breeding
Mating systems, mating designs, genotype X environmental interaction, Inbreeding depression and heterosis, genetic variance, heritability, biometrical techniques in plant breeding, assessment of variability, aids to selection; quantitative trait loci analysis in plant breeding, choice of parents and breeding procedures, varietal adaptation

MCS712 Population Genetics in Plant Breeding

Introduction to Population Genetics; Hardy-Weinberg equilibrium Genetic Variation Linkage Disequilibrium, Population Subdivision, Inbreeding, and Gene Flow The binomial coefficient and binomial equation. Application of binomial equation to HWE, LD, and drift. The Poisson distribution and the impact of variance in reproductive success and the effective population size. Gene Flow and Population differentiation, fitness, migration, founder effect, mutation, selection theory, breeding values, linkage relationships. Theory of selection, testers and combining ability

MCS714 Molecular Plant Breeding
Students will learn the application of the latest and emergent biotechnology options in crop improvement such as application of molecular markers in plant breeding, and principles of genetic engineering and modification, molecular tools for seed quality assessment such as genetic finger printing. It will also cover bioinformatics for crop improvement and production.

MCS715 Plant Genetic Resources Conservation
Centres of origins, domestication and plant introduction. Importance of germplasm conservation. Germplasm conservation organisations, role of national, regional and international bodies in germplasm conservation. Selection, collection and conservation of germplasm including in-situ, ex-situ and invitro conservation. Maintenance of germplasm.

MCS 720 Plant Breeding for stress & nutritional Quality
The module will have an in-depth concepts and procedures for breeding for abiotic stress, biotic stress and nutritional quality. The student should be able to Initiate a breeding programme; and - Formulate strategies for stress and quality breeding in selected crops.

MCS 721 Statistical Analysis for Plant Breeders
Students will learn methods of research and experimentation in plant breeding. Data Collection and Organisation. Experimental designs and data analysis in plant breeding. Regression models and analysis. Stability analysis models, augmented designs and incomplete designs. Principle component and cluster analysis. Intensive practicals on use of statistical software for managing plant breeding programme and data processing. Scientific Communication.

MCS722 Integrated Agricultural Systems and Rural Development
Students will learn concepts of various agricultural systems integrating cropping production systems i.e intensive, semi-intensive, semi-extensive, extensive, shifting cultivation, livestock production, agroforestry systems, organic farming, conservation farming and agroecological farming practices for sustainable agricultural development. Analysis of case studies on sustainable development initiatives at local, national, regional and international level will be done.

MCS723 Crop Simulation Modelling
Students will learn the practical method for integrating the effects of soil, weather, management and genetic factors for sustainable crop production, using crop simulation

models. They will analyse biological and environmental aspects of crop production using these models, evaluate the results/outputs of the models and use them for making management decisions. Selected growth and crop systems models (e.g. APSIM, DSSAT; FAO Aqua Crop) will be used. Laptops and data sets of soil, crop and weather are essential for execution of this module.

MCS724 Practical Applications of Climate Smart Agriculture (CSA)

Students will acquire practical knowledge and skills needed to inform interventions and innovations to address challenges of climate change and food insecurity. Topics covered include: Climate change and Agriculture (Contribution of Agriculture to GHG emissions; climate change adaptation and resilience; climate change and agrobiodiversity) and Definition and importance of CSA. CSA practices and technologies to mitigate, adapt and become resilient to climate change; Application of GIS in CSA. CSA applications apply to crop production, water management, soil and land management.

MCS 870 Dissertation

The dissertation provides an opportunity for the students to undertake and contribute to a piece of original research work, in an area related to crop science. The students are required to design an experiment (or investigation, survey or other means) to test a hypothesis or proposition, to plan and execute the research work, to evaluate the outcomes and draw valid conclusions. The research work is carried out individually, under guided supervision.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN CROP PROTECTION (MCPR)

Duration	:	2.5 years
Actual Credit Load	:	384
Minimum Credit Load	:	360
Maximum Credit Load	:	450
Total MBKS Credit Load	:	288
Level		9

1. PURPOSE OF THE PROGRAMME

This programme aims to give graduates in various areas of Crop Protection and related fields an opportunity to broaden their knowledge in Crop Protection. Emphasis will be placed on the scientific aspects of Biology, Ecology and Management of Plant Pathogen, Insects Pest, Weeds and Nematodes; Pesticide Science, Biotechnological Techniques; Climatology and Environmental Issues and Experimental Design and Analysis.

On completion of this programme, graduates will be able to:

a. Develop an understanding of the disciplines related to crop protection; make them familiar with chemical, biological, and physical principles that relate to crop plants and their environment. Develop their skills to apply the principles related to crops, soils, and environment to solve crop protection problems.
b. Develop their leadership, communication, and technical skills in crop protection sciences.
c. Manage agricultural enterprises and develop entrepreneurship skills to create employment and improve the economy of the country.

2. ENTRY REQUIREMENTS

To be eligible for the programme, a candidate must normally be a holder of a first degree with an upper second (2.1) class or better in Crop Protection, Agronomy, Crop Science, Horticulture or related fields from a recognized University.

3. PROGRAMME CHARACTERISTICS

Areas of study: Crop Protection - Plant Pathology; Weed Science and Management; Nematology; Insect Pest Sciences; Pesticide Science and Climate Change and Environmental Issues.
Specialist Focus: Sustainable Crop Protection Practices and Technologies.
Orientation: Research, Innovation and Application of Theory to Solve Practical Crop Protection-Related Problems.
Distinctive Features: Knowledge and skills development, Integrated Pest Management (IPM), Resistance Management, Plant Health Diagnosis, Precision Agriculture.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability

- **Private Sector** – Crop Protectionists, Plant Breeders, Agronomists Technical and Sales Representatives, Farm Managers, and Consultants.
- **Public Sector** – Researchers (Crop Protectionists, Plant Breeders, Agronomists), Public Policy Advisors, Lecturers.
- **Entrepreneurs** – Farmers, Consultants.

FURTHER STUDIES

PhD and DPhil degrees in Crop Protection, Biotechnology, Plant Breeding, Agronomy, Crop Processing etc.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, Tutorials, Laboratory Practical, Group Work, Industrial and Field Visits, Dissertation.

Assessment Methods: Written and Oral Examinations, Tests, Quiz, Group and Individual Presentations, Dissertation, Continuous Assessment (40%), Final Exam (60%).

6. INTENDED LEARNING OUTCOMES

- To equip graduates with specialized training so that they can better position their own businesses, or the organizations in which they work, for growth in a global marketspace.
- Enable students to conceptualize, formulate and implement solutions to organizational challenges using new and existing technologies.
- To develop new practices methods and technologies that enhances production efficiency.
- Create e-Policies that address ethical use of information and devices in the agribusiness environment.

7. GENERAL PROVISIONS

- Refer to Section 8 of General Regulations.
- The programme shall normally be studied over a minimum of four semesters on a block release basis.
- This programme is a taught Masters programme with modular work in the first two semesters and research and dissertation writing in the third and fourth semesters.
- Continuous assessment consists of lectures, presentations, seminars and tests.
- Lectures and taught modules shall normally be at the Midlands State University
- The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.
- Each student has to identify at least two supervisors who shall be approved by the Department of Agronomy and Horticulture
- There shall be oral presentations for the Project Proposal and Research Findings. Weighting of these, together with the final written document, shall be as follows:
    - Dissertation Proposal 15%
    - Presentation of Research Findings 15%
    - Final Written Document 70%

Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

Candidates shall be expected to submit at least three executive bound copies of the dissertation.

The format of the dissertation shall be provided for by the Department of Agronomy and Horticulture

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations, laboratory and or field practicals

8.2. Formal examinations shall be held at the end of each semester.

8.3. Continuous assessment shall account for 30-50 % of the overall assessment, while the examination shall account for 50-70% of the overall assessment.

8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 120 notional credit hours in the first level in order to proceed to the second level.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

For the purposes of degree weighting, the parts of the degree programme will be weighed as follows:

Level 1 (Semester 1 and 2)	50 %
Level 2 (Semester 1 and 2) & level 3 (Semester 1)	50 %

13. PROGRAMME STRUCTURE

The following structure would allow candidates to major in Crop Production, Crop Improvement or Crop Protection.

Module	**Credits**
Level I	
MCPR701 Advanced Experimental Design and Analysis	24
MCPR702 Advanced Weed Biology and Ecology	24
MCPR703 Advanced Bacteriology, Virology and Nematology	24
MCPR705 Advanced Mycology and Disease Epidemiology	24
MCPR706 Advanced Weed Management	24
MCPR707 Advanced Insect Pest Ecology and Biology	24
MCPR708 Advanced Pesticide Science and Insect Pest Management	24
Choose one	
MCPR709 – Agro- Climatology 7 Environmental Issues	24
MCPR710 – Entrepreneurship and Agribusiness Management	24
Level II	
MCPR 870 Dissertation	192

Module Synopses

MCPR701 Advanced Experimental Design and Analysis

Methods of research and experimentation in crop protection science. Data collection and organisation. Experimental designs and data analysis in crop production. Regression models and analysis. Intensive practicals on the use of statistical software for data processing. Principle component analysis. Survey methodology and scientific communication.

MCPR702 Advanced Weed Biology and Ecology

The weed problem, Weed identification and classification, Zimbabwe's problem weeds; Weed reproduction and dispersal; weed dormancy; genetics and evolution of weeds; patterns of evolutionary development; Weed demography and population dynamics; Associations of

weeds and crops; competition, response of crops and weeds to competition, competition thresholds; allelopathy. Biology of the problem of Zimbabwe's problem weeds, Biology of common poisonous weeds in the tropics, Biology and ecology of Aquatic weeds.

MCPR703 Advanced Bacteriology, Virology and Nematology
Bacteria: Structure, classification, characteristics, diversity and identification techniques; Symptomatology/Effects of plant pathogenic bacteria; Survival, reproduction, multiplication and dispersal. Viruses: Characteristics, classification, reproduction, transmission diversity and identification techniques; Symptoms. Viroids and mycoplasmalike organisms. Plant disease management tactics: cultural, biological, host plant resistance, chemical; the role of biotechnology in plant disease management. Classification, Structure, General characteristics, identification, ecology. Symptomatology, dispersal. Major nematode pests of economic importance in agriculture. Nematode management tactics: cultural, biological, host plant resistance, chemical and integrated nematode control in specific crops.

MCPR704 Basic Biotechnological Techniques
This module focuses on selected aspects of biotechnology that provide the non- specialist with the principles for understanding techniques in biotechnology including microscopy, separation of biomolecules and organs and their extraction and characterization, aseptic techniques, molecular based diagnostic techniques, tissue and organ culture techniques, DNA manipulation, sequencing, cloning, subcloning, library construction, screening, RNA isolation and characterization, analysis of expression, cDNA synthesis (RTPCR) and analysis, microarrays and gene chips, and Real Time PCR.

MCPR705 Advanced Mycology and Disease Epidemiology
Fungi: Classification, Structure, characteristics, and diversity; Symptoms incited by plant pathogenic fungi; Survival, reproduction. Plant disease management tactics: cultural, biological, host plant resistance, chemical; the role of biotechnology in plant disease management. Management of specific fungal diseases in selected crops. Factors in disease epidemics, disease progress curves, disease forecasting and modelling, relating disease severity and incidence to yield loss.

MCPR706 Advanced Weed Management
Concept and methods of weed management: preventive, cultural, biological, mechanical, and chemical and the role of genetic engineering in weed management. Integrated weed management. Herbicide classification, properties of major herbicide classes, herbicide movement in soil and plants, herbicide persistence; application; formulations; safety and environmental issues. Weed management system design and implementation; weed management in different systems (annual crops, aquatics, perennial cropping, conservation agriculture etc).

MCPR707 Advanced Insect Pest Ecology and Biology
General characteristics and effects of insect pests and mites; systematics of agriculturally important orders; general reproduction and development; insect and mite ecology; insect surveillance and population dynamics. Diversity of vertebrate and other invertebrate plant pests, biology, ecology. Effects and management of migratory and storage pests.

MCPR708 Advanced Pesticide Science and Insect Pest Management
Classification and diversity of pesticides Pesticide toxicity; pesticides and the environment; movement in plants; pesticide resistance causes, mechanisms and management; implications for pest management; use regulations and registration. Pesticide application technologies:

nozzles, calibration of sprayers. Safe and effective use of pesticides. Quantitative basis of pest management, systems analysis and modelling. Pest threshold levels, cost of pest management, estimation of damage by insects, birds, diseases. Ecological basis of pest management. Management techniques cultural, physical, biological, genetic, host plant resistance, chemical, for specific insect pests and nematodes.

MCPR709 Agro-Climatology and Environmental Issues
Definition and concepts of agro-climatology. Economic significance and importance of weather; Introduction to agro-meteorology: greenhouse effect, energy budgets; Atmospheric moisture and precipitation; weather teleconnections, Global climatic change and variability and agriculture, vulnerability adaptation and mitigation. International conventions on climate Weather patterns over Zimbabwe. Extreme weather events and management. Use of weather data for irrigation scheduling. Weather effects on pest and disease incidence. Crop weather modelling. Environmental impact assessment, agricultural pollution.

MCPR710 Entrepreneurship and Agribusiness Management
Role of entrepreneurship in agricultural development and the economy. Theories of entrepreneurship. Processes of establishing a business. Sources of funding. Feasibility studies. Planning production and marketing of enterprise. Entrepreneurship and public policy. Characteristics, nature and problems of small agricultural ventures. Advantages and disadvantages of big versus small. Management techniques. The agribusiness finance function; financial statements. Financial analysis; financial control; cash budgeting; capital budgeting; credit management; financial planning and management; short–term financing; long term financing; capital markets. Mergers and acquisitions.

MCPR870 Dissertation
The dissertation provides an opportunity for the students to undertake and contribute to a piece of original research work, in an area related to crop science. The students are required to design an experiment (or investigation, survey or other means) to test a hypothesis or proposition, to plan and execute the research work, to evaluate the outcomes and draw valid conclusions. The research work is carried out individually, under guided supervision.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN HORTICULTURE SCIENCES (MHORT)

Duration	:	2.5 years
Actual Credit Load	:	384
Minimum Credit Load	:	360
Maximum Credit Load	:	450
Total MBKS Credit Load	:	288
Level	:	9

1. PURPOSE OF THE PROGRAMME

1.1. Develop an understanding of the disciplines related to Horticulture Sciences; make them familiar with chemical, biological, and physical principles that relate to crop plants and their environment.

1.2. Develop their skills to apply the principles related to crops, soils, and environment to solve crop production, and protection problems.

1.3. Develop their leadership, communication, and technical skills in crop sciences.

1.4. Manage agricultural enterprises and develop entrepreneurship skills to create employment and improve the economy of the country.

2. ENTRY REQUIREMENTS

To be eligible in the programme, a candidate must normally be a holder of a good first degree in Agronomy, Horticulture or Crop Science or related fields from a recognised University.

3. PROGRAMME CHARACTERISTICS

Areas of study	:	Horticulture
Specialist Focus	:	Oriented towards innovation and application of theory to solve real life challenges
Distinctive Features	:	Knowledge and skills development in Horticulture Sciences, Horticultural research,

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Research and Academia: Research Scientist, Professor/Lecturer, Postdoctoral Researcher, Agronomist, Crop consultant, Policy Analysit, Extension specialist

Further Studies: Doctoral studies in plant breeding, genetics, agronomy, plant pathology, soil science, and sustainable agriculture.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory and field practicals classes, group work, industrial and farm/field visits, dissertation

Assessment Methods: Written and oral examinations, tests, quizzies, group presentations, dissertation. Continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- To equip graduates with specialized training so that they can better position their own businesses, or the organizations in which they work, for growth in a global marketspace.
- Enable students to conceptualize, formulate and implement solutions to organizational challenges using new and existing technologies.
- To develop new practices methods and technologies that enhance production efficiency.
- Create e-Policies that address ethical use of information and devices in the agribusiness environment.

7. GENERAL PROVISIONS

Refer to Section 8 of General Regulations.

7.1 The programme shall normally be studied over a minimum of four semesters on a block release basis.

7.2 This programme is a taught Masters programme with modular work in the first two semesters and research and dissertation writing in the third and fourth semesters.

7.3 Continuous assessment consists of lectures, presentations, seminars and tests.

7.4 Lectures and taught modules shall normally be at the Midlands State University

7.5 The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.

7.6 Each student has to identify at least two supervisors who shall be approved by the Department of Agronomy and Horticulture

7.7 There shall be oral presentations for the Project Proposal and Research Findings. Weighting of these, together with the final written document, shall be as follows:

- Dissertation Proposal 15%
- Presentation of Research Findings 15%
- Final Written Document 70%

Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

Candidates shall be expected to submit at least three executive bound copies of the dissertation.

The format of the dissertation shall be provided for by the Department of Agronomy and Horticulture

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations, laboratory and or field practicals

8.2. Formal examinations shall be held at the end of each semester.

8.3. Continuous assessment shall account for 30-50% of the overall assessment, while the examination shall account for 50-70 % of the overall assessment.

8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 120 notional credit hours in the first level in order to proceed to the second level.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Level 1 (taught modules) will contribute 50% of total weight and the dissertation 50%.

13. PROGRAMME STRUCTURE

MODULES

Level 1 Semester 1

MHORT 701	Advanced Research Methods and Scientific Communication	24
MCS 702	Advanced Crop Physiology	24
MHORT 703	Advanced Vegetable and Fruit Production	24

Optionals (choose 1)

MHORT 704	Horticulture Business, Trade and Logistics	24
MHORT 705	Herbs, Spices and Traditional Vegetables	24

Level 1 Semester 2

MHORT 706	Advanced Crop Protection	24
MHORT 707	Postharvest Physiology, Processing and Value Addition	24

Optionals (Choose 2)

MHORT 708	Horticulture Biotechnology	24
MHORT 709	Aquaponics, Greenhouse and Irrigation Technology	24
MHORT 710	Genetics, Seed Production and Breeding of Horticultural Crops	24
MHORT 711	Viticulture and Oenology	24

Level 2 Semester 1 and 2 and level 3 semester 1

MHORT 870	Dissertation	192

14. MODULE SYNOPSIS

MHORT 701 Advanced Research Methods and Scientific Communication

Students will learn methods of conducting research in horticulture and related crop science disciplines. The module will help students understand statistical principles, designs of experiments and surveys in agriculture. Experimental design principles, randomisation, replication, error control. Simple designs, randomised designs, randomised complete block designs, factorial designs, split plot designs, Latin square designs, incomplete block designs. Survey methodologies, questionnaire design and sampling methods (random, stratified, purposive sampling), data analysis. Proper techniques of writing a scientific publication. Data analysis techniques and soft wares.

MCS 702 Advanced Crop Physiology

The module will cover the following; Effects of environmental factors (temperature, photoperiod, rainfall, etc.) on crop growth and development. Adaptation of crop plants to stress factors.Analysis of growth and development. Maximisation of crop yield through optimisation of photosynthetic potential: C3, and CAM plants. Photosynthate partitioning in

relation to yield, modification of yield potential by chemical and cultural means. Phytohormones and plant, growth regulators in crop production. Physiological role of mineral nutrition. Crop geometry and competition. Photosynthesis, phototropism and photo-morphogenesis; and their relationship with photo-synthetically active radiation (PAR). Artificial lighting and light quality for crop production.

MHORT 703 Advanced Vegetable and Fruit Production
Students will learn the following; Global production and consumption of vegetables. Physiology and biochemistry of vegetable and fruit tree production. Controlled environment and outdoor production. Soil and soilless culture. Nutrient requirement and fertility management. Global Good Agricultural Practices (Global GAP).

MHORT 704 Horticulture Business, Trade and Logistics
The module will introduce students to concepts of running horticultural business at the local and international levels. The module will introduce students to concepts of horticulture product distribution, methods of transportation and traffic management. Supply chain strategies, global transportation, warehouse management, international logistics, and horticulture product handling. Students will study packaging, palletisation, transportation, refrigeration of horticulture products. Students will also familiarise with international rules, regulations and requirements for importing and exporting horticulture products. International trade regulations and phytosanitary considerations in agricultural trade, technical trade barriers, world trade organisation (WTO), Doha agreement. National plant protection organisations, codex standards, industry standards.

MHORT 705 Herbs, Spices and Traditional Vegetables
The module will cover; an introduction to herbs, spices and indigenous vegetables and their contribution to food and income security. Case studies of utilisation of herbs, spices and indigenous vegetables [Tiens, Herbalife, Forever Living, Robertsons Spices, etc]. Propagation of herbs, spices and indigenous vegetables. Utilisation and value addition of herbs, spices and indigenous vegetables. Growing herbs and spices in controlled environments and hydroponic cultures. Postharvest handling of spices, herbs and indigenous vegetables (drying, grinding, packaging, preservation and value addition). Marketing of herbs, spices and indigenous vegetables.

MCS 703 Advanced Crop Protection
Students will learn cultural, physical, chemical, biological, host-plant resistance, genetic and biotechnological methods of pest, disease, nematode & weed control. Environmental aspects of chemical control of pests; Economics pest control, Pest scouting; Integrated Pest Management. Applications of molecular methods in crop protection.

MHORT 707 Postharvest Physiology, Processing and Value Addition
The module will cover principles and practices of harvesting, handling, grading, packaging and marketing horticultural produce and food crops. Preservation methods and drying. Pre-cooling, controlled and modified atmosphere storage, mechanical refrigeration, construction of cold rooms. Packaging and product innovation. Issues to do with package designs, functions of packaging, packaging materials and controlled atmosphere packaging. Fruit, vegetable and flower packaging. Physiology of freshly harvested produce. Postharvest diseases and pests of horticultural products. Technology for handling harvested produce.

MHORT 708 Horticulture Biotechnology
This module will cover; social and ethical issues in biotechnology. Policy issues around biotechnology. Use of modern biotechniques to improve quality of horticulture plants and products. In this module students will be exposed to techniques such as tissue culture, genetic

engineering, molecular diagnostic techniques, molecular markers and the development of beneficial microbes. Identification and introduction of genes for resistance to diseases and pests, as well as improvement of quality (eg transgenic tomato with reduced pectin methyl esterase). Case studies of genetic biotechniques. Techniques such as polymerase chain reaction, gene amplification.

MHORT 709 Aquaponics, Greenhouse and Irrigation Technology
The module will expose students to understand production of crops in soilless cultures and controlled environments. Students will study principles of controlled environment, hydroponic production systems, greenhouse design and construction, greenhouse materials, irrigation systems in greenhouses, planting mediums used in greenhouses, the physiology of plants grown in controlled environments, planting mechanisms and trellising systems in greenhouses. The module also exposes students to irrigation methods used in greenhouses and the practical design and construction of the irrigation. Water quality and greenhouse irrigation application and scheduling.

MHORT 710 Genetics, Seed Production and Breeding of Horticultural Crops
Learners will appreciate genetics, breeding and seed production of horticultural crops. Seed production and technology. Harvesting, drying of processing of vegetables, ornamentals and fruit trees. Routing and advanced methods of seed testing. Seed certification and legislation. Breeding for disease tolerance and pest resistance. Breeding rights and regulations. Procedures for variety selection. Breeding vegetatively propagated plants (controlled crosses between clones, gene introgression and marker assisted selection). Conventional and novel methods of fruit breeding (in vitro protoplast culture, pollen culture, somatic embryogenesis). Selection and maintenance of scions. Problems in fruit breeding (juvenility and precocious flowering). Seed production of ornamental crops. Students to appreciate the global horticultural seed industry.

MHORT 711 Viticulture and Oenology
The module will cover basic grape morphology and production directions. Wine grape cultivars. An introduction to the composition of grapes, must and wine, as well as micro-organisms in winemaking. The fundamentals of alcoholic fermentation, cellar equipment and production methods. An introduction to wine styles and wine evaluation. Fining and clarification of wine: fining trials, filtration of wine. Bottling principles. Wine faults. Blending of wines and evaluation.

DEPARTMENT OF ANIMAL AND WILDLIFE SCIENCES

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN ANIMAL AND WILDLIFE SCIENCES (HAWS)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load:		480
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To promote the productive and sustainable management of livestock and wildlife resources and in so doing contribute towards sustainable food security, self-reliance and overall wellbeing of the entire Zimbabwean population and SADC region.

2. ENTRY REQUIREMENTS

Normal Entry

These will normally be as specified in the Faculty Regulations

Five 'O' level subjects including Science, Maths, English and any 2 'A' Level subjects with at least one science subject from the following; Biology, Chemistry, Agriculture, Geography, Physics. Applicants with good 'O' level Biology, Chemistry, Physics, Physical Science and Mathematics and any other 'A' level subjects can be considered.

Special Entry

For special entry, students with outstanding passes in a diploma with a livestock or wildlife bias can be admitted.

Mature Entry

Refer to Section 3.3 of the General Regulations.

3. PROGRAMME CHARACTERISTICS

Areas of study: Anatomy, Physiology, Nutrition, Health, Biotechnology, Microbiology, Genetics and Breeding, Ecology, Biochemistry, Statistics, Conservation, Aquaculture, Apiculture, Livestock and Wildlife Management, GIS and Remote Sensing.

Specialist Focus: Animal Production and Health

Orientation Oriented towards innovation and application of theory to solve real life problems

Distinctive Features: Knowledge of the role of animal production and health in agriculture, agribusiness and agro-industrial development

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability : In the private sector - consultants and specialists.
In government - extension and research officers and consultants.
In academic institutions – lecturers and consultants.

Self-employment - farmers and consultants.

Further Studies: Students can also further their studies in the following fields: Animal Nutrition, Animal Breeding, Animal Health, Wildlife Ecology, Biotechnology and Natural Resources Management.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- Understand and apply key principles and theories in animal and wildlife sciences and production
- Apply practical skills in the management of sustainable animal and wildlife production practices and systems
- Analyse problems in animal and wildlife production
- Develop solutions to problems in animal and wildlife production
- Design efficient animal and wildlife production systems
- Create and innovate new knowledge and practices in animal and wildlife production

7. GENERAL PROVISIONS

7.1. The BSc in **Animal and Wildlife Sciences** is a four-year full-time programme.

7.2. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
AGPR 131	Agricultural Practice 1	6
CS 131	Communication Skills	12
HSC 135	Introduction to Information Technology	12
HAWS 134	Principles of Ecology	12
AGRO131	Plant Biology and Physiology	12
LWR 135	Introduction to Soil Science	12
HAWS 133	Introduction to Microbiology	12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
AGPR 132	Agricultural Practice 2	AGPR131	6
AGRO 132	Introduction to Genetics		12
AGRO 136	Introduction to Statistics		12
HAWS 135	Introduction to Entomology and Parasitology		12
HAWS 136	Principles of Wildlife Management	HAWS 134	12
HAWS 132	Introduction to Biochemistry	AGRO135	12
HAWS 131	Anatomy and Physiology of Animals		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
GS 231	Gender Studies		12
HAWS 231	Statistical Methods and Experimental Designs	AGRO 136	12
ENT 235	Entrepreneurship		12
HAWS 232	Animal Breeding and Reproductive Technologies	AGRO 132	12
HAWS 233	Grassland Science		12
HAWS 234	Wildlife Nutrition	HAWS136	12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 235	Animal Health and Welfare	HAWS 133, HAWS135	12
HAWS 236	Animal Nutrition	HAWS 132, HAWS234	12
HAWS 237	Geographic Information Systems and Remote Sensing		12
HAWS 238	Natural Resources Management	HAWS134	12

Level 3 Semester 1: Work-Related Learning

AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 431	Rangeland Ecology and Management	HAWS233	12
HAWS 432	Beef, Goats and Sheep Production	HAWS 232, HAWS 233, HAWS 231, HAWS 236	12
HAWS 433	Biotechnology in Animal Science	HAWS 132, HAWS 232	12
HAWS 434	Pig and Poultry Production	HAWS 232, HAWS 231, HAWS 236	12
HAWS 435	Intensive Wildlife Production	HAWS 136, HAWS 131 HAWS 234	12
HAWS 436	Applied Terrestrial Ecology	HAWS134	12

Optional Module

HAWS 437	Vertebrate Faunas of Zimbabwe	HAWS134	12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 438	Animal Products Processing	HAWS 131, HAWS 132	12
HAWS 439	Research Project	AGRO 134, HAWS231	24
HAWS 440	Dairy Production	HAWS 236, HAWS 233, HAWS 231, HAWS 232	12
HAWS 441	Wildlife Resources Management	HAWS 131	12
HAWS 442	Applied Aquatic Ecology	HAWS134	12

Optional Module

AGED 439	Agricultural Extension		12

MODULE SYNOPSES

AGPR101 Practices 1

This module gives first-year students in the field of Agriculture, a broad understanding of the fundamental entrepreneurial skills essential in starting a farming business, screening of opportunities, developing a business concept and translating ideas into business opportunities. The module will also cover certain animal production enterprise such as ruminant production systems (beef, dairy, small ruminants);non-ruminant production systems (broilers, layers, piggery and rabbitry) and intensive wildlife management enterprises.. A field trip to an animal farm shall take place during each semester.

CS131 Basic Communication Skills

The module is aimed at assisting students to achieve full academic potential through equipping them with the necessary communication skills essential for their degree studies.

HCS135 Introduction to Information Technology

Introduction: uses of information technology, computer hardware and software. Data processing, computer networks and internet, network typologies, the internet, uses of the internet, E-business, word processors and spread sheets, information systems, security and ethical challenges in Information Technology.

HAWS 131 Anatomy and Physiology of Animals

Types of anatomy, general anatomy of organs and systems, descriptive terms for relative positions of animal structures, the skeletal system (osteology)- axial and appendicular skeletons, joints of the skeleton (Arthrology), muscles (myology), digestive system of monogastrics and ruminants, reproductive organs of the male and female, circulatory system, respiratory system, excretory system, nervous system and the endocrine system; Homoeostasis, blood, lymph and other body fluids-composition and their function; Blood circulation, cardiac cycle; Internal and external respiration; Digestion in ruminants and monogastric animals, rumen function; Physiology of kidneys: the nephron, rennin-angiotensin systems, acid base balance; Reproductive physiology of the male and female – foetal development of the gonads, puberty, folliculogenesis and mammary gland, milk secretion and composition, lactation cycle; Characteristics and functions of the nervous systems.

HAWS 102 Introduction to Biochemistry
Energy flows in biological systems; Structure of cells; Stereochemistry: positional, geometrical and optical isomerism; Chemistry of carbohydrates: classification, structure, functions, mobilization and transport; Chemistry of proteins: amino acids their classification, structure, functions, mobilization and transport; Chemistry and functions of nucleic acids; and DNA and RNA, their functions and roles in life processes.

HAWS 103 Introduction to Microbiology
Characteristics of eukaryotic and prokaryotic cells: micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; Classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens); Use of micro-organisms in industrial microbiology: cheese, beer, wine making, and in silage making; Immunology; Techniques of isolating and identifying micro-organisms; and Microscopy.

HAWS 134 Principles of Ecology
Definitions and examples of ecological parameters. Hierarchy of organisation (molecular – biosphere). Population and community ecology – structure and development in relation to environmental factors, interaction, e.g. competition and other biotic factors Agricultural ecosystems, evolution, succession and climax. Stability and productivity of ecosystems. Major ecosystem types of the world.

AGPR132 Agricultural Practices 2
In addition to entrepreneurial skills learnt in AGPR101, students shall learn to finance a farming business, do cash flow budgets and apply for a successful bank loan. Production enterprises shall include the production of maize, wheat, leafy vegetables, onions, tomatoes, field beans, tobacco, cotton, soyabean, citrus, mango and banana. Students shall also be exposed to the production of major flowers e.g. carnations and roses. Students shall be exposed to aspects of marketing, human resources, pricing, government policies and taxation. Students shall visit a crop production farm during each semester.

AGRO 132 Introduction to Genetics
Cell division: mitosis and meiosis; Nucleic acids, replication and protein synthesis; Mendelian principles: segregation and independent assortment. Environmental effects and gene expression; Gene interaction and lethality; sex determination and sex linkage; Gene mutation and induced genetic change (biotechnology); Gene structure and regulation; and Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources. Computing Relationships and the inbreeding coefficient, uses of inbreeding. Quantitative genetics: the quantitative traits model, components of phenotypic and genetic variance and their estimation, genotype value, additive genetic or breeding value, dominance genetic value, epistemic genetic value, influence of selection, influence of inbreeding, the selection response, heritability and repeatability, genotype *environment interaction, phenotypic, genetic and environmental correlations.

AGRO 136 Introduction to Statistics
Fundamentals of basic statistics and probability theory. Introduction to statistics: definition, uses of statistics in
research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability,
tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial
distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean; Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; Tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; and Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HAWS 135 Introduction to Entomology and Parasitology
Position of insects in the world: reproductive capacity, beneficial and injurious insects. Insect external anatomy: the exoskeleton, ecdysis. Internal anatomy and physiology: The nervous system-sight, sound and vibration detection, chemical communication (pheromones), the digestive system, locomotion, insect reproduction, metamorphosis; Insect classification, life cycles, population ecology, population monitoring and economic decision making; Mites, ticks and spiders; Nematodes-morphology, classification, reproduction and economic importance.

LWR 135 Introduction to Soil Science
Refer to the Department of Land and Water Resources Management.

HAWS 136 Principles of Wildlife Management
Important wildlife definitions- wildlife management, conservation, game ranching etc. Wildlife management compared to adaptive management. Zoogeography and factors affecting animal distributions. Wildlife values and perceptions. Wildlife biology: reproduction, fecundity, natality, recruitment, biotic potential, sex relations, density dependent mortality, starvation and malnutrition, predation and disease, nutrition, nutritional strategy, feeding habits and preference, water requirements. Wildlife behaviour and environmental physiology. Wildlife research and role in management

AGRO 101 Plant Biology and Physiology

Refer to the Department of Agronomy and Horticulture.

GS231 Gender Studies
The module empowers the students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include, Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, and Gender Issues in Zimbabwe, Redressing Gender Imbalances, Empowerment and Strategies for creating a gender responsive environment. Every student has to pass the module in order to graduate.

HAWS 231 Statistical Methods and Experimental Designs

Module will introduce students to application of statistics in animal and wildlife research by covering the following aspects: The Whys and Wherefores of Statistics; Descriptive statistics; Probability and Probability distribution; Sampling and sampling Distribution; Experimental and Quasi Experimental Designs; Hypothesis Testing (t-test, F-test, Chi-squared test); Linear Correlation and regression; Further Regression Analysis; Non-parametric statistical methods; Further Aspects of design and Analysis; Additional techniques (Bioassay and Diagnostic Analysis).Data entry, analysis and interpretation using statistical packages(STATA, GENSTAT, R and EPI Info) for all designs: Regression and Correlation. Fitting statistics in research proposal and dissertation write-ups.

ENT 131 Entrepreneurship
The module looks at business development, barriers encountered and issues related to marketing, human resources management, pricing, government policies, taxation of businesses. Case studies, visits to business premises and presentations by entrepreneurs will be used to understand the running of small business. The preparation of business plans will also be covered.

HAWS 232 Animal Breeding and Reproductive Technologies
Breeding programmes; selection objectives, animal model evaluations, multiple trait selection; prediction of breeding values, selection response; mating systems; traits of economic importance in livestock, genetic diversity in animal populations, marker assisted selection and genomic selection. Mating systems: random mating, positive and negative assortative mating. Breeding Systems: crossbreeding. Inbreeding, line breeding. Principles and practice in nucleus breeding schemes; records and data processing, computer applications in animal breeding. Biotechnology and animal improvement: Synchronization of oestrus, AI, MOET, sperm sexing, embryo sexing, Blood typing and DNA mapping, Cryopreservation, cloning, transgenic animals. Practical interaction in livestock improvement units/schemes.

HAWS 233 Grassland Science
Forage plants: - Morphology and identification of species, growth physiology and defoliation effects, Herbage quality. Principles of planted pasture production, Pasture management and productivity, Veld Management, Fodder crops, Fodder conservation, Forages in communal area, Fodder flows

HAWS 234 Wildlife Nutrition
Physiology and fermentation, the role of secondary chemicals and toxins in plants, measures against toxic plants and symptoms of poisoning; energy balance, mineral shortages, supplementary feeding and nutritional requirements of animals; diseases associated with poor nutrition

HAWS 235 Animal Health and Welfare
Veterinary methods and techniques in the care and treatment of injured, sick and vulnerable animals. Immunology: types of immunity; origin, development and functions of the leucocytes, the primary and secondary immune responses; theory of vaccination. Antibiotics; types and mechanisms of action. Causes of disease, transmission, host specificity, host response and defence, evasion of immunity, diagnostic methods, disease prevention and control with reference to common viral, bacterial, rickettsial, protozoal, fungal diseases and endo/ecto-parasitism of livestock common in the in the tropical environment. Environmental influences on disease incidence and spread of disease. Statutory regulations and the movement of animals and trade in animal products. Importance of animal welfare; Marketing; Animal handling; Role of wildlife and humans in disease dynamics; Knowledge of the National, Regional and International legislation and rules in livestock production.

HAWS 236 Animal Nutrition
Classification and chemical value of animal feeds, dry forage and roughages, pastures, range plants, silages, energy contribution fees, protein supplementary feed, minerals and vitamin supplements, and feed additives; Chemical composition of feeds; biochemistry of feed constituents; Analysis of feeds- proximate analysis scheme, Van Soest and Moore systems; Stock-feed manufacturing and legislation in the animal feeds industry legislation, improving forage feed value; Digestibility will feedstuffs; Biochemical theories of voluntary feed intake; Kinetics of digestion and passage ; and Metabolic functions and roles of micronutrients (vitamins and minerals).

HAWS 237 Geographic Information Systems and Remote Sensing
Introduction to GIS; Software and Hardware: GIS Applications; Sources of GIS data; Global Positioning Systems; Introduction to Remote Sensing; Imagery and Photogrammetry; GIS Acquisition and Platforms; Electromagnetic spectrum; Spectral signatures/patterns of objects: water, soil, vegetation; Normalized Difference Vegetation Index (NDVI); Raster and Vector; GIS Data Input; Digitizing; Data quality; Data Output; Geo-databases; Usage of Unmanned Aerial Vehicle (UAVs) Technologies in Cartography; Map projections; Geo-processing; Data queries; Spatial Data analysis; and Cartographic modelling.

HAWS 238 Natural Resources Management
Ecological concepts and sustainable use of natural resources; Concepts and principles of renewable and non-renewal resources; Management of resources: degradation of resources, depletion of resources, types of natural resources; Policies in Natural Resources Management; Appraisal of the relationship between resources, ecology and the environment; Management of heritage resources: land, water, animals and vegetation, cultural and related gene banks; Cultural property: Tangible and intangible attributes; and Identification, preservation and safeguarding of Zimbabwean values (ubuntu/hunhu) and heritage in its diverse forms.

AGPR 331 Work-Related Learning
This module assesses the reports submitted by the student during the time he is learning at the workplace in relation to the degree program.

AGPR 332 Academic Supervisors Report
This module is based on the academic's field visits to the employers and assesses the effectiveness of the student in the work environment.

AGPR 333 Employer's Assessment Report
This module assesses the role of the student in the work environment based on the employer's assessment of the student.

HAWS 431 Rangeland Ecology and Management
Forage plants: - Morphology and identification of species, growth physiology and defoliation effects, nutritive attributes and habitat description. Animal foraging behaviour. Grazing management terminology. Herbage quality, productivity and animal performance in tropical grasslands. Veld types; vegetation, productivity, and nutritive value. Basic veld ecology: Principles of grazing management: - Grass and legume forage/pasture species:- Grassland research methods: Beef herd dynamics, herd composition, separation of herds, productivity in terms of growth rate or milk production etc , reproductive performance, target market age and weight, and the cost of supplementary feeding. Veld management planning; Economic and ecological comparison of single versus multi-use options. Planning of communal area

grazing systems:- The grazing schemes concept versus alternative grazing management. Veld reinforcement. Veld rehabilitation.

HAWS 432 Beef Cattle, Goat and Sheep Production
The beef industry in Zimbabwe. Beef breeds and their characterization. Principles of beef production: Handling facilities, routine management tools and practices, health, reproduction, breeding, nutrition and grazing management. The beef management calendar. Beef herd dynamics, factors guiding choice of production policy. Beef Production systems. Pen and off-veld finishing systems. Transport and marketing of cattle. Meat quality, carcass grading and the economics of beef production. Practical interaction in beef enterprises. Characterisation of small ruminant breeds. Contribution to livestock industry. Reproduction, breeding, nutrition management. Handling facilities and routine management tools and activities. Production systems. Constraints and opportunities. Review of small ruminant research in Zimbabwe. Grading and processing of small ruminant products. Economics of small stock production. Practical interaction in production enterprises.

HAWS 433 Biotechnology in Animal Science
Methods in Biotechnology- traditional and modern: Ethical issues or Biopolitics. Module will also cover application of the latest and emergent options in animal science and improvement ie in animal health, breeding for resistance/ tolerance, nutrition, drug production, gene therapy recombinant DNA technology and other examples. Biosafety and legislation.

HAWS 434 Pig and Poultry Production
Biology of the Chicken; Breeds & their origin, Reproductive Physiology, Respiratory Physiology. Poultry production systems. Principles of poultry production in Broiler, Layer, Breeder stock management; Routine management practices, tools and equipment, Housing and environmental management, Factors affecting, and the management for reproductive efficiency in layers and breeder stock, Applied Breeding, Applied Nutrition, Poultry Health. Practical interaction in meat, egg, and breeding enterprises. Characterization of pig breeds. Pig production systems, and production options. Reproduction, health, breeding and nutritional management. Handling facilities, routine management tools and practices. Carcass quality, grading and marketing. Economics of pig production. Practical interaction in pig production enterprises.

HAWS 435 Intensive Wildlife Production Systems
Intensive apiculture; the biology of bees, apiculture equipment, harvesting, processing, yield, by-products and marketing. Ostrich farming; handling facilities, nutrition, disease, reproduction, egg collection and hatchery management, ostrich products and marketing. Fish farming; the biology of fish in the pond, large fresh water bodies, and controlled environments, transport of live fish. Pond management; construction, fertilisation, weed control, stocking and predator control. Fish harvesting; growth cycles, netting. Principle and practice of fish production in artificially controlled environment systems. Principle and practice in integrated beef-wildlife systems. Crocodile farming- setting up farm, crocodile biological characteristics, reproduction, management, collection and incubation, types of incubators, sex determination during incubation, hatching, care for hatchlings, dietary needs, diseases, skinning, grading and preserving. Processing products.

HAWS 436 Applied Terrestrial Ecology
Adaptive management of wildlife. The influence of body size of large mammal ecology. Foraging Theory. Optimal Foraging: population regulation, survival and extinction of a

species, conservation strategies, case studies, determinants of community structure, large mammal biomass, density, carrying capacity and culling, census method, wildlife utilisation.

HAWS 437 Vertebrate Faunas of Zimbabwe
Vertebrate fauna of Zimbabwe and Africa. Description, composition, distribution, biological characteristics, ecology and conservation significance. Special attention to those groups that illustrate evolutionary principles. Primate evolution and conservation.

HAWS 438 Animal Products Processing
Food borne diseases, with reference to handling of meat, eggs, milk and milk products, fish, and water. Water quality and processing. Milk; handling, quality, processing: pasteurisation, sterilisation, microbiology of sour milk production, yoghurt and cheese making, skimmed milk and butter making. Meat science; processing physiology, composition and other qualitative factors. Hides; quality, pre-treatment, tanning. Wool, mohair, cashmere, feathers; quality and processing products. Other minor animal products.

HAWS 439 Research Project
Explore research issues, research area selection, research concept development, research proposal writing, data collection and organization, data analyses and statistical indication, data interpretation and presentation, prototyping; presentation of research findings; research project write up.

HAWS 440 Dairy Production
Economic importance of dairy industries in Zimbabwe and globally; Objectives of dairy enterprises; Breeds and types of cattle; Production systems; Factors limiting production; Breeding, genetic improvement of dairy cattle, breeding systems-crossbreeding, role of artificial insemination and other reproductive techniques; Reproductive management–bull, cow, heifer and calf rearing feeding and feeding systems; Milking management (dairy cattle only); Disease control, herd health and routine management operations; Marketing of live animals and products; Labour and financial management of operations, and economics of production.

HAWS 441 Wildlife Resources Management
Wildlife Values. Consumptive and non-consumptive exploitation of wildlife. Ecological and zoological principles in game park management with emphasis on environmental protection, assessment of animal impact, impact of ecotourism, harvest intensity and conservation of biodiversity in wildlife species. Case studies on environmental degradation and rehabilitation strategies, on endangered species and strategies of restoration. An analysis of national and international statutory instruments guiding the sustainable utilisation of wildlife resources. Studies on the impact of sustainable use initiatives in Zimbabwe and the Southern African region; Campfire, Intensive Protection Zones, The Conservancy Concept and Cattle-wildlife integrated or multi-use systems, Trans-frontier Parks.

HAWS 442 Applied Aquatic Ecology
Introduction to limnology, hydrological cycle and nature or aquatic environments physical and chemical properties of water in streams and lakes. Nature and function of communities, benthos, plankton, epiphytes, macrophytes and fish, invasive drainage and siltation, fishing and exploitation. Effect of men on systems – pollution eutrophication, exploitation.

AGED 439 Agricultural Extension
Refer to the Department of Agricultural Economics and Development

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN ANIMAL PRODUCTS PROCESSING TECHNOLOGY (HAPPT)

DURATION	:	4 years
Credit Load	:	504
Minimum Credit Load		480
Maximum Credit Load		520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To impart knowledge and practicals in value addition of animal products through agro-processing technologies.

2. ENTRY REQUIREMENTS

Normal Entry

These will normally be as specified in the Faculty Regulations

Five 'O' level subjects including English and any 2 'A' Level subjects with at least one science subject from the following; Biology, Chemistry, Agriculture, Geography, Physics. Applicants with good 'O' level Biology, Chemistry, Physics, Physical Science and Mathematics and any other 'A' level subjects can be considered.

Special Entry

For special entry, students with outstanding passes in a diploma with a livestock or wildlife bias can be admitted.

Mature Entry

Refer to Section 3.3 of the General Regulations.

3. PROGRAMME CHARACTERISTICS

Areas of study: Biochemistry, Microbiology, Food Chemistry, Animal Production, Biotechnology

Specialist Focus: Meat Technology, Dairy Science, Animal Products Processing Technologies

Orientation Oriented towards innovation and application of theory to solve real-life problems

Distinctive Features: Practical application of animal products processing technologies

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Process Development Scientist:

Role: Optimize manufacturing systems and goods.

Responsibilities: Devise and refine processes for product manufacture.
Implement process controls to ensure high-quality, replicable production.
Scale up production processes via trials.
Improve efficiency and quality.
Validate new processes.
Work with product pipelines at various stages of development.

Prospects: Opportunities in manufacturing companies across various sectors, including food, medicine, and cosmetics.

Consultants:
Roles: Nutritionists, Geneticists, Biotechnologists, Housing and Environment experts, and Management Quality Control professionals.
Responsibilities:
Provide expertise in their respective fields.
Advise on nutrition, genetics, biotechnology, and environmental aspects.
Ensure quality control in food processing.
Prospects: Diverse roles in animal feed, health, and environmental industries

Further Studies: Masters and Doctoral studies in animal feed, health, and environment.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,
Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- Understand and apply knowledge on the production and processing methods of different animal products and by products
- Analyse and provide solutions to problems in animal products processing and preservation
- Develop practical and theoretical skills required to establish and manage any animal products processing plant.
- Apply practical skills in value addition of animal products animal by products through application of modern and traditional technologies and research

7. GENERAL PROVISIONS

7.1. The BSc in **Animal Products Processing** is a four year full-time programme.
7.2. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.
7.3.A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.
7.4. Students can include in their programme of study, modules offered by other departments.
7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
HAWS 137	Fundamentals of Food Chemistry	12
HAWS 131	Introduction to Chemistry	12
HSC 135	Introduction to Information Technology	12
HAWS 134	Principles of Ecology	12
HAWS 133	Introduction to Microbiology	12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 139	Cell Biology and Technology		12
AGRO 132	Introduction to Genetics		12
AGRO 134	Introduction to Statistics		12
HAWS 132	Introduction to Biochemistry		12
HAWS 131	Anatomy and Physiology of Animals		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 240	Introduction to Animal Products Processing and Preservation		12
HAWS 231	Statistical Methods and Experimental Designs	AGRO 134	12
ENT 235	Entrepreneurship		12
HAWS 232	Animal Breeding and Reproductive Technologies	AGRO 132	12
HAWS 239	Principles of Food Microbiology		12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 235	Animal Health and Welfare	HAWS 133, HAWS133	12
HAWS 236	Animal Nutrition	HAWS 132	12
HAWS 432	Beef, Goats and Sheep Production		12
HAWS 434	Pig and Poultry Production		12
HAWS 440	Dairy Production		12

Level 3 Semester 1: Work-Related Learning

AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 430	Meat Technology		12
HAWS 433	Biotechnology in Animal Science	HAWS 132, HAWS 232	12
HAWS 443	Dairy Science and Technology		12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 438	Animal Products Processing Technology	HAWS 131, HAWS 132	12
HAWS 439	Research Project	AGRO 134, HAWS231 HAWS 236, HAWS 233, HAWS 231, HAWS 232	24
HAWS 444	Processing of Poultry and Fishery Products		12

14. MODULE SYNOPSIS

HAWS 131 Anatomy and Physiology of Animals

Types of anatomy, general anatomy of organs and systems, descriptive terms for relative positions of animal structures, the skeletal system (osteology)- axial and appendicular skeletons, joints of the skeleton (Arthrology), muscles (myology), digestive system of monogastrics and ruminants, reproductive organs of the male and female, circulatory system, respiratory system, excretory system, nervous system and the endocrine system; Homoeostasis, blood, lymph and other body fluids-composition and their function; Blood circulation, cardiac cycle; Internal and external respiration; Digestion in ruminants and monogastric animals, rumen function; Physiology of kidneys: the nephron, rennin-angiotensin systems, acid base balance; Reproductive physiology of the male and female – foetal development of the gonads, puberty, folliculogenesis and mammary gland, milk secretion and composition, lactation cycle; Characteristics and functions of the nervous systems.

HAWS 132 Introduction to Biochemistry

Energy flows in biological systems; Structure of cells; Stereochemistry: positional, geometrical, and optical
isomerism; Chemistry of carbohydrates: classification, structure, functions, mobilization, and transport; Chemistry of proteins: amino acids their classification, structure, functions, mobilization and transport; Chemistry and functions of nucleic acids; and DNA and RNA, their functions and roles in life processes.

AGRO 132 Introduction to Genetics

Cell division: mitosis and meiosis; Nucleic acids, replication, and protein synthesis; Mendelian principles: segregation and independent assortment. Environmental effects and gene expression; Gene interaction and lethality; sex determination and sex linkage; Gene mutation and induced genetic change (biotechnology); Gene structure and regulation; and Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources. Computing Relationships and the inbreeding coefficient uses of inbreeding. Quantitative genetics: the quantitative traits model, components of phenotypic and genetic variance and their estimation, genotype value, additive genetic or breeding value, dominance genetic value, epistemic genetic value, influence of selection, influence of inbreeding, the selection response, heritability and repeatability, genotype *environment interaction, phenotypic, genetic and environmental correlations.

HAWS 133 Introduction to Microbiology
Characteristics of eukaryotic and prokaryotic cells: micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; Classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens); Use of micro-organisms in industrial microbiology: cheese, beer, wine making, and in silage making; Immunology; Techniques of isolating and identifying micro-organisms; and Microscopy.

HAWS 139 Cell Biology and Biotechnology
Introduction to cell biology. History of cell Biology, cell theory, types of cells (prokaryotic and eukaryotic cells), Microscopy- Principle and Application of the Electron microscopy, cell division (amitosis, mitosis, meiosis), Structure and transportation across cell membrane. In the Biotechnology section the students should have a theoretical and practical knowledge of the instrumentation and techniques employed in a biotechnology laboratory. Students should gain an understanding of biological concepts used in the field of biotechnology such as recombinant DNA and production of biological molecules. Introduction to laboratory techniques currently used in the field of biotechnology such as, aseptic technique, pipetting and measurement, DNA extraction, gel electrophoresis and PCR.

HAWS 130 Introduction to Chemistry
The module will cover the basic chemical and scientific concepts and application in chemistry. Topics to be covered include states of matter and energy, atomic structure, stoichiometry involving reactions, thermochemistry, atomic structure, periodicity, chemical bonding, compounds, states of matter, colligative properties, redox, basic chemical calculations, acids and bases, and different types of reactions.

HAWS 137 Fundamentals of Food Chemistry
The module looks at the chemical, physical and functional properties of food constituents and the chemical changes these constituents undergo during handling, processing, and storage (food shelf life). The module will focus on the major food components (water, lipids, proteins, and carbohydrates) and their performance under different conditions of relevance to food processing. The role of food components in overall food quality. Students should be able to evaluate and explain the high complex nature of food that may result in desired and undesired reactions which are controlled by a variety of parameters.

HAWS 134 Principles of Ecology
Definitions and examples of ecological parameters. Hierarchy of organisation (molecular – biosphere). Population and community ecology – structure and development in relation to environmental factors, interaction, e.g. competition and other biotic factors Agricultural ecosystems, evolution, succession, and climax. Stability and productivity of ecosystems. Major ecosystem types of the world.

HAWS 239 Principles of Food Microbiology
The scope of this module is to understand the role of micro-organisms in food processing and quality. Knowledge of important micro-organisms associated with food and their characteristics, understand the role of micro-organisms in food spoilage, food-borne illness, intoxication and general food processing and quality. Microbiology and spoilage of animal products such as meat, milk etc. The function of beneficial micro-organisms in food manufacturing processes. Understanding of microbiological quality and control. Understanding of microbiological risk

assessment (MRA) tools for evaluating microbiological risks in the food processing sector. Laboratory techniques useful in research, regulation, prevention and control of micro-organisms in food.

HAWS 240 Introduction to Food Processing, Preservation, and Safety
An integrated introduction to dairy and meat science and technology. An overview of the principles of food processes and preservation methods (for longer storage) that maintain the quality, edibility, or nutritional value of food. Meat preservation techniques. Application of various preservation techniques for preserving food (traditional to advanced) such as refrigeration and freezing, heat processing, dehydration, fermentation, high pressure, irradiation, pulsed electric field and packaging. Introduction of food safety, environmental issues, and food law. Water and waste management, Food microbiology, Good Manufacturing Practices (GMPs), Hazard Analysis Critical Control Points (HACCP) and food labelling.

HAWS 232 Animal Breeding and Reproductive Technology
Breeding programmes; selection objectives, animal model evaluations, multiple trait selection; prediction of breeding values, selection response; mating systems; traits of economic importance in livestock, genetic diversity in animal populations, marker-assisted selection and genomic selection. Mating systems: random mating, positive and negative assortative mating. Breeding Systems: crossbreeding. Inbreeding, line breeding. Principles and practice in nucleus breeding schemes; records and data processing, computer applications in animal breeding. Biotechnology and animal improvement: Synchronization of oestrus, AI, MOET, sperm sexing, embryo sexing, Blood typing and DNA mapping, Cryopreservation, cloning, transgenic animals. Practical interaction in livestock improvement units/schemes.

HAWS 235 Animal Health and Welfare
Veterinary methods and techniques in the care and treatment of injured, sick and vulnerable animals. Immunology: types of immunity; origin, development and functions of the leucocytes, the primary and secondary immune responses; theory of vaccination. Antibiotics; types and mechanisms of action. Causes of disease, transmission, host specificity, host response and defence, evasion of immunity, diagnostic methods, disease prevention and control with reference to common viral, bacterial, rickettsial, protozoal, fungal diseases and endo/ecto-parasitism of livestock common in the in the tropical environment. Environmental influences on disease incidence and spread of disease. Statutory regulations and the movement of animals and trade in animal products. Importance of animal welfare; Marketing; Animal handling; Role of wildlife and humans in disease dynamics; Knowledge of the National, Regional and International legislation and rules in livestock production.

HAWS 236 Animal Nutrition
Classification and chemical value of animal feeds, dry forage and roughages, pastures, range plants, silages, energy contribution fees, protein supplementary feed, minerals and vitamin supplements, and feed additives; Chemical composition of feeds; biochemistry of feed constituents; Analysis of feeds- proximate analysis scheme, Van Soest and Moore systems; Stock-feed manufacturing and legislation in the animal feeds industry legislation, improving forage feed value; Digestibility will feedstuffs; Biochemical theories of voluntary feed intake; Kinetics of digestion and passage; and Metabolic functions and roles of micronutrients (vitamins and minerals).

HAWS 432 Beef Cattle, Goat and Sheep Production
The beef industry in Zimbabwe. Beef breeds and their characterization. Principles of beef production: Handling facilities, routine management tools and practices, health, reproduction,

breeding, nutrition and grazing management. The beef management calendar. Beef herd dynamics, factors guiding choice of production policy. Beef Production systems. Pen and off-veld finishing systems. Transport and marketing of cattle. Meat quality, carcass grading and the economics of beef production. Practical interaction in beef enterprises. Characterisation of small ruminant breeds. Contribution to livestock industry. Reproduction, breeding, nutrition management. Handling facilities and routine management tools and activities. Production systems. Constraints and opportunities. Review of small ruminant research in Zimbabwe. Grading and processing of small ruminant products. Economics of small stock production. Practical interaction in production enterprises.

HAWS 440 Dairy Production

Economic importance of dairy industries in Zimbabwe and globally; Objectives of dairy enterprises; Breeds and types of cattle; Production systems; Factors limiting production; Breeding, genetic improvement of dairy cattle, breeding systems-crossbreeding, role of artificial insemination and other reproductive techniques; Reproductive management–bull, cow, heifer and calf rearing feeding and feeding systems; Milking management (dairy cattle only); Disease control, herd health and routine management operations; Marketing of live animals and products; Labour and financial management of operations, and economics of production.

HAWS 434 Pig and Poultry Production

Biology of the Chicken; Breeds & their origin, Reproductive Physiology, Respiratory Physiology. Poultry production systems. Principles of poultry production in Broiler, Layer, and Breeder stock management; Routine management practices, tools and equipment, Housing and environmental management, Factors affecting, and the management for reproductive efficiency in layers and breeder stock, Applied Breeding, Applied Nutrition, Poultry Health. Practical interaction in meat, egg, and breeding enterprises. Characterization of pig breeds. Pig production systems, and production options. Reproduction, health, breeding and nutritional management. Handling facilities, routine management tools and practices. Carcass quality, grading and marketing. Economics of pig production. Practical interaction in pig production enterprises.

HAWS 430 Meat Technology

This module covers theory and practice of meat processing from domestic animals (cattle, sheep, goats) and wildlife. Pre-slaughtering handling of animals, animal slaughter and slaughtering procedures; Evaluation of meat quality (inspection); Analysis of meat and meat products. Factors that affect meat quality (animal genetics, nutrition, and production system). Post mortem metabolism. The principles of meat processing, meat pigments, flavour and colour change. Meat preservation, storage, packaging, and transportation.

HAWS 443 Dairy Science and Technology

Production, processing and handling of milk and dairy products Physical, biochemical, and nutritional properties of milk components. Physical, enzymatic, and microbial transformation of milk. Understanding of milk processing procedures that include raw milk handling, homogenisation, membrane processing, fermentation, evaporation, dehydration, clarification, and heat treatment. Processing methods that are involved in the conversion of milk to other dairy products such as cheese, fermented foods, butter, milk powders, protein products and nutritional formulations.

HAWS 444 Processing of Poultry and Fishery Products

Production, processing, handling, and transportation of poultry and fishery products; Microbiology and spoilage of fish and fishery products; Fish science and technology. Basic principles in

producing eggs, egg products, egg preservation (egg powder) and muscle-based products. Handling, storage, and grading of eggs. Measurement of egg quality and factors that affect egg quality. The role of chemical and physical processes in producing high quality fish and poultry products.

HAWS 438 Animal Products Processing Technology
The students should be able understand and identify the specific processing and preservation technologies (Industrial and traditional) used for animal products. Meat, fish, and meat products processing and preservation technologies. Features of modern abattoirs and slaughter houses for domestic animals. Technologies for egg and egg products production. Machinery and methods used in the dairy industry.

HAWS 445 Animal By-products Processing
This course discusses the potential use and types of by-products (blood, manure, hooves, bones, fat, egg shell, hide, sheep wool, feathers and brain) along with physical and chemical properties; management and processing technology that produces food, non-food and handcrafts of economic value to support environmentally-friendly industries of livestock. By products from wildlife such as horns, hides etc. Hides and skin processing (utilisation)- flaying, curing, tanning. Processing of abattoir by-products.

HAWS 439 Research Project
Explore research issues, research area selection, research concept development, research proposal writing, data collection and organization, data analyses and statistical indication, data interpretation and presentation, prototyping; presentation of research findings; research project write up.

REGULATIONS FOR THE OF BACHELOR OF SCIENCE HONOURS DEGREE IN DAIRY SCIENCES (HDS)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME
To promote the productive and sustainable management of livestock resources and in so doing contribute towards sustainable food security, self-reliance and overall wellbeing of the entire Zimbabwean population and SADC region.

2. ENTRY REQUIREMENTS
Normal Entry
These will normally be as specified in the Faculty Regulations

Five ‘O‘ level subjects including English and any 2 ‘A‘ Level subjects with at least one science subject from the following; Biology, Chemistry, Agriculture, Geography, Physics. Applicants with good ‘O’ level Biology, Chemistry, Physics, Physical Science and Mathematics and any other ‘A’ level subjects can be considered.

Special Entry
For special entry, students with outstanding passes in a diploma with a livestock or wildlife bias can be admitted.

Mature Entry

Refer to Section 3.3 of the General Regulations.

3. PROGRAMME CHARACTERISTICS

Areas of study: Anatomy, Physiology, Nutrition, Health, Biotechnology, Microbiology, Genetics and Breeding, Ecology, Biochemistry, Statistics, Conservation, Dairy waste and Effluent Management, GIS and Remote Sensing.

Specialist Focus: Dairy Production, Dairy Products Processing, and Dairy Breeding

Orientation: Oriented towards innovation and application of theory to solve real-life problems

Distinctive Features: Dairy production and product value addition.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:
- In the private sector - consultants and specialists, farm managers and AI technicians, Allied agribusiness industries, Public relations.
- In government - extension and research officers, consultants and Dairy inspectors.
- In academic institutions – lecturers and consultants, agric communications.
- Self-employment - farmers and consultants.

Further Studies: Students can also further their studies in the following fields: Dairy Nutrition, Dairy Breeding, Animal Health, Dairy farm management, Biotechnology and Natural Resources Management.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), Final exam (60%).

6. INTENDED LEARNING OUTCOMES

- Understand and apply key principles and theories in dairy sciences and milk production
- Apply practical skills in management of dairy production practices and systems
- Analyse problems in dairy animal and milk production
- Develop solutions to problems in animal and milk production
- Design efficient dairy animal and milk production systems
- Create and innovate new knowledge and practices in dairy animal and milk production

7. GENERAL PROVISIONS

7.1. The Bsc. in Dairy Science Honours degree is a four year full-time programme.

7.2. In Level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

10. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 11 of Faculty Regulations.

13. PROGRAMME STRUCTURE

12.1 The degree programme shall have at least 36 taught modules spread over the six teaching semesters.

Level 1 Semester 1

Code	Module Description		Credits
AGPR 131	Agricultural Practice 1		6
HAWS 137	Animals in Society		12
HSC 135	Introduction to Information Technology		12
HAWS 134	Principles of Ecology		12
AGRO131	Plant Biology and Physiology	HAWS132	12
LWR 135	Introduction to Soil Science		12
HAWS 133	Introduction to Microbiology		12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
AGPR 132	Agricultural Practice 2	AGPR131	6
AGRO 132	Introduction to Genetics		12
AGRO 134	Introduction to Statistics		12
HAWS 135	Introduction to Entomology and Parasitology		12
HAWS 138	Animal Reproduction		12
HAWS 132	Introduction to Biochemistry		12
HAWS 131	Anatomy and Physiology of Animals		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
GS 231	Gender Studies		12
HAWS 231	Statistical Methods and Experimental Designs	AGRO 134	12
ENT 235	Entrepreneurship		12
HAWS 232	Animal Breeding and Reproductive Technologies	AGRO 132	12

HAWS 233	Grassland and Pasture Science		12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 235	Animal Health and Welfare	HAWS 133, HAWS133	12
HAWS 236	Animal Nutrition	HAWS 132	12
HAWS 239	Applied Cattle Nutrition		12
HAWS 240	Physiology of Lactation		12

Level 3 Semester 1: Work-Related Learning

AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 431	Rangeland Ecology and Management	HAWS233	12
HAWS 443	Replacement Animal Rearing and Management		12
HAWS 433	Biotechnology in Animal Science	HAWS 132, HAWS 232	12
HAWS 444	Intensive Dairy Production		12
HAWS 445	Dairy Effluent Management		12
HAWS 446	Dairy Products Value Addition		12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 438	Animal Products Processing	HAWS 131, HAWS 132	12
HAWS 439	Research Project	AGRO 134, HAWS231	24
HAWS 447	Applied Dairy Cattle Genetics		12
HAWS 448	Dairy Milking Machine Management		12
HAWS 449	Dairy Enterprise Management		6

14. MODULE SYNOPSIS

HAWS 131 Anatomy and Physiology of Animals

Types of anatomy, general anatomy of organs and systems, descriptive terms for relative positions of animal structures, the skeletal system (osteology)- axial and appendicular skeletons, joints of the skeleton (Arthrology), muscles (myology), digestive system of monogastrics and ruminants, reproductive organs of the male and female, circulatory system, respiratory system, excretory system, nervous system and the endocrine system; Homoeostasis, blood, lymph and other body fluids-composition and their function; Blood circulation, cardiac cycle; Internal and external respiration; Digestion in ruminants and monogastric animals, rumen function; Physiology of kidneys: the nephron, rennin-angiotensin systems, acid base balance; Reproductive physiology of the male and female – foetal development of the gonads, puberty, folliculogenesis and mammary

gland, milk secretion and composition, lactation cycle; Characteristics and functions of the nervous systems.

HAWS 132 Introduction to Biochemistry
Energy flows in biological systems; Structure of cells; Stereochemistry: positional, geometrical and optical isomerism; Chemistry of carbohydrates: classification, structure, functions, mobilization and transport; Chemistry of proteins: amino acids their classification, structure, functions, mobilization and transport; Chemistry and functions of nucleic acids; and DNA and RNA, their functions and roles in life processes.

HAWS 133 Introduction to Microbiology
Characteristics of eukaryotic and prokaryotic cells: micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; Classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens); Use of micro-organisms in industrial microbiology: cheese, beer, wine making, and in silage making; Immunology; Techniques of isolating and identifying micro-organisms; and Microscopy.

HAWS 134 Principles of Ecology
Definitions and examples of ecological parameters. Hierarchy of organisation (molecular – biosphere). Population and community ecology – structure and development in relation to environmental factors, interaction, e.g. competition and other biotic factors Agricultural ecosystems, evolution, succession and climax. Stability and productivity of ecosystems. Major ecosystem types of the world.

AGRO 132 Introduction to Genetics
Cell division: mitosis and meiosis; Nucleic acids, replication and protein synthesis; Mendelian principles: segregation and independent assortment. Environmental effects and gene expression; Gene interaction and lethality; sex determination and sex linkage; Gene mutation and induced genetic change (biotechnology); Gene structure and regulation; and Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources. Computing Relationships and the inbreeding coefficient, uses of inbreeding. Quantitative genetics: the quantitative traits model, components of phenotypic and genetic variance and their estimation, genotype value, additive genetic or breeding value, dominance genetic value, epistemic genetic value, influence of selection, influence of inbreeding, the selection response, heritability and repeatability, genotype *environment interaction, phenotypic, genetic and environmental correlations.

AGRO 134 Introduction to Statistics
Fundamentals of basic statistics and probability theory. Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean; Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; Tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; and Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HAWS 135 Introduction to Entomology and Parasitology
Position of insects in the world: reproductive capacity, beneficial and injurious insects. Insect external anatomy: the exoskeleton, ecdysis. Internal anatomy and physiology: The nervous system-sight, sound and vibration detection, chemical communication (pheromones), the digestive system, locomotion, insect reproduction, metamorphosis; Insect classification, life cycles, population ecology, population monitoring and economic decision making; Mites, ticks and spiders; Nematodes-morphology, classification, reproduction and economic importance.

LWR 135 Introduction to Soil Science
Refer to the Department of Land and Water Resources Management.

AGRO 131 Plant Biology and Physiology

Refer to the Department of Agronomy and Horticulture.

HAWS 231 Statistical Methods and Experimental Designs
Module will introduce students to application of statistics in animal and wildlife research by covering the following aspects: The Whys and Wherefores of Statistics; Descriptive statistics; Probability and Probability distribution; Sampling and sampling Distribution; Experimental and Quasi Experimental Designs; Hypothesis Testing (t-test, F-test, Chi-squared test); Linear Correlation and regression; Further Regression Analysis; Non-parametric statistical methods; Further Aspects of design and Analysis; Additional techniques (Bioassay and Diagnostic Analysis).Data entry, analysis and interpretation using statistical packages(STATA, GENSTAT, R and EPI Info) for all designs: Regression and Correlation. Fitting statistics in research proposal and dissertation write-ups.

HAWS 232 Animal Breeding and Reproductive Technologies
Breeding programmes; selection objectives, animal model evaluations, multiple trait selection; prediction of breeding values, selection response; mating systems; traits of economic importance in livestock, genetic diversity in animal populations, marker assisted selection and genomic selection. Mating systems: random mating, positive and negative assortative mating. Breeding Systems: crossbreeding. Inbreeding, line breeding. Principles and practice in nucleus breeding schemes; records and data processing, computer applications in animal breeding. Biotechnology and animal improvement: Synchronization of oestrus, AI, MOET, sperm sexing, embryo sexing, Blood typing and DNA mapping, Cryopreservation, cloning, transgenic animals. Practical interaction in livestock improvement units/schemes.

HAWS 233 Grassland Science and Pasture Science
Forage plants: - Morphology and identification of species, growth physiology and defoliation effects, Herbage quality. Principles of planted pasture production, Pasture management and productivity, Veld Management, Fodder crops, Fodder conservation, Forages in communal area, Fodder flows

HAWS 235 Animal Health and Welfare
Veterinary methods and techniques in the care and treatment of injured, sick and vulnerable animals. Immunology: types of immunity; origin, development and functions of the leucocytes, the primary and secondary immune responses; theory of vaccination. Antibiotics; types and mechanisms of action. Causes of disease, transmission, host specificity, host response and defence,

evasion of immunity, diagnostic methods, disease prevention and control with reference to common viral, bacterial, rickettsial, protozoal, fungal diseases and endo/ecto-parasitism of livestock common in the in the tropiucal environment. Environmental influences on disease incidence and spread of disease. Statutory regulations and the movement of animals and trade in animal products. Importance of animal welfare; Marketing; Animal handling; Role of wildlife and humans in disease dynamics; Knowledge of the National, Regional and International legislation and rules in livestock production.

HAWS 236 Animal Nutrition
Classification and chemical value of animal feeds, dry forage and roughages, pastures, range plants, silages, energy contribution fees, protein supplementary feed, minerals and vitamin supplements, and feed additives; Chemical composition of feeds; biochemistry of feed constituents; Analysis of feeds- proximate analysis scheme, Van Soest and Moore systems; Stock-feed manufacturing and legislation in the animal feeds industry legislation, improving forage feed value; Digestibility will feedstuffs; Biochemical theories of voluntary feed intake; Kinetics of digestion and passage ; and Metabolic functions and roles of micronutrients (vitamins and minerals).u

HAWS 239 Applied Dairy Cattle Nutrition
Application of basic principles of nutrition in developing rations for dairy herds. Emphasis placed on the appropriate use of forages, ration formulation techniques, development of profitable rations. Development, function and use of dairy nutritional information systems for dairy feeding.

HAWS 148 Dairy Chemistry
Recent advances in the area of the chemistry of milk constituents. Bioactive role of milk constituents in relation to dairy ingredients. New analytical techniques particularly rapid test kits and bio sensors.

Physiology of Lactation

Industrial Attachment
This module assesses the reports submitted by the student during the time he is learning at the workplace in relation to the degree program.This module is based on the academic's field visits to the employers and assesses the effectiveness of the student in the work environment. This module assesses the role of the student in the work environment based on the employer's assessment of the student.

HAWS 431 Rangeland Ecology and Management
Forage plants: - Morphology and identification of species, growth physiology and defoliation effects, nutritive attributes and habitat description. Animal foraging behaviour. Grazing management terminology. Herbage quality, productivity and animal performance in tropical grasslands. Veld types; vegetation, productivity, and nutritive value. Basic veld ecology: Principles of grazing management: - Grass and legume forage/pasture species:- Grassland research methods: Beef herd dynamics, herd composition, separation of herds, productivity in terms of growth rate or milk production etc , reproductive performance, target market age and weight, and the cost of supplementary feeding. Veld management planning; Economic and ecological comparison of single versus multi-use options. Planning of communal area grazing systems:- The grazing schemes concept versus alternative grazing management. Veld reinforcement. Veld rehabilitation.

HAWS 446 Dairy Products Value Addition
Process scheduling of heat dessicated, coagulated and fermented traditional dairy products. Process improvement in production of milk sweets. New products based on fruits, vegetables and cereals. Application of membrane technologies and microwave heating for industrial production of dairy products. Convenience treatment of dairy products, use of natural and permitted synthetic preservatives and new packaging systems.

HAWS 433 Biotechnology in Animal Science
Methods in Biotechnology- traditional and modern: Ethical issues or Biopolitics. Module will also cover application of the latest and emergent options in animal science and improvement ie in animal health, breeding for resistance/ tolerance, nutrition, drug production, gene therapy recombinant DNA technology and other examples. Biosafety and legislation.

HAWS 240 Physiology of Lactation
Anatomy and physiology of the mammary gland and lactation in dairy cattle. Mammary gland health and factors affecting lactation. Principles and techniques in dairy herd milking management. Immunology as related to the diseases of the mammary gland, concepts of the mammary gland immunity, disease etiology, immunopathology, diagnosis and therapy.

HAWS 443 Replacement Animal Rearing and Management
The art and science of calf rearing, feeding and health. Weaning, weighing and selection of replacement heifers. Various feeding programs and computerized feed formulation programs for calves as well as growing heifers. Milk weaning rations, weighing, and grouping of heifers. Various housing designs for calves and heifers.

HAWS 445 Dairy Effluent Management
Advanced technologies for dairy effluent treatment on farms I.e chemical and biological treatment. Effluent management guidelines, design of dairy shed effluent management systems and their operation. On farm benefits of good effluent management, maintenance of environmental values of surface and ground waters including their ecology by minimizing the discharges of effluents. Effective recycling and reuse of effluent where possible.

HAWS 438 Animal Products Processing
Food borne diseases, with reference to handling of meat, eggs, milk and milk products, fish, and water. Water quality and processing. Milk; handling, quality, processing: pasteurisation, sterilisation, microbiology of sour milk production, yoghurt and cheese making, skimmed milk and butter making. Meat science; processing physiology, composition and other qualitative factors. Hides; quality, pre-treatment, tanning. Wool, mohair, cashmere, feathers; quality and processing products. Other minor animal products.

HAWS 439 Research Project
Explore research issues, research area selection, research concept development, research proposal writing, data collection and organization, data analyses and statistical indication, data interpretation and presentation, prototyping; presentation of research findings; research project write up.

HAWS 444 Dairy Production
Economic importance of dairy industries in Zimbabwe and globally; Objectives of dairy enterprises; Breeds and types of cattle; Production systems; Factors limiting production; Breeding, genetic improvement of dairy cattle, breeding systems-crossbreeding, role of artificial insemination and other reproductive techniques; Reproductive management–bull, cow, heifer and calf rearing feeding and feeding systems; Milking management (dairy cattle only); Disease control, herd health

and routine management operations; Marketing of live animals and products; Labour and financial management of operations, and economics of production.

HAWS 449 Dairy Enterprise Management
Decision strategies for modern dairy businesses. Emphasis on relationships of enterprises and technologies for evaluation of business alternatives, efficiency of production and profit. Use of computer software to support management decisions. Concentration on herd replacement, personnel, facilities and issue of management associated with rapidly changing national and international markets. Environmental regulations and computer applications. Group projects and hands on management of dairy cattle.

HAWS 447 Applied Dairy Cattle Genetics
Application of basic principles of genetics to dairy cattle improvement. Setting goals for genetic improvement, characteristics of traits included in selection, methods of estimation of breeding values, Role of Artificial Insemination and Breed Association s in genetic improvement of dairy cattle

HAWS 448 Dairy Milking Machine Management
Types and designs of milking palours and milking machines. Installation and assembly of milking machines and associated units. Development and function of computerized dairy milking systems. Clean milk production, servicing and part replacement of vacuum pumps and main lines. Maintenance of bulk tanks, inter cooling and refrigeration systems.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN WILDLIFE ECOLOGY AND MANAGEMENT (HWEM)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To promote the productivity of wildlife resources and through an understanding of wildlife management and the techniques used in habitat and wildlife populations management and their consequences on wildlife resources. This will contribute towards sustainable wildlife management in Zimbabwe's ecosystems and SADC region.

2. ENTRY REQUIREMENTS

Five 'O' level subjects including English, Maths and Science and any 2 'A' Level subjects with at least one science subject from the following; Biology, Chemistry, Agriculture, Geography, Physics. Applicants with good 'O' level Biology, Chemistry, Physics, Physical Science and Mathematics and any other 'A' level subjects can be considered.

Articulation as provided by ZNQF.

Recognition criteria for foreign credentials and international students.

3. PROGRAMME CHARACTERISTICS

Areas of study	:	Animal welfare, behaviour and ethics Diversity of Life Principles of Ecology Principles of Wildlife Management Vertebrate Biology
Specialist Focus	:	Ecotourism Planning Geographic information systems and remote sensing Natural Resources Management
Orientation		Oriented towards innovation and application of theory to solve real life problems
Distinctive Features	:	Ecology, and Natural Resources Management.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability : Wildlife Biologist, Natural Resources Managers, Park Rangers, Consultants, Ecologists, Researchers.

Further Studies : Masters and Doctoral studies

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (40%), Final exam (60%).

6. INTENDED LEARNING OUTCOMES

- Understand and apply key ecological principles and theories in wildlife sciences and conservation.
- Acquire knowledge on current best practices in wildlife and rangeland management.
- Communicate basic wildlife resources management concepts for the establishment of viable wildlife-based businesses.
- Apply practical skills in management of sustainable wildlife production practices and systems
- Analyse problems in wildlife production and management in Zimbabwe.
- Analyse major international, regional, national and local natural resources policies.
- Develop solutions to problems in wildlife production.

7. GENERAL PROVISIONS

7.1. The Bachelor of Science Honours Degree in Ecology and Wildlife Management is a four year full-time programme.

7.2. In Level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAME ASSESSMENT

Refer To Section 6 Of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description		Credits
HAWS 146	Diversity of Life		12
HAWS 147	Animal welfare, behaviour and ethics		12
HSC 135	Introduction to Information Technology		12
HAWS 134	Principles of Ecology		12
AGRO131	Plant Biology and Physiology	HAWS132	12

HAWS 133	Introduction to Microbiology		12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 148	Vertebrate Biology	AGPR131	6
AGRO 132	Introduction to Genetics		12
AGRO 136	Introduction to Statistics		12
HAWS 135	Introduction to Entomology and Parasitology		12
HAWS 136	Principles of Wildlife Management	HAWS 134	12
HAWS 131	Anatomy and Physiology of Animals		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
GS 231	Gender Studies		12
HAWS 231	Statistical Methods and Experimental Designs	AGRO 134	12
ENT 235	Entrepreneurship		12
HAWS 240	Conservation Biology		12
HAWS 233	Grassland Science		12
HAWS 234	Wildlife Nutrition	HAWS132	12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 235	Animal Health and Welfare	HAWS 133, HAWS133	12
HAWS 241	Ecotourism Planning		12
HAWS 237	Geographic Information Systems and Remote Sensing		12
HAWS 238	Natural Resources Management	HAWS136	12

Level 3 Semester 1: Work-Related Learning

AGPR330	Work-Related Preliminary Report	30

Level 3 Semester 2: Work-Related Learning

AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 431	Rangeland Ecology and Management	HAWS233	12
HAWS 432	Wildlife Products Processing		12
HAWS 434	Natural Resources Economics		12
HAWS 435	Intensive Wildlife Production	HAWS 136, HAWS 131	12
HAWS 436	Applied Terrestrial Ecology	HAWS134	12
HAWS 437	Vertebrate Faunas of Zimbabwe	HAWS134	12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 448	Natural Resources Management and Climate Change		12
HAWS 439	Research Project	AGRO 134, HAWS231	24

HAWS 441	Sustainable Wildlife Resources Management	HAWS 131	12
HAWS 442	Applied Aquatic Ecology	HAWS134	12
HAWS 449	Social Ecology and Policy		12

Module Synopsis

HAWS 147 Animal Welfare, Behaviour and Ethics
This module will introduce students to the science of animal behaviour and the importance of behaviour in our understanding of animal welfare. It will analyse diversity in animal behaviour and relate this to the processes of evolution and natural selection. It will also consider the ethics of society's usage of different types of animals and the role of legislation and different organisations in the promotion of the interests of animals. Examples will be drawn from a range of diverse species and scenarios to illustrate the principles and practices discussed. An appreciation of the science of animal behaviour and welfare and how underlying ethical values may influence the acceptability of animal use, undertaking welfare assessments of animals and an overview of legislation governing animal welfare. Critical analysis of zoo establishments

HAWS 146 Diversity of Life

Theories for the origin of life, taxonomy and phylogeny, goals of classification, Leys and phylogenetic trees. Viruses and mineral, Protista, fungi, plants and kingdom animalia. Look at the diversity among the kingdoms. A brief outline of the grades of organization and the evolution of the two main lines of animals (vertebrates and invertebrates). The process of evolution through Darwinian natural selection and relating it to the diversity of life seen on Earth. Historic theories of evolution are evaluated, and the mechanisms underpinning evolution are explored, from microevolution, through speciation to macroevolution. The evolution of humans is considered, along with the history and process of animal domestication. The effects of evolution and domestication on animal physiology and behaviour are explored.

HAWS 134 Principles of Ecology
Understanding the dynamic changes in individuals, populations, communities and ecosystems in relation to each other and the physical environment. Explores the essential processes that determine the distribution and abundance of organisms and the variety of complex biotic and abiotic interactions that take place, ecology of living systems and basic ecological theory. Definitions and examples of ecological parameters. Hierarchy of organisation (molecular – biosphere); structure and development in relation to environmental factors and interactions. Agricultural ecosystems, evolution, succession and climax. Stability and productivity of ecosystems. Major ecosystem types of the world, practical elements of identification, sampling and analysis of data collected.

HAWS 136 Principles of Wildlife Management
Important wildlife definitions- wildlife management, conservation, game ranching etc. Wildlife management compared to adaptive management. Zoogeography and factors affecting animal distributions. Wildlife values and perceptions. Wildlife biology: reproduction, fecundity, natality, recruitment, biotic potential, sex relations, density dependent mortality, starvation and malnutrition, predation and disease, nutrition, nutritional strategy, feeding habits and preference, water requirements. Wildlife behaviour and environmental physiology. Wildlife research and role in management

HAWS 148 Vertebrate Biology
Overview, classification and identification of vertebrate animals. A comparative study of the function and structure of the muscle and skeletal structure and locomotion. Circulatory and immune

systems, reproductive and endocrine systems. Life history strategies, Feeding and nutrition, respiration and excretion. Description, composition, distribution, biological characteristics, ecology and conservation significance. Special attention to those groups that illustrate evolutionary principles.

AGRO 136 Introduction to Statistics

Fundamentals of basic statistics and probability theory. Introduction to statistics: definition, uses of statistics in
research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability,
tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial
distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean; Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; Tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; and Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HAWS 231 Statistical Methods and Experimental Designs

Module will introduce students to the application of statistics in animal and wildlife research by covering the following aspects: The Whys and Wherefores of Statistics; Descriptive statistics; Probability and Probability distribution; Sampling and sampling Distribution; Experimental and Quasi Experimental Designs; Hypothesis Testing (t-test, F-test, Chi-squared test); Linear Correlation and regression; Further Regression Analysis; Non-parametric statistical methods; Further Aspects of design and Analysis; Additional techniques (Bioassay and Diagnostic Analysis). Data entry, analysis and interpretation using statistical packages(STATA, GENSTAT, R and EPI Info) for all designs: Regression and Correlation. Fitting statistics in research proposal and dissertation write-ups.

HAWS 240 Conservation Biology

This module is designed to develop the appropriate skills and knowledge concerned with the science of conservation biology, analysis of current and historical factors contributing to biodiversity loss and the strategies that conservation biologists use for management, and protection of vulnerable species and their associated ecosystems. Local, national and international examples will be used to highlight key concepts and practices.

HAWS 233 Grassland Science

Forage plants: - Morphology and identification of species, growth physiology and defoliation effects, Herbage quality. Principles of planted pasture production, Pasture management and productivity, Veld Management, Fodder crops, Fodder conservation, Forages in the communal area, Fodder flows.

HAWS 234 Wildlife Nutrition

Physiology and fermentation, the role of secondary chemicals and toxins in plants, measures against toxic plants and symptoms of poisoning; energy balance, mineral shortages, supplementary feeding and nutritional requirements of animals; diseases associated with poor nutrition

HAWS 241 Ecotourism Planning
Ecotourism, wildlife, environmental interpretation and community development based on Sustainable Tourism. Challenges related to recreational and tourism activities and how these can be managed. The hierarchy of planning, the process of planning, the economic value of ecosystem services, assess the direct and indirect benefits to local stakeholders, and benefit sharing.

HAWS 237 Geographic Information Systems and Remote Sensing
Introduction to GIS; Software and Hardware: GIS Applications; Sources of GIS data; Global Positioning Systems; Introduction to Remote Sensing; Imagery and Photogrammetry; GIS Acquisition and Platforms; Electromagnetic spectrum; Spectral signatures/patterns of objects: water, soil, vegetation; Normalized Difference Vegetation Index (NDVI); Raster and Vector; GIS Data Input; Digitizing; Data quality; Data Output; Geo-databases; Usage of Unmanned Aerial Vehicle (UAVs) Technologies in Cartography; Map projections; Geo-processing; Data queries; Spatial Data analysis; and Cartographic modelling.

HAWS 238 Natural Resources Management
Ecological concepts and sustainable use of natural resources; Concepts and principles of renewable and non-renewal resources; Management of resources: degradation of resources, depletion of resources, types of natural resources; Policies in Natural Resources Management; Appraisal of the relationship between resources, ecology and the environment; Management of heritage resources: land, water, animals and vegetation, cultural and related gene banks; Cultural property: Tangible and intangible attributes; and Identification, preservation and safeguarding of Zimbabwean values (ubuntu/hunhu) and heritage in its diverse forms.

Industrial Attachment
This module assesses the reports submitted by the student during the time he is learning at the workplace in relation to the degree program. This module is based on the academic's field visits to the employers and assesses the effectiveness of the student in the work environment. This module assesses the role of the student in the work environment based on the employer's assessment of the student.

HAWS 442 Applied Aquatic Ecology
Introduction to limnology, hydrological cycle and nature or aquatic environments physical and chemical properties of water in streams and lakes. Nature and function of communities, benthos, plankton, epiphytes, macrophytes and fish, invasive drainage and siltation, fishing and exploitation. Effect of men on systems – pollution eutrophication, exploitation.

HAWS 436 Applied Terrestrial Ecology
Adaptive management of wildlife. The influence of body size of large mammal ecology. Foraging Theory. Optimal Foraging: population regulation, survival and extinction of a species, conservation strategies, case studies, determinants of community structure, large mammal biomass, density, carrying capacity and culling, census method, wildlife utilisation.

HAWS 431 Rangeland Ecology
Definition and description of rangelands, their characteristics and economic importance of rangelands. Biological cycle, physiology, growth and morphogenesis of rangeland plants. Phenology. Structure, function and productivity of rangeland ecosystems. Effects of grazing and the abiotic environment on rangeland ecosystems. Temporal changes of rangeland vegetation and succession. Drivers of rangeland changes, role of fire, decomposition, rainfall and herbivores in shaping rangelands. Human effects, disturbance and degradation of rangeland ecosystems -

desertification. Description and identification of the most important rangeland species. Plant diversity - indicators Forage plants, Animal foraging behaviour.

HAWS 435 Intensive Wildlife Production Systems
Intensive apiculture; the biology of bees, apiculture equipment, harvesting, processing, yield, by-products and marketing. Ostrich farming; handling facilities, nutrition, disease, reproduction, egg collection and hatchery management, ostrich products and marketing. Fish farming; the biology of fish in the pond, large fresh water bodies, and controlled environments, transport of live fish. Pond management; construction, fertilisation, weed control, stocking and predator control. Fish harvesting; growth cycles, netting. Principle and practice of fish production in artificially controlled environment systems. Principle and practice in integrated beef-wildlife systems. Crocodile farming- setting up farm, crocodile biological characteristics, reproduction, management, collection and incubation, types of incubators, sex determination during incubation, hatching, care for hatchlings, dietary needs, diseases, skinning, grading and preserving. Processing products.

HAWS 432 Wildlife Products Processing
Handling of wildlife products and by-products, harvesting for meat, eggs, and processing. Differences amongst species, effects of body condition, slaughter of animals, preservation, processing and storage. Game meat processing, marketing and illegal harvesting. Hides; quality, pre-treatment, tanning. Wool, mohair, cashmere, feathers; quality and processing products. Other minor animal products

HAWS 434 Natural Resources Economics
This module develops a rationale for and explains the methodologies used in, the application of economic theory to the allocation of natural resources. Using a series of natural resource sectors as examples, several economic models are then explained in detail in a way that is intended to raise the learner's confidence in the interpretation and assessment of various policy insights that are derived from the models. Whilst developing and applying the microeconomic applications deviations from basic models are explored in the context of the 'sustainable development' debate. Resource scarcity will be explored against the rationale for the use of natural resources economics theory, resource allocation challenges and limitations of the neo-classical economics theory in the allocation of resources.

HAWS 448 Natural Resources Management and Climate Change
The module investigates the many aspects of the natural environment which impact our communities. This will include the study of rainfall patterns in Zimbabwe and the SADC region and causes of climatic change, which has a marked effect on the range of natural habitats. The variety and nature of soils and how these affect the land quality will be considered. Conservation and the assessment, creation and management of habitats commonly found in, and around agricultural lands and the impact of pollution from agricultural sources will be investigated. The effects of climate change will be analysed in different habitats and how they can be managed to benefit the natural environment.

HAWS 439 Research Project
Explore research issues, research area selection, research concept development, research proposal writing, data collection and organization, data analyses and statistical indication, data interpretation and presentation, prototyping; presentation of research findings; research project write-up.

HAWS 449 Social Ecology and Policy
Philosophical issues in ecological science, its relationship with conservation biology, and non-anthropocentric values. Critical assessment of linkages amongst the natural environment, scientists

and policymakers and express their implications. A holistic approach to environmental problems by locating issues in their wider environmental, social, economic and political context. Building upon these concepts, students are introduced to the fields of ecological and environmental economics and the application of this knowledge to examine real-life problems from micro and macro levels. An analysis of various policy interventions that address diverse environmental sustainability problems and their effectiveness and efficiency.

HAWS 441 Sustainable Wildlife Resources Management

Wildlife Values. Consumptive and non-consumptive exploitation of wildlife. Ecological and zoological principles in game park management with emphasis on environmental protection, assessment of animal impact, impact of ecotourism, harvest intensity and conservation of biodiversity in wildlife species. Case studies on environmental degradation and rehabilitation strategies, on endangered species and strategies of restoration. An analysis of national and international statutory instruments guiding the sustainable utilisation of wildlife resources. Studies on the impact of sustainable use initiatives in Zimbabwe and the Southern African region; Campfire, Intensive Protection Zones, The Conservancy Concept and Cattle-wildlife integrated or multi-use systems, Trans-frontier Parks.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN POULTRY SCIENCE (HPS)

Duration	:	4 years
Credit Load	:	504
Minimum Credit Load	:	480
Maximum Credit Load	:	520
Maximum MBKS Credit Load	:	420
Level	:	ZNQF-LEVEL 8

1. PURPOSE OF THE PROGRAMME

To promote the productivity of wildlife resources and through an understanding of wildlife management and the techniques used in habitat and wildlife populations management and their consequences on wildlife resources. This will contribute towards sustainable wildlife management in Zimbabwe's ecosystems and SADC region.

2. ENTRY REQUIREMENTS

Five 'O' level subjects including English, Maths and Science and any 2 'A' Level subjects with at least one science subject from the following; Biology, Chemistry, Agriculture, Geography, Physics. Applicants with good 'O' level Biology, Chemistry, Physics, Physical Science and Mathematics and any other 'A' level subjects can be considered.

Articulation as provided by ZNQF.

Recognition criteria for foreign credentials and international students.

3. PROGRAMME CHARACTERISTICS

Areas of study	:	Poultry Breeding
		Poultry Diseases and Control
		Poultry Nutrition
		Poultry Immunology and Vaccination
Specialist Focus	:	Hatchery Management
		Broiler Production
		Indigenous Poultry Management
		Layer Production
		Duck, Turkey and Game-birds Production
		Poultry Welfare
Orientation		Oriented towards innovation and application of theory to solve real life problems
Distinctive Features	:	Poultry production and products value addition.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Research and Development Lab Technician, Food Inspector, Breeding Supervisor, Extension Agent.

Further Studies: Masters and Doctoral studies in hatchery management, nutrition, health, biosecurity, economics, and product processing.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (30%), Final exam (70%)

6. INTENDED LEARNING OUTCOMES

- Understand and apply key principles and theories in poultry science and production
- Apply practical skills in management of sustainable poultry production practices and systems
- Develop solutions to problems in poultry production
- Design efficient poultry production systems
- Create and innovate new knowledge and practices in poultry production

7. GENERAL PROVISIONS

7.1. The Bachelor of Science Honours Degree in Poultry Science is a four-year full-time programme.

7.2. In Level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4. Students can include in their programme of study, modules offered by other departments.

7.5. Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAME ASSESSMENT

Refer To Section 6 Of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
AGPR 131	Agricultural Practice 1	6
CS 131	Communication Skills	12
HSC 135	Introduction to Information Technology	12
HAWS 133	Introduction to Microbiology	12
HAWS 140	Introduction to Poultry Science	12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
AGPR 132	Agricultural Practice 2	AGPR131	6
AGRO 132	Introduction to Genetics		12
AGRO 134	Introduction to Statistics		12
HAWS 135	Introduction to Entomology and Parasitology		12
HAWS 132	Introduction to Biochemistry		12
HAWS 139	Poultry Anatomy and Physiology		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
GS 231	Gender Studies		12
HAWS 231	Statistical Methods and Experimental Designs	AGRO 134	12
ENT 235	Entrepreneurship		12
HAWS 250	Poultry Breeding		12
HAWS 253	Poultry Immunology and Vaccination		12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 251	Poultry Disease and Control	HAWS 133, HAWS133	12
HAWS 252	Poultry Nutrition	HAWS 132	12
HAWS 254	Poultry Housing and Equipment		12

Level 3 Semester 1: Work-Related Learning

AGPR330	Work-Related Preliminary Report		30

Level 3 Semester 2: Work-Related Learning

AGPR331	Work-Related Learning Report		45
AGPR332	Employer's Assessment Report		30
AGPR333	Academic Supervisor's Report		45

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits
HAWS 450	Hatchery Management		12
HAWS 451	Broiler Production		12
HAWS 433	Biotechnology in Animal Science	HAWS 132, HAWS 232	12
HAWS 453	Indigenous Poultry Management		12
HAWS 454	Layer Production		12
HAWS 455	Duck, Turkey and Game-birds Production		12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
HAWS 456	Poultry Products Processing	HAWS 131, HAWS 132	12
HAWS 439	Research Project	AGRO 134, HAWS231	24
HAWS 457	Poultry Welfare		12
HAWS 458	Environmental Management in Poultry Production		12
HAWS 459	Poultry Products and Technology		12
HAWS 460	Marketing of Poultry Products		12

14.0 MODULE SYNOPSES

HAWS 140 Introduction to Poultry Science
Definition of Poultry. History of poultry. Present status and future prospects. Poultry industry in Zimbabwe. Importance of Poultry farming. Market capitalisation, size of the industry, contribution to the economy of Zimbabwe, employment opportunities. Principles of domestication behavioural factors favouring domestication. Adaptations of poultry in different regions. Breeds of poultry. Poultry production systems

HAWS 139 Poultry Anatomy and Physiology
Descriptive terms for relative positions of animal structures, the skeletal system (osteology)- axial and appendicular skeletons, joints of the skeleton (Arthrology), muscles (myology), digestive system of poultry, reproductive organs of the male and female, circulatory system, respiratory system, excretory system, nervous system and the endocrine system; Homoeostasis, blood, lymph and other body fluids-composition and their function; Blood circulation, cardiac cycle; Internal and external respiration; Physiology of kidneys: the nephron, rennin-angiotensin systems, acid-base balance; Reproductive physiology of the male and female. Characteristics and functions of the nervous systems.

HAWS 132 Introduction to Biochemistry
Energy flows in biological systems; Structure of cells; Stereochemistry: positional, geometrical and optical isomerism; Chemistry of carbohydrates: classification, structure, functions, mobilization and transport; Chemistry of proteins: amino acids their classification, structure, functions, mobilization and transport; Chemistry and functions of nucleic acids; and DNA and RNA, their functions and roles in life processes.

HAWS 133 Introduction to Microbiology
Characteristics of eukaryotic and prokaryotic cells: micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; Classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens); Use of micro-organisms in industrial microbiology: cheese, beer, wine making, and in silage making; Immunology; Techniques of isolating and identifying micro-organisms; and Microscopy.

AGRO 132 Introduction to Genetics
Cell division: mitosis and meiosis; Nucleic acids, replication and protein synthesis; Mendelian principles: segregation and independent assortment. Environmental effects and gene expression; Gene interaction and lethality; sex determination and sex linkage; Gene mutation and induced genetic change (biotechnology); Gene structure and regulation; and Population genetics, genetic biodiversity of agriculturally important plants and animals and conservation of genetic resources. Computing Relationships and the inbreeding coefficient, uses of inbreeding. Quantitative genetics: the quantitative traits model, components of phenotypic and genetic variance and their estimation, genotype value, additive genetic or breeding value, dominance genetic value, epistemic genetic value, influence of selection, influence of inbreeding, the selection response, heritability and repeatability, genotype x environment interaction, phenotypic, genetic and environmental correlations.

AGRO 134 Introduction to Statistics
Fundamentals of basic statistics and probability theory. Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of

central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean; Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; Tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; and Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA.

HAWS 135 Introduction to Entomology and Parasitology
Position of insects in the world: reproductive capacity, beneficial and injurious insects. Insect external anatomy: the exoskeleton, ecdysis. Internal anatomy and physiology: The nervous system-sight, sound and vibration detection, chemical communication (pheromones), the digestive system, locomotion, insect reproduction, metamorphosis; Insect classification, life cycles, population ecology, population monitoring and economic decision making; Mites, ticks and spiders; Nematodes-morphology, classification, reproduction and economic importance.

HAWS 231 Statistical Methods and Experimental Designs
Module will introduce students to the application of statistics in animal and wildlife research by covering the following aspects: The Whys and Wherefores of Statistics; Descriptive statistics; Probability and Probability distribution; Sampling and sampling Distribution; Experimental and Quasi Experimental Designs; Hypothesis Testing (t-test, F-test, Chi-squared test); Linear Correlation and regression; Further Regression Analysis; Non-parametric statistical methods; Further Aspects of design and Analysis; Additional techniques (Bioassay and Diagnostic Analysis).Data entry, analysis and interpretation using statistical packages(STATA, GENSTAT, R and EPI Info) for all designs: Regression and Correlation. Fitting statistics in research proposal and dissertation write-ups.

HAWS 250 Poultry Breeding
Origin and history of poultry species. Poultry genetic resources: evolution, diversity, conservation. Qualitative genetics. Mutations and major variants in poultry. Cytology and cytogenetics. Pleiotropic and associated effects of major genes. Linkage relationships and gene mapping. Immunogenetics, Molecular genetics, genetic engineering. Quantitative genetics and selection. Genetics of growth and meat production in poultry. Genetics of egg production. Industrial breeding and selection

HAWS 251 Poultry Diseases and Control
Definition, symptoms and signs of diseases in poultry. Management of hygienic conditions. Importance of sanitation. Predisposing factors of diseases. Common diseases of poultry and their causative agents – symptoms and treatments. Viral, bacterial, fungal, parasitic and protozoan diseases. Drugs and antibiotics

HAWS 252 Poultry Nutrition
Importance of nutrition in poultry production. Classification and chemical value of poultry feeds, feed additives; Chemical composition of feeds; biochemistry of feed constituents; Analysis of feeds- proximate analysis scheme, Van Soest and Moore systems; Stock-feed manufacturing and legislation in the poultry feed industry, improving forage feed value; Digestibility, feed efficiency. Biochemical theories of voluntary feed intake; Kinetics of digestion and passage ; and Metabolic functions and roles of micronutrients (vitamins and minerals) in poultry. Nutrient requirements of different types of poultry. Feed formulation

HAWS 253 Poultry Immunology and Vaccination
Avian immune system, innate and acquired immune systems. Antigens- determinants of antigenicity, biological classes of antigens. Primary and secondary lymphoid organs. Production and role of different types of antibodies, IgM, IgA, IgG, Bile. Complement system. Immunodeficiency diseases. Live and killed vaccines. Vaccination programmes for poultry. Vaccination procedures.

HAWS 254 Poultry Housing and Equipment
Importance and principles of poultry housing. Housing considerations for different seasons. Heat loss and heat production in poultry. Methods of housing. Floor space requirements. Brooder house equipment. Grower house equipment. Layer house equipment. Water equipment, feeding equipment, Heaters and brooders, Incubator, chick box, egg scale, egg washer

Poultry Entrepreneurship
Economics of egg production, economics of broiler production, economics of indigenous (backyard) poultry production. Poultry farm records, inventory receipts and expenditure. Pricing, marketing, advertising. Government policy on poultry products. Taxation. Poultry insurance and financial management.

HAWS 450 Hatchery Management
Hatchery design, incubators and hatchers. Incubation basics, fertilised egg, temperature and humidity control, ventilation. Breeding stock selection and management. Egg handling – picking, grading, storage and transportation. Incubator settings. Egg turning. Feather sexing broiler chicks. Hatchery sanitation. Hatchery waste disposal. Record keeping

HAWS 451 Broiler Production
Housing Design. Stocking density. Insulation. Brooding Chamber. Equipment. Feeding Systems. Heating Systems. Ventilation Systems. House Preparation. Litter Management. Important Functions of Litter. Litter Alternatives. Minimum Litter Requirements. Chick Quality. Brooding Management. Internal chick temperature. Growing Phase. Lighting Programs. Ventilation Management. Management Techniques in Hot Conditions. Water Management. Water Sanitation and System Cleanout. Nutrition Management. Bio-Security and Farm Sanitation. Bird Health. Vaccination. Record Keeping

HAWS 453 Indigenous Poultry Management
Breeds. Selecting breeding stock and housing. Laying nests, Management practices – Feeding, Collecting the eggs, Hatching, Brooding, Heat Sources, Brooder Preparation, Rearing chicks, Scavenging, Record keeping, Diseases, Vaccination, Biosecurity.

HAWS 454 Layer Production
Brooding Management. Lighting During Brooding Period. Drinking Systems. Beak Treatment / Trimming. Growth and Development. Feed Consumption and Uniformity. Space requirements. Laying House. Vaccination programme. Lighting Practices. Light Program for Light-Controlled Housing. Lighting Programs for Open-Sided Housing. Heat Stress. Water Quality. Air Quality. Nutrition Management. Disease Control. Egg Standards. Egg Size Distribution. Moulting.

HAWS 455 Duck, Turkey and Game-birds Production
Duck – Breeds and varieties. Duck production systems, Feeds and Feeding, water requirements, Nutrient requirements. Management of Ducks- Brooder, Grower and Layer Ducks. Duck egg incubation. Diseases. Turkey -Biosecurity -Hygiene procedures, Vehicles and equipment, Birds,

rodents, insects and mammals. Cleaning and disinfection. Growing systems. Brooding and growing management. Growing. Water. feeding and Nutrition-Feed form, Addition of whole grain, Addition of insoluble grit, Feed composition, Feed management. Weighing. Marketing. Game birds - Guinea fowl, pheasant, squab (or pigeon) and quail management

HAWS 457 Poultry Welfare
Poultry welfare and ethics – Diseases and stress affecting poultry welfare. Recent advances in genetics impacting on poultry welfare. Welfare issues in husbandry issues and transport- Handling, care and treatment of poultry in confinement and extensive systems.

HAWS 439 Research Project
Explore research issues in poultry, research concept development, research proposal writing, data collection and organization, data analyses and statistical indication, data interpretation and presentation, prototyping; presentation of research findings; research project write up.

HAWS 458 Environmental Management in Poultry Production
Poultry litter materials. Poultry farm pollution. Value of poultry manure. Moisture management methods. Litter reutilisation. manure storage within and outside of the poultry house (including solids systems and liquid, slurry, and semisolids systems); waste treatment and utilization (e.g., composting, anaerobic lagoons, aerobic digestion, and direct incineration); nutrient management; dead-bird management (rendering, composting, and incineration); and alternative uses for manure.
Poultry and Egg Processing
Characteristics of Poultry, Fat content, Microbial contamination. Slaughter procedures – preslaughter handling, slaughtering, chilling. Processing of poultry – raw poultry products. Processed poultry products-battering and breading, tumbling and massaging, smoking, deboning and grinding. Egg processing

HAWS 459 Poultry Products and Technology
Poultry meat and egg quality identification and quality maintenance. Chemical and nutritive characteristics. Measuring yields and characteristics. Packaging and refrigerated storage. Canning poultry products and processing soups. Preservation by radiation. Cooking and barbecuing poultry. Inedible by-products. Eggs: physical, chemical, nutritional, and functional characteristics. Methods of analyses of eggs and egg products

HAWS 460 Marketing of Poultry Products
Definition. Marketing strategies of poultry and eggs. Importance of marketing. Direct marketing, Marketing channels. Marketing intermediaries. Evaluation of the marketing channel.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN ANIMAL BREEDING AND BIOTECHNOLOGY (MABB)

Duration	:	2 years
Credit Load	:	270
Minimum Credit Load	:	330
Maximum Credit Load	:	234
Total MBKS Credit Load	:	270
Level	:	ZNQF-LEVEL 9

1. PURPOSE OF THE PROGRAMME

The aim of the programme is to produce a horticultural graduate who can think outside the box and is not restrained by conventional wisdom and practices but rather delves into the less travelled paths to create new value in the form of products and services within the confines of horticultural sciences.

2. ENTRY REQUIREMENTS

To be admitted to this programme a prospective student should satisfy the following conditions:

- Have an Honours degree (class 2.2/GPA 2.0 or better) in Agriculture or the biological sciences in the following disciplines; Animal Science, Biological Science, Biotechnology and Natural Resources, or related fields from an accredited University. Candidates with a third class will be required to attend an interview to assess their competence.
- Have a General degree in Animal Science, Biological Science, Biotechnology, and Natural Resources plus a Postgraduate Diploma in Agricultural Studies from an accredited institution of higher learning.

3. PROGRAMME CHARACTERISTICS

Areas of study: Applied population and quantitative genetics, molecular biology, genomics

Specialist Focus: Applied Animal Breeding

Orientation Oriented towards innovation and application of theory to solve real life challenges

Distinctive Features: Applied Animal Breeding and Biotechnology

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: After completing the programme, graduates should be able to:

- Create their own business in animal breeding.
- Work as animal breeders, biotechnologists and farm managers.Work as researchers in public and private institutions.
- Work as consultants in Animal Breeding Industry.
- Work as Lecturers at colleges and universities.
- Work in Non-Governmental Organisations.
-

Further Studies: Doctoral studies in Animal Breeding / Biotechnology

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/field visits, dissertation

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, dissertation. Continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- On completion of the programme the graduate should be able to:
- Exhibit in-depth knowledge of animal breeding and biotechnology required to formulate breeding programmes.
- Demonstrate advanced techniques used in enhancing animal reproduction.
- Show the ability to use advanced computational approaches to analyze big data.
- Apply biotechnology to enhance the reproduction and growth of farm animals.
- Integrate knowledge and research skills to address challenges facing agriculture through commercializing research and innovations.
- Demonstrate the ability to connect theory, research, and practice in problem-solving.
- Effectively present research work to a broad spectrum of audiences using various platforms and media.
- Practice good research, animal care, and use ethics

7. GENERAL PROVISIONS

7.1. The programme shall normally be studied over a minimum of four semesters on a block release basis.

7.2. This programme is a taught Masters programme with modular work in the first two semesters and research and dissertation writing in the third and fourth semesters.

7.3. Continuous assessment consists of lectures, presentations, seminars and tests.

7.4. Lectures and taught modules shall normally be at the Midlands State University

7.5. The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.

7.6. Each student has to identify at least two supervisors who shall be approved by the Department of Agricultural Economics and Development.

7.7. There shall be oral presentations for the Project Proposal and Research Findings. Weighting of these, together with the final written document, shall be as follows:

- Dissertation Proposal 15%
- Presentation of Research Findings 15%
- Final Written Document 70%

7.8. Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

7.9. Candidates shall be expected to submit at least three executive bound copies of the dissertation

7.10. The format of the dissertation shall be provided for by the Department of Agricultural Economics and Development.

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, seminar presentations.

8.2. Formal examinations shall be held at the end of each semester.

8.3. Continuous assessment shall account for 40 % of the overall assessment, while the examination shall account for 60 % of the overall assessment.

8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 60 credits in the first level in order to proceed to the second level (dissertation).

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

The degree programme shall have at least 32 taught modules spread over the six teaching semesters.

Level 1 Semester 1		**Prerequisites**	**Credits**
*MABB 701	Biometry for Animal Science		18
*MABB 702	Reproductive Anatomy and Physiology		18
*MABB 703	Molecular Biotechnology		18
*MABB 704	Animal Genetics and Genetic Resource Conservation		18
*MABB 705	Population and quantitative genetics		18
*MABB 706	Seminar (Research Ethics, Biosecurity and Animal Welfare)		18
Level 1 Semester 2			
*MABB711	Animal Biotechnology	MABB 703	18
*MABB712	Bioinformatics and Proteomics	MABB 701	18
*MABB713	Microbial and Pharmaceutical Biotechnology	MABB 703	18
*MABB 14	Genomics	MABB704/705	18
*MABB715	Applied Animal Breeding	MABB 705	18
Level 2 Semester 1 & 2			
*MABB 800	Dissertation	MABB701/706	180

14. MODULE SYNOPSES

MABB701 Biometry for Animal Science

This module will include the following concepts in research methods; research process and research planning, site selection, data collection, statistical and data management software, sampling, and surveys. It will also include the use of statistical software (R, SAS, GEN STAT) in dealing with the following concepts: statistical inference; one-way and two-way analysis of variance; experimental design: single factor, multifactorial experiments. It will also cover fixed, random, and mixed models. Nested experiments and split-plot design; confounding systems.

MABB702 Rreproductive Anatomy and Physiology

The module covers the structure and function of the reproductive systems of farm animals in detail. It also covers the effects of abnormalities of the reproductive systems on animal health and productivity possible solutions. Enhancing the success of reproduction through techniques such as synchronizing oestrus, multiple ovulation and embryo transfer (MOET), and artificial insemination (AI) is emphasized. The use of hormones in enhancing animal growth and development are covered. Practicals include

reproductive tract evaluations, semen quality analysis, oestrus synchronization protocols, and estimating pregnancy stages through ultrasound.

MABB703 Molecular Biotechnology
The module covers replication, transcription, translation and cell function. The interaction between deoxyribonucleic acid (DNA), ribonucleic acid (RNA), and proteins; mechanisms regulating the interactions are covered. Application of molecular techniques is emphasized, including polymerase chain reaction (PCR), gel electrophoresis, macromolecule blotting and probing, microarrays, and allele-specific oligonucleotides. Effects of naturally and artificially induced mutations in micro molecules on animal products, health, and welfare are covered.

MABB704 Animal Genetics and Genetic Resource Conservation
The module covers the genomes of ruminants, poultry, and pigs. It also covers web-based genome browsers and databases (EMBL and NCBI). The course also explores methods used to study genomes, transcriptomes, and proteomes: high throughput microarray technologies for sequence variation and expression, tiling arrays, next-generation sequencing, quantitative trait loci (QTL) mapping. Genetic diversity based on molecular data and genetic variability within and between indigenous livestock is covered. *In situ* and *in vivo* conservation of genetic resources is discussed. Practicals involve phylogenetic analysis using computational approaches, morphological and molecular characterization of livestock.

MABB705 Population and Quantitative Genetics
The module covers Mendelian variation, Hardy-Weinberg law, changes in population gene frequency: migration, mutation, and selection. It also covers genetic parameters of quantitative traits: heritability, variance, breeding value, selection response, inbreeding coefficient, correlation coefficient. Methods of mapping quantitative trait loci (QTL): marker loci, QTL genotypes, single-marker analysis, and interval mapping analysis are discussed. Practicals include estimation of (co)variance components and resulting genetic parameters using data sets from livestock improvement programs.

MABB706 Seminar on Research Ethics, Biosecurity and Animal Welfare
Moral principles in the planning, conducting, analyzing, and reporting research work are discussed. Social issues that arise when dealing with human and animal subjects; the potential legal, social, and political issues, views on the issue of genetically modified organisms are also covered. The security measures required in conducting research using potentially harmful reagents and microbes are discussed. Focus is also on animal welfare with the aim of allowing normal animal behavior, minimizing stress, and providing hygienic environments. The impact of animal welfare on productivity, health, and end products is discussed. Students will be introduced to the Prevention of Animal Cruelty Act and World Organisation for Animal Health (OIE) guidelines on animal welfare.

MABB711 Animal Biotechnology
The module covers gene cloning, restriction endonucleases; cloning vectors: methods of gene transfer, chromosome walking; site-specific mutagenesis; potentials of recombinant DNA technology. Animal models in research, genetic manipulation in animals, and reproductive cloning in animals are also covered. Aspects of production of genetically modified animals including stages of production, advantages, and disadvantages of genetically modified organisms will be dealt with. Facilitated reproduction (*in vitro* fertilization), production of human proteins in genetically modified animals is looked into. Gene therapy, biotechnology applications in the diagnosis and treatment of diseases, biosafety, and bioethical issues of biotechnology will be covered.

MABB712 Bioinformatics and Proteomics
The module covers sequence analysis: DNA sequencing, sequence assembly, genome annotation, and computational evolutionary biology; analysis of gene and protein expression; microscopy and image analysis: protein localization and chromosome topology. Protein structure prediction and network system biology are dealt with. Post-translational modification of proteins; phosphorylation, ubiquitination; high-

throughput proteomic technologies: mass spectrometry and protein profiling, and reverse-phased protein microarrays are covered. Practicals include the collection of proteomic data using mass spectrometry, microarrays, and using computational approaches to analyze the data.

MABB713 Microbial and Pharmaceutical Biotechnology
The module covers the diversity of microbes, and their potential in microbial and pharmaceutical biotechnology as well as microbial gene and genome structure and function. Bacteriophage structure and biology; transduction particles; transposable elements, virulent and virulent phages, and microbial metabolism and their manipulation are dealt with. The module explores microbial culture preservation methods such as refrigeration, paraffin method, cryopreservation, and lyophilization. Production of heterologous proteins (e.g. insulin and recombinant porcine and bovine somatotropins) using microorganisms is also covered.

MABB714 Genomics
The module covers genomes of domestic animal species and the use of databases to obtain molecular data related to animal breeding and health. Current laboratory techniques used to obtain segments of genetic information and computational approaches in genome mapping are included. The module integrates genome information, biotechnology, and bioinformatics to uncover the molecular basis of complex genetic traits of importance to animal breeders. It also looks into the genomic selection of morphometric traits utilizing genetic markers spanning the whole genome. The use of molecular genetics in phylogenetic and evolutionary process analysis of different populations and nutrigenomics is emphasized.

MABB715 Applied Animal Breeding
The module covers the application of genetics (Mendelian, population, quantitative and molecular genetics) in the improvement of livestock productivity. It looks into the evaluation of traits of economic importance in livestock (e.g. yield estimate, growth rate, disease resistance, body parameters, meat, and milk quality) and pedigree analysis. Genetic improvement methods are covered, that is, crossbreeding and inbreeding. Application of multiple ovulation and embryo transfer, and artificial insemination. The module looks into the economic evaluation of different breeding plans. Practicals include developing breeding programs that are climate-smart and account for climate change.

MABB800 Research Project
In this module students are expected to review published literature, summarise and give an overview of methodologies used in consonance with their intended area of research. Students are expected to demonstrate the ability to undertake original research in animal breeding or biotechnology through: formulating a hypothesis, designing an experiment, gathering data, statistically analyzing the data, scientific writing, and preparing a manuscript. Emphasis will be on commercializing innovations and research. The oral presentations of the proposal, interim results, and final defense of their research findings are required. The departmental board shall ensure that the student has sufficient supervision. Where a student is permitted to carry out their research work in another institution or country, an Associate Supervisor in that institution or country will be required.

DEPARTMENT OF LAND AND WATER RESOURCES MANAGEMENT

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN BIOSYSTEMS ENGINEERING (HBE)

Duration:	4years
Minimum Credit Load:	480
Maximum Credit Load:	540
Total MBKS Credit Load:	384
ZNQF Level:	8

1. PURPOSE OF THE PROGRAMME

The aim of this programme is to contribute towards the raising of standards of living of Zimbabweans, and other citizens of the world directly or indirectly, through teaching, research, development and application of sustainable technical skills in biosystems engineering.

2. ENTRY REQUIREMENTS

Normal Entry

▪ At least five 'O' level subjects including English Language, Mathematics and Science plus: Advanced Level passes in Mathematics or Physics and any one of the following or their recognized equivalents: Biology, Agriculture, Chemistry, Geography, Engineering Drawing

▪ Articulation as provided by the ZNQF

▪ Recognition criteria for foreign credentials of international students

Special Entry

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the program.

3. PROGRAMME CHARACTERISTICS

3.1 Biosystems engineering provides an essential link between the biological sciences and the engineering profession. This linkage is necessary for the development of production and processing systems involving biological materials that preserve our natural resource base. The programme gives students the latitude to do either an Open option or develop an area of specialization relating to Bio-environmental engineering, Food engineering or Bio-renewable resources engineering.

3.2 The program is oriented towards research, innovation and application of theory to solve practical problems in the sustainable production and processing of biological materials.

3.3 Knowledge and skills development is a key feature of the programme.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: On successful completion of the degree programme, graduates can be formally employed in the private or public sector. They may also be self - employed. The following career prospects are available to graduates:

- Industry – as designers, research officers, technicians, field and sales officers.

- Government Departments, Parastatals, Local Authorities and Municipalities – as lecturers, technical, extension or research officers, water resource managers, land use planners, laboratory officers, environmental officers, Catchment management officers.
- Agricultural enterprises – as managers, technical and development officers and consultants.
- Self-employment – as farmers, contractors, entrepreneurs, and consultancy.
  Research and Academia - Graduates can also continue into higher degree programmes, which provide a basis for research and academic careers.

Further Studies: Students who excel in this degree programme can advance to do Masters and Doctoral studies in areas such as environmental engineering, biotechnology, food processing, machine systems, or controlled environment engineering.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/filed visits, work-related learning, dissertations,

Assessment Methods: Written and oral examinations, tests, quiz, group presentations, work-related learning reports, final year dissertations, continuous assessment (30%), Final exam (70%)

6. INTENDED LEARNING OUTCOMES

- Understand and apply key principles and theories in biosystems engineering.
- Understand contemporary issues and challenges in biosystems engineering and work towards solutions.
- Apply knowledge and practical skills in sustainable production and processing of biological materials.
- Research and innovate in biosystems engineering related issues and promote knowledge transfer
- Create new business ventures in biosystems engineering related fields

7. GENERAL PROVISIONS

7.1. The BSc in Biosystems Engineering is a four year full-time programme.

7.2. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.3. A student shall be enrolled for at least five modules at any time during the semester, two of which should be core/compulsory.

7.4.Students can include in their programme of study, modules offered by other departments.

7.5.Students may be allowed to change registration for modules taken in any semester only during the first two weeks of registration.

8. PROGRAME ASSESSMENT

Refer to Section 6 of General Regulations and Section 5 of Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

13.1. The degree programme shall have at least 36 taught modules spread over the six teaching semesters.

Module code	Module Narration	Credits
	Level I	
BSE 102	Engineering Graphics and Introductory Design	12
BSE 103	Systematic Problem Solving and Computer Programming	12
BSE 104	Chemistry for Engineering	12
BSE 105	Calculus	12
BSE 106	Fundamentals of Biosystems Engineering	12
BSE 108	Organic Chemistry I	12
BSE 110	Physics for Engineering	12
Level II		
BSE 201	Project Management & Design in Biosystems Engineering	12
BSE 204	Elementary Differential Equations and LaplaceTransforms	12
BSE 205	Applied Numerical Methods for Biosystems Engineering	12
BSE 206	Mechanics of Fluids	12
BSE 208	Engineering Thermodynamics	12
BSE 209	Biology of Microorganisms	12
Level III		
Level 3.1	**Work-Related Learning**	
AGPR330	Work-Related Preliminary Report	30
Level 3.2	**Work-Related Learning**	
AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45
Level IV		
BBSE403	Instrumentation for Biosystems Engineering	12
BSE 401	Biosystems Engineering Design	12
BSE 405	Engineering Analysis of Biological Systems	12
BSE 400	Design Project	24
BSE 408	Agri-Industrial Applications of Electric Power and Electronics	12
BSE 413	Engineering Economic Analysis	12

MODULE SYNOPSES

BSE 102 Engineering Graphics and Introductory Design

Applications of multi-view drawings and dimensioning. Techniques for visualizing, analyzing, and communicating 3-D geometries. Application of the design process including written and oral reports

BSE103 Systematic Problem Solving and Computer Programming
Introduction to algorithmic thinking; use of spreadsheet programs and computer programming language(s) to solve engineering problems

BSE104 Chemistry for Engineering
The module covers basic concepts of chemistry, atomic structure, chemical bonding, thermodynamics, chemical equilibrium, acids and bases, electrochemistry, and chemical kinetics. The course is designed to provide students with a fundamental understanding of chemistry that is relevant to engineering applications

BSE105 Calculus
Differential calculus, applications of the derivative, introduction to integral calculus. Integral calculus, applications of the integral, parametric curves and polar coordinates, power series and Taylor series.

BSE106 Fundamentals of Biosystems Engineering
Application of mathematics and engineering sciences in agricultural and biosystems engineering. Emphasis is on solving engineering problems.

BSE108 Organic Chemistry I
A survey of modern organic chemistry including nomenclature, structure and bonding, and reactions of hydrocarbons and important classes of natural and synthetic organic compounds.

BSE110 Physics for Engineering
Elementary mechanics including kinematics and dynamics of particles, work and energy, linear and angular momentum, conservation laws, rotational motion, oscillations, gravitation. Heat, thermodynamics, kinetic theory of gases; waves and sound.

BSE201 Project Management & Design in Biosystems Engineering
Engineering design process with emphasis on criteria and constraints, ideation, and analysis. Fundamental principles of project management including project management software. Open-ended project(s) to apply core principles using concepts from prerequisite courses.

BSE204 Elementary Differential Equations and Laplace Transforms
Solution methods for ordinary differential equations. First order equations, linear equations, constant coefficient equations. Eigenvalue methods for systems of first order linear equations. Introduction to stability and phase plane analysis including Laplace transforms and power series solutions to ordinary differential equations.

BSE205 Applied Numerical Methods for Biosystems Engineering
Computer aided solution of engineering problems by use of numerical techniques and mathematical models. Systems analysis and optimization applicable to agricultural and biological systems.

BSE206 Mechanics of Fluids
Properties of fluids. Fluid statics. Kinematics and kinetics of fluid flow. Mass, momentum, and energy conservation laws; dimensional analysis; flow in pipes and channels. Selected laboratory experiments.

BSE208 Engineering Thermodynamics
Fundamental concepts based on zeroth, first and second laws of thermodynamics. Properties and processes for ideal gases and solid-liquid-vapor phases of pure substances. Applications to vapor power cycles.

BSE209 Biology of Microorganisms
Basic cell biology, physiology, metabolism, genetics and ecology of microorganisms, with an emphasis on prokaryotes and viruses, as well as the roles of microorganisms in the environment, disease, agriculture, and industry.

BSE403 Instrumentation for Biosystems Engineering
Interfacing techniques for computer-based data acquisition and control systems. Basic interfacing components including A/D and D/A conversion, signal filtering and process control. Sensors and theory of operation applied to practical monitoring and control problems. Data collection, analysis, and calibration of sensors and data acquisition systems. Individual project required for graduate credit.

BSE401 Biosystems Engineering Design
Engineering design process with emphasis on team delivery of: clearly defined deliverables; criteria and constraints; wide-field ideation; discipline-appropriate analysis methods; identification and application of relevant standards. Final execution of the engineering design process with emphasis on team delivery of: oral and written communication in completion of the client-agreed deliverables.

BSE405 Engineering Analysis of Biological Systems
Systems-level quantitative analysis of various biological systems, including applications in foods, feeds, biofuels, bioenergy, and other bio-based systems. Introduction to techno-economic analysis and life-cycle assessment of these systems at multiple production scales. Applying these tools to evaluate and improve cost and sustainability performance.

BSE400 Design Project
Each student is expected to carry out a design project that culminates in the submission of a dissertation. The project should be carried out on biosystems engineering related areas.

BSE408 Agri-Industrial Applications of Electric Power and Electronics
Single phase and three phase circuit design. Electrical safety. Electric motors and controls. Programmable logic controllers. Digital logic, instrumentation and sensors.

BSE413 Engineering Economic Analysis
Economic analysis of engineering decisions under uncertainty. Financial engineering basics including time value of money, cash flow estimation, and asset evaluation. Make versus buy decisions. Comparison of project alternatives accounting for taxation, depreciation, inflation, and risk.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN ENVIRONMENTAL SCIENCE AND TECHNOLOGY (HEST)

Duration:	4 years
Minimum Credit Load:	480
Maximum Credit Load	540
Total MBKS Credit Load:	384
ZNQF Level:	8

1. PROGRAM OBJECTIVES

The aim of this program is to contribute towards improved environ**me**ntal quality at regional, national and global scales through teaching, research and by developing appropriate technologies to identify, control, or eliminate sources of pollutants affecting the environment and public health. The department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical knowledge and practical skills in environmental science and technology.
b. Producing competitive innovative graduates to take up or create opportunities in industry, government departments and the non-governmental organizations including small and medium scale enterprises or own initiatives. The graduates will be equipped with essential technical knowledge and skills for conducting research including: Selecting data collection methods; collecting and analysing air, water and soil samples; analyzing environmental data collected by others; analysing data for correlations to anthropogenic effects; and with skills in advocacy, problem solving, report writing and presentation.
c. Departmental participation in societal development through research and outreach programs
d. Collaborating with industry and other relevant stakeholders.

2. ENTRY REQUIREMENTS

Normal Entry

- At least five 'O' level subjects including English Language, Mathematics and Science plus: Advanced Level passes in any two of the following subjects or their recognized equivalents: Biology, Chemistry, Mathematics and Geography or a relevant National Certificate from an accredited Institution
- Articulation as provided by the ZNQF
- Recognition criteria for foreign credentials of international students

Special Entry

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the program.

3. PROGRAM CHARACTERISTICS

3.1. Broad areas of study include understanding components of the physical environment, environmental chemistry; soil, air and water pollution, soil biology and biochemistry science; solid waste management, waste water treatment, Climate change and variability;
3.2. Land use planning; Soil and water conservation; GIS & Remote sensing; Water resources management; Watershed management; Surveying; techniques and technologies for environmental science.
3.3. The program is oriented towards research, innovation and application of scientific knowledge and skills to prevent, identify, eliminate or control environmental problems u see usually associated with anthropogenic activities.
2.1 Knowledge and skills development is a key feature of the program

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

On successful completion of the degree program, graduates can be formally employed in the private or public sector. They may also be self - employed. The following career prospects are available to graduates:

- ***Industry*** – as designers, research officers, technicians, field and sales officers.
- ***Government Departments, Parastatals, Local Authorities and Municipalities*** – as lecturers, technical, extension or research officers, water resource managers, land use planners, laboratory officers, environmental officers, Catchment management officers.
- ***Agricultural enterprises*** – as environment specialist or managers, technical and environmental officers and consultants.
- ***Self-employment*** – environment analysts, contractors, entrepreneurs, and consultants.

Research and Academia - Graduates can also continue into higher degree programs, which provide a basis for research and academic careers.

Students who excel in this degree program can advance to do Masters and Doctoral studies in Environmental management, water resources management; water quality management; climate change and variability, Soil and water conservation, natural resources management and land use planning.

5. PROGRAM DELIVERY

5.1. Teaching and Learning Methods:

Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, research projects**.**

5.2. Assessment Methods:

▪ ***Formative***

− Coursework comprising written tasks, presentations, demonstrations, practice, production

▪ ***Summative***

− Written examinations, practical examinations in appropriate modules, research project report

6. LEARNING OUTCOMES

On completion of this program a graduate will be able to:

- Understand trends, processes, impacts that shape the Earth systems.
- Develop frameworks suitable for designing technological and sustainable environmental interventions for the mitigation of anthropogenic effects on the environment.
- Communicate effectively, orally and in writing, with audiences of diverse backgrounds.
- Work effectively with, and within, interdisciplinary and diverse groups to resolve environmental problems and achieve environmental management objectives.
- Research and innovate in environment management-related technologies and practices and promote knowledge transfer.
- Conduct an environmental management study, and interpret outcomes

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.2. Students can include in their program of study, modules offered by other departments.

7.3. A student shall be enrolled for at least five modules at any time during the semester

7.4. The module combination should be feasible in terms of the timetable.

7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAM ASSESSMENT

Continuous assessment*:*	Continuous assessment shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30% of the final mark.
Written Examinations:	A three-hour final examination, which shall contribute 70% of the final mark.
Practical Examination:	The duration of a practical exams, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 9 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

In the first year, learners are expected to acquaint themselves with introductory modules of the degree program. Choice of modules should be done with guidance of Departmental staff.

The degree program shall have at least 36 taught modules spread over the 6 teaching semesters.

MODULE CODE	MODULE NAME	CREDITS
Level 1Semester 1		
EST101	Introduction to Environmental Science	12
AGRO135	Plant Biology	12
LWR135	Introduction to Soil Science	12
HORT131	Introduction to Microbiology	12
CS131	Communication Skills	12
HCS135	Introduction to Information Technology	12
Level 1 Semester 2		
AGRO136	Introduction to Statistics	12
LWR131	Environmental Chemistry	12
HAWS134	Principles of Ecology	12
EST103	Air Pollution	12

EST102	Introduction to Hydrology	12
Level 2 Semester 1		
ENT131	Entrepreneurship	12
AGRO239	Statistical Methods and Experimental Designs	12
EST205	Heritage Based Natural Resources Management	12
EST206	Soil Chemistry	12
AEGS231	Introduction to Gender Studies	
LWR237	Geographical Information Systems and Remote Sensing	12
EST203	Environmental and Social Impact Assessment	12
EST204	Environmental Monitoring and Evaluation	12
Level 2 Semester 2		
EST201	Risk and Disaster management	12
LWR245	Water Quality	12
LWR246	Climate Change Adaptation and Mitigation	12
EST202	Soil Biology and Biochemistry	12
Level 3 Semester 1		
AGPR330	Work-Related Preliminary Report	30
Level 3 Semester 2		
AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45
Level 4 Semester 1		
EST401	Environmental Economics	12
EST403	Environmental Biotechnology and Bioremediation	12
EST404	Environmental Law	12
EST405	Life Cycle Assessment and Sustainability	12
EST406	Waste water treatment	12
EST409	Solid & Hazards Waste management	12
Level 4 Semester 2		
EST407	Environmental Auditing	12
EST402	Environmental Education and Extension	12
EST408	Environmental Management Systems	12
EST410	Environmental Modeling	12
EST400	Research Project	24

14. MODULE SYNOPSES

EST101 Introduction to Environmental Science

The module will cover basic concepts, history, nature and scope of Environmental Science and its contribution to society. Different aspects of the environment: physical, ecological, socio-economic, ethical, and philosophical. Major components of environment: physico-chemical, biological and social, and their relationships with various environmental factors. Human environment and its problems: global, national, regional. Environmental challenges for sustainable development:

current and future trends in population growth, environmental pollution, development in industry and agriculture, urbanization, poverty and resource depletion

AGRO136 Introduction to Statistics
The module covers an introduction to statistics definition and uses of statistics in research; Data types, presentation and summarization techniques; Probability and distributions: random variables, discrete, continuous, binomial distribution, and normal distribution; Measures of central tendency and dispersion; Sampling techniques; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA

LWR135 Introduction to Soil Science
The content of the module will include- Soil formation, Soil minerals, Soil classification, Soil texture and structure in relation to tillage, Soil colour, Other physical properties of soil – e.g. drainage, permeability, infiltration, Land capability classification, Soil profiling, Soil sampling.

HORT131 Introduction to Microbiology
The module will cover Characteristics of eukaryotic and prokaryotic cells and a range of micro-organisms viz. Bacteria, Fungi, Viruses, Viroid's and Protozoa; their classification, morphology, nutrition, growth, replication and metabolism; Micro-organisms as agents of diseases (pathogens), use of micro-organisms in industrial microbiology (cheese, beer and wine making) and in silage making; Immunology; techniques of isolating and identifying micro-organisms; Microscopy

AGRO135 Plant Biology
The plant cell and organelles; Plant water relations; Photosynthesis – the ecological significance of photosynthesis in primary production of organic molecules, the Light Reaction, The Calvin Cycle, C3/C4 photosynthesis, CAM photosynthesis. Photoperiodism and vernalization. Fruit and seed formation: double fertilization and formation of seed tissues, Seed structure, chemical composition of seed, Germination: Phases in germination: seed hydration, genetic and metabolic events during germination, role of plant growth regulators in germination. Seed dormancy: Mechanisms of dormancy: Innate, induced and enforced dormancy, Significance of dormancy in attuning cycles of plant growth to seasonal changes in environment. Plant growth and development: meristems, cell expansion, cell differentiation, acid growth theory, tropisms, cell differentiation, apical dominance, bud dormancy; Plant growth regulators – synthesis and transport and physiological effects of auxins, gibberellins, cytokines, abscisic acid and ethylene on plant growth and development; Use of artificial plant growth regulators in Agriculture.

AWS134 Principles of Ecology
The module will cover the topics Biological organization; Ecosystem concept; Concept of the Biosphere; interrelationships between biotic and abiotic factors. Energy flow: nutrient cycles in ecosystems. Ecology of communities: Terrestrial Biomes and Aquatic Biomes. Plant and Animal succession. Species diversity: Measures of species diversity; Calculations of species diversity. Wildlife biology: Biological potential; Reproduction factors; Mortality; Ageing criteria; Wildlife

diseases; Types of carrying capacities; Behavioral interaction: Home range; Territorialism; Dispersal; Niche selection; Guilds; Indicator species. Wildlife conservation

AGRO239 Statistical Methods and Experimental Designs

Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; Statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two factor experiments and multi-factor experiments; Multiple comparison of means; Partitioning of sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for goodness of fit and independence; and Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS) for all designs: Regression and Correlation.

LWR237 Geographical Information Systems and Remote Sensing

Introduction to GIS, Software and Hardware, Applications, Sources of GIS data, Global Positioning Systems, Introduction to Remote Sensing, Imagery and Photography, Acquisition and Platforms; the Electromagnetic Spectrum, Spectral Signatures/Pattern of Objects (water, soil, vegetation), NDVI. Raster and Vector, Data Input, Digitizing, Data Quality, Data Output, Geo-databases, Cartography, Map Projections, Geo-processing, Data Queries, Spatial Analysis, Cartographic Modelling

EST202 Soil Biology and Biochemistry

Methods for studying soil organisms in the environment; Components of the soil bio data: Occurrence and distribution in the soil environment; Growth and Metabolism of soil organisms; Carbon Cycling and Soil Organic Matter; Dynamics of residue decomposition; Soil biological nitrogen transformations: Nitrogen mineralization and immobilization; Nitrogen transformations: Denitrification and Nitrogen transport in soil; Symbiotic relations: Biological Nitrogen Fixation (BNF) and Mycorrhizal Symbiosis; Biological Phosphorus and Sulphur transformations in soil; Biological transformation of metals in the soil environment; Soil organisms and novel pollution problems.

EST203 Environmental and Social Impact Assessment

Concepts and issues in environmental management: Objectives of EIAs/EHIAs; terminologies etc. Statutory provisions of EIA in Zimbabwe; Brief on the Environmental and Natural Resources Act, etc. Introduction to the EIA process: Screening; Scoping; Impact identification (forms of impacts, data sources; sources of impacts etc.); Impact assessment (techniques and analysis tools); Migratory measures; EIA report; Draft EIA reviews (assessing the quality of draft Environmental Impact statement etc.); Project implementation; Decommissioning. Case Studies such as EIA of water resources, energy development, mining, paper industry etc.

EST204 Environmental Monitoring and Evaluation

The module will provide students with theoretical and practical knowledge in various aspects of environmental monitoring and an understanding of the role of environmental monitoring in environmental studies; Students develop analytical skills including defining a problem, selecting appropriate variables, and evaluating the data, as well as they can use the appropriate procedures for monitoring various environmental compartments. The course content will include Basics about

monitoring (definition, tasks, scheme, types and errors of monitoring); Monitoring of chemical and physical parameters; Monitoring of biological parameters; Monitoring of air, natural waters, soil, Land treatment; Emission factors (definition, application, source of errors); Importance of sampling; quality (limit of detection, limit of quantification, linear range, precision, trueness, recovery, selectivity); Equipment for monitoring and Legislation in the field of monitoring

LWR246 Climate Change Adaptation and Mitigation

Principles of meteorology: Introduction, Meteorological measurements, Meteorological terms, the sun and the earth, and the earth's atmosphere. **Radiative transfer:** electromagnetic spectrum, radiation laws, solar radiation; atmospheric influence on radiation, atmospheric windows, Beer's law, earth's energy budget, radiation balance. **Climate variability:** climate variability processes, drivers – externally induced and internally induced (variations in solar radiation, lunar tides, interaction of the atmosphere with oceans and biosphere) **Climate change:** definitions, causes, the science of climate change, impacts, adaptations and mitigation

EST400 Research project

Scientific report on a research project undertaken by the student during the work-related learning period. Assessed through a proposal presentation and a final presentation as well as a final written document.

EST401 Environmental Economics

Basic economic principles; Scope, nature and methodology of economics; Economics systems: traditional, planned/command; Liaise Faire enterprise/free market, mixed; Theory of price determination: definition of a market, supply and demand. Relationship between society and economic development, public and private choices, and their impact on the environment. Theories of development economics. The factors of economic growth: land labour, capital and entrepreneurship. Gross National Product; Economic growth and the environment; Paying for the privilege to pollute; Measurement of cost-benefit analysis (CBA). Comparison of cost-benefit analysis and Environmental Impact Assessment (EIA) as tools for appraisal. Linkage between economic planning, development, environmental management and legislature. Risks, uncertainties, and impact on resource use**.**

EST402 Environmental Education and Extension

Defining the concept environment; Environmental problems, crises and emerging responses; From Stockholm to Rio: Stockholm Declaration, Tbilisi Principles, Sustainable development, Rio Declaration and Agenda 21 (particularly Chapter 36); What is environmental education; Environmental and health education theories and processes; Methods of identifying environmental and health issues; Values and benefits; Issues for the educators and facilitators; Communication and delivery tools; Social marketing; Curriculum development in environmental and health education; Evaluation and selected fieldwork (including visiting environmental education and health centres); Environmental Education in the SADC and the Gaborone Declaration, UN Decade on Education for Sustainable Development; Environmental Education in Zimbabwe (particularly the Environmental Education Policy of 2003).

EST403 Environmental Biotechnology and Bioremediation

The module will cover Molecular biotechnology revolution and biological systems: DNA, RNA and protein synthesis, recombinant DNA technology; Chemical synthesis, sequencing, and amplification of DNA, manipulation of genes in prokaryotes and eukaryotes; Protein engineering,

synthesis of commercial products by recombinant microorganisms; Utilization microbial insecticides, genetic engineering of plants, transgenic animals, human gene therapy; Biotechnology and environmental applications, bio-treatment and bioremediation methods, bioremediation and biomass; Biological processes, activated sludge or biofilm in wastewater treatment, in situ and ex-situ bioremediation technologies and strategies; Environmental monitoring: bio-indicators, biomarkers, biosensors, biotechnological processes

EST404 Environmental Law

The module will cover Law, policy and ethics as keys to understanding how we use and manage our environment. An introduction to the concepts and principles which underpin environmental law from the international to the local level. Constitutional responsibilities and roles relating to the environment; Sustainable development and the law; Environmental planning through environmental impact assessment and land-use law; Environmental protection principles, climate change water resources law; Heritage issues and the protection of biological diversity

EST405 Life Cycle Assessment and Sustainability

This course is aimed at introducing basic concepts; Analytical frameworks and quantitative techniques for systematically and holistically evaluating the environmental trade - offs presented by different alternatives to enable more informed decision - making. Methodology of Lifecycle Analysis or Lifecycle Assessment (LCA), a well-established technique to compute the various material inputs and environmental releases from all activities associated with the lifecycle i.e. raw material extraction, processing, end-use, and disposal, of a product or service, approaches to LCA - their advantages and disadvantage; Strengths and limitations of LCA as a tool for decision-making compared to alternative approaches such as cost-benefit analysis and cost-effectiveness analysis.

EST406 Waste Water Treatment

Sources of waste water: Waste water characteristics (Industrial, domestic and municipal), wastewater conveyance systems. Methods of treating industrial waste water: Water-borne systems without connection to sewerage systems: septic tanks, construction of septic tanks, disposal of septic tank effluent. Water - borne sewerage systems: Biological filters, activated sludge systems, biological nutrients removal systems, wetland systems, waste stabilization ponds. Advanced waste water treatment: pathogen removal, toxic substance removal. Waste water re-use: forms of waste water re-use, barriers to waste-water re-use

EST407 Environmental Auditing

Relationship between organizations and the environment, Public awareness, Sustainable development, Environmental management, Environmental auditing-Definition, terminology, history of environmental auditing, Objectives and scope, the types of audit, General EA methodology, EA Basic structure and steps, Elements of the Audi t process, Benefits and costs of environmental auditing, Role of an environmental audit within an environmental management system, Evolution of environmental audits, Waste Audits and Pollution Prevention Assessments, EA in industrials Projects, Audit of environmental management systems- audit objectives, roles &responsibilities; scoping and review, preparing the audit, conducting the audit, content of the audit report, Case studies of environmental disasters and their management

EST408 Environmental Management Systems

Environmental Management Standards (to include the British BS 7750, European Union EMAS and the ISO 14000 series), Cleaner Production; Industrial ecology; Eco-efficiency; Ecological footprints; Introduction to SHE; Sustainability indicators; Corporate Governance and Sustainability reporting (introduction to the Global Reporting Initiative, UN, King II Report and the Johannesburg Stock Exchange Sustainability Indicators); Critical Review of Corporate Reporting and Trends in Environmental Management (review of selected company reports), and Field trip to selected companies resembling good and bad practice corporate responsibility in terms of sustainability reporting.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN WATER AND SANITATION TECHNOLOGY (HWST)

Duration:	4 years
Minimum Credit Load:	480
Maximum Credit Load:	540
Total MBKS Credit Load:	384
ZNQF Level:	8

1. PROGRAM OBJECTIVES

1.1. The aim of this program is to contribute towards improved environmental quality at regional, national and global scales through teaching, research and by developing appropriate technologies to identify, control, or eliminate sources of pollutants affecting the environment and public health.

1.2. The department will accomplish its mission by:

a. Developing curricula that provide appropriate theoretical knowledge and practical skills in environmental science and technology.
b. Producing competitive innovative graduates to take up or create opportunities in industry, government departments and the non-governmental organizations including small and medium scale enterprises or own initiatives. The graduates will be equipped with essential technical knowledge and skills for conducting research including: Selecting data collection methods; collecting and analysing air, water and soil samples; analyzing environmental data collected by others; analysing data for correlations to anthropogenic effects; and with skills in advocacy, problem-solving, report writing and presentation.
c. Departmental participation in societal development through research and outreach programs
d. Collaborating with industry and other relevant stakeholders.

2. ENTRY REQUIREMENTS

Normal Entry

- At least five 'O' level subjects including English Language, Mathematics and Science plus: Advanced Level passes in any two of the following subjects or their recognized equivalents:

Biology, Chemistry, Mathematics and Geography or a relevant National Certificate from an accredited Institution

- Articulation as provided by the ZNQF
- Recognition criteria for foreign credentials of international students

Special Entry

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the program

3. PROGRAM CHARACTERISTICS

3.1. Broad areas of study include understanding components of the physical environment, environmental chemistry; soil, air and water pollution, soil biology and biochemistry science; solid waste management, waste water treatment, Climate change and variability; Land use planning; Soil and water conservation; GIS & Remote sensing; Water resources management; Watershed management; Soil fertility and nutrient management; Surveying; techniques and technologies for environmental science.

3.2. The program is oriented towards research, innovation and application of scientific knowledge and skills to prevent, identify, eliminate or control environmental problems u see usually associated with anthropogenic activities.

3.3. Knowledge and skills development is a key feature of the program

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

On successful completion of the degree program, graduates can be formally employed in the private or public sector. They may also be self - employed. The following career prospects are available to graduates:

- ***Industry*** – as designers, research officers, technicians, field and sales officers.
- ***Government Departments, Parastatals, Local Authorities and Municipalities*** – as lecturers, technical, extension or research officers, water resource managers, land use planners, laboratory officers, environmental officers, Catchment management officers.
- ***Agricultural enterprises*** – as environment specialist or managers, technical and environmental officers and consultants.
- ***Self-employment*** – environment analysts, contractors, entrepreneurs, and consultants.

Research and Academia - Graduates can also continue into higher degree programs, which provide a basis for research and academic careers.

Students who excel in this degree program can advance to do Masters and Doctoral studies in Environmental management, water resources management; agricultural water management; water quality management; irrigation; agricultural metrology and climate change and variability, Soil and water conservation, natural resources management and land use planning.

5. PROGRAM DELIVERY

5.1. Teaching and Learning Methods:

Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, research projects

5.2. Assessment Methods:

- ***Formative***

− Coursework comprising written tasks, presentations, demonstrations, practice, production

▪ ***Summative***

− Written examinations, practical examinations in appropriate modules, research project report

6. LEARNING OUTCOMES

On completion of this program a graduate will be able to:

- Understand trends, processes, and impacts that shape the Earth and its environment.
- Develop frameworks suitable for designing technological and sustainable environmental interventions for the mitigation of anthropogenic effects on the environment.
- Communicate effectively, orally and in writing, with audiences of diverse backgrounds.
- Work effectively with, and within, interdisciplinary and diverse groups to resolve environmental problems and achieve environmental management objectives.
- Research and innovate in environment management-related technologies and practices and promote knowledge transfer.
- Conduct an environmental management study, and interpret outcomes

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.2. Students can include in their program of study, modules offered by other departments.

7.3.A student shall be enrolled for at least five modules at any time during the semester

7.4. The module combination should be feasible in terms of the timetable.

7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAM ASSESSMENT

Continuous assessment*:*	Continuous assessment shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30% of the final mark.
Written Examinations:	A three-hour final examination, which shall contribute 70% of the final mark.
Practical Examination:	The duration of a practical exams, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 9 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, learners are expected to acquaint themselves with the introductory modules of the degree program. The choice of modules should be done with the guidance of Departmental staff.

13.2. The degree program shall have at least 36 taught modules spread over the 6 teaching semesters.

Module Code	Module Narration	Credits
Level I		
AGRO104	Introduction to Statistics	12
HORT131	Principles of Microbiology	12
WST101	Health & safety issues in laboratories and treatment plants	12
BSE103	Engineering Mathematics 1	12
LWR134	Principles of Hydrology	12
BSE105	Engineering Mechanics	12
WST102	Water Governance & Community Participation	12
Level II		
BSE201	Engineering Mathematics 2	12
WST201	Health and Epidemiology	12
WST202	Introduction to Sanitation Systems	12
WST204	Fluid Mechanics	12
WST205	Hydraulics	12
BSE208	Engineering Mathematics 3	12
BSE301	Statistical Methods & Experimental Designs	12
Level III	**Work-Related Learning**	
AGPR330	Work-Related Preliminary Report	30
Level III	**Work-Related Learning**	
AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45
Level IV		
WST401	Hydraulic Design	12
WST402	Water Quality Monitoring	12

WST403	Water Tariff Systems	12
LWR414	Hydrogeology	12
WST404	Water treatment technologies	12
WST405	Waste Water Treatment Technologies	12
WST400	Research Project	24

14. MODULE SYNOPSES

AGRO104 Introduction to Statistics
This module introduces students to the fundamental concepts involved in using sample data to make inferences about populations. Included are the study of measures of central tendency and dispersion, finite probability, probability distributions, statistical inferences from large and small samples, linear regression, and correlation.

HORT131 Principles of Microbiology
History and development of microbiology; Discuss key concepts in microbiology; Laboratory types and legal requirements; Distinctive characteristics of the major groups of microorganisms including viruses; Procaryotic and eucaryotic cell structures; Nutritional requirements and microbiological media; Cultivation and growth of microorganisms; Physical and chemical control of microorganisms; Components and functions of the different parts of a typical bacterial cell (cell wall, cell membrane, flagella, endospores, etc.); Bacterial toxins; Methods of identification; Classification of major groups of bacteria.

WST101 Health and Safety Issues in laboratories and treatment plants
Working conditions in laboratories, working conditions in water/wastewater treatment plants, health and safety issues associated with gas, chemicals, electricity, radiation sources, personal protective equipment, documentation of laboratory and water/wastewater treatment plants activities, laboratory hazards and risks, risks and hazards associated with water/wastewater treatment plants, hazard identification and risk assessment in water/wastewater treatment plants, technical protection measures

BSE103 Engineering Mathematics I
Calculus in one variable; Limits and continuity of functions; Differentiation, derivatives of single variables, Leibniz's rule, L'Hospital's rule; Elementary functions including hyperbolic functions and their inverses. Integration techniques including reduction formulae; Applications-arc-length, area, volumes, moments of inertia, and centroids. Series tests of absolute, uniform and conditional convergence, Plane polar coordinates; Complex numbers Elementary Set Theory, Relations and Inequalities, Mathematical Induction.

LWR134- Principles of Hydrology
The module covers the hydrological cycle; meteorological parameters, their definitions and measurements; rainfall-intensity/duration/frequency relationships; evaporation; infiltration; surface runoff-processes, rainfall/runoff correlations; hydrograph analysis; flood routing surface runoff – processes, rainfall/runoff correlations; hydrograph analysis flood routing. Interception of depression storage. Hydrological statistical analysis.

BSE105 Engineering Mechanics
The module covers General Principles; Force Vectors; Equilibrium of a Particle; Force System resultants; Equilibrium of a Rigid Body; Structural Analysis; Friction; Centre of Gravity and Centroid; Moments of Inertia; and Virtual Work.

WST102 Water Governance & Community Participation
Introduction to peace and political stability, sociopolitical nexus, governance, management and water security, water management mapping, transboundary water management, political, legal and regulatory frameworks, conflict analysis and mapping tools, indigenous conflict resolution

BSE201 Engineering Mathematics II
The module covers Symbolic Logic, Boolean Algebra, Matrix Algebra, Scalar and vector products. Equations of lines and planes. Ordinary Differential Equations; Fourier analysis; Laplace Transforms; Z Transforms.

WST201 Health and Epidemiology
This module deals with Principles and practices of sanitation and hygiene; Sanitation regulations and standards; Safety in my environment, water borne diseases, health and lifestyle, detection of possible pathogens in water, methods for tracing disease pathogens to water sources; Sanitation infrastructure planning; design of on-site sanitation facilities; problems and solutions for sanitation systems; tools for design, building and operating domestic sanitation systems, environmental health basis of working in the water and sanitation, key principles, approaches and technologies for environmental sanitation including decentralized excreta disposal, simplified sewerage and solid waste management, operation and maintenance of sanitation infrastructure, tools for demand creation and management and cost benefit analysis of options.

WST202 Introduction to Sanitation Systems
This module deals with an overview of faecal sludge management which will include sludge production methods, sludge collection, characterization of faecal sludge, operational factors that impact the variability of faecal sludge, sludge treatment objectives, physical-chemical constituents of sludge, sludge treatment mechanisms.

WST204 Fluid Mechanics
The module focuses on basics of fluid mechanics and hydraulics: Static pressure and head, fluid pressure on surfaces, pressure measurement, buoyancy and stability of floating bodies, liquids in relative equilibrium, liquids in motion, fluid friction and viscosity, open channel flow and Mannings formula, pipe flow and Darcy-Weisbach, Hazen Williams, Scobey formulae, flow through orifices and mouthpieces, flow measurement; notches, weirs, propeller flow meters, pitot tubes, constriction flow meters etc.

WST205 Hydraulics
The course serves as a quantitative introduction to the principles of hydraulics, and water resource engineering. The course covers the fundamentals of hydraulics, including properties of water, hydrostatic forces/pressures, fluid statics/dynamics, head losses, and related phenomena in closed conduit flow. Topics to be covered include Energy equation, friction losses, minor losses, types of pipe flow & Reynolds number, series piping, parallel piping, pump's power, unsteady pipe flow, classification of free-surface flow, Froude number, uniform flow, critical flow, basics of channel design, specific energy, non-uniform rapidly varied flow (hydraulic jump), introduction to non-uniform gradually varied flow.

BSE208 Engineering Mathematics 111

The module covers: Introduction to numerical methods for solving problems in mathematics, science and finance; Computer arithmetic and rounding errors; Numerical methods for root-finding; Polynomial interpolation and splines; Solution for linear algebraic equations; Numerical integration and differentiation; Numerical integration of ODE's; Euler and second order; Runge-Kutta methods.

BSE301 Statistical Methods and Experimental Design
Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; Statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs (Including biological and engineering experiments) and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two-factor experiments and multi-factor experiments; Multiple comparisons of means; Partitioning of the sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for the goodness of fit and independence; and Data entry, analysis and interpretation using statistical packages (e.g. MINITAB, GENSTAT, SAS R, MATLAB, SPSS) for all designs: Regression and Correlation.

Industrial Attachment (Work-Related Learning)

Work-Related Learning Report; Employer's Assessment Report; Academic Supervisor's Report

WST401 Hydraulic design
The module is structured to equip learners with the requisite knowledge and skills in designing hydraulic structures and conveyance systems
Structural design aspects of hydraulic structures, hydraulic design of energy dissipation structures, spillways, surge tanks, gates and valves, dam and reservoir hydraulics, hydraulic design of irrigation structures, design of sanitation structures, hydraulic design of sewers and storm water drains, design of water/wastewater treatment plants

WST402 Water Quality Monitoring
Aquatic Chemistry, microbiology, standards, engineering and environmental problems, testing techniques and equipment sampling methodology; Laboratory procedures; processes for waste water; oxygen demand tests – biological oxygen demand, chemical oxygen demand; specific oxygen uptake rate; solids tests – total and volatile suspended solids; process control calculations; chlorination and de-chlorination; faecal coliform bacteria; ultra violet disinfection systems

WST403 Water Tariff Systems
Economic principles in water supply and management, water tariff regimes, types of tariffs, problems of water pricing, water charges, subsidies, water security financing mechanisms and models

LWR414 Hydrogeology
The module covers: fundamentals of subsurface flow and transport, emphasizing the role of groundwater in the hydrological cycle, the relation of groundwater flow to geologic structure, and the management of contaminated groundwater; groundwater in the hydrological cycle and recharge calculation; aquifers; Surface water groundwater connectivity; Groundwater resource evaluation.

WST404 Water Treatment Technology
Water treatment technologies for farm and rural water supply, Operation and maintenance of water treatment plants; coagulation; flocculation; sedimentation; filtration; disinfection processes;

drinking water regulations; control of taste and odours; daily operation procedures; chemical use and handling, records and reports; plant maintenance

WST405 Waste Water Treatment Technology
Characterization of effluent; calculation of BOD and reaction rate coefficients; parameters used in urban waste water description; sewerage systems (types of sewers, estimation of flow in sewers, waste water treatment design flow rates); waste water treatment (physical, chemical and biological operations, layout of waste water treatment plants); preliminary treatment units (designing of screening units, comminutors, grit removal, classification of settling behaviours, the ideal settling tank); design of
sedimentation tanks, loading rate methods; biological treatment (process microbiology and kinetics of microbial processes, aerobic biological treatment processes, anaerobic biological treatment processes, anoxic processes; fixed film reactors; suspended culture treatment systems

WST400 Research Project
Students are expected to undertake independent studies in any branch of Water Sanitation Technology, and summarize results in a dissertation. The dissertation is examined in the final year, but preparation starts during the fourth year by developing suitable topics and a preliminary literature search. During the final year, the student devotes 90 hours to data collection and/or experimentation, data analyses and dissertation write-up, for submission before the start of formal final University examinations, and may be required to appear for an oral examination.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN LAND AND WATER RESOURCES MANAGEMENT (HLWRM)

Duration	4 years
Minimum Credit Load:	480
Maximum Credit Load	540
Total MBKS Credit	384
ZNQF level	8

1. PROGRAM OBJECTIVES

The aim of this program is to produce a cadre equipped with essential technical skills and knowledge for effective and efficient utilization and management of land and water resources in the face of climate change impacts on the environment and agricultural water availability. The Department therefore also aims to contribute towards the raising of standards of living of Zimbabweans, directly or indirectly, through teaching, research, development and application of sustainable current technical skills in land and water resources management technologies.

2. ENTRY REQUIREMENTS

Normal Entry

At least five 'O' level subjects including English Language, Mathematics and any Science subject plus: Advanced Level passes in any Mathematics or Physics and one of the following subjects or their recognized equivalents: Biology, Agriculture, Chemistry, Engineering Drawing and Geography or a relevant National Diploma from an accredited Institution as articulated in the ZNQF.

Recognition criteria for foreign credentials of international students

Special Entry

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the program.

3. PROGRAM CHARACTERISTICS

3.1. Broad areas of study include understanding components of Climate change and variability; Land use planning; Soil and water conservation; GIS & Remote sensing; Water resources management; Watershed management; Soil fertility and nutrient management; Surveying; techniques and technologies for environmental science.

3.2. The program is oriented towards research, innovation and application of scientific knowledge and skills to enhance water use efficiency, sustainable agriculture and prevent soil, water and environmental pollution.

3.3. Knowledge and skills development is a key feature of the program

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

On successful completion of the degree program, graduates can be formally employed in the private or public sector. They may also be self - employed. The following career prospects are available to graduates:

- ***Industry*** – as irrigation system designers, research officers, technicians, field and sales officers. Hydrologists, environmental specialists, soil conservation specialists, river inspectors
- ***Government Departments, Parastatals, Local Authorities and Municipalities*** – as lecturers, technical, extension or research officers, water resource managers, land use planners, laboratory officers, irrigation officers, and Catchment management officers.
- ***Agricultural enterprises*** – as environment specialists or managers, technical and environmental officers and consultants.
- ***Self-employment*** – irrigation analysts, contractors, entrepreneurs, and consultants.

Research and Academia - Graduates can also continue into higher degree programs, which provide a basis for research and academic careers.

5. PROGRAM DELIVERY

5.1. **Teaching and Learning Methods:**

Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, and research projects**.**

5.2. **Assessment Methods:**

▪ ***Formative***

− Coursework comprising written tasks, presentations, demonstrations, practice, production

▪ ***Summative***

− Written examinations, practical examinations in appropriate modules, research project report

6. LEARNING OUTCOMES

On completion of this program a graduate will be able to:

- Understand and apply key principles and theories in land and water resources management
- Understand contemporary issues and challenges in land and water resources and work towards solutions
- Apply knowledge and practical skills in sustainable land and water resources management
- Research and innovate in land and water resources-related issues and promote knowledge transfer
- Create new business ventures in land and water resources management

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.2. Students can include in their program of study, modules offered by other departments.

7.3.A student shall be enrolled for at least five modules at any time during the semester

7.4. The module combination should be feasible in terms of the timetable.

7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAM ASSESSMENT

Continuous assessment*:*	Continuous assessment shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30% of the final mark.
Written Examinations:	A three-hour final examination, which shall contribute 70% of the final mark.
Practical Examination:	The duration of practical examinations, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 9 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, learners are expected to acquaint themselves with the introductory modules of the degree program. The choice of modules should be done with the guidance of Departmental staff.

13.2. The degree program shall have at least 36 taught modules spread over the 6 teaching semesters.

Module code	Modules	Credits
Level I		
AGPR131	Agricultural Practice 1	12
LWR131	Environmental Chemistry	12
LWR134	Principles and Applications of Hydrology	12
HAWS134	Principles of Ecology	12
LWR135	Principles to Soil Science	12
AGRO136	Introduction to Statistics	12
LWR144	Principles and Applications of Surveying	12
AGRO135	Plant Biology	12
LWR139	Engineering Drawing and Topographic Mapping	12
AGED131	Introduction to Agricultural Economics	12
AGPR132	Agricultural Practice 2	12
HCS135	Introduction to Information Technology	12
Level II		
LWR137	Soil fertility Management	12
LWR236	Principles of Irrigation	12
LWR240	Soil and Water Conservation	12
AGRO239	Statistical Methods and Experimental Designs	12
LWR241	Land Resource and Survey Methodology	12
LWR236	Irrigation Systems Design	12
LWR245	Water Quality	12
LWR246	Meteorology and Climate Change	12
Level III		
Level III	**Semester 1**	
AGPR330	Work-Related Preliminary Report	30
Level III	**Semester 2**	
AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45
Level IV		

LWR400	Research Project	24
LWR436	Watershed Management	12
LWR445	Rainwater Harvesting	12
LWR442	Land Degradation and Reclamation	12
LWR432	Land Use Planning	12
LWR440	Integrated Water Resources Management	12

14. MODULE SYNOPSES

AGPR131 Agricultural Practice

Agricultural equipment and machinery, maintenance, service and repair, the farm workshop, crop, livestock and other farm activities calendars; cropping programme; land preparation; input requirements and acquisition, planting, weeding and other agronomic practices, harvesting and post-harvest operations. Land capability classification, special crops production (e.g. tobacco, paprika), irrigated crops and pasture production, grazing and veld management; animal handling and restraint; management of large and small ruminants, feeding, routine practices including dipping dosing and disease control; pen fattening of livestock; and marketing and preparation for shows and public field days and displays

AGRO135 Plant Biology

The plant cell and organelles; Plant water relations; Photosynthesis – the ecological significance of photosynthesis in primary production of organic molecules, the Light Reaction, The Calvin Cycle, C3/C4 photosynthesis, CAM photosynthesis. Photoperiodism and vernalization. Fruit and seed formation: double fertilization and formation of seed tissues, Seed structure, chemical composition of seed, apomixis. Germination: What is germination? Phases in germination: seed hydration, genetic and metabolic events during germination, role of plant growth regulators in germination. Seed dormancy**:** Mechanisms of dormancy: Innate, induced and enforced dormancy, Significance of dormancy in attuning cycles of plant growth to seasonal changes in the environment. Plant growth and development: meristems, cell expansion, cell differentiation, acid growth theory, tropisms, cell differentiation, apical dominance, bud dormancy; Plant growth regulators – synthesis and transport and physiological effects of auxins, gibberellins, cytokines, abscisic acid and ethylene on plant growth and development; Use of artificial plant growth regulators in Agriculture

AGRO136 Introduction to Statistics

Introduction to statistics: definition, uses of statistics in research, business, tourism and agriculture; Probability: multiplicative law, addition law, conditional probability, tree diagram, law of total probability; Probability distributions: random variables, discrete, continuous, binomial distribution, normal distribution; Measures of central tendency and dispersion: mean, median, mode, range, variance, standard deviation, standard error of the mean. Sampling technique: simple random, stratified, cluster, systematic; Data types, presentation and summarization techniques; tables, graphs, charts; Regression and correlation: regression parameters, correlation coefficient, coefficient of determination; Simple statistical inference: hypothesis testing, confidence intervals, t-tests, chi-square tests and one-way ANOVA

LWR131 Environmental Chemistry

Sampling and analysis of an industrial sample, interpreting the analytical data and accounting for patterns and trends observed; air and water pollution; ozone generation and ozone removal reactions in the atmosphere; concentrations and biogeochemical cycles of that pollutant elements; The Environmental and Health Consequences of Polluted Air—Outdoors and Indoors, greenhouse effect; Chemistry of natural waters, Pollution and Purification of Water; Toxic Heavy Metals; Pesticides; Toxic Organic Compounds of Environmental Concern; Wastes, Soils, and Sediments; development and components of soil; sorptive properties of soils, and the effect of pH on surfaces charges; soil acidification and environmental effects; sediment transport and depositional processes; flocculation, precipitation, coagulation and dispersion; environmental cycling processes between sediment, water and biota at the sediment-water interface; global cycling of the major elements water quality

LWR135 Principles to Soil Science

Soil formation: Parent material characteristics; Influence of factors of soil formation. Physical properties of soils: soil separates and soil textural classes, inter-particle forces, flocculation and dispersion, and soil structure. Mineralogy of soils: particle mineralogy and its effects on physical and chemical properties (1:1, 2:1 and 2:2 phyllosilicates, Fe oxides), crystalline and amorphous clay colloids, origin and magnitude of permanent and pH- dependent charge, soil water relations; soil air and temperature; soil organic matter. Soil chemical environment: ion adsorption and exchange, ion exchange capacities, base cations, exchangeable aluminium, hydroxy aluminium. Soil microbiology and biochemistry: Plant mineral nutrition; cycles; movement, and uptake by plants; soil pH; Soil classification, land capability classification; soil management; management of saline and sodic soil

HAWS134Principles of Ecology

Definition and scope of ecology, source and concept of energy flow, laws of thermodynamics, concept of limiting factors, ecosystems, biotic and abiotic factors of ecosystem, food chain and food web, trophic levels, ecological pyramids of numbers, biomass and energy, biogeochemical cycles, community ecology, species diversity, diversity indices, successions and ecological niche

LWR134 Principles and Applications of Hydrology

The hydrological cycle; meteorological parameters, their definitions and measurements; rainfall-intensity/duration/frequency relationships; evaporation; infiltration; groundwater; surface runoff-processes, rainfall/runoff correlations; hydrograph analysis; flood routing surface runoff – processes, rainfall/runoff correlations; hydrograph analysis flood routing. Interception of depression storage. Hydrological statistical analysis

LWR144 Principles and Applications of Surveying

Introduction and history of Surveying. Survey measurements: Errors: Gross, Systematic, Random. Distance measurements: Taping, corrections/reductions, chain surveys. Levelling: dumpy, tilting and automatic levels, field procedures, booking and reduction, applications, grid levelling and contouring: Plane table surveys, Coordinate systems: plane, spherical. The Theodolite; angular measurements, traversing: field procedures, reductions, computations, Tacheometry: (i) Theodolite and staff, (ii) tacheometers, (iii) Theodolite and tape (iv) Theodolite and EDM, (v) Total Station, Plotting tachy details

LWR139 Engineering Drawing and Topographic Mapping

Engineering Drawing: -Projections; Isometric Drawing: AutoCAD:-Creating and manipulating drawings in AutoCAD, Topographic Mapping:- Topographic maps production, Google earth and Map production; Plotting, Setting out/ Geodetic Positioning; Production of drawings for water structures such as weirs, reservoirs and canals

LWR137 Soil Fertility Management

Factors controlling the growth of plants; Elements involved in plant nutrition and their sources; Soil organic matter and plant nutrient availability; Factors controlling various nutrient availabilities in soils; Soil plant relations affecting nutrient up-take and growth; Nutrient use efficiency; Types of fertilizers (mineral versus organic) and their use; Compost manure making; Biological Nitrogen Fixation; Agro-forestry; Soil fertility management in crop production; Conditions and processes of plant nutrient uptake, distribution and assimilation; Nutrient requirements of various crops; Design of fertilizer trials; Survey of fertilizer usage; Research findings on important soils and crops

LWR236 Principles of Irrigation

Role and history of irrigation development in Zimbabwe; Soil Water Plant Relations- Physical properties of soil and water, water retention and availability to plants, soil water movement and measurement; Evapotranspiration and crop water requirements; Basic principles of irrigation hydraulics- open channel flow, pipe flow, flow measurement; Irrigation System Types-Sprinkler irrigation systems and components, Surface irrigation systems and components, Drip irrigation systems and components; Planning irrigation projects; Land levelling and land levelling methods for irrigation; Issues of salinity in Irrigation

LWR241 Land Resource and Survey Methodology

Remote sensing: principles and applications of aerial photo interpretation and satellite imagery; Soil survey: principles, mapping methods, organization and implementation of a soil survey, the soil survey process; Irritability assessment of soils; Water quality for suitability for irrigation and environmental impact of irrigation projects; Vegetation survey: principles and mapping methods; Land evaluation: Zimbabwe capability classification systems, constraints of the methods, advantages and uses

LWR240 Soil and Water Conservation

The soil erosion system; water erosion; wind erosion; Classes, types and forms of soil erosion. Classification, processes, factors, analysis and measurements of water erosion and wind erosion. Engineering principles of soil and water conservation: erosion control, soil conservation planning. soil water and nutrient conservation; soil management and reclamation; open channel flow, principles, hydraulic grade stabilization, storm water management, design of structures for floodwater routing, culvert design, design of water ways and agricultural reservoirs stream bank protection. Erosion assessment experimentation and modelling, field methods, physical models, empirical models, wind studies, and conservation schemes. Students will do case studies and visits to erosion-affected areas

LWR246 Introduction to Meteorology and Climate Change

Meteorology; Climate change impacts; Adaptation and mitigation options in different developmental sectors; and International responses to global warming and climate change: treaties, conventions and policies. Weather watching basics; Climate- seasons; Effects of geography on weather; Clouds: precipitation; Fronts and Air Masses: air pressure and humidity;

Weather Forecasting: methods; Bodies of Water: impact of circulation and evaporation; Weather Conservation; Climate Change: history of earth's climate, the science behind climate and weather; global warming and greenhouse gases; Climate dynamics: Climate forcing factors natural & anthropogenic, Ecological balance of natural systems: Biogeochemical cycles; Climate change impacts in different development sectors: Vulnerability of African systems; Climate change impacts on agriculture; Environment: endangered species, ecosystems, and biological resources; Health; Tourism; Aquatic systems; Industry; Adaptation and mitigation: Understanding adaptation and mitigation; Case studies in adaptation and mitigation: carbon credits; REDD+; WWF; clean development mechanism; aviation industry; Fundamental requirements for adaptation in Africa; and Sustainable development concept

AGRO239 Statistical Methods and Experimental Design

Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two-factor experiments and multi-factor experiments; Multiple comparison of means; Partitioning of sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for goodness of fit and independence; Data entry, analysis and interpretation using statistical packages (MINITAB, GENSTAT, SAS) for all designs: Regression and Correlation

LWR236 Irrigation Systems Design

Design of Sprinkler systems; Design of drip systems; Design of surface systems; Irrigation Water Management Principles, Impacts of Irrigation, Organisation of Irrigation schemes, Performance Monitoring and Evaluation of Irrigation Systems, Irrigation scheduling, Operation and Maintenance of irrigation systems and equipment

LWR245Water Quality

Aquatic Chemistry, microbiology, standards, engineering and environmental problems, testing techniques and equipment sampling methodology

LWR430 Research Project

Research proposal writing, data collection and organization, data analyses and statistical competence, data interpretation and presentation, quality scientific writing, verbal presentation of research proposals

LWR436 Watershed Management

Introduction; Watershed Management Principles; Characteristics of a Watershed; Land-use Impacts on watersheds; Potential problems in a watershed and Watershed Degradation; Data Sources for Watershed Management; Strategies for integrated watershed Resources Conservation and Management; Tools for Watershed Management -Watershed Delineation, Participation Principles, Rapid Rural Appraisals and Participatory Rural Appraisals, Soil Loss Estimation - SLEMSA, USLE, Wind Equation, Environmental Impact Assessments; Watershed Management plan; Wetlands- Occurrence, geological characteristics, importance and functions of wetlands, Sustainable utilization, conservation and management of wetlands

LWR445 Rain Water Harvesting

Need for RWH; Planning and designing of RWH systems- RWH for domestic purposes, RWH in Urban Areas. RWH in dry land agriculture and irrigation; RW Quality management and control; Financial aspects and Economic analysis of RWH systems

LWR442 Land Degradation and Reclamation
An overview of the major forms and processes of land degradation, impacts of land degradation on food security, biodiversity, and water resources. Reclamation of degraded lands through agronomic, chemical, mechanical and biological remediation

LWR440 Integrated Water Resources Management
Strategic issues in water management; Integrated water resources management; Stakeholder participation; Water scarcity and concept of virtual water; Sustainability and intergenerational equity; Water using activities; Water demand projections; Principles of water allocation; Legal and institutional issues in water management; Zimbabwe water law and policy; International rivers; SADC protocol on shared water courses; Water economics: Water supply and demand, price-elasticity of demand of water, demand management, water pricing; and Water quality management

LWR432 **Land Use Planning**
Land use planning in early civilization, UN contributions to concepts of land use planning, definition of land use planning, functions of land, concepts of sustainability, the cycle of land use planning, degrees of limitation, land quality indicators

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN AGRICULTURAL WATER MANAGEMENT (HAWM)

Duration:	4 years
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

1. PROGRAM OBJECTIVES

The aim of this program is to produce a cadre equipped with essential technical skills and knowledge for effective management of agricultural water, land and water resources management in the face of climate change impacts. The Department therefore also aims to contribute towards the raising of standards of living of Zimbabweans, directly or indirectly, through teaching, research, development and application of sustainable current technical skills in agricultural water resources management technologies.

2. ENTRY REQUIREMENTS

Normal Entry

▪ At least five 'O' level subjects including English Language, Mathematics, and Science plus: Advanced Level passes in Mathematics or Physics and any one of the following or their recognized equivalents: Biology, Agriculture, Chemistry, Geography, Technical graphics.
▪ Articulation as provided by the ZNQF
▪ Recognition of foreign credentials of international students as provided by the SADC-QF.

Special Entry

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in a relevant Diploma, subject to such a qualification meeting part of the requirements of the program.

3. PROGRAM CHARACTERISTICS

Broad areas of study include understanding components of agricultural water management; Soil and water conservation; GIS & Remote sensing; Water resources management; Watershed management; Surveying; techniques and technologies for agricultural water management.

The program is oriented towards research, innovation and application of scientific knowledge and skills to enhance sustainable agricultural water management.

Knowledge and skills development is a key feature of the program

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

On successful completion of the degree program, graduates can be formally employed in the private or public sector. They may also be self - employed. The following career prospects are available to graduates:

- ***Industry*** – as designers, research officers, technicians, field and sales officers.
- ***Government Departments, Parastatals, Local Authorities and Municipalities*** – as lecturers, technical, extension or research officers, water resource managers, land use planners, laboratory officers, irrigation officers, and Catchment management officers.
- ***Agricultural enterprises*** – as irrigation specialists or managers, technical and agricultural water management officers and consultants.
- ***Self-employment*** – irrigation analysts, contractors, entrepreneurs, and consultants.

Research and Academia - Graduates can also continue into higher degree programs, which provide a basis for research and academic careers.

5. PROGRAM DELIVERY

5.1. Teaching and Learning Methods:

Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, research projects**.**

5.2. Assessment Methods:

▪ ***Formative***

− Coursework comprising written tasks, presentations, demonstrations, practice, production

▪ ***Summative***

− Written examinations, practical examinations in appropriate modules, research project report

6. LEARNING OUTCOMES

On completion of this program a graduate will be able to:

- Understand and apply key principles and theories in agricultural water management
- Understand contemporary issues and challenges in agricultural water resources management and work towards solutions
- Apply knowledge and practical skills in sustainable land and water resources management
- Research and innovate in agricultural water resources management-related issues and promote knowledge transfer
- Create new business ventures in agricultural water resources management

7. GENERAL PROVISIONS

7.1. In level 3, all students will be on Work-Related Learning for a minimum period of ten months. During the same period, students will be expected to carry out a research project.

7.2. Students can include in their program of study, modules offered by other departments.

7.3.A student shall be enrolled for at least five modules at any time during the semester

7.4. The module combination should be feasible in terms of the timetable.

7.5. A student may change his/her registration for modules to be taken in any semester only during the first two weeks of the beginning of each semester.

8. PROGRAM ASSESSMENT

Coursework:	Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 30% of the final mark.
Written Examinations:	A three-hour final examination, which shall contribute 70% of the final mark.

Practical Examination:	The duration of practical examinations, where they are deemed necessary, and their contribution to the final mark, will depend on the module.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 9 of the Faculty Regulations and Section 5 of General Academic Regulations for Undergraduate Degrees.

13. PROGRAMME STRUCTURE

13.1. In the first year, learners are expected to acquaint themselves with the introductory modules of the degree program. The choice of modules should be done with the guidance of Departmental staff.

13.2. The degree program shall have at least 36 taught modules spread over the 6 teaching semesters.

PROGRAMME STRUCTURE (MBKS)

Module code	Module name	Prerequisite	Credits
Level I			
LWR134	Principles of Hydrology		12
LWR144	Principles and Applications of Surveying		12
AGRO136	Introduction to Statistics		12
LWR131	Environmental Chemistry		12
LWR135	Principles of Soil Science		12
AWM135	Plant Biology and Physiology		12
AWM131	Materials Science		12
AWM 132	Engineering Mechanics 1		12
Level II			
AWM233	Engineering Mathematics 1		12
LWR245	Water Quality		12
AWM234	Fluid Mechanics and hydraulics		12
LWR 203	Principles of Irrigation		12
AWM236	Electrical Principles		12
AWM235	Heritage-Based Natural Resource Management		12
AGRO239	Statistical Methods and Experimental Designs	Introduction to Statistics	12
LWR236	Irrigation Systems Design	Principles of Irrigation	12
Level III	**Semester 1**		
AGPR330	Work-Related Preliminary Report		30

Level III	**Semester 2**	
AGPR331	Work-Related Learning Report	45
AGPR332	Employer's Assessment Report	30
AGPR333	Academic Supervisor's Report	45
Level IV		
AWM430	Dissertation	24
AWM430	Irrigation Water Management	12
AWM434	Geographical Information Systems & Remote Sensing	12
AWM435	Watershed Management	12
AWM436	Engineering Project Management	12

14. **MODULE SYNOPSES**

LWR134 Principles of Hydrology
The module covers the hydrological cycle; meteorological parameters, their definitions and measurements; rainfall-intensity/duration/frequency relationships; evaporation; infiltration; surface runoff-processes, rainfall/runoff correlations; hydrograph analysis; flood routing surface runoff – processes, rainfall/runoff correlations; hydrograph analysis flood routing. Interception of depression storage. Hydrological statistical analysis.

LWR144 Principles and Applications of Surveying
Introduction and history of Surveying. Survey measurements: Errors: Gross, Systematic, Random. Distance measurements: Taping, corrections/reductions, chain surveys. Levelling: dumpy, tilting and automatic levels, field procedures, booking and reduction, applications, grid levelling and contouring: Plane table surveys, Coordinate systems: plane, spherical. The Theodolite; angular measurements, traversing: field procedures, reductions, computations, Tacheometry: (i) Theodolite and staff, (ii) tacheometers, (iii) Theodolite and tape (iv) Theodolite and EDM, (v) Total Station, Plotting tachy details. Global Position System (GPS), GPS surveying techniques, Collecting and processing GPS data, GPS applications.

AGRO136 Introduction to Statistics
This module introduces students to the fundamental concepts involved in using sample data to make inferences about populations. Included are the study of measures of central tendency and dispersion, finite probability, probability distributions, statistical inferences from large and small samples, linear regression, and correlation.

LWR131 Environmental Chemistry
This module introduces the student to the components and natural resources of the environment. It explores atmospheric chemistry: air pollution, ozone depletion and global warming; aquatic chemistry and water pollution; and soil chemistry and pollution including bioremediation of soils.

LWR135 Principles of Soil Science
The module introduces students to soil science and covers: Concepts of soil science; Soil formation, Soil minerals, Soil classification, Soil texture and structure in relation to tillage, Soil colour, and Other physical properties of soil – e.g. drainage, permeability, infiltration, Land capability classification, Soil profiling, Soil sampling.

AWM135 Plant Biology and Physiology
The module introduces the student to classification and nomenclature in plant biology and physiology. General anatomy and morphology of plants. Plant growth and development. Plant growth regulators: nature and function. Seed germination. Plant-water and nutrient relations. Photosynthesis (pathways) (C3/C4 plants). Photoperiodism and vernalisation.

AWM131 Material Science
The module introduces the subject of material science. The following areas are also covered: Structure of atoms; Crystal structures; Non-crystalline structures; Imperfections in Solids; Mechanical properties; Failure Analysis; and Corrosion.

AWM132 Engineering Mechanics I
The module covers General Principles; Force Vectors; Equilibrium of a Particle; Force System resultants; Equilibrium of a Rigid Body; Structural Analysis; Friction; Centre of Gravity and Centroid; Moments of Inertia; and Virtual Work.

AWM233 Engineering Mathematics I
Calculus in one variable; Limits and continuity of functions; Differentiation, derivatives of single variables, Leibniz's rule, L'Hospital's rule; Elementary functions including hyperbolic functions and their inverses. Integration techniques include reduction formulae; Applications-arc-length, area, volumes, moments of inertia, and centroids. Series tests of absolute, uniform and conditional convergence, Plane polar coordinates; Complex numbers Elementary Set Theory, Relations and Inequalities, Mathematical Induction.

LWR245 Water Quality
The module covers Aquatic Chemistry, microbiology, standards, engineering and environmental problems, testing techniques and equipment sampling methodology.

AWM234 Fluid Mechanics and Hydraulics
The module focuses on basics of fluid mechanics and hydraulics: Static pressure and head, fluid pressure on surfaces, pressure measurement, buoyancy and stability of floating bodies, liquids in relative equilibrium, liquids in motion, fluid friction and viscosity, open channel flow and Mannings formula, pipe flow and Darcy-Weisbach, Hazen Williams, Scobey formulae, flow through orifices and mouthpieces, flow measurement; notches, weirs, propeller flow meters, pitot tubes, constriction flow meters etc.

LWR203 Principles of Irrigation
Role and history of irrigation development in Zimbabwe; Soil Water Plant Relations- Physical properties of soil and water, water retention and availability to plants, soil water movement and measurement; Evapotranspiration and crop water requirements; Basic principles of irrigation hydraulics- open channel flow, pipe flow, flow measurement; Irrigation System Types-Sprinkler irrigation systems and components, Surface irrigation systems and components, Drip irrigation systems and components; Planning irrigation projects; Land levelling and land levelling methods for irrigation; Issues of salinity in Irrigation.

AWM236 Electrical Principles
The module covers fundamental concepts of electrical and magnetic energy; Electrostatics and electric fields; Capacitance and Dielectric; Magnetic circuits; Electromagnetism; and Electromagnetic induction. Electric Circuits; Resistive Network Analysis; AC Network Analysis; Transient Analysis; Frequency Response and System Concepts.

AWM235 Heritage-Based Natural Resource Management
Ecological concepts and sustainable use of natural resources; concepts and principles of renewable and non-renewal resources, management of resources, degradation of resources, depletion of resources, types of natural resources and policies in natural resource

management; appraisal of the relationship between resources, ecology and the environment; management of heritage resources such as land, water, animals and vegetation, traditional food and their gene banks and products derived from natural resources; cultural property, thus the tangible and intangible attributes of society, inherited from the past, maintained in the present and bestowed for social, economic and political advancement. Identification, preservation and safeguarding of Zimbabwean values (ubuntu/hunhu) and heritage in its diverse forms

AGRO239 Statistical Methods and Experimental Design
Data collection: populations and random sampling; Introduction to experimental designs, data analysis and interpretation; Statistical methods related to variances; Statistical methods related to comparison of means; Experimental designs (Including biological and engineering experiments) and layout (randomized complete block designs, split plots, factorial, Latin squares); Analysis of variance for one factor, two-factor experiments and multi-factor experiments; Multiple comparisons of means; Partitioning of the sum of squares, transformations; Analysis of covariance; Repeated measures analysis; Regression, Correlation Multi-variate regression; Tests for the goodness of fit and independence; and Data entry, analysis and interpretation using statistical packages (e.g. MINITAB, GENSTAT, SAS R, MATLAB, SPSS) for all designs: Regression and Correlation.

LWR236 Irrigation Systems Design
The module covers: Planning irrigation projects, Design of sprinkler systems, design of localised systems, and design of surface systems.

AGPR330-3 Work-Related Learning
Work-Related Learning Report; Employer's Assessment Report; Academic Supervisor's Report

AWM430 Dissertation
Students are expected to undertake independent studies in any branch of agricultural water management and summarize results in a dissertation. The dissertation is examined in the final year, but preparation starts during the fourth year by developing suitable topics and a preliminary literature search. During the final year, the student devotes 90 hours to data collection and/or experimentation, data analyses and dissertation write-up, for submission before the start of formal final University examinations and may be required to appear for an oral examination.

AWM430 Irrigation Water Management
The module covers issues in the national and global context in irrigation water management principles, impacts of irrigation, irrigation scheme organisation and types of irrigation schemes, scheme operation and maintenance; Salinity management.

AWM434 Geographical information systems and remote sensing
The module covers the following major areas: **Geographical Information Systems:** Introduction to GIS; Cartography and Map Projections; Vector Data Structure; Databases; and Data Capture. **Remote Sensing:** Introduction to Remote Sensing; Electromagnetic Radiation and the Electromagnetic Spectrum; Energy Interactions with the Atmosphere. EMR interactions with the Earth's surface; Sensor Technology; Major Remote Sensing Systems; Sensors and applications; Image Processing;

AWM435 Watershed Management

Introduction; Watershed Management Principles; Characteristics of a Watershed; Land-use Impacts on Watersheds; Potential Problems in a Watershed and Watershed Degradation; Data Sources for Watershed Management; Strategies for Integrated Watershed Resources Conservation and Management; Tools for Watershed Management -Watershed Delineation, Participation Principles, Rapid Rural Appraisals and Participatory Rural Appraisals, Soil Loss Estimation - SLEMSA, USLE, Wind Equation, Environmental Impact Assessments; Watershed Management plan; Wetlands- Occurrence, geological characteristics, importance and functions of wetlands, Sustainable utilization, conservation and management of wetlands

AWM436 Engineering Project Management

The module covers the necessary skills for managing engineering projects: Decision analysis and network analysis; Project concept; logical framework (Logframe), concept of cost-benefit analysis; and financial analysis at the farm and project levels. Risk and uncertainty. Procurement procedures and managing contracts; Project control, Monitoring and evaluation; project report writing, quality management. Stakeholder roles and responsibilities.

REGULATIONS FOR THE MASTER OF SCIENCE DEGREE IN LAND AND WATER RESOURCES MANAGEMENT (MLWR)

Duration	:	2 years
Credit Load	:	279
Minimum Credit Load	:	279
Maximum Credit Load	:	248
Total MBKS Credit Load	:	257
ZNQF Level:		9

1. PURPOSE OF THE PROGRAMME

The programme aims to develop sound conceptual, technical, analytical and practical skills required for land resources assessment; land use planning and management and water resources management and development planning in dry areas in the face of climate change.

2. ENTRY REQUIREMENTS

To be eligible for the programme, a candidate must normally be a holder of at least a lower second class (2.2) honours degree in Land and Water Resources Management, Environmental Sciences, Irrigation Management or any other related disciplines from a recognised university.

3. PROGRAMME CHARACTERISTICS

Areas of study: Land and Water Resources Management
Specialist Focus: Land and Water Resources Management
Orientation Oriented towards innovation and application of theory to solve real agricultural challenges
Distinctive Features: Knowledge development in Land and Water Resources Management

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:MSc graduates in Land and Water Resources Management have career opportunities in a wide array of areas that include the following: university teaching, private sector, agricultural consultancy, government, UN agencies and NGOs.

Further Studies: Doctoral studies in land and Water Resources Management, Climate Change, and Climate Smart Agriculture.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial and farm/field visits, dissertation

Assessment Methods: Written and oral examinations, tests, quizzes, group presentations, dissertations. Continuous assessment (40%), Final exam (60%)

6. INTENDED LEARNING OUTCOMES

- Deepen understanding of key principles and theories in land and water resources management
- Deepen understanding and proffering of solutions to challenges in land and water resources and work towards solutions
- Apply knowledge and practical skills in sustainable land and water resources management
- Research and innovate in land and water resources-related issues and promote knowledge transfer
- Create new business ventures in land and water resources management

7. GENERAL PROVISIONS

7.1. The programme shall normally be studied over a minimum of four semesters on a block-release basis.

7.2. This programme is a taught Masters programme with modular work in the first three semesters, including a Field Survey in the third semester. In addition, students develop their dissertation proposals in the third semester. In the fourth and final semester, students engage in research seminars and dissertation writing.

7.3. Continuous assessment consists of lectures, presentations, seminars and tests.

7.4. Lectures and taught modules shall normally be at the Midlands State University

7.5. The dissertation shall constitute two semesters in level II during which the student is expected to undertake research for a period which may not exceed twelve months.

7.6. Each student has to identify at least two supervisors who shall be approved by the Department of Land and Water Resources Management.

7.7. There shall be oral presentations for the Project Proposal and Research Findings. The weighting of these, together with the final written document, shall be as follows:

- Dissertation Proposal 15%
- Presentation of Research Findings 15%
- Final Written Document 70%

Each candidate is expected to submit a dissertation of between 15,000 and 20,000 words

Candidates shall be expected to submit at least three executive-bound copies of the dissertation.

The format of the dissertation shall be provided for by the Department of Land and Water Resources Management.

8. PROGRAMME ASSESSMENT

8.1. Continuous assessment shall be in the form of tests, quizzes, assignments, and seminar presentations.
8.2. Formal examinations shall be held at the end of each semester.
8.3. Continuous assessment shall account for 40 % of the overall assessment, while the examination shall account for 60 % of the overall assessment.
8.4. The aggregate pass mark shall be 50 %

9. PROVISION FOR PROGRESSION

A student must have passed modules worth at least 144 credits in the first level in order to proceed to the second level (dissertation).

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 9 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to section 10 of Faculty Regulations.

13. PROGRAMME STRUCTURE

The degree programme shall have 12 taught modules spread over the three teaching semesters.

Level 1 Semester 1

Code	Description	Credit
MLWRM 701	Soils and Soil surveys	18
MLWRM 702	Remote Sensing and GIS Principles	18
MLWRM 703	Environmental Water Quality	18
MLWRM 704	Climate Change and Variability	18
MLWR 705	Integrated Research Methods	18

Level 1 Semester 2

MLWR 706	Land Evaluation	18
MLWR 707	Irrigation Planning and Evaluation	18
MLWR 707	Irrigation Planning and Evaluation	18
MLWR 708	Environmental Impact Assessments (EIA)	18
MLWR 709	Land and Agrarian Reform	18

Level 2 Semester 1

MLWR 710	Land Use Models and Development Planning	18
MLWR 711	Water Resources Assessment and Modelling	18
MLWR 712	Soil and Water Conservation	18
MLWR713	Field Survey	18

Level 2 Semester 2

MLWR 800	Dissertation	90
MLWRM 801	Seminars	18

14. PROGRAMME SYNOPSES

MLWR 701 Soils and Soil Survey
Soil formation; Soils of tropical environments; soil classification; soil catena; Zimbabwe soil classification system; USDA soil taxonomy; FAO Soil classification; other international soil classification systems; Soil properties; morphological, physical, chemical and biological; clay mineralogy; Pedological description of soils; soil survey and soil maps; Methods of soil surveys; Planning soil surveys; types of soil surveys; scales; accuracy; cost and returns; Strengths and weaknesses of soil surveys; case study.

MLWR 702 Remote Sensing and GIS
Principles; definitions and concepts; types of remote sensing imagery; Electromagnetic spectrum; Properties of Remote sensing imagery; Remote sensing systems; Interpretation of Remote sensing imagery; Vegetation and remote sensing; Energy interaction; digital image analysis; visual image analysis; Applications; GIS processes and application; GIS Database management; GIS GPS Application; GIS mapping; GIS and Remote sensing interactions**.**

MLWR 703 Environmental Water Quality
Principles and Procedures; Water law; Water flow and solutes; Water ecosystems; Water quality monitoring; bio monitoring; Water physical-chemistry; ecotoxicology; Ecological indicators and indices; Water resource protection.

MLWR 704 Climate Change and Variability
Principles; definitions; concepts; climate change and variability both temporal and spatially; climate variability and changes in relation to natural resources and agroecosystems management; vulnerability and adaptations to climate change and variability.

MLWR 705 Integrated Research Methods
Principles and procedures; concepts; Integrated research methodologies and techniques (both qualitative and quantitative) in natural resources assessment and management; Applied statistics in natural resources management. Biophysical resource and social survey methods; including participatory appraisal methods; Communication of research findings to influence policy**.**

MLWR 706 Land Evaluation
Planning environment; Principles and procedures; definitions and concepts; purpose of land evaluation; Types of land evaluation; Systems of land evaluation and their applications; Requirements of major kinds of land use; Productive potential of land; Problems of Land evaluation systems; Case study**.**

MLWR 707 Irrigation Planning and Evaluation
Definition and concepts; Irrigation project planning and public policy; Project appraisal; monitoring and evaluation; analytical techniques in project appraisal; Social cost-benefit analysis; Environmental impact analysis; Human health and irrigation projects.

MLWR 708 Environmental Impact Assessments (EIA)
Concepts of environmental assessment (EA) and Environmental Impact Assessment (EIA), Benefits of EA and EIA; Timing, Scope and compositions; Role of EIA in local development projects; Valuing environmental services and impacts; Principles and procedures in EIA; EIA in Zimbabwe; Socio-economic Impact assessment; Ecological Impact Assessment; Impact monitoring and auditing; Impact mitigation and management; Case study

MLWR 709 Land and Agrarian Reform
Historical background; land husbandry and land tenure laws; land ownership and land productivity; land reform and land redistribution; trends in land reform; agrarian change; trends in tillage system; evolution of crop and livestock cultures; crop and livestock breeding institutions; land reform and rural development; technological challenges and advancements; local and regional constraints.

MLWR 710 Land Use Models and Development Planning
Agriculture policy in Zimbabwe; Land use planning; principles and procedures; guidelines for land use planning; Land use planning in Zimbabwe; Land use planning in the SADCC Region; Surveys and analyses; Participatory Planning processes; Planning for development; Project cycle and management; Farming systems approaches; Risk analysis in dry-land farming systems; Agroforestry, Concepts of community involvement and participation; Self-Help promotion; PRA, Conflict management; Village land use planning; national and Regional land use planning; Farm planning; Resettlement Planning; Case study.

MLWR 711 Water Resources Assessment and Modelling
Principles, concepts, methods, approaches and strategies; Sources of water, Rainfall intensity, groundwater/surface runoff relationships; Rainfall/runoff relationships; Hydrograph analysis; Flood routing; Computer-based simulation models; Model selection for water resources management; Model parameterization; Floods and flood management.

MLWR 712 Soil and Water Conservation
Principles, concepts, methods, approaches and strategies. State of soil and water conservation in the SADC. Erosion assessment experimentation and modelling; Field methods, Physical models, Empirical models Wind studies, Conservation schemes, Case study

MLWR713 Field Survey Synopses
Students will embark on a field visit trip and carry out a field survey on a selected theme of study. a survey report and presentation will be the mode of assessment for the module.

MLWR 800 Dissertation
Students are required to identify a research topic, develop a research proposal, design instruments for data collection, engage in research work, and write a dissertation under the guidance of supervisors. Both internal and external examiners will evaluate the dissertation.

MLWR 801 Seminars
It is compulsory for students to attend all seminars as required.

FACULTY OF THE BUILT ENVIRONMENT, ART AND DESIGN

FACULTY REGULATIONS FOR UNDERGRADUATE DEGREES

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the Faculty Regulations for undergraduate Degrees hereinafter referred to as General Regulations.

1.2 These Regulations are in respect of programmes for the following degrees offered by the Faculty of the Built Environment, Art and Design:

1.2.1 Bachelor of Science Geomatics Honours Degree, specialising in Land Surveying or Geographical Information Science

1.3 The appropriate Degree will be awarded to a student who has successfully completed an approved programme.

2. DEFINITION OF TERMS

2.1.1 A Practical theory module is a module which contains both practical and theoretical components in the formal examination.

2.1.2 A Practical module is a module in which a student is required to carry out, under the direction and supervision of a member of staff in a laboratory, a series of practical exercises related to particular topics within a subject.

2.1.3 A Field Trip is an excursion outside the University Campus under the supervision of an academic and/or technical staff of the University. It may last for part of a day or be conducted over a longer period. The field trip can form part of a module or if it is of sufficient duration, form a 10-credit module in its own right.

2.1.4 A Project module is a module in which a student is required to carry out under the direction and supervision of a member of the academic staff, private studies or investigations related to a particular topic within a module. Each Project module shall be weighted based on the amount of time it occupies compared with a theory module.

2.1.5 A pre-requisite is a module the departmental board requires the student to pass before the student registers for any other related higher-level modules.

3. ENTRY REQUIREMENTS

3.1 ***Normal Entry***

For one to qualify for normal entry to an Undergraduate Degree programme in the Faculty of Built Environment, Art and Design, a prospective student must satisfy the entry Regulations specified in Section 3.1 of the General Regulations and in so doing, must also meet the following Faculty requirements:

Must have obtained a pass at 'O' Level in Mathematics and English Language and have obtained a pass at 'A' Level in at least two of the following subjects or their recognized equivalents: Mathematics, Physics, Geography, Chemistry, Biology, Computer Science, Management of Business, Accounting and Economics. Each department may prescribe specific combinations of these 'A' Level subjects.

3.2 Special Entry

3.2.1 Special entry would be granted to students who, in the opinion of the Departmental Board, have met the conditions set out in Section 3.2 of the General Regulations, subject to such a qualification meeting part of the requirements of the respective programme.

3.3 Mature Entry

3.3.1 Refer to Section 6.2 of the General Regulations.

4. GENERAL PROVISIONS

4.1 In Level 1, normally a student shall register for at least 12 modules from at least two subjects of which a maximum of 4 modules shall be from other Faculties.

4.2 For Level 2 and higher modules, all prerequisites that apply to the modules shall be listed in the special regulations for the subject to which the modules form a portion. A student shall not be admitted to Level 2 or higher level modules unless he/she has passed all prerequisites for the module prior to the start of the semester in which he intends to study that module.

4.3 A student may include in his/her programme, modules offered by other Faculties up to a total equivalent of at most 8 modules.

4.4 A student's selection of modules of a Degree programme is subject to the approval of the Executive Dean of Built Environment, Art and Design and the Chairpersons of all relevant Departments and the following conditions:

4.4.1 The student shall register for at least 12 modules at any time during the academic year.

4.4.2 The module combination is feasible in terms of timetable,

4.4.3 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and

4.4.4 A student may apply for exemptions during the first four weeks of the semester.

4.5 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme. A department shall not designate more than 8 modules as core modules per level.

5. ASSESSMENT

5.1 The assessment of a practical module shall be based on either written reports submitted during the semester or according to a schedule laid down by the appropriate Departmental Board.

5.2 The assessment of a Field module or a Project module shall be based on either the written reports or dissertation submitted at a date determined by the Departmental Board and the students shall be notified of this date at the start of the module.

5.3 The assessment of a Practical-Theory module shall be based on a formal theory examination, a formal practical examination and continuous assessment.

5.4 All other modules shall be assessed by continuous assessment and formal theory examination.

5.5 Each formal examination shall normally be held during an examination period at the end of a semester, in which the module is completed.

5.6 The formal theory examination for a module shall normally be of three hours duration.

5.7 A Practical-Theory module shall contain a practical component in the continuous assessment. The continuous assessment may be entirely practical or contain theory and practical components, but the relative weightings shall be stated in the special Regulations for the programme.

5.8 When in Level 3 of a four-year degree program, the student will undertake Work-Related Learning for a minimum period of 10 months.

5.9 When in Level 4 of a five-year degree program, the student will undertake Work-Related Learning for a minimum period of 10 months.

5.10 For each module that contains a theory and a practical component, the

Departmental Board of Examiners shall determine for each student a theory continuous assessment mark, a practical continuous assessment mark, a formal examination mark, a formal practical examination mark (where appropriate), and a final mark for the module. If a student obtains a final mark of at least 50%, an overall continuous assessment mark of at least 45%, and an overall practical mark of at least 40%, that student passes the module.

5.11 In a module other than the Project module, Practical module and Practical-Theory module, if a student obtains less than 40% in the practical continuous assessment the student may at the discretion of the Departmental Board of Examiners, be assessed by a special practical examination, which must be held before the end of the formal examination period of the semester in which the module is completed.

5.12 For each practical module, fieldwork and project module the Departmental Board of Examiners shall determine a final mark, for that student, for that module based on the assessment of the written reports or the dissertation submitted by that student. The Board shall determine whether the student has passed or failed the module.

5.13 The Departmental Board shall submit, for each module under its control and for each student enrolled in that module, the final mark to the Faculty Board of Examiners. For each student, it shall indicate the results of the modules under its control.

5.14 The final mark will be calculated as follows:

Final mark = (60% to 75%) examination mark + (25% to 40%) continuous assessment.

6. FAILURE TO SATISFY THE EXAMINERS

6.1 Refer to Section 11 of the General Regulations.

7. PROVISION FOR PROGRESSION

7.1 In order to proceed from Level 1 to Level 2, a student must have attained at least 80 credits.

7.2 To proceed from Level 2 to Level 3 the student must have attained at least 80 credits from the modules in that level including the core modules of the subject.

7.3 To proceed from Level 3 to Level 4, in a five-year program, the student must have attained at least 80 credits from the modules in that level including the core modules of the subject.

7.4 The student must have passed Work-Related Learning in order to proceed to the Final Year of study.

8. WORK-RELATED LEARNING PROGRAMME GENERAL GUIDELINES

8.1 The student is subject to Section 12 of the General Regulations and the company regulations during the Work-Related Learning period.

8.2 Once a student is on Work-Related Learning with an organisation approved by the Department/Faculty, he/she is not allowed to change/move to another organisation without the approval of the Department/ Faculty.

8.3 The student is expected to;

8.3.1 conform to the company's regulations, working time and discipline,

8.3.2 fulfil the supervisor's instructions concerning the training process and carry out the Work-Related research project.

8.3.3 write a log book on a daily basis and submit a report after finishing the training in a given department (or training unit).

8.3.4 take part only with educational purposes in mind according to the ultimate instructions of the supervisor.

8.3.5 put his/her best effort into acquiring extensive knowledge and skills in order to achieve the required standard of training.
8.3.6 keep good relations with all staff of the company and
8.3.7 promote the good image of Midlands State University.

8.4 The student is expected to always bear in mind that his/her conduct during Work-Related Learning will reflect not only on him/her but also on Midlands State University and that it may also affect considerably, future placements and the relationship between the University and that particular company.

9. GRADING AND DEGREE CLASSIFICATION

9.1 To be awarded a four-year Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 360 of the 516 credits which should be at Level 2 and higher.

9.2 For a five-year Bachelor of Science Honours degree, a student must pass each core module listed in that programme and must have accrued at least 480 of the 660 credits which should be at Level 2 and higher.

9.3 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

10. DEGREE WEIGHTING

10.1 For the purpose of degree classification, for a four-year degree program, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

10.2 For the purpose of degree classification, for a five-year degree program, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	15%
Level 3	20%
Level 4	25%
Level 4	30%

DEPARTMENT OF SURVEYING AND GEOMATICS

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN GEOMATICS (specialising in Land Surveying or Geographical Information Science) (BEAD01)

Duration	4 years
Minimum Credit Load	480
Maximum Credit Load	528

Credit Load for program	516
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

Over the last decade, there have been 'spatial turns' and an explosion of Geodata and information products through the widespread availability of the internet and ubiquity of location-capable data-creating technologies. This Geomatics degree as created through consultation with multiple stakeholders, will develop the next generation of scientists with the creative intelligence and skills to synthesize diverse datasets from multiple sources as part of improving decision making. Our graduates will be in high demand due to the growing need for new geospatial skills for a wide range of applications.

2. ENTRY REQUIREMENTS

Normal Entry

Minimum 5 O Levels including Mathematics, Geography and English,

Minimum of 2 A level passes in Mathematics and Physics

Special Entry

Candidates with a minimum of a related National Diploma and at least 2 years relevant working experience gain acceptance at first-year level into the programme

3. PROGRAMME CHARACTERISTICS

Areas of Study: Geographic Information Science (GISc), Remote Sensing, Photogrammetry, Mine Surveying, Cadastral Surveying, Engineering Surveying, Land Management

Specialist Focus: Mapping, Land Surveying Geographic Information Science, Mining and Engineering

Orientation: Integrated geomatics principles, knowledge and practical skills for research, innovation, industrialization and commercialization.

Distinctive Features: Geomatics is an earth observation science that involves collecting, processing, analyzing, storing and disseminating geospatial data. It encompasses the fields of Surveying, Geodesy, Remote Sensing, Geographic Information Systems, Geostatistics, Geospatial big data science, and Artificial Intelligence. Earth observation is an essential prerequisite for the sustainable development of any nation as it plays a crucial role in quantifying and managing natural and manmade resources. Geomaticians are at the forefront of fighting the grand global challenges including climate change, epidemics and natural disasters through their unique ability to synthesize geo-enabled information from diverse sources and use it for critical decisions and action.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Cadastral surveyors, mining surveyors, engineering surveyors, Cartographers, Geodesists, Photogrammetrists, and Project Managers. GIS specialists, Remote Sensing specialists, cartographers, photogrammetrists, research scientists, Spatial Data Scientists, Systems engineers

Further Studies: The graduate from the GISc stream will be qualified for a Master's degree in GISc or any of its specialized disciplines including but not limited to Remote Sensing, GIS, and Data science. The graduate from the survey stream is typically

prepared for professional registration in the discipline. They are also qualified for relevant Masters programs such as in Photogrammetry or Geodesy. However, they could also take a postgraduate diploma in GISc to qualify for entry into the MSc GISc. The graduate programs on offer in the department will provide innovative students with access to the University's incubation hub, further providing entrepreneurial opportunities.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, seminars, group work, industrial visits, industrial placements, research projects, individual independent study

Assessment Methods: Written and oral examinations, self-evaluation, tests, field reports, seminar presentations, WRL report, mini-research project report, final year research project report, continuous assessments

6. INTENDED LEARNING OUTCOMES

- Comprehensive understanding and knowledge of Geomatics fundamentals
- Ability to collect geospatial data using various appropriate methods and tools for different applications in Geomatics
- Ability to design experiments, analyse geospatial data and interpret resulting geo-information products
- An ability to think critically and independently as part of individual tasks or in the context of a team
- Demonstrate good leadership qualities coupled with good communication and other soft skills
- Adequate foundational knowledge to qualify for further postgraduate studies and research
- Ability to synthesize knowledge from various sources in order to design and develop innovative solutions to geospatial problems

7. GENERAL PROVISIONS

Refer to Section 5 of the Faculty Regulations

8. PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

8.1 EXAMINATIONS

Refer to the relevant section of the Faculty regulations.

9. PROVISIONS FOR PROGRESSION

Refer to Section 7 of the Faculty regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 6 of the Faculty regulations.

11. GRADING AND DEGREE CLASSIFICATION.

Refer to Section 9 of the Faculty regulations.

12. DEGREE WEIGHTING

Refer to Section 10 of the Faculty regulations.

13. PROGRAMME STRUCTURE

Module Code	Module Name	Credits
Level 1 Semester 1		
*HSVG131	Surveying I	12
*HSVG130	Introduction to Geomatics	12
*HMAT131	Calculus 1	12
*HCSCI132	Principles of Programming Language	12
*CS131	Communication Skills	12
*HAPI131	Mechanics	12
Level 1 Semester 2		
*GS231	Introduction to Gender Studies	12
*HSVG133	Cartography and Introduction to GIS	12
*HSVG139	Geomatics Camp	12
*HSTA139	Probability Theory and Statistics	12
*HAPI132	Waves and optics	12
*HAPI133	Electricity and magnetism	12

GEOGRAPHICAL INFORMATION SCIENCE SPECIALISATION

Level 2 Semester 1		
*HSVG230	Spatial Analysis and Modeling HSVG133	12
*HSVG121	Principles of Photogrammetry HAPI132	12
*HCSE135	Databases Design and Development	12
*HCSCI232	Systems Analysis and Design	12
*TCNP201	Technoprenuership	12
Level 2 Semester 2		
*HSVG247	GIS data structures and algorithms HSVG133	12
*HSVG238	Multi-Dimensional Modelling	12
*HSVG249	GIS Camp 1	12
*HSVG250	Geovisualisation HSVG133	12
*HSVG245	Programming for Geomatics Applications HCSCI132	12
*HSVG203	Principles of Remote Sensing	12
Level 3 Semester 1		
*HSVG341	Work-Related Learning 1	40
Level 3 Semester 2		
*HSVG342	Work-Related Learning 2	80
Level 4 Semester 1		
*HSVG430	Geostatistics HSTA139	12
*HSVG431	Geospatial Big Data and Artificial Intelligence	12
*HSVG432	GIS Camp 2 HSVG249	12
*HSVG424	Research Methods	12
*HSVG425	The Geoweb HSVG250	12
*HSVG429	Cadastres and Land Information Systems	12
Level 4 Semester 2		

*HSVG426	Professional ethics and business matters		12
*HSVG427	Project Planning and Management		12
*HSVG428	Rural, Urban and Spatial Planning		12
*HSVG435	Final Year Project		24
*HSVG434	Digital Image processing	HSVG203, HSVG121	12
HSVG433	Land Economics and Land Use Planning		12

LAND SURVEYING SPECIALIZATION

Level Two Semester 1

*HSVG231	Surveying 2	HSVG131	12
*HSVG234	Photogrammetry and Remote Sensing	HAPI132	12
*HSVG239	Linear Algebra	HMAT131	12
*HSVG240	Engineering Surveying		12
*HSVG241	Satellite Geodesy		12
*HSVG242	Numerical Methods		12

Level Two Semester 2

*HSVG232	Surveying 2b		12
*HSVG243	Adjustment computations		12
*HSVG244	Cadastral Surveying		12
*TCNP201	Technoprenuership		12
*HSVG238	Survey Camp 2	HSVG139	12
*HSVG245	Programming for Geomatics Applications		12

Level 3 Semester 1

*HSVG341	Work-Related Learning 1	40

Level 3 Semester 2

*HSVG342	Work-Related Learning 2	80

Level Four Semester 1

*HSVG421	Surveying 3	HSVG231	12
*HSVG422	Land Law	HSVG244	12
*HSVG423	Physical Geodesy		12
*HSVG424	Research Methods		12
*HSVG425	Survey Camp 3	HSVG238	12
*HSVG426	Differential Equations		12

Level 4 Semester 2

*HSVG426	Professional ethics and business matters	12
*HSVG427	Project Planning and Management	12
*HSVG428	Rural and Urban Spatial Planning	12
*HSVG434	Final Year Project	24
Choose two electives:		
HSVG430	Cadastres and Land Information Systems	12
HSVG431	Geology and Mine Planning	12
HSVG433	Land Economics and Land Use Planning	12
HSVG434	Digital Image Processing	12

Synopses

HSVG130 Introduction to Geomatics
Overview of Geomatics, Fundamentals of Geomatics, Spatial Data capture, data delivery and storage, quality control, future of Geomatics

HSVG133 Cartography and Introduction to GIS
Introduction to GIS and GISc, GIS as models of reality, GIS data capture techniques, Data integration, spatial relationships, GIS Analysis Functions, GIS modelling, Maps and Map analysis, Basics of cartography, cartographic output from GIS processes

HSVG131 Surveying I
Introduction & History of Surveying. Survey measurements: Errors: Gross, Systematic, Random. Distance Measurements: Taping, corrections/reductions, chain surveys. Leveling: dumpy, tilting and automatic levels, field procedures, booking and reduction, applications, grid leveling and contouring: Plane table surveys, Coordinate systems: plane, spherical; The Theodolite; angular measurements, Traversing: field procedures, reductions, computations, Tacheometry: (i) Theodolite and staff, (ii) tacheometers, (iii) Theodolite and tape, (iv) Theodolite and EDM, (v) Total Station, Plotting tachy details.

HMAT131 Calculus 1
Calculus, Differentiation, Leibniz's Rule, L'Hopital's Rule. Elementary functions, Integration, Series tests, Plane polar coordinates. Complex Numbers, De Moivre's theorem. Complex exponentials.

HSTA139 Probability Theory and Statistics
Data Representation. Average. Spread, Experiments, Outcomes, Events, Probability, Permutations and Combinations, Random Variables. Probability Distributions Mean and Variance of a Distribution, Binomial, Poisson, and Hypergeometric Distributions, Normal Distribution, Distributions of Several Random Variables. Random Sampling, Point Estimation of Parameters, Confidence Intervals, Testing Hypotheses. Decisions, Quality Control, Acceptance Sampling, Reliability, Goodness of Fit. Nonparametric Tests, Regression. Fitting Straight Lines. Correlation.

HCSCI132 Principles of Programming Languages
Writing simple programs using assignments, Variables, Simple types, Operators and expressions, Conditional and repetitive statements, Input and output, Simple functions, Program Structure and organization, Definition and scope, and Introduction to classes and objects.

HAPI132 Waves and optics
Refer to the Department of Applied Physics and Instrumentation.

HAPI133 Electricity and magnetism
Refer to the Department of Applied Physics and Instrumentation.

HAPI131 Mechanics
Refer to the Department of Applied Physics and Instrumentation.

GS231 Introduction to Gender Studies
Refer to the Department of Gender Studies.

TCNP201 Technoprenuership
Refer to regulations from the Relevant Department

HSVG139 Geomatics Camp

Two weeks of practical Geomatics work covering all the aspects studied in level one. Consolidates the theory of plane surveying with practice and gives experience in organization and team working on surveying projects. Includes basic GIS data capture processes and analyses and application of land administration and management principles

HSVG230 Spatial Analysis and Modeling
Spatial Analysis functions, proximity functions, neighbourhood functions, network, raster functions, heuristic analysis, models of reality vs conceptual models, GIS modelling, 3D modelling, GIS analysis project

HSVG234 Photogrammetry and Remote Sensing
Principles of photography, Stereoscopic viewing, Flight planning, Relationships between image and object space, Coordinate transformations, Collinearity and Coplanarity, Orientation procedures, Error analysis, photogrammetric instruments and products, Graphics superimposition, map revision and terrestrial and close-range Photogrammetry, panoramic imaging, Aerial triangulation.
Basic Physics of Remote Sensing; Kepler's Laws and Orbital Parameters, Platforms and Sensors, Principles of Image Interpretation, Visual Analysis, Integrating Remote Sensing and GIS.

HCSE135 Database Design and Development
Relational data model and Distributed database. DBMS architectures. Data dependencies, decomposition algorithms. Data dictionaries. Concurrency., integrity, security and reliability issues. Query decomposition, optimization, and evaluation strategies. Physical database design.

HSVG251 Programming for Geomatics Applications
Programming of surveying and GIS problems using Pascal, Visual Basic, C or C++, python, Matlab. The student should be able to write a suite of programs solving the following problems: (i) Angle/distance/area/volume conversions, (ii) Tape and EDM distance reductions, (ii) Joins (single, consecutive and radial), (iii) polars (single, consecutive and radial), (iv) intersection, (v) resection, (vi) linear coordinate transformations, (vii) traversing, (viii) Levelling reductions, (ix) network adjustment, (x) Topological line intersections

HCSCI232 Systems Analysis and Design
Refer to the Department of Computer Science

Digital Image Processing
An introduction to digital image processing (IP) and computer vision (CV) concepts, methods and algorithms which will enable the students to implement IP/CV systems or use IP/CV software with emphasis on remote sensing and photogrammetry applications and problem-solving

HSVG425 The Geoweb
History of the Web and Geoweb, Web computing architectures, Geo-spatial web services and standards, Spatial Data Infrastructures, location-based services (LBS), mobile GIS, The geoweb as a commodity, geoweb and commerce, g-governance

HSVG247 GIS data structures and algorithms
Storage of complex objects, Line Storage Techniques, Raster Storage Techniques, Vector structuring; Raster Structuring; Vector Algorithms; Raster Algorithms, integrating data from different sources.

HSVG429 Cadastres and Land Information Systems

Critical issues of this module include types of LIS, GIS taxonomy, relations of GIS and LIS, types of cadastres (fiscal cadastres, legal cadastres, multipurpose cadastres) and their aims and functions. The module also focuses on Land Information Systems and ther cadastral applications, taking into consideration the importance of cadastral records and land information systems and GIS issues in land records and LIS key players. Mine Cadastres will also be explored in this module

HSVG241 Satellite Geodesy

Fundamentals of Satellite orbits, Coordinate System, GNSS technologies, Processing of GNSS data, Principle of Doppler effect, Range measurements, Satellite Altimetry, Signal Structure, Time reference systems, VLBI, GNSS procedures, Error sources

HSVG249 GIS Camp 1

Assessment will be made on the basis of contact during the camp and on written reports and plans submitted. The survey camp will be undertaken in two weeks. Guided Study Topic (GIS Applications area). Student submits a detailed report on the application of GIS to an area of interest. Prototype database development.

HSVG231 Surveying 2

Instrumentation: (1) Optical theodolite; construction, errors, tests and adjustment, (2) Electronic Distance Measurement (EDM): construction, errors, tests, adjustment and standardization. The Gauss projection; t-T correction, meridian convergence and sea-level and projection correction. Linear Coordinate transformations. Precise traversing: Theodolite and tape, theodolite and EDM, total station, advanced traversing theory. Point positioning: Intersection and Resection; error figures, Precise Leveling, Trigonometrical heighting: reciprocal and simultaneous levelling: curvature and refraction corrections.

HSVG239 Linear Algebra

Vector Calculus Linear Algebra: Matrices, Vectors, and Determinants. Linear Systems, Matrices, Vectors: Addition and Scalar Multiplication, Matrix Multiplication, Linear Systems of Equations. Gauss Elimination, Linear Independence. The rank of a Matrix. Vector Space, Solutions of Linear Systems: Existence, Uniqueness. Matrix Eigenvalue Problems, the Matrix Eigenvalue Problem. Determining Eigenvalues and Eigenvectors, Some Applications of Eigenvalue Problems, Symmetric, Skew-Symmetric, and Orthogonal Matrices, Eigen bases. Diagonalization. Quadratic Forms, Complex Matrices and Forms. Vector Differential Calculus; Green's Theorem in the Plane, Surfaces for Surface Integrals, Surface Integrals, Triple Integrals. Divergence Theorem of Gauss.

HSVG240 Engineering Surveying

Route location surveys, profiles (longitudinal and cross-sections), curve ranging and setting out (circular, transition and vertical), calculation of areas and volumes of regular and irregular shapes or figures (the planimeter, trapezoidal rule, Simpson's rule, cross sections, spot-heights, contours), mass haul diagrams and applications, construction surveying (alignment, grade, setting out of buildings, tunnels, bridges, culverts, dams, railroads, canals, underground surveys), Use of Lasers in Civil Engineering, Metrology. Deformation and Hydrographic Surveying; Offshore and onshore surveys to determine the underground profile using different soundings, control surveys, measurement of river flow, discharge calculations, cross sectional sedimentation, control for deformation surveys, monitoring of structures, sea-level variations, land and satellite-based navigation, ocean mapping and charting.

HSVG422 Land Law

Concepts of Law, Property law. Land Tenure systems, Acquisition, transfer and disposition of land and rights in land. Servitudes, usufruct, usus, habitation, fideiscommissum, lease, tenancy, mortgage and lien. Conveyancing. Land disputes. Encroachment. Acts pertaining to land, land development and conservation in Zimbabwe. The Land Survey Act. The Land Surveyors Act. Introduction to international law on ownership. International boundaries.

HSVG244 Cadastral Surveying

Forms of land tenure. Dominium. Land tenure types in Zimbabwe. Partial rights in land. Acquisition and disposal of land and rights in land. Land registration. The cadastre. Boundary demarcation, delineation and restoration. Case Law. Cadastral surveys. Survey records. Cadastral systems and LIS. Land parcel appellation. Partitioning of exclusive rights of occupation. The Land Survey Regulations and the functions of the Department of the Surveyor-General, Township Layout, standards and field techniques. Relocation of beacons. **Automated survey Draughting**; each student is expected to produce a survey diagram, a layout plan, a general plan, a working plan and a topographic map. The software to be used for the above tasks will depend on what will be available in the department

HSVG238 Survey Camp 2

Two weeks of practical survey work covering all the aspects studied in the second year. Consolidates the theory of surveying with practice and gives experience in organisation and team working on surveying projects

HSVG430 Geostatistics

Probability Theory Review, Spatial Analysis, Variogram Modeling, Spatial Continuity Analysis, Basic Permissible Models, Model-Fitting “Rules of Thumb”, Nested models, Nongeostatistical (Deterministic) Estimation, Estimation Criteria, Geostatistical (Probabilistic) Estimation, Geostatistical Simulation

HSVG250 Geovisualization

Geovisualisation Overview, Geo-visual representation methods and techniques, Geo-visual analytics, Usability and Usefulness Analysis, Developing Geo-visual tools, Map interface design.

HSVG431 Geospatial Big Data and Artificial intelligence

Big Data Computational Methods, Geospatial data preprocessing, Overlay analysis, Land-use change prediction, Global scale terrain analysis, Human mobility (pattern discovery), Big Data Mining, Missing road generation, Knowledge Representation, Big Data Search, cognitive techniques

HSVG432 GIS Camp 2

Assessment will be made on the basis of contact during the camp and on written reports and plans submitted. The survey camp will be undertaken in two weeks. Guided Study Topic (GIS Applications area). Student submits a detailed report on the application of GIS to an area of interest. Prototype database development.

HSVG424 Research Methods

Research Basics, Theory of Research, Writing a research proposal, Literature review, Methodologies and Methods, Referencing, Reference managers, tools that aid research

HSVG426 Professional ethics and business matters

Social responsibilities of surveyors and engineers, Land Surveyors Act, the Engineering Act and code of ethics, occupational health and safety, sustainable development, environmental

stewardship, employment equity, legal duties and liabilities of the professional engineer/surveyor, contracts, the tort of negligence, labour law, intellectual and industrial property, conflict resolution.

HSVG427 Project Planning and Management

Introduction to project planning and implementation: defining a project, the project cycle, policies and programmes; Financial and economic analysis: discounting, project feasibility, economic analysis, sensitivity analysis, cost/benefit analysis; Operational planning: budgeting, network planning and bar charts, procurement of inputs.

HSVG428 Rural and Urban Spatial Planning

Land use planning and urban design, theoretical and practical background in the design and development of human settlement and land use planning in Zimbabwe, planning regulations and enforcement, conceptualise, design, and implement engineering, construction and services that will ensure sustainable environmental management. Principles of Rural and Urban Planning; Spatial Planning Models and Concepts, Spatial Planning Tools and Enforcements; Housing Models; Spatial Planning and Local Economic Development; Integrating Spatial Planning and Infrastructure; Social Equity and Spatial Planning Climate Change and Spatial Planning; Spatial Planning Regulations in Zimbabwe (RTCP Act)

HSVG421 Surveying 3

Geodetic leveling. Leveling datum. Triangulated heights. Classical geodetic positioning techniques, Network densification, Geodetic networks of Zimbabwe, The Arc 1950 datum and its transformation, Gauss and UTM coordinates, Overview of 3D networks.

HSVG243 Adjustment computations

Observables and Observations, Significant Digits of Observations, Concepts of Observation, Concepts of Stochastic Model. Needs for Adjustment, Introductory Matrices, Covariance, Cofactor, and Weight Matrices. Analysis and Error Propagation of Survey Observations, Statistical Distributions and Hypothesis Tests, Probability Functions, Sampling Distribution; Student's t-Distribution; Chi-square and Fisher's F-distributions; Joint Probability Function. Concepts of Statistical Hypothesis Tests; The Method of Least Square; Comparison of Different Adjustment Methods Confidence Region Estimation; Mean Squared Error and Mathematical Expectation, Population Parameter Estimation, General Comments on Confidence Interval Estimation, Error Ellipse and Bivariate Normal Distribution, Error Ellipses for Bivariate Parameters.

HSVG242 Numerical Methods

Numerics in General; Introduction, Solution of Equations by Iteration, Interpolation, Spline Interpolation, Numeric Integration and Differentiation. Numeric Linear Algebra; Linear Systems: Gauss Elimination, Linear Systems: LU-Factorization, Matrix Inversion, Solution by Iteration, Ill-Conditioning, Norms, Least Squares Method, Matrix Eigenvalue Problems: Inclusion of Matrix Eigenvalues, Power Method for Eigenvalues, Tridiagonalization and QR-Factorization. Runge Kuta Algorithms

HSVG426 Differential Equations

Numerics for ODEs and PDEs; Methods for First-Order ODEs, Multistep Methods, Methods for Systems and Higher Order ODEs Methods for Elliptic PDEs, Neumann and Mixed Problems. Irregular Boundary, Methods for Parabolic PDEs, Method for Hyperbolic PDEs.

HSVG423 Physical Geodesy

Definition of Geodesy, The Problem of Geodesy, Historical Development of Geodesy, the Spherical Earth Model, The Ellipsoidal Earth Model, The Geoid, Arc Measurements and

National Geodetic Surveys,3D Geodesy,4D Geodesy. Reference Systems; The Gravity Field of the Earth; The Geodetic Earth Model; the Rotational Ellipsoid, the Normal Gravity Field, Geodetic Reference Systems. Gravimetric Measurements; Terrestrial Geodetic Measurements; Computation and Evaluation; Geodetic and Gravimetric Networks; Structure and Dynamics of the Earth; the Geophysical Earth Model, Geodesy and Geodynamics.

HSVG433 Land Economics and Land Use Planning

The module introduces students to 1) the principles of micro and macroeconomics and their roles in understanding land markets in particular impact on land and estate values and 2) to ways in which different methods of valuation are applicable in different circumstances. Students shall be able to evaluate and synthesize estate management and economic development policy for public and private development.

HSVG431 Geology and Mine Planning

Introduction to geology and mineralisation, classification of resource and reserve; resource estimation algorithms; linear, nonlinear, and indicator kriging; stochastic simulation; variogram modelling; block-variance relationship; recoverable reserve. basics of mine planning, feasibility study, block modeling, economic analysis, cost estimation and price forecasting, mining method selection algorithms.

HSVG434 Final Year Project

Research Methodology. Each student will undertake a project in their specialist field: A research proposal will be generated in the Research methods module.

FACULTY OF BUSINESS SCIENCES

FACULTY REGULATIONS FOR UNDERGRADUATE AND POSTGRADUATE DEGREES

FACULTY REGULATIONS FOR UNDERGRADUATE DEGREES

1. INTRODUCTION

1.1 The Faculty of Business Sciences regulations shall be read in conjunction with the General Academic Regulations for undergraduate degrees of the Midlands State University hereinafter referred to as the General Regulations.

1.2 The Faculty of Business Sciences offers the following Undergraduate degree programmes at Honours level:

Bachelor of Commerce Honours Degree in Applied Accounting (BS02)
Bachelor of Commerce Honours Degree in Audit and Risk Management (BS03)
Bachelor of Commerce Honours Degree in Finance and Digital Banking (BS10)
Bachelor of Commerce Honours Degree in Economics (HECON)
Bachelor of Commerce Honours Degree in Econometric Modelling (BS09)
Bachelor of Commerce Honours Degree in Financial Economics and Investment (BS11)
Bachelor of Commerce Honours Degree in Cloud Computing and Internet of Things (BS06)
Bachelor of Commerce Honours Degree in Consumer and Marketing Intelligence (BS07)
Bachelor of Commerce Honours Degree in Data Science and Informatics (BS08)
Bachelor of Commerce Honours Degree in Information Systems (BS31)
Bachelor of Commerce Honours Degree in Information Security (BS33)
Bachelor of Commerce Honours Degree in Business Management (BS05)
Bachelor of Commerce Honours Degree in Business Health Care Management (BS04)
Bachelor of Commerce Honours Degree in Risk Management and Insurance (BS13)
Bachelor of Commerce Honours Degree Logistics and Supply Chain Management (BS12)
Bachelor of Commerce Honours Degree in Agri-tourism Management (BS01)
Bachelor of Commerce Honours Degree in Sustainable Tourism Management and Planning (BS14)
Bachelor of Commerce Honours Degree Tourism and Hospitality Management (BS15)

1.3 The Centre for Entrepreneurship offers Entrepreneurship 1 a module which is compulsory to all Level 2 Semester 1 undergraduate students across all faculties. To this end the 12-credit module Entrepreneurship 1, ENT131 is listed along with all Level 2 Semester 1 programme structures in all departments.

2. ENTRY REQUIREMENTS

2.1 **Normal Entry**

To be admitted into the programmes of the Faculty of Business Sciences, a candidate must have satisfied the minimum requirements for entry to degree

studies at the Midlands State University as stipulated in the General Regulations for undergraduate degrees. For entry to all Faculty of Business Sciences programmes the student must have passed English at the Ordinary level with at least a "C" or better and for Mathematics refer to Departmental entry requirements. The Faculty of Business Sciences' departments may further specify special requirements as entry qualifications.

Prospective students from academic jurisdiction whose prior qualifications do not match the normal entry requirements to a programme in the Faculty of Business Sciences will undergo a foundation course for a duration of one year to be able to enrol for a programme in the faculty.

Prospective students from academic jurisdiction whose prior qualifications do not match 'A' Level qualification, will undergo a bridging programme for a period of one year before they are enrolled into a programme in the Faculty

2.2 **Visiting School Undergraduate degree programmes**

2.2.1 To qualify for entry into the Bachelor of Commerce Honours Degree in Applied Accounting Degree (Visiting School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma (complete qualification) in Accounting, or any equivalent tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects

and:

Confirmation of employment in relevant accounting departments.

2.2.2 Anyone below the age of 23 years must have two years of working experience and 2 A-level passes.

2.3 **Mature Entry**

Refer to Sections 3.2 and 3.3 of the General Academic Regulations.

2.4 **Special Entry**

Refer to Section 3.2 of the General Academic Regulations.

2.4 **Special Honours Degree**

Refer to Section 2.3 of the General Academic Regulations.

3. DEFINITION OF TERMS

3.1 **Core Module**

A core module is a module described and included under the degree structure

3.2 **Non-Core/ Optional Module**

A non-core/ Optional module is a module which is not compulsory and not included under the provided degree structure of a programme. A student is also encouraged to pass a non-core module. A student can graduate when he/she meets the required number of credits even if he/she has failed some non-core modules.

3.3 **Elective Module**

An elective module is a prescribed module a student selects from a pool of modules from other faculties/ departments of the University. Pursuance of the module is subject to the availability of staff and approval by the respective department. A student cannot graduate without passing an elective module.

3.4 **Pre-requisite Module**

It is a compulsory module that must be passed before a candidate is allowed to register for a related subsequent module.

3.5 **Dissertation/ Project/ Research Assignment**

It is a defined practical assignment, which is separately examinable. It is related to independent studies or investigations. The investigation should be related to a particular topic within a module. A student under the direction and supervision of a member of the academic staff or an approved specialist undertakes the research project.

3.5.1 All dissertations/projects for undergraduate students shall be subjected to anti-plagiarism software and anti-plagiarism software reports will be produced as evidence in both Departmental and Faculty Boards of Examiners.

3.5.2 ***Dissertation Assessment***

The final written dissertation report shall account for at most 30% and 40% for undergraduate and postgraduate degrees respectively, of the overall dissertation assessment mark.

The dissertation oral examination accounts for at most 70% and 60% for undergraduate and postgraduate degrees respectively of the overall dissertation assessment mark.

3.6 **Field Trip**

A field trip is an excursion undertaken by a student(s) outside the University campus. The field trips should be undertaken under the supervision of the Academic and/or Technical Staff of the University. It may last for a part of a day or more. The field trip forms part of the studies.

3.7 **Other Terms**

For the definitions of level, semester, honours, programme, module, Work-Related learning, continuous assessment, retake, carryover, repeat, discontinue, withdraw, special examination, aegrotat. Refer to Section 1.7 of General Regulations.

4. GENERAL PROVISIONS

4.1 The Bachelor of Commerce Honours degree programme is normally studied over four levels and shall normally include a period of Work-Related Learning during Level Three.

4.2 The Visiting School/ Harare Weekend programme caters for students who are in full-time employment and is studied over three levels.

4.2.1 The Work-Related learning for the Visiting School/ Harare Weekend programme will run concurrently with the second level.

4.3 The Special Departmental Regulations shall prescribe the total number of modules per programme per semester and the degree structures.

4.4 In Level Four (Level Three for Bachelor of Commerce Visiting School/ Harare weekend), each student shall be required to write a Dissertation/ Project on a chosen subject or topic within his /her field programme of study and this work may be based on tasks and research performed during the Work-Related Learning period.

4.5 Exemption is a process of crediting a student with the module (s) already passed in programmes similar or equivalent to the one on offer. A student may be exempted from studying such modules, provided the application is being done within 3 years of completion of the study period.

However, the student has to apply for a credit transfer. In his/her application the following should be attached:

(a) A transcript of the module(s) to be credit transferred.

(b) A synopsis of the module(s) from the University/College where the module was attained.

(c) A support letter from the Registrar's Office of the institution where the module(s) were attained.

4.6 The respective Departmental Boards will sit and assess applications for credit transfers and will recommend accordingly to the Senate. A candidate applying for exemptions shall pay an exemption fee for each module.

4.7 No student shall be exempted from more than 50% of the modules offered in the programme.

5. ASSESSMENT

5.1 Evaluation of students shall be based on continuous assessment as well as formal university examinations that are held during or at the end of each semester.

5.1.1 Continuous Assessment will account for 30% of the overall assessment for undergraduate programmes. No continuous assessment shall be carried over to the next semester.

5.1.2 The final examination accounts for 70% of the overall assessment for undergraduate programmes.

5.1.3 Departments shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

5.2 To be admitted to the examinations, a candidate must:

5.2.1 Be a registered student of the University in accordance with the General Regulations

5.2.2 Have attended all and completed the approved modules of study. Satisfactory completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

5.2.3 Have attended and participated in seminars, tutorials and practical classes, work-related placement and other activities as stated in the Departmental Regulations.

5.2.4 Candidates who have attained credits from other reputable institutions shall be allowed to transfer their credits to Midlands State University. It is the prerogative of the Senate to determine if such credits would be added to students' academic records.

5.2.5 For contact hours refer to General Regulations.

6. PROVISION FOR PROGRESSION

6.1 Students shall progress from Semester One to Semester Two at each Level after they have passed at least 50% of modules taken in a semester.

6.2 Students who fail four modules and above in a semester will not be allowed to proceed to the next level.

6.3 The Faculty Board of Examiners shall meet at the end of each examination session to ratify the marks obtained by candidates and considered by the Departmental Panel and make recommendations to the Senate on issues of progression.

6.4 For a student to proceed from one level to the other, he/she must have passed at least 75% of the number of modules normally scheduled in a particular level of a programme.

7. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 10 of the General Academic Regulations.

8. WORK-RELATED LEARNING GENERAL GUIDELINES

8.1 Honours degree programmes in the Faculty of Business Sciences shall normally include twelve (12) months of supervised Work-Related Learning during the third level.

8.2 The Visiting School/ Harare weekend Programme will have the Work-Related Learning period running concurrently with the second Level of the programme.

8.3 The departments shall normally source Work-Related Learning placements for students and the Faculty Office shall provide the necessary administrative assistance.

8.4 Whilst on Work-Related Learning, students shall be supervised by an Academic Supervisor who shall visit the students at least once per semester during the Work-Related Learning period.

8.5 No student shall be allowed to change employment without the approval of the Department.

9. DEFERMENT OF STUDIES

Refer to Section 3.7 of the General Regulations.

10. REMARKING OF EXAMINATIONS

10.1.1 Marked scripts will be kept for six (6) months before they are destroyed

10.1.2 For remarking of examinations refer to Section 11.10 of the General Regulations.

11. PLAGIARISM

11.1 The dissertation shall be subjected to anti-plagiarism software. If the final dissertation has a similarity index of more than 50%, the student shall not submit it for marking and will redo the dissertation.

11.2 A dissertation shall be graded if the similarity index is not more than 20% for undergraduate and not more than 15% for postgraduate degrees.

12. GRADING AND DEGREE CLASSIFICATION

12.1 To be awarded the degree, a student must pass all core modules at all levels and must have accrued at least 540 credits.

12.2 The classification of all modules in a degree programme shall be as stated in Section 5 of the General Regulations.

13. DEGREEWEIGHTING

13.1 The results for Levels 1,2,3 and 4 shall be weighted as follows:

Level1	10%
Level2	20%
Level3	30%
Level4	40%

13.2 For the Visiting School Programme the weighting shall be as follows:

Level 1	10%
Level 2	50%
Level 3	40%

FACULTY REGULATIONS FOR POSTGRADUATE DEGREE PROGRAMMES

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with General Academic Regulations for Postgraduate Degrees hereinafter referred to as the General Regulations.

1.2 The Faculty of Business Sciences also offers the following Masters programmes:

Master of Commerce Degree in Forensic Auditing and Accounting (BS16)
Master of Commerce Degree in Finance and Digital Banking (BS17)
Master of Commerce Degree in Public Sector Accounting and Finance (MPSA)
Master of Commerce Degree in Economics (MECON)
Master of Commerce Degree in Financial Economics and Investment (BS18)
Master of Commerce Degree in Cloud Computing and Internet of Things (MCI)
Master of Commerce Degree in Information and Knowledge Management (BS23)
Master of Commerce Degree in Information Systems Management (BS30)
Master of Commerce Degree in Strategic Marketing (BS35)
Master of Commerce Degree in Strategic Brand Management (BS20)
Master of Commerce Degree in Strategy and Innovation (BS21)
Master of Commerce Degree in Strategic Management and Corporate Governance (BS27)
Master of Commerce Degree in Supply Chain Management (BS26)
Master of Commerce Degree in Tourism and Hospitality Management

Master in Business Administration Degree (BLEAD01)
Executive Master in Business Administration Degree (BLEAD02)

1.3 The Faculty of Commerce also offers the following Doctoral Programme:
Doctor of Business Leadership Degree (DBL)

2. ENTRY REQUIREMENTS

To be eligible for the programme, a candidate must normally be a holder of a good first degree in an appropriate field of study from any accredited/ recognized institution.

3. GENERAL PROVISIONS

3.1 For the period of study refer to Departmental regulations.

4. EXAMINATIONS

4.1 For a student to be allowed to write examinations they must:

4.1.1 Be a registered student of the University in accordance with the General Regulations,

4.1.2 Have attended all and completed the approved modules of study. Satisfactory completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment,

4.1.3 Have attended and participated in seminars, tutorials and practical lectures.

5. ASSESSMENT

5.1 Formal examinations will normally be written papers, but in some cases, the examiner may test the candidate orally.

5.2 Continuous assessment shall account for 40% of the overall assessment, while the final examination shall account for 60% of the overall assessment.

5.3 The aggregate pass mark shall be 50%.

6. PROVISIONFORPROGRESSION

6.1 Students shall progress from Semester One to Semester Two at each Level after they have passed at least 50% of modules taken in a semester.

6.2 The Faculty Board of Examiners shall meet at the end of each examination session to ratify the marks considered by Departmental Panels and make recommendations to the Senate on issues of progression.

7. DEFERMENT OF STUDIES

Refer to Section 3.7 of the General Regulations.

8. REMARKING OF EXAMINATIONS

8.1 Marked scripts will be kept for six (6) months before they are destroyed.

8.2 For remarking of examinations refer to Section 11.10 of the General Regulations.

9. PLAGIARISM

The dissertation shall be checked for originality with Anti-plagiarism software. A dissertation shall be accepted for grading if the similarity index is not more than 15%.

10. GRADINGANDDEGREECLASSIFICATION

10.1 To be awarded the degree, a student must pass all core modules at all levels and must have accrued at least 336 credits.

DEPARTMENT OF ACCOUNTING SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN APPLIED ACCOUNTING (BS02)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The Bachelor of Commerce Honours in Applied Accounting places emphasis more on the application of accounting skills and developing an accountant who is ready for the market. The program aims to:

- develop knowledge, skills, and competencies in the fields of Financial Accounting, Auditing, Taxation, Management Accounting, Public Sector Accounting, Corporate Governance and Financial Management relevant to various employment capabilities and careers in the world of work and society
- To prepare students for further studies and lifelong learning in the Accounting Profession and in financial resource management.

2. ENTRY REQUIREMENTS

2.1. Normal Entry

To qualify for entry into the Bachelor of Commerce Honours Degree in Applied Accounting programme, a candidate, in addition to satisfying the minimum conditions as prescribed under the General Regulations for English and Mathematics at "O" Level, must have obtained a pass in "A" Level Accounting and any of the following "A" Level subjects: Mathematics; Economics, Management of Business, Statistics, Computing [Computer Studies]

2.2. Visiting / Harare Weekend School Programme

To qualify for entry into the Bachelor of Commerce Honours Degree in Applied Accounting (Visiting / Harare Weekend School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma/ HND (complete qualification) in Accounting, or any equivalent tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects

and:

2 years of working experience.

Confirmation of employment in relevant accounting departments.

2.3. Special Entry

Candidates who have successfully completed a Professional Accounting Diploma or have obtained equivalent qualifications within three years may normally be exempted from level 1 on a module-by-module module-basis. No candidate may complete the

degree in less than three academic levels. Successful completion of the Work-Related Learning component at level three is compulsory for all candidates.

2.4. Mature Entry
Refer to faculty regulations

3. PROGRAMME CHARACTERISTICS

Areas of study: Financial Accounting, Cost and Management Accounting, Auditing, Taxation, Financial Management, Public Sector Accounting, Corporate Governance.

Specialist Focus: Applied Accountancy

Orientation: Research and innovation oriented.

Distinctive Features: The programme is driven so much by the requirements of Professional bodies who demand and expect compliance with the International Financial Reporting Standards (IFRS), International Accounting Standards (IAS) and International Standards on Auditing (ISA). Consultations are made time and again to maintain programme relevance to the ever-changing business world.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates with a Bachelor of Commerce honours degree in Applied Accounting can pursue careers as financial accountants, finance managers, project officers, auditors, tax accountants, financial analysts, cost and management accountants, financial consultants, revenue and expenditure officers, budget officers, financial loss control officers, treasurers of local authorities, financial administrators. The programme also opens opportunities for graduates to pursue further education within the accounting field. Students can enrol for Master of Philosophy in Accounting, Master of Commerce in Accounting, Master of Commerce in Forensic Auditing and Accounting, Master of Commerce in Public Sector Accounting, Master of Commerce in Finance and other related higher qualifications
In addition, with the emphasis today on small businesses, many graduates find great success in starting their businesses.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods**: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation, and individual independent study.

5.2 **Assessment Methods:** Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning reports, final-year research project/dissertation reports.

6 LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Apply Financial Accounting, Cost and Management Accounting, Auditing, Taxation and Finance knowledge and skill in resolving practical problems
- Advise internal and external stakeholders on financial matters like tax advice among others
- Demonstrate an understanding of the economic context and systems within which organisations operate and link them to business opportunities.
- Analytically assess accounting concepts and principles and create practical solutions through their application in solving practical accounting problems.
- Prepare financial statements of various organisations, comprising groups of companies,

using applicable financial information, accounting techniques and standards; and evaluate and interpret such financial statements.

- Applying Digital and Information Literacy to explain the organisational framework of the accountant and of the progress of accounting information systems; to better comprehend the need for the well-organized use of resources within an entity.
- Apply applicable management accounting techniques with cost information, for planning, decision-making, performance evaluation and control, within diverse business settings.
- Apply financial management techniques in financing, investment, and dividend policy decisions of an entity.
- Apply overall legal framework in decision-making, and employ precise legal principles relating to corporations, incorporating taxation legislation as relevant to persons, distinct companies and groups of companies.
- Carrying out the assurance audit engagement in line with the professional audit regulatory framework.
- Independently carry out Research Projects relating to a single organisation or industry sector.
- Apply information technology to document and analyse financial data, gather information from a range of sources and present research findings.
- Develop a written report which satisfies academic requirements of information gathering and referencing of information sources.
- Appraise the limitations of acquired knowledge of accounting issues and acknowledge the need to seek additional specialist advice where necessary from others in the professional community. This will be made possible through industrial visits that the students will partake in during the tenure of the program
- Resolve problems through the appraisal of fundamental issues, the creation of information and analytical tools, and the demonstration and critical assessment of a range of opportunities.
- Create employment through setting up advisory or consultancy companies within the economy in the various accounting facets.

7 GENERAL PROVISIONS

Refer to faculty and general regulations

8 PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 The final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline for submission of the dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by the Senate, the maximum mark allowed for such work should be 50%

9 PROVISION FOR PROGRESSION

In addition to general and faculty regulations for provision for progression:

- students at level 1 should have passed ACC 140 and ACC143 as prerequisites to ACC241.
- prerequisite modules to be passed before proceeding to level 3 are ACC 241 and ACC252

10 FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11 GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12 DEGREE WEIGHTING

Refer to faculty regulations

13 AWARD OF DEGREE

Refer to general regulations

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
EC131	Economic Principles I	12
BM131	Principles of Management	12
HCS135	Introduction to Information Technology	12
CS131	Basic Communication Skills	12
ACC140	Accounting Concepts, Principles and Procedures	12
ACC142	Accounting Systems in a Computer Environment	12

Level 1 Semester 2
Quarter 1

Code	Module Description	Credits
ACC143	Introductions to Applied Financial Reporting	12
ACC245	Corporate Governance and Ethics	12
BM135	Commercial Law	12
ACC144	Quantitative Analysis for Business	12
EC134	Economic Principles II	12
ACC145	Practical Accounting Data Processing	12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits

ACC246	Applied Taxation Law and Practice1	12
GSB 211	Gender Studies for Business	12
ENT131	Entrepreneurship 1	12
ACC241	Applied Financial Reporting for Companies [ACC140& ACC143]	12
ACC243	Applied Audit Skills and Process1	12
ACC244	Company Law	12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
ACC252	General and Distinctive Financial Reporting	[ACC140& ACC143]	12
ACC250	Applied Financial Management I		12
ACC251	Accounting Research Methods		12
ACC247	Applied Tax Law and Practice 11		12
ACC248	Applied Audit Skills and Process II		12
ACC249	Applied Cost and Management Accounting I		12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Prerequisites	Credits
ACC340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Prerequisites	Credits
ACC341	Work-Related Learning Continuous Assessment		40
ACC342	Work-Related Learning Report		40

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
ACC452	Applied Cost and Management Accounting II	[ACC249]	12
ACC453	Applied Public Sector Accounting and Finance I		12
ACC454	Applied Financial Management II		12
ACC450	Applied Corporate Reporting I	[ACC241&ACC252]	12
ACC451	Applied Audit Skills, Theory and Practice	[ACC243 &ACC248]	12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
ACC459	Applied Public Sector Accounting and Finance II		12
ACC460	Strategic Management Accounting and Management Information Systems		12
ACC456	Applied Corporate Reporting II	[ACC241&ACC252]	12
ACC457	Applied Advanced Auditing	[ACC243 &ACC248]	12
ACC458	Dissertation/Research Project	[ACC251)	24

15. MODULE SYNOPSES

EC131 Economic Principles I

The module will introduce economic analysis of aggregate employment, income, and prices. Topics may include major schools of economic thought; aggregate supply and demand; economic measures, fluctuations, and growth; money and banking; stabilization techniques; and international trade. Upon

completion of the module, students should be able to evaluate national economic components, conditions, and alternatives for achieving socioeconomic goals. The students should be able to analyse the economic situation in Zimbabwe as well as in different global markets.

BM131 Principles of Management
The module deals with the history and development of management thought, functions of management, organisational structure, decision-making, globalization, leadership and motivation, controlling, budgeting and non-budgetary controls, and change management.

HCS135 Introduction to Information Technology
The Information and Technology module aims to give students an understanding of the basics of the computer and its processes. How to use computer systems to make their work easier in their learning process and even after. It is also combined with the information literacy training program at the MSU Library which is aimed at making students aware of the limitless possibilities of information available to them. It also entails basic Computer Concepts, Data Processing Cycle, Number System, Computer Arithmetic, Types of computers, Computer Peripherals, System Software and Application Software, Data Communications, Introduction to Application Packages, Word Processors, Spreadsheet, Graphics, etc.

CS131 Basic Communication Skills
This module is designed to equip students with the communication skills required in the employment world. It consists of fundamental documents and methodologies for handling challenges in organisations, for example, conflict and human capital management. The module will also deal with concerns of meetings, time management, research and report writing. The students shall be equipped with the reference techniques that are applied when carrying out research or writing assignments and the use of proper written English.

ACC140 Accounting Concepts, Principles and Procedures
The module focuses on principles and techniques of financial accounting and their application to the preparation of financial statements of sole traders, partnerships, limited companies and non-profit organisations culminating in the analysis and interpretation of financial data. It is intended to equip the learners with the necessary skills to enable them to apply the basic principles of accounting and keep proper records. This knowledge is necessary to ascertain the financial results and financial position of various organisations, as well as the preparation of proper books from incomplete records.

ACC142 Accounting Systems in a Computer Environment
The module is intended to introduce the student to information technology with emphasis on its significance to accounting. It also covers communication in a technological environment as well as the role of accounting systems and their functions. The module will equip learners with knowledge of accounting systems ensure students are aware of their components and database and examine how accounting systems improve the processes in the accounting environment.

ACC143 Introduction to Applied Financial Reporting
The module builds on the foundation laid in accounting concepts, principles and procedures. The module aims at developing a thorough understanding of the practical framework of accounting and an ability to prepare, analyse and interpret financial statements. It covers the following areas of study: Accounting for limited companies, published accounts analysis and interpretation of final accounts, cash flow statements as well as the latest developments of International Financial Reporting Standards (IFRS) applications in relationship to the areas covered.

ACC245 Corporate Governance and Ethics
The module delves into the tenets of corporate governance and corporate social responsibilities. The key issues covered include the role of moral values, external institutions, internal structures, incentive systems, organisational culture, and means of guiding managerial behaviour among other things. The module will also cover local and international codes.

BM135 Commercial Law
The module is designed to equip students with basic but concise legal knowledge to enable students to make legally sound and informed business decisions at work or as managers. The module is an introduction to general principles of law and its interpretation, law of contract, purchase and sales, agency, negotiable instruments, insurance, partnership, law of derelict, insolvency and consumer protection.

ACC144 Quantitative Analysis for Business
The module introduces students to the mathematical tools relevant for application to the business environment. Students shall be expected to recognize the value of quantitative methods in analysing data and interpreting it in order to make relevant business decisions. The purpose of the module is the provide of a comprehensive exposition of basic mathematical concepts that are used in the business world. At the end of the module, learners will be in a position to apply mathematical tools to business decision-making processes.

EC134 Economic Principles II
The module is a continuation of the Economic Principles I. It further looks at the economic analysis of aggregate employment, income, and prices. Topics may include major schools of economic thought; aggregate supply and demand; economic measures, fluctuations, and growth; money and banking; stabilization techniques; and international trade. Upon completion of the module, students should be able to evaluate national economic components, conditions, and alternatives for achieving socioeconomic goals. The students should be able to analyse the economic situation in Zimbabwe as well as in different global markets.

ACC145 Practical Accounting Data Processing
The module is intended to enable the students to acquire knowledge about the significance of computers in a business environment. The learners will be exposed to accounting packages such as QuickBooks, Pastel Evolution, etc. The main thrust of the module will be the use of spreadsheet software in an accounting environment and the use of computerized accounting software packages to generate financial reports. Students will also learn the application of general accounting principles when processing data using accounting software packages.

ACC246 Applied Taxation Law and Practice I
The module introduces students to the role of tax in an economy and goes on to the applications of the Income Tax and Finance Acts sections relevant to the taxation of an individual with respect of employment income. It also covers application in nature looking at the practical preparation of tax returns, ZIMRA tax requirements, Companies' tax returns, how to file them through e-filling etc. In addition, the module will cover taxation of estates, tax administration and allowable deductions as well as the third schedule of the Income Tax Act. The core intention is to equip the students with the taxation of individual income.

GSB211 Gender Studies for Business
The module empowers the students with the knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, Gender Issues in Society, Redressing Gender Imbalances, Empowerment and Strategies for Creating a Gender Responsive Environment. Every student has to pass the module in order to graduate.

ENT131 Entrepreneurship I
The module will look at business development, barriers encountered and issues related to marketing, human resources management, pricing, government policies, and taxation of businesses. Case studies, visits to business premises and presentations by entrepreneurs will be used to understand the running of the business. The preparation of business plans will be covered.

ACC241 Applied Financial Reporting for Companies
The module builds on the foundation laid in financial accounting in the first year. The module aims at providing students with application of knowledge befitting applied accounting as related to accounting

for specialized transactions, interpretation of financial statements, and published accounts among others. It helps students to apply relevant IAS/IFRS to basic company accounts.

ACC243 Applied Audit Skills and Process1
The module introduces the learners to the application of auditing basic concepts, which includes basic auditing concepts, approaches and generally accepted auditing standards. It lays a foundation for more advanced studies in Auditing as it covers the fundamental topics in auditing. The module also focuses on the performing of the audit process and its application in the context of the regulatory framework and for business control and development.

ACC244 Company Law
The module aims to enable students to obtain a legal working knowledge of corporate enterprises applying acquired knowledge, and skills and displaying professional competence as expected of accountants. It covers basic principles of law related to Limited Liability Companies, Partnerships, Company Secretarial Practices, Meetings and Elements of Labour Law and Labour relations. Such knowledge is essential in the decision-making process relating to company secretarial functions and advice on the legality of actions and decisions taken by management and the board of directors.

ACC252 General and Distinctive Financial Reporting
The module aims to help students develop a thorough understanding of the practical framework of accounting and imbibe an ability to apply the International Financial Reporting Standards (IFRS), Financial Reporting Standards as adopted by the Institute of Chartered Accountants of Zimbabwe and PAAB. The module also introduces the student to different types of financial instruments and their characteristics. It covers the framework for the preparation and presentation of financial statements, introduction to international financial reporting standards, presentation of financial statements (IAS 1) and other accounting standards.

ACC250 Applied Financial Management I
The aim of the module is to identify the objective that Finance managers pursue or ought to pursue in order to satisfy the needs of corporate stakeholders and to develop, in students, concepts and corporate analytical tools that shall enable them to meet this objective. To this end, the module shall cover the following critical areas: Goals of a firm and the agency theory; Time value concepts and valuation of bonds and shares; Capital Budgeting under certainty; Operating and financial leverage; Introduction to portfolio theory and capital asset pricing; the stock market and other sources of long-term capital; innovations in Corporate Finance.

ACC251 Accounting Research Methods
This module enables the student to develop basic research skills focusing on a micro-unit such as an organisation or a firm. The student's mind is developed through observation to identify a research problem and its investigation leading to major findings and recommendations. It develops the student's capacity to logically gather data, present it and logically analyse it in an ethical manner. The student eventually writes up a project based on their research topic before completing their programme of study in line with ACC458.

ACC247 Applied Taxation Law and Practice II
The module introduces individual taxation, culminating in the computation of tax payable by individuals in employment, as well as the tax position of landlords. Aspects of taxation such as PAYE and double taxation agreements are covered. Administration provisions of the Income Tax Act are also dealt with. The module deals with the taxation of persons other than individuals, including businesses, companies, trusts and deceased estates. Taxes other than income tax are dealt with viz. capital gains tax, indirect taxes, tax planning, tax incentives as well as capital allowances relating to growth points, export processing zones and farmers and miners.

ACC 248 Applied Audit Skills and Process II
The module is meant to provide an understanding of the nature, purpose and scope of Auditing, the principles of Auditing and the Application of International Standards on Auditing (ISA). It also addresses the role of the external audit, planning, current issues in Auditing and its regulatory framework. The Module covers topics such as the practical approach to audit, audit evidence, audit risk, fraud and error, internal audit, the audit of transactions and balances, auditing materiality, substantive testing, management representation, quality control for audit, going concern.

ACC249 Applied Cost and Management Accounting I
The module aims to provide an understanding of the principles, concepts and techniques of Cost and Management Accounting and helps students develop an ability to apply this knowledge to practical situations related to cost ascertainment, cost control and planning. This Module will examine, in-depth, topical areas from the discipline of management accounting. The emphasis is on why and how the various concepts of Cost and Management accounting are applied in a business environment. It covers various costing techniques such as absorption costing, marginal costing and activity-based costing.

ACC340 Work-Related Learning Preliminary Report
The module is a preliminary report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of IFRSs and IASs, any gaps, conclusions and recommendations to the company and the University.

ACC341 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third year students with a practical exposure to all facets of the accounting functions. The student is expected to rotate into all sections of the Accounting Department to which thy will be attached to enable them to be exposed to all accounting areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

ACC342 Work-Related Learning Report
The module is a report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of International Financial Reporting Standards (IFRS) and International Accounting Standards (IAS), any gaps, conclusions and recommendations to the company and the University.

ACC452 Applied Cost and Management Accounting II
The module is a continuation of the Management and Cost Accounting at Part II level and seeks to consolidate students' knowledge and understanding of behavioural aspects of Management and Cost Accounting, covering Management & Cost Accounting, Budgetary Control, Standard Costing and current issues in Management Accounting. This module will examine, in-depth, topical areas from the discipline of management accounting. The emphasis is on why and how the various concepts learned in Cost and Management Accounting I are applied in a business environment.

ACC453 Applied Public Sector Accounting and Finance I
The aim of this module is to ensure that students gain knowledge and skills relating to accounting for public sector entities in compliance with the requirements of International Public Sector Accounting Standards [IPSAS]. The module covers the theoretical substructures which cover the administration as well as institutions that deal with a country's public financial management. Students will also gain an understanding of special funds created in Municipal accounting as well as be able to prepare financial statements for accounting for the funds in accordance with International Public Sector Accounting Standards.

ACC454 Applied Financial Management II
This module is a follow-up to ACC250, it introduces students to advanced issues of financial management. It covers the following areas portfolio theory and Capital Asset Pricing Model (CAPM), cost of capital, working capital management, advanced investment appraisal, dividend policy and theory, international risk management and mergers and acquisitions. Its emphasis is on how finance managers can maximize the wealth of shareholders through the management of key finance variables.

ACC450 Applied Corporate Reporting I
The module focuses on Accounting Theory and current issues relating to financial reporting. It concentrates on the role of Accounting Theory, users and objectives of corporate reports, Accounting for price level changes, analysis and interpretation of financial statements, cash flow statements and application of International Financial Reporting Standards (IFRS).

ACC451 Applied Audit Skills, Theory and Practice
The module builds on the basic principles and techniques acquired in Part II of the Auditing modules. It seeks to consolidate students' grasp of Auditing theory and the latest Auditing techniques and practices as enunciated in the International Standards on Auditing (ISAs). The aim of the module is to instruct students systematically regarding the environment in which auditing practice is conducted, in the theory of auditing and the execution of the audit process and in conducting an audit practice, to enable them to integrate various topics at the subject level.

ACC459 Applied Public Sector Accounting and Finance II
The module is a continuation of the Applied Public Sector Accounting and Finance I. The aim of this module is to ensure that students gain knowledge and skills relating to accounting for public sector entities in compliance with the requirements of International Public Sector Accounting Standards [IPSAS]. It module covers the theoretical substructures which cover the administration as well as institutions that deal with a country's public financial management. Students will also gain an understanding of special funds created in Municipal accounting as well as be able to prepare financial statements for accounting for the funds in accordance with International Public Sector Accounting Standards.

ACC460 Strategic Management Accounting and MIS
The module is a continuation of Applied Management and Cost Accounting II and discusses the characteristics of strategic Management Accounting decisions including performance evaluation, control quantitative techniques, TQM, ABC and non-financial performance measures. The Module builds up on the previous two modules on Cost and Management Accounting and seeks to consolidate the student's knowledge in behavioural aspects of management accounting, covering budgeting policies and procedures, performance evaluation, investment appraisal and strategic decision-making.

ACC456 Applied Corporate Reporting II
The module aims at students having a thorough knowledge of Advanced Accounting principles and practices as they apply to the rapidly changing international business environment. It focuses on group financial statements, accounting for foreign currency transactions deferred taxation and relevant International Financial Reporting Standards (IFRS). The module also provides knowledge and skills in preparation for the Annual Financial Statements of entities in accordance with IAS, IFRS and the Companies and Other Business Entities Act [24:31].

ACC457 Applied Advanced Auditing
The module is a continuation of the Part II Auditing modules. The module seeks to provide a comprehensive knowledge of the financial reporting provisions of the Companies and Other Business Entities Act Chapter 24:31 and focuses on Auditing issues and disclosure aspects of Financial Statements and specialized Audits and Investigations. It also covers Computer Auditing

ACC458 Dissertation/Research Project
A student is required to submit a dissertation of approximately 10 000 words on a topic relating to accounting, or auditing, or finance within a time period as may be prescribed by the department. This is an independent project in content areas related to accounting areas. Students work under the

supervision of an assigned departmental lecturer to address comprehensive company-based or research-orientated problems with documented results. The completed project should clearly present the problem of the research subject investigated or applied project undertaken, its significance to theory or business practice, the research or project background, a well-defined method, results or findings, and their implications.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN AUDIT AND RISK MANAGEMENT (BS03)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The purpose of the programme is to develop knowledge, skills, and competencies in the Auditing and Risk management field as well as to discover the fundamentals of cultivating a strong risk culture and managing risks within an organisation. This is in response to the ever-recurring external and internal business/enterprise risks being faced by organisations as well as providing insight into the organisation`s culture, policies, and procedures, and aids board and management oversight by verifying internal controls such as operating effectiveness, risk mitigation controls, and compliance with any relevant laws or regulations. This also includes policies and controls which the emerging indigenous business such as Small to Medium Enterprises should take into account for management of risks and survival. To this effect, the University Industrial Hub will also be used by students with innovative and industrialisation ideas in the field of auditing and risk management for the benefit of the communities and the nation at large. In formulating the regulations, the department therefore undertook extensive consultations with relevant stakeholders such as audit firms, professional bodies in accounting, Public Accountants and Auditors Board as well as government departments such as the Office of the Auditor General for their input and appropriateness of the program.

Upon completion, the student should therefore be able to understand the risk-based auditing standards and the core frameworks, principles and methodology of occupational fraud prevention, investigation, internal controls, and detection measures. The programme is also earmarked to prepare students for further studies and lifelong learning in the auditing and risk profession and in financial resource management and enable qualifying students to pursue careers as internal or external auditors in industry and commerce as well as gaining competent skills and competencies to form and run their audit and risk consulting firms. This would also create new job opportunities in the country.

2. ENTRY REQUIREMENTS

2.1. Normal Entry

To qualify for entry into the Bachelor of Commerce Audit and Risk Management Honours Degree programme, a candidate, in addition to satisfying the minimum conditions as prescribed under the General Regulations for English and Mathematics at "O" Level, must have 2 'A' Level in any of the following subjects: Accounting, Mathematics, Economics, Management of Business, Statistics, Computing [Computer Studies].

2.2. Visiting School Programme

To qualify for entry into the Bachelor of Commerce Audit and Risk Management Honours Degree (Visiting School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General

Regulations must have:

either:

A National Diploma (complete qualification) in Auditing, Accounting, Risk Management or any equivalent tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects

and:

Proof of employment in relevant departments.

2.3. Special Entry

Candidates who have successfully completed a Professional Auditing, Accounting or Risk Management Diploma or have obtained equivalent qualifications within three years may apply for direct entry into Level 2 of the degree programme, subject to availability of places. This is also in line with Zimbabwe Credit Accumulation and Transfer Systems (ZIMCATS) guidelines. Candidates admitted under the above regulation will normally be exempted from level 1 on a module-by-module module-basis. No candidate may complete the degree in less than three academic levels. Successful completion of the Work-Related Learning component at level three is compulsory for all candidates.

2.4. Mature Entry

Refer to faculty regulations

3. PROGRAMME CHARACTERISTICS

Areas of study: Auditing, Corporate Reporting, Forensic Auditing, Digital Ledgers & Artificial Intelligence, Financial Accounting, Risk Management, Computer Auditing & Big Data Analytics, Credit Risk and Derivatives, Technological risk Management, Macroeconomic and Financial Risk, Tax Law and Practice, Company and Business law

Specialist Focus: Auditing and Risk Management
Orientation: Research and innovation oriented. Teaching and learning are professionally oriented in business-related practices in the entities within the business environment
Distinctive Features: The programme is driven so much by the requirements of Professional bodies who demand and expect compliance with ISAs, IFRSs & IASs. Consultations are made time and again to maintain programme relevance to the ever-changing business world

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates of the Bachelor of Audit and Risk Management can pursue careers as Auditors, Financial Risk Managers and Advisors, tax accountants, financial analysts, financial consultants, revenue & expenditure officers, budget officers, financial loss control officers, treasurers of local authorities, financial administrators.

Career prospects are not limited to those types of companies traditionally associated with auditing i.e. consumer goods sellers; banks, insurance companies, retailers, market research houses, the public sector and many other organizations that can employ Audit and Risk Management graduates on an increasingly greater scale. In addition, with the emphasis today on small businesses, many graduates find great success in starting their businesses.

5. PROGRAMME DELIVERY

5.1. **Teaching and Learning Methods:** Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project, and individual independent study. During disruptive periods such as the COVID-19 pandemic, blended learning will be used, that is, online and face to face lectures and assessments. In fact, Open Distance Learning ODL (Block-Release, visiting school, e-learning and blended models) will also be availed.

6. AIMS

The aim of the syllabus is to develop knowledge and understanding of the process of carrying out the audit assurance engagement and its application in the context of the professional regulatory framework as well as to capacitate students with risk management and identification skills that enable them to anticipate and prepare for potential problems and risks before they occur, or, in the case of opportunities, to try to leverage them to cause them to occur.

6.1 LEARNING OBJECTIVES

The objectives of the program are to

- Describe the work and evidence obtained by the auditor and others required to meet the objectives of audit engagements and the application of the International Standards on Auditing (ISAs)
- Evaluate and manage business risks associated with non-compliance
- Explain the concept of audit and assurance and the functions of audit, and corporate governance, including ethics and professional conduct.
- Explain how consideration of subsequent events and the going concern principle can inform the conclusions from audit work and are reflected in different types of auditor's reports, written representations and the final review and report.
- Design strategies to eliminate harmful threats in business operations
- Describe and evaluate internal controls, techniques and audit tests, including IT systems to identify and communicate control risks and their potential consequences, making appropriate recommendations.
- Appraise better communication of risks within organisations
- Design audit and risk plans for organizations
- Formulate appropriate audit opinions and risk strategies in given scenarios
- Demonstrate ability to extract and analyze big data using contemporary technologies to formulate appropriate audit opinions and avert organizational risks
- Demonstrate how the auditor obtains and accepts audit engagements, obtains an understanding of the entity and its environment, assesses the risk of material misstatement (whether arising from fraud or other irregularities) and plans an audit of financial statement

6.2 LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Analyse, evaluate and solve problems in core areas of auditing and risk management.
- Assess risk across industries and create appropriate preparatory and coping strategies
- Distinguish between the respective roles and responsibilities of directors and internal and external auditors
- Describe the key basic component elements of the Auditor's Report, and identify and analyse matters that impact the wording of audit reports, including those that affect the auditor's opinion and those that do not affect the auditor's opinion
- Describe the nature of internal control systems of relevance to the audit, including IT systems
- Examine factors that influence risk management, including the roles employees and stakeholders have in a company's approach to risk
- Demonstrate a broad scope of management knowledge and how it applies to the discipline of audit and risk management.
- Demonstrate an understanding of the code of ethical behaviour in the field of audit and risk-based decisions and actions on appropriate business management values.
- Produce and communicate audit and risk management reports.
- Demonstrate how the auditor obtains and accepts audit engagements and conducts a detailed investigation into an audit client to identify the risks of material misstatement (whether arising from fraud, error or other irregularities)
- Devise an overall audit strategy, identify and formulate the audit tests and evidence required to meet the objectives of audit assignments to comply with the International Standards on Auditing

- Analyse how subsequent events review and going concern assessment informs the conclusions in the auditor's report
- Evaluate the nature, purpose and scope of an audit and the legal, regulatory and ethical framework for auditing
- Conduct a comprehensive assessment of associated physical, reputation, emotional, financial, and facilities risks
- Conduct a comprehensive exploration and examination of options for mitigating actions
- Develop contingency and crisis response plans
- Judge and provide a realistic assessment of those risks along the risk management matrix (probability and severity)

7. GENERAL PROVISIONS

7.1 A candidate is required to undertake a minimum of five (5) modules per semester unless one is carrying over or retaking certain modules from the previous semester.

7.2 Each module is worth twelve (12) credits except for Work-Related Learning Level, which is worth one hundred and twenty (120) credits and the dissertation which is worth twenty-four (24) credits.

7.3 The department may accept students discontinued from other degree programmes on condition they meet the entry requirements and subject to availability of places.

7.4 A student requires 540 credits to complete this programme and graduate.

8. PROGRAMME ASSESSMENT

8.1 Written and oral examinations, tests, Work-Related learning report, final year research project reports, and continuous assessment

8.2 Normally, the evaluation shall be based on the continuous assessment as well as. University examinations. Unless otherwise approved by the Senate, continuous assessment will contribute 30% of the overall assessment and the summative examination will contribute 70% of the overall assessment.

8.3 Students shall take University examinations at the end of each semester as prescribed in the Faculty and/or Departmental Regulations.

8.4 External Examiners shall be appointed to moderate all the examinations (except supplementary/special examinations).

8.5 All matters relating to the conduct of program examinations shall be the responsibility of the Registrar.

8.6 To be admitted to the program examinations, a candidate must:-

8.7 Be registered as a student of the University in accordance with the General Academic Regulations;

8.8 Have satisfactorily completed continuous assessment for approved modules of the program. 'Satisfactory Completion' of continuous assessment shall require submission of written work, attendance of lectures, seminars, tutorials, Work-Related learning placement, sandwich modules, and any other activities as stated in the Faculty and Departmental Regulations.

8.9 Exclusion from a University Examination shall require the authority of the Senate.

8.10 The Examiners may require any candidate to attend an oral examination and/or write a special examination.

8.11 Candidates studying this degree programme will be required to have attained a minimum of 480 credits acquired during their period of study in order to graduate.

9. PROVISION FOR PROGRESSION

In addition to general and faculty regulations for provision for progression:

- students at level 1 should have passed ACC 140 and ACC143 as prerequisites to

ACC241.

- prerequisite modules to be passed before proceeding to level 3 are ACC 241 and ACC252

10. FAILURE TO SATISFY EXAMINERS

Refer to General Academic Regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to General Academic Regulations

12. DETERMINATION OF CANDIDATES RESULTS

Refer to General Academic Regulations

13. Degree weighting

Refer to General Academic Regulations

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits
EC131	Economic Principles 1		12
CS131	Basic Communication Skills		12
HCS135	Introduction to Information Technology		12
BM131	Principles of Management		12
ARM101	Risk Management Fundamentals		12
ACC 140	Accounting Concepts, Principles and Procedures		12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credit
ACC143	Introduction to Applied Financial Reporting		12
BM 135	Commercial Law		12
ENT131	Entrepreneurship 1		12
ARM103	Macroeconomics and Financial Risk		12
ARM106	Auditing Principles and Procedures		12
ACC144	Quantitative Analysis for Business		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
ACC246	Applied Tax Law and Practice 1		12
ARM202	Practical Computer Auditing		12
GSB211	Gender Studies for Business		12
ARM201	Technological Risk Management		12
ACC241	Applied Financial Reporting for Companies	(ACC140 & ACC143)	12
ACC244	Company Law	(BM135)	12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits

ACC252	General and Distinctive Financial Reporting (ACC140& ACC143)	12
ACC250	Applied Financial Management I	12
ACC251	Accounting Research Methods	12
ACC249	Applied Cost and Management Accounting I	12
ACC248	Applied Audit Skills and Process II (ARM106)	12
ARM210	Big Data Analytics	12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Prerequisites	Credits
ARM300	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Prerequisites	Credits
ARM301	Work-Related Learning Continuous Assessment		40
ARM302	Work-Related LearningReport		40

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
ARM401	Credit Risk and Derivatives		12
ARM402	Forensic auditing		12
ACC450	Applied Corporate Reporting I	(ACC241&ACC252)	12
ARM404	Audit and Assurance	(ARM106 &ACC248)	12
ACC453	Public Sector Accounting and Finance I		12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
ARM406	Corporate Governance and Ethics		12
ACC456	Applied Corporate Reporting II	(ACC240 &ACC246)	12
ARM408	Auditing and Risk Management in Financial Technologies		12
ARM409	Enterprise Risk Management		12
ARM410	Dissertation	(ACC251)	24

14. MODULE SYNOPSES

EC131 Economic Principles

This module focuses on the principles of microeconomics. It covers the basic economic questions; scarcity, choice and opportunity cost; demand and supply; market equilibrium; consumer behaviour; production and costs; and market structures (perfect competition, monopoly, monopolistic competition, and oligopoly).

CS 131 Basic Communication Skills

The module intends to equip students with communication skills which involve listening, speaking, observing and emphasising. It is also designed to help students understand the differences in how to communicate through face-to-face interactions, phone conversations and digital communications like emails and social media.

HCS135 Introduction to Information Technology

The Information Technology module aims to give students an understanding of the basics of the computer and its processes. How to use computer systems to make their work easier in their learning processes and even after. It is also combined with the information literacy training

BM131 Principles of Management

The module deals with the history and development of management thought, functions of management, organizational structure, decision-making, globalization, leadership and motivation, controlling, budgetary and non-budgetary controls as well as change management in organisations.

ARM101 Risk Management Fundamentals
The module introduces students to understand concepts and fundamentals of risk management and modern techniques for analysing and identifying financial risk. Areas covered include sources of financial risk, foundation to Value at Risk (VaR) techniques, RAROC (Risk-Adjusted Return on Capital), sensitivity tests, scenario analysis, stress testing, and back-testing of risk models. Statistical models in basic ideas from probability theory such as probability distributions, expected values, higher moments, skewness, and kurtosis and more advanced topics such as nonlinear models, non-normal distributions, and tail index estimation, among others are also covered.

ACC140 Accounting Concepts, Principles and Procedures
The module focuses on principles and techniques of financial accounting and their application to the preparation of financial statements of sole traders, partnerships, limited companies and non-profit making organisations culminating in the analysis and interpretation of financial data. It is intended to equip the learners with the necessary skills to enable them to apply the basic principles of accounting and keep proper records. This knowledge is necessary to ascertain the financial results and financial position of various organisations, as well as the preparation of proper books from incomplete records.

ACC143 Introduction to Applied Financial Reporting
The module builds on the foundation laid in Accounting concepts, principles and procedures. The module aims at developing a thorough understanding of the practical framework of Accounting and an ability to prepare, analyse and interpret financial statements. It covers the following areas of study: Accounting for limited companies, published accounts analysis and interpretation of final accounts, cash flow statements as well as the latest developments of International Financial Reporting Standards (IFRS) applications in relationship to the areas covered.

BM135 Commercial Law
The module is designed to equip students with basic but concise legal knowledge to enable students to make legally sound and informed business decisions at work or as managers. The module is an introduction to general principles of law and its interpretation, law of contract, purchase and sales, agency, negotiable instruments, insurance, partnership, law of derelict, insolvency and consumer protection.

ENT131 Entrepreneurship I
The module will look at business development, barriers encountered and issues related to marketing, human resources management, pricing, government policies, and taxation of businesses. Case studies, visits to business premises and presentations by entrepreneurs will be used to understand the running of the business. The preparation of business plans will also be covered.

ARM103 Macroeconomic and Financial Risk
The module examines macroeconomic diagnostics that link interest rates, growth rates, inflation rates and exchange rates and helps identify macro shocks and policies that condition the risk environment of financial and nonfinancial firms worldwide. Topics include sources of variability in national macroeconomic performance across countries and over time, the role of government policy as a source (and dampener) of risk. The module assesses the global macroeconomic context through the cross-border assessment of risk associated with international transactions across jurisdictions and currencies. It provides an intensive view of country risk parameters.

ARM 106 Auditing Principles and Procedures
The module is meant to provide an understanding of the nature, purpose and scope of Auditing, the principles and procedures of Auditing and the application of ISAs (ISA 200, ISA 210, ISA 240-The Auditor's responsibilities relating to fraud in an audit of financial statements, ISA 135 Risk Assessment, ISA320-Audit Materiality It also addresses the role of the external audit, planning, current issues in Auditing and its regulatory framework

ACC144 Quantitative Analysis for Business
The module introduces students to the mathematical tools relevant for application to the business environment. Students are expected to recognize the value of quantitative methods in analysing data and interpreting it in order to make relevant business decisions. The purpose of the module is to provide of a comprehensive exposition of basic mathematical concepts that are used in the business world. At the end of the module, learners will be in a position to apply mathematical tools to business decision-making processes.

ACC246 Applied Taxation Law and Practice I
The module introduces students to the role of tax in an economy and goes on to the applications of the Income Tax and Finance Acts sections relevant to the taxation of an individual with respect to employment income. It also has application thrust looking at the practical preparation of tax returns, ZIMRA tax requirements, Companies' tax returns, how to file them through e-filling etc. In addition, the module will cover taxation of estates, tax administration and allowable deductions as well as the third schedule of the Income Tax Act. The core intention is to equip the students with the taxation of individual income.

ARM202 Practical Computer Auditing
The module is intended to enable the students to acquire knowledge about the significance of computers in a business environment. The learners will be exposed to auditing packages such as IDEA, Teamate, ACL etc. The main thrust of the module will be the use of spreadsheet software in an auditing environment and the use of computerized auditing software packages to generate audit reports.

GSB211 Gender Studies for Business
The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, the historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

ARM201 Technological Risk Management
Due to the complex and ever-evolving digital environment worldwide, the thrust of the module is to introduce students to various technological risks in business for them to have a better understanding and knowledge of managing these risks. This module covers among others areas such as Information Systems Risk and Cyber Risk for protecting enterprises against cyber-attacks falls largely on corporations since the regulatory environment is generally slower to close loopholes than criminals are to find new ones.

ACC241 Applied Financial Reporting for Companies
The module builds on the foundation laid in financial accounting in the first year. The module aims at providing students with application of knowledge befitting applied accounting as related to accounting for specialized transactions, interpretation of financial statements, and published accounts among others. It helps students to apply relevant IAS/IFRS to basic company accounts.

ACC244 Company Law
The module aims to enable students to obtain a legal working knowledge of corporate enterprises applying acquired knowledge, and skills and displaying professional competence as expected of accountants. It covers basic principles of law related to Limited Liability Companies, Partnerships, Company Secretarial Practice, Meetings and Elements of Labour Law and Labour Relations. Such knowledge is essential in the decision-making process relating to company secretarial functions and advice on the legality of actions and decisions taken by management and the board of directors.

ACC252 General and Distinctive Financial Reporting
The module aims to help students develop a thorough understanding of the practical framework of accounting and imbibe an ability to apply the International Financial Reporting Standards (IFRS),

Financial Reporting Standards as adopted by the Institute of Chartered Accountants of Zimbabwe and PAAB. The module also introduces the student to different types of financial instruments and their characteristics. It covers the framework for the preparation and presentation of financial statements, introduction to international financial reporting standards, presentation of financial statements (IAS 1) and other accounting standards.

ACC250 Applied Financial Management I
The aim of the module is to identify the objective that Finance managers pursue or ought to pursue in order to satisfy the needs of corporate stakeholders and to develop, in students, concepts and corporate analytical tools that shall enable them to meet this objective. To this end, the module shall cover the following critical areas: Goals of a firm and the agency theory; Time value concepts and valuation of bonds and shares; Capital Budgeting under certainty; Operating and financial leverage; Introduction to portfolio theory and capital asset pricing; the stock market and other sources of long-term capital; innovations in Corporate Finance.

ACC251 Accounting Research Methods
This module enables the student to develop basic research skills focusing on a micro-unit such as an organisation or a firm. The student's mind is developed through observation to identify a research problem and its investigation leading to major findings and recommendations. It develops the student's capacity to logically gather data, present it and logically analyse it in an ethical manner. The student eventually writes up a project based on their research topic before completing their programme of study in line with ACC458

ACC249 Applied Cost and Management Accounting I
The module aims at providing an understanding of the principles, concepts and techniques of Cost and management Accounting and helps students develop an ability to apply this knowledge to practical situations related to cost ascertainment, cost control and planning. This Module will examine, in-depth, topical areas from the discipline of management accounting. The emphasis is on why and how the various concepts of Cost and Management accounting are applied in a business environment. It covers various costing techniques such as absorption costing, marginal costing and activity-based costing.

ACC248 Applied Audit Skills and Process1
The module introduces the learners to the application of auditing basic concepts, which includes basic auditing concepts, approaches and generally accepted auditing standards. It lays a foundation for more advanced studies in Auditing as it covers the fundamental topics in auditing. The module also focuses on the performing of the audit process and its application in the context of the regulatory framework and for business control and development.

ARM210 Big Data Analytics
The module introduces students to the fundamentals of big data analytics, including the history of big data and its elements. The use of big data in business as well as the technologies used for handling big data are covered. Upon completion of this module, students should be able to process, test and analyse financial data using different technologies to help in the auditing process by acquiring hands-on experience in handling and analysing large, complex data structures.

ARM300 Work-Related Learning Preliminary Report
The module is a preliminary report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of IFRSs and IASs, any gaps, conclusions and recommendations to the company and the University.

ARM301 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third-year students with practical exposure to all facets of the accounting functions. The student is expected to rotate into all sections of the Accounting Department to which they will be

attached to enable them to be exposed to all accounting areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

ARM302 Work-Related Learning Report
The module is a report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of International Financial Reporting Standards (IFRS) and International Accounting Standards (IAS), any gaps, conclusions and recommendations to the company and the University.

ARM401 Credit Risk and Derivatives
This module focuses on credit and counterparty risk management in financial institutions and nonfinancial firms. This includes an analysis of the specialness of financial intermediaries and an overview of the types, operations, and regulatory structures that exist today. A key feature is their exposure to credit and counterparty risk. Consequently, the bulk of the module is devoted to analysing the nature of these risk exposures in different derivative instruments and strategies to manage those risks. A key part of the module covers the role of regulators in controlling such risks through mechanisms such as deposit insurance and capital requirements.

ARM402 Forensic auditing
By the end of the module, students are expected to explain fraud and forensic accounting, implement fraud controls to decrease the likelihood of fraud in the organization, utilize computerised tools to find fraudulent transactions and report on them, apply statistical and forensic accounting tools to analyse data, assist internal audit and senior management in testing data and finding anomalies, utilize advanced excel skills efficiently to identify suspicious information that needs further fraud investigation and identify fraud and anomalies using Excel and computer software.

ACC450 Applied Corporate Reporting I
The module focuses on Accounting Theory and current issues relating to financial reporting. It concentrates on the role of Accounting Theory, users and objectives of corporate reports, Accounting for price level changes, analysis and interpretation of financial statements, cash flow statements and application of International Financial Reporting Standards (IFRS).

ARM404 Audit and Assurance
The module builds on the basic principles and techniques acquired in Part II of the Auditing modules. It seeks to consolidate students on the application and audit skills and the latest auditing techniques and practices as enunciated in the International Standards on Auditing (ISAs). Standards to be covered includes ISA500, ISA610, ISA620, ISA570, ISA700, ISA701, ISA705, and ISA706.

ACC453 Public Sector Accounting and Auditing I
The aim of the module is to ensure that students gain knowledge and skills relating to accounting for public sector entities in compliance with the requirements of International Public Sector Accounting Standards [IPSAS]. The module covers the theoretical substructures which cover the administration as well as institutions that deal with a country's public financial management. Students will also gain an understanding of special funds created in Municipal accounting as well as be able to prepare financial statements for accounting for the funds in accordance with International Public Sector Accounting Standards.

ARM406 Corporate Governance and Ethics
Corporate governance systems across the world are covered in this module, for example, governance systems in market-based economies. The module includes theories in corporate governance and ethics. It also looks at the challenges of owner-managed companies vis-à-vis separation of powers, the chairman, board, shareholders and executive management. The Kings Report III & IV code, SOX Act and the Zimbabwe Code on Corporate Governance are also covered clearly articulating and evaluating the practices prescribed by these codes.

ACC456 Applied Corporate Reporting II

The module aims at students having a thorough knowledge of Advanced Accounting principles and practices as they apply to the rapidly changing international business environment. It focuses on group financial statements, accounting for foreign currency transactions deferred taxation and relevant International Financial Reporting Standards (IFRS). The module also provides knowledge and skills in preparation for the Annual Financial Statements of entities in accordance with IAS, IFRS and the Companies and Other Business Entities Act [24:31].

ARM408 Auditing and Risk Management in Financial Technologies
This module introduces students to broad trends in financial technology (FinTech) and to the emerging risks produced by technological innovation in financial services. The module further focuses on various topics including P2P Lending, cryptocurrencies and other digital assets that have emerged since 2009. There is also an emphasis on the underlying blockchain technology that supports them. The module also considers other blockchain applications (beyond cryptocurrencies) in diverse areas such as property registration, accounting and auditing, voting, smart contracts, and financial derivatives.

ARM409 Enterprise Risk Management
In this module, students gain a holistic and globally relevant view of integrated risk management. Students learn about the types of risk involved in an organisation and modern risk metrics and their evident limitations when being applied to different types of risk events, including, for example, market crashes and pandemics; the students acquire an appreciation for the importance of low-probability high-impact events and correlation spikes; discuss practical alternatives to create effective, efficient, and robust risk-transfer structures; and analyse issues related to the implementation of risk management systems and compliance with key regulatory requirements in risk-sensitive industries.

ARM410 Dissertation/Research Project
A student is required to submit a dissertation of approximately 10 000 words on a topic relating to accounting, auditing, or finance within a time period as may be prescribed by the department. This is an independent project in content areas related to accounting areas. Students work under the supervision of an assigned departmental lecturer to address comprehensive company-based or research-orientated problems with documented results. The completed project should clearly present the problem of the research subject investigated or applied project undertaken, its significance to theory or business practice, the research or project background, a well-defined method, results or findings, and their implications.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN FINANCE AND DIGITAL BANKING (BS10)

Duration	4 Years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The programme a comprehensive foundation in finance and digital banking and equips students with relevant skills to excel in the banking and finance industry.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce Honours Degree in Finance and Digital Banking programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at Ordinary Level, should have “A” Level Mathematics or Economics and any other 2 commercially related subjects such as Accounting, Business Studies or Computer Science.

2.2 Mature Entry

Over and above the requirements stipulated in the General Regulations, students who get into the Bachelor of Commerce Honours Degree in Finance and Digital Banking through mature entry should at least have a 'B' in Ordinary Level Mathematics.

2.3 Visiting School/Harare Weekend School Programme

To qualify for entry into the Bachelor of Commerce Honours degree in Finance and Digital Banking (Visiting School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have

either:

A Higher National Diploma or National Diploma in Banking and Finance related field, Education, or any equivalent tertiary qualification

or:

At least two (2) passes in relevant "A" level subjects

and

Proof of employment in the relevant field.
Two (2) years of working experience.

3. PROGRAMME CHARACTERISTICS

3.1 Areas of study: Development finance, financial risk management, Investment analysis, bank regulation and supervision, marketing of financial services, international banking and corporate finance, financial mathematics.

3.2 Specialist focus: Banking and Finance

3.3 Orientation: Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 Distinctive features: The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4 CAREER OPPORTUNITIES AND FURTHER EDUCATION

The Bachelor of Commerce Honours degree in Finance and Digital Banking provides employment opportunities in the following institutions; Commercial Banks, Merchant Banks, Building Societies, Discount Houses, Finance Houses, Asset Management Companies, Stock Exchanges, Stock Broking Firms, Insurance Companies, Microfinance Institutions, Central Banks and Academic institutions. The programme hence enables careers in the above institutions as Executive Bank Managers, Retail Banking Managers, Treasury Managers, Treasury Dealers, Financial Advisors and Financial Analysts, Bank Examiners, Risk Managers, Personal Bankers, Customer Relationship Managers, Credit Analysts, Business Development Managers and Lecturers. There is also a possibility for entrepreneurship in the form of micro-financing and financial consultancy.

5 PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research projects and individual independent study.

5.2 Assessment Methods: Written and oral examinations, tests, term papers, seminar

presentations, Work-Related learning reports, final-year research project report, and continuous assessments.

6 LEARNING OUTCOMES

By the end of the program, students will be able to:

- Develop internationally accepted core banking systems using the latest technologies
- Apply current banking and financial management practices
- Develop and apply internal control systems in compliance with the banking sector regulations
- Develop an efficient and effective approach to researching, interpreting, and analysing financial statements for risk management purposes.
- Create employment in banking and financial consultancy firms, foreign currency trading, internet/digital and mobile banking.
- Analyse money markets and the instruments used to manage liquidity and market risk in the banking sector.
- Be competent with the Basel framework capital requirements for the Institution in line with the Basel rules and regulations.
- Explain the ethical and social responsibilities of bankers and financial analysts in ensuring the integrity of financial information.

7 GENERAL PROVISIONS

7.1 A candidate is required to undertake a minimum of five (5) modules per semester excluding modules carried from the previous academic level or semester.

7.2 Each module is worth twelve (12) credits, except for the Work-Related Learning level which is worth one hundred & twenty (120) credits and the dissertation which is worth twenty-four (24) credits.

7.3 The department may accept students who have been discontinued from other programs on condition that they meet the entry requirements of the department and are subject to the availability of places.

7.4 A student requires 540 credits to complete this program.

8 PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9 PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations

10 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11 GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the Faculty Regulations.

12 DEGREE WEIGHTING

Refer to Section 13 of the Faculty Regulations.

13AWARD OF DEGREE

Refer to general regulations

12 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
EC131	Economics Principles 1		12
CS131	Basic Communication Skills		12
BM131	Principles of Management		12
HCS135	Introduction to Information Technology		12
FDB101	Money, Banking and Blockchain Technologies		12
FDB102	Financial Mathematics		12
ACC140	Accounting, Concepts, Principles and Procedures		12

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
FDB103	Financial Markets and Regulation		12
FDB104	Marketing of Financial Services		12
ACC143	Introduction to Applied Financial Reporting		12
FDB105	Foundations of Fintech		12
ACC144	Quantitative Analysis for Business		12
EC134	Economics Principles II		12

Level 2 Semester 1

Code	Module Description	Prerequisite	Credits
FDB203	Corporate Finance 1		12
ENT131	Entrepreneurship 1		12
GSB211	Gender Studies for Business		12
FDB201	Personal Financial Management		12
FDB202	Digital Banking and Financial Technology		12
ACC241	Applied Financial Reporting for Companies		12

Level 2 Semester 2

Code	Module Description	Prerequisite	Credits
FDB204	Corporate Finance II		12
FDB206	Programming in Finance and Banking		12
FDB209	Research Methods		12
ACC252	General and Distinctive Financial Reporting	[ACC140, ACC143]	12
FDB205	Banking Law and Practice		12
FDB207	Financial Econometrics and Data Analysis		12
FDB208	Foreign Exchange Markets and International Finance		12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Prerequisite	Credits
FDB340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Prerequisite	Credits
FDB341	Work-Related Learning Continuous Assessment		40
FDB342	Work-Related Learning Report		40

Level 4 Semester 1

Code	Module Description	Prerequisite	Credits

FDB401	Corporate Finance III [FDB203, FDB]	12
FDB402	Treasury Management [FDB208]	12
FDB403	Financial Risk Management	12
FDB404	Development Finance	12
FBD405	Financial Modelling	12

Level 4 Semester 2

Code	Module Description	Prerequisite	Credits
FDB408	Investment Analysis and Portfolio Management		12
FDB407	Bank Lending and Credit Risk Management		12
FDB406	Financial Engineering and Asset Pricing		12
FDB409	Dissertation/Research Project		12

13 MODULE SYNOPSES

FDB101 Money, Banking and Blockchain Technologies

The module deals with the role played by money and financial institutions within Zimbabwe. It serves as an introductory module to the broader field of banking and finance. It deals with practical aspects of the economy such as the role of money in economic activity, the role of the central banks, money supply and money demand, interest rates, monetary policy, exchange rates the issues relating to banking structures and systems, the Zimbabwean financial system and issues in the international banking business.

FDB102 Financial Mathematics

This module will examine theories and concepts of Financial Mathematics. The emphasis of this module will be placed on the computations of simple and compound interest, nominal and effective rates, future value, present value, discount factors, interpolation and extrapolation and cash flow analysis techniques, money market calculations, discount instruments, CDs and general pricing of assets. It also covers bond markets, equity markets, derivative markets, Macaulay duration and the Capital Asset Pricing model.

FDB103 Financial Markets & Regulation

The module covers the financial system framework, the role and uses of money markets, the stakeholders in money markets, money market securities and their derivative products. It also considers the nature and scope of capital markets as well as the role of the following capital market participants: Investors, depository institutions, non-depository institutions, investment banking firms, central banks, government and other stakeholders. The module further equips students to be conversant with the regulation of financial institutions and markets. Topics covered include elements of financial supervision, theories and structure of regulation banking risk, capital regulation, banking crises, lender of last resort and deposit insurance, banking and financial crises, insolvency and systemic risk, central banking, central banking law, the independence and accountability of central banks, institutional organization of supervision, regulation of securities markets; regulation of insurance companies, pension funds, mutual funds, hedge funds and alternative investment funds, regulation of derivatives markets, takeovers and mergers, international financial regulation and cross-border bank insolvency.

FDB104 Marketing of Financial Services

The module examines the need for marketing by Financial Institutions, and the formulation of marketing strategies that take into consideration the special circumstances and problems of financial services. The module looks at Services Marketing in general and Financial Services Marketing in particular focusing on the uniqueness of financial services products and hence the need to have a special approach to their marketing. It examines the marketing building blocks such as advertising, distribution, product development and pricing and how

these are dealt with in the context of financial services marketing.

FDB105 Foundations of Fintech
The module focuses on fintech transformation and evolution as well as fintech typology. It looks at the collaboration between financial institutions and start-ups. The foundations of the fintech module also seek to explore the opportunities and challenges that emerging economies are likely to face. The module basically seeks to give an introduction and solid background on the programme.

ACC140 Accounting Concepts, Principles and Procedures
This module is intended to equip students with the necessary skills and should enable the student to apply the basic principles of accounting and keep proper records required to ascertain the financial results and financial position of sole proprietors and non-profit organizations, as well as prepare proper books for incomplete records.

ACC144 Quantitative Analysis for Business
The module introduces students to the mathematical tools relevant for application to the business environment. Students are expected to recognize the value of quantitative methods in analysing data and interpreting it in order to make relevant business decisions. The purpose of the module is to provide a comprehensive exposition of basic mathematical concepts that are used in the business world. At the end of the module, learners will be in a position to apply mathematical tools to business decision-making processes.

ACC143 Introduction to Financial Reporting
The module builds on the foundation laid in accounting concepts, principles and procedures. The module aims at developing a thorough understanding of the practical framework of Accounting and an ability to prepare, analyse and interpret financial statements. It covers the following areas of study: Accounting for limited companies, published accounts analysis and interpretation of final accounts, cash flow statements as well as the latest developments of International Financial Reporting Standards (IFRS) applications in relationship to the areas covered

FDB201 Personal Financial Management
This module empowers students with knowledge and skills pertaining to personal finance which consists of the following: financial planning, budgeting, managing personal finances, financial assets, protecting assets, retirement planning, investing and taxes. It also includes a range of retirement concepts, principles and theories.

FDB202 Digital Banking and Finance
This course is about digital economics and how the digital economy influences markets and society. It looks at how the internet, mobile communications, the sharing economy, social media and cryptocurrencies impact global businesses. It looks at basic theory in digital economics covering network effects, value creation models and market modelling. It also looks at how the digital economy influences privacy, regulations and strategy.

FDB203 Corporate Finance I
The module examines corporate decisions pertaining to the utilization of long-term financial resources for the benefit of the ultimate risk-takers. It specifically examines the concepts of the agency problem, inflation, risk, value additively, present value, internal rate of return, payback period, and accounting rate of return and relates them to the Corporate Capital Budgeting Process. It further examines the valuation techniques for various financial assets and develops the efficient market theory.

ACC241 Applied Financial Reporting for Companies
The module is intended to enable students to gain knowledge and insight into the drafting of financial statements of companies so as to comply with the requirements of the Companies Act Chapter 24:03 and International Financial Reporting Standards [IFRSs]. The module also introduces the student to different types of financial instruments and their characteristics.

FDB204 Corporate Finance II
This module, which builds on Corporate Finance A, further examines the Corporate Capital budgeting Process valuation in the presence of Leverage by looking at the Cost of Capital, Single Asset and Portfolio Evaluation, Dividend Policy, and Real Options. It also evaluates the Capital structure theories and working capital management issues.

FDB205 Banking Law and Practice
This module examines the law and practice relating to Banking in Zimbabwe and at the international level. It covers the history and development of the Zimbabwean banking system, the role of regulators, the relationships between banker and customer, accounts of customers, banking instruments, different types of banks, principles of lending, deposits, advance securities, exchange controls and bank supervision and control are covered in the module.

FDB206 Programming in Finance and Banking
This module concentrates on helping students to appreciate and use programming languages important in the banking and finance industry. It concentrates on the following languages: Python, Java, Scala, C++ and SQL

FDB207 Financial Econometrics and Data Analysis
This module aims to introduce students to the fundamental techniques used in the analysis of financial data and to provide the necessary econometric background to carry out empirical investigations. It covers the classical linear regression model, multiple regression models, econometric problems, time series analysis, distributed lag and autoregressive models, and interpretation and presentation of key econometric findings.

FDB208 Foreign Exchange Markets and International Finance
The module provides students with a forum for understanding the theoretical and practical issues in international finance foreign exchange markets. It specifically looks at foreign exchange quotations, foreign exchange transactions, foreign exchange regimes, foreign exchange forecasts, parity conditions in money and forex markets, forex foreign exchange risk management, arbitraging and speculating in the forex market and the role played by both the Central and other financial institution in the forex market as well as in import and export financing.

FDB209 Research Methods
This course covers how they are used to examining relating to their studies. Quantitative, qualitative and mixed measures are all differentiated in this course. Students will be practically exposed to the main components of a research framework, i.e., problem definition, research design, data collection, ethical issues in research, report writing, and presentation. Once equipped with this knowledge, participants would be well-placed to conduct disciplined research under supervision in an area of their choosing.

ACC252 General and Distinctive Financial Reporting
The module is intended to equip the student with the knowledge and skills required to enable the student to prepare consolidated financial statements of a simple group of companies. By the end of the module, the student should also be able to analyse financial statements in a way useful for decision-making and be able to value financial instruments.

FDB301 Work-Related Learning Continuous Assessment
Approved industrial attachment of not less than eight (8) months and not more than twelve (12) months is aimed at providing third year students with a practical exposure to Banking and Finance Industry. The candidate is expected to rotate the various departments of the organization to which

he/she will be attached to enable them to be exposed to all functions of that organisation. The candidate must be appraised by each departmental supervisor who assigns the student day to day duties and a percentage score should be given before leaving for the next department.

FDB302 Work-Related Learning Report
The module is a problem-based report developed by the candidate during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a schematic report layout covering issues such as company background, departments attached, challenges encountered, new issues learnt, any gap between information covered in the University, review of related literature to issues covered, comparison of practical issues and reviewed literature, any gap and conclusions and recommendations to the company and the University. A guideline for the development of the report is issued to the candidates as a module outline.

FDB401 Corporate Finance III
The module endeavours to advance students' knowledge of corporate finance. The following aspects will be covered: Equity valuation, mergers and acquisitions, corporate restructuring, business and financial risks, private equity or venture capital, corporate financial distress or bankruptcies, structured finance and corporate financial strategies.

FDB402 Treasury Management
The module covers the treasury function with a particular focus on treasury strategies. These include money market; foreign exchange; capital market and derivatives trading strategies. Students are equipped with the pricing of securities in the various markets. In addition, the module aims to enable students to identify key customer relationships, internally and externally and develop strategies for improving them. Important aspects of liquidity management and operational risk are also presented. The module takes both strategic and perspectives of treasury management.

FDB403 Financial Risk Management
The module examines risks faced by the financial institution: market risk, operational risk, credit risk, and risks in the regulatory environment. Risk management frameworks, policies and implementation systems, in the financial and non-financial corporate sectors. Measuring financial risks and the implications of model risk for their measurement. Portfolio analysis, and the measurement of risks. Measuring risks in detail: market risk and the VaR and CFaR approaches. The management of risks: banks, non-bank financial institutions and non-financial corporations. It also includes infamous risk management disasters, options hedging strategies and interest rate option strategies as well as the future of risk management and the convergence of insurance risk management with financial risk management.

FDB404 Development Finance
The module's main focal areas are on the international finance for development covering innovative approaches to enhance economic growth and development finance to contribute towards realizing global sustainable development goals. Development Finance from a 'Financing for Development' Perspective, domestic resource mobilization, debt relief, microfinance, financial sustainability, International Private Capital Flows, Remittances for Development for Development tax buoyancy, Foreign Direct Investment, foreign capital flows, and labour productivity, Microfinance roots, mechanisms, competition, legal, regulatory and governance issues, The Impact of Microfinance on Household Livelihoods, Debt Relief.

FDB405 Financial Modelling
This module equips students with skills to make projections of revenue and costs and on the basis of such projections, evaluate investment decisions. In a stock market analysis, financial modelling helps students to track the past performance of a company and project what the revenue, costs and profit of the company are likely to be and therefore if we should buy, sell or hold that share. The module concentrates on the following models; Discounted Valuation Model, Leveraged Buyout

Model, Merger and Acquisition Model, Comparable Company Analysis and Credit Rating Model.

FDB406 Financial Engineering and Asset Pricing
Topics covered include major valuation techniques in a variety of contexts including arbitrage pricing, interest rate futures, forward rate agreements, swaps, equity options, fixed income and other exotic derivatives, and other fundamentals of financial innovation in quantitative finance.

FDB407 Bank Lending and Credit Risk Management
The module seeks to equip students with a thorough understanding of the principles of bank lending and credit risk management. To achieve this, the module covers lending policies used by banks and other financial institutions in the business of lending; credit risk analysis; credit execution and administration; loan syndication; securities in lending; loan pricing methodologies; credit review and finally how to deal with problems and non-performing loans.

FDB408 Investment Analysis and Portfolio Management
The module sets out to examine the design of investment policy statements, the creation of efficient portfolios guided by the modern portfolio theory and equilibrium capital market models such as the Capital Asset Pricing Model, the Single Index Model, Arbitrage Pricing Theory and other multifactor models; the Investment process, applying the fundamental and the technical techniques, Investment Management Strategies, distinguishing between Strategic Asset Allocation (SAA), Tactical Asset Allocation (TAA) and asset selection strategies. Investment performance measures for both broad portfolio and portfolio categories from both a theoretical and practical point of view will also be covered.

FDB409 Dissertation/Research Project
Students will be expected to complete a dissertation on a topic of their choice but limited to the taught modules. The dissertation is a consolidation of the theoretical knowledge gained in the taught modules and the practical experience gained from industrial attachment. This module aims to give students the opportunity to solve practical entrepreneurial problems through research. a research project of at least 10000 words on an approved banking and risk management-related topic is required. The dissertation will commence in 4th year's first semester and the final document is expected to be submitted in final year second semester. The study must address contemporary/ topical issues in banking, finance and risk. The dissertation consists of 5 chapters and must meet the requirements set by the department.

CS131 Communication Skills
Refer to the Centre for Communication Skills for the module synopsis.

BM131 Principles of Management
Refer to the Department of Business Management for the module synopsis.

HCS135 Introduction to Information Technology
Refer to the Department of Information Systems for the module synopsis.

EC131 Economic Principles I
Refer to the Department of Economics for the module synopsis.

EC134 Economics Principles II
Refer to the Department of Economics for the module synopsis.
ENT131 Entrepreneurship I
Refer to the Centre for Entrepreneurship for the module synopsis.

GSB211 Gender Studies for Business
Refer to the Centre for Gender Studies for the module synopsis.

REGULATIONS FOR MASTER OF COMMERCE DEGREE IN FORENSIC AUDITING AND ACCOUNTING (BS16)

Duration	1 ½ years
Credit load	306
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBK's Credit Load	216
ZNQF level	9

1. PURPOSE OF THE PROGRAMME

The programme explores investigation elements in forensic auditing and equips students with relevant critical thinking skills to collect and interpret Financial Fraud and Forensics Audit data in the contemporary world along with the legal mechanism to counter corporate fraud and understand Forensic Audit methods. The program will empower students to recognize the root causes of fraud and white-collar crime in the current economy, analyse the categories of fraud, equip them with methodologies of fraud detection and prevention, and heighten their ability to detect potential fraudulent situations.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Master of Commerce Degree in Forensic Auditing and Accounting programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations, for "O" level English and Mathematics, must be a holder of a first degree in Accounting, Forensic Accounting, Auditing, Banking and Finance or any other related degree.

3. PROGRAMME CHARACTERISTICS

3.1 Areas of study: Financial Analysis, Internal and External Fraud Investigation, Auditing, Communication Computer Skills, Asset Protection, Digital Forensics, Legal Evidence Methodologies, Expert Witnessing, Investigations and Criminal Law

3.2 Specialist focus: Forensic Auditing and Forensic Accounting

3.3 Orientation: Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 Distinctive features: The programme builds the research-technology-innovation continuum and focuses on Forensic accounting which includes conducting investigations to uncover financial fraud, often acting as expert witnesses during trials. It will basically be advanced training in areas related to investigation and auditing using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

The Master of Commerce Degree in Forensic Auditing and Accounting provides employment opportunities in Government or Private Firms. Institutions may include the following: Accounting Firms, Mining organisations, Consultancy Firms, Banks, Law Enforcement Agencies, Universities, and Telecommunication Companies. Roles will include conducting criminal and civil investigations into crimes such as illegal financial transactions and embezzlement. Forensic Auditors can also work as fraud examiners and certified public accountants (CPAs), reviewing financial documents to resolve civil disputes. There are also possibilities for entrepreneurship in the form of Forensic accounting and auditing Consultancy.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, research projects and individual independent study.

5.2 Assessment Methods: Written and oral examinations, tests, semester papers, hands-on, seminar presentations, continuous assessments and final-year research projects. Formal examinations will normally be held at the end of each semester.

For a student to be admitted to the examinations they must have:

5.1.1 Satisfactorily attempted approved modules of study including submission of required written assignments and term papers.

5.1.2 Attended compulsory classes.

5.1.3 Participated in prescribed seminars, tutorials and practical classes.

5.1.4 Paid the prescribed fees in accordance with the General Regulations.

5.1.5 Formal examinations will normally be written papers, but in some cases, the examiner may test the candidate orally.

5.1.6 Continuous assessment shall account for 40% of the overall assessment, while the final examination will account for 60% of the overall assessment.

5.1.7 The aggregate pass mark shall be 50% and the candidate must have obtained at least 50% in the examination.

6. LEARNING OUTCOMES

By the end of the program, students will:

- Apply forensic accounting and investigations expertise, tools, models, software and techniques in order to examine different complex fraud schemes.
- Utilise advanced Big Data Analytics tools, software and models in order to extract admissible evidence.
- Investigate complex economic & financial crimes and recommend suitable remedial action.

- Prepare court-ready expert forensic investigative reports.
- Design and implement anti-corruption and integrity management programmes. Students should appreciate the fundamentals of accounting practices and financial statement analysis.
- Develop knowledge and interest in applying skills in civil, corporate and criminal litigation procedures.
- Support dispute and litigation, expert witnessing, fraud examination and regulatory investigations.
- Provide expert skills in business valuation, assessment of economic damages, proceeds of crime and fraud.
- Investigate real-life case studies and learn the processes involved in compiling own case for example gathering and presenting evidence and preparing expert reports.
- Examine corporate integrity and management control, fraud and forensic analysis, cybercrime and interviewing and testimony.

7. GENERAL PROVISIONS

7.1 A candidate is required to undertake a minimum of five (5) modules per semester excluding modules carried from the previous academic level or semester.

7.2 A student requires a minimum of 306 credits to complete this programme.

8. PROGRAMME ASSESSMENT

8.1 Normally evaluation shall be based on continuous assessment as well as University Examinations. Unless otherwise approved by the Senate, continuous assessment will contribute 40% of the overall assessment.

8.2 The department shall determine which items of continuous assessment and practical work will be included in the continuous assessment. The department will maintain written records of each student's performance in these elements of continuous assessment items incorporated in continuous assessment may include assignments tests, essays, projects or any other as shall be determined by the department.

8.3 Students shall normally take examinations at the end of each semester as prescribed in the Faculty and or Departmental regulations

8.4 External examiners shall be appointed to moderate examinations.

8.5 For more detail refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9. PROVISION FOR PROGRESSION

9.1 A student shall progress from Semester One to Semester Two at each level after they have passed more than 50% of the modules taken in a semester.

9.2 The Faculty Board of Examiners shall meet at the end of each quarter of the semester to ratify the marks obtained by candidates and considered by Departmental Boards.

9.3 The Faculty Board of Examiners shall meet at the end of the second semester of a level to ratify the marks obtained by candidates and make recommendations to Senate on issues of progression by students to the next

level. In the case of repeat-level students, the Faculty Board of Examiners may, however, make a recommendation to withdraw or discontinue the student at this stage.

9.4 For a student to proceed from one level to the other, he/she must have passed at least 75% of the number of modules normally scheduled in a particular level of a programme

10. FAILURE TO SATISFY EXAMINERS

A candidate who fails to satisfy the Examiners in terms of the General Academic Regulations and Faculty regulations may be required by the Senate to:

(a) Retake
(b) Proceed to the next level of the program carrying one or more modules from a preceding level
(c) Repeat
(d) Discontinue
(e) Withdraw and reapply after two years

For more detail refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

The following grading scale shall be used for all the modules and grading of the degree:

80% - 100%	Distinction
70% - 79%	Merit
60% - 69%	Credit
50% - 59%	Pass
49% and below	Fail

12. DEGREE WEIGHTING

For purposes of degree classification, the parts of the degree programme will be classified as follows:

Level 1	50%
Level 2	50%

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MFA700	Forensic Accounting and Auditing	18
MFA701	Money Laundering and Asset Tracing	18
MFA702	Compliance and Governance	18
MFA703	Big Data Analytics in Forensic Accounting	18
MFA704	Professional Ethics in Forensic Auditing	18

Level 1 Semester 2

Code	Module Description	Credits

MFA707	Financial Crimes and the Law	18
MFA708	Cyber-crime and Digital Forensics	18
MFA705	Tax Audit and Investigation	18
MFA706	Fraud Risk Management	18
MFA709	Financial Statements and Analytical Techniques	18
MFA710	Research Methodology	18

Level 2 Semester 1

Code	Module Description	Credits
MFA800	Advanced Forensic Litigation and Expert Witnessing	18
MFA801	Dissertation	90

14. MODULE SYNOPSIS

MFA700 Forensic Accounting and Auditing
The module introduces students to investigative techniques for planning, gathering, interpreting, and documenting evidence in areas such as valuation of closely held businesses, lost profits analyses, and various types of litigation support services. Legal concepts governing expert witness testimony are also examined.

MFA701 Money Laundering and Asset Tracing
The module introduces students to money laundering, international controls and regulations to prevent and detect potential sources of money laundering, various methods and techniques of tracing assets

MFA702 Compliance and Governance
The module introduces students to the Zimbabwean regulatory environment, international regulatory environment (UK and USA), compliance framework, compliance function, compliance risk, corporate governance, professional ethical standards, professional conduct and integrity management.

MFA703 Big Data Analytics in Forensic Accounting
The module introduces students to data, methods of quantitative analysis, and concepts of data mining, data warehousing, predictive, descriptive and prescriptive analytics. Tools such R-Studio, F-sharp, Python, Open Source Toolkit – WEKA and Hadoop shall be used as hands on practical's.

MFA704 Professional Ethics in Forensic Auditing
The module introduces students to the fundamental principles underlying the IFAC code of ethics, the ethical responsibilities of accountants in business in relation to Fraud, Money laundering, theft, over-exaggeration or suppression of information or values, unethical behaviour to individuals, the accountancy profession and the public.

MFA705 Tax Audit and Investigation
The module introduces students to civil and criminal tax fraud matters, methodologies of investigation, prosecution, and defence, as well as issues surrounding the resolution of certain types of cases. The module covers introduction to Tax Fraud, tax fraud patterns,

offences, tax due diligence, tax audit and investigations, tax regulations, prosecution and defence.

MFA706 Fraud Risk Management
The module introduces students to risk identification, assessment & evaluation, fraud risk management, development and implementation of risk response strategy, risk monitoring and communication, development, measuring and monitoring of KRIs & KPIs, and designing and implementation of information systems controls.

MFA707 Financial Crimes
The module introduces students to financial crimes, motivations and typologies: the drivers of financial crime, offending and the types of financial crime commonly encountered, the laws and statutes governing financial crimes, financial crime investigation modelling, and managing a financial crime investigation.

MFA708 Cyber-crime and Digital Forensics
The module introduces students to database forensics, network forensics, computer forensics, investigative tools, digital evidence, evidence collection, forensic standards, search and seizure, e-discovery and network intelligence, types of cybercrimes, cybercrimes law, legal restrictions on the movement and use of cryptographic technology.

MFA709 Financial Statements and Analytical Techniques
The module focuses on Analysis of financial statements and audit reports, Horizontal and vertical analysis, Financial ratio analysis and Anomaly detection to identify nonconformities.

MFA710 Research Methodology
The module introduces students to advanced topics in research design and statistical analysis. The topics include underlying philosophies to research-based methods, observational and experimental research designs, strengths and weaknesses of each design, internal validity for different research designs, research proposal, advanced analysis and overall thesis writing.

MFA800 Advanced Forensic Litigation and Expert witnessing
This module introduces students to Dispute Resolution (DR) theory and practice, providing an overview of contemporary DR systems (including litigation, negotiation, mediation, conciliation, arbitration, and blended processes). Legal and professional issues arising out of DR developments are analysed.

MFA801 Dissertation
The dissertation is carried out by students in an area of their choice related to the discipline of forensic accounting and forensic auditing. Each student is required to apply their acquired knowledge and skills to a project that is motivated by a real-world challenge(s) in forensic accounting and fraud investigations.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN PUBLIC SECTOR ACCOUNTING AND FINANCE [MPSA]

Duration	1 ½ years
Credit Load	306
Minimum Credit Load	270
Maximum Credit Load	340

Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

The degree programme is a result of extensive consultations that the department undertook with the public sector and the accounting profession. Consultations with the public sector included the Ministry of Finance, Public Accountants and Auditors Board (PAAB), and Public Accountants Organisations (PAOs). Public Sector Accounting and Finance focuses on how the public sector and its entities of various kinds make the best use of the vast resources entrusted to them in creating public value. Public sector accounting is a core body of knowledge supporting public financial management. The purpose of this programme is to provide students with the advanced public sector accounting and financial management skills necessary to function in public sector environments of medium to high complexity and low uncertainty. Modules availed to that effect address governance and decision-making processes, government budgeting and planning, government financial reporting, financial analysis of government, administration and public sector auditing.

By the end of the program, students will be able to:

• Apply International Public Sector Accounting Standards (IPSASs) correctly when preparing financial statements.

• Process and treat complex accounting transactions according to the requirements of the IPSAS accounting standards

• Demonstrate leadership skills in diverse organizational situations through application of the best managerial practices.

• Research on work-related problems and provide solutions to the problems

• Reduce fraudulent and improper reporting by public sector enterprises by upholding the virtues of integrity, a sense of responsibility, ubuntu and a positive attitude, required for such a highly regarded professional discipline.

• Be innovative and be able to solve public sector accounting and finance-related organizational problems.

2. ENTRY REQUIREMENTS

Normal Entry

To qualify for entry into the Master of Commerce Degree in Public Sector Accounting and Finance Programme, a candidate must have obtained a minimum of a first degree in Accounting, a first degree in Public Sector Accounting and Finance or any other relevant first degree.

Special Entry

N/A

Mature Entry

N/A

3. PROGRAMME CHARACTERISTICS

3.1 Areas of study: Public Sector Auditing, Public Sector Financial Reporting, Public Sector Financial Management, Public Sector Performance Management,

Public Sector Ethics and Governance and Public Sector Administration and Law.

3.2 Specialist Focus: Public Sector Accounting and Finance

3.3 Orientation: Research and innovation-oriented. Teaching and learning are professionally oriented in organisational-related practices.

3.4 Distinctive Features: The programme is driven much by the problems faced in the public sector. Consultations are made time and again to maintain programme relevance to the ever-changing world.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION.

The graduates have a wide range of career paths to follow within the accounting profession. They can be employed as:

- Accountant (public, professional and certified)
- Finance Manager/Director
- Auditor
- Budget analyst
- Treasurer

Further studies can be undertaken through Doctoral Studies in any area within the public sector accounting and finance discourse.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods will include; Lectures, workshops, seminars, group work, research project, individual independent study and field reports

5.2 **Assessment Methods:** Written and oral examinations, tests, semester papers, hands on, seminar presentations, final year research project report, and continuous assessments. Formal examinations will normally be held at the end of each semester

5.1.1 For a student to be admitted to the examinations they must have:

5.1.2 Satisfactorily attempted approved modules of study including submission of required written assignments and term papers.

5.1.2 Attended compulsory classes.

5.1.3 Participated in prescribed seminars, tutorials and practical classes.

5.1.4 Paid the prescribed fees in accordance with the General Regulations.

5.1.5 Formal examinations will normally be written papers, but in some cases, the examiner may test the candidate orally.

5.1.6 Continuous assessment shall account for 40% of the overall assessment, while the final examination will account for 60% of the overall assessment.

5.1.7 The aggregate pass mark shall be 50% and the candidate must have obtained at least 50% in the examination.

6. LEARNING OUTCOMES

By the end of the program, students will be able to:

- Prepare financial statements that are consistent with the requirements of the International Public Sector Accounting Standards (IPSAS's)
- Evaluate public financial management policies and strategies.
- Analyse, process and treat complex accounting transactions according to the requirements of the international public sector accounting standards.
- Demonstrate leadership skills in diverse organizational situations through application of the best managerial practices.
- Review and assess the economic impact of fraudulent and improper reporting by public sector enterprises by upholding the virtues of integrity, a sense of responsibility, ubuntu and a positive attitude, required for such a highly regarded professional discipline.
- Demonstrate critical thinking essential for problem-solving in the public sector.

And the Learning outcomes read as follows;

By the end of the programme the students should be able to:

- Apply the theories, concepts and methods that underlie public sector accounting knowledge and skill in resolving practical problems.
- Apply taxation, financial reporting, financial management and performance management in the public sector in public policy and strategy development.
- Review techniques and methods for accounting, reporting, measurement and control as applicable to government and public enterprises.
- Analyse the economic condition of government and its institutions.
- Advise internal and external stakeholders on public sector financial and accounting matters like tax advice among others.
- Demonstrate an understanding of the economic context and systems within which the public sector institutions operates and link them to opportunities for economic development.
- Resolve problems through the appraisal of fundamental issues, creation of information and analytical tools, and the demonstration and critical assessment of a range of opportunities.
- Create employment through setting up advisory or consultancy companies within the economy in the various public sector accounting and finance facets.

7. GENERAL PROVISIONS

Refer to faculty and general regulations.

8. PROGRAMME ASSESSMENT

8.1 Normally evaluation shall be based on continuous assessment as well as University Examinations. Unless otherwise approved by the Senate, continuous assessment will contribute 40% of the overall assessment.

8.2 The department shall determine which items of continuous assessment and practical work will be included in the continuous assessment. The department will maintain written records of each students' performance in these elements of continuous assessment items incorporated in continuous

assessment may include assignments tests, essays and projects or any other as shall be determined by the department.

8.3 Students shall normally take examinations at the end of each semester as prescribed in the Faculty and or Departmental regulations

8.4 External examiners shall be appointed to moderate examinations.

8.5 For more detail refer to the General Regulations and Section 5 of the Faculty Regulations.

9. PROVISIONS FOR PROGRESSION

Students shall progress from Semester One to Semester Two at each Level after passing more than 50% of modules taken in a semester. The Faculty Board of Examiners shall, at the end of the second semester of a level, make recommendations to Senate on issues of progression by students to the next level. In the case of repeat-level students, the Faculty Board of Examiners may, however, make a recommendation to withdraw or discontinue the student at this stage. For a student to proceed from one level to the other, he/she must have passed at least 75% of the number of modules normally scheduled in a particular level of a programme.

Exemption is a process of crediting a student with the module(s) already passed in programmes similar or equivalent to the one on offer. A student may be exempted from studying such modules. However, the student has to apply for a credit transfer. In his/her application the following should be attached:

a) A transcript of the module(s) to be credit transferred.

b) A synopsis of the module(s) from the University/College where the module was attained.

c) A support letter from the Registrar's Office of the institution where the module(s) were attained.

The Departmental Board will sit and assess applications for credit transfers and will recommend accordingly to the Senate.

No student shall be exempted from more than 50% of the modules offered in the programme.

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations.

12. DEGREE WEIGHTING

Refer to faculty regulations.

13. AWARD OF DEGREE

Refer to general regulations.

14. PROGRAMME STRUCTURE

All the modules in this programme are core courses.

Level 1 Semester 1

Code	Module Description	Credits
MPSA700	Advanced Public Sector Auditing Practice and Assurance Services	18
MPSA701	Advanced Public Sector ethics and governance application	18
MPSA702	Advanced Public Sector Administration and Law	18
MPSA703	Advanced Public Sector Financial Management	18
MPSA704	Advanced Research Methods	18

Level 1 Semester 2

Code	**Module Description**	**Credits**
MPSA705	Applied Public Sector Financial Reporting	18
MPSA706	Strategic Public Procurement	18
MPSA707	Applied Strategy and Policy Development	18
MPSA708	Applied Taxation	18
MPSA709	Applied Public Sector Performance Management	18

Level 2 Semester 1

Code	**Module Description**	**Credits**
MPSA710	Public Debt Management	18
MPSA711	Revenue and Expenditure Management	18
MPSA712	Dissertation	90

MODULE SYNOPSES

MPSA700 Advanced Public Sector Auditing and Assurance Services

On successful completion of this module, students will be able to appraise, assess and evaluate the scope and legal, regulatory and ethical environment within which public sector audits are performed. The module covers audit risk assessment and planning procedures required by relevant auditing standards, requirements of audit programmes, including the identification of internal controls and design of audit tests, in order to obtain sufficient appropriate audit evidence, preparation of working papers, audit finalisation procedures performed, and application of the concepts and techniques of value for money auditing. The module also covers the audit reports.

MPSA701 Advanced Public Sector Ethics and Governance

This module provides students with the analytical and decision-making skills and knowledge to identify and resolve professional and ethical issues. An understanding of ethics and governance is essential to those in leadership roles, and to those who support their leaders. This module, through case studies in governance, within Zimbabwe and internationally will enhance critical thinking and develop an awareness of corporate governance. It will also help members (and those whom they support) in discharging their stewardship functions.

MPSA702 Advanced Public Sector Administration and Law

The module is designed to expose students' proper skills in public sector administration in the context of an increasingly globalised world, in which public and private actions intersect in hybrid and multilevel governance regimes. The focus is to enable students to demonstrate skills in managing and leading public organizations in ways that are legal, efficient, effective and economical. The module will also prepare students to correctly apply relevant statutory provisions in carrying out their duties through the use of scenarios

and case studies. The scenarios and case studies would require demonstration of deep understanding of the legal framework that regulates public sector financial operations in Zimbabwe includes; the Constitution of the Republic of Zimbabwe 2013; the Public Finance Management Act (chap 22.19); the Audit Office Act (chap 22.18); the Reserve Bank of Zimbabwe Act and the Anti-Corruption Act.

MPSA703 Advanced Public Sector Financial Management

The focus of the module is applied financial management. It deals with financial management issues in the full range of public service organisations – those funded by taxation, and those funded by other means including national government departments and agencies, local government, health bodies, universities and other educational institutions. The course prepares students to apply financial management skills in managing funds of public enterprises. It also focuses on evaluating the performance of the public service.

MPSA704 Advanced Research Methods

The module is intended to equip students with practical skills in formulating research designs that seek to solve specific problems and or provide innovative solutions to issues affecting an individual, group or society. The module would mostly teach scientific methods of inquiry or contractual research because it involves the practical application of scientific methods to everyday problems. The module thus, promotes non-systematic inquiries because of its direct approach in seeking solutions to practical problems. The types of applied research to be taught include evaluation research, research and development and action research.

MPSA710 Public Debt Management

This module focuses on the objectives of government debt management and familiarizes students with choices about risk exposure. It outlines why debt management is a critical public policy function and describes asset liability management and tax smoothing as a background for designing a debt management strategy. The concept of cost and risk, their role in debt management strategy development, risk exposure indicators, quantitative models in cost and risk analysis , links between debt management and fiscal and monetary policies, emphasizing the need for policy coordination, link between debt management and domestic capital market development, as well as describing the role of the debt manager in developing an efficient government securities market, need for clear and strong governance arrangements; both surrounding the debt management organization and the functionalities within the internal organization. It concludes with a debt management strategy.

MPSA705 Applied Public Sector Financial Reporting

The module is intended to make students demonstrate understanding and application of skill in the practice of Public Sector Financial Accounting and Reporting through proficient application of Public Sector International Financial Standards and other accounting conventions. The aim is to equip students with the key competencies necessary to successfully apply IPSAS in various given scenarios and or settings.

MPSA706 Strategic Public Procurement

Public procurement is the process by which public sector organisations acquire products, services and public works. The Constitution of Zimbabwe requires that procurement be effected in a manner that is transparent, fair, honest, cost-effective and competitive. Procurement in Zimbabwe is governed by statutes and this course focuses on the public procurement process. At the end of this course, students should be able to understand the legal and procedural requirements for the strategic procurement of goods and services,

discuss the methods for procuring services, works and consultancy and critically evaluate the management of procurement contracts.

MPSA707 Advanced Strategy and Policy Development
This module explores public sector strategy formulation and helps students to critically examine the main analytical tools available for strategic planning and evaluation and show how they might be applied to a variety of planning activities to support effective decision-making. Public policy consists of political decisions for implementing programs to achieve societal goals. This course focuses on public policy development and students will be able to discuss institutional arrangements for policy setting in central government and at decentralized government levels noting the different forms of administrative decentralization.

MPSA708 Applied Taxation
Taxation as a source of revenue generation and domestic resource mobilization is a distinctive feature of public sector financial management. While in the private sector the accountant's role comprises mostly tax calculation, return preparation, and advisory services; for the public sector context, a deeper understanding of taxation as a source of state finances is required. At the end of this course students should be able to apply taxation concepts and skills within the Zimbabwean tax framework.

MPSA709 Applied Public Sector Strategic Performance Management
This module is designed to expose the student to the necessary knowledge, skills, attitudes and competencies to analyse and solve problems using management accounting, with specific reference to relevant costing, budgeting, basic variance analysis and new perspectives in management accounting. They should also consider how the internal and external factors affecting the environment in which an organisation operates will affect the strategic decisions that are made.

MPSA711 Revenue and Expenditure Management
The module focuses on sources of revenues for the different forms of public sector organizations, and describe ways of collecting and custody of tax and non-tax revenues. It explains the objectives of taxation within the public sector context, types of taxes and their implications, different types of non-tax revenues, and methods of accounting for revenue in the public sector. Preparation of an extract of financial statements for exchange and non-exchange revenue transactions, internal controls in the public sector, types of controls over Government expenditure and the institutions and departments that play very big roles to ensure that there is compliance to the controls established. It goes on to explain the processes of preparing salaries and pensions and the controls that have to be in place. It discusses the internal controls over general expenditures, salaries, and pensions.

MPSA712 Dissertation
The dissertation shall be done in twelve (12) months. Each student will be assigned a supervisor whose role will be to guide and facilitate the student's research. The supervisor shall review and mark each chapter before giving a nod to the student to progress to the next chapter. The completed dissertation shall be submitted to the department two months before the end of the final semester. The submitted write-up will be marked by an examiner other than the assigned supervisor. The department shall conduct viva oral presentations to cross-examine the student who would have submitted his or her dissertation write-up. The submitted write-up will also be subjected to plagiarism checks. The aggregate of the viva mark and the mark for the submitted write-up will make the final mark for the MPSA711 Dissertation.

The dissertation is the culmination of the master's study and recognition of the student's capability to conduct a research project independently. Working with an organisation,

students can structure their project around a real business issue or challenge. Some of the outcomes here are publications in recognised journals, practical management solutions, policy recommendations, and models/frameworks to address issues of corporate reporting and auditing.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN FINANCE AND DIGITAL BANKING (BS17)

Duration:	1.5 years
Credit Load:	324
Minimum Credit Load:	270
Maximum Credit Load:	340
Maximum MBK/S Credit Load:	216
ZNQF Level:	9

1. PURPOSE OF THE PROGRAMME

The programme aims to develop sound conceptual, technical, analytical and problem-solving skills that are required to succeed in the Finance and Banking profession. Students will be expected to exercise their analytical abilities and develop effective verbal and written communication skills.

2. ENTRY REQUIREMENTS

To qualify for admission into the Master of Commerce degree in Finance and Digital Banking programme, a candidate should have an honours degree in Finance, Banking, Economics, Accounting or their equivalents.

3. PROGRAMME CHARACTERISTICS

b) **Areas of study:** Development finance, financial risk management, Investment analysis, bank regulation and supervision, marketing of financial services, international banking and corporate finance, financial mathematics.

c) **Specialist focus:** Finance and Digital Banking

d) **Orientation:** Research and innovation-oriented. Teaching and learning are professionally-oriented and focused on practical aspects.

e) **Distinctive features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 The Master of Commerce degree in Finance and Digital Banking provides employment opportunities in the following institutions; Commercial Banks, Merchant Banks, Building Societies, Discount Houses, Finance Houses, Asset Management Companies, Stock Exchanges, Stock Broking Firms, Insurance Companies, Microfinance Institutions, Central Banks and Academic institutions. The programme hence enables careers in the above institutions as Executive Bank Managers, Retail Banking Managers, Treasury Managers, Treasury Dealers, Financial Advisors and Financial Analysts, Bank Examiners, Risk Managers, Personal Bankers, Customer Relationship

Managers, Credit Analysts, Business Development Managers and Lecturers. There is also a possibility for entrepreneurship in the form of micro-financing and financial consultancy.

4.2 Holders of the Master of Commerce in Finance and Digital Banking Degree are eligible for Doctoral studies in specialised areas Finance and Digital Banking.

5 PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, colloquiums, research projects and individual independent study.

5.2 Assessment Methods: Written and oral examinations, tests, term papers, seminar presentations, Work-Related learning report, final-year research project report, and continuous assessments.

6 LEARNING OUTCOMES

By the end of the program, students will be able to:

- Solve complex banking, investment and financial economics problems creatively and innovatively.
- Apply knowledge of banking, finance, and financial economics specialty to solve complex economic problems.
- Effectively design and conduct systematic investigations in banking and financial economics.
- Apply appropriate investment and financial analysis methods, skills and tools, including those based on information technology.
- Communicate effectively, both orally and in writing, with financial audiences and the community at large.
- Work effectively as an individual, in teams and in multidisciplinary environments.
- Apply professional and ethical and judgment and take responsibility within own limits of competence.
- Create new business ventures based on knowledge and new thinking paradigms.

7 GENERAL PROVISIONS

7.1 Formal examinations will be held at the end of each semester.

7.2 For a student to be admitted to the examinations they must have:

7.2.1 Satisfactorily attempted approved modules of study including submission of required written assignments.

7.2.2 Attended compulsory classes

7.2.3 Participated in prescribed seminars, tutorials and practical classes.

7.2.4 Paid the prescribed fees in accordance with the General Regulations.

7.3 Formal examinations will normally be written papers, but in some cases the examiner may test the candidate orally.

7.4 Continuous assessment shall account for 40% of the overall assessment, while the final examination will account for 60% of the overall assessment.

7.5 The aggregate pass mark shall be 50%.

7.5 A student requires a minimum of 360 credits to complete this programme.

8 PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

7. 9. PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations.

9 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

8. GRADING AND DEGREE CLASSIFICATION

The following marking scheme shall be used for all the modules and grading of the degree:

80% - 100%	Distinction
70% - 79%	Merit
60% - 69%	Credit
50% - 59%	Pass
49% and below	Fail

9. DEGREE WEIGHTING

For the purposes of degree classification, the parts of the degree programme will be weighted as follows:

Level 1	50%
Level 2	50%

12. 0 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MFDB732	Advanced Financial Econometrics and Data Analysis	18
MFDB733	Advanced Corporate Financial Strategy	18
MFDB734	Digital Banking and New Customer Experience	18
MFDB731	Global Financial Markets	18
MFDB735	Monetary Systems and Digital Finance	18
MFDB736	Research Methodologies	18

Level 1 Semester 2

Code	Module Description	Credits
MFDB738	Financial Modelling and Forecasting	18
MFDB739	Strategic Risk Management	18
MFDB740	Applied Treasury Management	18
MFDB737	Strategic Investment Management	18
MFDB741	Financial Analytics and Machine Learning	18

Level 2 Semester 1

Code	Module Description	Credits

MFDB831	Corporate Governance and Ethics	18
MFDB832	SQL For Financial Data	18
MFDB833	Dissertation	90

13.0 MODULE SYNOPSES

MFDB731 Global Financial Markets
This module aims to provide an in-depth knowledge of financial markets and the main types of securities traded in these markets. The module will focus on money and capital markets products, futures, swaps and options. The module will strike a balance between the theoretical paradigms and the empirical literature and the important links between theory and the real world. The emphasis will be on both principles and problem-solving.

MFDB732 Advanced Financial Econometrics and Data Analysis
This module aims to explain how econometric methods can be used to learn about the future behaviour of the prices of financial assets by using information on the history of asset prices and the prices of derivative securities. The specific objectives of the course are to introduce a range of statistical techniques and indicate criteria by which one might judge the appropriateness of each method, identify the decision situation in a problem, formulate and solve econometrics problems and formulate and solve multi-stage problems. The following shall be covered, linear regression, multiple linear regression, qualitative response regression models, panel data regression time series analysis, etc. Emphasis will also be put on the ability of students to interpret the statistical results and make decisions based on the results.

MFDB733 Advanced Corporate Financial Strategy
The module is an extension of advanced corporate finance. The module aims to develop and apply the fundamental ideas and tools for corporate finance to real-world decisions. The module will focus on corporate strategy, capital structure, dividend policy, capital budgeting, real options, mergers, acquisition and restructuring. After completion of this module, students are expected to be able to value, analyse and make informed financial decisions for business organisations.

MFDB734 Digital Banking and New Customer Experience
This module intended to provide students with a critical and in-depth understanding of digital banking products. This module will enable the students to understand the significance and application of various digital banking products such as credit and debit cards, mobile money services, internet banking and ATMs. The emphasis will be on how electronic banking platforms can facilitate a good customer experience, the risks involved with such platforms and the benefits that such platforms have to the financial sector and the economy.

MFDB735 Monetary Systems and Digital Finance
The module aims to provide students with mastery of global monetary systems, the role of the Central Banks, and the evolution of modern payment systems, which includes digital currencies and the future of monetary systems, crypto-currencies and central bank digital currencies (CBDC): digital assets and blockchain technology. The module will also look at the global financial system's history and how they currently function. Students will also understand the theoretical and empirical underpinnings of money, central banking and monetary policies.

MFDB736 Research Methodologies
This course covers how they are used to examining relating to their studies. Quantitative, qualitative and mixed measures are all differentiated in this course. Students will be practically exposed to the main components of a research framework, i.e., problem definition, research design, data collection, ethical issues in research, report writing, and presentation. Once equipped with this knowledge, participants would be well-placed to conduct disciplined research under supervision in an area of their choosing.

MFDB737 Strategic Investment Management
This course focuses on the theory and practice of modern portfolio management. In addition to providing in-depth discussions of portfolio construction, monitoring and evaluation will allow students to gain hands-on experience through case studies and portfolio simulation. Topics covered in this course include Adaptive Market Efficiency, Black-Litterman Portfolio Selection, Asset Price Bubbles and Systemic Risk, International Equity and Bond Investing, Derivatives and Portfolio Management.

MFDB738 Financial Modelling and Forecasting
The module aims to provide hands-on specialisation to students with sound knowledge of modelling techniques and typical applications for investment analysis, company valuation, forecasting, and spreadsheet models. The module teaches how spreadsheets work and how they may be used to build scenarios, predict performance and make informed business and financial decisions. It also teaches how to manage large datasets efficiently, extract meaningful information from them, and present information effectively. By the end of the module, students should be able to use different computer software to solve corporate finance problems such as optimisation of lifetime investments, asset pricing, Capital Budgeting, Decision Tree Analysis, Sensitivity Analysis, Computer Simulations, and Business and Securities Valuations.

MFDB739 Strategic Risk Management
Given the volatile global economic environment, organisations are vulnerable to financial risk, particularly funding and liquidity, market, credit, and operational risk. This module covers one of the core functions of finance, namely risk management. On successful completion of this module, the student should be able to understand risk and a risk management framework, identify the types of financial risk faced by an organisation, apply a practical approach to assessing, monitoring and managing an organisation's financial risk, understand the funding, liquidity, interest rate, foreign exchange, commodity price, credit and operational risk faced by organisations, advise an organisation on managing long term investments, understand the nature and characteristics of derivatives, advise an organisation on the types of financial instruments that could be used to manage best manage an organisation's financial risk best, demonstrate the practical elements of accounting for derivatives for both embedded derivatives and derivatives used for hedging purposes.

MFDB740 Applied Treasury Management
This module concerns how the Treasury function operates in financial and non-financial environments. The key objective of the module is to equip students with an understanding of a wide range of theoretical financial concepts, tools and techniques as applied to treasury activities, including the key functions of an Active Treasury Department and the latest trends in Treasury Management. The module covers concepts such as the trade-off between risk and return, asset allocation, and security analysis in the investment

management process. It further examines the design and implementation of formal performance measurement and management control systems in a business-, particularly a banking institution. Furthermore, the module synthesizes the theory and practice of treasury management.

MFDB741 Financial Analytics and Machine Learning
This course teaches you why, when, and how to apply financial analytics in real-world situations. You will explore techniques to analyse time series and panel data and how to evaluate the risk-reward trade-off expounded in modern portfolio theory. The course delves deeper and explores best practices in understanding time-series data, creating forecasts, and determining the estimates' efficacy using concepts such as the Receiver Operator Curve and the Kolmogorov-Smirnov tests, to mention just a few.

MFDB831 Corporate Governance and Ethics
Governance is a continually developing subject, and promising candidates will be aware of any major developments that have occurred at the time they take their examination. This course introduces students to corporate governance principles with a strong Zimbabwe emphasis, and it is expected that Zimbabwe's corporate governance will be the focus of the study. Students are expected to demonstrate a good knowledge of the principles and provisions of corporate governance as regulated by the Companies Act (Chapter 24:03) and Zimbabwe Stock Exchange Act (Chapter 24:18) (ZSE) listing requirements, Public Finance Management Act (Chapter 22:19) (PFMA) as well as the rules of various professional bodies such as the Institute of Directors of Zimbabwe (IoDZ). Moreover, good knowledge and understanding of the code of corporate governance in other countries will be acceptable, particularly the King III report of South Africa.

MFDB832 SQL for Financial Data
This course is intended to build on the financial modelling course and introduces students to topics such as SQL basics, data wrangling, SQL analysis, AB testing, distributed computing using Apache Spark, and more. These topics will prepare students to apply SQL creatively to analyse and explore data; demonstrate efficiency in writing queries; create data analysis datasets; conduct feature engineering; use SQL with other data analysis and machine learning toolsets; and use SQL with unstructured data sets.

MFDB833 Dissertation
The dissertation shall be undertaken once the research method course has been completed. The student will submit a project report, which will be evaluated (end-semester assessment) by the duly appointed examiner(s). This evaluation will be based on the

DEPARTMENT OF ECONOMIC SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN ECONOMICS (HECON)

Duration	4 years

Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The programme provides an opportunity to obtain a more comprehensive foundation in Economics and widen the horizons in the field of business.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce Economics Honours Degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at 'O' Level, must have obtained at least two passes at 'A' level drawn from Commercials, Sciences or Arts.

2.2 Mature Entry

Over and above the requirements stipulated in the General Regulations, students who get into the Bachelor of Commerce Economics Honours Degree through mature entry should at least have a 'B' in Ordinary Level Mathematics.

2.3 Visiting/Harare Weekend School

To qualify for entry into the Bachelor of Commerce Economics Honours degree (Visiting/Harare Weekend School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma in a Business-related field, Education, or any equivalent tertiary qualification.

or:

at least two (2) passes in relevant "A" level subjects

and:

Proof of employment in the relevant field.
Two (2) years of working experience.

3. PROGRAMME CHARACTERISTICS

3.1 Areas of study:

Microeconomics, macroeconomics, mathematics and statistics, econometrics, trade, monetary economics, public sector, agriculture, ecology and environmental issues and industry.

3.2 Specialist focus:

Economics

3.3 Orientation:

Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 Distinctive features:

The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using student-centred approaches.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Bachelor of Commerce in Economics Honours Degree graduates can be absorbed in some of the following organizations and areas; Financial Institutions, Government Departments, Parastatals, private sector firms, non-governmental organizations, regional and international organizations, institutions of higher learning, consultancy firms and entrepreneurship.

After completing the Bachelor of Commerce in Economics Honours Degree, students are eligible to enrol for a Master of Commerce Degree in Economics, Finance, Investment and related programmes.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation, and individual independent study.

5.2 **Assessment Methods:** Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project report.

6. LEARNING OUTCOMES

- On completion of the programme, a graduate will be able to:
- Design sustainable and inclusive economic policy.
- Explain fundamental economic behaviour and implications to economies and society.
- Evaluate the role of institutions in economic outcomes.
- Forecast and propose optimal economic paths.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 13 of the Faculty Regulations.

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Pre-requisites	Credits
EC131	Economic Principles I		12
CS131	Communication Skills		12
HCS135	Introduction to Information Technology		12
GSB211	Introduction to Gender Studies		12
EC132	Statistics for Economists I		12
EC133	Mathematics for Economists I		12
ACC135	Financial Accounting IA		12
BM131	Principles of Management		12

Level 1 Semester 2

Code	Module Description	Pre-requisites	Credit
MM131	Principles of Marketing		12
ACC136	Financial Accounting IB		12
BM135	Business Law I		12
EC135	Statistics for Economists II		12
EC136	Mathematics for Economists II	(EC133)	12
EC134	Economic Principles II		12

Level 2 Semester 1

Code	Module Description	Pre-requisites	Credit
EC231	Microeconomics I	(EC131)	12
EC235	International Economics		12
ENT131	Entrepreneurship		12
EC233	Quantitative Methods	(EC136)	12
EC236	Computer & Statistical Analysis		12

Level 2 Semester 2

Code	Module Description	Pre-requisites	Credit
EC232	Monetary Economics		12
EC238	Research Methods		12
EC239	Environmental Economics		12
EC234	Macroeconomics I	(EC134)	12
EC237	Econometrics I		12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Pre-requisites	Credit
EC340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Pre-requisites	Credit
EC341	Work-Related Learning Continuous Assessment		40
EC342	Work-Related Learning Report		40

Level 4 Semester 1

Code	Module Description	Pre-requisites	Credit
EC431	Microeconomics II	(EC231)	12
EC436	Public Finance		12
EC433	Industrial Economics		12
EC432	Econometrics II		12
EC437	Investment Analysis		12

Level 4 Semester 2

Code	Module Description	Pre-requisites	Credit
EC434	Macroeconomics II	(EC234)	12
EC439	Development Economics		12
EC435	Agricultural Economics		12
EC445	Disseratation/Research Project		24
EC438	Managerial Economics		12

MODULE SYNOPSES

EC131Economic Principles I

This module focuses on the principles of microeconomics. It covers the basic economic questions; scarcity, choice and opportunity cost; demand and supply; market equilibrium; consumer behaviour;

production and costs; and market structures (perfect competition, monopoly, monopolistic competition, and oligopoly).

EC132 Statistics for Economists I
This is an introductory module in statistics designed to provide students with the basic concepts of data analysis and statistical computing. It is designed to prepare students for the Statistics for Economist II, econometrics and other modules taken in second and fourth year. It also complements the economics modules taken in by students in their first and second years. It provides an introduction to the topic of statistics, with reference to economics examples. Topics covered include basic descriptive measures, measures of association, and probability theory. This module provides students with pragmatic tools for assessing statistical claims and conducting their own statistical analyses.

EC133 Mathematics for Economists I
This is an introductory module in mathematics for economists. It gives students basic skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module acquires the students' knowledge in the field of mathematics and makes them ready to analyse simulated as well as real economic situations. The module covers properties of sets, vector spaces, functions, equations, differentiation and uses them to describe economic environments. Accordingly, this module systematically presents the underlying theory in each of these core areas, at an introductory level. The presentation of the theory is supported with several examples on their application to problem-solving.

CS131 Communication Skills (Refer to Centre for Communication Skills for the module synopsis)

BM131 Principles of Management (Refer to Department of Management Sciences for the module synopsis)

ACC135 Financial Accounting IA (Refer to Department Accounting Sciences for the module synopsis)

HCS135 Introduction to Information Technology (Refer to the Department of Information and Marketing Sciences for the module synopsis)

EC134 Economic Principles II
This module introduces an economic analysis of aggregate employment, income, and prices. Topics include major schools of economic thought; aggregate supply and demand; economic measures, fluctuations, and growth; money and banking; stabilization techniques; and international trade. Upon completion, students should be able to evaluate national economic components, conditions, and alternatives for achieving socioeconomic goals.

EC135 Statistics for Economists II
The module is designed to prepare students for the Econometrics and Computer & Statistical Analysis modules taken in the second and fourth years. It also complements the Economics modules taken by students in the previous semester. The topics include sampling methods, probability and distributions, point & interval estimation, and hypothesis testing - large sample and small sample test, t-test, correlation and regression analysis, time series analysis and index numbers. The module will mainly emphasize the various statistical methods which can be used to solve business/economic problems.

EC136 Mathematics for Economists II
This module gives students advanced skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module builds on the foundation provided in the first-level module in Mathematics for Economists I. It concentrates on the mathematical methods that are required to understand current economics and to investigate economic models. Students learn how to use and apply mathematics by working with concrete examples and exercises. By the end of the module, students should have a complete understanding of integration techniques and optimisation and should be able to apply the Lagrange multipliers approach to constrained optimization problems.

ACC136 Financial Accounting IB
(Refer to Department of Accounting Sciences for the module synopsis)

MM131 Principles of Marketing
(Refer to Department of Information and Marketing Sciences for the module synopsis)

BM135 Commercial Law
(Refer to Department of Management Sciences for the module synopsis)

EC231 Microeconomics I
This module is designed to extend the knowledge of the basic microeconomic principles that provide the foundation for future work in economics. It gives insight into how economic models can help us think about important real-world phenomena. Microeconomics is the study of rational choice behaviour on the part of individual consumers and firms. In general, economists are interested in how market mechanisms solve extremely complex resource allocation problems. This module presents a logical and coherent framework in which to organize observed economic phenomena. Topics include supply and demand interaction, utility maximization, profit maximization, elasticity, perfect competition, monopoly power, imperfect competition, game theory and economics of information.

EC232 Monetary Economics
The module aims to introduce students to some tools to understand the operations of monetary policy in the economy. By so doing the student after completing the module must be in a position to make an analysis of existing monetary policies within the context in which they have been formulated. The module therefore explains the nature, definitions and value of money, the role of money in economic activity, the nature and role of financial intermediation, demand of and supply for money; theoretical and empirical issues, monetary policy and money supply control; Zimbabwe monetary policy since 1980, the role of the Central Bank and its independence particularly commitment and credibility, monetary policy transmission mechanisms, theory of and empirics on inflation, optimal stabilisation policy and optimal inflation rate and inflation persistence, open and closed economy monetary policies including efficiencies.

EC233 Quantitative Methods
It is a pre-requisite for students taking this module to have done basic mathematics for economists. Quantitative methods seek to strengthen the use of mathematics in analysing and interpreting economic phenomena. The module covers areas like comparative statistics, optimization theory, difference and differential equations, matrix algebra etc. At the end of this module, the student will be able to identify real economic problems and apply the learnt mathematical principles to solve them. This module will ensure that students will gain the critical 21st Century skills of problem-solving, creativity and innovation.

EC234 Macroeconomics I
Macroeconomics studies the aggregate behaviour of the economy. This module provides an introduction to the economic analysis of key macroeconomic variables such as output, employment, inflation, interest rates and exchange rates. The intent of the module is to provide the student with a basic level of economic literacy essential for a well-informed citizenship in the years ahead. It capacitates the students to develop competence in using macroeconomic models to explain the determination of aggregate output, prices, employment, and growth. In addition, students should learn to analyse the impact of monetary and fiscal policy on the macroeconomy. Extensive references to current macroeconomic developments and applications of macroeconomic models are essential elements of the module.

EC235 International Economics
The module intends to help students to understand theories and policies related to international trade. The theories will illustrate how gains or losses from international trade arise thereby giving rise to policies that encourage or restrict trade. Also focus will be made on the world economy institutions that deal with international trade. Therefore, the module covers an understanding of trade theories that form the basis for trade policy analysis. These theories include: classical theories, neoclassical theories and 'new' trade theories. The theory of trade protection, exchange rates and the balance of payments are also covered in this theory section. The policy section covers such topics as terms of trade and economic development, the global trading environment, regional co-operation and

integration and the role world economy institutions in international trade. Current controversies in international trade policies are considered as well.

EC236 Computer & Statistical Analysis
The module forms an enhanced strong prelude to Econometrics I. It covers issues such as probability models, univariate probability distributions, multivariate probability distributions, non-parametric tests, sampling and sampling distributions, asymptotic distributions, Analysis of Variance. The module also infuses the use of computer softwares for inputting and analysis of statistical data. Computer-aided techniques of data analysis shall include descriptive, computational techniques and simulations.

EC237 Econometrics I
The module forms the foundation for Econometrics II. It covers methodology of econometrics, linear regression model assumptions, properties of estimators, dummy variables, non-linear regression models hypothesis testing procedures, bi-variate and multivariate analysis, Gauss-Markov theorem.

EC238 Research Methods
The module seeks to provide students with the basic tools to undertake independent research as well as understand how research can be communicated to end users of analysis. It will focus on pertinent issues in research in Economics including elements such as literature reviews and data sources; the status and growth of economic knowledge; ethics of economic research; and overall design of a research project - aims, philosophy and methods, evaluating existing research (review of articles), and writing up and disseminating findings.

EC239Environmental Economics
This module seeks to equip students with skills to utilize the techniques covered in core modules such as microeconomics, macroeconomics and quantitative methods to analyse environmental problems. The module explains the role of environmental economics in the development process in general and in the integrated management of ecological–economic systems in particular. It covers topics such as the history of environmental economics, sustainable development, environmental accounting and valuation of natural resources.

GSB211Gender Studies for Business
The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, the historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

ENT131 Entrepreneurship 1
The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation-building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan.
Evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; and identify the challenges being faced by entrepreneurs.

EC340 Work-Related Learning Preliminary Report
The module is a preliminary report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, conclusions and recommendations to the company and the University.

EC341 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third year students with a practical exposure to all facets of the economic functions.

The student is expected to rotate into all sections of related to economic sciences to which thy will be attached to enable them to be exposed to all economics areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

EC342 Work-Related Learning Report
The module is a report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University versus what is at the work-place, any gaps, conclusions and recommendations to the company and the University.

EC431 Microeconomics II
This module introduces the students to Advanced Microeconomics. It builds on the concepts of microeconomics covered in lower level. Some appreciation of basic economic calculus is greatly expected. The module links theoretical underpinnings with application and it teaches to; describe the technical progress, appreciate decision-making under imperfect markets, understand and appreciate game theory models, analyse the equilibrium between production and consumption, identify and understand welfare economics and understand market failure and to measure welfare changes.

EC432 Econometrics II
The modules cover univariate and bi-variate analysis and related topics. The topics covered are econometric problems and their formal tests, such as non-stationarity, heteroscedasticity, autocorrelation, model specification errors, etc. Also to be included are the weaknesses of regression models, problems of single equation models and simultaneous equation models. The module content focuses on the application of data analysis softwares to econometric techniques that provide students with the understanding they need in modern economics. Students will be given an opportunity to use modern statistical packages like STATA, PC Give, E-Views, SPSS – and these form part of their methodological toolkit for economic and development research.

EC433 Industrial Economics
This module presents the most traditional and recent principal concepts and methods of Industrial Economics/ Organisation. The module aims to provide an understanding of how theories from Industrial Economics can help one comprehend the behaviour of firms in perfect and imperfect competitive markets. The module covers the following issues: theory of the firm, structural and non-structural market assessment approaches, firm conduct and market structure, game theory, market concentration, market structure and profitability, technology and network markets, pricing and industrial strategies, anti-competitive strategies, competition policy and regulation, diversification and growth. More practical applications pertinent to Industrial Economics are covered through global and country-specific case studies, legal proceedings and rulings/outcomes.

EC434 Macroeconomics II
The module covers the closed economy models, determination of aggregate demand and aggregate supply, the inflation unemployment nexus, formation of expectations, overview of consumption theory, traditional macroeconomic policy models, theories of economic growth, structural adjustment models, open economy analysis, balance of payments approaches. The module also explores the government budget dynamics, including the strategies of funding government budget deficits.

EC435 Agricultural Economics
The module looks at the definitions of agricultural economics and related ideas, agriculture development in the African context, agricultural production theory, agriculture peasant household models (Chayanov and share tenancy models). The module also looks at agricultural policies; agricultural technology (innovation, transfer) and policy, food policy (food security and drought management), and land policy with particular reference to Zimbabwe.

EC436 Public Finance
The economic basis for government activity – efficiency markets and governments, externalities and government policy, provision of public goods, government expenditure and policy, budget deficits and

the government taxation – theory and structure, taxes on consumption and sales, state and local government fiscal issues and inter-governmental fiscal relations.

EC437 Investment Analysis
The module is designed to cover an understanding of financial markets with reference to Zimbabwe. Topics covered include the financial system, financial mathematics, investment appraisal, valuation of financial securities, portfolio theory, trading of financial securities, and the efficiency of capital markets.

EC438 Managerial Economics
The module considers business objectives: optimization, the principle of margin, decision analysis, demand measurement and forecasting, cost measurement, economies of scale, the learning curve, pricing policies: joint product pricing, transfer pricing, loss leaders, price discrimination, public utility pricing, advertising and promotion decisions: Dorfman – Steiner theorem, the advertising – sales relationship, corporate strategies: vertical integration, diversification, growth and market share, the location decision, profit planning and control, game theory for managers.

EC439 Development Economics
The module covers institutional and political economy of development in both developed and developing countries, mainly focusing on the latter. Focus is directed toward drivers of development such as institutions, human capital, international trade, aid, bilateral & multilateral development financing and financial development. It also dwells on theories/models of development shaping developed, emerging and developing economies. Concepts covered include development in the short-run and long-run. The following topics are studied: Economic Growth and Development; Industrialization and Modernization: Agrarian Change and Agricultural Revolutions, poverty trap, Technological Change and Innovation; Infrastructural Developments; Demographic Change, Commercial and Financial Changes.

EC445 Research project
During the final year of study, students are required to write and submit a research project on a topic of their choice. The research project gives students an opportunity to seek solutions to problems affecting households, businesses, government, and society in general. It also enhances the student's skills in terms of time management, academic writing, data analysis, and results interpretation. The grade of the research project will be based on oral examination (viva voce) before a panel of examiners, and a mark awarded for the write-up submitted to the department. The length of the research project shall not be less than 10000 words (excluding preliminary pages and appendices), and the department shall provide guidelines on the structure that should be followed by students.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN ECONOMETRIC MODELLING (BS09)

Duration	4 Years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540

Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The programme provides an opportunity to obtain a more comprehensive foundation in Economics and Econometric Modelling and widen the horizons in the field of business.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce in Economics and Econometric Modelling Honours Degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English at 'O' Level, must have obtained at least a B in 'O' level Mathematics, and at least two passes at 'A' level drawn from Commercials, Sciences or Arts.

2.1 Mature Entry

Over and above the requirements stipulated in the General Regulations, students who get into the Bachelor of Commerce in Economics and Econometric Modelling Honours Degree through mature entry should at least have a 'B' in Ordinary Level Mathematics.

2.3 Visiting School

To qualify for entry into the Bachelor of Commerce in Economics and Econometric Modelling Honours degree (Visiting School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma in a Business-related field or any equivalent tertiary qualification.

or:

At least two (2) passes in relevant "A" level subjects

and

Two (2) years of working experience **plus**

Confirmation of employment in the relevant department.

3. PROGRAMME CHARACTERISTICS

3.1 Areas of study: Microeconomics, Macroeconomics, Statistics, Mathematics for Economists, Economic Modelling, Blockchain Technology, Econometrics, Dynamic Stochastic General Equilibrium modelling

3.2 Specialist focus: Economics and Econometric Modelling

3.3 Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 Distinctive features: The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Computer programming, Data Analysis, Economic modelling, and Economic policy analysis.

After completing the Bachelor of Commerce in Economics and Econometric Modelling

Honours Degree, students are eligible to enrol for a Master of Commerce Degree in Economics, Financial Economics and Investment, Econometrics, Econometric Modelling and related programmes.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation, and individual independent study.

5.2 Assessment Methods: Written, practical and oral examinations, in-class tests, term papers, assignments, presentations, Work-Related learning report, final year research project report.

6 LEARNING OUTCOMES

On completion of the programme, a graduate will be able to:

- Develop econometric models used for forecasting sales, revenue, bank deposits, credit risk, etc
- Analyse the impact of government policy (for example, monetary and fiscal policy) on businesses using econometric techniques.
- Apply data-generating processes (DGP) to provide solutions to social and economic problems.
- Apply computer-based software programs in solving elementary computational problems in econometrics.
- Formulate independently a complex economic problem into a simulation model.

7 GENERAL PROVISIONS

Refer to faculty and general regulations.

8 PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9 PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations.

10 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11 GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the Faculty Regulations.

12 DEGREE WEIGHTING

Refer to Section 13 of the Faculty Regulations.

13 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module	Pre-requisites	Credits

EC131	Economic Principles I		12
HCS135	Introduction to Information Technology		12
CS131	Communication Skills		12
GSB211	Gender Studies for Business		12
EC132	Statistics for Economists I		12
EC133	Mathematics for Economists I		12
ACC135	Financial Accounting 1A		12
BM131	Principles of Management		12

Level 1 Semester 2

MM131	Principles of Marketing		12
ACC136	Financial Accounting 1B		12
BM135	Commercial Law		12
EC134	Economic Principles II		12
EC135	Statistics for Economists II		12
EC136	Mathematics for Economists II	(EC133)	12

Level 2 Semester 1

EC231	Microeconomics I	(EC131)	12
ENT131	Entrepreneurship		12
EC233	Quantitative Methods	(EC136)	12
EC236	Computer & Statistical Analysis		12
CIT140	Principles of Machine Learning		12

Level 2 Semester 2

EC238	Research Methods		12
EC433	Industrial Economics		12
CIT234	Artificial Intelligence and Expert Systems	(CIT140)	12
EC234	Macroeconomics I	(EC134)	12
EC237	Econometrics I		12

Level 3 Semester 1: Work-Related Learning

EC340	Work-Related Learning Preliminary Report	40

Level 3 Semester 2: Work-Related Learning

EC341	Work-Related Learning Continuous Assessment	40
EC342	Work-Related Learning Report	40

Level 4 Semester 1

EC431	Microeconomics II	(EC231)	12
EC241	Programming and Data Analytics		12
EC432	Econometrics II		12
EC441	Economic Modelling		12

Electives (Choose any 1)

EC232	Monetary Economics		12
EC239	Environmental Economics		12
EC436	Public Finance		12

Level 4 Semester 2

EC434	Macroeconomics II	(EC234)	12
EC444	Blockchain Technology	12	
EC446	Dynamic Stochastic General Equilibrium Modelling		12
EC445	Dissertation/Research Project		24

Electives (Choose any 1)

EC439	Development Economics	12
EC438	Managerial Economics	12
EC235	Agricultural Economics	12

14 MODULE SYNOPSES

EC131 Economic Principles I

This module focuses on the principles of microeconomics. It covers the basic economic questions; scarcity, choice and opportunity cost; demand and supply; market equilibrium; consumer behaviour; production and costs; and market structures (perfect competition, monopoly, monopolistic competition, and oligopoly).

EC132 Statistics for Economists I

This is an introductory module in statistics designed to provide students with the basic concepts of data analysis and statistical computing. It is designed to prepare students for the Statistics for Economists II, econometrics and other modules taken in second and fourth year. It also complements the economics modules taken in by students in first and second year. It provides an introduction to the topic of statistics, with reference to economics examples. Topics covered include basic descriptive measures, measures of association, and probability theory. This module provides students with pragmatic tools for assessing statistical claims and conducting their own statistical analyses.

EC133 Mathematics for Economists I

This is an introductory module in mathematics for economists. It gives students basic skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module acquires the students' knowledge in the field of mathematics and to make them ready to analyse simulated as well as real economic situations. The module covers properties of sets, vector spaces, functions, equations, and differentiation and uses them to describe economic environments. Accordingly, this module systematically presents the underlying theory in each of these core areas, at an introductory level. The presentation of the theory is supported with several examples of their application to problem-solving.

EC134 Economic Principles II

This module introduces an economic analysis of aggregate employment, income, and prices. Topics include major schools of economic thought; aggregate supply and demand; economic measures, fluctuations, and growth; money and banking; stabilization techniques; and international trade. Upon completion, students should be able to evaluate national economic components, conditions, and alternatives for achieving socioeconomic goals.

EC135 Statistics for Economists II

The module is designed to prepare students for the Econometrics and Computer and Statistical Analysis modules taken in the second and fourth years. It also complements the Economics modules taken by students in the previous semester. The topics include sampling methods, probability and distributions, point & interval estimation, and hypothesis testing - large sample and small sample test, t-test, correlation and regression analysis, time series analysis and index numbers. The module will mainly emphasize the various statistical methods which can be used to solve business/economic problems.

EC136 Mathematics for Economists II

This module gives students advanced skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module builds on the foundation

provided in the first level module in Mathematics for Economists I. It concentrates on the mathematical methods that are required to understand current economics and to investigate economic models. Students learn how to use and apply mathematics by working with concrete examples and exercises. By the end of the module, students should have a complete understanding of integration techniques and optimization, and should be able to apply the Lagrange multipliers approach to constrained optimization problems.

CS131 Communication Skills

(Refer to Centre for Communication Skills for the module synopsis)

BM131 Principles of Management

(Refer to Department of Management Sciences for the module synopsis)

HCS135 Introduction to Information Technology

(Refer to Department of Information and Marketing Sciences for the module synopsis)

ACC135 Financial Accounting 1A

Refer to the Department of Accounting Sciences for the module synopsis.

ACC136 Financial Accounting 1B

Refer to the Department of Accounting Sciences for the module synopsis.

CIT140 Principles of Machine Learning

(Refer to Department of Information and Marketing Sciences for the module synopsis)

BM135 Commercial Law

(Refer to Department of Management Sciences for the module synopsis)

EC231 Microeconomics I

This module is designed to extend the knowledge of the basic microeconomic principles that provide the foundation for future work in economics. It gives insight into how economic models can help us think about important real-world phenomena. Microeconomics is the study of rational choice behaviour on the part of individual consumers and firms. In general, economists are interested in how market mechanisms solve extremely complex resource allocation problems. This module presents a logical and coherent framework in which to organize observed economic phenomena. Topics include supply and demand interaction, utility maximization, profit maximization, elasticity, perfect competition, monopoly power, imperfect competition, game theory and economics of information.

EC233 Quantitative Methods

It is a pre-requisite for students taking this module to have done basic mathematics for economists. Quantitative methods seek to strengthen the use of mathematics in analysing and interpreting economic phenomena. The module covers areas like, comparative statistics, optimization theory, difference and differential equations, matrix algebra etc. At the end of this module the student will be able to identify real economic problems and apply the learnt mathematical principles to solve them. This module will ensure that students will gain the critical 21st century skills of problem-solving, creativity and innovation.

EC236 Computer & Statistical Analysis

The module forms an enhanced strong prelude to Econometrics I. It covers issues such as probability models, univariate probability distributions, multivariate probability distributions, non-parametric tests, sampling and sampling distributions, asymptotic distributions, Analysis of Variance. The module also infuses the use of computer software

for inputting and analysis of statistical data. Computer-aided techniques of data analysis shall include descriptive, computational techniques and simulations.

EC 241 Programming and Data Analytics
The module starts by introducing the fundamental data types and operations. Other topics covered in this module are: Control Statements, i.e., decision and repetition, are presented, Methods, as computations consisting of multiple steps, Arrays, representing a data structure storing multiple values and, the main concepts of object-oriented programming paradigm, i.e., class and instance. By the end of the module students should be able to understand the basic concepts of imperative programming understand the basic concepts of object-oriented programming.

EC234 Macroeconomics I
Macroeconomics studies the aggregate behaviour of the economy. This module provides an introduction to the economic analysis of key macroeconomic variables such as output, employment, inflation, interest rates and exchange rates. The intent of the module is to provide the student with a basic level of economic literacy essential for well-informed citizenship in the years ahead. It capacitates the students to develop competence in using macroeconomic models to explain the determination of aggregate output, prices, employment, and growth. In addition, students should learn to analyse the impact of monetary and fiscal policy on the macroeconomy. Extensive references to current macroeconomic developments and applications of macroeconomic models are essential elements.

EC237 Econometrics I
The module forms the foundation for Econometrics II. It covers methodology of econometrics, linear regression model assumptions, properties of estimators, dummy variables, non-linear regression models hypothesis testing procedures, bi-variate and multivariate analysis, Gauss-Markov theorem.

EC238 Research Methods
The module seeks to provide students with the basic tools to undertake independent research as well as understanding how research can be communicated to end users of analysis. It will focus on pertinent issues in research in Economics including elements such as literature reviews and data sources; the status and growth of economic knowledge; ethics of economic research; and overall design of a research project - aims, philosophy and methods, evaluating existing research (review of articles), and writing up and disseminating findings.

EC433 Industrial Economics
This module presents the most traditional and recent principal concepts and methods of Industrial Economics/Organisation. The module aims at providing an understanding of how theories from Industrial Economics can help one comprehend the behaviour of firms in perfect and imperfect competitive markets. The module covers the following issues: theory of the firm, structural and non-structural market assessment approaches, firm conduct and market structure, game theory, market concentration, market structure and profitability, technology and network markets, pricing and industrial strategies, anticompetitive strategies, competition policy and regulation, diversification and growth. More practical applications pertinent to Industrial Economics are covered through global and country specific case studies, legal proceedings and rulings/outcomes.

GSB211 Introduction to Gender Studies
(Refer to Centre for Centre for Gender Studies for the module synopsis)

ENT131 Entrepreneurship
(Refer to Centre for Centre for Entrepreneurship and Innovation for the module synopsis)

CIT234 Artificial Intelligence and Expert Systems
(Refer to Department of Information and Marketing Sciences for the module synopsis)

EC340 Work-Related Learning Preliminary Report

The module is a preliminary report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, conclusions and recommendations to the company and the University

EC341 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third year students with a practical exposure to all facets of the economic functions. The student is expected to rotate into all sections of related to economic sciences to which thy will be attached to enable them to be exposed to all economics areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

EC342 Work-Related Learning Report
The module is a report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University versus what is at the workplace, any gaps, conclusions and recommendations to the company and the University.

EC431 Microeconomics II
This module introduces the students to Advanced Microeconomics. It builds on the concepts of microeconomics covered in the lower level. Some appreciation of basic economic calculus is greatly expected. The module links theoretical underpinnings with application and it teaches to; describe the technical progress, appreciate decision-making under imperfect markets, understand and appreciate game theory models, analyse the equilibrium between production and consumption, identify and understand welfare economics, understand market failure and measure welfare changes.

EC432 Econometrics II
The modules cover univariate and bi-variate analysis and related topics. The topics covered are econometric problems and their formal tests, such as non-stationarity, heteroscedasticity, autocorrelation, model specification errors, etc. also to be included are the weaknesses of regression models, problems of single equation models and simultaneous equation models. The module content focuses on application of data analysis softwares to econometric techniques that provide students with the understanding they need in modern economics. Students will be given an opportunity to use modern statistical packages like STATA, PC Give, E-Views, SPSS – and these form part of their methodological toolkit for economic and development research.

EC441 Economic Modelling
The module provides the framing skills for modelling the economy; and to develop learners' problem-solving abilities in the context of both macroeconomics and microeconomics. A structured series of core economic models are introduced to the learners, familiarising them with modelling techniques and the intellectual demands of constructing economic models with which to analyse the economy. students develop skills in constructing and using the IS-LM framework for analysing the closed economy in the short run; and using micro-models to analyse situations in which economic agents both produce and consume, such as the general equilibrium model. By the end of the module the students should be able to understand some of the complexities of interdisciplinary policy problems, particularly in the areas of sustainable development.

EC434 Macroeconomics II
The module covers the closed economy models, determination of aggregate demand and aggregate supply, the inflation-unemployment nexus, formation of expectations, overview of

consumption theory, traditional macroeconomic policy models, theories of economic growth, structural adjustment models, open economy analysis, balance of payments approaches. The module also explores the government budget dynamics, including the strategies of funding government budget deficits.

EC445 Research project/Dissertation
During the final year of study, students are required to write and submit a research project on a topic of their choice. The research project gives students an opportunity to seek solutions to problems affecting households, businesses, government, and society in general. It also enhances the student's skills in terms of time management, academic writing, data analysis, and results interpretation. The grade of the research project will be based on oral examination (viva voce) before a panel of examiners, and a mark awarded for the write-up submitted to the department. The length of the research project shall not be less than 10000 words (excluding preliminary pages and appendices), and the department shall provide guidelines on the structure that should be followed by students.

EC435 Agricultural Economics
The module looks at the definitions of agricultural economics and related ideas, agriculture development in the African context, agricultural production theory, agriculture peasant household models (Chayanov and share tenancy models). The module also looks at agricultural policies; agricultural technology (innovation, transfer) and policy, food policy (food security and drought management), and land policy with particular reference to Zimbabwe.

EC239 Environmental Economics
This module seeks to equip students with skills to utilize the techniques covered in core modules such as microeconomics, macroeconomics and quantitative methods to analyse environmental problems. The module explains the role of environmental economics in the development processing general and in the integrated management of ecological–economic systems in particular. It covers topics such as the history of environmental economics, sustainable development, environmental accounting and valuation of natural resources.

EC436 Public Finance
The economic basis for government activity – efficiency markets and governments, externalities and government policy, provision of public goods, government expenditure and policy, budget deficits and the government taxation – theory and structure, taxes on consumption and sales, state and local government fiscal issues and inter-governmental fiscal relations.

EC232 Monetary Economics
The module aims to introduce students to some tools to understand the operations of monetary policy in the economy. By so doing the student after completing the module must be in a position to make an analysis of existing monetary policies within the context in which they have been formulated. The module therefore explains the nature, definitions and value of money, the role of money in economic activity, the nature and role of financial intermediation, demand of and supply for money; theoretical and empirical issues, monetary policy and money supply control; Zimbabwe monetary policy since 1980, the role of the Central Bank and its independence particularly commitment and credibility, monetary policy transmission mechanisms, theory of and empirics on inflation, optimal stabilisation policy and optimal inflation rate and inflation persistence, open and closed economy monetary policies including efficiencies.

EC4438 Managerial Economics
The module considers business objectives: optimization, the principle of margin, decision analysis, demand measurement and forecasting, cost measurement, economies of scale, the learning curve, pricing policies: joint product pricing, transfer pricing, loss leaders, price

discrimination, public utility pricing, advertising and promotion decisions: Dorfman – Steiner theorem, the advertising – sales relationship, corporate strategies: vertical integration, diversification, growth and market share, the location decision, profit planning and control, game theory for managers.

EC4439 Development Economics
The module covers the institutional and political economy of development in both developed and developing countries, mainly focusing on the latter. Focus is directed toward drivers of development such as institutions, human capital, international trade, aid, bilateral & multilateral development financing and financial development. It also dwells on theories/models of development shaping developed, emerging and developing economies. Concepts covered include development in the short-run and long-run. The following topics are studied: Economic Growth and Development; Industrialization and Modernization: Agrarian Change and Agricultural Revolutions, poverty trap, Technological Change and Innovation; Infrastructural Developments; Demographic Change, Commercial and Financial Changes.

EC444 Blockchain Technology
This module lays the groundwork for understanding blockchain technology. It will offer a low-level entry into the technology, by discussing its origins, its key technological features and its different potential applications. Central features such as wallets and transactions will be discussed and crucial notions such as hashes, blocks and chains will be explained. After completing this module, students will have a basic understanding of the origins and the key technological aspects of blockchain technology as well as a basic understanding of different architectures and myriad potential applications of blockchain technologies.

EC 445 Dissertation/Research Project
A student is required to submit a dissertation of approximately 10 000 words on a topic relating to accounting, or auditing, or finance within a period as may be prescribed by the department. This is an independent project in content areas related to accounting areas. Students work under the supervision of an assigned departmental lecturer to address comprehensive company-based or research-orientated problems with documented results. The completed project should clearly present the problem of the research subject investigated or applied project undertaken, its significance to theory or business practice, the research or project background, a well-defined method, results or findings, and their implications.

EC446 Dynamic Stochastic General Equilibrium Modelling
The module explains and predicts co-movements of aggregate time series over the business cycle and to perform policy analysis. Emphasis is on explaining how to use DSGE models for all three purposes – forecasting, storytelling, and policy experiments – and review their forecasting record. Module content includes introduction to DSGE Models, New Keynesian Models, Small Open Economy Models, Extensions to New Keynesian Models, Kalman Filter and Estimation Methods, and Practical Application of DSGE Models. By the end of this module, students should be able to address a wide range of theoretical research questions, analyse, forecast, and simulate policy issues in response to exogenous shocks.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN FINANCIAL ECONOMICS AND INVESTMENT (BS11)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

The programme provides an opportunity to obtain a more comprehensive foundation in Financial Economics and widen the horizons in the field of business. To prepare the learners for further studies in Economics and related disciplines.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce in Financial Economics and Investment Honours Degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at 'O' Level, must have obtained at least two passes at 'A' level drawn from Commercials, Sciences or Arts.

2.2 Mature Entry

Over and above the requirements stipulated in the General Regulations, students who get into the Bachelor of Commerce in Financial Economics and Investment Honours Degree through mature entry should at least have a 'B' in Ordinary Level Mathematics.

2.4 Visiting /Harare Weekend School

To qualify for entry into the Bachelor of Commerce in Financial Economics and Investment Honours degree (Visiting/Harare Weekend School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma in a Business-related field, or any equivalent tertiary education qualification.

or:

At least two (2) passes in relevant "A" level subjects.

And

Two (2) years of working experience **plus**

Proof of employment in a commercial environment

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Microeconomics, Financial Markets and Institutions, Macroeconomics Policy & Financial Markets, Investment & Portfolio Management, Financial Econometrics, International Trade, Monetary Economics, Taxation, Development Finance and Marketing of Financial Services.

3.2 **Specialist focus:** Financial Economics

3.3 **Orientation:** Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 **Distinctive features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach. Practitioners will be guest lecturers(presenters) to maintain the relevance of the programme and changes will be made based on their inputs.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates can be absorbed in the following organizations and areas: Investment Banks, Commercial Banks, Corporate advisory firms, Corporate development firms, Equity Research,

Financial Planning and Analysis, Stock Exchange Analysts

After completing the Bachelor of Commerce in Financial Economics and Investment Honours Degree, students are eligible to enroll for a Master of Commerce Degree in Economics, Master's Degree in Financial Economics and Investment and related programmes. Students are also being prepared for their own small business operations.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation, and individual independent study.

5.2 Assessment Methods: Written and oral examinations, tests, term papers, seminar presentations, Work-Related learning report, final year dissertation, and continuous assessments.

6. LEARNING OUTCOMES

On completion of the programme, a graduate will be able to:

- Understand and analyze the fundamental operations of financial markets and instruments.
- Apply financial economics models and formulae in solving problems in financial markets and institutions.
- Apply information technology to analyse financial data.
- Apply utility theory to describe and analyze investment decisions under risk aversion.
- Examine the causes of financial instability and financial crises.
- Set up advisory or consultancy services/ Advise the corporate bodies on financial matters.

7. GENERAL PROVISIONS

Refer to faculty and general regulations.

8. PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9. PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the Faculty Regulations.

12. DEGREE WEIGHTING

Refer to Section 13 of the Faculty Regulations.

13. AWARD OF DEGREE

Refer to General Regulations

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
EC131	Economic Principles I		12
HCS135	Introduction to Information Technology		12
CS131	Communication Skills		12
GSB211	Gender Studies for Business		12
EC132	Statistics for Economists I		12
EC133	Mathematics for Economists I		12
ACC135	Financial Accounting IA		12
BM131	Principles of Management		12

Level 1 Semester 2

EC151	Financial Markets & Institutions		12
ACC136	Financial Accounting IB		12
BM135	Commercial Law		12
EC134	Economic Principles II		12
EC135	Statistics for Economists II		12
EC136	Mathematics for Economists II	(EC133)	12

Level 2 Semester 1

EC231	Microeconomics I	(EC131)	12
ENT131	Entrepreneurship		12
EC235	International Economics		12
EC233	Quantitative Methods	(EC136)	12
EC236	Computer & Statistical Analysis		12

Level 2 Semester 2

EC234	Macroeconomics I	(EC134)	12
EC232	Monetary Economics		12
EC238	Research Methods		12
EC253	Investment Analysis and Portfolio Management		12
EC237	Econometrics I		12

Level 3 Semester 1: Work-Related Learning

EC340	Work-Related Learning Preliminary Report	40

Level 3 Semester 2: Work-Related Learning

EC341	Work-Related Learning Continuous Assessment	40
EC342	Work-Related Learning Report	40

Level 4 Semester 1

EC451	Risk Management	12
EC453	Financial Modelling and Forecasting	12
EC454	Financial Econometrics	12
Electives (Choose any 2)		
EC452	International Finance	12
EC455	Development Finance	12
EC435	Agricultural Economics	12
EC239	Environmental Economics	12

Level 4 Semester 2

EC456	Financial Derivatives	12
EC444	Blockchain Technology	12
Electives (Choose any 1)		
EC458	Corporate Restructuring	12
EC460	Natural Resource Economics	12
EC445	Dissertation/Research project	24
Electives (Choose any 1)		
EC438	Managerial Economics	12
EC457	Taxation	12

15. Module Synopses

EC131Economic Principles I (This is different from the one offered by Accounting Sciences, but the code is the same)

This module focuses on the principles of microeconomics. It covers the basic economic questions; scarcity, choice and opportunity cost; demand and supply; market equilibrium; consumer behaviour; production and costs; and market structures (perfect competition, monopoly, monopolistic competition, and oligopoly).

EC132 Statistics for Economists I

This is an introductory module in statistics designed to provide students with the basic concepts of data analysis and statistical computing. It is designed to prepare students for the Statistics for Economists II, econometrics and other modules taken in the second and fourth years. It also complements the economics modules taken in by students in their first and second years. It provides an introduction to the topic of statistics, with reference to economics examples. Topics covered include basic descriptive measures, measures of association, and probability theory. This module provides students with pragmatic tools for assessing statistical claims and conducting their own statistical analyses.

EC133 Mathematics for Economists I

This is an introductory module in mathematics for economists. It gives students basic skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module acquires the students' knowledge in the field of mathematics and makes them ready to analyse simulated as well as real economic situations. The module covers properties of sets, vector spaces, functions, equations, differentiation and uses them to describe economic environments. Accordingly, this module systematically presents the underlying theory in each of these core areas, at an introductory level. The presentation of the theory is supported with several examples of their application to problem-solving.

EC134 Economic Principles II

This module introduces economic analysis of aggregate employment, income, and prices. Topics include major schools of economic thought; aggregate supply and demand; economic measures,

fluctuations, and growth; money and banking; stabilization techniques; and international trade. Upon completion, students should be able to evaluate national economic components, conditions, and alternatives for achieving socioeconomic goals.

EC135 Statistics for Economists II
The module is designed to prepare students for the Econometrics and Computer &Statistical Analysis modules taken in the second and fourth years. It also complements the Economics modules taken by students in the previous semester. The topics include sampling methods, probability and distributions, point & interval estimation, and hypothesis testing - large sample and small sample test, t-test, correlation and regression analysis, time series analysis and index numbers. The module will mainly emphasize the various statistical methods which can be used to solve business/economic problems.

EC136 Mathematics for Economists II
This module gives students advanced skills for implementation of the mathematical knowledge and expertise to the problems of economics. The module builds on the foundation provided in the first level module in Mathematics for Economists I. It concentrates on the mathematical methods that are required to understand current economics and to investigate economic models. Students learn how to use and apply mathematics by working with concrete examples and exercises. By the end of the module, students should have a complete understanding of integration techniques and optimisation and should be able to apply the Lagrange multipliers approach to constrained optimization problems.

EC151 Financial Markets & Institutions
This module focuses on the operations of the financial system. It explores the workings of financial markets such as capital markets, money markets, foreign exchange markets, and derivative markets. The products traded in these markets are also discussed. The module also looks at the functions of financial institutions like the central bank, commercial banks, microfinance institutions, and building societies. Financial mathematics topics such as the time value of money, investment appraisal, and valuation of financial securities are also covered.

CS131 Communication Skills

(Refer to the Centre for Communication Skills for the module synopsis)

BM131 Principles of Management

(Refer to Department of Management Sciences for the module synopsis)

ACC135 Financial Accounting IA

(Refer to Department of Accounting Sciences for the module synopsis)

HCS135 Introduction to Information Technology

(Refer to Department of Information and Marketing Sciences for the module synopsis)

ACC136 Financial Accounting IB
(Refer to Department of Accounting Sciences for the module synopsis)

BM135 Commercial Law
(Refer to Department of Management Sciences for the module synopsis)

EC231 Microeconomics I
This module is designed to extend the knowledge of the basic microeconomic principles that provide the foundation for future work in economics. It gives insight into how economic models can help us think about important real-world phenomena. Microeconomics is the study of rational choice behaviour on the part of individual consumers and firms. In general, economists are interested in how market mechanisms solve extremely complex resource allocation problems. This module presents a logical and coherent framework in which to organize observed economic phenomena. Topics include

supply and demand interaction, utility maximization, profit maximization, elasticity, perfect competition, monopoly power, imperfect competition, game theory and Economics of information.

EC232 Monetary Economics
The module aims to introduce students to some tools to understand the operations of monetary policy in the economy. By so doing the student after completing the module must be in a position to make an analysis of existing monetary policies within the context in which they have been formulated. The module therefore explains the nature, definitions and value of money, role of money in economic activity, the nature and role of financial intermediation, demand of and supply for money; theoretical and empirical issues, monetary policy and money supply control; Zimbabwe monetary policy since 1980, the role of the Central Bank and its independence particularly commitment and credibility, monetary policy transmission mechanisms, theory of and empirics on inflation, optimal stabilisation policy and optimal inflation rate and inflation persistence, open and closed economy monetary policies including efficiencies.

EC233 Quantitative Methods
It is a pre-requisite for students taking this module to have done basic Mathematics for Economists (EC133 and EC136). Quantitative methods seek to strengthen the use of mathematics in analysing and interpreting economic phenomena. The module covers areas like comparative statistics, optimization theory, difference and differential equations, matrix algebra, and vector calculus. At the end of this module the student will be able to identify real economic problems and apply the learnt mathematical principles to solve them. This module will ensure that students will gain the critical 21st Century skills of problem-solving, creativity and innovation.

EC235 International Economics
The module intends to help students to understand theories and policies related to international trade. The theories will illustrate how gains or losses from international trade arise thereby giving rise to policies that encourage or restrict trade. Also focus will be made on the world economy institutions that deal with international trade. Therefore, the module covers an understanding of trade theories that form the basis for trade policy analysis. These theories include classical theories, neoclassical theories and 'new' trade theories. The theory of trade protection, exchange rates and the balance of payments are also covered in this theory section. The policy section covers such topics as terms of trade and economic development, the global trading environment, regional co-operation and integration and the role world economy institutions in international trade. Current controversies in international trade policies are considered as well.

EC236 Computer & Statistical Analysis
The module forms an enhanced strong prelude to Econometrics I. It covers issues such as probability models, univariate probability distributions, multivariate probability distributions, non-parametric tests, sampling and sampling distributions, asymptotic distributions, Analysis of Variance. The module also infuses the use of computer softwares for inputting and analysis of statistical data. Computer-aided techniques of data analysis shall include descriptive, computational techniques and simulations.

EC234 Macroeconomics I
Macroeconomics studies the aggregate behaviour of the economy. This module provides an introduction to the economic analysis of key macroeconomic variables such as output, employment, inflation, interest rates and exchange rates. The intent of the module is to provide the student with a basic level of economic literacy essential for a well-informed citizenship in the years ahead. It capacitates the students to develop competence in using macroeconomic models to explain the determination of aggregate output, prices, employment, and growth. In addition, students should learn to analyse the impact of monetary and fiscal policy on the macroeconomy. Extensive references to current macroeconomic developments and applications of macroeconomic models are essential elements of the module.

EC237 Econometrics I
The module forms the foundation for Econometrics II. It covers the methodology of econometrics, linear regression model assumptions, properties of estimators, dummy variables, non-linear

regression models hypothesis testing procedures, bi-variate and multivariate analysis, Gauss-Markov theorem.

EC238 Research Methods
The module seeks to provide students with the basic tools to undertake independent research as well as understanding how research can be communicated to end users of analysis. It will focus on pertinent issues in research in Economics including elements such as literature reviews and data sources; the status and growth of economic knowledge; ethics of economic research; and overall design of a research project - aims, philosophy and methods, evaluating existing research (review of articles), and writing up and disseminating findings.

EC253 Investment Analysis and Portfolio Management
The primary objective of the course is to study the theory and empirical evidence relevant for investing, particularly in the context of portfolio management and covers the major issues currently of interest to investors. The major topics will include: Global investment environment security markets and the investment industry, optimal portfolio selection, the relation between risk and return, market efficiency. In one way or another, most of the course is geared towards the understanding and implementation of "modern portfolio theory," which is a general approach for maximizing the expected return of a portfolio given a certain amount of risk.

ENT131 Entrepreneurship
The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation-building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan.

Evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; identify the challenges being faced by entrepreneurs.

GSB211Gender Studies for Business
The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, the historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

EC340 Work-Related Learning Preliminary Report
The module is a preliminary report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, conclusions and recommendations to the company and the University.

EC341 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third year students with a practical exposure to all facets of the economic functions. The student is expected to rotate into all sections of related to economic sciences to which thy will be attached to enable them to be exposed to all economics areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

EC342 Work-Related Learning Report
The module is a report by the student during internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as, company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University versus what is at the work-place,any gaps, conclusions and recommendations to the company and the University.

EC438 Managerial Economics

The module considers business objectives: optimization, the principle of margin, decision analysis, demand measurement and forecasting, cost measurement, economies of scale, the learning curve, pricing policies: joint product pricing, transfer pricing, loss leaders, price discrimination, public utility pricing, advertising and promotion decisions: Dorfman – Steiner theorem, the advertising – sales relationship, corporate strategies: vertical integration, diversification, growth and market share, the location decision, profit planning and control, as well as game theory for managers.

EC451 Risk Management
The module focuses primary sources of risks, risk and uncertainty, risk management processes Enterprise risk management framework, market risk management, interest risk management, liquidity risk management, credit risk management, strategic risk management, case study of risk management failure.

EC452 International Finance
This module deals with International financial markets and financial decision-making in the global economy. Topics include Exchange Rates, Foreign Exchange Markets, and Determinants of Exchange Rates; Foreign Exchange Derivatives (Futures, Forwards, Options) Government Role and Intervention in Foreign Exchange Markets Arbitrage and Equilibrium in the Foreign Exchange Market; Direct Foreign Investment (DFI), International Diversification Multinational Capital Budgeting, Managing political and Country Risk in foreign investment. Discount Rates Cost of Capital for MNCs; International financial crises.

EC453 Financial Modelling and Forecasting
This module focuses on some key statistical techniques relating to the analysis of financial data in order to understand key financial models and to forecast financial variables. Financial variables to be forecasted include operating income, earnings, share prices, interest rates, foreign exchange, etc. It covers financial time series modelling, or broadly time series econometrics as applied in finance, financial economics, securities and investments. Theoretical as well as practical issues are considered. The module is designed to provide hands on training in the use of the main econometric packages, EViews, and Excel.

EC454 Financial Econometrics

The module builds on the concepts covered in Econometrics 1 (EC237). It covers topics on time series analysis, panel data analysis and limited dependent variable models as they apply to financial economics and investment. The module also looks at the formal tests and remedies to econometric problems prevalent in estimating regression models in financial economics, among them, non-stationarity of data, non-normality of errors, serial correlation, multicollinearity, heteroscedasticity and model specification. Issues in micro econometrics, including the analysis of survey data using probit and logit models, are also covered.

EC445 Dissertation/Research project
During the final year of study, students are required to write and submit a research project on a topic of their choice. The research project gives students an opportunity to seek solutions to problems affecting households, businesses, government, and society in general. It also enhances the student's skills in terms of time management, academic writing, data analysis, and results interpretation. The grade of the research project will be based on oral examination (viva voce) before a panel of examiners, and a mark awarded for the write-up submitted to the department. The length of the research project shall not be less than 10000 words (excluding preliminary pages and appendices), and the department shall provide guidelines on the structure that should be followed by students.

EC455 Development Finance
The module will provide students with a thorough understanding of specific aspects of development finance. It will equip students with the necessary skills to make a meaningful contribution to policy formulation and implementation, by focusing on financial development policy and financial management, with specific reference to developing countries. Topics covered include microfinance, remittances, foreign direct investment, financing of infrastructure, foreign debt.

EC435 Agricultural Economics
The module looks at the definitions of agricultural economics and related ideas, agriculture development in the African context, agricultural production theory, agriculture peasant household models (Chayanov and share tenancy models). The module also looks at agricultural policies; agricultural technology (innovation, transfer) and policy, food policy (food security and drought management), and land policy with particular reference to Zimbabwe.

EC239 Environmental Economics
This module seeks to equip students with skills to utilize the techniques covered in core modules such as microeconomics, macroeconomics and quantitative methods to analyse environmental problems. The module explains the role of environmental economics in the development processing general and in the integrated management of ecological–economic systems in particular. It covers topics such as history of environmental economics, sustainable development, environmental accounting and valuation of natural resources.

EC456 Financial Derivatives
The module covers various types of derivatives: forwards, futures, swaps and options. It examines the use of derivatives for arbitrage, hedging and speculation. The module also looks at the pricing/ valuation of derivatives using the binomial approach and the Black-Scholes model. The advantages and limitations of derivatives are also examined.

EC444 Blockchain Technology
This module lays the groundwork for understanding blockchain technology. It will offer a low-level entry into the technology, by discussing its origins, its key technological features and its different potential applications. Central features such as wallets and transactions will be discussed and crucial notions such as hashes, blocks and chains will be explained (explored). After completing this module, students will have a basic understanding of the origins and the key technological aspects of blockchain technology as well as a basic understanding of different architectures and myriad potential applications of blockchain technologies.

EC457 Taxation
The module introduces students to taxation and the tax system in Zimbabwe. The module covers topics that include taxation system of Zimbabwe, Types of Taxes income tax, capital gains tax, inheritance tax, corporation tax, VAT customs duties and other special taxes that are applicable in Zimbabwe. It also covers tax planning, tax incentives as well as capital allowances relating to growth points, export processing zones and farmers and miners. Transfer pricing. Double taxation agreements are also covered. Aspects of taxation such as Administration provisions of the Income Tax Act and Jurisdictional Issues in Business Taxation are also dealt with.

EC460 Natural Resource Economics
Natural resource economics focuses on the supply, demand, and allocation of the scarce natural resources that we have. It aims to gain a better understanding of the role of natural resources in the economy and how to efficiently use these. Learning about the role of natural resources allows for the development of more sustainable methods to manage resources and make sure that they are maintained for future generations. The goal of natural resource economics is to develop an efficient economy that is sustainable in the long run. The module focuses on economic models for optimal management of renewable resources such as forestry, fishery, wildlife, and non-renewable as well as recyclable resources.

EC458 Corporate Restructuring
This module introduces students to up-to-date debates and issues of critical importance in modern finance. Specific topics covered include the cost of capital and market efficiency, various improvements made to a firm's financial capital structure in line with their cash flow needs, accretion vs dilution analysis, the P/E problem in acquisitions, strategic vs financial acquisition, mergers and amalgamation, reconstruction and demerger of companies, takeovers, funding of mergers and takeovers, post-merger reorganisation, modern theories relating to mergers, acquisition and other forms of restructuring.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN ECONOMICS (MECON)

Duration	1.5 years
Credit Load	324
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

The programme aims to develop the sound conceptual, technical, analytical and communication skills that are required to succeed in the Economics profession. Students will be expected to exercise their analytical abilities and develop effective verbal and written communication skills.

2. ENTRY REQUIREMENTS

To qualify for entry into the Master of Commerce Economics degree programme a student should have an honours degree in Economics, Agricultural Economics, Banking & Finance, Fiscal Studies or a related field.

3. PROGRAMME CHARACTERISTICS

Areas of study: Microeconomics, macroeconomics, mathematics and statistics, econometrics, trade, monetary economics, public sector, agriculture, ecology and environmental issues and industry.
Specialist Focus: Economics

Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects.

Distinctive Features: The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates can be absorbed in the following organizations and areas; Financial institutions, Government Departments, Parastatals, private sector firms, non-governmental organizations, regional and international organizations, institutions of higher learning, and consultancy firms among others.
Holders of the Master of Commerce in Economics Degree are eligible for Doctoral studies in specialised areas of Economics.

5. PROGRAMME DELIVERY

f) **Teaching and Learning Methods**: Lectures, tutorials, seminars, group-work, colloquiums, research projects and individual independent study.

g) **Assessment Methods:** Written and oral examinations, tests, term papers, seminar presentations, final year research project report, continuous assessments.

6. LEARNING OUTCOMES

On completion of the programme, a graduate will be able to:

- Critically evaluate economic policies.
- Come up with innovative solutions to economic problems.

- Undertake economic research and other tasks in economics with minimal monitoring and supervision.
- Supervise others in conducting economics and related research.

7. GENERAL PROVISIONS

Refer to faculty and general regulations.

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 40% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for 60% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment.

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%.

9. PROVISION FOR PROGRESSION

Refer to section 6 of the Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the general regulations.

12. DEGREE WEIGHTING

Refer to section 13 of the faculty regulations.

13. AWARD OF DEGREE

Refer to section 13 of the general regulations.

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MECO737	Advanced International Economics	18
MECO731	Advanced Microeconomic Analysis	18
MECO732	Advanced Mathematics for Economists	18
MECO735	Investment Analysis & Corporate Finance	18
MECO734	Advanced Monetary Economics	18

Level 1 Semester 2

Code	Module Description	Credits
MECO733	Advanced Public Finance	18
MECO739	Advanced Macroeconomic Analysis	18
MECO740	Environmental & Ecological Economics	18
MECO736	Advanced Econometrics	18
MECO738	Health Economics	18

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
MECO831	Agricultural Economic Policy		18
Electives (choose any one module)			
MECO832	Mineral Economics		18
MECO833	Contemporary Issues in Economic Development		18
MECO845	Dissertation/Research project		90
Electives (choose any one module)			
MECO834	Real Estate Economics		18
MECO835	Regional Economics		18

15. MODULE SYNOPSES

MECO731 Advanced Microeconomic Analysis
The module emphasizes topics such as choice under uncertainty, economics of information, game theory, general equilibrium analysis, and welfare economics. Emphasis will be on application of microeconomics in analysing real-world problems.

MECO732 Advanced Mathematics for Economists
The module enriches students with some economic insights, intuitions and rudimentary understanding of Economics. It allows the students to master beforehand the relevant mathematical tools. This requires the student to command elementary algebra. The module extensively reviews elementary algebra and also treats a number of other rather elementary topics. It also covers calculus for functions, multivariable optimization problems, linear programming, difference and differential equations and optimal control theory.

MECO733 Advanced Public Finance
The module takes an in-depth analysis of the following issues: public goods and externalities, political economy/ public choice, tax theory, taxes and efficiency, equitable taxation, Zimbabwean tax system.

MECO734 Advanced Monetary Economics
This module presents a comprehensive overview of the pertinent issued in advanced monetary economics. It enables students to acquire sufficient knowledge of monetary theory and policy. It focuses on the dynamic relationships that exist between monetary aggregates and the macro-economy. Topics covered include money in models of economic growth, money supply functions, portfolio choice, and money in the open economy, central banking and monetary policy.

MECO739 Advanced Macroeconomic Analysis
The module explores recent developments in macroeconomic theory and relationships to economic policy. It covers the modern interpretations of Keynes, overlapping generations models adopting the dynamic and general equilibrium analysis. The issues of gap models, IMF model, World Bank Model, neoclassical growth theories and endogenous growth theories are also covered. The content also extends to business cycle theories, global economic crises, open economy analysis, balance of payments approaches and theories of exchange rate determination as well as currency crises.

MECO736 Advanced Econometrics
The module covers important tools in modern econometrics and economic statistics, including topics in applied econometric time series, and topics concerning discrete and limited dependent variable analysis. It also covers modern econometric techniques that are frequently used in applied macroeconomic research and practical application of econometric methods to the analysis of real-world data using statistical packages such as E-views and Stata.

MECO737 Advanced International Economics
This module takes a theoretical and empirical approach to the study of international trade and investment. It focuses on core trade theories and empirical applications based on these theories. Furthermore, it dwells on recent theories that focus on the firm's decision to export and the role of heterogeneity in firm productivity on patterns of trade. Other topics will focus on multinational activity and investment abroad, including different types of foreign direct investment. The focus will also be on institutional frictions and their effects on patterns of trade and investment.

MECO738 Health Economics
The module covers the following topics: demand and supply of health services, systems of health care delivery, health care financing, health policy and planning, health sector reforms, and economics of disease.

MECO735 Investment Analysis & Corporate Finance
The module focuses on advanced theoretical and empirical aspects of corporate financial management and capital market behaviour. It provides students with techniques for evaluating investments from both the perspectives of individual assets as well as portfolios. Techniques for investment analysis focus on maximising expected returns while minimising risk. Topics covered include modern portfolio theory, equilibrium capital models, capital budgeting, discounted cash flow valuation models, capital structure and dividend payout decisions.

MECO740 Ecological & Environmental Economics
Selected issues such as market failures, externalities, pollution control, preservation of natural resources, sustainable development, common property resources, global environmental and natural resources, conservation of renewable and non-renewable resources; and evaluation techniques.

MECO831 Agricultural Economic Policy
The module aims to develop the skill to critically analyse and evaluate agricultural policy to understand the origin of current agricultural policy issues. Areas to be covered in this module include: policy setting, problems of food and agriculture policy process, particular interest groups (pigs), role of trade and macroeconomics on agriculture, the farm problem, historic farm programs, structure of agriculture resource policy food price policy nutrition, food safety, and quality rural development.

MECO832 Mineral Economics
The module is expected to cover the following aspects of mineral economics; environmental impact of mining, resource curse, mining project evaluation criteria, mine taxation, ore reserve estimation, Zimbabwe mining policy, sustainable extraction of mineral resources, and market for minerals

which include the competitive and monopoly markets, contribution of mining to economic growth, the welfare implications of mining among others.

MECO833 Contemporary Issues in Economic Development
The module aims to provide a thorough review and critical assessment of contemporary issues in economic development, covering the main analytical approaches, empirical evidence and policy issues in development economics. Topics to be covered include an overview of contemporary issues in economic development, the Growth-Inequality-Poverty triangle, rural development; rural-urban migration, industrialization and urban labour market informality; and population issues (demographic trends and human capital development).

MECO834 Real Estate Economics
This module focuses on urban economic processes, price determination in land and real property markets, and their interrelation with investment markets and portfolio choice, as well as the impact of land use regulations, transport, public policy and socio-economic change. It also covers the main tools and concepts of real estate economics, in particular that of housing economics, models addressing central topics within real estate economics, such as real estate in the wider economy (internationally and nationally) and the relationship between the property and capital markets.

MECO835 Regional Economics
The module looks at the regional economy & the regional level of economic activity, the impact of economic policy on the regions, the concepts of convergence, divergence & regional growth, methods in regional economics and in particular for analysing the impact of economic activity on the regional economy, the impact of innovation and research and development on regional development and an effective regional policy.

MECO845 Dissertation/Research project
The programme has the dissertation as an integral component where students are expected to use analytical methodology in handling a topic in any of the chosen fields of economics. Students are required to defend their dissertations in a viva voce. The final mark for the dissertation is composed of the viva voce and the mark awarded for the submitted dissertation. The length of the dissertation should not be less than 15000 words (excluding preliminary pages and appendices). The dissertation runs for both quarters but will only be graded in the last quarter of the final semester.

REGULATIONS FOR MASTER OF COMMERCE DEGREE IN FINANCIAL ECONOMICS AND INVESTMENT (BS18)

Duration	1.5 Years
Credit Load	324
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

The programme aims to equip students with sound conceptual, technical, and analytical and communication skills that are required to succeed in the Economics and investment profession. Students will be expected to exercise their analytical abilities and develop effective verbal and written communication skills.

2. ENTRY REQUIREMENTS

To qualify for entry into the Master of Commerce Financial Economics and Investment degree

programme, a student should have an honours degree in Economics, Financial Economics and Investment, Economics and Econometric Modelling, Agricultural Economics, Banking & Finance, Fiscal Studies or related field.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Microeconomics, macroeconomics, public finance and debt management, mathematics and statistics, econometrics, International trade, monetary economics, applied financial econometrics, financial engineering, agriculture, ecology and environmental issues as well as contemporary issues in development.

3.2 **Specialist focus:** Financial Economics and Investment

3.3 **Orientation:** Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 **Distinctive features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Financial planners/ advisors, Investment bankers, Equity researchers, Financial analysts, Economists and stock brokers.

Holders of the Master of Commerce in Financial Economics and Investment Degree are eligible for Doctoral studies in specialised areas of Economics, Finance, and Investment.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, colloquiums, research projects and individual independent study.

5.2 Assessment Methods: Written and oral examinations, tests, term papers, seminar presentations, final year dissertations and continuous assessments.

6 LEARNING OUTCOMES

On completion of the programme, a graduate will be able to:

- Critically evaluate economic, investment and financial policies.
- Come up with innovative solutions to financial, investment and economic problems.
- Undertake financial, investment and economic research and other related tasks in financial and investment economics with minimal supervision.
- Analyse the fundamental operations of financial markets.
- Supervise others conducting financial and investment economic research.

7 GENERAL PROVISIONS

Refer to section 4 of the General Regulations.

8 PROGRAMME ASSESSMENT

Refer to section 5 of the General Regulations.

9 PROVISION FOR PROGRESSION

Refer to Section 6 of the Faculty Regulations.

10 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the General Regulations.

12 DEGREE WEIGHTING

Refer to section 13 of the General Regulations.

13 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MECO731	Advanced Microeconomic Analysis	18
MECO737	Advanced International Economics	18
MECO754	Applied Financial Econometrics	18
MECO734	Advanced Monetary Economics	18
MECO735	Investment Analysis & Corporate Finance	18

Level 1 Semester 2

Code	Module Description	Credits
MECO739	Advanced Macroeconomic Analysis	18
MECO756	Agribusiness Finance and Risk Management	18
MECO753	Blockchain and Crypto-currencies	18
MECO755	Financial Engineering	18
MECO852	Public Finance and Debt Management	18

Level 2 Semester 1

Code	Module Description	Credits
MECO752	Advanced Financial modelling and forecasting	18
MECO832	Mineral Economics	18
MECO845	Dissertation/Research Project	90
Electives (Choose any 1)		
MECO833	Contemporary Issues in Economic Development	18
MECO740	Environmental & Ecological Economics	18
MECO738	Health Economics	18
MECO851	Sustainable Finance and Investment	18

14 MODULE SYNOPSES

MECO731 Advanced Microeconomic Analysis

The module emphasizes on topics such as choice under uncertainty, economics of information, game theory, general equilibrium analysis, and welfare economics. Emphasis will be on application of microeconomics in analysing real-world problems.

MECO734 Advanced Monetary Economics

This module presents a comprehensive overview of the pertinent issued in advanced monetary economics. It enables students to acquire sufficient knowledge of monetary theory and policy. It focuses on the dynamic relationships that exist between monetary aggregates

and the macro-economy. Topics covered include money in models of economic growth, money supply functions, portfolio choice, and money in the open economy, central banking and monetary policy.

MECO735 Investment Analysis & Corporate Finance
The module focuses on advanced theoretical and empirical aspects of corporate financial management and capital market behaviour. It provides students with techniques for evaluating investments from both the perspectives of individual assets as well as portfolios. Techniques for investment analysis focus on maximising expected returns while minimising risk. Topics covered include modern portfolio theory, equilibrium capital models, capital budgeting, discounted cash flow valuation models, capital structure and dividend payout decisions.

MECO737 Advanced International Economics
This module takes a theoretical and empirical approach to the study of international trade and investment. It focuses on core trade theories and empirical applications based on these theories. Furthermore, it dwells on recent theories that focus on the firm's decision to export and the role of heterogeneity in firm productivity on patterns of trade. Other topics will focus on multinational activity and investment abroad, including different types of foreign direct investment. The focus will also be on institutional frictions and their effects on patterns of trade and investment.

MECO752 Advanced Financial Modelling and Forecasting
The module is designed to help students with ways of preparing and determining expectations of future results in companies/institutions, taking forecast assumptions and making calculations using a company's financial statements. It provides the required skills in financial mathematics and statistics. The module will cover topics as matrix algebra, differential calculus, probability theory, descriptive statistics, hypothesis testing, regression analysis, forecasting and modelling volatility among others.

MECO739Advanced Macroeconomic Analysis
The module explores recent developments in macroeconomic theory and relationships to economic policy. It covers the modern interpretations to Keynes, overlapping generations models adopting the dynamic and general equilibrium analysis. The issues of gap models, IMF model, World Bank Model, neoclassical growth theories and endogenous growth theories are also covered. The content also extends to business cycle theories, global economic crises, open economy analysis, balance of payments approaches and theories of exchange rate determination as well as currency crises.

MECO753 Blockchain and Crypto-currencies
This module covers how crypto-currencies work, how to use them, their level of security and how they differ from fiat currencies. The module will include economics and financial topics such as game theory and cryptography, bitcoin and theories of money, wallets and exchange, pricing, initial coin offerings (ICOs,) risk analysis and volatility, market manipulation and trading instruments and basic strategies as well as other aspects such as legal and regulatory issues, such as data protection, privacy and securities.

MECO754 Applied Financial Econometrics
The module provides a practical but strong theoretical approach to econometrics in the field of finance. The module will extend students' knowledge of financial econometrics by enabling them to assess advanced research literature in finance. Students will be equipped with skills that will enable them to develop, conduct, and evaluate econometric analyses for practical investment and portfolio decisions. The module will cover areas that include analysing financial returns, modelling financial time series, estimation of risk frontier, workings of asset pricing models, ARCH and GARCH Models and behavioural Finance empirical foundations.

MECO755 Financial Engineering
The complexity of the financial world requires understanding of financial engineering. This module is meant to inculcate, in the students, the mathematical techniques to solve financial problems. This involves knowledge from various disciplines such as statistics, applied mathematics, economics and computer science, among others. Areas of focus may include, but are not limited to, cashflow engineering, interest rate swap engineering and options engineering. Hedging strategies and pricing tools and approaches must also be known.

MECO756 Agribusiness Finance and Risk Management
The module focuses on providing learners with skills to use finance and investment tools in agricultural development. Learners will get insights on how investors approach the agricultural sector and how to alleviate constraints as well as managing risks. Topics in the module include finance and investment in agriculture, trading in agribusiness, rural investment, agri-business lending and risk management for agricultural investment.

MECO852 Public Finance and Debt Management
The module looks at the management of domestic and external public debt. Sound public debt management is important, especially in low- and middle-income countries (LMICs) that are highly indebted. The issues covered include public expenditure theory, taxation theory, debt crises, economic and social impact of debt crises, public debt sustainability, debt restructuring and rescheduling.

MECO833 Contemporary Issues in Economic Development

The module aims to provide a thorough review and critical assessment of contemporary issues in economic development, covering the main analytical approaches, empirical evidence and policy issues in development economics. Topics to be covered include an overview of contemporary issues in economic development, the Growth-Inequality-Poverty triangle, rural development; rural-urban migration, industrialization and urban labour market informality; and population issues (demographic trends and human capital development).

MECO845 Dissertation/Research project

The programme has the dissertation as an integral component where students are expected to use analytical methodology in handling a topic in any of the chosen fields of investment and financial economics. Students are required to defend their dissertations in a viva voce. The final mark for the dissertation is composed of the viva voce and the mark awarded for the submitted dissertation. The length of the dissertation should not be less than 15000 words (excluding preliminary pages and appendices).

MECO851 Sustainable Finance and Investment

The module covers topics on financing investments that promote environmentally sustainable development. Topics to be covered include fundamentals of sustainable finance, the sustainable finance market, green, social and sustainable bonds, responsible investment, environmental and social risk analysis (ESRA), corporate eco-efficiency in financial institutions, sustainable finance for savings and credit cooperatives, climate change: risk and opportunities for finance sector

MECO738 Health Economics

The module covers the following topics: demand and supply of health services, systems of health care delivery, health care financing, health policy and planning, health sector reforms, and economics of disease.

MECO740 Ecological & Environmental Economics

Selected issues such as market failures, externalities, pollution control, preservation of natural resources, sustainable development, common property resources, global environmental & natural resources, conservation of renewable &non-renewable resources; evaluation techniques.

MECO832 Mineral Economics

The module is expected to cover the following aspects of mineral economics; environmental impact of mining, resource curse, mining project evaluation criteria, mine taxation, ore reserve estimation, Zimbabwe mining policy, sustainable extraction of mineral resources, and market for minerals which includes the competitive and monopoly markets, contribution of mining to economic growth, the welfare implications of mining among others.

DEPARTMENT OF INFORMATION AND MARKETING SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN DATA SCIENCE AND INFORMATICS (BS08)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

Data science and informatics is a skills-based programme which emphasizes data analytics, machine learning, algorithm design, computer vision and data protection. The programme aims to:

- Develop knowledge, skills, and competencies in the field of Data Science and Informatics, data analytics, machine learning, and computer vision relevant to various employment capabilities and careers in the world of work and society.
- To prepare students for further studies and lifelong learning in the Data Science Profession.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for entry into the Bachelor of Commerce Honours Degree in Data Science and Informatics programme, a candidate, in addition to satisfying the minimum conditions as prescribed under the General Regulations for English and Mathematics at 'O' Level, must have obtained a pass in 'A' Level Mathematics or Statistics and a pass at 'A' level in at least two of the following subjects or their equivalents: Accounts, Economics, Management of Business and Computing [Computer Studies/Software engineering]

2.2 Visiting / Harare Weekend School Programme

To qualify for entry into the Bachelor of Commerce Honours Degree in Data Science and Informatics programme (Visiting / Harare Weekend School), a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma in an Information Technology related field or any equivalent Tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects.

and:

Confirmation of employment in Data analytics or relevant Information Technology departments.

2.3 Special Entry

Candidates who have successfully completed a Professional Diploma in Data Science or have obtained equivalent qualifications within three years may apply for direct entry into Level 2 of the degree programme, subject to availability of places. Candidates admitted under the above regulation will normally be exempted from level 1 on a module-by-module basis. No candidate may complete the degree in less than three academic levels. Successful completion of the Work-Related Learning component at level three is compulsory for all candidates.

2.4 Mature Entry

Should be at least 23 years old for females and 25 years old for males AND should have at least 2 years relevant industrial experience.

3 PROGRAMME CHARACTERISTICS

Areas of study: Programming, Data Science fundamentals, Human-Computer Interaction, Computer and Information Ethics, Machine Learning, Data Warehousing, Data Mining, Enterprise Architectures, Web Mining, Database Administration

Specialist Focus: Data Science and Informatics

Orientation: Research and innovation oriented.

Distinctive Features: The programme is driven by an ever-changing data-intensive environment. The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach. Consultations are made time and again to maintain programme relevance to the ever-changing business world.

4 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates with a Bachelor of Commerce Honours Degree in Data Science and Informatics can pursue careers in Data Science and Informatics including Data Science Engineer, Data Analysts, Big Data Consultant, Data Warehouse Specialist, Database Administrator, and Business Intelligence Analysts.

The programme also opens opportunities for graduates to pursue further education within the data science and informatics fields. Students can enroll for Master's and Doctoral studies in Data Science and Informatics or in interdisciplinary programmes related to Artificial intelligence, Cloud Computing, the Internet of Things and other related higher qualifications.

5 PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods**: Lectures, tutorials, computer practicals, group work, work-related learning reports, mini WRL research and innovation projects, and individual independent study.

5.2 **Assessment Methods:** Written and oral examinations, tests, computer projects, presentations, work-related learning reports, mini WRL research projects, final year research project reports and continuous assessments.

6 LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Apply knowledge of Data Science and Informatics to solve real-life problems.

- Use and understand classical and modern data analytics techniques, statistical machine learning techniques, AI, and parallel and distributed computing, to solve practical problems characterised by large-scale data.
- Employ algorithmic problem-solving skills to the problem at hand, including defining clear requirements for a problem, decomposing the problem, using efficient strategies to arrive at an algorithmic solution, and implementing solutions through programming in a suitable high-level language.
- Build expertise in a range of appropriate software, including the widely used Python and R languages for a career in Data Science and Informatics.
- Apply modern data science methods to one or more domains of application (e.g., business analytics, finance, biotechnology, and public health).
- Demonstrate professional and ethical responsibility in areas such as data ownership, security and sensitivity of data, consequences and privacy concerns of data analytics, and the professionalism of transparency and reproducibility.

7 GENERAL PROVISIONS

Refer to faculty and general regulations

8 PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 Project documentation will account for 60% and 40% of project implementation (actual system)

8.6 To be admitted to the examinations, a candidate must;

8.6.1 be a registered student at the University in accordance with the General Regulations

8.6.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment.

8.6.3 have attended and participated in seminars, tutorials practical classes, presentations and other activities as required by the Department.

8.6.4 Candidates shall be informed in advance of the deadline of submission of the dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by the Senate, the maximum mark allowed for such work should be 50%.

9 PROVISION FOR PROGRESSION

In addition to general and faculty regulations for provision for progression:

- students at level 1 should have passed DSI 134, DSI 135, DSI 136, and DSI 139 prerequisites for DSI 235
- prerequisite modules to be passed before proceeding to level 3 are DSI 235 and DSI 239

10 FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11 GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12 DEGREE WEIGHTING

Refer to faculty regulations

13 AWARD OF DEGREE

Refer to general regulations

14 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
CS131	Basic Communication Skills	12
GSB211	Gender Studies for Business	12
DSI 132	Foundations of Data Science	12
DSI 136	Data Structures and Algorithms	12
DSI 131	Introduction to Informatics	12
DSI 133	Human-computer interaction principles and practices	12
DSI 134	Principles of Programming Languages	12
DSI 135	Computer Architecture and Organisation	12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
DSI 137	Computer-information ethics, social informatics, and data governance		12
DSI 139	Introduction to Python	(DSI 134 & DSI 136)	12
DSI 142	Operating systems	(DSI 135)	12
DSI 140	Information Infrastructure 1	(DSI 139)	12
DSI 141	Natural Language Processing	(DSI 139)	12
DSI 143	Fundamentals of Data Warehouse and Data Mining		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
ENT 131	Entrepreneurship 1		12
DSI 232	Information Systems Analysis, Design and Development Methodologies		12
DSI 234	Information Representation		12
DSI 231	Introduction to Research in Data Science and Informatics		12
DSI 233	Enterprise Architecture		12
DSI 235	Machine Learning I: Using Python	(DSI 139)	12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
DSI 237	Statistical analysis		12
DSI 240	R Programming and Computer Vision		12
DSI 236	Introduction to Media Application Development	(DSI 133)	12
DSI 238	Organisational Informatics		12
DSI 239	Data Science and Informatics Project 1		12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Prerequisites	Credits
DSI 340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Prerequisites	Credits
DSI 341	Work-Related Learning Continuous Assessment		40
DSI 342	Work-Related Learning Report		40

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
DSI 434	Optimisation Techniques with NumPy &SciPy	(DSI 235)	12
DSI 435	Applied cloud computing for data-intensive sciences		12
DSI 431	Data Science and Informatics Project Management		12
DSI 432	Information Infrastructure 2: OOP	(DSI 140)	12
DSI 433	Advanced Data Warehouse and Data Mining	(DSI 143)	12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
DSI 436	Data analytics and visualisation using matplotlib	& seaborn [DSI 240]	12
DSI 437	Machine Learning II: Using JAVA	(DSI 235)	12
DSI 438	Data Science and Informatics Project 2		24

15 MODULE SYNOPSES

CS131 Basic Communication Skills

This module is designed to equip students with the communication skills required in the employment world. It consists of fundamental documents and methodologies for handling challenges in organisations, for example, conflict and human capital management. The module will also deal with concerns of meetings, time management, research and report writing. The students shall be equipped with the reference techniques that are applied when carrying out research or writing assignments and the use of proper written English.

GSB211 Gender Studies for Business

The module empowers the students with the knowledge and skills that enable them to be gender sensitive in the university, workplace and all their social interactions. Topics covered include Understanding Gender, Theories of Gender Inequalities, Historical Development of Gender, Gender Analysis, Gender Issues in Society, Redressing Gender Imbalances, Empowerment and Strategies for Creating a Gender Responsive Environment. Every student must pass the module to graduate.

DSI 131 Introduction to Informatics

The module introduces students to foundational Informatics concepts. Topics to be covered include basic information representation and processing; searching (including fundamental search algorithm) and organisation; evaluation and analysis of information; internet-based information access tools; cutting edge technologies; informatics tools for various human domains; and problem-solving with information technology.

DSI 132 Foundations of Data Science

The module is designed to introduce students to a range of topics and concepts related to the Data Science process. The following topics will be covered: introduction to big data concepts, tools, and techniques; gathering and sorting data; data pre-processing; understanding structured and unstructured data types; data analysis and visualisation; probability and statistics; decision theory; estimation theory; coordinate systems; linear transformations; effects of computation on data graph theory; and algorithms; pattern recognition and basic machine learning algorithm; different clustering approaches.

DSI 133 Human-computer interaction principles and practices

This module introduces students to the fundamental principles and practices of human-computer interaction (HCI) and evaluation. The module will cover the Foundations of Human-Computer Interaction; Interaction Design; Understanding Users, Cognitive and Affective Factors; The Computer and Human-Computer Interaction; Web Interfaces; Introduction to Interactive System Design; discussion data Gathering and Requirements Analysis with a focus on HCI aspects; Interfaces Design and Prototyping; Usability Testing & Analytic Evaluation.

DSI 134 Principles of Programming for Data Science and Informatics

The module explores the concepts and structures governing the design and implementation of programming languages. It presents an introduction to the concepts behind compilers and runtime representations of programming languages; features of programming languages supporting abstraction and polymorphism; and the procedural, functional, object-oriented, and concurrent programming paradigms.

DSI 135 Computer Fundamentals: Architecture and Organisation

This module introduces students to the basic instruction set architecture and organisation of a modern computer. Topics include an introduction to computer systems; instruction set architecture and design; assembly language programming; computer arithmetic; processing unit design; memory system design; input-output design and organisation; pipelining design techniques; reduced instruction set computers (RISCs); and introduction to multiprocessors.

DSI 136 Data Structures and Algorithms

This module covers the analysis and design of fundamental data structures and engages learners to use data structures as tools to algorithmically design efficient computer programs that will cope with the complexity of actual applications. The module focuses on basic and essential topics in data structures and algorithms including basic concepts, Arrays, Lists, Stacks, Queues, Trees, Hash Tables, Graphs, heaps, sorting algorithms, and recursion.

DSI 137 Computer-information Ethics, Social Informatics, and Data Governance

This module introduces students to ethical and professional issues that occur when designing and using networked information technologies and information resources. It examines frameworks for making ethical decisions, emergent technologies and their ethical implications, and information and computer professionalism. Topics include data protection, privacy; intellectual property; cybercrime; codes of professional ethics; key principles and objectives of data governance in an organisation and how it impacts other parts/functions of the organisation that create, consume, or present data/information/insights; the importance of data design and architecture in future-proofing an organisation's decision support capabilities.

DSI 139 Introduction to Python

This module is designed as an introduction to the Python programming language. Students are introduced to core programming concepts like data structures, conditionals, loops, variables, functions, Python modules and libraries. The module includes an overview of the various tools available for writing and running Python and gets students coding quickly. It also provides hands-on coding exercises using commonly used data structures, writing custom functions, and reading and writing to files.

DSI 140 Information Infrastructure 1

This module introduces students to software architectures of information systems and basic concepts and procedures of system and application development. Topics include principles of developing

dynamic and database-driven applications for the World Wide Web using python; relational database concepts; and basic MySQL statements.

DSI 141 Natural Language Processing

This module is intended to teach the principles and methods of statistical natural language processing as well as provide hands-on experience with Python text analysis. Students will be able to conduct authoritative large-scale statistical analysis of textual data and discover relevant patterns using NLP. Natural language processing knowledge is also highly sought after in the industry. Topics include Markov Models, Neural Networks, RNN, LSTM Architectures, Sequence Labelling, Syntactic Parsing, Statistical Parsing, Neural Parsing, Semantic Parsing, Information Extraction, Machine Translation and Conversational Agents.

DSI 142 Operating Systems

This module explores the design and implementation of computer operating systems. Topics include historical aspects of operating systems development; systems programming; process scheduling; synchronisation of concurrent processes; deadlocks; virtual machines; memory management; virtual memory and paging; I/O and file systems; system security; OS/architecture interaction; and distributed operating systems.

DSI 143 Fundamentals of Data Warehouse and Data Mining

The aim of the module is for students to study the basic principles of data mining and data warehousing architecture. It intends to introduce more advanced topics in databases such as data mining and data warehousing. This module introduces methods and theory for the development of data warehouses and data analysis using data mining; Data quality and methods and techniques for pre-processing of data; Modelling and design of data warehouses; Algorithms for classification, clustering, and association rule analysis; and Practical use of software for data analysis.

ENT131 Entrepreneurship I
The module will look at business development, barriers encountered, and issues related to marketing, human resources management, pricing, government policies, and taxation of businesses. Case studies, visits to business premises and presentations by entrepreneurs will be used to understand the running of the business. The preparation of business plans will be covered.

DSI 231 Introduction to Research in Data Science and Informatics

This module presents a broad overview of research philosophy, designs, and methods. The module covers a wide range of applied and theoretical research, and the content is specifically tailored to reflect the rapidly emerging field of Data Science and Informatics. The module will include major methods that are at the core of contemporary approaches to research in Data Science and Informatics.

DSI 232 Information Systems Analysis, Design and Development Methodologies

The module introduces students to information systems development procedures. This module covers information systems analysis, designed, and developed to conform to the needs of users or use requirements. The module also looks at different types of systems development methodologies with particular emphasis being placed on how the human component and other stakeholders are involved or catered for. The development methodologies will include Waterfall; Agile; Lean; Iterative; Prototyping; DevOps; Spiral; and V-model.

DSI 233 Enterprise Architecture

This module provides students with a quick survey of enterprise architecture tools and business modelling and application development. The module will cover the following topics: Introduction

and evolution of Enterprise Architecture; Types of, and the relation between architectures; Software architecture to technology architecture continuum; Systems thinking and business thinking; Solution and Enterprise Architecture - business and IT strategies; Capturing architectures – use of models; Architectural styles; Documenting architectures; Quality attributes; Overview of EA frameworks and practices; Capturing enterprise application and data architectures; and the human side of architecting.

DSI 234 Information Representation

This module introduces students to information representation in computer systems. Topics include relational databases in the ER model; SQL commands; database design, implementation, and normalisation; database triggers; backup, security, and other data management tasks; data extraction from JSON and XML and their use for transmitting objects between browser and server; and social and ethical issues.

DSI 235 Machine Learning I: Using Python

This module has been prepared for students aspiring to develop applications involving machine learning techniques using the powerful yet simple programming language, Python, and its packages. Topics include Core concepts and algorithms used for supervised and unsupervised learning; Application of machine learning algorithms for the tasks of classification and prediction; Methods for data preprocessing including normalisation, feature extraction and selection, dimensionality reduction, and re-balancing; Practical recommendations for applying machine learning algorithms; and Evaluation and analysis of the performance of machine learning algorithms.

DSI 236 Introduction to Media Application Development

This module introduces students to concepts and skills related to the design of interactive multimedia applications for the web, desktop, and mobile devices. Topics covered include JavaScript fundamentals; Program flow control; User input and functions; DOM manipulation and CSS; Arrays and objects; Events; Introduction to P5; P5 graphics and animation; P5 graphics and animation: arrays and loops; P5 graphics and animation: functions, objects, and classes.

DSI 237 Statistical Analysis

This module is designed to provide students with the basic concepts of data analysis and statistical computing. Topics to be covered include Bayesian analysis, Judging conditional probability, Data classification, Linear regression, Resampling, Subset selection, Shrinkage, Dimension reduction, Nonlinear models and Tree-based analysis, Support vector machines and Unsupervised learning. Tools required are SPSS, NumPy, SciPy, Matplotlib and Pandas

DSI 238 Organisational Informatics

This module examines the various needs, uses, and consequences of information in organisational contexts. Topics include organisational types and characteristics; functional areas and business processes; information-based products and services; the use of and redefining the role of information technology; the changing character of work-life and organisational practices; socio-technical structures; and the rise and transformation of information-based industries.

DSI 239 Data Science and Informatics Project 1

This module puts knowledge gained into practice as students work in teams to develop an information system. Students will embark on a development project that combines their computing skills and the fundamental concepts learnt with available tools and metrics into an application or dashboard for measuring, analysing and interpreting data within the context of specific business goals and objectives.

DSI 240 R Programming

The module teaches students how to program in R and how to use R for effective data analysis and visualisation. The topic includes the R working environment; arithmetic and logical operators; salient functions for manipulating data, and getting help using R; data structures, variables, and data types used in R.

DSI 340 Work-Related Learning Preliminary Report
The module is a preliminary report by the student during the internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of data science and Informatics, any gaps, conclusions and recommendations to the company and the University.

DSI341 Work-Related Learning Continuous Assessment
Approved work-related learning of not less than ten (10) months and not more than twelve (12) is aimed at providing third-year students with practical exposure to all facets of Data analytics and Visualisation functions. The student is expected to rotate into all sections of the company that implement data science and informatics to which they will be attached to enable them to be exposed to all Data science and Informatics areas. The student is supposed to produce and/ or facilitate the production of a logbook, assessment form and employer assessment booklet.

DSI 342 Work-Related Learning Report
The module is a report by the student during the internship. The thrust of the report is to relate theory covered in the university to practice. It follows a formal report layout covering new concepts such as company background, organisational structure, sections attached to, challenges encountered, new concepts learnt, any gaps between information covered in the University, application of data science and Informatics, any gaps, conclusions and recommendations to the company and the University.

DSI 431 Data Science and Informatics Project Management

This module will focus on project management in a Data Science and Informatics setting. Students will become conversant in the tools and techniques of project management, such as project selection methods, work breakdown structures (WBS), network diagrams, critical path analysis (CPA), critical chain scheduling, cost estimates, earned value management, motivation theory and team building.

DSI 432 Information Infrastructure 2: OOP

This module will explore the systems architecture of distributed and Web-based applications in greater depth. The topics covered are practices of developing well-designed; reusable software components; advanced object-oriented programming (OOP) knowledge and practice; the use of Model/View/Controller (MVC), Factory and Abstract software architectural pattern; and application development with Rapid Application Development (RAD) method.

DSI 433 Advanced Data Warehouse and Data Mining

The module gives students advanced concepts of different statistical techniques to learn practical skills for learning useful patterns in data. The module will be covered by applying data mining

concepts using R Programming. The topics included are Statistical Learning, assessing model accuracy, Association, Classification, Clustering and Regression, and mining the web, text, and social media.

DSI 434 Optimisation Techniques with NumPy & SciPy

The module explores the optimisation process from a real planning problem to the interpretation of solutions to the underlying optimisation problem. The module will cover the following: development of advanced optimisation models and solution methods; discrete elements; classes of optimisation problems and the properties; deterministic and stochastic modelling; interpretation of the solution; advanced use of commercial software for modelling and solving optimisation problems.

DSI 435 Applied cloud computing for data-intensive sciences

This module provides students with programming models and tools of cloud computing to support data-intensive science applications. The module will cover the following topics: data-intensive sciences and the data centre model; clouds with infrastructure, platform, and software as a service; virtualisation technologies and tools; introduction to FutureGrid; parallel programming using MapReduce vs. Message Passing Interface (MPI); MapReduce and data-parallel applications using Hadoop; iterative MapReduce and data mining algorithms using Twister; MapReduce on multicore/graphics processing unit; NoSQL databases and parallel query processing; high-level language (Hive and Pig) Amazon Elastic Compute Cloud (EC2) and Microsoft Azure and their applications.

DSI 436 Data analytics and visualisation using matplotlib & seaborn

This module helps students develop the understanding that they will need to make informed decisions using data and to communicate the results effectively. Topics include descriptive statistics; correlation and simple regression; probability; point and interval estimation; hypothesis testing; multiple regression; time series analysis and index numbers; human visual perception; visualisation design, interaction techniques, and evaluation methods. Data analytic tools include Microsoft Excel; R Programming; and Python.

DSI 437 Machine Learning II: Using JAVA

This module teaches students how to perform various Data Science and Informatics tasks using Java. The following topics will be covered: ML algorithms for creating ML models; creating deep learning models using the Eclipse Deep Learning 4j Java library; concepts for building ML models such as pattern recognition, classification, and regression; and implementing NLP techniques in building ML and DL models.

DSI 438 Data Science and Informatics Project 2

The students design and develop a project in Data Science and Informatics. The project is a substantial multi-chapter document and a carefully designed and evaluated application, based on well-planned research methods and Information systems analysis, design and development methodologies. A student is required to submit a system with at least five functionalities [specific objectives] and documentation of approximately 10,000 words on a topic relating to data analytics, virtual reality, augmented reality, Natural language processing, gaming, computer vision, machine learning, detection systems or any other related to data science and informatics within a period as may be prescribed by the department. This is an independent project in multidisciplinary fields. Students work under the supervision of an assigned departmental lecturer to address comprehensive company-based or research-orientated problems with documented results. The completed project should present the problem of the research subject investigated or applied project undertaken, its significance to theory or business practice, the research or project background, a well-defined methodology, results or findings, and their implications.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN INFORMATION SECURITY (BS33)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBK's Credit Load	384
ZNQF Level	8

1. PURPOSE OF PROGRAMME

By the end of the programme a graduate will be able to:

- Apply knowledge of Information security to solve real-life problems.
- Demonstrate a sound educational basis to cope with future technological developments.
- Utilize a wide of range of information security tools available to the software engineer
- Exhibit a sound knowledge of the current state of Information security technology across a broad spectrum

2. ENTRY REQUIREMENTS

2.1 Normal Entry

Minimum of five O level passes including Mathematics and English language and at least two A level passes in Mathematics AND Computer Science (Computing) OR any other science-related subjects.

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2.2 Visiting / Harare Weekend School Programme

To qualify for entry into the Bachelor of Commerce Honours Degree in Information Security Degree (Visiting / Harare Weekend School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have the:

either:

A National Diploma/ HND (complete qualification) in Information Technology, or any equivalent tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects

and:

Confirmation of employment in relevant Information Technology departments.

2.3 Special Entry

Candidates who have successfully completed a Professional Technology Diploma or have obtained equivalent qualifications within three years may apply for direct entry into Level 2 of the degree programme, subject to availability of places. Candidates admitted under the above regulation will normally be exempted from level 1 on a module-by-module-basis. No candidate may complete the degree in less than three academic levels. Successful completion of the Work-Related Learning component at level three is compulsory for all candidates.

2.4 Mature Entry

Refer to faculty regulations.

3. PROGRAMME CHARACTERISTICS

Areas of Study: Ethical hacking, Information security, Programming, Network security, Cryptography, Digital forensics, Cyber security, IT Auditing and Assurance.
Specialist Focus: Information security and cyber security.

Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects.

Distinctive Features: The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Careers in Information Security include the following: Information Security Analyst, Software Security Engineer, Penetration Tester, Chief Information Security Officer, Information Security Crime Investigator/Forensics Expert. Further studies can include and not limited to Master's and Doctoral studies in Cloud Computing and Internet of Things or in interdisciplinary programmes related to Cloud Computing and Internet of Things.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, e-learning, tutorials, seminars, group work, computer laboratory classes, industrial visits, Work-Related Learning, research project and individual independent study.

Assessment Methods: Written and oral examinations, tests, computer project, presentations, Work-Related learning reports, mini WRL research project, final year research project report, and continuous assessments.

6. LEARNING OUTCOMES

Generic:

- **Multidisciplinary:** Ability to draw appropriately from multiple academic disciplines to define and solve problems based on the understanding of complex phenomena in Information Security
- **Quantitative and innovative reasoning:** Capability to draw on testing, designing, big data and use analytics for informed decision-making and strive to seek new ways of doing things in the Information Security landscape.
- **Communication skills:** Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences.
- **Analysis and synthesis:** Capacity for analysis and synthesis using logical arguments and proven facts.
- **Ethical commitment:** Professional integrity and awareness of the impact of science and technology on society and the environment.
- **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.

Discipline-specific:

- **Deep knowledge:** Ability to analyse security issues in terms of underlying principles and knowledge.
- **Production skills:** Ability to design and develop information security models and frameworks for real-world problems.
- **Technology development skills:** Ability to develop and implement security models and frameworks.
- **Problem-solving skills:** Ability to solve a wide range of problems in the business environment by identifying their fundamental aspects and using both theoretical and experimental methods.
- **Analytical and computational skills:** Ability to use data and apply security concepts.

- Become familiar with fundamentals of various science and technology subjects and thus gain knowledge with a global perspective in Information Security and synthesize new knowledge as well as investigate problems and formulate security requirements with appropriate solutions.
- Be capable of solving and designing optimal solutions by using various techniques and have the ability to do research through literature work, design, conduct experiments and analyse cyber security problems.
- Engage in lifelong learning and demonstrate professional, ethical, legal and security issues and responsibilities.
- Familiarize with the fundamentals of solving and designing optimal solutions by using various techniques.
- Develop skills to be effective members of a team and be able to work independently with reflective learning.
- Demonstrate effective professional communication skills.
- Become equipped with the knowledge and skill necessary for entry-level placement in cyber security industry as well as IT companies.
- Develop capacity to understand professional and ethical responsibility and will display skills required for continuous and lifelong learning.
- Become employers rather than jobseekers.
- Be expected to have knowledge of contemporary issues and modern practices in cyber security.

7. GENERAL PROVISIONS

Refer to faculty and general regulations.

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for70% of the overall assessment.

9.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment.

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%

9. PROVISION FOR PROGRESSION

Refer to general and faculty regulations for provision for progression:

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations.

12. DEGREE WEIGHTING

Refer to Faculty regulations.

13. AWARD OF DEGREE

Refer to general regulations.

14. PROGRAMME STRUCTURE

Module Distribution:

Level 1 Semester 1

Module Code	**and Title**	**Credits**
IS131	Introduction to Information Security	12
INFO135	Ethics and Professionalism	12
IS133	Data Structures and Algorithms	12
CS131	Communication Skills	12
IS134	Database Design & Security	12
IS135	Web Technologies	12
IS136	Ethical Hacking and Penetration Testing	12

Level 1Semester 2

Module Code	**and Title**	**Credits**
IS137	Introduction to Programming	12
INFO140	Operating Systems	12
IS139	Discrete Mathematics	12
IS140	Cyberspace Ethics and Laws	12
IS141	Research Methods and Statistics	12
IS142	Organizational Behaviour	12

Level 2 Semester 1

Module Code	**and Title**	**Credits**

IS231	Secure Coding	12
GSB211	Gender Studies	12
ENT205	Entrepreneurship	12
IS234	Network Security	12
INFO234	Data Communications and Networks	12
IS233	Systems Administration & Security	12

Level 2 Semester 2

Module Code and Title		**Credits**
IS235	Forensics & Incidence Response	12
IS236	Secure Software Management	12
IS237	Cryptography and Security	12
IS238	Firewall Fundamentals	12
IS240	Group Project	12

Level 3 Semester 1

Module Code and Title		**Credits**
IS340	Work-Related Learning Preliminary Report	40

Level Three Semester Two

IS341	Work-Related Learning Continuous Assessment	40
IS342	Work-Related Learning Report	40

Level 4 Semester 1

Module Code and Title		**Credits**
IS431	Cryptanalysis and Development of Cryptosystems	12
IS432	Web Security	12
IS433	I T Security Planning Strategies and Project Management	12
IS434	HPC Cluster & Cloud Computing Security	12
IS435	Penetration Testing and Vulnerability Assessment	12
IS 436	Industrial Control Systems Security	12

Level Four Semester Two

Module Code and Title		**Credits**
IS437	Digital Forensics	12
IS438	IT Auditing and Assurance	12
IS439	Information Systems Risk Management	12
IS 440	Cybercrime Investigations	12
IS441	Research Project/Dissertation	24

MODULE SYNOPSES

IS131 Introduction to Information Security

This module covers the security systems development life cycle, and critical characteristics of information (CIA). Legal, ethical, and professional issues in Information Security. Risk Management: an overview of risk management (know yourself/the enemy), Risk identification, Risk assessment, Risk control strategies, and Selecting a risk control strategy (feasibility studies). Benchmarking and best practices, Baselining, Residual risk. Information security policy, standards, and practices. EISP, ISSP, SysSP,

INFO 135 Ethics and Professionalism
This module covers ethical and social issues related to the development and use of computer technology. Topics include ethical theory, social, political, intellectual property (IP) and other legal considerations. Scenarios in problem areas: privacy, reliability and risks of complex systems, responsibility of professionals for applications and consequences of their work.

1S133 Data Structures and Algorithms
This module teaches the concepts skills for effective problem-solving skills using algorithmic techniques such as the greedy algorithm, brute force algorithm etc. It focuses on data structures such as arrays, stacks, queues/priority, linked lists, trees, graphs, hash tables and strings to solve programming challenges. Algorithms for searching and sorting, including common algorithm design techniques, are introduced.

IS134 Database Design and Security
This module introduces students to a basic understanding of databases and their management systems. The following topics will be covered Database system architecture, normal forms, query system (SQL) and database security, Information Storage security, data integrity and data authenticity as well as data masking. The assignments are implemented using SQL Server 2000 or Oracle.

IS135 Web Technologies
Students will cover Web design fundamentals and internet technologies. They will have an appreciation of Web communication protocols and methods. They will learn how to use Scripting languages, interface design tools, Application Programming Interface (API),Web Services and design patterns in Web Application Development

IS136 Ethical Hacking and Penetration Testing
This module teaches the latest trends and advances in ethical hacking and penetration testing. Securing information systems against cyber-attacks. Application of concepts learned in previous classes to both defend and compromise e-system. Various advanced tools form an aging and compromising system, safeguarding ancillary systems to prevent collateral damage during testing procedures. Legal and ethical issues associated with penetration testing.

CS131 Communication Skills
Refer to Communication Skills Department.

IS136 Introduction to Programming
This module introduces students to elementary computer algorithms. Topics to be covered include the following introduction to Programming using procedural paradigm: introduction to hardware, software, and application software: solving problems using programming; Program development process. Flow charts, algorithms and pseudocode. Compiling and running programs.

INFO140 Operating Systems
This module explores the design and implementation of computer operating systems. Topics include historical aspects of operating systems development; systems programming; process scheduling; synchronisation of concurrent processes; deadlocks; virtual machines; memory management; virtual memory and paging; I/O and file systems; system security; OS/architecture interaction; and distributed operating systems.

IS137 Discrete Mathematics
The module teaches fundamentals of algebraic, logical and combinatorial concepts. Set, relational algebra, algorithmic closure. Boolean algebra and prepositional logic, Mathematics related to computing. It will cover elements of calculus and discrete mathematics necessary to the field of computing. Integers, well-ordering principle, induction, Fibonacci numbers; Divisibility; prime numbers and distribution of primes.

IS138 Cyberspace Ethics and Laws
This module's focus is on the need for suitable legislation, and appropriate resourcing of law enforcers to address cybercrime. Students learn about the global nature of computer crime and examine national

and international laws, regulations and agreements that govern computer crime, including computer investigations, prosecution and defence. The module encourages analytical thinking and reasoning about computer crime topics and relevant legal issues.

IS139 Research Methods and Statistics
The module introduces the fundamentals of research. The module is intended to prepare students to find research topics formulate a hypothesis, perform the data collection and perform the required statistical analysis on the data using various software packages. This creates the research basis for students produce survey and research papers. Simple statistics for data analysis will be also covered.

IS140 Organisational Behaviour
This module will teach concepts of organizational behaviour. Individual Differences at work: Personality, attitude and intelligence. Motivation: Importance of motivation in work behaviour, approaches to motivation, content theories, and process theories. Job analysis and Design: Approaches, job rotation, job enlargement, job design models. Communication: Types, transaction analysis, Johari windows. Training and Development: Training needs assessment, training techniques and training evaluation.

IS231 Secure Coding
This module provides a detailed explanation of common programming errors in programming languages and describes how these errors can lead to software systems that are vulnerable to exploitation. The module will also cover defensive programming and identify its benefits and challenges. The module will also address the best practices for creating secure code, how to include defensive programming techniques into your software development process.

INFO 234 Data Communications and Networks
The objective of this book is to introduce students of computer engineering, computer sciences and pure and applied sciences to basic concepts, principles, and practice of data communication and network.

This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

IS233 Systems Administration and Security
This module introduces concepts of systems administration on servers such as Unix/Linux and Microsoft amongst other. As well making sure these servers are secured. It covers systems security, component security and organizational security are covered with a focus on system thinking, management, access and control as well as testing and retirement. System architectures are also covered.

IS234 Network Security
This module teaches network security fundamentals. Topics to be covered include Network trace route and packet filtering project. Protocol designs, Security Concepts and Terminology, TCP/IP and OSI Network Security, Access Control Issues (Packet Filters, Firewalls), Communication Security (OSI Layer Security Protocols), Security Tools, Cryptography, System Security - Intruders and Viruses, E-mail and Web Security. Implementation will be done using Appropriate Networking Programming.

ENT205 Entrepreneurship
Refer to the Entrepreneurship Department.

GS211 Gender Studies for Business
Refer to the Gender Studies Department.

IS235 Forensics and Incidence Response
This module explores forensic analysis of data from computer systems and response after an incident. It also covers Incident Response Overview, Incident Response Phases, Digital Forensics Overview,

Digital Forensics Evidence Acquisition, Digital Forensics Evidence Analysis, and Digital Forensics Reporting. Societal security, organizational security and human security are also covered.

IS236 Secure Software Management
The module involves learning principles of designing and implementing secure software, sufficiently to be able to apply those principles. Focus is also on requirements, testing, evaluation, and development processes, and their impact on implementing secure software. Prerequisites for this module are knowledge of software engineering principles, software development processes, design, programming techniques, computer technology. Human security and organizational security are also covered.

IS237 Cryptography and Security
This module introduces students to the nature, purposes and history of cryptography; Classical cryptography: transposition ciphers, substitution ciphers, Affine cipher, the Vigenère cipher. Entropy and equivocation; Perfect secrecy and one-time pads, Unicity distance. Fields: Field extensions, Finite fields, Fields over irreducible polynomials. Modern encryption algorithms will be covered also.

IS238 Firewall Fundamentals
The module highlights firewall theory and architecture, technology and the implementation in routed IPv4based TCP/IPnetworks.IPv6basedprotocolsinrelationto IPv4. Firewall fundamentals: firewall network architecture, the firewall's role in a network, firewall types, firewall performance attributes, and firewall protection. Firewall installation will also be covered.

IS239 Group Project
The students working a group to produce a working Product including a Project Proposal, a Project Plan, a System Specification, User Documentation and perform Project Management Activities to ensure the product is delivered on time.

IS331 and 332 Work-Related Learning
The internship programme exposes students to a real-life work environment and facilitates a structured and integrated learning programme for them. Students apply the knowledge and skills that they have acquired to work practice and new skills acquired in industry and come up with a consolidated report. This helps them to deepen their relevant skills for them to be well-placed to pursue a career in their chosen discipline.

IS302 Academic Supervisor and Employer's Assessment Report
A consolidated report with weighting of 60% from the Lecturer and 40% from the Industry supervisor will be made during the period of work-related learning. The academic supervisor produces a WRL Report for the student on Work-Related learning. The same applies, the employer periodically assesses the student on Work-Related learning and produce a report. Students are posed to an environment of being innovative.

IS431 Cryptanalysis and Development of Cryptosystems
This module introduces students on how to break codes, explicitly describe the real-world constraints and realities of Cryptanalysis, critic, design, and build secure cryptographic systems, explicitly the real-world constraints and realities of cryptography, critic, design, and build secure cryptographic systems, understand current implementations of cryptography algorithms and protocols in real life and teach emerging technologies.

IS432 Web Security and Secure E-Commerce
The module explores the browser security model including same-origin policy and threat models in web security and understand security related issues in Web-based systems and applications. The module will also cover web sessions, secure communication channels such as TLS and importance of secure certificates, authentication including single sign-on such as OAuth and SAML. The module will also examine the common types of vulnerabilities and attacks in web applications.

IS433 IT Security Planning Strategies and Project Management
This module allows students to have knowledge of Advances in Project Management strategies:

project planning, resources, costs, as well as project execution and progress tracking. The IT security project management process; project management methodologies and tools; IT project management standards; change management; Case studies: current issues in IT security project management; Disaster management and business continuity.

IS434 HPC Cluster and Cloud Computing Security
This module examines advances in High Performance Computing (HPC)technology: Virtualization, HPC Cluster, and Cloud Computing and security of data on the cloud. Virtualization technology: building HPC Cluster mimicking supercomputers. Design and build private and public cloud computing solutions using Ubuntu Enterprise Cloud Computing, Eucalyptus Cloud Systems and Open Nebula Cloud Computing solutions.

IS435 Penetration Testing and Vulnerability Assessment
This module covers concepts of penetration testing and vulnerability assessment, Footprinting and social engineering tools and methods, methods of ensuring and countering attacks on data security, network protection systems and security against attacks on the web, ethical hacking, laws and methodologies and emerging technologies in the field of penetration testing. Data, software, systems and human security are key in this module.

IS436 Industrial Control Systems Security
This module explores aspects of component security including component design, component fabrication, component procurement, component testing and reverse engineering. The module also covers understand the working of ICS, learning about SCADA ICS Vulnerabilities and emerging techniques to defend SCADA systems. Data, component, software and connection security are covered.

IS437 Digital Forensics
This module introduces students to mobile phone security and forensics, investigating wireless and web attacks, network-based forensics, anti-forensics methods, tools and techniques emerging technologies in the field of digital forensics. The module will also involve the reconstruction of web browsing history from web artifacts, capture and interpret network traffic, discuss the challenges associated with mobile device forensics, malware identification.

IS438 IT Auditing and Assurance
The module will introduce the fundamentals of IT auditing and assurance services, core reasons why IT auditing is a specialized area of auditing, evolution of IT assurance, and the principles objectives of IT auditing and assurance services. The module will emphasize business management issues regarding the security and control of IT and the achievement of value through managed IT processes.

IS439 Information Systems Risk Management Disaster Recovery Planning and Business Continuity/
This module is designed to teach best practices for conducting vulnerability assessments and countermeasures techniques. Advances in risk assessment tools; Establish cost-benefit analysis for specific safeguards to organization's assets, confidentiality, availability and integrity of data and network resources. Trends and latest advances in risk management plan and incidence response techniques via capstone research.

IS440 Cybercrime Investigations
The module explores the advances in Cybercrimes investigations: unauthorized access, mischief to data, possession of hacking tools, illegal drug trades; money laundry, human trafficking. The module will also cover the Legal aspects, Investigation process, Reporting process, and case studies with a hands-on capstone research project on cybercrime activities.

IS441 Research Project/Dissertation
The Research project involves supervision. This project is intended for the student to demonstrate skills acquired to develop systems for an organization of their choice. Candidates must be able to

demonstrate the highest level of innovation. A level of research is expected to ensure some level of originality and critical thinking in the project design and implementation.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN CLOUD COMPUTING AND INTERNET OF THINGS (BS06)

Duration	4 Years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

Cloud Computing and the Internet of Things place emphasis more on the application of computing in distributed environments as the world is globalized and continues to intersect. The program aims to:

- to merge Cloud Computing and IoT in industry and commerce.
- generate a huge amount of data or big data.
- manage the flow and storage of data in different enterprises.
- manage and analyze this data using different cloud computing models and implementation platforms that help companies to enhance the overall efficiency and working of IoT systems.
- Design, model and implement solutions that allow data transfer and storage through the internet or with a direct link that enables uninterrupted data transfer between devices, applications, and the cloud.
- To prepare students for further studies and lifelong learning in the Computing Profession and in IT-centred service provision organisations.

2 ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for entry into the Bachelor of Commerce Honours Degree in Cloud Computing and Internet of Things programme, a candidate, in addition to satisfying the minimum conditions as prescribed under the General Regulations for English and Mathematics at O' Level, must have obtained a pass in A'Level Mathematics or Computer Studies and any other Commercial or Science Subject

2.2 Visiting / Harare Weekend School Programme

To qualify for entry into the Bachelor of Commerce Honours Degree in Cloud Computing and Internet of Things (Visiting / Harare Weekend School) programme, a candidate, in addition to satisfying the minimum conditions prescribed under the General Regulations must have:

either:

A National Diploma/ HND (complete qualification) in Information Technology/Computer Science, or any equivalent or related tertiary qualification

or:

At least two (2) passes in relevant "A" Level subjects

and:

Confirmation of employment in relevant Information Technology/ Computing departments.

2.3 Special Entry

Candidates who have successfully completed a Professional Information Technology/Computing Diploma or have obtained equivalent qualifications within three years may apply for direct entry into Level 2 of the degree programme, subject to availability of places. Candidates admitted under above regulation will normally be exempted from level 1 on a module-by module-basis. No candidate may complete the degree in less than three academic levels. Successful completion of Work-Related Learning component at level three is compulsory for all candidates.

2.4 Mature Entry

Refer to faculty regulations

3. PROGRAMME CHARACTERISTICS

Areas of study:

Programming Fundamentals; Physical Computing and the Internet of Things; Applied Calculus; Internet Emerging Technologies; Data Structures and Algorithms; Operating Systems and Computer Architecture; Web and mobile computing; Database Design Concepts; Software Engineering;

Principles of Machine Learning; IoT Sensors and Devices; Object Oriented Programming; Wireless Communication Emerging Technologies; Artificial Intelligence and Expert Systems; Discrete Mathematics; Data Communications and Computer Networks; Internet of Things & Augmented Reality Emerging Technologies; Analogue Electronic Devices + Circuitry; Business Considerations for 5G, IoT and AI; Cloud and IoT Technopreneurship; Cloud Computing Infrastructure; Intelligent Signal Processing; IoT Networks and Protocols; Cloud Security; Big Data Technologies; Interaction Design; Smartphone Emerging Technologies; Big Data, Cloud Computing, & CDN Emerging Technologies; AWS IoT: Developing and Deploying and IoTs.

Specialist Focus: Cloud Computing and Internet of Things in business and industry

Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects

Distinctive Features: Programme is driven so much by the requirements of Professional bodies who demand and expect compliance with the Cloud Computing Association (CCA), Institute of Electrical and Electronics Engineers Computer Society (IEEE CS), Association for Computing Machinery (ACM), International Association of Computer Science and Information Technology (IACSIT). Consultations are made time and again to maintain programme relevance to the ever-changing technological world. The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates with the Bachelor of Commerce in Cloud Computing and Internet of Things can pursue careers as IoT/Cloud Software Developer; Cloud/IoT Infrastructure Architect; IoT Systems Administrator; Vulnerability/Cyber Engineer; IoT Embedded Systems Designer; IoT Solutions Engineer; Test Engineer; Data analytics; Cell and UI development. In general, careers include research scientists, consultants, and lecturers. The programme also opens opportunities for graduates to pursue further education within the computing field. Students can enrol for Master's and Doctoral studies in Cloud Computing and Internet of Things or in interdisciplinary programmes related to Cloud Computing and Internet of Things. In addition, with the emphasis today on small businesses, many graduates find great success in starting their own businesses and innovations.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project, practicalsand innovations/ dissertation, and individual independent study.

Assessment Methods: Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, projects and final year research project report.

6. LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Ability to demonstrate knowledge and understanding of Cloud Computing and IoT fundamentals in the business environment.
- Ability to use principles of Cloud Computing and IoT to understand software design, development and analysis, and hardware design and configuration.
- Ability to identify and define different Cloud Computing and IoT concepts and practices for the corporate world.
- Understanding of the Cloud Computing and IoT in terms of the practices, technology applications and security systems.
- Ability to identify Cloud Computing and IoT security aspects.
- Use appropriate research methods to create new IT solutions for business.
- Develop computerised solutions that enhance efficiencies and outputs of the corporate sector.
- Create new business solutions in the corporate world.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. .PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University following the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by the Senate, the maximum mark allowed for such work should be 50%

9. PROVISION FOR PROGRESSION

In addition to general and faculty regulations for provision for progression:
Students should have passed CIT132; CIT131; CIT140, CIT133 as pre-requisites to CIT213; CIT232; CIT234; CIT235 respectively

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12. DEGREE WEIGHTING

Refer to faculty regulations

13. AWARD OF DEGREE

Refer to general regulations

PROGRAMME STRUCTURE

Level 1 Semester 1

Quarter 1

Code	Module Description	Prerequisites	Credits
CIT131	Programming Fundamentals		12
CIT132	Physical Computing and the Internet of Things		12
CS131	Communication Skills		12
GSB 211	Gender Studies for Business		12
CIT134	Fundamentals of Cloud Computing		12
CIT135	Operating Systems and Computer Architecture		12
CIT133	Applied Calculus		12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
CIT136	Data Structures and Algorithms		12
CIT139	Software Engineering		12
CIT140	Principles of Machine Learning		12
CIT141	Cloud Networks and Programming		12
CIT137	Web and mobile computing		12
CIT138	Database Design Concepts		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits

ENT131	Entrepreneurship 1	12
CIT231	IoT Sensors and Devices (CT132)	12
CIT234	Artificial Intelligence and Expert Systems (CT140)	12
CIT232	Object Oriented Programming (CT131)	12
CIT233	Wireless Communication Emerging Technologies	12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
CIT237	Internet of Things & Augmented Reality Emerging Technologies		12
CIT238	Analogue Electronic Devices + Circuitry		12
CIT235	Discrete Mathematics	(CT133)	12
CIT239	Business Considerations for 5G, IoT and AI		12
CIT236	Data Communications and Computer Networks		12
CIT207	Group Project		12

Level 3 Semester 1 Work-Related Learning

Code	Module Description	Prerequisites	Credits
CIT340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Description	Prerequisites	Credits
CIT341	Work-Related Learning Continuous Assessment		40
CIT342	Work-Related Learning Report		40

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
CIT431	Cloud Computing Infrastructure		12
CIT432	Intelligent Signal Processing		12
CIT433	IoT Networks and Protocols		12
CIT434	Cloud Security		12
CIT435	Big Data Technologies		12
CIT436	Interaction Design		12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
CIT437	Smartphone Emerging Technologies		12
CIT438	Big Data, Cloud Computing, & CDN Emerging Technologies		12
CIT439	AWS IoT: Developing and Deploying and IoTs		12
CIT440	Project/Dissertation		24

- **MODULE SYNOPSES**

CIT131 Programming Fundamentals

This module covers the application of programming to problems that require structured thinking. Students learn to think through different kinds of problems and formalise the process of problem-solving strategy by defining the problem, looking for alternatives as solutions, and presenting the solutions. It also covers program testing, debugging, and some program design tools and techniques used to express the solutions needed to solve the impending problem.

CIT132 Physical Computing and the Internet of Things

This module introduces IoT and a structured approach to problem-solving independent of computer type or language. Students are taught to plan and describe program logic using flow charts and pseudo codes. This module introduces students to core techniques used in physical computing. The module complements other parts of the programme, especially regarding programming, networking and the analysis of sensor data.

CIT133 Applied Calculus
This module equips students with the fundamentals of computing mathematics and calculus and forms the foundation for more advanced mathematics topics. This course introduces Calculus Equations, Polar Coordinates and Vector, Vector-Valued Functions, Partial Derivatives, and Multiple Integrals (derivatives and integrals). Appropriate software is used by students to implement some of these ideas in practice (problems in Cloud/IoT using the concept of calculus).

CIT134 Fundamentals of Cloud Computing
This module equips students with basic knowledge of Cloud Technologies in use today. They will be able to achieve a fundamental understanding of what is cloud computing. The module gives students skills to make a clear and strategic plan to move applications and services to the Cloud. Cloud Segments Cloud Deployment Models, and key Cloud Companies are covered including gaining knowledge of Cloud Services and Cloud Security.

CIT135 Operating Systems, Computer Applications and Architecture
This module introduces the fundamental concepts of operating systems, computer applications and computer architecture to students. It provides an overview of the different types of operating systems such as Windows and Linux, as well as their general functions. It discusses in detail essential operating system components and concepts in the areas of process management, memory management, storage management and file management.

CS131 Communication Skills
Refer to the Communication Skills Department.

CIT136 Data Structures and Algorithms
This module teaches the concepts skills for effective problem-solving skills using algorithmic techniques such as the greedy algorithm, brute force algorithm, etc. It focuses on data structures such as arrays, stacks, queues/priority, linked lists, trees, graphs, hash tables, and strings to solve programming challenges. Algorithms for searching and sorting, including common algorithm design techniques, are introduced.

CIT137 Web and Mobile computing
This module familiarises students with the mobile computing world. It exposes students to current trends in mobile services and applications and focuses on new trends and various applications available in the mobile services arena. It also introduces various concepts and ideas of in-trend applications running on mobile devices. Elements of web and mobile computing are also covered.

CIT138 Database Design Concepts
This module covers the components of a database system together with its database management system from a relational model perspective. It also covers data modelling and design of a relational database as well as SQL in data definition and data manipulation. The module also introduces the concept of data and database administration and transaction management. Two database products will be used practically in illustrating concepts.

CIT139 Software Engineering
The module introduces concepts of software development lifecycle Related issues such as software quality and configuration management are introduced to provide students with a good understanding of producing a high-quality software system. Various software testing techniques will also be introduced to identify the test cases for software validation. Project management techniques such as project planning and control are taught.

CIT140 Principles of Machine Learning
This module examines the underlying technologies that are transforming business services. This module provides an introduction to foundational aspects of the field. The module covers the basics of learning theory, which provides a formal framework to assess the conditions under which machine learning algorithms can successfully achieve their tasks. It further covers the fundamentals of machine learning algorithms widely used in practice.

CIT141 Cloud Networks and Programming
The module equips students in supporting and tuning the cloud software and hardware networking and cloud infrastructure connections. Elements of networking protocols like the TCP, IP, DNS, NAT, layer 2 protocols, routing, and layer 3 protocols are explored and applied using programming languages like Perl, Python or Shell. The OSI model will be practically simulated in different environments.

CIT231 IoT Sensors and devices
This module offers learners practical experience to explore and interact with the IoT bridge between the cyber- and physical worlds, to create efficiencies or solve business problems. Students learn about the 'things' that get connected in the Internet of Things to sense and interact with the real-world environment – from something as simple as a smoke detector to a robotic arm in manufacturing.

CIT232 Object-Oriented Programming
The module aims at providing a solid foundation in Object Oriented Programming. Object-oriented programming will be taught by making reference to an OOP Languages widely used at that point in time. Some of these languages include Java, C#, C++ and Python. Students also learn how to program with graphical user interfaces and learn how to design programs capable of managing large amounts of data.

CIT233 Wireless Communication Emerging Technologies
This module covers the services and specifications of the most popular wireless communication technologies used around the world. These include mobile communication, WLAN, and WPAN, which are all included in smart phones. Wi-Fi and Bluetooth specifications are covered. In addition, further details on the state-of-the-art LTE and LTE-Advanced. Evolution of Mobile Wireless Technology from 0G to 5G and emerging technologies are covered.

CIT234 Artificial Intelligence and Expert Systems
This module provides an understanding of the relationship between Expert Systems and the wider field of artificial intelligence. The module enables students to solve problems using explicit knowledge and reasoning. It enables them (1) to express knowledge of a domain in propositional and/or first order predicate calculus, (2) design plans, (3) design solutions to problems where AI techniques can be employed and (4) write programs in Prolog

CIT235 Discrete mathematics
This module introduces and discusses the fundamentals of Discrete Mathematics as applied to Computer Science. Students are exposed to logic and proof techniques, set theory, elementary number theory, functions and relations, graphs, trees, modelling computations and abstract algebra. This module integrates symbolic tools, graphical concepts, and numerical calculations. Techniques of counting: permutations, combinations, recurrences, and algorithms are also covered.

CIT236 Data Communications and Computer Networks
This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing. The focus on error control techniques that include parity check, polynomial code, and Internet checksum is explored.

GSB211 Gender Studies for Business
Refer to the Gender Studies Department.

ENT231 Entrepreneurship
Refer to Entrepreneurship Department.

CIT237 Internet of Things & Augmented Reality Emerging Technologies
This module deals with the new emerging technologies of IoT (Internet of Things) and AR (Augmented Reality). Projects are explored. The Big Data project uses IBM SPSS Statistics, the AR (Augmented Reality) smart phone project analyzes limits of AR applications, the Cloud Computing

project uses AWS (Amazon Web Service) EC2 (Elastic Compute Cloud), and the Smartphone project analyzes mobile communication, Wi-Fi, and Bluetooth wireless IoT networks.

CIT238 Analogue Electronic Devices + Circuitry
This module equips students with the development of knowledge and application of fundamental electronic circuits and physical electronics of some key semiconductor devices, including the design, construction and testing of experimental electronic circuits. Electronic devices: construction and operation of discrete devices, p-n junction diodes, special diodes. Diode circuit applications. Power Supplies: rectification, filtering, regulation, IC regulators.

CIT239 Business considerations for 5G, IoT and AI
The module equips students with the knowledge needed to put 5G to work in their businesses, empowering them to lead their teams into the future. Students will redefine networks to support ***5G*** world and what is required of ***businesses*** to take advantage of new technologies like ***IoT***, Edge and ***AI*** *using different tools. Students* builds the next great innovations based on the capabilities of 5G. Technologies covered include Edge computing, IoT and AI.

CIT207 Group project
The students work in a group to produce a working Product including a Project Proposal, a Project Plan, a System Specification, System Design, System Implementation, User Documentation and perform Project Management Activities to ensure the product is delivered on time. The Implementation includes Developing the Product, Testing and Evaluation and Presentation. The resulting product should be functional and usable.

CIT340 Preliminary Work-Related Learning Report
The internship programme exposes students to a real-life work environment and facilitates a structured and integrated learning programme for them. Students apply the knowledge and skills that they have acquired to work practice and new skills acquired in industry and come up with a consolidated report within the first six months of placement. This helps them to deepen their relevant skills for them to be well-placed to pursue a career in their chosen discipline.

CIT341 Work-Related Learning Continuous Assessment
A consolidated report with weighting of 60% from the Lecturer and 40% from the industry supervisor will be made during the period of work-related learning. The academic supervisor produces a WRL Report for the student on work-related learning. The same applies, the employer periodically assesses the student on work-related learning and produce a report. Students are posed to an environment of being innovative.

CIT342 Work-Related Learning Report
Based on CIT340, students apply the knowledge and skills that they have acquired to work practice and new skills acquired in industry and come up with a consolidated report in the four to six months of placement. This helps them to deepen their relevant skills for them to be well-placed to pursue a career in their chosen discipline.

CIT431 Cloud Computing Infrastructure
This module provides a historical overview of data centres, along with design considerations. Students applies methods to evaluate data centres, cloud management techniques, software deployment considerations and IT management issues for clouds, cloud economics, cloud compliance, existing standards and industry practices including issues to do with cloud dependability.

CIT432 Intelligent Signal Processing
This module builds on other topics in the programme such as Artificial Intelligence and Expert Systems. It provides students with an advanced skill set for writing computer programs that can work with a range of digital signals. it also aims to provide students with a broad experience of digital signal processing techniques and applications. Areas covered also include how audio and video signals can be captured and processed by a computer program.

CIT433 IoT Networks and Protocols

This module focuses on how the IoT works. Students will learn about IoT networks and explore how 'things' connect to it, including whether the connection and processing is local (fog) or remote (cloud). Data networks, connection types, layer models and analysis of IoT protocols and standards are covered. Evaluation of different infrastructure components and network systems, and designing of a basic network for personal IoT ideas is covered.

CIT434 Cloud Security
This module introduces students to cyber security for the cloud. Students learn and apply classic security techniques to current cloud security problems. The module starts with a deceptively simple and secure web service and addresses the problems arising as the services are improved. It also looks at the analysis of recent cloud security vulnerabilities using standard, systematic techniques.

CIT435 Big Data Technologies
The module covers technologies, theories and techniques, along with their application, implementation and integration with legacy systems in big data systems. Students analyse new demands and the application of technologies in the management of data and information resources, and examine big data technologies shaping the way data is stored and utilised including the use of cloud-stored massive datasets, distributed systems in business processes.

CIT436 Interaction Design
The module teaches the students to put the users' needs and requirements first in digital interactive environments and user interfaces. The module provides the students with knowledge and awareness of human factors and conditions in interaction and interface design. The module also focuses on system design methodologies and strategies for the usability optimisation of products and services.

CIT437 Smartphone Emerging Technologies
In this module, the start-of-the-art smartphone and smart watch technology and components in addition to the global market trends and future forecasts are introduced. This helps students in all aspects of new product and app design & development, as well as business planning. The module also introduces students Mobile to Business Rankings & Analysis, mobile communications, Mobile Technology like 4G and 5G, mobile OSs and Smartphone & Mobile Network Project

CIT438 Big Data, Cloud Computing, & CDN Emerging Technologies
This module through Cloud Computing, will enable students to use software as well as processing platforms and computing infrastructure from any location through Internet services. The Big Data part provides students with new intelligence from massive data sets, which can help in situation/condition/status analysis and decision-making. In addition, CDN part provide students with a know-how on how Internet users access information/contents seamlessly.

CIT439AWS IoT: Developing and Deploying and IoTs
This module explores Amazon Web Services' IoT services. Elements explored include developing and deploying IoT devices that are integrated with services such as AWS IoT Core, AWS IoT Device Management, device gateway, AWS IoT Analytics (including how to collect, process and analyze data from IoT devices), the device registry, and shadows, using the AWS IoT APIs and SDKs.

CIT440 Project/Dissertation
The programme finishes with a supervised project which allows the student to design systems of specific interest to him/her. Its weight is equivalent to that of two core modules. The dissertation should be presented to a panel comprising of at least 2 members from the Departmental Board who determine the score. The project must be an innovative solution in the domain of cloud computing and IoT.

REGULATIONS FOR BACHELOR OF COMMERCE HONOURS DEGREE IN INFORMATION SYSTEMS (BS31)

Duration	4 years
Credit Load	528 credits
Minimum Credit Load	480 credits
Maximum Credit Load	540 credits
Maximum MBKs Credit Load	380 credits
ZNQF level	Level 8

1. PURPOSE OF THE PROGRAMME

The aim of this unique programme is to combine an in-depth coverage of computing and its applications with a systematic approach to the design and construction of high-quality software and systems.
By the end of the degree programme the student should be able to: -

- design and implement a programming task
- use a wide of range of tools available to the software engineer
- show knowledge of the current state of technology across a broad spectrum.
- demonstrate a sound educational basis to cope with future technological developments.
- pursue postgraduate degree(s) in the field of Information Systems
- analyse, design, implement and maintain information systems

2. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry

A pass at 'A' level in at least two of the following subjects or their equivalents: Mathematics, Accounts, Economics, Management of Business and Computer Studies.

Special Entry

National Diploma or Higher National Diploma in Information Technology or any related fields.

Mature Entry

Should be at least 23 years old for females and 25 years old for males AND should have at least 2 years relevant industrial experience.

Visiting School e*ither*:

A National Diploma in an Information Technology related field or any equivalent

Tertiary qualification

or : At least two (2) passes in relevant "A" level subjects *and* proof of employment

3. PROGRAMME CHARACTERISTICS

Areas of Study:	Programming; Computer networking of hardware-software; Computer graphics; artificial intelligence; Software development; Information Systems Security; Managing information systems; Ethics and Professionalism
Specialist Focus:	Information systems and technology in business and industry with a special focus on computer applications in business.
Orientation:	Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects
Distinctive Features:	The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Careers in computer programming, research scientists, software developers and consultants, systems analysts, business analysts, forensic auditors. Further Studies: Master's and doctoral studies in Information Systems or interdisciplinary programmes related to Information Systems

5. PROGRAMME DELIVERY

Teaching and Learning Methods:

Lectures (face-to-face and online), tutorials, computer practicals, group projects, work-related learning reports, mini WRL research projects, individual independent study

Assessment Methods: Written and oral examinations, tests, computer projects, presentations, work-related learning reports, mini WRL research projects, final year research project reports, continuous assessments

6. LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Model organizational processes and data
- Utilize different cloud services in their computing environments
- Apply artificial intelligence to optimize processes at workplaces
- Apply techniques for acquiring, converting, transmitting, and storing data and information including those related to data quality
- Design and implement systems to solve a business need
- Evaluate and possibly act on current ethical issues in the IS field by applying professional codes of conduct

- Extract raw data and uncover patterns to extract valuable insights for the business

- Apply critical thinking and problem-solving skills to detect current and future attacks on an organization's computer systems and networks

- Demonstrate an ability to engage in critical thinking by analysing situations and constructing and selecting viable solutions to solve problems

7. GENERAL PROVISIONS

The Bachelor of Commerce Honours degree programmes are normally studied over four levels and shall normally include a period of Work-Related Learning during Level Three.

7.1 The Visiting School programme caters for students who are in fulltime employment and is studied over three levels.

7.1.1 The Work-Related Learning for the Visiting School programme will run concurrently with the second level. The period that students are at their work places during the second level (outside the one week per month during the semester when the students attend lectures at the University) shall constitute the Work-Related Learning.

7.2 The Special Departmental Regulations shall prescribe the total number of modules per programme per semester, and the degree structures.

7.3 In level Four (Level Three for Bachelor of Commerce Visiting School), each student shall be required to write a Dissertation on a chosen subject or topic within his/her field programme of study and this work may be based on tasks and research performed during the Work-Related Learning period.

7.4 Exemption is a process of crediting a student with the module(s) already passed in programmes similar or equivalent to the one on offer. A student may be exempted from studying such modules.

However, the student has to apply for the credit transfer. In his/her application the following should be attached:

(a) A transcript of the module(s) to be credit transferred.

(b) A synopsis of the module(s) from the University/College where the module was attained.

(c) A support letter from the Registrar's Office of the institution where the module(s) were attained.

7.5 The respective Departmental Boards will sit and assess applications for credit transfers and will recommend accordingly to the Senate. No student shall be exempted from more than 50% of the modules offered in the programme.

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for70% of the overall assessment.

9. PROVISION FOR PROGRESSION

9.1 Students shall progress from Semester One to Semester Two at each Level after they have passed more than 50% of modules taken in a semester.

9.2 The Faculty Board of Examiners shall meet at the end of the first semester of the level to ratify the marks obtained by candidates and considered by Departmental Boards.

9.3 The Faculty Board of Examiners shall meet at the end of the second semester of a level to ratify the marks obtained by candidates and make recommendations to the Senate on issues of progression by students to the next level. In the case of repeat-level students, the Faculty Board of Examiners may, however, make a recommendation to withdraw or discontinue the student at this stage.

9.4 For a student to proceed from one level to the other, he/she must have passed at least 75% of the number of modules normally scheduled in a particular level of a programme.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. GRADING AND DEGREE CLASSIFICATION

All Bachelor's degrees shall be classified in the following classes: First Class, Upper Second Class, Lower Second Class and Pass.

The following Grading and Degree Classification System shall be adopted for all modules and programmes

75% and above	1	(First Class)
65%- 74%	2.1	(Upper Second
60%-64%	2.2	(Lower Second
50%-59%	3	(Pass)
40%-49%	Fail	(Supplementable)
39% and below	Fail	

12. DEGREE WEIGHTING

12.1 The results for Levels 1, 2, 3, and 4 shall be weighted as follows:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

12.2 For the Visiting School Programme, the weighting shall be as follows:

Level 1	10%
Level 2	50%%
Level 3	40%

13. PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed.

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits

CS131	Communication Skills	12
BM 131	Principles of Management	12
INFO 132	Foundations of Information Systems	12
INFO 134	Principles of Programming Languages	12
GSB211	Gender Studies for Business	12
INFO 133	Computer Architecture and Organization	12
INFO 135	Ethics and Professionalism	12
BM131	Principles of Management	12

ELECTIVES: (choose 1 module from the following)

ACC 135	Financial Accounting	12
INFO 141	Introduction to Computers and Computer Applications	12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
INFO 138	Software Engineering		12
INFO 140	Operating Systems	INFO 132	12
INFO 137	Database Systems	INFO 132	12
INFO 136	Data Structures and Algorithms	INFO 134	12
INFO 139	Applied Statistics		12

ELECTIVES: (choose 1 module from the following)

MM 131	Principles of Marketing	12
BM 138	E-Commerce	12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
GS201	Gender Studies		12
ENT 205	Entrepreneurship		12
INFO 232	System Analysis and Design		12
INFO 231	Object Oriented Programming		12
INFO 233	Information Systems Research Methods		12
INFO 234	Data Communications and Computer Networks		12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
INFO 237	Group Project	INFO 232	12
INFO 238	Simulation and Modelling		12
INFO 239	Cyber security		12
INFO 235	Web Technologies		12
INFO 236	Project Management	INFO 232	12

ELECTIVES: (choose 2 modules from the following)

INFO 240	Data Analytics	12
INFO 241	Network Management	12
INFO 242	Embedded Systems Design	12

***Student will not proceed for Work-Related Learning if INFO 237 is not passed.**

Code	Module Description	Prerequisites	Credits
Level 3 Semester 1			
INFO 340	Work-Related Preliminary Report		40

Level 3 Semester 2

INFO 341	Work-Related Learning continuous Assessment	40
INFO 342	Work-Related Learning report	40

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
INFO 431	Human Computer Interaction		12
INFO 432	Artificial Intelligence		12
INFO 433	Information Security		12
INFO 434	Management Information Systems		12
INFO 435	Decision Support Systems		12
INFO 442/INFO410	Object Oriented Programming II JAVA		12

ELECTIVES: (choose 2 modules from the following)

INFO 436	Introduction to Data Mining	12
INFO 443	Distributed Systems	12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
INFO 438	Information Systems Audit and Control		12
INFO 440	Research Project		24
INFO 441	Cloud Computing Fundamentals		12
INFO 437	Enterprise Architecture		12
INFO 439	Computer Graphics		12

ELECTIVES: (choose 1 module from the following)

INFO 444	Strategic Management	12

14. MODULE SYNOPSES

INFO 132 Foundations of Information Systems

This module is designed to introduce students to contemporary information systems and demonstrate how these systems are used throughout global organizations. The focus of this module will be on the key components of information systems - people, software, hardware, data, and communication technologies, and how these components can be integrated and managed to create competitive advantage. Through the knowledge of how IS provides a competitive advantage students will gain an understanding of how information is used in organizations and how IT enables improvement in quality, speed, and agility. This module also introduces systems and development concepts, technology acquisition, and various types of application software that have become prevalent or are emerging in modern organizations and society.

INFO 133 Computer Architecture and Organization

This module studies the basic instruction set architecture and organization of a modern computer. Topics include Assembly language, The Von Neumann Machine Instructions, Registers, translating high level arithmetic into Assembly, Memory and Registers Addressing Modes, Logic Gates, Truth Tables, implementing Truth Tables, Latches and Memory Reading, Multicycle implementations, the concept of a Cycle. Finite State Machines, Balancing the work into Single Cycles, ROMs, PLAs, Microcode, RISC/CISC, Pipelines, Principle of Locality, Direct

INFO 134 Principles of Programming Languages
This module examines the concepts and structures governing the design and implementation of programming languages. It presents an introduction to the concepts behind compilers and runtime representations of programming languages; features of programming languages supporting abstraction and polymorphism; and the procedural, functional, object-oriented, and concurrent programming paradigms. programs are required in languages illustrating each of these paradigms.

INFO 135 Ethics and Professionalism
This module covers ethical and social issues related to the development and use of computer technology. Topics include ethical theory, social, political, and intellectual property (IP) and other legal considerations. Scenarios in problem areas: privacy, reliability and risks of complex systems, and responsibility of professionals for applications and consequences of their work.

ACC 135 Financial Accounting
The module is intended to introduce the student to information technology with emphasis on its significance to accounting. It also covers communication in a technological environment as well as the role of accounting systems and their functions. The module will equip learners with knowledge on accounting systems and ensure students are aware of their components and database and examine how accounting systems improve the processes in the accounting environment

INFO 136 Data Structures and Algorithms
This module introduces and discusses the fundamentals of the discrete as applied to computer science, focusing on providing a basic theoretical foundation for further work. Students are exposed to logic and proof techniques, set theory, elementary number theory, functions and relations, graphs, trees, modelling computations and abstract algebra. This module integrates symbolic tools, graphical concepts, and numerical calculations. Techniques of counting: permutations, combinations, recurrences, and algorithms are also covered. Mapped Caches, I/O Polling and Interrupts.

INFO 137 Database Systems
The module focuses on concepts and principles related to database management systems and link these to Relational Database Systems. Topics covered include Database Systems Evolution, Database Systems in the Organization, Principles of Conceptual Design, Database Models, The Relational Data Model, Data Modelling, Database Design Theory, Data Definition and Manipulation Languages, Storage and Indexing Techniques, Query Processing and Optimization, Concurrency Control and Recovery and Database Programming Interfaces.

INFO 138 Software Engineering
The module gives the students practical experience in applying Software Life Cycle Models, standards and technologies. Topics covered include Software Processes and Problems, Requirements, Specifications and Software Design approaches, Modular Designs, CASE tools, Case Studies. Object Oriented Analysis and Design using the Unified Modelling Language (UML) to capture Requirements, Model User Interactions and Business Processes, Verification and Validation, Implementation and Integration Issues, Reuse Concepts and Maintenance.

INFO 139 Applied Statistics

This module covers Introduction to statistics: definition, uses of statistics, Probability, Probability distributions, Measures of central tendency and dispersion, Sampling techniques, Data types, Presentation and Summary techniques, Regression and Correlation, and Simple Statistical Inference.

INFO 140 Operating Systems

This module explores the design and implementation of computer operating systems. Topics include historical aspects of operating systems development, systems programming, process scheduling, synchronization of concurrent processes, deadlocks, virtual machines, memory management, virtual memory and paging, I/O and file systems, system security, OS/architecture interaction, and distributed operating systems.

INFO 141 Introduction to Computers and Computer Applications

The module addresses the following: Hardware: Computer Systems – micros, mini, mainframes, networked computers; computer system components - types and trends; computer peripherals – input, output, and storage. Software: System Software- operating systems (command, GUI, network); application software – end user applications; packages- word processing, spreadsheets, desktop publishing

GSB211 Gender Studies

This module examines gender issues including all aspects and concerns related to women's and men's lives and situations in society, to the way they interrelate, their differences in access to and use of resources, their activities, and how they react to changes, interventions, and policies. Gender equality is one of the objectives of the United Nations Universal Declaration of Human Rights.

INFO 231 Object-Oriented Programming

This module introduces students to the fundamentals of object orientation and equips them with object-oriented programming skills. Students will understand, write, modify, debug and assess the design quality of simple C++ applications at the beginner's level. Students will learn various object-oriented programming topics such as classes, objects, abstraction, encapsulation, inheritance and polymorphism.

INFO 232 System Analysis and Design

This module will cover the principles of Information Systems including analysis and design. Students will learn techniques in data requirements collection and analysis along with methods to modelling data needs. Modelling of data will occur at the conceptual, logical, and physical levels along with an ability to compare the different approaches. Students will understand the importance and constraints imposed by the domain of the information system along with business rules that guide the design. Functional dependencies and domain normalization will also be discussed as part of the requirements analysis. Object-oriented information system modelling and User-centred design techniques will be explored.

INFO 233 Research Methods

This module equips students with research techniques including the definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation. research ethics, report writing.

INFO 234 Data Communications and Computer Networks
This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signalling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

INFO 235 Web Technologies
Students will cover Web design fundamentals and internet technologies. They will have an appreciation of Web communication protocols and methods. They will learn how to use Scripting languages, interface design tools, Application Programming Interface (API), Web Services and design patterns in Web Application Development.

INFO 236 Project Management
Software Project Management is the practice of controlling the cost, time, manpower, hardware and software resources involved in the development activity that commences with a problem statement and ends with the delivery of a complete software product to the customer. It will include planning, monitoring and controlling, Project evaluation, scope management, team building, stakeholder management, risk assessment, scheduling, quality and conflict management.

INFO 237 Group Project
The students work in a group to produce a working product including a Project Proposal, a Project Plan, a System Specification, and User Documentation and perform Project Management Activities to ensure the product is delivered on time.

INFO 238 Simulation and Modelling
This module gives a brief introduction to models, applications areas of simulation and modelling, discrete event simulation, input analysis, output analysis, random number generation, tests for randomness, Queuing models. It also allows students to simulate queuing systems using ARENA. Simulations are also done using different programming languages.

INFO 239 Cyber security
The module covers the introductory concepts in cyber security. These concepts include cyber security theory and basic techniques for optimizing security on personal computers and small networks. It will include differentiating the various types of security from a computer systems perspective, describing the various basic security practices, file backup techniques, basic incident response techniques, and threats to wireless networks and creating a risk analysis for a network in a small business.

INFO 240 Data Analytics
The module covers the foundational skills in data analytics, including preparing and working with data; abstracting and modelling an analytic question and using tools from statistics, learning and mining to address the question. Concepts to be covered include introduction to analytics, basic tools (command line tools, plotting tools, programming tools), statistics probability recap, distributions, significance tests, hypothesis testing), data quality, data cleaning, clustering, classification and graphs.

INFO 241 Network Management
This module teaches students how to plan, configure, maintain, and scale a routed network. It focuses on using routers connected in LANs and WANs typically found at medium-to-large network sites. IP routing concepts and routing protocols are explored in depth. Students spend considerable time implementing laboratory networks that utilize IP

routing. Topics include OSPF, EIGRP, IS-IS, BGPv4, route summarization, route redistribution, route filtering, VLSM, NAT, and advanced Access Control Lists. After completing this course, students will be able to select and implement the appropriate IOS services required to build a scalable, routed network.

INFO 242 Embedded Systems Design
Micro controllers, microprocessors, history, current trends, assembly programming, iteration, looping, selections, interrupts, interfacing, inter device communication,

INFO 340, INFO 341, INFO 342 Work-Related Learning
Work-related learning is a "work-based experience programme" providing a real-life organizational context for students to develop specific or generic skills, valuable to their professional development. Students can apply and enhance their skills, contribute to the Organization, and, at the same time, obtain invaluable guidance from their mentors. Work-Related learning is also an excellent way to learn more about a career, find out what it is like to work in one's potential career, gain valuable experience to build a resume, get to know employers and make a solid network.

INFO 431 Human Computer Interaction
This module introduces the behavioural concepts and technology necessary to manage the design and implementation of "user-friendly" software. Students learn how to implement user-centred design which identifies the goals, tasks, and skills of the eventual end-users of the system and ensures that these are successfully considered in the development cycle.

INFO 432 Artificial Intelligence
The module develops among students a working knowledge of expert systems as well as an understanding of the application of artificial reasoning in industrial processes. Topics covered include An introduction to intelligence, historical and current trends and characterization of knowledge-based systems, Search, logic and deduction, Knowledge representation, production system, expert systems, architecture of expert systems, criteria for selecting expert system shells, end-user interface, developer interface, system interface, inference engine, knowledge base, data interface.

INFO 433 Information Security
This module covers Computer security issues such as Policy goals and mechanisms, Security services, mechanisms, and countermeasures, Cyber-attacks and detection, High assurance systems, Vulnerabilities, threats, and risk, Anonymity systems, Usable security, Cryptography overview, Malware fundamentals, Mitigation and recovery, Personal information, Operational issues, Reporting requirements.

INFO 434 Management Information Systems
The module focuses on developing managerial skills in students, equip them with knowledge of basic management tools and techniques to solve practical IT application related problems and develop IT project management skills. Management objectives, responsibilities, functions, styles, principles and environment. Management tools: element of management economics, financial and management accounting, costing, budgeting and budgetary control. Marketing management, business management, HRM, production and

contract management. Project planning and management. Management of expectations, change, system evaluation and selection, vendors and consultants.

INFO 435 Decision Support Systems
Introduction and overview of Decision Support Systems, need for computerised decision support and supporting technologies, role of models, decision-making concepts, development of decision support systems, group decision support systems, GDSS software and scenarios, business intelligence concepts and tools, data marts, data warehousing and data warehouse design methodologies, data mining, data visualisation and multi-dimensionality.

INFO 436 Introduction to Data Mining
The module is an introductory module on data mining. It introduces the basic concepts, principles, methods, implementation techniques and applications of data mining with a focus on two data mining functions: pattern discovery and cluster analysis. Concepts to be covered include why pattern discovery is important, how to apply pattern discovery in applications, methodologies for cluster analysis, algorithms and applications for cluster analysis.

INFO 437 Enterprise Architecture
This module explores the design, selection, implementation and management of enterprise IT solutions. The focus is on applications and infrastructure and their fit with the business. Students learn frameworks and strategies for infrastructure management, system administration, data/information architecture, content management, distributed computing, middleware, legacy system integration, system consolidation, software selection, total cost of ownership calculation, IT investment analysis, and emerging technologies. These topics are addressed both within and beyond the organization, with attention paid to managing risk and security within audit and compliance standards. Students also hone their ability to communicate technology architecture strategies concisely to a general business audience.

INFO 438 Information Systems Control and Audit
This module introduces the fundamental concepts of the information technology audit and control function. The main focus of this module is on understanding information controls, the types of controls and their impact on the organization, and how to manage and audit them. The concepts and techniques used in information technology audits will be presented. Students will learn the process of creating a control structure with goals and objectives, audit an information technology infrastructure against it, and establish a systematic remediation procedure for any inadequacies. The challenge of dealing with best practices, standards, and regulatory requirements governing information and controls is addressed.

INFO 439 Computer Graphics
The module covers characteristics of graphics I/O devices, 2D/3D transformations including scaling, translation, and rotation. It also explores Graphics Pipeline, Data Structures for Graphics, Geometrical representation, OpenGL programming, vertex processing, lighting and shading, rasterization including line and polygon drawing, ray casting, ray tracing, spline curves and surfaces, colour models, fractals, computer graphics in games and Computer Animation; visualization.

INFO 440 Research Project
The Research project involves supervision. This project is intended for the student to demonstrate skills acquired to develop systems for an organization of their choice.

Candidates must be able to demonstrate the highest level of innovation. A level of research is expected to ensure some level of originality and critical thinking in the project design and implementation.

INFO 441 Cloud Computing Fundamentals
The module provides a primer to help students understand cloud computing and the concepts behind cloud computing. Concepts to be covered include an introduction to the cloud computing terminology, cloud deployment models, and cloud service models. Cloud computing use cases, compute virtualization, network basics for cloud computing, storage basics for cloud computing and cloud computing trends.

INFO 442 Object-Oriented Programming II
Concept and techniques of object-oriented programming. Object classes, data abstraction, encapsulation, methods, inheritance, virtual function, polymorphism, memory management of objects and windows systems; Software reusability concepts, Classes inheritance and class categories. Foundation and collection classes. Design and implementation techniques, OOP and databases.

INFO 443 Distributed Systems
Detailed coverage of the distributed systems, with particular focus on operating systems issues: client-server paradigm, remote-procedure call as OS support for client-server, distributed shared memory, distributed memory coherency; distributed file system; distributed process management, including load sharing and process migration; concurrency control; fault tolerance, recoverability and distributed transactions; naming; industry standards, case studies.

INFO 444 Strategic Management
Concept of Strategy, Analysis of the Environment, Competitive Advantage, Strategy formulation, organisational culture, change, resistance to change.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE CONSUMER AND MARKETING INTELLIGENCE (BS07)

Duration of Degree	4years
Credit Load	528
Minimum credit Load	480
Maximum credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level:	8

1. PURPOSE OF THE PROGRAMME

The degree equips students with consumer and marketing analytical skills relevant to informed business planning and sound decision-making that drive smarter marketing and business strategies. The degree helps students gain the knowledge, understanding and ability to apply a combination of skills related to data science, consumer psychology and marketing communications in order to deliver cutting-edge multichannel marketing solutions.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce Consumer and Marketing Intelligence Honours Degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at 'O' Level, must have obtained at least two passes in any 'A' Level subjects.

2.2 Visiting/ Harare /Bulawayo Weekend School Programme

To qualify for entry into the Bachelor of Commerce Consumer and Marketing Intelligence Honours Degree (Visiting / Harare /Bulawayo Weekend School) programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations, should normally have:

Either: a relevant Diploma from an accredited and recognized institution.

Or

- Passes in "A" Level subjects

And:

- supported by proof of employment

2.3 Special Entry

This applies to students who may wish to transfer from other universities while undertaking a Consumer and Marketing Intelligence related degree programme.

An applicant may be exempted from certain level modules and shall be required to register additional modules to make up for the full component of that level. Successful completion of the Work-Related Learning component is compulsory for all candidates.

2.4 Mature Entry

Refer to Section 3.3 of the General Regulations.

3 PROGRAMME CHARACTERISTICS

Areas of study: Consumer intelligence, marketing intelligence, interactive marketing, consumer psychology, data science.

Specialist Focus: Consumer Intelligence, Marketing Intelligence

Orientation: Analysing consumer data, psychological and sociological factors influencing consumer behaviour, integrated marketing communications, research-based problem-solving, using knowledge and practical skills for research, innovation, industrialization and commercialization.

Distinctive Features: The programme focuses on developing consumer-orientated marketers who can solve contemporary consumer and marketing problems by employing research, data analytical skills and innovative marketing strategies.

4 CAREER OPOPPORTUNITIES AND FURTHER EDUCATION

Graduates of the Bachelor of Commerce (Honours) Consumer and Marketing Intelligence can pursue careers as brand managers, marketing/consumer data analysts, export marketing managers, distribution managers, advertising agents, marketing researchers, new product managers, marketing strategists, retail managers, key account managers, marketing consultant, public relations manager, digital

marketing managers, e-commerce managers, creative designers, creative director, technical sales person, media planners, event managers, business development managers, lecturers among others.

The qualification is articulated horizontally into related qualifications at ZNQF Level 8 and vertically into Marketing or related Masters programmes.

5. PROGRAMME DELIVERY

5.1. Teaching and Learning Methods: Lectures, tutorials, seminars, group work, hands-on, field trips, work-related learning, research projects, and independent study.

5.2. Assessment Methods: Written examinations, tests, field trips, seminar presentations, Work-Related Learning reports, research reports, and continuous assessments.

6. LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Develop and demonstrate competencies in consumer psychology.
- Conduct and apply marketing intelligence in decision-making.
- Develop and communicate marketing messages.
- Demonstrate competence in consumer and marketing data analysis.
- Formulate and implement marketing programs and strategies based on consumer and marketing intelligence.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8: PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each quarter.

8.2 Continuous Assessment shall account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 The final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must:

8.5.1 Be a registered student of the University in accordance with the General Regulations

8.5.2 Have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 Have attended and participated in seminars, tutorials practical sessions, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline for submission of the dissertation or project. Unless the department, Faculty or Senate has granted prior permission for an extension of this deadline, any candidate who fails to meet this submission deadline shall normally fail the dissertation or project.

8.5.5 A submitted dissertation shall be examined and awarded the following

grades namely Pass with First class division, Upper second class, Lower second class, Third class, Fail and a resubmission within three months where the candidate has a score between 40 and 49. Unless otherwise determined by the Academic Board, the maximum mark allowed for resubmission shall not be above 50.

9. PROVISION FOR PROGRESSION

Refer to Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to General Regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to General Regulations

12. DEGREE WEIGHTING

Refer to Faculty Regulations

13. AWARD OF DEGREE

Refer to General Regulations

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Descriptions	Prerequisites	Credits
CS131	Basic Communication Skills		12
BM131	Principles of Management		12
EC131	Economic Principles 1		12
HCS135	Introduction to ICT		12
GSB211	Gender Studies for Business		12
CMI131	Principles of Marketing 1		12
BM131	Principles of Management		12
ACC135	Financial Accounting for Business 1A		12

Level 1 Semester 2

Code	Module Descriptions	Prerequisites	Credits
CMI132	Principles of Marketing 2	(CMI131)	12
CMI134	Consumer Decision Sciences		12
CMI133	Consumer Intelligence		12
CMI135	Artificial Intelligence		12
EC134	Economic Principles 2		12

Level 2 Semester 1

Code	Module Descriptions	Prerequisites	Credits
CMI232	Market Analytics & Decision-making		12
CMI233	Consumer Psychology		12
ENT131	Entrepreneurship 1		12
CMI231	Marketing Communications		12

CMI235	Database Management Systems	12

Level 2 Semester 2

Code	Module Descriptions	Prerequisites	Credits
CMI241	Services Marketing		12
CMI236	Consumer Data Analytics		12
CMI238	Marketing Creativity and Innovation		12
CMI239	Digital Consumer Intelligence		12
CMI240	Brand Management		12
CMI237	Marketing Management		12

Level 3 Semester 1 Work-Related Learning

Code	Module Descriptions	Prerequisites	Credits
CMI340	Work-Related Learning Preliminary Report		40

Level 3 Semester 2 Work-Related Learning

Code	Module Descriptions	Prerequisites	Credits
CMI341	Work-Related Learning Continuous Assessment (employers assessment and academic supervisor report)		40
CMI342	Work-Related Learning Final Report		40

Level 4 Semester 1

Code	Module Descriptions	Prerequisites	Credits
CMI402	New Product and Service Innovation		12
CMI403	International Marketing Intelligence		12
CMI404	Ethics and Consumer Intelligence		12
CMI401	Quality Management		12
CMI405	Green Marketing		12

Level 4 Semester 2

Code	Module Descriptions	Prerequisites	Credits
CMI406	Customer Relationship Management		12
CMI408	Strategic Consumer and Marketing Intelligence		12
CMI407	Research Project		24

15 MODULE SYNOPSES

CMI131 Principles of Marketing 1

This module is designed to introduce basic marketing principles concepts to students. Key topic covered in the module are; introduction to marketing, evolution of marketing concept, the 4Ps of Marketing in detail: (Product, Price, Place and Promotion), the extended marketing mix, segmentation, targeting and positioning. This module is a prerequisite for CMI132 Principles of Marketing 2.

EC131 Economic Principles 1

Refer to the Department of Economic Sciences for the module synopsis.

HCS135 Introduction to ICT
Refer to the Department of Information and Marketing Sciences for module synopsis.

BM131 Principles of Management
Refer to the Department of Management Sciences for the module synopsis.

ACC135 Financial Accounting 1A
Refer to the Department of Accounting for the module synopsis.

CS131 Basic Communication Skills
Refer to the Faculty of Arts for the module synopsis.

CMI132 Principles of Marketing 2
The module is designed to be an extension of Principles of Marketing 1. Topics covered are; Consumer Markets and Buyer Behaviour, Business Markets and Buyer Behaviour, The Micro Marketing environment, The Macro Marketing Environment, Creating competitive Advantage, Sustainable Marketing, International Marketing, Marketing Research and Marketing Planning. To be eligible for CMI132 Principles of Marketing 2, candidates must have passed CMI131 Principles of Marketing 1 in level 1.1

CMI133 Consumer Intelligence
This module aims to provide a theoretical basis and practical techniques that help modern marketers to better understand contemporary consumers. Contemporary consumer behaviour within a collective social media environment, theories underpinning contemporary consumer behaviour as well as application domains for consumer intelligence will be examined. Students shall be oriented on the need to understand contemporary social phenomena and tap into these virtual discussions to generate a critical understanding of the connected consumer.

ACC136 Financial Accounting 1B
Refer to the Department of Accounting for the synopsis.

CMI134 Consumer Decision Sciences

This module focuses on understanding and predicting consumer behaviour by integrating theories from psychology, sociology, anthropology and economics. Topics include the importance of consumer behaviour and research; internal influences such as motivation and involvement, personality, self-image, life-style, perception, attitude formation and change, and communication; external influences such as culture, subculture, social class, reference groups and family, and the diffusion of innovations; and consumer decision-making.

EC131 Economic Principles 2
Refer to the Department of Economics for the module synopsis.

CMI135 Artificial Intelligence
The module aims to introduce the students to the basics of Artificial Intelligence. The main areas of focus will include an Introduction to Artificial Intelligence (AI). Rule-based expert system (RBES), architecture of RBES, development of RBES, advantages and disadvantages

of expert systems. Problem-solving through uninformed search. Problem-solving through informed search. Heuristics. Introduction to knowledge representation. First-order logic and Prolog. Machine learning. Neural Networks and Robotics. Intelligent agents and application of AI and e-expert systems to marketing.

CMI231 Marketing Communications
The module introduces the students to communication tools and media an organisation uses to convey designed messages about the products and the brands it sells to target stakeholders. It will help them to understand how the environment and characteristics of the target audience determine the objectives, models, theories and strategies the firm uses for the communication process in order to achieve favourable attitudinal and behavioural responses and enhance corporate image and identity.

CMI232 Market Analytics and Decision-making
This module introduces students to the concept and current practices in market analytics. The metrics and data sources used for evaluating business and marketing performance and gathering competitor and customer insights – both in offline and online contexts. Students will explore and apply computer-based approaches and techniques to gather customer data and generate market intelligence. Students will be introduced to both quantitative and qualitative methods of data collection, and analysis techniques.

CMI233 Consumer Psychology
The aim of the module is to develop the students' ability to analyse, critically evaluate and apply theories of Psychology to consumption-related behaviours. Once an understanding of the basic psychological processes has been gained, the following areas will be explored: Consumption; Consumerism; Self Identity; Buying Behaviour models; Shopper Typologies; Semiotics; Involvement; Rebel Consumers; Experiential marketing and consumption; Choice; Conspicuous consumption and Signaling strategies.

CMI234 Neuropreneurship
The module helps students to develop entrepreneurial skills that help them understand what is *really* happening inside the minds of consumers. The module uses theories and concepts from neuroscience to gain knowledge of how the human brain processes information to better serve customers. The module further explores how consumers make impulsive, emotionally charged, irrational decisions. The module helps students design marketing communication tools, products and services that help the consumer's subconscious brain make decisions in favour of a firm's offerings.

ENT131 Entrepreneurship 1
The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation-building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan
Evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; and identify the challenges being faced by entrepreneurs.

GSB211Gender Studies for Business

The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module to graduate.

CMI235 Database Management Systems
This module emphasises database concepts, developments, use and management in three main sections: database concepts, practice, and emerging trends. Relational database systems are the main focus, but other types, including object-oriented databases, are studied. Practical design of databases and developing database applications using modern software tools will be emphasized. The module also helps students understand the theories and techniques in developing database applications and be able to demonstrate the ability to build databases using enterprise DBMS products.

CMI236 Consumer Data Analytics
The module aims to equip learners with skills in translating immense volumes of consumer data into meaningful information which improves the quality of marketing decisions. Students will learn how to use different exploratory, descriptive and predictive analytics to identify, target and engage customers. The module also equips learners with the knowledge required to use data to inform customer relationship management practices in order to enhance their effectiveness.

CMI237 Marketing Management
The module is intended to ensure students develop organizational competitive advantage for maximum shareholder value. Topics covered in this module are; The Strategic Marketing Process, Understanding the Company and environment and Establishing Direction (Strategic Marketing Planning, Company Vision, Mission and Objectives, Environmental Scanning, Industry Analysis, Evaluating Company Resources and Capabilities), Designing Market Driven Strategies (Market Growth Strategies, Porter's Generic Strategies).

CMI238 Marketing Creativity and Innovation

This module focuses on equipping students with new ways of thinking necessary for the success of innovative organisations. Real-world cases discussed during the lectures will enable the students to understand in depth the intricacies of innovation, customer experience, solution development and the problems faced by managers in this area. Specifically, the module is designed to help students stimulate creativity and understand the impact of innovation on market share growth.

CMI239 Digital Consumer Intelligence
This module is meant to help students appreciate the impact of technology on marketing. It helps them understand the nature of E-Marketing in business, society and technology. Topics covered encompass the following; Introduction to Digital Marketing and E-Market Places, E-Marketing Business Models and Concepts, Digital Marketing Infrastructure, The Internet Today, Building an E-Commerce Presence, E-Payment Systems and Security and E-Marketing Communications.

CMI240 Brand Management
The module explores the art of creating a unique brand, brand name, term, sign, symbol, design, values, and extensions; building brand identity; launching the brand and maintaining the brand position in the market through managing the tangible and intangible characteristics of the brand. It helps students to identify models and theories that underpin the management of the brand. The students appreciate the importance of brand positioning strategies and tools that evaluate brand strength in terms of brand value, personality, market position and image.

CMI241 Services Marketing
The module aims to help students understand the practical implications of the unique characteristics of services and service provision and to help them cope with the challenges of marketing and managing a service. To achieve this aim, the module builds upon and expands, marketing management concepts and models, demonstrating how they apply to the services sector. This module also focuses on the distinctive characteristics and problems of marketing in service organizations and for any organization developing and marketing services as part of its business portfolio.

Work-Related Learning (CMI340, CMI341, and CMI342)
During their third year, students are expected to be on Work-Related Learning at an organization where they engage in the practical aspects of the programme under supervision. Work-Related Learning helps students to apply concepts they studied during the first two years of their programme, in a real-world context. Students are expected to add value to their academic and professional profiles whilst on attachment while contributing maximum stakeholder value to the organisations to which they are attached.

CMI401 Quality Management
The module introduces the student to quality management and the quest for best business practices in a changing environment. Students are guided on how they may assist an organisation embrace quality management as a competitive advantage in the business world where product and service excellence is shifting from mere possession of proprietary knowledge, access to raw materials etc. to a culture of conformance to specifications, conscientious value chain management and employment and employee training and empowerment.

CMI402 New Product and Service Innovation
This module covers product creation, service delivery, quality, service innovation, concept generation and development, problem-based ideation, analytical attribute approaches, pre-technical valuation and concept evaluation systems. Financial analysis, service/product protocol, development structure, development team management, knowledge work systems and applications in product design and special managerial needs, product testing, strategic launch, strategic action dimensions, implementation of strategy are also covered.

CMI403 International Marketing Intelligence.
The module introduces students to the fast-growing domain of international marketing intelligence. The module aims at equipping students with skills for generating, analysing and interpreting marketing intelligence to enable the crafting of data-driven international marketing strategies. The module also seeks to create awareness among marketing students concerning the relevance of marketing intelligence in enhancing international marketing decisions such as segmentation, positioning, and product development.

CMI404 Ethics and Consumer Intelligence
Computer security pervades every aspect of the modern online experience, now reaching into mobile phones and game consoles as well as conventional computers. This course covers some of the fundamental principles of computer security. After identifying some different aspects of security, several practical challenges will be presented. For each one, the aim is to specify the requirements of a solution, explain an appropriate (mathematics-based) tool, and then discuss pitfalls, attacks, and countermeasures.

CMI405 Green Marketing
This module aims to help students understand the following: What it means for a firm to have a "green product" or good "environmental reputation?" Who are green consumers? How much added value can environmental reputation add to a product? How can firms promote their green products? How important are environmental issues in the global marketplace?

CMI406 Customer Relationship Management.
This module aims at equipping students with knowledge of central areas of CRM. It seeks to enable marketing students to appreciate why an organisation should establish and maintain relationships with its key stakeholders. The module provides students with a solid understanding of the principles involved in managing customer relationships and why customer relationship management tactics and actions are important in attaining sustainable competitive advantage.

CMI407 Research Project
During the final year, students will undertake a Research & Innovation Project of interest to the researcher under supervision of designated mentors vested with knowledge and expertise in that specialisation. This project is expected to equip final-year students with practical skills and knowledge to tackle real-life problems and exploit opportunities as they join the world of work. Students are moulded, not only to be employment seekers but to own the capacity to produce goods and services in various economic sectors.

CMI408 Strategic Consumer and Marketing Intelligence
The module is intended to ensure students develop organizational competitive advantage for maximum shareholder value. The overall objective is to introduce the students to strategic decision-making in an organization and thus develop them to think like marketers. Topics covered in this module include: The strategic marketing process, understanding the company, environment and e designing market-driven strategies (market growth strategies, Porter's generic strategies, cooperative strategies, alliances and blue ocean strategies).

REGULATIONS FOR MASTER OF COMMERCE DEGREE IN STRATEGIC BRAND MANAGEMENT (BS20)

Duration of Degree	2 years
Credit Load	414
Minimum credit Load	360
Maximum credit Load	432
Maximum MBKs credit Load	288
ZNQF Level:	9

1. PURPOSE OF THE PROGRAMME

The programme is aimed at equipping students with applied competencies in the formulation and implementation of brand management strategies. The programme enables students to develop applied knowledge and competencies in brand management, brand valuation, legal aspects of branding, corporate branding, brand storytelling, brand management analytics and services branding. The programme also promotes critical thinking in a competitive branding environment.

2. ENTRY REQUIREMENTS

Applicants should normally have a good honours degree (1, 2.1 or 2.2) in Marketing, Business Management, Tourism and Hospitality Management, Insurance and Risk Management, Economics and Entrepreneurship, OR and any other relevant commercial degree.

3. PROGRAMME CHARACTERISTICS

Areas of Study: Integrated brand communications, Global brand strategies, digital brand management, neuro-marketing, brand architecture, brand valuation, Branding ethics and sustainability.

Specialist Focus: Strategic Brand Management

Orientation: Developing integrated brand communication plans, developing brand management plans, formulating global brand management strategies, developing competencies in brand development, innovation, industrialization and commercialization.

Distinctive Features: The programme focuses on strategic brand management with a thrust on developing brand management strategies, understanding global branding strategies, application of branding ethics and sustainability in brand strategy formulation and implementation

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Graduates for the Masters of Commerce in Strategic Brand Management can pursue careers in brand management, brand marketing communications, brand research, new brand design, business development or lecturing.

Further Studies: The qualification articulates horizontally into related qualifications at ZNQF Level 9 and vertically into a Doctoral marketing or related programmes.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation, and individual independent study.

5.2 Assessment Methods: Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project report.

6. LEARNING OUTCOMES

A graduate of this programme should be able to:

- Gain a broad understanding of strategic brand management in local and global

markets.

- Design, implement and evaluate brand management strategies including digital, global branding, integrated brand communications and service brand management strategies.
- To enrich knowledge on brand architecture, brand valuation and legal aspects of branding.
- Appreciate brand management as a critical aspect of marketing
- Gain a broad understanding of the strategic role of brands as corporate assets.
- Undertake research in brand development, innovation and commercialisation

7. GENERAL PROVISIONS

Refer to Faculty and general regulations.

8. PROGRAMME ASSESSMENT

The student is expected to write formal examination at the end of each quarter.

8.1 To be admitted to the examination the candidate must have:

8.1.1 Writtenallassignmentsandtestsgivenforthemoduleduringthesemester.

8.1.2 Attended compulsory classes.

8.1.3 Participated in prescribed seminars, tutorials and practical classes.

8.1.4 Paid the prescribed fees in accordance with the General Regulations.

8.2 Continuous assessment shall account for 40% of the overall assessment, while the formal examination will account for 60%.

8.3 The aggregate pass mark shall be 50% comprising of final examination mark and continuous assessment mark.

9. PROVISION FOR PROGRESSION

A student shall progress automatically from one level to the other provided he is carrying less than 50% of the modules from the previous semester or level.

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12. DEGREE WEIGHTING

Refer to faculty regulations

13. AWARD OF DEGREE

Refer to general regulations

14. PROGRAMME STRUCTURE

Level1 Semester 1

Code	Module Description	Credits
MSBM731	Integrated Brand Communications	18
MSBM732	Corporate Brand Management	18
MSBM738	Strategic Quality Management	18
MSBM734	Digital Brand Management	18

MSBM735	Legal Aspects of Branding	18

Level 1 Semester 2

Code	**Module Description**	**Credits**
Quarter 1		
MSBM736	Brand Architecture	18
MSBM737	Global Brand Management Strategies	18
MSBM733	Business Research Methods	18
MSBM739	Branding Ethics & Sustainability	18
MSBM740	Neuro-Marketing Aspects of Branding	18
MSBM741	Brand Innovation and Commercialisation	18

Level 2 Semester 1

Code	**Module Description**	**Credits**
MSBM833	Service Branding Strategies	18
MSBM834	Nation Branding	18
MSBM835	Brand Storytelling	18
MSBM832	Brand Management Analytics	18
MSBM842	Advanced Corporate Governance and Business Ethics	18

Level 2 Semester 2

Code	**Module Description**	**Credits**
MSBM836	Strategic Brand Management	18
MSBM837	Dissertation	90

*** Select one elective**

MSBM838	Brand Design Strategies	18
MSBM839	Political Branding Strategies	18
MSBM840	Personal Branding Strategies	18
MSBM841	Agro-branding Strategies	18

15. MODULE SYNOPSES

MSBM731 Integrated Marketing Communications

This module introduces students to communication strategies that can be used to position a brand and sustain brand leadership. It focuses on developing integrated brand communications plans, brand communication strategies, development of effective value propositions through the use of advertising, sales promotions, direct marketing, brand packaging and personal selling. It also covers branding techniques such as perceptual mapping, brand repositioning strategies and evaluation of integrated brand communications strategies.

MSBM732 Corporate Brand Management

The module focuses on understanding product brands and corporate brands, the value of corporate brands, characteristics of corporate branding- complexity, interactions with all stakeholders and responsibility, the role of leadership in corporate brand orientation-corporate identity, positioning, corporate design, vision-culture-image alignment model, corporate brand personality, corporate image dimensions, and building a corporate brand-sustainable brand value proposition.

MSBM733 Business Research methods

Introduction to brand management research methodologies, research philosophies and research approaches, primary and secondary research, business research and marketing research, research briefs and research proposals, literature review, research design and research strategy, sampling, research methods, research ethics, qualitative data analysis, quantitative data analysis, and writing up a branding research report.

MSBM734 Digital Brand Management

This module focuses on the application of branding concepts from an online point of view and assists branding personnel to understand branding strategies, challenges and online branded experiences and how best they can be able to distinguish their offering from competition since the internet technologies tend to commoditise brands, formulating brand strategy for an organisation in simulated environments and tracing online global trends in the branding of products and services.

MSBM 735 Legal Aspects of Branding

The module is designed to give a clear understanding of the role of law and its impact on the branding. It covers the national and international legal branding framework, brand protection; intellectual property law, copy right law, trade mark, patents, brand protection strategy and protecting, legal remedies, intellectual property, and legal aspects of brand registration.

MSBM736 Brand Architecture

This module focuses on developing a brand architecture strategy, defining brand potential, identifying brand extension, branding new products and services, brand hierarchies, designing a brand hierarchy, brand architecture guidelines, brand architecture models-branded house (monolithic brand architecture), house of brands, endorsed architecture, hybrid architecture) and the benefits of a strong brands architecture.

MSBM737 Global Brand Management Strategies

This module focuses on developing effective strategies for global brands. It covers global brand development strategies, positioning global brands, global brand equity, sustaining global brand leadership, global brand market entry strategies-franchising and foreign direct investment, global brand strategies-standardisation, customisation, thinking-global-act local and global customer-based brand equity. The module also analysis the philosophies and competitive profiles of leading global brands.

MSBM738 Strategic Quality Management

The module focuses on quality management as a strategy of creating and sustaining brand leadership. The module is a blend of both the theoretical framework and the practical component. The theoretical framework covers the evolution of the total quality management (TQM), TQM Gurus and their quality philosophies, TQM best practices, quality costing, TQM as a change programme, comparison between TQM and Business Process Re-engineering (BPR) as well as strategies for continuous improvement.

MSBM739 Branding Ethics & Sustainability

This module equips students with knowledge, strategies and tactical plans to implement in order to influence the perception of the brand as being good or bad, ethical or unethical. Ethical brands are conscious of the societal values, laws and all living beings. This module also covers the role of an ethical brand in creating value addition for the organisation, customers, stakeholders and ethical branding realities across different industries are also examined. The role of sustainability metrics and green marketing is also covered.

MSBM740 Neuro-Marketing Aspects of Branding
The module will help students to understand psychology and neuroscience in order to accurately predict consumer behaviour. This module equips students with knowledge of neuro-marketing tools as applied across industry sectors at all stages of the brand journey with a focus on knowledge of the application of psychological and neuroscientific techniques at all stages of the brand journey such as attention and consciousness, sensory neuro-marketing, emotions & feelings, wanting & liking, learning & memory.

MSBM741 Brand Innovation and Commercialisation
This course examines the importance of brand innovation and commercialisation to the achievement of an organization's objectives. Areas covered include new brand development process, strategic opportunity identification, how to generate new brand concepts and ideas, mapping customer perceptions, segmentation, product positioning, forecasting market demand, brand design, market entry strategies, value creation, technology and the new brand development and commercialisation process.

MSBM842 Advanced Corporate Governance and Business Ethics
The student is prepared for a leadership role within organisations through an exploration of the importance of social and moral values in business and management. This exploration takes place at an organisational level, where the critical role of the Board of Directors in shaping and defining corporate conduct is examined. At an individual level, the module examines the dilemmas marketers face in seeking to balance and integrate the often-conflicting demands of personal and professional life.

MSBM832 Brand Management Analytics

The module provides students with comprehensive knowledge of business Intelligence techniques that can be used to leverage brand performance. It focuses on brand data analysis, brand decision support systems, competitor data analytics and intelligence, data mining, web mining, web analytics, social analytics, and social network analysis. It also covers ethical and legal issues of brand management analytics.

MSBM833 Service Branding Strategies

The module aims to provide students with a sound knowledge of service branding, characteristics of services (intangibility, inseparability, heterogeneity and perishability) and how to apply them to service branding; Understanding of service brands as they differ from product brands and how to manage the difference; Critical service branding models important in building successful service brands, Service brand strategies, building Service Brand Equity and customer experience management.

MSBM834 Nation Branding

The focus of the module is to help students understand that Nations are brands and as such their unique endowments when properly packaged create differential and comparative advantages. Theories that support nation branding and strategies to achieve this will be covered. Other areas to be covered include relevance, scope and evolution of nation branding; tourism branding, nation -brand image, identity, positioning and equity; nation branding and the country of origin (COO) effect; ethical imperatives in nation branding and challenges of nation branding.

MSBM835 Brand Storytelling

This module aims to equip students with skills to position product or service brands using attitudinal and emotional states such as cognitive, affective and conative, brand anticipated feelings-utilitarian, hedonism. It also covers brand engagement, brand visibility, brand semiotics, brand culture, and digital brand storytelling and brand e-portfolios.

MSBM836 Strategic Brand Management

This module focuses on developing and managing brand strategies. It covers brand management, customer-based brand equity; brand positioning and brand values; selecting brand elements for brand equity building; designing marketing strategies to build brand equity; application of marketing communications in building brand equity; leveraging secondary brand knowledge to build brand equity; designing a brand equity measure and management framework and measuring sources of brand equity.

MSBM837 Dissertation

Dissertation clinic; students' research topics, generation and submission of research proposal; allocation of research project supervisors, implementing the research project and interacting and role of supervisors, writing up, editing and submitting the final project report—dissertation.

MSBM838 Brand Design Strategies

This module applies the theories of strategic brand management and communication to answer real-life client briefs. The focus is on the development and crafting of an innovative brand building strategy and plan that will address complex business and societal challenges. Students will implement brand research to investigate situational, environmental and commercial scenarios.

MSBM839 Political Branding Strategies

This module focus on positioning a political organisation or individual in order to create a favourable feeling, impression, association or image. The module covers principles of political branding, political brand identity and image, segmentation and political brand development, political branding strategies, political brand equity, interactivity and branding, use of political brand satire and innuendos.

MSBM840 Personal Branding Strategies

This module equips students with skills to market themselves, their work, their skills, and their careers. The aspects covered in the module include: importance of a personal brand, developing a personal brand, maximising personal impact, opinion leadership, ethics and credibility, brand style assessment and business etiquette.

MSBM841 Agro-Branding Strategies

The objective of this module is to develop students understanding of skills required to successfully manage the marketing and branding of agricultural produce. The module equips students with the branding concepts, branding of agricultural commodities and products, structuring of brand mix elements such as brand design, brand image, brand architecture and brand evaluation.

Duration	2 years
Credit Load	396
Minimum Credit Load	360
Maximum Credit Load	430
Maximum MBKs Credit Load	288
ZNQF Level	9

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the Faculty Business Science Regulations and the General Academic Regulations for Postgraduate Diplomas and taught Masters degrees herein referred to as General Regulations.

1.2 The degree will be awarded to candidates who have successfully completed the programme and passed the examinations in accordance with regulations set in below. The particular focus of the Master of Commerce in Strategic Marketing is to enable the manager to operate at a strategic level and manage effectively within a complex and uncertain environment.

2. OBJECTIVES

Students should be able to:

2.1 Gain a broad understanding of the managerial techniques of decision-making.

2.2 Design, implement and evaluate marketing plans including digital, international marketing, integrated marketing communication, and service marketing strategies.

2.3 Enrich the development and managerial prospects through learning within the context of their own experience.

2.4 Appreciate the marketing discipline as ubiquitous in all business institutions.

2.5 Gain a broad understanding of the strategic role of marketing in any organisation.

2.6 Undertake marketing consultancy in specialised areas of marketing.

3. ENTRY REQUIREMENTS

Applicants should have a good honours degree (first or upper second) in Marketing, Business Management, Tourism and Hospitality Management, Insurance and Risk Management, Economics and Entrepreneurship, or any other relevant degree programme.

4. GENERAL PROVISIONS

Registration will take place in accordance with the arrangements prescribed for each semester through the Registrar's Office. A student's registration shall not be confirmed until she/he has fulfilled the requirements for payment of fees. A student registered for the programme is expected to attend classes prescribed per each module. The programme shall not commence with fewer than ten students. The programme shall be studied over a minimum period of four (4) semesters and maximum period of eight (8) semesters. The students shall undergo orientation on aspects of registration and progression. A student requires a minimum of 396 credits to complete the programme.

5. ASSESSMENT

Students are expected to write formal examinations in the two quarters of the semester.

5.1. To be admitted to the examination the candidate must have:

5.1.1 Written all assignments and tests given for the module during the semester.

5.1.2 Attended all the lectures.

5.1.3 Participated in prescribed seminars, tutorials, and practical classes.

5.1.4 Paid the prescribed fees in accordance with the General Regulations.

5.2 Continuous assessment shall contribute 40% of the overall assessment, while the final examination contributes 60%. No continuous assessment shall be carried over to the next semester.

5.3 The aggregate pass mark shall be 50%, comprising of final examination mark and continuous assessment mark.

5.4 To be admitted to the examinations, a candidate must:

5.4.1 be a registered student of the Midlands State University in accordance with the General Regulations;

5.4.2 have satisfactorily attended lectures of the modules on offer in the quarter. Satisfactory attendance to lectures of the modules shall inclde participation and submission of all written assignments, presentations, and tests that constitute the continuous assessment.

5.5 Candidates shall normally submit a dissertation in level 2.2 provided that:

5.5.1 the dissertation was done by the student under the supervision of the allocated supervisor.

5.5.2 the dissertation was done in two semesters (Level 2.1 – level 2.2), unless there are special circumstances as can be directed by the Department, Faculty, Senate orAcademic Board.

5.5.3 where a student fails to submit a dissertation, it is an automatic **FAIL** and the student will be required to redo using a new topic and under the supervision of a different supervisor from the one allocated in the previous semester.

6. PROVISION FOR PROGRESSION

A student shall progress automatically from one level to the other provided s/he is carrying less than 50% of the modules from the previous semester or level.

7. DEGREE CLASSIFICATION

For purposes of degree classification, the parts of the degree programme shall be classified as follows:

Level 1 (Semesters 1 and 2)	50%
Level 2 (Semesters 1 and 2)	50%

8. GRADING SCHEME

The following grading scheme shall be used for all the modules.

80% - 100%	Distinction
70% - 79%	Merit
60% - 69%	Credit
50% - 59%	Pass
Below 50%	Fail

For the Dissertation, a mark between **40 – 49**, is a **RESUBMIT**, in which the student is required to resubmit the dissertation within **THREE (3) MONTHS** from the date of the examination's publication.

9. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MMRK731	Integrated Marketing Communications	18
MMRK735	Consumer Behaviour	18
MMRK732	International Marketing Strategy	18
MMRK734	Accounting for Managers	18

Level 1 Semester 2

Code	Module Description	Credit
MMRK738	Strategic Quality Management	18
MMRK736	Strategic Marketing	18
MMRK740	Entrepreneurial Marketing Strategy	18
MMRK737	Global Economy	18
MMRK739	Advanced Corporate Governance & Ethics	18
MMRK733	Business Research Methods	18

Level 2 Semester 1

Code	Module Description	Credits
MMRK831	Digital Marketing	18
MMRK833	Service Marketing & Consultancy	18
MMRK835	Corporate Strategy	18
MMRK834	Relationship Marketing Strategy	18
MMRK832	Advanced Data Analysis Techniques	18

Level 2 Semester 2

Code	Module Description	Credits
MMRK836	Strategic Brand Management	18
MMRK837	Social marketing	18
MMRK838	Dissertation/Research project	90

10. MODULE SYNOPSES

MMRK731 Integrated Marketing Communications

Integrated marketing communications and corporate strategy; marketing communications environment, marketing communications theories; segmentation, targeting, and positioning in promotional strategy; consumer and organisational buyer behaviour; determination of marketing communications objectives; promotional budget appropriation methods; the marketing communications industry agencies, roles, structures, and remuneration methods; the media mix and characteristics of media, press, broadcast, outdoor, cinema, the Internet and social media, mobile phone, supplemental media; the marketing communications mix sales promotion, public relations, sponsorship, personal selling, direct marketing, exhibitions, advertising, packaging, merchandising, and word of mouth; globalisation and international marketing communications; legal and ethical issues in marketing communications strategy; evaluation of integrated marketing communications strategy.

MMRK732 International Marketing Strategy

Strategic importance, scope, and challenge of international marketing; international economic environment; political-legal issues in international marketing; socio-cultural issues in international marketing; technological environment and internationalisation of businesses; international consumer and buyer behaviour; international marketing information and research; international market entry strategies; product and service decisions in international marketing; international marketing communications; international distribution strategy; pricing in an international marketing context; organisational structure in international marketing; and measuring the success of international marketing strategy.

MMRK733 Business Research Methods

Introduction to business research; research philosophies and research approaches; primary and secondary research; business research and marketing research; research briefs and research proposals; literature review; research design and research strategy; sampling; research methods; research ethics; qualitative data analysis, quantitative data analysis; writing up a research report.

MMRK734 Accounting for Managers

Roles of management accounting in corporate strategic decision-making; management accounting and decision-making (cost/volume/profit analysis~assumptions and limitations, application to profit planning and price/outcome decisions); budgeting and budgetary controls (fixed and flexible budgets, responsibility accounting, standard costs and variance analysis ratios for control); financial implications of marketing management decisions on corporate resources and performance including cash flow, profitability and risk; introduction to profit and loss and balance sheet analysis (financial ratios, financial interpretation; integrating results of financial analysis into marketing analysis and planning.

MMRK735 Consumer Behaviour

Consumer buyer behaviour and marketing strategy; consumer and customer research; market segmentation, targeting and positioning; consumer motivation; personality; perception; learning; attitude; social class; culture and subculture; family; international consumer behaviour; consumer influence and diffusion of innovations; and consumer decision-making.

MMRK736 Strategic Marketing

Role of marketing in society, marketing management philosophies and applications of marketing; marketing environment and marketing planning; competitor analysis; consumer and organisational behaviour; segmentation, targeting and positioning; international marketing; product management; new product development; service marketing; brand management; pricing decisions; promotion decisions (advertising, public relations, sales promotion, direct marketing, sponsorship, etc.); place decisions; and measuring marketing performance.

MMRK737 Global Economy

Globalisation trends and philosophy; motivations for globalisation; global strategy framework; basic competitive strategy profiles; pitfalls of global marketing; globalisation from a macro perspective; growth of Asian competitors.

MMRK738 Strategic Quality Management

The module defines the key concepts in quality management and further identifies the quality phases. It also explores how excellence in organisation and delivery of unparalleled service can be achieved through people, re-engineering and benchmarking. Costs of non-conformance to quality, costs of quality and quality perceptions will be studied.

MMRK739 Advanced Corporate Governance and Ethics

The student is prepared for a leadership role within organisations through an exploration of the importance of social and moral values in business and management. This exploration takes place at organisational level, where the critical role of Board of Directors in shaping and defining of corporate conduct is examined. At individual level, the module examines the dilemmas marketers face in seeking to balance and integrate the often conflicting demands of personal and professional life.

MMRK740 Entrepreneurial Marketing Strategy

The module seeks to raise students' awareness of the theoretical and philosophical thinking in the development and interrelationship of entrepreneurship and entrepreneurial marketing strategies. It adopts a deliberate practical approach that enables student practitioners at masters' level to fuse the philosophical concepts, tools and procedures to apply to their organizations so as to remain competitive and going forward. The module uses case analysis from a global perspective and Zimbabwean experiences to redefine and rejuvenate their organisations through key learning obtained.

MMRK831 Digital Marketing

The module covers the background, current state and future of e-marketing or digital marketing; e-marketing strategic planning and the e-marketing plan; international markets, ethical and legal issues in e-marketing; the e-marketing environment; marketing knowledge and marketing research in e-marketing; consumer behaviour and e-marketing; e-marketing and segmentation, targeting, and positioning strategies; managing e-marketing mix decisions—product, pricing, e-marketing communications, place and the Internet; and relationship marketing in the new economy.

MMRK832 Advanced Data Analysis Techniques

The module covers Descriptive Statistics, Probability, Random variables, The Normal Distribution, Sampling and Sampling Distribution, Confidence Intervals, Hypothesis Testing, Analysis of Variance, Simple Linear, Regression and Correlation, Multiple Regression, Exploratory factor Analysis, Non Parametric methods and Chi-Square Test, Bayesian Statistics and Decision Analysis, Sampling Methods, Univariate and Multivariate Techniques using SPSS, Data Mining for Business Intelligence, Data Visualisation, Dimension Reduction and Thematic content analysis. Enhance competencies in softwares / packages used in research such as ATLAS, SPSS, SMARTPLS and AMOS.

MMRK833 Services Marketing & Consultancy

Understanding service products, customers and markets (fundamentals of services marketing, customer behaviour in service contexts and segmentation, targeting and positioning services in competitive markets); Critical elements in service marketing (designing the service product, designing

marketing communications strategy for services, managing pricing policy and revenues in service contexts, managing services distribution policy); Managing the service value chain (designing and 368 managing service processes, balancing service demand and operations capacity, auditing and reengineering the service environment, managing people for service excellence); Implementing service marketing strategies (relationship marketing in the service context, customer feedback and service recovery, managing service quality and employee productivity, organising for service leadership); Consultancy (theory of consultancy, marketing consulting and the research process; strategic marketing models and marketing consultancy; client brief and/or Terms of Reference, interpreting the client consultancy brief, generating and agreeing the consultancy proposal; implementing the consultancy proposal: research and/or strategic marketing decision-making; reporting findings or recommendations of a consultancy assignment, structure and content of consultancy reports, presenting a consultancy report to a client(s), and agreeing the way forward, immediate versus delayed exit.

MMRK834 Relationship Marketing Strategy
The module identifies the need to market products and services to both internal and external customers. It describes and examines ways of developing and maintaining customer relationships.

MMRK835 Corporate Strategy
The module deals with the process of formulating and implementing strategy in modern business enterprises. Case discussions and class debates challenge students to develop interpersonal skills that they require to communicate effectively in the business world, and to analyse markets and select strategies that allow firms to achieve superior long-term performance.

MMRK836 Strategic Brand Management
Branding, brand management and marketing management; customer-based brand equity; brand positioning and brand values; selecting brand elements for brand equity building; designing marketing strategies to build brand equity; application of marketing communications in building brand equity; leveraging secondary brand knowledge to build brand equity; designing a brand equity measure and management framework; measuring sources of brand equity capturing customer mindset; measuring outputs of brand equity, capturing market performance; developing and implementing brand strategies; introducing and naming new products and brand extensions; managing brands over their brand life cycles; and managing brands across international boundaries.

MMRK837 Social Marketing
Introduction to social marketing; social marketing planning process, cases in social marketing campaigns; analysing the social marketing environment; application of research theory in social marketing; analysing competition, barriers and needs within social marketing settings; audience segmentation and targeting; defining social marketing goals and/or objectives; identifying barriers, benefits, competition and significant others; designing social marketing strategies; developing the desired positioning strategy; product strategy; pricing strategy; place strategy; promotion strategy messages, messengers and creative strategies; promotion strategy selecting communication channels; managing social marketing campaigns; developing a monitoring and evaluation plan for social marketing campaigns; budgeting and fund raising for social marketing programmes; designing implementation plans and sustaining behaviours; and ethical issues in social marketing

MMRK838 Dissertation/Research Project
A student is required to submit a dissertation of approximately 20 000 words on a topic relating to marketing within a period as may be prescribed by the department. This is an independent project in content areas related to marketing management areas. Students work under the supervision of an assigned departmental lecturer to address comprehensive industry/sector-based or research-orientated problems with documented results. The completed project should clearly present the problem of the

research subject investigated or applied project undertaken, its significance to theory or business practice, the research or project background, a well-defined method, results or findings, and their implications.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN INFORMATION SYSTEMS MANAGEMENT (BS30)

Duration	1,5 years
Credit Load	306
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBKs Credit Load	206
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

The goal of the programme is to educate students to become Information Systems Engineers and Managers who have the ability to conceive IT solutions in the widest context. The programme aims to develop the sound conceptual, technical, analytical and communication skills that are required to succeed in the I.T. profession. To this end, the programme employs a rigorous case analysis approach, which requires students to exercise their analytical abilities and develop effective verbal and written communication skills

2. ENTRY REQUIREMENTS

To qualify for entry into this programme, prospective students must hold a good pass in an honours degree in Information Systems, Computer Science or relevant qualification with a strong programming background.

3. PROGRAMME CHARACTERISTICS

Areas of study: Programming; Artificial Intelligence; Database Systems; Management; Emerging Technologies; Security; Networking

Specialist Focus: The goal of the programme is to help meet the increasing demand for knowledgeable personnel who possess a balanced combination of technical and managerial skills.

Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects

Distinctive Features: The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Programmer; Project Manager; Chief Information Officer; Consultant; Database Designer; Systems Designer; Chief Technology Officer

Further studies

Doctoral studies in Information Systems or in interdisciplinary programmes related to Information Systems

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods**: Lectures, tutorials, group work, dissertation,

individual independent study

5.2 **Assessment Methods:** Written and oral examinations, tests, final year dissertation, presentations, term papers, continuous assessments.

6. LEARNING OUTCOMES

By the end of the program, students will be able to:

- use computing and mathematics skills in solving real-time problems.
- develop models for development using the computing and mathematics application
- design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs of commerce and administration
- create companies and businesses in data analytics to reduce production costs, crime and detection of risks.
- innovate in the field of business computing, connectivity and data security
- Design affordable data analytics packages and business support for entrepreneurships.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

Coursework marked out of 40 (Test 15%, Assignments 5%, Practicals 15%, Term paper 10%) and

Written Examinations marked out of 60

9. PROVISION FOR PROGRESSION

9.1. Students shall progress from Semester One to Semester Two at each Level after they have passed more than 50% of modules taken in a semester.

9.2. The Faculty Board of Examiners shall meet at the end of the first semester of the level to ratify the marks obtained by candidates and considered by Departmental Boards.

9.3. The Faculty Board of Examiners shall meet at the end of the second semester of a level to ratify the marks obtained by candidates and make recommendations to Senate on issues of progression by students to the next level. In the case of repeat level students, the Faculty Board of Examiners may, however, make a recommendation to withdraw or discontinue the student at this stage.

9.4. For a student to proceed from one level to the other, he/she must have passed at least 75% of the number of modules normally scheduled in a particular level of a programme.

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12. DEGREE WEIGHTING

Refer to faculty regulations

13. AWARD OF DEGREE

Refer to general regulations

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits
MIM 731	Operating Systems and Microcomputer Technologies		18
MIM 732	Advanced Databases		18
MIM 733	Artificial Intelligence & Expert Systems		18
MIM 734	Object Oriented Programming- Java		18
MIM 735	E-Commerce Strategies		18

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
MIM 736	Software Engineering		18
MIM 737	Information Security		18
MIM 738	E-Corporate Governance and Ethics		18
MIM 740	Change Management		18
MIM 739	Applied Research Methods		18

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
MIM 833	Advanced Project Management		18
MIM 831	Advanced Data Communication and Networking		18
MIM 832	Dissertation/Research Project		90

15. SYNOPSES

MIM 731 Operating Systems and Microcomputer Technologies

Review of fundamental Operating systems concepts. The history of design of operating systems. Detailed study of the internal functioning of operating systems. Optimization of modern operating systems performance. Case studies of Linux and Windows. Introduction to microcomputer system software and a comparison of microcomputer systems vs. embedded systems.

MIM 732 Advanced Databases

The module builds upon the student's general understanding of database systems acquired in the prerequisite undergraduate module(s). Concepts such as database architectures, parallelism, distributed databases, speciality databases and applications, data warehousing, data mining and its algorithms, XML data and web databases will be discussed.

MIM 733 Artificial Intelligence and Expert Systems

This module seeks to consolidate the basics of artificial intelligence and expert systems and move to the application of these aspects in practical industrial situations. It also seeks to show how knowledge and reasoning are demonstrated through the use of intelligent agents. Knowledge representation techniques such as rules, frames, cases and semantic nets are also explored.

MIM 734 Object Oriented Programming - Java

This module offers full coverage of the theory and practice of object-oriented programming in a project-based environment. Detailed concepts such as classes, objects, inheritance, and encapsulation are covered. Java language concepts such as inner classes, exceptions, coupling, and cohesion will be described. The use of threads in Java will be looked at. Students should be able to develop real-life applications using Object Oriented Programming aspects.

MIM 735 E-Commerce Strategies

The module seeks to develop an understanding and appreciation of the significance of e-commerce concepts and issues in the contemporary business world. The student will be able to examine the business implications of e-commerce in the following areas Corporate Strategy, Marketing Strategy,

Customer Experience, Customer Relationship Management, E-Services, Marketing channels, distribution, Online Product Promotion and Payment Systems among others.

MIM 736 Software Engineering

The module aims to develop a broad understanding of the discipline of software engineering (gained in the earlier Software Engineering course) by considering the wider systems engineering context in which software plays a role. It aims to examine the concepts and techniques associated with a number of advanced and industrially relevant topics, relating to both the product and processes of software engineering. It seeks to complement this with an account of the associated practical and professional issues in software engineering. The module will also provide an on-going project clinic to directly support the group project work.

MIM 737 Information Security

Information security pervades every aspect of the modern online experience, now reaching into mobile phones and games consoles as well as conventional computers. This course covers some of the fundamental principles of computer security. After identifying some different aspects of security, a number of practical challenges will be presented. For each one, the aim is to specify the requirements of a solution, explain an appropriate (mathematics-based) tool, and then discuss pitfalls, attacks, and countermeasures. Topics covered include will one-way functions, symmetric and asymmetric block ciphers, keyed hashes, digital signatures, and simple key exchange protocols. Principles of digital forensics: Digitization (storage, interchange, digital objects, composites, and packages) . The module has been aligned to the ACM/IEE curriculum. The topics that dealt with theoretical aspects of digital computer signal processing have been reviewed to reflect a Solid approach rather than abstract approach

MIM 738 E-Corporate Governance and Ethics

The module looks at corporate governance theories, corporate governance codes, Board of Directors, Corporate governance in Zimbabwe in private Companies and public sector companies. The relationship between IT governance and corporate governance, IT governance focus areas, IT governance frameworks, IT Governance Risk Management, Information Security Governance, Problems in implementing IT Governance in Zimbabwe.

MIM 739 Applied Research Methods

Research, research types, Research planning and design, Project Proposal, Data collection techniques, Literature review, Research techniques, Methodology and Methods, Sampling techniques, Validity and reliability, Research report writing, Ethical issues in Information Systems Research.

MIM 740 Change Management

Understanding the Changing Environment, Drivers of Change in the World around Us, Living in the Comfort Zone and Resistance to Change, Typical Change Initiatives, Understanding the Need for Change, Building Advocacy - Diminishing Complacency, Tactics for Success.

MIM 831 Advanced Data Communication and Networking

Presentations and detailed analysis of computer/data networking technologies. Topics include ISOOSI layers 2 and above networking technologies, such as asynchronous transfer mode (ATM), frame relay, Ethernet networks, multi-protocol label switching (MPLS), and Internet protocol technologies, and their applications. Network architectures, protocol stacks, routing algorithms, quality of service (QoS), flow control and traffic management techniques, router/switch design, and data network applications/services will be studied.

MIM 832 Dissertation/Research Project

A completed Dissertation shall be submitted to the department two months before exams, A dissertation is carried out by the students in an area of their choice related to the discipline of Information Systems. The Dissertation applies the concepts and tools taught in the applied research module.

MIM 833 Advanced Project Management

The module seeks to equip students with the technical knowledge in areas of project management and the ability to work in a pro-active manner to minimise threats and maximise opportunities. Concepts to be covered include strategic project management, managing project integration, stakeholders and initiation, project quality management, project procurement management, project risk management and project HR management.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN CLOUD COMPUTING AND INTERNET OF THINGS (MCI)

Duration	1½ years
Credit Load	306
Minimum Credit Load	270
Maximum Credit Load	342
Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

Cloud Computing and Internet of Things places emphasis more on the application of computing in distributed environments as the world is globalised and continue to intersect. The program aims to:

- Develop cloud IoT managers and engineers
- Design and develop IoT applications and services adapted to industrial needs
- Manage and analyse data produced by Cloud IoT systems

2. ENTRY REQUIREMENTS

An Honours degree with a degree class of at least 2.1 in Computing (Computer Engineering, Informatics, Information Systems, Computer Science, Software Engineering) or Electronic Engineering or related supported with a strong programming background.

3. PROGRAMME CHARACTERISTICS

Areas of study:

Programming; Cloud Computing; IoT; Emerging Technologies in Cloud IOT; Security and Networking in Cloud IoT

Specialist Focus: The goal of the programme is to help meet the increasing demand for knowledgeable personnel who possess a balanced combination of technical and managerial skills in the field Cloud IOT.

Orientation: Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects

Distinctive Features: The programme is driven so much by the requirements of Professional bodies who demand and expect compliance with the Cloud Computing Association (CCA), Institute of Electrical and Electronics Engineers Computer Society (IEEE CS), Association for Computing Machinery (ACM), International Association of Computer Science and Information Technology (IACSIT). Consultations are made time and again to maintain programme relevance to the ever-changing technological world. The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Careers in Cloud IOT include Managers in Data Science Engineering, Data Analytics, Big Data Consultancy, Data Warehousing, Database Management and Business Intelligence, Scholars and Academia. Doctoral studies in Cloud IOT or in interdisciplinary programmes related to Cloud Computing and Internet of Things and Professional Certifications in related field.

5. PROGRAMME DELIVERY

5.1. **Teaching and Learning Methods**: Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research project, practicals and innovations/ dissertation, and individual independent study.

5.2. **Assessment Methods:** Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, projects and final year research project report.

6. LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Describe the IoT and Cloud architectures
- Deploy Cloud Services using different cloud technologies.
- Implement security features to protect data stored in the cloud
- Research, publish, and develop several Cloud IOT based projects
- Gain Knowledge and skill in data analytics which is essential for getting value from IoT

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 40% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for60% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%

9. PROVISION FOR PROGRESSION

Refer to general and faculty regulations for provision for progression

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12. DEGREE WEIGHTING

Refer to faculty regulations

13. AWARD OF DEGREE

Refer to general regulations

14 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits
MCI131	Fundamentals of Cloud IoT		18
MCI132	Programming for Data Science and IOT		18
MCI133	Cloud IOT Systems Management		18
MCI134	Database Technologies for Cloud IoT		18
MCI135	Cloud IOT infrastructure and operating systems		18

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
MCI136	Cloud services and Virtualization		18
MCI137	Research Methods		18
MCI138	Cyber Security in Cloud IOT		18
MCI139	Cloud networks and Internetworking		18
MCI140	Human-Computer interaction		18

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
MC1232	Optimization techniques in Cloud IOT		18
MCI233	Containerization in Cloud IOT		18
MCI231	Dissertation/Research Project		90
MCI234	Emerging Technologies for Cyber-Physical Systems		18

- **MODULE SYNOPSES**

MCI 131 Fundamentals of Cloud IoT

The module provides an overview of the Internet of Things (IoT) and Cloud Computing concepts, infrastructures and capabilities with emphasis on the architecture and design of IoT systems, the different technologies (wireless/mobile/sensor) governing system implementation, the migration of the data to the Cloud for processing and the commercial and business implications of technical advances in this area

MCI 132 Programming for Data Science and IOT

The module covers IOT applications. An IoT system may involve several programs in several different languages. This module covers the following IOT applications SOA Distributed Systems, Artificial Intelligence, Web Sockets and Big Data, Web Stacks and Service Platforms. The applications are supported by several languages that can be used namely C++/C, Swift, Raspbery PI, Block Based languages, Node-RED, LUA, Python, JavaScript, Java and Kotlin.

MCI 133 Cloud IOT Systems Management
An appraisal of reasons behind need for Cloud IOT systems management. Considerations of fundamentals of Cloud IOT management and requirements for IOT device management. An integrated approach of systems management in both physical and IT systems audits. Trust-based Cloud IOT service management and Cloud IOT deployment management. The wider IOT ecosystem and associated management challenges.

MCI 134 Database Technologies for Cloud IOT
The module is driven towards the latest IOT database technologies. The functionalities and implementation of IOT based databases and IOT database architectures. Enabling technologies for embeddable databases, comparison of Time series and Common database technologies for IOT. Requirements and considerations taken to adopt an IOT database. IOT database modularity and other compliance issues and opportunities and challenges for IOT databases.

MCI 135 Cloud IOT infrastructure and operating systems
The module recognizes the evolutionary step of Cloud based IOT. Cloud architecture layers which augment IOT through remote computing and power infrastructure. Cloud IoT infrastructure and OS service characteristics such as increased scalability, automatic deprovisioning, secure integration, Pay-as-you-go, performance monitoring and measuring. Based on IaaS, PaaS and/or SaaS, while maintaining cloud infrastructure convergence.

MCI 136 Cloud services and Virtualization
This module provides a hands-on comprehensive study of Cloud concepts and capabilities across the various Cloud service models including IaaS, PaaS, SaaS and BPaaS, how CPU, memory, and I/O resources are virtualized and present real use cases such as Amazon EC2. Virtualisation will look at evolution of infrastructure migration approaches (virtualization-VMWare/Xen/KVM virtualization, adaptive virtualization, Cloud Computing and on-demand resource provisioning).

MCI137 Research Methods
The module covers research, research types, Research planning and design, Project Proposal, Data collection techniques, Literature review, Research techniques, Methodology and Methods, Sampling techniques, Validity and reliability, Research report writing, Ethical issues in Information Systems Research.

MCI 138 Cyber Security in Cloud IOT
The module covers Cloud security concepts, security and protection architectures for cloud and IoT, threat models and attack methodologies for cloud and IoT, security applications and management of cloud and IoT, challenges related to security and privacy for cloud and IoT, cloud security approaches: encryption, tokenization/obfuscation, cloud security alliance standards, cloud security models and related patterns, data analysis of IoT and cloud for security

MIC 139 Cloud networks and internetworking

The module covers enabling communication technologies for IoT and cloud, IoT and Cloud Networks, Web services, REST and authorization protocols, Wireless Communication for LANS, PANS and IOT, Network Security and Mechanisms, Mobile & Wireless Networks, Wireless, Sensor & Actuator Networks, Connecting things: lower layers, IPv6, transport protocols, Web of things: REST, MQTT, and CoAP

MIC 140 Human Computer interaction
The module covers Principles (Feedback Cycles and Direct Manipulation, Design Principles and Heuristics, Mental Models and Distributed Cognition), Methods (Need finding and Requirements Gathering, Low-Fidelity to High-Fidelity Prototyping, Predictive, Empirical, and Qualitative Evaluation) and Applications (Virtual and Augmented Reality, and others, Ubiquitous and Context-Sensitive Computing, and others, Healthcare, Education, and Security, and others**)**

MIC 231 Dissertation/Research Project
The module is the final year dissertation that involves in-depth research in a particular area of interest in Cloud Computing and IOT. The research findings are expected to be published in peer-reviewed journals or conference proceedings. The deliverable for this module is a dissertation.

MIC 232 Containerization in Cloud IOT
The module covers how containerization works, determinants of selection of a platform for containerisation, types of commercial container management solutions, security architecture of containers, containerising applications and managing containers**,** container models, containerisation in Paas clouds, container orchestration and clustering, engines used by enterprise organizations such as Docker, CoreOS'rkt, Google Kubernete, Amazon AWS and Cloud Foundry

MIC 233Optimization Techniques in Cloud IoT
The modules focus on optimization of performance and efficiency of different cloud IOT architectures, comparisons of traditional cloud IOT models to newer models for optimizing performance and efficiency, optimization of energy consumption in IOT applications and cloud resources. It involves assessment of proposed algorithms and deducing their effectiveness in dynamic and heterogeneous environment and adoption algorithms for reducing the energy consumption of task requests.

MCI 234 Emerging Technologies for Cyber Physical Systems
This module will cover contemporary issues in relation to emerging technologies that can be used to realise cyber-physical systems. Cyber-Physical Systems (CPS), are collections of physical and software components that communicate and interact with users via networks.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN INFORMATION AND KNOWLEDGE MANAGEMENT (BS23)

Duration	1½ years
Credit Load	306
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME
The goal of the programme is to educate students to manage organisation's knowledge assets. The programme aims to develop the sound conceptual, technical, analytical and communication skills that are required to succeed in the specification, design, construction and Information and Knowledge systems. In addition, students will learn techniques to foster learning and knowledge communities within the organisation. To this end, the programme employs a rigorous case analysis approach, which requires students to exercise their analytical abilities and conduct independent research. Furthermore, they must be able to integrate technologies, theories, concepts, and models from diverse range of modules.

2. ENTRY REQUIREMENTS
Normal Entry: A candidate must normally possess a minimum of honours degree with a 2.2 better in Information System or relevant degree programme.

Special Entry: For special entry a student must satisfy one of the criteria prescribed in the General Regulations and is approved by the Faculty Board.

3. PROGRAMME CHARACTERISTICS

Areas of Study:	Knowledge Management Strategies; Information Management, Knowledge Discovery, Business Intelligence; Systems Development; Management; Emerging Technologies
Specialist Focus:	The goal of the programme is to help meet the increasing demand for knowledge personnel who possess a balanced combination of technical and managerial skills.
Orientation:	Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects
Distinctive Features:	The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:	Knowledge Manager, Data Analyst, Web Content Manager, Records Manager, Competitive Intelligence Analyst, Electronic Resource Librarian, Information Architect, Information Officer Programmer; Project Manager; Chief Information Officer; Consultant; Database Designer; Systems Designer; Chief Technology Officer
Further Studies:	Doctoral studies in Information and Knowledge Management or in interdisciplinary programmes related to Information Systems

5. PROGRAMME DELIVERY

Teaching and Learning Methods:	Lectures, tutorials, group work, dissertation, individual independent study
Assessment Methods:	Written and oral examinations, tests, final year dissertation, presentations, term papers, continuous assessments

6. LEARNING OUTCOMES

- Understand the theoretical frameworks that integrate the creation, sharing, and utilization of information and knowledge
- Identify and apply principles of knowledge management in all types of organizations.
- Develop, implement and evaluate operational and strategic knowledge plans.
- Build a strategy for effective knowledge management for individual organizations.
- Design and deploy data analytics in corporate, marketing and related functional strategies.
- Identify and analyse scholarly literature relating to Information and Knowledge Management
- Publish research in scholarly journals of international repute

7. GENERAL PROVISIONS

The Master of Commerce degree programme is studied over 1½ years.

7.1 The Special Departmental Regulations shall prescribe the total number of modules per programme per semester, and the degree structures.

7.2 Exemption is a process of crediting a student with the module(s) already passed in programmes similar or equivalent to the one on offer. A student may be exempted from studying such modules.

However, the student has to apply for the credit transfer. In his/her application the following should be attached:

a) A transcript of the module(s) to be credit transferred.
b) A synopsis of the module(s) from the University/College where the module was attained.
c) A support letter from the Registrar's Office of the institution where the module(s) were attained.

7.3 The respective Departmental Boards will sit and assess applications for credits transfers and would recommend accordingly to Senate.

7.4 No student shall be exempted more than 50% of the modules offered in the programme.

8. PROGRAMME ASSESSMENT

Coursework	40 (Test 15%, Assignments 5%, Practicals
Written Examinations	60

9. PROVISION FOR PROGRESSION

9.1 Students shall progress from Semester One to Semester Two at each Level after they have passed more than 50% of modules taken in a semester.

9.2 TheFaculty Board of Examiners shall meet at the end of the first semester of the level to ratify the marks obtained by candidates and considered by Departmental Boards.

9.3 The FacultyBoard of Examiners shall meet at the end of the second semester of a level to ratify the marks obtained by candidates and make recommendations to Senate on issues of progression by students to the next level. In the case of repeat level students, the Faculty Board of Examiners may however, make a recommendation to withdraw or discontinue the student at this stage.

9.4 For a student to proceed from one level to the other, he/she musthavepassed at least75% of the number of modules normally scheduled in a particular level of a programme.

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11.GRADING AND DEGREE CLASSIFICATION

Refer to general regulations

12. DEGREE WEIGHTING

Refer to faculty regulations

13. AWARD OF DEGREE

Refer to general regulations

14. PROGRAMME STRUCTURE

Code	Module Description	Credits
Level 1 Semester 1		
MIKM 701	The Information and Knowledge Economy	**18**
MIKM 702	Organisational Learning	**18**
MIKM 703	Information and Knowledge Assets	**18**
MIKM 704	Knowledge Management	**18**
MIKM 705	Business Intelligence	**18**
Level 1 Semester 2		
MIKM706	Knowledge Management Strategies & Policies	**18**
MIKM707	Research Methods, Mapping and Modelling	**18**
*MIKM708	Intellectual Capital Management	**18**
*MIKM709	Semantic Analysis Methods and Technologies	**18**
MIKM710	Knowledge Assessment and Evaluation	**18**

Level 2 Semester 1		
MIKM80	Statistics and Data Analytics	**18**
MIKM80	Strategic Information Management	**18**
MIKM80	Dissertation	**90**

*Non core modules

15. MODULE SYNOPSES

MIKM701: The Information and Knowledge Economy

This module focuses on information and the new information and digital economy, consumer behaviour, production theory and the demand for information; information as a factor of production; information costs and pricing. Case studies in information industry are explored

MIKM 702: Organisational Learning

This module will address organizational learning in fast-paced, complex and changing organisational environments. Knowledge organisation must learn through its people, groups and organisation as a whole to survive and grow. Covers the fundamental theory of organisation learning and organisation learning types

MIKM703: Information and Knowledge Assets

The module will develop thorough understanding of the internal and external knowledge sources important for organizational effectiveness. This module outlines knowledge assets in organizations; using the I-Space model to map organizational knowledge assets. Social learning cycle. Understanding the knowledge worker: Schumpeterian and Newtonian learning, Career Anchors, Gardner's Theory of Multiple Intelligences, Sternberg's Balance Theory of Wisdom.

MIKM704: Knowledge Management

This module introduces historical roots of knowledge and knowledge management: theories/ definition of knowledge; theories, application tools and practices of knowledge management; knowledge management life-cycle framework and models; knowledge management-best practices, culture, economics strategy, intellectual capital, sustainable innovation.

MIKM705: Business Intelligence

This module introduces students to strategic intelligence consisting of competitive and business intelligence. Business and governments require effective programs, process and tools to track business, competitors, markets and trends by acquiring, creating, managing, packaging and dissemination intelligence knowledge

MIKM706: Knowledge Management Strategies & Policies

This module defines and develops knowledge fusion strategies. Issues and challenges for knowledge economies and enterprises. It also looks into knowledge creation strategies, management of knowledge flows in alliance, subsidiaries, Management and Strategies as well as outsourcing contexts. Creating a knowledge index or expertise locator, knowledge audit and good practices repository

MIKM707: Research Methods, Mapping and Modelling

This module reviews techniques and tools that support strategic development and decision-making through phases of knowledge discovery, knowledge capture and knowledge analysis. Relational databases and machine learning to organizational entities. Identification knowledge gaps, ethical issues surrounding the manipulation of data.

MIKM708: Intellectual Capital Management

The module encompasses the new knowledge–intensive economic units: human capital and social capital intellectual asserts, property, brands and trademarks. It looks into "A Life cycle" approach for intellectual asserts, capital measurement framework. Major knowledge assets valuation approaches. Parameters for managing intellectual capital, knowledge workers and intellectual capital. Intellectual property protection and exploitation. Disclosure and corporate governance

MIKM709: Semantic Analysis Methods and Technologies

This module introduces students to the practical contexts, methods and tools associated with semantic analysis. Focuses on early life cycle semantics, including identification and modelling of semantic problems, design of semantic solution, and the identification and implementation of appropriate sematic technology. Covers natural language processing, rule based and grammar based concept extraction, rule based and dynamic classification and automated summarization.

MIKM710: Knowledge Assessment and Evaluation

The module explores all aspects of effective knowledge management within all types of organisations. Lays out the context for knowledge management at each multiple levels and across all competences. Covers theory and practice and methods of assessments and evaluation for all levels and competencies. Provides an introduction to knowledge strategy formulations.

MIKM801: Statistics and Data Analytics

This module provides a comparative analysis of statistical methods and tools, demonstrating how different analytic methods can be used to address the critical data issues faced by organisations in digital age. Students analyse a range of issues in data ethics to ensure responsible capture and use of data; ensuring cultural appropriateness and subject privacy; and to root analytics initiatives in a responsible and trustworthy regimen

MIKM802: Strategic Information Management

The module develops understating of information technology, information and knowledge management governance frameworks and strategy perspectives with particular emphasis on the regulatory environment, legislative and organisational controls. The module builds on intellectual capital theory -: social and emotional capital.

MIKM803: Dissertation/Research Project

The dissertation captures the details of the practical work carried out, in which the student demonstrates the ability to transfer the theoretical knowledge gained during the course of study. The student should demonstrate an ability to present Information and Knowledge in a manner that can be easily understood as well as capturing and analysing the experimental outcomes of one's project activity.

DEPARTMENT OF MANAGEMENT SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN BUSINESS MANAGEMENT (BS05)

Duration	4 years
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. OBJECTIVES (PURPOSE OF THE PROGRAMME)

- The Departmental Regulations shall be read in conjunction with the Faculty Regulations and the General Academic Regulations for Undergraduate degrees herein referred to as General Regulations.
- The degree will be awarded to candidates who have successfully completed the programme and passed examinations in accordance with the Faculty Regulations.
- The duration of the degree program shall normally be a minimum of 4 levels including a period of Work-Related Learning in a relevant industrial/commercial setting during the third level.

2. PURPOSE OF THE PROGRAMME

The programme provides knowledge which develops the students' intellectual and critical thinking ability as well their managerial and entrepreneurial skills. It fosters ability to behaviour, analyse and interpret information for decision-making while instilling and indicating high moral standards of personal and professional behaviour which are consistent with positions of trust and leadership in society and ever-changing environment.

3. ENTRY REQUIREMENTS

1. **Normal Entry**
   English and Mathematics at 'O' Level, must have obtained at least two passes at 'A' level in Management of Business, Accounting, Economics, Mathematics and Geography.
2. **Visiting School Programme**
   Diploma (Institute of Bankers), ND in Business Studies, Accountancy, Purchasing, Diploma in Education, Institute of Administration and Commerce (IAC), Institute of Personnel Management (IPMZ), Diploma in Business Administration (ZIM), Diploma in Marketing and any other relevant tertiary qualifications deemed to an equivalent or passes in relevant 'A' Level subjects**,** and must be at least 22 years old with 2 years relevant working experience and proof of employment.

3.3 Mature Entry

Refer to faculty regulations.

3.4 Special Entry

3.4.1 This applies to students who may wish to transfer from other universities while undertaking Business Management related degree programme.

3.4.2 An applicant may be exempted from certain level modules and shall be required to register additional modules to make up for the full component of that level.

3.4.3 Successful completion of the Work-Related Learning component at Level 3 is compulsory for all candidates.

4. PROGRAMME CHARACTERISTICS

1. **Areas of study:** Business research, investment management, mercantile law, entrepreneurship, small business management, taxation corporate governance
2. **Specialist Focus:** Marketing, finance, business management and entrepreneurship
3. **Orientation:** Training people with potential to get into Operations positions and leadership
4. **Distinctive Features:** The programme builds on enterprising student and focuses on knowledge development and application of concepts.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

1. **Employability:** Careers in small business, large organisation listed and non-listed in marketing, finance, academia, and research
2. **Further Studies:** Masters and doctoral studies in strategic management and corporate governance, marketing, entrepreneurship, financial engineering

6. PROGRAMME DELIVERY

1. **Teaching and Learning Methods:** Lectures (face to face and online), tutorials, seminars, group work, industrial visits, industrial attachment, research project, individual independent study.
2. **Assessment Methods:** Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project report.

7. LEARNING OUTCOMES

- Use relevant research methods to create new insights into the field of business.
- Ability to demonstrate knowledge and understanding of Business Management fundamentals.
- Ability to apply principles of Business Management to solve problems in business and commerce.
- Create new businesses
- Produce a holistic graduate able to integrate in any business form.

8. GENERALPROVISIONS

Refer to faculty and general regulations

9. FAILURE TO SATISFY THE EXAMINERS.

Refer to General Regulations

10. WORK-RELATED LEARNING GENERAL GUIDELINES

Refer to the Faculty Regulations

11. DEGREE WEIGHTING

Refer to the Faculty Regulations

12. GRADING AND DEGREE CLASSIFICATION

Refer to the Faculty Regulations.

13. PROVISIONFORPROGRESSION

Refer to the Faculty Regulations.

14. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Descriptions	Prerequisites	Credits
HCS135	Introduction to Information Technology		12
CS131	Basic Communication Skills		12
GSB211	Gender Studies for Business		12
EC131	Economic Principles 1		12

BM133	Financial Accounting for Business 1A	12
CMI131	Principles of Marketing 1	12
BM131	Principles of Management	12

Level 1 Semester 2

Code	Module Description	Credits
BM135	Commercial Law	12
BM136	Financial Accounting for Business1B	12
BM138	E-Commerce	12
BM132	Business Communication	12
BM134	Business Ethics	12
BM137	Quantitative Analysis for Business	12
EC134	Macroeconomics	12

Level 2 Semester 1

Code	Module Description	Credits
BM231	Human Resource Management	12
BM233	Company Law and Practice	12
ENT131	Principles of Entrepreneurship	12
BM235	Financial Accounting IIA	12
BM240	Management Appreciation of Taxation	12
BM249	Quality Management	12

Level 2 Semester 2

Code	Module Description	Credits
BM239	Industrial and Labour Relations	12
BM246	Digital Marketing	12
BM238	Business Research Methods	12
BM236	Business Investment Analysis	12
BM241	Financial Accounting for Business IIB	12
ACC145	Practical Accounting Data Processing	12

LEVEL 3 SEMESTER 1 WORK-RELATED LEARNING

BM 340	Work-Related Learning Preliminary Report	40

LEVEL 3 SEMESTER 2 Work-Related Learning

Code	Module Description	Credits
BM341	Work-Related Learning Continuous Assessment	40
BM342	Work-Related Learning Report	40

Level 4 Semester 1

Code	Module Description	Credits
BM432	Project Management	12
BM434	Organisational Behaviour	12
BM445	Public Relations Management	12
BM431	Change Management	12

BM435	Financial Management	12
BM436	Management Accounting	12

Level 4 Semester 2

Code	Module Description	Credits
BM437	Production and Operations Management	12
BM438	Strategic Management	12
BM439	Corporate Governance	12
BM440	Research project	12
BM440	Research project	12
BM446	Strategic Sourcing	12

13. MODULE SYNOPSES

BM131 Principles of Management

The module is intended towards giving the students a sound grounding in the history and development of management thought and functions of management. Emphasis is given to leadership, change management, decision-making and all the other aspects of managing in an agile environment.

BM132 Business Communication

The objective of the module is to expose students to concepts and theories of business communication and to assist them in developing into good and competent communicators. It focuses on the development of a student's ability to communicate effectively and concisely in written, oral and any other forms of communication. It also considers different styles in the use of English in business settings, communication skills, letter writing, business reports, notetaking, meetings, minutes, summarizing and referencing.

BM133 Financial Accounting for Business IA

This module is designed to introduce students with no accounting background to accounting principles and basics. The module is intended to develop the learner's awareness, skills and understanding of the principles of double entry book-keeping and how these principles are applied to produce the financial information required in business.

BM134 Business Ethics

The module examines the history and development of business Ethics by highlighting some models of moral behaviour and how they could be related to various situations in the business world. It explores the variant ethical theories such as utilitarianism, deontological ethics, and egoism. The Module also seeks to expose students to critical thinking necessary in resolving moral dilemmas vis-àvis the most pressing ethical issues in business such as fraud, insider trading, whistle blowing and many others.

BM135 Commercial Law

The module is designed to equip students with basic but concise legal knowledge to enable students to make legally sound and informed business decisions at work or managers. The module is an introduction to general principles of law and its interpretation, law of contract, purchase and sales, agency, negotiable instruments, insurance, partnership, law of derelict, insolvency and consumer protection.

BM136 Financial Accounting for Business IB

This module is a follow up to Financial Accounting for Business 1A. It introduces students to the following areas: incomplete records, control accounts, not-for-profit making organisations, manufacturing entities, departmental and partnership accounts. It prepares students for more advanced accounting studies in Financial accounting for business 2A and 2B. The module is meant for non-accounting students who should appreciate key accounting issues that they will interface with in the world of business.

BM137 Quantitative Analysis for Business

This module introduces students to the fundamental concepts of data presentation and analysis in management decision-making. The approach will emphasize practical management problem-solving approach using quantitative reasoning skills rather than the theoretical development of statistics. Students should develop quantitative research techniques for research writing.

BM138 E-Commerce

The module considers the benefits of the Internet, Website creation and Management, business-to-business transactions, e-banking, electronic order purchasing, Automated Teller Machines, Point of Sales, management information systems, electronic transfer of funds, and data security.

GSB211 Gender Studies for Business

The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module to graduate.

HCS135 Introduction to Information Technology

Refer to the Department of Information and Marketing Sciences for module synopsis.

CS131 Basic Communication Skills

Refer to the Faculty of Arts for the module synopsis**.**

EC131 Economic Principles 1

Refer to the Department of Economics for the module synopsis.

EC134 Macroeconomics

Refer to the Department of Economics for the module synopsis.

CMI131 Principles of Marketing 1

Refer to the Department of Information and Marketing Sciences for module synopsis.

BM231 Human Resource Management

The module covers the evolution of the Human Resources Management function; the emergence of the Human Resources Management philosophy; similarities and differences between traditional Personnel Management and Human Resources Management; approaches to Human Resources Management; Human Resources models and human resources Management inaction focusing on strategic integration, commitment, quality, flexibility and team working.

BM233 Company Law and Practice

This module gives a thorough grounding in the regulation of companies by law including statutes and case law. It covers knowledge of the application of statutory and other provisions including a grasp of the administration procedures relating to meetings, registration of companies, flotation of shares, duties of directors and auditors, judicial management winding up and liquidation.

BM235 Financial Accounting IIA

The module intends to equip learners with the necessary knowledge and skills which enable them to understand the preparation of financial statements of partnership as well as introduce them to accounting by companies. This will additionally equip students with necessary competencies for the preparation of financial statements according to the requirements of International Financial Reporting (IFRS) and understanding of the preparation of the statement of cash flow as well as interpretation of financial statements.

BM236 Investment Analysis

The module aims to develop an understanding of the financial theory of investment as well as its use in the practical world. The module shall revolve around the distinctive themes of information efficiency in financial markets, asset allocation, derivative security markets, elements of investments, portfolio theory, fixed-income securities, security analysis, derivative assets and active asset management.

ENT131 Entrepreneurship 1

The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation-building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan, and evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; identify the challenges being faced by entrepreneurs.

BM239 Industrial and Labour Relations

This module is targeted to equip students with the following: understanding of employment relationships, an understanding and appreciation of the origins of industrial relations and the worker problem, understanding of theoretical issues behind the collective bargaining role of trade unions. The course will endeavour to build on the basic knowledge and background needed to prepare you for a career in the dynamic field of Labour Relations.

BM240 Management Appreciation of Taxation

This module aims at equipping students with basic knowledge, skills and understanding of the Zimbabwe Tax system and its administration on individuals and companies. Taxation of individuals and companies, the basic laws governing taxation and tax payments, make up this module.

BM241 Financial Accounting for Business IIB

The module intends to equip the learners with the skills and knowledge which enable them to understand the preparation of consolidation accounts, consignment accounts, royalty accounts, branch accounts and joint venture accounts. This will equip students with the necessary competencies for the preparation and presentation of accounts in line with International Financial Reporting Standards (IFRS).

AC145 Practical Accounting for Data Processing

Refer to the Department of Accounting Sciences for the module synopsis

BM246 Digital Marketing

The module goes beyond e-commerce to explore the marketing process in the online arena. It assists students in understanding how to use digital channels to create valuable online market spaces, how dynamic technology allows firms to connect with customers through electronic platforms, and how digital marketing can be used to augment traditional marketing.

BM238 Business Research Methods

Business research concepts and methods, the traditional and scientific methods, the research process, problem definition, research design, questionnaire construction and administration, sampling basis, sources of research data, collection and interpretation, research application and uses, data presentation and analysis, ethics, and research proposal will be covered.

BM249 Quality Management

The module introduces the student to quality management and the quest for business practices in changing environment. Students are guided on how they may assist an organisation embrace quality management as a competitive advantage in the business world where product and service excellence is shifting from mere position of proprietary knowledge, access to raw material to a culture of conformance to specifications, value chain management and employment and employee training and empowerment. Issues covered include inspection quality control, quality assurance, total quality management and international standards association. Attention will also be paid to quality function Deployment, benchmarking, quality gaps, management commitment, education and training, employee empowerment, teamwork, and management of change and the search for excellence. Business processes reengineering, quality costs, scatter diagrams, Parato analysis, Fishbone diagrams, histogram, control charts, check sheets, Stratification service surveys.

BM431 Change Management

The module focuses on deepening the understanding of the challenges and opportunities associated with managing change and to equip students with the knowledge of how the process and implantation of change management unfolds within companies. It will dwell more on why organisations change, models of change, resistance to change and organisational development.

BM432 Project Management

The module involves a detailed study of network analysis and project scheduling techniques, project structuring, project finance, hiring and management of personnel, control of materials, monitoring the project, budgeting and project completion.

BM434 Organisational Behaviour

This module covers behavioural theory and research, human resources administration, models relating to human productivity, satisfaction, retention, earning in the organisational setting, organisational structures and leadership styles.

BM435 Financial Management

The module seeks to provide students with knowledge on the determination of financial objectives of business organisations, analysis of financial statements, and overview of financial planning, value, risk and return (share, debentures and options), the management of working capital, cash, debtors and inventories, investment decisions, financing decisions, optimal capital budgets and dividend theory.

BM436 Management Accounting

The module is an overview of the necessity of managerial accounting for management process, evaluation of the influence of globalization on management, managerial accounting systems and application to managerial operations, managerial accounting for management purposes, managerial accounting systems, techniques for application in managerial decisions, analysis of business situations and solving problems.

BM437 Production and Operations Management

Topics covered in the module include developing production/operations management strategies, the strategic role and objectives of production/operations, production/ operations management strategies - designing, planning and controlling the production/operations management system, process technology, materials requirements planning, JIT planning and control, quality planning and control and project planning.

BM438 Strategic Management

The module seeks to examine the nature and value of strategic management, the business environment, strategic planning, functional goals and strategies. It also examines strategy implementation, the role of leadership in implementing strategy, resource deployment in implementing strategy, designing organization to implement strategy effectively as well as factors influencing strategy implementation.

BM439 Corporate Governance

The module seeks to develop an understanding and appreciation of significant corporate governance concepts and issues in the contemporary business world, definition of corporate governance, key concepts associated with best practices, powers and rights of, and relations between directors and shareholders, history of corporate governance etc.

BM 440 Research Project/Dissertation

Students carry out an independent research project under the supervision of a tutor /supervisor. The length of the dissertation should be 10000 words.

BM445 Public Relations Management

The module is designed to guide the student in examining the scope, importance and characteristics of public relations in both private and public sectors. The student is shown how to study the elements of public relations, how to develop a Public Relations plan and the role of public relation in crisis and event management, the module will give the students an opportunity to apply their theoretical knowledge to practical situations.

BM446 Strategic Sourcing

The module is designed to create a managerial perspectiveof the core tasks and challenges required to effectively manage logistical operations and supply chains. It deals with the integration of information and material flows across multiple organisations in the supply chain. By managing functional areas of logistics such as customer service, transportation and order processing as an integrated unit, a firm can gain a strategic advantage in logistics competency and be a strong contributor to successful supply chain management.

BM340, BM341, BM342: Work-Related Learning

The modules are meant to enable the student to integrate theory and practice through exposure to an actual working environment.

REGULATIONS FOR THE BACHELOR OF COMMERCE DEGREE IN HEALTH CARE MANAGEMENT (BS04)

Duration	4years
Credit Load	480
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	216
ZNQF Level	8

1. OBJECTIVES (PURPOSE OF THE PROGRAMME)

1.1 The Departmental Regulations shall be read in conjunction with the Faculty Regulations and the General Academic Regulations for Undergraduate degrees herein referred to as General Regulations.

1.2 The degree will be awarded to candidates who have successfully completed the programme and passed examinations in accordance with the Faculty Regulations.

1.3 The duration of the degree program shall normally be a minimum of 4 levels including a period of Work-Related Learning in a relevant industrial/commercial setting during the third level.

2. PURPOSE OF THE PROGRAMME

2.1 The programme provides knowledge which develops the students' intellectual and critical thinking ability as well their managerial and entrepreneurial skills. It fosters ability to behaviour, analyse and interpret information for decision-making while instilling and indicating high moral standards of personal and professional behaviour which are consistent with positions of trust and leadership in society and ever changing environment

3. ENTRY REQUIREMENTS

3.1 Normal Entry

English and Mathematics at 'O' Level, must have obtained at least two passes at 'A' level in Management of Business, Accounting, Economics, Mathematics and Geography.

3.2 Visiting School Programme

Diploma (Institute of Bankers), ND in Business Studies, Accountancy, Purchasing, Diploma in Education, Institute of Administration and Commerce (IAC), Institute of Personnel Management (IPMZ), Diploma in Business Administration (ZIM), Diploma in Marketing and any other relevant tertiary qualifications deemed to an equivalent or passes in relevant 'A' Level subjectsand must be at least 22 years old with 2 years relevant working

experience and proof of employment.

3.3 Mature Entry

Refer to faculty General Regulations.

3.4 Special Entry

3.4.1 This applies to students who may wish to transfer from other universities while undertaking Healthcare management related degree programme.

3.4.2 An applicant may be exempted from certain level modules and shall be required to register additional modules to make up for the full component of that level.

3.4.3 Successful completion of the Work-Related Learning component at Level III is compulsory for all candidates.

4. PROGRAMME CHARACTERISTICS

- **Areas of study:** Healthcare systems, financial issues in Healthcare,
- **Specialist Focus:** Healthcare management systems
- **Orientation:** Training people with potential to get into Operations positions and leadership in Healthcare institutions
- **Distinctive Features:** The programme builds on enterprising student and focuses on knowledge development and application of concepts.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

5.1 **Employability:** Administrators in private and public health institutions, Healthcare procurement personnel.

5.2 **Further Studies:** Master's and doctoral studies in strategic management and corporate governance, Master's and doctoral studies in Healthcare management.

6 PROGRAMME DELIVERY

6.1 **Teaching and Learning Methods:** Lectures (face to face and online), tutorials, seminars, group work, industrial visits, industrial attachment, research project, individual independent study.

6.2 **Assessment Methods:** Written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project report.

7 LEARNING OUTCOMES

7.1 Use relevant research methods to identify, analyse and solve management and commercial problems in a systematic, critical and creative way as well as create new insights into the healthcare field.

7.2 Ability to demonstrate knowledge and understanding of Healthcare management fundamentals.

7.3 Produce a holistic graduate able to integrate in any business form.

7.4 Contribute towards the extension of knowledge of business management and administration in a healthcare context.

7.5 Acquire a critical understanding of how to manage people, resources and processes in a healthcare context.

7.6 Apply the techniques and methods used to evaluate health information systems and identify current threats and opportunities associated with the management of information systems.

8 **GENERAL PROVISIONS**
Refer to faculty and general regulations

9. ASSESSMENT
Refer to faculty and general regulations.

10. FAILURE TO SATISFY THE EXAMINERS.
Refer to General Regulations

11. WORK-RELATED LEARNING GENERAL GUIDELINES
Refer to the Faculty Regulations

12. DEGREE WEIGHTING
Refer to the Faculty Regulations

13. GRADING AND DEGREE CLASSIFICATION
Refer to the Faculty Regulations.

14. PROVISIONFORPROGRESSION
Refer to the Faculty Regulations.

15. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Descriptions	Prerequisites	Credits
HCS135	Introduction to Information Technology		12
CS131	Basic Communication Skills		12
GSB211	Gender Studies for Business		12
BHC131	Introduction to Healthcare Management		12
BM133	Financial Accounting for Business 1A		12
CMI131	Principles of Marketing 1		12
BM131	Principles of Management		12

Level 1 Semester 2

Code	Module Description	Credits
BHC132	Current Issues in Healthcare Policy & Practice	12
BHC135	Concepts of Health Promotion	12
BHC136	Medical Law and Risk Management	12
BM132	Business Communication	12
BHC134	Behavioural and Cultural Issues in Healthcare	12
BHC133	Health Insurance and Reimbursement	12

Level 2 Semester 1

Code	Module Description	Credits
BHC231	Human Resource Management in Healthcare	12
BHC232	Strategic Planning & Marketing in Healthcare	12
ENT131	Principles of Entrepreneurship	12

BHC233	Ethical and Legal Issues in Healthcare	12
BHC234	Healthcare policy and evaluation	12

Level 2 Semester 2

Code	Module Description	Credits
BHC235	Contemporary Health Systems	12
BHS239	Healthcare Supply Chain Systems	12
BM238	Business Research Methods	12
BHC236	Healthcare Informatics	12
BHC238	Healthcare Economics from a Population Perspective	12
BHC239	Occupational Health and Safety	12

LEVEL 3 SEMESTER 1 WORK-RELATED LEARNING

BHC 340	Work-Related learning Preliminary Report	40

LEVEL 3 SEMESTER 2 Work-Related Learning

Code	Module Description	Credits
BHC341	Work-Related Learning Continuous Assessment	40
BHC342	Work-Related learning Report	40

Level 4 Semester 1

Code	Module Description	Credits
BHC431	Sustainable Healthcare Systems	12
BM432	Project Management	12
BHC432	Healthcare Organisation theory & Behaviour	12
BM431	Change Management	12
BHC434	Healthcare Quality Management	12

Level 4 Semester 2

Code	Module Description	Credits
BM437	Production and Operations Management	12
BM439	Corporate Governance	12
BHC433	Research project	12
BHC436	Health data and Analysis	12

16. MODULE SYNOPSES

BHC131 Introduction to Healthcare Management

In this module, students examine management concepts and theories designed to influence and improve the performance of healthcare organizations. They identify and examine the external and internal environments of organizations as well as key management functions, roles, and responsibilities. Exploring essential aspects of healthcare management, students engage in a variety of conceptual and practical activities, such as profiling a healthcare manger, assessing the value of leadership in decision-making, and comparing strategic plans.

BHC132 Current Issues in Healthcare Policy and Practice

In this module, students examine the nature of healthcare facilities, major factors influencing the quality of care, and the impact of policy initiatives on all stakeholders, including providers and consumers. They explore the major issues in acute and long-term healthcare policy and practice from the perspective of the patient and the provider.

BHC133 Health Insurance and Reimbursement

Students in this module will examine the history of third-party reimbursement, the culture of patient expectations for healthcare, and **Medicare and Medic-aid** reimbursement funding and management. Students will consider how healthcare policy, insurance, and reimbursement impact on quality and population health.

BHC134 Behavioural and Cultural Issues in Healthcare

In this module, students examine the cultural and societal factors within communities affect the management and delivery of healthcare services. Students will also identify and analyse health disparities attributable to diverse cultural and behavioural factors and discuss their implications for healthcare policy.

BM131Principles of Management

The module deals with the history and development of management thought, functions of management, organizational structure, decision-making, globalization, leadership and motivation, controlling, budgetary and non-budgetary controls, and change management.

BM133 Financial Accounting for Business 1A

Students take a top-down approach to understanding introductory accounting documents and procedures by exploring a business's financial statements, including the balance sheet, income statement, and cash flow statement in this module. Students explore the practical uses for information that can be gleaned from these statements, individually and as a whole, through a detailed examination of the properties and characteristics of each statement.

BHC135 Concepts of Health Promotion

In this module, students formulate a definition of health and discuss the many influences that shape our individual and collective perceptions of health. Students consider the health-wellness continuum, including a number of factors, such as behavioural, demographic, psychological, and social forces. Additionally, students reflect on ways to shape their future careers in health and to promote positive change.

BHC 136 Medical Law and Risk Management

The module is designed to equip students with basic but concise legal knowledge to enable students to make legally sound and informed business decisions at work or as managers. The module is an introduction to general principles of law and its interpretation, law of contract, purchase and sales, agency, negotiable instruments, insurance, partnership, law of derelict, insolvency and consumer protection.

HCS135 Information Technology & Computer Applications

Refer to the Department of Information and Marketing Sciences for module synopsis.

ENT 131 Introduction to Entrepreneurship

The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan, and evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; identify the challenges being faced by entrepreneurs.

CMI131 Principles of Marketing

Refer to the Department of Information and Marketing Sciences for module synopsis.

BM132 Business Communication

The objective of this module is to expose students to concepts and theories of business communication and to assist them develop into competent communicators. It focuses on the development of student's ability to communicate effectively and concisely in written, oral and any other forms of communication. It also considers different styles in the use of English in business settings, interpersonal communication skills as well as effective business writing.

BHC136 Medical Law and Risk management

The module is designed to equip students with basic but concise legal knowledge to enable students to make legally sound and informed business decisions at work or managers. The module is an introduction to general principles of law and its interpretation, aw of contract, purchase and sales, agency, negotiable instruments, insurance, partnership, law of derelict, insolvency and consumer protection.

ENT131 Entrepreneurship 1

The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan

Evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; identify the challenges being faced by entrepreneurs.

BHC231 Human Resource Management in Healthcare

This module enables students to explore and discuss the role of HR in healthcare organizations as related to the recruitment, retention, and management of the healthcare workforce. They examine workforce development models, employee benefits, and compensation strategies. Students also explore methods used by HR professionals to develop goals and expectations for evaluating employee performance and promoting effective employee relations, and they

investigate HR problem-solving strategies, such as conflict resolution, collective bargaining, and arbitration methods.

BHC232 Strategic Planning and Marketing in Healthcare

The module covers important strategic planning issues in healthcare institutions. Key concepts related to strategic planning, including the relationship of the plan to the organization's mission, values, and vision, are presented in this module. In this module, students examine and apply assessment techniques and methodologies for evaluating the strengths, weaknesses, opportunities, and threats (SWOT analysis) of a healthcare organization. They also explore the relationship between strategic planning, marketing, and organizational performance.

BHC233 Ethical and Legal Issues in Healthcare

In this module, students examine the legal and ethical issues that are fundamental to the practice of healthcare and the conduct of health-related research. They also investigate and assess issues of privacy and confidentiality, informed consent, licensing, and malpractice, among others. Additionally, students consider ethical, decision-making models for assuring the quality, safety, and appropriateness of healthcare and services.

BHC 234 Healthcare policy and Evaluation

This module teaches qualitative research skills for project planning, monitoring, and evaluation activities. Students will analyse In collaboration with an organization in their local community, students will complete practical assignments that require the application of several data collection techniques.

GSB211Gender Studies for Business

The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender responsive environment. Every student has to pass the module in order to graduate.

BHC235 Contemporary Health systems

The purpose of this module is to enlighten students' knowledge and comprehension of current challenges and deficiencies in health systems and provide recommendations for proper planning, management, direction and control of resources; personnel and dynamics of healthcare institutions to provide responsive and appropriate healthcare.

BHC236 Health Care Informatics

The students are prepared to examine data sets so that they discover and analyse patterns of activities pertaining disease control and distribution of supplies. The module develops students' ability to make decisions relating to internal control systems with in organisations, and an understanding of analytical tools, strategies and technology to detect and deal information.

BHC 237 Healthcare Supply Chain Systems

After exploring enablers for successful and ethical health supply chain management (SCM) and how to develop effective strategies, the module will examine key management issues and priorities. These include achieving efficient operations and logistics, harnessing data and technology, ensuring compliance, monitoring and evaluating performance, and tackling risks within the supply chain, including corruption.

BHC238 Healthcare Economics from a Population Health Perspective

Students are provided with the foundation for economic evaluation from a population health approach. Students can develop an understanding of healthcare as a market, compare supply-and-demand principles in healthcare versus a typical business, discuss how economic losses or gains impact healthcare policy changes, sectors of the population, physician staffing, nursing staffing, and stability. Students in the module also explore epidemiological, social, global, and future economic impact of healthcare.

BHC 239 Occupational Health and safety

The modules looks at Current Occupational Health and Safety Legislation in Zimbabwe Who is responsible for workplace safety programs Critical connections and work site analysis Basic definitions of workplace safety programs and their purpose Challenges for those practicing safety management Safety and Special Hazards

BM238 Research Methods

In this module, students examine the basic components required for the conduct of health-related research and provides students with the analytic tools needed to understand and assess research methods described in the scientific literature. Basic research methods are described, including surveys, observational studies, experimental and quasi-experimental design, use of primary and secondary data, and statistical techniques for analyzing and interpreting data.

BHC 431 Sustainable Healthcare Systems

This module looks at the key pillars of sustainability in health; i.e. energy management, waste management, air quality management and water management. Emphasis is made on how to enhance sustainability, i.e. healthcare without harm through reduce, reuse, recycle, rethinking and repairing. The principles of sustainable healthcare are also explained in this module.

BHC432 Healthcare Organization Theory and Behaviour

The individual and group behaviours within healthcare organizations often have a direct impact on organizational success and the ability to deliver quality care. In this module, students examine the theories of behaviour of healthcare organizations at the macro (organization-wide) level and micro (individual and team performance) level.

BM431ChangeManagement

The module focuses on the management of change in organisations.It will also dwell on the types of organisations that are in existance, and how the differing structures impact on change management to include: performance, designing strategies, concept of power, organisational politics, change model, resistance, models for growth and decline, structural forms and the management of change.

BHC434 Occupational Health, Quality and Safety

A lack of quality service in healthcare, including medical errors, unnecessary surgery, and neglect, accounts for preventable deaths and a continual increase in the cost of delivery. Students in this module explore major quality and safety issues within healthcare organizations. They also examine methods of assessing quality and techniques to improve quality as well as opportunities to prevent adverse events.

BHC 436 Health and Data Analysis

This module explores quantitative methods in project planning, monitoring, and evaluation. It provides opportunities for students to design quantitative evaluations and apply statistical measures to test hypotheses. Students will explore the use of statistical software in managing and manipulating data and the production of descriptive and analytical reports that meet the guidelines and expectations of professional practitioners in the field. They will develop an understanding of an often intimidating and difficult subject with an approach that is informative, personable, and clear as they are guided through various statistical procedures, beginning with descriptive statistics, correlation, and graphical representation of data through inferential techniques and variance.

BM432 Project Management

The module involves a detailed study of network analysis and project scheduling techniques, project structuring, project finance, hiring and management of personnel, control of materials, monitoring the project, budgeting and project completion

BM437 Production and Operations Management

Topics covered in the module include developing production/operations management strategies, the strategic role and objectives of production/operations, production/ operations management strategies- designing, planning and controlling the production/operations management system, process technology, materials requirements planning, JIT planning and control, quality planning and control and project planning.

BM439 Introduction Corporate Governance

The module examines the history and development of business Ethics by highlighting some models of moral behaviour and how they could be related to various situations in the business world. It explores the variant ethical theories such as utilitarianism, deontological ethics, and egoism. The module seeks to develop an understanding and appreciation of significant corporate governance concepts and issues in the contemporary business world, definition of corporate governance, key concepts associated with best practices, powers and rights of, and relations between directors and shareholders, history of corporate governance etc.

BHC433 Research Project

Students carry out an independent research project under the supervision of a tutor /supervisor. The length of the dissertation should be 10000 words.

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN STRATEGIC MANAGEMENT AND CORPORATE GOVERNANCE (BS27)

Duration 2 years

Credit Load	396
Minimum Credit Load	360
Maximum Credit Load	430
Maximum MBKs Credit Load	288
ZNQF Level	9

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the Faculty of Commerce Regulations and General Academic Regulations for Postgraduate Diplomas and Taught Masters Degrees offered by The Midlands State University herein referred to as General Regulations.

1.2 The degree shall be awarded to candidates who have successfully completed the programme and passed the examinations in accordance with regulations set in the Faculty Regulations.

2. OBJECTIVES (PURPOSE OF THE PROGRAMME)

2.1 The Master of Science in Strategic Management degree at the Midlands State University is a two-year programme for students with an undergraduate back ground in accounting, management, human resources and other equivalent qualifications. The programme is for students interested in careers in all facets of strategic management and corporate governance. The programme provides an opportunity to obtain a more comprehensive foundation in strategic management an added knowledge in specific business-related areas.

2.2 The programme aims to develop sound conceptual, technical, analytical and communication skills that are required to succeed in the strategic management and corporate governance. To this end the programme employs a rigorous case analysis approach, which requires students to exercise their analytical abilities and develop effective verbal and written communication skills. For the crucial role managers play in a free enterprise economy and ever-changing business environment.

3. ENTRY REQUIREMENTS

3.1 To qualify for entry into the Master of Science in Strategic Management degree programme a student should have a good Honours degree in management, accounting, human resource management and/or other tertiary qualification approved by the Department.

3.2 Before the commencement of the programme, the department will offer one week bridging module sessions covering the following areas: Financial Accounting, Statistics, Information systems, Management overview and assignment research and presentation

4.PROGRAMME CHARACTERISTICS

4.1 Areas of study:	Business research, Corporate Governance and Strategic Management
4.2 Specialist Focus:	Strategy Formulation and Implementation; Strategic Leadership; Corporate Innovation; ethics and Corporate Governance; Decision-making
4.3 Orientation:	Training people with the potential to get into Executive Management levels and board leadership.
4.4 Distinctive Features:	The programme is for students interested in careers in all facets of strategic management and corporate governance. The programme provides an opportunity to obtain a more comprehensive foundation in strategic management and

added knowledge in specific business-related areas.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

5.1 Employability: Consultancy, academia, research, Operations Management, Executive management positions and boards.

5.2 Further Studies: Doctoral studies in strategic management, corporate governance, leadership.

6. PROGRAMME DELIVERY

Formal examinations will normally be held at the end of each semester.

6.1 For a student to be admitted to the examinations they must have:

6.1.1 Satisfactorily attempted approved modules of study including submission of required written assignments and term papers.

6.1.2 Attended compulsory classes.

6.1.3 Participated in prescribed seminars, tutorials and practical classes.

6.1.4 Paid the prescribed fees in accordance with the General Regulations.

6.2 Formal examinations will normally be written papers, but in some cases the examiner may test the candidate orally.

6.3 Continuous assessment shall account for 40% of the overall assessment, while the final examination will account for 60% of the overall assessment.

6.4 The aggregate pass mark shall be 50% and the candidate must have obtained at least 50% in the examination.

7 LEARNING OUTCOMES

7.1 Use relevant research methods to create new insights into the field of business.

7.2 Ability to demonstrate knowledge and understanding of strategic management and corporate governance

7.3 Ability to make decisions at strategic levels

8. GENERAL PROVISIONS

A student requires 330 credits to complete this programme.

9. SUPPLEMENTARY EXAMINATIONS

6.1 A student who fails to satisfy the examiners and obtains a mark less than 50% or fails the examination but not by less than 40% may, on **the recommendation of the Board of Examiners and subject to the** approval of the Senate, be permitted to sit for a supplementary examination(s) in the irrespective module(s) failed in the fourth semester.

6.2 Overall assessment of the Supplementary examination will not incorporate; continuous assessment. Supplementary exanimations shall be marked as ‘Pass’ or ‘Fail’ and the overall maximum mark awarded in a supplementary examination shall be 50%.

6.3 No student shall be allowed to sit for more than two supplementary examinations in the final semester.

10. PROVISION FOR PROGRESSION

7.1 A student shall progress automatically from one level to the other even carrying the modules from the previous level.

11. DEGREE WEIGHTING CLASSIFICATION

For purposes of degree classification, the parts of the degree programme will be classified

as follows:

Level	1 Semester 1 and 2	50%
Level2		50%

12. CLASSIFICATION

The following marking scheme shall be used for all the modules and grading of the degree.

80%-100%	Distinction
70%-79%	Merit
60%-69%	Credit
50%-59%	Pass
Below50%	Fail

13. PROGRAMME STRUCTURE

Level 1 Semester 1

CODE	MODULE DESCRIPTION	CREDITS
MBM732	Strategic Planning and Management	18
MBM734	Strategic Change Management	18
MBM735	Management of Organisational Behaviour	18
MBM 771	Management Accounting	18
MBM731	Corporate Governance and Business Ethics	18

Level 1 Semester 2

CODE	MODULE DESCRIPTION	CREDITS
MBM737	Financial Accounting	18
MBM739	Business Research Methods	18
MBM740	Learning Organization	18
MBM743	Managerial Economics	18
MBM738	Strategic Information Management and E-Business	18

Level 2 Semester 1

CODE	MODULE DESCRIPTION	CREDITS
MBM 772	Corporate Entrepreneurship	18
MBM 773	Financial Statement analysis	18
MBM745	Quality Management	18
MBM742	Strategic Financial Management	18
MBM744	Production and Operations Management	18

Level 2 Semester 2

MBM762	Strategic Leadership		18
MBM 774	Corporate Governance II		18
MBM770	Dissertation/Research Project	all quarters	90

MODULESYNOPSES

MBM731 Corporate Governance and Business Ethics

The course aims to enable students to define ethics, articulate what ethics entails, tapping from the various perspectives posited by the ethical theories, interrogate ethical relationships in business and the social responsibility of business and dissect the various corporate governance frameworks such as the American Round Table, King 111 and the Ubuntu Framework in business. The course will cover topics and aspects such as definition of ethics, utilitarian ethics, deontological ethics, virtue theory, theories of justice, ethical discussion and ethical relationships in business and social responsibility definition of corporate governance, separation of Ownership and control, corporate scandal and reaction, shareholder, corporate ownership around the world ,managers and boards of directors, independent directors compensation family, owned firms, corporate governance and the media, the king 111 report on corporate governance and ethical guidelines in Zimbabwe

MBM732 Strategic Planning and Management

This course aims to make the student know how an organization is managed in a strategic sense, as well as, bridge for the student, the gap between theory and actual business practice, develop the student awareness of analytical techniques, managerial models and academic theories, which contribute to strategic planning and management. The course will cover topics such as the concept of strategy, strategic planning and formulation concepts, crafting strategy: environmental assessment techniques, crafting strategy: internal analysis, objective setting, strategic analysis, strategic choices/options and valuation of options, strategic implementation, administering and organizing for the strategic planning function

MBM734 Strategic Change Management

The course aims to acquaint students with the management of change, how to develop successful organizations, lead change: introducing strategic change, issues in the management of change, strategies for the implementation and change, transformational change and strategic regeneration, power and politics in strategic change. The course will cover topics and aspects such as change management, adapting to change, systems approach, planned change, changing of culture, corporate culture, cultural resistance and how to overcome it, issues in the management of change, strategies for the implementation of strategic change, transformational change and strategic regeneration, power and politics.

MBM735 Management of Organisational Behaviour

The course will cover topics such as explanation of key concepts, why study organizational behaviour, people's entry into organization, leadership, communication, conflict and change and discussion issues.

MBM743 Managerial Economics

This course aims to bridge the gap between economic theory and the challenges faced by modern managers, offering a holistic approach to strategic decision-making. The course will cover topics such as introduction to managerial economics, demand analysis, supply analysis, market structures, pricing strategies, production and cost analysis, profit analysis, game theory, managerial decision-making, global managerial economics, ethical considerations in managerial economics and lastly emerging trends and issues.

MBM737 Financial Accounting

This course aims to equip students to be able to use financial accounting information to make informed judgments and economic decisions. The course will cover topics such as financial reporting and accounting cycle, financial distress prediction and valuation of companies

MBM738 Strategic Information Management and E-business

This course is aimed at reinforcing the strategic role of information in the survival of organizations in the modern and dynamic environment of technology. It also equips students with adequate knowledge and skills of e-business to understand and appreciate its critical role in the world of business. The course will cover topics such as the strategic importance of information, and indigenous and endogenous knowledge systems. Information processing systems, the computer as a basis for modern information systems, data storage and file organization, importance of e-business, opportunities offered by e-business, developing e-business systems, the environment for e-business, web server hardware and software, electronic information networks and security and ethical challenges to e-business.

MBM739 Business Research Methods

This course is designed to enable students to develop a deeper understanding of the research, broaden and strengthen understanding and application of the research process, apply research findings in decision-making processes, conduct meaningful and useful research and explain different research approaches available and applicable to Business sciences and management. The course will cover topics such as research concepts, research topic, research process, literature review, research designs, data collection methods and techniques, data presentation, analysis and interpretation, call for a proposal, research report writing, methodology and assessment.

MBM740 Learning Organisation

The module will also explore the dynamics of how organisations learn through the empowerment of individuals in the organisation. It goes beyond the incidence of mistakes as a punishable offence and treats these as opportunities to learn and enhance organizational performance. It will also examine the leadership regimes necessary for such organizational learning.

MBM762 Strategic Leadership

This module emphasizes the role of leaders in the twenty-first century organisation and their role in leading strategy formulation and implementation, role modelling, mentoring, training and performance management.

MBM742 Strategic Financial Management

The course aims to acquaint students with the contents and tools of financial management in a corporate context in particular and social setting in general. The course will cover Page 297 of 443 topics and aspects such as basic concepts of financial management, long-term investment decision, working capital management, long-term financial and dividend decisions, financial analyses, planning and forecasting, special issues and/or topics in financial management.

MBM770 Dissertation

The dissertation shall constitute the second semester of level two. Each candidate is required to submit a dissertation of approximately 20000 words on a topic relating to his/her area of managerial interest and approved in writing by the dissertation supervisor.

MBM744 Production and Operations Management

This course introduces students to the concepts of production and operations in an organization production and operations management help the students to be familiar with the production process and the operation strategies that any organization can use. Students should be able to plan and control as a component of production and operation management. The following topics need to be included as a way of helping students familiarize themselves with the production industry, production design, service production, productivity, layout facilities, location and design of the plant, inventory management and quality management among others. The students need to know how to calculate productivity as this will help them make wise decisions in the industry. Students also need to be familiar with calculations on process flow, forecasting and capacity planning. Students need to understand that any organization has its own production and operations systems and these need to be managed with knowledge.

MBM745 Quality Management

The aim of this module is to introduce the concept of Total Quality Management and Best Practice, examine and explore the philosophies and concepts, best practice tools and macro techniques as well as the detailed stages, requirements, structures and the processes in total quality management and best practice. The course will cover topics and aspects such as global competitiveness and quality costs, quality gurus, their philosophies and the new paradigm, strategic quality management planning, leadership and change, education and training empowerment, business excellence models using self-assessment as a strategy for achieving business excellence, benchmarking, business process reengineering, balanced scorecard and statistical process control- Total Quality Management Seven Old Tools.

MBM 772 Corporate Entrepreneurship
This module aims to equip students with corporate entrepreneurship skills. The module covers principles of corporate entrepreneurship, processes, benefits, challenges and models of corporate entrepreneurship in line with changes in the corporate environment.

MBM773 Financial Statement Analysis
This module is designed for those who expect to read, interpret and analyse Financial Statements. The module adopts a modern approach to the topic of Financial Statement Analysis. Although some attention is devoted to the mechanic of dissecting Financial Statement, primary emphasize lines on development of an understanding of the market environment in which Financial information is used

MBM771 Management Accounting
This **module** aims to equip students with decision-making skills in the corporate world. This module covers costing, budgeting, standard variance, and other aspects of management accounting,

MBM774 Corporate Governance II
This module equips students with corporate governance skills to enable the candidates to implement the concept of governance. The module covers, corporate governance practices, disclosure and transparence, disputes, corruption, company distress, governance in SMEs and Family businesses.

REGULATION FOR MASTER OF COMMERCE DEGREE IN STRATEGY AND INNOVATION (BS21)

Duration	2 years
Minimum Credit load	360
Maximum Credit Load:	430
Maximum MBKs Credit Load	288
ZNQF Level	9

1. ENTRY REQUIREMENTS

Entry requirements: A minimum of a good first degree in Business Management or equivalent and two (2) years post graduate work experience and a minimum degree class of 2.1.

2. OBJECTIVES (PURPOSE OF THE PROGRAMME)

2.1 The Master of Commerce in Strategy and Innovation at the Midlands State University is a two-year programme for students with an undergraduate background in accounting, management, human resources and other equivalent qualifications. The programme is for students interested in careers in all facets of strategic management and business innovation. The programme provides an opportunity to obtain a more comprehensive foundation in strategic management and added knowledge in specific business-related areas.

2.2 The programme aims to develop sound conceptual, technical, analytical and communication skills that are required to succeed in the strategic management and corporate governance. To this end the programme employs a rigorous case analysis approach, which requires students to exercise their analytical abilities and develop effective verb and written communication skills.

2.3 The programme also aim to instil a deep appreciation and respect for the crucial role managers play in a free enterprise economy and ever-changing business environment.

3. ENTRY REQUIREMENTS

3.1 To qualify for entry into the Master of Commerce in Strategy and Innovation degree programme a student should have a good Honours degree in management, accounting, human resources management and/or other tertiary qualification approved by the Department.

4.PROGRAMME CHARACTERISTICS

4.1 Areas of study:	Business Innovation, New venture creation, Digital transformation strategies
4.2 Specialist Focus:	Innovation; Leadership and creativity; Corporate Innovation; Global technoprenuership and Data Analytics.
4.3 Orientation:	Training people with potential to get into Executive Management levels and board leadership.
4.4 Distinctive Features:	The programme is for students interested in careers in all facets of strategic management. The programme provides an opportunity to obtain a more comprehensive foundation in strategic management and added knowledge in specific business-related areas.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

5.1 Employability:	Consultancy, academia, research, Operations Management, Executive management positions and boards.
5.2 Further Studies:	Doctoral studies in strategic management, Innovation management and business leadership.

6. PROGRAMME DELIVERY

Formal examinations will normally be held at the end of each semester.

6.1 For a student to be admitted to the examinations they must have:

- 6.1.1 Satisfactorily attempted approved modules of study including submission of required written assignments and term papers.
- 6.1.2 Attended compulsory classes.
- 6.1.3 Participated in prescribed seminars, tutorials and practical classes.
- 6.1.4 Paid the prescribed fees in accordance with the General Regulations.

6.2 Formal examinations will normally be written papers, but in some cases the examiner may test the candidate orally.

6.3 Continuous assessment shall account for 40% of the overall assessment, while the final examination will account for 60% of the overall assessment.

6.4 The aggregate pass mark shall be 50% and the candidate must have obtained at least 50% in the examination.

7. **LEARNING OUTCOMES**

By the end of the programme students will be able to:

7.1 Create new businesses and improve existing processes through application of acquired strategy and innovation expertise.

7.2 Advance expertise in the design and construction of innovative ideas and products that meet the needs of enterprises and economies both locally and internationally.

7.3 Exhibit advanced practical and theoretical understanding of technology in the formulation, implementation and evaluation of corporate strategy.

7.4 Use understanding of modern businesses operations in fostering strategic improvements as well as new venture creations.

7.5 Apply the techniques used in strategy and innovation in order to design strategic plans, business proposals, and construct innovative strategies and products.

7.6 Utilise key intellectual capital which includes Intellectual Property and organizational

8. GENERAL PROVISIONS

A student requires 360 credits to complete this programme.

9. SUPPLEMENTARY EXAMINATIONS

10. PROVISION FOR PROGRESSION

A student shall progress automatically from one level to the other even carrying the modules from the previous level.

11. DEGREE WEIGHTING CLASSIFICATION

For purposes of degree classification, the parts of the degree programme will be classified as follows:

Level 1 Semester 1 and 2	50%
Level 2	50%

12. CLASSIFICATION

The following marking scheme shall be used for all the modules and grading of the degree.

80%-100%	Distinction
70%-79%	Merit
60% - 69%	Credit
50%-59%	Pass
Below50%	Fail

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MSI730	Strategic Management and Innovation	18
MSI 731	Global Technoprenuership	18
MSI 734	Intellectual Property Law	18
MSI 732	Innovation and Commercialization	18
MSI 831	Product and Service Innovation	18

Level 1 Semester 2

MSI 736 Sustainable and Responsible Innovation	18
MSI 739 Digital Transformation Strategies	18
MSI 735 Innovation and Design Thinking	18

MSI 737 Research Methods	18
MSI 738 New Venture Creation and Simulation	18

Level 2 Semester 1

MSI 832 Leadership and Managerial Creativity	18
MSI 834 Knowledge, Innovation and Technological change	18
MSI830 Strategic Project Management	18
MSI 833 Corporate Financial Strategy	18
MSI 835 Business Model Innovation	18

Level 2 Semester 2

MSI 836 Data Analytics	18
MSI 837 Dissertation	90

MSI730 Strategic Management and Innovation

This course intends students to comprehend innovation strategically as a powerful means of competitive differentiation, enabling firms to penetrate new markets and achieve higher margins. As students race to better understand strategic management and innovation, a wide range of work will include the following topics: environmental analysis, strategic management schools of thought/perspectives, functional strategy, business strategy, corporate strategy and innovation, leadership and innovation, core competences and innovation, patents, and standards and innovation.

MSI731 Global Technoprenuership

This module equips students with knowledge and understanding of the processes involved in new venture creation, growth and management in foreign countries. The module takes into account the peculiarities of different economies, societies and regulatory environments together with the opportunities and challenges the environments expose to global technopreneurs.

MSI732 Innovation and Commercialisation

the module explores the role innovation plays in development of new products and services, and in the improvement of existing ones within a technological context. areas to be covered include the appreciation of the commercial, financial and marketing aspects of innovation projects, the requirements for successful commercialisation, funding of commercialisation as well as valuing intellectual property.

MSI734 Intellectual Property Law

The module exposes students to the treatment of laws relating to patents, trademarks, confidential information, designs and copyright. These concepts are used to examine the protection measures provided by the law in regard to ideas, inventions, information and other forms of creative effort. The module explores how law balances interests and protects investment, taking into account public welfare and technological developments.

MSI735 Innovation and Design Thinking

the module aims to equip students with creative strategies in designing business processes. design thinking in business uses the designer's sensibility and methods to match people's needs with what is technologically feasible. the module uses a set tools and skills that inform and facilitate the innovation process, from visual tools like sketches, mind maps and prototypes to mental processes like brainstorming, building on the ideas of others, and creating scenarios.

MSI736 Sustainable and Responsible Innovation

This module is designed to give students a comprehensive understanding of the sustainability challenges facing organisations and to highlight business risks and opportunities emerging as a consequence of global environmental change. The module seeks to use frame-breaking approaches to problem-solving and the development of organisational strategies. It provides a rigorous conceptual

understanding of sustainable and responsible innovation, as well as an understanding of the practical challenges for innovation management.

MSI737 Research Methods
Business research concepts and methods, the traditional and scientific methods, the research process, problem definition, research design, questionnaire construction and administration, sampling basis, sources of research data, collection and interpretation, research application and uses, data presentation and analysis, ethics, and research proposal will be covered.

MSI738 New Venture Creation and Simulation
The module equips students with a comprehensive understanding of the entrepreneurial process and the development of new ventures. It combines theory and experiential assignments to introduce the principles of creativity and innovation. Areas covered include the role of innovation for growth and wealth creation, effective innovation processes and the associated management issues.

MSI739 Digital Transformation Strategies
The module provides research-based insights on disruptive innovations and what it takes to build and manage successful business models in this complex era Disruption is a force that is changing the corporate landscape, and organisations must transform at a rapid pace to meet the demands of digital. This unprecedented disruption is happening fast, as many well-known incumbents will attest. Succeeding in this volatile environment requires not only a thorough understanding of what disruption is and how to deal with it but also the knowledge of how to disrupt others.

MSI 830 Strategic Project Management
This module seeks to provide solutions to strategic projects management challenges experienced in various organizations. Course participants will be exposed to the "business side" of strategic project management, including how to make fact-based decisions about the short and long-term projects, how to create and foster a project culture that works to achieve rapid outcomes, and how to work within the organization to break down communications barriers and create relationships that are vital to the strategic project environment. Topics to be covered include project strategy, strategic planning for project management, project management theories, project management approaches, project monitoring and evaluation.

MSI831 Product and Service Innovation
This module examines the importance of new product and service development to the achievement of an organization's objectives and how the process of innovation can be managed. Areas covered include new product development process, strategic opportunity identification, how to generate new product concepts and ideas, mapping customer perceptions, segmentation, product positioning, forecasting market demand, product design, market entry strategies, and testing.

MSI832 Leadership and Managerial Creativity
This module focuses on ways to manage creativity through effective leadership and teambuilding by introducing flexible and iterative innovation management process models. Leadership theories and concepts will improve the ability to apply the theories through critical thinking, and developing leadership skills is a major focus in the module.

MSI833 Corporate Financial Strategy
The module focuses on financing, investment, and risk management decisions of corporations in creating shareholder value. The module analyses the mixture of debt versus equity decision in detail considering the impact of taxes, agency problems, and asymmetric information. It analyses the different types of financial products such as callable and convertible debt and financial derivatives. Different methods of raising external capital and payout policies such as dividends and share repurchases will be covered.

MSI834 Knowledge, Innovation and Technological Change

This module aims to build an appreciation of the importance of innovation and to build an awareness of the role of technology, both strategically and operationally, within organisations. The module specifically looks at the opportunities enabled by technology.

MSI835 Business Model Innovation
This module moves away from the heroics to take a scholarly view of what is involved in business model innovation drawing lessons from new business models exemplified by firms such as AirBnB and Uber. The students will have an opportunity to learn how technological changes and their implications on new ways of creating value proposition.

MSI 836 Data Analytics
The module will build upon your hard-won experience and prepare you for the challenges of management and big data. Covering fields including data design, data handling, and decision-making, An understanding of how data-driven models can improve your ability to make decisions in a fast-paced and uncertain world, and the ability to use modelling to predict future trends.

MSI837 Dissertation/Research Project
Students will carry on with their independent research in a business discipline area of choice started in the previous semester. Students will work hand in hand with a designated academic advisor who will mentor the student throughout the research process. Frequent consultations between the student and their mentor will substitute lectures and tutorials as held in other modules.

DEPARTMENT SUPPLY CHAIN, INSURANCE AND RISK SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN RISK MANAGEMENT AND INSURANCE (BS13)

Duration:	4 years
Credit Load:	528
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384

1. PURPOSE OF THE DEGREE

To develop a graduate who is well versed in all aspects of risk management and the role of insurance in risk financing.

2. ENTRYREQUIREMENTS

1.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce in Risk Management and Insurance Honours Degree programme, a candidate, in addition to satisfy the minimum conditions as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at 'O' Level, must have obtained a pass in 'A' Level subjects: Economics, Management of Business, Accounting or Mathematics.

2.2 Visiting School Programme

To qualify for entry into the Bachelor of Commerce in Risk Management and Insurance Honours Degree (Visiting School) programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English Language at 'O' Level must have: -

Either:
-A Certificate of Proficiency in Insurance,
-A Certificate of Insurance Practice,
-A Diploma in Insurance,
-An Advanced Diploma in Insurance,
-National Diploma in Business Studies,
-A Higher National Diploma in Business Studies,
-Any other equivalent tertiary qualification
Or
–Passes in **2** any 'A' Level subjects
And
–At least two years working experience in a relevant field
–Proof of employment

2.3 **Mature Entry**
Refer to Section 3.3 of the General Regulations

2. PROGRAMME CHARACTERISTICS

2.1 **Areas of study:** The broad perspective on risk management with a special focus on enterprise risk management, financial risk management, Investment analysis, insurance regulation and supervision, marketing of financial services, reinsurance and corporate finance, and financial mathematics.

2.2 **Specialist focus:** Risk Management and Insurance

2.3 **Orientation:** Research and innovation-oriented. Teaching and learning are professionally oriented and focused on practical aspects.

2.4 **Distinctive features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centered approach.

3. CAREER OPPORTUNITIES

Graduates for the Bachelor of Commerce Risk Management and Insurance can pursue careers as Insurance brokers, risk managers, risk management officers, pension's consultants, pension administrator, compliance officers, and loss adjustors.

4. PROGRAMME DELIVERY

4.1 Teaching and Learning Methods:

Lectures, tutorials, seminars, group work, industrial visits, Work-Related learning, research projects and individual independent study.

4.2 Assessment Methods:

Written and oral examinations, tests, term papers, seminar presentations, Work-Related learning report, final-year research project report, and continuous assessments.

5. LEARNING OUTCOMES

By the end of the programme a graduate will be able to:

- Apply analytical skills in the day-to-day operations of businesses and solve problems in a logical manner, particularly in claims investigations.
- Explain technical information to various internal and external stakeholders.
- Make sound decisions in underwriting insurance policies.
- Negotiate policy terms with clients, brokers and reinsurers.

6. GENERAL PROVISIONS

6.1 A candidate is required to undertake a minimum of five (5) modules per semester unless one is carrying over or retaking certain modules from the previous academic level or semester, respectively.

6.2 Each module is worth twelve (12) credits except for the Work-Related Learning level, which is worth one-hundred and twenty (120) credits and the dissertation which is worth 24 credits.

6.3 Successful completion of Work-Related Learning component at Level III is compulsory for all candidates.

6.4 A student requires 528 credits to complete this programme.

8 PROGRAMME ASSESSMENT

Refer to Section 6 of the General Regulations and Section 5 of the Faculty Regulations.

9 PROVISION FOR PROGRESSION

Refer to section 6 of the faculty regulations.

10 FAILURE TO SATISFY EXAMINERS

RefertoSection9of the General Regulations.

11 GRADING AND DEGREE CLASSIFICATION

RefertoSection12FacultyRegulations.

12. PROGRAMMESTRUCTURE

Level 1 Semester 1

CODE	DESCRIPTION	CREDITS
CS131	Communication Skills	12
BM 131	Principles of Management	12
HCS135	Introduction to Information Systems	12
EC 131	Principles of Economics 1	12
BM 132	Business Communication	12
ACC135	Financial Accounting for Business 1	12
RMI 130	Risk and Insurance	12

Level 1Semester 2

RMI 131	Introduction to Risk Management	12
BM 135	Commercial Law	12
ACC136	Financial Accounting for Business 2	12
CMI 131	Introduction to Marketing	12
BF 134	Financial Mathematics	12
ACC144	Quantitative Analysis for Business 1	12
EC 134	Principles of Economics 2	12

Level 2 Semester 1

RMI 132	Insurance Law	12
RMI 232	Operational Risk Management	12
GS211	Gender Studies	12
BF233	Corporate Finance A	12
RMI234	Health and Disability Insurance	12
ENT 131	Introduction to Entrepreneurship	12
RMI 233	Liability Insurance	12

Level 2 Semester 2

RMI235	Life Insurance	12
RMI239	Introduction to Reinsurance	12
BF234	Corporate Finance 1B	12
RMI238	Insurance Broking and Client Services	12
RMI237	Retirement Funding	12
RMI 230	Property and Pecuniary Insurance	12
RMI 240	Transportation Insurance	12

Level 3 Semester 1

RMI 331	Work-Related Learning Continuous Assessment	40

Level 3 Semester 2

RMI 332	Work-Related Learning Employers' Assessment	40
RMI 333	Work-Related Learning Report	40

Level 4 Semester 1

RMI430	Enterprise Risk Intelligence	12
RMI434	Micro-insurance	12
RMI431	Advanced Reinsurance Practice	12
BF433	Financial Risk Management	12
RMI432	Occupational Health and Safety	12

Level 4 Semester 2

MM436	Services Marketing	12
BM438	Strategic Management	12
RMI435	Finance and Management Accounts for Insurance	12
RMI437	Research and Innovation Project	24

13. MODULE SYNOPSES

The synopsis provided is for RMI-coded modules. For the rest, reference is made to the respective departments where the modules are taught from.

RMI 130 Risk and Insurance

The aim of the module is to familiarize students with risk financing with special emphasis placed on insurance and self-insurance. It deals with guidelines for insurable risks, risk assessment and measurement of insurable risks, principles and practice of insurance and self-insurance, the insurance industry, including the role of the government. It introduces the students to international insurance markets and provides an overview of short-term and long-term insurance policies, claims management and the marketing of insurance services.

RMI 131 Introduction to Risk Management
The aim of this module is to introduce students to an early appreciation of basic concepts of risk and risk management. It deals with the concepts of risk and uncertainty, risk definitions and classifications, and the application of principles of management to risk management. Emphasis is placed in the generic risk management model with special focus in risk identification/awareness, risk measurement and techniques for dealing with risk. This module provides a foundation for the rest of the risk management and insurance modules to be taken later in the study programme.

RMI 132 Insurance Law
This module reviews the development of insurance as an introduction to Insurance Law. It explores the relationships between the insurer, agent and insured as created by the insurance contract. It also discusses other agency legal issues while providing a basic understanding of the principles of contract law and how they relate to insurance. It also looks into the interpretation of insurance contracts; conditions precedent; representations; warranties; terms; conditions; insurable interest; rights of beneficiaries; exemptions; excess liabilities; waiver and estoppels; subrogation; and controls on the insurance industry. The module also examines principles governing insurance claims and their application.

RMI 233 Liability Insurance
The aim of this module is to familiarize students with concepts, principles and practices for dealing with risks connected to legal liability. The module also examines the latest enactments and insurance products arising out of social and economic, specimen wordings of policy, insurance proposals, risk assessment and rating and claims management. Different types public liability, professional liability, product liability, and environmental liabilities insurances form part of the module content.

RMI 230 Property and Pecuniary Insurance
The aim of this module is to provide students with the key concepts, principles and underwriting criteria applicable to property and pecuniary portfolios handled by the majority of insurers domestically and internationally. Particular emphasis will Page 128 of 443 be placed on the practical application of the concepts through case studies and group assignments wherever appropriate

RMI 232Operational Risk Management
The aim of the module is to expand on the knowledge gained in RMI131 Introduction to Risk Management and RMI130 Risk and Insurance. The Operational risk management module impacts knowledge that will help students in identifying, assessing, and managing operational risks that can affect a firm and the different strategies to minimize their impact on an organization. Students will be taught how to design an effective Operational Risk Management Framework that includes the governance and oversight of all operational processes

RMI235 Life Insurance
The module seeks to develop in the student: knowledge and understanding of the scope of life assurance, different types of life insurance policies, appropriate uses and administration of life insurance; underwriting of life insurance policies including medical underwriting, financial underwriting, genetic underwriting and technological underwriting, claims management; legal and regulatory aspects of the conduct of life insurance business.

RMI239 Introduction to Reinsurance
The aim of this module is to give students an in-depth understanding of how reinsurance operates. Areas to be covered include reinsurance methods; life, non-life and liability reinsurance programmer design, legal issues relating to reinsurance, reinsurance contract wordings and other key issues associated with this business in today's world insurance markets. Attention will also be paid to financial reinsurance.

RMI 237 Retirement Funding
The aim of this module is to equip students with knowledge and skills in designing retirement plans. Emphasis will be placed on the development of analytical and critical abilities in relation to retirement rules, appropriate packages for different stages of life preparation for retirement, different strategies for funding retirement, taxation, regulations and techniques for assessing, measuring and managing risks, and costs that can affect the provision of financial security at retirement.

RMI 240 Transportation Insurance
This module provides an overview of transportation insurance, which is designed to protect businesses and individuals from financial losses arising from damage to or loss of the vehicle/ship/plane and/or goods during transportation. This module covers the various types of transportation insurance, that is, marine insurance, motor insurance and aviation insurance. The module will deal with the different types of insurance covers available to each class of insurance and the underwriting and claims procedures.

RMI 238 Insurance Broking and Client Services
The aim of this module is to equip students with knowledge of the role played by intermediaries in the provision of risk management and insurance services. Emphasis will be placed on the functions, roles, duties, and regulations of insurance brokers. The module will also include marketing and competition in the risk management and insurance intermediaries sector.

RMI234 Health and Disability Insurance
The aim of this module is to equip students with knowledge and skills of designing, marketing and managing health insurance, disability insurance and managed care plans provided under private sector, non-profit making and government programs. Emphasis will be placed on funding of the different schemes, regulations, tax, fraud and costs.

RMI433 Enterprise Risk Intelligence
Enterprise Risk Intelligence is a crucial aspect of modern-day business operations. The module will give students an appreciation of the different Enterprise-wide risk management processes that are used globally. The different risk management frameworks include COSO and ISO31000. The students will be able to choose and design a suitable risk management framework and then the different strategies on how to implement the framework. The module will also deal with risk governance and risk culture promotion techniques.

RMI431 Advanced Reinsurance Practice

Specific issues in property; casualty, marine, aviation and life reinsurance and their application to the various forms of reinsurance; underwriting and transaction of reinsurance business and renewals in these classes of reinsurance; and claims management.

RMI432 Occupational Health and Safety

The module provides an outline of the Principles of Health and Safety Management. it covers the following aspects: history of occupational health and safety, occupational health hazards, occupational exposure limits, job stress, shift work and extended work shifts, evaluation of chemical hazards, emergency response, principles of toxicology and epidemiology, risk perception, reality and risk control, material handling and storage, crane and hoist safety.

RMI434 Micro-insurance
This module aims to provide an introduction to the concept of micro-insurance, including its history and an overview of the different types of micro-insurance products available in life, health, and property classes of insurance. The module will include the development and designs of micro-insurance products, implementation as well as benefits and limitations of micro-insurance to different stakeholders.

RMI435 Finance and Management Accounts for Insurance
Analysing insurance accounting and financial statements for insurance companies is very different from analysing those of other companies and thus presents unique challenges and industry-specific issues. This module is therefore designed to provide an excellent foundation for evaluating insurance company financial statements, industry ratios, capital adequacy, regulatory issues, rating agency assessments, valuations and early warning signs of financial distress. Students will be equipped to recognize the impact of differing accounting standards, reserving policies and changes in external variables such as interest rates and asset prices on the financial statements. Attention will be paid to the risks inherent in the main products offered by life, non-life and reinsurance companies to help students recognize how these are reflected in the financial statements.

RMI 437 Research Project
Students will be expected to complete a research project on a topic of their choice but limited to the taught modules. The project is a consolidation of the theoretical knowledge gained in the taught modules and the practical experience gained from Industrial Attachment.

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT (BS12)

Duration	4 years
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the Faculty Regulations and the General Academic Regulations for Undergraduate degrees herein referred to as General Regulations.

1.2 The degree will be awarded to candidates who have successfully completed the programme and passed examinations in accordance with regulations set in the Faculty Regulations.

1.3 The duration of the degree programme shall normally be a minimum of four years including a period of Work-Related Learning in a relevant industrial/ commercial setting during the third level.

2. LEARNING OUTCOMES

By the end of the study students should be able to: -

a) Develop new product ideas and systems to improve the operations of supply chain systems

b) Advise businesses, states and communities on ways in which supply chains can enhance innovation and industrialization.

c) Demonstrate understanding of and appreciation for the importance of the impact of globalization and diversity in modern organizations.

d) Display an ability to engage in critical thinking by analysing situations and constructing and selecting viable solutions to solve problems.

e) Collaborate effectively with various groups and stakeholders.

f) Exhibit knowledge of current information, theories and models, and techniques and practices in supply chain management, logistics and retail environments

3. CAREER PROSPECTS

The Bachelor of Science Honours degree in Logistics and Supply Chain Management offers students a broad understanding of general management, accounting and economics as a foundation for the appreciation of end-to-end supply chain issues. On completion of the degree graduates can pursue careers as procurement and merchandising managers and supply chain operatives and managers. Opportunities also arise as supervisors of Transport, Logistics, Distribution and Supply Chain Management operations as well as various levels in the management of ports of entry, customs warehouse management, inventory, and brand management.

4. ENTRY REQUIREMENTS

4.1 Normal Entry

a) To qualify for normal entry into the Bachelor of Commerce Honours Degree in Logistics and Supply Chain Management programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English Language at 'O' Level, such candidate must have obtained at least two passes in any 'A' Level subjects.

b) To qualify for entry into the Bachelor of Commerce Honours Degree in Logistics and Supply Chain Management (Visiting School) programme a candidate, in addition to satisfying the minimum requirements prescribed under the General Regulations and the Faculty Regulations for English Language at 'O' Level must have: -

Either:

–Diploma of the Institute of Bankers (IOBZ)
–HND/ND in Business Studies
–HND/ND in Accountancy
–HND/ND in Purchasing Management
–Diploma in Education
–Institute of Administration and Commerce (IAC)
–Institute of Personnel Management Zimbabwe (IPMZ) Diploma
–Diplomas of the Institute of Real Estate Management

–Diploma in Business Administration (ZIM)
–Any other relevant tertiary qualifications
Or
–Passes in **2** any 'A' Level subjects
And
–At least two years working experience in a relevant field
–Proof of employment

4.2 Mature Entry

Refer to Section 3.3 of the General Regulations.

4.3 Special Entry

a) Candidates who have successfully completed Higher National Diploma in Business Studies or have obtained equivalent qualifications may apply for direct entry into Level II of the degree programme.
b) Direct entry into Level II of the programme is subject to availability of places.
c) Candidates admitted under Regulation 3.2 above will normally be exempted from Level I on a module-by module basis.
d) No candidate may complete the degree in less than three academic levels.
e) Successful completion of the Work-Related Learning component at Level III is compulsory for all candidates.

5. GENERAL PROVISONS

5.1 A candidate is required to undertake a minimum of five (5) modules per semester unless one is carrying over or retaking certain modules from the previous academic level or semester, respectively.

5.2 Each module is worth twelve (12) credits, Work-Related Learning level which is worth one hundred and twenty (120) credits and the dissertation which is worth twenty-four (24) credits.

5.3 The department may accept students who have been discontinued from other programmes on condition that they meet the entry requirements of the department and subject to availability of places.

5.4 A student requires 504 credits to complete this programme.

6. PROGRAMME ASSESSMENT

Course work	30%
Written examinations	70%

7. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Regulations.

8. PROVISIONFORPROGRESSION

Refer to Section 6 of the Faculty Regulations.

9. WORK-RELATED LEARNING GENERAL GUIDELINES

Refer to Section 8 of the Faculty Regulations.

10. GRADING AND DEGREE CLASSIFICATION

Refer to Section12 of the Faculty Regulations.

11. DEGREE WEIGHTING

Refer to Section 13of the Faculty Regulations.

12. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
BM131	Principles of Management	12
EC131	Economic Principles 1	12
HCS135	Introduction to Information Technology	12
CS131	Basic Communication Skills	12
LCS131	Principles of Supply Chain Management	12
LCS132	Retail Management	12
ACC135	Financial Accounting for Business 1A	12

Level 1 Semester 2

Code	Module Description	Credits
LSC133	Principles of Logistics and Transport Management	12
LSC135	Business Law	12
LSC136	Introduction to Business Statistics	12
ACC136	Financial Accounting for Business 1B	12
LSC134	Principles of Procurement	12
EC134	Economic Principles 2	12

Level 2 Semester 1

Code	Module Description	Credits
LSC231	Public Procurement	12
ENT131	Entrepreneurship1	12
GSB211	Gender Studies for Business	12
LSC232	Supply Chain Risk Management	12
LSC233	Transport Economics	12

Level 2 Semester 2

Code	Module Description	Credits
LSC234	Buyer and Consumer Behaviour	12
LSC 236	Promotions Management	12
LSC 238	Passenger and Cargo Transport Management	12
LSC 235	Customs Practice and Documentation	12
LSC 237	Business Research Methods	12

Level 3 Semester 1

Code	Module Description	Credits
LSC330	Work-Related learning placement report	40

Level 3 Semester 2

Code	Module Description	Credits
LSC331	Work-Related Report	40
LSC332	Academic Supervisor Report	40

Level 4 Semester 1

Code	Module Description	Credits
LSC431	Brand Management	12
LSC432	Inventory, Stores and Warehouse Management	12
LSC433	Global Sourcing	12
LSC434	Project Management	12

Level 4 Semester 2

Code	Module Description	Credits
LSC435	Corporate Governance and Business Ethics	12
LSC436	Contract and Relationship Management	12
LSC437	Research Project	24
LSC438	Consumer Protection	12

13. MODULE SYNOPSES

LSC131 Principles of Supply Chain Management

The module aims to cover foundational concepts and principles that form the bedrock for the management of firm and inter-firm activities. Topics cover definitions of logistics and supply chain management; types of supply chains; design and mapping. Sustainability, risks and resilience. Efficiency, coordination and responsiveness.

LSC132 Retail Management

Retail Management aims to equip candidates with the principles and managerial aspects of retailing. It explores the domain of retailing which is marketing to individuals and households. Emphasis is placed on management aspects of retailing such as the retail environment and its relationship to consumer demographics and trends, the retail mix strategy, merchandising, vendor relations and locations as they apply to both goods and services. The trends and developments and their impact in retailing are examined. Candidates are given real-life retailing cases upon which they make retail decisions. Other major areas covered include the determination of a target market, site selection decisions, staffing and training employees, outlet promotion, customer services, and control of outlet operations.

LSC133 Principles of Logistics and Transport Management

This module aims to provide an understanding of the principles, concepts and techniques of logistics and transport management. The module outline includes fundamentals such as definition and evolution of logistics and transport, research activities in the transport sector, stakeholders in the transport sector, transport modes and their economic operating characteristics, containerization, the role of transport in business, technology and transportation, total quality of transport in light of total quality management (TQM), logistics organization design and development, customer response principles and systems, inventory planning and management, warehouse operations, transportation and distribution management, supply management, logistics and supply chain information systems, logistics performance, cost and value measures.

LSC134 Principles of Procurement

This module aims at providing an understanding of the principles, concepts and techniques of procurement. Fundamentals in Principles of Procurement should focus on the purchasing environment and globalization purchasing drivers, theoretical frameworks, tools and

techniques used in international purchasing supply chain management, Trade Promotion Organizations [Common Market for Eastern Southern Africa (COMESA), Southern African Development Countries (SADC), World Trade Organisation (WTO), United Nations Conference on Trade and Development (UNCTAD), International Trade Conference (ITC), Zimbabwe Trade (ZIMTRADE), European Union (EU), Government, African Union, Export Processing Zone Association (EPZA), Zimbabwe Investment Centre (ZIC), Zimbabwe Chamber of Commerce (ZNCC)]. The new supplier research and Research and Development (R&D) the five rights of purchasing, supplier research and appraisal, negotiation, tendering and contract management to be included. The procurement process strategies, tactics and plans, ethics and social responsibility in procurement and global purchasing are key.

LSC135 Business Law

The module aims to develop knowledge and understanding of the general legal framework within which an accountant operates and to develop an awareness of and an ability to understand common law. Students will learn to appreciate the legal framework, and sources of Zimbabwean law and enable them to appreciate the legal implications of business transactions. The module will cover the nature of law, law of contract, sale and purchase, lease, agency, employment and delict.

LSC136 Introduction to Business Statistics

The module focuses on mathematical and statistical concepts, tools, skills and techniques in the analysis, interpretation and application of numerical information and testing of theoretical fundamentals. It encompasses basic concepts of statistics, representation of statistical results, measures of central tendency and dispersion including probability concepts, confidence intervals, hypothesis testing, index numbers, time series analysis, regression and correlation analysis and financial calculations (simple and compound interest, annuities, sinking funds and amortization schedules.)

LSC231 Public Procurement

Government expenditure on non-pay-related areas covers a vast range of equipment, goods and services: weapons systems, stationery, furniture, uniforms, food, capital projects including consultancy services, banking services, information systems and services, the management of facilities, as well as medical services, road building and maintenance and utilities. This requires an insight into the background and context of government procurement and the changing and challenging nature of such procurement in the light of commercial best practices. As such Public Procurement law is an area of interest for procurement practitioners both from the public and private sectors. There are major developments in most jurisdictions in regard to public procurement. The module will look into the objectives of Public Procurement system, Government Procurement v Private Sector Procurement, an overview of UNCITRAL Model Law, an overview of EU Procurement Directives, World Bank Procurement Rules, an overview of WTO GPA, an overview of Public Procurement Laws: such as COMESA, the Public Procurement and Disposal of Public Asset Act 2018. Consequently, the module covers current Procurement Law and Policy from international perspective.

LSC233 Transport Economics

The module aims to offer a clear understanding of the basics of transport economics. Fundamentals in Transport Economics to include financial and economic implications and types of transport costs, financial implications of transport regulations, and productivity standards. Distribution planning, security in transport and warehousing activities, reverse logistics, distribution requirements planning, cost versus service trade-offs in investing for transport, outsourcing transport and transport broking.

LSC234 Buyer and Consumer Behaviour
Buyer and Consumer Behaviour aims to introduce students to the principles of buyer and consumer behaviour. It touches on issues such as characteristics of consumers and buyers and principles of how people make choices of what to buy. Furthermore, it deals with factors which influence buyers and consumers such as economic, social, political and psychological factors affecting buyer behaviour, Industrial buyers versus domestic buyers. How marketers can benefit from understanding consumer behaviour

LSC235 Customs Practice and Documentation
The module aims at ensuring that students understand and appreciate customs practices and related documentation. Documents used in international trade, customs regulations customs, clearing procedures and Practice the use of Customs Asycuda Manual, the Harmonised System (HS) Tariff Book, Removal in Bond (RIB), Removal in Transit (RIT), Bonded Warehouses, Reserve Bank of Zimbabwe (RBZ) exchange control measures, trade facilitation, permits & licences practice of customs clearing (a lot of practicals have to be done), Trade Agreement (Export Processing zones, World Trade Organisations (WTO) International Regulations and Conventions to be included. The Vienna Conventions International Maritime Regulations, the Customer Rules of Containerization (1992 with amendments), principles and practice of Letters of Credit (including the UCP500 and the International Chamber of Commerce (ICC) Documentary Credits Handbook), Electronic Data Interchange and case studies on shipping, the law of international trade and World Customs Union to be covered.

LSC236 Promotions Management

Meaning, role, impact and scope of Public Relations. Promoting and protecting corporate image and building corporate brand and products. Public relations vehicles/ tools. Major types of media used. Managing stakeholder relations. Role of Internet and use of modern ITCs. Press relations, product publicity, corporate communications; dealing with legislators, government and others stakeholders.

LSC237 Business Research Methods
This module aims to give the students theoretical knowledge of business research and enabling the students to carry out their research projects in their specialized area. The scientific approach, research types and the role of research in business environment are covered. It deals with the general research process which includes problem identification and definition, topic formulation, proposal development, literature review, research methodology, data analysis and interpretation, conclusions and recommendations as well as the entrepreneurial thrust of the research. Computer packages such as Statistical Package for Social Sciences (SPSS) which aid research data handling and analysis should also be covered in the module.

LSC238 Passenger & Cargo Transport Management
The module aims at imparting knowledge and understanding of the principles of passenger and cargo transport management. This module covers: stakeholders in passenger transport acquisition, vehicle routing and scheduling, transport functionality, principles and participates, fuel management as well as breakdown, vehicle maintenance and replacement policy, methods of servicing vehicles, transport regulations on local and cross boarder transport operators, training and development of staff, traffic control, legislation, licensing and permits, preventive maintenance and route viability and transport planning and investment, national transport policy

LSC331 Work-Related Report
The module is a report developed by the candidate during attachment. The thrust of the report is to relate theory covered in the university to practice. It follows a schematic report layout covering issues such as, company background, departments attached, challenges encountered, new issues learnt, any gap between information covered in the University, review of related literature to issues covered, comparison of practical issues and reviewed literature, any gap and conclusions and recommendations to the company and the University. A guideline to the development of the report is issued to the candidates as a module outline.

LSC332 Academic Supervisor Report
Approved industrial attachment of not less than 8 months and not more than 12 months is aimed at providing exposure to the students in the third year. The purpose of industrial attachment is to expose students to professional working life and give them adequate exposure to relate theory to real life situations in industry. The candidate is expected to rotate within the company in order to be exposed to all functions of that organisation. The candidate must be appraised by each departmental supervisor who assigns the student day to day duties. A percentage score should be given before leaving for the next department. The scores for each department supervisor are aggregated at the end to give a final industrial supervisor's mark which will contribute to Industrial Attachment Continuous Assessment mark. During the attachment the student is expected to complete a logbook which will be utilised by the Academic Supervisor. The academic supervisor must assess the student at least twice and award a mark on the designated instruments on each visit and the average mark will contribute to the Industrial Attachment Continuous Assessment mark.

LSC431 Brand Management
The main objective of Brand Management is to provide the student with a comprehensive and up-to-date treatment of the subject of brands, brand equity and brand marketing strategies. At the end of the module the student should be able to develop insight into how to create profitable branding strategies by building, measuring and managing brand equity. Although it is easy for competition to often duplicate manufacturing processes and factory designs, it is not easy to reproduce strongly held beliefs and altitudes established in the minds of consumers. The student will learn about the concept of brand equity, family brands, corporate brands, national versus private brands and generic brands which can provide marketers with a valuable perspective on interpreting the potential effects of various strategies on their brands. It stresses the importance of branding in marketing strategies. At the end of the module the student should be able to develop insight into how to create profitable branding strategies by building, measuring and managing brand equity.

LSC432 Inventory, Stores and Warehouse Management
The module aims at offering an understanding of the principles, concepts and techniques of inventory and stores management. Inventory, Stores and Warehouse Management covers managing inventories under different demand situations, managing inventory for different organizational contexts, statistical analysis as an inventory management tool, warehouse management, automated warehouse and inventory management systems, materials handling facilities and equipment. Material Requirements Planning and Control, warehouse location and design principles, stores procedures, stocktaking, health and safety of staff and security of stocks, materials handling and the use of warehousing software programmes to be included.

LSC433 Global Sourcing
The module aims at offering a clear understanding of the basics of global sourcing, concepts and techniques. Developing foreign sourcing**,** the global sourcing model**,** landed costs**,** post

9/11 and the effect of Global Purchasing, international procurement strategies and plans, risk management and Insurance, look east policy, developing an inbound supply chain risk management strategy and low-cost centres and their impact in Zimbabwe to be covered.

LSC434 Project Management
This module provides a systematic and thorough introduction to all aspects of project management. Projects are an increasingly important aspect of modern business. Therefore, the module underlines the importance of understanding the relation between projects and the strategic goals of the organisation. The module also discusses the technical, cultural, and interpersonal skills necessary to successfully manage projects from start to finish. It emphasises that project management is a professional discipline with its own tools, body of knowledge, and skills. Concepts are reinforced by case studies covering a wide variety of project types and industries

LSC435 Corporate Governance and Business Ethics
The aim of the module is to equip students with the necessary principles and knowledge on how organizations are run paying attention to best practices and ethical behaviours. Areas covered includes principles of corporate governance, Corruption, whistle blowing and industrial espionage, Principles of business ethics in international context Pricing, product, promotion, distribution ethics, Environmental ethics, industrial relations ethics.

LSC436 Contract and Relationship Management
The module aims at imparting knowledge and understanding of the principles, concepts and techniques of contract management. Contract Management to cover definition of terms, evolution and role of tenders and contracts in business. Ethics in procurement (public and private), principles of tendering in Zimbabwe and beyond, public and private procurement procedures, tendering process, problems associated with tenders a lot of practicals, the dangers and risks of tenders, the formation of a contract, administration of contracts, termination of contracts, dispute settlement – Arbitration and litigation and the paradigm shift in Tender Management. It will provide a comprehensive understand of the legal issues that relate to the formation of contracts, type of contracts, analyse the documentation that can comprise a commercial agreement for the supply of goods or services, analyse the legal issues that relate to the creation of commercial agreements with customers or suppliers, compare types of contractual agreements made between customers and suppliers, appraise examples of key performance indicators (KPIs) in contractual agreements, contract variations, responsibilities of contracting parties, understand the dynamics of relationships in supply chains, differentiate between different types of commercial relationships in supply chains, appraise portfolio analysis techniques to assess relationships in supply chains, identify the competitive forces that impact on relationships in supply chains, identify the processes for terminating relationships.

LSC437 Research Project
Project in International Procurement/Logistics/and Transport to deal on: A case-based research project of at least 12000 words and not more than 12 500 words on an approved Procurement, Logistics and Transport topic is required at the end of semester 8. It starts in semester 7 with the submission of approved project proposal by the student to the department. Students are required to use the theoretical framework gained throughout their studies and the research skills taught to uncover the causes behind problems in international procurement and create practicable solutions to the problems in an analytical manner.

LSC438 Consumer Protection

The module acknowledges that the consumer is the principal focus of retail and business operations in general and so attempts to explore the legal implications of that relationship. The main objectives of the module include examining the meaning of consumerism and its impact on consumer rights; defining and describing the rights of consumer in the marketplace as well as the obligations of providers of goods and services towards consumers; protecting of consumers; identifying how the law protects consumers in the marketplace against malpractice and exploring international practice for the protection of consumers.

ACC135 Financial Accounting for Business 1A and ACC136 Financial Accounting forBusiness1B,
Refer to the Department of Accounting Sciences

BM131 Principles of Management
Refer to the Department of Management Sciences

EC131Economic Principles 1andEC134, Economic Principles 2
Refer to the Department of Economic Sciences

HCS135IntroductiontoInformationTechnology
Refer to the Department of Information and Marketing Sciences

CS131CommunicationSkills
Refer to English and Communication

GSB211Gender Studies for Business

REGULATIONS FOR THE MASTER OF COMMERCE DEGREE IN SUPPLY CHAIN MANAGEMENT (BS26)

Duration	18 months
Minimum Credit load	270
Maximum Credit Load:	342
Maximum MBKs Credit Load	216
ZNQF Level	9

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the Faculty of Commerce Regulations and the General Academic Regulations for Postgraduate Diplomas and Taught Masters degrees offered by the Midlands State University.

1.2 The degree shall be awarded to candidates who have successfully completed the programme and passed the examinations in accordance with Faculty Regulations.

2. ENTRY REQUIREMENTS

2.1 To qualify for entry into the Master of Commerce degree in Supply Chain Management, an applicant should have 5 O Level passes including English Language and at least 2 A Level

passes or a relevant Diploma, an Honours degree in either Logistics, Supply Chain Management, Procurement, Purchasing, Retailing, Marketing or other business management discipline and/or having a substantial or practical orientation in those fields.

2.2 Other undergraduate degrees can be considered at the discretion of the department, provided the applicant has at least three years practical experience in a position in which logistics, procurement, purchasing supply chain management, retailing or marketing play a prominent role.

3. OBJECTIVES

3.1 The Master of Commerce degree in Supply Chain Management at Midlands State University seeks to give students an opportunity to develop and promote critical, analytical and creative thinking in the dynamic fields of logistics and supply chain management.

3.2 The programme aims to develop sound conceptual, technical, analytical and problem-solving skills that are required for success in the logistics and supply chain management profession.

3.3 The programme will be guided by Education 5.0 through collaboration with the private and public sectors, as a result, transform learners to be fully equipped with the appropriate knowledge performance delivery.

4. PROGRAMME DELIVERY

4.1 **Teaching and Learning Methods**: via Lectures, tutorials, group work, dissertation, individual independent study with emphasis on practical aspects, community service, industrialisation and innovation. Programme to offer blended learning (Online classes and consultations).

4.2 **Assessment Methods:** Written and oral examinations, tests, dissertation, business research, consultancy work, presentations, term papers, continuous assessments

4.3 **Programme Assessment:** Coursework 40% and Written Examination 60%.

5. LEARNING OUTCOMES

By the end of the study students should be able to: -

5.1 Design appropriate techniques to support the development and operation of supply chain systems through innovation and development of networks **profitably and/or cost-effectively.**

5.2 Demonstrate knowledge and attitude required in encouraging involvement in sustainable development of professional supply chain management in Zimbabwe and the international community.

5.3 Demonstrate ethical behaviour in the business environment and the importance of professional and personal development in the rapidly developing and complex field of logistics and supply chain management.

5.4 Build networks and relationships that enhance exploitation of business opportunities, promote teamwork, professionalism and effective communication.

5.5 Analyze business environments and create economic and effective networks and chains for competitive advantage in the globalized economy.

6. CAREER PROSPECTS

Graduates in the Master of Commerce degree in Supply Chain Management can work for the following organisations: private sector firms, government departments or regional and international organisations, consultancy and non-governmental organisations.

7. GENERAL PROVISIONS

7.1 Formal examinations will be administered at the end of each semester.
7.2 The programme shall be completed in a minimum of 18 months (three semesters) on a block release basis.

8. PROGRAMME ASSESSMENT

For a student to be allowed to write examinations, the student must have:

8.1 Satisfactorily studied approved modules including submission of required written assignments, business research, presentations, and exhibits.
8.2 Attended compulsory classes.
8.3 Participated in prescribed seminars, tutorials and practical classes.
8.4 Formal examinations will normally be written papers, but in some cases the examiner may test the candidate orally.
8.5 Continuous assessment shall account for 40% of the overall assessment, while the final examination shall account for 60% of the overall assessment.

9. SUPPLEMENTARY EXAMINATIONS

9.1 A student who fails and obtains a mark less than 50%, but not less than 40%, may, on the recommendation of the Board of Examiners, and subject to the approval of the Senate, be permitted to sit for a supplementary examination in the respective module(s) failed.

9.2 Overall assessment of the supplementary examination will not incorporate continuous assessment. Supplementary examinations shall be marked as "Pass" or "Fail" and the overall maximum mark awarded in a supplementary examination shall be 50%.

10. FAILURE TO SATISFY EXAMINERS

Refer to the General Academic Regulations.

11. PROVISION FOR PROGRESSION

11.1 A student will retain credits for modules passed.

11.2 A student who after six semesters has not completed the requirements for the degree shall be required to apply for re-registration, which will only be permitted on receipt of a satisfactory motivation.

11.3 Students shall progress from Semester One to Semester Two at each Level after they have passed more than 50% of modules taken in a semester. In the case of repeat-level students, the Faculty Board of Examiners may, however, make a recommendation to withdraw or discontinue the student at this stage.

12. GRADING AND DEGREE CLASSIFICATION

The following Grading and Degree Classification System shall be adopted for all modules and the programme:

80% and above	Distinction
70%- 79%	Merit
50%-69%	Pass
Below 50%	Fail

13. PROGRAMME STRUCTURE

Level 1 Semester 1

CODE	DESCRIPTION	CREDITS
MSCM703	Procurement Theory and Practice	18
MSCM704	Research Methods	18
MSCM705	Statistics and Data Analysis	18
MSCM701	Supply Chain Management	18
MSCM702	Project Management and Evaluation	18
MSCM706	Sustainability Management	18

Level 1 Semester 2

MSCM708	Customs and Regional Trade	18
MSCM709	Transport Economics	18
MSCM710	Public Sector Procurement	18
MSCM707	Strategic Fleet Management	18
MSCM711	Value Chain Analysis	18

Level 2 Semester 1

MSCM802	Humanitarian Logistics	18
MSCM801	Supply Chain Risk Management	18
MSCM803	Dissertation/Research project	90

14. MODULE SYNOPSES

MSCM701 Supply Chain Management
This module aims to give students an understanding of concepts in strategic supply chain management like supply chain drivers and strategic alignment and fit, value chain management, supply chain management and company strategy, customer value, planning demand and supply in a supply chain ,supply chain relationships, supply chain outsourcing in supply chains, management of international/global supply chains, and sourcing, managing service supply chains, performance measurement in supply chains, supply chain evaluation and contemporary issues in SCM. Key supply chain issues designed and consideration in the Zimbabwe economy.

MSCM702 Project Management and Evaluation
This module aims to give students an understanding of project planning techniques including critical path method (CPM)earned value management (EVM), resource and budgeting, project scheduling, linear programming, replacement of items, computer projects, assignment problems, problem identification, project control techniques, project performance management, team building, teamwork and team. leadership, project post audit and evaluation.

MSCM703 Procurement Theory and Practice
This module aims to give students detailed understanding of procurement as reflected in the following; procurement management, the changing role of purchasing and supply, procurement concepts, strategic procurement and supply chain management, outsourcing, quality management, inventory management, negotiations, price and total cost of ownership, procurement of commodities, capital procurement, service procurement, green procurement, international and global sourcing.

MSCM704 Research Methods
This module aims to give students an understanding of Research and its philosophies; Research Methodologies; Research Problem; Research Hypothesis; The Research Process; Guidelines on Writing a Research Dissertation; Ethical Issues in Research; Univariate and Multivariate Techniques using SPSS, NVIVO, STATA, E-Views, Minitab, Statistics, Epi-Info, Data Mining for Business Intelligence, Data Visualisation, Dimension Reduction. Application of concepts to research in supply chain management, enriching problem-solving skills. Topical research agendas in supply chain management in Zimbabwe and other developing countries. This module provides a foundation for business research in MSCM803 aligned to Zimbabwe's developmental goals, priorities and policies.

MSCM705 Statistics and Data Analysis
This module aims to give students tools, skills and techniques in analysis, interpretation and application of numerical information. This covers methods that describe the characteristics of random variables, quantifying uncertainty, inferential statistics methods, forecasting and related topics plus financial calculations (GDP, GNP, CPI).

MSCM706 Sustainability
Defining sustainability. Profit, people and planet. Economic, social and environmental performance. Circular economy. Managing networks and supply chains. Evolution of sustainability principles. Costs

and benefits for sustainable practices. Reverse and green logistics. Approaches to sustainability. Managing impacts: Sustainable Development Goals, Global sustainability agenda, suppliers and customers. Sustainability and value. Sustainability challenges and economic development in Zimbabwe. Climate change and sustainability for developing countries.

MSCM707 Strategic Fleet Management
This module aims to give students an understanding of best practices for private and public fleet management. It further looks at shop management for vehicles and equipment, fleet facilities in addition to specifics for procuring vehicles and equipment. Health and safety own- and third-party accident management, management of shipping documentation and customs procedures, training & development of staff. Fleet management and sustainability. Use of ICT. Asset management, routing, acquisition and disposal.

MSCM708 Customs and Regional Trade
This module aims to give students an understanding of the policy, principles and purpose of customs administration and international trade that impact on customs, including social expectations, emerging technologies and the global economy in the context of COMESA, PTA, UNCTD, WTO, SADC, GSP, EU. Industrial parks, Export Processing Zones, Special Economic Zones and the promotion of regional and International trade. Bonded warehouses, their various forms and functions.

MSCM709 Transport Economics and Planning
This module aims to give the students an understanding of basics of transport economics, with a closer look at transportation models, regulations, distribution planning, transport planning, network design, fleet management, freight and documentation, outsourcing, reverse logistics and fleet management. The various modes of transportation and their implications to costing, in addition, transport costing of cargo and passenger is looked at in the context of SADC, COMESA and beyond.

MSCM710 Public Sector Procurement Management
This module aims to give students an understanding of public sector procurement. To be included is the Zimbabwe Treasury and Public Procurement and Disposal of Public Assets Act. Also, to look at are multi-lateral financial institutions and the donor world tender procedures.

MSCM711 Value Chain Analysis
This module aims to give students an understanding of value chain in the context of Porter's value chain and competitive advantage. Primary and support activities in a value chain. Process diagrams, maps and flow charts, global and regional value chains, cost drivers of value chain analysis, sustainability and resilience as value creation model (VCM). Concepts of value, competitive strategies and value addition in Zimbabwe.

MSCM801 Supply Chain Risk Management
This module aims to give students an understanding of the nature of risks affecting supply chains, processes in managing risk in supply chains, risk mitigation strategies, risk analysis, risk control, sources of risk as the internal and external environments (STEEPLED) plus tools for assessing this risk.

MSCM802 Humanitarian Logistics
This module aims to give the student an understanding of History and status of Humanitarian Logistics (HL), Humanitarian Supply Chains (HSC), Collaboration and partnerships with business and government agencies, Disaster management cycle, role of IT, Long-term development as an aspect of sustainability in the humanitarian context. Exit strategies. Delivering the right assistance to the right beneficiaries at the right time versus skill and understanding of the supply chain. Last mile logistics.

MSCM803 Dissertation/Research project
Each student carries out business research on a topic in supply chain management. Research topics chosen must be aligned to Zimbabwe's developmental goals, priorities and policies with an emphasis in promoting innovation in the provision of goods and services. Each student works under a supervisor appointed by the Department. Research topics must also be approved by the Department. Research for

this module builds on skills acquired in MSCM704. The dissertation should have a minimum of 20 000 words.

DEPARTMENT OF HOSPITALITY, TOURISM AND LEISURE SCIENCES

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN AGRI-TOURISM MANAGEMENT (BS01)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

- To develop knowledge, skills and competencies in the field of agritourism management relevant to tourism entrepreneurship and various careers in tourism and hospitality organisations world-wide.
- To provide a diversified tourism product base as enunciated in Zimbabwe's tourism recovery and growth plan.
- To operationalize Education 5.0 through industrialisation and commercialisation of agricultural tourism products in Zimbabwe rural and urban spaces.
- To prepare students for further studies and lifelong learning in agritourism management.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce in Agritourism Management degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at `O' Level, must have obtained at least two passes at `A' level from the following subjects: Mathematics, Accounting, Economics, Management of Business, Geography, Food Science and Computer Science

2.2 Visiting School/Harare Weekend School Programme

To qualify for the visiting school programme, a candidate in addition to satisfying the minimum requirements as prescribed under the general and faculty regulations for English and Maths at `O' Level, must have the following:

Either:

passes in at least 2 `A' Level subjects in any commercial subjects and/or any related subjects in the Science or humanities

Or:

A National Diploma in Travelling and Tourism or equivalent relevant qualification

And:

Confirmation of employment in a Tourism or hospitality organisation

2.3 Special Entry

Candidates who have successfully completed Higher National Diploma in Tourism and Hospitality or have obtained equivalent qualifications may apply for direct entry into Level 2 of the degree programme. Candidates admitted under regulation 3.2 above will normally be exempted from Level I on a module-by-module basis. No candidate may complete the degree in less than three academic levels. Successful completion of Work-Related Learning component at Level III is compulsory for all students

2.4 Mature Entry

Refer to faculty regulations.

3 PROGRAMME CHARACTERISTICS

Areas of study: Tourism, Hospitality, Agritourism operations, Destination Marketing, Rooms Management, Foreign languages, Aquaculture, Landscaping, Food and Beverages, Rural tourism management

Specialist Focus: Tourism and hospitality management

Orientation: Research and innovation-oriented

Distinctive features: The degree programme is industry driven and wide consultations are made to ensure programme relevance in the ever-changing and competitive business environment. The agri-tourism management degree programme is the only learning option in the country and is designed to meet Education 5.0's goals.

4 CAREER OPPORTUNITIES AND FURTHER EDUCATION

The most important career prospect is in agritourism entrepreneurship. Students can also be employed in the following sub-sectors: commercial farms, conservancies, national parks, hotels, lodges, game parks, botanical gardens, safari operations, tour operating companies, and environmental organisations. The programme paves way for prospective students to further their academic ambitions in the same field.

5 PROGRAMME DELIVERY

5.1 **Teaching and Learning methods: lectures**, tutorials, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation and individual independent study.

5.2 **Assessment methods: w**ritten, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project.

6 LEARNING OUTCOMES

By the end of the program, students will be able:

- To produce innovative and patentable agritourism products and services that are marketable worldwide.
- To apply skills learnt to change and industrialise rural landscapes in Zimbabwe
- To develop and manage agritourism operations.

- To develop new agritourism product designs and concepts.
- To effectively manage costs in order to make the agritourism product more competitive.
- To develop new and sustainable agritourism models and technologies.
- Apply research skills to develop, and enhance agritourism market intelligence and undertake research in agritourism.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. **PROGRAMME ASSESSMENT**

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%.

9. **PROVISION FOR PROGRESSION**:

Refer to faculty and general regulations

10. **FAILURE TO SATISFY EXAMINERS**

Refer to general regulations

11. **GRADING AND DEGREE CLASSIFICATION**

Refer to faculty regulations.

12. **DEGREE WEIGHTING-**

Refer to faculty regulations.

13. **AWARD OF DEGREE**

Refer to general regulations

14. **PROGRAMME STRUCTURE**

Module code	**Module Description**	**Credits**

Level 1 Semester 1

HAM101	Introduction to Agri-tourism	12
THM136	Zimbabwe Tourism Geography	12

Level 1 Semester 2

HAM102	Principles of Agri-tourism Marketing	12
HAM103	Tour Guiding	12
HAM104	Agri-tourism Operations	12
THM140	Rooms Division	12
HAM105	Adventure Tourism	12

Level 2 Semester 1

ENT131	Entrepreneurship	12
THM 231	Food and Beverage Operations	12
HAM 201	Agritourism Development and Planning	12
HAM 202	Natural Resources and Conservation Management	12
GSB211	Gender Studies for Business	12

Level 2 Semester 2

HAM203	Agritourism Enterprise Management	12
THM236	Tourism Research Methods	12
THM232	Food and Beverage Production	12
HAM204	Geographical Information Systems	12
THM238	Innovation and Entrepreneurship	12

Level 3 Semester 1: Work-Related Learning

HAM340	Work-Related Learning Preliminary Report	40

Level 3 semester 2: Work-Related Learning

HAM341	Work-Related learning continuous assessment	40
HAM342	Work-Related learning report	40

Level 4 Semester 1

THM431	Sustainable Tourism	12
HAM401	Spatial and Community Dimensions of Agritourism	12
HAM402	Principles of Agribusiness Management	12
THM435	French 1	12
HAM403	Rural Tourism	12

Level 4 Semester 2

HAM 404	Creative Tourism (Food, wine and festivals)	12
HAM405	Aquaculture and Tourism	12
THM438	French 2	12
HAM 406	Dissertation/Research project* (THM 236 prerequisite)	24

15. MODULE SYNOPSES

HAM 101 Introduction to Agritourism

The module introduces students to the foundation for the Agritourism Management Degree programme. The module imparts an understanding of the nature, structure and historical development of the agritourism industry. The economic and social aspects of the agrotourism industry are also a vital component of the module. In addition, the module also explores the key issues and trends affecting the agrotourism industry in Zimbabwe.

THM 136 Zimbabwe Tourism Geography

The module introduces students to the natural and man-made tourist attractions in Zimbabwe and the structure of Zimbabwe`s tourism industry. In addition, marketing destination Zimbabwe, the physical and social impact of tourism, and people-park conflicts are also explored in the module. The overall aim of the module is to give students a firm understanding of key issues and trends affecting Zimbabwe's tourism industry.

HAM 102 Principles of Agritourism Marketing

The module introduces students to the principles of marketing agrotourism products and services. The module covers marketing functions, the broad marketing environment, marketing information systems and marketing research. The marketing mix, consumer behaviour, the social responsibility of marketing, internal marketing, relationship marketing, e-marketing, segmentation and branding are also covered in the module.

HAM 103 Tour Guiding

The module equips students with the knowledge and skills of tour guiding. On completion of this module, students will be able to design innovative itineraries and provide insightful destination interpretive guides to tourists. Students will be able to start their own tour-guiding entities upon completion of the module.

HAM 104 Agrotourism Operations

The module introduces students to the operations and functions of agrotourism as a sub-sector of the travel and tourism industry. Students also examine management issues at both the strategic and tactical levels in order to develop critical thinking.

THM 140 Rooms Division

The module introduces students to the front-of-house operations as an integral part of the lodging sector. As a professionally oriented module, emphasis is placed on its traditional role as the nerve centre of the hotel as well as its principal operational functions. The module covers practical aspects of accommodation management, personnel issues and operation structures, guest reception; manning front desk and front office etiquette.

HAM105 Adventure Tourism

The module introduces students to the growth and development of adventure tourism. Aspects covered include an appreciation of the history of adventurous leisure practices and their relationships to the environment, including models of adventure tourism. Theories and issues associated with adventure tourism are illustrated using case study materials from both the local and international context.

ENT131 Entrepreneurship 1

The aim of this module is to develop entrepreneurial acumen in students. Students will appreciate the importance of entrepreneurship to the individual and in nation building and will be able to analyse and explain how PESTLEG factors affect the entrepreneur and to design a bankable business plan.

GSB211 Gender Studies for Business

The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, historical development of gender, gender analysis, gender issues in

society, redressing gender imbalances, empowerment and strategies for creating a gender responsive environment. Every student has to pass the module in order to graduate.

THM231 Food and Beverage Operations

The module introduces students to the structure and organization of the functional areas of the food and beverage department of the hotel. The module also includes other non-hotel based food and beverage segments. Students are provided with the knowledge required to manage any food service outlet. The module focuses on the basic principles of these operations including planning, organizing, leading and control of food service operations.

THM232 Food and Beverage Production

The module introduces students to the practical aspects of food and beverage production. Students are expected to have a deep appreciation of kitchen designs, food production organization and menu planning for different types of menus. Students are also expected to prepare various types of local and international cuisines and present them.

HAM201 Agrotourism Development and Planning

The module introduces students to the concept of agrotourism planning and development. The module critically examines tourism's contribution to national economic development goals. The module provides a thorough grounding in planning and places emphasis on skills development for the enterprise. The module covers the components of strategic planning, and the techniques available to the tourism planner.

HAM202 Natural Resources and Conservation Management

The module introduces students to the strategic management approaches for physical resources and biodiversity. The module reviews natural resources from a global resource appraisal point of view and how this relates to international conventions and their implementation. This module also focuses on the challenges facing businesses, communities, administrations and countries as they attempt to protect, maintain and enhance the natural resources they rely on.

HAM 203 Agrotourism Enterprise Management

The module introduces students to the managerial skills required in the various types of agrotourism facilities which include operational planning and site layout; work scheduling; occupational health, safety and security; products and services design; facilities location and design and many more. Students are also expected to develop and present a comprehensive business plan for a prototype agrotourism business.

THM 236 Tourism Research Methods

The module introduces students to the critical competencies required for them to be able to undertake tourism research. The module aims to help students understand the research process and undertake research to solve tourism related problems. In order to fulfill this objective, the module covers the formulation of a research project, literature review, tourism research methodologies, data analysis and presentation of results as well as research ethics.

THM 238 Innovation and Entrepreneurship

The module introduces students to the creative and innovative skills required to solve societal problems and develop higher order thinking. A specific problem in the field of agrotourism is addressed. This involves identification of the problem guided by user needs, planning and managing the process, evaluating alternatives, analysing the economic and sustainable performance of the prototype. Legal issues in designing and making products and prototypes are also explored.

HAM 204 Geographical Information Systems

The module introduces students to the theory and practice of geographic information systems (GIS). It introduces some of the basic concepts of GIS including; input of data, storage and management of data and modelling output from GIS. Importantly, practical sessions are offered in this module to build basic skills in GIS such as adding, visualising, analysing and modelling data and creating effective map layouts.

HAM 301, 302,303 Work-Related Learning

The purpose of the Work-Related learning is to expose students to professional working life over a period of 8 to 12 months. This also gives them adequate exposure to relate theory to real life situations in the industry. The student is expected to rotate within the company in order to be exposed to all functions of that organisation. The student is also expected to produce a Work-Related learning report at the end of the Work-Related learning period.

THM 431 Sustainable Tourism

The module introduces students to the critical insights of the evolution of sustainable tourism and the impact of tourism on a variety of environments. The emergence of more sustainable forms of tourism like ecotourism, green tourism, culinary tourism, among others are also explored. The module also aims to develop an understanding of a range of strategies used to develop and monitor sustainable tourism in the global, national and local contexts.

HAM 401 Spatial and Community Dimensions of Agrotourism

The module introduces students to spatial and community dimensions of agrotourism. Distinguishing aspects of agrotourism supply and economic development dimensions that target tourism demand enhancement are also covered in the module.

HAM 402 Principles of Agribusiness Management

The module introduces students to the skills and equips them with knowledge on how an ordinary farm operation can be turned into a viable commercial venture that combines both tourism and agriculture. Areas focused on in the module include forms of business and analysis of business environment, value chain in agricultural operations and management principles. The module also includes issues of cooperate governance, operations research, agribusiness viability and analysis.

HAM 403 Rural Tourism

The module introduces students to the factors that determine successful and sustainable rural tourism development and management of sites in environmentally and culturally sensitive areas/settings. Demand and supply of rural tourism products and services including planning and management, sustainable rural tourism development, community involvement in rural tourism development and the future of rural tourism are also explored in the module.

THM 435 French 1

The module introduces the students to the basics of communicating in French. The module is for students who have no previous knowledge of French. Emphasis is on achieving a basic level of communication in all four skills (listening, speaking, reading and writing) while developing confidence and a degree of accuracy when using the language in a limited range of situations. The module thus enables students to adapt to French-speaking environments.

HAM 404 Creative Tourism (Food, Wine and Festivals)

The module introduces the students to food, wine and festivals as cultural phenomena from multiple perspectives including gastronomy, sociology, politics, tourism and hospitality. The particular focus of the module is on analysing creative ideas and emerging characteristics of tourist markets, products and services offered by food, wine and festival tourism. The module provides learners with hands-on experiences of food and wine trails, and festivals.

HAM 405 Acquaculture and Tourism

The module introduces students to the multi-use of the space (MUS) in agrotourism and aquaculture settings. The module equips the students with knowledge and skills to innovative and creative. The also covers restaurant services and outdoor catering activities where tourists can be served their own catch. Ornamental fish production linked to marine reserves or aquariums are also studied in this module.

THM 438 French 2

The module introduces students to the advanced communication skills required in French. Building on the concepts learnt in French 1, the module is intended to enable students to write and fluently speak French. Due to the great diversity in the nature and origin of tourists, it is imperative that tourism students be multi-lingual in order to cater for this diversity. The module also enables students to adapt to French speaking environments.

HAM 406 Dissertation/Research Project

The module introduces students and equips them with the competences in formulating research proposals, that is, review of literature, identifying research problems, formulating a research design, design of data collection instruments, data collection, data cleaning, data analysis, interpretation and report writing. The students apply the knowledge learnt in tourism research methods to solve an existing agrotourism or related problem.

ACC135 Financial Accounting 1A- Refer to Accounting Sciences Department

ACC138 Financial Accounting 1B- Refer to Accounting Sciences Department

BM131 Principles of Business Management- Refer to Management Sciences Department

BM438 Strategic Management- Refer to Management Sciences Department

BM436 Organisational Behaviour- Refer to Management Sciences Department

CS131 Basic Communication Skills- Refer to Communication Skills Centre

HCS135 Introduction to Information Technology- Refer to Information and Marketing Sciences Department

EC134 Economic Principles 1- Refer to Economic Sciences Department

MM135 Quality Management- Refer to Information and Marketing Sciences Department

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN SUSTAINABLE TOURISM MANAGEMENT AND PLANNING (BS14)

Duration	4 years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8

1. PURPOSE OF THE PROGRAMME

- To develop knowledge, skills and competencies in the field of Sustainable Tourism.
- To prepare students to be policy makers and experts in current and future urban and rural tourism planning. The programme equips students with knowledge and skills in establishing sustainable tourism businesses at local, regional and international levels.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce Honours in Sustainable Tourism Management degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at `O' Level, must have obtained at least two passes at `A' level in Business and Science-related subjects.

2.2 Visiting School/Harare Weekend School Programme

To qualify for the visiting school programme, a candidate in addition to satisfying the minimum requirements as prescribed under the general and faculty regulations for English and Maths at `O' Level, must have the following:

Either:

passes in at least 2 `A' Level subjects in any commercial subjects and/or any related subjects in the Science or humanities

Or:

A National Diploma in Travelling and Tourism or equivalent relevant qualification

And:

Confirmation of employment in a Tourism or hospitality organisation

2.3 Special Entry

Candidates who have successfully completed Higher National Diploma in Tourism and Hospitality; Conservation; Environmental Management or have obtained equivalent qualifications may apply for direct entry into Level 2 of the degree programme. Candidates admitted under above regulation will normally be exempted from Level I on a module-by module basis. No candidate may complete the degree in less than three academic levels. Successful completion of Work-Related Learning component at Level III is compulsory for all students

2.4 Mature Entry

Refer to Section 3.3 of the General Regulations.

3. PROGRAMME CHARACTERISTICS

Areas of study: Tourism, Environmental management, Game parks, Destination marketing, Conservation management, Ecotourism, Community development, Community-based tourism, Natural resources governance, Sustainable tourism and Tourism project management.

Specialist focus: Sustainable Tourism management

Orientation: Research and innovation oriented

Distinctive features: The degree programme is industry driven and wide consultations are made to ensure programme relevance in the ever changing and competitive business environment. The sustainable tourism management degree programme is the only learning option in the country.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

The following are possible areas of employment for the Bachelor of Commerce Honours in Sustainable Tourism graduates; National Parks, Museums, Destination Marketing, Safari Operations, Environmental Organisations, Conservation Organisations, Community Development Organisations; Non-Governmental Organisations and the academic world. Students can enrol for a Master's degree in Sustainable Tourism management and other related higher qualifications. In addition, with the emphasis today on small businesses, many graduates find great success in starting their own businesses.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning methods: lectures**, tutorials, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation and individual independent study

5.2 **Assessment methods: w**ritten, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project.

6. LEARNING OUTCOMES

By the end of the programme, a graduate will be able to:

- demonstrate in depth knowledge and understanding of management and planning fundamentals in tourism organisations and destinations.
- Understand the global tourism value chain in terms of the practices, technology applications and services.
- Identify challenges in nature-based tourism, sustainable tourism, environmental management and develop solutions to mitigate them.
- research on sustainable tourism, environmental and development issues.
- Develop new and sustainable tourism practices and technologies that enhances tourism development while promoting environmental management.

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the

dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%

9. PROVISION FOR PROGRESSION

Refer to faculty and general regulations

10. FAILURE TO SATISFY EXAMINERS

Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION

Refer to faculty regulations.

12. DEGREE WEIGHTING

Refer to faculty regulations.

13. AWARD OF DEGREE

Refer to general regulations

14. PROGRAMME STRUCTURE

CODE	MODULE DESCRIPTION	CREDITS
LEVEL 1 Semester 1		
THM136	Introduction to the Tourism and Hospitality Industry	12
STM101	Principles of Sustainable Tourism Management	12
LEVEL 1 Semester 2		
THM138	Principles of Tourism Marketing	12
STM103	Global Environmental Issues and Tourism	12
STM102	Natural Resources Governance in Tourism	12
STM104	Principles of Ecotourism	12
LEVEL 2 Semester 1		
THM233	Regional and International Tourism	12
STM202	Political Ecology	12
STM203	Tourism Policy and Development	12
STM204	Community Tourism Management	12
LEVEL 2 Semester 2		
THM 236	Tourism Research Methods	12
STM205	Culture and Heritage Management	12
STM 206	Tourism and Environmental Law	12
STM207	Risk and Disaster Management	12
STM208	Destination Management	12
Level 3 Semester 1 Work-Related Learning		
Quarter 2		
STM 340	Work-Related Learning Preliminary Report	30
Level 3 Semester 2 Work-Related Learning		

Quarter 2

STM 341	Work-Related Learning continuous assessment	45
STM 342	Work-Related report	45

LEVEL 4 Semester 1

THM 434	Hospitality and Tourism Innovation	12
STM403	Earth, Climate and Ecological Systems in Tourism	12
STM404	Environmental Impact Assessment	12
STM405	Conflict Management in Tourism	12

LEVEL 4 Semester 2

STM406	Parks and Wildlife Management	12
STM407	Rural and Urban Development Management	12
STM408	Political Tourism Geography and Development	12
STM409	Research Project/ Dissertation (THM236 pre-requisite)	24

MODULE SYNOPSES

THM 136 Introduction to the Tourism and Hospitality Industry
An introductory module, which sets the foundation for the Sustainable Tourism management degree program. The module explores the nature, structure, historical development, and economic and social aspects of the tourism and hospitality Industry. The module also traces key issues and trends affecting the tourism industry such as the new tourist, globalisation, climate change, global pandemics, digital marketing and virtual tourism.

STMP 101 Principles of Sustainable Tourism
In this module students learn how to plan, assess and implement sustainable tourism. The role of government, development agencies, tourism industry, private sector, and local communities in sustainable tourism management is investigated from a conservation perspective. The module provides a firm understanding of tourism that takes full account of its current and future economic, social and environmental impacts, addressing the needs of visitors, the industry, and the environment and host communities.

THM 138 Principles of Tourism Marketing
The module introduces students to the marketing principles of tourism and hospitality products in a competitive marketplace. The module covers marketing functions, the broad marketing environment, marketing information systems and marketing research, the marketing mix, consumer behaviour, the social responsibility of marketing, internal marketing, relation marketing, e-marketing, segmentation, branding, and the major marketing methods.

THM 233 Regional and international tourism
This module analyses key tourism trends and impacts at sub-regional, continental and global scale. It seeks to familiarize learners with regional tourism product/resource bases, and introduces concepts such as regional cooperation, integration and tourism development initiatives. In addition, the module proffers an understanding of the international dimension of tourism within the delimitation framework of the United Nations World Tourism Organization through a thorough examination of principal players in the regional and international tourism system.

STMP 103 Global Environmental Issues and Tourism

This module explores the relationship between global environmental issues and tourism. The relationship between the tourism industry and natural disasters like floods, cyclones, earthquakes, volcanoes, climate change, deforestation, land degradation, overfishing, and air pollution is examined in this module,

STMP 104 Principles of Ecotourism
The module provides a detailed description and analysis of ecotourism in the context of environmental conservation and sustainability. It also equips students with the analytical skills and methodologies required to effectively manage ecotourism in natural areas. The module also provides theoretical and practical issues relevant to community-based nature tourism by exploring relationships between hosts and guests from cultural and socio-economic perspectives.

ENT131 Entrepreneurship 1

The aim of this module is to develop entrepreneurial acumen in students, to appreciate the importance of entrepreneurship to the individual and in nation-building, to analyse and explain how PESTLEG factors affect the entrepreneur, design a bankable business plan

Evaluate various options available to the entrepreneur to go global. The emphasis of this module is to define entrepreneurship, entrepreneurship and innovation; understand the nature and importance of entrepreneurship for economic development; identify business opportunities and generate business ideas; develop a bankable business plan; understand the characteristics of a successful entrepreneur; identify the challenges being faced by entrepreneurs.

GSB211 Gender Studies for Business
The module empowers students with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all social interactions. Topics covered include understanding gender, theories of gender inequalities, historical development of gender, gender analysis, gender issues in society, redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

STMP 201 Natural Resources Governance in Tourism
The module explores the concepts of natural resources development and conservation. The module aims to provide an understanding of current issues in tourism and environmental policy and the development of resource management systems. Students are expected to understand the use of natural resource management policy and legislation.

STMP 202 Political Ecology
This module introduces students to the study of political ecology, and its usefulness in understanding the geography of human-environment relations, particularly in Sub-Saharan Africa. The module examines alternative approaches and strategies to local or community development in Africa. The political ecology of conservation and tourism will be used to understand how key actors shape environmental activities and their spatial configurations and outcomes.

STMP 203 Tourism Policy and Development

This module provides students with an understanding of the role and function of tourism policy in the development of tourism. It also discusses empirical examples of tourism policies with the aim of drawing out best practices. Students will explore and debate the key pillars of arguments on various policy frameworks and implementation gaps.

STMP 204 Community Tourism Management
This module provides a basic understanding of community management as it relates to tourism by outlining the main approaches taken to date and describing the organisations involved. The module covers components of management and approaches to planning. The module will discuss the role of communities in tourism planning, the impacts of tourism on communities and communal leadership.

STMP 205 Cultural and Heritage Management
The module exposes learners to culture and heritage as a finite tourism product highly sought after globally. The module also focuses on the economic significance of cultural and heritage tourism to national development, evaluation of cultural and heritage tourism resources of the regions, the development of master plans and conservation plans for cultural and heritage resources and the diversification of tourism attractions.

THM 236 Tourism Research Methods
The module is designed to give students the necessary competencies that will allow them to undertake tourism research. In order to fulfil this objective, the module covers the formulation of a research project, literature review, tourism research methodologies, data analysis and presentation of results as well as research ethics. Students will be exposed to a variety of research papers and exercises in this module.

STMP 206 Tourism and Environmental Law
The focus of this module is not only on the law as written rules but also on customary law, norms, practices and issues of enforcement in relation to development. It discusses human rights such as the right to a clean environment as third generation rights. The module also analyses prevailing environmental legislation and its implications to the tourism industry.

STMP 207 Risk and Disaster Management
The module provides a firm understanding of the risk and disaster management processes in the tourism industry in light of the ever-changing internal and external operating environment. The module covers the importance of information sharing and experiences of crises amongst tourism destination operators, crises management frameworks, tourist destination post-disaster recovery strategies, investment in risk and disaster mitigation strategies.

THM 239 Sustainable tourism
Tourism relies on the quality of the environment so the careful management of businesses, people and places is a critical issue for the sector. This module offers critical insights into the evolution of sustainable tourism; the impact on tourism on some variety environments; and the emergence of more sustainable forms of tourism like ecotourism, green tourism, culinary tourism, among others. The module offers the student an opportunity to develop a detailed and critical awareness of the theoretical and practical issues that influence how the principles of sustainable development are applied in tourism.

STMP 301, 302& 303 Work-Related Learning

Work-Related learning exposes students to professional working life and give them adequate exposure to relate theory to real-life situations in industry. The student is expected to be exposed to all functions of an organisation. The student is also expected to produce two reports during the Work-Related learning period. In addition, the employer is also required to assess the student's performance.

STMP 401 Destination Management
This module provides knowledge on how destinations compete to provide the highest quality of experience for visitors and to manage the impacts of tourism on host communities and environments. The module helps students to appreciate best practices in Destination management by drawing on empirical examples and relevant theories. The roles of private sector and public sector organisations in destination marketing are also explored in this module,

STMP 403 Earth, Climate and Ecological Systems in Tourism
The module explores issues related to climate change. It informs students of the impacts of climate change and earth changes on tourism. Ecological, social and economic aspects related to climatic

change, modelling, projections, and impacts of future climatic on tourism are also a focus of this module as are the. impacts of climate change on community-based tourism and land resources.

STMP 404 Environmental Impact Analysis
This module equips students with knowledge of environmental monitoring techniques. The module covers environmental appraisal and management; assessment, planning, management and response strategies; environmental policies and practices and risk assessment, impact assessment, audits, environmental information systems and environmental data analysis techniques. Emphasis will be placed on environmental impact analysis (EIA), strategic environmental assessment (SEA), sustainability assessments and sustainability indicators, cost-benefit analysis and multi-criteria decision analysis.

STMP 405 Conflict Management in Tourism
This module explores why conflict emerges in communities and community groups, how to minimise, deal with, 'manage', or solve community and group conflict. Students are expected to understand how conflict is used by groups as a strategy to attain desired changes.

STMP 406 Parks and Wildlife Management
This module provides students with the skills to determine and consider ways in which environments and ecosystems can be protected. The module provides students with information on the status of wildlife populations in national parks and devising effective conservation strategies. Empirical examples of best practises are emphasised in this module so as to equip students with the requisite skills.

STMP 407 Rural and Urban Development Management
The module introduces students to aspects of urban and rural geography, with the aim to enable sustainable planning of rural and urban spaces as tourist destinations. It discusses trends of urban and rural development in the context of tourism, urban and rural theories, functional differentiation within the city/rural, inner-city problems; urban and rural governance and waste management.

STMP 408 Political Tourism Geography and Development
The module examines elements of political geography: the spatial dimension of politically organised units from the local state and international scales; colonial policies and their impact on the development of emergent African political boundaries. Problems of national integration; regional and international organisations as instruments of political liberation and stability; and how politics informs tourism geographies.

STMP 409 Research Project /Dissertation (THM 236 pre-requisite)
The supervised dissertation will be at least 15000 words. This research project allows the student to explore in depth subjects of specific interest to him/her. The main requirement of the dissertation is that it should engage with a problem of relevance in the sustainable tourism domain. Students should draw out practical, and feasible recommendations that are resonate with their findings in this research project.

THM 434 Hospitality and tourism innovation

This module examines the concepts and processes in evaluating and determining the importance of entrepreneurship and innovation within the Tourism and Hospitality sector. This module gives students the opportunity to address the key issues of business start-up growth and long term survival, considering the concept of innovation and creativity in relation to the tourism industry. The module examines the internal and external factors influencing change in the tourism industry and how experience with the industry impacts demand and the need for New Product Development. The importance of innovation in the tourism and hospitality industry as well as an assessment of new forms of tourism will also be explored.

ACC135 Financial Accounting 1A- Refer to the Accounting Department

ACC138 Financial Accounting 1B- Refer to the Accounting Department
BM131 Principles of Business Management- Refer to the Business Management Department
BM438 Strategic Management- Refer to the Business Management Department
CS131 Basic Communication Skills- Refer to the Communication Skills Centre
HSC135 Introduction to Information Technology- Refer to Information Systems Department
EC131 Economic Principles 1- Refer to the Economics Department
EC134 Economic Principles 11- Refer to the Economics Department
MM135 Quality Management- Refer to the Marketing Department

REGULATIONS FOR THE BACHELOR OF COMMERCE HONOURS DEGREE IN TOURISM AND HOSPITALITY MANAGEMENT (BS15)

Duration	4 Years
Credit Load	540
Minimum Credit Load	480
Maximum Credit Load	540
Maximum MBKs Credit Load	384
ZNQF Level	8
Accreditation Organisation	Zimbabwe Council of Higher Education (ZIMCHE)

1. PURPOSE OF THE PROGRAMME

- To develop knowledge, skills and competencies in the field of Tourism and Hospitality management relevant to tourism entrepreneurship and various careers in tourism and hospitality organisations world- wide.
- To prepare students for further studies and lifelong learning in tourism and hospitality management.

2. ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for normal entry into the Bachelor of Commerce Tourism and Hospitality Management degree programme, a candidate, in addition to satisfying the minimum requirements as prescribed under the General Regulations and the Faculty Regulations for English and Mathematics at `O' Level, must have obtained at least two passes at `A' level from the following subjects: Mathematics, Accounting, Economics, Management of Business, Geography, Food Science and Computer Science

2.2 Visiting School/Harare Weekend School Programme

To qualify for the visiting school programme, a candidate in addition to satisfying the minimum requirements as prescribed under the general and faculty regulations for English and Maths at `O' Level, must have the following:

Either:

passes in at least 2 `A' Level subjects in any commercial subjects and/or any related subjects in the Science or humanities

Or:

A National Diploma in Travelling and Tourism or equivalent relevant qualification

And:

Confirmation of employment in a Tourism or hospitality organisation

2.3 Special Entry

Candidates who have successfully completed Higher National Diploma in Tourism and Hospitality or have obtained equivalent qualifications may apply for direct entry into Level 2 of the degree programme. Candidates admitted under regulation 3.2 above will normally be exempted from Level I on a module-by-module basis. No candidate may complete the degree in less than three academic levels. Successful completion of Work-Related Learning component at Level III is compulsory for all students.

2.4 Mature Entry

Refer to faculty regulations.

3. PROGRAMME CHARACTERISTICS

Areas of study: Tourism, Hospitality, Tour operations, Travel agency, Destination Marketing, Hotel Management, Rooms Management, Sustainable Tourism, Safari, Foreign languages., Food and Beverages, Food Preparation and Services, Service Quality, Hotel and Institutional Management

Specialist focus: Tourism and hospitality management

Orientation: Research and innovation-oriented

Distinctive features: The degree programe is industry-driven and wide consultations are made to ensure programme relevance in the ever-changing and competitive business environment. The agritourism and sustainable tourism management degree programmes are the only learning options in the country including the uniquely entrepreneurial oriented Masters in Tourism degree programme.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

The following are possible areas of employment for the Bachelor of Commerce Tourism and Hospitality Management graduates; Hotel Operations, Tour and Travel Agency, Lodge Management, Services Marketing, National Parks, Museums, Car Hire Companies, Airlines, Destination Marketing, Safari Operations, Conference Management, Civil Aviation, Environmental organisations and the academic world. Students can enrol for Master's degree in Tourism and hospitality management and other related higher qualifications. In addition, with the emphasis today on small businesses, many graduates find great success in starting their own businesses.

5. PROGRAMME DELIVERY

5.1 Teaching and Learning methods: lectures, tutorials, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation and individual independent study

5.2 Assessment methods: written, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project.

6. LEARNING OUTCOMES

By the end of the program, a graduate will be able to:

- Effectively manage and market hospitality and tourism products
- Prepare and produce innovative recipes, indigenous cuisines and food products
- Create new tourism and hospitality business ventures as well as efficiently run the existing ones
- Develop new and sustainable tourism and hospitality management models and technologies
- Apply research skills to develop and enhance tourism and hospitality market intelligence
- Develop new hospitality and tourism product designs and concepts
- Effectively manage costs in order to make the tourism product more competitive

7 GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 30% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 Final examination accounts for 70% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials and practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline of submission of dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by Senate, the maximum mark allowed for such work should be 50%.

9. **PROVISION FOR PROGRESSION**: refer to faculty and general regulations

10. FAILURE TO SATISFY EXAMINERS – Refer to general regulations

11. GRADING AND DEGREE CLASSIFICATION – Refer to Faculty Regulations.

12. DEGREE WEIGHTING - Refer to Faculty Regulations.

13. AWARD OF DEGREE - refer to general regulations

14. PROGRAMME STRUCTURE

CODE	MODULE DESCRIPTION	CREDITS
Level 1 Semester 1		
THM 136	Introduction to the Tourism and Hospitality Industry	12
THM 137	Zimbabwe Tourism Geography	12
Level 1 Semester 2		
THM 140	Rooms Division	12
THM138	Principles of Tourism Marketing	12
THM 139	Food Preparation	12
Level 2 Semester 1		
THM 232	Food and Beverage Production	12
THM 233	Regional and International Tourism	12
THM 234	Digital Tourism and Hospitality Marketing	12
THM 231	Food and Beverage Management	12
Level 2 Semester 2		
THM 235	Food and Beverage Service	12
THM 237	Hospitality and Tourism Law	12
THM 240	Tourism Operations	12
THM 239	Sustainable Tourism	12
THM 236	Tourism Research Methods	12
Level 3 Semester 1		
THM 340	Work-Related Learning Preliminary Report	40
Level 3 Semester 2		
THM 341	Work-Related learning continuous assessment	40
THM 342	Work-Related Learning Report	40
LEVEL 4 SEMESTER 1		
THM 433	Human Resource Management	12
BM438	Strategic Management	12
THM 432	Event Management	12
THM 434	Hospitality and Tourism Innovation	12
THM 435	French 1	12
LEVEL 4 SEMESTER 2		
THM 437	Tourism Transport Logistics	12
THM 438	French 2 (THM 435 pre-requisite)	12
THM 439	Research Project/ Dissertation (THM236 pre-requisite)	24
THM 436	Hotel Management	12

14. MODULE SYNOPSES

THM 136 Introduction to the Tourism and Hospitality Industry
An introductory module, which sets the foundation for the Tourism and Hospitality Management Degree Program. The module imparts an understanding of the nature, structure, historical development, and economic and social aspects of the tourism and hospitality Industry. The module also explores the key issues and trends affecting the tourism industry.

THM 137 Zimbabwe Tourism Geography
This module covers natural and man-made attractions in Zimbabwe, Zimbabwe`s weather patterns and their impact on tourism, the physical and social impact of tourism, and people-parks conflict. The overall motive of the module is to give students an understanding of the key trends and issues affecting the country's tourism industry.

THM 138 Principles of Tourism and Hospitality Marketing
The module introduces students to the marketing principles of tourism and hospitality products in a competitive marketplace. The module covers marketing functions, the broad marketing environment, marketing information systems and marketing research, the marketing mix, consumer behaviour, the social responsibility of marketing, internal marketing, relation marketing, e-marketing, segmentation, branding, a general survey of the major marketing methods, techniques, institutions and practices among many other related aspects. Learners are envisaged to develop competencies in marketing skills of tourism services.

THM 139 Food Preparation
The module focus on kitchen skills, different meat and vegetable products, menu setting, costing and gross profit margin various cooking methods, time management, nutritional value of foods, plate setting, hygiene and safety, food microbiology and taxonomy. In addition, learners are introduced to microbial growth in foods, influence of microbes in foods, spores and their significance, detection and enumeration of microbes in foods, control of micro-organisms in food, microbial food-borne diseases. Other aspects explored in the module encompasses fermentative organisms, spoilage organisms, beneficial use of micro-organisms in food, moulds, anti-microbial chemicals, biological based preservation and pro-biotic bacteria, physical methods of food preservation, industrial methods of food safety, spoilage, contamination and subsequent food poisoning; hazardous analysis critical control point (HACCP) for all larder commodities (dry, moist and frozen foods) and innovation in food preparation and microbiology.

THM 140 Rooms Division
This is a professionally oriented module, which introduces the student to the front-of-house as an integral part of the total operations. Emphasis will be placed on its traditional role as the nerve centre of the hotel as well as its principal operational functions. This module covers practical aspects of accommodation management such as personnel issues and operation structures, guest reception; manning the front desk and front office etiquette among others.

THM 231 Food and Beverage Management
A professionally oriented course that introduces students to the structure and organization of the functional areas of the food and beverage department of the hotel including other non-hotel based food and beverage segments. The module will focus on the basic principles of these operations. This module covers planning, organizing, leading and control of food service operations; marketing of food and beverages- marketing concepts, and hospitality marketing mix among others.

THM 232 Food and Beverage Production
The module contains both the practical and theoretical aspect of food and beverage production. Students are expected to have a deep appreciation of kitchen designs, food production organization, menu planning for table d'hôte, a'la carte, cocktail menu, and buffet menu for different events, menu and recipe costing and cost management, food production method and beverage production methods,

mise-en-place on advanced food production, measuring and weighing techniques, recipe conversion, and flavour building. Students will also be expected to have a thorough understanding of food commodities and principles of cooking different types of food commodities, innovation in food and beverage production. Students are expected to apply knowledge gained from the (THM139) module to prepare the following advanced dishes; soups, sauces; hors d'oeuvres, egg and pasta dishes, fish and shellfish dishes, meat and game dishes, poultry dishes, vegetable and pulse dishes, potato dishes, international cuisine, indigenous dishes, vegetarian dishes, breakfast, mass food production, pastry and baking, production of larder kitchen items. Students also learn advanced modern food presentation, garnishing and display work for the dishes produced

THM 233 Regional and International Tourism
This module analyses key tourism trends and impacts at sub-regional, continental and global scale. It seeks to familiarize learners with southern Africa`s tourism product/resource base, and introduces concepts such as regional cooperation, integration and tourism development initiatives. In addition, the module proffers an understanding of the international dimension of tourism within the delimitation framework of the United Nations World Tourism Organization through a thorough examination of principal players in regional and international tourism systems, international tourist destinations/regions, global tourism trends, leading international tourist destinations, airline carriers, international tourism forecasts and future projections.

THM 234 Digital Tourism and Hospitality Marketing
This module focuses on digital marketing, digital marketing strategies, types of digital marketing tools and their effectiveness in marketing tourism products. It covers aspects such as mobile digital marketing, managing tourism and hospitality digital marketing tools, social media marketing, and digital marketing plan. In addition, the management of digital tourism and hospitality booking and reservations systems, display and interactive digital media, impact of digital marketing on tourism and hospitality marketing mix, impact of digital marketing in tourism and hospitality industry, digital marketing practices to B2C and B2B, evolution of Web1.0, Web2.0, Web3.0 and uses in Tourism and Hospitality Industry, risk management and cyber security in digital marketing, ethics in digital marketing, managing online reviews and service recovery, customer relationship management are other aspects explored in the module.

THM 235 Food and Beverage Service
This introductory module provides students with skills and competences in food and beverage service. It involves both practical and theoretical aspects of food and beverage service. The modules covers introduction to food and beverage service, current trends in food and beverage service, the food and beverage department and hierarchy, duties and responsibilities of food and beverage personnel, relationship of food and beverage department to other departments, job descriptions, grooming and etiquette of service personnel, uniform, types and characteristics of restaurants, type of customers, reasons for eating out, food and beverage equipment, safety handling and cleaning of equipment, service areas, menu and classes of menu among many other aspects.

THM 236 Tourism Research Methods
The module is designed to give students the necessary competencies that will allow them to undertake tourism research. In order to fulfil this objective, the module covers the formulation of a research project, literature review, tourism research methodologies, data analysis and presentation of results as well as research ethics.

THM 237 Hospitality and Tourism Law
The module is designed to introduce students to the basic functions of law and legislation relating to the operation of hospitality and tourism business. The legal requirements for the hospitality and tourism industry are explored in detail, in particular the rights and responsibilities that the law imposes upon an innkeeper, tourism operators and food service. It therefore allows learners to develop a full appreciation of the legal consequences of failure to oblige to these legal obligations. This module also applies those basic principles and concepts of law which affect day to day transactions that will be delivered with a local community and international focus. Students are expected to identify and examine the legal issues which govern the environment in which international tourism and hospitality services are provided. The other objective is to provide the students with a working knowledge of the

legal environment in the international hospitality and tourism industry and the responsibilities in the implementation of law for guests and staff alike. The module covers the following aspects: tourism company registration and grading of hotels, Liquor Licensing, Entertainment Permissions Admission, Law governing Supply of Accommodation, liability for Personal Injury Law and governing Liability Law for theft of customer property.

ENT 131 Entreprenuership
In this module, students will use their creative and innovative skills to solve societal problems and develop higher-order thinking. Students will use these newly developed skills to develop solutions to problems by designing prototype event designs or technologies, working individually or in a team. This involves: identification of the problem guided by user needs, planning and managing the process, evaluating, alternatives and analysing the economic, sustainable performance of a prototype.

THM 239 Sustainable Tourism

Tourism relies on the quality of the environment so the careful management of businesses, people and places is a critical issue for the sector. This module offers critical insights into the evolution of sustainable tourism; the impact on tourism on a variety of environments; and the emergence of more sustainable forms of tourism like ecotourism, green tourism, and culinary tourism, among others. The module offers the student an opportunity to develop a detailed and critical awareness of the theoretical and practical issues that influence how the principles of sustainable development are applied in tourism. The module aims to develop an understanding of a range of strategies used to develop and monitor sustainable tourism in global, national and local contexts.

THM 240 Tourism Operations
The module unpacks the tour operators industry , including different types of operators, their products and services, the scale of the industry and how it has been affected by global trends and developments. Learners will explore the stages involved in creating a holiday and develop skills associated with determining a selling price for a holiday from given information. The role of brochures, new methods of promoting holidays and product distribution methods such as the internet are other key aspects covered by the module.

GSB211 Gender Studies for Business Sciences
This module seeks to provide students in the Faculty of Business an understanding of why gender equality matters in the issues of economic development and why it should be pursued. The module will endeavour to explore ways in which a gendered lens is fast becoming a necessity in the global economic development front. The module will assist students to be able to analyse the effects of gender gaps and reduction of gender inequalities have a positive impact on the sustainable economic growth of a country.

THM 331; 333; 334 Work-Related Learning
The purpose of the Work-Related learning is to expose students in 8 to 12 months to professional working life and give them adequate exposure to relate theory to real life situations in industry. The student is expected to rotate within the company in order to be exposed to all functions of that organisation. The student is also expected to produce report at the end of the Work-Related learning period that covers the organisation background, organisational structure, departments attached, work carried out during attachment, new issues learnt and any gap between information covered in the University. In addition, the employer is also required to assess the student at the workplace.

THM 432 Event Management
This module gives learners the opportunity to complete the full event management cycle from event planning, design and to the actual delivery and evaluation. Event management tests a wide range of different skills and abilities. A key element of the module is each student's personal development through their interaction with the content, with other students and through the planning, delivery and evaluation of an event. Students will analyse and critically reflect upon their skill base and plan how

best to improve identified skills. Students have the chance to participate in live projects and undertake work placements to gain practical hands-on event experience. The module enables students to physically schedule and plan for an event effectively, troubleshoot event plans to identify potential problem areas before they emerge. The module provides a complete framework of how popular world class events where planned and the possible post-mortem curve determined. The module covers the following areas: events budgeting, cost management, marketing media, media coverage and post event management.

THM 433 Human Resource Management
This module covers human resource management in the tourism context. The module explores best practices in human resources management in employee recruitment and retention as well as human resources challenges and trends shaping the industry. Other aspects explored in the module include employee training, development and performance evaluation; compensation and safety; and employee and labour relations.

THM 434 Hospitality and Tourism Innovation
This module examines the concepts and processes in evaluating and determining the importance of entrepreneurship and innovation within the Tourism & Hospitality Sector. This module gives Tourism and Hospitality students the opportunity to address the key issues of business start-up growth and long term survival, considering the concept of innovation and creativity in relation to the tourism industry. The module examines the internal and external factors influencing change in the tourism industry and how experience with the industry impacts demand and the need for New Product Development. The importance of innovation in tourism and hospitality industry as well as an assessment of new forms of tourism will also be explored. The module also aims to equip students with a wide range of processes in terms of ideas generation for products, modern services processes, both within existing organisations as an entrepreneur or an entrepreneur in the development of new business ventures.

THM 435 French 1
The module is for students who have no previous knowledge of French. The emphasis is on achieving a basic level of communication in all four skills (listening, speaking, reading and writing) while developing confidence and a degree of accuracy when using the language in a limited range of situations. The module also aims to stimulate students' interest in the francophone world and deepen their knowledge and understanding of the diversity of "la francophone" society and culture. Owing to the great diversity in the nature of the tourism and hospitality industry, it is imperative that tourism students be multi-lingual in order to cater for this diversity. The module will also enable students to work comfortably in French speaking countries.

THM 436 Hotel Management
The module outlines the structure, design and trends in the accommodation services sector. It enables the student to have a clear understanding of the management of hotels and other similar establishments such as lodges, guesthouses, hostels etc. Using a practical and case approach, the module covers the following areas: introduction to organisations and management, principles and theories of management, organisational designs, management tools and techniques in the hospitality industry and control systems in hospitality management. The course also covers profitability, efficiency and effectiveness in the hotel, organizational cultures and leadership in the hotel.

THM437 Tourism Transport Logistics
Transport forms a very important component and link in the tourism sector. This course provides students with knowledge about transport; modes, systems, organisation and planning in relation to tourism operations. It equips students with skills in how to manage transport systems to enhance tourism development. This module also covers the following themes: civil aviation, basics of the airline industry, car hire business, fleet management, tourist cargo movement, ground handling services, immigration and customs policies and practices.

THM438 French 2
Building on the concepts learnt in French 1, the module is intended to enable students to write and fluently speak French. Arising from the great diversity in the nature and origin of tourists, it is imperative that tourism students be multi-lingual in order to cater for this diversity.

THM 439 Research Project/ Dissertation (THM236 pre-requisite).
The module seeks to equip students with competencies in formulating and undertaking research. Key themes that are covered in the module include research proposals, literature review, identifying research problems, formulating a research design, data collection, data cleaning, data analysis, interpretation and report writing.

ACC134 Financial Accounting 1A - Refer to the Accounting Department
ACC138 Financial Accounting 1B - Refer to the Accounting Department
BM131 Principles of Business Management - Refer to the Business Management Department
BM438 Strategic Management - Refer to the Business Management Department
BM436 Organisational Behaviour - Refer to the Business Management Department
CS131 Basic Communication Skills - Refer to the Communication Skills Centre
HSC135 Introduction to Information Technology - Refer to the Information Systems Department
EC131 Economic Principles 1 - Refer to the Economics Department
EC134 Economic Principles 11 - Refer to the Economics Department
MM135 Quality Management - Refer to the Marketing Department

REGULATIONS FOR THE MASTER OF COMMERCE IN TOURISM AND HOSPITALITY ENTERPRISES MANAGEMENT DEGREE (MTHM)

Duration	2years
Credit Load	340
Minimum Credit Load	270
Maximum Credit Load	340
Maximum MBKs Credit Load	216
ZNQF Level	9

1. PURPOSE OF THE PROGRAMME

- To develop advanced knowledge, skills and competencies in the field of Tourism and Hospitality enterprise management.
- To prepare students for further studies, entrepreneurial roles and lifelong learning in tourism and hospitality management.

2 ENTRY REQUIREMENTS

2.1 Normal Entry

To qualify for entry into the Master of Commerce in Tourism and Hospitality Enterprises Management degree, a student should have a minimum of a good Bachelor of Commerce Honours degree in Tourism and Hospitality Management with a (2,1or better) or an equivalent qualification. The student should also have a minimum of two years working experience in the industry.

2.2 Mature Entry

Refer to faculty regulations

3. PROGRAMME CHARACTERISTICS

Areas of study: Tourism and hospitality innovations, tourism accommodation, destination marketing, human resources management, e-commerce, crisis management, food preparation and services, tourism entrepreneurship and advanced tourism research methods.
Specialist focus: Tourism and hospitality enterprises management
Orientation: Research and innovation-oriented

Distinctive features: The degree programme is industry driven and wide consultations are made to ensure programme relevance in the ever-changing and competitive business environment. The uniquely entrepreneurial oriented Masters in Tourism and hospitality enterprises management degree programme is in line with Education 5.0's goals as it culminates in a bankable and innovative tourism business idea.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

The following are possible areas of employment for the Master's degree in Tourism and hospitality enterprises management graduates hotels, lodges, consultancy, casinos, national parks, airlines, tour operations and pursuing doctoral studies. In addition, with the emphasis today on small businesses, many graduates find great success in starting their own businesses.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning methods: lectures**, tutorials, group work, industrial visits, Work-Related learning, research project and innovation/ dissertation and individual independent study

5.2 **Assessment methods: w**ritten, practical and oral examinations, in-class tests, assignments, presentations, Work-Related learning report, final year research project.

6. LEARNING OUTCOMES

By the end of the program, a graduate will be able to:

- Effectively manage and market hospitality and tourism products
- Prepare and produce innovative recipes, indigenous cuisines and food products
- Create new tourism and hospitality business ventures as well as efficiently run the existing ones
- Develop new and sustainable tourism and hospitality management models and technologies
- Apply research skills to develop and enhance tourism and hospitality market intelligence
- Develop new hospitality and tourism product designs and concepts
- Effectively manage costs in order to make the tourism product more competitive
-

7. GENERAL PROVISIONS

Refer to faculty and general regulations

8. PROGRAMME ASSESSMENT

8.1 Evaluation of students shall normally be based on continuous assessment as well as formal university examinations that are held at the end of each semester.

8.2 Continuous Assessment will account for 40% of the overall assessment. No continuous assessment shall be carried over to the next semester.

8.3 The final examination accounts for 60% of the overall assessment.

8.4 The department shall determine which items will be included in the continuous assessment and shall define the relevant weighting to each item.

8.5 To be admitted to the examinations, a candidate must;

8.5.1 be a registered student of the University in accordance with the General Regulations

8.5.2 have satisfactorily attended and completed the approved modules of study. Completion of modules shall require submission of all written assignments and tests that constitute the continuous assessment

8.5.3 have attended and participated in seminars, tutorials practical classes, presentations and other activities as required by the Department.

8.5.4 Candidates shall be informed in advance of the deadline for submission of the dissertation or project. Unless the Academic Board has granted prior permission for an extension of this deadline any candidate who fails to

meet this submission deadline shall normally fail the dissertation or project. In such cases, on the recommendation of the Examiners, candidates may be permitted to resubmit the dissertation or project at a later date, normally within three months of the original submission deadline. Unless otherwise determined by the Senate, the maximum mark allowed for such work should be 50%.

9. **PROVISION FOR PROGRESSION**

Refer to faculty and general regulations

10. **FAILURE TO SATISFY EXAMINERS**

Refer to general regulations

11. **GRADING AND DEGREE CLASSIFICATION**

Refer to faculty regulations.

12. **DEGREE WEIGHTING**

Refer to faculty regulations.

13. **AWARD OF DEGREE**

Refer to general regulations

14. **PROGRAMME STRUCTURE**

Level 1 Semester 1

Code	Module Description	Credits
MTHE 709	Tourism and hospitality enterprises management	18
MTHE710	Financial management for Tourism Enterprises	18
MTHE711	Tourism and Hospitality Enterprises Innovations	18
MTHM 712	Tourism and Hospitality Enterprises e-commerce	18

Level 1 Semester 2

MTHE 713	Tourism and hospitality enterprises entrepreneurship	18
MTHE714	Advanced Research Methods	18
MTHE 715	Marketing Tourism and hospitality enterprises	18
MTHE716	Corporate governance for tourism and hospitality enterprises	18

Level 2 Semester 1

MTHE 807	Tourism and Hospitality Enterprises Crisis Management	18
MTHE 808	Strategic management of tourism accommodation enterprises	18
MTHE 809	Strategic human resources management for tourism and hospitality enterprises	18
MTHE810	Problem-solving live Tourism and Hospitality Enterprises projects	45

Level 2 Semester 2

MTHE811	Business Research Project/Dissertation (*MTHE 810 pre-requisite)	90

15. MODULE SYNOPSES

MTHE709 Tourism and Hospitality Enterprises Management

This module creates a comprehensive understanding of the nature and historical development of Tourism and Hospitality enterprises. The module raises students' awareness of the global importance

of Tourism and Hospitality enterprises and their unique structural and behavioural characteristics. The module therefore lays the foundation of this postgraduate degree programme.

MTHE 710 Financial Management for Tourism Enterprises
The aim of this module is to provide students with the requisite financial skills and knowledge needed to manage Tourism and Hospitality Enterprises. The students will be required to apply financial management skills to both live and theoretical Tourism and Hospitality Enterprises.

MTHE 711 Tourism and Hospitality Enterprises Innovations
Tourism and Hospitality Enterprises are renowned for their innovations. Therefore, students are expected to demonstrate how innovation has created sustainable competitive advantage for some Tourism and Hospitality Enterprises. Students also need to be aware of managerial strategies that are required to foster a culture of innovation among Tourism and Hospitality Enterprises.

MTHE 712 Tourism and Hospitality Enterprises E-commerce
The main objective of this module is to provide education and training for current and future tourism managers around the important role of ecommerce in today's business world. The module will cover how digital solutions can be deployed by Tourism and Hospitality Enterprises to create formidable value chains.

MTHE713 Tourism and Hospitality Enterprises Entrepreneurship
The module seeks to build the necessary competencies and motivation to transform students from employees to entrepreneurs. Students will explore theories around business entrepreneurship and best managerial practices among leading Tourism and Hospitality Enterprises entrepreneurs.

MTHE 714 Advanced Research Methods
The major objective of this module is to equip students with the relevant skills needed to undertake tourism research at postgraduate level. Additionally, the course will draw heavily on research that is relevant to Tourism and Hospitality Enterprises. It is expected that this course will give students further competencies needed to publish in peer reviewed platforms.

MTHE 715 Marketing Tourism and Hospitality Enterprises

This module will examine the marketing of tourism and hospitality businesses on a global scale. Students will explore best practices among Tourism and Hospitality Enterprises in terms of marketing management. This is expected to raise awareness and understanding of the vital role of marketing in creating sustainable competitive advantage for Tourism and Hospitality Enterprises.

MTHE 716 Corporate Governance for Tourism and Hospitality Enterprises
The modules will cover issues of corporate governance in Tourism and Hospitality Enterprises. The module covers such topics as the applicability of the King Code III, stakeholders in the governance structures, board composition, performance and remuneration management, risk governance, succession planning and ethics.

MTHE 807 Tourism and Hospitality Enterprises Crisis Management
This module is designed to give students advanced skills in handling tourism crises. The tourism industry is today affected by a range of natural and man-made crises. To this end, the module examines how Tourism and Hospitality Enterprises are responding to these crises and explores contemporary disaster management strategies and models.

MTHE 808 Strategic Management for Tourism Accommodation Enterprises
Investment in tourism accommodation is the major entry point for many entrepreneurs. It is with this background in mind that this module is designed to focus on the theory, principles and practices of managing tourism accommodation. The module considers key issues, current trends, and challenges among tourism accommodation providers.

MTHE 809 Strategic Human Resources Management for Tourism and Hospitality Enterprises

The module explores how Tourism and Hospitality Enterprises can be reconfigured in order to derive inimitable and sustainable competitive advantage from their human capital. The main themes explored in the module are employee recruitment, induction, training and development, retention and motivation. Students are expected to articulate leading and appropriate human resources strategies that can be utilized by Tourism and Hospitality Enterprises.

MTHE 810 Problem-solving live Tourism and Hospitality Enterprises Projects
This module brings to the fore, the learnt knowledge to the world of business in order to solve practical business problems. To this end, students will prescribe, recommend and implement feasible measures to address challenges faced by selected Tourism and Hospitality Enterprises. Students will therefore be working closely with tourism entrepreneurs in this module.

MTHE 811 Business Research Project
In this module, students will initiate a Tourism and Hospitality Enterprise business plan in association with the university hub and the department. The Tourism and Hospitality Enterprises business plan should reflect innovative and enterprising ideas in its conception and possible operation.

GRADUATE SCHOOL OF BUSINESS LEADERSHIP

REGULATIONS FOR THE MASTER IN BUSINESS ADMINISTRATION DEGREE (BLEAD01)

Duration	2 years
Credit Load	378
Minimum Credit Load	360
Maximum Credit Load	430
Maximum MBK's Credit Load	288
ZNQF Level	9

1. INTRODUCTION

These Regulations shall be read in conjunction with the General Regulations for Postgraduate Studies, hereinafter referred to as the General Postgraduate Regulations.

2. DEFINITION OF TERMS

2.1 **Faculty Post Graduate Studies Committee** – Means the Committee comprising the Dean of the Faculty of Business Sciences (who shall be the Chairperson) the GSBL Director, all Chairpersons in the Faculty of Business Sciences, all Graduate School of Business Leadership lecturers and the External Examiners.

2.2 For the definition of Special Examinations, Programme, Level, Semester, Module, Project, Continuous Assessment, Retake, Carryover, Repeat, Withdraw, Aegrotat, Credit, Dissertation refer to Section 1.9 of the General Post Graduate Regulations.

3. ENTRY REQUIREMENTS

3.1 To be eligible for admission to the MBA Programme, a candidate must normally:

3.1.1 Be at least 25 years of age.

3.1.2 Be a holder of at least a second-class degree or its equivalent awarded by Midlands State University or any other recognized institution.

3.1.3 Be a holder of a third-class degree or its equivalent and may be admitted only after providing evidence of academic maturity in the field of study as shall be judged by the Faculty Post Graduate Studies Committee.

3.1.4 Be required to provide two references from reputable people certifying their ability and suitability to undertake the programme successfully.

3.1.5 Have at least two (2) years relevant working experience.

3.1.6 Undergo an interviewing process that shall be conducted by the Faculty Post Graduate Studies Committee.

4. GENERAL PROVISIONS

4.1 The MBA programme shall normally be studied over a minimum period of two academic years on a full-time basis.

4.2 The programme maximum duration shall be four academic years.

4.3 Each module shall have 60 contact hours.

4.4 The candidate shall complete the programme after passing a minimum of 15 modules which comprise 13 core modules and 2 non-core modules. The core modules have asterisks in the programme structure.

4.5 Each candidate is required to submit a dissertation of approximately 20 000 words on a topic relating to his/her area of managerial interest and approved in writing by the dissertation supervisor.

4.6 Candidates shall be required to submit three hard bound copies of dissertation, one of which shall be deposited in the University Library, the other shall be submitted to the GSBL Office and the last copy should be returned to the student.

4.7 The format for the dissertation shall be provided by the GSBL Office.

4.8 Fees for the MBA programme shall be determined by the University Fees Revision Committee from time to time.

4.9 Fees shall be paid per semester.

5. ASSESSMENT

5.1 Evaluation of students shall normally be based on continuous assessment as well as formal examinations that are held at the end of each semester.

5.2 Continuous Assessment shall account for 30% of the overall assessment.

5.3 The final examination shall account for 70% of the overall assessment.

5.4 Items that should constitute the continuous assessment, and the weighting attached to each item, shall depend on the module in question.

5.5 With the exception of dissertation, sessional examination shall be conducted in all modules.

5.6 To be admitted for examinations, a candidate must:

5.6.1 Be a registered student of the University in accordance with the General Postgraduate Regulations.

6. PROVISIONS FOR PROGRESSION

6.1 The Faculty Post Graduate Studies Board of Examiners shall meet at the end of each semester to ratify the marks obtained by candidates.

6.2 A student shall progress automatically from one level to the other even carrying the modules from the previous level.

7. FAILURE TO SATISFY THE EXAMINERS

A candidate who fails to satisfy the examiners in terms of the Masters in Business Administration Regulations, read in conjunction with the General Post Graduate Regulations, shall retake failed module(s) in the subsequent semester(s).

8. REMARKING OF EXAMINATIONS

8.1 No requests for remarks of examinations shall be entertained unless the Senate directs such requests.

8.2 However, a student who has been given a withdrawal decision may appeal against such a decision to the Registrar, in writing, within 21 days after the publication of Examination results.

9. DEGREE CLASSIFICATION

The MBA degree shall be classified as follows:

80%-100%	Pass with Distinction
70%-79%	Pass with Merit
55%-69%	Credit
50%-54	Pass
0%- 49%	Fail

10. DEGREE WEIGHTING

In the overall degree classification, all Levels shall be equally weighted as follows:

Level One	-	50%
Level Two	-	50%

11. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MBA731	Financial Accounting	18
MBA732	Management Information Systems	18
MBA733	The Economic Environment	18
MBA734	Organisational Behaviour	18
MBA739	Business Statistics	18

Level 1 Semester 2

Code	Module Description	Credits
MBA736	Human Resource Management	18
MBA737	Financial Management	18
MBA738	Marketing Management	18
MBA849	Corporate Governance	18
MBA835	E-Commerce	18

Level 2 Semester 1

Code	Module Description	Credits
MBA740	Business Research Methods	18
MBA845	Strategic Management	18
MBA832	Entrepreneurship	18
MBA841	Change Management	18
MBA844	Project Management	18

Level 2 Semester 2

MBA851	Business Leadership	18
MBA850	Dissertation	90

12. MODULE SYNOPSES

MBA731 Financial Accounting
The module deals with accounting principles and conventions. It will attempt to develop accounting skills and enhance appreciation of accounting related information. The main topics include record keeping and double entry, the preparation of the profit and loss account and balance sheet, profit measurement, valuation of assets, company accounts, legal and regulatory framework, international accounting standards, some aspects of auditing, some aspects of taxation.

MBA732 Management Information Systems
The module exposes students to modern information management strategies. Topics to be covered include; Concepts of systems and organisation, strategic uses of information technology computer system resources (computer, hardware, computer software, file and data base management systems). The internet and office communication, applications of information systems to functional business areas, tactical and strategic information systems to business applications, decision support and expert systems, planning and the development of information systems, management of information systems and end user computing, security and ethical issues of information systems.

MBA733 The Economic Environment

The module analyses the economic fundamentals that drive the behaviour of the macro economy. It seeks to explain the behaviour of key macroeconomic variables such as output, employment and inflation. It also analyses the environment in which the macroeconomic policy in Zimbabwe is conducted.

MBA734 Organisational Behaviour

The module is intended to equip post graduate students and /or practitioners with concepts and models that provide home-grown solutions to organizational behaviour related challenges facing organizations. The emphasis is in-depth understanding of the under-pining OB concepts /theories and their application in the real world of work.

Contemporary issues to be covered include, inter-alia;

Foundations of behaviour in organizations, Practical implication of OB in the 21st century, counterproductive and productive workplace behaviours, organizational culture &climate, organizational ethics, organizational structures & designs, the learning organizational model, leadership and power, managing organizational change, motivating employees, decision-making, managing conflicts, stress and politics in organizations.

MBA736 Human Resource Management

This module is targeted at everyone in a managerial position or who will someday be in a managerial position. It explains why the human resource of an organization is the most important asset and the need for every line manager to appreciate and embrace some HRM duties in their organizations. It clearly outlines the role of HR practitioners and the need to adopt the appropriate HR practices, systems, policies and procedures which are well linked with the overall strategy of the organization. This leads to the implementation of HR systems that result in efficiency and effectiveness in the management of the flow of people into, within and out of the organization is discussed. How a strategic approach to the various HR practices like recruitment and selection, training and development, knowledge management, etc results in employee self commitment and loyalty leading to better performance outcomes for the organization is discussed. The evolution of the HR profession is given with emphasis being on the current drivers that continue to make the profession dynamic.

MBA737 Financial Management

The module is intended to offer the participants the principles and major analytical approaches that are applied in financial management to facilitate decision-making.

The module should facilitate the formulation, implementation and evaluation of financial management practices or strategies by businesses within their environments that are dynamic. Having studied the module, students should be equipped with decision-making skills that should enable them to perform completely in an ever-changing business environment.

MBA738 Marketing Management

The module aims to provide strategic insights into the broader understanding of marketing management and its critical role to the success of a business. It is, therefore, necessary to understand marketing from its basics such as the several definitions from several distinguished marketing gurus, and schools of thought, the development of marketing as a science, the main functions of marketing and concepts, the marketing environment, analysis of consumer and industrial markets, market segmentation, matching company capabilities and market opportunities (using SWOT and Portfolio tools), establishing the competitive edge through the marketing mixes and the marketing implementation and control (through the development and implementation of marketing plans).

MBA739 Business Statistics

This module introduces students to the fundamental concepts of data analysis useful in a management decision-making environment. The approach will emphasize practical management problem-solving using quantitative reasoning skills rather than the theoretical development of statistics. This understanding is also intended to assist students in developing a rigorous quantitative research methodology for the research report. Topics include Basic Concepts of Statistics, Graphical and tabular descriptive techniques, Numerical descriptive techniques, Probability and sampling Distributions, Data collection & Sampling, Confidence Interval Estimation, Inference about comparing two populations, Chi-Squared tests, Simple Linear Regression and Correlation Analysis and Time series analysis

MBA740 Business Research Methods

This module will provide an understanding of integrated research methodology which will enable students to undertake a study commensurate with the master's level.
The module will show how academic theory is developed and will provide research training in the individual components of research design, implementation and reporting.
The module will outline the basic steps in the formulation of research questions and the way in which this should be used to derive a data specification and method of data collection. Strengths and weaknesses of different data collection methods will be emphasized and students will be encouraged to develop their own research skills.

MBA845 Strategic Management

The students are initially introduced to the basic concepts, principles and the main analytical approaches that are used in strategic management. The emphasis is on formulation, implementation and evaluation of long term strategies utilized by business organizations. The issues that will be factored in include the environment, the constant flux of society and the strategies required for the management of that flux. Main topics include: the role of analysis and planning, mission, objectives, values, ethics and goals. Also analysed will be issues relating to markets, industries, competitors, stakeholder's internal capabilities, value chains, resources and portfolios core and generic strategies, structure, culture and organizational design, strategy evaluation, implementation and control.

MBA832 Entrepreneurship

Entrepreneurship is a module that is all about starting and running your own business. It is much more than a set of knowledge or skills, and a set of attitudes that everyone can benefit from. This module is a very dynamic, practical, and hands-on type of course where students are allowed to both practice what they have learned and even practice what they have not yet learned. A great attitude, work ethic, creativity, and sense of adventure are required for success in this course.

In addition to providing students with an understanding of entrepreneurial activity, the aim of this module is also to give student participants a rigorous and honest insight into the realities of entrepreneurial activity by focusing on the manifold requirements necessary to turn a rudimentary idea into a feasible and attractive new enterprise. By the end of the module, students will have a clear idea of the nature of the Entrepreneurial environment and the risks and rewards of self-employment. They will also be well equipped to write a credible and informed business plan; assess whether or not the entrepreneurial route could be an appropriate career choice; and, if it is, be able to embark with some greater confidence on the process of converting an idea into commercial reality. If students already have or are working on an idea for a new business, this module provides an ideal opportunity to develop, refine and test the idea or concept. For those students wishing to take their idea further and to raise external resources, the module provides direct experience in how to conceive, present and communicate an attractive and credible business plan

MBA844 Project Management
This module involves a detailed study of the entire Project through the Project Life Cycle from initiation to termination. This involves coming up with a business case, resource loading and leveling, monitoring and evaluation up to project commissioning. Emphasis is put on the management of the quadruple constraints namely time, budget, quality and scope in achieving project objectives. The module involves a detailed study of network analysis and scheduling techniques. The main topics include the following: Strategic Project Management, Project Life Cycle, Project Planning and Scheduling, Project Team Dynamics, Project Organisational Structures, Project Feasibility Studies, Project Appraisal and Risk Management, Earned value Analysis, Project Budgeting and Procurement, Code of Ethics and Responsibility code, Contracts Management, Project Leadership and Integration and Management and Control of Donor Funded Projects and the application of Project Software in Project Management with emphasis on MS Project, MS Access, MS Visio and Excel. The module gives emphasis on the alignment of the Project Strategy to the overall organisational strategy which is commonly referred to as project maturity. The EMBA module gives emphasis on managerial application of theoretical concepts.

MBA841 Change Management
The module focuses on the management of change in organizations. It will also swell on the types of organizations that are in existence, and how the differing structures of these organizations impact on change management. The main topics covered include the following: Performance, design strategies, concept of power, organizational politics, change, renaissance, Organisational theory, models of growth and decline, structural forms, organizational and the management of change.

MBA835 E-Commerce
The module looks at computer-based transactions. It includes E- Business i.e. the transaction of business via electronic gadgets. It analyses the usage and effect of the internet, Electronic Fund transfers, computer based transactions, consumer responses e.g. as we look at the future, the hole of the computer on business transactions is increasing in importance.

MBA849 Corporate Governance
Corporate governance deals with the complex set of relationships between the corporation and its board of directors, management, shareholders, and other stakeholders. In the recent years, the regulators and legislators have intensified their focus on how businesses are being run. They are endeavoring to create a template for new corporate governance and disclosure measures, which is beneficial for both the stakeholders and controllers. Students will be exposed to challenges of good Corporate Governance by comparing different systems and approaches. This will enhance their understanding of the roles and responsibilities of direction setting and strategic thought process. The module will provide cross-functional insight on the effective integration of corporate performance and conformance roles. In the process we hope to create a core team of trained directors with determination, focused intelligence, creativity and innovativeness, to lead the wealth creating organs of society. Some of the key areas include: history of corporate governance; powers and rights of the relations between directors and shareholders' financial reporting and auditing.

MBA851 Business Leadership
It is more than ever that organizations are appreciating the relationship between leadership and performance. Leadership ultimately determines the kind of organizations that we have and the quality of life that we enjoy. Thus the module seeks to impact leadership skills to managers for the betterment of the organizations and society at large. The areas of focus

include; Framework for studying Business Leadership; Leadership theories; Leadership Styles; Leadership roles; measurement of effective leadership.

MBA850 Dissertation

The dissertation is carried out by a student in his/her chosen area of interest in the field of business administration. It equips the student with competencies in conducting independent research, thus encompasses identifying a research problem, conducting a literature review, writing the research methods section, conducting data analysis and report writing. The candidate is expected to apply knowledge acquired from the business research methods module to solve business administration problems. The dissertation shall be supervised by one or more lecturers, from one chapter to the next, and be approximately 20 000 words, excluding preliminary pages and appendices.

REGULATIONS FOR THE EXECUTIVE MASTER IN BUSINESS ADMINISTRATION DEGREE PROGRAMME (BLEAD02)

Duration	1.5 years
Credit Load	378
Minimum Credit Load	360
Maximum Credit Load	430
Maximum MBK's Credit Load	288
ZNQF Level	9

1. INTRODUCTION

These Regulations shall be read in conjunction with the General Regulations for Postgraduate Studies, hereinafter referred to as the General Postgraduate Regulations.

2. DEFINITION OF TERMS

2.1 **Faculty Post Graduate Studies Committee** – Means the Committee comprising the Dean of the Faculty of Business Sciences (who shall be the Chairperson) the GSBL Director, all Chairpersons in the Faculty of Business Sciences, all EMBA lecturers and the External Examiners.

2.2 For the definition of Special Examinations, Programme, Level, Semester, Module, Project, Continuous Assessment, Retake, Carryover, Repeat, Withdraw, Aegrotat, Credit, and Dissertation (refer to Section 1.9 of the General Post Graduate Regulations).

3. ENTRY REQUIREMENTS

3.1 To be eligible for admission to the EMBA Programme, a candidate must normally:

3.1.1 Be at least 35 years of age.

3.1.2 Be an Executive of their own business or has been a manager for at least five (5) years.

3.1.3 Hold a first degree from a recognised institution in a recognised area of study.

3.1.4 Be required to provide two references from reputable people certifying their ability and suitability to undertake the programme successfully.

3.1.5 Undergo an interviewing process that shall be conducted by the Faculty Post Graduate Studies Committee.

4. GENERAL PROVISIONS

4.1 The EMBA programme shall normally be studied over a minimum period of two academic levels, or 18 months on a full-time basis.

4.2 The candidate shall complete the programme after passing a minimum of 13 modules which comprise 11 core modules and 2 non-core modules. The core modules have asterisks in the programme structure.

4.3 Each module shall be studied over 60 contact hours.

4.4 Each candidate is required to submit a dissertation of approximately 20 000 words on a topic relating to his/her area of managerial interest and approved in writing by the dissertation supervisor.

4.5 Candidates shall be required to submit three hard-bound copies of the dissertation, one of which shall be deposited in the University Library, the other shall be submitted to the GSBL Office and the last copy should be returned to the student.

4.6 The format for the dissertation shall be provided by the GSBL Office.

4.7 Fees for the EMBA programme shall be determined by the University Fees Revision Committee from time to time.

4.8 Fees shall be paid per semester.

5. ASSESSMENT

5.1 Evaluation of students shall normally be based on continuous assessment as well as formal examinations that are held at the end of each semester.

5.2 Continuous Assessment shall account for 40% of the overall assessment.

5.3 Final examination shall account for 60% of the overall assessment.

5.4 Items that shall constitute the continuous assessment, and the weighting attached to each item, depend on the module in question.

5.5 With the exception of dissertation, sessional examination shall be conducted in all modules.

5.6 To be admitted for examinations, a candidate must:

5.6.1 Be a registered student of the University in accordance with the General Postgraduate Regulations.

5.6.2 Have satisfactorily attended and completed the approved modules of study. Satisfactory completion of modules shall require submission of all written assignments and tests and any other work that constitutes the continuous assessment.

6. PROVISIONS FOR PROGRESSION

The Faculty Post Graduate Studies Board of Examiners shall meet at the end of each semester to ratify the marks obtained by candidates.

A student shall progress automatically from one level to the other even carrying the modules from the previous level.

7. FAILURE TO SATISFY THE EXAMINERS

A candidate who fails to satisfy the examiners in terms of the Executive Masters in Business Leadership Regulations read in conjunction with the General Post Graduate Regulations, shall retake module(s) failed in the following semester(s)

8. REMARKING OF EXAMINATIONS

No requests for remarks of examinations shall be entertained unless Senate directs such requests.

9. DEGREE CLASSIFICATION

The EMBA degree shall be classified as follows:

80%-100%	Pass with Distinction
70%-79%	Pass with Merit
55%-69%	Credit
50%-54%	Pass
0%- 49%	Fail

10. DEGREE WEIGHTING

In the overall degree classification, all Levels shall be equally weighted as follows:

Level One - 50%
Level Two - 50%

11. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
EMBA738	Marketing Management	18
EMBA736	Human Resources Management	18
EMBA734	Organisational Behaviour	18
EMBA740	Business Research Methods	18
EMBA749	Corporate Governance and Business Ethics	18

Level 1 Semester 2

Code	Module Description	Credits
EMBA731	Financial Accounting and Management	18
EMBA845	Strategic Management	18
EMBA832	Entrepreneurship	18
EMBA841	Change Management	18
EMBA849	Managerial Economics	18

Level 2 Semester 1

Code	Module Description	Credits
EMBA835	E-Commerce	18
EMBA851	Leadership	18
EMBA850	Dissertation	90

12. MODULE SYNOSPES

EMBA731 Financial Accounting and Management
The module is meant to assist practicing managers from various disciplines who are not necessarily well grounded in financial accounting make sound financial decisions. An overview of the different branches of accounting which are Bookkeeping, Financial Accounting, Management Accounting and Financial Management as well as how these differ is given including how auditing and taxation fits in. Users of financial statements and their information interests are discussed. The components of financial reports, which consist of the Statement of Comprehensive Income, Statement of Financial Position, Statement of Changes in Equity, Statement of Cash Flows and the Notes to financial statements, are explained. Special emphasis is placed on the interpretation of financial statements and how ratios help in interpreting financial statements and making investment decisions. The limitations of financial statements and ratio analysis are also covered. The module also facilitates the formulation, implementation and evaluation of financial management practices or strategies by business.

EMBA734 Organisational Behaviour
The module is intended to equip postgraduate students and/or practitioners with concepts and models that provide home-grown solutions to organizational behaviour related challenges facing organizations. The emphasis of the module is on the practical application of organizational behaviour concepts and theories. The focus will be on contemporary issues which include, inter-alia;
Foundations of behaviour in organizations, Practical implication of OB in the 21st century, counterproductive and productive workplace behaviours, organizational culture & climate, organizational ethics, organizational structures & designs, the learning organizational model, leadership and power, managing organizational change, motivating employees, decision-making, managing conflicts, stress and politics in organizations.

EMBA736 Human Resource Management
This module is targeted at everyone in a managerial position or who will someday be in a managerial position. It explains why the human resource of an organization is the most important asset and the need for every line manager to appreciate and embrace some HRM duties in their organizations. It clearly outlines the role of HR practitioners and the need to adopt the appropriate HR practices, systems, policies and procedures which are well linked with the overall strategy of the organization. This leads to the implementation of HR systems that result in efficiency and effectiveness in the management of the flow of people into, within and out of the organization is discussed. How a strategic approach to the various HR practices like recruitment and selection, training and development, knowledge management, etc results in employee self commitment and loyalty leading to better performance outcomes for the organization is discussed. The evolution of the HR profession is given with emphasis being on the current drivers that continue to make the profession dynamic.

EMBA738 Marketing Management
The module aims to provide strategic insights into the broader understanding of marketing management and its critical role to the success of a business. It is, therefore, necessary to understand marketing from its basics such as the several definitions from several distinguished marketing gurus, and schools of thought, the development of marketing as a science, the main functions of marketing and concepts, the marketing environment, analysis of consumer and industrial markets, market segmentation, matching company capabilities and market opportunities (using SWOT and Portfolio tools), establishing the competitive edge through

the marketing mixes and the marketing implementation and control (through the development and implementation of marketing plans).

EMBA740 Business Research Methods
This module will provide an understanding of integrated research methodology which will enable students to undertake a study commensurate with masters level. The module will show how academic theory is developed and will provide research training in the individual components of research design, implementation and reporting. The module will outline the basic steps in the formulation of research questions and the way in which this should be used to derive a data specification and method of data collection. Strengths and weaknesses of different data collection methods will be emphasized and students will be encouraged to develop their own research skills.

EMBA749 Corporate Governance and Business Ethics
Corporate governance deals with the complex set of relationships between the corporation and its board of directors, management, shareholders, and other stakeholders. In the recent years, the regulators and legislators have intensified their focus on how businesses are being run. They are endeavoring to create a template for new corporate governance and disclosure measures, which is beneficial for both the stakeholders and controllers. Students will be exposed to challenges of good Corporate Governance by comparing different systems and approaches. This will enhance their understanding of the roles and responsibilities of direction setting and strategic thought process. The module will provide cross-functional insight on the effective integration of corporate performance and conformance roles. In the process we hope to create a core team of trained directors with determination, focused intelligence, creativity and innovativeness, to lead the wealth creating organs of society. Some of the key areas include: history of corporate governance; powers and rights of the relations between directors and shareholders' financial reporting and auditing.

EMBA845 Strategic Management
The students are initially introduced to the basic concepts, principles and the main analytical approaches that are used in strategic management. The emphasis is on formulation, implementation and evaluation of long term strategies utilized by business organizations. The issues that will be factored in include the environment, the constant flux of society and the strategies required for the management of that flux. Main topics include: the role of analysis and planning, mission, objectives, values, ethics and goals. Also analysed will be issues relating to markets, industries, competitors, stakeholder's internal capabilities, value chains, resources and portfolios core and generic strategies, structure, culture and organizational design, strategy evaluation, implementation and control.

EMBA832 Entrepreneurship
Entrepreneurship is a module that is all about starting and running your own business. It's much more than a set of knowledge or skills, but it's also a set of attitudes that everyone can benefit from developing. This module is a very dynamic, practical, and hands-on type of course where students are given the opportunity to both practice what they've learned - and even practice what they haven't yet learned! A great attitude, work ethic, creativity, and sense of adventure are required for success in this course.
In addition to providing students with an understanding of entrepreneurial activity, the aim of this module is also to give student participants a rigorous and honest insight into the realities of entrepreneurial activity by focusing on the manifold requirements necessary to turn a rudimentary idea into a feasible and attractive new enterprise. By the end of the module, students will have a clear idea of the nature of the Entrepreneurial environment and the risks

and rewards of self-employment. They will also be well equipped to write a credible and informed business plan; assess whether or not the entrepreneurial route could be an appropriate career choice; and, if it is, be able to embark with some greater confidence on the process of converting an idea into commercial reality. If students already have or are working on an idea for a new business, this module provides an ideal opportunity to develop, refine and test the idea or concept. For those students wishing to take their idea further and to raise external resources, the module provides direct experience in how to conceive, present and communicate an attractive and credible business plan

EMBA841 Change Management
The module focuses on the management of change in organizations. It will also swell on the types of organizations that are in existence, and how the differing structures of these organizations impact on change management. The main topics covered include the following: Performance, design strategies, concept of power, organizational politics, change, renaissance Organisational theory, models of growth and decline, structural forms, organizational and the management of change.

EMBA849 Managerial Economics
This module basically deals with topics that relate to a managerial decision–making process. Some of the topics deal with business objectives, decision analysis, demand measurement and forecasting, pricing policies, advertising and promotion decisions, corporate strategies and game theory for managers.

EMBA 835 E-Commerce
The module looks at computer-based transactions, internationally. It includes E- Business i.e. the transaction of business via electronic gadgets. It analyses the usage and effect of the internet, Electronic Fund transfers, computer based transactions, consumer responses e.g. as we look at the future, the hole of the computer on business transactions is increasing in importance.

EMBA851 Leadership
It is more than ever that organizations are appreciating the relationship between leadership and performance. Leadership ultimately determines the kind of organizations that we have and the quality of life that we enjoy. Thus the module seeks to impact leadership skills to managers for the betterment of the organizations and society at large. The key issues to be discussed include the following: Framework for studying leadership; Leadership theories; leadership styles; leadership roles; measurement of effective leadership.

EMBA850 Dissertation
The dissertation is carried out by a student in his/her chosen area of interest in the field of business administration. It equips the student with competencies in conducting independent research, thus encompasses identifying a research problem, conducting a literature review, writing the research methods section, conducting data analysis and report writing. The candidate is expected to apply knowledge acquired from the business research methods module to solve business administration problems. The dissertation shall be supervised by one or more lecturers, from one chapter to the next, and be approximately 18 000 words, excluding preliminary pages and appendices.

REGULATIONS FOR THE DOCTOR OF BUSINESS LEADERSHIP DEGREE (DBL)

Duration	3 years
Credit Load	400
Minimum Credit Load	340
Maximum Credit Load	440
Maximum MBKs Credit Load	300
ZNQF Level	10

1. INTRODUCTION

These Regulations shall be read in conjunction with the General Academic Regulations for Postgraduate Diplomas and Taught Masters Degrees and Doctor of Philosophy degrees, hereinafter referred to as the General Regulations.

2. TARGET GROUP

The Doctor of Business Leadership (DBL) is a doctoral degree programme for entrepreneurs, directors, general managers, management educators and consultants in both the public and private sectors as well as those not in conventional businesses, such as leaders in religion, politics, clubs and sports. The programme will provide an opportunity to practicing managers to investigate the ongoing practices in their organisations and develop new contextual theories, or theories that will compliment those built around the practices of organisations in more developed markets.

3. ENTRY REQUIREMENTS

To be eligible for admission to the DBL programme, a candidate must normally:

3.1 Have a minimum of five (5) years of working experience of which at least three (3) must be in senior positions.

3.2 In addition to a relevant business-related Master's degree from a recognized institution, the applicant must have passed the first degree with a 2.1 or its GPA equivalent classification. Applicants with 2.2 may be considered if they have more than 5 years of managerial experience and must undergo an assessment interview.

3.3 Be required to provide two references from reputable people certifying their ability and suitability to undertake the programme successfully.

3.4 Submit an acceptable Application Proposal or business portfolio**.**

4. GENERAL PROVISIONS

4.1 The degree shall be awarded to candidates who have successfully completed the programme and passed examinations in accordance with the General Regulations.

4.2 The duration of the degree programme shall normally be a minimum of 6 semesters or three (3) years and a maximum of 12 semesters or six (6) years. An extension of one year may be granted depending on the supervisor's progress reports.

4.3 A candidate is required to undertake ten (10) modules in the first three (3) semesters of which three (3) modules are research-oriented and the remaining seven (7) are discipline specific.

4.4 At the beginning of the third semester, the candidate is expected to submit a Concept Note to the school. The school will review the Concept Note and recommend supervisor(s) once it is found to meet the minimum requirements for this level of study. Thereafter, the Concept Note will be reviewed by the Faculty Higher Degrees Committee and eventually submitted to the Deans' Committee on Academic Affairs (DCAA) for approval. Once approved by DCAA, the Research and Innovation Directorate, in liaison with the Human Resources Department, shall process contracts for the supervisor(s) so that the supervision process can begin.

4.5 The candidate, in liaison with the supervisor(s), shall be given one (1) semester to develop and complete the DBL Research Proposal. Upon completion, the candidate shall submit the proposal to the Faculty Higher Degrees Committee, through the Graduate School of Business Leadership, for approval. The Faculty Higher Degrees Committee shall arrange for an Oral Presentation of the DBL Research Proposal through the Faculty Seminar Series platform, as part of the approval process. If the candidate does not submit the proposal after two (2) semesters, the Faculty Higher Degrees Committee may recommend that the candidate's enrolment be terminated. Such candidates shall be awarded a Postgraduate Diploma in Business Leadership.

4.6 Once the Research Proposal has been approved by the Faculty Higher Degrees Committee, the candidate can then proceed with thesis writing.

4.7 From the third semester, colloquia shall be held every semester to provide open discussions on candidates' research progress. It is strongly recommended that all registered DBL candidates attend these colloquia since they provide opportunities for constructive interaction between academics and candidates.

4.8 Each taught module will be studied over 60 contact hours.

4.9 Each candidate should get at least 400 credits in order to pass the DBL programme**. The thesis on its own carries 200 credits.**

4.10 Each candidate is required to submit a doctoral thesis of **40 000 – 60 000 words**, excluding preliminary pages and appendices.

4.11 Fees for the DBL programme shall be determined from time to time by the University Fees Revision Committee.

4.12 Fees shall be paid per semester.

5. SUPERVISION OF DBL THESIS

Refer to Section 8 of the General Academic Regulations for Master of Philosophy degrees and Doctor of Philosophy Degrees.

6. ASSESSMENT OF TAUGHT MODULES

6.1 Evaluation of students shall normally be based on continuous assessment **or prescribed assignments to be completed within a specific period** as well as formal examinations that are held at the end of each semester.

6.2 Continuous Assessment shall account for 50% of the overall assessment.

6.3 The final examination shall account for 50% of the overall assessment.

6.4 Items that shall constitute continuous assessment, and the weighting attached to each item, depend on the module in question.

6.5 With the exception of the doctoral thesis and colloquia, sessional examinations shall be conducted in all modules.

6.6 To be admitted for examinations, a candidate must:

6.6.1 Be a registered student of the University in accordance with the General Regulations.

6.6.2 Have satisfactorily attended and completed the approved modules of study. Satisfactory completion of modules shall require submission of all written assignments and tests and any other work that constitutes the continuous assessment.

7. DBL THESIS EXAMINATION

Refer to Section 10 of the General Academic Regulations for Master of Philosophy Degrees and Doctor of Philosophy Degrees.

8. PROVISIONS FOR PROGRESSION

8.1 All applicants to this programme register for DBL from the onset.

8.2 To progress to the next level, the candidate shall be required to pass at least 80% of the modules offered.

8.3 For deferment of studies, refer to Section 14 of the General Academic Regulations for Master of Philosophy Degrees and Doctor of Philosophy Degrees.

8.4 A candidate must pass all research related modules before embarking on Doctoral thesis writing.

9. FAILURE TO SATISFY THE EXAMINERS

A candidate who fails to satisfy the examiners in terms of the DBL Regulations, read in conjunction with the General Regulations, shall retake module(s) failed in the semester when they will be on offer.

10. REMARKING OF EXAMINATIONS

No requests for remarks of examinations shall be entertained unless the Senate directs such requests.

11. PROGRAMME OF STUDY

Refer to Section 13 of the General Academic Regulations for Master of Philosophy degrees and Doctor of Philosophy Degrees.

12. PROGRAMME STRUCTURE

LEVEL 1 SEMESTER 1

Code	Module Description	Credits
DBL931	Communication for Management Research	20
DBL932	Philosophy of Science and Business Research Methods I	20
DBL933	Advanced Leadership for Excellence in Global and Multicultural Organisations & Organisational Development	20
DBL934	Advanced Corporate Governance and Ethics in Business	20

LEVEL 1 SEMESTER 2

DBL935	Philosophy of Science and Business Research Methods II	20
DBL936	Fiscal Leadership in the Global Environment	20
DBL937	Advanced Strategic Marketing & Innovation in Dynamic Markets	20
DBL938	Business Innovation & Entrepreneurship	20

LEVEL 2 SEMESTER 1

DBL939	Advanced Strategic Management in Action	20
DBL940	Contemporary Issues in Leadership	20
DBL941	Colloquia: Proposal Writing, Data Collection and Data Analysis	--

LEVEL 2 SEMESTER 2

DBL942 Doctoral Thesis 1
Continuing

LEVEL 3 SEMESTER 1

DBL943 Doctoral Thesis 2
Continuing

LEVEL 3 SEMESTER 2

DBL944	Final Doctoral Thesis	200

13. MODULE SYNOPSIS

DBL931 Communication for Management Research

This module aims to develop and enhance students' ability to communicate their research data and results in both written and oral form. The communication of scholarly research is somewhat different from managerially-oriented writing. This module has the objective of

helping new DBL students to improve their writing and presentation skills so that they become successful at publishing and presenting their research at conferences.

DBL932 Philosophy of Science and Business Research Methods I
This module introduces students to the philosophy of science. The justification, interpretation and application of scientific theories and models is discussed. It covers the generation of scientific theories with an emphasis on qualitative research techniques and includes topics on observational field research, case histories, narratives, case studies, interviewing, principles of action research, evaluation and grounded theory. It examines the strengths of alternative methods and the connection between qualitative and quantitative research methods.

DBL933 Advanced Leadership for Excellence in Global and Multicultural Organizations and Organisational Development
This module aims at developing a deeper level of appreciation for the issues and trends in Organizational Leadership. It is assumed that students have previously had exposure to the basic issues through formal exposure as well as their own experiences of organizational practices. This overview of the field is designed to enable students to discover connections between theory and practice and help them develop an integrated perspective on management. This module also covers the major topics in managing multicultural organizations such as comparative leadership and management issues, the formulation and implementation of appropriate strategies, building strategic alliances, negotiations, cross-culture communications, international human resource management, and business ethics.

DBL934 Advanced Corporate Governance and Ethics in Business
This module will help students to understand and apply ethical decision-making models within an organizational management model. These models will introduce students to good corporate governance and ethical skills, vocabulary, and other tools necessary to behave in an ethical manner, not only within organizations but in society at large.

DBL935 Philosophy of Science and Business Research Methods II
The module aims at providing doctoral students with statistical techniques and skills for analyzing quantitative data that may be required for their research. The emphasis is on the practical uses of these tools in real databases in order to discover and prove new scientific facts and relationships. Some of the mistakes most commonly committed in the process will be discussed.

DBL936 Fiscal Leadership in the Global Environment
The objectives of the module are to explore the important elements and research issues in corporate finance decision-making, to present strategic finance issues integrating with strategy and other functional areas; and to develop in students the competence to manage complex strategic decisions, through the interplay of corporate finance and its linkage to competitive strategies.

DBL937 Advanced Strategic Marketing and Innovation in Dynamic Markets
The module is designed to evaluate and practice case research in the field of strategic marketing and innovation. It provides students with the skills to identify and relate theoretical concepts to the strategic marketing practices of a company. In particular, the module examines the impact of local marketing management practices compared to those in developed and emerging markets and looks at how we can incorporate contextual factors in marketing research.

DBL938 Business Innovation and Entrepreneurship

This module examines the nature of learning, creativity and innovation and through entrepreneurial leadership the ability to identify market opportunities based on thinking new things. Detailed attention is given to the entrepreneurial process: the concepts, skills, know-how and know-who, information, attitudes, alternatives and resources that entrepreneurs need to manage creatively in the process of developing something with tangible economic value. The module relies on classroom discussion, participation, guest speakers, case analysis, company visits, the creation of a feasibility plan, and building a business plan to develop a strategy for launching and managing a business.

DBL939 Advanced Strategic Management in Action
The module is designed to develop an understanding of the context and practices of strategy and the nature of theoretical traditions in the field of strategic management. Through seminars and case studies the students discover the external and internal challenges faced by leaders and entrepreneurs in the rapidly changing and increasingly complex environments in strategizing and implementing strategic change. The real-life exercises help the students to acquire and develop their strategic skills and competences.

DBL940 Contemporary Issues in Leadership
This module **is designed to a**nalyse the modern business landscape, together with the challenges and opportunities currently being experienced by organisations. Through seminars and case studies, students will, among other things, **identify the five dimensions of trust, define the qualities of a charismatic leader, Identify the skills that visionary leaders exhibit and explain how to find and create effective leaders.**

DBL941 Colloquia: Proposal Writing, Data Collection and Data Analysis
This is a compulsory non-examinable module, with the purpose of assisting doctoral students acquire relevant skills for writing their theses. In this module, students will focus on developing a research topic, conducting literature review, developing an argumentative essay of significant length concerning their thesis proposals, and document sources appropriately while focusing on the critical thinking and writing skills established in DBL901. Fundamental elements of data collection, to help students understand and become proficient in the appropriate methods for collecting data will be covered. Issues to be covered will include research designs, sampling techniques, data collection methods and instruments, negotiating access, data collection procedures, testing for data validity and reliability and ethical issues of research. Both qualitative and quantitative data analysis techniques will be covered, together with the use of appropriate data analysis packages. Students use this module as a forum for the ongoing exchange of ideas, input, and feedback with peers and their mentor.

DBL944 Doctoral Thesis
Ultimately, students shall write a doctoral thesis of 40 000 – 60 000 words, excluding preliminary pages and appendices, in which they make a fresh contribution to the field of practice in the professional business environment. The final doctoral study should demonstrate students' scholarly ability to examine, critique, and synthesize knowledge, theory, and experience, so that new ideas can be tested; best practices identified, established, and verified; and theories, practice, or policy constructs evaluated and advanced. Students engage in rigorous inquiry that results in new knowledge, insight, or practice, demonstrating efficacy in the world of business and management.

FACULTY OF ENGINEERING AND GEOSCIENCES

FACULTY REGULATIONS FOR UNDERGRADUATE DEGREE PROGRAMMES

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the General Academic Regulations for undergraduate Degrees hereinafter referred to as General Regulations.

1.2 These Regulations are in respect of programmes for the following degrees offered by the Faculty of Engineering and Geosciences:

Bachelor of Engineering Honours Degree in Electrical and Electronic Engineering (ENG01)
Bachelor of Engineering Honours Degree in Fuels and Energy Engineering (ENG02)
Bachelor of Engineering Honours Degree in Mechanical Engineering (ENG03)
Bachelor of Engineering Honours Degree in Materials Engineering (ENG05)
Bachelor of Engineering Honours Degree in Metallurgical Engineering (ENG04)
Bachelor of Engineering Honours Degree in Mining Engineering (ENG08)
Bachelor of Science Honours Degree in Applied Geology (ENG06)
Bachelor of Science Honours Degree in Exploration Geophysics (ENG07)

1.3 The appropriate Degree will be awarded to a student who has successfully completed an approved programme.

2. DEFINITION OF TERMS

2.1 A Practical-Theory module is a module which contains both practical and theory components in the formal examination.

2.2 A Practical module is a module in which a student is required to carry out, under the direction and supervision of a member of staff in a laboratory or in the field or in the workshop, a series of practical exercises related to particular topics within a subject.

2.3 A Field Trip is an excursion outside University Campus under the supervision of academic and/or technical staff of the University. It may last for part of a day or be conducted over a longer period. The Field trip can form part of a module or if it is of sufficient duration, form a 12-credit module in its own right.

2.4 A Project module is a module in which a student is required to carry out under the direction and supervision of a member of the academic staff, private studies or investigations related to a particular topic within a module. Each Project module shall be weighted based on the amount of time it occupies compared with a theory module.

2.5 A Pre-requisite is a module the departmental board requires the student to pass prior to the student registering for any other related higher level modules.

2.6 Approved First Aid Certificate means any First Aid Certificate duly issued by the Midlands State University, St. John Ambulance Services, Zimbabwe Red Cross Society or any other Institutions recognized by the Republic of Zimbabwe.

2.7 Mine Blasting Licence means a mine blasting licence granted in terms of the Government Explosive Regulations.

2.8 Field camp means a practical module normally conducted and examined outside campus for a period of up to two weeks to afford the student a real practical field experience.

3. ENTRY REQUIREMENTS

3.1 Normal Entry

For one to qualify for normal entry to an Undergraduate Degree programme in the Faculty of Engineering and Geosciences, a prospective student must satisfy the entry requirements specified in the General Regulations and in so doing, must also meet the following Faculty requirements:

3.1.1 have obtained a pass at 'A' Level in at least two of the following subjects or their recognized equivalents: Mathematics, Chemistry, Physics, and Geography. A department may prescribe specific combinations of these 'A' Level subjects. (Refer to Departmental regulations)

3.1.2 have obtained a pass at 'O' Level in a third subject chosen from those listed in 3.1.1.

3.1.3 have obtained a pass at 'O' Level in Mathematics and English

Language.

3.2 Special Entry

Candidates with recognized diplomas in the respective areas of study may be admitted into relevant programmes subject to departmental approval.

3.3 Mature Entry

Refer to Section 3.3 of the General Regulations.

3.4 Visiting School Undergraduate Degree programmes

In order to qualify for entry to the Visiting School programmes offered by the Faculty of Engineering and Geosciences, a candidate must, in addition to satisfying the entry requirements described above, be employed within a related field for the duration of the programme.

4. STRUCTURE OF PROGRAMMES

4.1 The minimum duration of the Bachelor of Engineering Honours Degree shall include a minimum of six taught semesters plus at least six months of Work-related Learning.

4.2 Bachelor of Engineering Honours Degree programmes will be taught and examined at least at 4 different levels and there are prescribed modules for each level.

5. GENERAL PROVISIONS

5.1 In Level 1, normally a student shall register for at least 12 modules from at least two departments of which a maximum of 4 modules shall be from other Faculties.

5.2 For Level 2 and higher modules, all pre-requisites that apply to the modules shall be listed in the special regulations for the modules to which the modules form a portion. A student shall not be registered to level 2 or higher level modules unless he/she has passed all pre-requisites for the particular module prior to the start of the semester in which he intends to study that module.

5.3 A student may include in his/her programme, modules offered by other Faculties up to a total equivalent of at most 8 modules.

5.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of the Faculty of Engineering and Geosciences and the

Chairpersons of all relevant Departments and to the following conditions:

5.4.1 A student shall register for at least 12 modules at any time during

the academic year.

5.4.2 The module combination is feasible in terms of timetable.

5.4.3 A student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester.

5.4.4 A student may apply for exemptions during the first four weeks of the

semester.

5.5 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme. A department shall not designate more than 8 modules as core modules per level.

6. ASSESSMENT

6.1 The assessment of a module shall be based on a written examination and continuous assessment (assignments, tests, practicals and field trips) submitted during the semester according to a schedule laid down by the appropriate Departmental Board. The examinations shall normally contribute 60% of the final mark while continuous assessment accounts for 40%. A student may not be allowed to sit an examination if he/she has no continuous assessment mark.

6.2 Field Trips, Practicals and Tutorials are compulsory.

6.3 Departments shall determine which items shall be included in the continuous assessment and shall define the relevant weight of each item.

7. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Regulations.

8. PROVISION FOR PROGRESSION

8.1 Students shall earn above 75% per level to be allowed to progress to the next level.

8.2 A student who fails any module shall repeat that module or take an equivalent module specified in the programme regulations.

8.3 A student who is allowed to proceed with his/ her studies but has failed modules at a given level of study is allowed to register for the next higher-level modules provided he/she has passed the prerequisites for the modules and does not exceed the maximum allowed modules in that year of study. Repeat modules take priority over any other modules should there be a clash on the timetable.

8.4 A module shall normally have at least 4 lecture hours per week and a minimum of 40 hours per semester.

8.5 A module with laboratory component shall normally have at least 2 hours per week for laboratory exercises in addition to the hours in 8.4 above and a minimum of 60 hours per semester.

9. WORK-RELATED LEARNING PROGRAMME GENERAL GUIDELINES

Refer to Section 10 of the General Regulations

10. REMARKING OF EXAMINATIONS

Refer to Section 11.10 of the General Regulations.

11. GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Engineering/Science Honours degree a student must pass all core modules and must have accrued at least 480 credits.

11.2 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

12. DEGREE WEIGHTING

12.2 For the purpose of degree classification, the weighting of the results of the Bachelor of Engineering Honours Degree levels differs per degree programme:

DEPARTMENT OF FUELS AND ENERGY ENGINEERING

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN ELECTRICAL AND ELECTRONIC ENGINEERING (ENG01)

Duration of Degree:	5 years
Credit Load:	627
Minimum Credit Load	600
Maximum Credit Load	660
Maximum MBKs Credit Load	480

1. PURPOSE OF THE PROGRAMME

The purpose of the degree is to develop knowledge, skills, and competencies in the field of Electrical and Electronic Engineering that are relevant to various entrepreneurial and employment opportunities in Zimbabwe, regionally and globally.

It is also to prepare students for specialist higher degree studies within the field and to be able to achieve a high level of environmental consciousness in the field.

2. ENTRY REQUIREMENTS

Normal Entry

For normal entry the candidates should have:

(i) Passed at least 5 subjects at Ordinary Level including Mathematics, English and Combined Science or equivalent.

(ii) At least two Advanced Level passes; Mathematics, Physics and any other relevant science subject.

Special Entry

For special entry candidates should:

(i) Have passed at least 5 subjects at Ordinary Level including Mathematics, English and Combined Science or equivalent.

(ii) Hold at least a National Diploma in an approved Engineering field or any other recognized equivalent. Candidates may be invited for an interview or entrance examination.

3. PROGRAMME CHARACTERISTICS

Areas of Study:	Electrical Power Engineering, Electronics Engineering, Communication and Network Engineering, Environmental Sustainability, Energy Management and Auditing
Specialist Focus:	Electrical power engineering and Electronic engineering Orientation:

Integrated Electrical/Electronic problem-solving, using knowledge and practical skills for research, innovation, industrialization, commercialization and community service Distinctive Features: The programme focuses on the production of research and development orientated engineers who are able to solve electrical or electronic engineering problems to achieve sustainable economic development, and empowerment of society through the application of sound theoretical and practical knowledge.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:	Electrical and Electronic engineering graduates find employment as Electrical engineers, Electronic engineers, Energy Auditors, Energy Consultants, Power Plant Engineers, Communication Engineers, Renewable Energy Engineers, Control system Engineers, Electrical/Energy Projects Managers, Process Design Engineers, Electrical Transmission and Distribution Engineers, Electrical/Electronic Techno-preneurs.
Further Studies:	Masters' and Doctoral studies applied to the Electrical, Electronic and Energy Engineering

5. PROGRAMME DELIVERY

Teaching and Learning

Methods: Lectures, tutorials, laboratory experiments, seminars, group work, hands-on, site-based learning, site visits, work-related learning, research projects, and independent study. Online teaching is an option where physical contact is not possible.

Assessment Methods: Written examinations, tests, laboratory reports, field reports, seminar presentations, Work-Related Learning report, project reports, and continuous assessments

6. INTENDED LEARNING OUTCOMES

A graduate of this programme should be able to:

- Demonstrate knowledge and understanding of electrical and or electronic engineering principles and practices, assess viability of electrical and or electronic projects, design projects, maintain projects, operate and sustainably abandon projects.
- Collect and synthesize multiple electrical or electronic project data sets to make interpretations that account for scientific uncertainty.
- Effectively communicate and disseminate findings of scientific investigations.
- Manage time and other resources efficiently and effectively in executing electrical or electronic project tasks.
- Effectively deploy appropriate technologies to collect and interpret electrical or electronic project data as part of a scientific investigation.
- Independently acquire and utilize new electrical or electronic project knowledge from diverse sources.
- Use appropriate research methods to create new insights into electrical or electronic projects.
- Develop new electrical or electronic project practices and technologies that enhance efficiencies and outputs of the electrical or electronic industry.
- Create new business ventures in the electrical or electronic industry.
- Plan, design and operate safe and viable electrical or electronic projects.
- Evaluate and assess viability of electrical or electronic projects.
- Assess environmental impacts of electrical or electronic projects and acceptably manage, rehabilitate and reclaim electrical or electronic properties.
- Apply remediation measures to counter hazards due to electrical or electronic engineering activity.

7. GENERAL PROVISIONS

Refer to Section 5 of the Faculty Regulations

8. PROGRAMME ASSESSMENT

Refer to Section 6 of the Faculty Regulations.

9. PROVISIONS FOR PROGRESSION

Refer to Section 8 of the Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 11 of the Faculty Regulations.

12. DEGREE WEIGHTING

Level 1	10%
Level 2	15%
Level 3	20%
Level 4	25%
Level 5	30%

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
HMTE111	Engineering Mathematics I		12
HFEE112	Engineering Drawing		15
CSENG131	Communication Skills		6
HFEE113	Computer Applications for Engineers		12
HEEE111	Introduction to Electrical Engineering		12
HEEE112	Introduction to Electronic Engineering		12

Level 1 Semester 2

HMTE122	Engineering Mathematics II	HMTE111	12
HFEE122	Computer Programming		12
HEEE121	Workshop Practice		12
HEEE122	Power Electronics	HEEE112	12
HEEE123	Electrical Engineering Technology	HEEE111	12
HEEE124	Power Generation, Transmission & Distribution Networks	HEEE111	12

Level 2 Semester 1

HMTE211	Engineering Maths III	HMTE111	HMTE122	12
HEEE211	Electrical Engineering Design and Materials	HEEE111	HEEE123	12
HEEE212	Analogue Electronics	HEEE112		12
HEEE213	Digital Electronics	HEEE112		12
ENT215	Entrepreneurship			12
GS212	Introduction to Gender Studies			6

Level 2 Semester 2

HFEE221	Engineering Statistics	HMTE111	12
HEEE221	Process Control and Instrumentation Systems		12
HEEE222	Embedded Systems I	HEEE213	12
HEEE223	Signals and Systems I	HEEE212	12
HEEE224	Electronics Innovation Project		12

Level 3 Semester 1

HEEE311	Embedded Systems II	HEEE222	12
HEEE312	Signals and Systems II	HEEE223	12
HEEE313	Electromagnetic Engineering		12
HEEE314	Electrical Machines and Drives		12

HFEE314	Research Methods and Technical Writing		12

Level 3 Semester 2

HEEE321	Control Systems Engineering	HEEE221	12
HEEE322	Electrical Power Systems	HEEE314	12
HEEE323	Advanced Electrical Networks	HEEE111	12
HEEE324	Communication Engineering		12
HEEE325	Electrical Power Innovation Project		12

Level 4 Semester 1 and 2: Work-Related Learning

Level 4 Semester 1

HEEE 400	Work-Related Learning Placement Report	10

Level 4 Semester 2

HEEE 401	Academic Supervisors' Report	45
HEEE 402	Employer's Assessment Report	30
HEEE 403	Student's Industrial Report	35

Level 5 Semester 1

HFEE512	Project Management		12
HFEE513	Power Systems Analysis		12
HEEE511	Modelling and Simulation of Electrical and Electronic Systems		12
HEEE512	Digital Signal Processing	HEEE213	12
HEEE513	Communication and Network Engineering		12

Level 5 Semester 2

HEEE521	Mobile and Wireless Networks	12
HEEE522	Microwave Engineering	12
HEEEXXX	Elective	12
HEEE525	Design and Make Project	

ELECTIVES

HEEE523	Power Plant Technology	12
HEEE524	Advanced Power Electronics	12
HFEE511	Energy Management and Auditing	12

MODULE SYNOPSES

HMTE111 Engineering Mathematics I

This module seeks to enhance basic Mathematics concepts: Linear algebra, specifically, matrices manipulations, linear independence; linear transformations, projection, orthogonality, systems of linear equations and Gaussian elimination, among other components. It also explores a deeper understanding of vector calculus and the three major engineering mathematics theorems: Green's theorem, Gauss's theorem, and Stoke's theorem.

HFEE112 Engineering Drawing

This module is designed to familiarise students with Engineering Drawing standards as a means of communication. It covers the use of drawing instruments and lettering; drawing standards and conventions, geometrical construction and geometrical tolerances. It also introduces projection methods: First and third angle orthographic projections, sectional views, ISO conventions in sectioning and elementary assembly drawing.

CSENG131 Communication Skills
This module focuses on equipping students with communication skills. In particular, this module teaches pre-writing skills, writing skills, essay construction, scientific, and technical writing skills. In the process, students are equipped with reflective, critical evaluation, and teamwork skills.

HFEE113 Computer Applications for Engineers
This module focuses on design and development of solvers and their application using spreadsheet (Microsoft Excel). It covers knowledge and practice of the production of system integration, production planning, database design and process optimization. Students are also exposed to MATLAB fundamentals and MATLAB Operations (variable, plotting, polynomials, interpolation, derivatives and integrals, matrices, programming and optimization). The module will also familiarise students to computer systems; microcomputers and network environments including introduction to computer programming.

HEEE111 Introduction to Electrical Engineering
This module introduces the students to the fundamentals of electricity, theoretical solutions and practical applications of network theorems on DC circuits, symbolic notation and AC circuits, magnetic field theory and their significance in engineering. It also aims at introducing the principle of single phase and three phase ac systems. The module equips students with skills to apply DC circuit transients, Single phase transformers, the tariff system, DC and AC machine technology.

HEEE112 Introduction to Electronic Engineering
This module introduces students to the fundamentals of electronic engineering. It covers the importance of the electron in the creation of electricity, to include current, voltage and power. It also gives students an opportunity to explore the design and application of passive components, which include resistors, capacitors and inductors. Furthermore, the theory and application of semiconductor devices is covered. Students also learn and apply digital and analogue integrated circuits.

HMTE122 Engineering Mathematics II
This module focuses on theory and practical application of first and second order Differential Equations (ODEs) to engineering, Fourier analysis: Functions, coefficients, integrals, multiple integrals, frequency shift and the Convolution theorem, Laplace Transforms, solution of first and second order ODEs. The module also introduces students to periodic functions.

HFEE122 Computer Programming
This module fosters students' in-depth knowledge and practice of object-oriented programming (Programming fundamentals, Operators, Condition statements, Loops, Functions, Pointers, Arrays and Data Structures). In the process, students develop energy-related programs encompassing program structure, input, output and organization, and execution of conditional and repetitive operations. Students are also exposed to the use of functions, classes, objects, and numerical and non-numerical algorithms for sorting and searching of data.

HEEE121 Workshop Practice
This module equips students with knowledge and skills in SHEQ issues, HIRA procedure, employer/employee and the engineer's responsibilities. It also develops design skills in

analogue, digital and power electronic circuits, including advanced analytical, CAD, simulation and design software. It also equips students with skills on pcb development using modern methods. The module also covers domestic and industrial electrical circuit design, protection and electrical management systems. Students are also exposed to PLC programming and wiring including SCADA systems.

HEEE122 Power Electronics
This module focuses on the concepts and applications of power electronics in the electrical and electronic field. It covers the characteristics of power electronic devices, power losses and thermal design. It also covers types of power converters and their principle operation, to include AC-DC, AC -AC, DC-DC, switched mode converters, Inverters, Voltage and current source converters. It also explores power supplies, commutation techniques, advanced energy efficient motor drives, modulation methods and system design and control.

HEEE123 Electrical Engineering Technology
This module further develops students' knowledge and skills in electrical engineering technology. It covers symbolic notation and its application on electrical ac circuits, application of advanced network theorems and circuit analysis. The module further develops students' understanding in electric fields, electromagnetic principles, transient analysis, complex waveforms and measuring instruments. The module also introduces reliability of components, transmission lines, attenuators, filters and two port networks. It also covers solar PV technology.

HEEE124 Power Generation, Transmission and Distribution Networks
This module aims to develop an advanced understanding of power generation, transmission and distribution networks. It covers in detail, generation, transmission and distribution, electrical loads and load forecasting. It also covers the design of overhead lines, cables and substations. Furthermore, it covers renewable energy generation, distributed generation, smart grids, power system protection, high voltage engineering, power system reliability and power quality, delivery process and pricing.

HMTE211 Engineering Mathematics III
The module covers inverse transforms and Z-Transforms: Sequences, inverse transformations and their application to control engineering problems. Introduction to numerical methods; Computer arithmetic and rounding errors; Numerical methods for root-finding; Polynomial interpolation and splines; Solution for linear algebraic equations; Numerical integration and differentiation; Numerical integration of ODEs; Euler and second order; Runge-Kutta methods.

HEEE211 Electrical Engineering Design and Materials
This module explores phases of the design process and considerations, alongside electrical materials technology. The module explores crystal structure of materials, types of electrical materials, dielectric and magnetic properties, and characteristics of electrical materials.

It covers computer aided design of electrical and electronic machines, transmission and distribution lines, insulators, lightning arrestor systems, surge diverters and impulse generators. It also covers domestic and industrial circuit design.

HEEE212 Analogue Electronics
This module aims to give students a sound background in analogue electronics design which help students understand, analyse and design circuits involving analogue electronic components and parts. It also covers rectifier circuits, wave-shaping circuits, linear small-signal equivalent circuits and bipolar junction transistors. The module also provides an analysis of amplifier circuits, current and voltage relationships, integrators and differentiators,

Wheatstone Bridge, frequency response and active filter RLC circuits to include Butterworth filter design.

HEEE213 Digital Electronics
This module develops understanding of digital electronics to include logic families, numerical systems, Boolean algebra, logic gate behaviour, representation of logic expressions and simplifying methods. It also covers combinational logic: decoders, multiplexers and adders. The module also covers sequential logic: Flip Flops, registers, counters, RAM and ROM. The application of digital electronics to computer data path control is also covered.

ENT215 Entrepreneurship
This module seeks to equip students with entrepreneurship skills to guide them on business fundamentals which include businesses development, marketing, management, human resources, pricing, taxation of business, and government policies. Students acquire skills on preparation and presentation of business plans. The module concludes by examining case studies to assist students appreciate the content of the module.

GS212 Introduction to Gender Studies
This module focuses on gender sensitivity and gender inequalities. An in-depth examination of gender issues in Zimbabwe is done by tracing the historical development of gender issues through gender analysis. The module concludes by redressing gender imbalances, empowerment, and strategies for gender responsive environment.

HFEE221 Engineering Statistics
This module explores probability and statistical methods for quality control and inspection such as correlation and regression (sums of squares and sums of products), scatter plots, correlation matrix, predictions, Pearson's and Spearman's correlation coefficients including residuals and transformations. It also enhances students' ability to estimate, test hypotheses: t-test, z-test, Chi-Square test and F-tests, multiple linear regression and inference in multiple linear regression.

HEEE221 Process control and Instrumentation Systems
This module imparts students with skills to model, analyse and develop practical solutions to typical situations in industrial control, automation and instrumentation engineering. The students design circuits using passive components, semiconductor devices, transducers, and sensors. Measuring instruments, to include their errors, accuracy, precision, sensitivity and linearity will also be covered. In addition, students will be equipped with knowledge and skills on functional operation of PID control systems, PLC programming and SCADA systems.

HEEE222 Embedded Systems I
This module aims to give students a strong foundation in embedded systems by introducing them to the main generic and distinctive features of embedded computing. It also develops students' practical skills in designing firmware for microcontrollers by use of programming languages and techniques. These include information representation, algorithmic state machines and interfacing of microcontrollers.

HEEE223 Signals and Systems I
This module provides students with the basic skills required for understanding linear systems and deterministic signals. Students are equipped with techniques to characterise and manipulate linear time-invariant systems in terms of input-output relationships for both time and frequency domain methods. It also includes concepts related to signal representation, linear convolution, Fourier theory, sampling of continuous-time signals, and Laplace transforms to solve continuous, linear, time-invariant systems and obtain transfer functions.

HEEE224 Electronics Innovation Project
This module equips students with skills to solve industrial or community problems relevant to the Electronic Engineering discipline. Students are expected to look for an electronic challenge or problem that requires application of electronic engineering solution and solve it. This entails designing, testing or commissioning a prototype or sub-system or simulation that addresses the problem. The project will provide insight to understand what is involved in real-life electronic engineering field.

HEEE311 Embedded Systems II
This course fosters knowledge and skills on embedded systems architecture, firmware and software tool stacks. It includes Internet of Things (IoT) in the form of design and project-oriented learning through modelling and simulation. The module introduces Linux command line, source code version control, Hardware Description Language (HDL) programming and computer architecture fundamental techniques.

HEEE312 Signals and Systems II
This module aims at developing an understanding of Random signals and processes in continuous /discrete time. It also covers time and frequency domain signal processing for electronic systems, continuous-time Fourier theory, sampled signals and use of the discrete Fourier transform. The module also explores the propagation of signals and noise through linear systems, complex analytic signal representation, PSD functions, modulation and demodulation.

HEEE313 Electromagnetic Engineering
This module provides in-depth knowledge and skills on electromagnetic engineering. It covers static electric and magnetic fields, radiation mechanism by antennas and the nature of fields they produce. The module also covers the propagation of plane waves in space and in lossy media including calculations. One-dimensional models for TEM transmission lines are constructed and their application is explored. The module also covers waveguide structures and solutions to EM problems using computational methods and Fourier analysis.

HEEE314 Electrical Machines and Drives
This module equips students with knowledge and skills on the design, operation, and application of electrical machines which include single and three phase induction machines, synchronous machines, and DC machines. In addition, the module enables students to perform machine steady state and transient stability analysis, and load frequency control. The module provides an introduction to reference frame theory and dq-machine modelling.

HFEE314 Research Methods and Technical Writing
This module equips students with scientific research and report writing skills to prepare them for writing dissertations, work-based reports, and scientific publications. The module provides guidelines on scientific research methods and prepares students to conduct independent research, report and correctly cite published work to support their research.

HEEE321 Control Systems Engineering
This module equips students with theoretical and practical skills on open and closed loop system configurations and block diagrams. It also focuses on feedback of control systems. Time and frequency domain analysis to include errors, P, PI, PD and PID controllers are covered. The module explores the stability of linear control systems to include Routh Hurwitz stability criterion, Root locus, Nyquist and Bode plots. It also covers controllability and observability, compensation, sensitivity functions, state variables and state space models.

HEEE322 Electrical Power Systems
This module provides students with the theory and design of power system components including transformers, synchronous machines, switchgear and different types of supply systems. It covers design of transmission lines inclusive of overhead line insulators, overhead transmission line mechanical design, inductance and capacitance. The module also covers HVAC and HVDC voltage transmission, protection devices and effects of corona interference. Furthermore, it covers earthing, insulated cable types and their characteristics.

HEEE323 Advanced Electrical Networks
This module aims to develop advanced understanding and skills on network analysis. It covers advanced AC circuit analysis, frequency response circuits including resonance and harmonics, Electromagnetic principles, reliability and transmission lines. It also covers advanced filter circuits and transfer functions, application of Laplace transforms and Fourier series/transforms to circuit analysis. The module also explores two port networks inclusive of interconnection of networks, Z, Y, H and transmission parameters.

HEEE324 Communication Engineering
This module equips students with the fundamentals of digital communication engineering. It covers probability and random signals, band limited channels and elements of information theory. It also covers channel capacity and reliability function, source coding, channel coding and error control coding. The module extends to baseband signalling, bandpass modulation and demodulation in noise and data detection.

HEEE325 Electrical Power Innovation Project
This module equips students with skills to solve industrial or community problems relevant to the Electrical power engineering discipline. Students are expected to look for an electrical challenge or problem that requires an electrical solution and solve it. This entails designing, testing or commissioning a prototype or sub-system or simulation that addresses the problem. This will provide insight to understand what is involved in real-life electrical engineering field.

HEEE 400 Work-Related Learning Placement Report

The module covers: The report of the student's understanding of an organization's purpose, functions, operations and linkages with relevant stakeholders i.e. suppliers, clients, government agencies etc.

HEEE401 Academic Supervisors' Report
In this module the academic supervisor provides a comprehensive report of the student's Work-Related learning progress. The academic supervisor will visit the student at least twice for the whole period to get primary evidence on what the student has been exposed to and been working on during the attachment period. The supervisor also engages the student in discussions on possible project proposals in the organisation. This will be done through interviews and feedback from the company.

HEEE402 Student's Report
This module provides students an opportunity to carry out a project during work-related learning period. Students will record and maintain a log-book of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report. A work-related learning research project supported by appropriate engineering drawings, and design charts or process charts is required. Students will be visited at their work place at least twice by their lecturers.

HEEE403 Employer's Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises of the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HFEE512 Project Management
The module provides students with project management skills and methodologies. It provides students with an understanding of a project and project management. The module also equips students with practical tools and techniques to plan, deliver, and monitor a project throughout its life cycle. The basic four-step project management life cycle, planning, implementation, evaluation and closure are analysed and demonstrated.

HFEE513 Power Systems Analysis
This module fosters knowledge and practice in the modelling of power system elements, symmetrical components and fault analysis. It also covers power system planning, forecasting and load flow analysis. The students will gain knowledge and skills on basic stability and instability concepts and their control, protection, compensation, and regulation. Furthermore, the analysis of travelling waves and power optimization systems are explored.

HEEE511 Modelling and Simulation of Electrical and Electronic Systems
This module equips students with theory and skills on modelling and simulation of electrical and electronic systems. It covers inductance in machine airgaps, the dynamic equation of induction and synchronous electrical machines and steady state models for electrical machines. It also explores the design methodologies, analysis of dynamic properties and synthesis of drive systems employing various simulation methods including virtual modelling and simulation software.

HEEE512 Digital Signal Processing
This module aims to develop an understanding of digital signal processing. It also provides further knowledge and skills on the application of discrete time signals and systems, the discrete Fourier transform properties and fast algorithms. Students are exposed to digital filter design including FIR and IIR. Furthermore, it covers the z-transform and the frequency response from z-plane.

HEEE513 Communication and Network Engineering
This module aims to develop theoretical and practical skills on computer networks and the internet. It also fosters knowledge and skills on application and transport layers which include principle of network applications, socket programming, transport layer services and multiplexing/DE multiplexing. The module also covers network layer design, data link layer and physical layer. Emerging communication networks including the fundamentals of mobile networks and the smart grid communication networks are explored.

HEEE521 Mobile and Wireless Networks
This module focuses on developing knowledge and skills on mobile and fixed networks. Topics to be covered include wireless network fundamentals, wireless and wireline access technologies. It also develops an understanding on radio resource and mobility management, traffic and congestion control, QOS in packet network, IP traffic engineering, network convergence and network services. The module also covers data plane and control plane technologies alongside with performance modelling and simulation.

HEEE522 Microwave Engineering
This module equips students with theoretical and practical skills on the operation of RF systems, microwave and millimetre wave frequency systems, including communication systems, radar systems and radio-astronomy receivers. It also includes antennas and antenna

array theory, propagation in space and urban environments. A selection of radar, radio astronomy and communications system architecture are covered in detail. System design principles and practical computational EM modelling are also a major part of this module.

HEEE523 Power Plant Technology
This module covers different power plant technologies. This includes renewable power sources inclusive of generation of solar energy, wind energy, geothermal energy, tidal power, bioenergy and MHD power generation. It also covers thermal power plants, hydropower plants, gas power plants, diesel power plants and nuclear power plants. The module also gives students' knowledge and skills in substation layout and power plant economics.

HEEE524 Advanced Power Electronics
The module covers advanced ac-dc converters, power factor improvement methods including line and forced commutation. It also covers switched mode dc-dc converters, mode operations and flexible ac transmission systems (FACTS). The module further explores advanced power electronics devices which include static var compensators, thyristor controlled series capacitor, static phase, electronic braking systems and unified power flow control

HEEE525 Design and Make Project
The module covers the design, development and assembly of prototype projects in the area of Electrical and Electronic field. The students apply the knowledge and skills attained during their learning or through research, to design and make a working practical project or simulated project model. The project should be problem-solving oriented to address community or industrial needs.

HFEE511 Energy Management and Auditing
This module equips students with knowledge and skills on the management and auditing of energy for improvement in availability, quality, cost, and efficiency. This include the supply and demand side, energy management and condition monitoring control systems, energy conservation measures, and energy audits. The module employs the concept of life cycle cost analysis as a tool to evaluate the economic viability of energy-efficiency improvement or energy conservation measures.

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN FUELS AND ENERGY ENGINEERING (ENG02)

Credit Load:	642
Minimum Credit Load	600
Maximum Credit Load	660
Maximum MBKs Credit Load	480
Level: ZNQF	8

1. PURPOSE OF THE PROGRAMME
The purpose of the degree is to develop knowledge, skills, and competencies in the field of Fuels and Energy Engineering that are relevant to various entrepreneurial and employment opportunities in Zimbabwe, regionally and globally. It is also to prepare students for specialist higher degree studies within the energy field and to be able to achieve a high level of environmental consciousness in the sustainable exploitation of energy resources.

2. ENTRY REQUIREMENTS
Normal Entry

For normal entry the candidates should have:

(i) Passed at least 5 subjects at Ordinary Level including Mathematics, English and Combined Science or equivalent.
(ii) Two Advanced Level passes; Mathematics and Physics or Chemistry
Special Entry

For special entry candidates should:

(i) Have passed at least 5 subjects at Ordinary Level including Mathematics, English and Combined Science or equivalent.
(ii) Hold at least a National Diploma in an approved Engineering field or any other recognized equivalent.
Candidates may be invited for an interview or entrance examination.

3. PROGRAMME CHARACTERISTICS

Areas of Study: Electrical Power Systems, Renewable Energy, Fossil Fuels, Environmental Sustainability, Energy Management and Auditing

Specialist Focus: Production and supply of sustainable energy

Orientation:

Integrated energy -problem-solving, using knowledge and practical skills for research, innovation, industrialization, commercialization and community service

Distinctive Features: The programme focuses on the production of research and development-orientated engineers who are able to solve energy problems to achieve sustainable economic development, and empowerment of society through the application of sound theoretical and practical knowledge.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Fuels and Energy engineering graduates find employment as Energy Auditors, Energy Consultants, Power Plant Engineers, Fuel Cells Engineers, Renewable Energy Engineers, Petroleum Engineers, Energy Projects Managers, Process Design Engineers, Electrical Transmission and Distribution Engineers, Energy Techno-preneurs.

Further Studies: Masters' and Doctoral studies applied to the Energy industry (Electrical, Chemical, Fossil Fuels and Renewable energy)

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory experiments, seminars, group work, hands-on, site-based learning, site visits, work-related learning, research projects, and independent study. Online teaching is an option where physical contact is not possible.
Assessment Methods: Written examinations, tests, laboratory reports, field reports, seminar presentations, Work-Related Learning report, project reports, and continuous assessments

6. INTENDED LEARNING OUTCOMES

A graduate of this programme should be able to:

- Demonstrate knowledge and understanding of energy engineering principles and practices, assess viability of energy projects, design projects, maintain projects, operate and sustainably abandon energy projects.
- Collect and synthesize multiple energy project data sets to make interpretations that account for scientific uncertainty.

- Effectively communicate and disseminate findings of scientific investigations.
- Manage time and other resources efficiently and effectively in executing energy project tasks.
- Effectively deploy appropriate technologies to collect and interpret energy project data as part of a scientific investigation.
- Independently acquire and utilize new energy project knowledge from diverse sources.
- Use appropriate research methods to create new insights into energy projects.
- Develop new energy project practices and technologies that enhance efficiencies and outputs of the energy industry.
- Create new business ventures in the energy industry.
- Plan, design and operate safe and viable energy projects.
- Evaluate and assess viability of energy projects.
- Assess environmental impacts of energy projects and acceptably manage, rehabilitate and reclaim energy properties.
- Apply remediation measures to counter hazards due to energy production activity.

7. GENERAL PROVISIONS

Refer to Section 5 of the Faculty Regulations

8. PROGRAMME ASSESSMENT

Refer to Section 6 of the Faculty Regulations.

9. PROVISIONS FOR PROGRESSION

Refer to Section 8 of the Faculty Regulations.

10. FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. GRADING AND DEGREE CLASSIFICATION

Refer to Section 11 of the Faculty Regulations.

12. DEGREE WEIGHTING

Level 1	10%
Level 2	15%
Level 3	20%
Level 4	25%
Level 5	30%

13. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Prerequisite	Credits
HMTE111	Engineering Mathematics I		12
HFEE111	Engineering Physics		15
HFEE112	Engineering Drawing		15
CSENG131	Communication Skills		6
HFEE113	Computer Applications for Engineers		12

Level 1 Semester 2

HMTE122	Engineering Mathematics II	HMTE111	12

HFEE121	Introduction to Manufacturing Processes			15
HFEE122	Computer Programming		HFEE113	12
HFEE123	Electrical and Electronic Principles		HFEE111	15
HFEE124	Chemistry for Engineers			12
HFEE125	Workshop Practice			

Level 2 Semester 1

HFEE211	Engineering Design principles		HFEE112	12
HFEE212	Fluid Mechanics		HMTE122	12
HFEE213	Thermodynamics and Heat Transfer		HMTE122	12
ENT215	Entrepreneurship			12
GS212	Introduction to Gender Studies			6
HMTE211	Engineering Mathematics III	HMTE111	HMTE122	12

Level 2 Semester 2

HFEE221	Engineering Statistics HMTE111			12
HFEE222	Engineering Materials HFEE111			12
HFEE223	Mass Transfer HFEE213			12
HFEE224	Instrumentation and Embedded Control		HFEE122	12
HFEE225	Electrical Machines HFEE122			12
HFEE226	Innovation Project I			12

Level 3 Semester 1

HFEE311	Solar Energy Technology			12
HFEE312	Thermal Power Plants		HFEE213	12
HFEE313	Combustion Technology		HFEE213	12
HFEE314	Research Methods and Technical Writing			12
HFEE315	Bioenergy			12
HFEE316	Hydrogen and Fuel Cells Technology			12

Level 3 Semester 2

HFEE321	Coal Processing and Handling		HFEE123	12
HFEE322	Petroleum Processing and Handling		HFEE123	12
HFEE323	Electrical Power Systems		HFEE122	12
HFEE324	Hydro and Wind Power Technologies			12
HFEE325	Innovation Project			12

Level 4 Semester 1 and 2: Work-Related Learning

Level 4 Semester 1

HEEE 400	Work-Related Learning Placement Report			10

Level 4 Semester 2

HFEE 401	Academic Supervisors' Report			45
HFEE 402	Employer's Assessment Report			30
HFEE 403	Student's Industrial Report			35

Level 5 Semester 1

HFEE511	Energy Management and Auditing			12
HFEE512	Project Management			12
HFEE513	Power Systems Analysis		HFEE323	12
HFEE514	Modelling and Simulation of Energy Systems			12
HFEE515	Nanotechnology for Energy Applications			12

Level 5 Semester 2

HFEE521	Energy Market, Legislation and Policy	12
HFEE522	Energy, Environment and Sustainable Development	12
HFEEXXX	Elective	12
HFEE526	Design and Make Project	

ELECTIVES

HFEE523	Nuclear Power Technology	12
HFEE524	Control Systems	12
HFEE525	Petroleum Geology and Reservoir Engineering	12

14. MODULE SYNOPSES

HMTE111 Engineering Mathematics I
This module seeks to enhance basic Mathematics concepts: Linear algebra, specifically, matrices manipulations, linear independence; linear transformations, projection, orthogonality, systems of linear equations and Gaussian elimination, among other components. It also explores a deeper understanding of vector calculus and the three major engineering mathematics theorems: Green's theorem, Gauss's theorem, and Stoke's theorem.

HFEE111 Engineering Physics
This module provides fundamentals of engineering physics to provide background to future modules. It covers vectors, motion in one and two dimensions, Newton’s laws of motion, rotational motion, mechanical equilibrium, momentum, energy, work and power, periodic motion, conservation laws and fluid mechanics. The module also covers electrostatics, electric field, electric circuits, magnetism, electromagnetic induction, light and optics

HFEE112 Engineering Drawing
This module is designed to familiarise students with Engineering Drawing standards as a means of communication. It covers the use of drawing instruments and lettering; drawing standards and conventions, geometrical construction and geometrical tolerances. It also introduces projection methods: First and third angle orthographic projections, sectional views, ISO conventions in sectioning and elementary assembly drawing.

CSENG131 Communication Skills
This module focuses on equipping students with communication skills. In particular, this module teaches pre-writing skills, writing skills, essay construction, scientific, and technical writing skills. In the process, students are equipped with reflective, critical evaluation, and teamwork skills.

HFEE113 Computer Applications for Engineers
This module focuses on design and development of solvers and their application using spreadsheet (Microsoft Excel). It covers knowledge and practice of the production of system integration, production planning, database design and process optimization. Students are also exposed to MATLAB fundamentals and MATLAB Operations (variable, plotting, polynomials, interpolation, derivatives and integrals, matrices, programming and optimization). The module will also familiarise students to computer systems; microcomputers and network environments including introduction to computer programming.

HMTE122 Engineering Mathematics II
This module focuses on theory and practical application of first and second order Differential Equations (ODEs) to engineering, Fourier analysis: Functions, coefficients, integrals, multiple integrals, frequency shift and the Convolution theorem, Laplace Transforms, solution of first and second order ODEs. The module also introduces students to periodic functions.

HFEE121 Introduction to Manufacturing Processes

This module aims to establish relevant technical knowledge for the manufacturing processes and systems. The students will be exposed to basic physical and mechanical properties of metals, polymers, ceramics and composites, behaviour, and their relevance to manufacturing processes. Heat treatment methods for metals, standards of accuracy, tolerances, limits and fits, surface finishes and all the machining and joining process will be covered. Students will also be exposed to engineering materials assembling methods, metal casting and metal forming.

HFEE122 Computer Programming
This module fosters students' in-depth knowledge and practice on object-oriented programming (Programming fundamentals, Operators, Condition statements, Loops, Functions, Pointers, Arrays and Data Structures). In the process, students develop energy related programs encompassing program structure, input, output and organization, execution of conditional and repetitive operations. Students are also exposed to the use of functions, classes, objects, numerical and non-numerical algorithms for sorting and searching of data..

HFEE123 Electrical and Electronic Principles
This module introduces the students to the fundamentals of electricity, theoretical solutions and practical applications of DC circuits, the electric and magnetic field theory and their significance in engineering. The module equips students with skills to apply single and three phase AC circuits, Star- Delta transformations and transients response systems. The module also covers principle of operation and application of semiconductor devices and integrated circuits. Students get exposure on combinational logic circuits and binary number system. The module also imparts skills on circuit board development and nanotechnology.

HFEE124 Chemistry for Engineers
This module imparts students with the theoretical and practical knowledge of Chemistry for Engineering. It covers stoichiometry, the mole concepts, bonding, molecular structure, nomenclature and synthesis. Furthermore, it familiarizes students with areas such as reaction of organic compounds, separating and purification techniques. This module gives students a strong background to understand different facets of fuel technology including different analytical techniques used in industry.

HFEE125 Workshop Practice
This module equips students with basic theoretical and practical skills in healthy and safety regulations (SHE), electrical machines overhaul (single and three phase) induction starters (DoL, STAR – DELTA and Resistance starting. Pumps installation and maintenance, heat exchangers operations, miniature of steam power plants will be covered. Basic solar PV systems components and bio digester types, working principles and construction. Energy storage- batteries, batteries connection, testing, safety and maintenance will be covered.

HFEE211 Engineering Design Principles
This module covers mainly Computer Aided Drawing and Design of engineering equipment and machinery in 2D and 3D isometric drawing, sectioning, assembly drawings and materials analysis. It also focuses on design principles covering morphology of design, introduction to concurrent engineering, quality function deployment and methods decision-making. The module also covers product design processes, life cycle, service life reliability, and selection of component production processes.

HFEE212 Fluid Mechanics
This module is designed to introduce and apply the concepts of fluid mechanics and to solve problems related to engineering practice in fluid mechanics. It encompasses fluid statics, fluid dynamics, the boundary layer theory and the classification of flows. Flow-in pipe systems and open channels are analysed to enable students to construct related systems. It also introduces dimensional analysis and similarity, pumps and pump sizing.

HFEE213 Thermodynamics and Heat Transfer
The module aims to give students the knowledge required to design, analyze, operate, and assess thermodynamic and heat transfer systems. It concentrates on the understanding of thermodynamic laws and their application to open and closed systems, reversible and irreversible processes, ideal gasses, vapor, and gas power cycles. The module also focuses on heat transfer laws (Fourier, Newton's law of cooling, Stefan- Boltzmann) that govern steady-state conduction, convection (natural and forced), and radiation. It also covers their application in heat exchanger's operation and design.

ENT215 Entrepreneurship
This module seeks to equip students with entrepreneurship skills to guide them on business fundamentals which include business development, marketing, management, human resources, pricing, taxation of business, and government policies. Students acquire skills on preparation and presentation of business plans. The module concludes by examining case studies to assist students to appreciate the content of the module.

GS212 Introduction to Gender Studies
This module focuses on gender sensitivity and gender inequalities. An in-depth examination of gender issues in Zimbabwe is done by tracing the historical development of gender issues through gender analysis. The module concludes by redressing gender imbalances, empowerment, and strategies for gender-responsive environment.

HMTE211 Engineering Mathematics III
The module covers inverse transforms and Z-Transforms: Sequences, inverse transformations and their application to control engineering problems. Introduction to numerical methods; Computer arithmetic and rounding errors; Numerical methods for root-finding; Polynomial interpolation and splines; Solution for linear algebraic equations; Numerical integration and differentiation; Numerical integration of ODEs; Euler and second order; Runge-Kutta methods.

HFEE221 Engineering Statistics
This module explores probability and statistical methods for quality control and inspection such as correlation and regression (sums of squares and sums of products), scatter plots, correlation matrix, predictions, Pearson's and Spearman's correlation coefficients including residuals and transformations. It also enhances students' ability to estimate, test hypotheses: t-test, z-test, Chi-Square test and F-tests, multiple linear regression and inference in multiple linear regression.

HFEE222 Engineering Materials
This module explores changes in the shape of a body and the internal state of the body due to the effects of external force systems focusing on bending, plane stress, stress transformation for plane stress, principal stresses and maximum shear stress including plane strain and principal strains. The students will be exposed to basic physical and mechanical properties of metals, polymers, ceramics and composites, material behaviour, and failure modes, focusing on combined loading, failure theories, thin-wall pressure vessels and buckling of columns.

HFEE223 Mass Transfer
The module focuses on principles regarding diffusion and convective mass transfer. Students are equipped with skills and knowledge to operate and analyse mass transfer systems in porous media, with chemical reactions, across a phase boundary and for multi component mixtures. It covers the knowledge of mass transfer coefficients and their applications to unsteady state mass transfer problems. The module also covers mass transfer to a sphere, dispersion of a slug in a pipe flow, diffusion into a slab and gas absorption

HFEE224 Instrumentation and Embedded Control
This module imparts students with skills to model, analyse and develop practical solutions to typical situations in control engineering. The students design circuits using passive components, semiconductor devices, transducers, and sensors. Measuring instruments, to include their errors, accuracy, precision, sensitivity and linearity will also be covered. In addition, students will be equipped with knowledge and skills on functional operation of PID control systems and PLC programming.

HFEE225 Electrical Machines
This module equips students with knowledge and skills on the design, operation, and application of electrical machines which include single and three phase induction machines, synchronous machines, and DC machines. In addition, the module enables students to perform machine steady state and transient stability analysis, and load frequency control. The students will also acquire the capability to design turbines and governors for specific applications.

HFEE226 Innovation Project I
This module equips students with skills to solve industrial or community problems relevant to electrical, electronic and mechanical engineering disciplines. Students are expected to look for an engineering challenge or problem that requires application of an electrical or electronic or a mechanical engineering solution and solve it. This entails designing, testing or commissioning a prototype or sub-system or simulation that addresses the problem. The project will provide insight to understand what is involved in real-life engineering field.

HFEE311 Solar Energy Technology
This module focuses on both theoretical and practical design of solar systems. Specifically, concepts for increasing efficiency based on loss analysis, module sizing and PV cells are explored. The module also equips students with theoretical and practical skills to wire solar PV modules and their system components. The module also covers solar thermal system design and its installation.

HFEE312 Thermal Power Plants
This module explores the theoretical and practical knowledge on heat power plants; steam turbine power plant; design calculations, steam generators and heat balance. Characteristics of furnaces, super heaters, cooling towers, gas turbine power plants and internal combustion engines will also be explored. The module also cover water supply and treatment on thermal power plants and vapour and power cycles.

HFEE313 Combustion Technology
This module covers analysis and design of modern combustion systems to account for fuel properties, maximise output and minimise air pollution. Aspects covered include combustion stoichiometry, combustion chemistry, flame characterisation, thermodynamics of combustion, thermochemical modelling and energy balances. The design of industrial burners including boiler design; anatomy, maintenance of industrial boilers and design principles involved in reducing pollutant emissions will also be covered.

HFEE314 Research Methods and Technical Writing
This module equips students with scientific research and report writing skills to prepare them for writing dissertations, work-based reports, and scientific publications. The module provides guidelines on scientific research methods and prepares students to conduct independent research, report and correctly cite published work to support their research.

HFEE315 Bioenergy
This module explores fundamental concepts of bioenergy technology, which include feedstock, municipal solid wastes, agricultural waste and animal waste, resource assessment, and bioenergy production. It also focuses on biochemical conversion to fuel, including

biodiesel production and bio methanation. The module also explores environmental impacts of biofuels production.

HFEE316 Hydrogen and Fuel Cells Technology
The module focuses on hydrogen and fuel cell technology. Students are equipped with knowledge and skills in the application of hydrogen and production, storage and transportation. Fuel cell power plants. Electro- Chemistry applied to Fuel Cells. Operations and Types of Fuel Cells Performance Characteristics of Fuel Cells. Types of Fuels Cells Systems Commercial Fuel Cell Power Plants and their Applications. Prospects of Fuel Cell Power Plants.

HFEE321 Coal Processing and Handling
This module explores the theoretical and practical knowledge of coal conversion technologies, coal handling systems, size reduction and classifications. Some of the technologies to be covered include coal to liquid fuels and chemicals (Coal to X), methods and mechanisms on coal pyrolysis and carbonization. Technologies on coal uploading devices, coal firing units, coal blending, storage and transportation will also be explored.

HFEE322 Petroleum Processing and Handling
This module equips the students with theoretical and operational knowledge about the modern, integrated petroleum refinery; its feedstocks; product slate and the processes employed to convert crude oil and intermediate streams into finished products. Highlights of the processes to be covered include atmospheric distillation; vacuum distillation; coking; catalytic cracking and reforming; hydro-treating; isomerization and alkylation. Gas processing: separation from liquids, treatment in amine units, gas liquefaction, and gas conversion technologies.

HFEE323 Electrical Power Systems
This module provides students with the theory and design of the electricity grid system and its application in the transmission and distribution of AC, HVDC and LVDC, Tap changers, overvoltage and overcurrent protective devices, and electrical energy converters. The module also focuses on the methods employed in maintaining the stability of electrical power systems. The students will acquire the knowledge and skills needed for power system planning and forecasting including load flow analysis, economic operation and optimization of power systems.

HFEE324 Hydro and Wind Power Technologies
This module covers the harvesting of energy from water and wind. It addresses the availability of the resources, the types of systems and machines, their capabilities and limitations, and the processes of setting up such systems. The module introduces the basic operation of wind and hydro turbines used in generating electrical power. It also equips students with skills to perform site assessments and planning strategies for the installation of wind and hydropower systems.

HFEE325 Innovation Project II
This module equips students with skills to solve industrial or community problems relevant to Fuels and Energy. Students are expected to look for a fuels and energy challenge or problem that requires the application of energy engineering solutions and solve it. This entails designing, testing or commissioning a prototype sub-system or simulation that addresses the problem. The project will provide insight to understand what is involved in the real-life energy engineering field.

HFEE 400 Work-Related Learning Placement Report

The module covers: The report of the student's understanding of an organization's purpose, functions, operations and linkages with relevant stakeholders i.e. suppliers, clients, government agencies etc.

HFEE401 Academic Supervisors' Report
In this module the academic supervisor provides a comprehensive report of the student's work-related learning progress. The academic supervisor will visit the student at least twice for the whole period to get primary evidence on what the student has been exposed to and has been working on during the attachment period. The supervisor also engages the student in discussions on possible project proposals in the organisation. This will be done through interviews and feedback from the company.

HFEE402 Student's Report
This module provides students an opportunity to carry out a project during a work-related learning period. Students will record and maintain a logbook of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report. A work-related learning research project supported by appropriate engineering drawings, and design charts or process charts is required. Students will be visited at their workplace at least twice by their lecturers.

HFEE403 Employer's Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HFEE511 Energy Management and Auditing
This module equips students with knowledge and skills in the management and auditing of energy management for improvement in availability, cost, and efficiency. This includes the supply and demand side, energy management and control systems, energy conservation measures, and energy audits. The module employs the concept of life cycle cost analysis as a tool to evaluate the economic viability of energy-efficiency improvement or energy conservation measures.

HFEE512 Project Management
The module provides students with a project management skills and methodologies. It provides students with an understanding of a project and project management. The module also equips students with practical tools and techniques to plan, deliver, and monitor a project throughout its life cycle. The basic four-step project management life cycle, planning, implementation, evaluation and closure are analysed and demonstrated.

HFEE513 Power Systems Analysis
This module fosters knowledge and practice in the modelling of power system elements, symmetrical components and fault analysis. It also covers power system planning, forecasting and load flow analysis. The students will gain knowledge and skills on basic stability and instability concepts and their control, protection, compensation, and regulation. Furthermore, the analysis of travelling waves and power optimization systems is explored.

HFEE514 Modelling and Simulation of Energy Systems
This module provides a clear insight into complex energy systems by the provision of safe and efficient real-world solutions to problems through modelling, simulation and optimization of computational, mathematical, and physical energy models. Furthermore, computer

software including PVsyst, HOMER, Excel, MATLAB, ANSYS, Aspen, and HYSYS are used to provide process system engineering (PSE) and energy economics optimization of energy systems.

HFEE515 Nanotechnology for Energy Applications
The module covers the theory and practical applications of nanotechnology in fuels and energy production. The topics to be covered include an introduction to nanotechnology, synthesis techniques, characterization techniques, nano fuels, and applications in energy systems. The module also covers the impact of nanotechnology on the market, environment, and society. Students are exposed to research on the future of nanotechnology in renewable energy and fuel processing technology.

HFEE521 Energy Market, Legislation and Policy
The module provides students with basics of energy markets, regulation, and policy. The module examines both the traditional monopoly and evolving competitive alternatives. The module also covers an in-depth exposure to energy resource planning, pollution management, green markets, demand side management, renewable energy portfolios, climate change, and carbon management. Furthermore, it explores administrative federal law and practical issues in the field.

HFEE522 Energy, Environment and Sustainable Development
The module explores the effects of energy production and exploitation on the environment. It covers fossil fuels, nuclear fuels, and combustion-generated pollution to water, air, and the environment. The greenhouse effect, global warming, and climate change will be covered. Energy conservation techniques, environmental regulations standards, international agreements, energy SDGs, and their purposes will also be covered. Furthermore, it explores principles of energy management and control emission of fossil fuels, and thermal pollution.

HFEE523 Nuclear Power Technology
The module provides students with in-depth knowledge of nuclear energy production (the concepts, systems, and applications). The module explores the design of nuclear power plants using the basic principles of reactor physics, thermodynamics, fluid flow, and heat transfer. Students will also acquire knowledge and skills in reactor design, thermal conversion cycles, nuclear safety, and nuclear waste management.

HFEE524 Control Systems
This module equips students with knowledge and skills of control system components and their representation. The module covers various methods for analyzing time response (P, PI,PD and PID), stability of linear control systems and steady state variable analysis. The module also covers frequency-domain techniques: Root-locus methods, Frequency responses, Bode-plots, Gain-margin and phase-margin, Nyquist plots, and compensator design.

HFEE525 Petroleum Geology and Reservoir Engineering
This module gives an introduction to the technical aspects of exploration and production of oil and gas reservoirs. Students will gain working knowledge of petroleum geology, petroleum systems and reservoir fluid phase behaviour and reservoir fluid properties. The module also covers reserves and their estimation techniques, material balance and the different forms of the Material Balance Equation (MBE), drilling, unconventional reservoirs, and lastly, gaining a global perspective on the industry.

HFEE526 Design and Make Project
The module covers the design, development and assembly of prototype projects in the area of fuels and energy. The students apply the knowledge and skills attained during their learning or through research to design and make a working practical project or simulated project model. The project should be problem-solving oriented to address community or industrial needs.

DEPARTMENT OF MECHANICAL ENGINEERING

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN MECHANICAL ENGINEERING DEGREE (ENG03)

Duration	5 years
Actual Credits	660
Minimum Credits	600
Maximum Credits	660
Total MBK Credits	477

1.0 PURPOSE OF THE PROGRAMME

To produce hands-on mechanical engineering graduates who possess integrated theory and certified skills competencies. The programme further aims to develop self-starter mechanical engineers who are able to establish and effectively run their own engineering companies for self-sustenance and for the sustainable development and economic growth of Zimbabwe.

2.0 ENTRY REQUIREMENTS

Normal Entry: At least two 'A' level passes in Mathematics and Physics. A pass in any one of the following 'A' Level subjects will be an added advantage: Chemistry, Computer Science, Metal Technology and Design, Wood Technology and Design, Design and Technology, Technical Graphics and Design, and Software Engineering with an 'O' level pass in Mathematics, Physics and English Language with `C 'grade or better.

Special Entry: Refer to Section 3.2 of the General Academic Regulations. Candidates with a recognized National Diploma in Mechanical Engineering, or equivalent may be admitted into the programme subject to department approval

3.0 PROGRAMME CHARACTERISTICS

Major Fields of Study Fundamentals of Mathematics and Natural Sciences
Mechanical Engineering Skills Training
Applied Information and Communication Technology
Thermo-fluids

Mechanical Engineering Research and Innovation

Design and Manufacture

Engineering and Technology Management

Mechanics & Materials

Maintenance Engineering

Manufacturing Processes and Systems

On The Job Training

Control Systems

Non-Technical Skills

Specialist Focus:	Producing competent hands-on mechanical engineering graduates in the following areas: Thermo-fluids, Mechanical Engineering Research and Innovation, Engineering and Technology Management, Mechanics & Materials, Maintenance Engineering, Computational Fluid Dynamics, Numerical Thermo flow, Finite Element Methods.
Orientation:	Integrated mechanical engineering, principles, knowledge and practical skills for research, innovation, industrialization and commercialization.
Distinctive Features:	The programme focuses on producing innovative and skilled mechanical engineering graduates.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Employment opportunities exist in Construction, Energy, Automotive, Agriculture, Consultancy, Aircraft, Electrical, Electronics, Biotechnology, Manufacturing, Mining, Production, Water, Automation, Biomedical, Clothing, Chemical, Oil and Gas, Combustion engines, Defense, Design, Heating Ventilation and Air Conditioning, Maintenance, Marine, Robotics, Warehousing, Welding, Zeppelins.

Further Studies: Masters and Doctoral studies in Mechanical Engineering in the fields of Mechanical Engineering Sciences, Maintenance Engineering, Engineering Management, Machine Design, Vehicle Engineering, Automation Engineering and Manufacturing Engineering.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, seminars, group work, skills training, industrial tours, work-related learning, research projects, and independent study.

Assessment Methods: • Continuous Assessments: tests, laboratory experimental reports, industrial tour reports, seminar presentations, design/model/simulation and manufacture projects and mechanical engineering skills practical tests

- Work-Related Learning
- Written examinations

5.1 PROGRAMME COMPETENCIES

Generic:

- Multidiscipline: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on an understanding of complex phenomena
- Quantitative and innovative reasoning: Capability to draw on big data and use analytics for informed decision-making and strive to seek new ways of doing things

- Communication skills: Ability to communicate effectively and to present information orally, in writing and using ICTs to both expert and non-expert audiences
- Analysis and synthesis: Capacity for analysis and synthesis using logical arguments and proven facts
- Ethical commitment: Professional integrity and awareness of impact of science and technology on society and the environment
- Entrepreneurial skills: Capability to identify and create new business ventures based on knowledge and skills acquired

Discipline-specific:

- Knowledge: Ability to conceptualize, design and innovate through inventive imagination coupled with applied principles of mathematics, engineering sciences and technologies
- Production skills: Ability to complement, develop and implement engineering design processes for manufacture, installation, commissioning, operation, maintenance and repair
- Technology development skills: Ability to develop and utilize new technologies across the mechanical value chain with a view to enhancing productivity
- Problem-solving skills: Ability to solve a wide range of problems in mechanical-related fields by creative thinking through logically developed and reasoned solutions to discovering how things work so as to operate, maintain and repair
- Analytical and computational skills: Ability to develop and use engineering software to improve engineering technologies

6.0 LEARNING OUTCOMES

The hands-on mechanical engineering graduate of this program should be able to:

- Identify, assess, formulate, interpret, analyze and solve engineering problems creatively and innovatively using principles of mathematics, engineering sciences and technology
- Plan and manage small engineering projects, demonstrating fundamental knowledge, understanding and insight into the principles, methodologies and concepts that constitute socially responsible engineering practice to local and other communities
- Work effectively, individually or with others, as a member of a team, group, organization, community or in multi-disciplinary environments
- Organize and manage self and activities responsibly, effectively, professionally and ethically, accept responsibility within limits of competence, and exercise judgment based on knowledge and expertise
- Plan and conduct limited investigations, research and experiments by applying appropriate theories and methodologies, and perform appropriate data analysis and interpretation
- Communicate effectively, both orally and in writing, with engineering audiences and the community at large, using appropriate structure, style and graphical support
- Use and assess appropriate research methods, skills, tools and information technology effectively and critically in engineering practice, and show an understanding and a willingness to accept responsibility for the impact of engineering activities on society and the environment

- Perform procedural and non-procedural design and synthesis of components, systems, works, products or processes as a set of related systems and assess, where applicable, their social, legal, health, safety and environmental impacts and benefits
- Apply various learning strategies and skills for lifelong learning using mathematics, engineering sciences, engineering design research and management
- Participate as a responsible citizen in the life of local, national and global communities by acting professionally and ethically
- Demonstrate cultural and aesthetic sensitivity across a range of social contexts in the execution of engineering activities
- Explore further education and career opportunities
- Organize and develop entrepreneurial opportunities through engineering problem-solving, design, technical research and managerial skills

7.0 GENERAL PROVISIONS

The rest of the modules carry 12 credits except for: On The Job Training Project which carries 60 credits, (On The Job Training Report; Industrial Supervisor's Assessment Report and Mechanical Engineering Skills Training) each with 30 credits, (Engineering Drawing and Computer Aided Design I; Workshop Technology and Practice; Engineering Drawing and Computer Aided Design II; Stress Analysis and Finite Element Methods; Computational Fluid Dynamics and Numerical Thermoflow; Mechanical Engineering Innovation Project I and Mechanical Engineering Innovation Project II) each with 15 Credits, (Engineering Physics; Engineering Chemistry; Electrical and Electronics Technology; Research Methods for Engineers; Instrumentation and Control; Maintenance Engineering; Product Design and Development; Engineering Economics and Accounting; and Project Management for Engineers) each with 9 Credits, and (Communication Skills; Entrepreneurship; Introduction to Gender Studies and Computer Programming) each with 6 Credits.

8.0 PROGRAMME ASSESSMENT

Refer to Section 6 of the Faculty Regulations.

9.0 PROVISION FOR PROGRESSION

Refer to Section 8 of the Faculty Regulations.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to Section 11 of the Faculty Regulations.

12.0 DEGREE WEIGHTING

For the purposes of classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	15%
Level 3	20%
Level 4	25%
Level 5	30%

13.0 PROGRAMME STRUCTURE

Code	Module Description	Prerequisite	Credits

Level 1 Semester 1

HMECH111*	Engineering Drawing and Computer-Aided Design I		12
HMECH101*	Mathematics for Engineers I		12
HMECH112*	Workshop Technology and Practice		15
CSENG131	Communication Skills		06
HMECH113*	Engineering Physics		09
HMECH110*	Engineering Chemistry		09

Level 1 Semester 2

HMECH121*	Engineering Drawing and Computer Aided Design II	HMECH111	15
HMECH102*	Mathematics for Engineers II	HMECH101	12
HMECH123*	Fluid Mechanics I	HMECH113	12
HMECH124*	Thermodynamics	HMECH110	12
HMECH125*	Strength of Materials	HMECH113	12
HMECH126*	Electrical and Electronics Technology		09

Level 2 Semester 1

HMECH214	Applied Thermodynamics	HMECH124	12
HMECH213	Fluid Mechanics II	HMECH123	12
HMECH215	Applied Strength of Materials	HMECH125	12
HMECH201*	Mathematics for Engineers III	HMECH102	12
HMECH216*	Electrical Machines and Drives	HMECH126	12
HMINE124*	Computer Programming		06

Level 2 Semester 2

HMECH221*	Mechanical Engineering Software and Numerical Methods	HMECH113	12
HMECH222*	Mechanics of Machines	HMECH113	12
HMECH224*	Heat and Mass Transfer	HMECH214	12
HMECH202*	Mathematics for Engineers IV	HMECH201	12
HMECH225	Design of Machine Elements	HMECH215	12
HMECH223*	Hydraulic and Pneumatic Machines	HMECH213	12

Level 3 Semester 1

HMECH311*	Manufacturing Processes and Systems	HMECH112	12
HMECH314*	Fluid Dynamics	HMECH213	12
HMECH315*	Machine Design	HMECH122	12
ENTENG205	Entrepreneurship		06
GSENG211	Introduction to Gender Studies		06
HMECH310*	Mechanical Engineering Skills Training	HMECH112	30

Level 3 Semester 2

HMECH321*	Research Methods for Engineers	HMECH102	12
HMECH323*	Heating, Ventilation and Air Conditioning	HMECH214	12
HMECH324	Advanced Heat and Mass Transfer	HMECH224	12
HMECH325*	Turbo Machinery	HMECH314	12
HMECH327	Product Design and Development		09
HMECH322*	Instrumentation and Control	HMECH126	09

Level 4 Semester 1: Work-Related Learning

HMECH410	Work-Related Learning Placement Report	15

Level 4 Semester 2: Work-Related Learning

HMECH401*	On the Job Training Project		60
HMECH402	On the Job Training Report		30
HMECH403	Industrial Supervisor's Assessment Report		15

Level 5 Semester 1

HMECH510*	Maintenance Engineering		9
HMECH511	Structural and Fracture Mechanics		12
HMECH512*	Stress Analysis and Finite Element Methods	HMECH215	15
HMECH514	Computational Fluid Dynamics and Numerical Thermo-flow	HMECH224, HMECH221	15
HMECH515*	Mechanical Engineering Innovation Project I:		
Option 1: Design and Manufacture:	Technical Report		15
Option 2: Modelling and Simulation:	Technical Report		15

Level 5 Semester 2

HMECH521*	Mechanical Engineering Innovation Project II:		
Option 1: Design and Manufacture:	Functional Product		
Option 2: Modelling and Simulation:	Validation Product/System	HMECH515	15
HMECH523	Engineering Economics and Accounting		09
HMECH524*	Managing Engineering and Technology		12
HMECH525	Computer Aided Design and Manufacturing (CAD/CAM)		12
HMECH526*	Project Management for Engineers		9

Note: All Modules are Core Modules

14.0 SYNOPSES

HMECH111 Engineering Drawing and Computer Aided Design I
The module covers: Drawing equipment and standards; Free-hand sketches; Geometry; Loci; Conics and curves; Orthographic projection; Isometric, Oblique, Perspective and auxiliary views; Fabrication drawings; Sectioning; Fasteners; Detail and Assembly drawings. CAD software; Editing and Plotting; Geometric Constructions; Template Drawings; Plotting; 2D Orthographic Drawings; Dimensioning; Section and Auxiliary Views; Blocks; Design Center; and Tool Palettes.

HMECH101 Mathematics for Engineers I
The module covers Algebra; Trigonometry; Calculus; Complex numbers; Coordinate geometry; Analytic geometry; Matrices and Determinants; and Applications to the Mechanical Engineering field.

HMECH112 Workshop Technology and Practice
The module covers: Workshop Layout; Industrial Safety; Ferrous and Non-Ferrous Materials; Furnaces; Properties of Metals; Heat Treatment; Foundry Technology; Forging; Cold and Hot Working of Metals; Welding; Sheet Metal Fabrication; Fitting; Metal Cutting; Lathe; Drilling; Slotting and Milling Machines; Powder Metallurgy; Inspection and Quality Control; Factories Safety Act and Statutory instruments; Measuring; Maintenance and repair of Industrial machines.

CSENG101 Communication Skills
The module covers: Introduction to communication; Pre-writing skills, Writing skills; Whole essay construction; Scientific and technical writing, Reflective; Critical and evaluative thinking; Teamwork skills CVs; Technical report writing; Citations; Referencing; Internet; e-books and journals; Reporting structures; Organizational structure; Grievance procedures and decision-making.

HMECH113 Engineering Physics
The module covers: Vectors and scalars; Forces; Equilibrium; Friction; Particle kinematics and kinetics; Area properties; Stress and strain; Torsion; Flexural loading; Stress-strain transformation; Combined loading; Columns; Hydrostatics; Hydrodynamics. Straight line motion; 3-D Vectors; 2-D and 3D Motion; Kinetic energy and work; Potential energy; Collisions; Rotation; Oscillations and waves; Temperature; Heat and the first law of thermodynamics, Energy sources, Circuit elements, Circuit theorems, Basic DC and AC, Electrical conductivity in metals, Dielectric and Magnetic properties of materials, introduction to 3 phase circuits, star delta transformation and current relations.

HMECH110 Chemistry and Engineering Materials
The module covers: Units; Elementary inorganic, analytical and physical chemistry; Stoichiometry; Molecular and atomic structure; periodicity; Bonding; Electrochemistry; Reactions; Entropy; Free energy; Solid-state; Coordination chemistry; Chemical and phase equilibrium; Colligative properties; Chemical kinetics; Organic compounds; Polymer chemistry; Mixtures; Solutions; Gas laws; Water chemistry & treatment; Air pollution; Effluent; Waste disposal; Basic Thermodynamics, Engineering materials, Corrosion, Lubricants, Materials characterization.

HMECH121 Engineering Drawing and Computer-Aided Design II
The module covers: Advanced working, detailed and assembly drawings; Limits; Tolerances and Fits; Machining drawings; Geometry of sheet metal work and developments; Solid and Surface Modelling; Changing and Plotting Solid Models; Creating Assembly Drawings from Solid Models; Solid Modelling for Section and Auxiliary Views; Rendering; Engineering fabrication - surface developments and structural assembles; CAD drawings.

HMECH102 Mathematics for Engineers II
The module covers: Differentiation, Partial differentiation, Series, Integration, Solutions of first-order differential equations, Numerical methods, Introduction to statistics; Applications to Mechanical Engineering field.

HMECH123 Fluid Mechanics I
The module covers: System of measurements; Fluids and their properties; Pressure and Head; Fluid pressure on surfaces and Buoyancy; Motion of fluid particle and streams; The momentum equation and its application; The energy equation and its application; Dimensional Analysis and Similitude.

HMECH124 Thermodynamics
The module covers: System; Control volume; Property; State; Process; Cycles; Mass; Volume; Density; Pressure; pure Substances, Ideal gases. Work and heat; Internal energy; Enthalpy; Specific heat capacity; First and second law of Thermodynamics; Conservation of mass; Processes; Compressible and incompressible gases; Entropy and enthalpy; Third Law of Thermodynamics; Introduction to vapor power; Cooling and gas cycles; Experimental techniques in thermodynamics.

HMECH125 Strength of Materials
The module covers: Mechanical properties of materials; Stress-normal and shear; Strain; Equilibrium in shear; shear strain; stress/strain, Hooke's Law; Poisson's ratio and the shear

stress-strain diagram; Axial loads; Elastic deformation; Displacements; Statically determinate and indeterminate structures and thermal effects; Torsion; Transverse shear; Combined loads; Thin cylinders; Stress transformation; Strain transformation; Moment of inertia; Deflection of beams.

HMECH126 Electrical and Electronics Technology
The module covers: Electrical terms and units; Sources of energy; Electrical power systems; Generation; transmission and distribution; Steam; Hydel; Gas and Nuclear power generation; Electrostatics and the Electron theory; DC Networks; Single and Three phase AC Circuits; Single phase EMF generation; Magnetic Circuits; Electromagnetism; Capacitance; Transformers; Induction Motor; DC Machines; Electrical Measuring Instruments; Analog Electronic Circuits; Digital Electronics & Integrated Circuits; Electronic Measurements & Instrumentation; Transducer & Sensors; Amplifiers; Microprocessors and microcontrollers, smart sensors, and Industrial Internet of Things, Laboratory Component.

HMECH214 Applied Thermodynamics
The module covers Applications of laws of thermodynamics; Work, heat; Heat capacity; Reversibility; Enthalpy; Entropy; Heat Engines and Heat Pumps; Carnot Cycle for Ideal Gas; Internal energy; Ideal and real gases; Compressible flow; Power cycles; Mixtures of gases psychrometry; Heating and cooling; Refrigeration and air-conditioning systems; Combustion; Mathematical Relations of Thermodynamics; Reversible Processes; Phase-change process; Cycles in Engineering.

HMECH213 Fluid Mechanics II
The module covers Scalars; Vectors; Tensor algebra; Green's lemma; Divergence; Eularian and Lagrangian; Function derivative, Reynolds transport; Governing equations; Conservation of mass; Momentum; Navier Stokes, Boundary conditions; Non-dimensionalisation; Viscous compressible and incompressible flow; Uniform and Steady Flow; Real and Ideal Fluids; Dimensional flow; Laminar & Turbulent Flow; Discharge and Mean Velocity; Continuity; Energy; Applications.

HMECH215 Applied Strength of Materials
The module covers the deflection of beams; Energy methods; Unsymmetrical bending; Theories of failure; Shear in thin-walled sections; Inelastic behaviour; Struts; Stress transformation; Strain Transformation; Thick Cylinders; Continuous beams; Transverse shear; Combined loading; 3-D equilibrium equations.

HMECH201 Mathematics for Engineers III
The module covers Laplace transforms; Fourier Series and Fourier Analysis; Linear algebra; First order differential equations; Higher order linear differential equations; Numerical solutions of differential equations; Applications to Mechanical Engineering field.

HMECH216 Electrical Machines and Drives
The module covers Single phase transformers, 3 phase transformers, 3 phase induction machines, single phase ac machines, Synchronous machines, DC machines, special motors, converter techniques for motor drive applications, Control techniques of electrical machines, selection and sizing of drives, IEE regulations pertaining to various motors.

HMINE124 Computer Programming
The module covers programming in BASIC, PASCAL, C, C++ and Visual Basic; Conditional and repetitive statements; Input and output; Simple functions; Program structure and organization; Introduction to classes and objects.

HMECH221 Mechanical Engineering Software and Numerical Methods
The module covers: Matlab; Python; Fortran; ANSYS; Linear algebraic equations; Numerical solutions of linear and non-linear equations; Eigenvalue problems; Numerical

approximation; Numerical differentiation; Numerical Integration; Numerical solutions to initial-value problems; Boundary-value problems for ordinary differential equations and Partial-differential equations; Direct methods of linear equations.

HMECH222 Mechanics of Machines
The module covers: Principles of mechanics as applied to velocity, acceleration and forces in mechanisms; Balancing of rotating masses and engines; Kinematics and Kinetics of mechanisms; Cams and Followers; Flywheel; Gears and Gears trains; Epicyclic gears; Friction Clutches; Belt drives and Band Brakes; Power Screw; Universal Joint; Gyroscope; Speed Governors; Steering mechanism.

HMECH224 Heat and Mass Transfer
The module covers Convection correlations; Convection; Evaporation and Boiling; Thermal radiation; Heat exchangers; Mass transfer; and Porous catalysts. Mass exchangers; Conduction; Natural and forced external and internal Convection; Velocity Boundary Layer; Thermal Boundary Layer; Laminar and Turbulent Flows; Heat and Momentum Transfer; Cooling law; Similarity and Dimensional Approach; Internal Flow and External Flow.

HMECH202 Mathematics for Engineers IV
The module covers: Convergence of series; Power series; Complex analysis; Laplace transforms revisited; Z-transforms and difference equations; transfer functions; State space representation; Applications to Mechanical Engineering field.

HMECH223 Hydraulic and Pneumatic Machines
The module covers: Principles of hydraulic systems; Types of Hydraulic Fluids; Hydraulic System Components; Hydraulic System Maintenance; Water Turbines; Centrifugal Pumps; Hydraulic Machines and Dimensional Analysis and Model Studies; Principles of pneumatic systems; Pneumatic Gases; Heavy-Duty Air Compressors; Air Compressor Maintenance and Safety; Contamination Control; Fans and Fan Systems.

HMECH225 Design of Machine Elements
The module covers: Machine Design basics; Engineering materials; Defects in materials; Manufacturing considerations in machine design; Simple stresses; Torsional and bending stresses and variable stresses in machine parts; Design against static and fluctuating loads; Pressure vessels; Pipes and pipe joints; Riveted and welded joints; Gears; Cams; Gear boxes; Brakes; Clutches; Engine head; Shafts; Keys; Couplings; Helical and Leaf springs; Bearings.

HMECH311 Manufacturing Processes and Systems
The module covers: Conventional manufacturing processes; Modern manufacturing process; Powder metallurgy; Quality control; measurement; inspection and testing; Manufacturing automation; Rapid prototyping and freeform fabrication; Manufacturing systems design concepts; Arc and oxy-gas welding techniques; Resistance welding processes, Power beam processes and Solid-state welding.

HMECH314 Fluid Mechanics
The module covers: Exact solutions: potential flow; Couette flow; Poiseuille flow and combined Couette-Poiseuille flow; laminar boundary layers; Stability of laminar flows: linearized stability; transition to turbulence; approximate prediction of transition; Turbulent flow: Reynolds averaged equations; Two-dimensional turbulent-boundary-layer equations; Velocity profiles; Turbulent flow in ducts; Flat plate flow; and turbulence modelling.

HMECH315 Machine Design
The module covers: Open-ended subsystem design using the following elements: Beams; Shafts; Bolts; Bearings; Rivets; Welds; Springs; Couplings; Clutches; Brakes; Gears and

gear systems; Static and fatigue design fundamentals; Code design: Pressure vessels; Structural steel design, Hoisting systems and ropes; Welding ASTM, ASM, BS, SAZ codes.

HMECH310 Mechanical Engineering Skills Training
Option 1: Boilermaking / Plating Welding

The module covers: Safety; Tools and equipment; Materials; Measuring and marking out; Job planning; Development and fabrication; Drawings and sketching; Manual arc welding; Gas cutting and welding; Tungsten Inert Gas welding; Silver soldering / Brazing (High cost materials); Gouging; Welding; Assembly; Repair and Testing; Machine tools.

Option 2: Fitting Including Machining
The module covers: Safety; Hand tools and equipment; Measuring and marking off; Power tools; Materials; Drawings and sketches; Assemblies; Lubrication; Pipe work; Welding; Lifting devices; Pneumatics and hydraulics; Measuring and Inspection; Installation of machines; Machining.

Option 3: Turning Including Machining

The module covers: Safety; Hand tools and equipment; Measuring and marking off; Materials; Lubrication and machine care; Job planning; Drawings and sketches; Quality control; Balancing; Power drilling; Shaping; Turning; Milling; Slotting; Grinding-surface, cylindrical, tool and cutter; Boring machines

Option 4: Maintenance Fitting

The module covers: Safety; Hand tools and equipment; Measuring and marking off; Power tools; Materials; Drawings and sketches; Assemblies; Removal lubrication; Pipe work; Repairing hydraulic brakes; Welding; Lifting devices; Pneumatics and hydraulics; Measuring and Inspection; Installation of machines.

ENTENG 205 Entrepreneurship
The module covers Business development; Marketing; Human resources management pricing; Government policies; Taxation of business; Case studies based on site visits; and Preparation of a business plan.

GSENG 201 Introduction to Gender Studies
The module covers Gender sensitivity; Gender inequalities; Historical development of gender issues; Gender analysis; Gender issues in Zimbabwe; Redressing gender imbalances; Empowerment and strategies for gender responsive environment.

HMECH321 Research Methods for Engineers
The module covers Technical writing; Simple Comparative Experiments; Single Factor Experiments; Analysis of Variance; Randomized Blocks, Latin Squares, and Related Designs; Introduction to Factorial Designs; 2k Factorial Design; Blocking and Confounding in the 2k Factorial Design; Two-Level Fractional Factorial Designs; Additional Design and Analysis: Factorial and Fractional Factorial Designs; Fitting Regression Models; Response Surface Methods and Designs.

HMECH322 Instrumentation and Control
The module covers: Measurement system elements; Elements of an instrumentation system; Signal conditioning and processing; Data presentation; Choosing appropriate measuring instrument; Errors; Elements of Automatic Control; Governors; Gyroscopes; Transducers and System design; Introduction to control systems engineering; Control systems mathematics.

HMECH323 Heating, Ventilation and Air Conditioning
The module covers Comfort and indoor air quality; Psychrometrics; System types and selection; Cooling and heating load calculations; Conduction; Radiation, Convection; Internal loads and thermal storage; Design of air handling unit; Ducts; Plant and reticulation; Control systems; Introduction to integrated system simulation.

HMECH324 Advanced Heat and Mass Transfer
The module covers Convection correlations; Thermal radiation; Convection, Evaporation; Boiling: Film condensation, Film evaporation, Pool boiling; Forced-convection boiling and condensation; Flow regime maps; phase change at low pressures; Heat pipes; Heat exchangers; Mass transfer: Fick's Law; Mass diffusion; Mass convection; Simultaneous heat and mass transfer; Porous catalysts; High mass transfer rate theory; Mass exchangers.

HMECH325 Turbo Machinery
The module covers Introduction: Dimensional Analysis: Similitude 1; Basic Thermodynamics, Fluid Mechanics: Definitions of Efficiency; Two dimensional cascades; Axial-flow turbines: Two-dimensional Theory; Axial-flow compressors and Fans; Three-Dimensional Flows in Axial Turbomachines; Centrifugal Pumps, Fans and Compressors; Radial Flow Gas Turbines; Hydraulic Turbines and Wind Turbines.

HMECH327 Product Design and Development
This module covers Morden strategies for material pre-selection in product design; Product: Development Process and Organisations, Life Cycle, Architecture, Planning and Specifications; Customer needs; Quality function deployment; Total Quality Management, Environmental Engineering, Concept generation, selection, and testing; Visual Design, Branding, Illustration, Animation, Tools for Product Design on the Internet for Designers, Product Design Software, Design for Manufacturing, Design for the environment, Prototyping, Industrial design, Robust Design, Patents and Intellectual Property, Product Development Economics.

Level 4 Semester 1: Work-Related Learning

HMECH410 Work-Related Placement Report
The module covers The report of the student's understanding of an organisation's purpose, functions and relations, operations and relationship between the organisation and other organisations. The report for the brief student's experience, observations and recommendations. Areas of coverage include but are not limited to Drawing Office, Energy generation equipment, Heat exchangers, Ventilation and air conditioning equipment, safety, gearboxes, crushers, compressors, machining equipment conveyor systems, maintenance management of equipment, budgeting etc

Level 4 Semester 2: Work-Related Learning

HMECH401 On the Job Training Project
The module covers: Industrial application and appreciation of knowledge and skills gained during the 6 semesters. The student is expected to solve a particular problem at the industry of placement as part of his/her contribution to the industry under the guidance of the industrial supervisor. In a mini dissertation format, the student shall present a supervised write-up of the problem formulation, how it was solved and the achieved results. In addition to areas stipulated in the students' Work-Related Learning Log book, the project forms part of the day to day Engineer in Training duties of the student in line with the industry of placement's daily business.

HMECH402 On the Job Training Report
The module covers: The reports of the assigned day to day tasks undertaken by the student for the whole stint of Work-Related Learning. The reports for conducted tasks should be frequently signed by the industrial supervisor and will be assessed by the academic supervisor who shall frequently visit the student during Work-Related Learning.

HMECH403 Industrial Supervisor's Assessment Report

The module covers The assigned industrial supervisor's overall assessment of the student's performance during Work-Related learning.

HMECH510 Maintenance Engineering
The module covers: Statistical concepts; Mathematics of failure; Reliability concepts; Fitting distribution to failure data; Maintenance management; Maintenance structure; Asset inventory management; Data analysis; Renewable, repairable systems, Laplace trend test; Optimizing maintenance strategies; Replacement/overhaul age; inspection frequencies; Capital replacement, and simulation; Reliability-Centered Maintenance; Maintenance systems; Components; structure; Tribology; Maintenance Practice.

HMECH511 Structural and Fracture Mechanics
The module covers: Statically determinate and indeterminate systems; Buckling; Stress and strain transformation; Plane kinematics and kinetics of rigid bodies; Particle kinetics. Single and Multi-degree of freedom systems; Undamped and damped free vibration and harmonic motion; discretisation; Eigen-problem; Linear Elastic and Elasto-plastic Fracture Mechanics; Fatigue crack growth; Paris Law; Life prediction; Combined mode fracture, Strain energy density methods.

HMECH512 Stress Analysis and Finite Element Methods
The module covers: The finite element procedure; stiffness matrix method; space trusses; beam element equation, loading and boundary conditions; beam analysis; formulation of 2D elements; Analysis of cylindrical and spherical elements; Isoparametric formulation; Numerical procedures and systems of equations; eigenvalue problems; integration; Approximation and boundary values; Finite element analysis; Modelling procedures and results processing.

HMECH514 Computational Fluid Dynamics and Numerical Thermoflow
The module covers: Continuum mechanics; Continuity; Momentum; Navier-Stokes; Energy; Thermo-fluids; Mesh generation and Boundary conditions; Finite difference and volume method; FVM for Diffusion; convection-diffusion problems and Pressure-velocity coupling; Simple algorithm; Inviscid flow; Heat conductions; Governing equations, discretization, finite approximation, solution methods; Gauss-Seidel, Tri-diagonal matrix algorithm; CFD.

HMECH515 Mechanical Engineering Innovation Project I:
The student can choose a project of their own choice, select from the list of departmental project topics or get a project from the industry. The student shall submit a written and supervised dissertation and be subjected to oral presentation.

Option 1: Design and Manufacture: Technical Report

The module covers: A wide range of innovative project topics that stem from the major fields of study or their combinations. The project should be problem-solving centered, addressing community or industrial needs. The student is expected to employ the knowledge and skills attained during the 9 semesters of both university and industrial training to generate a concept that shall be developed into a functional product in the follow-up module.

Option 2: Modelling and Simulation: Technical Report

The module covers: Areas of modelling and simulation of products or systems that stem from the major fields of study or their combinations, using the learnt mechanical engineering software and the knowledge and skills attained during the 9 semesters of both university and industrial training. The student is expected to design and develop a model that simulates real-world system(s) that are community or industrial centered. The designed model shall be verified and validated through a functional product or system in the follow-up module.

HMECH521 Mechanical Engineering Innovation Project II:
The student builds and orally present a functional product or system based on the design presented and submitted for examination in the preceding module. The product or system can be a prototype or full size scale. Depending on its novelty, the product or system and its design is published, commercialized, industrialized or patented.

Option 1: Design and Manufacture: Functional Product

The student manufactures the product as designed in the preceding module and proves its functionality. Among others, the student is expected to generate a detailed document on functionality of the product, its standard operating procedure, ease of assembly and dismantling, expected life cycle, maintenance schedule, product/system evaluation and costing, manufacturing time, attained output, payback period and return on investment.

Option 2: Modelling and Simulation: Validation Product/System

The student builds the product/system modelled in the preceding module. The product/system shall be used to verify and validate the model results. Among others, the student is expected to generate a detailed document on functionality of the product, its standard operating procedure, expected life cycle, maintenance schedule, product/system evaluation and costing, manufacturing time, attained output, how the product/system output measures to or varies from the model results and perform the required optimization.

HMECH523 Engineering Economics and Accounting
The module covers: Money-time relationships and equivalence; Bases for comparison of alternatives; Decision-making among alternatives; The influence of inflation on engineering economic calculations; Decision-making among alternatives on an after-tax basis; Replacement analysis; Risk analysis of cash flows; Accounting Principles and the Balance Sheet; Introduction to a new business venture; Constructing a Profit and Loss, and Cash Flow.

HMECH524 Managing Engineering and Technology
The module covers: Introduction to Engineering Management; Functions of Technology Management Managing Technology; Managing Projects; and Managing Engineering Career; Industrial ethics and legislation; Intellectual property, Professional studies.

HMECH525 Computer-Aided Design and Computer-Aided Manufacturing (CAD/CAM)
The module covers Solid Works; Computer Aided Manufacturing; Computer Aided Process Planning; Computer Numerical Controlled (CNC) machining; Lathe and milling, Robotics and Smart Manufacturing.

HMECH526 Project Management for Engineers
The module covers: Introduction to Project Management; Responsibilities and Roles; Practices and Scheduling; Scope of Projects; Communication; Management; Legal aspects for Project Managers; Project procurement and Contract management; Time management and Project techniques; Project risk management, Critical chain project management;

Quality; Project costing; Project reporting, control and closing; Resource utilization; Project management software.

Department of Metallurgical and Materials Engineering

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN MATERIALS ENGINEERING (ENG05)

Duration:	5 years
Actual Credit Load:	660
Minimum Credit Load	600
Maximum Credit Load	660
Total MBK Credits	480

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in the field of Materials Engineering relevant to various entrepreneurial and employment possibilities in Zimbabwe and other economies in the regions, Africa, and abroad.

To prepare students for higher degree studies in Materials, and to be environmental stewards required for sustainable manufacturing.

2.0 ENTRY REQUIREMENTS

Normal Entry: At least two A' Level passes, with Mathematics and Chemistry, including passes in O' Level Mathematics and English

Special Entry: Refer to Section 3.2 of the General Academic Regulations

Candidates with diplomas in Metallurgical Assaying, Industrial Metallurgy, or similar may be considered

Mature Entry: Refer to Section 3.2 of the General Academic Regulations

3.0 PROGRAMME CHARACTERISTICS

Areas of Study: Manufacturing Technologies, Ceramic Materials, Polymers, Composites, Physical Metallurgy

Specialist Focus: Materials science and engineering for understanding value addition through manufacturing, and developing new materials

Orientation: Biased towards hands-on components, including laboratory and industry visits, and employing relevant software and other computer-aided methods

Distinctive Features: The emphasis of the programme is on the practical application of materials science and engineering to value addition and goods manufacture for wealth creation, sustainable economic development and empowerment of society

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Graduates can be employed as process engineers, corrosion engineers, welding engineers, production managers, consultants, solicitors, analysts, third-party inspectors, lecturers and researchers in foundries, manufacturing companies, law firms, banks, investment and insurance companies, research and commercial laboratories, universities and colleges, in both government and privately owned entities.

The programme also allows for the development of entrepreneurial skills, presenting the possibility of self-employment.

Further Studies: Possible avenues for further education include postgraduate diplomas in Engineering Education, Law and Finance, Masters and Doctoral studies in Welding Engineering, Physical Metallurgy, Materials Science and Engineering, Biomaterials Science and Engineering, and similar areas

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Face-to-face and online lectures, laboratory classes and online simulations, seminars, group work, industry visits, Work-Related learning, research projects and independent study

Assessment methods:

1. Summative assessment in the form of a written examination at the end of the semester. The examination shall contribute no less than 60% of the final mark.
2. Formative assessment in the form of tests, laboratory reports, seminar presentations, and assignments. These may be individual or group work, and shall contribute no more that 40% to the final mark.
3. Assessment of research projects, Work-Related learning, product design and development, shall be based on written reports, oral presentations and the physical product.

5.1 PROGRAMME COMPETENCIES

5.1.1 Generic competences:

- Multidisciplinary- Ability to draw from diverse academic disciplines to define and solve problems
- Quantitative and innovative reasoning- Capability to analyse big data for informed decision-making, as well as seeking new and creative ways of solving problems
- Communication skills- Ability to effectively engage expert and non-expert audiences through oral and written presentations using appropriate ICTs
- Analysis and synthesis- Capacity for analysis and synthesis using logical arguments and proven facts
- Ethical commitment- Professional integrity and awareness of impact of science and technology on society and the environment
- Entrepreneurial skills- Capacity to identify and create new business venues based on knowledge and new thinking paradigms

5.1.2 Discipline specific competencies:

- Deep knowledge- Ability to analyse materials behaviour and performance in terms of underlying principles and knowledge, and by means of appropriate research
- Production skills- Ability to design, implement and monitor manufacturing operations and practices for competitiveness and profitability
- Technology development skills- Ability to develop new technologies across the manufacturing value chain with a view to enhancing production efficiencies and outputs
- Problem-solving skills- Ability to identify and solve problems in materials and manufacturing using theoretical and experimental methods
- Analytical and computational skills- Ability to analyse materials' behaviour, and performance using appropriate ICT packages

6.0 INTENDED LEARNING OUTCOMES

Graduates from the BEng (Hons) in Materials Engineering programme should:

- be able to apply knowledge of mathematics, science and engineering principles to identify and solve complex materials engineering problems;
- .demonstrate understanding of fundamental structure-processing-property-performance relations in materials;
- demonstrate competence in using logical arguments to design and conduct experiments to solve complex materials engineering problems cost-effectively;
- demonstrate ability to effectively communicate materials science and engineering phenomena and manufacturing processes, both orally and in writing, to engineering and lay communities;
- be able to utilize a range of hardware and software systems to model, design and implement manufacturing processes sustainably and profitably;
- demonstrate awareness of the impact of materials engineering and other engineering activities on society, economy and the environment;
- demonstrate ability to work effectively as an individual, in teams, and in multidisciplinary environments;
- demonstrate critical awareness of the need to act professionally and ethically, and to exercise and take responsibility within the limits of their competencies;
- demonstrate competence to engage in independent learning, and to operate independently in complex contexts that require personal responsibility, initiative, and accurate self-evaluation;
- demonstrate ability to use appropriate engineering methods, skills and tools to solve problems and to enhance efficiencies and outputs of manufacturing processes;
- identify and create sustainable business ventures based on acquired knowledge and skills;
- recognise the need to engage in life-long learning, and to identify this as a vehicle for developing new materials and manufacturing practices and technologies that can enhance efficiencies and outputs of the industry;
- demonstrate the ability to draw logic conclusions from data, and use analytical tools including those based on information and communication technology, for informed decision-making and innovation.

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for Work-Related Learning, and Final Year Project.

Student is expected to have obtained an approved First Aid certificate by the end of second semester of Level 3.

8.0 ASSESSMENT
Refer to section 6 of the Faculty Regulations

9.0 FAILURE TO SATISFY EXAMINERS
Refer to section 9 of the General Academic Regulations

10.0 GRADING AND DEGREE CLASSIFICATION
Refer to section 11 of the Faculty Regulations

11.0 DEGREE WEIGHTING

For the purpose of classification, weighting of the results of the different levels shall be:
Level 1: 5%
Level 2: 10%
Level 3: 15%
Level 4: 30%
Level 5: 40%

PROGRAMME STRUCTURE

Code	Module Description	Pre-requisite	Credits
Level 1 Semester 1			
HMETE103	Introduction to Computers and Computer Applications		8
HMTE111	Engineering Mathematics I		12
HMINE119	Workshop Practise		12
HMETE118	Chemistry for Engineering		12
CSENG131	Communication Skills		6
HMINE117	Engineering Drawing and Design		12
HMETE112	Physics for Engineering		12
Level 1 Semester 2			
HMINE126	Engineering Mechanics		12
HMATE140	Organic Chemistry		12
HMETE140	Materials Science		12
HMINE128	Computer Science for Engineers		12
HMINE127	Electrical Principles		12
HMTE122	Engineering Mathematics II		12
Level 2 Semester 1			
HMETE213	Fluid Mechanics		12
HMETE214	Nanoscience I		12
HMETE215	Thermodynamic Principles		12
HMETE216	Mineral Beneficiation		12
HMTE211	Engineering Mathematics III		12
ENT205	Entrepreneurship		12
GS201	Introduction to Gender Studies		8

Level 2 Semester 2

HMETE225	Heat and Mass Transfer	12
HMATE223	Microstructural Analysis Techniques	12
HMATE224	Fuels and Energy	12
HMATE225	Introduction to Extractive Metallurgy	12
HMATE226	Nanoscience II	12
HMETE227	Engineering Mechanics II	12
HMTE221	Engineering Mathematics IV	12

Level 3 Semester 1

HMATE315	Materials Technology	12
HMATE311	Design of Materials	12
HMATE312	Materials Degradation and Surface Engineering	12
HMETE318	Safety, Health and Environment	12
HMETE319	Research Methods	12

Level 3 Semester 2

HMETE326	Foundry Metallurgy	12
HMATE321	Polymers and Polymer Matrix Composites	12
HMATE331	Energy Materials	12
HMETE328	Mechanical Metallurgy	12
HMATE323	Ceramics and Ceramic Matrix Composites	12
HMATE325	Third Year Project	20

Level 4 Semester 1 and 2: Work-Related Learning

HMATE400	Work-Related Learning Placement Report	10

Level 4 Semester 2

HMATE404	Work-Related Learning Report	40
HMATE405	Academic Supervisor's Report	40
HMATE406	Employer's Assessment Report	20

Level 5 Semester 1

HMATE512	Manufacturing Process Design, Control and Simulation	12
HMETE514	Technological Entrepreneurship and Product Development	12
HMINE516	Professional Ethics, Practise and Engineering Management	12

Level 5 Semester 2

HMETE523	Corrosion Engineering(elective)	12
HMATE524	Welding and Joining of Materials(elective)	12
HMATE522	Gemstones (elective)	12
HMATE525	Final Year Project	30

12. Module Synopses

HMINE117 Engineering Drawing and Design
This module covers use of drawing instruments, lettering, drawing standards and conventions; projection methods, engineering graphics, and measurement of areas; introduction to CAD, machine drawing, sectional views, ISO conventions and elementary assembly drawing

HMTE111 Engineering Mathematics I

This module covers calculus in one variable; differentiation and integration; functions of several variables; multiple integrals; sequences and series; complex numbers

HMETE112 Physics for Engineering
The module covers fundamental concepts of electrical and magnetic energy; electrostatics and electric fields; capacitance and dielectrics; magnetic circuits; electromagnetism; electromagnetic induction; magnetic field; electromagnetic induction, conservation of energy; lateral and volumetric expansion; laws of thermodynamics; ideal gas laws, equilibrium of forces; simple beams and trusses; shear force diagrams and bending moment diagrams; simple stress and strain

HMETE118 Chemistry for Engineering
The module covers basic concepts of physical, organic, analytical and inorganic chemistry; experiments involving the mole concept, stoichiometry, percent composition and empirical formula determination; synthesis, analysis, purification and separation techniques; reactions of functional groups in hydrocarbons and industry-relevant analytical techniques

CSENG131 Communication Skills
The module covers introduction to communication; pre-writing skills; writing skills; whole essay construction; scientific and technical writing; reflective, critical and evaluative thinking; teamwork skills

HMETE103 Introduction to Computers and Computer Applications
The module focuses on concepts of computer technology and practical use of computers; computing software, computer communications; computer-aided drawing; computer systems organization; number systems; legal and ethical considerations; security; use of the internet to develop research skills

HMINE119 Workshop Practice
The module covers bench fitting; measuring instruments; forging; wood-working; metal forming; foundry practice; safety and care

HMINE126 Engineering Mechanics
The module covers general principles; force vectors; equilibrium of a particle; force system resultants; work; energy; power; momentum; friction; equilibrium of a rigid body; structural analysis; friction; centre of gravity and centroid; moments of inertia; virtual work; hydraulics

HMATE140 Organic Chemistry
This module introduces basic principles in organic chemistry covering bonding and structural representation, classification, IUPAC system of nomenclature, fundamentals of reaction mechanisms, isomerism and stereochemistry. Methods of preparation and quantitative analysis of organic compounds are also included. The chemistry of aliphatic and aromatic compounds covering alkanes, alkenes, alkynes, alcohols, halogen compounds, carbonyl compounds, carboxylic acids, derivatives of carboxylic acids and amines is also covered.

HMETE140 Materials Science
The module covers the classification of materials; material properties; structure-property-performance relationships in materials; atomic structure of materials; bonding, crystal structures and defects in solids; electronic materials; failure of materials; corrosion and corrosion protection

HMINE128 Computer Science for Engineers

The module covers programming in BASIC, PASCAL, C, C++ and Visual Basic; conditional and repetitive statements; input and output; simple functions; program structure and organization; introduction to classes and objects

HMINE127 Electrical Principles
The module covers the fundamentals of electric circuits; resistive network analysis; AC network analysis; transient analysis; frequency response and system concepts

HMTE 122 Engineering Mathematics II
The module covers ordinary differential equations; application of differential equations to engineering; Fourier analysis; Laplace transforms; Z transforms

HMINE213 Fluid Mechanics
The module covers the introduction to fluid mechanics and fluid statics; general fluid motion; descriptive treatment of boundary layer concepts; conservation of mass and energy; flow through pipes and open channels; steady uniform flow in a rectangular open channel

HMETE214 Nanoscience I
The module covers a basic introduction to nano technology; technological applications such as physico-chemical considerations: band structures, typical and useful nano effects; colloids, typical syntheses of nanoparticles, fullerenes, synthesis, forms, variants, properties, applications of carbon nanotubes; synthesis, forms, variants, properties, applications oxide-nanotubes and fibres; bio-nano-link, risk and future perspectives

HMETE215 Thermodynamic Principles
The module focuses on thermodynamic and electrochemical principles applied to metallurgical processes; phase rule, heat of reaction, free energy, activity, thermodynamic equilibrium; thermodynamics of aqueous solutions, thermochemistry of gas mixtures, solution thermochemistry, inter-action parameters

HMINE216 Mineral Beneficiation
The module introduces concepts in mineral beneficiation. It covers basic mineral processing unit operations and related equipment. Unit operations covered are comminution: rock size reduction by crushing and grinding; size classification: screening, de-sliming, cycloning; concentration techniques and dewatering

HMTE211 Engineering Mathematics III
The module covers introduction to numerical methods for solving problems in mathematics, science and engineering; computer arithmetic and rounding errors; numerical methods for root-finding; polynomial interpolation and splines; solution for linear algebraic equations; numerical integration and differentiation; numerical integration of ODEs; Euler and second order; Runge-Kutta methods

ENT205 Entrepreneurship
The module focuses on business development; marketing; human resources management pricing; government policies; taxation of business; case studies based on site visits; preparation of a business plan

GS201 Introduction to Gender Studies
The module covers gender sensitivity; gender inequalities; historical development of gender issues; gender analysis; gender issues in Zimbabwe; redressing gender imbalances; empowerment and strategies for gender responsive environment

HMETE225 Heat and Mass Transfer
The module focuses on conduction, convection, phase change heat transfer and heat exchangers; radiation; application to metallurgical processes; mass transfer, Fick's law; Knudsen diffusion coefficient; mass transfer at phase boundaries; application to metallurgical processes

HMATE223 Microstructural Analysis Techniques
The module focuses on techniques in the microstructural characterisation of materials, including sampling; sample handling and storage; and metallographic sample preparation. Analytical techniques covered Optical Microscopy, X-ray Diffraction (XRD), Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), Transmission Electron Microscope (TEM) and high-energy electron analysis

HMETE224 Fuels and Energy
The module covers forms of energy, renewable and non-renewable energy; fuels, fuel systems and combustion of fuels; fuels and energy in metallurgical and manufacturing processes; energy efficiency, energy management, sustainable and safe use; energy generation and storage; energy system innovations

HMETE225 Introduction to Extractive Metallurgy
The module covers techniques used in extractive metallurgy: pyrometallurgy, hydrometallurgy, electrometallurgy; case studies

HMETE226 Nanoscience II
The module covers epitaxy, beam lithographs, self-assembly, bio-catalytic synthesis, atom optics, and scanning probe lithography. The unique size-dependent properties, mechanical, thermal, chemical, optical, electronic, and magnetic that result from nanoscale structure will be explored in the context of technological applications including computation, magnetic storage, sensors, and actuators

HMINE227 Engineering Mechanics II
The module covers stress and strain; combined torsion and bending stress; continuous beams; stiffness matrix and nodal force vector; three-dimensional stress; computer analysis; Lagrangian formulation method; Hamiltonian method and D'alambert's principle; rotating coordinate system; analysis and design of structures

HMTE222 Engineering Mechanics IV
The module covers correlation and regression, scatterplots, and correlation matrix; residuals and transformations; outliers; Pearson's and Spearman's correlation coefficients and use in predictions; regression in terms of sums of squares and sums of products; estimation and testing; t and f-tests; multiple linear regression; matrices in simple and multiple linear regression; testing and inference in multiple linear regression using matrices; partial correlation; analysis of variance (ANOVA)

HMATE315 Materials Technology
The module focuses on technologies used to process metals into useful products. It covers forming processes such as forging and extrusion; solidification theory, phase equilibria and microstructure; heat treatment of steels, microstructure-property-performance relations; strengthening mechanisms; introduces welding, powder metallurgy and metal matrix composites

HMATE311 Design of Materials
The module looks at point, linear and surface defects in solids: how they are formed, associated energies, and how they influence mechanical, electrical, magnetic and optical properties of material; controlling and quantifying population of intrinsic and extrinsic defects; doping; alloy design; semiconductors; case studies

HMATE312 Materials Degradation and Surface Engineering

The module focuses on degradation mechanisms of materials; surface engineering techniques used to reduce susceptibility to degradation including coatings produced by vapour deposition including deposition and electrodeposition, polymer and ceramic coatings

HMETE318 Safety, Health and Environment
The module covers integrated environmental, quality, safety and health management systems; safety, health and environmental hazards and concerns in mineral processing and manufacturing plants; applicable local and international legislations; psychology and acceptability of risk; probability theory and its application to risk, risk assessment and management; economic implications of safety, health and environment issues; local and international case studies

HMINE319 Research Methods and Technical Writing
The module aims to equip students with scientific research and report writing skills. It covers scientific research methods; design of experiments; writing ethics, citation, and the importance of intellectual property

HMETE326 Foundry Metallurgy
The module covers the economic importance of foundries; types of furnaces used in foundries; melting practises and melt quality; casting processes and mould-making materials; types and prevention of casting defects; quality control in foundries; casting ferrous and nonferrous alloys

HMATE331 Energy Materials
The module covers materials used on materials for fuel cells, batteries, organic solar cells, and electrochemical capacitors; the effect of microstructure; electrochemical characterisation

HMETE328 Mechanical Metallurgy
The module looks at the tensile response of metals; response to 2D and 3D stress states; stress-strain relations for elastic and plastic behaviour; fracture and fracture mechanics; effect of temperature on flow properties; fatigue; environmental assisted cracking; failure analysis

HMATE323 Ceramics and Ceramic Matrix Composites
The module covers classification of ceramics; structure, properties and processing of ceramic materials; reinforced ceramics; case studies

HMATE321 Polymers and Polymer Matrix Composites
The module covers the structure and properties of polymeric materials; polymerization methods; the effect of additives; reinforced polymers and polymer matrix composites with emphasis on their engineering properties and applications

HMATE325 Third Year Project
The module builds on skills acquired in HMINE314. Students are expected to demonstrate understanding of scientific research and technical writing by undertaking practical research projects with no more than 3 objectives and writing structured reports. Projects must be relevant to metallurgy

Level 4 Semester 1
HMETE400 Work-Related Learning Placement Report
The module covers: The report of the student's understanding of an organization's purpose, functions, operations and linkages with relevant stakeholders i.e. suppliers, clients, government agencies etc.
Level 4 Semester 2

HMETE404 Work-Related Learning Report
This module provides students an opportunity to identify and carry out project(s) during a work-related learning period. Students will record and maintain a logbook of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report with appropriate design charts or process charts are required.

HMETE405 Academic Supervisor's Report
This module provides an assessment done by the academic supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student. Students will be visited at their workplace at least twice by their academic supervisors.

HMETE406 Employer's Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HMATE512 Manufacturing Process Design, Control and Simulation
The module focuses on manufacturing process development, data collection, analysis, optimization and control. It includes computer simulation in energy, development of process flow diagram, plant equipment selection and plant layout, development of process control scheme, cost estimation and economic evaluation. The module also includes environmental impact assessment, development of occupational health and safety plan, consideration of sustainability and social impact issues

HMETE514 Technological Entrepreneurship and Product Development
The module covers entrepreneurship and enterprise development; developing a business plan and the maximum utilization of the financial, marketing and human resources; challenges in entrepreneurship, product development and marketing; steps in developing a product

HMINE516 Professional Ethics, Practise and Engineering Management
The module covers professionalism; professional ethics and social responsibility; professional practices, partnerships and client relationships; national and international associations and societies in the industry; continuous professional development; professional practice legislation; engineering management in enterprises; project management; capital project economics; breakeven point; structure responsibility charts; modern trends- intranet; virtual teams; strategic partnering; ASP's; quality assurance and management; operational research

HMETE523 Corrosion Engineering
The module covers the economic impact of corrosion; corrosion principles; forms of corrosion; corrosion testing techniques; materials used for corrosion applications corrosion control through materials selection, design, environment control, cathodic and anodic protection, coatings; high temperature corrosion; case studies

HMATE524 Welding and Joining of Materials

The module covers welding processes, welding joints and welding symbols; welding defects; welding of ferrous and nonferrous alloys; mechanical joints; brazing and soldering; adhesives

HMATE522 Gemstones
The module covers, the following aspects of gemstones: definitions; mineralogy; crystallography; identification; classification; valuation; cutting and polishing; jewellery; synthesis; marketing

HMATE525 Final Year Project
Students undertake research projects that involve solving a problem related to materials. Emphasis is on research design; use of logical argument, learned materials science and ICT in designing and conducting experiments to aid problem-solving. Economic value of the project and/or the proposed solution must be highlighted. Projects that demonstrate innovation and that may lead to commercialisation are encouraged

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN METALLURGICAL ENGINEERING (ENG04)

Duration:	5 years
Actual Credit Load:	660
Minimum Credit Load	600
Maximum Credit Load	660
Total MBK Credits	480

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in the field of Metallurgical Engineering relevant to various entrepreneurial and employment possibilities in Zimbabwe and other economies in the regions, Africa, and abroad.

To prepare students for higher degree studies in Metallurgy, and to be environmental stewards required for sustainable exploitation of mineral resources.

2.0 ENTRY REQUIREMENTS

Normal Entry: At least two A' Level passes, with Mathematics and Chemistry, including passes in O' Level Mathematics and English

Special Entry: Refer to Section 3.2 of the General Academic Regulations

Candidates with diplomas in Metallurgical Assaying, Industrial Metallurgy, or similar may be considered

Mature Entry: Refer to Section 3.2 of the General Academic Regulations

3.0 PROGRAMME CHARACTERISTICS

Areas of Study: Mineral Beneficiation, Pyrometallurgy, Hydrometallurgy, Physical Metallurgy, Manufacturing Technologies

Specialist Focus: Metallurgy for understanding value addition from ore dressing, to mineral extraction and manufacture of metallic products

Orientation: Biased towards hands-on components, including laboratory and field activities, and employing relevant software and other computer aided methods

Distinctive Features: The emphasis of the programme is on the practical application of metallurgical principles to value addition of mineral resources for wealth creation, sustainable economic development and empowerment of society

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Graduates can be employed as process metallurgists, corrosion engineers, welding engineers, production managers, consultants, solicitors, analysts, third party inspectors, lecturers and researchers in mines, foundries, manufacturing companies, law firms, banks, investment and insurance companies, research and commercial laboratories, universities and colleges, in both government and privately owned entities.

The programme also allows for the development of entrepreneurial skills, presenting the possibility of self-employment.

Further Studies: Possible avenues for further education include postgraduate diplomas in Metallurgy, Engineering Education, Law and Finance, Masters and Doctoral studies in Metallurgy, Materials Science and Engineering, Geometallurgy and similar areas.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Face-to-face and online lectures, laboratory classes and online simulations, seminars, group work, industry visits, Work-Related learning, research projects and independent study

Assessment methods: 4. Summative assessment in the form of a written examination at the end of the semester. The examination shall contribute no less than 60% of the final mark.

Formative assessment in the form of tests, laboratory reports, seminar presentations, and assignments. These may be individual or group work and shall contribute no more than 40% to the final mark.

Assessment of research projects, Work-Related learning, product design and development, shall be based on written reports, oral presentations and the physical product.

5.1 Programme Competences

5.1.1 Generic competences:

- Multidisciplinary- Ability to draw from diverse academic disciplines to define and solve problems
- Quantitative and innovative reasoning- Capability to analyse big data for informed decision-making, as well as seeking new and creative ways of solving problems
- Communication skills- Ability to effectively engage expert and non-expert audiences through oral and written presentations using appropriate ICTs
- Analysis and synthesis- Capacity for analysis and synthesis using logical arguments and proven facts
- Ethical commitment- Professional integrity and awareness of the impact of science and technology on society and the environment

- Entrepreneurial skills- Capacity to identify and create new business venues based on knowledge and new thinking paradigms

5..1.2 Discipline-specific competencies:

- Deep knowledge- Ability to analyse metallurgical phenomena in terms of underlying principles and knowledge, and by means of appropriate research
- Production skills- Ability to design, implement and monitor metallurgical operations and practices for competitiveness and profitability
- Technology development skills- Ability to develop new technologies across the mineral value chain with a view to enhancing production efficiencies and outputs
- Problem-solving skills- Ability to identify and solve problems in metallurgy using theoretical and experimental methods
- Analytical and computational skills- Ability to analyse metallurgical phenomena and trends using appropriate ICT packages

6.0 INTENDED LEARNING OUTCOMES

Graduates from the BEng (Hons) in Metallurgical Engineering programme should:

- be able to apply knowledge of mathematics, science and engineering principles to identify and solve complex metallurgical problems;
- demonstrate competence in using logical arguments to design and conduct experiments to solve complex metallurgical problems cost-effectively;
- demonstrate ability to effectively communicate metallurgical phenomena and processes, both orally and in writing, to engineering and lay communities;
- be able to utilize a range of hardware and software systems to model, design and implement metallurgical processes sustainably and profitably;
- demonstrate awareness of the impact of metallurgical and other engineering activities on society, economy and the environment;
- demonstrate ability to work effectively as an individual, in teams, and in multidisciplinary environments;
- demonstrate critical awareness of the need to act professionally and ethically, and to exercise and take responsibility within the limits of their competencies;
- demonstrate competence to engage in independent learning, and to operate independently in complex contexts that require personal responsibility, initiative, and accurate self-evaluation;
- demonstrate the ability to use appropriate engineering methods, skills and tools to solve problems and to enhance efficiencies and outputs of metallurgical processes;
- identify and create sustainable business ventures based on acquired metallurgical knowledge and skills;
- recognise the need to engage in life-long learning, and identify this as a vehicle for developing new metallurgical practices and technologies that can enhance efficiencies and outputs of the industry;
- demonstrate the ability to draw logical conclusions from data, and use analytical tools including those based on information and communication technology, for informed decision-making and innovation.

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for Work-Related Learning, and Final Year Project.

Student is expected to have obtained an approved First Aid certificate by the end of second semester of Level 3.

8.0 ASSESSMENT

Refer to section 6 of the Faculty Regulations

9.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Academic Regulations

10.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 11 of the Faculty Regulations

11.0 DEGREE WEIGHTING

For the purpose of classification, weighting of the results of the different levels shall be:

Level 1:	5%
Level 2:	10%
Level 3:	15%
Level 4:	30%
Level 5:	40%

12.0 PROGRAMME STRUCTURE

Code	Module Description	Pre-requisite	Credits
Level 1 Semester 1			
HMINE117	Engineering Drawing and Design		12
HMTE111	Engineering Mathematics I		12
HMETE118	Chemistry for Engineering		12
CSENG131	Communication Skills		6
HMINE119	Workshop Practice		12
HMETE103	Introduction To Computers and Computer Applications		12
HMETE112	Physics for Engineers		12
HMETE114	Fundamentals of Geology		12

Level 1 Semester 2

Code	Module Description	Pre-requisite	Credits
HMINE126	Engineering Mechanics		12
HMETE140	Materials Science		12
HMETE141	Mineral Beneficiation I		12
HMINE128	Computer Science for Engineers		12

HMINE127	Electrical Principles	12
HMTE122	Engineering Mathematics II	12
HMETE124	Mining Geology	12

Level 2 Semester 1

Code	Module Description	Pre-requisite	Credits
HMETE213	Fluid Mechanics		12
HMETE214	Nanoscience I		12
HMETE215	Thermodynamic Principles		12
HMTE211	Engineering Mathematics III		12
ENT205	Entrepreneurship		12
GS201	Introduction to Gender Studies		8

Level 2 Semester 2

Code	Module Description	Pre-requisite	Credits
HMETE225	Heat and Mass Transfer		12
HMETE226	Mineral Beneficiation II		12
HMETE227	Plant Machinery and Materials Handling		12
HMETE228	Fuels and Energy		12
HMETE229	Engineering Mechanics II		12
HMTE221	Engineering Mathematics IV		12

Level 3 Semester 1

Code	Module Description	Pre-requisite	Credits
HMETE311	Ferrous Pyrometallurgy		12
HMATE315	Materials Technology		12
HMETE316	Hydrometallurgy I		12
HMETE317	Metallurgical Laboratory Techniques		12
HMETE318	Safety, Health and Environment		12
HMETE319	Research Methods		12

Level 3 Semester 2

Code	Module Description	Pre-requisite	Credits
HMETE326	Foundry Metallurgy		12
HMETE327	Hydrometallurgy II		12
HMETE328	Mechanical Metallurgy		12
HMETE329	Non-Ferrous Pyrometallurgy		12
HMETE320	Refractory Materials		12
HMETE321	Third Year Project		20

Level 4 Semester 1

HMETE400	Work-Related Learning Placement Report	10

Level 4 Semester 2

HMETE404	Work-Related Learning Report	40
HMETE405	Academic Supervisor’s Report	40
HMETE406	Employer’s Assessment Report	20

Level 5 Semester 1

Code	Module Description	Pre-requisite	Credits
HMETE513	Metallurgical Process Design, Accounting, Control and Simulation		12
HMETE514	Technological Entrepreneurship and Product Development		12
HMETE515	Engineering and Mineral Economics		12
HMETE516	Professional Ethics, Practise and Engineering Management		12

Level 5 Semester 2

Code	Module Description	Pre-requisite	Credits
HMETE523	Corrosion Engineering (elective)		12
HMETE524	Welding and Joining of Materials (elective)		12
HMETE525	Final Year Project		30

13.0 Module Synopses

HMINE117 Engineering Drawing and Design
This module covers the use of drawing instruments, lettering, drawing standards and conventions; projection methods, engineering graphics, and measurement of areas; introduction to CAD, machine drawing, sectional views, ISO conventions and elementary assembly drawing

HMTE111 Engineering Mathematics I
This module covers calculus in one variable, differentiation and integration; functions of several variables, multiple integrals, complex numbers, sequences and series.

HMINE112 Physics for Engineering
This module covers fundamentals concepts of electrical and magnetic energy; electrostatics and electric fields; capacitance and dielectrics; magnetic circuits; electromagnetism; electromagnetic induction; magnetic field; electromagnetic induction, conservation of energy; lateral and volumetric expansion; laws of thermodynamics; ideal gas laws, equilibrium of forces; simple beams and trusses; shear force diagrams and bending moment diagrams; simple stress and strain

HMINE118 Chemistry for Engineering
This module covers basic concepts of physical, organic, analytical and inorganic chemistry; experiments involving the mole concept, stoichiometry, per cent composition and empirical formula determination; synthesis, analysis, purification and separation techniques; reactions of functional groups in hydrocarbons and industry-relevant analytical techniques

CSENG131 Communication Skills
The module covers introduction to communication; pre-writing skills; writing skills; whole essay construction; scientific and technical writing; reflective, critical and evaluative thinking; teamwork skills

HMETE103 Introduction to Computers and Computer Applications
The module deals with concepts of computer technology and practical use of computers; computing software, computer communications; computer-aided drawing; computer systems organization; number systems; legal and ethical considerations; security; use of the internet to develop research skills

HMETE114 Fundamentals of Geology

The module provides a general introduction to the field of Geology and examines earth material processes and products. Practical examination of rocks and minerals will be undertaken in the field and laboratory.

HMINE119 Workshop Practice
The module covers bench fitting; measuring instruments; forging; wood-working; metal forming; foundry practice; safety and care

HMINE126 Engineering Mechanics I
The module covers general principles; force vectors; equilibrium of a particle; force system resultants; work; energy; power; momentum; friction; equilibrium of a rigid body; structural analysis; friction; centre of gravity and centroid; moments of inertia; virtual work; hydraulics

HMETE124 Mining Geology
The module covers the theory and practice of ore deposit geology, mineral exploration, including geophysics and hydrogeology; introduces students to mining engineering and metallurgical concepts; covers practical aspects through laboratory work, mine visits and fieldwork

HMETE140 Materials Science
This module covers the classification of materials; material properties; structure-property-performance relationships in materials; atomic structure of materials; bonding, crystal structures and defects in solids; electronic materials; failure of materials; corrosion and corrosion protection

HMETE141 Mineral Beneficiation I
The module introduces concepts in mineral beneficiation. It covers basic mineral processing unit operations and related equipment. Unit operations covered are comminution: rock size reduction by crushing and grinding; size classification: screening, de-sliming, cycloning; concentration techniques and dewatering

HMINE128 Computer Science for Engineers
The module covers programming in BASIC, PASCAL, C, C++ and Visual Basic; conditional and repetitive statements; input and output; simple functions; program structure and organization; introduction to classes and objects

HMINE127 Electrical Principles
The module covers the fundamentals of electric circuits; resistive network analysis; AC network analysis; transient analysis; frequency response and system concepts

HMTE 122 Engineering Mathematics II
The module covers ordinary differential equations; application of differential equations to engineering; Fourier analysis; Laplace transforms; Z transforms

HMINE213 Fluid Mechanics
The module covers the introduction to fluid mechanics and fluid statics; general fluid motion; descriptive treatment of boundary layer concepts; conservation of mass and energy; flow through pipes and open channels; steady uniform flow in a rectangular open channel

HMETE214 Nanoscience I
The module covers a basic introduction to nano technology; technological applications such as physico-chemical considerations: band structures, typical and useful nano effects; colloids, typical syntheses of nanoparticles, fullerenes, synthesis, forms, variants, properties, applications of carbon nanotubes; synthesis, forms, variants, properties, applications oxide-nanotubes and fibres; bio-nano-link, risk and future perspectives

HMETE215 Thermodynamics Principles

The module covers thermodynamic and electrochemical principles applied to metallurgical processes; phase rule, heat of reaction, free energy, activity, thermodynamic equilibrium; thermodynamics of aqueous solutions, thermochemistry of gas mixtures, solution thermochemistry, inter-action parameters

HMTE211 Engineering Mathematics III
The module covers the introduction to numerical methods for solving problems in mathematics, science and engineering; computer arithmetic and rounding errors; numerical methods for root-finding; polynomial interpolation and splines; solution for linear algebraic equations; numerical integration and differentiation; numerical integration of ODEs; Euler and second order; Runge-Kutta methods

ENT205 Entrepreneurship
The module covers business development; marketing; human resources management pricing; government policies; taxation of business; case studies based on site visits; preparation of a business plan

GS201 Introduction to Gender Studies
The module covers gender sensitivity; gender inequalities; historical development of gender issues; gender analysis; gender issues in Zimbabwe; redressing gender imbalances; empowerment and strategies for gender-responsive environment

HMETE225 Heat and Mass Transfer
The module focuses on conduction, convection, phase change heat transfer and heat exchangers; radiation; application to metallurgical processes; mass transfer: Ficks law; Knudsen diffusion coefficient; mass transfer at phase boundaries; application to metallurgical processes

HMETE226 Mineral Beneficiation
The module covers gravity concentration by heavy media separation, jigging flowing film concentration methods, centrifugal concentrators, shaking tables, spirals; froth flotation including bulk floatation, and differential floatation; magnetic and electrostatic separation; and optical sorting. The module also looks at fluid mechanics-fluid properties practical applications

HMETE227 Plant Machinery and Materials Handling
The module focuses on material handling systems in the mining and minerals industries. It covers topics on material storage facilities such as stockpiles, bins, and hoppers; feeder design principles, operation and monitoring of conveyance; blending principles and equipment; loading and unloading facilities; pumps, principles of slurry pumping and pump selection calculations; pneumatic systems; flowability and other characteristics of materials

HMETE228 Fuels and Energy
The module covers forms of energy, renewable and non-renewable energy; fuels, fuel systems and combustion of fuels; fuels and energy in metallurgical and manufacturing processes; energy efficiency, energy management, sustainable and safe use; energy generation and storage; energy system innovations

HMINE229 Engineering Mechanics II
The module covers stress and strain; combined torsion and bending stress; continuous beams; stiffness matrix and nodal force vector; three dimensional stress; computer analysis; Lagrangian formulation method; Hamiltonian method and D'alambert's principle; rotating coordinate system; analysis and design of structures

HMTE221 Engineering Mechanics IV
The module covers correlation and regression, scatterplots, and correlation matrix; residuals and transformations; outliers; Pearson's and Spearman's correlation coefficients and use in predictions; regression in terms of sums of squares and sums of products; estimation and testing; t and f-tests; multiple linear regression; matrices in simple and multiple linear regression; testing and inference in multiple linear regression using matrices; partial correlation; analysis of variance (ANOVA)

HMETE311 Ferrous Pyrometallurgy
The module covers types, occurrence, and beneficiation of iron ores; iron and steel production: agglomeration processes, pyrometallurgical calculations, fluxes and fuels for the production of iron, steel and ferroalloys

HMATE315 Materials Technology
The module focuses on technologies used to process metals into useful products. It covers forming processes such as forging and extrusion; solidification theory, phase equilibria and microstructure; heat treatment of steels, microstructure-property-performance relations; strengthening mechanisms; introduces welding, powder metallurgy and metal matrix composites

HMETE316 Hydrometallurgy I
The module introduces the basic principles of hydrometallurgy. It covers thermodynamic and kinetic aspects of leaching, separation, extraction and recovery of metals; purification and recovery of metals by precipitation, ion exchange, solvent extraction and electro-winning; leaching techniques, pros and cons; introduction to leaching of gold

HMETE317 Metallurgical Laboratory Techniques
The module focuses on analytical techniques commonly used in mineral processing, materials characterisation and failure analysis. Topics covered include metallurgical sampling; sample handling and storage; and metallographic sample preparation. Analytical techniques covered include Optical Microscopy, X-ray Diffraction (XRD), Scanning Electron Microscopy (SEM), Atomic Absorption Spectrophotometry (AAS), Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES), UV-VIS Spectrophotometry and Infrared Spectrophotometry

HMETE318 Safety, Health and Environment
The module covers integrated environmental, quality, safety and health management systems; safety, health and environmental hazards and concerns in mineral processing and manufacturing plants; applicable local and international legislations; psychology and acceptability of risk; probability theory and its application to risk, risk assessment and management; economic implications of safety, health and environment issues; local and international case studies

HMINE319 Research Methods and Technical Writing
The module aims to equip students with scientific research and report-writing skills. It covers scientific research methods; design of experiments; writing ethics, citation, and the importance of intellectual property

HMETE326 Foundry Metallurgy
The module covers the economic importance of foundries; types of furnaces used in foundries; melting practises and melt quality; casting processes and mould-making materials; types and prevention of casting defects; quality control in foundries; casting ferrous and nonferrous alloys

HMETE327 Hydrometallurgy II
The module covers hydrometallurgy of gold, nickel and copper: reactions, process parameters, and improving process efficiencies; designing of leaching circuits; mineral valorisation and sustainable practises; bioleaching; local and international case studies

HMETE328 Mechanical Metallurgy
The module looks at the tensile response of metals; response to 2D and 3D stress states; stress-strain relations for elastic and plastic behaviour; fracture and fracture mechanics; effect of temperature on flow properties; fatigue; environmental assisted cracking; failure analysis

HMETE329 Non-Ferrous Pyrometallurgy
The module covers pyrometallurgy of platinum group metals, gold, chrome, tantalite, and the refining of cobalt, copper and nickel. Other topics include process control, process instrumentation, P & ID cascade and feed-forward control, Laplace transforms and troubleshooting

HMETE320 Refractory Materials
The module focuses on the classification of refractory and ceramic materials; application of refractory materials metallurgical processes; and phase diagrams for refractory systems: silica, fireclay, high alumina, magnesia, chromite, magnesia-chromite, doloma and magnesia-carbon systems

HMETE321 Third Year Project
The module builds on skills acquired in HMINE314. Students are expected to demonstrate understanding of scientific research and technical writing by undertaking practical research projects with no more than 3 objectives and writing structured reports. Projects must be relevant to metallurgy

The student has to undertake a minimum of 6 months of work-related learning at a metallurgical engineering organisation where they are expected to augment theoretical knowledge with practical experience. At the end of this period, the student submits a report detailing their experience

HMETE400 Work-Related Learning Placement Report
The module covers: The report of the student's understanding of an organization's purpose, functions, operations and linkages with relevant stakeholders i.e. suppliers, clients, government agencies etc.

HMETE404 Work-Related Learning Report
This module provides students an opportunity to identify and carry out project(s) during work-related learning period. Students will record and maintain a log-book of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report with appropriate design charts or process charts is required.

HMETE405 Academic Supervisor's Report
This module provides an assessment done by the academic supervisor during the period of work-related learning. The report comprises of the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student. Students will be visited at their work place at least twice by their academic supervisors.

HMETE406 Employer's Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises of the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HMETE513 Metallurgical Process Design, Accounting Control and Simulation
The module focuses on metallurgy process development, data collection, analysis, optimization and metallurgical accounting. It includes computer simulation in energy and mass balances, development of process flow diagram, plant equipment selection and plant layout, development of process control scheme, cost estimation and economic evaluation. The module also includes environmental impact assessment, development of occupational health and safety plan, consideration of sustainability and social impact issues

HMETE514 Technological Entrepreneurship and Product Development
The module covers entrepreneurship and enterprise development; developing a business plan and the maximum utilization of the financial, marketing and human resources; challenges in entrepreneurship, product development and marketing; steps in developing a product

HMINE515 Engineering and Mineral Economics
The module covers micro- and macro-economics; banks and their role in the economy; employment and inflation; project evaluation and project control; sensitivity and risk analysis techniques; project planning; cost estimation and control; commodity marketing; price stabilization and control; capital estimates in mineral projects; cash flow estimates and budgeting over project cycles; project funding and impacts; project financing; organisation and management of projects; feasibility studies

HMINE516 Professional Ethics, Practice and Engineering Management
The module covers professionalism; professional ethics and social responsibility; professional practices, partnerships and client relationships; national and international associations and societies in the industry; continuous professional development; professional practice legislation; engineering management in enterprises; project management; capital project economics; breakeven point; structure responsibility charts; modern trends- intranet; virtual teams; strategic partnering; ASP's; quality assurance and management; operational research

HMETE523 Corrosion Engineering
The module covers economic impact of corrosion; corrosion principles; forms of corrosion; corrosion testing techniques; materials used for corrosion applications corrosion control through materials selection, design, environment control, cathodic and anodic protection, coatings; high temperature corrosion; case studies

HMATE524 Welding and Joining of Materials
The module covers welding processes, weld joints and welding symbols; welding defects; welding of ferrous and nonferrous alloys; mechanical joints; brazing and soldering; adhesives

HMETE525 Final Year Project
Students are to undertake research projects that involve solving a metallurgical problem. Emphasis is on research design; the use of logical argument, learned metallurgical concepts, mathematical and science principles, and ICT in designing and conducting experiments to aid problem-solving. The economic value of the project and/or the proposed solution must

be highlighted. Projects that demonstrate innovation and that may lead to commercialisation are encouraged

DEPARTMENT OF MINING ENGINEERING

REGULATIONS FOR THE BACHELOR OF ENGINEERING HONOURS DEGREE IN MINING ENGINEERING (ENG08)

Duration	4 years
Actual Credit Load	532
Minimum Credit Load	480
Maximum Credit Load	540
Total MBK Credits	384
ZNQF Level	8

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in the field of Mining Engineering relevant to various entrepreneurial and employment possibilities in Zimbabwe and other mining economies in the region, Africa and abroad. To prepare students for higher degree studies within the minerals field, and to be able to achieve top level environmental stewardship in sustainable exploitation of mineral resources.

2.0 ENTRY REQUIREMENTS

Normal Entry: 'A' Level subjects, Mathematics, Physics and any relevant science subject. Should have passed 'O' Level Mathematics and English.

Special Entry: A National Diploma in Mining from a recognized Institution.

Other (indicate)

Visiting School: 'A' Level subjects, Mathematics, Physics and any relevant science subject. Should have passed 'O' Mathematics and English. Plus employment within the minerals industry.

3.0 PROGRAMME CHARACTERISTICS

Areas of Study: Engineering principles, minerals beneficiation, surface and underground mining practices and strategies for different deposits, minerals legislation, engineering geology, geological processes and deposits, planning and design of mines, ventilation and environmental engineering of subsurface plus surface environments, geotechnical engineering, assessment of minerals projects and feasibility, mineral resources management practices, safe and professional ethics and practice.

Specialist Focus: Development, operation and technical assessment of minerals projects.

Orientation: Strong hands-on components, including laboratory and on-site activities. A special emphasis on efficient computer-aided methodologies.

Distinctive Features: The programme focuses on production of practically orientated engineers who are able to create wealth, achieve sustainable economic development and empowerment of society through application of sound theoretical foundations.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Employment and entrepreneurships in mining and minerals projects at all levels including top management and research. Opportunities for graduates with the Bachelor of Engineering (Honours) Degree in Mining Engineering exist in quarrying, tunneling and mining among many possible areas. Mining engineering graduates find employment as tunneling engineers, explosives engineers, mine planning engineers, mining superintendents, mine managers, rock mechanic engineers, scheduling engineers, mine ventilation and refrigeration engineers, earth moving specialists, consultants, chief executive officers and investment analysts among many opportunities. Some graduates engage in their own ventures.

Further Studies: Master's and doctoral studies applied to the minerals industry.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, seminars, group work, hands on site-based learning, mine visits, work-related learning, research projects, and independent study.

Assessment Methods: Written examinations, tests, laboratory reports, field reports, seminar presentations, work-related learning report, project reports, and continuous ass essments.

Programme Competences

Generic:

- Multidisciplinarity: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on the understanding of complex phenomena
- Quantitative and innovative reasoning: Capability to draw on big data and use analytics for informed decision-making and strive to seek new ways of doing things
- Communication skills: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences
- Analysis and synthesis: Capacity for analysis and synthesis using logical arguments and proven facts.
- Ethical commitment: Professional integrity and awareness of the impact of science and technology on society and the environment
- Entrepreneurial skills: Capability to identify and create new business ventures based on knowledge and new thinking paradigms

Discipline-specific:

- Deep knowledge: Ability to analyse minerals projects and production phenomena in terms of methods

- Production skills: Ability to design and implement minerals projects practices and tools for a competitive and profitable mining industry
- Technology development skills: Ability to develop new technologies across the mining value chain with a view to enhance production efficiencies and outputs
- Problem-solving skills: Ability to solve a wide range of problems in minerals projects by identifying their fundamental aspects and using both theoretical and experimental methods
- Analytical and computational skills: Ability to use data to analyse minerals phenomena and trends using appropriate ITC packages

6.0 INTENDED LEARNING OUTCOMES

- Demonstrate knowledge and understanding of mineral deposits, identification of mineral deposits.
- Demonstrate knowledge and understanding of mining engineering principles and build projects, maintain projects, and operate mines sustainably.
- Ability to analyse minerals project and production phenomena in terms of underlying principles and knowledge and by means of appropriate research and technological methods
- Employ appropriate research and technologies to collect and interpret minerals project data leading to planning, designing of viable solutions for the industry.
- Create new business ventures in the mining industry
- Exploit minerals in an environmentally sustainable way.
- Effectively communicate and disseminate findings of investigations
- Should be able communicate effectively at all levels.
- Should be able to start projects and manage them effectively using professional project management tools.
- Use formal decision-making tools to appraise, evaluate, select and promote projects.
- Participate in due diligence teams in assessing the viability and desirability of mining ventures
- Understand and utilize the law pertaining to the acquisition and maintenance of mineral rights and the handling of mining disputes.
- Demonstrate skill in utilizing economic intelligence to decide strategies for success in mineral and mining projects.
- Use modern ICT tools to develop quantitatively determined solutions for optimal mining projects.
- Show skill in managing the human resource that mines use to achieve their objectives.
- Demonstrate necessary technical and resource sensitivity in environmental protection and sustenance in all projects undertaken.
- Always exhibit utmost ethical behaviour and provide an upright moral compass in the leadership of all projects and operations.

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for Work-Related Learning, and Final Year Project.

8.0 ASSESSMENT

Refer to section 6 of the Faculty Regulations

9.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Academic Regulations

10.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 11 of the Faculty Regulations

11.0 DEGREE WEIGHTING

For the purposes of classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

PROGRAMME STRUCTURE

All modules are core modules

Code	Module Description	Pre-Requisites	Credits
Level 1 Semester 1			
HMETE 118	Chemistry for Engineering		12
HMINE119	Workshop Practice		12
HMINE117	Engineering Drawing and Design		12
HMTE111	Engineering Mathematics I		12
CSENG131	Communication Skills		6
HMETE112	Physics for Engineers		12
Level 1 Semester 2			
HMINE126	Engineering Mechanics		12
HMINE127	Electrical Principles		12
HMINE128	Computer Science for Engineers		12
HMTE122	Engineering Mathematics II	HMTE111	12
HMETE140	Materials Science		12
Level 2 Semester 1			
HMINE214	Surface Mining Technology		14

HMINE215	Fundamentals of Geology		12
HMINE216	Computer applications in mining		12
HMTE221	Engineering Mathematics III	HMTE122	12
GS201	Introduction to Gender Studies		12
ENT131	Entrepreneurship		12

Level 2 Semester 2

HMINE228	Mine Surveying and GIS		12
HMINE225	Underground Mining Technology		14
HMINE226	Mining Geology		12
HMINE227	Thermodynamics and Fluid Mechanics		12
HMTE222	Engineering Mathematics IV	HMTE221	12

Level 3 Semester 1

HMINE316	Operations research		12
HMINE317	Rock & Soil Mechanics		15
HMINE318	Mining Machinery, Materials Handling and Services		15
HMINE319	Research Methods		12
HMINE331	Mine Environmental Management Health and Safety		12
HMINE332	Mine Ventilation Practice		15

Level 3 Semester 2: Work-Related Learning

HMINE 333	Work-Related Learning Report		25
HMINE 334	Academic Supervisor's Report		25
HMINE 335	Employer's Assessment Report		10

Level 4 Semester 1

HMINE413	Mineral Economics and Management		12
HMINE412	Mining Law		12
HMINE411	Mineral Beneficiation		12
HMINE414	Mineral Resources Evaluation		15
HMINE 415	Rock Engineering	HMINE317	12
HMINE416	Professional Ethics, Practice and Engineering Management		12

Level 4 Semester 2

HMINE421	Mine Design and Planning		15
HMINE422	Small Scale Mining		12
HMINE423	Final Year Project		27

12.0 MODULE SYNOPSES

HMETE 118 Chemistry for Engineering
The module covers basic concepts of physical, organic, analytical and inorganic chemistry; experiments involving mole concept, stoichiometry, percent composition and empirical formula determination; synthesis, analysis, purification and separation techniques; reactions of functional groups in hydrocarbons and industry-relevant analytical techniques

HMINE119 Workshop Practice
Module covers bench fitting; measuring instruments; forging; wood-working; metal forming; foundry practice; safety and care

HMINE 117 Engineering Drawing and Design
This module shall equip students with skills of using instruments and computer software in technical engineering designs. The students shall learn how to accomplish the following components of technical engineering design work using software; line work, dimensioning and layout, lettering and tangency, conics, ortho-graphic, isometric and oblique projections, sectioning of components, cycloids, involutes, spirals, helices and drawings of simple machine parts.

HMTE 111 Engineering Mathematics I
This module will cover calculus concepts as applied in engineering. Students will learn how to apply principles of calculus in functions of one/several variables, differentiation and integration, multiple Integrals, sequences and series and complex numbers.

CSENG 131 Communication Skills
Introduction to communication; pre-writing skills; writing skills; whole essay construction; scientific and technical writing; reflective, critical and evaluative thinking; teamwork skills

HMETE112 Physics for Engineers
The module covers fundamentals concepts of electrical and magnetic energy; Electrostatics and electric fields; Capacitance and Dielectrics; Magnetic circuits; Electromagnetism; Electromagnetic induction; Magnetic field; Electromagnetic induction, conservation of energy; lateral and volumetric expansion; laws of thermodynamics; ideal gas laws, equilibrium of forces; simple beams and trusses; shear force diagrams and bending moment diagrams; simple stress and strain.

HMINE 126 Engineering Mechanics
This module will introduce the mechanics of physics as applied in engineering. The module will initially cover the general principles of physics regarding the interaction of forces, their direction and magnitude. In addition, principles of work, energy and power together with their related components that include momentum and rigid body equilibrium will be introduced. Having laid the foundational principles, the module will then cover concepts in which the principles are applied and these include structural analysis, virtual work and hydraulics.

HMINE 127 Electrical Principles
This module will equip students with knowledge regarding the use and application of electrical energy. The module will introduce concepts related to circuit elements, circuit theorems, electric and magnetic field strength and potential induction, reluctance, alternating current theory and transient current.

HMINE 128 Computer Science for Engineers
This module will introduce students to concepts of computer technology and practical use of computers. The students will learn at least one programming language and associated critical components of computing science that is computing hardware and software, communication, computer systems organization, number systems, legal and ethical considerations and security. The use of the Internet to develop research skills will also be covered.

HMTE 122 Engineering Mathematics II
This module will give students an in-depth understanding of the advanced principles of calculus covered in Engineering Mathematics I. Topics covered will include Ordinary Differential Equations, Fourier analysis, Laplace Transforms and Z Transforms

HMETE140 Materials Science

The module introduces the subject of materials science. The following areas are also covered: Structure of atoms; Crystal structures; Monocrystalline structures; Imperfections in Solids; Mechanical Properties; Failure Analysis; Corrosion.

HMINE 214 Surface Mining Technology
This module provides an overview of the surface mining methods by which minerals are extracted. This module deals with the theoretical principles and practical engineering methodologies associated with surface mining. Following completion of this module, a student should have acquired knowledge and skills necessary for selection and evaluation of surface mining methods and exhibit adept knowledge of surface and subsurface mining systems. Students will able to design pits. Students will be able to describe basic operations and production methods utilized in surface mining (open pit and strip) and determine factor of safety of pit walls and slopes. Students will be proficient in designing safe, effective and economical drill and blast patterns. Students will also be able to design safe, stable, effective and economical mine haulage systems. Students will learn how to identify the potential environmental hazards and risks associated with surface mining and how to prevent and mitigate mining risks.

HMINE 216 Computer applications in mining
This module will cover the fundamentals of information systems, software development, database design and networking as well as applications of these techniques to solve some real-world problems in mining. The module will also cover the applications of cutting-edge digital technologies in the mining industry.

HMINE 215 Fundamentals of Geology
This module will cover the basic fundamentals of mineralogy and petrology with emphasis on identification of rocks in hand specimen. Students will study the geological history of Zimbabwe. There will be laboratory-based practicals introducing identification of minerals and rocks, geophysical site investigations, and exercises based on case studies.

HMTE 211 Engineering Mathematics III
This module will introduce students to numerical methods for solving problems in science and engineering. The module will specifically teach students how to undertake computer arithmetic and round errors, solve linear algebraic equations, use numerical methods to find roots and interpolate polynomials and splines. Students will also be equipped with knowledge to understand numerical integration and differentiation, integration of ODEs, the use of Euler and Runge-Kutta methods.

GS 201 Introduction to Gender Studies
Module covers gender sensitivity; gender inequalities; historical development of gender issues; gender analysis; gender issues in Zimbabwe; redressing gender imbalances; empowerment and strategies for gender responsive environment.

ENT 131 Entrepreneurship
Business development; marketing; human resources management pricing; government policies; taxation of business; case studies based on site visits; preparation of a business plan.

HMINE 228 Mine Surveying and GIS
This module covers the role played by surveyors in mining operations. It lays out surveying requirements in the mining industry, both in surface and underground operations. At the end of the module, students will have a basic understanding of the role played by a surveyor in mineral exploration. Students will acquire knowledge of the ethical and environmental concerns in mining and the surveyor's role in monitoring these. Students will demonstrate a

sound knowledge of the survey methods normally used to establish and control the development of underground mines. Students will acquire computer skills by using software to model mine workings. Geographical Information Systems software will be introduced and combined with surveying measurements captured by the student. This gives context to the connection between Surveying and GIS and opens the student to the myriad applications offered by intelligent use of geospatial data. Students will learn to be critical of the quality, precision and accuracy of data and how it is applied to a range of low and high precision tasks.

HMINE 225 Underground Mining Technology
This module will outline the principles of designing underground mining. Students will learn major underground mining methods and their related equipment, supporting infrastructure, key performance drivers, and constraints. Students will learn how to apply appropriate underground mining methods to orebodies to specific types of deposits. The student will be able to evaluate different mining methods with respect to productivity, safety, efficiency, risks, and sustainability.

HMINE 226 Mining Geology
Building on Fundamentals of Geology, this module covers the theory and practice of: ore deposit geology, mineral exploration (including geophysics) and hydrogeology. The module winds up by introducing students to the mining engineering and metallurgical concepts. Practical aspects are covered via laboratory work, mine visits and fieldwork.

HMINE 227 Thermodynamics and Fluid Mechanics
This module examines the role of thermodynamics and fluid mechanics in engineering. It develops an understanding of the basic properties of fluids, fluid statics, and simple analysis of fluid motion, the laws of thermodynamics, and the application of control volume techniques to engineering problems.

HMTE 222 Engineering Mathematics IV
The module will mainly cover statistical concepts as applied in engineering. Students will learn concepts related to correlation and regression analysis. They will learn all the components of regression which include residuals and transformations, outliers, Pearson's and Spearman's correlation coefficients and they should be able to predict and estimate trends using knowledge acquired. In addition, the module will cover statistical analysis and validation concepts that include t and F-tests and Analysis of variance (ANOVA).

HMINE 316 Operations research
This module shall cover quantitative methods such as linear programming, allocation models, transportation models, network analysis applied to planning of mines and processing plants, including critical path, cost-time schedule in construction and shaft sinking, decision theory, queuing theory and simulation applied to ore transportation and traffic problems, dynamic programming to optimise mining operations. The student will also learn the fundamental of project management.

HMINE 318 Mining Machinery, Materials Handling and Services
This module will expose students to ancillary functions for mineral production particularly in the arena of mining asset planning, acquisition, operating, renewal and disposal. In this module, students familiarise with mining equipment and services employed in the entire mining cycle. Students are expected to appreciate available technologies and as well apply digital skills in developing innovative asset and mining service management technologies.

HMINE 317 Rock & Soil Mechanics
This module equips students with an understanding of the engineering properties of rocks, geological and engineering rock classifications, rock failure theories, in-situ stresses in rock, and the fundamental concepts and principles of rock and soil mechanics. The students learn

to solve real world problems by applying rock mechanics principles in the design of foundations, slopes and underground openings in various rock formations.

HMINE 319 Research Methods
This module will prepare students for work-related learning to enable understanding and appreciation of the importance of multi-disciplinary and interdisciplinary partnerships in research and innovation; Appreciation of the different tools, software packages, methods and approaches in data analysis and be able to draw conclusion; and Capacitation of skills to implement context-specific 'hands-on' research activities under targeted conditions (e.g. archival, field, laboratory, industrial, civil and commercial conditions). Students will also learn Information Packaging and Scholarly/Academic Writing skills that include the ability to express their opinions logically and convincingly in acceptable scholarly and professional language. Students will also acquire skills to edit and proof-read their work and that of others.

HMINE 331 Mine Environmental Management Health and Safety
In this module, students will learn the impact of mining on the external environment. Students will learn how to carry out Environmental Impact Assessments (EIA), and come up with an Environmental Management Program (EMP) for mining projects. Students will also learn the legislative instruments related to the disposal of liquid, solid and gaseous wastes, surface restoration and protection. In addition, the module will also teach students how to deal with specific hazards of the internal mine environment that include rock falls, rock bursts and outbursts, explosives, mine fires and toxic gases, machinery and transportation accidents. Students should then be able to come up with risk management strategies in line with mining laws of health and safety.

HMINE 332 Mine Ventilation Practice
In this module, students will acquire knowledge on the designing of mine ventilation systems; understanding of mine air quality control, mine gases, dust and radiation: sources, detection, suppression, physiological effects and permissible limits. Students will be exposed to mine air conditioning principles including principles of thermodynamics, heat and moisture, sources, psychrometry and physiological reactions to climatic conditions. The module also covers mine ventilation: airflow network, circuit laws and natural ventilation. Finally the module is integrated with application of computer programs to ventilation systems such as Ventsim and VUMA3D.

Level 3 Semester 2: Work-Related Learning (Minimum 6 months)
HMINE 333 -Work-Related Learning Report
This module provides students an opportunity to carry out a project during the work-related learning period. Students will record and maintain a logbook of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report. A work-related learning research project supported by appropriate engineering drawings, and design charts or process charts is required. Students will be visited at their workplace at least twice by their lecturers

HMINE 334 Academic Supervisor's Report
This module provides an assessment done by the Academic supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the Academic supervisor regarding the performance and conduct of the student

HMINE 335 Employer's Assessment Report

This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises of the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student

HMINE 413 Mineral Economics and Management
In this module, students will learn the Basic Economic Concepts. They will learn how to appraise mining projects. They will acquire skills on how to reject or accept a mining investment project. Students will learn how to interpret financial reports and the fundamentals of mine management processes.

HMINE 412 Mining Law
This module equips students with the knowledge of laws and regulations that govern mining in Zimbabwe. Students are expected to appreciate and critique existing laws. Subsequent to appreciation of existing laws, students must be able to apply those laws in the acquisition and preservation of mining rights as well as the management of mining operations. The module further enables students to participate in mining policy formulation and policy advocacy. Students are also expected to develop skills in mining contract drafting and contract negotiation.

HMINE 411 Mineral Beneficiation
This module will cover the principles of crushing and grinding, screening, classification and flotation, sedimentation, thickening and filtration, gravity and magnetic separation, leaching, cementation, precipitation by gas, iron exchange techniques, solvent extraction, unit operations in ore dressing and hydrometallurgy.

HMINE 414 Mineral Resources Evaluation
This module covers a range of basic technical skills relating to the estimation of the grade and quantity of raw materials available for exploitation prior to mineral development and extraction. The module comprises four modules arranged in a continuous and linear body of essential technical knowledge, including: The sampling of mineral deposits; Statistical valuation of ore reserves; Geostatistical methods in mineral evaluation; and Geostatistical applications. Upon completion of the module the student should be able to: Analyse the compilation and implementation of a sampling protocol in order to identify points of weakness and suggest appropriate improvements; To statistically describe a set of 'real mine' data with the relevant population parameters and their levels of confidence; To identify normal, lognormal and three-parameter lognormal distributions; To compile cut-off grades and grade/tonnage curves for normal and lognormal distributions; To establish the relationships between spatially correlated variables in two dimensions by applying inverse distance and semi-variogram methods; To apply geostatistical estimation techniques to simple, two-dimensional ore bodies using the methods of Ordinary Kriging and verify the result using cross validation techniques, and to describe the variations amongst kriging techniques; To apply Mineral Resources and Reserves Reporting Codes to mineral resource and ore reserve classification; and To identify the influence and impact that cut-off grades will have on grade control.

HMINE415 Rock Engineering
The course focus on the ability to apply rock engineering design principles within a creative framework in which there is need to make necessary assumptions. The course further augments concepts introduced in Rock and Soil Mechanics and aids students in understanding different considerations (both, theoretical and application) made in the rock engineering design process.

HMINE 416 Professional Ethics, Practice and Engineering Management

Professionalism; professional ethics and social responsibility; Professional practices, partnerships and client relationships; National and International associations and societies in the industry; Continuous Professional Development; Professional practice legislation; Engineering Management in Enterprises; Project Management; Capital Project Economics; Breakeven Point; Structure Responsibility Charts; Modern Trends- Intranet; Virtual Teams; Strategic Partnering; ASP's; Quality Assurance and Management; Operational research.

Level 4 Semester 2

HMINE421 Mine Design and Planning
This module will focus on the design, planning, and scheduling of the exploitation of mineral deposits. The major goal is to explore ways in which mineral resources can be exploited optimally, that is in line with value maximisation or cost reduction objectives as enshrined in different minerals industry strategies.

HMINE422 Small Scale Mining
Definitions and scope; Global distribution; Target minerals; Traditional technologies and practices; Environmental issues; Legal aspects; Innovative mining and mineral technologies for the sector; Case studies; Small scale mining as a business; Entrepreneurial opportunities in small scale mining; Best practices in small-scale mining.

HMINE 423 Final Year Project
To fulfil the requirements of the final year project, students are required to identify a problem in a mining environment and solve it using engineering designing skills.

DEPARTMENT OF GEOSCIENCES

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED GEOLOGY (ENG06)

Duration:	4 years
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load	384
ZNQF Level	8

1.0 PURPOSE OF THE PROGRAMME
To develop knowledge, skills and competences in the field of Geology relevant to various entrepreneurial and employment possibilities in Zimbabwe and other mining economies in the region, Africa and abroad. To prepare students for higher degree studies in Geosciences, and to be environmental stewards required for sustainable exploitation of mineral resources.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry: At least 5 Ordinary Level passes including English Language and Mathematics plus two Advanced Level passes in any two of the following subjects/ National Foundation Certificates: Mathematics, Physics, Chemistry, Biology and Geography OR Relevant National Certificate.

Special Entry: National Diploma or Higher National Diploma in Geology

Mature Entry: At least 25 years of age plus 5 years relevant experience and at least 5 Ordinary subjects inclusive of English Language and Mathematics

3.0 PROGRAMME CHARACTERISTICS

Areas of Study: Mineralogy, Petrology, Structural Geology & Tectonics, Mineral Deposits, Mineral Exploration, Geoscience Mapping, Stratigraphy, Geochemistry & Geochronology

Specialist Focus: Geology for understanding the genesis and predictive discovery of mineral deposits for eventual sustainable exploitation for economic development.

Orientation: Strong hands-on components, including laboratory and field activities. A

special emphasis on efficient computer-aided methodologies.

Distinctive

Features: The 'applied' tag refers to Geology beyond mere academic curiosity, but geology for the creation of wealth, sustainable economic development and empowerment of society through the practical application of sound theoretical foundations.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Employment and entrepreneurships in mining and mineral exploration at all levels including top management.

Government; Mining companies; Engineering companies; Consulting companies; Research institutions and Universities.

Further Studies: Master's and doctoral studies in Geology or other Geoscience areas.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods:

Lectures, tutorials, laboratory classes, seminars, group work, field mapping, mine visits, work-related learning, research projects, independent study

Assessment

Methods: Written examinations, tests, laboratory reports, field reports, seminar presentations, work-related learning reports, project reports, continuous assessments

6.0 LEARNING OUTCOMES

By end of the programme a graduate will be able to:

- Identify and classify minerals, rocks and other earth materials
- Carry out geological mapping
- Interpret geological principles and earth processes at different temporal and spatial scales

- Apply geoscientific knowledge to address mining industry problems
- Effectively promote and demonstrate sustainable utilization of earth resources
- Communicate geoscientific information accurately in written, verbal and visual forms
- Design and execute mining and mineral exploration projects at local and international arenas
- Create business ventures in the minerals sector
- Exhibit professional ethics and teamwork skills in all geoscientific endeavours

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for the Work-Related Learning Level, which is worth forty (120) credits, and the dissertation which is worth 36 credits.

8.0 PROGRAMME ASSESSMENT

ASSESSMENT

Coursework: 40%: Test (15%), Practicals (20%), Assignments (5%)

Written

Examinations: 60%

Other a) Research Project is assessed on the basis of a research project(75%), Oral presentation (15%) and student conduct in the laboratory (10%).

b) Industrial Attachment is assessed based on Industrial Attachment report (30%), assessment by work supervisor (50%), and assessment by Academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 8 of the Faculty Regulations.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Academic Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 11 of the Faculty Regulations.

12.0 DEGREE WEIGHTING

For the purposes of classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits

CS101	Communication Skills	6
HGEOL 111	Introduction to Geology	12
HGEOL 112	Environmental Geoscience	12
HEGP113	Physics Foundations	8
HEGP114	Mathematics Foundations	8
HGEOL113	Principles of Chemistry	8
HGEOL116	Computer for Geoscientists	12

Level 1 Semester 2

Code	Module Description	Credits
HGEOL 121	Mineralogy and Crystallography	12
HGEOL 122	Igneous Petrology	12
HGEOL 123	Metamorphic Petrology	12
HGEOL 124	Structural Geology	12
HGEOL 125	Stratigraphy and Sedimentology	12
HGEOL 126	Earth Systems, Geomorphology and Paleontology	12
HMINE124	Computer Programming	12

Level 2, Semester 1

Code	Module Description	Credits
HGEOL 211	Geochemistry	12
HGEOL 212	Research Methods and Technical Writing	12
HGEOL 213	Mineral Deposit Geology	12
HEGP 116	Introduction to Geophysics	12
GS 201	Introduction to Gender Studies	6
ENT131	Entrepreneurship	6

Level 2, Semester 2

Code	Module Description	Credits
HGEOL 221	Mineral Exploration	12
HGEOL 222	Engineering Geology and Hydrogeology	12
HGEOL 223	Geological Field Mapping	12
HGEOL 224	Geomodelling and Mineral Resources Estimation	12
HGEOL 225	GIS and Remote Sensing for Geology	12
HGEOL 226	Isotope Geology and Geochronology	12

Level 3 Semester 1

Code	Module Description	Credits
HGEOL300	Work-Related Learning Placement Report	10

Level 3, Semester 2

Code	Module Description	Credits
HGEOL 301	Academic Supervisor's Report	40
HGEOL 302	Students' Report	45
HGEOL 303	Employers' Assessment Report	25

Level 4, Semester 1

Code	Module Description	Credits
HGEOL 411	Basin Analysis and Hydrocarbon Geology	12
HGEOL 412	Advanced Igneous and Metamorphic Petrology	12
HGEOL 413	Advanced Structural Geology and Tectonics	12
HGEOL 414	Geometallurgy and Mineral Beneficiation	12
HMINE 412	Mining Law	12

Level 4, Semester 2

Code	Module Description	Credits
HGEOL 421	Geology of Zimbabwe and Southern Africa	12
HGEOL 421	Mine Design and Planning	12
HGEOL 422	Final Year Research Project	36

14.0 MODULE SYNOPSES

CS101 Communication Skills
Introduction to communication; Pre-writing skills; Writing skills; Whole essay construction; Scientific and technical writing; Reflective, critical and evaluative thinking; Teamwork skills.

HGEOL 111 Introduction to Geology
Origin of the earth, the petrology of igneous, metamorphic and sedimentary rocks, and mineralogy; and the interaction of the environment with geological processes.

HGEOL 112 Environmental Geoscience
Ecosphere and environmental geochemistry; matter/ nutrient cycles; Metal and nutrient pollution in the environment; Geological hazards such as volcanoes, earthquakes and tsunamis; Climate change; Greenhouse effect; Air pollution; Geochemistry and health; Environmental and social effects of mining and energy production; Mine rehabilitation; Interaction between local communities and the resources industry.

HEGP113 Physics Foundations
This introductory module to physics is meant to introduce fundamental physics concepts especially meant for those who did not do A Level Physics and others to review physics principles critical for physical sciences. The topics include Dimensional Analysis, Classical Mechanics, Heat and Gases and Waves.

HEGP114 Mathematics Foundations
This introductory module to mathematics is meant to introduce fundamental mathematics concepts especially meant for those who did do A Level Mathematics and others to review mathematics principles critical for physical sciences. The topics include Functions and 2D coordinates, Complex numbers, Differential Calculus, Power Series, Integral Calculus, Linear Algebra and Ordinary Differential Equations.

HGEOL 113 Principles of Chemistry
This module is an introduction to the fundamental nature of matter. A range of chemical processes that underpin a broad array of disciplines will illustrate the importance of and differences between chemical and mathematical representations of reactions and reactivity. The physical phenomena resulting from bond types and their resulting intermolecular forces will explain the properties of natural and man-made processes and materials. Using examples from nature and biology the role of the certain groups in ascribing reactivity and function of organic molecules will be highlighted.

HGEOL116 Computer for Geoscientists
This course will introduce students to concepts of computer technology and the practical use of
computers. The students will learn at least one programming language and associated critical components of computing science that is computing hardware and software, communication, computer systems organization, number systems, legal and ethical considerations and security. The use of the Internet to develop research skills will also be covered.

HGEOL 121 Mineralogy and Crystallography
Formation of minerals, aggregates, rocks, and soils; The atomic structure and bonding of some of the minerals; Chemical formulae and compositions of common rock-forming minerals,
Diagnostic features of minerals; Simple mineralogical tests used to identify minerals; Essential mineral attributes/and physical properties that can be used to determine their identity; The crystal system to which each mineral belongs, Modes of occurrence of the minerals, Uses of minerals; Solid solution series-triangular diagrams and phases

HGEOL 122 Igneous Petrology
Processes by which igneous rocks are formed, classified and analyzed in hand specimen and thin sections, description, classification and interpretation; Primary magma generation and emplacement of magmas as intrusions and lavas; Magma differentiation; Plate tectonics and magma generation and the resulting major rock groupings, e.g.: basaltic rocks from mid-ocean ridges and hot spots, andesites and granites from subduction zones, and under saturated mafic rocks from continental rifts.

HGEOL 123 Metamorphic Petrology
Chemical thermodynamics and phase equilibria. It includes the identification of metamorphic rocks in hand specimens and microscopically, the use of mineral assemblages to identify the protolith and place the rock in a metamorphic facies framework.

HGEOL 124 Structural Geology
Geological structures; Stress and strain in rocks; Brittle and ductile deformation in rocks; Stress analysis with Mohr circles; Stereographic projections in structural geology; Kinematic analysis of shear zones; Fold mechanisms; Superposed deformation and refolded folds; Rheology; Deformation processes; Deformation mechanisms; Microstructure; Deformation histories operating in the earth's crust through the description, analysis and mapping of deformed rocks.

HGEOL 125 Stratigraphy and Sedimentology
Ancient and modern sedimentary deposits, types of sedimentary deposits, depositional processes and resulting deposits and sedimentary structures, as well as principles of facies analysis and models, basin analysis and geomorphological contributions. The module also includes the following: sediment cycle; Physical properties of sediments; Sediment classification; Sedimentary Processes; Sedimentary petrology; Stratigraphy - principles, nomenclature and procedures; Lithostratigraphy, biostratigraphy and sequence stratigraphy; Sedimentary depositional systems and facies; Sedimentary geochemistry and climate.

HGEOL 126 Earth Systems, Geomorphology and Paleontology
Physical and biological agents and processes that have shaped the Earth's surface through geological time; Evolutionary paths and mechanisms in paleontology; Distribution of fossils in space and time; Laboratory and field characterization of fossils.

HMINE124 Computer Programming
Programming in BASIC, PASCAL, C, C++ and Visual Basic; Conditional and repetitive statements; Input and output; Simple functions; Program structure and organization; Introduction to classes and objects.

HGEOL 211 Geochemistry
Distribution and geochemical classification of elements; Chemistry of the solid Earth, Thermodynamics; Equilibrium; Phase relationships; Kinetics; Major and trace element fractionation and partitioning; Stable and radiogenic isotopes and their applications; Fluid-rock interaction; Organic geochemistry and the carbon cycle; Analytical techniques and geochemical data presentation. The module also introduces isotope geology and geochronology.

HGEOL 212 Research Methods and Technical Writing
This module aims to equip students with scientific research and report writing skills to prepare them for dissertation writing, work-based report writing and writing for publications such as books, journals and magazines. The scientific research method and ethical aspects of writing and correct citation of published works are emphasized.

HGEOL 213 Mineral Deposit Geology
Definitions of mineral deposits, occurrences and resources; Mineral deposit classification; Mineral deposit genesis and models; Classification of mineral resources and mineral reserves; Mineral deposit examples from Zimbabwe and the world; Laboratory practicals on reflected light microscopy and ore textures and associations; Filed visits to selected mineral deposits

HEGP 116 Introduction to Geophysics
Basics of geophysics and various geophysical field techniques such as electromagnetics, gravity, DC, resistivity, IP, magnetics, and ground penetrating radar surveys.

GS201 Introduction to Gender Studies
This module covers Gender sensitivity; Gender inequalities; Historical development of gender issues; Gender analysis; Gender issues in Zimbabwe; Redressing gender imbalances; Empowerment and strategies for gender-responsive environment.

ENT201 Entrepreneurship

This module covers Business development; Marketing; Human resources management pricing; Government policies; Taxation of business; Case studies based on site visits; Preparation of a business plan; Technological entrepreneurship; innovations competitions.

HGEOL 221 Mineral Exploration
Mineral exploration planning and sequencing; Orientation surveys; Regional geochemical sampling; Soil sampling; Exploration drilling equipment and methods, Laboratory sample analysis instruments and techniques; Statistical interpretations of results; Methods of exploration geophysics (gravity; magnetics; magnetotellurics; radio detection and ranging; induced polarisation; radiometrics; seismics; borehole methods, etc.); Integration of geological, geochemical and geophysical exploration methods in detection and characterization of anomalies.

HGEOL 222 Engineering Geology and Hydrogeology
Engineering geology, classification and properties of rock masses, regolith and soil, geological maps, site investigations, geohazards etc. The role of groundwater in exploration and management, hydrological cycle, recharge and discharge phenomena, aquifer types, geophysical exploration for water, groundwater quality and contamination, and hydrogeology and the environment.

HGEOL 223 Geological Field Mapping
Geological mapping principles. Observation, measurement, recognition of major rock types and structures, and project management and report writing.

HGEOL 224 Geomodelling and Mineral Resources Estimation
Procedures and uncertainties inherent in 2D and 3D geological models, including computational geological modelling, mineral resources estimation, geostatistics, spatial data analysis, variogram models, the theory of krigging, in-situ resource estimation and classification, construction of data sets allowing visualization and analysis of key spatial relationships of geological elements.

HGEOL 225 GIS and Remote Sensing for Geology
This module equips students with principles and applications of geographic information systems (GIS) and remote sensing in geology and mineral exploration. GIS topics include Structure and format of GIS data; Data input and transformation, Database compilation, Spatial analysis and modelling in mineral exploration and target identification. The remote sensing part focuses on the capture and processing of satellite images and the application of data from various satellite platforms
in geology and mineral exploration. The module is computer-based, and students will experiment and use various GIS and remote sensing software packages.

HGEOL226 Isotope Geology and Geochronology
This module covers isotope geology and geochronology; Stable isotope geology and isotopic
fractionation of Li, Ca, Cr, Fe, Cu, Zn, Se and Mo in geologic; Radiogenic isotope geology and geochronology; U-Th-Pb geochronology and SHRIMP, Evolution and the isotopic composition of, Sr, Nd, Hf, Os and Pb; Geothermochronology; Ar-Ar metamorphic P-T-t paths; Isotopic cosmochemistry and evolution of the Solar System.

Level 3, Semester 1

HGEOL 300 Work-Related Learning Placement Report

The module covers The report of the student's understanding of an organisation's purpose, functions and relations, operations and relationship between the organisation and other organisations.

Level 3 Semester 2

HGEOL301 Academic Supervisor's Report
This module provides an assessment done by the academic supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the academic supervisor regarding the performance and conduct of the student.

HGEOL302 Students' Report
This module provides students an opportunity to carry out a project during the work-related learning period. Students will record and maintain a logbook of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report. A work-related learning research project supported by appropriate engineering drawings, and design charts or process charts is required. Students will be visited at their workplace at least twice by their lecturers.

HGEOL303 Employers' Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HGEOL 411 Basin Analysis and Hydrocarbon Geology
Principles and applications of geological basin analysis, Petroleum exploration, Hydrocarbon exploration

HGEOL 412 Advanced Igneous and Metamorphic Petrology
Treatment of igneous and metamorphic petrology and explores in detail igneous and metamorphic rock associations, their descriptions, occurrence and origin. It involves detailed petrographic and petrogenetic studies of these important rock types.

HGEOL 413 Advanced Structural Geology and Tectonics
Structural geology especially: micro and macrotectonics ,stress and strain in rocks ,kinematic analysis of fault and joint systems, rock deformation mechanisms and histories, superposed deformation, etc.

HGEOL 414 Geometallurgy and Mineral Beneficiation
Applied/process mineralogy in the recovery and process operations using various techniques such as XRD, XRF, SEM applications etc. Application of various advanced techniques to find solutions of process mineralogy. Mineral processing, comminution, screening, concentration, dewatering, tailings disposal, metallurgical accounting, and simulation of mineral beneficiation.

HMINE412 Mining Law
This module covers Mining legislation; its Impact on the mining industry and its stakeholders; Mine health and safety; Environmental Impacts; Exploration; Extraction; Duty of care; Safety management systems; Risk management; Legislation in developing countries; Industrial law issues. Students are expected to appreciate and critique existing laws. Subsequent to appreciation of existing laws, students must be able to apply those laws in the acquisition and preservation of mining rights as well as the management of mining

operations. The course further enables students to participate in mining policy formulation and policy advocacy. Students are also expected to develop skills in mining contract drafting and contract negotiation.

HGEOL 421 Geology of Zimbabwe and Southern Africa
The aims of the module are to make students: understand the physical conditions that existed during, and immediately after the formation of the earth; and to trace how such conditions have changed over time in Africa; to equip the students with a basic understanding of the tectonics, geological rock associations and major structures that characterize the Achaean, Proterozoic and Phanerozoic terrains in Africa; to relate the rock associations with those occurring in the rest of the world; to link the geodynamics in both space and time e.g. timing of cratonisation; to understand the economic significance of each terrain

HMINE421 Mine Design and Planning
This module covers mining method selection, mine life, principles of mine layout and design, mine scheduling of the exploitation of mineral deposits, mine optimization, capital and operating cost estimation, environment issues, feasibility studies, risk analysis, underground and surface mining methods, material handling and transport etc.

HGEOL 422 Final Year Research Project
This is the final year project module undertaken in the last two semesters. A student can do a project based on field (mapping) geology, mineral exploration, mine geology, any combination of these, or any other approved topic.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED GEOLOGY (ENG06) (VISITING)

Duration:	4 years (including 2 years of prior learning at ND/HND level)
Minimum Credit Load:	480 (including 240 credits from prior learning)
Maximum Credit Load:	540 (including 240 from prior learning)
Maximum MBK Credit Load:	192 (Degree-level MBKs only)
ZNQF – SADC Level:	8

1.0 PURPOSE OF THE PROGRAMME
To develop knowledge, skills and competencies in the field of Geology relevant to various entrepreneurial and employment possibilities in Zimbabwe and other mining economies in the region, Africa and abroad. To prepare students for higher degree studies in Geosciences, and to be environmental stewards required for sustainable exploitation of mineral resources.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and Chemistry or Combined Science at grade C or better.

Normal Entry: National Diploma or Higher National Diploma in Geology plus two years post-diploma working experience.

3.0 PROGRAMME CHARACTERISTICS

Areas of Study:	Mineralogy, Petrology, Structural Geology & Tectonics, Mineral Deposits, Mineral Exploration, Geoscience Mapping, Geoscience Modelling Stratigraphy, Geochemistry.

Specialist Focus:	Geology for understanding the genesis and predictive discovery of mineral deposits for eventual sustainable exploitation for economic development.
Orientation:	Strong hands-on components, including laboratory and field activities. A special emphasis on efficient computer-aided methodologies.
Distinctive Features:	The 'applied' tag refers to Geology beyond mere academic curiosity, but geology for the creation of wealth, sustainable economic development and empowerment of society through the practical application of sound theoretical foundations.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:	Employment and entrepreneurship in mining and mineral exploration at all levels including top management.

Government; Mining companies; Engineering companies; Consulting companies; Research institutions and Universities.

Further Studies:	Master's and doctoral studies in Geology or other Geoscience areas.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods:	Lectures, tutorials, laboratory classes, seminars, group work, field mapping, mine visits, work-related learning, research projects, independent study
Assessment Methods:	Written examinations, tests, laboratory reports, field reports, seminar presentations, work-related learning report, project reports, continuous assessments

6.0 LEARNING OUTCOMES

Upon successful completion of the programme a graduate will be able to:

- Identify and classify minerals, rocks and other earth materials
- Map geological materials, features and processes in the field using appropriate tools and techniques, including compass, hand drafting, GPS and GIS
- Interpret geological principles and earth processes at different temporal and spatial scales
- Apply geoscientific knowledge to address mining industry problems
- Effectively promote and demonstrate sustainable utilization of earth resources
- Clearly and concisely communicate geoscience concepts using written and spoken words in the form of reports, oral presentations, visual representations of data, maps, cross-sections, and technical posters

- Design and execute mining and mineral exploration projects at national and international level
- Create business ventures in the mining sector
- Exhibit professional ethics and teamwork skills in all geoscientific endevours
- Produce geologic models to explain earth phenomena by integrating geologic data, including maps, cross-sections, aerial photographs, and satellite imagery.
- Critically evaluate geoscience publications and data
- Generate new knowledge by employing appropriate modes of research inquiry to 1) identify and frame geoscience problems, 2) collect and organize data to test and refine hypotheses, 3) analyze and interpret data, and 4) develop logical conclusions with supporting rationale
- Integrate a current and comprehensive range of knowledge and methods to answer geoscience questions

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.
Each module is worth 12 credits except for the Work-Related Learning Level, which is worth forty (120) credits, and the dissertation which is worth 36 credits.
The duration of the Visiting School Programme shall be four Semesters

8.0 PROGRAMME ASSESSMENT

ASSESSMENT

Coursework: 40%: Test (15%), Practicals (20%), Assignments (5%)

Written Examinations: 60%

Other a) Research Project is assessed on the basis of a research project (75%), Oral presentation (15%) and student conduct in the laboratory (10%).

b) Industrial Attachment is assessed based on Industrial Attachment Mark obtained at their previous diploma or Higher Diploma studies (refer to Year 2 WRL, below).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 8 of the Faculty Regulations.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Academic Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 11 of the Faculty Regulations.

12.0 DEGREE WEIGHTING

For the purposes of classification, weighting of the results of the different levels shall be:

Level 1 0%
Level 2 0%
Level 3 40%
Level 4 60%

13.0 PROGRAMME STRUCTURE

RPL Exempted one year of study in Recognized Prior Learning (RPL) 120
Level 2

HGEOL220 Exempted Work-Related Learning in Recognized Prior Learningor/and current industrial experience 120

Level 3, Semester 1 (Level 3.1)

Code	Module Description	Credits
HEGP 114	Mathematics Foundations	12
HEGP 116	Introduction to Geophysics	12
HGEOL121	Mineralogy and Crystallography	12
HGEOL122	Igneous Petrology	12
HGEOL123	Metamorphic Petrology	12
HGEOL113	Principles of Chemistry	12

Level 3, Semester 2

Code	Module Description	Credits
HGEOL125	Stratigraphy and Sedimentology	12
HGEOL212	Geochemistry	12
HGEOL213	Mineral Deposit Geology	12
HGEOL321	Structural Geology and Tectonics	12
HGEOL225	GIS and Remote Sensing for Geology	12
HGEOL223	Geological Field Mapping	12
HMINE412	Mining Law	12

Level 4 Semester 1 (Level 4.1)

Code	Module Description	Credits
HGEOL221	Mineral Exploration	12
HGEOL 224	Geomodelling and Mineral Resources Estimation	12
HGEOL411	Basin Analysis and Hydrocarbon Geology	12
HGEOL412	Advanced Igneous and Metamorphic Petrology	12
HGEOL414	Geometallurgy and Mineral Beneficiation	12
HMINE421	Mine Design and Planning	12

Level 4, Semester 2 (Level 4.2)

Code	Module Description	Credits
HGEOL421	Geology of Zimbabwe and Southern Africa	12
HGEOL422	Final Year Research Project	36

14.0 MODULE SYNOPSES

Level 1 Semesters 1 and 2

Exempted one year of study in Recognized Prior Learning.
Applicants with National Diplomas (or equivalents) and/or Higher National Diplomas (or equivalents) are offered exemptions worth a total one year of study, thus recognizing their prior learning.

HGEOL220 Exempted Work-Related Learning in Recognized Prior Learning/ or and current industrial experience.

Work-Related Learning Work-related Learning (WRL) is considered covered as part of prior learning since (i) a one-year industrial placement is normally part of diploma level training, (ii) the entry requirements to this degree programme demands that applicants have a minimum of two (2) years' working experience, and are currently employed.

HEGP114 Mathematics Foundations
This introductory module is meant to introduce fundamental mathematics concepts especially for students without A Level Mathematics and to serve as a refresher module in mathematical principles critical for physical sciences. The topics include Functions and 2D coordinates, Complex numbers, Differential Calculus, Power Series, Integral Calculus, Linear Algebra and Ordinary Differential Equations.

HEGP 116 Introduction to Geophysics
Basics of geophysics and various geophysical field techniques such as electromagnetics, gravity, DC, resistivity, IP, magnetics, and ground penetrating radar surveys.

HGEOL 121 Mineralogy and Crystallography
Formation of minerals, aggregates, rocks, and soils; The atomic structure and bonding of some of the minerals; Chemical formulae and compositions of common rock forming minerals, Diagnostic features of minerals; Simple mineralogical tests used to identify minerals; Essential mineral attributes/and physical properties that can be used to determine their identity; The crystal system to which each mineral belongs, Modes of occurrence of the minerals, Uses of minerals; Solid solution series-triangular diagrams and phases

HGEOL 122 Igneous Petrology
Primary magma generation and emplacement of magmas as intrusions and lavas; Magma differentiation; Plate tectonics and magma generation and the resulting major rock groupings, e.g.: basaltic rocks from mid-ocean ridges and hot spots, andesites and granites from subduction zones, and under-saturated mafic rocks from continental rifts.

HGEOL 123 Metamorphic Petrology
Chemical thermodynamics and phase equilibria. It includes the identification of metamorphic rocks in hand specimen and microscopically, the use of mineral assemblages to identify the protolith and place the rock in a metamorphic facies framework.

HGEOL 113 Principles of Chemistry
This module is an introduction to the fundamental nature of matter. A range of chemical processes that underpin a broad array of disciplines will illustrate the importance of and differences between chemical and mathematical representations of reactions and reactivity. The physical phenomena resulting from bond types and their resulting intermolecular forces will explain the properties of natural and man-made processes and materials. Using examples from nature and biology the role of certain groups in ascribing reactivity and function of organic molecules are highlighted.

HGEOL 125 Stratigraphy and Sedimentology
Ancient and modern sedimentary deposits, types of sedimentary deposits, depositional processes and resulting deposits and sedimentary structures, as well as principles of facies analysis and models, basin analysis and geomorphological contributions. The module also includes the following: sediment cycle; Physical properties of sediments; Sediment classification; Sedimentary Processes; Sedimentary petrology; Stratigraphy - principles, nomenclature and procedures; Lithostratigraphy, biostratigraphy and sequence stratigraphy; Sedimentary depositional systems and facies; Sedimentary geochemistry and climate.

HGEOL 212 Geochemistry

Distribution and geochemical classification of elements; Chemistry of the solid Earth, Thermodynamics; Equilibrium; Phase relationships; Kinetics; Major and trace element fractionation and partitioning; Stable and radiogenic isotopes and their applications; Fluid-rock interaction; Organic geochemistry and the carbon cycle; Analytical techniques and geochemical data presentation. The module also introduces isotope geology and geochronology.

HGEOL 213 Mineral Deposit Geology
Definitions of mineral deposits, occurrences and resources; Mineral deposit classification; Mineral deposit genesis and models; Classification of mineral resources and mineral reserves; Mineral deposit examples from Zimbabwe and the world; Laboratory practicals on reflected light microscopy and ore textures and associations; Field visits to selected mineral deposits

HGEOL 321 Structural Geology and Tectonics
Geological structures; Stress and strain in rocks; Brittle and ductile deformation in rocks; Stress analysis with Mohr circles; Stereographic projections in structural geology; Kinematic analysis of shear zones; Fold mechanisms; Superposed deformation and refolded folds; Rheology; Deformation processes; Deformation mechanisms; Microstructure; Deformation histories operating in the earth's crust through the description, analysis and mapping of deformed rocks; micro and macrotectonics.

HGEOL 225 GIS and Remote Sensing for Geology
This module equips students with principles and applications of geographic information systems (GIS) and remote sensing in geology and mineral exploration. GIS topics include Structure and format of GIS data; Data input and transformation, Database compilation, Spatial analysis and modelling in mineral exploration and target identification. The remote sensing part focuses on the capture and processing of satellite images and the application of data from various satellite platforms in geology and mineral exploration. The module is computer-based, and students will experiment and use various GIS and remote sensing software packages.

HGEOL 223 Geological Field Mapping
Geological mapping principles. Students will undertake geological mapping in selected areas, with training and support to sharpen their skills in field observation, measurement, recognition of major rock types and structures, and project management and report writing.

HMINE412 Mining Law
This module covers Mining legislation; its impact on the mining industry and its stakeholders; Mine health and safety; Environmental Impacts; Exploration; Extraction; Duty of care; Safety management systems; Risk management; Legislation in developing countries; Industrial law issues. Students are expected to appreciate and critique existing laws. Subsequent to appreciation of existing laws, students must be able to apply those laws in the acquisition and preservation of mining rights as well as the management of mining operations. The course further enables students to participate in mining policy formulation and policy advocacy. Students are also expected to develop skills in mining contract drafting and contract negotiation

HGEOL 221 Mineral Exploration
Mineral exploration planning and sequencing; Orientation surveys; Regional geochemical sampling; Soil sampling; Exploration drilling equipment and methods, Laboratory sample analysis instruments and techniques; Statistical interpretations of results; Methods of exploration geophysics (gravity; magnetics; magnetotellurics; radio detection and ranging; induced polarisation; radiometrics; seismics; borehole methods, etc.); Integration of geological, geochemical and geophysical exploration methods in detection and characterization of anomalies.

HGEOL 224 Geomodelling and Mineral Resources Estimation
Procedures and uncertainties inherent in 2D and 3D geological models, including computational geological modelling, mineral resources estimation, geostatistics, spatial data analysis, variogram models, the theory of kriging, in-situ resource estimation and classification, construction of data sets allowing visualization and analysis of key spatial relationships of geological elements.

HGEOL 411 Basin Analysis and Hydrocarbon Geology
Principles and applications of geological basin analysis, Petroleum exploration, Hydrocarbon exploration

HGEOL 412 Advanced Igneous and Metamorphic Petrology
This module builds upon earlier treatment of igneous and metamorphic petrology and explores in detail igneous and metamorphic rock associations, their descriptions, occurrence and origin. It involves detailed petrographic and petrogenetic studies of these important rock types.

HGEOL 414 Geometallurgy and Mineral Beneficiation
Applied/process mineralogy in the recovery and process operations using various techniques such as XRD, XRF, SEM applications etc. Application of various advanced techniques to find solutions to process mineralogy. Mineral processing, comminution, screening, concentration, dewatering, tailings disposal, metallurgical accounting, and simulation of mineral beneficiation.

HMINE421 Mine Design and Planning
This module covers mining method selection, mine life, principles of mine layout and design, mine scheduling of the exploitation of mineral deposits, mine optimization, capital and operating cost estimation, environment issues, feasibility studies, risk analysis, underground and surface mining methods, material handling and transport etc.

HGEOL 421 Geology of Zimbabwe and Southern Africa
The aims of the module are to make students: understand the physical conditions that existed during, and immediately after the formation of the earth; to trace how such conditions have changed over time in Africa; to equip the students with basic understanding of the tectonics, geological rock associations and major structures that characterize the Achaean, Proterozoic and Phanerozoic terrains in Africa; to relate the rock associations with those occurring in the rest of the world; to link the geodynamics in both space and time e.g. timing of cratonisation; to understand the economic significance of each terrain

HGEOL 422 Final Year Research Project
This is the final year project module undertaken in the last two semesters. A student can do a project based on field (mapping) geology, mineral exploration, mine geology, any combination of these, or any other approved topic.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN EXPLORATION GEOPHYSICS (ENG07)

Duration of Degree: 4 years
Credit Load: 540 credits
Level: SADC-QF - 8

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in the field of Geophysics relevant to various entrepreneurial and employment possibilities in Zimbabwe and other mining economies in

the region, Africa and abroad. To prepare students for higher degree studies in Geosciences, and to be environmental stewards required for sustainable exploitation of mineral resources.

2.0 ENTRY REQUIREMENTS

Normal Entry: At least 5 Ordinary Levels including English Language and Physics; two Advanced level passes including Mathematics and Physics, or Mathematics and Chemistry.

Special Entry: N/A

Mature Entry: At least 25 years of age plus 5 years relevant experience and at least 5 Ordinary level subjects including English Language and Mathematics

3.0 PROGRAMME CHARACTERISTICS

Areas of Study: Potential Fields, Electrical Methods, Electromagnetic Methods, Seismic Methods, Borehole Geophysics, Reconnaissance Techniques, Structural Geology, Mineral Exploration, Geoscience Mapping

Specialist Focus: Geophysics for understanding the genesis and predictive discovery of mineral deposits, groundwater resources, environmental and geotechnical investigations for eventual sustainable exploitation of resources in a holistic manner leading to sustainable economic development.

Orientation: Strong hands-on components, including laboratory and field activities. A special emphasis on industrial related techniques and efficient computer-aided methodologies.

Distinctive Features: The 'Exploration' tag refers to Geophysics beyond mere academic curiosity, but geophysics for direct exploitation of resources, creation of wealth, sustainable economic development and empowerment of society through practical application of sound theoretical foundations.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:Employment and entrepreneurship in mining and mineral exploration at all levels including top management.

Government; Mining companies; Engineering companies; Consulting companies; Research institutions and Universities.

Further Studies: Master's and Doctoral studies in Geophysics or other Geoscience areas.

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, seminars, group work, filed mapping, mine visits, work-related learning, research projects, independent study

Assessment Methods: Written examinations, tests, laboratory reports, field reports, seminar presentations, work-related learning report, project reports, continuous assessments

6.0 LEARNING OUTCOMES

- By the end of the programme a graduate will be able to:
- Demonstrate knowledge and understanding of geophysical principles in scientific investigations, and collect and synthesize multiple geoscientific data sets to make interpretations that account for scientific uncertainty
- Work competently and innovatively as mineral exploration geophysicists, hydro geophysicists and entrepreneur consultants; meet the challenges of economic productivity and career advancement in a rapidly advancing and increasingly competitive world; and integrate diverse Earth science information for solving problems in mineral and groundwater exploration, environmental studies and geotechnical investigations
- Utilize a range of computer hardware and geoscience software for analysis and lmodelling of geoscientific phenomena; effectively communicate and disseminate findings of geoscientific investigations; and manage time and other resources efficiently and effectively in executing geoscientific tasks
- Effectively deploy appropriate technologies to collect and interpret geophysical and geological data as part of a scientific investigation; and independently acquire and utilise new geoscientific knowledge from a diversity of sources
- Use appropriate research methods to create new insights into geoscience; develop new geoscience practices and technologies that enhance efficiencies and outputs of the mining industry; and create new business ventures in the mining industry

7.0 GENERAL PROVISIONS

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for the Work-Related Learning Level, which is worth forty (120) credits, and the dissertation which is worth 36 credits.

8.0 PROGRAMME ASSESSMENT

Coursework: 40% (Tests 15%, 20% practicals, 5% assignments)

Written Examinations: End of semester examinations contribute 60% of the final mark

Other: a) Research Project is assessed on the basis of a research project (75%), Oral presentation (15%) and student conduct in the laboratory (10%).

b) Industrial Attachment is assessed based on Industrial Attachment report (30%), assessment by work supervisor (50%), and assessment by Academic supervisor (20%).

9.0 PROVISION FOR PROGRESSION

Refer to section 5 of the Faculty Regulations.

Each module is worth 12 credits except for the Work-Related Learning Level, which is worth forty (120) credits, and the dissertation which is worth 36 credits.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Academic Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 11 of the Faculty Regulations.

12.0 DEGREE WEIGHTING

For the purposes of classification, weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
CS101	Communication Skills	6
HGEOL116	Computers for Geoscientists	12
HEGP 113	Foundations of Physics	12
HEGP114	Fundamentals of Geology	12
HEGP 115	Mathematics for Geophysicists I	12
HEGP 116	Introduction to Geophysics	12

Level 1 Semester 2

Code	Module Description	Credits
HEGP121	GPS and Remote Sensing	12
HEGP122	Mathematics for Geophysicists II	12
HEGP123	Electricity & Magnetism	12
HEGP124	Thermodynamics	12
HGEOL124	Structural Geology	12
HMINE124	Computer Programming	12

Level 2 Semester 1

Code	Module Description	Credits
GS201	Introduction to Gender Studies	6
ENT131	Entrepreneurship	6
HEGP 210	Engineering Geology and Hydrogeology	12
HEGP 211	Gravity and GPS Methods	12
HEGP 212	Magnetic Methods	12
HEGP 214	Seismic Wave and Ray Theory	12

Level 2 Semester 2

Code	Module Description	Credits
HEGP 220	Field Geophysics	12

HEGP 221	Electromagnetic Methods	12
HEGP 222	Mathematics for Geophysicists III	12
HEGP 223	Induced Polarization and Resistivity Methods	12
HEGP 224	Seismic Exploration Methods	12
HGEOL 223	Geological Field Mapping	12

Level 3, Semester 1

Code	Module Description	Credits
HEGP 300	Work-Related Learning Placement Report	10

Level 3, Semester 2

Code	Module Description	Credits
HEGP 310	Academic Supervisor's Report	40
HEGP 311	Students' Report	45
HEGP 314	Employers' Assessment Report	25

Level 4 Semester 1

Code	Module Description	Credits
HEGP 410	Potential Fields & Electrical Methods Data Processing	12
HEGP 411	Research Methods and Technical Writing	12
HEGP 412	Reconnaissance Techniques	12
HEGP 413	Industrial Applications of Exploration Geophysics	12
HGEOL414	Petrophysics and Formation Evaluation	12

Level 4 Semester 2

Code	Module Description	Credits
HEGP 420	Seismic Data Processing	12
HMINE 412	Mining Law	12
HEGP 421	Dissertation	36

14.0 MODULE SYNOPSES

CS101 Communication Skills
Introduction to communication; Pre-writing skills; Writing skills; Whole essay construction; Scientific and technical writing; Reflective, critical and evaluative thinking; Teamwork skills.

HGEOL116 Computers for Geoscientists
This is an introductory module to computers for engineering students. The module focuses on basic computer concepts, data processing cycle, and number systems. Computer Arithmetic. Types of computers. Computer peripherals. System software and application software. Data communications. Introduction to IBM PCs. Windows operating system. Introduction to application packages; Word processors, Spreadsheet, Graphics. Word Processing using MS Word. Spreadsheet using MS Excel. The Internet.

HEGP113 Physics Foundations
This introductory module to physics is meant to introduce fundamental physics concepts especially meant for those who did not do A Level Physics and others to review physics principles critical for the geophysics modules. The topics include Dimensional Analysis, Classical Mechanics, Heat and Gases and Waves.

HEGP114 Fundamentals of Geology

Internal structure of the Earth, mineralogy, crystallography, petrology, geochemistry, structural geology, tectonics, and stratigraphy.

HEGP 115 Mathematics for Geophysicists 1
This module provides students with a critical understanding of Vector Algebra, Complex numbers, Introduction to Differential equations, Second order homogeneous linear differential equations with constant coefficients, First order differential equations, Inhomogeneous second order constant coefficient equations; Physics applications. Introduction to partial differential equations; and gradient vector of functions of 2 and 3 variables.

HEGP 116 Introduction to Geophysics
Basic characteristics of Earth, Earth geometry, Geochronology, Geological times, Radiometric dating, Seismology.

HEGP 121 GIS and Remote Sensing
Geographical Information Systems (GIS), Geographical Positioning System (GPS) and Remote Sensing (RS) techniques within the field of applications in geology, mining and environmental sciences, Map projections; datums and reference frames; cartographic principles; remotely sensed data, GIS software applications; surveying principles; GPS constellation and data structure; Differential GPS, ArcGIS basics.

HEGP 122 Mathematics for Geophysicists II
This module emphasizes fundamental mathematical principles required In understanding geophysics modules. The topics include Differentiation and integration; Ordinary Differential Equations; Vectors and Geometry; Matrices and Determinants; Solving Linear Equation Systems; Coordinate Transformations; Partial Derivatives; Fields and Differential Operators; Partial Differential Equations; Inverse problem

HEGP 123 Electricity & Magnetism
Field forces, Gauss' Theorem, Monopole and Dipole sources, Electrostatics, Electric field, Capacitance, Magnetostatics, Biot-Savart Law, Ampere's circuital law, Electromagnetic Induction, D.C. Circuits, and Transient Response.

HMINE124 Computer Programming
Programming in BASIC, PASCAL, C, C++ and Visual Basic; Conditional and repetitive statements; Input and output; Simple functions; Program structure and organization; Introduction to classes and objects.

GS201 Introduction to Gender Studies
The module empowers the student with knowledge and skills that enable them to be gender sensitive in the university, workplace and all their social interactions. Topics covered include understanding gender, theories of gender inequalities, the historical development of gender, gender analysis, and gender issues in Zimbabwe. The module will also look at redressing gender imbalances, empowerment and strategies for creating a gender-responsive environment. Every student has to pass the module in order to graduate.

ENT131 Entrepreneurship
This module covers Business development; Marketing; Human resources management pricing; Government policies; Taxation of business; Case studies based on site visits; Preparation of a business plan; Technological entrepreneurship; and innovations competitions.

HEGP 210 Engineering Geology and Hydrogeology
Engineering Geology: Engineering classification and properties of rocks, rock masses, regolith and soils; Engineering geology in the built environment; Geological maps; Site investigations; Geo-hazards; Predictive and mitigation techniques for geohazards.

Hydrogeology: Water as a resource; Role of geology in groundwater exploration and management; Hydrological cycle; Recharge-discharge phenomena, Aquifer types and their characteristics, Groundwater flow, Aquifer tests, Geology and geophysics of groundwater exploration; Groundwater quality and contamination, Hydrogeology in the mining environment.

HEGP 211 Gravity and GPS Methods
The theory and application of gravity surveys, the link between geophysical properties controlling these methods and subsurface geological parameters, field procedures, informed interpretation of survey datasets, processing and interpretation of potential field data. Laboratories will focus on the processing of gridded potential field data, Forward modelling.

HEGP 212 Magnetic Methods
This module provides students with a critical understanding of magnetic methods. The topics include the theory and application of magnetic surveys, the link between geophysical properties controlling these methods and subsurface geological parameters, field procedures, informed interpretation of survey datasets, and processing and interpretation of potential field data. Laboratories will focus on the processing of gridded potential field data and forward modelling.

HEGP 214 Seismic Wave and Ray Theory
Foundations of wave theory necessary for seismic processing, imaging, AVO analysis and structural interpretation. Topics include Elasticity theory, Wave Equation, Reflection geometry and wave path curvature, Surface waves, Scattering theory, Head waves, S-waves and C-waves, Anisotropy, Wave theory concepts in processing, migration and imaging, Earthquake waves, elastic waves in solids, electromagnetic and hydromagnetic radiation, Seismic Reflection and Seismic Refraction.

HEGP 220 Field Geophysics
Field-based module with an emphasis on environmental, groundwater and mineral exploration applications. It consists of data collection, processing and interpretation. Field techniques used include Ground Probing Radar, Magnetic surveying, Gravimetry, Electrical Resistivity and Electromagnetic Methods. For computer based processing, students make use of modern mapping and geophysical software.

HEGP 221 Electromagnetic Methods
This module provides students with a critical understanding of higher frequency methods (electromagnetic techniques) Electromagnetic induction methods and Ground penetrating radar (GPR). Topics include Electromagnetics (frequency-domain and time-domain), Time domain induction, Time domain loop systems; The Very Low Frequency (VLF) technique. Magnetotellurics MT, AMT, CSAMT, Ground Penetrating Radar. Laboratories will focus on the application of two electromagnetic geophysical methods: time domain reflectometry (TDR) and ground penetrating radar (GPR).

HEGP 222 Mathematics for Geophysicists III
Standard Integrals, Multiple Integrals, Introduction to line, surface and volume integrals in physics, Fourier series, Applications to solution of wave equation, Linear algebra, Linear maps and matrices, Application to simultaneous linear equations. Rank and inverse of matrix.

HEGP 223 Induced Polarization and Resistivity Methods
Resistivity, Induced Polarization, and Electromagnetics (frequency-domain and time-domain). Time domain induction, Time domain loop systems; The Very Low Frequency (VLF) technique. Magnetotellurics MT, AMT, CSAMT, Ground Penetrating Radar. Electrical surveys - Resistivity method, Induced polarisation (IP) and Self-potential (SP) method. Laboratories will focus on data acquisition, presentation and the processing of gridded geoelectric field data and forward modelling.

HEGP 224 Seismic Exploration Methods
Review of theories of seismic waves, recording, and data acquisition. Land refraction procedures. Data processing of refraction and reflection; Time signal analysis and inverse theory; Digital filtering and computer techniques in seismic data processing; Time and depth sections in model interpretation. Case histories.

HEGP300 Work-Related Learning Placement Report.
The module covers: The report of the student's understanding of an organisation's purpose, functions and their relations, operations and relationship between the organisation and other organisations

HEGP310 Work-Related Learning Report
This module provides students an opportunity to carry out a project during the work-related learning period. Students will record and maintain a logbook of daily activities done for at least eight months. They are required to submit a comprehensive industrial attachment report. A work-related learning research project supported by appropriate engineering drawings, and design charts or process charts is required. Students will be visited at their workplace at least twice by their lecturers.

HEGP311 Employer's Assessment Report
This module provides an assessment done by the industrial supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the industrial supervisor regarding the performance and conduct of the student.

HEGP314 Academic Supervisor's Report
This module provides an assessment done by the academic supervisor during the period of work-related learning. The report comprises the tasks performed and the assessment or marks allocated. This will also be an overall view of the performance of the student during the whole period of attachment. This gives an overview from the academic supervisor regarding the performance and conduct of the student.

HEGP410 Potential Fields and Electrical Methods Data Processing
Fourier Transforms (FT), Fast Fourier Transforms (FFT) and Time Series analysis. Fast Fourier transform application in window functions, Spectral analysis for global, regional and local data, Numerical solutions and digital applications of transforms and filters in the treatment of geophysics data. Geophysical data Imaging in gravity, magnetic, electromagnetic and electrical methods, modelling of different electromagnetic, gravity and magnetic datasets.

HEGP411 Research Methods and Technical Writing
This module aims to equip students with scientific research and report writing skills to prepare them for dissertation writing, work-based report writing and writing for publications such as books, journals and magazines. The scientific research method and ethical aspects of writing and correct citation of published works are emphasised. This module is linked with the dissertation module as candidates are expected to choose dissertation topics and submit a complete research proposal

HEGP412 Reconnaissance Techniques
Airborne geophysics (electromagnetic, radiometric, aeromagnetic) surveys (theory, collection, processing, interpretation), Geochemistry, Regional Gravity and GPS survey. Students are expected to complete a mini-regional survey.

HEGP413 Industrial Applications of Exploration Geophysics
This is a field-based module with an emphasis on environmental and mineral exploration applications. The module is designed to build advanced practical skills in geophysical software development, commercial project proposal, geophysical project design, implementation, data processing, interpretation and report writing. It is built around linked commercial practical projects requested by mining clients.

HEGP414 Petrophysics and Formation Evaluation
Physical properties of rocks and fluids and the geophysical response to materials with those properties with direct applications to exploitation and geophysical modelling. The topics include, Reservoir environment, Elasticity of porous media, Velocity of rocks, Velocity of poorly consolidated rock, Velocity of carbonate, Hydrocarbon fluids properties, Seismic Rock Physics: basic methods of open-hole well log analysis, and logging suite choices. Basic petrophysical logging methods.

HEGP420 Seismic Data Processing
Discrete Fourier Analysis, Statistical Analysis of Geophysical Data, Digital Filters, Deconvolution, Seismic Modelling and Inversion, Principles of Migration, introduction to seismic data processing. Topics to be covered include statics corrections, filtering, velocity analysis, deconvolution, stacking and migration. The module has a field component to record seismic data.

HMINE412 Mining Law
This module covers Mining legislation; its impact on the mining industry and its stakeholders; Mine health and safety; Environmental Impacts; Exploration; Extraction; Duty of care; Safety management systems; Risk management; Legislation in developing countries; and Industrial law issues. Students are expected to appreciate and critique existing laws. Subsequent to appreciation of existing laws, students must be able to apply those laws in the acquisition and preservation of mining rights as well as the management of mining operations. The course further enables students to participate in mining policy formulation and policy advocacy. Students are also expected to develop skills in mining contract drafting and contract negotiation.

HEGP421 Dissertation
Advanced laboratory or field research program involving application of geophysics and preparation of the dissertation. Project will generally involve a field component, both supervised and unsupervised. It involves the preparation and presentation of a research proposal, completion of an assigned research project including a focused literature review and progress report, the preparation of a manuscript describing the research results, and the presentation of an oral defence of the research outcomes.

FACULTY OF MEDICINE AND HEALTH SCIENCES

REGULATIONS FOR THE BACHELOR OF SCIENCE (BSc) GENERAL AND HONOURS DEGREES IN BIOMEDICAL SCIENCES

Duration:	3 Years (General), 4 Years (Honours)
Minimum Credit Load:	360
Maximum Credit Load:	540
Maximum MBKs Credit Load:	384
ZNQF Level:	8

Accreditation Organisations: Zimbabwe Council for Higher Education (ZIMCHE) and the Medical Dental Practitioners Council of Zimbabwe (MDPCZ)

1. PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in basic biomedical sciences in preparation for professional degrees in health sciences such as medicine, pharmacy, medical laboratory sciences, biomedical engineering and public health.

2. ENTRY REQUIREMENTS

2.1 Normal entry requirements

To be admitted into the Bachelor of Science Degree in Biomedical Sciences programme, an applicant must satisfy the general entry requirements of the University and the following specific requirements:

2.1.1 Six passes at Ordinary Level, with Mathematics, proficiency in English and local language and an approved science subject AND,
2.1.2 Passes in Advanced Level Chemistry and any other two subjects which shall be from the following: Biology, Mathematics, Physics.

2.2 Special Entry

Sciences degree graduates or those already done part or completed a relevant science program. (As guided by the Zimbabwe National Qualifications Framework).

3. PROGRAMME CHARACTERISTICS

Areas of study: Biochemistry; Anatomy; Physiology; Behavioural Sciences; Pathology; Biomedical Engineering; Data Science, Robotics and Artificial Intelligence; Medicinal

Indigenous Fruits and Herbs; District Health Services, School Health and Health Care Financing

Specialist Focus: Biochemistry, Anatomy, Physiology, Behavioural Sciences and Pathology

Orientation: Practice, research and innovation oriented. Teaching and learning are professionally oriented in health-related practices.

Distinctive Features: The programme is driven by the requirements of professional bodies such as the Medical and Dental Practitioners Council of Zimbabwe (MDPCZ). Consultations are made regularly to maintain programme relevance to the ever-changing health field.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Graduates of the Bachelor of Biomedical Sciences can pursue further undergraduate and graduate studies for careers as medical practitioners, dentists, biomedical engineers, radiographers, pharmacists, physiotherapists, occupational therapists, medical laboratory scientists, public health practitioners, optometrists, health researchers and other health disciplines.

5. PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, seminars, group work, industrial visits, work related learning, research project, and individual independent study. This also involves practical classes, laboratory and computer-based sessions, and via web-based virtual learning environment.

Assessment Methods: Written and oral examinations, tests, work related learning report, final year research project report, and continuous assessment

6. LEARNING OUTCOMES

By the end of this programme, successful students will be able to:

- Demonstrate strong and sound scientific base for all health science disciplines.
- Employ practical and analytical skills for innovations that lead to the development of drugs and appliances for the management of health problems in the country.
- Demonstrate competencies and skills to undertake research in health care.
- Demonstrate the competency to develop and produce technologies that contribute to the latest advances in medical science.
- Apply research methodologies and generate solutions in community-based projects that address local health needs.

7. GENERAL PROVISIONS

7.1 A candidate is required to undertake a minimum of five (5) modules per semester unless one is carrying over or retaking certain modules from the previous semester.

7.2 Each module is worth twelve (12) credits except for Work Related Learning Level, which is worth one hundred and twenty (120) credits and the dissertation which is worth twenty-four (24) credits.

7.3 The department may accept students discontinued from other degree programmes on condition they meet the entry requirements and subject to availability of places.

7.4 A student requires 540 credits to complete this programme and graduate.

8. PROGRAMME ASSESSMENT

8.1 Evaluation shall be based on continuous assessment and formal examinations.

8.2 Continuous assessment shall contribute between 50% of the overall mark as determined at the beginning of the course by the Departmental Board.

8.3 Assessment of Attachment in Level 3 of the degree programme shall be either satisfactory (P) or unsatisfactory (F). If unsatisfactory, the student shall repeat either the unsatisfactory Attachment.

8.4 The Dissertation assessment will include seminars, the written dissertation report of between 8 000 and 10 000 words and an oral examination. The dissertation undertaken must show evidence of its contribution to knowledge.

9. PROVISION FOR PROGRESSION

As stipulated in the General Academic Regulations.

10. FAILURE TO SATISFY EXAMINERS

As stipulated in the General Academic Regulations.

11. GRADING AND DEGREE CLASSIFICATION

The Bachelor of Science Degree programme will be classified in the following divisions, using the indicated designations and mark scheme in accordance with the GAR:

First Division	(1)	75%–100%
Upper Second Division	(2.1)	65%–74%
Lower Second Division	(2.2)	60%–64%
Third Division	(3)	50%– 59%
Fail	(F)	Below 50%

12. DEGREE WEIGHTING

10.2 The Bachelor of Science Degree programme shall be classified according to a candidate's overall performance in levels I, II, III core modules. University-wide modules shall not be used in weighing the degree.

10.3 The overall contribution of each part to the final general degree shall be as follows:

Level I	30%
Level II	30%
Level III	40%

13. PROGRAMME STRUCTURE

Duration of the Programmes: The programme for the Bachelor of Science Degree in Biomedical Sciences shall normally be three (3) years.

All core modules for the Bachelor of Science Degree in Biomedical Sciences will be compulsory in line with the Faculty of Medicine and Health Sciences Regulations for Undergraduate Degree Programmes.

Summary of Modules arranged in logical sequence and allocation of Notional Hours and Credits

Module Name **Total Notional Study Hours /Credits**

Level 1 Semester 1

Code	Module Description	Notional Hours/Credits
CSHS111	Communication Skills in Health Professions 1	10/1
BMS111	Introduction to Heritage Studies and Medical Sciences 1	10/1
BMS112	Biophysics and Biomedical Engineering	30/3
BMS113	Biochemistry 1	150/15
BMS114	Anatomy 1	170/17
BMS115	Physiology 1	170/17
BMS116	Behavioural Sciences (Sociology and Psychology for Health Sciences) 1	30/3
BMS117	Infections and diseases 1	10/1
LAIC111	Language acquisition and intercultural communication	10/1
BMS118	Medicinal Indigenous Fruits and Herbs 1	10/1

Level 1 Semester 2

CSHS121	Communication Skills in Health Professions 2	10/1
BMS121	Introduction to Heritage Studies and Medical Sciences 2	10/1
BMS123.	Biochemistry 2	150/15
BMS124	Anatomy 2	180/18
BMS125	Physiology 2	180/18

BMS126	Behavioural Sciences (Sociology and Psychology for Health Sciences) 2	50/5
BMS127	Infections and diseases 2	10/1
BMS128	Medicinal Indigenous Fruits and Herbs 2	10/1

Total credits (Part 1) **120**

Level 2 Semester 1

BMS214	Anatomy 3	200/20
BMS215	Physiology 3	200/20
BMS216	Advanced Behavioural Sciences (Sociology and Psychology for Health Sciences) and Filed Attachment 1	50/5
BMS217	Pharmacology 1	100/10
BMS218	Entrepreneurship, Professional Ethics, Communication and Philosophy	20/2
BMS219	Research Innovation and Development Skills	20/2
BMS220	Student Development and Life Skills	10/1

Level 2 Semester 2

BMS224	Anatomy 4	200/20
BMS225	Physiology 4	200/20
BMS226	Advanced Behavioural Sciences (Sociology and Psychology for Health Sciences) and Filed Attachment 2	50/5
BMS227	Pharmacology 2	100/10
BMS228	Heritage Studies	30/3
BMS229	Gender Studies	20/2

Total credits (Part 2) **120**

Level 3 Semester 1

BMS311	Histopathology and Tumour Sciences 1	100/10
BMS312	Chemical Pathology and Biochemical Diagnostic Methods 1	100/10
BMS313	Haematology and Blood Transfusion 1	100/10
BMS314	Medical Microbiology and Infectious Diseases 1	100/10
BMS315	Immunology and Diagnostic Methods 1	100/10
BMS316	Data Science, Robotics and Artificial Intelligence	100/10

Level 3 Semester 2

BMS321	Histopathology and Tumour Sciences 2	100/10
BMS322	Chemical Pathology and Biochemical Diagnostic Methods 2	100/10
BMS323	Haematology and Blood Transfusion 2	100/10
BMS324	Medical Microbiology and Infectious Diseases 2	100/10
BMS325	Immunology and Diagnostic Methods 2	100/10
BMS327	District Health Services Management and Health Care Financing	70/7
BMS328	Medicinal Indigenous Fruits and Herbs 3	30/3

Total credits (Part 3) **120**

Level 4 (BSc Honours Route)

26. Research Project **1000/100**

Options:
BMS401 Human Anatomy
BMS402 Human Physiology
BMS403 Biochemistry
BMS404 Neurophysiology
BMS405 Immunology
BMS406 Haematology
BMS407 Medical Microbiology
BMS408 Virology
BMS409 Parasitology
BMS410 Mycology
BMS411 Medical Physics
BMS412 Chemical Pathology
BMS413 Clinical Pharmacology
BMS414 Indigenous Fruits and Herbs

Elective Modules
Any 2 elective modules (100 hours/10 credits each)

BMS415 Primary Health Care and School Services 200/20
BMS416 Attachment at a Biomedical Institute
BMS417 Mental Health

Total notional hours / credits for level 4 **1200/120**

Total notional hours/credits for the BSc Honours Degree **4800/480**

MODULE SYNOPSES

CSHS 111 Communication Skills for Health Professions 1

The module provides students with skills and knowledge for effective writing, reading, speaking and listening in the health sector. The module also enables students to acquire information and literacy skills which are important in research.

BMS 111 Introduction to Heritage based Medicine and Medical Sciences 1

This module examines the practical implications of applying international, regional and national policy frameworks promoting the use of indigenous knowledge systems and heritage-based medicine. The module examines the implications of the 2003 UNESCO Convention for the Safeguarding of Intangible Heritage to heritage-based medicine, alongside the relationship of traditional medicine to the fields of human rights, public health and development. It considers the cultural significance of heritage-based medicine combined with the fundamental

principles of the Convention render the Convention significant in safeguarding heritage based medicine for the future.

BMS 112 Biophysics and Biomedical Engineering

Students are expected to have a holistic appreciation of the human being through application of properties of living matter and energy as well as the effects of the forces upon matter. The module also explores making of equipment and assistive devices for routine testing and preventative medicine. In this module students are introduced to medical equipment and how they work.

BMS 113 Biochemistry 1

This module introduces students to Medical Biochemistry which is the subspecialty of Biomedical Sciences that deals with biochemical analysis of body fluids and thereby helps in the screening, diagnosis, monitoring and management of most, if not all, human diseases. Medical Biochemistry includes the chemical basis of life, organ function test, biomarkers, Basic principles of nutrition, functions and chemistry of hormones and biocatalyst, metabolism of basic biomolecules with mineral, vitamins and xenobiotics. Medical Biochemistry helps medical and dental students understand the key biochemical processes of the human body such as metabolism and nutrition, cellular transport and signalling, enzyme activity, water, electrolyte and acid-base balance, blood coagulation, neuronal function, cellular homeostasis, growth, differentiation, cancer and gene expression. Medical Biochemistry also provides for the diagnosis of diseases and includes specialized investigations for the measurement of hormones, cancer markers, vitamins, trace elements, drugs and specific proteins.

BMS 114 Anatomy 1

On completion of the module, the learner is expected to be competent in recognising anatomical structures/organs, exhibit knowledge of their location and relations, blood, nerve, and lymphatic supply and/or drainage, and show evidence of applying this knowledge to novel scenarios. A successful learner is thus one who effectively communicates this knowledge using correct and appropriate anatomical terminology orally (in various fora with instructors and peers alike) and through written work (assignments and examinations) at all times during the module. The module contains the following

Human Gross Anatomy: The morphological anatomy of the human body and consists of lectures, practical dissections and tutorials of the following regions: upper limb, thorax, abdomen, pelvis and perineum, lower limb, head and neck. The clinical relevance of the gross anatomy is emphasized. The first semester covers the following: Upper limb anatomy, Thorax.

Histology: This consists of lectures, practical sessions and tutorials dealing with the microscopic anatomy and, where relevant, the ultrastructure of the primary tissues and organs systems. The first semester covers the following: Basic Histology; Introduction. cell, epithelial tissue, connective tissue, adipose tissue and cartilage, muscle, nervous tissue, glandular tissue, bone, blood and lymphatics, lymph nodes and thymus.

Embryology: This consists of a series of lectures in which the development of the embryo and foetus, and the development of all major systems of the human body are covered. The clinical relevance of congenital malformations and their examples in the clinic are emphasized. The first semester covers the following: Fertilization and Embryogenesis.

BMS 115 Physiology 1

The Physiology module introduces Biomedical Sciences students to the foundation upon which knowledge of what "life" is, is built upon. Physiology being the study of the function of the body forms a vital part of the curriculum for most of the undergraduate and postgraduate courses in the health and allied health professions. The focus of the physiology module is on how organisms, organ systems, organs, cells, and biomolecules carry out the chemical and physical functions that exist in a living system. Physiology is the fundamental module to medicine and biomedical sciences, focuses on the body functions in health, but also cover a few pathological states in the form of clinical correlates. The homeostatic nature and the critical physiological ranges of values are taught with respect to how the body copes with stresses imposed upon human bodies by different environments. Medical Physiology helps the biomedical, pharmaceutical, nursing and clinical students to understand how the body systems work in health and how they respond and adapts to the challenges of everyday life. The systems are studied in detail in order to enable the appropriate correctional measures to be applied in the different health professions. Physiology helps us to correctly determine what goes wrong in disease, facilitating the development of new treatments and guidelines for maintaining human and animal health. The emphasis on integrating molecular, cellular, systems and whole-body function distinguishes the physiology module. The first semester covers the following: introduction to physiology, cell physiology, homeostasis, blood, body fluids and immunity as well as excitable cell physiology.

BMS 116 Behavioural Sciences (Sociology and Psychology for Health Sciences) 1

This module is an introductory to the Sociology and Psychology of Health Sciences. Students will be exposed to the socio-behavioural-cultural perspectives to Health. Topics to be covered include Introduction to Behavioural Sciences, Genetics and Behaviour (Western perspectives on genetics -Galton, twin studies) and behaviour, African perspectives on genetics, constitution of the "seed" and behaviour) & Interviewing Skills. Students will also get an introduction to the key behavioural sciences theory including Psycho-social approach- Eric Erickson, Psychoanalytic approach &, Theories of Learning (Operant conditioning & Classical Conditioning & social learning theory. The following ethical topics will be covered (Respect for patients, Informed consent, Non-judgmental & Confidentiality) and Children, Law and Ethics in Zimbabwe. Topics to prepare students to interact with clients will include qualities of a good doctor & doctor patient relationship. Eight hours will be spent in the community for the family health study.

BMS 117 Infections and Diseases 1

This module will link concepts covered in medical microbiology and with communicable disease epidemiology.

LAIC 111 Language Acquisition and Intercultural Communication

This module enables students to learn national/ regional languages; operate as local and global citizens capable of communicating inter-culturally and appreciate cultural, historical and linguistic diversity; acquire communicative competencies in the languages chosen both in terms of writing and speaking in social, academic and professional contexts

BMS 118 Medicinal and Indigenous Fruits and Herbs 1

These modules enable students to learn and appreciate the diversity of fruits and herbs that are culturally important and have the potential for the production of nutraceuticals. The first module emphasises on the indigenous knowledge systems on indigenous fruits and herbs as

well as the nutritional aspects of these fruits and herbs. Subsequent modules enable students to appreciate the potential therapeutic properties of the fruits and herbs. Students assess the pharmacological and toxicological profiles of the indigenous fruits and herbs that are available in Zimbabwe. Students also develop potential products from the fruits and herbs.

CSHS121 Communication Skills in Professions 2

This module imparts professional communication skills that will help medical students to develop a sense of confidence and rapport to engage in their healthcare work. It will also help to build a healthy doctor - patient relationship within the students thereby resulting in better health care practices, good patient treatment, better health outcomes, fewer misunderstandings and a reduced risk of medical errors. In other words, the module equips the students with health professional skills that will enable them to work effectively with all members of medical care.

BMS 121 Introduction to Heritage Studies and Medical Sciences 2

This module explores the complexities of keeping healthy based on traditional and cultural practices. Traditional medicines are part of the cultural heritage of many people. This module also empowers students with knowledge about traditional health practices, approaches, knowledge and beliefs incorporating plant, animal and mineral based medicines, spiritual therapies, manual techniques and exercises, applied singularly or in combination to treat, diagnose and prevent illnesses or maintain well-being.

BMS 123 Biochemistry 2

This module focuses on the application of biochemistry principles in the health care industry. It unravels the principles and mechanisms behind some products and techniques used in the health such as immunology, hormone therapy, genomics, microbiology, pharmacogenomics, biotechnology and bioinformatics. This part of biochemistry is meant to give the student basic appreciation of the above subjects. This will lay an adequate baseline in the event the student chooses to specialize in any of these applied disciplines in the future. The practicals will include field visits to spaces where the topics learnt in the classroom will be applied.

BMS 124 Anatomy 2

Gross Anatomy: Abdomen and Pelvis
Histology: Systems Histology: Skin, respiratory systems, general structures of GIT, pancreas and spleen, liver and biliary system, small intestines, large intestines, appendix and entero endocrine, kidneys and adrenal glands, urinary bladder, ureters and urethra, prostate, seminal vesicles and epididymis.
Embryology: Foetal Life.

BMS 125 Physiology 2

The second module of Physiology builds up on the first module and includes the study of body systems that include the Endocrine Physiology, the Gastro-intestinal tract (GIT) Physiology, Metabolism and Liver. This module expands on the systems that are responsible for the uptake, digestion and absorption of nutrients. It also expands on the possible derangements of the systems.

BMS 126 Behavioural Sciences (Sociology and Psychology for Health Sciences) 2

In this module students will be exposed to Psychosocial factors in Health and illness as well as advanced theories in health. Topics to be covered include psycho-social factors in Health and illness in African cultures, Health Belief Model, Traditional Medical Models, The Bio-

psychosocial Model, Systems theory, Moral Development (Kohlberg); Humanistic Approach to Personality Development; Cognitive Development (Piaget), Challenges in Professionalism & Attachment. In Sociology the following will be covered Sociology of the Sick and the sick role as well as the following theories: symbolic interactionism; functionalism; Marxism; feminism & postmodernism as they relate to health. Eight hours will be spent in the community for the family health study.

BMS 127 Infectious Diseases 2

This module will link concepts covered in medical microbiology and with communicable disease epidemiology but with an emphasis on specific diseases affecting various systems such as infectious diseases of the respiratory tract, gastrointestinal tract, urinary tract, genital tract, skin, central nervous system and zoonotic diseases.

BMS 128 Medicinal and Indigenous Fruits and Herbs 2

These modules enable students to learn and appreciate the diversity of fruits and herbs that are culturally important and have the potential for the production of nutraceuticals. The first module emphasises on the indigenous knowledge systems on indigenous fruits and herbs as well as the nutritional aspects of these fruits and herbs. Subsequent modules enable students to appreciate the potential therapeutic properties of the fruits and herbs. Students assess the pharmacological and toxicological profiles of the indigenous fruits and herbs that are available in Zimbabwe. Students also develop potential products from the fruits and herbs.

BMS 214 Anatomy 3

Gross Anatomy: Lower Limb; Head and Neck Anatomy.

Histology:

Continuing with Systems Histology: Corpus luteum, penis and male genital system, uterus, placenta, salivary glands, tongue and taste buds, buccal mucosa and tonsils, thyroid and parathyroids, eye, tooth and gums, nose and ear.

Embryology: Systems Embryology: 1. Corpus luteum, penis and male genital system, uterus, placenta, salivary glands, tongue and taste buds, buccal mucosa and tonsils, thyroid and parathyroids, eye, tooth and gums, nose and ear

BMS 215 Physiology 3

The third module of Physiology includes the study of Reproductive Physiology, Renal Physiology and Respiratory Physiology. This module focuses on systems that are responsible for reproductive processes, excretion of waste products in urine and the gaseous exchange processes that maintain the normal functions of the body. It also expands on the possible derangements of the specific systems in the module.

BMS216 Advanced Behaviours Sciences (Sociology and Psychology for Health Sciences and Filed Attachment 1

The module exposes students to applied sociology and psychology topics. In addition, the module focuses on contemporary issues in health. Topics include, Introduction to Common Mental Disorders, Cognitive Behaviour Approach, Inter-professional Education and collaborative practice (IEP/CP), Group dynamics, Leadership, Health Risk Behaviours; Self-

care among health professionals; MH gap in LMICS, Dentistry; Principles of Social and Behaviour Change in health, class and health inequalities, Ethnicity, race and health, Gender and health and the determinants of health, substance use and abuse and the sociology of climate change and health.

BMS 217 Pharmacology 1

The basic pharmacology module introduces drugs to biomedical students. This module is the theoretical basis of drug application to different disease conditions, on which clinical departments then build up on practically in ward rounds and clinical rotations. In this module, students learn about drug classification by indication and mechanism of action of drugs, adverse effects, contraindications, and drug interactions.

BMS 218 Entrepreneurship, Professional Ethics Communication and Philosophy

This module introduces students to both theoretical and operational aspects of ethical standards; theories of communication and communication for development; participatory research methodologies and scenario planning or anticipation; production of certain goods and services; process of identifying business opportunities, evaluating business opportunities for their viability, selecting the most suitable business opportunity to pursue and most effective route of pursuing it

BMS 219 Research Innovation and Development Skills

This module capacitates students to develop skills in the research and innovation cycle with a particular focus on problem identification, subjecting the problem to rigorous intellectual scrutiny and diagnosis, critically appraising published literature, data collection and analysis, intellectual property processes, as well as producing patentable research outputs, in line with the Government's Education 5.0.

BMS 220 Student Development and Life Skills

This module informs students on development theories and how they inform teaching and learning at tertiary education institutions and soft skills that include effective communication, leadership, interpersonal, problem-solving, negotiation, networking and presentation skills, empathy and compassion, community consciousness, voluntarism, integrity, self-efficacy, self-confidence, self-actualisation, self-esteem, critical awareness, responsible citizenship, motivation and self-driven, healthy lifestyles.

BMS 224 Anatomy 4

Gross Anatomy: Neuroanatomy: The gross and functional anatomy of the nervous system and consists of lectures, practical dissections and tutorials of the following: the brain, spinal cord, spinal nerves, cranial nerves and the autonomic nervous system. The clinical relevance of neuroanatomy is emphasized.

Histology: Central nervous system

Embryology: Development of the brain.

BMS 225 Physiology 4

The Fourth module of Physiology includes the study of Cardiovascular Physiology, Medical Genetics and Neurophysiology. This module covers the complicated body functions that include the cardiovascular system set up and processes, the genetic processes that determine the integrity of an individual as well as the central nervous system and its functions. It also expands on the possible derangements of the specific systems in the module.

BMS 226 Advanced Behavioural Sciences (Sociology and Psychology of Health Sciences) and Field Attachment 2

This module consists of the following sections taught during two semesters:

a. Socio-cultural perspectives to Health: Students learn how culture affects different aspects of health, health seeking behaviour and health choices.

b. Health and Human behaviour 1: This introduces the commonest mental disorders such as addiction, anxiety and depression. The behaviours that contribute to vulnerability to and resilience and recovery from these disorders are interrogated.

c. Health social Science: During this course, students learn about key psychological concepts such as human development, motivation, education, emotion and how they contribute to wellness or disease.

d. Information gathering skills: In level 2, learners will acquire the skills to conduct medical enquiry into clients' beliefs and history in a culturally consonant manner to enable effective communication.

e. Community focused education: In level 2, learners will learn about and interact the primary health care facilities, informal and formal health services and their importance as the most widely utilized services in Zimbabwe. Learners will also be expected to learn how to develop public service information packets educating lay people about disease in the community.

f. Health and Human behaviour 2: In level 2, students are introduced to the socio-cultural and behavioural issues affecting the health of special populations specifically women, children and geriatric people. This module also covers health models in detail.

BMS 227 Pharmacology 2

The second semester will explore the interaction of various pharmacological agents on the various systems of the human body, such as the cardiac system, the respiratory system, the gastrointestinal system, the nervous system, the endocrine system and the skin. At the end of the course students will be able to apply and analyse basic knowledge of pharmacology in treatment interventions.

BMS 228 Heritage Studies

This module equips students with knowledge and understanding of the relationship that exists between people and tangible as well as intangible heritage. Heritage studies are an interdisciplinary field that focuses on the exploration, analysis, and preservation of tangible and intangible cultural heritage. The module empowers students with knowledge about the socio-cultural systems that heritage has towards society. The module also interrogates indigenous health systems in Zimbabwe and their preservation and promotes the building of Zimbabwe's national health systems from the foundation of national heritage.

BMS 229 Gender Studies

The module will explore the following topics:

Understanding Gender: Gender as a development concept, gender related concepts, social construction of gender,
Theories of gender: Liberal Feminist Theory, Marxist Feminist Theory, Socialist Feminist Theory, Radical Feminist Theory, Third world Political Economy, Eco – feminism.
Historical Development of Gender: Women in development Approach (WID), Women and Development Approach (WAD), Gender and Development Approach (GAD) (Moser), G.E.D
Gender Analysis: What is gender analysis? Gender analysis matrix, The rationale for Gender analysis, Gender Analysis Framework, The Harvard Analytical Model, The Women Empowerment Framework.
Gender issues and sustainable development: What is a gender issue? Gender and sustainable development related issues: HIV/AIDS; Poverty; Stereotypes; Relationships; Lack of empowerment; Cultural attitudes, beliefs and practices; Religious beliefs and practices; Sexual harassment; Patriarchy; Domestic violence; Human rights; OVC; Child abuse; Climate change.
Strategies for redressing Gender issues: Legislation, Policies, Empowerment, Use of leadership roles, Changing people's attitudes, Gender responsive environment, Community gender related aid projects, Gender responsive budgeting, Gender based research.

BMS 311 Histopathology and Tumour Science 1

This module bridges the gap between the theoretical knowledge acquired in the basic sciences and the application of it in clinical medicine thereby acting as the foundation of clinical medicine. At the end of the module students will have a clear understanding of the development of disease processes from their inception to their termination including: aetiology; pathogenesis; micro and macroscopic appearances of the lesion produced; the clinical manifestations of those lesions; an appreciation of the causes of death, both natural and unnatural

BMS 312 Chemical Pathology and Biochemical Diagnostic Methods 1

Chemical Pathology is the study of biochemical and molecular markers in the diagnosis and treatment of disease. It provides a bridge between concepts taught in the basic sciences and their application in clinical practice. In this biomedical science module, students receive instruction on the following aspects of biochemical disorders: aetiology; pathogenesis; biochemical and molecular markers used in the diagnosis and monitoring of disease; risk factors and complications of biochemical disorders.
The first semester introduces chemical pathology including specimen management and training on quality assurance activities to ensure the accuracy and reliability of results generated in chemical pathology. In the first semester, students are also trained on the general interpretation of chemical pathology laboratory data/ results to enable appropriate management of patients.

BMS 313 Haematology and Blood Transfusion 1

The first semester introduces students to basic haematology, how blood is formed and aspects of blood transfusion.

BMS 314 Medical Microbiology and Infectious Diseases 1

The module introduces basic medical microbiology to students, describing the classification, structure and physiology of microorganisms that cause infectious diseases in humans. Emphasis is first placed on the importance of pathogenic microbes in human disease with respect to infections and their pathogenesis resulting in disease presentation. It covers both

theoretical and practical aspects of virology, bacteriology, parasitology, mycology, molecular microbiology, antimicrobial resistance and infection prevention and control. In this module and the module on immunology, students learn aspects of infectious diseases.
(Semester 1 we covered the basics of microbiology i e. classification, morphology, general principles of diagnosis)

BMS 315 Immunology and Diagnostic Methods 1

This module introduces students to clinical immunological concepts. Students are introduced to and are expected to understand current advances in immunology of infectious diseases techniques and the application of immunological techniques in disease diagnosis and treatment. The diseases of main focus are HIV, TB, malaria, schistosomiasis, endocrine diseases, eye diseases, skin diseases, cardiovascular diseases, tumour immunology, gastrointestinal diseases, immunity to viruses, bacteria, fungi. Students develop a deeper understanding of basic immunology concepts focusing on humoral and cellular immunity.
The aim of the module is to provide advanced theoretical knowledge and practical training in the immunology of infectious diseases through a comprehensive range of teaching methods. It equips students with the range of specialised knowledge and skills in applying scientific concepts, evaluating scientific data and carrying out modern top of the range immunological techniques. Students to demonstrate specialist knowledge and understanding of the basic principles of host immunity to infection against the diverse range of pathogens which confront human populations. Students to apply this specialist knowledge to a range of practical skills and techniques, in particular modern molecular and cellular techniques for diagnosis and assessment of immune responses to pathogens. Students to critically assess, select and apply appropriate research methods to investigate basic immunological mechanisms and applied issues in the immunology of infection. Students to be able to critically evaluate primary scientific data and the published relevant scientific literature.
The first semester module focusses on basic immunology concepts of innate, humoral and cellular immunity Students are introduced to the role of the different immunological responses to infection against the diverse range of pathogens which confront human populations. Immunity to infection is categorised into 4 main areas as immunity to bacteria, immunity to viruses, immunity to fungi and immunity to parasites. Diseases of epidemiological importance are used as key examples and these include HIV, TB, malaria, schistosomiasis, Covid-19, Influenza, Cryptococcus, among others. Students are also introduced to immunological techniques that are applied in clinical investigations.)

BMS 316 Data Sciences, Robotics and Artificial Intelligence

The course introduces data science as an intercept of programming, statistics, and machine learning. Students will be introduced to data manipulation tools such as using spreadsheets and R/Python tools. Data Extraction, Transformation and Loading techniques will be explored. Statistics foundations using R or Python will be introduced. Robotics and Artificial Intelligence Modules introduces use of computers in modelling and or replicating intelligent behaviour useful in medical diagnosis and treatment.

BMS 321 Histopathology and Tumour Sciences 2

In the second semester there will be emphasis on abnormal presentations and tumours found in various organs and systems, such oral pathology, the respiratory tract, the gastro-intestinal tract, the uro-genital tract, the endocrine system, the central nervous system, the musculo-skeletal system and the skin.

BMS 322 Chemical Pathology and Biochemical Diagnostic Methods 2

The second semester focuses on the use of biochemical tests to diagnose system pathologies such as the renal, gastrointestinal, cardiovascular and endocrine systems. Learners are also taught on the utilisation of cutting-edge technology used in diagnosis such as molecular techniques.

BMS 323 Haematology and Blood Transfusion 2

Students learn therapeutics of patients with haematological diseases which include haematology, oncology, bleeding disorders, thrombophilia and non-haematological disease. They also recognise laboratory abnormalities and to some limited extent perform some of the bedside tests.

BMS 324 Medical Microbiology and Infectious Diseases 2

The second semester examines various communicable diseases, epidemiology, pathogenesis, laboratory diagnosis, infection prevention and control, antimicrobial treatment and resistance.

BMS 325 Immunology and Diagnostic Methods 2

The second semester module leverages on immunological concepts learnt in the first semester and introduces students to clinical immunology, that is the use of immunological techniques for diagnosis and assessment of immune responses in the management of diseases of the immune system. Diseases are discussed by organ involvement, and illustrative case histories are used to show the usefulness (or otherwise) of immunological investigations in the management of these patients. Organ systems include gastrointestinal, genitourinary, respiratory, skin, ocular, reproductive, endocrine, cardiovascular and neurological systems. Clinical immunology relating to non-specific organ involvement is taught as transplantation and tumour immunology. Students also learn disorders of the immune system under autoimmune, immune-deficiencies, hypersensitivities.

BMS 327 District Health Services Management, School Health Services and Health Care Financing

Students gain a broad understanding of how health systems are managed at district level using the primary health care approach including procurement systems, school health programmes and basic health care financing

BMS 328 Medicinal Indigenous Fruits and Herbs 3

Innovative Project that will produce a medical product

The students conduct a project which they can start early during their Biomedical training so that they develop medical products that can be exploited commercially and to improve health care of Zimbabweans

Field Attachment and Community Health Care

Students are given an option to choose any one of the following focus areas for the attachment. These are suggestions and recommendations to guide students' choice.

a. Rural Attachment. (Recommended for those who will pursue the MB ChB and Dentistry Degree programmes on completion).

b. Industrial Attachment to Pharmaceutical Industry. (Recommended for those who will pursue Pharmacy or Clinical Pharmacology Degree programmes on completion).
c. Industrial Attachment to a Medical Technology Innovation or Production Centre. (Recommended for those who will pursue Biomedical Engineering Degree programmes).
d. Private Laboratory. (Recommended for those who will pursue Laboratory Sciences or Radiography or Optometry Degree programmes).
e. Epidemiology Departments in various health related institutions. (Recommended for those who will pursue Public Health or Medical Bio-Analytics or Bio-Informatics Degree programmes).

Project / Optional Module
This project / optional module defines the field in which the Bachelor of Science Biomedical Sciences Honours Degree is being awarded. (Human Anatomy, Human Physiology, Clinical Pharmacology, Biochemistry, Medical Laboratory Sciences, Neuropsychology and others.)

REGULATIONS FOR THE BACHELOR OF MEDICINE AND BACHELOR OF SURGERY (MBChB) DEGREES

PREAMBLE
With the approval of the University of Zimbabwe (UZ) and the Medical and Dental Professions Council of Zimbabwe, the first two cohorts of MBChB students from the Midlands State University (MSU) will graduate with degrees from the University of Zimbabwe. The following MBChB Regulations are therefore those of the UZ which will be followed by the MSU students.

1. INTRODUCTION

These Regulations apply only to students who are in full-time attendance at the Midlands State University and should be read in conjunction with the University's General Regulations for Undergraduate Degrees, hereinafter referred to as the General Regulations.
The MBChB Degrees will be awarded to candidates who have successfully completed approved courses and examinations in accordance with the Regulations set out below. The Degrees are awarded in two categories - the MBChB Degrees with Honours and the MBChB Degrees.

2. QUALIFICATIONS FOR REGISTRATION

2.1 Entry to the University:
All prospective entrants must satisfy the entry requirements of the University.
2.2 Course Requirements:

2.2.1 To qualify for consideration for admission, a prospective student must normally have:
Passes at Ordinary Level (or approved equivalent) in English Language and Mathematics;
Passes at Advanced Level (or approved equivalent) in Chemistry and any two of the following three subjects:
Biology (or Zoology)
Mathematics
Physics
The subject not offered at A Level must normally have been passed at 'O' Level.

3. DATES OF EXAMINATIONS

3.1 University Examinations:
The professional Examinations for the Degrees of MBChB will normally take place at the following times:
First year, MBChB Part I - in or about the 14th week of the second semester.
(Biochemistry)
Second year, MBChB Part II - in or about the 16th week of the Part II Second Year
(Anatomy, Behavioural Sciences
and Community Health and Physiology.)
Third year, MBChB Part III - in March of the third semester, Third Year
(Chemical Pathology, Haematology,
Histopathology, Medical
Microbiology, Forensic Pathology, Immunology)
Fourth year, MBChB Part IV - in May
(Clinical Pharmacology,
Community Medicine and
Psychiatry) Clinical Pathology
Examinations to be incorporated
into the major clinical subjects
being examined.
Fifth year, MBChB Part V - in July
(Obstetrics & Gynaecology
Medicine, Paediatrics and Surgery)
Clinical Pathology
Examinations to be incorporated into major clinical subjects being examined.

3.2 University Supplementary Examinations
Normally Supplementary Examinations are taken in accordance with General Regulations except where specified.

4. ADMISSION TO EXAMINATION

To be admitted to the examination for the MBChB Degrees, a candidate must:

4.1 have registered as a student of the Midlands State University in accordance with the General Regulations for students;

4.2 have completed approved Courses of study at the Midlands State University to the satisfaction of the appropriate Chairmen of Departments.

5. PROGRAMME

5.1 Field Attachments
The Undergraduate medical students are required to undertake field attachment activities in rural areas of four weeks duration in each of the second, third and final years.

5.1.1 Behavioural Sciences and Community Health examination incorporating experiences gained in the field attachments and knowledge forthcoming from Clinical input shall be taken as part of Part II of the course, coinciding with the examinations in Anatomy and Physiology.

5.2 Pre-Clinical Division

5.2.1 The Pre-clinical Courses begin at the same time as all other University courses in the first year and much earlier in the Second Year.

5.2.2 Department of Anatomy
Gross Anatomy
Microscopic anatomy
Embryology

5.2.3 Department of Physiology
The course includes the study of
Cell Physiology
The Cardiovascular System
The Respiratory System
The Blood
The Alimentary System
Metabolism
The Kidney
The Hormones
Reproduction
The Nervous System
Genetics
Environmental Physiology

5.2.4 Department of Biochemistry
The Course covers all aspects of a basic introduction to Biochemistry and Molecular Biology.

5.2.5 Department of Linguistics
Communications Skills for Academic Purposes (CSHS101)
The course covers the Process of human communication, Models of learning, Reading, Academic writing, and Research skills.
Communication Skills for Professional Purposes (CSHS102)
The course covers Models of communication, Consultation techniques and establishing collaborative relationships, the Health professional as educator and counsellor, conflict management, oral and written messages, Non-verbal communication, Research and Report Writing.

5.2 Clinical Division

5.3.1 The clinical training commences in the first year of the programme and continues throughout the second year with introductory courses in Clinical Medicine, Community Medicine, Clinical Pharmacology, Chemical Pathology, Haematology, Histopathology, Medical Microbiology, Psychiatry and Behavioural Sciences being coordinated with the pre-clinical disciplines.

5.3.2 Behavioural Sciences and Community Health examination incorporating subject matter provided as part of the Clinical Input as well as that exercise gained in rural attachments shall be given at the end of Part II (See 5.1).

5.3.3 The formal Pathology Course (Chemical Pathology, Haematology, Histopathology, Medical Microbiology and Immunology) begins in the third term of the second year. Clinical Pathology shall continue to be taught in the fourth and fifth years and examined together with the major clinical subjects.

5.3.4 The formal Clinical Course begins in the 3rd year. The emphasis throughout is on the importance of the patient and not merely his presenting disease and the major portion of this training is on the wards, supplemented by formal lectures,

tutorials, discussions and topic teaching. A multi-disciplinary approach is encouraged by the close integration of the systematic pathology lectures with those of the clinical subjects. This attitude is further enhanced by topic teaching, where discussion may include any department within the Faculty.

5.3.5 During the 3rd and 4th years, ward clerking and teaching are based on the tutorial system, emphasising the elicitation of signs and symptoms and differential diagnosis. In the final year, the students act as assistant House Staff and present the cases they have admitted both on the ward rounds and in demonstrations to the whole class. In addition, there is a 5-week elective period in the final year, when students undertake a study of their own choice.

6. SCHEME OF EXAMINATION

6.1 General Regulations:

6.1.1 Examinations for degrees of MBChB will be run in accordance with the provisions in the General Regulations.

6.1.2 Faculty Board of Examiners

The Chairman of the Board of Examiners shall comprise the Dean and Deputy Dean of the Faculty, the Chairman of each Department and the External Examiner for the Department, plus normally, one other member of the Department nominated by the Departmental panel from each Department involved in the subjects for examination.

6.2 First and Second Years MBChB Part I

Part I: Biochemistry	BC1M
Communication Skills for Academic Purposes	CSHS101
Part II: Anatomy	AT2M
Physiology	PY2M
Behavioural Sciences and Community Health	BS2M
Communication for Health Sciences and Life Skills	CSHS103

6.2.1 Examiners

There shall be at least one External Examiner for each of the three subjects. Departmental Panels of Examiners shall be constituted in conformity with the General Regulations of the University.

6.2.2 Examinations

The examination, which shall be held in the University, shall be divided into two parts, Part I (Biochemistry (BC1M) and Communication Skills for Academic Purposes (CSHS101). Part II Anatomy (AT2M), Physiology (PY2M), Behavioural Sciences and Community Health (BS2M) and Communication Skills for Professional Purposes (CSHS102) shall be examined in the middle of the second year.

6.2.2.1 Anatomy (AT2M)

6.2.2.1.1 Anatomy Examinations (Special requirements)

The examinations shall comprise:

6.2.2.1.2 Written Examinations

Two two-hour essay papers (AT2M/01 and AT2M/02) and one two-hour multiple choice question paper (AT2M/03)

6.2.2.1.3 Oral Examinations (AT2M/90)

6.2.2.1.4 Practical Examinations (AT2M/50)

6.2.2.1.5 Continuous Assessment

Marking	
Paper I and Paper II (Essays)	300

Paper III (Multiple Choice Questions)	200
Orals and Practicals	100
Continuous Assessment	300
TOTAL	**900**
Pass Mark is 50% i.e.	450

6.2.2.2 Biochemistry

6.2.2.2.1 Biochemistry Examinations (Special Requirements)

The examinations shall comprise:

6.2.2.2.2 Written Examinations

One three-hour paper consisting of essays and short answer questions (BC1M/01).

6.2.2.2.3 Oral Examinations (BC1M/90)

6.2.2.2.4 Continuous Assessment of the year's work covering: practical laboratory work throughout the year, class tests in the first and second terms, and tutorials. BC1M/50 and BC2M/50) BC1M/80 and BC2M/80)

6.2.2.2.5 Marking

Paper I	500
Paper II	100
Continuous Assessment	300
TOTAL	**900**
Pass Mark is 50% i.e.	450 marks.

6.2.2.3 Physiology (PY2M)

6.2.2.3.1 Physiology Examinations (Special Requirements)

The examinations, which shall be of the same format for both Parts I and II, shall comprise:

6.2.2.3.2 Written Examinations

Two three-hour essay papers (PY2M/01and PY2M/02) and one -hour multiple choice question paper (PY2M/03)

6.2.2.3.3 Oral Examinations (PY2M/90)

6.2.3.3.4 Continuous Assessment (including Practicals) (PY2M/50) (PY2M/80)

6.2.2.3.5 Marking

Paper I and Paper II (Essays)	300
Paper III (Multiple Choice Questions)	200
Oral Examinations	100
Continuous Assessment (including Practicals)	300
TOTAL	**900**
Pass Mark is 50% i.e.	450

6.2.2.4 Behavioural Sciences and Community Health Examination (BS2M.) taken in Part Two only.

6.2.2.4.1 Written Examinations (BS2M/01)

Paper I

6.2.2.4.1.1 One three-hour essay paper consisting of Section A and Section B.

Section A will contain three compulsory questions (60% of the marks). One question will be on Rural Attachment (20%), the second will be on Behavioural Sciences (20%), and the third will be on Biostatistics (20%).

Section B will consist of three questions:

1. Community Health

2. Behavioural Sciences
3. Rural Attachment

Candidates will be required to answer any two questions (maximum 40%).

6.2.2.4.1.2 Paper II

One two-hour Multiple Choice Question paper consisting of 60 questions. The questions will be selected from the following topics:

Rural Attachment	15
Behavioural Sciences (Psychology, Anthropology and Sociology)	12
Biostatistics	10
Environmental Health/ Occupational Health	6
Family Health	5
Miscellaneous Community Health Sciences: (Community Organisation, Political Economy, Basic Epidemiology, Health Promotion)	7
TOTAL	**60**

6.2.2.4.1.3 Continuous Assessment (BS2M/80) will carry 30% of the marks and will be based on written assignments and projects in the Family Health case study, Rural Attachment, and Communication Skills presentation.

6.2.2.4.2 Marking

Paper I	400
Paper II	300
Continuous Assessment	300
TOTAL	**1000**
Pass Mark is 50% i.e.	500 marks

6.2.2.5 Part I: Communication Skills for Academic Purposes (CSHS101). The examination shall comprise:

6.2.2.5.1 Written Examination

One two-hour paper of short essay/answer questions; and

6.2.2.5.2 Continuous Assessment consisting of tests, assignments and presentations

6.2.2.5.3 Marking

Written paper	70%
Course Assessment	30%
Total	**100%**

6.2.2.6 Part II: Communication for Health Sciences and Life Skills (CSHS103). The examination shall comprise:

6.2.2.6.1 Written Examinations

One three-hour paper of short essay/answer questions; and

6.2.2.6.2 Continuous Assessment consisting of tests, assignments and presentations.

6.2.2.6.3 Marking

Written paper	70%
Course Assessment	30%
Total	**100%**

6.2.3 Examination Procedures

6.2.3.1 Setting of Question Papers

The setting of question papers shall be the responsibility of the Departmental Panel of Examiners.

6.2.3.2 Conduct:

The examinations shall be conducted in conformity with the Regulations prescribed by the University.

6.2.3.3 Failure

Candidates must normally satisfy the examiners in all subjects of the examination, namely;

(i) Anatomy
(ii) Biochemistry
(iii) Physiology
(iv) Behavioural Sciences and Community Health Examination
(v) Communication Skills for Academic Purposes

Candidates obtaining between 40% and 49% shall normally be required to write supplementary examinations in the subject in which they have failed. In the case of students who obtain less than 40%, in any of the subjects, the General Regulations of the University shall apply. Candidates who fail the supplementary examinations shall normally be required to discontinue the programme of studies in the College of Health Sciences.

A candidate may not proceed to the second year of the programme until he has passed all four subjects at Parts I and II

6.2.3.4 Supplementary Examinations:

Marking

The marking scheme for Biochemistry (Part I) Anatomy and Physiology (Part II) shall be as follows:

Theory Papers	500
Oral	100
TOTAL	**600**
Pass Mark 50% i.e.	300 marks.

The marking scheme for the Behavioural Sciences and Community Health, and Communication Skills examination shall be as for the June Examination.

6.2.3.5 Results

Results shall be determined in accordance with the provisions of the General Regulations.

6.3 Third Year MBChB Part III

(Chemical Pathology, Forensic Pathology, Haematology, Histopathology, Immunology and Medical Microbiology).

6.3.1 Pathology (Code PG3M)

6.3.1.1 Examiners:

There shall be five External Examiners, one each for Chemical Pathology, haematology, Histopathology, Forensic Pathology, Immunology and Medical Microbiology.

6.3.1.2 Examination

The examination, which shall be held in the University, shall comprise:

6.3.1.2.1 Written Examinations: 3 papers

Paper I Histopathology and Forensic Pathology (PG3M/01)
Paper II Haematology and Medical Microbiology (PG3M/02)
Paper III Chemical Pathology and Immunology (PG3M/03)
Each paper shall be arranged in two sections:

(a) A multiple-choice question paper of one hour twenty minutes containing forty questions;

(b) An essay paper (two hours) containing two sections with two questions to be answered in each section.

6.3.1.2.2 Practical examination (PG3M80) which shall consist of:

1) Histopathology/Haematology	1 hour
2) Medical Microbiology	1 hour
3) Chemical Pathology/Immunology	1 hour

6.3.1.2.3 Classwork (PG3M/50) including the performance in class examinations and during the period of Pathology attachment.

6.3.1.2.4 The examiners may require any candidate to attend an oral examination at which not less than two examiners shall attend.

6.3.1.3 Marking:

Paper I	75
Paper II	75
Paper III	75
Practical	75
Classwork	100
TOTAL	**400**

Pass Mark 50%, i.e. 200 marks

6.3.1.3.1 The Pathology Examination is a composite examination. The overall mark determines whether a candidate passes or fails.

6.3.1.4 Setting of Papers:

The setting of papers shall be the responsibility of the Departmental Panel of Examiners.

6.3.1.5 Conduct

The examinations shall be conducted in conformity with the Regulations prescribed by the University for this purpose.

6.3.1.6 Failure

Any candidate who fails the examination with an overall mark of between 40% and 49% may be permitted to write a Supplementary Examination. The marking schedule for this examination shall be:-

Marks	
Paper I	50
Paper II	50
Paper III	50
Oral IV	50
TOTAL	**200**

Pass Mark: 50% i.e. 100 marks

A candidate shall not be admitted to the Fourth Year (MBChB Part IV) until Part III has been completed.

6.3.1.7 Results

Results shall be determined in accordance with the provisions of the General Academic Regulations.

6.4 Fourth Year MBChB Part IV

Clinical Pharmacology, Community Medicine, Psychiatry)

6.4.1 Examiners:

There shall be three External Examiners, one each for Clinical Pharmacology, Community Medicine and Psychiatry.

6.4.2 Examinations

6.4.2.1 Clinical Pharmacology: (CP4M)

The examination, which shall be held in the University, shall comprise:

6.4.2.1.1 A written examination (three hours): CP4M/01

6.4.2.1.2 An oral examination: (CP4M/90)

Candidates selected by the Examiners may be required to attend an oral examination.

6.4.2.1.3 Marks obtained in classwork and examinations held throughout the course, shall be available to the Examiners and shall be taken into consideration in the final pass mark (CP4M/50). The final pass mark will be 50%.

6.4.2.2 Community Medicine: (CM4M)

The examination, which shall be held in the University, shall comprise:

6.4.2.2.1 A written examination (Three hours):

(CM4M/01) which shall consist of one compulsory question and five other questions, of which the candidate is to answer four questions.

6.4.2.2.2 An oral examination: (CM4M/90)

At which not less than two Examiners shall be present.

6.4.2.2.3 Marks obtained in classwork and examinations held throughout the course, including marks in Biometrics, shall be available to the Examiners and shall be taken into consideration in the final assessment (CM4M/50).

Marking: A maximum of 20 marks shall be awarded for each question in the written paper and 50 marks shall be the maximum attainable in the oral examination.

Pass mark: 50% i.e. 75 marks.

6.4.2.3 Psychiatry: (PZ4M)

6.4.2.3.1 A written examination (three hours):

(PZ4M/01) which shall contain 100 multiple-choice questions, one essay or a long Clinical Case History. Candidates are to attempt all questions.

6.4.2.3.2 An oral examination: (PZ4M/90)

At which not less than two Examiners shall be present.

6.4.2.3.3 Marks obtained in classwork and examinations held throughout the Course, shall be available to the Examiners and shall be taken into consideration in the final assessment (PZ4M/50).

Marking: The multiple-choice questions shall be allocated 100 marks. The essay question or long Clinical Case History shall be allocated 50 marks. The oral examination shall be allocated 50 marks and the continuous assessment shall be allocated 100 marks.

Pass mark: 50% i.e. 150

6.4.3 Setting of Papers

The setting of question papers shall be the responsibility of the Departmental Panel of Examiners.

6.4.4 Conduct

The examination shall be conducted in conformity with the Regulations prescribed by the University for this purpose.

6.4.5 Failure

Any candidate who fails the examination may be permitted to write Supplementary Examinations. Candidates who fail in one or more subjects may sit Supplementary Examinations in those subjects they have failed. A candidate shall not be admitted to the 5th year MBChB Part V until the whole of Part IV has been completed.

6.4.6 Results

Results shall be determined in accordance with the provisions of the General Regulations.

6.5 Fifth Year MBChB Part V

(Medicine, Obstetrics & Gynaecology, Paediatrics & Child Health and Surgery)

6.5.1 Examiners

There shall be four External Examiners, one in each of Surgery, & Gynaecology, Medicine and Paediatrics and Child Health.

6.5.2 Examinations

6.5.2.1 There shall be a written examination in Surgery, including surgical specialities, elementary anaesthetics and dentistry, and in Obstetrics and Gynaecology, and there shall be a written examination in Medicine, including therapeutics and medical specialities not otherwise designated, and in Paediatrics and Child Health.

6.5.2.2 Clinical and oral examinations shall be conducted in each of the subjects Surgery, Obstetrics and Gynaecology, Medicine and Paediatrics and Child Health.

6.5.3 Medicine, (MD5M) and Obstetrics & Gynaecology: (OG5M)

6.5.3.1 One case history paper with 6 case histories and 20 questions on each. Candidates are expected to answer all questions. The paper is marked out of 60 with 10 marks given to each case history. (MD5M/01) or (OG5M/01)

6.5.3.2 A multiple choice question paper of one and a half hours: a maximum of 60 marks shall be allotted for this paper. (MD5M/02) or (OG5M/02)

6.5.3.3 A clinical examination: a maximum of 150 marks shall be allotted for this section. (MD5M/80) or OG5M/80)

6.5.3.4 An oral examination: a maximum of 30 marks shall be allotted for this section. (MD5M//90) or (OG5M/90)

6.5.4 Surgery: (SG5M)

6.5.4.1 A multiple choice question paper of two hours: a maximum of 50 marks shall be allotted (SG5M/90).

The Department and College records of the work and assessment of each candidate shall be available to the Examiners at the Examiners' Meeting in each subject and at the combined Examiners' Meeting. (SG5M/50)

6.5.5 Paediatrics and Child Health (PD5M)

6.5.5.1 A written examination comprising two papers (PD5M/01 and 02); of which (PD5M/02) shall be a Multiple Choice paper.

6.5.5.2 A clinical examination, during which long and short cases shall be presented. (PD5M/80)

6.5.5.3 The examiners may require any candidate to attend an oral examination. (PD5M/90)

An assessment of studies undertaken during the student's attachment to the Department may be considered. (PD5M/50)

6.5.6 Setting of Papers:

The setting of question papers shall be the responsibility of the Departmental Panel of Examiners.

6.5.7 Pass Marks:

The pass mark in each Subject shall be 50%. Candidates must satisfy the Examiners in both theory and clinical, but candidates whose total mark is less than 150 out of 300 in each Subject, are not permitted to pass unless they achieve at least 50% in the clinical part of the examination in each subject. Candidates are required to present themselves in all Subjects.

6.5.8 Failure:

Any candidate who is attempting the examination for the first time and who fails one or more of the four Subjects shall normally be permitted to sit a Supplementary Examination in that subject or subjects.

6.5.9 Results:

Results shall be determined in accordance with the provisions of the General Academic Regulations.

7. CLASSIFICATION OF DEGREES

7.1 Aims

7.1.1 To group the twelve subjects in the Programme under two main headings and so give due weight to their clinical importance in the professional sense;

7.1.2 To maintain the present marking system.

7.2 Marking System:

75% +	=	Distinction
65%–74%	=	Honours
50%– 64%	=	Pass
0–49%	=	Fail

7.3 Conditions of the Award of Honours and Distinctions:

A candidate in the Examinations for the Degrees of MBChB may be awarded Honours or Distinctions in any one subject according to the regulations outlined in 7.2 above, provided that:

7.3.1 he/she is presenting the subject for the first time;

7.3.2 at the same time he/she obtains passes in all subjects which may be linked to any part of the examinations in which he/she is a candidate.

7.3.2.1 Pre-Clinical Course:

Biochemistry
Anatomy
Physiology
Behavioural Sciences and Community Health Examination
Communication Skills for Academic Purposes
Communication Skills for Professional Purposes

7.33. Clinical Course: Pathology
Clinical Pharmacology
Community Medicine
Psychiatry
Medicine
Obstetrics and Gynaecology
Paediatrics and Child Health
Surgery

7.4 Award of the Degrees 'With Honours'

Successful candidates for the Degrees of MBChB may be awarded degrees 'With Honours' providing that they meet the following requirements:

7.4.1 Subject Distribution

7.4.1.1Pre-Clinical Course:

Biochemistry] Professional Examination Part I
Communication Skills for Academic Purposes]

Anatomy]
Physiology]
Behavioural Sciences and Community Health] Professional Examination Part II
Communication Skills for Professional Purposes]

7.4.1.2 Clinical Course:

Pathology] Professional Examinations Part III

Clinical Pharmacology]
Community Medicine] Professional Examinations Part IV
Psychiatry]

Medicine]
Obstetrics and Gynaecology] Professional Examinations Part V
Paediatrics and Child Health]
Surgery

7.4.2 Regulation

In order to graduate 'With Honours' a student must obtain: -

7.4.2.1 either a total of SIX subject passes at Honours and/or Distinction standard out of the TWELVE separate subjects examined in the pre-clinical and clinical sections of the Programme providing that:

(i) at least ONE of these SIX passes at Honours and/or Distinction standard is gained in the pre-clinical examinations (i.e. Parts I and II) and;

(ii) at least FIVE of these SIX passes at Honours and/or Distinction standard are gained in the clinical examination (i.e. Parts III, IV, and V) ONE of which must be from among the four subjects comprising Part V Examinations.

FACULTY OF SCIENCE AND TECHNOLOGY

FACULTY REGULATIONS FOR UNDERGRADUATE AND POSTGRADUATE DEGREES

1. INTRODUCTION

1.1 These regulations shall be read in conjunction with the General Academic Regulations for undergraduate Degrees hereinafter referred to as General Regulations.

1.2 These Regulations are in respect of programmes for the following degrees offered by the Faculty of Science and Technology:

(HABB)
Bachelor of Science Biotechnology and Bioinformatics Honours Degree (HBB)
Bachelor of Science Chemical Technology Honours Degree (HCT)
Bachelor of Science Industrial Chemistry Honours Degree (HIC)
Bachelor of Science in Forensic Science Honours Degree (HFSC)
Bachelor of Science Chemical Laboratory Management Honours Degree (HCLM)
Bachelor of Science Honours in Medicinal and Pharmaceutical Chemistry (HMPC)
Bachelor of Science Computer Science Honours Degree (HCSCI)
Bachelor of Science Computer Systems Engineering Honours Degree (HCSE)
Bachelor of Science in Software Engineering Honours Degree (HSWE)
Bachelor of Science in Cybersecurity Honours Degree (HCSEC)
Bachelor of Science Honours Artificial Intelligence and Machine Learning Honours Degree (HAIML)
Bachelor of Science Honours in Applied Mathematics and Computational Science Honours Degree (HMTCS)
Bachelor of Science Degree in Actuarial Science Honours Degree (HACS)
Bachelor of Engineering in Telecommunication Engineering Honours Degree (HTENG)
Bachelor of Science Honours Space Science and Technology (HSS)
Bachelor of Science Honours Applied Sciences and Nanotechnology (HNN)
Bachelor of Science Honours Industrial Physics and Instrumentation (HIPI)
Bachelor of Science Honours Applied Statistics (HSTAT)
Bachelor of Science in Food Science and Technology Honours Degree (HFST)
Bachelor of Science in Nutrition Science Honours Degree (HNS)

1.3 The Faculty of Science and Technology also offers the following Masters programmes:
Master of Science in Biodiversity Conservation for Sustainable Development (MBSD)
Master of Science in Ecological Resources Management (MERM)
Master of Science in Biostatistics and Epidemiology (MSBE)
Master of Science in Pharmaceutical Chemistry (MPCH)
Master of Science in Analytical Chemistry (MACH)
Master of Science in Environmental Chemistry (MECH]
Master of Science in Organic Chemistry (MOCH)
Master of Science in Computer Science (MSC)
Master of Science in Software Engineering (MSE)
Master of Science in Applied Mathematical Modelling (MAMM)
Master of Science Degree in Industrial Physics (MIP)
Master of Science in Mobile and Wireless communications (MMW)
Master of Science Degree in Optic Fibre and Photonic Engineering (MOF)
Master of Science Degree in Food Science and Nutrition (MFSN)
Master of Science in Food Processing Technology (MFPT)
Master of Science in Food Safety and Quality Management (MFSQ)

1.4 The appropriate Degree will be awarded to a student who has successfully completed an approved programme.

2. DEFINITION OF TERMS

2.1 A Practical theory module is a module which contains both practical and theory components in the formal examination.

2.2 A Practical module is a module in which a student is required to carry out, under the direction and supervision of a member of staff in a laboratory, a series of practical

exercises related to particular topics within a subject.

2.3 A Field Trip is an excursion outside the University Campus under the supervision of an academic and/or technical staff of the University. It may last for part of a day or be conducted over a longer period. The field trip can form part of a module or if it is of sufficient duration, form a 12-credit module in its own right.

2.4 A Project module is a module in which a student is required to carry out, under the direction and supervision of a member of the academic staff, private studies or investigations related to a particular topic within a module. Each Project module shall be weighted based on the amount of time it occupies compared with a theory module.

2.5 A pre-requisite is a module the departmental board requires the student to pass prior to the student registering for any other related higher- level modules.

3. ENTRY REQUIREMENTS

3.1 ***Normal Entry***

For one to qualify for normal entry to an Undergraduate Degree programme in the Faculty of Science and Technology, a prospective student must satisfy the entry Regulations specified in the General Regulations and in so doing, must also meet the following Faculty requirements:

3.1.1 Must have obtained a pass at 'O' Level in Mathematics and English Language and have obtained a pass at 'A' Level in at least two science subjects or one requisite science subject and an approved equivalent as specified in the departmental regulations.

3.1.2 Each department may prescribe specific combinations of these 'A' Level subjects.

3.2 ***Special Entry***

Special entry would be granted to students who, in the opinion of the Departmental Board, have outstanding passes in the relevant Diploma/Professional Certificate, subject to such a qualification meeting part of the requirements of the respective programme.

3.3 ***Mature Entry***

Refer to Section 3.3 of the General Regulations.

4. GENERAL PROVISIONS

4.1 In level 1, normally a student shall register for at least 10 modules.

4.2 For level 2 and higher modules, all pre-requisites that apply to the modules shall be listed in the special regulations for the subject to which the modules form a portion. A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre- requisites for the module prior to the start of the semester in which he intends to study that module.

4.3 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons of all relevant Departments and to the following conditions:

4.3.1 The student shall register for at least 10 modules during the academic year.

4.3.2 The module combination is feasible in terms of timetable,

4.3.3 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and

4.3.4 A student may apply for exemptions during the first four weeks of the semester.

4.4 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that

programme.

5. ASSESSMENT

5.1 The assessment of a practical module shall be based on either written reports submitted during the semester or according to a schedule laid down by the appropriate Departmental Board.

5.2 The assessment of a Field module or a Project module shall be based on either the written reports or dissertation submitted at a date determined by the Departmental Board and the students shall be notified of this date at the start of the module.

5.3 The assessment of a Practical-Theory module shall be based on a written report, formal theory examination and continuous assessment.

5.4 All other modules shall be assessed by continuous assessment and formal theory examination.

5.5 Each formal examination shall normally be held during an examination period at the end of each quarter.

5.5.1 The formal theory examination for a module shall normally be of three hours duration.

5.5.2 A Practical-Theory module shall contain a practical component in the continuous assessment. The continuous assessment may be entirely practical or contain theory and practical components, but the relative weightings shall be stated in the special Regulations for the programme.

5.5.3 In Level 3 the student will be on Work-Related Learning for a minimum period of 10 months.

5.6 For each practical module, field work and project module the Departmental Board of Examiners shall determine a final mark, for that student, for that module based on the assessment of the written reports or the dissertation submitted by that student. The Board shall determine whether the student has passed or failed the module.

5.7 The Departmental Board shall submit, for each module under its control and for each student enrolled in that module, the final mark to the Faculty Board of Examiners. For each student, it shall indicate the results of the modules under its control.

5.8 In all modules that consist of both theory and practical, the practical mark is normally taken to be the average of the practical assessments throughout the quarter.

5.9 The final mark will be calculated as follows:

Final mark = (60%) examination mark + (40%) continuous assessment. NB: Note that continuous assessment is normally calculated as 20% for practicals, 15% for tests and 5% for assignments.

6. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

7. PROVISION FOR PROGRESSION

7.1 In order to proceed from level 1 to level 2 a student must have attained at least 96 credits.

7.2 To proceed from level 2 to level 3 the student must have attained at least 120 credits from the modules at that level including the core modules of the subject.

7.3 To proceed from level 3 to level 4 the student must have passed Work-Related Learning.

7.4 A student may not proceed to another semester carrying 4 modules and above from the previous semester.

8. WORK-RELATED LEARNING PROGRAMME GENERAL GUIDELINES

8.1 The student is subject to Section 12 of the General Academic Regulations and the company regulations during the Work-Related Learning period.

8.2 Once a student is on Work-Related Learning with an organization approved by the Department/Faculty, he/she is not allowed to change/move to another organization without the approval of the Department/ Faculty.

The student is expected to;

8.2.1 conform to the company's regulations, working time and discipline,

8.2.2 fulfil the supervisor's instructions concerning the training process and carrying out of the work-related research project.

8.2.3 Submit an examinable report at level 3.1, write a log book on a daily basis and submit a report after finishing the training in a given department (or training unit).

8.2.4 take part only with educational purposes in mind according to the ultimate instructions of the supervisor.

8.2.5 put his/her best effort into acquiring extensive knowledge and skills in order to achieve the required standard of training.

8.2.6 keep good relations with all staff of the company and

8.2.7 promote the good image of Midlands State University.

8.3 The student is expected to always bear in mind that his/her conduct during Work-Related Learning will reflect not only on him/her but also on Midlands State University and that it may also affect considerably, future placements and the relationship between the University and that particular company.

9. REMARKING OF EXAMINATIONS

Refer to Section 13 of the General Regulations.

10. GRADING AND DEGREE CLASSIFICATION

10.1 To be awarded a Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 480 of the 540 credits.

10.2 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

11. DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level3	30%
Level4	40%

12. MASTERS AND DOCTORATE DEGREES AWARDED

12.1 The Faculty of Science and Technology offers the following Postgraduate degrees:

12.1.1 Master of Science (MSc)

12.1.2 Master of Philosophy(MPhil)

12.1.3 Doctor of Philosophy (DPhil)

DEPARTMENT OF APPLIED BIOSCIENCES AND BIOTECHNOLOGY

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED BIOSCIENCES AND BIOTECHNOLOGY (HABB)

Duration:	4 years
Actual Credit Load	540 credits
Minimum Credit Load:	480 credits
Maximum Credit Load:	540 credits
Maximum MBKs Credit Load:	384 credits
ZNQF level:	8

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competences in the field of Biological Sciences and Biotechnology relevant to various employment capabilities and careers in world of work and society. To prepare students for further studies and lifelong learning in Biological Sciences and Biotechnology.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry: **P**asses in 'A' level Biology or equivalent, and any other Science subject at 'A' Level or Relevant National Certificate or equivalent.

Special Entry: National Certificate (NC), National Diploma (ND), or Higher National Diploma (HND) in Science and Technology.

Mature Entry: 5 'O' level passes including Mathematics and Science and at least 25 years of age with at least 5 years relevant work experience.

3.0 PROGRAMME CHARACTERISTICS

Areas of study: Molecular Biology, Cell Biology, Genetics, Ecology, Physiology, Microbiology, Biochemistry, Entomology, Invertebrate Biology, Parasitology, Biostatistics, Biodiversity Conservation, Biotechnology, Bioinformatics

Specialist Focus: Agricultural and Industrial Biotechnology, Biodiversity Conservation, Animal Parasitology, Agricultural and Medical Entomology, Aquatic Toxicology

Orientation Oriented towards innovation and application of theory to solve real life problems

Distinctive Features: Student-centred approach

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Biotechnologists at research institutions, manufacturing industries, food and pharmaceutical industries; Ecologists in biodiversity conservation organizations; Plant and animal breeders; Curators in natural history museums; Forensic technologists; Technopreneurs in bio-businesses in aquaculture, beekeeping, and mushroom production

Further Studies: Masters and Doctoral studies in Biotechnology, Medical Laboratory Sciences, Biostatistics, Biodiversity Conservation, Applied Biosciences

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, seminars, and dissertations

Assessment Methods: Written and oral examinations, tests, laboratory reports, seminar presentations, work-related learning reports, final year dissertations, continuous assessments

6.0 LEARNING OUTCOMES

6.1 Knowledge and skills in areas of Biosciences and Biotechnology

6.2 Application of dry-bench, wet bench, and field research skills in Biosciences and Biotechnology in research and related industries

6.3 Competencies and skills for creating bio-products and bio-business ventures (e.g. apiculture, aquaculture, mushroom production)

6.4 Effective communication of research findings using multi-media to both expert and non-expert audiences

6.5 Adherence to ethical and safety principles in conducting research and discharge of duties and responsibilities.

7.0 GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for at least 10 modules from at least two subjects of which a maximum of 4 modules shall be from other Faculties.

7.2 For level 2 and higher modules, all prerequisites that apply to a particular module shall be listed in the special regulations for the module to which the modules form a portion. A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other Departments up to a total equivalent of at most 8 modules.

7.4 A student's section of modules of the Degree programme is subject to the approval of the Dean of Science and Technology and the Departmental Chairperson to the following conditions:

7.5 The student shall register for at least 10 modules at any time during the academic year.

7.6 The module combination is feasible in terms of timetable.

7.7 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and

7.8 A student may apply for exemptions during the first four weeks of the semester.

7.9 The Departmental Board may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme. The department shall not designate more than 10 modules as core modules per level.

8.0 PROGRAMME ASSESSMENT

Continuous Assessment: Continuous assessment shall contribute 40 % towards the final mark of each module and shall consist of Practicals (20%), Written assignments (5%), and In-class tests (15%).

Written Examinations: The formal written examination shall contribute 60% to the overall assessment of a module.

Other:

a) Research Project is assessed on the basis of a research project (75%), Oral presentation (15%) and student conduct in the laboratory (10%).

b) Industrial Attachment/ Work-Related Learning is assessed based on the Industrial Attachment report (30%), assessment by the work supervisor (50%), and assessment by the Academic supervisor (20%).

9.0 PROVISION FOR PROGRESSION

9.1 A student will repeat all failed modules if he/she fails at least 4 modules in a semester.

9.2 To proceed from level 1 to level 2 a student must have accrued at least 96 credits, provided he/she may not carry over more than three modules.

9.3 To proceed from level 2 to level 3 the student must have accrued at least 96 credits from the modules in level 2, provided he/she may not carry over more than three core modules to the next level.

9.4 To proceed from level 3 to level 4 the student must have satisfied the requirements for Work-Related Learning, including Work-Related Learning modules 1 and 2.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the University General Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Science Honours in Applied Biosciences and Biotechnology, a student must pass each core module listed in the programme and must have accrued at least 480 credits.

11.2 The classification of all modules and the degree programme shall be as stated in Section 5 of the General Regulations.

12.0 DEGREE WEIGHTING

For the purpose of degree classification, weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0 PROGRAMME STRUCTURE

Indicated in parenthesis are pre-requisite modules

Asterisk * denotes non-core modules

Code	Description	Credits
Level 1 Semester 1		
HABB 131	Biodiversity 1	12
HABB 135	Molecular and Cell Biology	12
*HABB 138	Analytical and Laboratory Techniques	12
CS 131	Communication Skills	12
HCSCI 131	Introduction to Computers and Computer Applications	12
Level 1 Semester 2		
HABB 132	Biodiversity 2	12
HABB 133	Principles of Ecology	12
HABB 134	Principles of Genetics and Evolution	12
HABB 136	Microbiology	12
HABB 137	Experimental Design and Data Analysis for Biologists 1	12
*HBB 140	Principles of Biochemistry	12
Level 2 Semester 1		
HABB 231	Molecular Genetics (HABB 134)	12
HABB 232	Immunology	12
HABB 233	Population and Community Ecology (HABB 133)	12
HABB 234	Animal Biology (HABB 132)	12
TCNP 201	Technopreneurship	12
GS 231	Gender Studies	12

Level 2 Semester 2

HABB 235	Recombinant DNA Technology (HABB 135)	12
HABB 236	Plant Biology (HABB 131)	12
HABB 237	Entomology (HABB 132)	12
HABB 239	Fermentation Biotechnology	12
*HABB 248	Bioethics, Safety, Regulation and IPR	12

Level 3 Semester 1

HABB 330	Work-related Learning/Industrial Attachment I	40

Level 3 Semester 2

HABB 331	Work-related Learning/Industrial Attachment II	80

Level 4 Semester 1

HABB 441	Experimental Design and Data Analysis for Biologists 2 (HABB 137)	12
HABB 442	Generic Skills for Biologists	12
HABB 443	Bioinformatics	12
HABB 444	Gene Structure and Function (HABB 231)	12

Choose at least TWO from the following electives:

*HABB 446	Plant Biotechnology (HABB 235)	12
*HABB 447	Industrial Biotechnology (HABB 136/235239)	12
*HABB 448	Biotechnology of Biofuels and Bioenergy (HABB136/235)	12
*HABB 453	Plant Biosystematics	12

Level 4 Semester 2

HABB 460	Research Project	38

Choose at least TWO from the following electives:

*HABB 445	Environmental Biotechnology (HABB 136)	12
*HABB 449	Medical and Veterinary Entomology (HABB 237)	12
*HABB 450	Agricultural Entomology (HABB 237)	12
*HABB 451	Applied Virology (HABB 136)	12
*HABB 452	Aquatic Ecotoxicology (HABB 133/233)	12
*HABB 453	Plant Biosystematics	12
*HABB 454	Medical and Veterinary Parasitology (HABB 132)	12
*HABB 455	Veterinary Parasitology (HABB 132)	12
*HABB 456	Population Genetics (HABB 134)	12
*HABB 457	Conservation Biology (HABB131/132)	12

14.0 Synopses

HABB 131 Biodiversity 1

Origins of life; comparative metabolism; stromatolites and other fossils; the geological time scale; the primitive atmosphere and the evolution of the genetic code. Taxonomy and phylogeny: the goals of classification; the stages of classification; phenetic, cladistics, and

evolutionary systematics, taxonomic characters, homology; analogy; taxonomic keys, and phylogenetic trees. Viruses and Monera: the diversity among viruses and in the kingdom Monera. Protista: the diversity within Kingdom Protista. Fungi: the diversity among the Fungi, including the division eumycota and lichens. Plants: brown algae; red algae; green algae; bryophytes; psilopsids; lycopsids; sphenopsids; ferns; tracheophyte gymnosperms and angiosperms, and their adaptations to terrestrial life.

HABB 132 Biodiversity 2

Characteristics, origins and trends of Kingdom Animalia; A brief outline of the grades of organization and the evolution of the two main lines of animals (vertebrates and invertebrates); Major trends in embryonic development and body construction; Introduction to use of DNA based methods to assess phylogenetic relationships at higher taxonomic levels (Phyla, Class); Major Phyla (Cnidaria, Platyhelminthes, Nematoda, Annelida, Mollusca, Arthropoda, Echinodermata, Chordata.

HABB 133 Principles of Ecology

Definition of Ecology; Ecology and levels of organization in biological systems; Doing ecology; Population structures; Population growth; Concepts of habitat and niche; Methods of estimating population size and density; Intra- and inter-species interactions, including competition, commensalism, mutualism, parasitism, infectious diseases, and predation; Population regulation.

HABB 134 Principles of Genetics and Evolution

Genetics: Mitosis and meiosis; Mendelian genetics and the chromosome theory of heredity; mutation and variation; introduction to population genetics. Evolution: population genetics and variation; the concept of, and evidence for evolution; theory of evolution by natural selection; adaptation and adaptive radiation; the nature of species; specific mate recognition systems; isolating mechanisms; speciation.

HABB 135 Cell and Molecular Biology

Microscopy, preparation of specimens for light and electron microscope; Cell theory; Prokaryotic and eukaryotic cell structures; Structure and function of different cell organelles; Cytoskeleton and cell motility; Interactions between cells and their environment; Traffic across membranes; Cell growth and division in the cell cycle;
Cell differentiation; Cell signalling and Signal transduction; Cancer; Cytoplasmic membrane systems, Nature of the Gene and Genome, Molecular anatomy of genes and chromosomes.

HABB 136 Microbiology

The module explores microscopic life forms including bacteria, fungi and viruses. Emphasis is on the nature and behaviour of these microorganisms and their importance in manufacturing, agriculture, medical and food industries. Major topics include history of microbiology, aseptic techniques, nutrient media, classes of microorganisms, microbial growth, procedures for characterization, and microbial energetics. Applications of microbiology in bioremediation, wastewater treatment, bio-fertilizers, animal-microbial symbiosis, industrial microbiology, and control of microbial diseases. Practicals introduce the students to techniques involved in the cultivation, enumeration and purification of microbes.

HABB 137 Experimental Design and Data Analysis for Biologists 1

Introduction: definitions, and uses of statistics (research, business, tourism, agriculture). Probability: multiplication law, addition law, condition probability, tree diagram, law of total probability. Probability distribution: random variables; discrete, continuous. Binomial

distributions: normal distributions. Measures of central tendency and dispersion: mean, mode, range, variance, standard deviation, standard error of the mean. Parametric and non-parametric statistic. Sampling techniques: simple, random, stratified, cluster, systematic. Data types, presentation and summarization techniques: tables, graphs, charts. Regression and correlation: regression parameters, correlation coefficient, coefficient of determination. Simple statistical inference: hypothesis testing, confidence intervals, t-tests and chi-square tests. Practicals using appropriate statistical packages (e.g. SPSS, GENSTAT, SAS)

HABB 138 Analytical and Laboratory Techniques

Laboratory safety practices and procedures. Standard Operating Procedures. Preparation of solutions - understanding of the terms: molarity, molecular weight/mass, percentages, and basic units of volume and weight. Principles, techniques and biological applications of Microscopy; Spectrophotometry; Electrophoresis; Centrifugation; Chromatography; Polymerase Chain Reaction. Basics of writing a scientific report. Practicals will develop competence in preparing stock solutions and accurate dilutions, accurate pipetting and using standard laboratory equipment, techniques in biochemical evaluation, techniques in molecular biology and statistics of instrumental analysis.

HBB 140 Principles of Biochemistry

Carbohydrates: Classification - monosaccharide-structure, stereoisomers and structural isomers, and mutarotation. Oligosaccharides-Disaccharides -structure and importance of sucrose, lactose, maltose, cellobiose. Polysaccharides-structure and importance of homopolysaccharides and heteropolysaccharides. Lipids: definition and classification. Fatty acids: classification, nomenclature, structure and properties of saturated and unsaturated fatty acids. Essential fatty acids. Triacylglycerols: nomenclature, chemical properties and characterization of fats - hydrolysis, saponification value, acid value, rancidity of fats, Reichert-Meissel number. Biological significance of fats. Glycerophospholipids (lecithins, lysolecithins, cephalins. phosphatidylserine, phosphatidyl inositol, plasmalogens), sphingomyelins, glycolipids -cerebrosides, gangliosides. Amino acids: General properties, peptide bond, essential and non-essential amino acids. Protein chemistry: Classification, different levels of protein structure, forces stabilizing protein structure, and protein folding. Nucleic acids: Introduction, chemistry of nucleic acids, double helical structure and properties of DNA, RNA –types, structure and functions. Vitamins: Introduction, properties, functions and deficiency diseases of fat soluble and water soluble Vitamins. Enzyme: Classification, Nomenclature, Mechanism of enzyme action, derivation of Michaelis Menten equation, Enzyme inhibition, Factors affecting enzyme activity, Allorteric enzymes, Isoenzymes.

HABB 231 Molecular Genetics

Chemical nature of DNA, Replication, transcription and RNA processing, translation. Mutations and genetic basis of cancer. Control of gene expression in prokaryotes, and eukaryotes. Applications of molecular genetics in control of cancer, manufacturing of antibiotics and anti-retroviral drugs.

HABB 232 Immunology

Definition of immunology; Cellular participants in immune response; Types of immunity; Cytotoxic T lymphocytes; cytokines; Primary and secondary responses; Antigens and antigen processing; Structure and function of antibodies; Antibody and antigen interactions; Antibody antigen reactions; Theory of vaccination; Basis of cellular immunity; Theory of McFarlane Burnet; Monoclonal vs. polyclonal antibodies; AIDS and HIV- immunological basis.

HABB 233 Population and Community Ecology

How organisms adapt and interact with each other and with their physical environment; Nature and consequences of trophic interactions between species; The role of the abiotic environment, and the factors that affect stability and resilience at both small and large scales; The basis for sustainable management of natural and semi-natural systems; Introduction to Limnology.

HABB 234 Animal Biology
Main animal groups: their life cycles, development, body plan, functional organisation including metabolism and their adaptation to different environments. Adaptations to aquatic and terrestrial survival; Locomotion and structural support; Circulation and gas exchange in invertebrates and vertebrates; Control of the internal environment; Animal nutrition: nutritional requirements, digestion; Endocrine systems; Sexual and asexual reproduction; Aestivation, dormancy and cryptobiosis as survival strategies.

HABB 235 Recombinant DNA Technology
This module provides a background to recombinant DNA technology. Emphasis will be on techniques used to multiply, control, detect, screen, alter, express and propagate the recombinant DNA in foreign host systems. Topics include the enzymology of recombinant DNA technology, hosts and vectors, microbial genetics, DNA sequencing, DNA amplification by PCR, RNA labeling, site-directed mutagenesis, isolation of DNA/RNA, construction of DNA and cDNA libraries, screening and characterization of recombinant DNA libraries and methods used, in vitro translation, Northern/Southern blot analysis, restriction mapping, DNA cloning and expression and the principles behind the analytical techniques used in recombinant DNA technology.

HABB 236 Plant Biology
Form and function of higher plants. Transport in plants. Sexual and asexual reproduction and dispersal mechanisms. Adaptations to survival in different environments. Plant hormones and their function. Photosynthesis. Physiology of flowering. Plant nutrition and functions of macronutrients and micronutrients. In-plant and plant-to-plant communication. Regulation of plant growth and dormancy. Biological clocks in plants.

HABB 237 Entomology
Importance of insects; Insect systematics and taxonomy; External Anatomy of insects; Structural and functional modifications of head and thoracic appendages; Insect growth and development; Internal anatomy and physiology; Insect behaviour; Insect ecology; The application of DNA technology in insect taxonomy.

HABB 239 Fermentation Biotechnology
This module will cover alcoholic and non-alcoholic fermentation, industrial fermentation and brewing industry, construction of yeast, genetic systems with greater fermentation efficiency, fermentation design and control as well as downstream processing.

HABB 248 Bioethics, Safety, Regulation
The module covers aspects of handling and disposal of hazardous biological and chemical substances of laboratory origin. Bioethics explores controversial issues brought about by advances in biotechnology and medicine. Ethical questions that arise in the relationships among life sciences, biotechnology, medicine, politics, law, and philosophy are addressed. The module mainly focuses on laboratory set-ups to contain possible epidemic situations when working with microbes, GMOs, and biohazards which become the epicentre of biosafety issues.

TCNP 201 Technopreneurship

Introduction: nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur. Small business model and financial issues: developing a business model, basics of small business management, risks and stages of funding, sources of funding, financial funding for growth, product valuation, and how to form and register a small business in Zimbabwe. New Product Development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; the NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, the theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interactions. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology.

HABB 330 Work-related Learning/Industrial Attachment I

This internship allows the student to gain experience working in a professional environment and to apply knowledge gained in the first two years to acquire professional skills or competencies. Students will be able to expand their professional network, convert academic knowledge into industrial skills, narrow down their potential careers and gain unforgettable life experiences.

The main objectives of the training module are:

i) To help the student put into practice the basic experimental and theoretical skills which were learnt,

ii) To provide exposure to and experience of the industrial environment.

iii) To give the student an insight into the production activities of Zimbabwe's main industries,

iv) To embark on a research project under the joint supervision of a member of the department and a member of the training institution.

The Industrial attachment will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience.

HABB 331 Work-related Learning/Industrial Attachment II

This internship allows the student to gain experience working in a professional environment and to apply knowledge gained in the first two years to acquire professional skills or competencies. Students will be able to expand their professional network, convert academic knowledge into industrial skills, narrow down their potential careers and gain unforgettable life experiences.

The main objectives of the training module are:

v) To help the student to put into practice the basic experimental and theoretical skills which were learnt,

vi) To provide exposure to and experience of the industrial environment.

vii) To give the student an insight into the production activities of Zimbabwe's main industries,

viii) To embark on a research project under the joint supervision of a member of the department and a member of the training institution.

The second part of the internship will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience.

HABB 441 Experimental Design and Data Analysis for Biologists 2

Importance of statistics in Biosciences. Hypothesis testing. Principles of Experimental Design. Types of experimental design - completely randomized design, randomized block design, nested designs, factorial design, repeated measures design, Before-After-Control-Impact studies in Ecology. Linear regression and correlation. Analysis of variance - one way ANOVA, factorial ANOVA, repeated-measures ANOVA, random and fixed effects, checks for the assumptions of ANOVA/general linear models. Data Transformation. Non-parametric. Introduction to multivariate models - MANOVA for analysis of multiple response variables simultaneously; Principal Component and Factor Analysis for data reduction, Discriminate Function Analysis methods for discriminating between groups based on multiple variables. Competent use of at least one Statistical Software programme for data analyses.

HABB 442 Generic Skills for Biologists

Philosophy and ethics of Science. How to communicate effectively as a scientist. Effective reading, searching and evaluating literature. Students will receive training in scientific writing skills. Research and Grant proposal writing. GIS: An introduction to the use of GIS in life sciences, with the aim of providing the tools to access relevant spatial data sources, and construct basic maps of sampling sites etc.

HABB 443 Bioinformatics

This module explores the use of Bioinformatics databases and software as research and educational tools. Students will use data mining tools to extract DNA and protein sequences from primary and secondary databases. Software tools will be used to compare and analyze these sequences and construct gene and protein models for solving research problems related to molecular evolution, drug discovery and genetic bases for development and disease. Major topics include- Protein Analysis, Nucleic acid sequence analysis, Data Mining, Post-Transcriptional Modifications Functional & Structural Proteomics, sequence alignments, and primer design.

HABB 444 Gene Structure and Function

Control of gene expression in prokaryotes; Lysis and lysogeny in bacteriophage Lambda, Control of pathways; Mating behaviour in yeast; Control of gene expression in higher eukaryotes- Gene amplification, Gene rearrangement and mobile units, Regulation at the

transcriptional level, Post-transcriptional regulation, Processing transcripts, Differential mRNA processing, Changes in chromatin structure and gene regulation.

HABB 445 Environmental Biotechnology
Microbial diversity and its importance in biotechnology; Molecular methods for estimating the extent of microbial diversity; Environmental processes catalysed by microorganisms; Pollution control by microorganisms; Wastewater biotechnology; Bioremediation; Biomining; Environmental health and ecosystem sustainability e.g. bio-fuel production; Genetically modified organisms and the environment.

HABB 446 Plant Biotechnology
Overview of plant biotechnology; Model plants; Plant genomes; Gene structure; Control of gene expression; The 'omics' era; Plant breeding and crop improvement; Plant tissue culture; Recombinant DNA technology; Plant transformation systems; Plants as bioreactors; Applications of plant biotechnology; Diseases and pest resistance, Resistance to abiotic factors, Crop modification: society, ethics and regulatory issues.

HABB 447 Industrial Biotechnology
Use of microorganisms as cell factories - examples of bio-based products on the market: enzyme, bio-ethanol, fine chemicals i.e amino acids, citric acid, vitamins etc; Fermentation; Bioreactor and Bioprocess design; High throughput screening for novel metabolites; Strain improvement; Gene discovery; Protein expression and production; Protein engineering; Use of recombinant microorganisms.

HABB 448 Biotechnology for Biofuels and Bioenergy
Biofuels versus fossil fuels; Biofuels feedstock: lignocellulosic biomass, energy crops; Conversion and utilization of biofuels feedstock; First generation biofuels: bioethanol; Second generation biofuels: lignin modification and degradation, cell wall degrading enzymes, lignocellulosic bioethanol, biodiesel (Jatropha), processing technology for FAME, biogas, Third generation biofuels: Hydrogen and biological processes for hydrogen production, Algae-based hydrogen production and water splitting, Microbe fuel cells; Traditional and molecular breeding of energy crops; metabolic engineering of energy crops; Commercialization of biofuels; Biofuels and food security conflict;
Bioenergy Industry Development and Government Policy.

HABB 449 Medical and Veterinary Entomology
Nuisance insects and entomophobia; Epidemiology of vector-borne diseases; Major groups of medical and veterinary insects; Taxonomy, biology and ecology of each group, their medical and veterinary importance; Integrated disease management; Public health pest management; Forensic entomology; Medico-legal forensic entomology, Stored-product forensic entomology; Taxonomy, biology and ecology of each insect taxa of forensic importance.

HABB 450 Agricultural Entomology
Definition of a pest; Insects as pests; How pests arise; Economics of pest control; Methods for estimating pest population size or density; Life cycles and life history strategies of the most important agricultural pests in Zimbabwe; Pest control tactics; The problem of pesticide resistance; Integrated pest Management (IPM); Biotechnology in IPM - genetic manipulation of the pest population (e.g. SIR), and genetic manipulation of the crop.

HABB 451 Applied Virology

Advanced study of viruses; Virus molecular biology; Retrovirology; Virus detection and serology; HIV pathogenesis; Prevention and control of viral diseases; Drugs and vaccines; Viral vectors and their applications in biotechnology; Gene therapy.

HABB 452 Aquatic Eco-toxicology
Introduction to toxicology; Environmental stress; Exposure and Dose assessments - the exposure component of risk assessment; Experimental approaches in assessing biological effects of toxicants; Applications of toxicology in modern Industry; Legal aspects of aquatic toxicology

HABB 453 Plant Biosystematics
Angiosperm plant keying theory and practice; Taxonomic revisions; Species concepts; Theory of phylogenetic reconstruction; Morphological character analysis; Molecular systematics; Phylogenetics computing; Plant biogeography; Phenetics; Cladistics; Methods used to infer phylogenies; Modern taxonomy including DNA barcoding; Plant families of ecological and economic importance in the flora Zambesiaca; The weed flora of Zimbabwe.

HABB 454 Medical Parasitology
Meaning of commonly-used terms in human parasitology; Important human parasites-intestinal and systemic protozoa, taxonomic classification of helminths, intestinal and systemic nematodes, cestodes; How parasitic infections affect communities in developing countries; Life cycles and their necessity for effective prevention and control of human parasites; Epidemiology, clinical presentation, management and control of schistosomiasis with special reference to Zimbabwean schistosomes, lymphatic filariasis.

HABB 455 Veterinary Parasitology
Intimate associations; protoctistan parasites; life cycle strategies; categories of nematode parasitism; cestodes; digeneans; monogeneans and acanthocephalans; parasitic arthropods; host specificity; host response and defense; parasite evasion of immunity; adaptations to parasitism; parasite control; specimen collection and analysis.

HABB 456 Population Genetics
Measuring genetic variation; Mating systems and their genetic effects; Estimation of population parameters; The Hardy Weinberg principle; Microevolutionary forces; the neutral theory, synthesis of forces; Macroevolution - the species concept, reproductive isolation mechanisms, geographic models of speciation: allopatric speciation, sympatric speciation and parapatric speciation.

HABB 457 Conservation Biology
Science and practice of conservation biology with emphasis on Africa and Zimbabwe. Conservation issues; definition of conservation, definition of biodiversity, the value of biodiversity, extinction risks and the history and philosophy of conservation. Recognition and conservation of biodiversity at genetic, species, population and ecosystem levels. Conservation and society: incentives, access, who benefits from conservation, conservation and sustainable development, legal aspects and management policies.

HABB 460 Research project
The research project involves an experimental or observational investigation of a fundamental or practical problem in Applied Biosciences, Biotechnology or bio-product development. With

guidance from an academic supervisor, each student should choose and propose his/her own project theme. Each student is required to submit a proposal, carry out an independent research project or develop a product; submit a final written report, and to deliver an oral presentation.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN BIOTECHNOLOGY AND BIOINFORMATICS (HBB)

Duration:	4 years
Actual Credit Load	504
Minimum Credit load:	480
Maximum Credit Load:	516
Maximum MBKs Credit Load:	402
ZNQF level:	8

1. PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competencies in the field of Biotechnology and Bioinformatics relevant to careers in the world of work and society. To prepare students for further studies and lifelong learning in Biotechnology and Bioinformatics.

2. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry:	A pass in 'A' level Biology or equivalent, and Mathematics or Chemistry at 'A' Level or Relevant National Certificate or equivalent.
Special Entry:	National Certificate (NC) National Diploma (ND), or Higher National Diploma (HND) in Science and Technology.
Mature Entry:	5 'O' level passes including English, Mathematics and Science and at least 25 years of age with at least 5 years of relevant work experience.

3.0 PROGRAMME CHARACTERISTICS

Areas of study:	Molecular Biology, Cell Biology, Genetics, Physiology, Microbiology, Biochemistry, Biostatistics, Biotechnology, Bioinformatics, Programming, Research Ethics
Specialist Focus:	Agricultural and Industrial Biotechnology, Bioinformatics, Drug discovery, Vaccine Development
Orientation	Oriented towards innovation and application of theory to solve real-life problems
Distinctive Features:	Student-centred approach

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability:	Biotechnologists at research institutions, manufacturing industries, food and pharmaceutical industries; Plant and animal breeders; Forensic technologists; Technopreneurs in bio-businesses in aquaculture, beekeeping, mushroom production

Further Studies:	Masters and Doctoral studies in Biotechnology, Medical Laboratory Sciences, Bioinformatics, Molecular Biology

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods:	Lectures, tutorials, laboratory classes, group work, industrial visits, work-related learning, seminars, dissertations
Assessment Methods:	Written and oral examinations, tests, laboratory reports, seminar presentations, work-related learning reports, final-year dissertations, continuous assessments

6.0 LEARNING OUTCOMES

1.1 Knowledge and skills in the areas of Biotechnology and Bioinformatics

2.1 Application of dry-bench, wet bench, field research, and information technology skills in Biotechnology and Bioinformatics in research and related industries

3.1 Competencies and skills for creating bio-products and bio-business ventures (e.g. apiculture, aquaculture, mushroom production, production of pharmaceutical products)

4.1 Effective communication of research findings using multi-media to both expert and non-expert audiences

5.1 Adherence to ethical and safety principles in conducting research and discharge of duties and responsibilities.

7.0 GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for at least 10 modules from at least two subjects of which a maximum of 4 modules shall be from other Faculties.

7.2 For level 2 and higher modules, all prerequisites that apply to a particular module shall be listed in the special regulations for the module to which the modules form a portion. A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other Departments up to a total equivalent of at most 8 modules.

7.4 A student's section of modules of the Degree programme is subject to the approval of the Dean of Science and Technology and the Departmental Chairperson to the following conditions:

- 7.4.1 The student shall register for modules worth at least 120 credits at any time during the academic year.
- 7.4.2 The module combination is feasible in terms of timetable,
- 7.4.3 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and
- 7.4.4 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme. The department shall not designate more than 10 modules as core modules per level.

8.0 PROGRAMME ASSESSMENT

Coursework:	Coursework shall contribute 40 % towards the final mark of each module and shall consist of Practicals (20%), Written assignments (5%), and In-class tests (15%).
Written Examinations:	The formal written examination shall contribute 60% to the overall assessment of a module.
Other:	2.1 Research Project is assessed on the basis of a research project (75%), Oral presentation (15%) and student conduct in the laboratory (10%). 3.1 Industrial Attachment/ Work-Related Learning is assessed based on the Industrial Attachment report (30%), assessment by work supervisor (50%), and assessment by Academic supervisor (20%).

9.0 PROVISION FOR PROGRESSION

9.1 A student will repeat all failed modules if he/she fails at least 4 modules in a semester.

9.2 To proceed from level 1 to level 2 a student must have accrued at least 96 credits, provided he/she may not carry over more than three modules.

9.3 To proceed from level 2 to level 3 the student must have accrued at least 96 credits from the modules in level 2, provided he/she may not carry over more than three core modules to the next level.

9.4 To proceed from level 3 to level 4 the student must have satisfied the requirements for Work-Related Learning, including Work-Related Learning modules 1 and 2.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the University General Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Science Honours in Biotechnology and Bioinformatics, a student must pass each core module listed in the programme and must have accrued at least 480 credits.

11.2 The classification of all modules and the degree programme shall be as stated in Section 5 of the General Regulations.

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0 PROGRAMME STRUCTURE

Indicated in parenthesis are pre-requisite modules
Asterisk * denotes non-core modules

Code	Description	Credits
Level I Semester 1		
HBB 131	Introduction to Bioinformatics	12
HBB 135	Molecular and Cell Biology	12
*HBB 138	Analytical and Laboratory Techniques	12
HBB 139	Medicinal Chemistry	12
CS 131	Communication Skills	12
HCSCI 131	Introduction to Computers and Computer Applications	12
Level 1 Semester 2		
HBB 132	Programming Languages For Bioinformatics	12
HBB 133	Biocomputing	12
*HABB 134	Principles of Genetics and Evolution	12
HABB 136	Microbiology	12
HBB 137	Experimental Design and Data Analysis for Biologists 1	12
HBB 140	Principles of Biochemistry	12
Level 2 Semester 1		
HBB 231	Molecular Genetics	12
HBB 232	Immunology	12
HBB 243	Mathematical Methods for Bioinformatics	12
TCNP 201	Technopreneurship	12
GS 231	Gender Studies	12
*HBB 236	Introduction to Genomics and Proteomics	12
Level 2 Semester 2		
HABB 235	Recombinant DNA Technology	12
HBB 237	Biological Databases and Data Mining	12
HABB 248	Bioethics, Safety, Regulation and IPR	12
*HABB 239	Fermentation Biotechnology	12
HBB241	Molecular Modelling and Drug Design I	12
HBB 242	Introduction to Whole Genome Sequencing	12
Level 3 Semester 1		
HBB 330	Work-related Learning/Industrial Attachment I	40
Level 3 Semester 2		
HBB 331	Work-related Learning/Industrial Attachment II	80

Level 4 Semester 1

HABB 441	Experimental Design and Data Analysis for Biologists 2	12
HABB 442	Generic Skills for Biologists	12
HBB 443	Molecular Modelling and Drug Design II	12
HBB 461	Biological Databases and Data Mining II	12
HBB 462	Scripting Language and Website Development for Bioinformatics	12
HABB 446	Plant Biotechnology	12
	Choose at least ONE from the following electives:	
*HABB 445	Environmental Biotechnology	12
*HBB 449	Animal Biotechnology	12
*HABB 444	Gene Structure and Function	12

Level 4 Semester 2

HBB 460	Research Project	36
	Choose at least ONE from the following electives:	
*HBB 448	Pharmacogenomics & IPR	12
*HABB 447	Industrial Biotechnology	12

SYNOPSES

HBB 131 Introduction to Bioinformatics

What is bioinformatics and its relation with molecular biology? Examples of related Tools (FASTA, BLAST, BLAT, RASMOL), databases (GENBANK, Pubmed, PDB) and Software (RASMOL, Ligand Explorer). Data generation; Generation of large-scale molecular biology data. (Through Genome sequencing, Protein sequencing, Gel electrophoresis, NMR Spectroscopy, X-Ray Diffraction, and microarray). Applications of Bioinformatics.

HBB 132 Programming Languages for Bioinformatics

Programming basics. Programming in C: Pointers, pointers to functions, macro programming in C, graphs, data structure - linked list, stack, queue, binary trees, threaded binary trees. File and exception handling in C, Programming in Visual Basic: Introduction to Application Development using Visual Basic; Working with Code and Forms. Variables, Procedures and Controlling Program Executor; Standard Controls; Data Access Using Data Control; Connecting to Database using VB. Introduction to JAVA, variables, constants, control structures, input-output, and classes. Jar and Java applets. Introduction to Linux. Introduction to Perl, Perl subroutines and regular Expressions, Perl modules, BioPerl

HBB 133 Biocomputing
Introduction Meaning of sequence, sequence similarity, homology, meaning of alignment. Pairwise Sequence Alignment Different scoring models, Substitution matrices (PAM and BLOSUM), Pairwise Alignment: Concept of Global and Local Alignment, Dot matrix method, Dynamic programming (Needleman-Wunsch algorithm, Smith-Waterman algorithm, Choosing of best scoring matrix, gap penalties, Significance of score, EVD, FASTA and BLAST algorithms, Information theory and Shanon Entropy. Multiple Sequence alignment Multiple Sequence Alignment methods (MSA), Scoring of an MSA, Progressive (CLUSTALW and PILEUP), Iterative (Genetic) and Hidden Markov Model (HMM) methods of MSA, Local MSA (Profile and BLOCK analysis, and Pattern searching, and Expectation Maximization (EM) Algorithm (MEME) and Gibbs Sampler. Structural Alignment Tools and Protein Tertiary Structure Prediction Structure alignment algorithms & Homology modelling. Markov Chains and HMM Frequent words in DNA, Consensus word analysis, Transaction and emission matrix, Development of training set, CpG island prediction using HMM, Application of HMM in gene finding, and Multiple sequence alignment by HMM method. Phylogenetic Analysis Phylogenetic tree and terminology, different methods of Phylogenetic tree prediction: maximum parsimony, distance (UPGMA, NJ), maximum likelihood methods, bootstrapping, Jacknifing and Phylogenetic analysis by using Bayesian Network. RNA Structure Analysis Terminology of RNA secondary structure, inferring structure by comparative sequence analysis, RNA secondary structure prediction, Nussinov folding algorithm, energy minimization and Zuker folding algorithm.

HABB 134 Principles of Genetics and Evolution
Genetics: Mitosis and meiosis; Mendelian genetics and the chromosome theory of heredity; mutation and variation; introduction to population genetics. Evolution: population genetics and variation; the concept of, and evidence for evolution; theory of evolution by natural selection; adaptation and adaptive radiation; the nature of species; specific mate recognition systems; isolating mechanisms; speciation.

HABB 135 Cell and Molecular Biology
Microscopy, preparation of specimens for light and electron microscope; Cell theory; Prokaryotic and eukaryotic cell structures; Structure and function of different cell organelles; Cytoskeleton and cell motility; Interactions between cells and their environment; Traffic across membranes; Cell growth and division in the cell cycle; Cell differentiation; Cell signalling and Signal transduction; Cancer; Cytoplasmic membrane systems, Nature of the Gene and Genome, Molecular anatomy of genes and chromosomes.

HABB 136 Microbiology
The module explores microscopic life forms including bacteria, fungi and viruses. Emphasis is on the nature and behaviour of these microorganisms and their importance in manufacturing, agriculture, medical and food industries. Major topics include the history of microbiology, aseptic techniques, nutrient media, classes of microorganisms, microbial growth, procedures for characterization, and microbial energetics. Applications of microbiology in bioremediation, wastewater treatment, bio-fertilizers, animal-microbial symbiosis, and industrial microbiology, and control of microbial diseases. Practicals introduce the students to techniques involved in cultivation, enumeration and purification of microbes.

HABB 137 Experimental Design and Data Analysis for Biologists 1
Introduction: definitions, and uses of statistics (research, business, tourism, agriculture). Probability: multiplication law, addition law, condition probability, tree diagram, law of total probability. Probability distribution: random variables; discrete, continuous. Binomial distributions: normal distributions. Measures of central tendency and dispersion: mean, mode, range, variance, standard deviation, and standard error of the mean. Parametric and non-parametric statistic. Sampling techniques: simple, random, stratified, cluster, systematic. Data types, presentation and summarization techniques: tables, graphs, charts. Regression and correlation: regression parameters, correlation coefficient, coefficient of determ
ination. Simple statistical inference: hypothesis testing, confidence intervals, t-tests and chi-square tests. Practicals using appropriate statistical packages (e.g. SPSS, GENSTAT, SAS

HABB 138 Analytical and Laboratory Techniques
Laboratory safety practices and procedures. Standard Operating Procedures. Preparation of solutions - understanding of the terms molarity, molecular weight/mass, percentages, and basic units of volumes and weight. Principles, techniques and biological applications of Microscopy; Spectrophotometry; Electrophoresis; Centrifugation; Chromatography; Polymerase Chain Reaction. Basics of writing a scientific report. Practicals will develop competence in preparing stock solutions and accurate dilutions, accurate pipetting and using standard laboratory equipment, techniques in biochemical evaluation, techniques in molecular biology and statistics of instrumental analysis.

HBB139 Medicinal Chemistry
Pre-Regulatory Medicine and the Drug Approval Process: Early natural products and synthetic molecules used as drugs, Safety issues in the first drugs, Modern drug discovery and approval process, and Intellectual property issues with drugs. Drug Targets: Structure of proteins, Enzymes, enzymatic activity, and inhibition, Receptors and molecules that modulate their activity. Pharmacokinetics: Drug transport in the blood, Clearance, Volume of distribution, Compartment modelling. Metabolism: Types of metabolic reactions, Genetic and population effects, Prodrugs. Binding, Structure, and Diversity: Intermolecular forces, Drug-target complementarity, Molecular diversity and chemical libraries, Combinatorial chemistry. Lead Discovery: Drug screening, Filtering hits to find leads, Existing drugs and natural products as leads. Lead Optimization: Functional group replacements, Alkyl group replacements, Isosteres and bioisosteres. Peptidomimetic.

HBB 140 Principles of Biochemistry
Carbohydrates: classification, monosaccharide-structure, stereo isomers and structural isomers, and mutarotation. Oligosaccharides-Disaccharides -structure and importance of sucrose, lactose, maltose, cellobiose. Polysaccharides-structure and importance of homopolysaccharides and heteropolysaccharides. Lipids: definition and classification. Fatty acids: classification, nomenclature, structure and properties of saturated and unsaturated fatty acids. Essential fatty acids. Triacylglycerols: nomenclature, chemical properties and characterization of fats - hydrolysis, saponification value, acid value, rancidity of fats, Reichert-Meissel number. Biological significance of fats. Glycerophospholipids (lecithins, lysolecithins, cephalins. phosphatidylserine, phosphatidyl inositol, plasmalogens), sphingomyelins, glycolipids -cerebrosides, gangliosides. Amino acids: General properties, peptide bond, essential and non-essential amino acids. Protein chemistry: Classification, different levels of protein structure, forces stabilizing protein structure, and protein folding. Nucleic acids: Introduction, chemistry of nucleic acids, double helical structure and properties of DNA, RNA types, structure and functions. Vitamins: Introduction, properties, functions and deficiency diseases of fat soluble and water soluble Vitamins. Enzyme: Classification,

Nomenclature, Mechanism of enzyme action, derivation of Michaelis Menten equation, Enzyme inhibition, Factors affecting enzyme activity, Allorteric enzymes, Isoenzymes.

HABB 231 Molecular Genetics
Chemical nature of DNA, epilcation, transcription and RNA processing, and translation. Mutations and genetic basis of cancer. Control of gene expression in prokaryotes, and eukaryotes. Applications of molecular genetics in control of cancer, manufacturing of antibiotics and anti-retroviral drugs.

HABB 232 Immunology
Definition of immunology; Cellular participants in immune response; Types of immunity; Cytotoxic T lymphocytes; cytokines; Primary and secondary responses; Antigens and antigen processing; Structure and function of antibodies; Antibody and antigen interactions; Antibody antigen reactions; Theory of vaccination; Basis of cellular immunity; Theory of McFarlane Burnet; Monoclonal vs. polyclonal antibodies; AIDS and HIV- immunological basis.

HABB 235 Recombinant DNA Technology
This module provides a background to recombinant DNA technology. Emphasis will be on techniques used to multiply, control, detect, screen, alter, express and propagate the recombinant DNA in foreign host systems. Topics include the enzymology of recombinant DNA technology, hosts and vectors, microbial genetics, DNA sequencing, DNA amplification by PCR, RNA labelling, site-directed mutagenesis, isolation of DNA/RNA, construction of DNA and cDNA libraries, screening and characterization of recombinant DNA libraries and methods used, in vitro translation, Northern/Southern blot analysis, restriction mapping, DNA cloning and expression and the principles behind the analytical techniques used in recombinant DNA technology.

HBB 236 Introduction to Genomics and Proteomics
Genome Annotation, Genome Assembly, Structural and Functional Genomics.
Comparative Genomics. Metagenomics: Introduction, metagenome, shotgun metagenomics (pyrosequencing). Tools in metagenomics, MEGAN, MG- RAST, and SEED. Application: Gene survey, Enviornmental genomes, Microbial diversity. Microarray: technique, Design, Analysis, Drug target identification. Metabolic pathway database (KEGG pathway database), Concept of metabolome and metabolomics. Drug Discovery and Design: Target identification, Target Validation, Lead Identification, lead optimization, preclinical Pharmacology & Taxology, proteomics and transcriptomics

HBB 237 Biological Databases and Data Mining
Overview of database management- Advantages of database systems, the architecture of database systems, Levels of abstraction, data models- hierarchical network and relational models, and Entity elationships. Relational data models, relational algebra, SQL and commercial RDBMS-ORACLE, Views, Triggers, cursors, exceptions. Data independencies and Normalization-First normal form, second normal form, third normal form, Boyce-Codd normal form, security and integrity constraints. Biological databases: sequence databases, structural databases.

HABB 239 Fermentation Biotechnology
This module will cover alcoholic and non-alcoholic fermentation, industrial fermentation and

brewing industry, construction of yeast, genetic systems with greater fermentation efficiency, fermentation design and control as well as downstream processing.

HABB 248 Bioethics, Safety, Regulation and IPR

The module covers aspects of handling and disposal of hazardous biological and chemical substances of laboratory origin. Bioethics explores controversial issues brought about by advances in biotechnology and medicine. Ethical questions that arise in the relationships among life sciences, biotechnology, medicine, politics, law, and philosophy are addressed. The module mainly focuses on laboratory set-ups to contain possible epidemic situations when working with microbes, GMOs, and biohazards which become the epicentre of biosafety issues.

HBB 241 Molecular Modelling and Drug Design 1

Structure of proteins: Principles and Anatomy of proteins; Hierarchical organization of protein structure- Primary, Secondary, Super secondary, Tertiary, Quaternary structure, internal coordinates of Proteins: Theory, Derivation and significance of Ramachandran Plot. Structure of Nucleic Acids: DNA and RNA; Base pairing- Watson Crick and Hoogestein; Types of double helices A, B, Z and their geometrical as well as structural features; Types of RNA and geometrical parameters of each and their composition. Molecular Interactions: Protein-Protein interaction, Structural Analysis of domain interactions, Protein-DNA interactions, Types of interactions of DNA with proteins and small molecules. Protein structure prediction: Principles of protein structure prediction, Secondary structure prediction- Chou Fasman and GOR method. Homology Modelling- concepts, basic principles and protocol. Molecular modelling tool: SPDBV, Structure visualization tools such as Rasmol, Cn3D, VMD, MOLMOL, and Chime. Basic concepts in structural bioinformatics.

HBB 242 Introduction to Whole Genome Sequencing

The purpose of this module is to provide an overview of whole-genome sequencing and next-generation sequencing (NGS) technologies, including their historical, technical and utilization perspectives. Students will also be equipped with essential skills required to analyse next-generation sequencing data and carry out some of the most common types of analysis. Students will be exposed to the current best-practice workflows for Genome Assembling, Variant Calling, Trio Analysis, and Differential Expression Analysis as well as practical biological problems that motivate them.

HBB 243 Mathematical Methods for Bioinformatics

Introduction to sets, Set operations- union, intersection- complementation- symmetric difference, power sets (simple problems). Cartesian product- relations- functions, types of functions (definition and examples only). Matrices: types of matrices, addition and multiplication of matrices, the inverse of matrices, the rank of a matrix, non-singular and singular matrices, elementary transformations, the inverse of elementary transformations, and equivalent matrices. Mathematical logic: propositional calculus, basic logic operations- conjunction, disjunction, negation, conditional and bi-conditional, converse, inverse and contrapositive statements, tautologies, contradiction, equivalence and implications. Integral calculus: a quick review of indefinite integral as anti-derivative, the definite integral, the fundamental theorem of Calculus. Graph theory: definition, terminology, paths and circuits, representation of graphs, path matrix, adjacency matrix; exterior paths and circuits, Hamiltonian paths and circuits, spanning trees, minimum spanning trees. Markov models, Cluster analysis: hierarchical and non-hierarchical methods. Phylogenetic analysis tools: Maximum Likelihood, Parsimony methods.

TCNP 201 Technopreneurship

Introduction: nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur. Small business model and financial issues: developing a business model, basics of small business management, risks and stages of funding, sources of funding, financial funding for growth, product valuation, and how to form and register a small business in Zimbabwe. New Product Development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; the NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interactions. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HBB 330 Work-related Learning/Industrial Attachment I

This internship allows the student to gain experience working in a professional environment and to apply knowledge gained in the first two years to acquire professional skills or competencies. Students will be able to expand their professional network, convert academic knowledge into industrial skills, narrow down their potential careers and gain unforgettable life experiences.

The main objectives of the training module are:

i) To help the student put into practice the basic experimental and theoretical skills which were learnt,

ii) To provide exposure to and experience of the industrial environment.

iii) To give the student an insight into the production activities of Zimbabwe's main industries,

iv) To embark on a research project under the joint supervision of a member of the department and a member of the training institution.

The Industrial attachment will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience.

HBB 331 Work-related Learning/Industrial Attachment II

This internship allows the student to gain experience working in a professional environment and to apply knowledge gained in the first two years to acquire professional skills or competencies. Students will be able to expand their professional network, convert academic knowledge into industrial skills, narrow down their potential careers and gain unforgettable life experiences.

The main objectives of the training module are:

v) To help the student to put into practice the basic experimental and theoretical skills which were learnt,

vi) To provide exposure to and experience of the industrial environment.

vii) To give the student an insight into the production activities of Zimbabwe's main industries,

viii) To embark on a research project under the joint supervision of a member of the department and a member of the training institution.

The second part of the internship will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience.

HABB 441 Experimental Design and Data Analysis for Biologists 2

Importance of statistics in Biosciences. Hypothesis testing. Principles of Experimental Design. Types of experimental design - completely randomized design, randomized block design, nested designs, factorial design, repeated measures design, Before-After-Control-Impact studies in Ecology. Linear regression and correlation. Analysis of variance - one-way ANOVA, factorial ANOVA, repeated-measures ANOVA, random and fixed effects, checks for the assumptions of ANOVA/general linear models. Data Transformation. Non-parametric. Introduction to multivariate models - MANOVA for analysis of multiple response variables simultaneously; Principal Component and Factor Analysis for data reduction, Discriminate Function Analysis methods for discriminating between groups based on multiple variables. Competent use of at least one Statistical Software programme for data analyses.

HABB 442 Generic Skills for Biologists

Philosophy and ethics of Science. How to communicate effectively as a scientist. Effective reading, searching and evaluating literature. Students will receive training in scientific writing skills. Research and Grant proposal writing. GIS: An introduction to the use of GIS in life sciences, with the aim of providing the tools to access relevant spatial data sources, and construct basic maps of sampling sites etc.

HABB 443 Molecular Modelling and Drug Design

To understand the Modelling of small molecules; to understand the computational chemistry principles and to familiarize the role of computers in the drug-discovery process.

Concepts of Molecular Modelling, Molecular structure and internal energy, Application of molecular graphics, Energy minimization of small molecules, Use of Force Fields and MM

methods, Local and global energy minima. Techniques in MD and Monte Carlo. Simulation for conformational analysis, Ab initio, DFT and semi-empirical methods. Design of ligands, Drug-receptor interactions, Classical SAR/QSAR, Docking of Molecules; Role of computers in chemical research; Structure representation, SMILES; Chemical Databases, 2D and 3D structures, reaction databases, search techniques, similarity searches; Chemo-informatics tools for drug discovery.

HABB 444 Gene Structure and Function

Control of gene expression in prokaryotes; Lysis and lysogeny in bacteriophage Lambda, Control of pathways; Mating behaviour in yeast; Control of gene expression in higher eukaryotes- Gene amplification, Gene rearrangement and mobile units, Regulation at the transcriptional level, Post-transcriptional regulation, Processing transcripts, Differential mRNA processing, Changes in chromatin structure and gene regulation.

HABB 445 Environmental Biotechnology
Microbial diversity and its importance in biotechnology; Molecular methods for estimating the extent of microbial diversity; Environmental processes catalysed by microorganisms; Pollution control by microorganisms; Wastewater biotechnology; Bioremediation; Biomining; Environmental health and ecosystem sustainability e.g. bio-fuel production; Genetically modified organisms and the environment.

HABB446 Plant Biotechnology
Overview of plant biotechnology; Model plants; Plant genomes; Gene structure; Control of gene expression; The 'omics' era; Plant breeding and crop improvement; Plant tissue culture; Recombinant DNA technology; Plant transformation systems; Plants as bioreactors; Applications of plant biotechnology; Diseases and pest resistance, Resistance to abiotic factors, Crop modification: society, ethics and regulatory issues.

HABB 447 Industrial Biotechnology
Use of microorganisms as cell factories - examples of bio-based products on the market: enzyme, bio-ethanol, fine chemicals i.e amino acids, citric acid, vitamins etc; Fermentation; Bioreactor and Bioprocess design; High throughput screening for novel metabolites; Strain improvement; Gene discovery; Protein expression and production; Protein engineering; Use of recombinant microorganisms.

HBB 448 Pharmacogenomics & IPR
Bioinformatics companies, Genomes, transcriptomes and proteomes – their applications in medicine and agriculture, disease monitoring, and profile for therapeutic molecular targeting. Diagnostic drug discovery and genomics. Pharmacogenomics and its application. SNPs and their applications. Microarray and genome-wide expression analysis: Introduction to basic microarray technology, Bioinformatics in microarrays, Customised microarray design, Image processing and quantification, Normalization and filtering, Exploratory statistical analysis, Public. Microarray data resources. Patenting and data generation from patent literature for commercial benefits. IPR, and bioinformatics. Bioinformatics patents.

HBB 449 Animal Biotechnology
Structure of animal cell, history of animal cell culture, cell culture media and reagents, culture of mammalian cells, tissues and organs, primary culture, secondary culture, continuous cell lines, suspension cultures, somatic cell cloning and hybridization, transfection and transformation of cells, commercial-scale production of animal cells, application of animal cell culture for in vitro testing of drugs, testing of toxicity of environmental pollutants in cell culture, application of cell culture technology in production of human and animal viral vaccines and pharmaceutical proteins.

HABB 460 Research Project
The research project involves an experimental or observational investigation of a fundamental or practical problem in Applied Biosciences Biotechnology or bio-product development. With guidance from an academic supervisor, each student should choose and propose his/her own project theme. Each student is required to submit a proposal, carry out an independent research project or develop a product; submit a final written report, and deliver an oral presentation.

HBB 461 Biological Databases and Data Mining

To understand the biological databases – types and formats and to learn the retrieval, deposition and analysis of sequences and structures from biological databanks.

Data warehousing, data capture, data analysis; Introduction to Nucleic Acid and Protein Data Banks; Nucleic acid sequence data banks: Genbank, EMBL nucleotide sequence data bank. AIDS Virus sequence data bank, rRNA data bank, Protein sequence data banks: NBRF-PIR, SWISSPROT, Signal peptide data bank; Database. Similarity Searches. BLAST, FASTA, PSI-BLAST algorithms; Pairwise sequence alignment -NEEDLEMAN and Wunsch, Smith-Waterman algorithms; Multiple sequence alignments - CLUSTAL, Patterns, motifs and Profiles in sequences. Derivation and searching; Derived Databases of patterns, motifs and
profiles: Prosite, Blocks, Prints-S, Pfam, etc.; Primer Design.

HBB 462 Scripting Language and Website Development for Bioinformatics

Introduction to Internet and World Wide Web. An overview of scripting languages, with applications towards biological data and sequence analysis. Complexity of DNA problems and their computational implications and applications. Introduction to HTML, DHTML, XML. Accessing different objects of the HTML page, Dynamic page generation, Cascading Style Sheets (CSS). JAVASCRIPT: Document object model, Elements of the document object model, basic principles of JS, object-based programming using JavaScript; data types and structures, array and string handling, function implementations, XML: DTD, XML schemas, XML document structure, retrieving data from database in XML format; various bio-based versions of XML. PHP: PHP beginning to advanced level, data types, array and

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN BIODIVERSITY CONSERVATION FOR SUSTAINABLE DEVELOPMENT (MBSD)

Duration:	18 months
Actual Credit Load:	270
Minimum Load:	270
Maximum Credit Load:	288
Maximum MBKs Credit Load:	210
ZNQF level:	9

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competencies in the field of Biodiversity Conservation and Sustainable development relevant to various employment capabilities and careers in the world of work and society. To prepare students for further studies and lifelong learning Biodiversity Conservation and Sustainable development. To cultivate an entrepreneurial and innovative spirit in graduates of the programme.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry:	Normally a BSc Honours degree in Biological Sciences or Environmental Science or any related field with at least a 2.2 degree class.
Special Entry:	Applicants with a degree class of 3 but have considerable experience in a related industry will be considered
Mature Entry:	No provision for mature entry.

3.0 PROGRAMME CHARACTERISTICS

Areas of study:	Biodiversity Conservation, Molecular Biology, Sustainable Development, GIS and Remote sensing
Specialist Focus:	Biodiversity Conservation of Freshwater and Terrestrial Systems, Climate Change Impact Management, Sustainable development
Orientation:	Using Molecular Biological techniques to address challenges in Biodiversity Conservation
Distinctive Features:	Use of Molecular Biology to address Biodiversity conservation Problems

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Conservationists and Ecologists with Conservation organisations, Environmental Managers, Curators at natural history museums, Lecturers in

Technical colleges and universities, and Entrepreneurs in fisheries and related industries.

Further Studies: Doctoral studies in Biodiversity Conservation, Environmental Management, Sustainable Development

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, fieldwork, industrial visits, and dissertations

Assessment Methods: Written and oral examinations, tests, laboratory reports, seminar presentations,

research project, assignments

6.0 LEARNING OUTCOMES

1. Application of principles, strategies, and methods of biodiversity conservation to conserve and manage biodiversity
2. Competencies and skills to create new bio-business ventures (e.g. Consultancies in Biodiversity Management, Environmental Management)
3. Competencies and skills to develop and manage programmes for biodiversity conservation and ecological resources management
4. Effective communication of research findings using multi-media to both expert and non-expert audiences

7.0 GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for at least 9 modules including all compulsory modules.

7.2 At level 2, a student should register for the compulsory research project.

7.3 A student may include in his/her programme, modules offered by other Departments up to a total equivalent of at most 8 modules.

7.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons of all relevant Departments and to the following conditions:

7.4.1 The module combination is feasible in terms of timetable,

7.4.2 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and

7.4.3 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board may designate, in the special regulations for the programme, certain modules as core modules for the programme. Core modules are the modules, which are considered to be essential for qualification in the programme.

8.0 PROGRAMME ASSESSMENT

Coursework: Coursework shall contribute 40 % towards the final mark of each module and shall consist of Practicals (30%), Written assignments and/or tests (10%).

Written Examinations: The formal written examination shall contribute 60% to the overall assessment of a module.

Other: The Research Project is assessed on the basis of a Dissertation (85%) and oral presentation (15%).

9.0 PROVISION FOR PROGRESSION

9.1 In order to proceed from level 1 to level 2 a student must have attained at least 180 credits from the modules in that level including all the core modules.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Master of Science in Biodiversity Conservation and Sustainable Development, a student must pass each core module listed in the programme and must have accrued at least 270 credits.

11.2 The classification of all modules and the degree programme shall be as stated in Section 5 of the General Regulations.

12.0 DEGREE WEIGHTING

For the purpose of degree classification, weighting of the results of the different levels shall be:

Level 1 67%

Level 2 33%

13.0 PROGRAMME STRUCTURE

Code	Description	Credits
Level 1 Semester 1		
MBSD 731	Research Methods in Biodiversity Conservation	20
MBSD 732	Current Issues in Biodiversity Conservation	20
MBSD 733	Sustainable Development and Conservation	20
MBSD 734	Conservation Biology	20
MBSD 735	GIS and Remote Sensing in Biodiversity Conservation	20
Level 1 Semester 2		
MBSD 736	Systematics and Biodiversity	20
MBSD 737	Molecular Methods in Biodiversity	20
Choose TWO from the following electives:		
MBSD 738	Climate Change Impact and Mitigation	20
MBSD 739	Aquatic Pollution and Ecotoxicology	20

MBSD 740	Conservation Genetics	20
MBSD 741	Wildlife Parasitology	20
MBSD 742	Environmental Law and Policy	20

Level 2 Semester 1

MBSD 831	Research Project	90

14.0 SYNOPSES

MBSD 731 Research Methods in Biodiversity Conservation

Analytical and writing skills essential for careers in Ecology, Conservation and Environmental Management ranging from sampling & experimental design; exploratory data analysis; statistical inferences in regression analysis & univariate statistical analysis; multivariate techniques; estimating abundance in animal & plant populations; estimating community parameters; estimation of survival rates; database management, and scientific communication skills.

MBSD 732 Current Issues in Biodiversity Conservation

Introduction: definition of biodiversity. Current issues: Poverty, Limited attention to the management of Genetically Modified Organisms (GMOs) and Invasive Alien Species (IAS), Human-Wildlife conflict, exponential human population growth, habitat loss and fragmentation, poaching, knowledge gap, Inadequate research and development approaches for biodiversity initiatives, Limited and unsustainable funding for implementing biodiversity programmes, Corruption, Lack of political will, Blockage of migratory corridors, Climate change, Civil wars, Overexploitation of wildlife due to poaching and unregulated legal hunting, Habitat degradation and loss and associated risks of species extinction, Wildfires, Weak institutional and legal frameworks for carrying out biodiversity initiatives. Wildlife Diseases.

MBSD 733 Sustainable Development and Conservation

Introduction: meaning of Sustainable Development, history. Pillars of SD: environment, economy, and society. Principles of SD: conservation of ecosystem and biodiversity, production systems, population control, human resource management, conservation of progressive culture and people's participation. The sustainable development goals (SDGs). Key players: the United Nations, governments, private sector, and civil society organisations.

MBSD 734 Conservation Biology

Types of Biodiversity: genetic, species, ecosystem; alpha, beta, gama; Biodiversity of southern Africa. Importance of Biodiversity- Productive value, Consumptive value, Social value, Aesthetic value, Ethical value, Ecological value, Economic value, Uses of Biodiversity. Application of biological techniques to the specific problems of biodiversity, species preservation, ecological sustainability, and habitat fragmentation in the face of advancing human, social, economic, and industrial pressures. Social science dimensions of conservation and sustainable development, social economic, and political considerations in managing natural resources, policy-level aspects to project implementation. Strategies of Biodiversity conservation. Conservation methods: in situ vs ex-situ conservation.

MBSD 735 GIS and Remote Sensing in Biodiversity Conservation

This module focuses on the application of GIS and remote sensing in Biodiversity Conservation. Instruments, e.g. satellite sensors, provide data which can be used for studying

individual organisms, species assemblages or ecological communities on the ground. The image processing and GIS techniques developed to derive information from the captured satellite data are reviewed. Use of GIS and remote sensing techniques for mapping, monitoring and modelling organisms and their habitats.

MBSD 736 Systematics and Biodiversity
History of taxonomy and uses of classifications and phylogenies; Sources and handling of taxonomic data; Theory and practice of biodiversity assessment. Conceptual issues of biodiversity, analysis and practical application; Within country monitoring of biodiversity and conservation programmes; Analytical methods include diversity indices, species abundance models, species accumulation curves, comparative species richness and abundance.

MBSD 737 Molecular Methods in Analyses of Biodiversity
Principles of molecular genetics and their implications for molecular measurement of genetic diversity: DNA structure and organization, nuclear and cytoplasmic DNA, transcription and translation, protein structure, gene organization and its implication for sequence divergence, mutational mechanisms and mutation rate variation in different organisms and molecular markers and its implications in population genetic diversity, genetic diversity analyses of diploid and haploid genomes. Laboratory methods for DNA analyses: theoretical background for DNA isolation, detection methods, enzymatic reaction, restriction, ligation and DNA amplification: conditions, parameters of PCR, different PCR conventional strategies PCR and variation to conventional PCR, Polymerase errors and consequences, Primer design, Primer labelling, Southern and Northern blots, DNA Sequencing methods. Laboratory methods to detect genetic variation: a historical perspective from electrophoretic analysis of proteins to genomic analysis, the use of different marker systems using restriction enzymes: AFLPs and RFLPs, construction of microsatellite libraries; Mutation rates and evolutionary models of microsatellites, genotyping and applications; Development and genotyping of SNPs. The use of SNPs arrays, data analysis; Distinction between dominant and codominant markers. New developments in next generation sequencing and its potential use for evolutionary biology research: conceptual background of whole genome sequencing, laboratory workflow, Analysis principle, detection principle for different platforms, and potential applications for molecular evolutionary biology related studies using different genome downsizing strategies. RNA molecule based method: RNA extraction: methods and problems related to it; uses of RNA: gene characterization; gene expression analyses (microarray, qRT-PCR) RNA-seq (transcriptomics); fields of application of the above-mentioned techniques.

MBSD 738 Climate Change and Mitigation
The module addresses the main drivers of climate change and the challenges encountered in climate mitigation. The course addresses following issues: The greenhouse effect, global warming and the response of the climate system to anthropogenic emissions or land use changes. Possible consequences of climate change including on biodiversity, adaptation and associated costs. Biogeochemical cycles of carbon and other important greenhouse gases and biophysical properties of land-climate interactions. Analysis of the dynamics of emissions using scenario models (SSPs, RCPs) and quantification of climate change impacts and mitigation benefits (radiative forcing, temperature, CO2-equivalents). Technological strategies to reduce emissions (renewable energy, changes in lifestyle, negative emission technologies and solar radiation management) and land management strategies for climate change mitigation, their characteristics, costs, co-benefits and barriers. Climate mitigation status and experience in selected sectors, such as industry, transport, energy and households. Climate

policy instruments and framework conditions (Paris Agreements).

MBSD 739 Aquatic Pollution and Ecotoxicology
Introduction to ecotoxicology. Environmental contaminants. Contaminant fate and flow: bioavailability bioaccumulation and trophic transfer. Metabolism and excretion. Lethal effects-toxicity testing. Sublethal effects: biochemical, physiological, behavioural, reproductive. *In situ* biomonitoring. Effects on Populations. Ecological risk assessment. Case studies.

MBSD 740 Conservation Genetics
Application of principles from evolutionary and ecological genetics to conservation biology, fishery management, wildlife management, and aquaculture. Includes an overview of classical, molecular, population and quantitative genetics leading to an understanding of how managers can conserve genetic diversity and the evolutionary potential of natural and captive populations.

MBSD 741 Wildlife Parasitology
Fish parasites – protozoans, monogeneans, cestodes and nematodes, treatment procedures against fish parasites; water–borne diseases caused by protozoans and helminths; the relationship of adverse environmental factors and diseases caused by fish parasites; relationship between the severity of pollution in natural waters and parasitic diseases; fish health management in re-circulating aquaculture systems. Wildlife parasitology- diversity and complexity of host-parasite relationship in our Indigenous wildlife.

MBSD 742 Environmental Law and Policy
Sources of international law, General Principles of International Environmental Law. Principles and legislation governing, and techniques used in Environmental Management. The module content includes Definitions; Social Impact Assessment; Environmental management Policy (ISO 14000); Environmental Auditing; Product Life cycle assessment; Environmental Risk Management; and Environmental legislation in Zimbabwe.

MBSD 831 Research project
Each student will complete a one-semester research project in one of the following areas: Biodiversity Conservation, Environmental Management, Aquatic Ecology, Aquatic Toxicology, Fisheries Management, Environmental Biotechnology, Wildlife Parasitology, and Fisheries and Aquaculture. The research project involves an experimental or observational investigation of a fundamental or practical problem in Biodiversity Conservation, or bio-product development. With guidance from an academic supervisor, each student should choose and propose his/her own project theme. Each student is required to submit a proposal, carry out an independent research project or develop a product; submit a final written report, and deliver an oral presentation.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN ECOLOGICAL RESOURCES MANAGEMENT (MERM)

Duration:	18 months
Actual Credit Load:	270
Minimum Load:	270
Maximum Credit Load:	288
Maximum MBKs Credit Load:	210
ZNQF level:	9

1.0 PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competencies in the field of Biodiversity Conservation and Sustainable development relevant to various employment capabilities and careers in the world of work and society. To prepare students for further studies and lifelong learning Biodiversity Conservation and Sustainable development. To cultivate an entrepreneurial and innovative spirit in graduates of the programme.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry:	Normally a BSc Honours degree in Biological Sciences or Biodiversity Conservation or any related field with at least a 2.2 degree class.
Special Entry:	Applicants with a degree class of 3 but have considerable experience in a related industry will be considered
Mature Entry:	No provision for mature entry.

3.0 PROGRAMME CHARACTERISTICS

Areas of study: Development	Biodiversity Conservation, Molecular Biology, Sustainable
Specialist Focus:	Biodiversity Conservation of Freshwater and Terrestrial Systems, Climate Change Impact Management, Sustainable development, GIS and Remote Sensing
Orientation:	Using Molecular Biological techniques to address challenges in Biodiversity Conservation
Distinctive Features: Problems	Use of Molecular Biology to address Biodiversity conservation

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability: Conservationists and Ecologists with Conservation organisations, Environmental Managers, Curators at natural history museums, lecturers in Technical colleges, Universities, Entrepreneurs in fisheries and related industries.

Further Studies: Doctoral studies in Biodiversity Conservation, Environmental Management, Sustainable Development

5.0 PROGRAMME DELIVERY

Teaching and Learning Methods: Lectures, tutorials, laboratory classes, group work, field work, industrial visits, and dissertations

Assessment Methods: Written and oral examinations, tests, laboratory reports, seminar presentations, research project, assignments

6.0 LEARNING OUTCOMES

1. Application of principles, strategies, and methods of biodiversity conservation to conserve and manage biodiversity
2. Competencies and skills to create new bio-business ventures (e.g. Consultancies in Biodiversity Management, Environmental Management)
3. Competencies and skills to develop and manage programmes for biodiversity conservation and ecological resources management
4. Effective communication of research findings using multi-media to both expert and non-expert audiences

7.0 GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for at least 9 modules including all compulsory modules.

7.2 At level 2, a student should register for the compulsory research project.

7.3 A student may include in his/her programme, modules offered by other Departments up to a total equivalent of at most 8 modules.

7.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons of all relevant Departments and to the following conditions:

7.4.1 The module combination is feasible in terms of timetable,

7.4.2 The student may change his/her registration for modules to be taken in the second semester during the last week of the first semester or the first two weeks of the second semester and

7.4.3 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board may designate, in the special regulations for the programme, certain modules as core modules for the programme. Core modules are the modules, which are considered to be essential for qualification in the programme.

8.0 PROGRAMME ASSESSMENT

Coursework: Coursework shall contribute 40 % towards the final mark of each module and shall consist of Practicals (30%), Written assignments and/or tests (10%).

Written Examinations: The formal written examination shall contribute 60% to the overall assessment of a module.

Other: The Research Project is assessed on the basis of a Dissertation (85%) and oral presentation (15%).

9.0 PROVISION FOR PROGRESSION

9.1 In order to proceed from level 1 to level 2 a student must have attained at least 180 credits from the modules in that level including all the core modules.

10.0 FAILURE TO SATISFY EXAMINERS

Refer to Section 9 of the General Regulations.

11.0 GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Master of Science in Biodiversity Conservation and Sustainable Development, a student must pass each core module listed in the programme and must have accrued at least 270 credits.

11.2 The classification of all modules and the degree programme shall be as stated in Section 5 of the General Regulations.

2.0 DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	67%
Level 2	33%

13.0 PROGRAMME STRUCTURE

***Denotes a non-core module**

Code	Description	Credits
Level 1 Semester 1		
MERM 731	Scientific Research Methodology	20
MERM 732	Principles of Environmental Management	20
MERM 733	Conservation Biology	20
MERM 734	Systematics and Biodiversity	20
Choose ONE from the following electives:		
MERM 740	Aquaculture and Fisheries Management	20
MERM 741	GIS and Remote Sensing in Biodiversity Conservation	20
Level 1 Semester 2		
MERM 735	Environmental Biotechnology	20
MERM 736	Seminar and Review Paper	20

Choose TWO from the following electives:

MERM 737	Freshwater Resources Management and Environmental Sanitation	20
MERM 738	Wildlife Management	20
MERM 739	Fisheries and Wildlife Parasitology	20

Level 2 Semester 1

MERM 831	Research Project	90

Each student will complete one semester research in one of the following areas: Biodiversity and Conservation, Environmental Management, Aquatic Ecology, Fisheries Management, Environmental Biotechnology, Entomology, Animal Parasitology, and Aquaculture

14.0 MODULE SYNOPSES

MERM 731 Scientific Research Methodology
Analytical and writing skills essential for careers in Ecology, Conservation and Environmental Management ranging from sampling & experimental design; exploratory data analysis; statistical inferences in regression analysis & univariate statistical analysis; multivariate techniques; estimating abundance in animal & plant populations; estimating community parameters; estimation of survival rates; database management, and scientific communication skills.

MERM 732 Principles of Environmental Management
Principles and legislation governing, and techniques used in Environmental Management. The module content includes Definitions; Social Impact Assessment; Environmental management Policy (ISO 14000); Environmental Auditing; Product Life cycle assessment; Environmental Risk Management; and Environmental legislation in Zimbabwe.

MERM 733 Conservation Biology
Application of biological techniques to the specific problems of biodiversity, species preservation, ecological sustainability, and habitat fragmentation in the face of advancing human, social, economic, and industrial pressures. Social science dimensions of conservation and sustainable development, social economic, and political considerations in managing natural resources, policy-level aspects to project implementation.

MERM 734 Systematics and Biodiversity
History of taxonomy and uses of classifications and phylogenies; Sources and handling of taxonomic data; Theory and practice of biodiversity assessment. Conceptual issues of biodiversity, analysis and practical application; Within country monitoring of biodiversity and conservation programmes; Analytical methods include diversity indices, species abundance models, species accumulation curves, comparative species richness and abundance.

MERM 735 Environmental Biotechnology
Use of biotechnology in preventing and abating environmental pollution. It involves; an overview of biotechnology and the environment, environmental monitoring; and monitoring pollution. Water quality concept, Waste, Sewage and Sludge treatment. Waste minimization,

Bioremediation, Energy and environment, Natural Resources recovery, Biosafety and legislation, and Environmental Impact assessment.

MERM 736 Seminar & Review Paper
This module offers candidates an opportunity to sharpen their critical thinking skills through searching, analyzing, synthesizing and evaluating academic literature in the broad discipline of Biological Sciences. Additionally, candidates should demonstrate in-depth knowledge of a specific topic, and showcase his or her ability to identify research gaps. Candidates shall be assessed through an oral presentation and a review paper.

MERM 737 Freshwater Resources Management and Environmental Sanitation
Freshwater Resources Management and Environmental Sanitation Concepts will be developed and applied to conservation and management of aquatic ecosystems. Interbasin water Transfers, Ecological Reserve concept, Water demand management, introduced plant and animal species in southern Africa; Sources and causes of aquatic pollution.

MERM 738 Wildlife Management
Introduction to the wildlife of Zimbabwe - an overview of the present status of wildlife, and an understanding of both conservation problems and management solutions. Issues in wildlife management are exemplified using a broad range of vertebrate species occupying different environments. Introducing methods used in population pattern analysis, demographic analysis, threat and resource assessment, emphasizing the value of a coordinated and interdisciplinary approach to problem recognition and resolution.

MERM 739 Fisheries and Wildlife Parasitology
Fish parasites – protozoans, monogeneans, cestodes and nematodes, treatment procedures against fish parasites; water–borne diseases caused by protozoans and helminthes; the relationship of adverse environmental factors and diseases caused by fish parasites; relationship between the severity of pollution in natural waters and parasitic diseases; fish health management in re-circulating aquaculture systems. Wildlife parasitology- diversity and complexity of host-parasite relationship in our indigenous wildlife.

MERM 740 Aquaculture and Fisheries Management
Fisheries biology, ecology and habitats on global and regional (Southern Africa) scale. Principles of fisheries management in freshwater systems. Population sampling and estimation, statistical assumptions and robustness, length-frequency analysis, length-weight relationships, mark-recapture, growth and empirical modelling of populations; Larval Food Production and Larviculture, Algae/Fish Culture, Genetics and Nutrition, Health and Disease Control, Fish Farming Systems, Farm Management

MERM 741 GIS and Remote Sensing in Biodiversity Conservation
This module focuses on the application of GIS and remote sensing in Biodiversity Conservation. Instruments, e.g. satellite sensors, provide data which can be used for studying individual organisms, species assemblages or ecological communities on the ground. The image processing and GIS techniques developed to derive information from the captured satellite data are reviewed. Use of GIS and remote sensing techniques for mapping, monitoring and modelling organisms and their habitats.

MERM 831 Research project
The research project involves an experimental or observational investigation of a fundamental or practical problem in Applied Biosciences, Biotechnology, or bio-product development. With guidance from an academic supervisor, each student should choose and propose his/her own

project theme. Each student is required to submit a proposal, carry out an independent research project or develop a product; submit a final written report, and to deliver an oral presentation.

DEPARTMENT OF CHEMICAL SCIENCES

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN CHEMICAL TECHNOLOGY (HCT)

Duration:	4 years
Actual Credit load:	540
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

1.0 PURPOSE OF PROGRAMME

The programme aims to provide a broad range of chemical technologies that enable graduates to work effectively in industry and research and innovation institutes. There is an emphasis on the application of chemical knowledge and entrepreneurship. The programme also offers qualities and transferable skills necessary for employment requiring, the exercise of initiative and personal responsibility, decision-making in complex and unpredictable contexts and the learning ability needed to undertake appropriate further training of a professional or equivalent nature.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five 'O' Level subjects/National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry
In order to qualify for normal entry to a Bachelor of Science Chemical Technology Honours Degree, a prospective student must satisfy the Entry Requirements specified in the General Regulations and the Faculty Regulations.
In addition, the applicant must: Have obtained a pass at 'A' Level Chemistry and any one of the following subjects at 'A' Level; Mathematics, Physics, Computer Science, Biology, Food Science.

Special Entry
National Certificate, National Diploma, Higher National Diploma in relevant fields

Mature Entry
Five 'O' Level passes including Mathematics and English plus at least 25 years of age and 5 years of experience in the relevant field.

3.0 PROGRAMME CHARACTERISTICS

This degree programme is geared towards those students who envisage a career in chemical technology, applied chemical sciences, and related industries. The programme offers an opportunity for students to gain a foundation in the core chemistry concepts and related

technology and the manufacture of chemical products. The knowledge, skills and competencies attained give the individuals an opportunity to succeed in employment in a wide range of manufacturing and processing industries, consulting firms, government, research and educational institutions. The course exposes students to the development and design of chemical processes at different scales and creates a bridge between science and manufacturing, by applying the principles of chemistry, chemical technology and engineering to solve problems involving the production or use of chemicals. The programme also provides opportunities for learners to focus on the development of higher-level skills in a scientific and technological context, providing learners to develop a range of skill and attributes essential for successful performance in professional working life. The students are able to apply the methods and techniques that they have learned to review, consolidate, extend and apply their knowledge and understanding, and to initiate and carry out projects. The programme enables students to critically evaluate arguments, assumptions, abstract concepts and data to make judgements, and to frame appropriate questions to achieve a solution or identify a range of solutions to a problem and communicate information, ideas, problems and solutions to both specialist and non-specialist audiences.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities:-

– in chemical industry as technical business analysts, production managers (supervisors), product development chemists/managers, quality controllers, quality assurance managers

– in the public sector such as customs (chemical experts) environmental pollution control, industrial development organisations and industrial/research laboratories.

– in the Police Force as Forensic Scientists.

– as chemists in food, metallurgical, pharmaceutical, textile, paint, agrochemicals, plastics, rubber and allied, pulp and paper, mining industries and agriculture, water and effluent treatment plants.

–in research and development departments in industry and research institutes

–be an entrepreneur and an employer rather than an employee

–pursue Master of Science in Chemical Technology, Chemistry and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a four year period during which the student is engaged in a number of lectures, tutorials, practical investigations, field visits, work placements, research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

The programme hopes to achieve the following broad objectives:-

i. understand organic, inorganic, analytical and physical chemistry principles
ii. implement quality management systems in manufacturing industries and chemical laboratories
iii. apply chemical engineering concepts in the design of any chemical manufacturing process
iv. Develop new chemical technologies that enhance the efficiencies and outputs of the chemical industry
v. create new business ventures in the chemical industry.
vi. communicate effectively and present information using multi-media to both experts and non-experts in the chemical industry.
vii. develop skills and attitudes that enable graduates to participate actively in the development of the country
viii. produce professionals who can utilise natural resources (agriculture, mining, water, air) for the benefit of the country
ix. produce professionals who are innovative and capable of creating employment for themselves
x. positively impact on the life of the ordinary Zimbabwean

7.0 GENERAL PROVISIONS

Refer to Section 4 of the Faculty Regulations

8.0 PROGRAMME ASSESSMENT

Refer to Section 5 of the Faculty Regulations as well as the following:-

Continuous Assessment:	40% (Tests 15%, 20% practical, 5% assignments)
Written Examinations:	End of semester examinations contribute 60% of the final mark
Other:	a) The Research Project is assessed on the basis of a research project (75%), oral presentation (15%) and student conduct in the laboratory (10%). b) Work-Related Learning is assessed based on Work Related Learning report (30%), assessment by work supervisor (50%) and assessment by academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 7 of the Faculty Regulations

10.0 FAILURE TO SATIFTY EXAMINERS

Refer to section 9 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 10 of the Faculty Regulations

12.0 DEGREE WEIGHTING

Refer to Section 11 of the Faculty Regulations

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits
**HCT 131	Analytical Chemistry I		12
**HCT 132	Inorganic Chemistry I		12
**HCT 137	Mathematics for Chemists		12
*HCSCI 131	Introduction to Computers and Computer Applications		12
*CS 131	Basic Communication Skills		12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
**HCT 133	Organic Chemistry I		12
**HCT 134	Physical Chemistry I		12
**HCT 135	Laboratory Techniques		12
HCT 136	Material Chemistry		12
**HCT 138	Industrial Chemistry I		12
**HCT 139	Fundamentals of Chemical Engineering		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
**HCT 231	Analytical Chemistry II	**HCT 131	12

**HCT 232	Inorganic Chemistry II	**HCT 132	12
**HCT 235	Chemical Research Methods and Statistics		12
**HCT 236	Minerals Technology		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
**HCT 233	Organic Chemistry II	**HCT 133	12
**HCT 234	Physical Chemistry II	**HCT 134	12
HCT 237	Medicinal and Pharmaceutical Chemistry		12
**HCT 238	Industrial Chemistry II	**HCT 138	12
HCT 239	Cheminformatics and Drug Design		12
HCT 240	Quality Management Systems		12

Level 3 Semester 1

Code	Module Description	Credits
**HCT 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HCT 333	Work-Related Learning II	HCT 332	80

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
*HCT 431	Chemical Instrumentation	**HCT 231	12
*HCT 432	Inorganic Chemistry III	**HCT 232	12
HCT 435	Forensic Technology		12

**HCT 436	Nanotechnology	12
**HCT 437	Polymer Technology	12
**HCT 438	Environmental Management	12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
*HCT 433	Organic Chemistry III	**HCT 233	12
*HCT 434	Physical Chemistry III	**HCT 234	12
**HCT 439	Fuel Technology		12
**HCT 440	Research Project		36

14.0 MODULE SYNOPSES

HCT 131 Analytical Chemistry I
Ionic Equilibrium; Chemical composition of solutions; Chemical equations and stoichiometry; Elementary treatment of analytical data; Gravimetric Methods: Precipitation from homogeneous precipitation (PFHS); Volumetric (titrimetric) methods of analysis.

HCT 132 Inorganic Chemistry I
Atomic Structure; Electronic structure of atoms; Periodicity of atomic properties; Polyatomic molecules; Transition Metal Chemistry: Introduction to the magnetic properties of transitional metal complexes; Practical Inorganic Chemistry; Qualitative analysis.

HCT 133 Organic Chemistry I
Bonding and structure of organic reactions and their mechanisms; Stereochemistry; Projection formulae: Fischer, Sawhorse, Newman, D/L configurations, R/S systems, compounds with more than one chiral centre; Aromatic chemistry; Aromatic substitution reactions; Organic practicals: isolation of selected simple natural products from local plant materials.

HCT 134 Physical Chemistry I
Thermodynamics; Work; Heat and energy; First law of thermodynamics; Reversible and irreversible changes; Thermochemistry; Reaction kinetics; Electrochemistry; Thermodynamics of electrochemical cells; Ions in solution.

HCT 135 Laboratory Techniques
Sampling techniques; Separation and purification techniques; Concentration techniques; Stoichiometry; Common laboratory processes; Reporting analytical results; Interpretation of data; Keeping an organized lab notebook; Laboratory safety; Safety symbols; Materials safety data sheets; Handling of chemicals and their disposals; Strategies for procurement of chemicals; Good laboratory Practices (GLPs).

HCT 136 Material Chemistry

Structure, processing properties and performance. The range of engineering materials, structure of materials, growth of metal crystals, slip, dislocations, movement of dislocations. Structure of alloys: Gibbs phase rule, phase diagrams for binary alloys, properties and phase diagram, precipitation. Structure of non-metals, ceramics, structure of electric and magnetic materials, Materials of engineering: Ferrous alloys: Heat treatment of steels, Wood.

HCT 137 Mathematics for Chemists
The course is a survey of mathematical methods mostly for physical chemistry. Calculus: Differentials and Derivatives, Partial Derivatives, Integrals, Algebraic Functions, Transcendal Function; Coordinate Systems- Point in Space, Coordinate Systems for Molecules, Abstract Coordinate; Integration: Change of variables in integrand, Jacobian determinant, Gaussian integrals, Improper Integrals, Line Integrals; Numerical Methods: Interpolation, Numerical Differentiation, Random Numbers; Differential Equations in chemical reaction rates, thermodynamics, chemical reaction mechanism; Linear Algebra: Vector Spaces, Spaces of Functions, Matrices, Eigen Value Equations, Schrödinger's Equation; Fourier Analysis.

HCT 138 Industrial Chemistry I
Unit operations; Industrial finance; Industrial separation processes; Drying of solids; Industrial gases: Oxygen, Nitrogen, Argon, Helium, Hydrogen; Liquefied Natural Gases: Acetylene and acetylene chemistry (Reppe Chemistry), Carbon dioxide, Nitrous oxide, sulphur and sulphuric acid, Nitrogen compounds, synthetic fertilizers, pesticides, herbicides, sugar processing, food and beverages, soaps and detergents chlorine, (Weldon Process), Methane and methane products (synthesis gas, acetylene, HCN, chloromethane, CS_2, egg white); Cement: Wood: Paper Technology. Environmental impacts of industrial processes.

HCT 139 Fundamentals of Chemical Engineering
Definition of a chemical process; Process analysis and modelling; Physical quantities and process variables; Material balances with and without reactions; Single and multi-phase systems; Energy balances; Combined mass and energy balances; Process safety; Classification of unit operations; and Transport processes.

CS 131 Basic Communication Skills
Refer to Communication Skills Department regulations

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

HCT 231 Analytical Chemistry II
Introduction to Instrumental methods; Statistical treatment of analytical data; Spectroscopic methods of analysis; Chromatographic methods of analysis; Electroanalytical chemistry; and Practical.

HCT 232 Inorganic Chemistry II
Periodic trends of elements and compounds Groups I to VIII; Introduction to bonding in coordination compounds: valence bond theory, crystal field theory, ligand field theory and molecular orbital theory; Isomerism in coordination compounds; Jahn Teller distortions and geometry in complex compounds; Introduction to spectra Electron transfer reactions: inner

sphere mechanism and outer sphere mechanism; Practical inorganic chemistry: synthesis and characterization.

HCT 233 Organic Chemistry II
Introduction to organic spectroscopy; Synthetic Methods and Reaction mechanism; Formation of C-C bonds; Multi-centre reactions e.g. Wittig and Diels Alder reactions; oxidation reactions; Epoxidation of alkenes and ring opening; Reduction reactions; and Practical in organic chemistry.

HCT 234 Physical Chemistry II
Quantum Mechanics and Molecular Spectroscopy: Dawn of the Quantum Theory; Postulates of Quantum Mechanics; Molecular Spectroscopy; Raman spectra; General and Gas Phase Kinetics; Simple Collision theory of bimolecular reactions; Transition state theory; Unimolecular reactions; Lindermann mechanisms; RRKM Theory; Complex reactions; Unbranched chain reaction; and Explosion reactions.

HCT 235 Chemical Research Methods and Statistics
Understanding research: research methods and scientific methods; Ethics in research; Planning research proposal; Organizing data with descriptive statistics; Linear and nonlinear models; and Experimental designs.

HCT 236 Mineral Technology
Minerals and metallurgical Industry; Metallurgical Principles; Mineral dressing: Milling Size Control; Separation: Principal Exploitable characteristics, Refining, Specific industrial processes for the production of selected metals: iron, cobalt, nickel, sodium, copper, zinc, tin and manganese; Minerals processing operations: ore preparation, pre-blending, size reduction, separation and concentration, sorting, floatation, hydrometallurgy, pyrometallurgy and electrometallurgy; Fire Assaying; Calculation of gold and silver; Critical success factors; Limitations and limit of detection; Effect of impurities; and Analysis of the PGMS.

HCT 237 Medicinal and Pharmaceutical Chemistry
Introduction to biochemistry: important biological molecules (proteins, nucleic acids, carbohydrates, vitamins, enzymes and lipids) in the human body, medicinal drug metabolism, the search for drugs in plants, drug design and development. Selective toxicity, Strategies to select the plant material for study; main groups of natural products derived from plants; representative natural products derived from marine organisms, preparation of extracts and selection of active fractions, screening strategies; separation and purification of active components; benchtop bioassays, chemical assays and structure elucidation. Synthesis of some pharmaceutical products: analgesics, anti-allergy agents, antibacterial, antibiotics, antidepressants, and antiviral. Drug discovery, delivery, absorption and metabolism. Biomedical analysis, pharmacology, pharmacokinetics and pharmacodynamics, analytical techniques and quality assurance. Industrial preparation of drugs and marketing.

HCT 238 Industrial Chemistry II
Types of chemical industries: synthesis, processing, purification; Economics of industry: classification of costs, operating costs, direct and indirect expenses; Economic criteria and evaluation; Break-even point; investment criteria; Chemical conversions; Modelling and

process optimization: single multivariable optimization, constrained optimization, linear and nonlinear optimization, structural, profile optimization, solution of equations using optimization, search of global optimality; Choice of separators for homogeneous and heterogeneous mixtures: and distillation sequencing for azeotropic distillation, Catalysis and catalysts.

HCT 239 Cheminformatics and Drug Design
Introduction to principles and techniques used in cheminformatics as well as computational drug design especially in the area of Target Based Drug Design, Exploring protein and ligand databases, representation and manipulation of 2D structures, curation of chemical libraries, descriptor calculation, virtual screening and Machine Learning in drug design and analysis, prediction of ADMET properties, Introduction to tools and software in cheminformatics and drug design.

HCT 240 Quality Management Systems
Basic concepts, Statistical tools in Quality Control, Metrology, Inspection, Reliability and Dependability, New product quality, Quality specifications, Supplier relationship, Manufacture of quality, Customer relations, Quality Audit, Economics of Quality, Quality Data, Improving Quality, Human Factors in Quality, Quality Policy and Objectives, Quality Systems, Organisation of Quality, Quality Consulting, Quality Assurance in Chemical Laboratory, Total Quality Management Systems, Quality Control in Chemical Measurement Systems, Practical Work.

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven *vs* non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 INTRODUCTION TO GENDER STUDIES

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HCT 431 Chemical Instrumentation
Instrumental Method of analysis, Electrochemical methods, Modified voltammetric methods, Spectroscopic methods, Radiochemical methods, Chromatographic methods, Practical.

HCT 432 Inorganic Chemistry III
Further properties of transition metal compounds, reaction mechanisms, introduction to photochemical reactions, organometallic chemistry fundamental concepts. Practical Inorganic Chemistry.

HCT 433 Organic Chemistry III
Alicyclic Chemistry, Heterocyclic Chemistry, Further Aromatic Chemistry, Carbohydrate Chemistry

HCT 434 Physical Chemistry III
Ionic and Solution Kinetics: Kinetics and mechanism of thermal and photochemical reactions. Surface Chemistry: the liquid-air interface, the Kelvin equation liquid to liquid interface, the liquid-solid-air interface. Practical Physical Chemistry.

HCT 435 Forensic Technology
Investigating the Crime Scene. Investigating and Processing Physical Evidence, Trace evidence, Pattern evidence, Biological evidence, and Chemical evidence. Chemical toxicology, environmental toxicology, forensic aspects of toxicology, types of poisons, how poisons act, treatment of acute poisoning and control of poisonous material, Forensic anthropology.

HCT 436 Nanotechnology
This module covers the definition of nanomaterials, classification of nanomaterials, morphologies of nanomaterials, properties of nanomaterials relative to bulk materials, synthetic and fabrication approaches, characterisation of nanomaterials, uses/applications of nanomaterials, toxicological aspects of nanomaterials, current and emerging research in nanomaterials.

HCT 437 Polymer Technology
Polymerization methods; Bonding in polymers; Polymerization reactions and their mechanisms; Structure and stereochemistry; Polymerization practice/techniques; Molecular masses of polymers; Characterization of polymers; Ageing of polymers; Properties and applications of selected polymer; Polymer waste disposal and recycling; New trends in polymer chemistry; Practical polymer chemistry; Definition of terms used in plastic technology; Polymer structure; Testing of plastics; Plastics Rheology; Plastics and the environment; Polymer Production; Sources of pigments; Dyes; Inorganic and organic pigments; Analysis and design; Objectives of coatings surfaces; Preliminary treatment of surface; Classification of surface coatings; Paints and pigments-formulation; Composition and related properties; Oil paint; and Vehicle modified oils: Pigments, toners and lakes pigments, Fillers, Thinners, Enamels, emulsifying agents.

HCT 438 Environmental Management
Types of industrial wastes; solids and liquids, nuclear waste; Causes and effects of water, land and air pollution; Legislation on pollution; Industries responsibilities; Waste water treatment; Air pollution control technologies; Incineration of industrial wastes; Responsible use of landfills; Recycling; Sampling analysis and management of industrial effluent; Fate and transport of contaminants of emerging concern in air, water, solid waste, and soils; Investigate the fate, transport, and remediation of potentially harmful contaminants and their by-products;

Emerging contaminants; and International agreements (Montreal Protocol, Kyoto Protocol, Paris Agreement).

HCT 439 Fuel Technology

Fuel Chemistry: Review of energy sources both renewable and non-renewable; Classification of fuels and their calorific value; Coal: Uses of coal both fuel and non-fuel in various industries; Coal composition; Carbonization of coal; Coal gas, producer gas and water gas composition and uses; Fractionation of coal tar; Uses of coal tar bases chemicals; Requisites of a good metallurgical coke; Coal gasification (Hydro Gasification and Catalytic gasification); Coal liquefaction and Solvent Refining; Gas-To-Liquid (GTL) and Coal-To-Liquid (CTL) processes; Fischer-Tropsch process; Sasol process; Direct Coal Liquefaction: Bergius Process; Methanol-To-Gasoline Process (MTG); Petroleum and Petrochemical Industry: Composition of crude petroleum, Refining and different types of petroleum products and their applications; Ongoing or planned future projects; Renewable energy sources: Biomass & Biofuels overview, Biogas (anaerobic digesters), Bioenergy from wastes, dedicated bioenergy crops, Woody biomass, Liquid biofuels, (bioethanol, biooils), issues and future prospects, biodiesel-uses, production and process; Fuel cells, Transportation (hybrids, fuel cells, flexi-fuels); and Government, regional and international initiatives.

HCT 440 Research Project

Students carry out research projects on a topic of their choice concerning pertinent issues in chemical technology under the guidance of their supervisors.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN FORENSIC SCIENCE (HFSC)

Duration:	4 years
Actual Credit load:	540
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

With specializations in any of:
a) Forensic Biology and Forensic Biochemistry;
b) Forensic Biology and Forensic Chemistry;
c) Forensic Biochemistry and Forensic Chemistry, or
d) Forensic Physics and Forensic Chemistry

1.0 PURPOSE OF PROGRAMME

The programme aims to provide a broad range of forensic scientists capable of working in government and private sectors as well as research and innovation institutes. The programme also offers the exercise of initiative and personal responsibility, decision-making in complex and unpredictable contexts and the learning ability needed to undertake appropriate further training of a professional or equivalent nature as well as entrepreneurship skills in the field of forensic science.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five 'O' Level subjects/National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry

Pass at 'A' Level in two of the following subjects or their recognised equivalents:
Biology and Biochemistry Option: Biology and Chemistry
Chemistry and Biology Option: Biology and Chemistry
Chemistry and Biochemistry Option: Biology and Chemistry
Physics and Chemistry Option: Physics, Chemistry and Mathematics

Special Entry

Diploma in Forensic Science or equivalent qualification plus a minimum of 2 years relevant work experience.

3.0 PROGRAMME CHARACTERISTICS

Students specialise in either Forensic Biology and Biochemistry, Forensic Chemistry and Biology, Forensic Chemistry and Biochemistry or Forensic Physics and Chemistry. This degree programme is geared towards those students who envisage a career in forensic sciences

and related fields. The programme produces graduates specialised in forensic chemistry, forensic biochemistry, forensic physics or forensic biology. The programme offers an opportunity for students to gain a foundation in the core chemistry, physics, biology or biochemistry concepts and forensic technology. The knowledge, skills and competencies attained give the individuals an opportunity to succeed in employment as forensic scientists in government and private sector. The students are able to apply the methods and techniques that they have learned to review, consolidate, extend and apply their knowledge and understanding, and to initiate and carry out projects.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities:-

- In government departments and private sector as pathologists, serologists, DNA experts, toxicologists, forensic odontologists, forensic anthropologists, forensic psychologists, forensic entomologists, firearm examiners, bomb and arson experts, document and handwriting experts, and fingerprint experts.
- In the Police Service and Private investigations as Forensic Scientists. in research and development departments in industry and research institutes
- Entrepreneurship in the field of forensic science
- pursuit of Master of Science in Forensic Chemistry, Forensic Biology, Forensic biochemistry, Forensic Physics and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a four year period during which the student is engaged in a number of lectures, tutorials, practical investigations, field visits, work placements, research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

The programme hopes to achieve the following broad objectives:-

- Demonstrate competency in the collection, processing, analysis, and evaluation of evidence.
- Demonstrate competency in the principles of crime scene investigation, including the recognition, collection, identification, preservation, and documentation of physical evidence.
- Demonstrate an understanding of the scientific method and the use of problem-solving within the field of forensic science.
- Identify the role of the forensic scientist and physical evidence within the criminal justice system
- Demonstrate the ability to document and orally describe crime scenes, physical evidence, and scientific processes.
- Identify and examine current and emerging concepts and practices within the forensic science field
- Produce professionals who are innovative and capable of creating employment for themselves

7.0 GENERAL PROVISIONS

Refer to section 4 of the Faculty Regulations

8.0 PROGRAMME ASSESSMENT

Refer to section 5 of the Faculty Regulations as well as the following:-

Coursework:	40% (Tests 15%, 20% practical, 5% assignments)
Written Examinations:	End of semester examinations contribute 60% of the final mark
Other:	a) Research Project is assessed on the basis of a research project (75%), oral presentation (15%) and student conduct in the laboratory (10%). b) Industrial Attachment is assessed based on industrial attachment report (30%), assessment by work supervisor (50%) and assessment by academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 7 of the Faculty Regulations

10.0 FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 10 of the Faculty Regulations

12.0 DEGREE WEIGHTING

Refer to section 11 of the Faculty Regulations

13.0 PROGRAMME STRUCTURE

***Denotes core modules which are not Minimum Body of Knowledge and Skills**

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 131	Introduction to Forensic Science		12
*HFSC 132	Plant Diversity		12
**HFSC 133	Forensic Practical Techniques I		12

*HCSCI 131	Introduction to Computers and Computer Applications	12
*CS 131	Basic Communication Skills	12

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 131	Introduction to Forensic Science		12
**HFSC 134	Analytical Chemistry		12
**HFSC 133	Forensic Practical Techniques I		12
*HCSCI 131	Introduction to Computers and Computer Applications		12
*CS 131	Basic Communication Skills		12

Biology and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 131	Introduction to Forensic Science		12
**HFSC 132	Plant Diversity		12
**HFSC 134	Analytical Chemistry		12
*HCSCI 131	Introduction to Computers and Computer Applications		12
*CS 131	Basic Communication Skills		12

Physics and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 131	Introduction to Forensic Science		12
**HFSC 135	Mechanics		12
*HFSC 134	Analytical Chemistry		12
*HCSCI 131	Introduction to Computers and Computer Applications		12
*CS 131	Basic Communication Skills		12

Level 1 Semester 2

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 136	Statistics for Forensic Science I		12
HFSC 137	Cytology and Histology		12
**HFSC 138	Basic Entomology		12
**HFSC 139	Forensic Practical Techniques II		12
**HFSC 140	Forensic Genetics I		12
*HFSC 141	Cell Biology and Microbiology		12

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 136	Statistics for Forensic Science I		12
**HFSC 133	Forensic Practical Techniques I		12
*HFSC 140	Forensic Genetics I		12
**HFSC 142	Organic Chemistry I		12
HFSC 143	Materials Chemistry		12
*HFSC 144	Inorganic Chemistry		12

Biology and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 136	Statistics for Forensic Science I		12
**HFSC 142	Organic Chemistry I		12
HFSC 143	Materials Chemistry		12
**HFSC 137	Cytology and Histology		12
*HFSC 138	Basic Entomology		12
*HFSC 144	Inorganic Chemistry		12

Physics and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 136	Statistics for Forensic Science I		12
**HFSC 145	Waves and Optics		12
**HFSC 146	Fluid Mechanics and Thermal Physics		12
**HFSC 142	Organic Chemistry I		12
*HFSC 143	Materials Chemistry		12
HFSC 144	Inorganic Chemistry		12

Level 2 Semester 1

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 231	Crime/Incident Scene Investigation I		12
**HFSC 232	Forensic Botany I		12
**HFSC 233	Forensic Genetics II	*HFSC 140	12
**HFSC 234	Forensic Microbiology		12
*TCNP 201	Technopreneurship		12

*GS 231	Introduction to Gender Studies	12

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 231	Crime/Incident Scene Investigation I		12
**HFSC 233	Forensic Genetics II	*HFSC 140	12
**HFSC 235	Organic Chemistry II	**HFSC 142	12
**HFSC 236	Electrochemical and Chromatographic Methods I		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Biology and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 231	Crime/Incident Scene Investigation I		12
**HFSC 235	Organic Chemistry II	**HFSC 142	12
**HFSC 236	Electrochemical and Chromatographic Methods I		12
**HFSC 232	Forensic Botany I		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Physics and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 231	Crime/Incident Scene Investigation I		12
**HFSC 237	Electricity and Magnetism		12
*HFSC 235	Organic Chemistry II	**HFSC 142	12
**HFSC 236	Electrochemical and Chromatographic Methods I		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Level 2 Semester 2

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 238	Forensic Science Administration		12
**HFSC 239	Legal Aspects in Forensic Science		12
**HFSC 240	Forensic Entomology I		12
HFSC 241	Histology		12
HFSC 242	Immunology		12
*HFSC 243	Forensic Serology		12

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 238	Forensic Science Administration		12
**HFSC 239	Legal Aspects in Forensic Science		12
HFSC 242	Immunology		12
HFSC 243	Forensic Serology		12
*HFSC 244	Spectroscopic Methods of Analysis		12
**HFSC 245	Forensic Chemistry		12

Biology and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 238	Forensic Science Administration		12
**HFSC 239	Legal Aspects in Forensic Science		12
*HFSC 244	Spectroscopic Methods of Analysis		12
HFSC 245	Forensic Chemistry		12
**HFSC 240	Forensic Entomology I		12
HFSC 241	Histology		12

Physics and Chemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 238	Forensic Science Administration		12
**HFSC 239	Legal Aspects in Forensic Science		12
**HFSC 246	Forensic Physics I		12
**HFSC 247	Atomic and Nuclear Physics I		12
HFSC 244	Spectroscopic Methods of Analysis		12
HFSC 245	Forensic Chemistry		12

Biology and Biochemistry Specialisation

Level 3 Semester 1

Code	Module Description	Credits
**HFSC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HFSC 333	Work-Related Learning II	HFSC 332	80

Chemistry and Biochemistry Specialisation

Level 3 Semester 1

Code	Module Description	Credits
**HFSC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HFSC 333	Work-Related Learning II	HFSC 332	80

Biology and Chemistry Specialisation

Level 3 Semester 1

Code	Module Description	Credits
**HFSC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HFSC 333	Work-Related Learning II	HFSC 332	80

Physics and Chemistry Specialisation

Level 3 Semester 1

Code	Module Description	Credits
**HFSC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HFSC 333	Work-Related Learning II	HFSC 332	80

Level 4 Semester 1

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 431	Crime/Incident Scene Investigation II	**HFSC 231	12
**HFSC 432	Statistics for Forensic Science II	**HFSC 136	12
**HFSC 433	Forensic Entomology II	**HFSC 240	12

*HFSC 434	Applied Forensic Microbiology	**HFSC 234	12
HFSC 435	Clinical and Environmental Toxicology		12
**HFSC 436	Applied Forensic DNA analysis		12

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisite	Credits
**HFSC 431	Crime/Incident Scene Investigation II	**HFSC 231	12
**HFSC 432	Statistics for Forensic Science II	**HFSC 136	12
HFSC 435	Clinical and Environmental Toxicology		12
**HFSC 436	Applied Forensic DNA analysis		12
*HFSC 437	Environmental Chemistry		12
**HFSC 438	Electrochemical and Chromatographic Methods II	**HFSC 236	12

Biology and Chemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 431	Crime/Incident Scene Investigation II	**HFSC 231	12
**HFSC 432	Statistics for Forensic Science II	**HFSC 136	12
HFSC 437	Environmental Chemistry		12
**HFSC 438	Electrochemical and Chromatographic Methods II	**HFSC 236	12
**HFSC 433	Forensic Entomology II	**HFSC 240	12
*HFSC 434	Applied Forensic Microbiology	**HFSC 234	12

Physics and Chemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 431	Crime/Incident Scene Investigation II	**HFSC 231	12
**HFSC 432	Statistics for Forensic Science II	**HFSC 136	12
*HFSC 439	Mathematical Modelling		12
**HFSC 440	Forensic Science Instrumentation and Investigation Techniques I		12
HFSC 441	The Forensic Scientist in Court		12
**HFSC 442	Practical Applications in Forensic Science		12

Level 4 Semester 2

Biology and Biochemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 443	Forensic Botany II	HFSC 232	12
**HFSC 444	Forensic Toxicology		12
HFSC 445	Drug Metabolism		12
**HFSC 450	Research Project		36

Chemistry and Biochemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 444	Forensic Toxicology		12
HFSC 445	Drug Metabolism		12
**HFSC 446	Forensic Drug Analysis		12
**HFSC 450	Research Project		36

Biology and Chemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 446	Forensic Drug Analysis		12
**HFSC 443	Forensic Botany II	HFSC 232	12
HFSC 447	Forensic Microscopy		12
**HFSC 450	Research Project		36

Physics and Chemistry Specialisation

Code	Module Description	Prerequisites	Credits
**HFSC 448	Forensic Physics II	HFSC 246	12
**HFSC 449	Forensic Science Instrumentation and Investigation Techniques II	HFSC 440	12
HFSC 444	Forensic Toxicology		12
**HFSC 450	Research Project		36

14.0 MODULE SYNOPSES

Level 1 Semester 1

Biology and Biochemistry Specialisation

HFSC 131 Introduction to Forensic Science

Introduction to Forensic Science introduces the student to the principles and historical basis of forensic science and its application in legal practice. It consists of lectures, supervised field and laboratory practicals and tutorial sessions. The course is intended for BSc (Honours) Part I students in Forensic Science. It focuses on the securing and management of the incident scene, collection and handling of forensic evidence and its reporting. It lays the groundwork for all later courses in Forensic Science. The following topics are covered: History and Development of Forensic Science; Basic Deductive Reasoning; Organisation of a Crime Laboratory; Services of a Crime Laboratory; Functions of the Forensic Scientist; Forensics and Policing: Incidence Scenes (proper approach to crime and disaster scenes, their interpretation and preservation), Evidence Types (identity of articles of forensic value), Evidence Collection, (packaging, storage and transportation of articles from crime or disaster scene), Analysis (order of priority for analysis of articles; evidential or intelligence value of articles), Reporting (reports for criminal and investigation information). Upon completion of the course, students should be able to appreciate the role of forensic evidence in legal practice. They should be able to collect, secure and communicate forensic evidence.

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

HFSC 132 Plant Diversity
The course provides an understanding of plant biology through critical examination of facts and principles of organismal biology, and relates this to everyday life. It also lays the framework for specialized course options in plant biology. The content of this course will lay the foundation for later courses in plant biology and provide the basic understanding of plant biology. The course aims at developing students' understanding of plant biology through critical examination of facts and principles. Students will be expected to practice critical thinking and problem-solving skills, and learn theoretical and practical approaches to understanding origins, diversification and speciation of plants.

HFSC 133 Forensic Practical Techniques I
The main goal of this course is to impart laboratory practical techniques. Students will be taught good laboratory practice, approaches to biochemical investigations, preparation of buffer solutions, potentiometric and polarographic techniques, spectroscopic and immunochemical techniques. This course provides the student with the basic practical skills and experience in laboratory techniques used in science. The target students are those interested in pursuing careers in biochemistry, industry, forensics, research and government laboratories.

Chemistry and Biochemistry Specialisation

HFSC 131 Introduction to Forensic Science
Introduction to Forensic Science introduces the student to principles and historical basis of forensic science and its application in legal practice. It consists of lectures, supervised field and laboratory practicals and tutorial sessions. The course is intended for BSc (Honours) Part I students in Forensic Science. It focuses on the securing and management of the incidence scene, collection and handling of forensic evidence and its reporting. It lays the ground work for all latter courses in Forensic Science. The following topics are covered: History and Development of Forensic Science; Basic Deductive Reasoning; Organisation of a Crime Laboratory; Services of a Crime Laboratory; Functions of the Forensic Scientist; Forensics and

Policing: Incidence Scenes (proper approach to crime and disaster scenes, their interpretation and preservation), Evidence Types (identity of articles of forensic value), Evidence Collection, (packaging, storage and transportation of articles from crime or disaster scene), Analysis (order of priority for analysis of articles; evidential or intelligence value of articles), Reporting (reports for criminal and investigation information). Upon completion of the course, students should be able to appreciate the role of forensic evidence in legal practice. They should be able to collect, secure and communicate forensic evidence.

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

HFSC 134 Analytical Chemistry
Analytical Chemistryis a pure chemistry course that focuses on the chemical principles and techniques behind analysis of chemical compounds. It also includes the study of the statistical concepts valuable for validation of gathered chemical data. Analytical Chemistry is offered to Level 1 students pursuing the Bachelor of Science (Honours) degree in Forensic Science. It is a foundation for the study of advanced analytical chemistry concepts as used in all applied science disciplines.

HFSC 133 Forensic Practical Techniques I
The main goal of this course is to impart laboratory practical techniques. Students will be taught good laboratory practice, approaches to biochemical investigations, preparation of buffer solutions, potentiometric and polarographic techniques, spectroscopic and immunochemical techniques. This course provides the student with the basic practical skills and experience in laboratory techniques used in science. The target students are those interested in pursuing careers in biochemistry, industry, forensics, research and government laboratories.

Biology and Chemistry Specialisation

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Introduction to Forensic Science introduces the student to principles and historical basis of forensic science and its application in legal practice. It consists of lectures, supervised field and laboratory practicals and tutorial sessions. The course is intended for BSc (Honours) Part I students in Forensic Science. It focuses on the securing and management of the incidence scene, collection and handling of forensic evidence and its reporting. It lays the ground work for all latter courses in Forensic Science. The following topics are covered: History and Development of Forensic Science; Basic Deductive Reasoning; Organisation of a Crime Laboratory; Services of a Crime Laboratory; Functions of the Forensic Scientist; Forensics and Policing: Incidence Scenes (proper approach to crime and disaster scenes, their interpretation and preservation), Evidence Types (identity of articles of forensic value), Evidence Collection, (packaging, storage and transportation of articles from crime or disaster scene), Analysis (order of priority for analysis of articles; evidential or intelligence value of articles), Reporting (reports for criminal and investigation information). Upon completion of the course, students should be able to appreciate the role of forensic evidence in legal practice. They should be able to collect, secure and communicate forensic evidence.

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The course provides an understanding of plant biology through critical examination of facts and principles of organismal biology, and relates this to everyday life. It also lays the framework for specialized course options in plant biology. The content of this course will lay the foundation for later courses in plant biology and provide the basic understanding of plant biology. The course aims at developing students' understanding of plant biology through critical examination of facts and principles. Students will be expected to practice critical thinking and problem-solving skills, and learn theoretical and practical approaches to understanding origins, diversification and speciation of plants.

HFSC 134 Analytical Chemistry
Analytical Chemistry is a pure chemistry course that focuses on the chemical principles and techniques behind the analysis of chemical compounds. It also includes the study of the statistical concepts valuable for validation of gathered chemical data. Analytical Chemistry is offered to Level 1 students pursuing the Bachelor of Science (Honours) degree in Forensic Science. It is a foundation for the study of advanced analytical chemistry concepts as used in all applied science disciplines.

Physics and Chemistry Specialisation

HFSC 131 Introduction to Forensic Science
Introduction to Forensic Science introduces the student to principles and historical basis of forensic science and its application in legal practice. It consists of lectures, supervised field and laboratory practicals and tutorial sessions. The course is intended for BSc (Honours) Part I students in Forensic Science. It focuses on the securing and management of the incidence scene, collection and handling of forensic evidence and its reporting. It lays the ground work for all latter courses in Forensic Science. The following topics are covered: History and Development of Forensic Science; Basic Deductive Reasoning; Organisation of a Crime Laboratory; Services of a Crime Laboratory; Functions of the Forensic Scientist; Forensics and Policing: Incidence Scenes (proper approach to crime and disaster scenes, their interpretation and preservation), Evidence Types (identity of articles of forensic value), Evidence Collection, (packaging, storage and transportation of articles from crime or disaster scene), Analysis (order of priority for analysis of articles; evidential or intelligence value of articles), Reporting (reports for criminal and investigation information). Upon completion of the course, students should be able to appreciate the role of forensic evidence in legal practice. They should be able to collect, secure and communicate forensic evidence
CS 131 Basic Communication Skills
Refer to Communication Skills Department regulations

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

HFSC 135 Mechanics
The course introduces the students to basic physics principles and laws employed in solving problems in mechanics and practical applications.

HFSC 134 Analytical Chemistry
Analytical Chemistry is a pure chemistry course that focuses on the chemical principles and techniques behind the analysis of chemical compounds. It also includes the study of the statistical concepts valuable for the validation of gathered chemical data. Analytical Chemistry is offered to Level 1 students pursuing a Bachelor of Science (Honours) degree in Forensic Science. It is a foundation for the study of advanced analytical chemistry concepts as used in all applied science disciplines.

Level 1 Semester 2

Biology and Biochemistry Specialisation

HFSC 136 Statistics for Forensic Science I
This course introduces students to statistical techniques in a forensics context. The focus of the unit is the description and analysis of data with an emphasis on statistical methods used in forensic applications. Students are introduced to the statistical capabilities of Excel. Topics include graphs; measures of location and dispersion; probability; hypothesis testing; linear correlation and regression modelling; principal components analysis; Bayes Theorem and the prosecutors and defenders fallacies.

HFSC 137 Cytology and Histology
The target students are those interested in pursuing careers in forensics and pathology. This course introduces basic prokaryotic and eukaryotic cell structure, plasma membrane and membrane-bound organelles. It also introduces students to basic histology of epithelial and connective tissue, as well as equipping them with basic laboratory skills in tissue processing, microscopy, cytology and histology techniques.

HFSC 138 Basic Entomology
The purpose of this course is to introduce undergraduate students to the diverse field of entomology. The course meets the needs of undergraduate biology students who wish to pursue further studies in general and applied entomology, forensic entomology, medical and veterinary entomology. The primary aim of this course is to give the basic foundation needed by students pursuing further studies in forensic entomology.

HFSC 139 Forensic Practical Techniques II
The purpose of this course is to provide the student with advanced practical skills and experience in laboratory techniques used in Forensic science. The goal of this course is to impart laboratory practical skills to students and prepare them for good laboratory practice in biochemical and forensic investigation. Students will be introduced to advanced chromatographic techniques, spectroscopic, molecular and immunochemical analyses.

HFSC 140 Forensic Genetics I
The purpose of this course is to provide an in-depth background in all areas of classical Mendelian genetics, molecular genetics and population and evolutionary genetics. Students who successfully complete this course will gain sufficient basic knowledge to master advanced topics in genetic analysis for kinship, phylogeny and individualisation.

HFSC 141 Cell Biology and Microbiology
This course will introduce students to the structural differences between prokaryotes and eukaryotes. Microbial diversity will also be introduced, examining the similarities and differences between viruses (non-living), archaea and bacteria (prokaryotes), protists, fungi, helminths and algae (eukaryotes). In addition students will be introduced to the fundamental aspects of microbial genetics, growth and pathogenesis.

Chemistry and Biochemistry Specialisation

HFSC 136 Statistics for Forensic Science I
This course introduces students to statistical techniques in a forensics context. The focus of the unit is the description and analysis of data with an emphasis on statistical methods used in forensic applications. Students are introduced to the statistical capabilities of Excel. Topics include graphs; measures of location and dispersion; probability; hypothesis testing; linear correlation and regression modelling; principal components analysis; Bayes Theorem and the prosecutors and defenders fallacies.

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The purpose of this course is to provide the student with advanced practical skills and experience in laboratory techniques used in Forensic science. The goal of this course is to impart laboratory practical skills to students and prepare them for good laboratory practice in biochemical and forensic investigation. Students will be introduced to advanced chromatographic techniques, spectroscopic, molecular and immunochemical analyses.

HFSC 140 Forensic Genetics I
The purpose of this course is to provide an in-depth background in all areas of classical Mendelian genetics, molecular genetics and population and evolutionary genetics. Students who successfully complete this course will gain sufficient basic knowledge to master advanced topics in genetic analysis for kinship, phylogeny and individualisation.

HFSC 142 Organic Chemistry I
Organic Chemistry 1 is a sub discipline within Chemistry involving the study of carbon compounds, their structure, properties, composition, reactions, and preparations. Due to catenation these compounds are far more numerous than those of other elements. The course consists of 3 sections: Introduction to Stereochemistry which will cover various types of isomerism, projection formulae to R/S configuration; Systematic Aliphatic Chemistry; and basic Benzene Chemistry. Organic compounds are the basis of important constituents of plastics, drugs, petrochemicals, food, explosives, paints etc. They are the basis of living matter.

HFSC 143 Materials Chemistry
The course introduces the student to relationships among atoms, ions and molecules of a material and the bulk physical and structural properties of that material. The synthesis and characterisation of these materials comprises a significant part of the study. The course aims to develop the students' understanding of the different types of materials and their properties as well as to familiarise students with the synthesis and characterization of materials. Students' awareness of advanced materials and their use in modern technologies; as well as to encourage innovation through the design of new materials is expected at the end of the course.
HFSC 144 Inorganic Chemistry

Inorganic Chemistry 1 is a pure chemistry course that focuses on the properties of elements on the Periodic Table and their compounds, excluding those of carbon. It also includes the study of the Periodic Table itself and the general trends in periodic and group properties of the elements. Inorganic chemistry is a foundation for all applied disciplines.

Biology and Chemistry Specialisation

HFSC 136 Statistics for Forensic Science I
This course introduces students to statistical techniques in a forensics context. The focus of the unit is the description and analysis of data with an emphasis on statistical methods used in forensic applications. Students are introduced to the statistical capabilities of Excel. Topics include graphs; measures of location and dispersion; probability; hypothesis testing; linear correlation and regression modelling; principal components analysis; Bayes Theorem and the prosecutors and defenders fallacies.

HFSC 142 Organic Chemistry I
Organic Chemistry I is a sub-discipline within Chemistry involving the study of carbon compounds, their structure, properties, composition, reactions, and preparations. Due to catenation, these compounds are far more numerous than those of other elements. The course consists of 3 sections: Introduction to Stereochemistry which will cover various types of isomerism, projection formulae to R/S configuration; Systematic Aliphatic Chemistry; and basic Benzene Chemistry. Organic compounds are the basis of important constituents of plastics, drugs, petrochemicals, food, explosives, paints etc. They are the basis of living matter.

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The target students are those interested in pursuing careers in forensics and pathology. This course introduces basic prokaryotic and eukaryotic cell structure, plasma membrane and membrane-bound organelles. It also introduces students to basic histology of epithelial and connective tissue, as well as equipping them with basic laboratory skills in tissue processing, microscopy, cytology and histology techniques.

HFSC 138 Basic Entomology
The purpose of this course is to introduce undergraduate students to the diverse field of entomology. The course meets the needs of undergraduate biology students who wish to pursue further studies in general and applied entomology, forensic entomology, and medical and veterinary entomology. The primary aim of this course is to give the basic foundation needed by students pursuing further studies in forensic entomology.
Physics and Chemistry Specialisation

HFSC 144 Inorganic Chemistry
Inorganic Chemistry 1 is a pure chemistry course that focuses on the properties of elements on the Periodic Table and their compounds, excluding those of carbon. It also includes the study of the Periodic Table itself and the general trends in periodic and group properties of the elements. Inorganic chemistry is a foundation for all applied disciplines.

Physics and Chemistry Specialisation

HFSC 136 Statistics for Forensic Science I
This course introduces students to statistical techniques in a forensics context. The focus of the unit is the description and analysis of data with an emphasis on statistical methods used in forensic applications. Students are introduced to the statistical capabilities of Excel. Topics include graphs; measures of location and dispersion; probability; hypothesis testing; linear correlation and regression modelling; principal components analysis; Bayes Theorem and the prosecutors and defenders fallacies.

HFSC 145 Waves and Optics
This course introduces general concepts of wave propagation and optics. It covers the following major topics: Wave Concepts, Fraunhofer Diffraction, Huygen's-Fresnel Diffraction, Vector nature of Light and Polarization and Simple Optical Systems

HFSC 146 Fluid Mechanics and Thermal Physics
This introductory physics course addresses the following two major physics topics:
(1) Fluid Mechanics: Hydrostatic pressure, Buoyancy, Fluid flow continuity and Bernoulli's equation
(2) Thermal Physics: Temperature, Heat, Thermal properties of matter, the first law of thermodynamics, kinetic theory of gases, entropy and the second law of thermodynamics. This course provides a systematic introduction to the main principles of Fluid Mechanics and Thermal Physics and emphasizes the development of conceptual understanding and problem-solving ability in these aspects. It provides a foundation in physics for students in the life sciences, medicine, and some applied sciences, as well as other fields not directly related to science. Numerous research-based teaching strategies will be employed to facilitate student learning and to help students solve problems and relate to physical concepts. In addition, peer-tutoring and seminar presentations will allow students to work collaboratively to understand concepts and develop problem solving strategies.

HFSC 142 Organic Chemistry I
Organic Chemistry I is a sub discipline within Chemistry involving the study of carbon compounds, their structure, properties, composition, reactions, and preparations. Due to catenation these compounds are far more numerous than those of other elements. The course consists of 3 sections: Introduction to Stereochemistry which will cover various types of isomerism, projection formulae to R/S configuration; Systematic Aliphatic Chemistry; and basic Benzene Chemistry. Organic compounds are the basis of important constituents of plastics, drugs, petrochemicals, food, explosives, paints etc. They are the basis of living matter.

HFSC 143 Materials Chemistry

The course introduces the student to relationships among atoms, ions and molecules of a material and the bulk physical and structural properties of that material. The synthesis and characterisation of these materials comprise a significant part of the study. The course aims to develop the student's understanding of the different types of materials and their properties as well as to familiarise students with the synthesis and characterization of materials. Students' awareness of advanced materials and their use in modern technologies; as well as to encourage innovation through the design of new materials is expected at the end of the course.

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Level 2 Semester 1

Biology and Biochemistry Specialisation

HFSC 231 Crime/Incident Scene Investigation I
The purpose of the course is to explore the origins and main purposes of Rules of Evidence and its theoretical dimensions; impart a sound working knowledge of the practical application of this branch of law and its main principles and concepts; familiarize students with the general principles of rules of evidence with the defences to criminal liability; critically examine the nature and content of the law and to engage students in the contemporary debates relating to rule of evidence; explore ways of improving and strengthening this area of law using comparisons with systems of law in other countries
Biology and Biochemistry Specialisation

HFSC 232 Forensic Botany I
The course introduces the student to the use of botanical evidence in legal investigations. The course aims to raise students' awareness of the growing use and importance of botanical evidence (trichomes, fibres, pollen, plant tissues, seeds fruits, algae, phytoplankton, lichens and fungi) in modern forensic science; To provide students with practical experience of techniques involved in the enhancement, recovery and identification of forensically important plant material; To develop the student's understanding of the principles and applications of botanical information to legal practice; To encourage students to apply scientific principles in legal investigation

HFSC 233 Forensic Genetics II
The course gives students an in-depth understanding, experience and expertise in performing contemporary techniques used by forensic DNA testing laboratories. The targeted areas to be covered are population genetics, the legal aspects of DNA evidence, the ethics of DNA database, real-time quantitative PCR, detection and analyses of short tandem repeats (STR's) and mitochondrial DNA sequence data. The course objective is to give the student the fundamental tools to understand the scientific basis, detailed biochemistry, and current analytical methods for each technique. The course will also train the student in data analysis, quality assurance and quality control in forensic case file management.

HFSC 234 Forensic Microbiology
The course aims to give the student a broad overview of basic classical biochemical techniques used to isolate, identify and characterise microorganisms. Students will use the skills and knowledge acquired from the course in various laboratory setting in industry, medical labs, academia and research and development. The target students are those interested in pursuing careers in areas such as counter-bioterrorism and bio-piracy, law enforcement and laboratories.

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, and product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

Chemistry and Biochemistry Specialisation

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HFSC 235 Organic Chemistry II
The course introduces the student to carbon compounds by looking at their reactions, properties, structures, synthesis and compositions. The course covers stereochemistry and alicyclic chemistry, heterocyclic chemistry, further aromatic chemistry and rearrangement reactions. The above aspects are very important for understanding the chemistry of most chemical products at use in homes and industries. The course helps students to fully

comprehend the chemistry of many products in use such as drugs, petrochemicals and their derivatives, food and explosives. To achieve the above mentioned goal and be able to produce these chemical products in the right chemical form is the jurisdiction of the course.

HFSC 236 Electrochemical and Chromatographic Methods
Electrochemical and Chromatographic Methods serves to introduce student to the analytical techniques in chemistry. This course will cover the basic elements of electrochemistry. There will also be an introduction to separation methods of analysis mainly GC and HPLC.

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

Biology and Chemistry Specialisation

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Physics and Chemistry Specialisation

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Biology and Biochemistry Specialisation

HFSC 237 Electricity and Magnetism
This is a course to introduce students to the fundamentals of Electricity and Magnetism. The goal is for students to learn how to approach, solve, and understand a wide variety of physics problems on both qualitative and quantitative levels, and to relate "classroom physics" to real-world examples. Emphasis will be placed on conceptual understanding along with problem solving skills. The course will begin with a study of electromagnetism and the associated world-changing ideas and discoveries of Maxwell and Faraday. It will continue with the extension of these ideas which underpin our understanding of electromagnetic radiations/waves and conclude by examining the interaction of these radiations with matter and the associated quantum ideas that govern various processes. The goal of this course is to give a solid grounding in the core topics of classical electricity and magnetism, and to prepare students for further study in physics. The objective of the course is to improve the student's understanding of electric and magnetic phenomena, including basic circuits containing capacitors and inductors.

HFSC 235 Organic Chemistry II
The course introduces student to carbon compounds by looking at their reactions, properties, structures, synthesis and compositions. The course covers stereochemistry and alicyclic chemistry, heterocyclic chemistry, further aromatic chemistry and rearrangement reactions. The above aspects are very important for understanding the chemistry of most chemical products at use in homes and industries. The course helps students to fully comprehend the chemistry of many products in use such as drugs, petrochemicals and their derivatives, food and explosives. To achieve the above mentioned goal and be able to produce these chemical products in the right chemical form is the jurisdiction of the course.

HFSC 236 Electrochemical and Chromatographic Methods I
Electrochemical and Chromatographic Methods serves to introduce students to the analytical techniques in chemistry. This course will cover the basic elements of electrochemistry. There will also be an introduction to separation methods of analysis mainly GC and HPLC.

GS231 Introduction to Gender Studies
This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

Level 2 Semester 2

Biology and Biochemistry Specialisation

HFSC 238 Forensic Science Administration
Forensic science administration studies how to reduce error rates in forensic science by reorganizing forensic work. It studies the organization of forensics labour in the criminal justice system, using the tools of social science and business administration. Forensic science administration studies forensic science within its legal and political context. The forensic scientist does not work in isolation but has to interface with various segments of society, particularly law enforcement agencies. Forensic science students must also get a knowledge of the workings of a forensic unit and the key skills needed in forensic science administration. Students must also have an appreciation of how law enforcement agencies operate and interface and be guided on the importance of good ethics in forensic science. Forensic science laboratories' reputations have increasingly come under fire. Incidents of tainted evidence, false reports, allegations of negligence, scientifically flawed testimony, or perjury in
in-court testimony, have all served to cast a shadow over the forensic sciences.

HFSC 239 Legal Aspects in Forensic Science
The purpose of the course is to explore the origins and main purposes of Criminal Law in Zimbabwe and its theoretical dimensions; impart a sound working knowledge of the practical application of this branch of law and its main principles and concepts; familiarize students with the general principles of the criminal law with the defences to criminal liability; critically examine the nature and content of the law to engage students in the contemporary debates relating to the criminal law; explore ways of improving and strengthening this area of law using comparisons with systems of law in other countries. The goal of this course is to give students a full appreciation of criminal law in Zimbabwe upon which later courses will be built.

HFSC 240 Forensic Entomology I
Forensic entomology is the branch of forensic science concerned with the use of insects in legal matters. Its principal application is in death investigations where an assessment of insect succession on the body can be used to estimate the time of death. Other areas of its application include detection of exposure to drugs and toxins, and determination of the location of a crime. The course meets the needs of undergraduate biology students wishing to pursue postgraduate research, as well as those seeking careers as forensic science consultant entomologists. The primary aim of this course is to equip the student with knowledge of the types of insects which

are utilized in forensic investigations, their identification, biology and ecology, laboratory rearing techniques and curation. This knowledge makes it easier for the next stage of the degree programme where the focus is on the practical applications of entomology in forensic investigations.

HFSC 241 Histology
Through observation of the four basic tissue types, the way in which they are integrated to form organs and how organs are combined to form systems, this course will introduce the physical description of normal tissue at the light microscope level, while looking at research examples and techniques used to investigate abnormal tissues.

HFSC 242 Immunology
Students acquire knowledge of how adaptable the immune system in fighting pathogens and how pathogens try to evade the immune system. Knowledge on how the human body responds to foreign substances is important in forensic sciences.

HFSC 243 Forensic Serology
Bodily fluids are very important in criminal investigations as they can sometimes be the only evidence that a crime was committed. Bodily fluids are divided into two categories: excreted and secreted fluids. Excreted fluids include faeces, vomit and skin oil (perspiration), while secreted fluids include blood, plasma, semen, saliva and urine. These bodily fluids can be used to identify crime victims or suspects at crime scenes. To use bodily fluids as evidence for a crime, forensic scientists must first take samples of fluids present at crime scenes. To do this, forensic scientists swab, bag and collect the fluid samples in a vial to bring back to the crime lab. At the lab, the scientists run a series of tests on the fluid, first to determine if it is in fact a bodily fluid and if so, what type. If the bodily fluid is confirmed, further tests can identify whom the sample came from.

Chemistry and Biochemistry Specialisation

HFSC 238 Forensic Science Administration
Forensic science administration studies how to reduce error rates in forensic science by reorganizing forensic work. It studies the organization of forensics labour in the criminal justice system, using the tools of social science and business administration. Forensic science administration studies forensic science within its legal and political context. The forensic scientist does not work in isolation but has to interface with various segments of society, particularly law enforcement agencies. Forensic science students must also get a knowledge of the workings of a forensic unit and the key skills needed in forensic science administration. Students must also have an appreciation of how law enforcement agencies operate and interface and be guided on the importance of good ethics in forensic science. Forensic science laboratories' reputations have increasingly come under fire. Incidents of tainted evidence, false reports, allegations of negligence, scientifically flawed testimony, or perjury in
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HFSC 244 Spectroscopic Methods of Analysis
This course will introduce students to chemical instruments used in analytical chemistry, based on atomic spectroscopy and molecular spectroscopy. Spectroscopic Methods of Analysis is a Level 2 course and is required for the BSc Honours Degree in Forensic Science.

HFSC 245 Forensic Chemistry
This course approaches the challenges, methods and analyses of forensic science from a fundamental, chemical perspective. Topics include drug analysis, arson investigation, questioned document analysis, and the analysis of paint and gunshot residue samples. The course helps the student to develop approaches to understanding, correctly using and further developing current chemical tools that are used in the Forensic Sciences.

Biology and Chemistry Specialisation

HFSC238 Forensic Science Administration
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HFSC244 Spectroscopic Methods of Analysis
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HFSC240 Forensic Entomology I
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HFSC241 Histology
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Physics and Chemistry Specialisation

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HFSC246 Forensic Physics I
This course introduces the student to forensic applications of physics, via the study of selected topics. Students will study the physics behind investigative methods used to gather evidence and reconstruct crime events. Topics include Ballistics and physics underlying the motion of a bullet or other projectile, vehicular collisions and vehicular accident reconstruction, Bloodstain analysis and blood spatter analysis.

HFSC247 Atomic and Nuclear Physics I
This course will give an introduction to atomic and nuclear physics based on quantum mechanics. It will enable students to develop an understanding and knowledge of physical laws and principles of atomic and nuclear physics and their application to diverse areas of forensic science.

HFSC244 Spectroscopic Methods of Analysis
This course will introduce students to chemical instruments used in analytical chemistry, based on atomic spectroscopy and molecular spectroscopy. Spectroscopic Methods of Analysis is a Level 2 course and is required for the BSc Honours Degree in Forensic Science.

Level 4 Semester 1

Biology and Biochemistry Specialisation

HFSC 431 Crime/Incident Scene Investigation II
Crime scene management is the branch of forensic science concerned with the actions to be carried out at scenes of crime by investigators arriving at the scene first in order to protect the scene and preserve evidence. Therefore the purpose of this course is to take students through the process of crime scene management to ensure that evidence is not tempered with. The course meets the needs of undergraduate students wishing to pursue postgraduate research, and university teaching careers as well as those seeking careers as forensic scientists. The aim of this course is to introduce students to the theoretical and practical concepts of crime scene

management in forensic investigations, placing particular emphasis on its use in criminal investigations.

HFSC 432 Statistics for Forensic Science II
This course is an extension of Biometrics 1 which will provide further practical to analysis of biological data. Further statistical techniques will be covered, including linear regression, multiple regression, ANOVA and non parametric tests. Emphasis will be on how to choose appropriate statistical tests and how to assess statistical significance. To visualize data and carry out statistical testing, students will use R, a powerful programming language for statistical computing and graphics.

HFSC 433 Forensic Entomology II
In this course, students will learn the theory, concepts and practices of applying entomology in forensic investigations, placing particular emphasis on its use in criminal investigations. Upon successful completion of this course students will be able to apply different sampling methods for medico-criminal investigations, conduct analyses of arthropod faunal succession in carrion, to use Postmortem Interval estimation as entomological evidence, document entomological evidence at the scene of investigation, appreciate the applications of forensic entomology using case histories and understand the documentation and presentation of entomological evidence in court.

HFSC 434 Applied Forensic Microbiology
The course aims to give the student a broad overview novice molecular techniques used to characterize microbial metagenomes and microbial communities. The prerequisite for this course is Forensic Microbiology I. Student will use the skills and knowledge acquired from the course in various laboratories setting in industry, medical labs, academia and research and development. The target students are those interested in pursuing careers in areas such as counter bioterrorism and bio-piracy, law enforcement and laboratories.

HFSC 435 Clinical and Environmental Toxicology
This course examines basic concepts of environmental toxicology, including distribution, cellular penetration, metabolic conversion, and elimination of toxic agents, as well as the fundamental laws governing the interaction of foreign chemicals with biological systems. The course also focuses on the application of these concepts to the understanding and prevention of mortality and morbidity resulting from environmental exposure to toxic substances through a case study format. The target students are those interested in pursuing careers in clinical toxicology and environmental management.

HFSC 436 Applied Forensic DNA analysis
DNA is the chemical code that is found in every cell of an individual's body and is unique to each individual. Because it is unique, the ability to examine DNA found at a crime scene is a very useful forensic tool. It is important for forensic science students to develop a hands-on practical approach and to develop a real world appreciation of cases involving the use of DNA technologies. DNA is found in all living things and students need to develop an appreciation that its applications for Forensic science transcend human cases. Examples include the smuggling of illegal animal products; poaching of wildlife such as black rhinos and elephants; plant products suspected to have been stolen; and cattle & other livestock theft. Also included are paternity disputes in humans and other animals that could be of commercial importance.

Cases of intellectual property theft and patent violations can also find applications in DNA analysis.

Chemistry and Biochemistry Specialisation

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HFSC 437 Environmental Chemistry
This course will introduce students to issues pertaining the environment namely: Environmental chemistry and Green chemistry.

HFSC 438 Electrochemical and Chromatographic Methods II
This course will advance the student's knowledge of electrochemistry, which was introduced at Level 2. Mass spectrometry will also be introduced.

Biology and Chemistry Specialisation

HFSC 431 Crime/Incident Scene Investigation II
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HFSC 434 Applied Forensic Microbiology
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Physics and Chemistry Specialisation

HFSC 431 Crime/Incident Scene Investigation II
Crime scene management is the branch of forensic science concerned with the actions to be carried out at scenes of crime by investigators arriving at the scene first in order to protect the scene and preserve evidence. Therefore the purpose of this course is to take students through the process of crime scene management to ensure that evidence is not tempered with. The course meets the needs of undergraduate students wishing to pursue postgraduate research, university teaching careers as well as those seeking careers as forensic scientists. The aim of this course is to introduce students to the theoretical and practical concepts of crime scene management in forensic investigations, placing particular emphasis on its use in criminal investigations.

HFSC 432 Statistics for Forensic Science II
This course is an extension of Biometrics 1 which will provide further practical to analysis of biological data. Further statistical techniques will be covered, including linear regression, multiple regression, ANOVA and non-parametric tests. Emphasis will be on how to choose appropriate statistical tests and how to assess statistical significance. To visualize data and carry out statistical testing, students will use R, a powerful programming language for statistical computing and graphics.

HFSC 439 Mathematical Modelling
The overall goal of this course is to enable students to build mathematical models of real-world systems, analyze them and make predictions about behaviour of these systems. A variety of modelling techniques will be discussed with examples taken from physics, biology, chemistry, and other fields. The focus of the course will be on seeking the connections between mathematics and physical systems, studying and applying various modelling techniques to creating mathematical description of these systems, and using this analysis to make predictions about the system's behaviour.

HFSC 440 Forensic Science Instrumentation and Investigation Techniques I
This course introduces the principles of instrumentation and an overview of forensic laboratory instrumentation used in today's forensic laboratories. This course provides a systematic introduction to the main principles of electronic instrumentation and emphasizes the development of conceptual understanding and problem-solving ability in these aspects. Along with an overview of instrumentation principles, the physical principles and electrical characteristics for several common instrument transducers are studied. The electronic signal-conditioning circuits required to convert the electrical changes in the transducers to signal which can be interpreted accurately by a microprocessor or embedded controller, are analyzed and designed. Numerous research-based teaching strategies will be employed to facilitate student learning and to help students solve problems and relate to physical concepts. In addition, peer-tutoring and seminar presentations will allow students to work collaboratively to understand concepts and develop problem solving strategies.

HFSC 441 The Forensic Scientist in Court
The purpose of the course is to understand the Law of Criminal Procedure and its theoretical dimensions; impart a sound working knowledge of the practical application of this branch of law and its main principles and concepts; familiarize students with the general principles of Law of Criminal Procedure; critically examine the nature and content of the law and to engage students in the contemporary debates relating to Law of Criminal Procedure ; explore ways of improving and strengthening this area of the law using comparisons with systems of law in other countries.

HFSC 442 Practical Applications in Forensic Science
This course provides a thorough grounding in practical forensic science and prepares the student for employment. A notable feature of this course is the large amount of laboratory practical work associated with all of the scientific modules. Visits to external agencies, including forensic laboratories and courts of law, are incorporated, to give the student an opportunity to put learning into a practical context.

Level 4 Semester 2

Biology and Biochemistry Specialisation

HFSC 443 Forensic Botany II
The course introduces students to the analysis and interpretation of botanical evidence in forensic contexts and application of botanical evidence to legal and civil trials. The course aims to provide students with technical skills for preserving incident scenes; to develop students' skills and practical experience in analysing and interpreting botanical evidence for legal and civil procedures; to develop students' appreciation of the role of botanical traces evidence in crime investigation; to develop students practical skills in communicating evidence in court.

HFSC 444 Forensic Toxicology
This course examines fundamentals of forensic toxicology, including basic principles and procedures used in forensic toxicology, distribution, cellular penetration, metabolic conversion, and elimination of toxic agents, as well as the fundamental laws governing the interaction of foreign chemicals with biological systems. The target students are those interested in pursuing careers in forensic toxicology.

HFSC 445 Drug Metabolism
Upon completing this course, participants will have a basic knowledge of drug-drug interactions (DDI), bioactivation of foreign compounds and how they lead to toxicity, basic pharmacokinetics when a foreign compound interacts with tissue components. Students will learn about and understand the requirements for DME (absorption, distribution, metabolism, and elimination), pharmacokinetics (PK) and toxicokinetics (TK), free radical generation and drug metabolism (DM) and to compare and extrapolate metabolism profiles from animal models to humans (nonclinical and clinical evaluations). The target students are those interested in pursuing careers in biochemistry, pharmaceutical sciences, forensics and toxicology.

HFSC 450 Research Project
Students are required to do independent investigations of selected problems pertaining to thesis or dissertation in their chosen specialisation area. Students will be introduced to scientific research and will develop professional skills through forensic research, whilst extending their

knowledge by introducing them to current topics of research in forensic science as well as practice in the skills that are relevant to performing research. The research project is a core course, that is, a student has to pass the project in order to graduate. Generally, project proposals originate from the student or Department staff who will then supervise their respective projects. A written report on the project must be submitted by a specified deadline for assessment.

Chemistry and Biochemistry Specialisation

HFSC 444 Forensic Toxicology
This course examines the fundamentals of forensic toxicology, including basic principles and procedures used in forensic toxicology, distribution, cellular penetration, metabolic conversion, and elimination of toxic agents, as well as the fundamental laws governing the interaction of foreign chemicals with biological systems. The target students are those interested in pursuing careers in forensic toxicology.

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Upon completing this course, participants will have a basic knowledge of drug-drug interactions (DDI), bioactivation of foreign compounds and how they lead to toxicity, basic pharmacokinetics when a foreign compound interacts with tissue components. Students will learn about and understand the requirements for DME (absorption, distribution, metabolism, and elimination), pharmacokinetics (PK) and toxicokinetics (TK), free radical generation and drug metabolism (DM) and to compare and extrapolate metabolism profiles from animal models to humans (nonclinical and clinical evaluations).The target students are those interested in pursuing careers in biochemistry, pharmaceutical sciences, forensics and toxicology.

HFSC 446 Forensic Drug Analysis
This course provides an introduction to toxicological assays, including presumptive and confirmation testing in a variety of biological matrices, and a comprehensive pharmacological and analytical review of some major drug classes include alcohol, drugs for the treatment of mood disorders (anti-psychotics and anti-depressants), anxiolytics and hypnotics (barbiturates and benzodiazepines), anaesthetics, opioids, psychedelic drugs, amphetamine related compounds, cocaine, and cannabinoids. Other topics covered include the clandestine manufacture and profiling of illicit drugs, drug-facilitated sexual assault, and drugs in sport.

HFSC 450 Research Project
Students are required to do independent investigations of selected problems pertaining to thesis or dissertation in their chosen specialisation area. Students will be introduced to scientific research, and will develop professional skills through a forensic research, whilst extending their knowledge by introducing them to current topics of research in forensic science as well as practice in the skills that are relevant to performing research. The research project is a core course, that is, a student has to pass the project in order to graduate. Generally, project proposals originate from the student or Department staff who will then supervise their respective projects. A written report on the project must be submitted by a specified deadline for assessment.

Biology and Chemistry Specialisation

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HFSC 443 Forensic Botany II
The course introduces students to the analysis and interpretation of botanical evidence in forensic contexts and application of botanical evidence to legal and civil trials. The course aims to provide students with technical skills for preserving incident scenes; to develop students' skills and practical experience in analysing and interpreting botanical evidence for legal and civil procedures; to develop students' appreciation of the role of botanical traces evidence in crime investigation; to develop students practical skills in communicating evidence in court.

HFSC 447 Forensic Microscopy
The target students are those interested in pursuing careers in forensics, pathology and microscopy, as well as biomedical sciences. This course aims to impart knowledge on basic and advanced microscopy to assist in diagnostics, post-mortem pathological analysis and crime-scene investigation.

HFSC 450 Research Project
Students are required to do independent investigations of selected problems pertaining to a thesis or dissertation in their chosen specialisation area. Students will be introduced to scientific research, and will develop professional skills through a forensic research, whilst extending their knowledge by introducing them to current topics of research in forensic science as well as practice in the skills that are relevant to performing research. The research project is a core course, that is, a student has to pass the project in order to graduate. Generally, project proposals originate from the student or Department staff who will then supervise their respective projects. A written report on the project must be submitted by a specified deadline for assessment.

Physics and Chemistry Specialisation

HFSC 448 Forensic Physics II
This course provides an introduction to toxicological assays, including presumptive and confirmation testing in a variety of biological matrices, and a comprehensive pharmacological and analytical review of some major drug classes include alcohol, drugs for the treatment of mood disorders (anti-psychotics and anti-depressants), anxiolytics and hypnotics (barbiturates and benzodiazepines), anaesthetics, opioids, psychedelic drugs, amphetamine related compounds, cocaine, and cannabinoids. Other topics covered include the clandestine manufacture and profiling of illicit drugs, drug-facilitated sexual assault, and drugs in sport.

HFSC 449 Forensic Science Instrumentation and Investigation Techniques II
This course presents an overview of forensic laboratory instrumentation used in today's forensic laboratories and introduces the principles and analysis of crime scene materials using

a range of forensic instrumentation/equipment, including the physical principles and practical applications of electron microscopy and X-ray analysis, comparison and polarizing microscopy, chromatography, spectroscopy and chemical instrumentation/techniques. Students will develop the knowledge and skills to identify and process a number of examples of physical material evidence found at the crime scene. The science of dactyloscopy (fingerprinting) and blood spatter analysis will also be addressed. Specific application of these techniques for forensic investigation will be demonstrated, including the analysis of gunshot residues, metal alloys, indentation and scratch analysis, animal and human bone comparison, and paint chip and glass identification. Analysis of explosives by physics methods (Infrared Spectroscopy, Mass Spectrometry) will also be demonstrated. The course covers both the theoretical and practical aspects, using lectures and laboratory exercises.

HFSC 444 Forensic Toxicology
This course examines fundamentals of forensic toxicology, including basic principles and procedures used in forensic toxicology, distribution, cellular penetration, metabolic conversion, and elimination of toxic agents, as well as the fundamental laws governing the interaction of foreign chemicals with biological systems. The target students are those interested in pursuing careers in forensic toxicology. Drug Metabolism. Upon completing this course, participants will have a basic knowledge of drug-drug interactions (DDI), bioactivation of foreign compounds and how they lead to toxicity, basic pharmacokinetics when a foreign compound interacts with tissue components. Students will learn about and understand the requirements for DME (absorption, distribution, metabolism, and elimination), pharmacokinetics (PK) and toxicokinetics (TK), free radical generation and drug metabolism (DM) and to compare and extrapolate metabolism profiles from animal models to humans (nonclinical and clinical evaluations). The target students are those interested in pursuing careers in biochemistry, pharmaceutical sciences, forensics and toxicology.

HFSC 450 Research Project
Students are required to do independent investigations of selected problems pertaining to a thesis or dissertation in their chosen specialisation area. Students will be introduced to scientific research, and will develop professional skills through forensic research, whilst extending their knowledge by introducing them to current topics of research in forensic science as well as practice in the skills that are relevant to performing research. The research project is a core course, that is, a student has to pass the project in order to graduate. Generally, project proposals originate from the student or Department staff who will then supervise their respective projects. A written report on the project must be submitted by a specified deadline for assessment.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN CHEMICAL LABORATORY MANAGEMENT (HCLM)

Duration:	4 years
Actual Credit Load	540
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

1.0 PURPOSE OF THE PROGRAM

The program focuses on chemical laboratory management and provides career opportunities in laboratory-related occupations in a multitude of different industries and sectors such as environmental and mining. The program curriculum combines elements from industrial analytical chemistry and environmental testing while at the same time equipping students with the skills to design, implement, maintain and improve a quality management industrial laboratory. Modules will place emphasis on a combination of technical and management skills in quality control and assurance, data analysis, and instrumentation control and usage.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five 'O' Level subjects/National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry
In order to qualify for normal entry to a Bachelor of Science Chemical Laboratory Management Honours Degree, a prospective student must satisfy the Entry Requirements specified in the General Regulations and the Faculty Regulations.
In addition the applicant must: Have obtained a pass at 'A' Level Chemistry and any one of the following subjects at 'A' Level; Mathematics, Physics, Computer Science, Biology, Food Science.

Special Entry
National Diploma, Higher National Diploma in relevant fields

Mature Entry
Five 'O' Level passes including Mathematics and English plus at least 25 years of age and 5 years of experience in the relevant field.

3.0 PROGRAM CHARACTERISTICS

This degree program is geared towards those students who envisage a career in chemical laboratory management, chemical analysis, applied chemical sciences, and related industries. The program offers an opportunity for students to gain management skills in a chemical laboratory set-up as well as the technology used in chemical analysis. The knowledge, skills and competencies attained give the individuals an opportunity to succeed in employment in a

wide range of manufacturing and processing industries, consulting firms, government, research and educational institutions as laboratory managers.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the program graduates are open to the following opportunities;

- work as chemical analysts, quality controllers, research chemists, chemical laboratory managers and environmental analysts in relevant industries and government
- experts and consultants in industry and government institutions
- academia as researchers
- further education in laboratory management and related fields

5.0 PROGRAM DELIVERY

The program is delivered over a four year period during which the students are engaged in a number of lectures, tutorials, practical investigations, field visits, research projects, problem based learning, blended learning, student-led learning, seminar presentations and Work-Related learning.

6.0 LEARNING OUTCOMES

- Ability to create a healthy, safe, and productive workplace by appropriately hiring, training, supporting, and evaluating laboratory personnel
- Plan, organize, and direct the daily work activities of a laboratory setting by working independently and as a member of a team
- Communicate in a clear, well-organized manner that effectively persuades, informs, and clarifies ideas, information, and laboratory techniques and procedures to staff, the scientific community, and the public
- Practice ethical standards of integrity, honesty, and fairness as a laboratory manager
- Monitor and maintain laboratory-related documentation, equipment, and supplies necessary for conducting efficient, safe, cost-effective, and hygienic laboratory operations
- Manage scientific and laboratory practices and procedures by complying with and adhering to national, state, and local standards, policies, protocols, and regulations
- Acquire skills in assessing the risk, identifying the contaminants as well as mitigating pollution.

7.0 GENERAL PROVISIONS

Refer to section 4 of the Faculty Regulations

8.0 PROGRAM ASSESSMENT

Refer to section 5 of the Faculty Regulations as well as the following:-

Coursework:	40% (Tests 15%, 20% practical, 5% assignments)
Written Examinations:	End of semester examinations contribute 60% of the final mark
Other:	a) Research Project is assessed on the basis of a research project (75%), oral presentation (15%) and student conduct in the laboratory (10%). b) Industrial Attachment is assessed based on industrial attachment report (30%), assessment by work supervisor (50%) and assessment by academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION
Refer to section 7 of the Faculty Regulations

10. FAILURE TO SATISFY EXAMINERS
Refer to section 9 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION
Refer to section 10 of the Faculty Regulations

12.0 DEGREE WEIGHTING
Refer to section 11 of Faculty Regulations

13.0 PROGRAM STRUCTURE

***Denotes core modules which are not Minimum Body of Knowledge and Skills**

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits
**HCLM 131	Analytical Chemistry I		12
**HCLM 132	Inorganic Chemistry		12
**HCLM 137	Normative Standards I		12
*HCSCI 131	Introduction to Computers and Computer Applications		12
*CS 131	Basic Communication Skills		12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
**HCLM 133	Organic Chemistry		12
**HCLM 134	Physical Chemistry		12
**HCLM 135	Laboratory Techniques		12
**HCLM 136	Occupational Health and Safety		12
**HCLM 138	Industrial Chemistry I		12
HCLM 139	Fundamentals of Chemical Engineering		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
**HCLM 231	Analytical Chemistry II	**HCLM 131	12
**HCLM 232	Normative standards II	**HCLM 137	12
*HCLM 233	Chemical Research Methods and Statistic		12
**HCLM 234	Laboratory Quality Management		12
*TCNP 201	Technoprenuership		12
*GS 231	Introduction to Gender studies		12

Level 2 Semester 2

Code	Module Description		Credits
**HCLM 235	Performance Management		12
**HCLM 236	Industrial Chemistry II	**HCLM 138	12
HCLM 237	Mineral Technology		12
HCLM 238	Medicinal and Pharmaceutical Chemistry		12
**HCLM 239	Managing Operations and Processes		12
*HCLM 240	Finance for the non-financial managers		12

Level 3 Semester 1

Code	Module Description	Credits
**HCLM 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HCLM 333	Work-Related Learning II	**HIC 332	80

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
**HCLM 431	Management and Organizational Theory		12
**HCLM 432	Quality Management Systems		12
**HCLM 433	Chemical Instrumentation	**HCLM 231	12
HCLM 434	Forensic Technology		12
**HCLM 435	Planning and Decision Making		12
*HCLM 436	Project Management		12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
HCLM 437	Fuel Technology		12
**HCLM 438	Environmental Management		12
*HCLM 439	Green Chemistry		12
**HCLM 440	Research Project		36

14.0 MODULE SYNOPSES

HCLM 131 Analytical Chemistry
Ionic Equilibrium; Chemical composition of solutions; Chemical equations and stoichiometry; Elementary treatment of analytical data; Gravimetric Methods: Precipitation from homogeneous precipitation (PFHS); Volumetric (titrimetric) methods of analysis.

HCLM 132 Inorganic Chemistry
Atomic Structure; Electronic structure of atoms; Periodicity of atomic properties; Polyatomic molecules; Transition Metal Chemistry: Introduction to the magnetic properties of transitional metal complexes; Practical Inorganic Chemistry; Qualitative analysis.

HCLM 137 Normative standards 1

Students are introduced to the Normative Standards which are the internationally accepted documents that describe the current state of the art and science on how to manage and operate a modern analytical testing laboratory. The purpose of this module is to introduce and study these normative documents and Standard Operating Procedures for the purposes of managing laboratories.

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

CS 131 Basic Communication Skills
Refer to Communication Skills Department regulations

HCLM 133 Organic Chemistry
Bonding and structure of organic reactions and their mechanisms; Stereochemistry; Projection formulae: Fischer, Sawhorse, Newman, D/L configurations, R/S systems, compounds with more than one chiral centre; Aromatic chemistry; Aromatic substitution reactions; Organic practicals: isolation of selected simple natural products from local plant materials.

HCLM 134 Physical Chemistry
Thermodynamics; Work; Heat and energy; First law of thermodynamics; Reversible and irreversible changes; Thermochemistry; Reaction kinetics; Electrochemistry; Thermodynamics of electrochemical cells; Ions in solution.

HCLM 135 Laboratory Techniques
Sampling techniques; Separation and purification techniques; Concentration techniques; Stoichiometry; Common laboratory processes; Reporting analytical results; Interpretation of data; Keeping an organized lab notebook; Laboratory safety; Safety symbols; Materials Safety Data Sheets; Handling of chemicals and their disposals; Strategies for procurement of chemicals; Good laboratory Practices (GLPs).

HCLM 136 Occupational Health and Safety
Introduction to occupational health; Hazardous installations; Emergency exposure limits; Accidents causation and prevention; Stressors: Identification, evaluation and control: Physical (noise, heat, barometric, illumination), chemical, egonomical, biological, psychological; Risk assessment and management; Emergency preparedness and response; Job safety analysis (HAZOP: Hazard Operatability); Confined space management; Fire; Safety communication; Safety audit; Safety legislation; Safety systems (NOSA, OSHA18000); Practical; Threshold limit values; Safety Programme management; ILO Conventions; Occupational epidemiology and statistics; Practical and occupational hygiene fieldwork; Occupational hygiene legislation.

HCLM 138 Industrial Chemistry I
Unit operations; Industrial finance; Industrial separation processes; Drying of solids; Industrial gases: Oxygen, Nitrogen, Argon, Helium, Hydrogen; Liquefied Natural Gases: Acetylene and acetylene chemistry (Reppe Chemistry), Carbon dioxide, Nitrous oxide, sulphur and sulphuric acid, Nitrogen compounds, synthetic fertilizers, pesticides, herbicides, sugar processing, food and beverages, soaps and detergents chlorine, (Weldon Process), Methane and methane products (synthesis gas, acetylene, HCN, chloromethane, CS_2, egg white); Cement: Wood: Paper Technology. Environmental impacts of industrial processes

HCLM 139 Fundamentals of Chemical Engineering

Definition of a chemical process; Process analysis and modelling; Physical quantities and process variables; Material balances with and without reactions; Single and multi-phase systems; Energy balances; Combined mass and energy balances; Process safety; Classification of unit operations; and Transport processes.

HCLM 231 Analytical Chemistry II
Introduction to Instrumental methods; Statistical treatment of analytical data; Spectroscopic methods of analysis; Chromatographic methods of analysis; Electroanalytical chemistry. The module covers the theory and methodology for the technical analysis.

HCLM 232 Normative Standards II
Building on Normative Standards I, this module explores new items in the language of analytical chemistry. Students use documents for laboratory audits, assessments and management reviews of the laboratory quality system in small groups, discussing and formulating their own documents and Standard Operating Procedures for the purposes of managing a quality industrial laboratory. Students attend a local laboratory to discuss current practices with industry professionals.

HCLM 233 Chemical Research Methods and Statistics
Understanding research: research methods and scientific methods; Ethics in research; Planning research proposal; Organizing data with descriptive statistics; Linear and nonlinear models; and Experimental designs.

HCLM 234 Laboratory Quality Management
Quality Assurance in Chemical: The Chemical Laboratory: Quality Systems and Quality Assurance, Total Quality Management Systems. (TQM). Quality Control in Chemical Measurement Systems: Methods selection and validation, Equipment calibration and maintenance, Estimation of measurement uncertainty, Data handling, Quality Assessment of Chemical measurement systems

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven *vs* non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interactions. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HCLM 235 Performance Management
Managing individual performance, poor performance, performance appraisal versus performance management, rewarding performance, ethics of managing performance.

HCLM 236 Industrial Chemistry II
Types of chemical industries: synthesis, processing, purification; Economics of industry: classification of costs, operating costs, direct and indirect expenses; Economic criteria and evaluation; Break-even point; investment criteria; Chemical conversions; Modelling and process optimization: single multivariable optimization, constrained optimization, linear and nonlinear optimization, structural, profile optimization, solution of equations using optimization, search of global optimality; Choice of separators for homogeneous and heterogeneous mixtures: and distillation sequencing for azeotropic distillation, Catalysis and catalysts.

HCLM 237 Mineral Technology
Minerals and metallurgical Industry; Metallurgical Principles; Mineral dressing: Milling Size Control; Separation: Principal Exploitable characteristics, Refining, Specific industrial processes for the production of selected metals: iron, cobalt, nickel, sodium, copper, zinc, tin and manganese; Minerals processing operations: ore preparation, pre-blending, size reduction, separation and concentration, sorting, floatation, hydrometallurgy, pyrometallurgy and electrometallurgy; Fire Assaying; Calculation of gold and silver; Critical success factors; Limitations and limit of detection; Effect of impurities; and Analysis of the PGMS.

HCLM 238 Medicinal and Pharmaceutical Chemistry
Introduction to biochemistry: important biological molecules (proteins, nucleic acids, carbohydrates, vitamins, enzymes and lipids) in a human body, medicinal drug metabolism, the search for drugs in plants, drug design and development. Selective toxicity, Strategies to select the plant material for study; main groups of natural products derived from plants; representative natural products derived from marine organisms, preparation of extracts and selection of active fractions, screening strategies; separation and purification of active components; benchtop bioassays, chemical assays and structure elucidation. Synthesis of some pharmaceutical products: analgesics, anti-allergy agents, antibacterial, antibiotics, antidepressants, and antiviral. Drug discovery, delivery, absorption and metabolism. Biomedical analysis, pharmacology, pharmacokinetics and pharmacodynamics, analytical techniques and quality assurance. Industrial preparation of drugs and marketing

HCLM 239 Managing Operations and Processes
The context in which the laboratory operates; the role of the laboratory in achieving the organization's strategic goals. The formulation of the laboratory strategy; establish what the competitive advantage of the laboratory in which the person works, is. The operational management process in the laboratory. The potential inefficiencies in the process, the impact on productivity and how to plan to address the inefficiencies identified in the process flow. The value of delegation in managing and monitoring the plan to address the gap identified in the process flow.

HCLM 240 Finance for the non-financial managers
Development of the financial skills needed by functional experts in human resources, marketing, production, and general management. The objective is to interpret finance and accounting documents and apply that information to sound business decision-making. Topics include financial statements and forecasting, capital budgeting, project evaluation, working

capital management, stocks and bonds, time value of money, and international financial management. Emphasis is on practical applications to facilitate informed discussions with business professionals for financial decision-making. Students may receive credit for only one of the following courses

HCLM 431 Management and Organizational Theory
An examination of the four functions of management; planning, organizing, leading, and controlling; with emphasis on the application of management concepts and theories to achieve organizational goals. The aim is to develop strategies, goals, and objectives to enhance performance and sustainability. Topics include pollution, social responsibility, globalization, and change and innovation.

HCLM 432 Quality Management Systems
Basic concepts, Statistical tools in Quality Control, Metrology, Inspection, Reliability and Dependability, New product quality, Quality specifications, Supplier relationship, Manufacture of quality, Customer relations, Quality Audit, Economics of Quality, Quality Data, Improving Quality, Human Factors in Quality, Quality Policy and Objectives, Quality Systems, Organization of Quality, Quality Consulting.

HCLM 433 Chemical Instrumentation
Instrumental Method of analysis, Electrochemical methods, Modified voltammetric methods, Spectroscopic methods, Radiochemical methods, Chromatographic methods, Practical.

HCLM 434 Forensic Technology
Investigating the Crime Scene. Investigating and Processing Physical Evidence, Trace evidence, Pattern evidence, Biological evidence, Chemical evidence. Chemical toxicology, environmental toxicology, forensic aspects of toxicology, types of poisons, how poisons act, treatment of acute poisoning and control of poisonous material, Forensic anthropology.

HCLM 435 Planning and Decision-Making
Levels of planning, organisational goals, barriers to planning, management by objectives, decision-making, models of decision-making, decision-making process, group decision-making, strategic management and planning.

HCLM 436 Project Management
The module involves the application of knowledge, skills, tools and techniques to project activities to meet the project requirement.

HCLM 437 Fuel Technology
Fuel Chemistry: Review of energy sources both renewable and non-renewable; Classification of fuels and their calorific value; Coal: Uses of coal both fuel and non-fuel in various industries; Coal composition; Carbonization of coal; Coal gas, producer gas and water gas composition and uses; Fractionation of coal tar; Uses of coal tar bases chemicals; Requisites of a good metallurgical coke; Coal gasification (Hydro Gasification and Catalytic gasification); Coal liquefaction and Solvent Refining; Gas-To-Liquid (GTL) and Coal-To-Liquid (CTL) processes; Fischer-Tropsch process; Sasol process; Direct Coal Liquefaction: Bergius Process; Methanol-To-Gasoline Process (MTG); Petroleum and Petrochemical Industry: Composition of crude petroleum, Refining and different types of petroleum products and their applications; Ongoing or planned future projects; Renewable energy sources: Biomass & Biofuels overview, Biogas (anaerobic digesters), Bioenergy from wastes, dedicated bioenergy crops, Woody

biomass, Liquid biofuels, (bioethanol, biooils), issues and future prospects, biodiesel-uses, production and process; Fuel cells, Transportation (hybrids, fuel cells, flexi-fuels); and Government, regional and international initiatives.

HCLM 438 Environmental Management
Types of industrial wastes; solids and liquids, nuclear waste; Causes and effects of water, land and air pollution; Legislation on pollution; Industries responsibilities; Wastewater treatment; Air pollution control technologies; Incineration of industrial wastes; Responsible use of landfills; Recycling; Sampling analysis and management of industrial effluent; Fate and transport of contaminants of emerging concern in air, water, solid waste, and soils; Investigate the fate, transport, and remediation of potentially harmful contaminants and their by-products; Emerging contaminants; and International agreements (Montreal Protocol, Kyoto Protocol, Paris Agreement), ISO 14001 Environmental management and audits.

HCLM 439 Green Chemistry
The twelve principles of Green Chemistry are considered together with their application to chemical synthesis, whilst contributing to improved sustainability.

HCLM 440 Research Project
Students carry out research projects on a topic of their choice concerning pertinent issues in chemical industries under the guidance of their supervisors.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN INDUSTRIAL CHEMISTRY (HIC)

Duration:	4 years
Actual Credit load:	540
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

1.0 PURPOSE OF PROGRAMME

The programme aims to produce industrial chemists with the skills and knowledge needed to work in the chemical industry, education, research and innovation institutes. There is an emphasis on research and innovation entrepreneurship and commercialisation of goods and services. The programme also offers qualities and transferable skills necessary for further training of a professional or equivalent nature.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five 'O' Level subjects/National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry
In order to qualify for normal entry to a Bachelor of Science Industrial Chemistry Honours Degree, a prospective student must satisfy the Entry Requirements specified in the General Regulations and the Faculty Regulations.
In addition the applicant must: Have obtained a pass at 'A' Level Chemistry and any one of the following subjects at 'A' Level; Mathematics, Physics, Computer Science, Biology, Food Science.

Special Entry
National Diploma, Higher National Diploma in relevant fields

Mature Entry
Five 'O' Level passes including Mathematics and English plus at least 25 years of age and 5 years of experience in the relevant field.

3.0 PROGRAMME CHARACTERISTICS

This degree programme is geared towards those students who envisage a career in industrial chemistry and chemical process. The programme offers an opportunity for students to gain a foundation in the core chemistry concepts and related technology and the manufacture of various chemical products. The course exposes students to the development and design of

sustainable chemical processes at different scales and creates a bridge between science and manufacturing, by applying the principles of chemistry, chemical technology and engineering to solve problems involving the production or use of chemicals. The students are able to apply the methods and techniques that they have learned to review, consolidate, extend and apply their knowledge and understanding, and to initiate and carry out sustainable projects.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities:-

– in chemical industry as technical business analysts, production managers (supervisors), product development chemists/managers, quality controllers, quality assurance managers

– as chemists in food, metallurgical, coatings, soaps and detergents, pharmaceutical, textile, paint, agrochemicals, plastics, rubber and allied, synthetic fibre and artificial leather, ceramics, heavy chemicals, sugar, petroleum, pulp and paper, mining industries and agriculture, water and effluent treatment plants

–in research and development departments in industry and research institutes

–entrepreneurship in the manufacture of chemicals

–pursue Master of Science in Industrial Chemistry and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a four year period during which the student is engaged in a number of lectures, tutorials, practical investigations, field visits, work placements, research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

This curriculum is designed to impart knowledge, skills and training in order to prepare graduates to have:

Understanding of the needs of the society and demands of the 21st century and be able to improve quality of life

Understanding of the principles of industrial chemistry and their application for sustainable development

Ability to understand, analyze, interpret and solve problems of chemical and allied industries by using modern techniques, engineering tools, research and innovation

Understanding of safety principles and practices in process industries

Understanding of professional and ethical responsibilities

Knowledge of contemporary issues and ability to work in multidisciplinary teams– produce professionals who can utilise natural resources (agriculture, mining, water, air) for the benefit of the country

Produce professionals who are innovative and capable of creating employment for themselves

7.0 GENERAL PROVISIONS

Refer to section 4 of the Faculty Regulations

8.0 PROGRAMME ASSESSMENT

Refer to section 5 of the Faculty Regulations as well as the following:-

Coursework:	40% (Tests 15%, 20% practical, 5% assignments)
Written Examinations:	End of semester examinations contribute 60% of the final mark
Other:	a) Research Project is assessed on the basis of a research project (75%), oral presentation (15%) and student conduct in the laboratory (10%). b) Work-Related Learning is assessed based on Work-Related learning report (30%), assessment by work supervisor (50%) and assessment by academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 7 of the Faculty Regulations

10.0 FAILURE TO SATIFTY EXAMINERS

Refer to section 9 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 10 of the Faculty Regulations

12.0 DEGREE WEIGHTING

Refer to section 11 of the Faculty Regulations

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits

**HIC 131	Analytical Chemistry I	12
**HIC 132	Inorganic Chemistry I	12
**HIC 137	Mathematics for Chemists	12
*HCSCI 131	Introduction to Computers and Computer Applications	12
*CS 131	Basic Communication Skills	12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
**HIC 133	Organic Chemistry I		12
**HIC 134	Physical Chemistry I		12
*HIC 135	Laboratory Techniques		12
HIC 136	Materials Chemistry		12
**HIC 138	Introduction to Industrial Chemistry		12
HIC 139	Mineral Technology		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
**HIC 231	Chemical Instrumentation I	**HIC 131	12
**HIC 232	Inorganic Chemistry II	**HIC 132	12
*HIC 235	Chemical Research Methods and Statistics		12
**HIC 236	Fluid Flow, Heat and Mass Transfer		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
**HIC 233	Organic Chemistry II	**HIC 133	12
**HIC 234	Physical Chemistry II	**HIC 134	12
HIC 237	Medicinal and Pharmaceutical Chemistry		12
**HIC 238	Chemical Processes		12
**HIC 239	Unit Operations		12
HIC 240	Quality Management Systems		12

Level 3 Semester 1

Code	Module Description	Credits
**HIC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HIC 333	Work-Related Learning II	**HIC 332	80

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
*HIC 431	Chemical Instrumentation II	**HIC 231	12
*HIC 432	Inorganic Chemistry III	**HIC 232	12
**HIC 434	Industrial Microbiology		12
**HIC 435	Energy Technology		12
**HIC 436	Polymer Chemistry and Composites		12
**HIC 437	High Energetic Materials and Chemical Safety		12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
*HIC 433	Organic Chemistry III	*HIC 233	12
**HIC 438	Environmental Chemistry, Legislation and Sustainability		12
**HIC 439	Project Management		12
**HIC 440	Research Project		36

MODULE SYNOPSIS

HIC 131 Analytical Chemistry I
The module covers fundamental concepts of solution chemistry, the chemical composition of solutions, acid-base equilibrium, solubility and solubility equilibria, redox stoichiometry, gravimetric methods of analysis and elementary treatment of data. The module also covers some introduction to practical aspects of chemical analysis.

HIC 132 Inorganic Chemistry I
The module equips students with knowledge on atomic theory and elements of bonding. It also focuses on the study of the Periodic Table itself and the general trends in periodic and group properties of the elements. The introductory elements of coordination chemistry are also covered in this course.

HIC 137 Mathematics for Chemists
The course is a survey of mathematical methods mostly for physical chemistry. Calculus: Differentials and Derivatives, Partial Derivatives, Integrals, Algebraic Functions, Transcendal Function; Coordinate Systems- Point in Space, Coordinate Systems for Molecules, Abstract Coordinate; Integration: Change of variables in integrand, Jacobian determinant, Gaussian integrals, Improper Integrals, Line Integrals; Numerical Methods: Interpolation, Numerical Differentiation, Random Numbers; Differential Equations in chemical reaction rates, thermodynamics, chemical reaction mechanism; Linear Algebra: Vector Spaces, Spaces of Functions, Matrices, Eigen Value Equations, Schrödinger's Equation; Fourier Analysis.

HCSCI 131 Introduction to Computers and Computer Applications

Refer to Computer Science Department regulations

CS 131 Basic Communication Skills
Refer to the Communication Skills Department regulation

HIC 133 Organic Chemistry I
The module covers the study of carbon compounds, their structure, properties, composition, reactions, and preparations. The areas covered include Stereochemistry, Systematic Aliphatic Chemistry and basic Benzene Chemistry.

HIC 134 Physical Chemistry I
This module is an entry level module for university physical chemistry. Both kinetics and thermodynamics are the main elements constituting the bulk of physical chemistry

HIC 135 Laboratory Techniques
Sampling techniques; Separation and purification techniques; Concentration techniques; Stoichiometry; Common laboratory processes; Reporting analytical results; Interpretation of data; Keeping an organized lab notebook; Laboratory safety; Safety symbols; Materials safety data sheets; Handling of chemicals and their disposals; Strategies for procurement of chemicals; Good Laboratory Practices (GLPs).

HIC 136 Materials Chemistry
The module is focused on understanding the relationship between the arrangement of atoms, ions and molecules of a material and the bulk physical and structural properties of that material. Students will acquire skills to synthesize and characterize these materials.

HIC 138 Introduction to Industrial Chemistry
This module helps the student chemist to clarify the career areas and technical problems which are to be considered when chemical reactions are carried out on a large scale. It covers the research and development of consumer products based on chemical processes. Topics covered include the chemical industry and large-scale chemical manufacturing.

HIC 139 Mineral Technology
Minerals and metallurgical Industry; Metallurgical Principles; Mineral dressing: Milling Size Control; Separation: Principal Exploitable characteristics, Refining, Specific industrial processes for the production of selected metals: iron, cobalt, nickel, sodium, copper, zinc, tin and manganese; Minerals processing operations: ore preparation, pre-blending, size reduction, separation and concentration, sorting, floatation, hydrometallurgy, pyrometallurgy and electrometallurgy; Fire Assaying; Calculation of gold and silver; Critical success factors; Limitations and limit of detection; Effect of impurities; and Analysis of the PGMS.

HIC 231 Chemical Instrumentation I
This module will introduce students to chemical instruments used in Analytical chemistry, based on atomic spectroscopy and molecular spectroscopy. Methods covered in this module include flame and furnace atomic absorption spectrophotometry, flame emission spectroscopy,

plasma emission spectroscopy, ultraviolet visible, infrared, fluorescence, phosphorescence, chemiluminescence spectrophotometry and X-Ray spectrometry.

HIC 232 Inorganic Chemistry II
The module introduces students to the main group, d-block and lanthanide chemistry. In addition, students will acquire knowledge on bonding of coordination compounds in terms of the crystal field and molecular orbital theories as well as an interpretation of electronic spectra of transition metal complexes.

HIC 235 Chemical Research Methods and Statistics
Understanding research: research methods and scientific methods; Ethics in research; Planning research proposal; Organizing data with descriptive statistics; Linear and nonlinear models; and Experimental designs.

HIC 236 Fluid Flow, Heat and Mass Transfer
This module imparts knowledge of the concepts and laws of fluid mechanics, and their applications in pipe sizing and flow metering. In addition, it will impart in-depth knowledge about the selection and design of fluid flow systems in process industry. Students will develop an understanding of the concepts and laws of heat transfer for design of heat transfer equipment.

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven *vs* non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HIC 233 Organic Chemistry II
This module seeks to equip the student with further techniques that are required in synthesis and characterization of organic compounds. The module covers advanced stereochemistry and alicyclic chemistry, rearrangement reactions and introduction to organic spectroscopy.

HIC 234 Physical Chemistry II
The module is an advanced study from Physical Chemistry 1. Students will have a deeper appreciation of gas-phase and solution kinetics. In addition, concepts of phase equilibria, properties of simple mixtures in solution and surface studies will be imparted.

HIC 237 Medicinal and Pharmaceutical Chemistry
Introduction to biochemistry: important biological molecules (proteins, nucleic acids, carbohydrates, vitamins, enzymes and lipids) in a human body, medicinal drug metabolism, the search for drugs in plants, drug design and development. Selective toxicity, Strategies to select the plant material for study; main groups of natural products derived from plants; representative natural products derived from marine organisms, preparation of extracts and selection of active fractions, screening strategies; separation and purification of active components; benchtop bioassays, chemical assays and structure elucidation. Synthesis of some pharmaceutical products: analgesics, anti-allergy agents, antibacterial, antibiotics, antidepressants, and antiviral. Drug discovery, delivery, absorption and metabolism. Biomedical analysis, pharmacology, pharmacokinetics and pharmacodynamics, analytical techniques and quality assurance. Industrial preparation of drugs and marketing.

HIC 238 Chemical Processes
This module aims to familiarize students with conversion of raw materials into finished products on industrial scale using conventional and green technology. The module will cover areas such as, development of Chemical Process Industry; agrochemicals, petroleum products, mineral processing, pulp and paper, sugar, heavy chemicals, cement and ceramics, glass, food and beverages, soaps and detergents, water treatment and purification, coatings, synthetic fibre and artificial leather.

HIC 239 Unit Operations
This module aims to expose students to the fundamentals of various unit operations and their applications. Equipment for various types of operations is covered, for example, gas – liquid, liquid – liquid, and solid – fluid operations.

HIC 240 Quality Management Systems
Basic concepts, Statistical tools in Quality Control, Metrology, Inspection, Reliability and Dependability, New product quality, Quality specifications, Supplier relationship, Manufacture of quality, Customer relations, Quality Audit, Economics of Quality, Quality Data, Improving Quality, Human Factors in Quality, Quality Policy and Objectives, Quality Systems, Organisation of Quality, Quality Consulting, Quality Assurance in Chemical Laboratory, Total Quality Management Systems, Quality Control in Chemical Measurement Systems, Practical Work.

HIC 431 Chemical Instrumentation II
This module advances the student's knowledge of electrochemistry and chromatographic techniques, which were introduced at Level 2. Areas covered will include modified voltammetric techniques and high performance liquid chromatography. Students will acquire skills in state-of-the-art instrumental methods of analysis based on Fourier transform and hyphenated techniques.

HIC 432 Inorganic Chemistry III
Further properties of transition metal compounds, reaction mechanisms, introduction to photochemical reactions, organometallic chemistry fundamental concepts. Practical Inorganic Chemistry

HIC 434 Industrial Microbiology
This module aims to develop understanding of the large-scale and profit motivated production of microorganisms or their products for direct use, or as inputs in the manufacture of other goods, that is, in the production of high-value products such as drugs, chemicals, fuels and electricity

HIC 435 Energy Technology
The purpose of this module is to develop understanding of various energy sources and related technologies. The areas covered will include petrochemicals, hydrocarbon fuels and natural gas, coal technology, biomass conversion, electrochemical energy storage and renewable energy

HIC 436 Polymer Chemistry and Composites
This module gives knowledge of polymers, their raw materials, processing techniques, and uses. Student will gain skills on how to synthesize, characterize polymers and polymeric composite materials. Nanocomposites synthesis and fabrication approaches, characterisation and applications.

HIC 437 High Energetic Materials and Chemical Safety
This module aims to develop an understanding of the chemistry and ballistics of propellants, primary and high explosives and pyrotechnics. Students will acquire knowledge and skills for the safe production and use of energetic materials.

HIC 433 Organic Chemistry III
Alicyclic Chemistry, Heterocyclic Chemistry, Further Aromatic Chemistry, Carbohydrate Chemistry

HIC 438 Environmental Chemistry, Legislation and Sustainability
The module seeks to enlighten the students to environmental chemistry and how it relates to industrial operations. Nanotechnology and the environment. Students will gain knowledge and understanding on cleaner production technologies, sustainable practices and environmental legislation

HIC 439 Project Management
The module involves the application of knowledge, skills, tools and techniques to project activities to meet the project requirement.

HIC 440 Research Project
This is a double module that involves a research proposal, experimental work, and a dissertation. The student is required to make oral presentations on the above-mentioned stages.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN MEDICINAL AND PHARMACEUTICAL CHEMISTRY (HMPC)

Duration:	4 years
Actual Credit load:	540
Minimum Credit Load:	480
Maximum Credit Load:	540
Maximum MBKS Credit Load:	384
ZNQF Level:	8

1.0 PURPOSE OF PROGRAMME

The programme is designed to equip graduates with the skills needed to work in the pharmaceutical industry and allied sectors, research and innovation institutes with modules in drug design, synthesis and development alongside core chemistry. There is emphasis on research that leads to innovation and entrepreneurship as well as further studies.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five 'O' Level subjects/National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry

In order to qualify for normal entry to a Bachelor of Science Medicinal and Pharmaceutical Chemistry Honours Degree, a prospective student must satisfy the Entry Requirements specified in the General Regulations and the Faculty Regulations.

In addition the applicant must: Have obtained a pass at 'A' Level Chemistry and any one of the following subjects at 'A' Level; Mathematics, Physics, Computer Science, Biology, Food Science.

Special Entry

National Diploma, Higher National Diploma in relevant fields

Mature Entry

Five 'O' Level passes including Mathematics and English plus at least 25 years of age and 5 years of experience in the relevant field.

3.0 PROGRAMME CHARACTERISTICS

The search for new drugs to combat cancer, heart disease and infections remains an important challenge at the forefront of medical research. The medicinal and pharmaceutical industry has a need for highly skilled graduates with a strong background in organic chemistry, coupled with a broad understanding of pharmacology and related biochemical areas. This programme offers graduates an opportunity to study chemistry as a major subject along with subjects allied to the medicinal and pharmaceutical industries. The programme is designed to train scientists

to conceptualize, design, synthesize and optimize new molecules with enhanced therapeutic effects, which may become part of future pharmaceuticals. This degree programme combines the exactness of science with the creativity of innovation and discovery, providing graduates with strong skills in modern chemical techniques, such as computer-aided drug design, and a thorough understanding of biomedical issues.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities:-

- work in pharmaceutical industries as technical business analysts, production managers, product development chemists/managers, quality controllers, quality assurance managers.
- work in public and private research and innovation institutes focusing on drug design and development, identifying candidates for clinical studies, drug action, drug transport, drug delivery, drug testing, and targeting.
- work as academia in government institutions
- engage in entrepreneurship to develop and manufacture drugs
- pursue Master of Science in Medicinal Chemistry, Pharmaceutical chemistry, Biochemistry and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a four year period during which the student is engaged in a number of lectures, tutorials, practical investigations, field visits, work placements, research projects, problem-based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

The programme hopes to achieve the following broad objectives:-

- Understanding of the basic biological and pharmacological interactions by using both natural products and total synthesis of bioactive molecules
- Use of corresponding knowledge for the development of biologically and clinically active drugs.
- Designing, synthesizing and evaluating molecular products in a pharmacological context.
- Rationalizing and predicting, based on theoretical and computational models, structure-activity relationships in the field of medicine.
- Promoting innovation, identifying relationships between complex phenomena and modifying molecular structures to improve results.
- Using the most sophisticated analytical to obtain information of medical relevance.
- Dealing with complex issues in a multidisciplinary environment.
- Create new business ventures in the Medicinal and Pharmaceutical industry.
- Communicate effectively and present information using multi-media to both experts and non-experts in the Medicinal and Pharmaceutical chemistry industry.
- Develop skills and attitudes that enable graduates to participate actively in the development of the country

– produce professionals who are innovative and capable of creating employment for themselves and positively impacting the life of the ordinary Zimbabwean

7.0 GENERAL PROVISIONS

Refer to section 4 of the Faculty Regulations

8.0 PROGRAMME ASSESSMENT

Refer to section 5 of the Faculty Regulations as well as the following:-

Coursework:	40% (Tests 15%, 20% practical, 5% assignments)
Written Examinations:	End of semester examinations contribute 60% of the final mark
Other:	a) Research Project is assessed on the basis of a research project (75%), oral presentation (15%) and student conduct in the laboratory (10%). b) Industrial Attachment is assessed based on industrial attachment report (30%), assessment by work supervisor (50%) and assessment by an academic supervisor (20%).

9.0 PROVISIONS FOR PROGRESSION

Refer to section 7 of the Faculty Regulations

10.0 FAILURE TO SATIFTY EXAMINERS

Refer to section 9 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 10 of the Faculty Regulations

12.0 DEGREE WEIGHTING

Refer to section 11 of the Faculty Regulations

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Prerequisites	Credits

**HMPC 131	Analytical Chemistry	12
**HMPC 132	Inorganic Chemistry	12
*HMPC 137	Mathematics for Chemists	12
*HCSCI 131	Introduction to Computers and Computer Applications	12
*CS 131	Basic Communication Skills	12

Level 1 Semester 2

Code	Module Description	Prerequisites	Credits
**HMPC 133	Organic Chemistry I		12
**HMPC 134	Physical Chemistry I		12
**HMPC 135	Laboratory Techniques		12
**HMPC 136	Chemistry of Natural Products		12
**HMPC 138	Biochemistry		12
**HMPC 139	Introduction to Medicinal and Pharmaceutical Chemistry		12

Level 2 Semester 1

Code	Module Description	Prerequisites	Credits
**HMPC 231	Pharmaceutical Process Design		12
**HMPC 232	Organometallic Compounds and their Applications		12
**HMPC 235	Chemical Research Methods and Statistics		12
**HMPC 236	Principles of Drug Design		12
*TCNP 201	Technopreneurship		12
*GS 231	Introduction to Gender Studies		12

Level 2 Semester 2

Code	Module Description	Prerequisites	Credits
**HMPC 233	Organic Chemistry II	**HMPC 133	12
**HMPC 234	Physical Chemistry II	**HMPC 134	12
HMPC 237	Pharmaceutical Formulation Technology		12
HMPC 238	Structure and Spectroscopy		12
**HMPC 239	Cheminformatics		12
**HMPC 240	Quality Management Systems		12

Level 3 Semester 1

Code	Module Description	Credits
**HMPC 332	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Prerequisites	Credits
**HMPC 333	Work-Related Learning II	**HMPC 332	80

Level 4 Semester 1

Code	Module Description	Prerequisites	Credits
*HMPC 431	Chemical Instrumentation	**HMPC 131	12
**HMPC 432	Pharmacology and Toxicology		12
*HMPC 434	Drug Design and Development		12
**HMPC 435	Drug Stability and Metabolism		12
HMPC 436	Advanced Organometallics and Catalysis		12
HMPC 437	Nanotechnology in Medicinal Chemistry and Drug design		12

Level 4 Semester 2

Code	Module Description	Prerequisites	Credits
*HMPC 433	Organic Chemistry III	**HMPC 233	12
**HMPC 438	Heterocyclic Chemistry and Drug Synthesis		12
HMPC 439	Environmental Management		12
**HMPC 440	Research Project		36

14.0 MODULE SYNOPSES

HMPC 131 Analytical Chemistry
Ionic Equilibrium; Chemical composition of solutions; Chemical equations and stoichiometry; Elementary treatment of analytical data; Gravimetric Methods: Precipitation from homogeneous solution (PFHS); Volumetric (titrimetric) methods of analysis.

HMPC 132 Inorganic Chemistry
Atomic Structure; Electronic structure of atoms; Periodicity of atomic properties; Polyatomic molecules; Transition Metal Chemistry: Introduction to the magnetic properties of transitional metal complexes; Practical Inorganic Chemistry; Qualitative analysis.

HMPC 137 Mathematics for Chemists
The course is a survey of mathematical methods mostly for physical chemistry. Calculus: Differentials and Derivatives, Partial Derivatives, Integrals, Algebraic Functions, Transcendal Function; Coordinate Systems- Point in Space, Coordinate Systems for Molecules, Abstract Coordinate; Integration: Change of variables in integrand, Jacobian determinant, Gaussian

integrals, Improper Integrals, Line Integrals; Numerical Methods: Interpolation, Numerical Differentiation, Random Numbers; Differential Equations in chemical reaction rates, thermodynamics, chemical reaction mechanism; Linear Algebra: Vector Spaces, Spaces of Functions, Matrices, Eigen Value Equations, Schrödinger's Equation; Fourier Analysis.

CS 131 Basic Communication Skills
Refer to Communication Skills Department regulations

HCSCI 131 Introduction to Computers and Computer Applications
Refer to Computer Science Department regulations

HMPC 133 Organic Chemistry I
Bonding and structure of organic reactions and their mechanisms; Stereochemistry; Projection formulae: Fischer, Sawhorse, Newman, D/L configurations, R/S systems, compounds with more than one chiral centre; Aromatic chemistry; Aromatic substitution reactions; Organic practicals: Techniques in organic chemistry. Qualitative tests for functional groups and synthesis of simple organic compounds.

HMPC 13 Physical Chemistry I
Thermodynamics; Work; Heat and energy; First law of thermodynamics; Reversible and irreversible changes; Thermochemistry; Reaction kinetics; Electrochemistry; Thermodynamics of electrochemical cells; Ions in solution. Practical work.

HMPC 135 Laboratory Techniques
Sampling techniques; Separation and purification techniques; Concentration techniques; Stoichiometry; Common laboratory processes; Reporting analytical results; Interpretation of data; Keeping an organized lab notebook; Laboratory safety; Safety symbols; Materials safety data sheets; Handling of chemicals and their disposals; Strategies for procurement of chemicals; Good Laboratory Practices (GLPs).

HMPC 136 Chemistry of Natural Products
Introduction on identification and classification of various natural products. Methods of isolation, separation, purification, and structure determination of the natural products. Chemistry of terpenes, the structure, classification and biosynthesis of the terpenes, building blocks of monoterpenes, sesquiterpenes, diterpenes and triterpenes based on the combination of a given number of isoprene units. Chemistry of alkaloids, general properties of the alkaloids; the structure, classification and biosynthesis of alkaloids. Chemistry of flavonoids, general properties, the structure, biosynthesis and their classifications. Practical on the isolation and identification of common secondary metabolites from plants.

HMPC 138 Biochemistry
The module provides thorough knowledge and understanding of the molecules of life - nucleic acids, proteins, carbohydrates and lipids - and their building blocks, including their structure and composition, synthesis and degradation, as well as about energy metabolism in the human body. The course also provides knowledge and understanding of the composition and function of biological membranes, and puts special emphasis on pharmaceutically relevant elements within biochemistry, employing pharmacy-related examples. The module forms the basis for

further understanding of the chemistry of pharmaceuticals and other pharmacologically active compounds, their mode of action, and their turnover in the human body. Nucleic acid biochemistry: Replication, transcription, translation, gene regulation. Human intermediary metabolism: Digestion and absorption of nutrients, energy metabolism of cells, including oxidative phosphorylation, metabolism and biosynthesis of carbohydrates, lipids and proteins and regulation and pathological changes of these processes, integration and hormonal regulation of the intermediary metabolism, biochemical functions of vitamins, signal substances. The laboratory exercises include enzyme kinetics and analytical methods used in clinical chemistry.

HMPC 139 Introduction to Medicinal and Pharmaceutical Chemistry
Introduction to biochemistry: important biological molecules (proteins, nucleic acids, carbohydrates, vitamins, enzymes and lipids) in a human body, medicinal drug metabolism, the search for drugs in plants, drug design and development. Strategies to select the plant material for study; main groups of natural products derived from plants; representative natural products, preparation of extracts and selection of active fractions, screening strategies; separation and purification of active components; benchtop bioassays, chemical assays and structure elucidation. Introduction to drug discovery, delivery, absorption and metabolism. Biomedical analysis, pharmacology, pharmacokinetics and pharmacodynamics, analytical techniques and quality assurance.

HMPC 231 Pharmaceutical Process Design
The module covers the essential operations used in the manufacture of pharmaceutical products. The course discusses the pharmaceutical product life-cycle, variability, testing, and specifications of pharmaceutical ingredients. Unit operations including blending, granulation, fluidized bed operations, milling, capsule filling, compaction, tablet coating and other processes will be addressed. Students learn to recognize how the output of one process is the input to the next process, and how deviations can cascade along the production sequence until they cause process failures. The course emphasizes design, scale-up, trouble-shooting, and optimization.

HMPC 232 Organometallic Compounds and their Applications
Ligands and their typical bonding modes in organometallic compounds (structures and bonding); Main group organometallic compounds: general trends and bonding. Main group organometallic compounds: general trends, bonding, structures and reactivity; NMR and ESR for the study of organometallic compounds. Organolithium and organomagnesium and their uses in organic syntheses; Organozinc, organocadmium/organomercury and their uses in syntheses; Organoboron/organoaluminum and their uses in organic synthesis and materials science (non-linear optical materials and fluorescent sensors, co-catalysts for olefin polymerization); Organosilicon/organotin/organolead and their uses in synthesis and materials science. Organometallic compounds application in medicinal chemistry.

HMPC 235 Chemical Research Methods and Statistics
Understanding research: research methods and scientific methods; Ethics in research; Planning research proposal; Organizing data with descriptive statistics; Linear and nonlinear models; and Experimental designs.

HMPC 236 Principles of Drug Design
Drug Discovery/Development; Source of Drugs; Structural effects on drug action; Approaches to New Drug Discovery; Drugs Derived from Natural Products; Existing Drugs as a Source for New Drug Discovery; Screening for New Drug Leads; Modern "Rational Approach" to Drug Design; Homologation of alkyl chain(s) or alteration of chain branching, design of aromatic ring position; Enzymes as Targets of Drug Design; Receptors as Targets of Drug Design.

TCNP 201 Technopreneurship
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship. Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

GS231 INTRODUCTION TO GENDER STUDIES

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HMPC 233 Organic Chemistry II
Introduction to organic spectroscopy; Synthetic Methods and Reaction mechanism; Formation of C-C bonds; Multi-centre reactions e.g. Wittig and Diels Alder reactions; oxidation reactions; Epoxidation of alkenes and ring opening; Reduction reactions; and Practical in organic chemistry. Practical work.

HMPC 234 Physical Chemistry II
Quantum Mechanics and Molecular Spectroscopy: Dawn of the Quantum Theory; Postulates of Quantum Mechanics; Molecular Spectroscopy; Raman spectra; General and Gas Phase Kinetics; Simple Collision theory of bimolecular reactions; Transition state theory; Unimolecular reactions; Lindermann mechanisms; RRKM Theory; Complex reactions; Unbranched chain reaction; and Explosion reactions. Practical work.

HMPC 237 Pharmaceutical Formulation Technology
In this module, students shall gain insight on functions and pathways involved in product development**,** including the purpose and conduct of clinical trials in Phases I, II, and III**.** Better understand how decisions during drug development (i.e. dosage forms, batch size, production method, outsourcing**)** impact product lifecycle viability and success, and why physical and chemical attributes of products have implications in production. Also review the production process and the role of interactions of ingredients/materials employed in pharmaceutical

development and manufacturing, as well as the impact of processing, storage, and transport environments on ingredients/materials and semi- and finished goods.

HMPC 238 Structure and Spectroscopy
Interpretation of NMR and mass spectra to determine structures of small molecules; use of point groups to describe the symmetry of molecules and how this mathematical understanding of symmetry helps describe molecular orbitals and vibrations; use of infrared and electronic spectroscopy to probe transitions between vibrational and electronic states.

HMPC 239 Cheminformatics
Introduction to principles and techniques used in cheminformatics as well as computational drug design especially in the area of Target Based Drug Design, Exploring protein and ligand databases, representation and manipulation of 2D structures, curation of chemical libraries, descriptor calculation, virtual screening and Machine Learning in drug design and analysis, prediction of ADMET properties, Introduction to tools and software in cheminformatics and drug design.

HMPC 240 Quality Management Systems
Basic concepts, Statistical tools in Quality Control, Metrology, Inspection, Reliability and Dependability, New product quality, Quality specifications, Supplier relationship, Manufacture of quality, Customer relations, Quality Audit, Economics of Quality, Quality Data, Improving Quality, Human Factors in Quality, Quality Policy and Objectives, Quality Systems, Organisation of Quality, Quality Consulting, Quality Assurance in Chemical Laboratory, Total Quality Management Systems, Quality Control in Chemical Measurement Systems, Practical Work.

HMPC 431 Chemical Instrumentation
Instrumental Method of analysis, Electrochemical methods, Modified voltammetric methods, Spectroscopic methods, Radiochemical methods, Chromatographic methods. Practical work.

HMPC 432 Pharmacology and Toxicology
This module aims to introduce you to the general principles of pharmacological active ingredients and the effect on the major organs. You will develop an understanding of techniques applicable to research in the field of pharmacology. Toxicology studies the body's response to drugs, foods, and toxic substances. Fundamentals of pharmacology and mechanisms of action are examined for acute and chronic exposure derived from environmental, dietary, occupational and pharmaceutical sources. In this module, the general toxicological principles and overview of toxic substances shall be covered. The module includes basic description how substances are absorbed by, distributed and eliminated from the body. The part contains awareness about toxicokinetic models and the processes of biotransformation.

HMPC 434 Drug Discovery and Development
Introduction to drug discovery and target product profiles; High throughput screening methods; Medicinal chemistry; Pharmacology and ADME toxicology; Fragments, peptidomimetics and structure based drug design; IP considerations; Case studies in drug discovery.

HMPC 435 Drug Stability and Metabolism

This module focuses on stability of medicines, incorporating ageing, oxidation, hydrolysis, degradation mechanisms, and prevention of deterioration. It also deals with metabolism of drugs by proteins and that includes cytochromes, P450, metabolic pathways, conjugation reactions, enzyme induction and inhibition.

HMPC 436 Advanced Organometallics and Catalysis
Advanced organometallic chemistry; bonding and reactivity of metal carbon σ,π donor/π acceptor ligands; concepts of the 18 electron rule and metal-metal bonding; introduce aspects of transition metal cluster chemistry (especially synthesis, structure, bonding and reactivity); Introduction the isolobal principle; develop concepts and ideas in metal-metal bonded and metal-carbon (element) multiple bonded systems (for example metal carbenes carbynes, and higher metal cumulenes); describe the role of organometallic species in catalysis for selected examples.

HMPC 437 Nanotechnology in Medicinal Chemistry and Drug Design
This module covers the definition of nanomaterials, classification of nanomaterials, morphologies of nanomaterials, properties of nanomaterials relative to bulk materials, synthetic and fabrication approaches, characterisation of nanomaterials, uses/applications of nanomaterials in medicinal chemistry and drug design. Toxicological aspects of nanomaterials, current and emerging research in nanomaterials.

HMPC 433 Organic Chemistry III
This module covers the chemistry of biomolecules; carbohydrate chemistry, proteins, lipids, nucleic acids and vitamins. The section on further aromatic chemistry focuses on aromaticity and criteria for aromaticity, case studies of selected aromatic systems, systematic chemistry of naphthalene, phenanthrene and anthracene as well as illustrated syntheses of these compounds.

HMPC 438 Heterocyclic Chemistry and Drug Synthesis
Structural features and reactivity of heterocyclic compounds, including stereochemistry; major synthetic pathways in heterocyclic chemistry, involving carbon-carbon and carbon-heteroatom bond formation, functional group interconversions and ring substitution; importance of heterocycles as key components in major drug classes and combinatorial libraries; examples of modern synthetic methodology and the importance of three-dimensional structure in Organic Chemistry. Practical on synthesis of common heterocyclic drugs.

HMPC 439 Environmental Management
Types of industrial wastes; solids and liquids, nuclear waste; Causes and effects of water, land and air pollution; Legislation on pollution; Industries responsibilities; Wastewater treatment; Air pollution control technologies; Incineration of industrial wastes; Responsible use of landfills; Recycling; Sampling analysis and management of industrial effluent; Fate and transport of contaminants of emerging concern in air, water, solid waste, and soils; Investigate the fate, transport, and remediation of potentially harmful contaminants and their by-products; Emerging contaminants; and International agreements (Montreal Protocol, Kyoto Protocol, Paris Agreement).

HMPC 440 Research Project
Students carry out research projects on a topic of their choice concerning pertinent issues in Medicinal and Pharmaceutical Chemistry under the guidance of their supervisors.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN PHARMACEUTICAL CHEMISTRY (MPCH)

Duration:	2 years
Actual Credit load:	360
Minimum Credit Load:	360
Maximum Credit Load:	420
Maximum MBKS Credit Load:	288
ZNQF Level:	9

1.0 PURPOSE OF PROGRAMME

The programme is aimed at producing graduates who wish to tailor their expertise towards employment, research and innovation in the pharmaceutical and health sciences industries/institutes. The program includes the study of drugs and drug development and other disciplines like concepts and processes of drug discovery, delivery, absorption and metabolism. It also prepares graduates for further studies and research.

2.0 ENTRY REQUIREMENTS

Bachelor of Science in Chemical or Biological Sciences or related field from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

MSc Pharmaceutical Chemistry is a two year postgraduate chemistry degree course designed to explore the links between disease, mechanism of action and the development of safe, effective commercial drugs. This program disciplines at the intersection of chemistry, especially synthetic organic chemistry, pharmacology and various other biological specialities, where it is involved with the design, chemical synthesis and development for a market of pharmaceutical agents (drugs). The knowledge, skills and competencies attained give the individuals an opportunity to succeed in employment in a pharmaceutical industry/institute, consulting firms, government, research and educational institutions. The course exposes students to the development and design of drugs at different scales and creates a bridge between science and manufacturing, by applying the principles of pharmaceutical chemistry to solve problems involving the production or use of drugs.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities:-

– in pharmaceutical or any health products industry as technical business analysts, production managers (supervisors), product development pharmaceutical chemists/managers, quality controllers, quality assurance managers

– as chemists in pharmaceutical or health products industries

–in research and development departments in industry and research institutes

–venturing into entrepreneurship for drug discovery, development and synthesis

–pursue PhD in Chemistry, pharmaceutical sciences and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a two-year period during which the student is engaged in a number of lectures, practical investigations, field visits, journal article writing, research projects, problem-based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

The programme hopes to achieve the following broad objectives:-

- Understand pharmaceutical chemistry principles
- Apply learnt concepts in the design of drugs
- Develop chemists that enhance efficiencies and outputs of the pharmaceutical industry
- Create new business ventures in the pharmaceutical industry.
- Communicate effectively and present information using multi-media to both experts and non-experts in the chemical industry.

7.0 GENERAL PROVISIONS

The degree programme shall normally be studied over a minimum of four semesters on a block release basis. This programme is a taught Masters programme with modular work in the first three semesters. The research project is to be done in the second year.

7.1 Dissertation

7.1.2 The dissertation shall be done in level two semesters two during which time the student is expected to undertake research

7.1.3 Each candidate is expected to submit a dissertation of at least one hundred (100) pages.

7.1.4 The format of the dissertation shall be provided by the Department of Chemical Sciences

8.0 PROGRAMME ASSESSMENT

8.1 Formal examinations are held at the end of every semester

8.2 Continuous assessment shall account for 40% of the overall assessment, while the final examination accounts for 60% of the overall assessment.

8.3 The aggregate pass mark shall be 50%

8.4 The aggregate pass mark shall comprise a pass in the final examination and a pass in the continuous assessment.

9.0 PROVISIONS FOR PROGRESSION

9.1 A student must pass modules worth at least 180 credits in the first level in order to proceed to the second level

9.2 To graduate a student should have accumulated 360 credits

10.0 FAILURE TO SATIFTY EXAMINERS

Refer to section 21 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the Faculty Regulations

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the parts of the degree programme are weighted as follows:

Leve1 (semester 1and 2) and Level 2 semester 1 taught modules (70%)

Level 2.2 Dissertation (30%)

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Credits

**MPCH 801	Quantitative Analytical Methods	18
*MPCH 802	Biochemistry	18
**MPCH 803	Advanced Organic Chemistry	18
**MPCH 804	Pharmacology	18
**MPCH 805	Modern Analytical Techniques I	18

Level 1 Semester 2

Code	Module Description	Prerequisite	Credits
**MPCH 806	Physical Pharmacy and Biopharmaceutics		18
**MPCH 807	Chemistry of Natural Products		18
**MPCH 808	Modern Analytical Techniques II	*MMCH 805	18
*MPCH 809	Laboratory Management and Research Methods		18
**MPCH 810	Drug Design		18

Level 2 Semester 1

Code	Module Description	Credits
*MMCH 901	Chemotherapeutic agents	18
**MPCH 902	Pharmacodynamic agents	18
*MPCH 903	Pharmaceutical Quality and Regulation	18
**MPCH 904	Drug Delivery System and Biopharmaceutics	18
**MPCH 905	Laboratory Practical	18

Level 2 Semester 2

Code	Module Description	Credits
**MPCH 906	Dissertation	90

14.0 MODULE SYNOPSES

MPCH 801 Quantitative Analytical Methods

General: Computation of analytical results, significant figures, concept of error, precision and accuracy, standard deviation, rejection of doubtful values with special reference to volumetric and gravimetric analysis, calibration of analytical equipment. Fundamental of volumetric analysis: Methods of expressing concentrations, primary and secondary standards. Neutralization reactions: Theory of indicators and neutralization indicators. Oxidation-reduction titration: Principle of oxidation reduction filtrations, redox indicators and their use in pharmaceutical analysis. Precipitation titration: Theory of precipitation titrations and use of adsorption indicators. Gravimetric analysis: Method of gravimetric analysis. Complexometric titrations: Complexometric methods using EDTA, principle of complexometric titrations, chelating agents, indicators, titrations with disodium edetate. Nonaqueous titrations:General discussion and principle of titrations in non-equeous media, aprotic, protophilic, protogenic and amphiprotic solvents. Titrations with perchloric acid, potassium methoxide and tetrabutyl ammonium hydroxide. Ionic equations, numerical problems based on the above topics.

MPCH 802 Biochemistry

Enzyme, enzyme kinetics, enzyme action, biological oxidation and reduction. Energy metabolism, bioenergetics, Introduction to Intermediary metabolism and carbohydrate metabolism. Protein and nucleic acid metabolism, lipid, water and mineral metabolism. Biosynthesis of protein, transmission & expression of genetic information, DNA genetic role, Structure replication of m-RNA, transcription, gene protein relationship and control of gene. Immunoglobulins, structure classification, and their biological role. Vitamins, skeleton structure and their biological role.

MPCH 803 Advanced Organic Chemistry

Optical isomerism, configuration, Cahn-Ingold-Prelog rule for designation of configuration. Stereochemistry of carbon compounds with no chiral atom, Biphenyls, Allenes, Alkylidenecycloalkanes, spirans and Geometrical isomerism. Stereoisomerism of rings, stability of rings, ease of ring formation, Actual shape of six membered rings and its relation to properties and reactivity and Conformational analysis of Butane. Carbocation, Carbanions, free radicals, formation and stability. Mechanism of reaction and methods of determining them. Mechanism involving aliphatic nucleophilic and electrophilic substitution reactions. Mechanisms involving Aromatic eletrophilic and aromatic nucleophilic substitution reactions, free radical reactions. Addition to carbon-carbon multiple bonds and elimination reactions. Study of reactions such as: Fries Rearrangement Beckmann rearrangement, Hofmann rearrangement, Curtius reaction, Schmidt Reaction, Claisen's Condensation, Wittig Reaction, Oppenaueroxidation, Meerwein Pondroff Valery Reduction, Birch Reduction, Clemmensen reduction, Reimer- Tiemann Reaction, Wolf Kishner's Reduction, Pinacol-Pinacolone Rearrangement and Aldol Condensation,

MPCH 804 Pharmacology

History of development of Pharmacology, introduction and general principles of route of drug administration, pharmacokinitics (absorption, distribution, metabolism and excreation) a pharmacodynamics (general mechanism of drug action). Elementry introduction to adverse drug reactions and drug interactions, Drug allergy. Toxicity: General concepts of toxicity, Acute, subacute & chronic toxicity tests, teratogenicity & carcinogenicity, itrogenic diseases, LD50, ED50, tolerance, habituation and addiction. Bio-assays: General principles, general methods, biological variations & animal ethics. Bioassays of insulin, heparin, d-tubocurarin,

digitalis, acetylcholine, adrenaline, histamine. General principles of screening of drugs, general screening methods, clinical trial. Screening methods for evaluation of anti-inflammatory, analgesics, antipyretics and antiulcer, anticonvulsants, hepatoprotective, antidiabetic, diuretic and drugs acting on CNS.

MPCH 805 Modern Analytical Techniques I

Chromatography: Principles Instrumentation, Application of adsorption, partition, paper, thin layer, HPTLC, HPLC, Gel Electrophoresis. Gas and Ion exchange Chromatography. Colorimetry: theory, methods of colour measurements and instrumentation. Optical rotation, its significance, instrumentation. Optical rotatory dispersion-terminology, plain curve, rotatory dispersion and circular dichroism and octane rule. Atomic absorption and flame emission spectroscopy, theory, instrumentation, atomic absorption spectrophotometers and structure determinations. UV and infrared spectroscopy: Principle, techniques, instrumentation and application including interpretation of data.

MPCH 806 Physical Pharmacy and Biopharmaceutics

Preformulation studies: Factors affecting dissolution, diffusion – pH, pka, KO/W, particle size, solubility. Methods to increase solubility of poorly soluble drugs, drug release mechanisms. Factors affecting/necessitating preformulation of Drugs. Micromeritics: Introduction, pharmaceutical importance, particle size distribution, surface area and particle volume, derived properties of powder, flow properties of powder, angle of repose, factors affecting flow of powders. Different methods of particle size determination. Surface and interfacial tension: Measurement of surface and interfacial tension, spreading coefficient, general classification of surfactants, hydrophilic- lipophilic balance. Solubilization, factors affecting solubilization, micelle formation, factors affecting micelle formation, determination of critical micelle Concentration, detergency and wetting agents. Contact angle, flocculating agents, de-flocculating agents, foaming and antifoaming agents. Pharmaceutical applications of surfactants. Colloids: Pharmaceutical application of colloids, brief introduction to properties of colloids. Coarse dispersions: Physical stability of suspension and emulsions, types of suspension, controlled flocculation-flocculated suspension, types of emulsion, theories of emulsification, emulsifying agents mechanism of action, factors to improve physical stability of emulsions. Rheology: Fundamentals of rheology, types of flow, thixotropy in formulations, rheology of disperse system. Pharmaceutical applications of rheology. Types of viscometers. Chemical kinetics: Influence of temperature, light, oxygen acid base catalyst and methods to prevent drug degradation. Prediction of stability, accelerated stability studies. Disintegration: Disintegration, Disintegration time, factors affecting disintegration Dissolution: Dissolution, dissolution models, factors affecting dissolution rate, co-relation of dissolution with bioavailability

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MPCH 807 Chemistry of Natural Products

Heterocyclic Compounds: Five membered heterocycles: Furan, Thiophene, pyrole, thiazole, pyrazole, oxazole, imidazole, Six membered hydrocycles: Pyridine, pyrimidine. Benzfused heterocycles: Quinoline, isoquinoline and indole. Carbohydrate: Introduction, mutarotation, ring structure of glucose, configuration of monosaccharides, structure elucidation of disaccharides, sucrose and maltose. Polysaccharides, starch. Glycosides: general structural elucidation of glycoside including cardiac glycosides. Alkaloids: General introduction, distribution in plants, isolation & purification. General methods of structure determination. Structural elucidation of atropine, quinine and Nicotine. Structural features of morphine and Caffiene. Terpenoids: General introduction, isolation, purification and structure elucidation of

citral, menthol, and camphor. Steroids: General introduction and structure elucidation of sterols with special reference to cholesterol.

MPCH 808 Modern Analytical Techniques II

Molecular Emission Spectroscopy: Principle, molecule exhibiting fluorescence, Factors interfering with florescence intensity and Application, Raman Spectroscopy-Principle, Instrumentation and Application. Mass Spectroscopy: Principle, techniques, instrumentation, fragmentation pattern & structural elucidation of compounds. GC-MS and LC-MS Principle and Application. Proton Magnetic Resonanace, Principle, techniques, instrumentation, ^{1}HNMR signals, chemical shift, spin-spin coupling, shielding desheilding effect, diamagnetic anisotropy, geminal coupling, AMX, ABX, ABC system, shift reagents and interpretation of spectra. ^{13}C-NMR, introduction and interpretation of data. Application of spectroscopic techniques to structural elucidation, introduction aids to spectral intepretation exercises. Microbiology assays, Principles of microbiological assays, assays of vitamins and antibiotics.

MPCH 809 Laboratory Management and Research Methods

Laboratory management is important to the graduate, who gets an opportunity to join the chemical industry. They require skills to plan, design and manage laboratories, which they are expected to operate in. Laboratory and business management principles are also important in this module. Research Methods involve practical steps of a research proposal, experimental design, data collection, treatment and use of statistical packages. It also covers intellectual property and commercialization processes.

MPCH 810 Drug Design

Specific and non-specific drug action, Drug receptors, Basic concept and classification of receptors, Forces involved in drug receptors- interactions, Receptor agonism and ntagonism, Concept of Spare receptors, Simple kinetics of drug- receptor interaction, Receptor theories: Clark's occupancy theory, Ariens-Stephenson modification, Induced fit and macromolecular perturbation theories, Ion Channel receptors. Topographical study of the following receptors includes only preferred conformations, pharmacophores and modes of bindings/interactions: Adrenergic, Cholinergic, Opioid, receptors, H-1 and H-2 receptors, Diazepine, Serotonin. Concept of isosterism and bioiososterism and their applications in drug design, Antimetabolite approach to drug design, Analog drug design, Prodrugs and drug latentiation, Carrier-linked prodrugs, Bioprecursors, Role of functional groups in prodrug design, General pathways of drug metabolism and simple kinetics of drug metabolism, Pharmacokinetic oriented drug design. Stereochemical aspects of drug action: Setereoselectivity of optical isomers, Role of planarity in drug action, Stereoselectivity of conformational isomers. QSARincluding: Types of QSAR models, Classification of parameters utilized in QSAR studies, Statistical concept of QSAR, Hansch model of QSAR, De Novo model of QSAR, Hammett and Taft model of QSAR equations, Applications of QSAR in drug design. Basics of combinatorial chemistry, Rational approach to drug design, Basic strategies of drug discovery, Role of molecular docking/modelling in drug design, computer assisted drug design. Drug synthesis and manufacturing process.

MMCH 901 Chemotherapeutic Agents
Chemical Classifications, SAR Studies, Mode of actions and Therapeutic uses of the following; Beta lactam antibiotics: Penicillins, Cephalosprins including their semi-synthetic products. Monobactams. Tetracyclines, Semi-synthetic tetracyclines, Gentamycins, Neomycins, Kanamycins, Fluoroquinoolines, type of antibacterials. Sulfonamides, Antileprotics, Antimycobacterials, Antifungals. Antiprotozoal includes-Antimalarials, Antiamoebics, Antihelminths. Anticancer, Antivirals and Synthesis of following drugs involving not more than three steps. 5-Fluorouracil, 6-Mercaptopurine, Amoxycillin, Ampicilln, Benzathine, Benzidiazole, Busulfan, Cefrozil, Cepirone, Chloroquine, Ciprofloxacin, Clotrimazole, Cyclophosphamide, Dapsone, Deoxycycline, Diloxamide., Ethionamide, Fucytocine, Hydroxy urea, Idoxuridine, Isoniazid, Ketoconazole, Loracarbef, Mafenide Acetate, Melphalan, Metronidazole, Pamaquine, Penicillins G, Piperazine Citrate, Proguanil, Pyrimethamine, Sulfadioxime, Sulfasalazine, Thiabendazole, Thiotepa, Trimethoprin.

MMCH 902 Pharmacodynamic Agents
Chemical Classifications, SAR Studies, Mode of actions and Therapeutic uses of the following classes of drugs. Drugs acting on CVS – Antianginals, Antihypertensives, Antilipidemics, Antiarrhythmics, Diuretics. Analgesics – Narcotic and NSAIDs, Antipyretics, Uricosurics (Antigouts). Drug acting on CNS – Hypnotic Sedative only, Barbiturates and halogenated drugs, Antiepileptics, General anaesthetics, Antiparkinsonian drugs. Psychotherapeuticals – Antipsychotics, Anxiolytic drugs, Antidepressants and Antidiabetics. H- 1 – R blockers (Anti-histaminics), H –2 – R blockers (Anti-ulcers), Carbohydrate based drugs, Oligonucleotides and Synthesis of following drug involving not more than three steps: Acetazolamide, Amantidine HCl, Amyl nitrate, Amoexpine, Buspropion, Captopril, Carbamazepine, Cetrizine, Chlorocyclizine, Chloropheniramine, Chloropromazine, Chloropropamide, Cimetidine, Clofibrate, Clonidine HCl, Desipiramine, Diazepam, Diltiazem HCl, Diphenyhydramine, Disopyramide, Enalapril maleate, Enthimate, Ethacrynic acid, Ethosuximide, Felodipine, Fentanyl citrate, Fluvastatin, Fluoxetine, Furasemide, Gemfibrozil, Glipizide, Guanethidin monosulphate, Hydrothiazide, Imipramine, Ibuprofen, Isosorbide dinitrate, Labetolol, Loratidine, Lorazepam, Lovastatin, Meperidine, Methadone, Methyldopatate HCl, Nalorphine, Naloxone, Nitrazepam, Oxycodone, Phenoformin, Phenytoin, Polythiazide, Procainamide, Promazine, Propoxyphene napsylate, Propranolol, Pyrilamine maleate, Ranitidine, Salsalate, Sotalol, Sufindac, Tolbutamide, Triamterene, Triazolam, Trimeprazine, Thioridazine, Tripelennamine maleate, Verampril

MMCH 903 Pharmaceutical Quality and Regulation
Good manufacturing practices (GMP). Local and global legislation. Solutions to critical manufacturing defects. Production planning and inventory control, method development and validation, quality risk management and regulatory roles such as development and assessment of clinical trial dossiers in the pharmaceutical industry.

MMCH 904 Drug Delivery System and Biopharmaceutics
Types, advantages, disadvantages and formulation and evaluation/quality control of oral dosage forms including Liquid dosage forms like solution, syrups, suspension and emulsion, Solid dosage forms like tablets, capsules and Semisolid dosage like ointment, cream, gels.
Parenteral drug delivery system: Preparation, Evaluation and quality control. A brief introduction to new approach such as liposomes, neosomes, nanoparticles, Resealed erythrocytes as novel parenteral drug delivery. Controlled release drug delivery system: Advantages, drug properties relevant of controlled release formulation. Oral dosage form: diffusion system, dissolution system, osmotic pump ion exchange resin and prodrug. Parenteral

dosage form: Intramuscular injection and implants. Drug absorption: factors affecting drug absorption including physicochemical, biological and pharmaceutical. Passive and active diffusion. Drug disposition: Distribution in blood, plasma protein binding, cellular distribution, drug excretion, biotransformation of drugs. Bioavailability: Concept of bioavailability and comparative bioavailability, methods of estimation of bioavailability, bioequivalence studies.

MMCH 905 Laboratory Practical

Students to design and carry out a practical in any of the following areas; Analysis of pharmaceuticals; Analysis of dosage form, typical pharmacopial assays, including disintegration time, dissolution rate and friability. Synthetic medicinal chemistry; Synthesis of Compounds using 3 to 4 steps. Natural Products; Isolation and identification of compounds. Phytochemical test for Carbohydrate, Alkaloids, Steroids, Terpenoids, Amino acids, Phenolics, Proteins. Structure confirmation by spectroscopic methods. Interpretation of UV, IR, NMR and Mass Spectroscopy of compounds.

MMCH 906 Dissertation

This is a research project, based on a dissertation on an approved topic in the area of pharmaceutical chemistry. The module consists of project proposal, progress report and the final dissertation, together with oral presentations of each stage

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN ANALYTICAL CHEMISTRY (MACH)

Duration:	2 years
Actual Credit Load	360
Minimum Credit Load:	360
Maximum Credit Load:	420
Maximum MBKS Credit Load:	288
ZNQF Level:	9

1.0 PURPOSE OF THE PROGRAMME

The progamme aims to produce students who show originality in the application of knowledge and understanding how the boundaries of knowledge are advanced through research and innovation in analytical chemistry and related fields which results in production of goods and services. The students are able to deal with complex issues both systematically and creatively, and show originality in tackling and solving problems. The programme also prepares students for further studies and research.

2.0 ENTRY REQUIREMENTS

At least a good first degree in Chemistry or Chemical Technology or related fields from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

The programme offers in-depth and advanced knowledge and understanding of the subject and/or profession, informed by current practice, scholarship, research and innovation in the fields of analytical chemistry and other relevant areas. This includes a critical awareness of current issues and developments in the subject and/or profession; critical skills; knowledge of professional responsibility, integrity and ethics; and the ability for students to reflect on their own progress as learners. The programme also enables students to demonstrate self-direction and originality in tackling and solving problems, and act autonomously in planning and implementing tasks at a professional or equivalent level, continue to advance their knowledge and understanding, and to develop new skills to a high level.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities;

- work as analytical chemists industries
- work in the industry as technical business analysts, production managers, product development chemists/managers, quality controllers or quality assurance managers

- work in research and development in industry
- work as research chemists in universities and research institutes
- venture into entrepreneurship in the field of analytical chemistry and related fields
- PhD opportunities in analytical chemistry or related fields that lead to lectureship and research in universities.

5.0 PROGRAMME DELIVERY

The programme is delivered over a two year period during which the students are engaged in a number of lectures, tutorials, practical investigations, field visits, journal article writing, research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

Upon successful completion of the programme, graduates should be able to:

- Evaluate and apply classical analytical techniques in the determination of chemical substances.
- Design and validate analytical methods
- Apply and troubleshoot atomic spectroscopic instruments in chemical analysis.
- Use analytical instruments that employ principles of molecular spectroscopy.
- Make use of state-of-the-art electrochemical methods to determine analytes.
- Select and apply appropriate chromatographic techniques for the separation and characterization of compounds.
- Carry out determinations of chemicals in different environments, by use of thermal and radiochemical methods.
- Undertake a research project and publish the output
- Commercialize or industrialize research output
- Professionally manage a laboratory

7.0 GENERAL PROVISIONS

The degree programme shall normally be studied over a minimum of four semesters on a block release basis. This programme is a taught Masters programme with modular work in the first year. The research project to be done in the second year.

7.1 Dissertation

7.1.2 The dissertation shall be done in level two during which time the student is expected to undertake research

7.1.3 Each candidate is expected to submit a dissertation of at least one hundred (100) pages.

7.1.4 The format of the dissertation shall be provided by the Department of Chemical Sciences

8.0 PROGRAMME ASSESSMENT

8.1 Formal examinations are held at the end of every semester

8.2 Continuous assessment shall account for 40% of the overall assessment, while the final examination accounts for 60% of the overall assessment.

8.3 The aggregate pass mark shall be 50%

8.4 The aggregate pass mark shall comprise a pass in the final examination and a pass in the continuous assessment.

9.0 PROVISIONS FOR PROGRESSION

9.1 A student must pass modules worth at least 180 credits in the first level in order to proceed to the second level

9.2 To graduate a student should have accumulated at least 360 credits

10. FAILURE TO SATISFY EXAMINERS

Refer to section 21 of the General Academic Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the General Academic Regulations

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the parts of the degree programme are weighted as follows:

Leve1 (50%)

Level 2 (50%)

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Credits
**MACH 801	Classical Analysis and Chemometrics	18
**MACH 802	Analytical Atomic and Molecular Spectroscopy	18
**MACH 803	Pharmaceutical and Clinical analysis	18
*MACH 804	Electrochemical Methods	18
*MACH 805	Surface Analysis Methods	18

Level 1 Semester 2

Code	Module Description	Credits
**MACH 806	Separation Methods	18
**MACH 807	Thermal and Radiochemical Methods	18
*MACH 808	Environmental and Green Chemistry	18
**MACH 809	Laboratory Management and Research Methods	18
*MACH 810	Nuclear Chemistry	18

Level 2

Code	Module Description	Credits
**MACH 901	Dissertation	180

14.0 MODULE SYNOPSES

MACH 801 Classical Analysis and Chemometrics
This module covers theories, techniques and applications of gravimetric analysis, volumetric analysis, non-aqueous titrations and kinetic methods of analysis. It also incorporates validation techniques, calibration, sampling and statistical analysis.

MACH 802 Analytical Atomic and Molecular Spectroscopy
Analytical Atomic Spectroscopy encompasses principles, methods and applications of flame, spark, arc and furnace methods for atomic absorption, emission and fluorescence techniques. Analytical Molecular Spectroscopy encompasses theory, instrumentation and uses of ultraviolet-visible, luminescence and infrared spectrophotometry. Raman spectroscopy, refractometry, chiroptical and light scattering methods are also included.

MACH 803 Pharmaceutical and Clinical Analysis

The module looks at drug discovery and development: Pharmacokinetics and pharmacodynamics; drug approval process; drug stability testing; design and analysis of clinical trials; formulation development and evaluation; advanced pharmaceutical compound analysis: Sample handling, storage and preparation; medical nanotechnology techniques; biological methods of drug analysis; advanced methods in analysis of biopharmaceuticals and quality assurance; GLP and GMP.

MACH 804 Electrochemical Methods
Electrochemical methods incorporate electrogravimetry, coulometry, polarography, voltammetry, potentiometry, conductrometry, amperometric titrations, biosensors and electrode kinetics.

MACH 805 Surface Analysis Methods
Photon Probe Techniques: X-Ray Photoelectron spectroscopy - Principle, Instrumentation, applications. Electron Probe Techniques: Scanning electron microscopy (SEM) – Principle, Instrumentation, applications. Electron Probe X-ray analysis (EPXMA) - Principle, Instrumentation, applications. Auger electron spectroscopy (AES) - Principle, Instrumentation, applications. Ion Probe Techniques: Rutherford backscattering spectrometry (RBS) - Principle, Instrumentation, applications. Secondary ion mass spectrometry (SIMS) – Fundamental aspects of sputtering, Principle, Instrumentation (static & dynamic), applications Scanning probe microscopy Techniques: Scanning Tunneling Microscopy – Principle, Instrumentation, applications. Atomic Force Microscopy - Principle, Instrumentation, applications.

MACH 806 Separation Methods
This module is about analysis through separation techniques. It covers solvent extraction and instrumental methods that comprise gas chromatography, liquid chromatography, supercritical fluid chromatography, electrochromatography and hyphenated techniques.

MACH 807 Thermal and Radiochemical Methods
Thermal Methods include: Thermogravimetric analysis. Differential thermal analysis. Differential scanning calorimetry. Thermomechanical analysis. Thermometric titrations. Principles. Methods. Applications. Radiochemical Methods incorporate Isotope Dilution, Neutron Activation Analysis, X-ray Fluorescence, Absorption and Diffraction.

MACH 808 Environmental and Green Chemistry
Aspects of environmental monitoring, protection, theory and technologies are important to this module. The sustainability of chemical products defines green chemistry, where production processes that minimize pollutants and use of non-renewable resources are explored.

MACH 809 Laboratory Management and Research Methods
Laboratory management is important to the graduate, who gets an opportunity to join the chemical industry. They require skills to plan, design and manage laboratories, which they are expected to operate in. Laboratory and business management principles are also important in this module. Research Methods involve practical steps of a research proposal, experimental design, data collection, treatment and use of statistical packages. It also covers intellectual property and commercialization processes.

MACH 810 Nuclear Chemistry
The atomic nucleus-elementary particles, quarks, classification of nuclides based on Z and N values, nuclear stability, nuclear potential, binding energy. Nuclear structure: Shell model-salient features, forms of the nuclear potential, magic numbers, filling of orbitals, nuclear configuration, Liquid drop model, Fermi gas model, Collective model and Optical model. Nuclear reactors :- General aspects of reactor design, thermal, fast and intermediate reactors, reactor fuel materials, reactor moderators and reflects, coolants, control materials, shield, regeneration and breeding of fissile matter, types of research reactors. Nuclear reactions, fission and fusion, radio-analytical Radioactivity, radioactive decay kinetics, Parent-daughter decay-growth relationship-secular and transient equilibria, theories of α, β-, β+ and γ-decay, internal conversion, Auger effect. Radio isotopes its applications.

MACH 901 Dissertation
This is a research project, based on a dissertation on an approved topic. The module consists of project proposal, progress report and the final dissertation, together with oral presentations of each stage.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN ENVIRONMENTAL CHEMISTRY (MECH)

Duration:	2 years
Actual Credit Load	360
Minimum Credit Load:	360
Maximum Credit Load:	420
Maximum MBKS Credit Load:	288
ZNQF Level:	9

2.0 PURPOSE OF THE PROGRAMME

The programme aims to produce students who can address today's global challenges which are expedited by increasing industrialization and exploitation of natural resources. The students are able to manage resources sustainably, predict impact of activities and cleanup of past contaminated sites through systematic and creative thinking in problem solving.

2.0 ENTRY REQUIREMENTS

At least a good first degree in Chemistry, Chemical Technology or Environmental Chemistry from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

The programme provides an opportunity for students to analyse and process complex environmental interrelations through project-oriented learning provided by state of the art analytical approaches, modelling and data analysis techniques. Students are also equipped with problem-oriented education towards air, soil and water.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following opportunities;

- environmental chemists in relevant industries
- experts and consultants in government and industry
- national and state authorities regulatory bodies
- academia as researchers
- PhD opportunities in the area of Environmental Chemistry and related fields

5.0 PROGRAMME DELIVERY

The programme is delivered over a two year period during which the students are engaged in a number of lectures, tutorials, practical investigations, field visits, journal article writing,

research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

- Capability in analyzing fundamental chemical reactions in the atmosphere, aquatic and soil systems.
- Acquire skills in assessing the risk, identifying the contaminants as well as mitigating pollution.
- Competence in use and interpretation of data generated by the following analytical instruments
- Modelling and predicting toxic substances.
- Research and innovation that lead to commercialization

7.0 GENERAL PROVISIONS

The degree programme shall normally be studied over a minimum of four semesters on a block release basis. This programme is a taught Masters programme with modular work in the first three semesters. The research project to be done in the second year.

7.1 Dissertation

7.1.2 The dissertation shall be done in level two semester two during which time the student is expected to undertake research

7.1.3 Each candidate is expected to submit a dissertation of at least one hundred (100) pages.

7.1.4 The format of the dissertation shall be provided by the Department of Chemical Sciences

8.0 PROGRAMME ASSESSMENT

8.1 Formal examinations are held at the end of every semester

8.2 Continuous assessment shall account for 40% of the overall assessment, while the final examination account for 60% of the overall assessment.

8.3 The aggregate pass mark shall be 50%

8.4 The aggregate pass mark shall comprise a pass in the final examination and a pass in the continuous assessment.

9.0 PROVISIONS FOR PROGRESSION

9.1 A student must pass modules worth at least 180 credits in the first level in order to proceed to the second level

9.2 To graduate a student should have accumulated 360 credits

10. FAILURE TO SATISFY EXAMINERS

Refer to section 21 of the General Academic Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the General Academic Regulations

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the parts of the degree programme are weighted as follows:

Leve1 (semester 1and 2) and Level 2 semester 1 taught modules (70%)

Level 2.2 Dissertation (30%)

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Credits
**MECH 801	Introduction to Environmental Chemistry	18
**MECH 802	Aquatic Chemistry	18
**MECH 803	Environmental Analytical Chemistry	18
*MECH 804	Health, Safety and Environment	18
**MECH 805	Practical I	18

Level 1 Semester 2

Code	Module Description	Credits
**MECH 806	Research Methods and Statistics	18
*MECH 807	Atmospheric Chemistry	18
**MECH 808	Environmental Toxicology	18

**MECH 809	Green Chemistry	18
**MECH 810	Practical II	18

Level 2 Semester 1

Code	Module Description	Credits
*MECH 901	Modelling in Environmental Chemistry	18
**MECH 902	Pollution Control in Chemical Industry	18
*MECH 903	Environmental Pollution Risk Management	18
**MECH 904	Environmental Soil Chemistry	18
**MECH 905	Practical III	18

Level 2 Semester 2

Code	Module Description	Credits
**MECH 906	Dissertation	90

14.0 MODULE SYNOPSES

MECH 801 Introduction to Environmental Chemistry
The module introduces fundamental chemical, physical and biological mechanisms that explain the transport and chemical degradation of aquatic contaminants, such as sorption, biodegradation, redox mechanisms and photolysis.

MECH 802 Aquatic Chemistry
The module creates an understanding of the chemical reactions in natural waters by providing a comprehensive treatment of homogeneous and heterogeneous chemical equilibria in natural aquatic systems (acid/base reactions, precipitation/dissolution reactions, complex formation, redox reactions and solid-solution interface reactions). This provides a solid foundation for understanding and evaluating how human activities can influence natural water systems.

MECH 803 Environmental Analytical Chemistry
An introduction to environmental analytical chemistry and an overview of important analytical methods and their range of application within detection of inorganic and organic compounds in the environment is given. Quantitative techniques such as electrochemical methods, spectrochemical/spectrophotometric methods, and separation techniques are reviewed. Theory on sampling, analyses of real samples, risk assessment of chemical experiments, important

steps and procedures in analytical chemistry, and evaluation/interpretation of results is included.

MECH 804 Health, Safety and Environment
Regulations and guidelines concerning health, safety and the environment. Risk assessment and emergency preparedness are considered. Handling fire and hazardous chemicals with practical considerations is covered.

MECH 805 Practical I
Sampling and analysis of pollutants in water, soil, air and living organisms

MECH 806 Research Methods and Statistics
Principles of research, managing a research project; performing a literature review; organizing and performing data collection, analysis and presentation; writing and presenting a technical paper/report; understanding the dynamics of teamwork; critically evaluating a given research project and applying the findings appropriately; developing subject-specific skills. Major project: carried out in teams and reported in written and oral form.

MECH 807 Atmospheric Chemistry
The module provides a wide overview of the environmental impact of atmospheric constituents. Fundamental concepts of chemical reactions in troposphere and stratosphere are introduced. The importance and danger of greenhouse gases, the formation of photochemical smog, and the natural and disturbed ozone layer are considered. Students should are able to analyse fundamental chemical reactions in the atmosphere, and evaluating changes of the atmospheric composition.

MECH 808 Environmental Toxicology
The module gives an advanced understanding of uptake and distribution of environmental pollutants in an organism and how pollutants can affect molecular, cellular and organismal processes. Introduction to Environmental Toxicology, Cellular and Molecular Toxicology, Toxicokinetic and Toxicodynamic, General Toxicology and Introduction to Risk Assessment, Environmental Toxicology, Exposure Assessment and Biomonitoring, Reproductive and Development Toxicology, Toxic effects of pesticides, POPs, Metals and of Radiation and radioactive Materials, Food toxicology, Nanotoxicology, Ecotoxicology is considered. Emerging pollutants and environmental forensics is also covered.

MECH 809 Green Chemistry
The twelve principles of Green Chemistry are considered together with their application to chemical synthesis, whilst contributing to improved sustainability.

MECH 810 Practical II
Conversion of pollutants to harmless or useful products.
MECH 901 Modelling in Environmental Chemistry

Basic principles of cheminformatics for building, validating and applying models of chemical property prediction related to toxicology and environmental effects are covered.

MECH 902 Pollution Control in Chemical Industry
Overviews of chemical industries producing chemicals which impact the environment are presented. Efficient technologies and processes which save raw materials, energy and by-product use as well as recycling for process efficiency and environmental protection are considered. Application of nanotechnology in environmental remediation.

MECH 903 Environment Management and Risk Assessment
Principles of integrated environmental management; environmental impact assessment; environmental management systems (ISO 14000 series), water resource management; environmental legislation; site investigation guidelines; natural hazard risk assessment; Statutory and regulatory controls on the environment. Air, water and land pollutants in general, standards, monitoring systems, and prevention and control techniques.

MECH 904 Environmental Soil Chemistry
The module covers the development and management of human soils, together with their physical chemical and biological properties. The knowledge assists in understanding the regulation of soils.

MECH 905 Practical III
The module involves reviewing recent articles related to environmental chemistry.

MECH 906 Dissertation
This is a research project, based on a dissertation on an approved topic. The module consists of a project proposal, progress report and the final dissertation, together with oral presentations of each stage.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN ORGANIC CHEMISTRY (MOCH)

Duration:	2 years
Actual Credit load:	360
Minimum Credit Load:	360
Maximum Credit Load:	420
Maximum MBKS Credit Load:	288
ZNQF Level:	9

1.0 PURPOSE OF PROGRAMME

The programme is aimed at producing graduates ready for employment, research and innovation in industry, universities and research institutes. The program includes the study of structure, properties and reactions of organic compounds. The students are able to deal with complex issues both systematically and creatively and show originality in tackling and solving problems as well as gaining entrepreneurship skills in the manufacturing industry.

2.0 ENTRY REQUIREMENTS

Bachelor of Science in Chemical Sciences or related field from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

This is a two-year postgraduate chemistry programme that involves the scientific study of the structure, properties, reactions, composition, and arrangements of organic compounds like carbon-based compounds, hydrocarbons, and their derivatives. The application of the programme is found in various fields as most of the products we use in our daily to day life are made of organic compounds. It includes drugs, polymers, electronic equipment, pesticides, food additives, explosives, medicines, synthetic fibres and paints.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

After completing the programme graduates are open to the following roles:-
-in pharmaceuticals, agrochemical, plastic and polymer, cosmetic, petrochemical industry as technical business analysts, production managers (supervisors), product development chemists/managers, quality controllers or quality assurance managers

-in research and development departments in industry and research institutes

-in government and private laboratories as technicians, analysts, research and research and development scientists

- in universities as laboratory technicians, research scientists, research associates and teaching assistants

–pursue PhD in Organic Chemistry, Pharmaceutical Sciences and other relevant related areas

5.0 PROGRAMME DELIVERY

The programme is delivered over a two year period during which the student is engaged in a number of lectures, practical investigations, field visits, journal article writing, research projects, problem-based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

The programme hopes to achieve the following broad objectives:-

- Understand the organic synthesis methods and their applications
- Understand characterisation methods for organic compounds
- Understand the chemistry of natural products, synthetic drugs and bio-organic compounds
– Communicate effectively and present information using multi-media to both experts and non-experts in the chemical industry.
- Undertake innovation and commercializing of research output
- Produce professionals who can utilise natural resources for the benefit of the country
- Produce professionals who are capable of creating employment

7.0 GENERAL PROVISIONS

The degree programme shall normally be studied over a minimum of four semesters on a block release basis. This programme is a taught Masters programme with modular work in the first three semesters. The research project to be done in the second year.

7.1 Dissertation

7.1.2 The dissertation shall be done in level two semesters two during which time the student is expected to undertake research

7.1.3 Each candidate is expected to submit a dissertation of at least one hundred (100) pages.

7.1.4 The format of the dissertation shall be provided by the Department of Chemical Sciences

8.0 PROGRAMME ASSESSMENT

8.1 Formal examinations are held at the end of every semester

8.2 Continuous assessment shall account for 40% of the overall assessment, while the final examination account for 60% of the overall assessment.

8.3 The aggregate pass mark shall be 50%

8.4 The aggregate pass mark shall comprise a pass in the final examination and a pass in the continuous assessment.

9.0 PROVISIONS FOR PROGRESSION

9.1 A student must pass modules worth at least 180 credits in the first level in order to proceed to the second level

9.2 To graduate a student should have accumulated 360 credits

10.0 FAILURE TO SATIFTY EXAMINERS

Refer to section 21 of the General Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the Faculty Regulations

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the parts of the degree programme are weighted as follows:

Leve1 (semester 1and 2) and Level 2 semester 1 taught modules (70%)

Level 2.2 Dissertation (30%)

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

CODE	Module Description	Credits
**MOCH 801	General Chemistry	18
**MOCH 802	Inorganic Chemistry	18
**MOCH 803	Physical Chemistry	18
*MOCH 804	Research Methods and Statistics	18
**MOCH 805	Laboratory Practical 1	18

Level 1 Semester 2

CODE	Module Description	Credits
**MOCH 806	Organic Reaction Mechanisms I	18
**MOCH 807	Modern Organic Synthesis I	18
**MOCH 808	Organic Spectroscopy I	18
*MOCH 809	Chemistry of Natural Products	18
**MOCH 810	Laboratory Practical II	18

Level 2 Semester 1

CODE	Module Description	Prerequisite	Credits
**MOCH 901	Organic Reaction Mechanisms II	MOCH 806	18
**MOCH 902	Organic Spectroscopy II	MOCH 808	18
**MOCH 903	Modern Organic Synthesis II	MOCH 807	18
*MOCH 904	Bio-Organic Chemistry		18
*MOCH 905	Medicinal, Biogenesis and Green Chemistry		18

Level 2 Semester 2

CODE	Module Description	Credits
**MOCH 906	Dissertation	90

14.0 MODULE SYNOPSES

MOCH 801 GENERAL CHEMISTRY

Computation of analytical results, concept of error, precision and accuracy, standard deviation, rejection of doubtful values with special reference to volumetric and gravimetric analysis. Calibration of analytical equipment. Methods of purification- distillation, drying techniques, solvent extraction, crystallisation techniques. Adsorption and partition chromatography- introduction to chromatography, column chromatography, thin layer chromatography, paper chromatography, Gas chromatography and HPLC. Titrimetric analysis.

MOCH 802 INORGANIC CHEMISTRY

Introduction to Exact Quantum Mechanical Results, Approximate Methods, Metal π-complexes, Structure and Bonding, Metal-ligand bonding, Metal-ligand Equilibria in solutions. Non-metal cages and metal clusters, Organometallic chemistry of transition metals, Reaction mechanism of transition metal complexes, Electronic spectra of transition metal complexes, Bio-inorganic chemistry and magnetic properties of complexes.

MOCH 803 PHYSICAL CHEMISTRY
Classical thermodynamics, Conditions of equilibrium and spontaneity, Surface phenomena and phase equilibria, Surface active agents, Electrochemistry, Chemical kinetics, Reactions in solutions, Motion of molecules, Microwave Spectroscopy, Rotational Vibrational Spectroscopy, Third law and Statistical thermodynamics, Raman Spectroscopy, Electrodes, photo chemistry, Symmetry and Group theory in chemistry.

MOCH 804 RESEARCH METHODS AND STATISTICS
Principles of research, managing a research project; performing a literature review; organizing and performing data collection, analysis and presentation; writing and presenting a technical paper/report; understanding the dynamics of team working; critically evaluating a given research project and applying the findings appropriately; developing subject-specific skills. Major project: carried out in teams and reported in written and oral form.

MOCH 805 LABORATORY PRACTICAL I
General methods of separation and purification of organic compounds. Preparation of some chromatography and qualitative analysis.

MOCH 806 ORGANIC REACTION MECHANISMS I
Aliphatic Nucleophilic Substitution, Aliphatic Electrophilic Substitution, Principles of asymmetric synthesis, Pericyclic Reactions-I, Pericyclic Reactions-II.

MOCH 807 MODERN ORGANIC SYNTHESIS I
Formation of C-C single bonds, Formation of Carbon-Carbon double bonds, Reactions of unactivated C-H bonds and organoboranes, Protecting groups and Synthetic applications of PTC and crown ethers.

MOCH 808 ORGANIC SPECTROSCOPY I
UV-Visible Spectroscopy, Infrared spectroscopy, Nuclear Magnetic Resonance Spectroscopy and Mass Spectrometry.

MOCH 809 CHEMISTRY OF NATURAL PRODUCTS
Heterocyclic Compounds, Benzfused heterocycles, Carbohydrate, Alkaloids, Terpenoids, Steroids, Flavonoids and Isoflavonoids.

MOCH 810 LABORATORY PRACTICAL II
Multistep synthesis of organic compounds

MOCH 901 ORGANIC REACTION MECHANISMS II
Free Radical Reactions, Rearrangements, Methodologies in asymmetric synthesis, Photo Chemistry-I, Photochemistry-II.

MOCH 902 ORGANIC SPECTROSCOPY II
Optical Rotatory Dispersion (ORD) and CD spectroscopy, Improving the PMR spectrum, Simplification of complex spectra, 2D NMR spectroscopy, Solution of structural problems by joint application of UV, IR, NMR and MS.

MOCH 903 MODERN ORGANIC SYNTHESIS II

Organo Silanes, Oxidation, Reduction, Retro Synthetic Analysis.

MOCH 904 BIO-ORGANIC CHEMISTRY

Cell Structure and Functions, Carbohydrates, Lipids, Biopolymers, Antimalarials and Antibiotics, Vitamins, Nucleic Acids, Amino-acids, Peptides and Proteins, Enzymes and Enzyme Models, Biotechnological Applications of Enzymes

MOCH 905 MEDICINAL, BIOGENESIS AND GREEN CHEMISTRY

Drug discovery, design and development; Drug design, development and synthesis- Introduction to quantitative structure activity relationship studies; Introduction to modern methods of drug design and synthesis; Concept of prodrugs and soft drugs, Synthesis and application of the following drugs: Fluoxetine, cetrizine, esomeprazole, fluconazole, zidovudine, methotrexate, diclofenac, labetalol, fenofibrate. Biogenesis and biosynthesis of natural products- Primary and secondary metabolites and the building blocks, general pathway of amino acid biosynthesis; Acetate pathway; Shikimic Acid pathway; Mevalonate pathway. Green chemistry- Introduction, basic principles of green chemistry; Use of green synthesis with suitable examples; Comparison of traditional processes versus green processes in the syntheses; Green Catalysts.

MOCH 906 Dissertation

This is a research project, based on a dissertation on an approved topic in the area of pharmaceutical chemistry. The module consists of project proposal, progress report and the final dissertation, together with oral presentations of each stage

DEPARTMENT OF COMPUTER SCIENCE

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN COMPUTER SCIENCE (HCSC)

Duration:	4 Years (Full time)
	3 Years (Visiting)
Maximum Credit Load:	540
Minimum Credit Load:	480
Maximum MBKs Credit Load:	398
ZNQF Level:	8

1. PURPOSE OF THE PROGRAMME

1.1 This programme is designed to develop and produce graduates with a broad understanding of both computing principles and computing practice. The programme is designed to help graduates effectively communicate computing concepts and solutions to bridge the gap between computing industry experts and business leaders to create and initiate innovation.

2. ENTRY REQUIREMENTS

2.1 For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better:

2.1.1 **Normal Entry**

A minimum of 2 A' Level passes in Mathematics AND (Physics OR Computer Science OR any other relevant Science subject).

2.1.2 **Special Entry**

Special entry may be granted to applicants with a National Diploma in Computer Science or any related field from a recognized institution.

2.1.3 **Mature Entry**

Refer to Section 3.3 of the General Academic Regulations.

2.1.4 **Visiting School/Block Release**

Should have at least an ND or HND in Software Engineering or any relevant field AND have proof of employment in the field specifying the nature of duties.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Computer Architecture and Organization; Computational Mathematics; Computer Graphics and Visual Simulation; Computer Security; Intelligent Systems; Networking and Data Communication; Operating Systems; Programming Languages; Software Development Fundamentals, Website Development, Software Engineering and Software Project Management; Social Issues and Professional Practice; Database Systems.

3.2 **Specialist Focus:** Programming, Networking, Computer Hardware, Computer Software.

3.3 **Orientation:** Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 **Employability:** Careers in the Computer Science field which include:
Research Scientists, IT Managers, IT Consultants, Software Engineers, Analyst Programmers, Systems Analysts, Security Analysts, Website Designers, Website Developers, Database Administrators, Network and Systems Administrators, and Hardware Technicians.

4.2 **Further Studies:** Master's and doctoral studies in Computer Science or in interdisciplinary programmes related to computing practices.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, programming laboratory classes, seminars, group work, industrial visits, Work-Related learning, research project, individual independent study.

5.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, continuous assessments.

6. PROGRAMME COMPETENCIES

6.1 **Generic:**

6.1.1 **Multidisciplinary**: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

6.1.2 **Flexibility;** to succeed in a rapidly changing field.

6.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things

6.1.4 **Communication skills**: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences

6.1.5 **Analysis and synthesis**: Capacity for analysis and synthesis using logical arguments and proven facts.

6.1.6 **Ethical commitment:** Professional integrity and awareness of impact of science and technology on society and the environment

6.1.7 **Technopreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.

6.2 **Discipline specific:**

6.2.1 **Technical understanding of Computer Science;** Graduates should have a mastery of Computer Science as described by the core of the Body of Knowledge.

6.2.2 **Familiarity with common themes and principles;** Graduates need understanding of a number of recurring themes, such as abstraction, complexity,

and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Computer Science and should not consider them as relevant only to the domains in which they were introduced.

6.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Computer Science is understanding the interplay between theory and practice and the essential links between them. Graduates of a Computer Science programme need to understand how theory and practice influence each other.

6.2.4 **Problem-solving skills;** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

6.2.5 **Project experience;** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Computer Science programmes should have been involved in at least one substantial project.

6.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

6.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

6.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

7. INTENDED LEARNING OUTCOMES

7.1 By the end of the programme a graduate will be able to:

7.1.1 apply knowledge of computing and mathematics appropriate to the program's student outcomes and to the discipline.

7.1.2 analyze a problem, and identify and define the computing requirements appropriate to its solution Design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.

7.1.3 understand professional, ethical, legal, security and social issues and responsibilities.

7.1.4 function effectively on teams to accomplish a common goal.

7.1.5 communicate effectively with a range of audiences.

7.1.6 analyze the local and global impact of computing on individuals, organizations, and society.

7.1.7 learn new programming and computing environments on their own.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

9. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

10. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. PROVISION FOR PROGRESSION

11.1 Refer to Section 7 of the Faculty of Science and Technology Regulations.

11.2 In addition to Section 11.1, to proceed from Level 2 to Level 3, the student must have passed the Group Project.

12. WORK-RELATED LEARNING

Refer to Section 8 of the Faculty of Science and Technology Regulations

13. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty of Science and Technology Regulations.

14. DEGREE WEIGHTING

Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to the availability of personnel. * denotes core modules.

Level 1 Semester 1

Code	Module Description	Credits
HCSCI131*	Introduction to Computers and Computer Applications	10
HCSCI132*	Principles of Programming Languages	10
HCSCI133*	Operating Systems	10
HCSCI134*	Fundamentals of Digital Electronics	10
HCSE133*	Ethics and Professionalism	10
CS131*	Basics of Communication Skills	12

Level 1 Semester 2

Code	Module Description	Credits
HCSCI135*	Computer Architecture and Organization	10
HCSCI136*	Data Structures and Algorithms	10
HCSCI137*	Software Engineering	10
HCSE135*	Database Systems	10
HCSE138 *	Discrete Mathematics	10
HTENG133	Physics for Engineers	10

Level 2 Semester 1

Code	Module Description	Credits
HCSCI231*	Web Development	10
HCSCI232*	Systems Analysis and Design	10
HCSCI233	Computer Security	10
HCSCI234*	Research Methods	10
HSTA132*	Applied Statistics	12
GS231*	Gender Studies	12
TCNP201*	Technopreneurship	12

Level 2 Semester 2

Code	Module Description	Pre-Requisites	Credits
HCSCI235*	Software Project Management	(HCSCI137)	10
HCSCI237*	Data Communications and Computer Networks		10
HCSCI238*	Design and Analysis of Algorithms.	(HCSCI136)	10

HCSCI239	Digital Image Processing		10
HCSCI240*	Group Project	(HCSCI232)	10
HCSE224*	Object Oriented Programming	(HCSCI132)	10

Level 3 Semester 1

Code	**Module Description**	**Credits**
HCSCI311*	Work-Related Learning I	40

Level 3 Semester 2

Code	**Module Description**	**Pre-Requisites**	**Credits**
HCSCI321*	Work-Related Learning II	(HCSCI311)	80

Level 4 Semester 1

Code	**Module Description**	**Pre-Requisites**	**Credits**
HCSCI431*	Theory of Computation		10
HCSCI432*	Simulation and Modelling		10
HCSCI433*	Parallel and Distributed Computing	(HCSCI237)	10
HCSCI434*	Artificial Intelligence		10
HTENG432*	Communications Network Design	(HCSCI237)	10
Electives (Select 1 module from the following)			
HCSE433	IoT and Cloud Systems Engineering	(HCSCI237)	10
HCSE434	Machine Learning		10
HCSE435	Digital Media Engineering		10

Level 4 Semester 2

Code	**Module Description**	**Pre-Requisites**	**Credits**
HCSCI435*	Computer Graphics		10
HCSCI437*	Human Computer Interaction		10
HCSCI438*	Research Project	(HCSCI240)	20
HCSE439*	Network Security and Cryptography		10
Electives (Select 1 module from the following)			
HCSCI439	Fundamentals of Data Science and Big Data		10
HCSCI436	Information Security		10

MODULE SYNOPSES

HCSCI131 Introduction to Computers and Computer Applications
Basic Computer Concepts, Data Processing Cycle, Number System. Computer Arithmetic, Types of Computer, Evolution of Computers, Concepts of computer languages, Programming construct, Programming data structure, Programming algorithms. Data Communications (Wired, Hybrid and Wireless networks). Variations in Computer Architecture including buses, internal.

HCSCI132 Principles of Programming Languages
The module aims to introduce the basic principles of programming, programming paradigms, program constructs and implementation of algorithms using Python programming language. Modular programming through use of functions, modules will also be covered to enhance students' understanding of code reusability. Moreover, data persistence will be demonstrated in files and data structures. The module will equip students with knowledge on the basics of

Object-Oriented Programming (OOP) and databases. Lastly, students will get to explore fundamentals of object-based programming using the Tkinter library in Python.

HCSCI133 Operating Systems
This module explores the design and implementation of computer operating systems. Topics include historical aspects of operating systems development, systems programming, process scheduling, synchronization of concurrent processes, deadlocks, virtual machines, memory management, virtual memory and paging, I/O and file systems, system security, os/architecture interaction, and distributed operating systems.

HCSCI134 Fundamentals of Digital Electronics
Introduction to digital design, number systems and codes, digital circuits, combinational logic design principles and practices, combinational circuit design examples, sequential logic design principles and practices, sequential circuit design examples. Memory (ROM, RAM, static and dynamic RAMs), CPLDs and FPGAs, computer-aided design tools.

HCSE133 Ethics and Professionalism
This module covers ethical and social issues related to the development and use of computer technology. Topics include ethical theory, social, political, and intellectual property (IP) and other legal considerations. Scenarios in problem areas: privacy, reliability and risks of complex systems, and responsibility of professionals for applications and consequences of their work.

CS131 Basics of Communication Skills
Refer to Communication Skills Department regulations

HCSCI135 Computer Architecture and Organization
This module studies the basic instruction set architecture and organization of a modern computer. Topics include Assembly language, The Von Neumann Machine Instructions, Registers, translating high level arithmetic into Assembly, Memory and Registers Addressing Modes, Logic Gates, Truth Tables, implementing Truth Tables, Latches and Memory Reading, Multicycle implementations, the concept of a Cycle. Finite State Machines, Balancing the work into Single Cycles, ROMs, PLAs, Microcode, RISC/CISC, Pipelines, Principle of Locality, Direct Mapped Caches, I/O Polling and Interrupts.

HCSCI136 Data Structures and Algorithms
This module builds on the programming skills acquired in Principles of Programming Languages. It couples work on Program Design, Analysis, and Verification with an introduction to the study of Data Structures. Students are introduced to: Lists, Stacks, Queues, Trees, Hash Tables, and Graphs. Students are expected to write several programs, ranging from very short programs to more elaborate systems. Emphasis is placed on the development of clear, modular programs that are easy to read, debug, verify, analyze, and modify.

HCSCI137 Software Engineering
The module gives the students practical experience in applying Software Life Cycle Models, standards and technologies. Topics covered include: Software Process and Problems, Requirements, Specifications and Software Design approaches, Modular Designs, CASE tools, Case Studies. Object Oriented Analysis and Design using the Unified Modelling Language (UML) to capture Requirements, Model User Interactions and Business Process, Verification and Validation, Implementation and Integration Issues, Reuse Concepts and Maintenance.

HCSE135 Database Systems
The module focuses on concepts and principles related to database management systems and links these to Relational Database Systems. Topics covered include: Database Systems Evolution, Database Systems in the Organization, Principles of Conceptual Design, Database Models, The Relational Data Model, Data Modelling, Database Design Theory, Data Definition and Manipulation Languages, Storage and Indexing Techniques, Query Processing and Optimization, Concurrency Control and Recovery and Database Programming Interfaces.

HCSE138 Discrete Mathematics
This module introduces and discusses the fundamentals of discrete mathematics as applied to computer science, focusing on providing a basic theoretical foundation for further work. Students are exposed to logic and proof techniques, set theory, elementary number theory, functions and relations, graphs, trees, modelling computations and abstract algebra. This module integrates symbolic tools, graphical concepts, and numerical calculations. Techniques of counting: permutations, combinations, recurrences, algorithms are also covered.

HTENG133 Physics for Engineers
The physics of electricity and magnetism; application of electronic, semiconductor and magnetic materials, ferromagnetism, electric and magnetic fields. Forces on magnetized iron surfaces; forces of attraction and alignment, Waves and Optics. Mechanics: work and energy, momentum and collisions. Thermal physics: temperature, kinetic theory and the ideal gas, heat and the first law of thermodynamics.

HCSCI231 Web Development
The module will explain the basics of Web Design, Site Development Processes, Design Principles; Page Layout Navigation. Topics include; HTML, CSS and JavaScript, developing dynamic, database-driven web pages using frameworks and scripting languages. This module introduces students to web database connections.

HCSCI232 Systems Analysis and Design
This module will cover the principles of Information Systems including analysis and design. Students will learn techniques in data requirements collection and analysis along with methods 18 to modelling data needs. Modelling of data will occur at the conceptual, logical, and physical levels along with an ability to compare the different approaches. Students will understand the importance and constraints imposed by the domain of the information system along with business rules that guide the design. Functional dependencies and domain normalization will also be discussed as part of the requirements analysis. Object-Oriented information system modelling and User-Centered design techniques will be explored.

HCSCI233 Computer Security
Basic concepts in information security, including security policies, security models, and security mechanisms. Concepts related to applied cryptography, including plain-text, cipher-text, the four techniques for crypto-analysis, symmetric cryptography, asymmetric cryptography, digital signature, message authentication code, hash functions, and modes of encryption operations. concepts of malicious code, including virus, Trojan horse, and worms. Common vulnerabilities in computer programs, including buffer overflow vulnerabilities, time-of-check to time-of-use flaws, incomplete mediation. The requirements and mechanisms for identification and authentication. Issues about password authentication, include dictionary attacks (password guessing attacks), password management policies, and one-time password

mechanisms. Security mechanisms for conventional operating systems, including memory, time, file, and object protection requirements and techniques and protection in contemporary operating systems. The requirements for trusted operating systems, and describe the independent evaluation, including evaluation criteria and evaluation process. Security requirements for database security, and describe techniques for ensuring database reliability and integrity, secrecy, inference control, and multi-level databases. Threats to networks, and techniques for ensuring network security, including encryption, authentication, firewalls, and intrusion detection. The requirements and techniques for security management, including security policies, risk analysis, and physical threats and controls.

HCSCI234 Research Methods

This module equips students with research techniques including definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation, research ethics, report writing.

HSTA132 Applied Statistics

This module covers Introduction to statistics: definition, uses of statistics, Probability, Probability distributions, Measures of central tendency and dispersion, Sampling techniques, Data types, Presentation and Summarization techniques, Regression and Correlation, Simple Statistical Inference.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

TCNP 201 Technopreneurship

Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

HCSCI235 Software Project Management

Software Project Management is the practice of controlling the cost, time, manpower, hardware and software resources involved in the development activity that commences with a problem statement and ending with the delivery of complete software product to the customer. It will include planning, monitoring and controlling, Project evaluation, scope management, team building, stakeholder management, risk assessment, scheduling, quality and conflict management.

HCSCI237 Data Communications and Computer Networks

This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

HCSCI238 Design and Analysis of Algorithms.
This module investigates methods for designing efficient and reliable algorithms. It introduces several algorithm design strategies that build on data structures and programming techniques introduced in Data structures and Algorithms. These include induction, divide-and-conquer, dynamic programming, and greedy algorithms. Topics of study include graph theory, hashing, and advanced data structures.

HCSCI239 Digital Image Processing
The module introduces basic concepts and methodologies for the formation, representation, compression, enhancement and analysis of digital images. It introduces students to the fundamental techniques and algorithms used for acquiring, processing and extracting useful information from digital images. Particular emphasis will be placed on covering methods used for image sampling and quantization, image transforms, image enhancement and restoration, image encoding, image analysis and pattern recognition. In addition, the students will learn how to apply the methods to solve real-world problems in several areas including medical, remote sensing and surveillance and develop the insight necessary to use the tools of digital image processing (DIP) to solve any new problem.

HCSCI240 Group Project
The students work in a group to produce a working Product including a Project Proposal, a Project Plan, a System Specification, and User Documentation and perform Project Management Activities to ensure the product is delivered on time. The Implementation includes Developing the Product, Testing and Evaluation and Presentation.

HCSE234 Object Oriented Programming
The module aims at providing a solid foundation in Object Oriented Paradigm. Topics covered include: Objects Overview and Review, creating Class Instances within constructors, Object Analysis, creating Fields and Properties, Inheritance and specialized Classes, Base Class and Abstract Classes, Events and Exceptions, providing services using Interfaces and Abstract Classes, Polymorphism, Shared and Static members, Overloading Operators, Overriding, Multithreading

HCSCI311 Work-Related Learning I
Refer to Section 8 of the Faculty of Science and Technology Regulations

HCSCI321 Work-Related Learning II
Refer to Section 8 of the Faculty of Science and Technology Regulations

HCSCI431 Theory of Computation
The module explores Turing machines, universal computation, the halting problem and general undecidability, Rice's theorem, the recursion theorem, efficient computation models including finite automata, regular languages, context-free grammars, time and space (memory) bounds, deterministic and nondeterministic computation and their relationships, the P versus NP problem and hard problems for NP and beyond.

HCSCI432 Simulation and Modelling
The module presents general mathematical modelling and simulation principles by applying them to concrete problems. Topics covered include the Modelling Process, System Dynamics, Computational Error, Simulation techniques, Empirical Models, Cellular Automaton and High-Performance Computing.

HCSCI433 Parallel and Distributed Computing
The module covers Parallelism Fundamentals, Parallel Decomposition, Parallel Algorithms, Analysis, and Programming, Parallel Architecture, Parallel Performance, Formal Models and Semantics. The module will also cover key design principles of distributed systems including communication protocols, processes and threads, naming, synchronization, consistency and replication, fault tolerance, and security

HCSCI434 Artificial Intelligence
The module develops in students a working knowledge of intelligent systems as well as the understanding of the application of artificial reasoning in industrial processes. Topics covered include: An introduction to intelligence, historical and current trends and characterization of knowledge-based systems, knowledge representation, reasoning, problem solving, and learning. It then explores those further by surveying current applications in selected areas such as game playing and natural language processing

HTENG432 Communications Network Design
Business justification. System requirements. System analysis and network modelling. System design and implementation. System performance and evaluation. Market forecasts of demand on the network to the implementation of plans. Aspects of the design that influence the likely replacement of legacy telecommunications networks. Network performance parameters that impact on the quality of service perceived by customers. Layered view of network requirements, signalling, operation, service management, support systems and associated architectures.

HCSE433 IoT and Cloud Systems Engineering
IoT and Cloud Computing. IoT Physical Devices and Endpoints: IoT physical servers and cloud offerings: cloud storage models and communication Networks, Cloud technologies, Framework and Platforms. Internet of services, SOA, Grid computing, System types, architectures and models, applications in telecommunication systems, cloud systems examples, security management, Governance, legislation, economic environment.

HCSE434 Machine Learning
Introduction to machine learning, data mining, and statistical pattern recognition. Topics include: Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks); Unsupervised learning (clustering, dimensionality reduction, recommender systems, deep learning); Best practices in machine learning (bias/variance theory; innovation process in machine learning and AI); This module will also draw from numerous case studies and applications, so that learners will also learn how to apply learning algorithms to building smart robots (perception, control), text understanding (web search, anti-spam), computer vision, medical informatics, audio, database mining, and other areas.

HCSE 435 Digital Media Engineering

Audio Engineering: concepts and standards, Overview of audio in the broadcast context, Computer-based audio systems, Software engineering and the MAX programming environment, Audio quality assessment and monitoring. Video Engineering: Concepts and standards, Overview of video in the broadcast context, Compression standards for broadcast, internet distribution and streaming, video transmission over IP networks.

HCSCI435 Computer Graphics
The module covers characteristics of graphics I/O devices, 2D/3D transformations including scaling, translation, and rotation. It also explores Graphics Pipeline, Data Structures for Graphics, Geometrical representation, OpenGL programming, vertex processing, lighting and shading, rasterization including line and polygon drawing, ray casting, ray tracing, spline curves and surfaces, color models, fractals, computer graphics in games and Computer Animation; visualization.

HCSCI436 Information Security
The module covers concepts and applied methods in information security encompassing the confidentiality, integrity, and availability model of information assets. Topics include security threats, cryptography, security analysis (attack taxonomy), access control, threat protection, Security Policies, Laws, Computer Crimes and Cyber Security.

HCSCI437 Human Computer Interaction
Human characteristics: human and computer Graphic design basics, Interaction: input and output, Dialogue interactions and response times, Design process, Models of users in design, Dialog notations and design, Implementation support, Evaluation, Help and documentation, Hypertext, multimedia and WWW, Game design basics.

HCSCI438 Research Project (HCSCI238)
The Research project involves supervision. This project is intended for the student to demonstrate skills acquired to develop systems for an organization of their choice. Candidates must be able to demonstrate the highest level of innovation. A level of research is expected to ensure some level of originality and critical thinking in the project design and implementation.

HCSCI439 Fundamentals of Data Science and Big Data
Introduction: Big Data Overview, importance of data science, big data analytics in industry verticals. Data Analytics Lifecycle and methodology: Business Understanding, Data Understanding, Data Preparation, Modelling and Evaluation. Data exploration and pre-processing, Data Analytics: Theory and Methods, Unstructured Data Analytics, Data Visualization Techniques, Creating final deliverables

HCSE439 Network Security and Cryptography
Mathematical Foundations - Overview on Modern Cryptography, Number Theory, Probability and Information Theory. Classical Cryptosystems – Shannon theory I, II, III. Symmetric Key Ciphers, Cryptanalysis of Symmetric Key Ciphers, Stream Ciphers and Pseudo randomness, Hash Functions and MACs, Asymmetric Key Ciphers: Construction and Cryptanalysis, Digital Signatures, Modern Trends in Asymmetric Key Cryptography and Network Security.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN COMPUTER SYSTEMS ENGINEERING (HCSE)

Duration:	4 Years (Full time)
	3 Years (Visiting)
Maximum Credit Load:	540
Minimum Credit Load:	480
Maximum MBKs Credit Load:	384
ZNQF Level:	8

1. PURPOSE OF THE PROGRAMME

1.1 This programme is designed to create graduates who stand out with the skills required to solve 21st Century Computer Engineering problems. The aim is to combine an in-depth coverage of computing and its applications with a systematic approach to the design and construction of high-quality software and systems.

2. ENTRY REQUIREMENTS

2.1 For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better:

2.1.1 **Normal Entry**

A minimum of 2 A' Level passes in Mathematics AND (Physics OR Computer Science OR any other relevant Science subject).

2.1.2 **Special Entry**

Special entry may be granted to applicants with a National Diploma in Computer Science or any related field from a recognized institution.

2.1.3 **Mature Entry**

Refer to Section 3.3 of the General Academic Regulations.

2.1.4 **Visiting School/Block Release**

Should have at least an ND or HND in Software Engineering or any relevant field AND have proof of employment in the field specifying the nature of duties.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Systems development, Hardware design, Digital systems, Digital signal processing, Computer systems administration, Computer networks, Computational intelligence, Database Systems, Website development, Embedded systems

3.2 **Specialist Focus:** Hardware Design, Systems Development, Computer Networking, Machine Learning.

3.3 **Orientation:** Research, Innovation and Technopreneurial orientation. Teaching and learning is focused on producing graduates with the necessary knowledge, skills and competencies sufficient to start technology-based businesses.

3.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 **Employability:** Careers in the Computer Engineering field which include: Software Engineers, Network Engineers, Hardware Engineers, Website Designers, Analyst Programmers, Network and Systems Administrators, Database Administrators, IT Managers, and Systems Analysts.

4.2 **Further Studies:** Master's and doctoral studies in Computer Systems Engineering, Computer Science, Data Science, or interdisciplinary programmes related to computing practices.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, programming laboratory classes, seminars, group work, industrial visits, Work-Related learning, research project, individual independent study.

5.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, continuous assessments.

6. PROGRAMME COMPETENCIES

6.1 **Generic:**

6.1.1 **Multidisciplinary**: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

6.1.2 **Flexibility;** to succeed in a rapidly changing field.

6.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things

6.1.4 **Communication skills**: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences

6.1.5 **Analysis and synthesis**: Capacity for analysis and synthesis using logical arguments and proven facts.

6.1.6 **Ethical commitment:** Professional integrity and awareness of the impact of science and technology on society and the environment

6.1.7 **Technopreneurial skills:** Capability to identify and create new technology-based business ventures based on knowledge and new thinking paradigms.

6.2 **Discipline-specific:**

6.2.1 **Technical understanding of Computer Science;** Graduates should have a mastery of Computer Systems Engineering as described by the core of the Body of Knowledge.

6.2.2 **Familiarity with common themes and principles;** Graduates need an understanding of a number of recurring themes, such as abstraction, complexity, and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Computer Systems Engineering and should not consider them as relevant only to the domains in which they were introduced.

6.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Computer System Engineering is understanding the interplay between theory and practice and the essential links between them. Graduates of a

Computer Systems Engineering programme need to understand how theory and practice influence each other.

6.2.4 **Problem solving skills;** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

6.2.5 **Project experience;** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Computer Systems Engineering programmes should have been involved in at least one substantial project.

6.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

6.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

6.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

7. INTENDED LEARNING OUTCOMES

7.1 By the end of the programme a graduate will be able to:

7.1.1 apply knowledge of computing and mathematics to solve Computer Engineering problems.

7.1.2 use engineering concepts to design digital systems that drive the economy and boost productivity in industries.

7.1.3 design and develop quality software systems following sound engineering principles, and design, develop, configure and deploy a computer network using the right tools.

7.1.4 use machine learning tools to perform predictions and establish patterns from data sets.

7.1.5 conceptualize, design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.

7.1.6 exhibit Computer Engineering best practices and standards of professional and ethical responsibility as well as communicate technical matters effectively in oral, written or graphical form.

7.1.7 pursue postgraduate programmes in any of the identified thematic areas.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

9. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

10. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. PROVISION FOR PROGRESSION

11.1 Refer to Section 7 of the Faculty of Science and Technology Regulations.

11.2 In addition to Section 11.1, to proceed from Level 2 to Level 3, the student must have passed the Group Project.

12. WORK-RELATED LEARNING

Refer to Section 8 of the Faculty of Science and Technology Regulations

13. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty of Science and Technology Regulations.

14. DEGREE WEIGHTING

Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to availability of personnel. * denotes core modules.

Level 1 Semester 1

Code	Module Description	Credits
HCSE131*	Introduction to Computer Engineering	10
HCSE133*	Ethics and Professionalism	10
HCSCI132*	Principles of Programming languages	10
HTENG133*	Physics for Engineers	10
HMAT131*	Calculus 1	12
CS131*	Basics of Communication Skills	12

Level 1 Semester 2

Code	Module Description	Credits
HCSE134*	Electrical and Electronic Principles	10
HCSE135*	Database Systems	10
HCSCI136*	Data Structures and Algorithms	10
HCSE137*	Digital Electronics and Logic Design	10
HCSE138*	Discrete Mathematics	10
HCSCI137*	Software Engineering	10

Level 2 Semester 1

Code	Module Description	Credits
HCSE231*	Object Oriented Analysis and Design	10
HCSE232*	Engineering Statistics	10
HCSE233*	Linear Algebra	10
HCSCI133*	Operating Systems	10
HCSCI135*	Computer Architecture and Organization	10
GS231 *	Gender Studies	12
TCNP201*	Technopreneurship	12

Level 2 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSE224*	Object Oriented Programming	(HCSE231)	10
HCSE235*	Web and Mobile Application Development		10
HCSE236*	Group Project	(HCSE231)	10
HCSCI237*	Data Communications and Computer Networks		10
HCSCI238*	Design and Analysis of Algorithms	(HCSC136)	10
HTENG237*	Digital Signal Processing		10

Level 3 Semester 1

Code	Module Description	Pre-requisites	Credits
HCSE311*	Work-Related Learning I		40

Level 3 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSE321*	Work-Related Learning II	(HCSE311)	80

Level 4 Semester 1

Code	Module Description		Credits
HCSE431*	Microprocessors and Microcontrollers Systems	(HCSE134)	10
HCSE432*	Embedded Systems	(HCSCI135)	10
HCSCI234*	Research Methods		10
HCSCI433*	Parallel and Distributed Computing	(HCSE137)	10
HTENG432*	Communications Network Design	(HCSCI237)	10
Electives (Select 1 module from the following)			
HCSE433	IoT and Cloud Systems Engineering	(HCSCI237)	10
HCSE434	Machine Learning		10
HCSE435	Digital Media Engineering		10

Level 4 Semester 2

Code	Module Description		Credits
HCSE436*	Control Systems Engineering		10
HCSE439*	Network Security and Cryptography		10
HCSE437*	Project Management	(HCSCI137)	10
HCSE438*	Research Project	(HCSE236)	20
Electives (Select 1 module from the following)			
HCSCI439	Fundamentals of Data Science and Big Data	(HCSE135)	10
HCSCI436	Information Security		10

MODULE SYNOPSES

HCSE131 Introduction to Computer Engineering
This module provides the basic concepts of computer design and operation, both hardware and software, to new computer engineering students. Topics include the representation of information and computer arithmetic combinational logic sequential logic registers, counters, adders, multiplexers register transfer language, hardware description language basic computer organization assembly programming.

HCSE133 Ethics and Professionalism
This module covers ethical and social issues related to the development and use of computer technology. Topics include ethical theory, social, political, and intellectual property (IP) and other legal considerations. Scenarios in problem areas: privacy, reliability and risks of complex systems, and responsibility of professionals for applications and consequences of their work

HCSCI132 Principles of Programming languages
This module examines the concepts and structures governing the design and implementation of programming languages. It presents an introduction to the concepts behind compilers and runtime representations of programming languages; features of programming languages

supporting abstraction and polymorphism; and the procedural, functional, object-oriented, and concurrent programming paradigms. Programs are required in languages illustrating each of these paradigms.

HTENG133 Physics for Engineers

The physics of electricity and magnetism; application of electronic, semiconductor and magnetic materials, ferromagnetism, electric and magnetic fields. Forces on magnetized iron surfaces; forces of attraction and alignment, Waves and Optics. Mechanics: work and energy, momentum and collisions. Thermal physics: temperature, kinetic theory and the ideal gas, heat and the first law of thermodynamics.

HMAT131 Calculus 1

This module introduces the basic knowledge of limit theorem, including functional limit, continuity of function, and the calculation of limit; includes calculus methods such as derivative and differential, integral, ordinary differential equation and Improper integrals; series, convergence criteria, convergence domain and uniform convergence of function series, term-by-term limit, term-by-term summation, term-by-term integral. multivariate functions; derivative, integral, line/surface integrals of the first type, line/surface 8 integrals of the second type, plane vector field and Green formula, space vector field and Gauss formula, Stokes formula, path independent integral, linear ordinary differential equation of second order and system of linear ordinary differential equation of first order.

CS131 Basics of Communication Skills

Refer to Communication Skills Department regulations

HCSE134 Electrical and Electronic Principles

Circuit elements theorems. Analysis of DC and AC single-phase circuits, phase representation of AC quantities, complex representation of AC quantities, transients, series and parallel resonance. Power and energy factor, Magnetic fields, electric fields, behaviour of charges in Law. Magnetic and dielectric materials, forces in electric and magnetic media, energy stored Faraday's and Lenz laws. Motors and Generators, electrostatic generators, Magnetic circuits and flux measurement, self and mutual inductance, transformers. Introduction to three–phase circuits, star–delta transformations, current relations, power circuit calculations. Principles of operation of single-phase transformers. IC, basic operational characteristics and parameters, TTL circuits, practical consideration in the use of TTL, CMOS circuits, comparing CMOS and TTL characteristics, interfacing logic families, memory, IC applications, circuit board technology, technologies of the future, nano-technology

HCSE135 Database Systems

The module focuses on concepts and principles related to database management systems and link these to Relational Database Systems. Topics covered include Database Systems Evolution, Database Systems in the Organisation, Principles of Conceptual Design, Database Models, The Relational Data Model, Data Modelling, Database Design Theory, Data Definition and Manipulation Languages, Storage and Indexing Techniques, Query Processing and Optimization, Concurrency Control and Recovery and Database Programming Interfaces.

HCSCI136 Data Structures and Algorithms

This module builds on the programming skills acquired in Principles of Programming Languages. It couples work on Program Design, Analysis, and Verification with an introduction to the study of Data Structures. Students are introduced to: Lists, Stacks, Queues, Trees, Hash Tables, and Graphs. Students are expected to write several programs, ranging from

very short programs to more elaborate systems. Emphasis is placed on the development of clear, modular programs that are easy to read, debug, verify, analyze, and modify.

HCSE137 Digital Electronics and Logic Design
Introduction to digital design, number systems and codes, digital circuits, combinational logic design principles and practices, combinational circuit design examples, sequential logic design principles and practices, sequential circuit design examples. Memory (ROM, RAM, static and dynamic RAMs), CPLDs and FPGAs, computer aided design tools

HCSE138 Discrete Mathematics
This module introduces and discusses the fundamentals of Discrete Mathematics as applied to Computer Science, focusing on providing a basic theoretical foundation for further work. Students are exposed to logic and proof techniques, set theory, elementary number theory, functions and relations, graphs, trees, modelling computations and abstract algebra. This module integrates symbolic tools, graphical concepts, and numerical calculations. Techniques of counting: permutations, combinations, recurrences, algorithms are also covered.

HCSCI137 Software Engineering
The module gives the students practical experience in applying Software Life Cycle Models, standards and technologies. Topics covered include: Software Process and Problems, Requirements, Specifications and Software Design approaches, Modular Designs, CASE tools, Case Studies. Object Oriented Analysis and Design using the Unified Modelling Language (UML) to capture Requirements, Model User Interactions and Business Process, Verification and Validation, Implementation and Integration Issues, Reuse Concepts and Maintenance.

HCSE231 Object Oriented Analysis and Design
Introducing object orientation; classes; objects; abstraction; encapsulation; software modelling; aims of modelling; principles of modelling; overview of UML; object-oriented analysis and design process; user stories; analysis and design of artefacts; the conceptual model; identifying classes and their relationships; identifying class responsibilities; CRC cards; Domain Driven Design; Object-Oriented Design Principles; Object-Oriented Design Patterns.

HCSE232 Engineering Statistics
This module covers: Introduction to statistics: definition, uses of statistics, Probability, Probability distributions, Measures of central tendency and dispersion, Sampling techniques, Data types, Presentation and Summarization techniques, Regression and Correlation, Simple Statistical Inference.

HCSE233 Linear Algebra
This module will introduce basic concepts and techniques from linear algebra that will be required in later modules in areas such as computer graphics and machine learning. Topics include eigenvalues and eigenvectors, vector linear independence and bases for vector spaces, spaces and subspaces, orthogonal vector spaces and the Gram-Schmidt orthogonalisation process, matrices, inverse matrices, solution of linear systems, elementary matrix factorisations, determinants of matrices, linear transformations

HCSCI133 Operating Systems
This module explores the design and implementation of computer operating systems. Topics include historical aspects of operating systems development, systems programming, process scheduling, synchronization of concurrent processes, deadlocks, virtual machines, memory

management, virtual memory and paging, I/O and file systems, system security, os/architecture interaction, and distributed operating systems.

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This module studies the basic instruction set architecture and organization of a modern computer. Topics include Assembly language, The Von Neumann Machine Instructions, Registers, translating high level arithmetic into Assembly, Memory and Registers Addressing Modes, Logic Gates, Truth Tables, implementing Truth Tables, Latches and Memory Reading, Multicycle implementations, the concept of a Cycle. Finite State Machines, Balancing the work into Single Cycles, ROMs, PLAs, Microcode, RISC/CISC, Pipelines, Principle of Locality, Direct Mapped Caches, I/O Polling and Interrupts.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

TCNP 201 Technopreneurship

Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

HCSE224 Object Oriented Programming

The module aims at providing a solid foundation in Object Oriented Paradigm. Topics covered include: Objects Overview and Review, creating Class Instances within constructors, Object Analysis, creating Fields and Properties, Inheritance and specialized Classes, Base Class and Abstract Classes, Events and Exceptions, providing services using Interfaces and Abstract Classes, Polymorphism, Shared and Static members, Overloading Operators, Overriding, Multithreading.

HCSE235 Web and Mobile Application Development

Site development processes, Design principles; page layout navigation, HTML, CSS and PHP programming language. Database-driven web pages using PHP. PHP framework and syntax, connection to any ODBC-compliant database, and hands-on practice with a MySQL database to create database-driven HTML forms and reports. Software development models and processes, component parts, software development practices, requirements engineering, system analysis, high- and low-level design/architecture, coding, integration, design and code reviews, testing, maintenance, project management and configuration management. Mobile development, user interface design and object-oriented programming for mobile development; design constraints and choices for touch screens and small interface sizes; portability of code between mobile and stationary platforms.

HCSE236 Group Project

The students work in a group to produce a working Product including a Project Proposal, a Project Plan, a System Specification, System Design, System Implementation, User Documentation and perform Project Management Activities to ensure the product is delivered

on time. The Implementation includes Developing the Product, Testing and Evaluation and Presentation. The resulting product should be functional and usable.

HCSCI237 Data Communications and Computer Networks
This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

HCSCI238 Design and Analysis of Algorithms
This module investigates methods for designing efficient and reliable algorithms. It introduces several algorithm design strategies that build on data structures and programming techniques introduced in Data structures and Algorithms. These include induction, divide-and-conquer, dynamic programming, and greedy algorithms. Particular topics of study include graph theory, hashing, and advanced data structures.

HTENG237 Digital Signal Processing
The topics covered in this module include purpose of digital signal processing (DSP), theories and concepts, role of DSP in the context of computer engineering; analysis of digital spectra; application of discrete Fourier transforms, convolution types; filtering, digital filtering; transforms; discrete time signals; sampling issues; applications to include image processing, audio processing; use of relevant software tools.

HCSE311 Work-Related Learning I
Refer to Section 8 of the Faculty of Science and Technology Regulations

HCSE321 Work-Related Learning II
Refer to Section 8 of the Faculty of Science and Technology Regulations

HCSE431 Microprocessors and Microcontrollers Systems
The purpose of this module is to teach students the fundamentals of microprocessor and microcontroller systems topics include the history of microprocessors, microcomputers, microcontrollers, Microprocessor architecture, program development tools. Digital input-output. Interrupt system. Event manager. Analogue-to-digital and digital-to-analogue converters. Serial peripheral interface. Flash programming. Digital signal processors built-in operating system. Microprocessor's boot ROM

HCSCI433 Parallel and Distributed Computing
The module covers Parallelism Fundamentals, Parallel Decomposition, Parallel Algorithms, Analysis, and Programming, Parallel Architecture, Parallel Performance, Formal Models and Semantics. The module will also cover key design principles of distributed systems including communication protocols, processes and threads, naming, synchronization, consistency and replication, fault tolerance, and security.

HCSE432 Embedded Systems
This module covers Embedded Systems concepts. Topics include nature of embedded systems, specific problems, special issues; role in computer engineering; embedded microcontrollers, embedded software; real time systems, problems of timing and scheduling; testing and performance issues, reliability; low power computing, energy sources, leakage; design methodologies, software tool support for the development of such systems; problems of maintenance and upgrade; networked embedded systems.

HCSCI234 Research Methods
This module equips students with research techniques including definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation, research ethics, report writing.

HCSE433 IoT and Cloud Systems Engineering
IoT and Cloud Computing. IoT Physical Devices and Endpoints: IoT physical servers and cloud offerings: cloud storage models and communication Networks, Cloud technologies, Framework and Platforms. Internet of services, SOA, Grid computing, System types, architectures and models, applications in telecommunication systems, cloud systems examples, security management, Governance, legislation, and economic environment.

HCSE434 Machine Learning
Introduction to machine learning, data mining, and statistical pattern recognition. Topics include: Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks); Unsupervised learning (clustering, dimensionality reduction, recommender systems, deep learning); Best practices in machine learning (bias/variance theory; innovation process in machine learning and AI); This module will also draw from numerous case studies and applications, so that learners will also learn how to apply learning algorithms to building smart robots (perception, control), text understanding (web search, anti-spam), computer vision, medical informatics, audio, database mining, and other areas.

HCSE 435 Digital Media Engineering
Audio Engineering: concepts and standards, Overview of audio in the broadcast context, Computer-based audio systems, Software engineering and the MAX programming environment, Audio quality assessment and monitoring. Video Engineering: Concepts and standards, Overview of video in the broadcast context, Compression standards for broadcast, internet distribution and streaming, video transmission over IP networks.

HTENG432 Communications Network Design
Business justification. System requirements. System analysis and network modelling. System design and implementation. System performance and evaluation. Market forecasts of demand on the network to the implementation of plans. Aspects of the design that influence the likely replacement of legacy telecommunications networks. Network performance parameters that impact the quality of service perceived by customers. Layered view of network requirements, signalling, operation, service management, support systems and associated architectures.

HCSCI 436 Information Security
The module covers concepts and applied methods in information security encompassing the confidentiality, integrity, and availability model of information assets. Topics include security threats, cryptography, security analysis (attack taxonomy), access control, threat protection, Security Policies, Laws, Computer Crimes and Cyber Security.

HCSE436 Control Systems Engineering
This module introduces the students to the theory and practice of control systems engineering. Topics include Introduction to Feedback Control Systems, Review of the Laplace Transforms, System modelling and analysis, Modelling in the Frequency Domain, structural properties, Time Response Analysis, Frequency Response Analysis, Feedback system design, Optimum

feedback control, Introduction to the minimum principle, Planar and spatial kinematics, and motion planning; mechanism design for manipulators and mobile robots, multi-rigid-body dynamics, 3D graphic simulation; control design, actuators, and sensors; wireless networking, task modelling, human-machine interface, and embedded software.

HCSE437 Project Management
Project Management is the practice of controlling the cost, time, manpower, hardware and software resources involved in the development activity that commences with a problem statement and ending with the delivery of a complete product. It will include planning, monitoring and controlling, Project evaluation, scope management, team building, stakeholder management, risk assessment, scheduling, quality and conflict management

HCSE438 Research Project
The Research project involves supervision. This project is intended for the student to demonstrate skills acquired to develop systems for an organization of their choice. Candidates must be able to demonstrate the highest level of innovation. A level of research is expected to ensure some level of originality and critical thinking in the project design and implementation.

HCSCI439 Fundamentals of Data Science and Big Data
Introduction: Big Data Overview, importance of data science, big data analytics in industry verticals. Data Analytics Lifecycle and methodology: Business Understanding, Data Understanding, Data Preparation, Modelling and Evaluation. Data exploration and pre-processing, Data Analytics: Theory and Methods, Unstructured Data Analytics, Data Visualization Techniques, Creating final deliverables

HCSE439 Network Security and Cryptography
Mathematical Foundations - Overview on Modern Cryptography, Number Theory, Probability and Information Theory. Classical Cryptosystems – Shannon theory I, II, III. Symmetric Key Ciphers, Cryptanalysis of Symmetric Key Ciphers, Stream Ciphers and Pseudo randomness, Hash Functions and MACs, Asymmetric Key Ciphers: Construction and Cryptanalysis, Digital Signatures, Modern Trends in Asymmetric Key Cryptography and Network Security.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN SOFTWARE ENGINEERING (HSWE)

Duration:	4 Years (Full time) 3 Years (Visiting)
Maximum Credit Load:	540
Minimum Credit Load:	480
Maximum MBKs Credit Load:	384
ZNQF Level:	8

- **PURPOSE OF THE PROGRAMME**

6.1 The programme seeks to produce 21st century graduates with the ability to develop systematic models and reliable techniques throughout the production process of high-quality computer systems to be delivered on time and within budget.

1.2 The programme will also develop engineering professionals who have an ever-learning capability, and knowledge of modern software engineering theoretical concepts and principles applicable in crafting solutions which address national and global needs.

- **ENTRY REQUIREMENTS**

4.1 For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better:

- **Normal Entry**

A minimum of 2 A' Level passes in **Mathematics AND** Software Engineering OR Computer Science OR their equivalents.

- **Special Entry**

Should have at least an **ND** or **HND** in Software Engineering or any relevant field, OR any equivalent qualification acceptable according to the Institute's General Academic Regulations.

- **Mature Entry**

Refer to Section 3.3 of the General Academic Regulations.

- **Visiting School/Block Release**

Should have at least an ND or HND in Software Engineering or any relevant field AND have proof of employment in the field specifying the nature of duties.

- **PROGRAMME CHARACTERISTICS**
  - **Areas of study:** Computing Essentials, Mathematical and Engineering fundamentals, Professional Practice, Software Modelling and Analysis, Requirements Analysis and Specification, Software Design, Software Verification and Validation, Software Process, Software Quality, Security.
  - **Specialist Focus:** Systems Development, Artificial Intelligence, Software Testing, Security, Database Systems, Networking, Requirements Engineering and Data Analytics.
  - **Orientation:** Research, innovation and Technopreneurial orientation. Teaching and learning is focused on producing graduates with the necessary knowledge, skills and competences sufficient to start technology-based businesses.
  - **Distinctive Features:** The programme focuses on building high quality systems using international Software Process Methodologies and Requirements Engineering practices so as to address societal and industrial issues while promoting research and innovation.

- **CAREER OPPORTUNITIES AND FURTHER EDUCATION**
  - **Employability:** Careers in the Software Engineering field which include: Software Architect, Software Developer, Systems Analyst, Systems Engineer, Software Tester, Information Systems Managers, Cloud Manager, Database Engineer, Data Scientist, Network Engineer, Software Project Manager.
  - **Further Studies:** Masters and Doctoral studies in Software Engineering, Data Science, Cloud Computing, Information Security and any other relevant Computing discipline.

- **PROGRAMME DELIVERY**
  - **Teaching and Learning Methods:** Lectures, e-learning, tutorials, seminars, group work, computer laboratory classes, industrial visits, industrial attachment, research project and individual independent study.
  - **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, and continuous assessments.

- **PROGRAMME COMPETENCIES**
  - **Generic:**
    - **Multidisciplinary:** Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena.
    - **Flexibility:** To adapt and succeed in a dynamic field.
    - **Quantitative and innovative reasoning:** Capability to leverage emerging technologies e.g. Big Data, Cloud Computing etc. for informed decision making and providing innovative ways to address global needs.
    - **Communication skills:** Ability to communicate and present oral and written information effectively using ICTs.
    - **Analysis and synthesis**: Capacity for carrying out analysis and synthesis using logical arguments and proven facts.
    - **Ethical commitment:** Show professional integrity and awareness of Software Engineering impact on society and the environment.
    - **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.
  - **Discipline specific:**
    - **Deep knowledge:** Technical understanding of Software Engineering concepts, methodologies, practices, quality and security standards.
    - **Production skills:** Exhibit ability to carry out research and project work as an individual and in a team.
    - **Technology development skills:** Develop and use systematic models**,** reliable Software Engineering methodologies and modern project management principles to produce high quality software.
    - **Problem-solving skills:** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.
    - **Analytical and computational skills:** Ability to carry out an efficient analysis of a problem, proffer a suitable solution and select appropriate computational tools.
    - **Appreciation of the interplay between theory and practice:** Graduates need to understand the relationship between theory and practice and how they influence each other.
    - **Appreciation of domain specific knowledge:** Graduates should understand that Software Engineering discipline interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

- **Commitment to life-long learning:** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

- **INTENDED LEARNING OUTCOMES**
  - By the end of the programme a graduate will be able to:
    - develop software systems using universal Software Engineering principles.
    - identify a problem and proffer a suitable solution.
    - carry out Software Requirements Engineering and produce a Software Requirements Specification (SRS) document.
    - function effectively on teams to accomplish a common goal.
    - translate the SRS into a Software Design Document (SDD) using an evolving Software Design Process.
    - implement and Test a developed solution.
    - deploy and maintain the developed product.
    - apply international quality assurance standards throughout the Software Development Process.
    - acquire knowledge on professional ethics applicable to both national and international bodies.
    - demonstrate an awareness of current trends and emerging technologies.
    - pursue further studies in any of the identified thematic areas.

- **GENERAL PROVISIONS**

Refer to Section 4 of the Faculty of Science and Technology Regulations.

- **ASSESSMENT**

Refer to Section 5 of the Faculty of Science and Technology Regulations.

- **FAILURE TO SATISFY THE EXAMINERS**

Refer to Section 9 of the General Academic Regulations.

- **PROVISION FOR PROGRESSION**
  - Refer to Section 7 of the Faculty of Science and Technology Regulations.
  - In addition to Section 11.1, to proceed from Level 2 to Level 3, the student must have passed the Mini Project.
- **WORK-RELATED LEARNING**

  Refer to Section 8 of the Faculty of Science and Technology Regulations
- **GRADING AND DEGREE CLASSIFICATION**

  Refer to Section 10 of the Faculty of Science and Technology Regulations.
- **DEGREE WEIGHTING**

  Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to the availability of personnel. * denotes core modules.

Level 1 Semester 1

Code	Module Description	Credits
HSWE111*	Software Engineering Fundamentals	10
HCSCI132*	Principles of Programming Languages	10
HCSCI133*	Operating Systems	10

HCSCI136*	Data Structures and Algorithms	10
HCSE138	Discrete Mathematics	10
CS131*	Basics of Communication Skills	12

Level 1 Semester 2

Code	**Module Description**	**Credits**
HSWE121*	Visual Programming	10
HSWE122*	Computer Networks and Information Security	10
HSTA132*	Applied Statistics	10
HCSE133*	Ethics and Professionalism	10
HCSCI135*	Computer Architecture and Organization	10
HCSE135*	Database Systems	10

Level 2 Semester 1

Code	**Module Description**	**Pre-requisites**	**Credits**
HCSE224*	Object Oriented Programming	(HCSCI132)	10
HSWE212*	Requirements Engineering		10
HCSCI235*	Software Project Management	(HSWE111)	10
HCSCI234*	Research Methods		10
GS231*	Gender Studies		12
TCNP201*	Technopreneurship		12

Level 2 Semester 2

Code	**Module Description**	**Pre-requisites**	**Credits**
HSWE221*	Web Technologies		10
HSWE222*	Software Architecture and Design		10
HSWE223*	Advanced Database Systems	(HCSE135)	10
HSWE224*	Mini Project		10
Electives (Select 1 module from the following)			
HSWE225	Mobile Application Development	(HCSE224)	10
HSWE226	Game Programming	(HCSE224)	10

Level 3 Semester 1

Code	**Module Description**	**Pre-requisites**	**Credits**
HSWE311*	Work-Related Learning I		40

Level 3 Semester 2

Code	**Module Description**	**Prerequisites**	**Credits**
HSWE321*	Work-Related Learning II	(HSWE311)	80

Level 4 Semester 1

Code	**Module Description**	**Pre-requisites**	**Credits**
HSWE411*	Software Testing and Quality Assurance	(HSWE111)	10
HSWE412*	Software Evolution and Re-engineering	(HSWE212)	10
HCSCI437*	Human Computer Interaction		10
HCSCI432*	Simulation and Modelling	(HSTA132)	10
Elective 1 (Select 1 module from the following)			
HSWE415	Mobile Computing and Wireless Networks	(HSWE122)	10
HSWE416	IoT and Cloud Computing	(HSWE122)	10
HCSCI433	Parallel and Distributed Computing	(HSWE122)	10

Elective 2 (Select 1 module from the following)

HSWE418	Web Services and Service Oriented Architecture	(HSWE221)	10
HSWE419	Advanced Web Engineering	(HSWE221)	10
HSWE420	Image Processing and Machine Vision		10

Level 4 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSCI431*	Theory of Computation		10
HCSE439*	Network Security and Cryptography		10
HCSCI434*	Artificial Intelligence		10
HSWE424*	Data Science and Big Data Analytics		10
HSWE425*	Capstone Project	(HSWE214, HSWE224)	24

MODULE SYNOPSES

HSWE111 Software Engineering Fundamentals

This subject introduces the issues, concepts, techniques, tools, and methodologies related to large-scale software development covering the entire life cycle. Topics include definition, estimation, analysis, design, implementation, testing, deployment, maintenance, refinement of the software and use of CASE Tools.

HCSE138 Discrete Mathematics

In this course the discovery that logical propositions are the underlying model of discrete systems is made. From this modest beginning algorithms are developed and their efficacy proven. Topics include propositional and predicate logic, basic proof techniques, set algebra and Boolean algebra, recursion and induction, trees and graphs, introductory combinatorics, and matrix algebra.

HCSCI132 Principles of Programming languages

The course aims to introduce the basic principles of programming, programming paradigms, program constructs and implementation of algorithms using Python programming language. Modular programming through use of functions, modules will also be covered to enhance students' understanding of code reusability. Moreover, data persistence will be demonstrated in files and data structures. The course will equip students with knowledge on basics of Object-Oriented Programming (OOP) and databases. Lastly, students will get to explore fundamentals of object-based programming using the Tkinter library in Python.

HCSCI133 Operating Systems

Operating Systems Overview: Evolution of operating systems, Operating Systems Concepts, Operating System structure, Processes, CPU Scheduling, Interprocess Communication, Deadlocks, Memory Management, Virtual Memory and Paging, File Systems, Protection and Security.

HCSCI136 Data Structures and Algorithms

This module builds on the programming skills acquired in Principles of Programming Languages. It couples work on Program Design, Analysis, and Verification with an introduction to the study of Data Structures. Students are introduced to: Lists, Stacks, Queues, Trees, Hash Tables, and Graphs. Students are expected to write several programs, ranging from very short programs to more elaborate systems. Emphasis is placed on the development of clear, modular programs that are easy to read, debug, verify, analyze, and modify.

CS131 Basics of Communication Skills
Refer to Communication Skills Department regulations.

HSWE121 Visual Programming
The course covers Object-based programming concepts, Visual Basic .NET framework, Working with Windows Forms and Controls. It also includes topics like designing user friendly user interface (UI), Data Types, Variables, Creating and Using Data Structures, Creating and using windows Objects, Using Functions, Using Decision Structures and Conditional Loop Structures. Students will be exposed to restricting user input, validating field and form data, exception handling, creating menus, creating status bars and navigation panels, and creating toolbars. Additionally, students will work with Web Forms (ASP.Net) and Database Connection Objects (ADO.NET) and deploy their Windows-based Applications on an Application Server.

HSWE122 Computer Networks and Information Security
This course introduces the technologies related to computer communications and networks. Topics include computer network topologies, and protocols with special emphasis on ISO model, ATM and TCP/IP protocol, and communication media, framing, addressing, packet switching, virtual circuits, congestion and deadlock control. Fundamentals of network security, risk management, Internet and Emerging Technologies will also be covered.

HSTA132 Applied Statistics
The module introduces students to an overview of statistics, data description, measurement scales, adequate graphic display, measures of central tendency and variability, differentiation between population and sample. It also covers how to use a statistic to estimate a population's parameter, confidence interval and its interpretation, hypothesis testing, understanding Type I and Type II errors, comparing two population means, proportions or variances, independent data versus paired data and identifying relationships between two variables.

HCSE133 Ethics and Professionalism
This course covers ethical and social issues related to the development and use of computer technology. Topics include ethical theory, and social, political, and intellectual property (IP) and other legal considerations. Scenarios in problem areas: privacy, reliability and risks of complex systems, and responsibility of professionals for applications and consequences of their work.

HCSCI135 Computer Architecture and Organization
The course covers the fundamentals of computer architecture, Von Neumann and Harvard architecture, functions of control unit (CU), arithmetic logic unit (ALU), memory and I/O devices. Topics to be covered include number representations, fetch-decode cycle, machine instructions, memory and registers addressing modes, arithmetic operations, translating high level arithmetic into assembly, stack memory, heap memory, subroutines, recursive subroutines, negative numbers and two's complement. The subject will also explore the evils of floating point, latches and memory reading, multi-cycle implementations, finite state machines, balancing the work into single cycles, PLAs, Microcode, RISC, CISC, pipelines, principle of locality, Direct Mapped Caches, I/O polling and interrupts.
HCSE135 Database Systems

The module covers fundamentals of database system principles, database management systems (DBMS), database design concepts and implementation. It also includes topics in database models, database system architecture, normalization, structured query system (SQL), NoSQL database manipulation and database security. The assignments are implemented using any SQL database tool depending on the instructor.

HCSE224 Object Oriented Programming

The purpose of the module is to equip students with application experience of Object-Oriented (OO) principles (i.e. inheritance, encapsulation and polymorphism) using an Object-Oriented Programming (OOP) programming language like Java. This module will build on Object-Oriented Analysis and Design knowledge already gained. Students will be exposed to generic programming concepts, use of packages, input/output (IO) streams and exception handling techniques. The course will also enable students to build sound console and Graphical User Interface (GUI) applications which are integrated to the database.

HSWE212 Requirements Engineering

Students will be exposed to cover the fundamentals of requirements engineering; definitions, characteristics, process, evolution and management. Requirements elicitation. Requirements specification and documentation. Requirements validation.

HCSCI235 Software Project Management

The course will equip students with Project Management skills encompassing planning, tracking, people management, scheduling, estimation, quality and risk management during the development of software projects.

HCSCI234 Research Methods

This course equips students with research techniques including definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation. research ethics, report writing.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

TCNP 201 Technopreneurship

Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, and product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

HSWE221 Web Technologies

Students will cover Web design fundamentals and internet technologies. They will have an appreciation of Web communication protocols and methods. They will learn how to use

Scripting languages, interface design tools, Application Programming Interface (API), Web Services and design patterns in Web Application Development.

HSWE222 Software Architecture and Design
This will be a follow-up course on Software Engineering Fundamentals covered in Level I, introducing students to the architectural and design patterns used in the design of software systems. It covers principles and practices for structuring software. Topics include Design principles, Design patterns (Creational, Structural and Behavioural), Architectural structures and views, Quality attributes, Architectural styles and Reconstructing architecture. They will also consider the use of different libraries, frameworks, and APIs, so that they are better able to choose and integrate these tools when developing software.

HSWE223 Advanced Database Systems
This will be a follow up course on Database Systems Fundamentals in Level I and will cover advanced concepts in Database Systems with a focus on Data Mining and Emerging Technologies. Various techniques for implementing relational databases, object-oriented and distributed databases using MySQL, Oracle, Microsoft SQL and MongoDB. In depth analysis on ODBC and JDBC connectivity will be made, Database as a Service (DaaS), query optimization, data performance tuning, data mining process and visualization will also be covered.

HSWE224 Mini Project
Students will be expected to use Software Engineering principles and other Domain area knowledge in coming up with a Mini project which addresses a particular need. The project will be done as a team/group of 2 or 3 and interdepartmental collaboration is allowed.

HSWE225 Mobile Application Development
This module includes concepts used in developing mobile applications. The following concepts should be covered: Mobile application development environment, mobile software engineering, VUIs, Intents and Services, Storing and Retrieving data, communications via the net and web, Telephony, Notifications and Alarms, Graphics, Multimedia, Location, Security, Platform issues.

HSWE226 Game Programming
The module will seek to equip students with gaming concepts and development abilities. The students will be required to apply visual and OOP programming concepts in creating interactive games. The course will cover event driven programming, gaming design patterns, scene creation and game security. The practicals will be done using Unity3D and any OOP language as selected by the instructor.

HSWE311 Work-Related Learning I
Refer to Section 8 of the Faculty of Science and Technology Regulations.

HSWE321 Work-Related Learning II
Refer to Section 8 of the Faculty of Science and Technology Regulations.

HSWE411 Software Testing and Quality Assurance
The subject will cover Software Testing fundamentals, Static Testing, Structural Testing, Regression Testing, Software Quality concepts and culture, Software Quality Assurance (SQA) process, Quality metrics, Configuration Management, Impact of Software Quality, Verification and Validation (VandV) concepts and techniques.

HSWE412 Software Evolution and Re-engineering
Students will acquire knowledge on Software Maintenance concepts, the evolution processes and Program evolution dynamics. They will also cover Lehman's Laws of Software Evolution, Build Environment Management, Legacy System Management, Reverse engineering concepts, Reengineering process activities, Types of Re-engineering and Open source development.

HCSCI437 Human Computer Interaction
Students will cover Human Computer Interaction (HCI) concepts and best practices, Components of HCI, cognitive frameworks for HCI, Cognitive models in HCI, Interactive design methods and techniques, evaluation.

HCSCI432 Simulation and Modelling
Students will cover Software modelling and analysis including Modelling foundations, principles, mathematical models and formal notation, and types of models. Analysis fundamentals, dependability, Formal analysis. Statistical analysis includes collecting and reading data so that one can describe past behaviour, characteristics and predict future ones.

HSWE415 Mobile Computing and Wireless Networks
Students will cover basic mobile computing and wireless networking concepts including issues, standards and architectures. It will also cover Telecommunication Systems, Wireless LAN Standards, Mobile Network Layer, Mobile Transport Layer and Emerging Trends in Mobile Networks.

HSWE416 IoT and Cloud Computing
This module deals with concepts of IoT. It also looks at the basics of cloud, different services offered by cloud, deployment models, edge computing and security issues in cloud. It also covers fundamental aspects of the cloud environment, mobile edge devices, radio remote heads (RRH), hybrid cloud and edge deployments.

HCSCI433 Parallel and Distributed Computing
Common applications of parallel computing; parallel machine architectures including Single Instruction Multiple Data (SIMD) or short-vector processing; multi-core and multi-processor shared memory; custom co-processors including DSPs and GPUs, and cluster and grid computing; programming approaches including parallelizing compilers; explicit message-passing (such as MPI); and specialized co-processor programming (such as for GPUs).

HSWE418 Web Services and Service Oriented Architecture
The module introduces students to the world of web service principles, applications, best practices and the service-oriented paradigm. It covers the background on distributed systems, component-based development and the factors which influenced service-oriented architecture (SOA). The course also seeks to provide learners with sound knowledge on web services architecture, technologies, development, registration and discovery process. Web services security and management will conclude the module.

HSWE419 Advanced Web Engineering
The course advances the hands-on experience of students with enterprise web development tools like J2EE, JavaBeans, JFX and Struts. It also covers web design patterns which include MVC, Observer etc., use of application and session data, state in web applications, XML and JSON will be explored. Both server-side and client-side scripting technologies like JSP,

ASP.Net, JavaScript and Ajax, web application security and integration with Cloud Database will also be covered.

HSWE420 Image Processing and Machine Vision
Image Processing and Computer Vision Background, Image Processing and Computer Vision Applications Digital Image Processing Hierarchy: Human Perception of Pictures, Digital Image Processing Hardware. Image Model, Amplitude digitization: Intensity Quantization, Spatial co-ordinate digitization: Image Sampling, Image Quality, Image Pixel Relationships, Linear Operators, 2-D Transforms. Spatial Domain Methods, Frequency Domain Methods. Inverse Filtering. Image Compression, Redundancy Types, Lossless and Lossy Compression, Image Compression Standards. Object Detection Methods, Edge Liking and Boundary Detection, Thresholding Methods, Region Oriented Methods. Object Representation and Description, Representation schemes, Description. Pattern Recognition, Decision Theoretic Methods for Recognition.

HCSCI431 Theory of Computation
The course explores Turing machines, universal computation, the halting problem and general undecidability, Rice's theorem, the recursion theorem, efficient computation models including finite automata, regular languages, context-free grammars, time and space (memory) bounds, deterministic and nondeterministic computation and their relationships, the P versus NP problem and hard problems for NP and beyond.

HCSE439 Network Security and Cryptography
This is composed of Network traceroute, packet filtering, TCP/IP and OSI Network Security, Access Control Issues (Packet Filters, Firewalls), Communication Security (OSI Layer Security Protocols), Web Security, Messaging services security, Modern Encryption Algorithms, Cryptography Fundamentals, Cryptosystems and Techniques.

HCSCI434 Artificial Intelligence
Students are introduced to concepts of AI such as Turing test, Intelligent Agents and various techniques. They will also cover Problem-Solving, Knowledge Representation, Reasoning, Bayesian Networks, Fuzzy Logic Systems, Artificial Neural Networks (ANNs), Expert Systems and Advanced topics like Natural Language Processing (NLP) and Machine Learning.

HSWE424 Data Science and Big Data Analytics
This module covers the fundamentals of data science, Big Data systems organisation and Design, and Big Data Analytics (BDA). It also enables students to carry out practicals with Big Data analytics platforms including Hadoop, Spark, and cloud based Big Data services. The course will also cover Big Data Infrastructure management, Big Data systems for application domains and Big Data software architectures.

HSWE425 Capstone Project
Students will be expected to use modern Software Engineering principles, Emerging Technologies and other Domain area knowledge in coming with a final year research project which addresses a particular need and produce a working prototype.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN CYBERSECURITY (HCSEC)

Duration:	4 Years (Full time)
	3 Years (Visiting)
Maximum Credit Load:	540
Minimum Credit Load:	480
Maximum MBKs Credit Load:	384
ZNQF Level:	8

3. PURPOSE OF THE PROGRAMME

3.1 The BSc Honours degree in Cybersecurity is a programme that aims to provide students with sound theoretical and practical training in security of cyberspace as well as in-depth detail of the risks that users are likely to encounter within the cyberspace realm.

4. ENTRY REQUIREMENTS

4.1 For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better:

4.1.1 **Normal Entry**

A minimum of 2 A' Level passes in Mathematics AND (Physics OR Computer Science OR any other relevant Science subject).

4.1.2 **Special Entry**

Special entry may be granted to applicants with a National Diploma in Computer Science or any related field from a recognized institution.

4.1.3 **Mature Entry**

Refer to Section 3.3 of the General Academic Regulations.

2.1.4 **Visiting School/Block Release**

Should have at least an ND or HND in Software Engineering or any relevant field and have proof of employment in the field specifying the nature of duties.

5. PROGRAMME CHARACTERISTICS

5.1 **Areas of study:** Operating Systems Security, Software Engineering, Discrete Mathematics, Database Systems, and Cryptography, Cyber law and Computer Forensics, Digital Forensics.

5.2 **Specialist Focus:** Secure Coding, Coding Theory, Ethical Hacking, Cloud Architectures and Security, Penetration Testing and Vulnerability Assessment

5.3 **Orientation:** Research, innovation and Technopreneurial orientation. Teaching and learning is focused on producing graduates with the necessary knowledge, skills and competencies sufficient to start technology-based businesses.

5.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach. Impart practical training based on scientific knowledge and

methods-based training which qualifies for a self-sufficient professional activity as engineer in the field of Cybersecurity.

6. CAREER OPPORTUNITIES AND FURTHER EDUCATION

6.1 **Employability:** Careers in the Computer Science field which include:
Research Scientists, Information Security Analyst, Lead Software Security Engineer, Chief Information Security Officer, Analyst Programmers, Penetration Tester, Cybersecurity Crime Investigator/ Forensics Expert

6.2 **Further Studies:** Master's and doctoral studies in Cybersecurity or in interdisciplinary programmes related to computing practices.

7. PROGRAMME DELIVERY

7.1 **Teaching and Learning Methods:** Lectures, tutorials, programming laboratory classes, seminars, group work, industrial visits, Work-Related learning, research project, individual independent study.

7.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, continuous assessments.

8. PROGRAMME COMPETENCIES

8.1 **Generic:**

8.1.1 **Multidisciplinary**: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

8.1.2 **Flexibility;** to succeed in a rapidly changing field.

8.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things

8.1.4 **Communication skills**: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences

8.1.5 **Analysis and synthesis**: Capacity for analysis and synthesis using logical arguments and proven facts.

8.1.6 **Ethical commitment:** Professional integrity and awareness of impact of science and technology on society and the environment

8.1.7 **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms

8.2 **Discipline specific:**

8.2.1 **Technical understanding of Cybersecurity;** Graduates should have a mastery of Cybersecurity as described by the core of the Body of Knowledge.

8.2.2 **Familiarity with common themes and principles;** Graduates need understanding of a number of recurring themes, such as abstraction, complexity, and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Cybersecurity and should not consider them as relevant only to the domains in which they were introduced.

8.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Cybersecurity dwells on understanding the interplay between theory

and practice and the essential links between them. Graduates of a Cybersecurity program need to understand how theory and practice influence each other.

8.2.4 **Problem solving skills;** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

8.2.5 **Project experience;** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Cybersecurity programmes should have been involved in at least one substantial project.

8.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

8.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

8.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions too many problems require both computing skills and domain knowledge.

9. INTENDED LEARNING OUTCOMES

9.1 By the end of the programme a graduate will be able to:

9.1.1 apply knowledge of cybersecurity appropriate to the program's student outcomes and to the discipline.

9.1.2 select appropriate tools to secure an organisation's system

9.1.3 evaluate vulnerabilities present in various Information Systems

9.1.4 develop knowledge of cyber security and forensic auditing through research

9.1.5 develop secure systems for various organisations.

9.1.6 work independently and in groups.

9.1.7 understand professional, ethical, legal, security and social issues and responsibilities.

9.1.8 function effectively on teams to accomplish a common goal.

9.1.9 communicate effectively with a range of audiences.

9.1.10 learn new programming and computing environments on their own.

10. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

11. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

12. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

13. PROVISION FOR PROGRESSION

13.1 Refer to Section 7 of the Faculty of Science and Technology Regulations.

13.2 In addition to Section 11.1, to proceed from Level 2 to Level 3, the student must have passed the Mini Project.

14. WORK-RELATED LEARNING
Refer to Section 8 of the Faculty of Science and Technology Regulations
15. GRADING AND DEGREE CLASSIFICATION
Refer to Section 10 of the Faculty of Science and Technology Regulations.
16. DEGREE WEIGHTING
Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE
A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to availability of personnel. * denotes core modules.

Level 1 Semester 1

Code	Module Description	Credits
HCSCI132*	Fundamentals of Programming Languages	10
HSWE111*	Software Engineering Fundamentals	10
HCSCI133*	Operating Systems	10
HCSEC111*	Introduction to Information Security	10
HCSE138*	Discrete Mathematics	10
CS131*	Communication Skills	12

Level 1 Semester 2

Code	Module Description	Credits
HCSCI135	Computer Architecture and Organization	10
HCSEC122*	Cyberspace Ethics and Laws	10
HCSEC123*	Principles of Secure Coding	10
HCSE135	Database Systems	10
HCSCI136*	Data Structures and Algorithms	10
HCSEC125*	Forensics and Incidence Response	10

Level 2 Semester 1

Code	Module Description	Pre-requisites	Credits
HCSEC211*	Coding Theory	(HCSE138)	10
HCSEC212*	Number Theory	(HCSE138)	10
HCSCI234*	Research Methods		10
HCSCI237	Data Communications and Computer Networks		10
TNCP201*	Technopreneurship		12
GS231*	Gender Studies		12

Level 2 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSEC221*	Network Programming	(HCSEC123)	10
HCSEC222*	Cryptography	(HCSEC111)	10
HCSEC223*	Ethical Hacking		10
HCSEC224 *	Cybersecurity Mini Project		12
Elective (Select 1 from the following)			
HSWE221	Web Technologies		10
HCSEC225	Systems Administration and Security		10

Level 3 Semester 1

Code	Module Description	Credits
HCSEC311*	Work-Related Learning I	40

Level 3 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSEC321*	Work-Related Learning II	(HCSEC311)	80

Level 4 Semester 1

Code	Module Description	Pre-requisites	Credits
HCSEC411*	Blockchain Technology		10
HCSEC412*	Network Security	(HCSEC221)	10
HCSEC413*	Digital Forensics	(HCSEC125)	10
HCSEC414 *	Cryptanalysis and Development of Cryptosystems	(HCSEC222)	10
HCSEC415*	Cybercrime and Security Management		10
Elective (Select 1 from the following)			
HCSEC416	Secure Ecommerce		10
HSWE419	Advanced Web Engineering	(HSWE221)	10
HSWE418	Web Services and Service Oriented Architecture		10

Level 4 Semester 2

Code	Module Description	Pre-requisites	Credits
HCSEC421*	Android Security Design and Internals		10
HCSEC422*	Cloud Architectures and Security	(HSWE221)	10
HCSEC423*	IT Auditing and Assurance		10
HCSEC424 *	Penetration Testing and Vulnerability Assessment	(HCSEC125, HCSEC223)	10
HCSEC425*	Capstone Project	(HCSEC224)	24

MODULE SYNOPSES

HCSCI132 Programming Languages

The module aims to introduce the basic principles of programming, programming paradigms, program constructs and implementation of algorithms using Python programming language. Modular programming through the use of functions, and modules will also be covered to enhance students understanding of code reusability. Moreover, data persistence will be demonstrated in files and data structures. The module will equip students with knowledge on basics of Object-Oriented Programming (OOP) and databases. Lastly, students will get to explore the fundamentals of object-based programming using the Tkinter library in Python.

HSWE111 Software Engineering Fundamentals

This subject introduces the issues, concepts, techniques, tools, and methodologies related to large-scale software development covering the entire life cycle. Topics include definition, estimation, analysis, design, implementation, testing, deployment, maintenance, refinement of the software and use of CASE Tools.

HCSCI133 Operating Systems

Operating Systems Overview: Evolution of operating systems, Operating Systems Concepts, Operating System structure, Processes, CPU Scheduling, Interprocess Communication, Deadlocks, Memory Management, Virtual Memory and Paging, File Systems, Protection and Security.

HCSEC111 Introduction to Information Security

A review of the origins of information security, outline of the phases of the security systems development life cycle, critical characteristics of information (CIA). Examination of the various threats facing organizations and methods for ranking these threats (in order to assign them relative priority), examination of the types of attacks that could result from these threats, and how these attacks could impact the organization's information systems, secure software development. Legal, ethical, and professional issues in Information Security. Risk Management: an overview of risk management (know yourself/the enemy), Risk identification, Risk assessment, Risk control strategies, Selecting a risk control strategy (feasibility studies). Benchmarking and best practices, Baselining, Residual risk. Information security policy, standards, and practices. EISP, ISSP, SysSP, The information security blueprint, Security Education, Training, and Awareness Program (SETA), Continuity strategies. Security Technology: Security Technology: Intrusion Detection and Prevention Systems. Physical Security: Physical access controls, Mobile and portable systems, Remote computing security. Implementing Information Security: Information security project management, Technical aspects of implementation (Conversion strategies, The Bull's-Eye model). Personnel Security: Positioning and staffing the security function, Credentials of information security professionals, Employment policies and practices, Security considerations for non-employees. Information Security

HCSE138 Discrete Mathematics

In this module the discovery that logical propositions are the underlying model of discrete systems is made. From this modest beginning algorithms are developed and their efficacy proven. Topics include propositional and predicate logic, basic proof techniques, set algebra and Boolean algebra, recursion and induction, trees and graphs, introductory combinatorics, and matrix algebra

CS131 Communication Skills

Students will cover theory of technical communication and the requisite principles. Students learn how to communicate technically through writing and oral presentations while applying verbal and nonverbal communication skills with the use of ICT tools. They also critique visual and audio images or literature according to modern standards. Students learn how to design a technical report for laboratory experiments, research project reports and analyzing literature.

HCSCI135 Computer Architecture and Organization

The module covers the fundamentals of computer architecture, Von Neumann and Harvard architecture, functions of control unit (CU), arithmetic logic unit (ALU), memory and I/O devices. Topics to be covered include fetch-decode cycle, machine instructions, memory and registers addressing modes, arithmetic operations, translating high level arithmetic into assembly, stack memory, heap memory, subroutines, recursive subroutines, negative numbers and two's complement. The subject will also explore the evils of floating point, latches and memory reading, multi-cycle implementations, finite state machines, balancing the work into single cycles, PLAs, Microcode, RISC, CISC, pipelines, principle of locality, Direct Mapped Caches, I/O polling and interrupts.

HCSEC122 Cyberspace Ethics and Laws

Focus is on the need for suitable legislation, and appropriate resourcing of law enforcers to address cybercrime. Students learn about the global nature of computer crime and examine national and international laws, regulations and agreements that govern computer crime, including computer investigations, prosecution and defense. The module encourages analytical thinking and reasoning about computer crime topics and relevant legal issues so that students can identify, analyze, and solve problems in the continually emerging cybercrime and cyber law issues and trends. The module explores legal responses to cybercrime, including regulatory and law enforcement initiatives aimed at combating computer crime. This module is designed to acquaint students with electronic privacy, security and ethics. Students will gain an understanding of information ethics, existing and emerging cyber-laws, organizational liability issues, and explore several Code of Ethics. Students will learn about real and potential security issues, steps that can be taken to create environments of trust, how to evaluate the strengths and weaknesses of a firm's information resource environment, and risk management and operation feasibility issues. Cyber policies and privacy are also catered for.

HCSEC123 Principles of Secure Coding
This module provides a detailed explanation of common programming errors in programming languages and describes how these errors can lead to software systems that are vulnerable to exploitation. It will also cover defensive programming and identify its benefits and disadvantages, secure programming, its relationship to defensive programming, and its benefits and disadvantages, sources of risk that can negatively impact your software applications, best practices for creating secure code, how to include defensive programming techniques into your software development process, why testing should be performed and identify the major phases of the software testing process, what unit testing is and the benefits it provides, the benefits of employing defensive and secure programming and recognizing the phases of the software testing process. Software security, organizational security and connection security are covered.

HCSE135 Database Systems
The module covers fundamentals of database system principles, database management systems (DBMS), database design concepts and implementation. It also includes topics in database models, database system architecture, normalization, structured query system (SQL), NoSQL database manipulation and database security. The assignments are implemented using any SQL database tool depending on the instructor.

HCSCI136 Data Structures and Algorithms
This module covers data structures and algorithms fundamentals: algorithm correctness, algorithm analysis, the relationship between algorithms and data structures- containers, dictionaries, binary search trees and priority queues; specialized data structures; graph data structures and algorithm design. Applications and approaches to sorting: incremental insertion, divide and conquer, randomization, and bucketing techniques. Algorithm Design and Breaking Problems Techniques for increasing efficiency: dynamic programming, application, implementation, and limitations; divide and conquer. Graph traversal, minimum spanning trees, and shortest paths. Backtracking, search pruning, bandwidth minimization, heuristic methods, and parallel algorithms. Programming assignments will be implemented in python/C.

HCSEC125 Forensics and Incidence Response
This module focuses on the forensic analysis of data from computer systems and response after an incident. It also covers Incident Response Overview, Incident Response Phases, Digital Forensics Overview, Digital Forensics Evidence Acquisition, Digital Forensics Evidence

Analysis, Digital Forensics Reporting. Societal security, organizational security and human security are also covered.

HCSEC211 Coding Theory
The module covers the following areas: Introduction to Information Theory. Concept of information, units, entropy, marginal, conditional and joint entropies, relation among entropies, mutual information, and information rate. Source coding: Instantaneous codes, construction of instantaneous codes, Kraft's inequality, coding efficiency and redundancy. Noiseless coding theorem, construction of basic source codes, Shannon – Fano Algorithm, Huffman coding, Channel capacity – redundancy and efficiency of a channel, binary symmetric channel (BSC), Binary erasure channel (BEC) – capacity of band limited Gaussian channels. Continuous Sources and Channels: Differential Entropy, Mutual information, Waveform channels, Gaussian channels, Shannon – Hartley theorem, bandwidth, SNR trade off, capacity of a channel of infinite bandwidth, Shannon's limit. Introduction to rings, fields, and Galois fields. Codes for error detection and correction – parity check coding – linear block codes – error detecting and correcting capabilities – generator and parity check matrices – Standard array and syndrome decoding. Perfect codes, hamming codes, encoding and decoding Cyclic codes, polynomial and matrix descriptions, generation of cyclic codes, decoding of cyclic codes BCH codes, Construction and decoding, Reed Solomon codes. Convolutional Codes – encoding – time and frequency domain approaches, State Tree and Trellis diagrams – transfer function and minimum free distance – Maximum likelihood decoding of convolutional codes – The Viterbi Algorithm. Sequential decoding.

HCSEC212 Number Theory
The module covers areas such as prime numbers and divisibility in integers, G.C.D, L.C.M -- prime numbers – prime factorization-- Fundamental theorem of arithmetic – Euclidean division algorithm -- Fermat numbers. ARITHMETICAL FUNCTIONS The Mobius function $\mu(n)$- divisor sum formula for $\mu(n)$ – The Euler totient function $\varphi(n)$ - divisor sum formula for $\varphi(n)$– A relation connecting μ and φ- A product formula for $\varphi(n)$ – properties of $\varphi(n)$–Multiplicative functions–completely multiplicative function. CONGRUENCES Basic properties – Residue classes and complete residue systems – linear congruences – Reduced residue systems and Euler Fermat theorem – Simultaneous linear congruences – The Chinese remainder theorem. PRIMITIVE ROOTS Primitive roots and reduced residue system – Non-existence of p-roots mod 2α ($\alpha \geq 3$) - existence of p-roots mod p for odd primes p – Existence of p-root mod $p\alpha$ - p- root mod 2 $p\alpha$ -Non-existence of p-roots in other cases. QUADRATIC RECIPROCITY AND ELLIPTIC CURVES - Quadratic Residues – Legendre's symbol and its properties – Evaluation of $(-1 \mid p)$ and $(2 \mid p)$ – Gauss' lemma – The Quadratic Reciprocity law – Applications – The Jacobi symbol – continued fractions – elliptic curves –applications.

HCSCI234 Research Methods
This module equips students with research techniques including definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation. research ethics, report writing.

HCSCI237 Data communications and Computer Networks
Introduction and Background to Data Communications. LANS (Wired and Wireless). Internetworking and Routing. Routing Fundamentals. Multicast and multicast routing.

Application Layer Protocols. Non-real-time and real-time Internet applications: Telnet, FTP, SMTP, DNS, WWW, VoIP, SIP and H.323. Network security. IPsec, TLS and PGP.

TCNP 201 Technopreneurship

Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HCSEC221 Network Programming

Modules covers Linux Utilities- File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking utilities, Filters, Text processing utilities and Backup utilities. Bourne again shell(bash) – Introduction, pipes and redirection, here documents, running a shell script, the shell as a programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples. Files- File Concept, File types File System Structure, Inodes, File Attributes, file I/O in C using system calls, kernel support for files, file status information-stat family, file and record locking-lockf and fcntl functions, file permissions. File and Directory management – Directory contents, Scanning Directories- Directory file APIs. Process- Process concept, Kernel support for process, process attributes, process control – process creation, replacing a process image, waiting for a process, process termination, zombie process, orphan process.: Signals- Introduction to signals, Signal generation and handling, Kernel support for signals, Signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions. Interprocess Communication – Introduction to IPC mechanisms, Pipes- creation, IPC between related processes using unnamed pipes, FIFOs-creation Shared Memory- Kernel support for shared memory, UNIX system V APIs for shared memory, client/server example. Network IPC – Introduction to Unix Sockets, IPC over a network, Client-Server model, Address formats (Unix domain and Internet domain), Socket system calls for Connection Oriented – Communication, Socket system calls for Connectionless-Communication, Example-Client/Server Programs- Single Server-Client connection, Multiple simultaneous clients, Socket options – setsockopt, getsockopt, fcntl. Network Programming in Java- Network basics, TCP sockets, UDP sockets (datagram sockets), Server programs that can handle one connection at a time and multiple connections (using multithreaded server

HCSEC222 Cryptography

The nature, purposes and history of cryptography; Classical cryptography: transposition ciphers, substitution ciphers, Affine cipher, the Vigenere cipher. Entropy and equivocation; Perfect secrecy and one-time pads, Unicity distance. Fields: Field extensions, Finite fields, Fields over irreducible polynomials. Modern encryption algorithms: stream ciphers vs. block

ciphers (e.g. RC4, AES), Symmetric (Secret key systems) vs. Asymmetric (Public key systems) – RSA (history, construction and applications), ElGamal. Information flow controls, Inference controls, Access controls, Cryptography techniques. Component security and organizational security.

HCSEC223 Ethical Hacking
Areas to be covered include: Introduction: Hacking Impacts, The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, Deliverable, Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security, Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis and Ethical Hacking. The Business Perspective: Business Objectives, Security Policy, Previous Test Results, Business Challenges Planning for a Controlled Attack: Inherent Limitations, Imposed Limitations, Timing is Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attacks, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement. Preparing for a Hack: Technical Preparation, Managing the Engagement Reconnaissance: Social Engineering, Physical Security, Internet Reconnaissance. Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration, Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, Rootkits, applications, Wardialing, Network, Services and Areas of Concern. Deliverable: The Deliverable, The Document, Overal Structure, Aligning Findings, Presentation Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy, Conclusion.

HCSEC224 Cyber Security Mini Project
After having gained knowledge in modules undertaken earlier on, students will be equipped enough to conceive a project idea in any of the thematic areas within the cyberspace realm exposed to earlier on. The expectation is that the student will design and implement a solution under the supervision of a lecturer and submit a suitable report on the work carried out. An artifact commensurate with the level of study is expected at this point. Projects can be done in groups depending upon approval by the departmental board.

HSWE221 Web Technologies
Students will cover Web design fundamentals and internet technologies. They will have an appreciation of Web communication protocols and methods. They will learn how to use Scripting languages, interface design tools, Application Programming Interface (API), Web Services and design patterns in Web Application Development.

HCSEC225 Systems Administration and Security
This module introduces concepts of systems administration on servers such as Unix/Linux and Microsoft amongst others. As well as making sure these servers are secured. It covers systems security, component security and organizational security with a focus on system thinking, management, access and control as well as testing and retirement. System architectures are also covered.

HCSEC311 Work-Related Learning I
Refer to Section 8 of the Faculty of Science and Technology Regulations.

HCSEC321 Work-Related Learning II

Refer to Section 8 of the Faculty of Science and Technology Regulations.

HCSEC411 Blockchain Technology
Areas to be covered include: Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. • Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof. Blockchain: Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft and Hard Fork, Private and Public blockchain. Distributed Consensus: Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternate. Cryptocurrency: History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin. Cryptocurrency Regulation: Stakeholders, Roots of Bitcoin, Legal Aspects-Crypto currency Exchange, Black Market and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain. Tutorial and Practical: Naive Blockchain construction, Memory Hard algorithm - Hashcash implementation, Direct Acyclic Graph, Play with Go-ethereum, Smart Contract Construction, Toy application using Blockchain, Mining puzzles

HCSEC412 Network Security
Topics include Network traceroute and packet filtering project. Protocol designs, Security Concepts and Terminology, TCP/IP and OSI Network Security, Access Control Issues (Packet Filters, Firewalls), Communication Security (OSI Layer Security Protocols), Security Tools, Cryptography, System Security - Intruders and Viruses, E-mail and Web Security. Implementation will be done using Appropriate Networking Programming. Areas to be looked at include Network and web basics, Socket Programming. Distributed objects. Connection security and component security as well as systems security are also catered

HCSEC413 Digital Forensics
This module covers mobile phone security and forensics, investigating wireless and web attacks, network-based forensics, anti-forensics methods, tools and techniques emerging technologies in the field of digital forensics. It will also involve the reconstruction of web browsing history from web artifacts, capture and interpret network traffic, discuss the challenges associated with mobile device forensics, inspect a system (network, computer, or application) for the presence of malware or malicious activity and identify anti-forensic methods. Data security and societal security are catered for along with elements of component and connection security.

HCSEC414 Cryptanalysis and Development of Cryptosystems
This module covers teaching students how to break codes, explicitly describe the real-world constraints and realities of Cryptanalysis, critic, design, and build secure cryptographic systems, explicitly the real-world constraints and realities of cryptography, critic, design, and build secure cryptographic systems understand current implementations of cryptography algorithms and protocols in real life and teach emerging technologies. Data security and software security as well as component security are catered for.

HCSEC415 Cybercrime and Security Management

This module covers types of computer crimes and various modes of attacks, investigating cybercrimes, Identity management, social engineering, social and behavioural privacy, usable security and privacy, business continuity and management with respect to information security, Security patch management and Security Incident and Event Management (SIEM), management of computer security and emerging technologies in security management and investigating and handling cybercrimes.

HCSEC416 Secure E-commerce
The module introduces the concepts and principles of electronic commerce, security in E-commerce transactions, Internet technologies and tools for implementing E-commerce applications; and to implement some basic secure E-commerce applications to gain hands-on experience. The following topics will be covered: Introduction to basic concepts of E-commerce and security, introduction to the underlying network infrastructure of Internet, Introduction and implementations on the operations and security of Client Side and Server Side. Introduction and implementations on secure web-based application development using various technologies such as XML, Perl, PHP, ASP, JSP and JavaScript. Implementing cryptographic tools to support confidentiality, integrity, authentication, digital signature, non-repudiation services for electronic transmissions and transactions. Familiarity with business, legal, and ethical issues in the area of E-commerce, and the interaction of these issues with technical issues. Data and connection security as well as organizational security are covered.

HSWE419 Advanced Web Engineering
The module advances the hands-on experience of students with enterprise web development tools like J2EE, JavaBeans, JFX and Struts. It also covers web design patterns which include MVC, Observer etc., use of application and session data, state in web applications, XML and JSON will be explored. Both server-side and client-side scripting technologies like JSP, ASP.Net, JavaScript and Ajax, web application security and integration with Cloud Database will also be covered.

HSWE418 Web Services and Service Oriented
The module introduces students to the world of web service principles, applications, best practices and the service-oriented paradigm. It covers the background on distributed systems, component-based development and the factors which influenced service-oriented architecture (SOA). The module also seeks to provide learners with sound knowledge on web services architecture, technologies, development, registration and discovery process. Web services security and management will conclude the module.

HCSEC421 Android Security Design and Internals
This module will look at the following areas: Android Security Model - Linux Kernel- Native User space – Dalvik VM- Java Run Time Libraries- System Services- IPC- Binder's- Framework Libraries- Applications- Sandboxing- Code Signing and Platform Key- SELinux- System Updates- Verified Boot. Permissions - Nature of Permission- Request for permission- Management- Protecting Levels- Assignment- Enforcement- System Permission- Shared User ID- Custom Permission – Broadcast Permissions- Content Provider Permission- Pending Intents. Introduction to Secure Coding - Building a Secure Smartphone Society - Developer's Context- Steps to Install Sample Codes into Android Studio- Android Application Security- Handling Input Data Carefully and Securely. Application Development - Creating/Using Activities- Receiving/Sending Broadcasts. -Creating/Using Content Providers- Creating/Using Services- Using SQLite- Handling Files- Using Browsable Intent- Outputting Log to LogCat- Using WebView- Using Notifications. Secure Functions - Building custom Android ROM-

Steps and Tools, Creating Password Input Screens- Permission and Protection Level- Add In-house Accounts to Account Manager- Communicating via HTTPS- Handling privacy data- Using Cryptography- Using fingerprint authentication features- Risk of Information Leakage from Clipboard.

HCSEC422 Cloud Architectures and Security

The module will cover the following areas: CLOUD COMPUTING FUNDAMENTALS - Cloud Computing definition, private, public and hybrid cloud. Cloud types; IaaS, PaaS, SaaS. Benefits and challenges of cloud computing, public vs private clouds, role of virtualization in enabling the cloud; Business Agility: Benefits and challenges to Cloud architecture. CLOUD APPLICATIONS - Technologies and the processes required when deploying web services- Deploying a web service from inside and outside a cloud architecture, advantages and disadvantages- Development environments for service development; Amazon, Azure, Google App. SECURING THE CLOUD - Security Concepts - Confidentiality, privacy, integrity, authentication, non-repudiation, availability, access control, defence in depth, least privilege- how these concepts apply in the cloud and their importance in PaaS, IaaS and SaaS. e.g. User authentication in the cloud. VIRTUALIZATION SECURITY - Multi-tenancy Issues: Isolation of users/VMs from each other- How the cloud provider can provide this- Virtualization System Security Issues: e.g. ESX and ESXi Security, ESX file system security- storage considerations, backup and recovery- Virtualization System Vulnerabilities. CLOUD SECURITY MANAGEMENT - Security management in the cloud – security management standards- SaaS, PaaS, IaaS availability management- access control- Data security and storage in cloud.

HCSEC423 IT Auditing and Assurance

The module is designed to provide the student with a foundation for the study and professional career development in the field of information technology (IT) auditing and assurance services. The module will introduce the fundamentals of IT auditing and assurance services, core reasons why IT auditing is a specialized area of auditing, evolution of IT assurance, and the principle objectives of IT auditing and assurance services. The module will also address career and skill and knowledge development. Also involves management's goals and concerns regarding the security, integrity and availability of IT systems. The module will emphasize business management issues regarding the security and control of IT and the achievement of value through managed IT processes. Over the module, students will gain a working familiarity with COBIT, which is the principle control model for IT governance and control (or an equivalent and comparable platform). Students will also be introduced to control evaluation techniques and a number of the primary references used by IT professionals and IT auditors regarding IT

HCSEC424 Penetration Testing and Vulnerability

This module covers concepts of penetration testing and vulnerability assessment, Footprinting and social engineering tools and methods, methods of ensuring and countering attacks on data security, network protection systems and security against attacks on the web, ethical hacking, laws and methodologies and emerging technologies in the field of penetration testing. Data, software, systems and human security are key in this module.

HCSEC425 Capstone Project

Students will be expected to use modern Cybersecurity principles, Emerging Technologies and other Domain area knowledge in coming with a final year research project which addresses a particular need and produce a working prototype.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (HAIML)

Duration:	4 Years (Full time)
	3 Years (Visiting)
Maximum Credit Load:	540
Minimum Credit Load:	480
Maximum MBKs Credit Load:	384
ZNQF Level:	8

1. PURPOSE OF THE PROGRAMME

The BSc Honours degree in Artificial Intelligence (AI) and Machine Learning (ML) is a programme that aims to provide students with sound theoretical and practical training in artificial intelligence and in-depth detail of the risks that users are likely to encounter within the artificial intelligence realm.

2. ENTRY REQUIREMENTS

2.1 For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better:

2.1.1 **Normal Entry**
A minimum of 2 A' Level passes in Mathematics AND (Physics OR Computer Science OR any other relevant Science subject).

2.1.2 **Special Entry**
Special entry may be granted to applicants with a National Diploma in Computer Science or any related field from a recognized institution.

2.1.3 **Mature Entry**
Refer to Section 3.3 of the General Academic Regulations.

2.1.4 **Visiting School/Block Release**
Should have at least an ND or HND in Software Engineering or any relevant field AND have proof of employment in the field specifying the nature of duties.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Artificial Intelligence, Programming, Data structures, Machine Learning, Algorithms, computational systems, data management systems, advanced AI algorithms and codes, Systems Security, Software Engineering, Discrete Mathematics and Database Systems

3.2 **Specialist Focus:** Artificial Intelligence, AI Programming, Data structures, Machine Learning, computational systems, data management systems, advanced AI algorithms and codes

3.3 **Orientation:** Research and innovation oriented. Teaching and learning are professionally oriented and focused on practical aspects.

3.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 **Employability:** Careers in the Careers in the AI and Machine Learning field include:
AI Data Science Experts, AI Business Development Manager, AI Team Leader, AI Engineer, Machine Learning Engineer, AI Scientist, ML Data Engineer, ML Architect, AI Specialist Developer, ML Intelligent System Researcher and ML Applied Research Scientist and AI Technopreneur.

4.2 **Further Studies:** Masters in AI, Machine Learning, Data Science, or in interdisciplinary programmes related to computing practices.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, programming laboratory classes, seminars, group work, industrial visits, Work-Related learning, research project, individual independent study.

5.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, and continuous assessments.

6. PROGRAMME COMPETENCIES

6.1 **Generic:**

6.1.1 **Multidisciplinary**: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

6.1.2 **Flexibility;** to succeed in a rapidly changing field.

6.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things

6.1.4 **Communication skills**: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences

6.1.5 **Analysis and synthesis**: Capacity for analysis and synthesis using logical arguments and proven facts.

6.1.6 **Ethical commitment:** Professional integrity and awareness of impact of science and technology on society and the environment

6.1.7 **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.

6.2 **Discipline-specific:**

6.2.1 **Technical understanding of Computer Science;** Graduates should have a mastery of Artificial Intelligence and Machine Learning as described by the core of the Body of Knowledge.

6.2.2 **Familiarity with common themes and principles;** Graduates need an understanding of a number of recurring themes, such as abstraction, complexity, and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Artificial Intelligence and Machine Learning, and should

not consider them as relevant only to the domains in which they were introduced.

6.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Computer Science is understanding the interplay between theory and practice and the essential links between them. Graduates of an Artificial Intelligence and Machine Learning programme need to understand how theory and practice influence each other.

6.2.4 **Problem solving skills;** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

6.2.5 **Project experience;** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Artificial Intelligence and Machine Learning programmes should have been involved in at least one substantial project.

6.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

6.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

6.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

7. INTENDED LEARNING OUTCOMES

7.1 By the end of the programme a graduate will be able to:

7.1.1 apply knowledge of computing and mathematics appropriate to the program's student outcomes and to the discipline.

7.1.2 analyze a problem, and identify and define the computing requirements appropriate to its solution Design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.

7.1.3 understand professional, ethical, legal, security and social issues and responsibilities.

7.1.4 function effectively on teams to accomplish a common goal.

7.1.5 communicate effectively with a range of audiences.

7.1.6 analyze the local and global impact of computing on individuals, organizations, and society.

7.1.7 learn new programming and computing environments on their own.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

9. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

10. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. PROVISION FOR PROGRESSION

11.1 Refer to Section 7 of the Faculty of Science and Technology Regulations.

11.2 In addition to Section 11.1, to proceed from Level 2 to Level 3, the student must have passed the Mini Project.

12. WORK-RELATED LEARNING

Refer to Section 8 of the Faculty of Science and Technology Regulations

13. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty of Science and Technology Regulations.

14. DEGREE WEIGHTING

Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite (defined as Pre.) if the pre-requisite is not passed. All the modules under this programme are core. Electives will be offered subject to availability of personnel.

Level 1 Semester 1

Code	Module Description	Credits
HAIML111*	Fundamentals of Artificial Intelligence	10
HAIML112*	Machine Learning Fundamentals	10
HAIML113*	Intelligent Signal Processing	10
HCSCI132*	Principles of Programming Languages	10
HCSE138*	Discrete Mathematics	10
CS131*	Basics of Communication Skills	12

Level 1 Semester 2

Code	Module Description	Credits
HAIML121*	Natural Language Processing	10
HCSEC111*	Introduction to Information Security	10
HCSC1135 *	Computer Architecture and Organization	10
HCSE121	Web and Multimedia Development	10
HCSE135*	Database Systems	10
HCSCI136*	Data Structures and Algorithms	10

Level 2 Semester 1

Code	Module Description	Credits
HCSCI237	Data Communications and Networks	10
HCSE231*	Object Oriented Analysis and Design	10
HAIML212*	Deep Learning	10
HAIML211*	Data Science and Analytics	10
TNCP201*	Technopreneurship	12
GS231*	Gender Studies	12

Level 2 Semester 2

Code	Module Description	Credits
HAIML222*	Artificial Intelligence and AI programming	10
HAIML221*	Programming with Big Data and Data Analytics	10

HAIML223*	Mini Project	10
HCSCI234*	Research Methods	10
HCSEC221*	Network Programming (Pre. HCSEC123)	10
HCSE224*	Object Oriented Programming (Pre. HCSE231)	10

Level 3 Semester 1

Code	**Module Description**	**Credits**
HAIML311*	Work-Related Learning I	40

Level 3 Semester 2

Code	**Module Description**	**Credits**
HAIML321*	Work-Related Learning II (Pre. HAIML311)	80

Level 4 Semester 1

Code	**Module Description**	**Credits**
HCSEC411*	Blockchain Technology	10
HCSE432	Embedded Systems	10
HCSE433*	IoT and Cloud Systems Engineering	10
HAIML411*	Neural Networks	10
HAIML412*	Metaverse	10
HAIML413*	Robotics	10

Level 4 Semester 2

Code	**Module Description**	**Credits**
HCSE422*	Image Processing and Machine Vision	10
HAIML421*	Human AI Interaction	10
HAIML422	Games Development	10
HAIML423*	Robot Programming (Pre HAIML413)	10
HAIML430*	Capstone Project	24

Electives (Choose one)

HAIML424	Introduction to Industry 5.0	10
HAIML425	Expert systems and Recommender systems	10

MODULE SYNOPSES

HAIML111 Fundamentals of Artificial Intelligence

This module is focused on Artificial Intelligence techniques. Students will become familiar with the foundations of agent-based approaches to software design, decision making and problem solving including under uncertainty through the topics covered in this module. This module also covers the basic underpinnings of AI that provide a foundation for degree level study. It will give a broad overview of the field that will support learning in other modules and will be expanded further within the degree.

HAIML112 Machine Learning Fundamentals

This module covers the basic underpinnings of Machine Learning that provide a foundation for degree level study. It will give a broad overview of the field that will support learning in other modules and will be expanded further within the degree.

HAIML113 Intelligent Signal Processing

This module aims to provide students with a broad experience of digital signal processing techniques and applications on how audio and video signals can be captured and processed by a computer program. Students will learn about time domain and frequency domain representations and processing. Students will learn how students can extract information from

audio signals. Students will implement movement and face detection systems that work with live camera input

HCSCI132 Principles of Programming Languages
The module aims to introduce the basic principles of programming, programming paradigms, program constructs and implementation of algorithms using Python programming language. Modular programming through use of functions, modules will also be covered to enhance students' understanding of code reusability. Moreover, data persistence will be demonstrated in files and data structures. The module will equip students with knowledge on basics of Object-Oriented Programming (OOP) and databases. Lastly, students will get to explore fundamentals of object-based programming using the Tkinter library in Python.

HCSE138 Discrete Mathematics
This module introduces and discusses the fundamentals of Discrete Mathematics as applied to Artificial Intelligence, focusing on providing a basic theoretical foundation for further work. Students are exposed to logic and proof techniques, set theory, elementary number theory, functions and relations, graphs, trees, modelling computations and abstract algebra. This module integrates symbolic tools, graphical concepts, and numerical calculations. Techniques of counting: permutations, combinations, recurrences, and algorithms are also covered.

CS131 Basics of Communication Skills
Refer to Communication Skills Department regulations.

HAIML121 Natural Language Processing
This module will provide students with a grounding in both rule-based and statistical approaches to NLP combining theoretical study with hands-on work. The module focuses on text processing, and by taking this module, students will learn about how they can work with text-based natural language in their computer programs. Students will learn about grammars and how they can be used to analyse text. Students will learn how statistical analysis can be used to extract information from and classify text. They will work in an appropriate programming environment for NLP, using libraries to implement NLP workflows.

HCSEC111 Introduction to Information Security
A review of the origins of information security, outline of the phases of the security systems development life cycle, critical characteristics of information (CIA). Examination of the various threats facing organizations and methods for ranking these threats (in order to assign them relative priority), examination of the types of attacks that could result from these threats, and how these attacks could impact the organization’s information systems, secure software development. Legal, ethical, and professional issues in Information Security. Risk Management: an overview of risk management (know yourself/the enemy), Risk identification, Risk assessment, Risk control strategies, Selecting a risk control strategy (feasibility studies). Benchmarking and best practices, Baselining, Residual risk. Information security policy, standards, and practices. EISP, ISSP, SysSP, The information security blueprint, Security Education, Training, and Awareness Program (SETA), Continuity strategies. Security Technology: Security Technology: Intrusion Detection and Prevention Systems. Physical Security: Physical access controls, Mobile and portable systems, Remote computing security. Implementing Information Security: Information security project management, Technical aspects of implementation (Conversion strategies, The Bull’s-Eye model). Personnel Security: Positioning and staffing the security function, Credentials of information security

professionals, Employment policies and practices, Security considerations for non-employees. Information Security

HCSCI135 Computer Architecture and Organization
This module studies the basic instruction set architecture and organization of a modern computer. Topics include Assembly language, The Von Neumann Machine Instructions, Registers, translating high level arithmetic into Assembly, Memory and Registers Addressing Modes, Logic Gates, Truth Tables, implementing Truth Tables, Latches and Memory Reading, Multicycle implementations, the concept of a Cycle. Finite State Machines, Balancing the work into Single Cycles, ROMs, PLAs, Microcode, RISC/CISC, Pipelines, Principle of Locality, Direct Mapped Caches, I/O Polling and Interrupts.

HCSE121 Web and Multimedia Development
Site development processes, Design principles; page layout navigation, HTML, CSS and PHP programming language. Database-driven web pages using PHP. PHP framework and syntax, connection to any ODBC-compliant database, and hands-on practice with a MySQL database to create database-driven HTML forms and reports.

HCSE135 Database Systems
The module focuses on concepts and principles related to database management systems and links these to Relational Database Systems. Topics covered include: Database Systems Evolution, Database Systems in the Organization, Principles of Conceptual Design, Database Models, The Relational Data Model, Data Modelling, Database Design Theory, Data Definition and Manipulation Languages, Storage and Indexing Techniques, Query Processing and Optimization, Concurrency Control and Recovery and Database Programming Interfaces.

HCSCI136 Data Structures and Algorithms
This module builds on the programming skills acquired in Principles of Programming Languages. It couples work on Program Design, Analysis, and Verification with an introduction to the study of Data Structures. Students are introduced to: Lists, Stacks, Queues, Trees, Hash Tables, and Graphs. Students are expected to write several programs, ranging from very short programs to more elaborate systems. Emphasis is placed on the development of clear, modular programs that are easy to read, debug, verify, analyze, and modify.

HCSEC123 Principles of Secure Coding
This module provides a detailed explanation of common programming errors in programming languages and describes how these errors can lead to software systems that are vulnerable to exploitation. It will also cover defensive programming and identify its benefits and disadvantages, secure programming, its relationship to defensive programming, and its benefits and disadvantages, sources of risk that can negatively impact software applications, best practices for creating secure code, how to include defensive programming techniques into software development process, why testing should be performed and identify the major phases of the software testing process, what unit testing is and the benefits it provides, the benefits of employing defensive and secure programming and recognizing the phases of the software testing process. Software security, organizational security and connection security are covered.

HCSCI237 Data Communications and Computer Networks
This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

TCNP 201 Technopreneurship
Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, and product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HCSE231 Object Oriented Analysis and Design
Introducing object orientation; classes; objects; abstraction; encapsulation; software modelling; aims of modelling; principles of modelling; overview of UML; object-oriented analysis and design process; user stories; analysis and design of artefacts; the conceptual model; identifying classes and their relationships; identifying class responsibilities; CRC cards; Domain Driven Design; Object-Oriented Design Principles; Object-Oriented Design Patterns.

HAIML211 Data Science and Analytics
This module deals with areas that involve extraction of meaningful information and insights by applying various algorithms, processes, and scientific methods from structured and unstructured big data. The data science part also comprises mathematics, computations, statistics, programming, etc to gain meaningful insights from the large amount of data provided in various formats. On the other hand, the Data analytics is then used to get conclusions by processing the raw data through converting a large number of figures in the form of data into Plain English i.e., conclusions which are further helpful in making the decisions. Areas covered also include: Data and Visualizations - Introduction to Data: What it is and What's Driving it, Emerging Issues On Data, Types of Data. Data Collection Methods: Traditional and Modern. Data wrangling and pre-processing. Predictive Data Analysis. Data Predictions and Models. Statistical Data Analysis including distributions, probability, and simulations. Data mining: finding similar items, mining data streams, frequent itemsets. Machine learning: k nearest neighbor, decision trees, naive Bayes, regression. Ecological and Environmental Data Analysis. Application can be done using Excel for Data Analysis and Data Visualisation. R: The Statistical Programming Language or any other relevant software at the time.

HAIML212 Deep Learning
Deep learning principles and fundamentals, the practical knowledge of TensorFlow, DialogFlowpython, AI for DevOps and Python will be covered

HCSEC221 Network Programming
Modules covers Linux Utilities- File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking utilities, Filters, Text processing utilities and Backup utilities. Bourne again shell(bash) – Introduction, pipes and redirection, here documents, running a shell script, the shell as a programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples. Files- File Concept, File types File System Structure, Inodes, File Attributes, file I/O in C using system calls, kernel support for files, file status information-stat family, file and record locking-lockf and fcntl functions, file permissions. File and Directory management – Directory contents, Scanning Directories- Directory file APIs. Process- Process concept, Kernel support for process, process attributes, process control – process creation, replacing a process image, waiting for a process, process termination, zombie process, orphan process.: Signals- Introduction to signals, Signal generation and handling, Kernel support for signals, Signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions. Interprocess Communication – Introduction to IPC mechanisms, Pipes- creation, IPC between related processes using unnamed pipes, FIFOs-creation Shared Memory- Kernel support for shared memory, UNIX system V APIs for shared memory, client/server example. Network IPC – Introduction to Unix Sockets, IPC over a network, Client-Server model, Address formats (Unix domain and Internet domain), Socket system calls for Connection Oriented – Communication, Socket system calls for Connectionless-Communication, Example-Client/Server Programs- Single Server-Client connection, Multiple simultaneous clients, Socket options – setsockopt, getsockopt, fcntl. Network Programming in Java- Network basics, TCP sockets, UDP sockets (datagram sockets), Server programs that can handle one connection at a time and multiple connections (using multithreaded server).

HAIML222 Artificial Intelligence and AI Programming
Problem solving, reasoning, planning, natural language understanding, computer vision, automatic programming, machine learning, pattern recognition, search algorithms for problem solving; knowledge representation and reasoning; pattern recognition; fuzzy logic; and neural networks.

HCSE224 Object Oriented Programming
The module aims at providing a solid foundation in Object Oriented Paradigm. Topics covered include: Objects Overview and Review, creating Class Instances within constructors, Object Analysis, creating Fields and Properties, Inheritance and specialized Classes, Base Class and Abstract Classes, Events and Exceptions, providing services using Interfaces and Abstract Classes, Polymorphism, Shared and Static members, Overloading Operators, Overriding, Multithreading.

HAIML221 Programming with Big Data and Data Analytics
Programming with Big Data in R is a series of R packages and an environment for statistical computing with big data by using high-performance statistical computation. It uses the same programming language as R with S3/S4 classes and methods which is used among statisticians and data miners for developing statistical software. It mainly focuses on distributed memory systems, where data are distributed across several processors and analyzed in a batch mode, while communications between processors are based on MPI that is easily used in large high-performance computing (HPC) systems.

HCSCI234 Research Methods
This module equips students with research techniques including definition of research objectives, research framework, design, research problem, experimental research, experiment data acquisition and processing, population and sampling methods, research methods and instruments, data processing and analysis, descriptive statistics, inferential statistics, data presentation and interpretation, research ethics, report writing.

HAIML223 Mini Project
After having gained knowledge in modules undertaken earlier on, students will be equipped enough to conceive a project idea in any of the thematic areas within the cyberspace realm exposed to earlier on. The expectation is that the student will design and implement a solution under the supervision of a lecturer and submit a suitable report on the work carried out. An artifact commensurate with the level of study is expected at this point. Project can be done in groups depending upon approval by the departmental board.

HAIML311 Work-Related Learning I
Refer to Section 8 of the Faculty of Science and Technology Regulations

HAIML321 Work-Related Learning II
Refer to Section 8 of the Faculty of Science and Technology Regulations

HCSEC411 Blockchain Technology
Areas to be covered include: Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof. Blockchain: Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft and Hard Fork, Private and Public blockchain. Distributed Consensus: Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternate. Cryptocurrency: History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin. Cryptocurrency Regulation: Stakeholders, Roots of Bit coin, Legal Aspects-Crypto currency Exchange, Black Market and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain. Tutorial and Practical: Naive Blockchain construction, Memory Hard algorithm - Hashcash implementation, Direct Acyclic Graph, Play with Go-ethereum, Smart Contract Construction, Toy application using Blockchain, Mining puzzles

HCSE432 Embedded Systems
This module covers Embedded Systems concepts. Topics include the nature of embedded systems, specific problems, special issues; role in computer engineering; embedded microcontrollers, embedded software; real time systems, problems of timing and scheduling; testing and performance issues, reliability; low power computing, energy sources, leakage; design methodologies, software tool support for the development of such systems; problems of maintenance and upgrade; networked embedded systems.

HCSE433 IoT and Cloud Systems Engineering
IoT and Cloud Computing. IoT Physical Devices and Endpoints: IoT physical servers and cloud offerings: cloud storage models and communication Networks, Cloud technologies, Framework and Platforms. Internet of services, SOA, Grid computing, System types, architectures and models, applications in telecommunication systems, cloud systems examples, security management, Governance, legislation, economic environment.

HAIML411 Neural Networks
Introduces machine learning fundamental principles, supervised and unsupervised machine learning algorithms, linear regression, logistic regression, decision trees, k-nearest neighbour, Bayesian learning, naïve Bayes algorithm, support vector machines and kernels and artificial neural networks (Deep Learning), Ensemble methods, clustering algorithms (K-means).Data preprocessing, feature extraction and dimensionality reduction.
Model selection, generalization, over fitting and optimization of training models. Provides a hands-on problem-solving experience with programming in Python. Python, anaconda package installation and other relevant software Jupyter note book, google colab. It also covers applications of machine learning computer vision in self driving cars, spam filtering in social networks etc.

HAIML412 Metaverse
Computer graphics basics , implementation of various algorithms to scan, convert the basic geometrical primitives, transformations, fundamentals of animation, augmented reality, virtual reality, 3D display systems, additive manufacturing concepts, display optics and electronics, IMUs and sensors, tracking, haptics, rendering pipeline, multimodal human perception and depth perception, strengths and limitations of VR technology in order to be able to construct simple immersive environments as well as to understand the human factors and cognitive issues that should be considered when using this medium.

HAIML413 Robotics
This module introduces the basic concepts of robotics, focusing on: Robot Anatomy-Definition, law of robotics, History and Terminology of Robotics, Accuracy and repeatability of Robotics, Specifications of Robots, Speed of Robots, Robot joints and links, Robots classifications, Architecture of robotic systems, Robot Drive systems, Hydraulic, Pneumatic and Electric systems, construction and programming of autonomous mobile robots

HCSE422 Image Processing and Machine Vision
Image Processing and Computer Vision Background, Image Processing and Computer Vision Applications Digital Image Processing Hierarchy: Human Perception of Pictures, Digital Image Processing Hardware. Image Model, Amplitude digitization: Intensity Quantization, Spatial co-ordinate digitization: Image Sampling, Image Quality, Image Pixel Relationships, Linear Operators, 2-D Transforms. Spatial Domain Methods, Frequency Domain Methods. Inverse Filtering. Image Compression, Redundancy Types, Lossless and Lossy Compression, Image Compression Standards. Object Detection Methods, Edge Liking and Boundary Detection, Thresholding Methods, Region Oriented Methods. Object Representation and Description, Representation schemes, Description. Pattern Recognition, Decision Theoretic Methods for Recognition.

HAIML421 Human AI Interaction

Human characteristics: human and computer Graphic design basics, Interaction: input and output, Dialogue interactions and response times, Design process, Models of users in design, Dialog notations and design, Implementation support, Evaluation, Help and documentation, Hypertext, multimedia and WWW, Game design basics.

HAIML422 Games Development
The module will seek to equip students with gaming concepts and development ability. The students will be required to apply visual and OOP programming concepts in creating interactive games. The course will cover event driven programming, gaming design patterns, scene creation, game security and basic game programming. The practicals will be done using the Unity3D and any OOP language as selected by the instructor.

HAIML423 Robot Programming
This module is a fundamental programming module that teaches students how to safely manipulate the robot through proper use of the robot controller and teach pendant. This module is designed to enable students to set-up, program, edit and operate robots. It is a lab-based module that uses a hands-on approach to introduce the basic concepts of robotics, focusing on the construction and programming of autonomous mobile robots. Module information will be tied to lab experiments and students will work in groups to build and test increasingly more complex mobile robots. Students will: 1. Explore the broad scope of robotic applications 2. Learn the basic components and building blocks of robots 3. Develop the robot construction skills 4. Learn to program the robots 5. Program autonomous mobile robots to achieve challenging tasks.

HAIML430 Capstone Project
Students will be expected to use AI and Machine Learning principles, Emerging Technologies and other Domain area knowledge in coming with a final year research project which addresses a particular need and produce a working prototype.

HAIML424 Introduction to Industry 5.0
Looks at trending and emerging AI technologies like edge AI

HAIML425 Expert systems and Recommender systems
Looks at the Fundamentals and the basics of expert systems, rule-based systems and recommender systems.

REGULATIONS FOR MASTER OF SCIENCE DEGRE IN COMPUTER SCIENCE (MSC)

Duration:	18 months
Maximum Credit Load:	360
Minimum Credit Load:	270
Maximum MBKs Credit Load:	216
ZNQF Level:	9

1. PURPOSE OF THE PROGRAMME

8 This programme is designed to develop and produce graduates with a broad understanding of the state-of-the-art miniaturized and computer-based systems, smart device technology and emerging technologies. Hence allowing them greater specialization, that enables them to explore topics that align more closely with their professional goals.

2. ENTRY REQUIREMENTS

2.1 **Normal Entry**

An Honours degree with a degree class of at least 2.2 in Computing (Computer Engineering, Informatics, Information Systems, Computer Science, Software Engineering), Electronic Engineering, Cyber security or Data analytics.

2.2 **Special Entry**

Special entry may be granted to applicants with any equivalent qualification acceptable according to the Institute's General Academic Regulations.

2.3 **Mature Entry**

Refer to Section 3.3 of the General Academic Regulations.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Image Processing, Architecture Programming, Database Engineering, Data Mining, Advanced Machine Learning, Artificial Intelligence, Mobile Computing

3.2 **Specialist Focus:** Hardware Design, Systems Development, Cyber Security, Computer Networking, Machine Learning, Parallel Distributed Processing, Artificial Intelligence

3.3 **Orientation:** Research, innovation and Technopreneurial orientation. Teaching and learning is focused on producing graduates with the necessary knowledge, skills and competencies sufficient to start technology-based businesses.

3.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach. The programme is designed to allow students to learn to apply systematic computing principles and impart practical training based on scientific knowledge and methods-based training which qualifies for a self-sufficient professional activity

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 **Employability:** Careers in the Computer Science field which include:
Embedded Software Engineers, Senior Network Engineers, Senior Hardware Engineers, Network and Systems Administrators, Database Managers, Senior Analyst Programmers, Senior Systems Analysts, Project Manager, Cyber Security Manager, Software Architect, PhD opportunities that lead to lectureship and research in universities.

4.2 **Further Studies:** Doctoral studies in Computer Science, Computer System Engineering, Software Engineering, Data Science, or in interdisciplinary programmes related to computing practices.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, blended learning through e-learning, programming laboratory classes, seminars, group work, industrial visits, industrial attachment, research project, individual independent study

5.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, mini-research project report, final year research project report, and continuous assessments.

6. PROGRAMME COMPETENCIES

6.1 **Generic:**

6.1.1 **Multidisciplinary**: Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

6.1.2 **Flexibility;** to succeed in a rapidly changing field.

6.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things.

6.1.4 **Communication skills**: Ability to communicate effectively and to present information orally and in writing and using ICTs to both expert and non-expert audiences.

6.1.5 **Analysis and synthesis**: Capacity for analysis and synthesis using logical arguments and proven facts.

6.1.6 **Ethical commitment:** Professional integrity and awareness of impact of science and technology on society and the environment.

6.1.7 **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.

6.2 **Discipline specific:**

6.2.1 **Technical understanding of** Computer Science**;** Graduates should have a mastery of Computer Science as described by the core of the Body of Knowledge.

6.2.2 **Familiarity with common themes and principles;** Graduates need understanding of a number of recurring themes, such as abstraction, complexity, and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Computer Science and should not consider them as relevant only to the domains in which they were introduced.

6.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Computer Science is understanding the interplay between theory and practice and the essential links between them. Graduates of a Computer Science program need to understand how theory and practice influence each other.

6.2.4 **Problem solving skills;** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

6.2.5 **Project experience;** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Computer Science programmes should have been involved in at least one substantial project.

6.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

6.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

6.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

7. INTENDED LEARNING OUTCOMES

7.1 By the end of the programme a graduate will be able to:

7.1.1 Use software development tools to produce effective software systems.

7.1.2 Communicate computer science concepts design and solutions effectively.

7.1.3 Configure, manage and troubleshoot computer systems.

7.1.4 Identify and analyze scholarly literature relating to Computer Science.

7.1.5 Publish research in scholarly journals of international repute.

7.1.6 Deploy and administer selected computing technologies that meet specific needs of industry.

7.1.7 Document and develop solutions that solve specific problems in the industry.

7.1.8 Select computing technologies that meet specific needs of the industry.

7.1.9 Work as an effective team player when undertaking software projects.

7.1.10 Apply knowledge of contemporary issues and modern practices to address ever-evolving business needs in computer science.

7.1.11 Write and present technical and non-technical research papers in Computer Science and enter into higher degree programmes such as PhDs.

7.1.12 Design and evaluate solutions for complex computing problems and evaluate systems or components.

7.1.13 Acquire social and methodological skills in addition to professional competence, international aspects designed to prepare and empower students to face increasingly global challenges.

7.1.14 Model, design, implement, test, evaluate and integrate computer hardware and software to create advanced computing systems.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

9. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

10. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. PROVISION FOR PROGRESSION

Refer to Section 7 of the Faculty of Science and Technology Regulations.

12. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty of Science and Technology Regulations.

13. DEGREE WEIGHTING

Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to availability of personnel. * denotes core modules.

Level 1 Semester 1

Code	Module Description	Credits
*MSC111	Pattern Recognition and Image Processing	18
*MSC112	Advanced Enterprise Architecture Programming	18
*MSC113	Advanced Database and Data Mining	18
*MSC121	Computational Discrete Mathematics	18
MSE114	Cyber Security and Cryptography	18

Level 1 Semester 2

*MSE124	Research Methods	18
*MSC211	Evolutionary Computing and Parallel Distributed Processing	18
*MSC212	Simulation and Modelling	18

Electives (Select 2 modules from the following)

MSC122	Software Methodology	18
MSC123	Machine Learning and Artificial Intelligence	18
MCSE122	Mobile Computing	18
MCSE125	Real-Time Systems	18
MCSE127	Advanced Operating System Design	18
MSE111	Advanced Data Structures and Algorithms	18
MSE215	Enterprise Computing In Java	18

Level 2 Semester 1

*MSC221	Dissertation	90

MODULE SYNOPSES

MSC111 Pattern Recognition and Image Processing

Introduction to pattern recognition. Fundamental problems in pattern recognition. Foundations of pattern recognition algorithms and machines, including statistical and structural methods. Data structures for pattern representation, feature discovery and selection, classification vs. description, parametric and non-parametric classification, supervised and unsupervised learning, use of contextual evidence, clustering, recognition with strings, small sample-size problems biological object recognition, Bayesian decision theory.

MSC112 Advanced Enterprise Architecture Programming

Introduction to application server programming and business logic programming. Transaction processing, concurrency control, Event-driven programming, asynchronous method invocation, job scheduling, Inter process communication. Deployment of software components in an application server. Business Interface development and deployment.

MSC113 Advanced Database and Data Mining

Data Models; The Enhanced Entity Relationship (EER) Model, EER Models to Relational Databases, Database Design and Implementation; design methodologies, implementation methodologies, Physical Database design and Tuning, Query processing and Optimization. Algorithms for Query Processing and Optimization, Transaction Processing, Concurrency

Control Techniques. Database Security and Distribution, Distributed Databases, Mobile Databases Machine Learning and Pattern Recognition, Data Mining.

MSE114 Cyber Security and Cryptography

Cryptography is a key component in computer network security engineering. This course gives a broad overview of methods of implementing security services based on cryptography in data networks. Topics to be covered include the fundamentals of contemporary cryptography and its application to security engineering in software services, such as confidentiality, integrity, authentication, and non-repudiation. The course will also introduce relatively new ideas in cryptology, such as public key cryptography and zero-knowledge protocols, permitting efficient solutions to the problems of digital signature, electronic cash, key exchange, and access control.

Students will also learn how to analyse the strength of today's ciphers and their implementations, and discuss the best known crypto analytical techniques used to break security systems. During the course we will analyse the most popular implementations of cryptography used on the Internet, including systems for electronic mail protection, secure WWW, and electronic payment protocols. We discuss the ongoing work on the development of international standards for secure communications and present the most recent research trends in cryptology.

MSC121 Computational Discrete Mathematics

Discrete models. Foundations. Basic concepts of sets and functions. Finite series. Logic. Propositional logic. Predicate logic. Combinational circuits. Induction. Finite probability space, events. Conditional probability, Bayes' theorem. Integer random variables. Expectations. Variance Analysis and verification. Searching algorithms. Recursive algorithms. Relations. Basic concepts. Properties of relations. Operations on relations. Undirected graph, directed graph, weighted graph, Euler circuits and Hamiltonian cycles. Graph isomorphism and representations. Planar graphs. Trees. Different state machines. Input, Output, Initial state. Transition table.

MSC122 Software Methodology

Overview of Software Engineering, the Software Development Process; requirements analysis and specification phase Design phase; implementation phase, maintenance; Engineering with a Programming Language; Software Engineering Paradigms; Engineering with existing software. Software Engineering Project.

MSC123 Machine Learning, and Artificial Intelligence

The module is intended to combine the theory with the hands-on practice of solving modern industry problems with an emphasis on image processing and natural language processing. Topics include outlier detection, advanced clustering techniques, deep learning, dimensionality reduction methods, frequent itemset mining, and recommender systems. Topics also considered include reinforcement learning, graph-based models, search optimization, and time series analysis. The module uses Python as the primary language, although later projects can include R and other languages. The module also introduces some industry standard tools to prepare students for artificial intelligence jobs.

MSE124 Research Methods

This module serves as a fundamental building block in equipping students with the knowledge and competencies to conduct research in the computing field. Qualifying students have an appreciation of the research process and the reason for conducting research according to a

specific research paradigm, to solve a problem. They determine appropriate sources to consult for a literature review. Students understand appropriate research strategies, data gathering and analysis techniques to obtain answers to the stated research problem. The module will cover the following; Problem identification, Objective formulation, Anatomy of a literature review, Writing a captivating abstract, Methodology, Results and discussion, conclusion, Research contributions. Students are taught to write research project proposals, use of bibliography management software and explore quantitative and qualitative software**.**

MCSE122 Mobile Computing

The module covers Mobile Communications, Mobile Computing – Paradigm, Architecture; Mobile and Handheld Devices. GSM – Services, System Architecture, Radio Interfaces, Protocols, Security, New Data Services, GPRS, CSHSD, DECT. (Wireless) Medium Access Control (MAC) Motivation for a specialized MAC, SDMA, FDMA, TDMA, CDMA, Wireless LAN/(IEEE 802.11) Mobile Network Layer IP and Mobile IP Network Layers, Mobile Transport Layer Conventional TCP/IP Protocols, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Database Issues, Client-Server Computing and Adaptation, Transactional Models, Query processing, Data Recovery Process and QoS Issues. Data Dissemination and Synchronization Mobile Ad hoc Networks (MANETs), Applications and Challenges, Routing, Classification of Routing Algorithms, Algorithms such as DSR, AODV, DSDV, Mobile Agents, Service Discovery. Protocols and Platforms for Mobile Computing WAP, Bluetooth, XML, J2ME, JavaCard, PalmOS, Windows CE, SymbianOS, Linux for Mobile Devices, Android.

MCSE125 Real-Time Systems

The module focuses on Real-time systems models and types, structure, Performance measures, Examples of real-time systems and real-world applications, Modelling and Designing real-time systems. Real-Time Process Management: Task scheduling for Uniprocessor systems, handling priorities with critical section, interrupts, task allocation and scheduling for multiprocessor systems, adaptive scheduling. Programming Environment: In depth Knowledge of RTOS programming languages, tools and techniques. Real-Time System Design: Design techniques for Reliability, Fault Tolerance and other application specific quality considerations. Study of Micro C/OS-II or Vx Works or Any other popular RTOS - RTOS System Level Functions - Task Service Functions - Time Delay Functions - Memory Allocation Related Functions Trends in Real-Time System- Design and Development in fields such as Robotics. Introduction to research topics.

MCSE127 Advanced Operating System Design

Introduction and History of OS, OS Concepts, System Calls, OS Structure, PROCESSES: Process Concept, Interprocess Communication, Classical IPC Problems and Scheduling. Principles of I/O Hardware and I/O Software, Deadlocks, RAM Disks, Disks and Terminals. Basic Memory Management, Swapping, Virtual Memory, Page Replacement Algorithms, Design Issues for Paging Systems and Segmentation. Files, Directories, File System Implementation, Security and Protection Mechanisms. MINIX 3. Overview and implementation of processes, the system task and the clock task in minix 3. Overview and implementation of I/O, Block Devices, Process Manager, and File System in Minix3.

MSC211 Evolutionary Computing and Parallel Distributed Processing

Fundamentals of genetic algorithms, genetic programming. Conceptual simplicity and broad applicability of genetic algorithms. Features of evolutionary computation, evolutionary strategies, evolutionary programming. Hybridization and Optimization techniques. Heuristic

level: knowledge representation, inference strategies. Man-machine interfaces. Fuzzy set theory. Decision: Classical, nonstandard and fuzzy logic. Data representation. Network configurations: single layer non-recurrent networks. Multilayer non-recurrent networks. Recurrent networks. Application for artificial neural networks: character and speech recognition, image analysis Parallel distributed processing. General framework. Distributed representation. Basic mechanisms and formal analysis.

MSC212 Simulation and Modelling
Advances in simulation and modelling methodology. All students are expected to have completed an introductory module in simulation. Modelling complexities and decision-making simulation using system dynamics. Applied statistical functions, Experimentation, Applied statistical methods for analysis and modelling. Approaches to structuring simulations. Contrasting discrete, continuous and agent-based simulation. The module explores the power of the use of computers in solving complex problems using modelling and simulation, by substituting physical experiments that are costly and time consuming by computations. This allows for fast extraction of useful information from experimental data in science and engineering.

MSE111 Advanced Data Structures and Algorithms
Topics include Algorithm analysis covering RAM model–nations, Recurrence analysis – Master's Theorem, Amortized analysis, Disjoint sets, Randomized Algorithms - n-queen problem. Sorting and Order Statistics covering -Heap Sort, Selection sort, Quick sort, Heap Structures, Min-Max Heaps, Leftist trees, Binomial Heaps, Fibonacci heaps. Search Structures covering OBST, AVL, 2-3 trees, Red-Black trees, B-trees, Digital trees, tries, Splay trees. Graph Algorithms covering Minimum spanning trees, Single source shortest path, All pairs shortest path, Maximum flow, Planar graph colouring. Advanced design and Analysis Techniques covering Dynamic Programming, Greedy Algorithms, Amortized Analysis, Linear programming, Job Scheduling, TSP.NP-Hard and NP-complete: Non-deterministic algorithms, clique problem, Vertex cover problem, TSP.

MSE215 Enterprise Computing in Java
This course will equip students with knowledge and development skills in J2EE, Hibernate, Spring and Web Services. The course will include topics in building J2EE Applications using Servlets, Java Server Pages (JSP), Enterprise Java Beans (EJB) and Java Database Connectivity (JDBC). It will also give students experience with Model View Control (MVC) frameworks which include Hibernate, Spring and Struts. The Service Oriented Architecture (SOA) principles and implementations will also be covered with a special focus on Java Web Services (JAX), SOAP, JAX-RPC and Web service clients in Java and C Sharp.

MSC221 Dissertation
The Final year dissertation involves in-depth research in a particular area of interest in Computer Science. The dissertation will capture the details of the practical work carried out by a master's student, in which the student demonstrates the ability to transfer the theoretical knowledge gained during the course of study into some tangible computer science product that will be evaluated by the department board of examiners. The research finding is expected to be published in peer reviewed journals or conference proceedings. Finally, all the work should be presented in a thesis.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN SOFTWARE ENGINEERING (MSE)

Duration:	18 Months
Maximum Credit Load:	360
Minimum Credit Load:	270
Maximum MBKs Credit Load:	216
ZNQF Level:	9

1. PURPOSE OF THE PROGRAMME

1.1 The purpose of the Masters in Software Engineering programme is to allow students to delve deeper into topics like software solutions and emerging technologies. Hence allowing them greater specialization, that enables them to explore topics that align more closely with their professional goals.

2. ENTRY REQUIREMENTS

2.1 **Normal Entry**
To qualify for acceptance to the Software Engineering Master's Degree, the candidate must have obtained an Honours degree in a Software Engineering or any equivalent computing science or engineering qualification with a minimum of a 2.1degree class.

2.2 **Special Entry**
Any equivalent qualification acceptable according to the Institute's General Academic Regulations.

2.3 **Mature Entry**
Refer to section 3.3 of the General Academic Regulations.

3. PROGRAMME CHARACTERISTICS

3.1 **Areas of study:** Software Development Methodologies, Advanced Software Engineering, Software Project Management, Ethics and principles in Software Engineering, Big Data Analytics and Artificial Intelligence.

3.2 **Specialist Focus:** Software Development Principles, Secure systems, Software Modelling and Simulation, Development Ethics.

3.3 **Orientation:** Research, innovation and Technopreneurial orientation. Teaching and learning is focused on producing graduates with the necessary knowledge, skills and competences sufficient to start technology-based businesses.

3.4 **Distinctive Features:** The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach. The programme is designed to allow students to learn to apply systematic software engineering principles to design, construction, and maintenance of larger software applications.

4. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 **Employability:** Careers in the Software Engineering field which include:

Embedded Software Engineers, Lead Software Engineers, Senior Analyst Programmers, Senior Systems Analysts, Project Managers, Cyber Security Manager, Software Architect.

4.2 **Further Studies:** Doctoral studies in Software Engineering, Computer Science, Data Science, or in interdisciplinary programmes related to computing practices.

5. PROGRAMME DELIVERY

5.1 **Teaching and Learning Methods:** Lectures, tutorials, programming laboratory classes, seminars, group work, industrial visits, industrial attachment, research project, individual independent study.

5.2 **Assessment Methods:** Written and oral examinations, tests, laboratory reports, seminar presentations, mini-research project report, final year research project report, continuous assessments.

6. PROGRAMME COMPETENCIES

6.1 **Generic:**

6.1.1 **Multidisciplinary:** Ability to draw appropriately from multiple academic disciplines to define and solve problems based on understanding of complex phenomena. Should be flexible work across many disciplines.

6.1.2 **Flexibility:** To succeed in a rapidly changing field.

6.1.3 **Quantitative and innovative reasoning:** Capability to draw on big data and use analytics for informed decision making and strive to seek new ways of doing things.

6.1.4 **Communication skills:** Ability to communicate and present oral and written information effectively and use ICTs to both expert and non-expert audiences.

6.1.5 **Analysis and synthesis**: Capacity for carrying out analysis and synthesis using logical arguments and proven facts.

6.1.6 **Ethical commitment:** Show professional integrity and awareness of Software Engineering impact on society and the environment.

6.1.7 **Entrepreneurial skills:** Capability to identify and create new business ventures based on knowledge and new thinking paradigms.

6.2 **Discipline specific:**

6.2.1 **Deep knowledge:** Graduates should have a mastery of Software Engineering as described by the core of the Body of Knowledge.

6.2.2 **Familiarity with common themes and principles;** Graduates need understanding of a number of recurring themes, such as abstraction, complexity, and evolutionary change, and a set of general principles, such as sharing a common resource, security, and concurrency. Graduates should recognize that these themes and principles have broad application to the field of Software Engineering and should not consider them as relevant only to the domains in which they were introduced.

6.2.3 **Appreciation of the interplay between theory and practice**; A fundamental aspect of Software Engineering is understanding the interplay between theory and practice and the essential links between them. Graduates of a computer systems program need to understand how theory and practice influence each other.

6.2.4 **Problem-solving skills:** Graduates need to understand how to apply the knowledge they have gained to solve real problems, not just write code and move bits. They should be able to design and improve a system based on a quantitative and qualitative assessment of its functionality, usability and performance.

6.2.5 **Project experience:** To ensure that graduates can successfully apply the knowledge they have gained, all graduates of Software Engineering programmes should have been involved in at least one substantial project.

6.2.6 **Commitment to life-long learning;** Graduates should realize that the computing field advances at a rapid pace, and graduates must possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves.

6.2.7 **Communication and organizational skills;** Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams.

6.2.8 **Appreciation of domain-specific knowledge;** Graduates should understand that computing interacts with many different domains. Solutions to many problems require both computing skills and domain knowledge.

7. INTENDED LEARNING OUTCOMES

7.1 By the end of the programme the student should be able to:

7.1.1 Use software development tools to produce effective software systems.

7.1.2 Communicate computer science concepts design and implement software solutions effectively.

7.1.3 Configure, manage and troubleshoot computer systems.

7.1.4 Identify and analyze scholarly literature relating to Computer Science Software Systems.

7.1.5 Lead projects on the analysis, design, and implementation of software information systems in various development environments and platforms.

7.1.6 Publish research in scholarly journals of international repute.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty of Science and Technology Regulations.

9. ASSESSMENT

Refer to Section 5 of the Faculty of Science and Technology Regulations.

10. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 9 of the General Academic Regulations.

11. PROVISION FOR PROGRESSION

11.1 Refer to Section 7 of the Faculty of Science and Technology Regulations.

12. GRADING AND DEGREE CLASSIFICATION

Refer to Section 10 of the Faculty of Science and Technology Regulations.

13. DEGREE WEIGHTING

Refer to Section 11 of the Faculty of Science and Technology Regulations.

PROGRAMME STRUCTURE

A student will not be allowed to register for a module with a pre-requisite if the pre-requisite is not passed. Electives will be offered subject to availability of personnel. * *Denotes core module.*

Level 1 Semester 1

Code	Module Description	Pre-requisite	Credits
*MSE111	Advanced Data Structures and Algorithms		18
*MSE112	Software Development Methodologies		18

*MSE113	Software Project Management and Ethics	18
*MSE114	Cyber Security and Cryptography	18
*MSE124	Research Methods	18

Level 1 Semester 2

Code	Module Description	Pre-requisite	Credits
*MSE121	Artificial Intelligence and Expert Systems		18
*MSE122	Business Intelligence and Big Data Analytics		18
MSE211	Research Seminar		18
Electives (Select 1 module from the following)			
MSE123	Web Technologies and Services		18
MSE115	Software Testing		18
MSE212	Software Modelling and Simulation		18
MSE125	Pervasive Computing		18
MSE126	Distributed Systems		18
MSE127	HCI for Information Systems		18
Electives (Select 1 module from the following)			
MSE213	Bioinformatics	18	
MSE214	Cloud and Mobile Applications		18
MSE215	Enterprise Computing in Java		18
MSE216	Digital Image Processing		18

Level 2 Semester 1

Code	Module Description	Pre-requisites	Credits
MSE221*	Dissertation	(MSE124)	90

MODULE SYNOPSES

MSE111 Advanced Data Structures and Algorithms

This module will focus on presenting general techniques for designing correct and efficient advanced algorithms. Topics to be covered include data structures (lists, stacks, queues, trees and graphs), Sorting- selection sort and quicksort, flow of networks, algorithm design (searching algorithms, sorting algorithms, streaming algorithms and graph algorithms), recursive algorithms, complexity analysis, linear programming, dynamic programming, NP Hard and NP completeness. It will also cover RAM model – notations, Recurrence analysis – Master's Theorem, Amortized analysis, Disjoint sets, Randomized Algorithms - n-queen problem.

MSE112 Software Development Methodologies

The module will cover both traditional and modern software development life cycles, requirements engineering, software design paradigms, computer-aided software engineering, system documentation, configuration management tools, common scripting languages and debuggers.

MSE113 Software Project Management and Ethics

The module focuses on Software Maturity Framework, Principles of Software Process Change, Software Process Assessment, and Process Framework. Process Reference Models; CMM, CMMi, PCMM, PSP, TSP, Conventional Software Management, Evolution of Software Economics, Improving Software Economics, Life-Cycle phases and Process artifacts; Engineering and Production stages, artifact sets, model-based software architectures. Other key knowledge areas are: Software process workflows, Iteration workflows, Major milestones,

Minor milestones, Periodic status assessments. Process Planning; WBS, Planning guidelines, cost and schedule estimating process, iteration planning process, Pragmatic planning; Line-of-business organizations, project organizations, evolution of organizations, process automation. Project Control and process Instrumentation; Seven Core Metrics, Management indicators, Quality indicators, Life-Cycle expectations, Pragmatic software metrics, metrics automation; Ethics and Good Governance in Software Project Management Practices and Modern Project Profiles, Next Generation Software Economics, and Modern Process Transitions.

MSE114 Cyber Security and Cryptography
Topics to be covered include the Cybersecurity basics, standards and laws, cyber threats and attacks, ethical hacking, cybersecurity protocols, fundamentals of contemporary cryptography and its application to security engineering in software services, cryptology (i.e. public key cryptography and zero-knowledge protocols), digital signature, key exchange, access control, Blockchain, Cryptocurrency, crypto-analytical techniques. Case studies of cryptography on the Internet.

MSE115 Software Testing
This module covers assessment phases, quality management, quality control tools, quality assurance plan, verification and validation. The module also assesses traditional software testing techniques applicable to all development phases. The techniques include functional, structural, regression, bottleneck, alpha, beta and graphical user interface testing. The agile development paradigm of test-driven development and modern enterprise testing strategies are also discussed. Concepts covered include test cycles, testing objectives, testing in the software development process, types of software errors, reporting and analyzing software errors, problem tracking systems, test case design, testing tools, test planning, test documentation, and managing a test group.

MSE121 Artificial Intelligence and Expert Systems
The module on AI defines the AI Problems and covers: AI Techniques, Basic Problem-solving methods, state space search, problem characteristics, Production systems characteristics, issues in design of intelligent search algorithm. State Space Search: Depth First Search, Breadth First Search, DFID. Heuristic Search: Best First Search, Hill Climbing, Beam Search, Tabu Search. Randomized Search: Simulated Annealing, Genetic Algorithms, Ant Colony Optimization. Problem Reduction, Constraint Satisfaction, Means-End Analysis. Knowledge Representation issues, Knowledge Representation using Predicate logic, Propositional logic. Theorem proving in First Order Logic. Soundness and Completeness, Forward and Backward chaining. Situation calculus. Semantic Nets, Semantic Frames, Conceptual Dependency, scripts, Knowledge representation with uncertainty. Building a Knowledge Base. Partial order planning. Uncertain Knowledge and Reasoning, Probabilities, Bayesian Networks. Overview of different forms of learning: Learning Decision Trees, Neural Networks. Concepts of Fuzzy Logic, Intelligent Agents, Machine Learning. The course focuses on: aspects of Machine Learning, Bayesian Decision Theory, Clustering, and Neural Networks.

MSE122 Business Intelligence and Big Data Analytics
The module covers: Basic concepts in Business Intelligence (BI), Big Data Analytics (BDA), data mining and warehousing; Data Warehouse and Mining Architectures, Design, Implementation and Maintenance. The course also looks at Data Collection and Cleaning, Analysis, Visualization based on query language. Topics to be covered also include; Classification and Prediction, types of data in Cluster Analysis, Outlier Analysis, Mining Spatial Databases, Mining Multimedia Databases, Mining Time Series and Sequence Data, and

Mining NoSQL Databases. Current trends in BI, Data mining, Spatial Mining, and Web Mining will also be discussed.

MSE123 Web Technologies and Services
This module will cover Web evolution, Web 2.0 and Web 3.0 differences, core web technologies (i.e. hypertext mark-up language, cascading style sheets and JavaScript). It will also look at the evolution of distributed computing and service oriented architecture (SOA), web service principles, enterprise web development tools which include J2EE (i.e. Servlets and Javaserver Page) and MVC ASP.Net, XML and SOAP web service architecture, development, registration and discovery process.

MSE124 Research Methods
This module serves as a fundamental building block in equipping students with the knowledge and competencies to conduct research in the computing field. Qualifying students have an appreciation of the research process and the reason for conducting research according to a specific research paradigm, to solve a problem. They determine appropriate sources to consult for a literature review. Students understand appropriate research strategies, data gathering and analysis techniques to obtain answers to the stated research problem. The module will cover the following; Problem identification, Objective formulation, Anatomy of a literature review, Writing a captivating abstract, Methodology, Results and discussion, conclusion, Research contributions.

MSE125 Pervasive Computing
The module will introduce students to Pervasive Computing Principles, Characteristics and Architecture. It also covers Pervasive devices, Embedded controls, Smart sensors and actuators and Open Protocols Service discovery technologies (SDP). The module includes topics in Context Aware Sensor Networks and Context aware security. Additionally, current trends in Pervasive Computing and case studies will be covered.

MSE126 Distributed Systems
The module covers topics in Distributed System Models, Inter-process Communication, Process and Synchronization, Transparency, Scalability, Resource Allocation And Distributed File Systems, Current Trends in Distributed Systems. Concepts to be covered include Middleware, Layered Protocol, Remote Method Invocation (RMI), Remote Procedure Call (RPC), Processes, Threads and Serialization. Students will also learn about Distributed Shared Memory, Process Scheduling, Load Balancing and Load Sharing, Mutual Exclusion, Election algorithms, Coordination and Agreement, Distributed Transaction, Consistency and Replication, Parallel and Distributed Programming and Case Studies in detail and research directions.

MSE127 HCI For Information Systems
This module focuses on major concepts of Human Computer Interaction as applied in Information Systems. It will cover user-centred design (UCD) principles and frameworks, Design Process and Screen Designing, Windows and Components, Prototyping techniques, Devices, Health Care and Health Informatics, HCI in Geographical Information Systems, HCI for E-Commerce, Methodological Issues and Reflections.

MSE211 Research Seminar

This module will include journal article reviewing/reading, plagiarism software interpretation, writing journal papers, peer-reviewing articles and presenting the papers using different styles like IEEE and ACM journal and conference formats. It will be mandatory for students to publish their work as a survey /design paper at a reputable conference or journal at the end of the semester.

MSE212 Software Modelling and Simulation
The module covers basics of Simulation and Modelling – mathematical modelling and the different types of models (Linear Models, Probabilistic Models and Regression Models), Modelling Complex Systems, Simulation Software – Simulation Packages with Programming Languages, Classification of Simulation Software, Experimentation, Sensitivity Analysis, and Optimization Simulation of Manufacturing Systems.

MSE213 Bioinformatics
The course will cover basics of Biology, Methods Of Gene Sequencing, Data acquisition, Data standardization. DNA chips, Protein targeting, Data normalization, Linear view and Statistics Approaches. It will also include Ontology of Genetics, Software availability, Software targets, text parsing, bio-inspired algorithms and systems.

MSE214 Cloud and Mobile Applications
This module covers the key services and application programming interfaces (APIs) offered by Microsoft Azure to build highly scalable, low cost cloud computing services that bring the power of the cloud to mobile, tablet, IOT and desktop devices. The module covers several platform as a service (PaaS) services such as those offered by Microsoft Azure including federated authentication, Azure app service, API apps, mobile apps, Azure web apps, Azure SQL, DocumentDB, cloud services, Notification Hubs, and storage, to name a few. Using these services and others, students learn how to build highly scalable, fine-grained multi-tenant, secure representational state transfer (REST) services in the cloud. Throughout the module, students learn how to build highly responsive and engaging mobile, IOT, tablet, and desktop applications that take full advantage of these services on Android, iOS, and Windows platforms including the Raspberry Pi. This module will also cover the essentials necessary to utilize Azure cloud services from Android, iOS, Windows Universal App platform. It will also cover architecture, design and usability guidance regarding how to decide what software and data executes/resides in the cloud vs on the client IOT and mobile device.

MSE215 Enterprise Computing in Java
This course will cover basics in Enterprise Computing, use and benefits of DevOps. Enterprise Cloud Platform and Web Technologies, Enterprise Infrastructure Solutions, Enterprise Network Solutions and Enterprise Security Products. Enterprise DevOps Integration Platforms. Enterprise Web Technologies like Enterprise Java Beans (EJB), Spring and Microsoft Azure DevOps will be exploited. IBM Infrastructure Solutions will also be investigated.

MSE216 Digital Image Processing
Digital Image Fundamentals: Digital Image Representation, Fundamental Steps in Image Processing, Elements of Digital image processing systems. Image Transforms: Discrete Fourier Transform, Some properties of the two-dimensional fourier transform, Fast fourier transform, Inverse FFT. Image Enhancement: Spatial domain methods, Frequency domain methods, Enhancement by point processing, filtering methods, Colour Image Processing. Image Restoration: Degradation model, Diagonalization of Circulant and Block-Circulant Matrices, Algebraic Approach to Restoration, filters, Image Segmentation: Image Compression: Coding,

Interpixel and Psychovisual Redundancy, Image Compression models, Error free comparison,. Representation and Description: Representation schemes like chain coding, Polygonal Approximation and Signatures.

MSE221 Dissertation

The dissertation involves in-depth research in a particular area of interest in software engineering. The student is expected to select a topic and come up with a research proposal which will be reviewed by the faculty project committee. The student will be given a choice to select a supervisor(s) or the department will choose one for him or her. The student will need to conduct a thorough research and present their research findings in the form of well-articulated and structured documents in line with the faculty guidelines under the guidance of the supervisor(s). The student is also expected to create a prototype showing their full understanding of the software engineering development life cycle and present their solution which will be evaluated by the department board of examiners.

DEPARTMENT OF APPLIED MATHEMATICS AND STATISTICS

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED MATHEMATICS AND COMPUTATIONAL SCIENCE (HMTCS)

Duration:	4 years
Actual Credit Load:	528
Minimum Credit Load:	510
Maximum Credit Load:	540
MBKS Credit Load:	408
ZNQF Level:	8

1. **INTRODUCTION**
   These regulations shall be read in conjunction with the General Regulations and the Faculty of Science and Technology Regulations.

2. **PURPOSE OF THE PROGRAMME**
   The purpose of this programme is to provide a solid foundation and skills in the basic subjects of mathematical knowledge and its applications in the big data world. In addition to the enhanced career prospects that can be gained by taking this programme, it also allows complementary scientific training in computational mathematics with further studies in other scientific fields, extending the range of knowledge and skills of the students and, consequently, increasing the possibility to enter the labour market.

3. **ENTRY REQUIREMENTS**
   For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

   Normal Entry: At least two Advanced Level passes including Mathematics or its recognized equivalents (Mechanics, Statistics, Pure Mathematics) plus at least five Ordinary Level passes/ National Foundation Certificates including Mathematics and English Language OR National Certificate.

   Special Entry: National Diploma or Higher National Diploma.

   Mature Entry: See General Academic Regulations

4. **PROGRAM CHARACTERISTICS**
   Areas of Study:
   Calculus; algebra; applied mathematics; mathematical programming; computational modelling and simulation; graph theory; complex numbers; differential equations; analysis; financial mathematics and mathematical modelling.
   Specialist Focus:
   Understanding fundamentals of computational mathematics and its application in big data and analytics for making informed decisions.
   Orientation:

Teaching and learning are professionally oriented and focused on the application of computational mathematics to solve real-world problems.

Distinctive Features:

The programme focuses on knowledge development and application using a student-centred approach.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

Employability

After students graduate, they are employable in the following areas: Actuarial sciences; Financial Institutions: (banks, building societies, insurance companies and pension funds) as market or financial analysts, Brokers, Business Advisors, Entrepreneurs; Meteorology (forecasting); Network Managers, Manufacturing industry (production/operation management); Industrial Research; Research and Project Management in NGOs, research stations, and many other areas of applications; Mining Sector; Schools and Universities and many other relevant areas of the economy.

Further Studies

Master's and doctoral studies in applied computational mathematics or in related interdisciplinary programmes.

6. PROGRAM DELIVERY

Teaching and Learning Methods:

Lectures, tutorials, practical exercises, group work, mini project, Work-Related learning (WRL) / Industrial Attachment report, mini WRL research project and individual independent study.

Assessment Methods:

Written examinations, tests, practical exercises, presentations, WRL/ Industrial Attachment report, mini research projects and final year research project report.

7. LEARNING OUTCOMES

Upon successful completion of the programme a graduate will be able to:

- Demostrate understanding of fundamentals of computational mathematics.
- Use big data and analytics for informed decision making.
- Apply mathematical and computing knowledge and skills to develop models and solve real world problems in various disciplines such as medicine, agriculture, ecology, epidemiology, actuarial science.
- Apply mathematical and computer skills in finance, risk analysis, actuary sciences, weather forecasting and engineering.
- Communicate concepts and mathematical ideas effectively and with coherence, both with peers and non-mathematics audiences, making use of relevant technology
- Analyze and synthesize ideas using logical arguments and proven facts.

8. GENERAL PROVISIONS

Refer to Section 4 of the Faculty Regulations

9. PROGRAMME ASSESSMENT

Module Assessment Marks (MAM):

40%: Test (15%), Practical works (20%), Assignments (5%)

Written Examinations:
60%: For each module a student will write a three-hour examination at the end of the semester. Final mark for each module is F = 0.4MAM + 0.6E.

Other:

a) Research Project is assessed on the basis of the written research project (75%), oral presentation (15%) and student conduct during the research period (10%).

b) WRL/Industrial Attachment is assessed based on WRL/Industrial Attachment report (30%), assessment by a work supervisor (50%), and assessment by an Academic supervisor 20%).

10. PROVISION FOR PROGRESSION

Refer to Section 7 of the Faculty regulations.

11. FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Regulations.

12. GRADING AND DEGREE CLASSIFICATION

Refer to Section 5 of the General Regulations.

13. DEGREE WEIGHTING

Refer to Section 11 of the Faculty Regulations.

14. PROGRAMME STRUCTURE

The MBK modules are in bold type and they comprise 80% minimum body knowledge and the rest comprise the 20% additional modules.

Level 1 Semester 1

Code	Module Description	Credits
HMAT131*	Calculus 1	12
HMAT132*	Linear Algebra 1	12
HMTCS131*	Introduction to Machine Learning	12
HMTCS132	Web Database systems	12
CS131*	Communication Skills	12
HCSCI132	Principles of Programming Languages	12

Level 1 Semester 2

HMAT139*	Mathematical Discourse and Structures	12
HMAT140*	Mathematical Computing and Simulation	12
HMAT133	Linear Algebra II	12
HSTA139*	Probability and Statistics	12
HMTCS 133	Web Programming	12
	Annual Credits	**144**

Level 2 Semester 1

Code	*Module Description*	***Credits***
HMAT231*	Ordinary Differential Equations	12
HMTCS231	Information Security and Visualisations	12
HMTCS232	Advanced Programming	12
HMAT241*	Analysis 1	12
HMAT223*	Algebra	12
GS231*	Gender Studies	12

Level 2 Semester 2

HMAT248*	Numerical Methods	12
HMAT210	Calculus II	12
HMAT233*	Introduction to PDEs and Fourier Series	12
HMAT249*	Complex Variables	12
HMAT244*	Vector Calculus	12
TCNP231*	Technoprenuership	12
HMTCS270*	Mini Project	12
	Annual Credit	**156**

Level 3 Semester 1

Code	*Module Description*	***Credits***
HMTCS331*	Work-Related Learning/Industrial Attachment Report	36

Level 3 Semester 2

HMTCS332*	Academic Supervisor's Report	24
HMTCS333*	Employer's Assessment Report	60
	Annual Credits	**120**

Level 4 Semester 1

Code	*Module Description*	***Credits***
HMAT424*	Optimization Techniques	12
HMAT433*	Partial Differential Equations	12
HMTCS440*	Mathematical Finance	12

Level 4 Semester 2

HMTCS441*	Mathematical Computing for Life Sciences	12
HMAT442*	Real Analysis	12
HMTCS470*	Research Project	36

Elective Modules for Level IV

HMTCS442	Numerical Computations for Financial Modelling	12

HMTCS443	Coding and Cryptography	12
HMTCS444	Database Management and Administration	12
HMTCS446	Applications of Graph Theory	12
HMTCS447	Human Computer Interaction and Design	12
HMTCS410	Stochastic Processes for Applied Mathematics	12
HMAT436	Mathematical Programming	12
HMAT444	Mathematical Modelling	12
	Annual Credits	**120**

NB: Students must choose at least two elective modules from the given list.

15. MODULE SYNOPSES

HMAT131 Calculus 1

Real number system, intervals, inequalities and their solutions, absolute value, functions, Limits and continuity of single variable functions, Differentiation: rules of differentiation, differentiation from first principles, L'Hopital's Rule, Rolle's Theorem, Mean Value Theorem of Differential Calculus. The module also covers: Integration - indefinite and definite integrals, integration techniques; substitution method, integration by parts, tabular integration, trigonometric substitutions, reduction formulae, Mean Value Theorem of Integral Calculus.

HMAT132 Linear Algebra 1
Vectors: Basic properties and operations. Real and complex vectors in n dimensions Linear independence, bases for R^n and C^n,the scalar product of two vectors, vector product, geometrical representation of real vectors. Complex numbers: geometric representation, algebra. De Moivre's theorem polynomials and roots of polynomial equations. Matrices and determinants: algebra of matrices, inverses, definition and manipulation of determinants, solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating factors, linear equation with constant coefficients. Application: Use of practical big data in linear equations about business problems, tax problems, economic planner's models, and input-output matrix for economic producing transportation problems, and interpret big data analytic problems. Explore real data using eigenvalues, and eigenvectors to discover information from the big data. Applications in structural engineering, control theory problem, vibration analysis problem, electric circuits problem, and advanced dynamic problem and others.

HMTCS131 Introduction to Machine Learning
Introduction to fundamental concepts and algorithms that enable computers to learn from experience, with an emphasis on their practical application to real life problems from the point of view of modelling and prediction: modelling and algorithmic techniques for machines to learn concepts from data extracting meaningful patterns from random samples of large data sets. Introduction to Python programming. Introduction to supervised learning (decision trees, logistic regression, support vector machines, Bayesian methods, neural networks and deep learning), unsupervised learning (clustering, dimensionality reduction), and reinforcement. The module includes formulation of learning problems and concepts of representation, over-fitting, and generalization. These concepts are exercised in supervised learning and reinforcement learning, with applications to images and temporal sequences. Application of Machine Learning (ML) to recent technologies such as autonomous vehicles, search engines, genomics, automated medical diagnosis, image recognition and social network analysis among others. Tasks: classification, regression, clustering, mixture models, neural networks, deep learning, ensemble methods and reinforcement learning. Optimization-based learning: loss minimization. regularization. Statistical learning: maximum likelihood, Bayesian learning. Algorithms: nearest neighbour, (generalized) linear regression, mixtures of Gaussians, Gaussian processes, kernel methods, support vector machines, deep learning, sequence learning, ensemble techniques. Large scale learning: distributed learning and stream learning. Common paradigms: neural networks, kernel methods and support-vector machines, boosting, nearest neighbour classifiers Applications to data mining, human computer interaction. Additionally, the module will discuss evaluation methodology and recent applications of ML including large scale learning for big data and network analysis.

HMTCS132 Web Database Systems
An Introduction to Information Systems. Information gathering and requirements specification. An introduction to the main software design cycle phases: understanding the difference between requirements specification, analysis and design. An introduction to systems development using a structured methodology. Basic systems development with Data driven approach. The Unified Modelling Language (UML). ER modelling concepts, introduction to databases and SQL. Object-oriented approach to system development. Basic concepts of objects and OOAD. SQL and database software.

HMAT139 Mathematical Discourse and Structures

Logic, set theory, relations, functions, algebraic structures and graph theory. Sets: formulae, propositions, Boolean algebra and its applications. Logic, mathematical reasoning and proof: examples taken from various areas of mathematics. Relations: binary, n-ary, reflexive, symmetric, transitive, equivalence relations and classes, partitions, order relations, inverse relations. Functions: one-to-one, onto, inverse functions. Operations: binary, n-ary closed, associative, distributive. Structures: sets with one or two binary operations: permutations, symmetry groups, modular Arithmetic

HMAT140 Mathematical Computing and Simulation

Introduction to ideas of mathematical modelling: testing and validating a model. Review of essential concepts in statistics. Deriving and solving models with deterministic and probabilistic demand. Models with Poisson arrival, steady-state analysis, and cost analysis. Application to simple problems of Operational Research including business and physical systems (e.g. queueing, inventory control). Using software such as spreadsheets for calculation and data presentation. Introduction to computer programming for mathematical work: routines, functions, control structures for selection and repetition. Designing a computer program, testing and debugging. Mathematical logic and truth tables. Introduction to mathematical software: MATLAB, Python, Mathematica, Minitab and R. Any other suitable mathematical software will be included. Introduction to software capable of performing simulation. Case Study: Group work to use Operational Research techniques to model and solve realistic business and analyse the output in a professional manner.

HMAT133 Linear Algebra II

Vector spaces, linear dependence and independence, bases and dimension. Inner product spaces. Basic definitions with examples, the notion of norm and distance, the Cauchy-Schwarz inequality, the Gram-Schmidt orthogonalisation process, orthonormal basis. Linear transformations. Operations on linear operators, algebra of operators. Eigenvectors, eigenvalues, orthogonality of eigenvectors, geometric and algebraic multiplicity of eigenvalues. Applications of diagonalisation of matrices, quadratic and bilinear forms, Jordan Normal form of a matrix, solutions of systems of differential equations. The Cayley Hamilton Theorem and its applications. Revision of the basic techniques for solutions of first and second order differential equations.

HSTA139 Probability and Statistics

Counting rules used in Probability: summation notation, product notation, random experiments, and basic principles of counting. Sets and events: operations with sets and subsets, De Morgan's rules, sample spaces and events. Probability: the concept of probability, conditionally probability, law of total probability, properties of probability measure. Random variables: random variables and probability, discrete and continuous random variables. Special probability distributions: Expectation and moments: Chebyshev's

inequality. Generating functions: probability generating function and its properties, moment generating function and its properties, Normal approximation, Laws of large numbers. Joint probability distributions: discrete and continuous random vectors, independence, conditional distributions, covariance, and correlation. Bayesian inference with known priors, probability intervals, conjugate priors, Bayesian inference with unknown priors, Frequentist significance tests and confidence intervals, Resampling methods: bootstrapping, Linear regression. Application: Classifying data sets into categories that describe the shape of the data distribution. Graphical visualization ranging from simple graphics such as histograms, boxplots,and scatterplots to advanced graphics such as PCA projection plots, trellis plots, maps, etc. Students must use available software packages to explore big data using graphics to discover information from real data. Basic statistics, the definition of terms, data presentations, data descriptive measures of centre, position and dispersion

HMTCS133 Web Programming
Review of HTML/CSS and HTML Forms. Client / Server relationship – HTTP . Client-side scripting and CSS frameworks. Relational databases: creating, updating and normalisation in MySQL, Server-side PHP PDO programming. Website security including basic data encryption. Understanding of Content Management Systems. Accessibility and user-testing. Review of JavaScript. Introduction to Node JS and server-side JS Intro to Mongo DB and NoSQL ExpressJS, VueJS. Client / Server relationship - HTTP Accessibility and user-testing. Ethical, Legal and Social Issues.

HMAT111 Ordinary Differential Equations
Modelling with first order ODEs in population dynamics, and second order ODEs (mass-spring systems, RLC circuits). Methods of undetermined coefficients, reduction of order and method of variation of parameters. Existence and uniqueness of solutions, revision of continuous functions and Lipschitz conditions. Series solutions of ODEs, solutions near ordinary and singular points. Systems of linear first order ODEs, solution and stability. Differential equations of special functions. Laplace transforms and inverse Laplace transforms, applications to solving IVPs, Heaviside and Dirac functions.
Application: Use the practical big data for differential equations such as simple chemical conversion problems, growth of population problems, price of commodities models, Newton's law of cooling problems, and many physical problems.

HMTCS231 Information Security and Visualisation
Fundamental principles of information security and concepts. Security risk assessment and controls for different systems and organizations. Algorithms and procedures covering cryptography and its applications. Principles of secure design. Approaches to cyber security. Fundamental concepts in information visualisation. Good design practices in information visualisation. Different types of information visualisation and the options for using them. Interactive tools for information visualisation and dashboards. Visual analytics for identifying trends and patterns in datasets. Practical experience of data exploration.

HMTCS232 Advanced Programming
Programming with threads; Using reflection; Creating software components; Test-driven development (e.g. with JUnit); Version control and build tools (e.g. Git); Creating Generic classes; Design patterns; Building libraries. Lambda expressions

HMAT241 Analysis 1

Revision of elementary functions including hyperbolic functions and their inverses. Limits and continuity of functions. Techniques of differentiation. Higher order derivatives. Applications to curve sketching and extreme value problems. Taylor series. L'Hopital's rule. Methods of integration. The fundamental theorem of calculus. Applications of the integral. Complex exponentials. De Moivre's theorem. Historical development of the real number system: Countability, cardinal numbers, existence of transcendental numbers. The real numbers as a complete, ordered field. Supremum axiom, Archimedean property, principle of monotone bounded convergence, nested interval theorem, Bolzano-Weierstrass theorem, convergence of Cauchy sequences. Limits and continuity of real functions. Boundedness theorem, intermediate value theorem, interval theorem, application to fixed –point theorem, Uniform continuity. Differentiability, Local extrema, Rolle's theorem, mean – value theorem, L'Hospital's rule, Leibnitz's theorem. Taylor's theorem. Applications to finding roots, curve sketching, classification of local extrema, and approximation by polynomials. The Riemann integral, Integrability, properties of the Riemann integral, the mean-value theorem for integrals, the fundamental theorem of calculus.

HMAT223 Algebra

Sets and relations, mapping, binary operation including closure of sets. Equivalence relations. Identities, inverse and zero divisors. *Group theory (An introduction)*: Axioms and examples of groups. Subgroups. Homomorphisms and isomorphisms between groups. Automorphisms of a group. *Vector Spaces* (A more abstract and subspaces over arbitrary fields) Linear dependence and independence. Basic and Dimension. Linear conformation. *Rings and Fields:* Axioms and examples of rings and fields.

HMAT248 Numerical Methods

Introduction to simple numerical methods for solving problems in Mathematics and Science; Computer arithmetic and rounding errors; Root location in one- and higher-dimension (Newton-Raphson, Bisection), Numerical Linear Algebra (Gaussian elimination, LU decomposition, Gauss-Seidel method, Eigenvalue and Eigenvector iterative methods). Polynomial interpolation and splines; Solution of linear algebraic equations, scaled partial pivoting; Solutions to Ordinary Differential Equations including Initial Value Problems (Euler's method, Runge-Kutta methods) and Boundary Value Problems (tridiagonal system and Thomas algorithm for tridiagonal matrices), Solutions to Partial Differential Equations (finite-difference methods). Examples from applications (e.g. engineering, economics, internet search engines, bio-sciences).

HMAT210 Calculus II

Theorems on differentiation, higher order derivatives and Leibnitz's formula. The Mean value theorems: Rolles theorem, the Mean value theorem, the generalised mean value theorem, Taylor s theorem. Applications to maxima and minima, curve sketching, approximations and Newton's Method. Leibnitz's Theorem, Functions of several variables: limits, continuity. Differentiation of functions of several variables, Tailors theorem. Applications of maxima and minima problems, Lagrange multipliers. Multiple and triple integrals: change of order of integration transformations, Applications to finding area, volume, arc length, centroid, moments of inertia, etc. Series: tests of convergence, absolute and conditional convergence, series of functions, uniform convergence.

HMTCS231 Introduction to Partial Differential Equations and Fourier Series
Partial differential equations of mathematical physics and introduction to their study. Classification of second order partial differential equations in two independent variables. Derivation of the wave, Laplace and Poisson equations, method of separation of variables and Laplace transform techniques. Orthogonal sets of functions in an inner product space. Introduction to Hilbert spaces. Fourier sine and Fourier Cosine series. Discussion of convergence theorem. Integration and differentiation of Fourier series. Application of Fourier series to boundary value problems. Fourier series in two variables. The Fourier transform and its inverse. The convolution theorem. Applications. Bessel functions J(x). The zeros of J(X). Orthogonal sets of Bessel functions. Fourier Bessel series. Applications of the theory to the solution of partial differential equations will be stressed throughout.

HMAT224 Complex Variables
Analytic Functions. Cauchy-Riemann Equations. Conformal Mappings. Line integrals. Cauchy's Integral Theorem and Formula. Power series. Taylor series, Laurent series. Zeros and singularities. The Residue Theorem. Evaluation of real integrals.

HMAT214 Vector Calculus
Brief review of line: surface and volume integrals and applications to work done, flax through surfaces etc. Derivatives of Vector Functions. Directional Derivatives. Gradient of Scalar Fields. Divergence and Curl of Vector Fields. Constrained extremal problems. Line and surface integrals. Green's Theorem, Gauss' Divergence Theorem, Stokes' Theorem and their applications. Orthogonal Curvilinear Coordinates. The Laplacian in polar, cylindrical and spherical coordinates.

HMTCS270 Mini Research Project
The mini research project involves the specification, design and implementation of a piece of software or hardware or the use of existing software to develop some computational mathematical application. Work may be drawn from one of several proposed research themes and is guided by an academic supervisor. The mini research project aims to develop research, analytic and report writing skills.

HMAT424 Optimization Techniques
Formulations of combinatorial optimization problems, greedy algorithms, dynamic programming, branch-and-bound, cutting plane algorithms, decomposition techniques in integer programming, approximation algorithms. Nonlinear Optimization A course on the fundamentals of nonlinear optimization, including both the mathematical and the computational aspects. Necessary and sufficient optimality conditions for unconstrained and constrained problems. Convexity and its applications. Computational techniques and their analysis. Deterministic OR Models**:** An applications-oriented approach illustrating how various mathematical models and methods of optimization can be used to solve problems arising in business, industry, and science. Computational optimization methodologies underlying portfolio problems in finance. Computational linear algebra, determining derivatives, quadratic, and nonlinear optimization. The efficient frontier problem. Applications: Use and interpret practical big data in business/finance optimization, economic optimization and other economic producing problems, and interpret big data analytic problems.

HMAT433 Partial Differential Equations
Second Order PDE's and Method of Separation of Variables derivation of the wave and diffusion equations, solutions of second order equations, initial and boundary conditions, solution of the wave equation, Laplace's equation, wave equation and the heat flow equation. Parabolic Partial Differential Equations- equation for time-dependent heat flow, explicit method, Crank-Nicolson method, the Theta method, parabolic equations in Two or Three dimensions. Hyperbolic Partial Differential Equations, wave equation. Introduction to solution of PDEs using finite differences and finite elements. Solution procedures: elliptic equations -- Green functions, conformal mapping; hyperbolic equations -- generalized d'Alembert solution, spherical means, method of descent; transform methods Fourier, multiple Fourier, Laplace, Hankel (for all three types of partial differential equations); Duhamel's method for inhomogeneous hyperbolic and parabolic equations.

HMTCS440 Mathematical Finance
Theory of interest rates: simple interest, compound interest, nominal rates of interest, accumulation factors, force of interest, present values, Stoodley's formula for the force of interest, present values of cash flows. Basic compound interest functions: the equation of value and yield of a transaction, annuities certain, present values and accumulations, deferred annuities, continuously payable annuities, the general loan schedule, the loan schedule for a level annuity. Nominal rates of interest: annuities payable $pthly$, annuities payable $pthly$, present values and accumulations, annuities payable at intervals of time r, where r>1, the loan schedule for a p$thly$ annuity. Discounted cash flow: Net cash flows, Net present values and yields, comparison of two investment projects, different interest rates for lending and borrowing, effects of inflation, the yield of a fund, measurement of investment performance. Capital redemption policies: Introduction to premium calculations, policy values, policy values when premiums are payable p$thly$, surrender values, paid up policy values and policy alterations, variations in interest rates, Stoodley's logistic model for the force of interest, reinvestment rates. Application: use of practical big data in the financial sector.

HMTCS441 Mathematics Computing for Life Sciences
Population modelling - predator-prey models, dynamic modelling, simulation, analysis of behaviours. Lotka-Volterra equations and other population models. Modelling the spread of disease. The mathematics of DNA. The topology of viruses. Evolutionary concepts: kin selection, group selection. Mathematics of evolution: game theory, Hawk-dove models, evolutionary steady states. The Prisoner's Dilemma as a model for the evolution of altruism. Tit for tat and other strategies for the Iterated Prisoner's Dilemma. The Tragedy of the Commons. Modelling co-operative behaviour. Cycles of behaviour patterns. Punishment and reward. Trust and reputation. "Game Theory can save the world" (Nowak and Highfield). Climate change as a case study.

HMAT442 Real Analysis
Sets. Algebra of sets, Borel sets. Topology of real line. Outer measure on R, its properties. Measurable sets and Lebesgue measure. Properties of Lebesgue measure. The sigma algebra of measurable sets. The sigma algebra of Borel sets. Example of a non-measureable set. Measurable functions, equivalent definitions, properties. Convergence almost everywhere, convergence in measure. Lebesgue integral. Properties. Classes of integrable functions. Comparison with Riemann integral. Bounded convergence theorem. Lebesgue theorem. Fatou's lemma. Monotone convergence theorem. Lebesgue convergence theorem. Absolutely continuous functions. Product measures. The Fubini theorem. Completion of product measures. Distribution functions. Integrating on product spaces. Introduction to linear vector spaces. Normed spaces. spaces, L∞space. The Holder and Minkowski

inequalities. Convergence in LP. Comparison with other kinds of convergence. Measure spaces.

HMTCS442 Numerical Computations for Financial Modelling
Introduction to the mathematical models used in finance and economics with emphasis on pricing derivative instruments. Financial markets and instruments; elements from basic probability theory; interest rates and present value analysis; normal distribution of stock returns; option pricing; arbitrage pricing theory; the multi-period binomial model; the Black-Scholes option pricing formula; proof of the Black-Scholes option pricing formula and applications; trading and hedging of options; Delta hedging; utility functions and portfolio theory; elementary stochastic calculus; Ito's Lemma; the Black-Scholes equation and its conversion to the heat equation.

HMTCS443 Coding and Cryptography
Introduction to coding theory. Parity checking, error detection and correction, code bounds. Examples of simple codes such as Hamming codes and Hadamard codes. Linear codes, polynomial codes and cyclic codes, including historical and modern examples. Modern applications such as Reed-Solomon codes. Introduction to cryptography. Historical ciphers, including substitution ciphers, stream ciphers (e.g. linear shift register) and block ciphers (e.g. DES/AES). Public-key cryptography, including the mathematical theory behind the ciphers (e.g. RSA, elliptic curves). Privacy, authentication, integrity and non-repudiation. Digital signatures and PKI infrastructures. Future developments, including quantum computing and quantum cryptography.

HMTCS444 Database Management and Administration
Exploring the database architecture; Creating the database; Managing the database instance; Managing database storage structures; Managing transactional processing and locking mechanism; Database security; Monitoring the database and using the advisors; Backup and recovery concepts; investigating, reporting, and resolving
problems.

HMTCS446 Applications of Graph Theory
Introduction to graphs and digraphs, walks, paths, trails and cycles. Definitions and characterization of classes of special graphs. Distance and connectedness measures. Eulerian and Hamiltonian graphs. Connectivity - Whitney's theorem and Menger's theorem. Planarity - Kuratowski's theorem and Euler's theorem. Colourability - 5-colour theorem, 4-colour theorem. Algorithms - Greedy algorithm, shortest path algorithms, Fleury's algorithm. NP completeness. Applications in computing, engineering, finance, mathematical biology, social networks.

HMTCS447 Human Computer Interaction and Design
Design Process, Prototyping, Prototyping Tools, User Research, Research Methods, Evaluation of HCI solutions, Usability testing, Cognitive Psychology, Information Processing, HCI methodologies, standards and guidelines, Multi-Modal Interactions, Multi-Sensory Interactions, Tangible User Interfaces, Brain Computer Interfaces, Natural User Interfaces, Dialog Systems, Metaphors, Conceptual Models, Relevant legal, social, ethical and professional issues.

HMTCS410 Stochastic Processes for Applied Mathematics
Random variables, expectations, conditional probabilities, conditional expectations, convergence of a sequence of random variables, limit theorems, minimum mean square error estimation, the orthogonality principle, random process, discrete-time and continuous-time Markov chains and applications, forward and backward equation, invariant distribution, Gaussian process and Brownian motion, expectation maximization algorithm, linear discrete stochastic equations, linear innovation sequences, Kalman filter, various applications.

HMAT436 Mathematical Programming
Introduction to mathematical programming problems: Linear programming problem formulation; simplex method; Chmens and two phase techniques; sensitivity analysis; duality in LP; Dual simplex method; transportation and assignment methods; integer programming; dynamic programming; quadratics and separable programming; K T conditions for optimality.

HMAT444 Mathematical Modelling
This module introduces mathematical modelling in a variety of contexts including using Newton's laws of motion, Newton's law of gravitation, population models, exponential growth, density dependent growth, and predator-prey models. Deriving differential equations from big data; dimensional analysis; discrete models and difference equations: steady states and their stability; continuous models and ordinary differential equations: steady states and their stability; the slope fields and phase lines; applications of Linear Algebra (in lower dimensions): systems of linear ordinary differential equations; linear phase plane analysis and stability; electrical networks; vector algebra, vector geometry, vector equations, coordinate systems and vector differentiation; application in mechanics: Newton's laws for a single particle in 3-D; conserved quantities; angular velocity, angular momentum, moment of a force; harmonic motion.

HCSCI132 Principles of Programming Languages

Refer to Department of Computer Science and Computer Engineering.

CS131 Communications Skills

Refer to Communication Skills Centre.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

TCNP231 Technoprenuership

Refer to Entrepreneurship Department.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN ACTUARIAL SCIENCES (HACS)

Duration:	5 years
Minimum Credit Load:	600
Maximum Credit Load:	660
Maximum MBKS Credit Load:	528
ZNQF Level	8

1. INTRODUCTION

These regulations shall be read in conjunction with the General Regulatic Undergraduate Programmes and the Faculty of Science and Technology Regulatio

2. PURPOSE OF THE PROGRAMME

The purpose of the programme is to equip students

- with actuarial and economic concepts, principles and processes in the areas of Scier Engineering, Industry and Commerce.
- with a background for post- graduate studies in actuarial and related areas.
- with skills to design and implement computer programming tasks.
- with a sound knowledge, of actuarial and of business studies which they can apply a broad spectrum of areas.
- for teaching secondary school mathematics and actuarial courses at tertiary instituti
- for conducting research at whatever level.

3. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ Nati Foundation Certificates including English Language, Mathematics and a Science subje grade C or better.

Normal Entry: At least 5 Ordinary level subjects/ National Foundation Cert inclusive of Mathematics and English Language AND they shoul passed "A" Level Mathematics and any other science or commercial or their recognized equivalents.

Special Entry: National Certificate, National Diploma or Higher National Dipl relevant fields**.**

Mature Entry: Refer to Section 3.3 of the General Regulations.

4. PROGRAMME CHARACTERISTICS:

Areas of Study: Actuarial science, mathematics, statistics, economics, operational research

Specialist Focus:
Life Insurance, Health Insurance, Pensions, Investment, General Insurance and Enterprise

Risk Management.

Distinctive Features: The programme is tailored to provide students with technical and analytical *actuarial* skills.

Orientation:
Teaching and learning are professionally oriented and focused on application of actuarial skills to solve real world problems.

5. CAREER OPPORTUNITIES AND FURTHER EDUCATION

5.1 Career Opportunities

- **Insurance:** investigating, analysing and explaining a wide range of numerical information to create and price polices, or to ensure they have sufficient funds to cover claims.
- **Pensions:** designing and advising on company pension schemes.
- **Investments:** involved in research and on the pricing and management of investments.
- **Banking:** calculating and quantifying an array of risks faced by these institutions such as the inability of some borrowers to repay their debt, or the risk of a fall in financial markets.
- **Healthcare:** investigating, analysing and explaining a wide range of health and medical information to price contributions to medical schemes or assess the impact to the industry of changes in regulation.
- **Risk management:** assisting businesses to better understand the multiple risks they face in a holistic and comprehensive manner, as well as provide assistance and guidance in terms of how to understand their impact and how they can best be managed.
- **Consultancy:** Actuaries in administrative positions have to explain technical matters to executives, government officials, shareholders, policyholders.

5.2 Further Education

Graduates could continue to an MSc in Applied Actuarial Science, an MSc in Mathematics and its applications or an MSc in Statistics.

6. PROGRAM DELIVERY

Teaching and Learning Methods: Lectures, tutorials, practical exercises, group work related learning (WRL) / Industrial Attachment report, mini WRL research project and ind independent study.

7. LEARNING OUTCOMES

Upon successful completion of the programme, a graduate will be able to:

- Apply Statistics and Mathematics to solve real life problems encountered in Actuarial Science.
- Use several of the technical tools, computer languages or software packages used by actuaries.
- Demonstrate mastery of data analysis and statistical concepts of Actuarial Science.
- Critically analyze statistics and mathematics related to Actuarial Science in journal articles.
- Write technical reports and make technical presentations containing statistical and actuarial results.

8. ASSESSMENT:

Assignments and tests will be given for each module that a student takes. The continuous assessment mark C = 0.25 (assignment average) + 0.75 (test average).

For each module a student will write a three hour examination at the end of the semester and the final mark for each module is F = 0.25C + 0.75E, where E is the examination mark.

Other:

a) Research Project is assessed on the basis of the written research project (75%), oral presentation (15%) and student conduct in the laboratory (10%).

b) WRL/Industrial Attachment is assessed based on WRL/Industrial Attachment report (30%), assessment by work supervisor (50%), and assessment by Academic supervisor 20%).

9. FAILURE TO SATISFY EXAMINERS

Refer to section 9 of the General Regulations.

10. PROVISION FOR PROGRESSION

Refer to Section 7 of the Faculty regulations.

11. WORK-RELATED LEARNING GENERAL GUIDELINES

Refer to Section 10.2 of the General Regulations.

12. GRADING AND DEGREE CLASSIFICATION

Refer to Section 5 of the General Regulations.

13. DEGREE WEIGHTING

Refer to Section 11 of the Faculty Regulations.

12. PROGRAMME STRUCTURE

N.B Modules in bold are MBK modules

Level 1 Semester 1

Module Description	Credits
HACS131 * Introduction to Actuarial Methods and Applications	12
HMAT131* Calculus 1	12
HSTA133* Probability 1	12
HSTA132* Introduction to Statistics	12
HACS132*Introduction to Information Technology	12
CS131*Communication Skills	12

Level 1 Semester 2

Module Description	Credits
HMAT132*Linear Mathematics 1	12
HSTA* Statistical Computing 1	12
HSTA107*Statistical Inference 1	12
HACS134* Micro Economics	12
HSTA105* Regression and ANOVA I	12
HCSCI132 Principles of Programming Languages	12
Total number of credits	*144*

Level 2 Semester 1

Module Description	Credits
HACS231* Actuarial Mathematics I	12
HACS232* Theory of Interest	12
HMAT241*Analysis	12
HSTA237*Time Series Analysis	12
HACS233*Fundamentals of Actuarial Mathematics 1	12
GS231*Gender Studies	12

Level 2 Semester 2

Module Description	Credits
HACS234* Fundamentals of Actuarial Mathematics 1I	12
HACS235*Corporate Finance 1	12
HMAT233*Partial Differential Equations and Fourier Series	12
HMAT424*Optimization Techniques	12
HCSCI Models of Database and Database Design	12
TCNP201*Technoprenuership	12
Total number of credits	*144*

Level 3 Semester 1	Credits
HSTA410*Stochastic Processes	12
HACS331* Actuarial Statistics I	12
HACS332*Risk Theory	12
HSTA210*Operations Research I	12
HACS333*Corporate Finance II	12

Level 3 Semester 2	Credits
HACS334* Financial Economics	12
HSTA408* Econometrics	12
HACS335* Actuarial Mathematics I	12
HACS336* Methods of Actuarial Investigations	12
HACSS337* Computational Finance	12
Total number of credits	*120*

Level 4 Semester 1 and 2

Module Description	Credits
HACS431* Work-Related Learning I	40

HACS432*	Work-Related Learning II	80
Total number of credits		*120*

Level 5 Semester 1

Module Description		Credits
HSTA412*	Survival Models	12
HACS531*	Principles of Financial Management	12
HACS532*	Health Indicators	12
HACS533*	Life Contingencies	12
HACS534*	Actuarial Mathematics II	12

Level 5 Semester 2

Module Description		Credits
HACS535*	Pensions and Benefits Insurance	12
HACS536*	Investment and Asset Liability Management	12
HACS537*	Financial Engineering	12
HACS538	Actuarial Statistics II	12
HACS570*	Dissertation	36
Total number of credits		*84*

13 MODULE SYNOPSES

HACS131: Introduction to Actuarial Applications

Elementary Mathematics, Statistics and multistate models. Principles of Mathematics of Finance, life contingencies, risk assessment and management; practice of investments, life insurance, general insurance and retirement provision; and current topics. The course culminates by addressing questions concerning professionalism and what it is to be an actuary.

HMAT131: Calculus 1

Limit of functions. One-sided and infinite limits. Continuity. Differentiation. Definition, basic properties. Rolle's Theorem, mean value theorem, Cauchy's mean value theorem. Leibniz' rule. Applications. Taylor series. Integration, Definite

integrals. Antiderivatives. Fundamental theorem of calculus. Improper integrals. Gamma and Beta functions. Definition of natural logarithm as integral of 1/x and exponential as inverse. Area, volume of revolution, arc length, surface area. Parametric equations. Arc length, surface area. Polar coordinates. Graph sketching. Area in polar coordinates. Complex numbers. Algebra of complex numbers. De Moivre's theorem, Exponential form.

HMAT132: Linear Mathematics I
Complex numbers: geometric representation, algebra. De Moivres theorem polynomials and roots of polynomial equations. Matrices and determinants: algebra of matrices, inverses, definition and manipulation of determinants, solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating factors, linear equation with constant coefficients.

HSTA132: Introduction to Statistics
Graphical techniques. Kinds of measures of central tendency. Measures of variability. Empirical distributions. Moments. Skewness and Kurtosis. Applications. Indicators. Contingency Tables. Introduction to Time Series trends. Sampling. Introduction to estimation procedures: Judgemental method and Method of moments. Introduction to Hypothesis testing. Ideas about non-parametric statistics. Chi-square contingency methods, Goodness of fit, Q-Q plots, using applications in agricultural and health statistics.

HSTA133: Probability Theory I
Axiomatic probability, sets and events, sample space, conditional probability, Independence, laws discrete and continuous random variables, probability density functions, mean, variance, expectation. Independence, Chebyshev's inequality, moments and moment generating functions. Common Discrete Distributions, Uniform, Bernoulli and Binomial, multinomial, hypergeometric, Poisson, Geometric and negative binomial. Use of tables. Common Continuous Distributions: Uniform, Normal, Exponential, gamma, beta. Joint Probability Distributions. Conditional and marginal distribution, expectation, covariance and correlation. Approximations, Law of large numbers, Central limit Theorem, Normal approximation to Binomial, Poisson, etc.

HSTA104: Statistical Computing
Introduction to use of statistical software. Training on database software including Microsoft Excel, Microsoft Access, SPSS, Epi-Data and Epi-Info. Use of these software for data entry, data management and analysis. Interpretation of analysis results and statistical report writing.

HSTA107: Statistical Inference 1
Deductive inference, population and sample concepts as the basis of statistical inference, parameters and statistics, review of probability theory. Central Limit Theorem, Chi-square, student-t and F distributions, distribution of min and max. Estimation: methods of estimation, properties of estimators and their sampling distributions. Interval estimation. and Hypothesis testing.

HACS133: Economic Principles
Economics as a science, the scope of economics. Introduction to microeconomics. Demand and supply analysis, effects of controls on prices and supply; classicity of demand and supply, production factors, cost analysis. Utility theory and consumer behaviour. Analysis of insurance problems in terms of utility. Market forms and income distribution, general equilibrium theory. The theory of firms. Introduction to Macro Economics and the role of government in Economics, public sector, finance and taxation. National Income measures; the circular flow of income; the multiplier and accelerator; aggregate demand and supply. Government fiscal policy and its effects. Government monetary policy and its effects. The money supply and credit creation by banking systems. The major factors affecting unemployment, inflation, and economic growth. Monetarist and Keynesian approaches. International trade, exchange rates and the balance of payments.

HMAT241: Analysis
Revision of elementary functions including hyperbolic functions and their inverses. Limits and continuity of functions. Techniques of differentiation. Higher order derivatives. Applications to curve sketching and extreme value problems. Taylor series. L'Hopital's rule. Methods of integration. The fundamental theorem of calculus. Applications of the integral. Complex exponentials. De Moivre's theorem. Historical development of the real number system: Countability, cardinal numbers, existence of transcendental numbers. The real numbers as a complete, ordered field. Supremum axiom, Archimedean property, principle of monotone bounded convergence, nested interval theorem, Bolzano-Weierstrass theorem, convergence of Cauchy sequences. Limits and continuity of real functions. Boundedness theorem, intermediate value theorem, interval theorem, application to fixed–point theorem, Uniform continuity. Differentiability, Local extrema, Rolle's theorem, mean – value theorem, L'Hospital's rule, Leibnitz's theorem. Taylor's theorem. Applications to finding roots, curve sketching, classification of local extrema, and approximation by polynomials. The Riemann integral, Integrability, properties of the Riemann integral, the mean-value theorem for integrals, the fundamental theorem of calculus.

HSTAT105: Regression and Analysis of Variance 1
Correlation and regression, scatterplots, correlation matrix. Method of least squares, associated lines, assumptions underlying regression. Checking validity of assumptions. Residuals and transformations. Outliers. Pearson's and Spearman's correlation coefficients, predictions. Regression in terms of sums of squares and sums of products. Estimation and testing, t and F-tests. Multiple linear regression: linear equations and matrices. Matrices in simple and multiple linear regression. Testing and inference in multiple linear regression using matrices. Partial correlation. Analysis of variance (ANOVA). Assumptions underlying ANOVA. One-way, balanced design ANOVA.

HACS231: Actuarial Mathematics I
Annuities: Level annuities-continuously payable annuities, accumulations and annuities payable pthly, deferred and increasing annuities(including compound and varying annuities).Definition and application of an equation of value .Loan payments and repayment of regular instalments of interest and capital:
Discounted cash flow techniques in investment project appraisal:
Investment and risk characteristics of the following types available for investment

purposes: Fixed interest government borrowings, Fixed-interest borrowing other bodies, Shares and other equity-type finance, Derivatives

Analyse elementary compound interest problems:

PV of payments from a fixed interest security where the coupon rate is constant and security is redeemed in one instalment; Calculate upper and lower bounds for the present value of fixed interest security that is redeemable on a single date within a given range at the option of the borrower; running yield and redemption yield from a fixed interest security, given price; PV of real rate of interest implied by equation in the presence of specified inflationary growth; PV or real yield from an index-linked bond, given assumptions about the rate of inflation; Calculate the delivery price and the value of a forward contact using arbitrage free pricing methods.

Arbitrage:

Show an understanding of term structure of interest rates;

Par yield and yield to maturity; discrete spot rates and forward rates; continuous spot rates and forward rates; duration and convexity of a cash flow sequence, and illustration of how these may be used to estimate the sensitivity of the value of the cash flow sequence to a shift in interest rates; Redington (immunisation.)

Show an understanding of simple stochastic models for investment returns;

The concept of a stochastic interest rate model and the fundamental distinction between this deterministic model; annual rates of returns which are iid and for simple models, expressions for the mean value and the variance of accumulated amount of a single premium; recursive relationship which permit the evaluation of the mean value and the variance of the accumulated amount of an annual premium; r.v (1+i) with

Log-normal distribution amount of a single premium and from the present value of a sum due at a given specified future time; applying the above results to the calculation of the probability that a simple sequence of payments will accumulate to a given amount at a specified future time.

HACS232: Theory of Interest

Measurement of simple and compound interest; accumulated and present value functions, annuities, yield rates, amortization. Schedules and sinking funds; bonds, securities and related funds; Application to mortgages and bonds.

HACS233: Fundamentals of Actuarial Mathematics 1

Various rates of simple and compound interest, present and accumulated values based on these, annuity functions and their application to amortization, sinking funds and bond values; depreciation methods; introduction to life tables, life annuity and life insurance values.

HSTA240: Operations Research I

Survey of continuous optimization problems; unconstrained optimization problems and methods of solution; introduction to constrained optimization. Linear Programming: formulation of LP problems, graphical solution of simple LPs; the simplex algorithm, duality and economic interpretation; post optimality/sensitivity analysis. Decision analysis: decisions under risk, decision trees, decisions under uncertainty. Markov decision processes and dynamic programming. Project scheduling; probability and cost considerations in project scheduling; project control, critical path analysis. Integer programming. Queuing models: types of queues, queues with combined arrivals and departures; queues with priorities of service.

Stochastic simulation: role of random numbers; simulation experiments; Monte Carlo calculus.

HACS234: Fundamentals of Actuarial Mathematics II

The single decrement model and calculations based on it; the stationary population model; present values and accumulations of stream of payments based on a single decrement model; equation of value for payments based on a single decrement model; annuity and assurance

Commutation functions and their relationships; assurance and annuity contracts; product pricing, reserving, surrender values, emergence of profit.

HACS235: Corporate Finance I

The aim of the course is to identify the objective that corporate finance managers pursue or ought to pursue in order to satisfy the needs of corporate stakeholders and to develop, in students, concepts and corporate analytical tools that will enable them to meet this objective. To this end, the course will cover the following critical areas: Goals of a firm and the agency theory; Time value concepts and valuation of bonds and shares; Capital Budgeting under certainty; Operating and financial leverage; Introduction to portfolio theory and capital asset pricing; the stock market and other sources of long-term capital; innovations in corporate finance.

HSTA237: Time Series Analysis

Time series models, estimation and elimination of trend and seasonal variation. Tests of randomness and normality. Introduction to projects. Model building strategy. Variance and covariance of linear combinations. Time series as a stochastic process, stationary stochastic process, white noise, and random walk. Variance of sample mean estimation of trends and seasonal variation. Sample ACP. Variance of sample autocorrelation and corresponding significance test for zero autocorrelation. General linear process. Auto-covariance generating function. Moving average process. Invertibility. Autoregressive processes. Yule-Walker equations. Solution of difference equations. AR(1) and AR(2) processes: stationarity conditions. ARMA (1, 1) process. General ARMA(p, q) process. ARIMA models for non-stationary processes: IMA(1,1), AR(1,1) IMA(2,2) models. Log transformation to stationarity. Identification of ARIMA models. Var (Z) for stationary processes. Partial ACF and applications to AR(1), AR(2) and MA(1) models. Parameter estimation: method of moments, least squares, and maximum likelihood. Properties of parameter estimators. Goodness of fit: Box Pierce statistics, overfitting, autocorrelation of residuals. Forecasting: minimum mean measure square error forecast; forecast errors. Applications to ARMA and ARIMA processes. MA(1)12, AR(1) 12, ARMA(1,1) 12 models. Multiplicative seasonal ARMA(p, q) x (P,Q)s model. Introduction to the frequency domain. Periodogram and spectral analysis.

HSTA410: Stochastic Processes

To include integer-valued variables: probability generating functions, convolutions. Markov chains: transition probabilities, classifications of states, stationary distributions, transient states. Gambler ruin, random walk. Markov processes: Chapman-Kolmogorov equations, transition rate matrix, forward and backward systems. Poisson process, normal equations, machine operation machinery breakdown, queuing model.

HACS331: Actuarial Statistics I

Sampling and statistical inference; t-distribution and F-distribution. Point estimation, method of moments, maximum likelihood method, least squares method, properties of estimators, score function. Unbiased estimators, minimum variance unbiased estimators, consistent estimators, asymptotic normality of estimators, confidence intervals; exact and asymptotic confidence intervals. Hypothesis testing; test statistics, critical regions, type I and type II errors, level of significance power test, Neyman- Pearson Lemma, Likelihood- Ratio tests. Analysis of variance, Fisher's transform of r, correlation analysis, residual analysis, inferences, Properties of MLE's, consistency of estimators, efficiency and relative efficiency. Correlation and regression; correlation analysis, linear and multiple linear regression model. Analysis of variance; one-way ANOVA, treatment means-analysis and confidence intervals.

HACS332: Risk Theory

Classical approaches to risk include the insurance principle and the risk-reward trade-off. Risk models, review of probability, Bachelier and Lundburg models of investment and loss aggregation. The fallacy of time diversification and its generalization. Loss distributions, geometric and Brownian motion and the compound Poisson process. Modelling of individual losses which arise in a loss aggregation process. Distributions for modelling size loss; statistical techniques for fitting data. Credibility theory. Economic rationale for insurance, problems of adverse selection and moral hazard. Utility theory; ruin theory. Capital asset pricing model, Black-Scholes option pricing model. Application of risk theory.

HSTA412: Survival Models

Survival models and the life table; Describe the future lifetime as a random variable Define probabilities of death and survival, Define the actuarial functions tpx, tqx, n/mqx, Define the complete and curtate expectations of future lifetime, Describe the life table functions lx and dx, Describe the simple laws of mortality, Define simple assurance and annuity contracts and develop formulae for means and variances.

Estimating the lifetime distribution Fx(t); Describe how lifetime data might be censored, Describe the estimation of empirical survival function, Describe the Kaplan- Meier estimate of the survival function in the presence of censoring, Describe the Nelson-Aalen estimate of the cumulative hazard rate in the presence of censoring, compute it from typical data and estimate its variance.

T**he Cox regression model;** Describe the Cox model for proportional hazards and derive the partial likelihood estimate

The two state Markov model; Describe the two state model of a single decrement and compare the assumptions with those of the random lifetime, Derive the MLE for the transition intensities in models of transfers between states with piecewise constant transition intensities, Define waiting time in a state.

The general Markov model; Describe the statistical models of transfers between multiple states, State the assumptions underlying the Markov model of transfers between a finite number of states in continuous time,

Binomial and Poisson Models of Mortality; Describe the Binomial model of mortality, derive a maximum likelihood estimator for the probability of death and compare the Binomial model with the multiple state models

Graduation and statistical tests; Describe how to test crude estimates for consistency with standard table or a set of graduated estimates, and describe the process.

Methods of graduation; Describe the process of graduation by the three common methods and state the advantages and disadvantages of each.
Exposed to risk; Define initial and central exposed to risk, and the various common rate intervals, Calculate the central exposed to risk in simple cases. State the principle of correspondence.

HACS333: Corporate Finance II
The aim of the course is to develop further, in students, concepts and corporate financial analytical tools. The areas covered will include the following: Introduction to capital structure theory and practice; Cost of capital and valuation; Introduction to capital budgeting under uncertainty; Dividend policy theory and practice; corporate working capital management; and innovations in corporate finance.

HACS334: Financial Economics
Utility theory - the theory and application; comparability, transitivity, Independence, certainty equivalence, common utility functions, attitude to risk.
Stochastic dominance and Absolute dominance; first order, second order and third order stochastic dominance in portfolio selection
Investment measures; mean, variance, short fall probability, mean shortfall, mean-variance portfolio theory; method of Lagrangian multiplier, Gaussian eliminator, the two fund theorem, Kuhn tucker approach, corner portfolios,
Models of asset returns; Markowitz efficient frontier, single index model, multi index model, orthogonal indices
Asset-liability modelling;
Equilibrium models (CAPM & APT); tangent portfolio/ market portfolio, opportunity set Efficient frontier with or without short sells, separation theorem, indifference curves, security, market line theorem, arbitrage pricing theory;
Efficient market hypothesis (EMH)
Strong for EMH, Semi-Strong EMH, weak for EMH, Technical analysis, Fundamental analysis

HACS335: Actuarial Mathematics I
Life assurance contracts; Define simple assurance contracts and develop formulae for the means and variances of the present values of the payments under these contracts, assuming constant deterministic interest.
Life annuity contracts; Define simple annuity contracts and develop formulae for the means and variances of the present values of the payments under these contracts, assuming constant deterministic interest.
The life table; Describe practical methods of evaluating expected values and variances of the simple assurance and annuity contracts. Describe the life table functions lx and dx and their select equivalents l[x]+r and d[x]+r and expressing the following life table probabilities in terms of the following functions: n px , nqx , n|mqx and their select equivalents n p[x]+r , nq[x]+r , n|mq[x]+r
Evaluation of assurances and annuities; Define simple assurance and annuity contracts, develop formulae for the means and variances of the present values of the payments under these contracts, assuming constant deterministic interest, Annuities paid in arrears, in advance and continuously, Assurance benefits paid at the end of the year and immediately, Use of commutation functions.
Net premiums and provisions; Describe and calculate, using ultimate or select mortality, net premiums and net premium reserves of simple insurance contracts,

Define the net random future loss under an insurance contract and state the principle of equivalence and reasons for setting up provisions, Define and calculate net premiums for the insurance contract benefits in cases where premiums and annuities may be payable annually, more frequently than annually, continuously and benefits may be payable at the end of the year of death, immediately on death, annually, more frequently than annually, or continuously, Describe
prospective and retrospective policy values, Define and evaluate prospective and retrospective net premium policy values, Derive Thiele's differential equation satisfied by net premium policy values for contracts with death benefits paid immediately on death,
Variable benefits and with-profit policies; Describe the calculation, using ultimate or select mortality, of net premiums and net premium reserves for increasing and decreasing benefits and annuities, Calculate the expected present value of an annuity, premium, or benefit payable on death, which increases or decreases by a constant
compound rate, Calculate net premiums and net premium reserves for contracts with variable premiums and benefits, Define with profits contracts including the types of bonuses that may be given to with profit contracts, Calculate net premiums and net premium reserves for with profits contracts.
Gross premiums and provisions for fixed and variable benefit contracts; Describe the calculation of gross premiums and reserves of assurance and annuity contracts, Define the gross future loss random variable for the benefits and annuities, Calculate gross premiums using the future loss random variable and the equivalence
principle including other criteria in cases where premiums and annuities may be payable annually, more frequently than annually, or continuously. Benefits may be payable at the end of the year of death, immediately on death, annually, more frequently than annually, or continuously, Prove that, under the appropriate conditions, the prospective reserve is equal to the retrospective reserve, with or without allowance for expenses, for all standard fixed benefit and increasing/decreasing benefit contracts, Derive a recursive relation between successive annual reserves for an annual premium contract, with allowance for expenses, for standard fixed benefit contracts.

HACS336: Methods of Actuarial Investigations

Capital redemption policies; determination of the rate of interest in a given transaction. Valuation of securities; effect of income and capital gains taxes. Cumulative sinking funds; yield curves; discounted mean terms; matching and immunization; consumer credit. Introduction to stochastic interest models. Decremental rates and other indices; analysis of data and derivation of exposed to risk formulae; calculation of mortality, sickness and other decremental rates, including multiple decremental rates; graduation methods and their applications; tests of graduation; features of principal tables in common use; national vital statistics and population projection; applications outside insurance, national social security and pension schemes.

HACS337: Computational Finance

Computational methods in finance and financial modelling; interest rate models and interest rate derivatives; derivative securities; Black-Scholes theory; no-arbitrage and complete markets theory; Hull and White models; Heath-Jarrow-Morton models;

Hedging and immunization; the stochastic differential equations and martingale approach, multinomial tree and Monte Carlo methods, the partial differential equations approach, finite difference methods.

HMAT424: Optimisation Techniques

Formulations of combinatorial optimization problems, greedy algorithms, dynamic programming, branch-and-bound, cutting plane algorithms, decomposition techniques in integer programming, approximation algorithms. Nonlinear Optimization A course on the fundamentals of nonlinear optimization, including both the mathematical and the computational aspects. Necessary and sufficient optimality conditions for unconstrained and constrained problems. Convexity and its applications. Computational techniques and their analysis. Deterministic OR Models**:** An applications-oriented approach illustrating how various mathematical models and methods of optimization can be used to solve problems arising in business, industry, and science. Computational optimization methodologies underlying portfolio problems in finance. Computational linear algebra, determining derivatives, quadratic, and nonlinear optimization. The efficient frontier problem. Applications: Use and interpret practical big data in business/finance optimization, economic optimization and other economic producing problems, and interpret big data analytic problems**.**

HSTA408: Econometrics

Role of Econometrics in Zimbabwe. Review of general linear model. Linear restrictions, Generalised least squares, GLS estimator, heteroscedasticity, pure and mixed estimation, group observations and grouping of equations. Autocorrelation. Heteroscedasticity. Multicollinearity. Stochastic regressors. Simultaneous equations systems. Restrictions on structural parameters. Two stage and three stage least squares.

HSTA410: Stochastic Processes

Multiple decrement tables; Describe the construction and use of multiple decrement tables, including the relationships with associated single decrement tables.

Pension funds; Use multiple decrement tables to evaluate expected cash flows dependent upon more than one decrement, including pension benefits, other salary related benefits, health and care insurance, Describe practical alternatives to the multiple decrement table in order to evaluate expected cash flows, Evaluate expected cash flows contingent upon risks other than human life.

Mortality, selection and standardization; Describe the principal forms of heterogeneity within a population and the ways in which selection can occur, Explain how selection can be expected to occur amongst individuals taking out each of the main types of life insurance contracts, or amongst members of large pension schemes, Explain why it is necessary to have different mortality tables for different classes of lives, Explain how decrements can have a selective effect, Explain the theoretical basis of the use of risk classification in life insurance.

HACS531: Principles of Financial Management

Objective of financial management. The annual financial statements; content, interpretation and application for planning and control. Budgets as a management tool. The time value of money, risk and return. Company structure and financing. Basic principles of taxation. Different types of taxation. The role of the main institutions in financial markets. Basic principles of group accounts. Calculation and use of accounting ratios. Limitations of company accounts.

HACS532: Health Indicators

Health indicators: Uses and criteria of selection; health policy indicators, social and economic indicators, indicators of provision of health care, health status indicators. Sources of health indicator data. General principles of a health statistics system. Classification of diseases and causes of death. Nutrition surveys.

HACS533: Life Contingencies

Multiple life models; joint life, last survivor, contingent insurance:- values of premiums for multiple life annuities and assurances and reversionary annuities and compound statuses. Multiple decrement models: disability, withdrawal, retirement etc. and reserving models for life insurance. The control cycle. Introduction to the stochastic approach to life and other contingencies.

HACS534: Actuarial Mathematics II

Simple annuities and assurances involving two lives; Define and use straightforward functions involving two lives.

Contingent and reversionary benefits; Extension of techniques covered in Actuarial Mathematics IIA (CIN 4110) in order to deal with cash-flows dependent upon the death or survival of either or both of two lives and functions dependent upon a fixed term as well as age.

Profit testing; Describe the technique of discounted emerging costs, for use in pricing, reserving, and assessing profitability, Define unit-linked contract, Evaluate expected cash-flows for whole life, endowment and term assurances, annuities, and unit-linked contracts including profit testing of these contracts and determine the profit vector, the profit signature, the net present value, and the profit margin, Show how the profit test may be used to price a product.

Determining provisions using profit testing; Show how the profit test may be used to determine reserves.

Competing risks; Describe methods which can be used to model cash flows contingent upon competing risks, Explain how the value of a cash flow, contingent upon more than one risk, may be valued using a multiple-state Markov Model, Derive dependent probabilities from given transition intensities, using the Kolmogorov equations.

HACS535: Pensions and Benefit Insurance

Principles of pension funds. Mathematical models for retirement income, retiree medical benefits, disability benefits and survivor benefits. Computer applications, simulation. Guarantees and options. Principles of pension valuation: actuarial cost methods, asset valuation methods, actuarial assumptions, gain and loss analysis.

HACS536: Investment and Asset Management

Covers part of subject (CA1) of the Institute and Faculty of Actuaries

The aim of the module is to develop necessary skills that will enable students to apply

the principles of actuarial control planning and control to the appraisal of investments, and to the selection and management of investments appropriate to the needs of the investors. The structure of the course mainly follows the expectations required for subject CA1 of the Faculty and institute of Actuaries (or old course 301 of the Faculty and Institute of Actuaries but not everything in this old course). Not everything pertaining to subject CA1 will be covered so no exemptions from the Faculty and Institute of Actuaries are expected. The following will be covered under this course

Introduction to Investments and Asset Management; The actuarial control cycle (ACC); Taxation; Cash and money markets; Bond markets; Equity markets; Property markets; Derivatives; Collective investment vehicles; Overseas markets; Economic influences on investments; Other factors affecting relative valuation; Relationships between returns on asset classes; the institutional framework; personal investment; Investment indices; Valuation of individual investments; Valuation of Asset Classes and portfolios; Developing an investments strategy; Regulation of financial services; Capital project appraisal

HACS537: Financial Engineering

The primary objective of financial engineering is to optimize financial outcomes by designing sophisticated financial products and strategies that enable investors to manage risk, maximize returns, and achieve their financial goals. Financial engineers work with insurance companies, asset management firms, hedge funds, and banks. Within these companies, financial engineers work in proprietary trading, risk management, portfolio management, derivatives and options pricing, structured products, and corporate finance departments. It entails:

- The use of mathematical techniques to solve financial problems.
- The test and issue new investment tools and methods of analysis.
- work with insurance companies, asset management firms, hedge funds, and banks.
- Led to an explosion in derivatives trading and speculation in the financial markets.
- It has revolutionized financial markets, but it also played a role in the 2008 financial crisis.

Types of Financial Engineering: Derivatives Trading and speculation
Criticism of Financial Engineering

HACS538: Actuarial Statistics II

Explain the concepts of decision theory and apply them; Fundamental concepts of Bayesian statistics; Calculate probabilities and moments of loss distributions both with and without simple reinsurance arrangements; Construct risk models appropriate to short term insurance contacts and calculate the moment generating functions for the risk models both with and without simple reinsurance arrangements; Calculate and approximate the aggregate claim distribution for short term insurance contracts; ruin for a risk model; Calculate the adjustment coefficient; effect on the probability of ruin of changing parameter values and simple reinsurance arrangements; Fundamentals of credibility theory; fundamentals of concepts of simple experience rating systems; Apply techniques for analysing a delay (or run-off) triangle and projecting the ultimate position; Fundamental concepts of a generalised linear model (GLM), and describe how a GLM may be applied.

Monte Carlo Simulation: The generation of pseudo-random variates from a

specified distribution: Inverse transform method, The generation of standard normal variates: Box-Muller algorithm and polar algorithm.
Time Series Analysis: Generalised linear process, White noise process, Auto covariance and Auto-Correlation Functions, Moving Average Process (MA (q)) process, Autoregressive processes (AR(p)) process, Yule walker equations, Mixed Autoregressive-Moving Average model (ARMA (p, q)), ARIMA models for non-stationary series, Stationarity and Invertibility of the different models above.

HACS570: Dissertation
Students will be expected to complete a research project on a topic of their choice but limited to the taught courses. The project is a consolidation of the theoretical knowledge gained in the taught courses and the practical experience gained from Industrial Attachment

Introduction to Management 1

The module deals with the history and development of management thought, functions of management, organizational structure, decision making, globalization, leadership and motivation, controlling, budgetary and non-budgetary controls, change management.
CS131: Communication Skills
Refer to Communication skills
HCSCI: Models of Database and Database Design
Review of the relational data models and introduction to distributed databases, DBMS architectures, data dependences, decomposition, algorithms, data dictionaries. concurrency, integrity, security and reliability issues. Query decomposition, optimization, and evaluation strategies. Physical database design
Gender Studies
Refer to Gender studies.

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED STATISTICS (HSTAT)

Duration:	4 years
Actual Credit Load:	504 credits
Minimum Credit load:	480 credits
Maximum Credit Load:	516 credits
Maximum MBKs Credit Load:	402 credits
ZNQF level:	8

1. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National

Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry:	*At least two Advanced Level passes including Mathematics or recognized equivalence (Mechanics, Statistics, Pure Mathematic plus at least five Ordinary Level passes/ National Foundatic Certificates including Mathematics and English Language C National Certificate.*
Special Entry:	*National Diploma and Higher National Diploma.*
Mature Entry:	*At least 25 years plus relevant work experience.*

2. LEARNING OUTCOMES

Upon successful completion of the programme a graduate will be able to:

- Understand statistics and how its various disciplines are integrated.
- Work effectively in a broad range of scientific, government, financial, analytic, health, technical and other related fields.
- Understand the relationship between theory and applications of statistics.
- Use statistical thinking, training, and approach to problem solving in a diverse
- variety of disciplines.
- Read and interpret statistical literature of various types, including online sources,
- survey articles and books.
- Modify and improve existing statistical models to solve real world problems.
- Use software and quantitative models to analyse statistical data.
- Communicate statistical information orally and in writing to a wide variety of
- audiences.

3. ASSESSMENT

Coursework:	Course-work shall contribute 40 % towards the final mark of e module and shall consist of Practical (20%), Written assignme (5%), and In-class tests (15%).
Written Examinations:	The formal written examination shall contribute 60% to overall assessment of a module.
Other:	5.1 Research Project is assessed on the basis of a resea project (75%), Oral presentation (15%) and student conduc the laboratory (10%). 6.1 Industrial Attachment is assessed based on Indust Attachment report (30%), assessment by work supervi (50%), and assessment by Academic supervisor (20%).

4. MINIMUM BODY OF KNOWLEDGE AND SKIILS MODULES

Module	**Credits**
Level I	
*HMAT131 Calculus 1	12
*HMAT132 Linear Mathematics 1	12

*HSTA132 Applied Statistics	12
*HSTA133 Probability Theory 1	12
HCSCI132 Principles of Programming Languages	12
HSTA134 Statistical Computing 1	12
HSTA105 Regression and ANOVA 1	12
HMAT110 CALCULUS 2	12
*HSTA107 Statistical Inference 1	12
HSTA106 PROBABILITY THEORY 2	12
HMAT109 Mathematical Discourse and Structures	12
Level II	
*HSTA231 Design and Design Issues 1	12
*HSTA235 Hypothesis Testing Techniques	12
*HSTA239 Estimation Theory Techniques	12
*HSTA236 Survey Techniques	12
*TNCP201 Technopreneurship	12
*HSTA237 Time Series Analysis	12
*HSTA240 Operations Research 1	12
HSTA231 Official, Social and Economic Statistics	12
HMAT231 ODEs	12
HSTA234 Statistical Computing 2	12
*GS231 Gender Studies	12
HSTA238 Statistical Inference 2	12
Level III	
Work-related Learning	120
Level IV	
*HSTA401 Linear Models	12
*HSTA402 Multivariate Analysis	12
*HSTA408 Econometrics	12
*HSTA409 Demography	12
*HSTA412 Survival Models	12
HSTA410 Stochastic Processes	12
HSTA405 Operations Research and Quality Control Techniques	12
HSTA406 Design and Design Issues 2	12
HMAT436 Mathematical Programming	12
HSTA470 Research Project	36
Maximum MBKS Credits	**432**

MODULE SYNOPSES

HMAT131 Calculus I

Number systems: The principle of mathematical induction. The real number system: Functions: Limits of functions. Continuity. Sequences: Differentiation: Derivatives of functions of a single variable. Integration: Method of substitution, integration by parts and reduction formulae, fundamental theorem of calculus.

HMAT132 Linear Mathematics I

Complex numbers: De Moivre's theorem polynomials and roots of polynomial equations. Matrices and determinants: solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating

factors, linear equation with constant coefficients**.**

HSTA132 Introduction to Statistics
Measures of central tendency. Measures of variability. Empirical distributions. Moments. Skewness and Kurtosis. Indicators. Contingency Tables. Introduction to Time Series and Trends. Sampling. Introduction to estimation procedures: Judgemental method and Method of moments. Introduction to Hypothesis testing. Non-parametric statistics. Chi-square contingency methods, Goodness of fit tests, Q-Q plots, Z-test and t-test.

HSTA133 Probability Theory I
Axiomatic probability, sets and events, sample space, conditional probability, Independence, laws discrete and continuous random variables, probability density functions, mean, variance, expectation. Independence, Chebyshev's inequality, moments and moment generating functions. Common Discrete and continuous Distributions. Joint Probability Distributions. Approximations, Law of large numbers, Central limit Theorem.

HSTA134 Statistical Computing 1
Introduction to use of statistical software. Training on database software including Microsoft Excel, Microsoft Access, SPSS, and STATA. Use of these softwares for data entry, data management and analysis. Interpretation of analysis results and statistical report writing.

HSTA105 Regression and Analysis of Variance I
Method of least squares. General linear model assumptions. Checking validity of assumptions. Outliers. Pearson's and Spearman's correlation coefficients, predictions. Regression in terms of sums of squares and sums of products. Estimation of parameters. Multiple linear regression. Testing and inference in multiple linear regression using matrices. Partial correlation. Analysis of variance (ANOVA). Assumptions underlying ANOVA. One-way, balanced design ANOVA

HSTA107 Statistical Inference 1
Deductive inference, population and sample concepts as the basis of statistical inference, parameters and statistics, review of probability theory. Central Limit Theorem, Chi-square, student-t and F distributions, distribution of min and max. Estimation: methods of estimation, properties of estimators and their sampling distributions. Interval estimation. and Hypothesis testing.

HMAT231 Ordinary Differential Equations
Modelling with first order ODEs in population dynamics, and second order ODEs (mass-spring systems, RLC circuits). Methods of undetermined coefficients, reduction of order and method of variation of parameters. Existence and uniqueness of solutions, revision of continuous functions and Lipschitz conditions. Series solutions of ODEs, solutions near ordinary and singular points. Systems of linear first order ODEs, solution and stability. Differential equations of special functions. Laplace transforms and inverse Laplace transforms, applications to solving IVPs, Heaviside and Dirac functions.

HSTA231 Design and design issues 1
Principles of experimentation. Completely randomized designs, randomized block designs, Latin square design. Ecological Studies, Cross Sectional Studies, Correlational studies, Case control studies, Cohort and randomized control trials. Epidemiology Methods, Framingham Heart Study, Measurement in Epidemiology, Binary Outcomes, Definition of Prevalence, Determinants of Prevalence, Incidence, Measures of Association, Odds Ratio, Attributable Proportions.

HSTA236 Survey techniques
Uses, scope and advantage of sample surveys. Types of surveys. The phases of a survey. Survey organisation. Questionnaire design, dummy tables, pre-tests, training of field workers. Report writing. Errors in surveys, monitoring reviews, quality control. Sampling techniques, sample design. Further sampling theory. Estimation of means, totals, proportions. Ratio estimation. Variance calculations. Practical work.

HSTA237 Time series analysis
Time series models. Box-Jenkins model building technique, white noise, and random walk. Autocovarince functions. Autocorrelation functions, Yule-Walker equations. General linear process. Autocovariance generating function. Stationarity and Inevitability. Autoregressive processes and Moving average process. Yule-Walker equation. ARIMA models for non-stationary processes. Parameter estimation. Goodness of fit tests. Forecasting. Application to SARIMA processes. Time series in frequency domain.

HSTA239 Estimation techniques
General Minimum Variance Unbiased Estimation, Cramer-Rao Lower Bound, Linear Models & Unbiased Estimators, Maximum Likelihood Estimation, Least squares estimation, Bayesian Estimation, Statistical Detection Theory, Deterministic Signals, Random Signals, Non-parametric and robust detection. Case studies and mini project.

HSTA240 Operations Research I
M-technique, Dual linear programming methods. Dynamic programming. Project scheduling: network construction, PERT-CPM methods, project control. Queuing theory: single and multi-queuing models, finite queue variation and P. K. formula. Inventory Control Models. Probability models. Decision Analysis: Bayesian methods, mini-max, maxi-maxi criteria, maximum likelihood, maximal opportunity criteria, introduction to Utility Theory.

HSTA238 Statistical inference II
Types of statistical data. Order statistics. Exact and asymptotic distribution of order statistics. Wilcoxon one-sample and two sample tests. Tests of location a tests of variability. Non-parametric tests. Test for extreme reactions. Hollander. Tests for dichotomised or cardinal data. Kendal's measures. Fisher's exact test. Chi-square based tests. Kolmogorov'-Smirnov, generation of random numbers. Bootstrap and Jackknife estimation. Resampling. M-, L- and R-estimator.

TCNP201 Technopreneurship
Introduction: nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur. Small business model and financial issues: developing a business model, basics of small business management, risks and stages of funding, sources of funding, financial funding for growth, product valuation, how to form and register a small business in Zimbabwe. New Product Development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; the NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurship Trade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe. Case studies of successful technopreneurs. Project.

HSTA401 Linear Models
Regression: Linear regression model, point and interval estimation of parameters. Pure error and lack of fit. Residual analysis. Multiple regression: estimation and confidence intervals. General linear hypothesis. Stepwise methods. Experimental design models: one factor models. Fixed and random effects. Two factor models, with and without interaction. Qualitative and quantitative contrasts.

HSTA402Multivariate Analysis
Multivariate data, descriptive statistics, graphical techniques. Random vectors and matrices. Multivariate normal distribution. Wishart distribution. Transformation to near normality. Inferences about mean vector. Comparison of several multivariate means: one-way MANOVA. Simultaneous confidence intervals for treatment effects, profile analysis, ideas of two-way MANOVA. Principal component analysis. Factor analysis, Canonical correlation and Discriminant analyses

HSTA470 Research project
The research project involves an experimental or observational investigation of a fundamental or practical problem in Applied Statistics, or product development. With guidance from an academic supervisor, each student should choose and propose his/her own project theme. Each student is required to submit a proposal, carry out an independent research project or develop a product; submit a final written report, and to deliver an oral presentation**.**

HSTA408 Econometrics
Role of Econometrics in Zimbabwe. Review of general linear model. Linear restrictions, Generalised least squares, GLS estimator, heteroscedasticity, pure and mixed estimation, group observations and grouping of equations. Autocorrelation. Heteroscedasticity. Multicollinearity. Stochastic regressors. Simultaneous equations systems. Restrictions on structural parameters. Two stage and three stage least squares.

HSTA409 Demography
Basic techniques of demographic analysis. Sources of data available for demographic research. Population composition and change measures will be presented. Measures of mortality, fertility, marriage and migration levels and patterns will be defined. Life table, standardization and population projection techniques. Case studies.

HSTA410 Stochastic processes
Integer-valued variables: probability generating functions, convolutions. Markov chains: transition probabilities, classifications of states, stationary distributions, transient states. Gambler ruin, random walk. Markov processes: Chapman-Kolmogorov equations, transition rate matrix, forward and backward systems. Poisson process, normal equations, machine operation machinery breakdown, queuing model.

HSTA412 Survival models
Survival time, survival function, hazard function, types of censoring and truncation. Methods (including life table, Kaplan Meier and Nelson Aalen) for estimating survival function, hazard function. Semi parametric (e.g., Cox-Proportional hazards model) survival models and parametric survival models. Evaluation of the proportional hazards assumption. Practical work on fitting semi parametric and parametric survival models in Stata or R software.

HSTA401: Linear models
Regression: Linear regression model, point and interval estimation of parameters. Pure error and lack of fit. Residual analysis. Multiple regression: estimation and confidence intervals.

General linear hypothesis. Stepwise methods. Experimental design models: one factor models. Fixed and random effects. Two factor models, with and without interaction. Qualitative and quantitative contrasts.

HSTA402: Multivariate analysis
Multivariate data, descriptive statistics, graphical techniques. Random vectors and matrices. Multivariate normal distribution. Wishart distribution. Transformation to near normality. Inferences about mean vector. Comparison of several multivariate means: one-way MANOVA. Simultaneous confidence intervals for treatment effects, profile analysis, ideas of two-way MANOVA. Principal component analysis. Factor analysis, Canonical correlation and Discriminant analyses**.**

HST405: Operations Research and Quality Control Techniques
OR techniques with a strong orientation towards computer based solution techniques and case studies. LP problem formulation as an illustration of alternatives and objectives. Transportation problems and algorithms, Assignment problems, Hungarian method with emphasis on formulation, structure and computer solution. Multi-objective programming making emphasis on goal programming and integer linear programming. Dynamic programming, Napsack problem, Advanced linear programming, non-linear programming algorithms, classical optimization theory, unconstrained and constrained problems with practical problems. Control charts, $\bar{X}$ charts, R charts, S charts, P chart. Average run length capability analysis or indices. Reliability of system, structures of the system, parallel and series system, system life as a function, expected system life and failure rate

HSTA406: Design and design issues 2
Issues in the design of Balanced incomplete designs, crossover designs, nested designs, split plot designs, repeated measures, factorial designs, fractional factorial designs. Open and Closed Cohort, Prospective and Retrospective Cohort Studies. Threats to validity of results; Bias, Confounding. Controlling for confounding at both design and analysis stage. Missing observations. Prospective Cohort Studies. Retrospective Cohort Studies.

HSTA407 Statistical Ecology
The course covers aspects on setting up an ecological study, spatial pattern analysis, species abundance relations, community classification and community ordination. Estimation of population parameters (population size, survival, detectability). Mark recapture and related methods. Population modelling, matrix models, integral population modelling, stochastic matrix model. Combining models and data, Bayesian hierarchical modelling, models with density dependence. Behavioural statistics especially analysis of event sequences.

HMAT436 Mathematical Programming
Introduction to mathematical programming problems: Linear programming problem formulation; simplex method; Chmens and two phase techniques; sensitivity analysis; duality in LP; Dual simplex method; transportation and assignment methods; integer programming; dynamic programming; quadratics and separable programming; K T conditions for optimality.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN APPLIED MATHEMATICAL MODELLING (MAMM)

Duration:	2 Years
Actual Credit load:	360
Minimum Credit load:	360
Maximum Credit load:	420
Maximum MBKs Credit Load:	254
ZNQF Level:	9

INTRODUCTION

These regulations shall be read in conjunction with the General Regulations and the Faculty of Science and Technology Regulations.

1.0. PURPOSE OF THE PROGRAMME

To develop knowledge, skills and competencies in the field of Applied Mathematics and Modelling relevant to various employment capabilities and careers in the world of work and society. To equip learners with research and analytical skills which can be used to solve real world problems.

2.0. ENTRY REQUIREMENTS

Normal Entry: Prospective students must hold a good first Degree in Applied Mathematics/Mathematics, Applied Statistics/Statistics, Computer Science, Operations Research or any other relevant mathematical discipline from any university recognized by Midlands State University.

3.0. PROGRAMME CHARACTERISTICS

3.1. ***Areas of Study:*** Mathematical modelling, epidemiology, research methodologies, advanced differential equations, financial mathematics, statistical analysis, numerical methods, report writing and mathematical computing.

3.2. ***Specialist Focus:*** Research in epidemiology, finance, or any of the areas of study as determined by the student's choice of electives.

3.3. ***Orientation:*** Research and computing oriented. Teaching and learning are focused on research design and the use of mathematical software.

3.4. ***Distinctive Features:*** The programme builds the research capacity and model analysis techniques.

4.0. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1. ***Employability:*** Research scientists in research institutions. Data analysts, software developers, risk analysts in financial institutions, actuarial analysts in insurance companies and epidemiologists in medical laboratories and lecturers in tertiary institutions.

4.2. ***Further Studies:*** Doctoral studies in Mathematical modelling, Epidemiology and/or any interdisciplinary programmes related to the modules offered in the programme.

5.0. PROGRAMME DELIVERY

5.1. *Teaching and Learning Methods*

Lectures, tutorials, seminars, group work, computer practicals, research project and individual independent study.

5.2. *Assessment Methods*

Written and oral examinations, tests, seminar presentations, final year research project report and continuous assessments.

6.0. LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- approach problems in an analytical and rigorous way, formulating theories and applying them to solve problems in the sciences, technology and industry, and other fields.
- apply mathematical techniques to problems arising from the planning, monitoring and management of large-scale systems such as health service, communication, energy distribution and transportation systems.
- analyse and interpret data, finding patterns and drawing conclusions to support and improve business decisions.
- develop mathematical and statistical models to solve current practical problems.
- breakdown complex systems into simple and understandable models.
- design and conduct observational and experimental studies.
- demonstrate knowledge and understanding of fundamental concepts in areas of applied mathematical modelling.
- use mathematical and statistical packages to model and solve problems in applied mathematical modelling.
- deal with abstract concepts, to think logically and to present mathematical arguments and conclusions with accuracy and clarity.
- Communicate effectively and present information methodically and accurately using multi-media.

7.0. GENERAL PROVISIONS

The programme shall normally be studied over a minimum period of 4 semesters on a block release basis. A student shall study at least three modules in a semester and is required to attend all lectures and tutorials unless there is substantiated proof of their reason for absence.

8.0. PROGRAMME ASSESMENT

8.1. *Continuous Assessment*

Continuous assessment shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical modules and 40% of final mark for practical modules.

8.2. *Written Examinations*

A three-hour final examination, shall contribute 75% of the final mark for theoretical modules and 60% for practical modules.

8.3. ***Practical Examination***
The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1 A student must pass at least eight (8) taught modules including all core modules in level 1 to proceed to level 2.

9.2. To be awarded a Master of Science degree in Applied Mathematics and Modelling, a student must pass all the core modules listed in the programme and must have accrued at least 360 credits.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 6 of the Faculty Regulations for Undergraduate and Postgraduate Degrees.

11.0. GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the General Regulations for Postgraduate Diplomas and Taught Masters Degrees.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, weighting of the results of the different levels shall be:

Level 1	40%
Level 2	60%

13.0. PROGRAMME STRUCTURE

Summary of Modules arranged in logical sequence, and allocation of Notional Hours

Module name	Minimum body of knowledge and skills	Credits
Level One		**200**
Semester 1		
MAMM8301 Introduction to Mathematical Modelling	Y	24

MAMM8302 Numerical Solution of Ordinary Differential Equations	Y	24
MAMM8303 Stochastic Probability Spaces	Y	24
Total number of credits		72

Semester 11

Students do all modules

MAMM8304 Advanced Dynamical Systems	Y	24
MAMM8305 Advanced Functional Analysis	Y	24
MAMM8306 Financial Mathematics (pre-requisite is Stochastic Differential Equations)	Y	24
		72

Level Two

Semester 1

Electives (Students select at least four modules)

MAMM8307 Industrial Statistics	24
MAMM8308 Mathematical Epidemiology	24
MAMM8309 Continuum Mechanics	24
MAMM8310 Integral Equations	24
MAMM8311 Variational Calculus	24
MAMM8312 Forecasting	24
MAMM8313 Advanced Fluid Dynamics	24
MAMM8314 Perturbation Methods	24
MAMM8315 Numerical Solution of Partial Differential Equations (pre-requisite is Numerical Solution of Ordinary Differential Equations)	24

The choice of electives will be offered subject to staff availability.

Semester 11

MAMM8370 Dissertation	96

14.0 MODULE SYNOPSES (For all the 80% Modules Threshold. **NB:** Synopses are very

central in that these are summaries of the key concepts to be taught in each module.

MAMM8301 Introduction to Mathematical Modelling:
General principles of mathematical modelling and modelling skills needed for abstraction, idealisation, identification of important factors such as variables and parameters. Students will study case studies from the following areas:
Case 1: *Simulation modelling*. Discrete event simulation. Systems dynamics. Simulation software. Sampling methods. Model testing and validation.
Case 2: *Materials Science Modelling*: Understand the micro-level molecular and sub-atomic effects and subtle engineering of special compounds. The behaviour of non-typical materials or new materials like semiconductors, polymer crystals, composite materials, piezoelectric materials, optically active compounds, optical fibres etc. create a multitude of research questions, some of which can be approached with mathematical models. Models will be formulated to design and control the manufacturing processes.
Case 3: *Traffic and Transportation Modelling*: Roads, railway networks and air traffic contain many challenges for modelling. In railway industry, mechanical models about the rail-wheel contact, explaining the phenomena of wear, slippage and braking functions. The train itself is a dynamical system with a lot of vibrations and other phenomena. Analysis of traffic flow, scheduling, congestion effects, planning timetables and derivation of operational characteristics.
Case 4: *Modelling in Food and Brewing Industry:* Mathematics has to do with butter packages, lollipop ice-cream, beer cans and freezing of meatballs. The food and brewing industry; biochemical processes, mechanical handling of special sorts of fluids and raw materials. The control of microbial processes, production chain.
Case 5: *Chemical Reactions and Processes Modelling*: Chemical processes modelled on various scales. The spatial structures and dynamical properties of individual molecules, to understand chemical bonding mechanisms etc. The chemical reactions are modelled using probabilistic and combinatorial methods.
Case 6: *Climate modelling*: Basic model equations, Global climate models and regional climate models.

MAMM8302 Numerical Solutions of Ordinary Differential Equations:
Introduction. Solutions of system of linear and non-linear differential equations, Linear and Non-linear initial value problems, Existence and uniqueness of solutions. Dependence of solutions on initial conditions. Numerical methods: - Euler, Runge-kutta methods. Multistep methods and variable step-size methods – Predictor-corrector methods. Refining of the step size convergence. Convergence and stability. Boundary value problems. Shooting methods for linear and nonlinear problems. Finite difference methods for linear and non-linear problems. Raleigh-Ritz method. Applications: growth models and epidemiological models.

MAMM8303 Stochastic Probability spaces:

Random variables and stochastic processes, Ito integrals, Ito's formula and martingale representation theorem. Stochastic differential equations. Diffusions, Boundary value problems. Optimal stopping. Stochastic control. Introduction to jump diffusions. Differential Equations.

MAMM8304 Advanced Dynamical systems:
Systems of differential equations. Two-dimensional linear and almost linear autonomous systems. Finite difference equations. Steady states and their stability. Stability of periodic orbits. Lyapunov methods. Bifurcation, one and two-dimensional systems. Discrete systems. Self-similarity and fractal geometry. Chaos.

MAMM8305 Advanced Functional Analysis:
Metric spaces. Definitions and examples. Rn, C[a,b]. Inequalities of Holder, Minkowski, Cauchy-Schwarz. Open and closed sets, neighbourhoods. Convergence, completeness. Contraction Mapping Theorem. Applications to linear systems, integrals equations, differential equations. Normed spaces. Definitions and examples. Banach space. Finite dimensional space. Compactness and Riesz Lemma. Linear operators and functionals. Dual space. Second dual. Reflexivity. Weak convergence. Hilbert spaces. Definitions and examples. Cauchy-Schwarz inequality, Pythagoras's theorem. Orthogonal complements and direct sums. Orthonormal sets. Fourier series and orthogonal polynomials. Hilbert adjoint operator. Self-adjoint operators. Eigenvalues and eigenfunctions. Operators. General measure theory. Lebesgue Integral and L^p spaces, with special emphasis on the case $p = 2$.

MAMM8306 Financial Mathematics:
An introduction to financial derivatives, the Cox-Ross-Rubinstein model. Finite security markets. Market imperfections. The Black-Scholes model. Foreign market derivatives. American options. Exotic options. Continuous-time security markets. Arbitrages and equivalent Martingale measures. The one period model. Multi period models. The continuous model. Hedging and completeness. Self financing portfolios. Attainability of a claim. Complete markets. Ito representation theorem. Girsanov's first theorem. Option pricing. European options. American options. The Black Scholes option pricing formula. Optimal portfolio and stochastic control. Stochastic control theory. The Hamilton-Jacobi-Bellman equation. Girsanov's second theorem. Numerical analysis in finance (solving nonlinear partial differential equations arising in finance. Use of appropriate computer packages in finance e.g. Matlab). Levy processes in finance.

MAMM8307 Industrial statistics:
Principles of experimental design. Completely randomised designs. Randomised Block designs. Balanced incomplete Block designs. Latin square and crossover designs. Factorial designs. Fractional factorial designs. Response surface methodology. Nested designs. Split-plot designs. Repeated measures designs. Analysis of covariance. Quality control. Reliability.

MAMM8308 Introduction to Mathematical Epidemiology:
Modelling Transmission Dynamics of Infectious Diseases: Basic concepts of epidemiological modelling. Epidemiological principles and concepts. Tools required to develop mathematical models to understand the transmission dynamics of infectious diseases and to evaluate potential control strategies. Topics to be covered include: history of mathematical epidemiology, Introduction to population

modelling, Basic models of disease transmission, SI, SIS, SIR, SIRS, SIE, SIER and SIERS. Epidemiological measures and their relationship to disease transmission models. The reproductive number. Use of models to plan clinical trials. Modelling of sexual, waterborne and vector borne transmitted diseases.
Immunological Modelling: focusing on modelling the pathogenesis of infectious diseases. The interaction of humans and pathogens. The biochemical, pharmacological, immunological, and molecular biological understanding of how infectious agents and the human body interact. Development of models to study host susceptibility to particular pathogens, development of models to study host resistance to chronic or acute disease, development of models for basic studies of infectious organisms, as long as they are oriented toward understanding how the organism interacts with the host, development of models to study virulence factors, immune mechanisms, and genetic studies in the host and in the pathogens. Work on modelling pathogenesis of HIV, malaria, and tuberculosis will be given higher priority.

MAMM8370 Dissertation:
Dissertations may be carried out on an individual basis. The dissertation normally involves work with some outside organization. The dissertations test students' ability to organise, complete and report on a significant piece of Mathematical modelling.

MAMM8309 Continuum Mechanics:
Rigid deformable bodies. Concept of stress. Deformation and kinetics. Balance equations. Constitutive equations. Examples of complex material. Solution of problems in elasticity and viscoelasticity.

MAMM8310 Integral Equations:
Iterative methods for linear systems. Initial and Boundary Value Problems for ODES. Methods for Fredholm Integral Equations of the second kind. Neumann series. Degenerate kernels. Quadrature methods. Expansion methods. Applications.

MAMM8311 Variational Calculus:
Calculus of Variations: Function of one variable and several variables, constrained extrema and Lagrange multipliers, Euler-Lagrange equations. Functions with higher-order derivatives and several dependent variables and independent variables. Applications.

MAMM8312 Forecasting:
Applications to business management. Multiple regression modelling. Binary choice models, multiple discrete choice models and limited dependent models. Time series analysis: ARIMA, ARMA and VARMA models.

MAMM8313 Advanced Fluid Dynamics:
Basic principles of fluid dynamics. Equations of continuity and motion. Dynamical similarity. Some solutions of viscous flow equations. Inviscid flow. Boundary layers. Instability and turbulent flows. Flow in rotating fluids. Geotropic flow, Ekman layer and Rossby waves. Stratified flow. Stratification and rotation. Applications to meteorology, aerodynamics and irrigation systems.

MAMM8314 Perturbation Methods:

Concept of asymptotic development. Elementary operations on asymptotic expansions. Equations containing a small parameter and or a region slightly perturbed from a regular figure. The solution in terms of the small perturbation parameter. Methods of regular perturbation. Examples of the possibility of non-uniform expansions. Methods of singular perturbation: Poincare-Lighthhill-Kuo, matched asymptotic expansion and multiple scales. All the methods will be illustrated by solving ordinary and partial differential equations.

MAMM8315 Numerical Solutions of Partial Differential Equations:
Elliptic Partial differential equations. Poisson Problem with Dirichlet, Neumann and Robin Boundary Conditions, finite difference method. Parabolic partial differential equations. Initial-boundary value problems, one-dimensional explicit and implicit methods, stability. First-order hyperbolic partial differential equations. Finite Element Methods and Variational Techniques: Introduction-functional, Green's theorem, divergence theorem, Euler-Lagrange equations, mixed boundary conditions, functionals for differential problems. Approximation of solution -Ritz method. Variational and weak forms in Hilbert (Sobolev) spaces. Finite Element Methods: Review of elliptic and partial differential equations, Laplace, Poisson, biharmonic and Lame's equations all with various types of boundary condition

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN BIOSTATISTICS AND EPIDEMIOLOGY (MSBE)

Duration	2 years
Minimum Credit Load	360
Maximum Credit Load	420
Maximum MBKs Credit Load	270
ZNQF Level	9

1. INTRODUCTION
These regulations should be read in conjunction with the General Academic Regulations and the Faculty of Science and Technology Regulations for the Postgraduate Degrees.

2. PURPOSE OF THE PROGRAMME

The major aim of the program is to provide students with an in-depth understanding of the design, analysis and interpretation of different epidemiological studies.
Study designs to be covered include cross-sectional, case-control, cohort and intervention studies.
The HIV/AIDS pandemic has also necessitated the mathematical statistics approaches in order to curb its unprecedented spread, through the design of intervention evaluation of new therapies for its treatment and prevention.

3.0 ENTRY REQUIREMENTS

Normal Entry: A good **Honours** degree in Mathematics, Applied mathematics, Statistics, Operations Research (or their equivalence), Medicine, Epidemiology and any other related programs, with **upper second class** or better. Applicants with a lower class and relevant experience may be considered. Such applicants may have to pass either an entrance examination or interview to determine their suitability for the programme.

4.0 PROGRAMME CHARACTERISTICS

3.1 ***Areas of Study:*** Biostatistics, epidemiology, research methodologies, advanced multivariate analysis, further time analysis, disease modelling, health insurance, population dynamics, spatial statistics, report writing and statistical computing

3.2 ***Specialist focus:*** Research and application of Biostatistics to Epidemiology, health insurance or any of the areas of study as determined by the student's choice of electives.

3.3 ***Orientation:*** Research and computing oriented. Teaching and learning are focused on research design and the use of statistical software.

3.4 ***Distinctive features:*** The program builds the research capacity and model data analysis techniques

5.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 ***Employability:*** Research scientists in health institutions. The disciplines of *epidemiology* and *biostatistics* are essential to achieving the goals of public health. Both fields consider population health concerns, and both fields have the ultimate goal of promoting overall health and disease prevention in a specific community, academic research and in the private medical and health care fields. Data analysts, software developers, risk analysts in health institutions, epidemiologists in medical laboratories and lecturers in tertiary institutions.

4.2 ***Further Studies:*** Doctoral studies in Statistical Modelling , Epidemiology and/or Biostatistics as well as in any disciplinary programmes related to the modules offered in this programme.

6.0 PROGRAMMME DELIVERY

5.1 **Teaching and Learning Methods:**

Lectures, tutorials, seminars, group work, computer practical, final year research project and individual independent study.

5.2 **Assessment Methods:**

Written and oral examinations, tests, seminar presentations, final year research project report and continuous assessment.

7.0 LEARNING OUTCOMES

8.1 Demonstrate the ability to conduct methodological and collaborative research to examine, develop, test, and apply established epidemiological and biostatistical methods for the purpose of addressing national health needs.

9.1 Capability to offer educational programs, training, and mentoring in the design and analysis of public health programs through the application of epidemiological and biostatistical methods.

10.1 Display ability to provide consultancy services to local communities and the country at large in a way that furthers the mission of the Department and the University.

11.1 Promotion of advanced statistical reasoning in environmental and health problem scenarios.

8.0 GENERAL PROVISIONS

The programme shall normally be studied over a minimum period of 4 years on a block-release basis. A student shall study at least three modules in a semester and is required to attend ALL lectures and tutorials unless there is substantial proof of their reason for absence.

9.0 PROGRAMME ASSESSMENT

9.1 *Continuous Assessment*:
Continuous assessment shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical modules and 40% of the final mark for practical modules.

***9.2 Written examination*:**
A four-hour final examination shall contribute 75% of the final for theoretical modules and 60% for practical modules.

9.3 Practical Examination:
The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark will depend on the module offered.

10.0 PROVISION FOR PROGRESS

10.1 A student must pass at least eight (8) taught modules including all core modules in level 1 to proceed to level 2.

10.2 To be awarded a Master of Science (MSc) in Biostatistics and Epidemiology, a student must pass all the core modules listed in the programme and must have accrued at least 360 credits.

10.0 FAILURE TO SATISFY THE EXAMINERS

Refer to section 6 of the Faculty Regulations for undergraduate and postgraduate Degrees.

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to Section 12 of the General Regulations for postgraduate diplomas and taught Masters Degrees.

12.0 DEGREE WEIGHTING

For the purpose of degree classification. Weighting of the results of the different levels shall be:

Level 1 – 40%
Level 2 - 60%

13.0 PROGRAMME STRUCTURE

N.B. *Denotes core modules which are not Minimum Body of Knowledge and Skills

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester I

Code	Module Description	Credits
** MSBE8301	Fundamentals of Biostatistics	18
** MSBE8302	Research Methods in Epidemiology	18
** MSBE8303	Principles of Epidemiology	18
* MSBE8307	Applied Multivariate Analysis	18
* MSBE8309	Further Time Series Analysis	18

Level 1 Semester II

**MSBE8304	Survival Analysis in Epidemiology	18
**MSBE8305	Experimental Design and Analysis of Clinical Studies.	18
* MSBE8306	Generalized Linear Models	18
* MSBE8308	Infectious Disease Modelling	18
* MSBE8312	Health Insurance	18

Level Two Semester I

**MSBE8370	Dissertation	180

MODULE SYNOPSES (For all the 80% Modules Threshold. NB: Synopses are very central in that these are summaries of the key concepts to be taught in each module.

MSBE8301 Fundamentals of Biostatistics
Review of basic statistics and its applications; univariate and bivariate descriptive statistics, probability and statistical distributions, statistical inference- confidence intervals and hypothesis testing, confidence intervals for case-control and cohort studies, sample size calculation, correlation and regression analysis, introduction to survival analysis,

standardization of rates, use of statistical software such as Excel, Epidata, EpiInfo and Stata for data management and analysis.

MSBE8302 Research Methods in Epidemiology
Review of statistical methodology from epidemiological viewpoint, including cohort and case-control designs, relative odds (odds ratio) and relative risks. Computational statistics using appropriate software packages. *S*ampling techniques, data collection strategies, questionnaire construction, sampling, health surveys.

MSBE8303 Principles of Epidemiology
Measures of disease frequency. Observational studies, Case control studies, Cross sectional surveys, outline data-based studies, individual level, Ecological studies. Logistic Regression; interpretation of parameter estimates; probabilities and odds; comparison of classical discriminant analysis and logistic regression. intervention studies, Clinical trials, Field trials, Individual and aggregated levels, sample size determination for each study design, biases and their control in study design, confounding and methods of control and effect modification.

MSBE8304 Survival Analysis in Epidemiology

Examples of Survival Data, Analysis of dichotomous data, Introduction to Survival Analysis, Censoring and Truncation, nonparametric estimation methods, Hypothesis Testing for survival data. Refinements to Proportional Hazards method, Regression Diagnostics and inference. Models Preliminaries – survival analysis in Stata. Shapes of hazard and survival functions, Estimation of the (integrated) hazard and survivor functions: Kaplan-Meier product-limit and life-table methods, Estimation: (i) continuous time (ii) discrete time models, Unobserved heterogeneity, Competing risk models.

MSBE8305 Experimental Design and Analysis of Clinical Studies
Theoretical, Methodological and Measurement Issues in the Clinical Research Model. Single-Subject Designs: AB and ABAB designs, Withdrawal or reversal designs, tailor-made designs for particular practice situations. Theoretical, Methodological and Measurement Issues in Program Evaluation, Evaluation Utilization and Reporting. Practice effectiveness studies in social work, Identifying and operationalizing goals, objectives, and interventions. Evaluation, Experimental and non-experimental designs. Data Analysis: Qualitative and Computer analysis using SPSS-X. Clinical versus statistical significance.

MSBE8306 Generalized Linear Models

Theory and applications - regression and ANOVA to non-normal data. Logistic, log-linear models, and gamma regression models. Review of the GLM for Normal Data: Linear regression, fixed- and mixed-model ANOVA, ANCOVA, Extending the GLM: Non-normal error structure, The exponential class, Linear and non-linear link functions, Theory of Estimation and Model Fitting: Likelihood functions and maximum likelihood, Iteratively reweighted least squares, Theory of Statistical Inference: Analysis of deviance, Likelihood ratio tests, Wald tests, Confidence regions, Classical normal-based models, Extending the exponential class, over-dispersed models, Quasi-likelihood models, Generalized estimating equations, polytomous response models.

MSBE8307 Applied Multivariate Analysis

P-Values, Nonparametric Hypothesis Tests and their properties (ARE's), Nonparametric Confidence Intervals, Fundamentals of Data Manipulation, Standardization, statistical

distance, Matrix, Sum of Squares; Covariance Structure, Multivariate Analysis of Variance, Violating Standard Parametric Assumptions. Power of statistical tests. Path Analysis, Direct and Indirect effects, partial correlation, coefficients; causal modelling, Covariates and repeated measures in experimental designs. Concomitant or covariates in ANOVA, parallel regression lines. Two Group Discriminant Analysis Classification, Multiple Group Discriminant Analysis, Dimensionality of discriminant functions, Quadratic discriminant, Stepwise discriminant analysis.

MSBE8308 Infectious Disease Modelling

Background: examples of problems and issues, types of models. Threshold behavior. Implications for disease control. Data: evidence for the limitations of mass-action model. Multi-group epidemiological models. Implications for disease control. Stochastic invasion theorems, final epidemic size distributions. Importance of population size; limitations of deterministic approaches. Bartlett's dynamics of childhood diseases. Small Community Models and Statistics. Stochastic models for disease outbreaks. Parameter estimation. Evolutionary and Spatial issues, Network models, Within-host models. Emergence of drug resistance. Case Studies.

MSBE8309 Further Time Series

The Seemingly Unrelated Regression and Multivariate Regression Models. Generalized Method of Moments, "Large Sample Properties of Generalized Method of Moments Estimators", Nonlinear Systems of Equations, Univariate Time Series Lagged Variables, Simultaneous Equations Models, Multiple Time Series, Panel Data Models, Models with Discrete and Limited Dependent Variables, Resampling and Simulation Techniques: Monte Carlo, Bootstrap, Gibbs Sampler, and Simulated Method of Moments.

MSBE8312 Health Insurance

Basic principles of Health Insurance and Managed Care, History Health Insurance and Managed Care, key trends in Health Insurance and Managed Care, definition of insurance, Organized Delivery System, Risk, Capitation, and Other Financial Issues in Managed Care, Risk adjustment, Health Insurance models – Traditional (Fee-for-Service) Indemnity, "Managed" Indemnity, Preferred Provider Organization (PPO), Health Maintenance Organization (HMO), Taxonomy for determining the type of Health Insurance Plan. Consumer-Directed Health Plans (CDHPs). Case studies on national and global trends as well as practises in health insurance.

Study Session 1: An introduction to epidemiology and implication of chronic non-communicable diseases

Study Session 2: Demographic, epidemiological and nutrition transition

Study Session 3: Social Determinants for non-communicable diseases

Session 1 explains what chronic diseases are, and the diseases that fall into this category. We also start to develop an overview of why they are a concern. We also highlight the burden associated with chronic disease globally and in developing countries, dispelling assumptions that certain parts of the world are completely unaffected by chronic diseases. Basic concepts in chronic disease epidemiology are also defined.

Session 2 discusses demographic, epidemiological and nutrition transition and how these relate to the development of chronic diseases.

Session 3 looks at how social determinants of health are associated with non communicable diseases

MSBE8370 Dissertation

The student undertakes a project based on a thorough study of some mathematical and/or statistical aspects of the theory with reference to applications. The project will be supervised by a departmental lecturing member of staff and should therefore be of high standard.

DEPARTMENT OF APPLIED PHYSICS AND TELECOMMUNICATIONS

REGULATIONS FOR BACHELOR OF ENGINEERING HONOURS DEGREE IN TELECOMMUNICATIONS ENGINEERING (HTENG)

Duration:	4 years
Credit load:	486
Minimum Credit load:	480
Maximum Credit load:	540
Maximum MBKs Credit Load:	380
ZNQF Level:	8

1.0. PURPOSE OF THE PROGRAMME

This program prepares students for exciting careers in the dynamic field of communication. Through classroom and laboratory experience, they gain in-depth knowledge of the components and systems that make up the global telecommunications network.
Students learn about the infrastructures (devices and media) that transport and direct communication signals through telecommunications networks.
The Telecommunication Engineering Programme includes a thorough teaching in the areas of communication systems, electromagnetic, antennas & propagation, digital & data communications, etc.

2.0. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better

2.1. ***Normal Entry***

Two Advanced level passes to include Mathematics (Pure Mathematics/Further Mathematics/Mechanics), and Physics.

2.2 Special Entry

At least a minimum of National Diploma in a related field.

2.3 Mature Entry

See General Academic Regulations

3.0. PROGRAMME CHARACTERISTICS

3.1. ***Areas of Study***

Telecommunication systems, Terrestrial Networks, Internet, Free space Optical communication systems, Telecommunication Media, RF & Microwave systems, Industrial telecommunication systems, Satellite & Mobile communication engineering,

3.2. ***Specialist Focus***

Programme is centred on the detailed Telecommunications engineering and communication principles of current technologies and the next generation networks. Telecommunication standards and regulatory aspects related to deployment of the communication infrastructure include the main course focus.

3.3. ***Orientation***

Innovative research including professional academia teaching and learning of theoretical and practical (including industry) aspects

3.4. ***Distinctive Features***

The course guides students through the processes of assessing the specification, performance, selection, and commissioning of telecommunications systems and network applications using a student-centred approach

4.0. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1. ***Employability***

Graduates with the BSc Telecommunications Engineering Degree can seek employment in Industry, Commerce and Research as

- Signal processing engineers,
- Network engineers,
- Mobile communication systems engineers,
- Satellite communication engineers,
- Software/hardware specialists,
- Design and development engineers in telecommunications

They are also developed to have the ability to start a telecommunications and/or electronics business

4.2. ***Further Studies***

Master's and Doctoral studies in Telecommunications Engineering, Electronic communications and related areas

5.0. PROGRAMME DELIVERY

5.1. Teaching and Learning Methods

Lectures, tutorials, practicals & individual lab projects, group presentations, group work, mini research projects, industrial visits, Work-Related learning, research project, and individual independent study

5.2. Assessment Methods

Written and oral examinations, tests, laboratory reports, laboratory project presentation, seminar presentations, Work-Related learning report, mini-research project report, final year dissertation, continuous assessments

6.0. LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Identify, formulate, analyse and solve complex telecommunication engineering problems creatively and innovatively.
- Apply knowledge of mathematics, natural sciences, engineering fundamentals and an engineering specialty to solve complex telecommunication engineering problems.
- Perform creative, procedural and non-procedural design and synthesis of components, systems, of telecommunication engineering works.
- Design and conduct investigations and experiments.
- Use appropriate engineering methods, skills and tools, including those based on information technology.
- Differentiate the different communication protocols and be able to apply the

protocols accordingly

- Communicate effectively, both orally and in writing, with engineering audiences and the community at large.
- Aware of the sustainability and impact of engineering activity on the social, industrial and physical environment.
- Work effectively as an individual, in teams and in multidisciplinary environments.
- Engage in independent learning through well-developed learning skills.
- Aware of the need to act professionally and ethically and to exercise judgment and take responsibility within own limits of competence.
- Understand engineering management principles and economic decision making.

7.0. GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for a minimum of 12 modules

7.2 A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other Faculties as long as the total credits for the degree does not exceed 540 credits

7.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons of all relevant Departments and to the following conditions:

7.4.1 The module combination is feasible in terms of timetable,

7.4.2 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme.

8.0. PROGRAMME ASSESSMENT

8.1. ***Coursework***

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of the final mark for practical courses.

8.2. ***Written Examinations***

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses

8.3. Practical Examination

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1. A student will repeat all failed modules if he/she fails at least 4 modules in a semester.

9.2. To proceed from level 1 to level 2 a student must have accrued at least 90 credits, provided he/she may not carry over more than three modules.

9.3 To proceed from level 2 to level 3 the student must have accrued at least 90 credits from the modules in level 2, provided he/she may not carry over more than three core modules to the next level.

9.4 To proceed from level 3 to level 4 the student must have satisfied the requirements for Work-Related Learning, including Work-Related Learning modules 1 and 2.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 480 credits

11.2 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0 PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
HTENG131*	Communications Principles	10
HTENG133	Physics for Engineers	10
HTENG134*	Engineering Mathematics	10
HTENG135*	Analogue Electronics	10
HCSE131*	Introduction to Computer Engineering	10
CS131*	Communication Skills	12

Level 1 Semester 2

HTENG136*	Engineering Drawing	10

HTENG137*	Digital Electronics and Systems	10
HTENG138*	Digital Communications	10
HTENG139*	Network Theory	10
HTENG140*	Analogue Communications	10
HTENG141	Engineering Mathematics 2	10

Level 2 Semester 1

HTENG231*	Transmission Technologies	10
HTENG232*	Mobile Communications	10
HTENG233*	Electromagnetic Theory	10
HCSCI234	Research Methods	10
TCNP201*	Technopreneurship	12
GS231*	Introduction to Gender Studies	12

Level 2 Semester 2

HTENG234	Telematics and Industrial Telemetry	10
HTENG235*	Fundamentals of Radio & TV	10
HTENG236*	Power Supplies	10
HTENG237*	Digital Signal Processing	10
HTENG238	Telecommunications Engineering Mini Project	10
HCSCI237*	Data Communication and Computer Networks	10

Level 3 Semester 1

HTENG332*	Work-Related Learning I	40

Level 3 Semester 2

HTENG333*	Work-Related Learning II	80

Level 4 Semester 1

HTENG431*	Telecommunications Planning & Management	10
HTENG432*	Communications Network Design	10
HTENG433*	Antenna Engineering	10
HTENG434*	Project Management	10
HTENG435	Satellite communication Engineering	10

Electives (choose any one from the following)

HCSE432	Embedded Systems	10
HCSE433	IoT and Cloud Systems Engineering	10
HCSE435	Digital Media Engineering	10

Level 4 Semester 2

HTENG437*	Next Generation Networks	10
HTENG439*	Radar Systems	10
HTENG440*	Cryptology & Security in Telecommunications	10
HCSCI439	Fundamentals of Data Science and Big Data	10
HTENG441*	Research Project	20

(Note: * denotes core modules)

14.0. SYNOPSES

HTENG131 Communications Principles

Fundamentals of telecommunications. Power measurements. Signals and their classifications. Time domain representation of signals. Periodic & non periodic, analog & digital, deterministic & random, energy & power signals. Elements of communication systems. Telephone circuit. Telephone networks. Basic call procedure. Transmission line basics matched and mismatched lines, reflection coefficient and standing wave ratio. Bandwidth and Spectral efficiency and their limitations on signal transmission.

HCSE131 Introduction to Computer Engineering

Introduction to Computers. History of computers. Functional structure and main components of computer hardware and software. Applications of computers. Computer components: CPU Harvard and Von Neumann Microprocessor architectures. Types of computer systems, system features, central processing unit, memory, the motherboard, the expansion bus, memory map. Memories and Storage devices: RAM ROM and their derivatives such as SRAM, DRAM, EEPROM. Input-Output (IO) devices and processing: Direct Memory Access DMA; Memory Mapped; Port Mapped and IRQ processes. Operating systems introduction: MSDOS, UNIX and Windows. Computer maintenance.

Introduction to programming using the C procedural language. Variables, data types, operators and statements; program flow control; if-then-else, switch-case, ternary operator and other language basics.

HTENG133 Physics for Engineers

The physics of electricity and magnetism; application of electronic, semiconductor and magnetic materials, ferromagnetism, electric and magnetic fields. Forces on magnetized iron surfaces; forces of attraction and alignment, Waves and Optics. Mechanics: work and energy, momentum and collisions. Thermal physics: temperature, kinetic theory and the ideal gas, heat and the first law of thermodynamics.

HTENG134 Engineering Mathematics

Limits and continuity of functions. Differentiation. Leibniz's Rule. L'Hopitals Rule. Elementary functions including hyperbolic functions and their inverses. Integration – techniques including reduction formulae. Applications – arc-length, area, volumes, moments of inertia, centroids.

Plane polar coordinates. Complex Numbers: Basic algebra. De Moivre's theorem. Complex exponentials. Linear Algebra: Vector algebra in 2 and 3 dimensions. Scalar and vector products. Equations of lines and planes. Systems of Linear Equations. Ordinary differential equations. First-order differential equations. Linear second order equations. Higher order linear differential equations. Practical applications of differential equations to engineering

Integration. Integrals of standard functions. Double and triple integrals. Multiple Integrals. Iterated integrals change of order. Change of variable. Polar, cylindrical and spherical coordinates. Applications in three dimensions. Vector Calculus. Scalar and vector fields. Directional derivatives. Gradient, divergence and curl. Line and surface integrals. Theorems of Green, Gauss and Stokes. Fourier Analysis. Fourier Series. Half range series. Fourier integrals and transformations. Laplace transforms of basic functions. Inverse Laplace transforms, derivation using partial fractions.

Probability and statistics: distributions, mean, variance, covariance, bounds, likelihood function, Bayes rule, entropy, divergence, mutual information Statistics. Concepts of Probability. Random variables. Distribution functions and distribution theory.

HTENG135 Analogue Electronics
Characteristics of an ideal operational amplifier. Practical operational amplifier – structure, parameters and characteristics. Operational amplifier circuits: zero crossing and voltage level detectors; inverting and non-inverting amplifier; summing amplifiers; integrating and differentiating circuits; logarithmic and antilogarithmic amplifier; monolithic differential amplifier; active filters – first, second and higher order types; oscillators and multi-vibrators. Special-purpose operational amplifiers: low input current, low drift, low noise, wideband and unit gain type. Analogue multipliers integrated circuits and applications. 555 Timer integrated circuit – operation, parameters, characteristics and applications. Instrumentation amplifiers. Analogue-to-Digital and Digital-to Analogue converters.

HTENG136 Engineering Drawing
Introduction. Sketching, discrete geometry, charts. Plane geometry. First and third angle projection. Dimensioning. Pictorial views. Freehand sketching. Drawing of common objects; sectioning. Intersections. Developments. Conventioning. Assembly Drawing. Exercises. Graphs of electrical and mechanical concepts.

HTENG137 Digital Electronics and Systems
Numerical systems and binary arithmetic. Binary Coded Decimal (BCD) and ASCII. Logic gates and truth tables. Boolean algebra theorems. Karno (K) - maps. Combinational logic - minimizing logic expressions using Boolean algebra theorems and K-maps. Universal gates (NAND and NOR) implementation of combinational logic circuits. Combinational logic applications: adders and subtractors, magnitude comparators, multipliers, encoders and decoders, multiplexers and de-multiplexers. Digital integrated circuits. Integrated circuit families: RTL, DTL, TTL, ECL, NMOS, PMOS, CMOS. Bi-Stable systems – flip flops: R-S type, T-type, D-type, J-K-type and Master-Slave type. Sequential logic circuits: counters and registers and applications. Finite State Machines (FSMs).
Memories: RAM, ROM, EAROM. PROM, EPROM. Programmable Logic Devices (PLDs).

HTENG138 Digital Communications
Digital communication systems overview. Review of probability and statistics. Review of linear systems and signals. Principles of information theory. Source coding and channel coding. Shannon's theorems and their implications. Formatting and modulation of baseband signal. Baseband demodulation/detection. Band pass modulation and de-modulation/detection. Communication link analysis. Synchronization. Multiplexing and multiple access. Spread-spectrum techniques. Fading channels. Optimal receivers for binary communications systems. Matched-filter and correlation receivers. Performance analysis of binary communication systems. Binary phase shift keying (BPSK), quadrature phase shift keying (QPSK) and quadrature amplitude modulation (QAM). Pulse shaping. Inter-symbol interference. Channel capacity. Error probabilities; signal-to-noise ratios; probability of error for digital modulation schemes. Digital transmission of analogue signals (PCM) and quantization. Noise. Forward error correction codes.

HTENG139 Network Theory
DC circuit analysis. First order circuits: The source free RC and RL circuits, step response of RC and RL circuits. Second order circuits: The source free series and parallel RLC circuits, step response of a series and parallel RLC circuit. AC circuits analysis: Kirchoff's law in the frequency domain. Sinusoidal steady analysis. Frequency response. Series and parallel resonance. Filters. Transfer functions.

Advanced circuit analysis: Applications of Laplace Transform, Fourier series and Fourier Transform to circuit analysis. Two-port networks: impedance parameters, admittance parameters, hybrid parameters, transmission parameters, relationship between parameters, interconnection of networks.

HTENG140 Analogue Communications
Introduction to analogue communications. Modulation. Amplitude modulation methods: Double Sideband with Suppressed Carrier, Single Sideband and Vestigial Sideband Modulation. Application of AM systems. AM transmitters and receivers. Angle modulation methods. Instantaneous frequency, frequency deviation, modulation index, significant sideband criteria, bandwidth of angle modulation. Power of an FM signal. Narrowband FM, wideband FM. FM generation. FM Transmitters and receivers. Phase modulation. Demodulation.

HTENG141 Engineering Mathematics 2
Matrix Algebra, Eigenvalues & eigenvectors and Calculus. Fourier series and integrals, Periodic functions, Euler formulae for Fourier coefficients & transformations, Linearity property, transform of derivatives, convolution theorem, Gamma and Beta functions, BVPs and systems of ODEs. Applied statistics and probability. Solving engineering problems using high-performance language for technical computing (e.g. MATLAB).
CS131: Communication Skills - Refer to Communication skills department

HTENG231 Transmission Technologies
Microwave passive components: behavior and construction of components at microwave. Resonant circuits, filter design, impedance matching. Smith Chart, impedance matching on the Smith Chart. Directional Coupler, Power Divider, Magic Tee, Attenuator, Resonator. Microwave oscillators, active components. Diodes, transistors, behavior and construction of microwave semiconductor diodes: Gunn, IMPATT, Schottky, PIN and Varactor diodes. Modes of operation and circuit realization. Microwave tubes: Klystron, Travelling Wave, Magnetron; behavior and construction performance. Application and characteristics of the devices. Design of small signal amplifiers, distributed amplifiers and power amplifiers. Y parameters and S parameters. Microwave mixers design. Control devices and their application. Microwave integrated circuits. Design of micro-strip and related applications in antennas. Scattering parameters. Microwave techniques using miniature technology.
FIBRE OPTIC TECHNOLOGY - Introduction to vector nature of light, propagation of light, propagation of light in a cylindrical dielectric rod, Ray model, wave model. Different types of optical fibres and their applications. Modal analysis of a step-index fibre. Signal degradation on optical fibre due to dispersion and attenuation. Fabrication of fibre and measurement techniques like OTDR. Optical sources - LEDs and Lasers, Photo-detectors - pin-detectors, detector responsivity, noise, optical receivers. Optical link design. Optical switches: coupled mode analysis of directional couplers, electro-optic switches. Nonlinear effects in fibre optic links. Optical amplifiers - EDFA, Raman amplifier, and Dense Wavelength Division Multiplexing (DWDM) systems. Broadcast-and-select networks: Topologies, medium access control and test-bed examples (LAMBDANET, RAINBOW, STARNET and LIGHTNING). Wavelength routing networks: Node design, the routing and wavelength assignment problem (RWA), virtual topology design, current testbeds (MONET, ONTC, MWTN).

HTENG232 Mobile Communications
Cellular system design and analysis. Limits of wireless transmission systems. Mobile radio

propagation. Modulation techniques for mobile radio. Analog and Digital Modulation schemes. Digital modulations applied in Mobile Communications. Frequency management and channel assignment. Equalization, diversity and channel coding. Multiple-Access Techniques (FDMA, TDMA, CDMA, SDMA, OFDM). Cellular radio architectures and capacity calculation. Wireless channel and system models. Fading and diversity, resource management and power control. Multiple antenna and MIMO systems. Space-time codes and decoding algorithms. Multiple access techniques and multi-user detection. Cellular and ad-hoc network topologies. Mobility management: roaming /handover. GSM case study.

HTENG233 Electromagnetic Theory

Maxwell's equations. Laplace and Poisson equations and their solution. Boundary conditions. Plane waves in a perfect dielectric; propagation in imperfect dielectric. Propagation in imperfect conductors, skin effect. Generalized wave equation, field distributions in rectangular waveguide. Radiation field, dipoles, radiation resistance, impedance, mutual impedance, linear arrays.

HTENG234 Telematics and Industrial Telemetry

Instrumentation Standard Protocols and automation networks, bus standards, third party interface, object linking and embedding for Process Control. Wireless instrumentation-Wireless HART,ISA100.11A, Foundation Fieldbus H1: frame structure, programming. Buses, Controlnet, Industrial Ethernet. PLC and SCADA/DCS.

TCNP201 Technopreneurship

Introduction:Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur.

Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation, How to form and register a small business in Zimbabwe.

New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization.

Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurshipTrade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe; Case studies of successful technopreneurs , Project.

HTENG235 Fundamentals of Radio & TV

Constituents of Radio. Radio communications. Radio transmitters and receivers. Radio wave propagation. Waveguides: higher-order modes on parallel-plate waveguides, rectangular waveguides, dielectric slab waveguides. Antenna properties. Principles of TV. Audio and video transmission. Scanning principle. TV broadcasting systems. Transmission & reception of TV signals. Aspect ratio, Resolution, Video bandwidth. TV broadcast studio. TV Transmitter: video modulation, vestigial sideband transmission. Standard TV channels.

TV Receiver: block diagram, receiver controls, RF tuners, Video channel and picture tube. Automatic Gain Control and synchronization circuits. Deflection circuits (horizontal and

vertical). Video detector. Sound section. Colour TVs. Digital TV (terrestrial, satellite and cable). Video display systems (LED, LCD, Plasma).

HTENG236 Power Supplies

AC Theory; power factor correction, Tariff philosophies , AC power supplies:- Single phase and three phase systems; Transformers; Characteristic, ratings and protection of Silicon Controlled Rectifiers (SCRs), Gate Turnoff Thyristors (GTTs), Bipolar Junction Transistors (BJTs), Metal Oxide Semi-Conductors (MOSFETs), Single phase AC –DC Conversion; Single and three phase AC –DC conversion using thyristors, include determination of size of filtering capacitors. Inverters (DC-AC conversion single and three phase systems) and UPS systems. Linear power supplies overview of switching mode power supplies (DC-DC converters- Buck, Boost and Buck –boost regulators). Sources of energy: - An overview of Photovoltaic and wind energy sources. Portable power supplies.

HTENG237 Digital Signal Processing

Analysis of continuous and discrete signals and systems. Discrete-time signals, Discrete-time systems, Linear Time-Invariant systems and their properties. Linear constant coefficient difference equations. Discrete Fourier transform. Z-transform. Definition of Z-transform, Region of Convergence, Properties of z-Transform, Inversion of the z-Transform, Analysis of Linear Time-Invariant Systems in the Z-Domain. Basics of image processing. Radar and sonar signal processing. Digital filters and their application.

HCSCI237 Data Communication And Computer Networks

This module explores the principles underlying the design of computer networks. Topics covered include: Computer network technologies and applications, Transmission Media, Signaling, Communication protocols, Communication architectures, Network connections, Network types, Routing and routing algorithms, spanning tree protocol and IP addressing.

HCSCI234 Research Methods

The module provides students with a grounding in the Philosophy of Science as a foundation and equips them with advanced knowledge in educational research. It provides them with concepts and skills in statistics and capacity to use current computer packages such as SPSS and NVIVO to analyse both quantitative and qualitative data pertaining to Philosophy of Science Education related research. Students are also required to relate their research to real issues that affect Philosophy of Science Education in different contexts and are required to validate their findings and recommendations at specified cites. Students are expected to understand and apply ethics considerations in line with laid out guidelines. Overall, the research module should prepare students with adequate knowledge and skills that enable them to embark on their Doctoral studies as the next stage of their academic pursuit.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

HTENG331 Industrial Attachment

Familiarization with actual plan organization and operations, training in practical engineering work for graduate engineers, exposure to as many of the following as possible; industrial management, plant maintenance, design and development, service/field engineering; working

with planning, manufacturing, production and marketing departments, industrial research.

HTENG431 Telecommunications Planning & Management
Telecommunications sector. Strategic Planning. Telecommunications Policies. By-Laws and Regulations governing telecommunications infrastructures. ITU Standards. Telecommunications Markets. Telecommunications Technologies. Telecommunications Costing. Project planning. Project management.

HTENG432 Communications Network Design
Business justification. System requirements. System analysis and network modelling. System design and implementation. System performance and evaluation. Market forecasts of demand on the network to the implementation of plans. Aspects of the design that influence the likely replacement of legacy telecommunications networks. Network performance parameters that impact on the quality of service perceived by customers. Layered view of network requirements, signalling, operation, service management, support systems and associated architectures.

HTENG433 Antenna Engineering
Fundamentals of antennas, arrays and mobile communications. Antenna types and design methods: dipoles, monopoles, loop antennas, small antennas, micro-strip antennas, slot antennas. Frequency independent antennas, horn and reflector antennas, earth station antennas. Antenna instrumentation and measurements. Direction finding antennas and systems. Propagation of radio waves, propagation mechanisms, propagation modelling Electromagnetics and RF propagation. Antenna fundamentals, RF safety, atmospheric effects. Antenna matching.

HTENG434 Project Management
Project proposal writing- types of proposals; Project definition, life cycle, and systems approach; Project scoping, work definition, and work breakdown structure (WBS); Project time estimation and scheduling using GANTT, PERT and CPM. Project costing, budgeting, and financial appraisal; Project control and management, using standard tools of cost and schedule variance analysis; project management use-case through practical, example projects; use of computers in project management, some software tools for PM e.g. MS Project; PM techniques e.g. PRINCE2.

HTENG435 Satellite Communication Engineering
Satellite communication fundamentals, Satellite orbits, orbit parameters, satellite types, ITU standards, satellite launching methods, orbital mechanics, radio wave propagation, satellite communication and interference, satellite lifetime and reliability, multiple access techniques, access protocols for data traffic, satellite applications

HCSE432 Embedded Systems
This module covers Embedded Systems concepts. Topics include nature of embedded systems, specific problems, special issues; role in computer engineering; embedded microcontrollers, embedded software; real time systems, problems of timing and scheduling; testing and performance issues, reliability; low power computing, energy sources, leakage; design methodologies, software tool support for the development of such systems; problems of maintenance and upgrade; networked embedded systems.

HCSE433 IoT and Cloud Systems Engineering
IoT and Cloud Computing. IoT Physical Devices and Endpoints: IoT physical servers and

cloud offerings: cloud storage models and communication Networks, Cloud technologies, Framework and Platforms. Internet of services, SOA, Grid computing, System types, architectures and models, applications in telecommunication systems, cloud systems examples, security management, Governance, legislation, economic environment.

HCSE435 Digital Media Engineering

Audio Engineering: concepts & standards, Overview of audio in the broadcast context, Computer-based audio systems, Software engineering and the MAX programming environment, Audio quality assessment and monitoring. Video Engineering: Concepts and standards, Overview of video in the broadcast context, Compression standards for broadcast, internet distribution and streaming, video transmission over IP networks.

HCSCI439 Fundamentals of Data Science and Big Data

Introduction: Big Data Overview, importance of data science, big data analytics in industry verticals. Data Analytics Lifecycle and methodology: Business Understanding, Data Understanding, Data Preparation, Modelling and Evaluation. Data exploration and pre-processing, Data Analytics: Theory and Methods, Unstructured Data Analytics, Data Visualization Techniques, Creating final deliverables

HTENG437 Next Generation Networks

Optical networks. Design and management of optical networks. Enabling technologies: Tuneable sources and tuneable filters, couplers, isolators, circulators, optical multiplexers, photonic switches, optical amplifiers, wavelength converters. Optical access networks: Optical access architecture, fibre-to-the-curb/home (FTTC/H), hybrid fibre coaxial (HFC) approaches, passive optical networks (PONs) for access. Photonic Packet Switching: Optical time division multiplexing (OTDM). Photonic switching node design, broadcast OTDM networks, OTDM testbeds (ATMOS, KEOPS, CORD, AON).

Optical network architectures: point to point, star, ring, mesh; system principles: modulation formats, link budgets, optical signal to noise ratio, dispersion, error rates, optical gain and regeneration. Wavelength division multiplexed networks; WDM components: optical filters, gratings, multiplexers, de-multiplexers, wavelength routers, optical cross-connects, wavelength converters, WDM transmitters and receivers; Wavelength switched/routed networks, ultra-high speed TDM, dispersion managed links, soliton systems; broadcast and distribution networks. Multiple access, subcarrier multiplexed light wave video networks. Optical LAN, MAN and WAN architectures: SDH/SONET, FDDI and Optical Synchronous Networks (OSN).

Next Generation Network (NGN) standards and protocols. NGN applications and architecture. NGN operations and management. NGN Network Case Study: Long Term Evolution (LTE): network architecture and interfaces, air interfaces and radio network. Backhaul considerations. Scheduling. Basic procedures. Mobility management and power optimization. Security architecture. Interconnection with other networks. Network planning aspects.

HTENG439 Radar Systems

Radar fundamentals: range, Doppler frequency, coherence, radar range equation; Continuous wave and pulsed radars. Radar detection, radar wave propagation, clutter and moving target indicator. Radar antennas and target tracking. Synthetic aperture radar: real versus synthetic arrays, design considerations, geometry.

HTENG440 Cryptology & Security in Telecommunications

Network security. Cryptography, cryptology and cryptanalysis, encryption, measures of

effectiveness of encryption algorithms, symmetric and asymmetric (public key) algorithms, standards, block ciphers and stream ciphers, public key algorithms, authentication, integrity and non-repudiation, key handling, multiple public key cryptography, secret sharing, SET (secure electronic transactions), e-commerce, cryptographic hardware/software requirements and tradeoffs.

HTENG441 Research Project

Each student must complete an approved design or research project during their final year, after which a comprehensive report on the project is submitted. The project must be of an investigative nature and the ability of the student to work independently will be assessed.
The process will involve selection of a problem, research on and planning of a possible solution. Modelling of possible solutions. Selection and testing of an optimal solution. Presentation of these preliminary results. Acquisition of components, testing of subsystems, construction of overall prototype. Writing the final report. Oral presentation of results.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN SPACE SCIENCE AND TECHNOLOGY (HSS)

Duration:	4 years
Credit load:	504
Minimum Credit load:	480
Maximum Credit load:	540
Maximum MBKs Credit Load:	380
ZNQF Level:	8

1.0 PURPOSE OF THE PROGRAMME

- The degree programme aims to strike a balance between the physical principles underlying Space Science Technology and its applications.
- Enhance knowledge in subjects covering the mechanics behind space stations, satellites, space crafts and transportation systems.
- The delivery of the degree programme is focused on teaching the use of mathematics, computing and experimentation to solve important real-world problems, and develop specialised knowledge of space science.
- Develops technical and professional skills according to individual needs and interests.
- Culminates with an in-depth individual research project in space science or space technology.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better.

2.1 Normal Entry

Two Advanced level passes to include Mathematics (Pure Mathematics/Further Mathematics/Mechanics), Physics.

2.2 Special Entry

At least a minimum of ND in a related field.

2.3 Mature Entry

See General Academic Regulations

3.0 PROGRAMME CHARACTERISTICS

3.1 Areas of Study

This includes classical mechanics, waves and fields, electromagnetism, special relativity, thermodynamics, quantum and atomic physics, space science and its technological aspects.

3.2 Specialist Focus

This is a unique programme that brings the understanding and applications of physics and the technology of space science.

3.3 Orientation

Research and innovation oriented, teaching and learning are academically and professionally oriented and focused on practical aspects.

3.4 Distinctive Features

The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 Employability

Graduates with the BSc Space Science and Technology Degree will seek employment as space scientists, aerospace engineers, research scientists, astrophysicists, Remote sensing scientist, geo/bio physicist, climatologists, teachers or professors, human space exploration consultancy, meteorologists and of course, astronauts.

4.2 Further Studies

Master's and doctoral studies in Space Science and Applied Physics, Cosmology and high-energy astrophysics,

Astroinformatics, Human space exploration, Solar physics, Planetary surfaces, Stellar astronomy.

5.0 PROGRAMME DELIVERY

5.1 Teaching and Learning Methods

Lectures, tutorials, practical work, group presentations, group work, research projects, industrial visits, Work-Related learning, research project, individual independent study.

5.2 Assessment Methods

Written and oral examinations, tests, laboratory reports, seminar presentations, industrial attachment report, final year research project report, and continuous assessments.

6.0 LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Identify the requirements and limitations of instrumentation for modern astrophysical observations.
- Apply cross-disciplinary theoretical knowledge and practical skills using the space scientific facilities such as satellites, rockets, landers, stratospheric balloons, radars, robots, space and remote sensing instruments.
- To carry out scientific investigations using space-based instrumentation, both individually and as a team.
- Demonstrate professional and ethical responsibility.
- Apply knowledge and skills to professionally manage projects.
- Utilize specialized computational software to analyse and solve problems related to space science and technological issues.

7.0. GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for a minimum of 11 modules.

7.2 A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other Faculties as long as the total credits for the degree does not exceed 540 credits.

7.4 A student's section of modules of a Degree programme is subject to the approval by the Dean of Science and Technology and the

Chairpersons of all relevant departments and to the following condition:

7.4.1 The module combination is feasible in terms of the timetable,

7.4.2 a student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme.

8.0 PROGRAMME ASSESSMENT

8.1 Coursework

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of final mark for practical courses.

8.2 Written Examinations

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses.

8.3 Practical Examination

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1. A student will repeat all failed modules if he/she fails at least 4 modules in a semester.

9.2. To proceed from level 1 to level 2 a student must have accrued at least 96 credits, provided he/she may not carry over more than three modules.

9.3 To proceed from level 2 to level 3 the student must have accrued at least 96 credits from the modules in level 2, provided he/she may not carry over more than three core modules to the next level.

9.4 To proceed from level 3 to level 4 the student must have satisfied the requirements for Work-Related Learning, including Work-Related Learning modules 1 and 2.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

To be awarded a Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 480 credits.

The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
HSS131	Atomic and Nuclear Physics*	12
HSS132	Introduction to Astronomy	12
HIPI133	Mechanics*	12
HMAT131	Calculus I*	12
HCSCI13	Principles of Programming Language*	12
CS131	Communication Skills	12

Level 1 Semester 2

HSS133	Electronic Circuits and Devices*	12
HSS134	Introduction to Meteorology*	12
HIPI135	Waves, Oscillations and Thermodynamics *	12
HIPI136	Electricity and Magnetism*	12
HMAT132	Linear Mathematics I*	12

Level 2 Semester 1

HSS231	Applied Optics and Photonics*	12
HSS232	Introduction to Space Science*	12
HSS233	Radio wave propagation and Antennas *	12
TCNP201	Technopreneurship*	12
GS231	Introduction to Gender Studies *	12

Level 2 Semester 2

HSS234	Orbital Mechanics	12
HSS235	Workshop Practice and Technology*	12
HSS236	Mathematical Modelling *	12
HSS237	Astrophysics*	12

Electives (Choose any two modules from any one option)

Option 1_Satellite Technology (Choose any two)

HSS238	Satellite Launch Systems	12
HTENG233	Electromagnetic Theory	10
HSS239	Navigation and space instrumentation	12

Option 2 _Space weather (Choose any two modules)

HSS240	Solar and stellar astronomy	12
HSS241	Space Weather and Space-Earth Interactions	12
HTENG233	Electromagnetic Theory	10

Level 3 Semester 1

HSS332	Work-Related Learning I*	40

Level 3 Semester 2

HSS333	Work-Related Learning 2	80

Level 4 Semester 1

HSS431	GIS, remote sensing and spatial analysis*	12
HSS432	Space Modelling and Software Development*	12
HSS433	Statistical Physics*	12
HIPI442	Quantum mechanics*	12

Level 4 semester 2

HSS434	Relativity and Particle Physics	12
HSS435	Flight Dynamics and Control	12
HSS438	Dissertation*	24

Electives (Choose any two modules from any one option: *Please note that if you have selected option one in level 2 electives then you must also choose option one here*)

Option one

HSS436	Aerospace Vehicle and Space Mission Design	12
HSS437	Satellite and Space Communication systems	12
HSS439	Space Mission Design and Development	12

Option two

HSS440	Astronomical spectroscopy	12
HTENG 434	Project Management	10
HTENG439	Radar Systems	10

MODULE SYNOPSES

HSS131 Atomic and Nuclear Physics

This module will introduce students to basic concepts in nuclear and atomic physics, and will provide an understanding of how the principles of quantum mechanics are used to describe matter at atomic and sub-atomic level. The following concepts will be covered: description of atoms, starting by the description of the hydrogen atom and covering other topics such as the effect of magnetic fields on an atom or X-ray spectra.

Properties of nuclei: Rutherford scattering. Size, mass and binding energy, stability and parity. Nuclear Forces: properties of the deuteron, magnetic dipole moment, spin-dependent forces* Nuclear Models: Semi-empirical mass formula M(A, Z), stability, binding energy B(A, Z)/A. Shell model, magic numbers, spin-orbit interaction, shell closure effects. Alpha and Beta decay: Energetics and stability, the positron, neutrino and anti-neutrino. Nuclear Reactions: Q-value. Fission and fusion reactions, chain reactions and nuclear reactors, nuclear weapons, solar energy and the helium cycle.

HSS132 Introduction to Astronomy

Naked-Eye Astronomy. Fundamental Astrophysical Concepts: The Nature of Energy and Matter, Space and Time. Solar Systems Galore: Our Own Solar System, and Other Planetary Systems. Stellar Structure: The Birth of Stars, and Stellar Alchemy5. Death of Stars: White Dwarfs, Neutron Stars, Black Holes. Extragalactic Astronomy: The Sea of Galaxies. Cosmology and the Early Universe. Life in the Universe

HIPI133 Mechanics

Inertia framework reference; motion in two dimensions, three dimensions. Particle dynamics, rotational dynamic systems, gravitation, mechanical oscillations, properties of matter, fluid mechanics. Special Relativity: Space-time frame reference Galilean transformation, simultaneity of event, Einstein special relativity theory and Lorentz transformation, time dilation and length contraction, velocity transformation, fluid mechanics.

HMAT131 Calculus I

The principle of mathematical induction. The real number system: Functions: Limits of functions. Continuity. Sequences: Differentiation: Derivatives of functions of a single variable. Integration: Method of substitution, integration by parts and reduction formulae, fundamental theorem of calculus

HSCI132 Principles of Programming Language

Introduction to operating System, Exploring print, building blocks, variables, input/output, operators, loops: for loops, while loop, do while loop, decisions, if statement, if else statement, else if construct switch statement, conditional statement, function, function that return a value using argument to pass data to a function, external variable, arrays and strings, pointers, structure, files and Introduction to C/C++ concepts, introduction to a database management system.

CS131 Communication Skills

Refer to the Communication Skills Department

HSS133 Electronic Circuits and Devices

The module introduces students to the analysis of DC circuits (including both steady state and transient behaviour) for resistors, capacitors and inductors. Techniques for the analysis of DC resistive circuits will be introduced including serial and parallel networks, mesh and node analysis and the principle of superposition. Equations for the response of a switched voltage across a capacitor and inductor will be developed considering an R-C and R-L circuits. The module then develops the analysis of AC circuits, introducing the more powerful methods associated with the use of complex numbers. The module is also designed to introduce the most common electronic devices and their applications in small-scale electronic systems. The module is based on formal seminar/lecture sessions followed by comprehensive practicals/tutorials, which provide an opportunity for students to gain experience in using and applying test and measurement equipment and simulators.

HSS134 Introduction to Meteorology

Definition of the atmosphere, atmospheric structure and composition, including trace gases, aerosols, dust and ash, and pollutants composition, vertical structure of the atmosphere in terms of its constituents, temperature and moisture content. Global mean seasonal patterns of atmospheric variables and their inter-relationships. The impacts of various weather and climate phenomena on society. Functions of meteorological service, service provision, key products and services, accuracy and quality of forecasts.

HIPI135 Waves, Oscillations & Thermodynamics

Oscillations and SHM; Wave motion and interaction, standing waves, sound propagation and effects; EM waves, ray model of light, the wave model of light. Temperature, thermal effects on solids, ideal gases; Kinetic theory of gases; 1st and 2nd law of thermodynamics. The Einstein model of a solid. The ideal gas. Temperature. Paramagnetism Mechanical equilibrium and pressure. Diffusive equilibrium and chemical potential. Engines and refrigerators. Free energy and chemical thermodynamics. Boltzmann- Maxwell statistics. Partition function. Elements of quantum statistics. Bose-Einstein and Fermi-Dirac distributions.

HIPI136 Electricity & Magnetism

This module introduces general concepts of electricity and magnetism to the undergraduate student. The module explores topics such as electrostatics; conductors, capacitors, and dielectrics; electric circuits; magnetic fields; and electromagnetism. Introductory differential and integral calculus is used throughout the module. The goal of this module is to get students acquainted with the basics of electricity and magnetism and its applications.

HMAT132 Linear Mathematics I

Complex numbers: De Moivre's theorem polynomials and roots of polynomial equations. Matrices and determinants: solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating factors, linear equation with constant coefficients.

HSS231 Applied Optics and Photonics

Introduction: Overview of light models: geometrical, electromagnetic and quantum. Basic concepts: refraction index, ray and optical length. Light propagation: rays in homogenous and heterogeneous media. Reflection and refraction laws. Fundamentals of Electromagnetic Optics: Electromagnetic waves characteristics. Electromagnetic spectrum. Plane and spherical waves. Intensity. Coherence. Polarization: Unpolarised, partially polarized and polarized lights. Types of polarized light: linear, circular and elliptical. Polarizers. Half- and quarter-wave plates (topic 4) Classical interaction of light with matter: Absorption. Chromatic dispersion. Scattering. Interferences and diffraction: Double-slit Young's experiment. Multiple-wave interferences. Difraction phenomena. Huygens-Fresnel Principle. Fresnel and Fraunhofer diffraction. Fraunhofer diffraction through different apertures: rectangular and circular apertures. Diffraction gratings. Imaging systems: Paraxial Optics. Principal planes and points. Focal planes and points. Spherical refractive surface. Mirrors. Prisms. Thin lenses. Thick lenses. Basic optical instruments. Quantum Optics: Photons. Basic processes between energy levels: absorption, spontaneous emission and stimulated emission.

HSS232 Introduction to Space Science

This module gives an overview of the contents and dimensions of space, and the basic tools for understanding it: celestial coordinate systems, celestial mechanics and satellite orbits, solar system physics and astronomy, and emission mechanisms relevant for these fields. This knowledge forms a solid foundation for all the space science and technology modules within the programme.

Solar system: the properties of the planets and moons, solar system physics including formation, comets and asteroids

Exoplanets: classification, detection methods, state-of-the-art

Stars: The Hertzsprung-Russell (H-R) diagram; H-R diagrams of star clusters; classification of stars from their position in the H-R diagram, The dependence of stellar properties and timescales on mass, The determination of the ages of clusters, Physical conditions within stars, Nuclear fusion, Stellar evolution, Core-collapse supernovae, White dwarfs, neutron stars and black-holes

Galaxies: Classification of galaxies by their appearance**,** Galaxy interactions**,** The

Milky Way**,** Active galaxies**,** Radio jets

Cosmology: The expansion of the Universe**,** The Cosmic Microwave Background**,** The Big Bang model**,** Measurement of the age and mass and the future evolution of the Universe

HSS233 Radio wave propagation and Antennas*

Introduction to antenna theory, Transmission Line theory related to antenna, Antenna design parameters, Antenna design parameters, Radiation integrals and auxiliary potential functions, Loop antennas, Classification of narrowband and broadband antennas, Aperture and reflector type antennas, Antenna arrays fundamentals, Micro strip antenna design, Smart antenna concept, Modes of propagation , Radio wave propagation in free space and types of waves, Propagation models

Maxwell's equations. Laplace and Poisson equations and their solution. Boundary conditions. Plane waves in a perfect dielectric; propagation in imperfect dielectric. Propagation in imperfect conductors, skin effect. Generalized wave equation, field distributions in rectangular waveguide. Radiation field, dipoles, radiation resistance, impedance, mutual impedance, linear arrays.

TCNP201 Technopreneurship

Introduction: Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur. Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation, How to form and register a small business in Zimbabwe. New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization. Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurshipTrade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe; Case studies of successful technopreneurs.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Science and Technology development contributions by gender. Students gain insight into accounts of gender studies in Science and Technology.

HSS234 Orbital Mechanics

Introduction to orbital mechanics, which is the basis for spacecraft mission design and is a key component of spacecraft engineering and operations. The basic principles of orbits and astrodynamics which informs the designer with options for selecting orbits, maneuvers, and mission profiles that impact the eventual spacecraft design are

explored. Introduction to the laws of Kepler and Newton; universal gravitation and integrals of motion. Fundamental concepts associated with the two-body problem and conics; orbital elements. Orbital maneuvers: orbit establishment; single impulse adjustments; multiple impulse transfers including Hohmann transfers, local gravity fields and flybys, Hoelker and Silber transfers, Lambert time-of-flight theorem, three-dimensional transfers; mission design issues. Orbital perturbations including Euler-Hill equations for two-close orbiters and some navigational issues.

HSS235 Workshop Practice and Technology

The module aims at imparting knowledge and skill components in the field of basic workshop technology. It deals with different hand and machine tools required for manufacturing simple mechanical and electronic components and articles. The module also aims to further the ability to communicate information by graphical means. This will be achieved through the ability to visualise and understand spatial relationships, and the competence to select and use appropriate graphical methods for representing design concepts.

HSS236 Mathematical Modelling

Infinite Series: Sequences of numbers and their convergence, algebra of convergent sequences, infinite series and their convergence, convergence tests for infinite series. *Fourier series*: Periodic and piecewise continuous functions, Fourier series, Fourier sine and cosine series, Fourier integrals and Laplace transform. *Differential Equations*: Introduction to differential equations, modelling and solution of first order differential equations, Bernoulli, Riccati and Clairaut equations, Application of first order differential equations and higher order differential equations with constant coefficients and their methods of solutions, Cauchy Euler equations, System of second order linear equations, method of indeterminate coefficient, method of variation of parameter, Second order differential equations with variable coefficient, series solution.

HSS237 Astrophysics

Physics of the Sun and Stars: blackbody radiation, the Planck, Stefan-Boltzmann and Wien laws, effective temperature, interstellar reddening; hydrogen spectral lines and Doppler effect; Hertzprung-Russell diagram; Freefall and Kelvin-Helmholtz time; nuclear fusion; basic stellar structure (hydrostatic equilibrium, equation of state); white dwarfs, neutron stars, and black holes. Planetary systems: Kepler's laws; Detection methods of extrasolar planets; search for life elsewhere; SETI. Galaxies: Star formation and the interstellar medium; stellar populations; galaxy rotation curves, mass and dark matter; Galaxy collisions; central engines.

Cosmology: Olber’s paradox, Hubble's Law; the age of the Universe; Evolution of the Universe: Madau diagram; Evidence for the Big Bang (blackbody radiation, nucleosynthesis); dark energy and the accelerating Universe.

HSS238 Satellite Launch Systems

Transmission channels, Linear modulation, Exponential modulation, Noise in analogue carrier wave transmission, Carrier wave systems, Base-band digital

transmission, Sampling and pulse coded modulation, Band-pass digital transmission, Error control coding, Information theory, Advanced system applications: Sampling, interpolation, and aliasing, Definition and Role of layered approach, SI and TCP/IP models, Physical layer: Digital transmission Systems. Data Link layer. Local Area Networks (Aloha, slotted Aloha, CSMA), Network layer: Packet Switching and routing.

Principles of Satellite Communication, Electromagnetic Spectrum, Interactions of Microwaves with Atmosphere, Communication Satellite System and Sub-Systems (Attitude and orbit control system), Telemetry, Tracking and Command (TT and C) Power Sub-System, Communication Sub-Systems, Spacecraft Antenna, Satellite Link Design (Basic transmission theory, RF Links, Optimization of RF Link, System Noise Temperature, Noise Figure, G/T Ratio, Down/Uplink Design and Complete Link Budget for LEO, MEO and GEO Satellites), Earth Station Parameters, Base Band Formation, Multiple Access Techniques, Post-launch space operations, Virtual Ground Station Development, Special Purpose Communication Satellites (DBS, INMARSAT, VSAT, SARSAT and LEO Satellites etc.).

HTENG233 Electromagnetic Theory

Maxwell's equations. Laplace and Poisson equations and their solution. Boundary conditions. Plane waves in a perfect dielectric; propagation in imperfect dielectric. Propagation in imperfect conductors, skin effect. Generalized wave equation, field distributions in rectangular waveguide. Radiation field, dipoles, radiation resistance, impedance, mutual impedance, linear arrays.

HSS239 Navigation and space instrumentation

Instruments and observing techniques used in solar system and astrophysical studies. Frequencies used by the different instruments from radio to gamma-rays are discussed. Different types of orbits, spacecraft, control of spacecraft, electromechanical system requirements for space-based applications, and space environment interactions with spacecraft systems. The space environment and the engineering approaches required to operate in it. The module equips the students with the knowledge of space missions design

HSS240 Solar and stellar astronomy

Introduction to the orbital mechanics, interior structure, surface features, atmosphere, and origin of the sun, planets, and solar system. Detection techniques and current census of extrasolar planets are studied as well as a dismodule on the possibility of life on other planets. Binary Systems and Stellar Parameters, The Classification of Stellar Spectra, Stellar Atmospheres. The Interiors of Stars, The Sun, The Process of Star Formation, Post-Main-Sequence Stellar Evolution, Stellar Pulsation, Supernovae, The Degenerate Remnants of Stars, Black Holes, and Close Binary Star Systems.

HSS241 Space Weather and Space-Earth Interactions

Introduction to the physical science of solar-terrestrial physics and technological impacts of space weather hazards. Concepts such as the Solar-terrestrial interactions and space weather which explores the sun, solar wind, and interplanetary magnetic field; space plasma physics and magneto- hydrodynamics; geomagnetic storms and auroral substorms; societal impacts of space weather; planetary magnetospheres. • Introduction • The Sun: Driver of Space

Weather • The Earth: Impact on its Environment • Technology: Basics and Impact • SPWX Sources: Institutes & Sensors • SPWX Practice: Forecasting & Products

HSS332 Work-Related Learning I

This year long internship allows the student to gain experience working in a professional environment and to apply knowledge gained in the first two years to acquire professional skills or competences. Students will be able to expand their professional network, convert academic knowledge into industrial skills, narrow down their potential careers and gain unforgettable life experiences.

The main objectives of the training module are:

i) To help the student to put into practice the basic experimental and theoretical skills which were learnt,
ii) To provide exposure to and experience of the industrial environment.
iii) To give the student an insight into the production activities of Zimbabwe's main industries,
iv) To embark on a research project under the joint supervision of a member from the department and a member of the training institution.

The Industrial attachment will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience and the industrial project.

HSS333 Work-Related Learning II

The Industrial attachment will be assessed jointly by the industrial supervisor and the academic supervisor as well as by written reports submitted by the student on the industrial attachment experience and the industrial project.

HSS431 GIS, remote sensing and spatial analysis

The science of remote sensing and its applications in meteorology, the basic principles of how radiation is emitted, transmitted and absorbed in the Earth's atmosphere and the techniques by which space-borne satellites can observe the chemical composition and properties of the Earth's atmosphere and surface. The course covers various aspects of Geoinformatics such as coordinate systems and projections, spatial data models and their structure, spatial database technology, data supply for geographic information systems: digital maps, digitizing, and surveying with geodetic instruments, basic cartographic methodology, basic geodesy; basic remote sensing, thematic classification of multispectral data.

HSS432 Space Modelling and Software Development

The module provides the students with enhanced programming and software development skills with emphasis on use of HPC for computational and analysis demands associated with space data analysis. Students are equipped with knowledge and techniques on how to analyze problems and scenarios related to the different space environments in order to provide and simplify application of space technologies and exploration. Different modelling techniques applicable to space exploration and data acquisition are discussed as well model optimizing techniques

HSS433 Statistical Physics

The module treats important aspects of Statistical Mechanics, based on the fusion of two approaches: Statistical Physics/Mechanics combined with Classical Thermodynamics. Both approaches are complementary and they can enforce and enlighten each other. The covered material will be useful for all advanced Physics students because it covers important subjects related to Statistical Mechanics, which any serious Physics student should know. Given its scope, the matter will be useful for several other branches in Physics: Atomic Physics, Nuclear Physics, and Solid State Physics. The module is part of the general training in the physical sciences. Students in Chemistry may find the module useful as well.

HIPI442 Quantum mechanics

An introduction to the quantum theory, as formulated in the 1920's and 1930's by Born, Bohr, Schrödinger, Heisenberg, and others. Discussion s include how quantum theory arose in the face of certain discrepancies between 19th-century classical theory and experiment. The module also include equipping the students with knowledge of a set of mathematical tools needed to formulate problems in quantum mechanics, introducing methods of theoretical physics required to solve them as needed. Topics covered include Recap of wave motion, Quantum superposition and probability, Heisenberg's uncertainty principle and applications, Indeterminacy, complementarity principle, Bohr's correspondence principle, Particles and atomic systems, Angular momentum, Bound systems. This module also covers quantum physics applications drawn from modern physics including the general formalism of quantum mechanics, harmonic oscillator, quantum mechanics in three-dimensions, angular momentum, spin, and addition of angular momentum**.**

HSS434 Relativity and Particle Physics

The module provides an overview of modern particle physics stressing fundamental concepts and processes: An introduction to the Standard Model and its components. Antiparticles. Symmetries and conservation laws and their significance in particle physics. Hadron-hadron interactions. The quark model including spectroscopy. Quantum Chromo Dynamics (QCD). Electromagnetic interactions - form factors. The Parton model and deep inelastic scattering - structure functions. Weak interactions including beta decay and Cabbibo- Kobayashi-Maskawa mixing. The unified electroweak interaction, W, Z and the Higgs boson. Beyond the Standard Model: the unification of strong and electroweak interaction, supersymmetry, neutrino oscillations and more.

HSS435 Flight Dynamics and Control

The module provides knowledge on the design and operations of the space vehicle. It covers topics such as Flight vehicle rigid-body equations of motion; linearization via small perturbation techniques. Trim analysis, static and dynamic stability, aerodynamic stability derivatives and control effectiveness. Vehicle transfer functions, stability augmentation, and aircraft handling qualities.

HSS438Dissertation

Provides students with opportunity to design, undertake or conduct an independent

piece of research study related to their programme of study under the guidance of a supervisor who is usually a member of academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

HSS436 Aerospace Vehicle and Space Mission Design

Design and fabrication of space vehicles and missions requires a wide-ranging approach including conceptual design, vehicle and missions' simulation, budgeting and operation analysis and sustenance and maintenance practices. The module provides knowledge and skills on designing and analyzing space vehicles and missions. This involves combined use of theoretical, computational (including simulations) and experimental techniques using space design tools. Mission design will range from micro satellites to complex systems such as national missions. The module will also equip students with knowledge of aircraft and avionic system design, propulsion as well as stability and performance assessments for both civil and military sectors.

HSS437 Satellite and Space Communication systems

Overview of the issues associated with the design and provision of modern satellite communications systems. Transmission channels, Linear modulation, Exponential modulation, Noise in analogue carrier wave transmission, Carrier wave systems, Base-band digital transmission, Sampling and pulse coded modulation, Band-pass digital transmission, Error control coding, Information theory, Advanced system applications: Sampling, interpolation, and aliasing, Definition and Role of layered approach, SI and TCP/IP models, Physical layer: Digital transmission Systems. Data Link layer. Local Area Networks (Aloha, slotted Aloha, CSMA), Network layer: Packet Switching and routing.

Principles of Satellite Communication, Electromagnetic Spectrum, Interactions of Microwaves with Atmosphere, Communication Satellite System and Sub-Systems (Attitude and orbit control system), Telemetry, Tracking and Command (TTandC) Power Sub-System, Communication Sub-Systems, Spacecraft Antenna, Satellite Link Design (Basic transmission theory, RF Links, Optimization of RF Link, System Noise Temperature, Noise Figure, G/T Ratio, Down/Uplink Design and Complete Link Budget for LEO, MEO and GEO Satellites), Earth Station Parameters, Base Band Formation, Multiple Access Techniques, Post-launch space operations, Virtual Ground Station Development, Special Purpose Communication Satellites (DBS, INMARSAT, VSAT, SARSAT and LEO Satellites etc.).

HSS439 Space Mission Design and Development

Introduction to space missions. Community science priorities, introductory and detailed science investigations. Introduction to science topics in Heliophysics, Astrophysics, and Planetary Science. Proposal science topic definition for semester mission/proposal projects. Science Traceability Matrix. Requirements definition, measurements and instrumentation. Science instrument selection and design.

Spacecraft design and subsystems. Space environment consideration, Space Mission Development. Spacecraft data reduction and analysis. Mission schedule and budget development. Team roles; proposal layout, responsibilities, and writing.

HSS440 Astronomical spectroscopy

A lab-based introduction to stellar spectral classification, spectroscopic parallax measurement, orbit analysis of spectroscopic binaries, redshift measurements of galaxies and CCD imaging techniques. Students will use telescopes to obtain spectroscopic data and will use digital CCD cameras to image star clusters and nebulae.

HSS441 Project Management

Project proposal writing- types of proposals; Project definition, life cycle, and systems approach; Project scoping, work definition, and work breakdown structure (WBS); Project time estimation and scheduling using GANTT, PERT and CPM. Project costing, budgeting, and financial appraisal; Project control and management, using standard tools of cost and schedule variance analysis; project management use-case through practical, example projects; use of computers in project management, some software tools for PM e.g. MS Project; PM techniques e.g. PRINCE2.

HTENG439 Radar Systems

Basic Principles: Radar equation, Radar Cross section. CW Radar, FMCW Radar. Pulsed Radar Principles. Clutter Analysis, MTI Improvement Factor, Pulsed Doppler Radar. Tracking Radar, Angular resolution, Monopulse Technique. Detection Theory: Match Filtering, Radar Ambiguity Function. Imaging Radar: Resolution Concept, Pulse Compression. Synthetic Aperture Processing, ISAR Imaging. Probability of false alarm and Detection, Modified Radar Range Equation with Swerling Models. Ground Penetrating Radar for close sensing. Radar Tomography and Radar based Microwave Imaging. Emerging and Modern Applications of Radar Principles.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN APPLIED SCIENCES AND NANOTECHNOLOGY (HNN)

Duration:	4 years
Credit load:	502
Minimum Credit load:	480
Maximum Credit load:	540
Maximum MBKs Credit Load:	380
ZNQF Level:	8

1.0 PURPOSE OF THE PROGRAMME

- The degree programme aims to explore the cluster of technologies that harness the unique properties and functions of nanoscale systems.
- The delivery of the degree programme is focused on teaching Nanotechnology and related topics through the how-and-why approach of a physicist. There is an emphasis on developing flexibility in thinking with the ability to synthesize new ideas whilst further advancing practical skills.
- The degree programme particularly suits students who are interested in careers which leads to devices and technologies that can increase the efficiency of energy consumption, help clean the environment, and solve major health problems.
- Applied Sciences and Nanotechnology, constitute the core of the programme and provide a systematic and comprehensive treatment on both fundamental principles and practical applications.

2.0 ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better

2.1 *Normal Entry*

Two Advanced level passes in Chemistry or Mathematics and Physics

2.2 *Special Entry*

At least a minimum of ND in the relevant field

2.3 *Mature Entry*

See General Academic Regulations

3.0 PROGRAMME CHARACTERISTICS

3.1 *Areas of Study*

Physics, NanoPhysics, Nanomaterials, Quantum Physics, Solid State Physics, Thermal Physics, Materials Science and Waves & Optics

3.2 *Specialist Focus*

Nanotechnology engineers are at the forefront of research and development related to a cluster of technologies that harness the unique properties and functions of nanoscale systems.

3.3 *Orientation*

Research and innovation oriented. Teaching and learning are academically and

professionally oriented and focused on practical aspects

3.4 *Distinctive Features*

The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 *Employability*

Graduates with the BSc in Applied Sciences and Nanotechnology will seek employment in Industry, Commerce and Research as:

- Applications Engineer,
- Director of Product Marketing,
- Manufacturing Technician,
- Director of Research,
- Optical Engineer,
- Technical Program Manager,
- Research Scientist R& D Scientists;
- Hi-Tech Consultants; and College/University Lecturers.

4.2 Further Studies

Master's and doctoral studies in Applied Physics, Nanotechnology, Engineering Science, Material Science and related.

5.0 PROGRAMME DELIVERY

5.1 Teaching and Learning Methods

Lectures, tutorials, practical, group presentations, group work, research projects, industrial visits, Work-Related learning, research project, individual independent
study

5.2 Assessment Methods

Written and oral examinations, tests, laboratory reports, seminar presentations, industrial attachment report, final year research project report, continuous assessments

6.0 LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Produce Nanomaterials and Nanostructures and their related technology
- Apply research skills and knowledge to solve industrial, manufacturing, complex problems
- Demonstrate professional and ethical responsibility
- Apply technopreneurship skills to start new technology-driven businesses

- Apply knowledge and skills to professionally manage projects
- Utilize specialized computational software to analyse and solve problems related to industrial production phenomena and technological issues.

7.0. GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for a minimum of 11 modules.

7.2 A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other departments/Faculties as long as the total credits for the degree does not exceed 540 credits

7.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons of all relevant departments and to the following conditions:

7.4.1 The module combination is feasible in terms of timetable,

7.4.2 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board responsible for a programme may designate, in the special Regulations for the programme, certain modules as core modules for that programme. Core modules are the modules, which are considered to be essential for qualification in a particular programme.

8.0 PROGRAMME ASSESSMENT

8.1 Coursework

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of the final mark for practical courses.

8.2 Written Examinations

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses

8.3 Practical Examination

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1. A student will repeat all failed modules if he/she fails at least 4 modules in a semester.

9.2. To proceed from level 1 to level 2 a student must have accrued at least 96 credits, provided he/she may not carry over more than three

modules.

9.3 To proceed from level 2 to level 3 the student must have accrued at least 96 credits from the modules in level 2, provided he/she may not carry over more than three core modules to the next level.

9.4 To proceed from level 3 to level 4 the student must have satisfied the requirements for Work-Related Learning, including Work-Related Learning modules 1 and 2.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 480 credits

11.2 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0. PROGRAMME STRUCTURE

Level 1 Semester 2

Module	**Credits**
HNN131 Introduction to Nanotechnology*	12
HNN132 Chemistry of Materials	12
HCSCI132 Principles of programming Language*	12
HIPI133 Mechanics*	12
HMAT131 Calculus I*	12
CS131 Communication Skills*	12

Level 1 Semester 2

HNN133Analytic Spectroscopy*	12
HIPI134 Modern Physics*	12
HIPI135 Waves, Oscillations & Thermodynamics*	12
HMAT132 Linear Mathematics I*	12

HIPI136 Electricity & Magnetism*	12
Level 2 Semester 1	
HNN231 Nanotechnology Practice*	12
HIPI231 Computational Physics*	12
HIPI236 Introduction to Condensed Matter Physics	12
HMAT111 Ordinary Differential Equations*	12
TCNP201 Technopreneurship*	12
GS231 Introduction to Gender Studies*	12
Level 2 Semester 2	
HNN232 Semiconductor device fabrication	12
HNN233 Computational Nanoscience*	12
HTENG233 Electromagnetic Theory*	10
HNN234 Solid State Physics	12
Electives (**choose any one module)**	
HNN235 Nanotechnology Laboratory	12
HNN236 Material Science	12
HIPI235 Optics and Optical Instrumentation	12
Level 3 Semester 1	
HNN332 Work-Related Learning I	40
Level 3 Semester 2	
HNN333 Work-Related Learning II	80
Level 4 semester 1	
HNN431 Nanomaterials and Nanotechnology*	12
HIPI442 Quantum mechanics*	12
HNN432 Fabrication of nanostructures*	12
HNN433 Characterization of Nanostructures and Nanomaterials*	12
HTENG434 Project Management	12
Level 4 semester 2	
HNN434 Carbon nanomaterials*	12
HNN435 Thermal properties at Nanoscale*	12
HNN438 Dissertation*	24
Electives (**choose any two modules)**	
HNN436 Nanotechnology in drug delivery	12
HIPI439 Lasers and Laser Technology	12
HNN437 Energy conversion and storage	12
HNN439 Environmental Nanotechnology.	12

Module Synopsis

HNN131 Introduction to Nanotechnology

Introduction to Nanotechnology aims to provide a broad overview of fundamental principles and current research directions in nanoscience and nanotechnology.
Specifically the course covers: Fundamental physical scaling laws applied to understanding the properties of materials at the nanometre scale. Experimental and computational characterisation of nanomaterials. Surfaces and interfaces in nanotechnology. Other specialist topics in nanotechnology.

HNN132 Chemistry of Materials

This course provides an introduction to Materials Chemistry. Scientific topics include: INORGANIC CHEMISTRY: Atomic structure and bonding. ORGANIC CHEMISTRY: Nomenclature, structure, bonding and reactions of hydrocarbons; reactions and properties of haloalkanes, alcohols, ethers and amines. PHYSICAL CHEMISTRY: Thermochemistry. Equilibria. Thermodynamics. Electrochemistry.

HIPI133 Mechanics

Inertia framework reference; motion in two dimensions, three dimensions. Particle dynamics, rotational dynamic systems, gravitation, mechanical oscillations, properties of matter, fluid mechanics. Special Relativity: Space-time frame reference Galilean transformation, simultaneity of event, Einstein special relativity theory and Lorentz transformation, time dilation and length contraction, velocity transformation, fluid mechanics

HCSCI132 Principles of Programming Language

This module examines the concepts and structures governing the design and implementation of programming languages. It presents an introduction to the concepts behind compilers and runtime representations of programming languages; features of programming languages supporting abstraction and polymorphism; and the procedural, functional, object-oriented, and concurrent programming paradigms. Programs are required in languages illustrating each of these paradigms

HMAT131 Calculus I

The principle of mathematical induction. The real number system: Functions: Limits of functions. Continuity. Sequences: Differentiation: Derivatives of functions of a single variable. Integration: Method of substitution, integration by parts and reduction formulae, fundamental theorem of calculus.

CS131 Communication Skills

Refer to communications department

HNN133 Analytical Spectroscopy

Analytical Spectroscopy, will enable you to identify the structures of simple organic molecules using chemical tests combined with NMR spectroscopy, IR spectroscopy and Mass Spectrometry.
It also introduces you to methods of elemental analysis using atomic spectroscopy

techniques. This is a very hands-on course and the material in the lectures and your laboratory work will be closely.

HIPI134 Modern Physics

The nuclear atom: Thompson and Rutherford; The Bohr Model of hydrogen atom; Quantisation of energy; Line spectra; Rydberg constant. Black Body radiation; radiation and quantization of light; Plank distribution; Compton effects; Continuous and discrete energy spectrum; Frank- Hertz experiment; spontaneous and stimulated emission; Wein and Plank distribution.Wave particle duality; Uncertainty principle; Nuclear Physics.

HIPI135 Waves, Oscillations & Thermodynamics

Oscillations and SHM; Wave motion and interaction, standing waves, sound propagation and effects; EM waves, ray model of light, the wave model of light. Temperature, thermal effects on solids, ideal gases; Kinetic theory of gases; 1st and 2nd law of thermodynamics. The Einstein model of a solid. The ideal gas. Temperature. Paramagnetism Mechanical equilibrium and pressure. Diffusive equilibrium and chemical potential. Engines and refrigerators. Free energy and chemical thermodynamics. Boltzmann- Maxwell statistics. Partition function. Elements of quantum statistics. Bose-Einstein and Fermi-Dirac distributions

HMAT132 Linear Mathematics I

Complex numbers: De Moivre's theorem polynomials and roots of polynomial equations. Matrices and determinants: solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating factors, linear equation with constant coefficients.

HIPI136 Electricity & Magnetism

Static charges: Electric charge, Coulomb's law, Electric fields, moving point charges in an electric field, lines of force, electric dipoles, Electric flux, Gauss's law, Electric potential, capacitors, energy in a capacitor and dielectrics. Charge distributions, force vector at a point; superposition. Current and Resistance: Conductors, current density, Kirchhoff's laws, DC network theorems, Wheatstone bridge, pd and resistance, capacitance and capacitors. Magnetic Fields: Force between currents, magnetic fields, magnetic flux density, magnetic intensity, magnets in magnetic fields, magnetic dipole moments, Torque on a current loop, motion of charge in a magnetic field, Bio-Savart law, Ampere's law, Induction and Inductance, Faraday's and lentz's law, self and mutual inductance, energy storage in inductors in magnetic fields, Superconductivity, Principle of electric motor and generator, Laurenz's force, Alternating currents: LR, LC and LCR circuits, ac circuts, phasor notation, power in ac circuits and ac networks, frequency filtering and tuning circuits, transformer, capacitor and inductor circuit in a generator, Eddy currents. Time constants and waveforms.

HNN231 Nanotechnology practice

The course will cover: Applications of neutron, X-ray and light scattering to nanotechnology. Advanced electron microscopy and surface analysis techniques. Computational methods in nanotechnology.

HIPI 231 Computational Physics

Computing has become part of the core toolset of the professional physicist. Through problem-solving, this module introduces their use for modelling and simulation of physical systems. Good programming practice is emphasised throughout the module so you should begin to appreciate the discipline of programming beyond that of developing simulations of simple physical systems. Assessment is based on written laboratory reports/mini projects and computer programming tasks. Hands-on application of Scientific Software for Numerial Analysis; Simulation and Modelling of Dynamic Systems and control processes- Programming, Simulation and Modelling of dynamic systems and control systems through Scientific Computing in Maple and Matlab, COMSOL, MULTIPHYSICS/ C++and other specialist Packages.

HIPI236 Introduction to Condensed Matter Physics

The course covers the structure of solids and other phases of condensed matter. The first part will give an introduction to materials which are not solid, or crystalline, and neither liquids, but are still condensed matter, such as colloidal fluids or glasses, gels and liquid crystals. The remainder of the course will be focused on providing a basic understanding of the electrical, thermal and mechanical properties of solids, and in particular metals and semiconductors. The aim is to give a firm grounding in the core concepts.

HMAT111 Ordinary Differential Equations

This module aims at providing a broad introduction to differential equations (ordinary and partial), First order differential equations: Second order differential equations: 3D coordinate systems: Introduction to 2nd order partial differential equations, their solutions written in terms of a Fourier series; the one dimensional wave equation; the time dependent Schrodinger equation; the Laplace equation in two dimensions; the Diffusion equation. Fourier Series:. Fourier transforms: mathematical definition, relationship with Fourier series, inverse Fourier transform, Fourier transform pair, Fourier integrals; definition of odd and even functions. Dirac delta-function; Fourier transform of a Gaussian; Fourier transforms applied to diffraction, Young's slits Lagrange's Equations including the following: Generalised coordinates and forces, Holonomic constraints, The Lagrangian and Lagrange's equations.

TCNP201 Technopreneurship

Introduction: Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur.

Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation, How to form and register a small business in Zimbabwe.

New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization.

Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven

technopreneurshipTrade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe; Case studies of successful technopreneurs.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Science and Technology development contributions by gender. Students gain insight into accounts of gender studies in Science and Technology.

HNN232 Semiconductor Device Fabrication

The course will focus on the fabrication procedures which are used in industry to produce silicon semiconductor devices. The topic material will be supported where practicable by laboratory/clean room demonstrations and student activities relevant to the various processes presented in lectures.

The subject material is divided into five modules as follows:

Introduction of basic fabrication processes, materials and clean room protocols. Photolithography, masking and patterning of materials. Semiconductor oxidation and diffusion. Ion implantation and thermal processing. Metal and dielectric deposition for contacts, gate and interconnect engineering.

HNN233 Computational Nanoscience

This course will provide the fundamentals of computational problem-solving techniques that are used to understand and predict properties of nanoscale systems. Emphasis will be placed on how to use simulations effectively, intelligently, and cohesively to predict properties that occur at the nanoscale for real systems.

HTENG233 Electromagnetic Theory

Maxwell's equations. Laplace and Poisson equations and their solution. Boundary conditions. Plane waves in a perfect dielectric; propagation in imperfect dielectric. Propagation in imperfect conductors, skin effect. Generalized wave equation, field distributions in rectangular waveguide. Radiation field, dipoles, radiation resistance, impedance, mutual impedance, linear arrays.

HNN234 Solid State Physics

A module in the physics underlying solid state electronic and optical devices. The module presents an introduction to the electrical and optical properties of insulators, semiconductors and metals, including crystal structure, band theory, and electron transport. Ewald’s sphere, interplanar distance of reciprocal lattice, Methods of x-ray diffraction lane’s method, Rotating crystal and powder method. Brillouin zones.

This is applied to obtain a physical understanding of the physics governing the behaviour of diodes, field effect and bipolar transistors, and other discrete optical and electronic devices.

HNN235 Nanotechnology laboratory

Nanotechnology Methodology provides a framework of laboratory sessions and tutorials which aims to further your understanding and give practical experience in

solving Nanotechnology problems. A series of lectures will also be given to provide the background knowledge to laboratory work into context.
Specifically the course covers: Preparation of nanoscale/nano-featured materials. Analysis of nanoscale/nano-featured materials using electromagnetic radiation and standard and sophisticated instrumental techniques.

HNN236 Material Science

Classification of crystalline solids into insulators. Interatomic forces, metallic, ionic, covelent and van der Waals forces. Crystal systems and brais lattices substitutional and intertial solutions. Smart material systems (piezoelectric; piezorestrictive;magnetoresistive; etc); Hume- Rotary rules for solid solutions. Phase diagrams, ceramic phase systems- (eutectic, hypher eutectic and hypho eutectic compositions. Alloys and their applications lattice defects, point defects. Deformation of metals: Fracture,Corrosion: Corrosion prevention methods:

HIPI235 Optics and Optical Instrumentation

Waves- snapshot and history graphs. wavelength, period, displacement, amplitude, speed, phase and phase difference, wave superposition, reflection at a boundaries and discontinuities, coherence, coherence length Interference in thin films, thin film equation. Huygen's Principle Single slit, Young's double slit experiment, and diffraction grating. Michelson Interferometer Fresnel and Fraunhofer Diffraction Polarisation, Malus's Law, polarising filter. Monochromator Spectrometer.

WRL332 WORK-RELATED LEARNING

WRL333 WORK-RELATED LEARNING

HNN431 Nanomaterials and Nanotechnology

Nanomaterials is a field of study that takes fundamental science approach to nanotechnology. It covers materials with special properties stemming from their nanoscale size regime, 1-100 nm.
This course covers Feynman’s vision. Synthesis and Fabrication: Top down vs. bottom up techniques, nucleation theory, surface energy and stabilization. Characterization: Composition, structure, porosity, crystallinity, single vs. ensemble measurements. Examples: General classification (zero – two dimensional and assembled nanostructures), materials composition/function (metals, metal oxides, semiconductors, carbon, biological). Size Dependent Chemical and Physical Properties: Electrical, optical, catalytic, magnetic, thermodynamic, why purification is needed Applications: Electrical, optical, catalytic, magnetic, thermodynamic, purification, sensing, biology, medicine, solar cells, etc. (literature) Implications: Environment, health, and safety as well as impacts on policy, society, and education.

HIPI442 Quantum Mechanics

Origins of Quantum Mechanics. Introductory Wave Mechanics: Schrodinger’s equation and its physical interpretation. Energy eigenvalues and eigenfunctions. Potential wells, harmonic oscillator Step potentials and barrier potentials & tunnelling. Operators, particle in a 2-D box. The Hydrogen Atom: Radial solutions, Radial probability densities, energy level. Separation of variables. Spherical harmonics & angular momentum.

HNN432 Fabrication of Nanostructures

The thermodynamic driving force and the kinetics of nucleation and growth of nanoparticles focusing on precipitation from solutions. Mechanisms for nucleation and crystal growth along with calculations of nucleation and growth rates define the basis for the design of different particle populations. The classical crystallization theory is the fundamental theoretical. Synthesis and functionalization of metallic and polymeric nanoparticles. An understanding of how growth can be controlled by tuning synthetic parameters. Functionalization of particle surfaces. Solution-based characterization techniques. Methods for the fabrication of catalysts and catalyst supports based on precipitation, e.g., sol-gel and colloid-based fabrication. Significance of particle and pore size (Au, Co, Ni- catalysts and carbon nanofibres (CNF)). An introduction to the catalytic model systems and surface science and their experimental and theoretical applications.

HNN433 Characterization of Nanomaterials and Nanostructures

Basic concepts. Synthesis of materials. Diffraction and scattering techniques (XRD, SAXS, DLS, SANS). Electron microscopy, Scanning Probe Microscopy
Spectroscopy: UV-Vis, IR, Fluorescence, ultrafast spectroscopy,
NMR, Raman, SERS Confocal technique: Confocal Raman,
Confocal FL, FCS
Surface Characterization:
BET, XPS Electrochemistry:
CV, Electrochemical analysis
Magnetic: Superparagnetism
Basic concepts of biochemistry: DSC,
ITC, TGA/DTA, CD Separation
techniques: HPLC, GC, GCMS
Mass spectroscopy, Applications of nanotechnology

HNN434 Carbon Nanomaterials

Introduction to Carbon nanomaterials and their physical properties: C-C bonding, Types of carbon fullerenes, Crystal structure of selected carbon nanomaterials: CNT, Graphene, nano crystalline diamond, Electronic Band structure of CNT and Graphene, Electrical transport properties of graphene & CNTs, Thermal transport and electron-phonon interaction, Effect of structural disorder, Elastic Properties, Optical response.

Synthesis and Characterizations: Brief introduction to various techniques of CNTs and Graphene- Arc Discharge, Thermal CVD, Microwave plasma CVD, Laser Ablation; Growth mechanism; Special synthesis techniques- Vertical aligned growth of CNT, Selective area growth of CNT, single walled CNT growth, Large area graphene synthesis, nano-patterning on graphene, Characterization of Carbon Nanomaterials: Raman spectroscopy, SEM, HRTEM, AFM, STM. Applications of CNTs and Graphene: Field emission, Electronic Devices, Optoelectronic devices, Chemical and Biological sensors.

HNN435 Thermal properties at Nanoscale

Heat transfer mechanism: Laws of macroscopic heat transfer and their limits, Electrons and Phonons, thermal properties of solid and fluids and thermal radiation.
Simulation methods: Solutions of the Boltzmann and Maxwell equations, Technique of Molecular dynamics simulations, First-principles based approaches.

Experimental techniques: Introductions to different forms of near-field microscopy and discuss their application in thermal science, Photothermal microscope and reflectometry, Hybrid technique and multipurpose microscopes.

HTENG434 Project Management

Project proposal writing- types of proposals; Project definition, life cycle, and systems approach; Project scoping, work definition, and work breakdown structure (WBS); Project time estimation and scheduling using GANTT, PERT and CPM. Project costing, budgeting, and financial appraisal; Project control and management, using standard tools of cost and schedule variance analysis; project management use-case through practical, example projects; use of computers in project management, some software tools for PM e.g. MS Project; PM techniques e.g. PRINCE2.

HNN436 Nanotechnology in drug delivery

Fundamentals of Nanotechnology in Drug Delivery, Nanotechnology in Drug Delivery: Past, Present and Future; Conventional delivery of pharmaceuticals and its limitations; Significance of nano-size in Pharmacy,

Pharmacokinetics and Bio distribution: Introduction, Definitions, concepts of Absorption, Distribution, Metabolism and excretion, concentration time profile,

Nano carriers for Drug Delivery: Design, fabrication Properties and applications Closed bilayered system:

Controlled and Targeted drug delivery: Programmed Drug Delivery Systems-principles of sustained, controlled, pulsatile and triggered release of drug from nanoparticles.

Nano toxicology and regulatory issues- Potential Risks and Remedies Human health and safety- Interaction of nanomaterials with biological systems, Toxicology of nanoparticles- in-vitro and in-vivo, Complications with Nanotoxicity studies, environmental impact of nano-materials, Regulatory Toxicology. Bio-imaging tools Overview of tools and tasks in various biological and biomedical imaging modalities, such as fluorescence microscopy and electron microscopy (SEM, TEM).

HIPI439 Lasers and Laser Technology

Introduction to laser physics: Classical electron oscillator model. Fundamental optical processes: absorption, spontaneous emission, stimulated emission. Basic principles of lasers: Gain processes, optical feedback, spectral linewidth. Laser performance: Frequency and intensity distribution, temporal behavior, pulsed systems. Laser systems: Solid, liquid and gaseous systems. Applications of lasers: CD players, holography, optical fibre communications, gas sensing and LIDAR, medical applications. Pumping processes: optical pumping, radiative and transfer efficiency, Quantum efficiencies for absorption and power, Electrical pumping: electron impact excitation, ionization balance equation, pump rate calculations. Optical systems; Matrix formulation of geometrical optics, Fabray-Parot interferometer, Fox and Lee treatment. Confocal resonator, Gaussion beam propagation and ABCD Law. Stability condition, unstable resonators, hard edge, unstable resonators. Resonators design, and transformation of impedance multilayer optical systems. Applications in optical instruments, metrology & medicine.

HNN437 Energy conversion and storage

Photovoltaic Cell Introduction, Semiconductor Junctions, Solar Cell Parameters and

Equivalent Circuit, Losses and Efficiency Limits,
Electrocatalysis Introduction; Hydrogen from different routes and viability, electrocatalysis & catalysis; effect of potential on rate of reaction; rates of complex processes;
Photocatalytic Water Splitting Introduction, Basic Principles of Photocatalytic water splitting, Experimental method for water splitting
Waste Energy Harvesting Introduction, Special topic on renewable energy, Energy sources, Mechanical energy harvesting: Piezoelectric effect, Nanopiezotronics, Triboelectric effect, Thermal energy harvesting: Thermoelectric effect, Pyroelectric effect, Nanogenerator, Self-powered Systems, Device fabrication.
Fuel Cell Design: Types of fuel cells, their operational principles and basic electrochemistry for understanding key processes in the fuel cell.
Energy Storage Batteries: Battery Applications and Parameters, History of Batteries, Principles of Batteries, Lead Acid Batteries, Lithium Ion Batteries, Lithium recycling, Beyond Lithium Ion Batteries.

HNN438 Dissertation

Provides students with opportunity to design, undertake or conduct an independent piece of research of study related to their programme of study under the guidance of a supervisor who is usually a member of academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

HNN439 Environmental Nanotechnology

Nanotechnology and the Environment: An
Overview Nanomaterials: Preparation,
Fabrication, and Characterization
Photocatalysts for Applications in Environmental Treatment and Sensing. Carbon Nanomaterials for Environmental Applications. Nanoiron and Nanoscale Bimetals for Subsurface Remediation.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN INDUSTRIAL PHYSICS AND INSTRUMENTATION (HIPI)

Duration:	4 Years
Credit load:	534
Minimum Credit load:	480
Maximum Credit load:	540
Maximum MBKs Credit Load:	380
ZNQF Level:	8

1.0. PURPOSE OF THE PROGRAMME

12 To explore the physical principles underlying modern industrialization and instrumentation and the use of industrial physics in problem solving, improvising existing systems and designing systems for modern industry.

13 To meet the rapidly changing demands of modern economy and society where highly skilled and adaptable graduates with a strong grounding in fundamental Physics concepts are equipped with industrially-focused training and expertise in areas such as Instrumentation and Automation**.**

14 Equip graduates with the skills needed for employment physics, with a particular emphasis on the applications of physics in industrial and environmental contexts – also with the inclusion of science underlying renewable energy, sustainable technologies, medical physics etc.

15 To capacitate graduates with industrial and instrumentation knowledge to synthesize new ideas, interact with wide range of industrial technologies whilst further advancing practical skills in the industry and application of knowledge for the betterment of industry and the country at large.

2.0. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects including English Language, Mathematics and a Science subject at grade C or better

2.1. *Normal Entry*

Three Advanced level passes to include Mathematics (Pure Mathematics/Additional Mathematics), Physics and a third science subject.

2.2. *Special Entry*

At least a minimum of NC, ND and HND

2.3. *Mature Entry*

See General Academic Regulations

3.0. PROGRAMME CHARACTERISTICS

3.1. *Areas of Study*

This includes Applied and Industrial Physics, Instrumentation, Optics and Photonics, electromagnetism and electronic devices, air and water quality instrumentation, Industrial Automation and process control, Thermodynamics and statistical mechanics, computational Physics.

3.2. *Specialist Focus*

This is a unique program that brings the applications of physics in industry and instrumentation Aspects.

3.3. *Orientation*

Research and innovation oriented. Teaching and learning are academically and professionally oriented and focused on practical aspects.

3.4. *Distinctive Features*

The program builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4.0. CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1. *Employability*

Industrial and Instrumentation Physics graduates work in a range of advanced manufacturing sectors where fundamental concepts in science are applied to manufacture high value products with precision and an advanced level of process control and automation. Sectors include Bio-Pharmaceutical, Energy, IT Hardware Manufacturing, food and beverage, water management and aerospace/space. They get employed as Instrumentation and process control Engineers, Design Scientists/Technologists; Metrologists; Radiation control Officers; Industrial Software/hardware specialists, R&D Scientists; Hi-Tech Consultants; and College/University Lecturers.

4.2. *Further Studies*

Master's and doctoral studies in Physics, Applied Physics, Industrial Physics, and Instrumentation

5.0. PROGRAMME DELIVERY

5.1. *Teaching and Learning Methods*

Lectures, tutorials, practicals, group presentations, group work, research projects, industrial visits, Work-Related learning, research project, individual

independent study

5.2. ***Assessment Methods***

Written and oral examinations, tests, laboratory reports, seminar presentations, industrial attachment report, final year research project report, and continuous assessments

6.0. LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

7 Innovate, implement and maintain industrial and instrumentation & control systems
8 Apply research skills and knowledge to solve industrial, manufacturing, medical and agricultural complex problems
9 Demonstrate professional and ethical responsibility
10 Apply technopreneurship skills to start new technology-driven businesses
11 Apply knowledge and skills to professionally manage projects
12 Utilize specialized computational software to analyse and solve problems related to industrial production phenomena and technological issues.

7.0. GENERAL PROVISIONS

7.1 In level 1, normally a student shall register for a maximum of 12 modules

7.2 A student shall not be admitted to level 2 or higher level modules unless he/she has passed all pre-requisites for the module prior to the start of the semester in which he intends to study that module.

7.3 A student may include in his/her programme, modules offered by other as long as the total credits for the degree does not exceed 540 credits

7.4 A student's section of modules of a Degree programme is subject to the approval of the Dean of Science and Technology and the Chairpersons all relevant Departments and to the following conditions:

7.4.1 The module combination is feasible in terms of timetable,

7.4.2 A student may apply for exemptions during the first four weeks of the semester.

7.5 The Departmental Board responsible for a programme may designate, in the special regulations for the programme, certain modules as core modules for that 1055
programme. Core modules are the modules, which are considered to be essential
for qualification in a particular programme.

8.0. PROGRAMME ASSESMENT

8.1. ***Coursework***

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of final mark for practical courses.

8.2. ***Written Examinations***
A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses

8.3. ***Practical Examination***
The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1 In order to proceed from 1 semester to the next a student must not fail more than 4 modules.
9.2. To proceed from level 2 to level 3 the student must not carry more than 4 core modules
9.3. To proceed from level 3 to level 4 the student must have passed Work-Related Learning.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

11.1 To be awarded a Bachelor of Science Honours degree a student must pass each core module listed in that programme and must have accrued at least 380 credits

11.2 The classification of all modules and degree programmes shall be as stated in Section 5 of the General Regulations.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	10%
Level 2	20%
Level 3	30%
Level 4	40%

13.0. PROGRAMME STRUCTURE

Level 1 Semester 1

Module	Credits
HIPI131 Instrument Measurement*	12
HIPI132 Industrial Automation*	12

HCSCI132 Principles of Programming Language	10
HIPI133/HAPI131 Mechanics*	12
HMAT131 Calculus I*	12
CS131 Communication Skills*	12
Level 1 Semester 2	
HIPI134 Modern Physics*	12
HIPI135 Waves, Oscillations & Thermodynamics*	12
HMAT132 Linear Mathematics I*	12
HIPI136 Electricity & Magnetism*	12
HIPI137 Fundamentals of Instrumentation & Metrology*	12
Level 2 Semester 1	
HIPI231 Computational Physics*	12
HTENG137 Digital Electronics and Systems*	10
HTENG233 Electromagnetic Theory*	10
HMAT231 Ordinary Differential Equations*	12
TCNP201 Technopreneurship*	12
GS211 Gender Studies*	12
Level 2 Semester 2	
HIPI232Industrial Communications and Networks*	12
HTENG234 Telematics and Industrial Telemetry	12
HIPI 238 Physics Mini Project*	12
Electives: (choose two modules)	
HIPI234 Digital Systems and Interfacing	12
HIPI240 Biophysics	12
HIPI241 Biomedical Instrumentation	12
HIPI233 Environmental Instrumentation	12
HIPI235 Optics & Optical Instrumentation	12
HIPI236 Introduction to Condensed Matter Physics	12
Level 3 Semester 1	
HIPI332 Industrial Attachment* (WRL1)	40
Level 3 Semester 2	
HIPI333 Industrial Attachment* (WRL2)	80
Level 4 semester 1	
Choose one area of specialization:	
Industrial Automation Specialization:	
HIPI431 Process Control Technology*	12
HIPI442 Quantum Mechanics*	**12**
HIPI433 Advanced Industrial Automation*	12
HTENG434 Project Management*	**12**
HCSE432 Embedded Systems	12
HIPI444 Advanced SCADA and PLCs	12

Medical Physics Specialization:

HIPI 442 Quantum Mechanics*	**12**
HTENG434 Project Management*	**12**
HIPI 434 Medical Physics and Biomedical Instrumentation*	**12**
HIPI 445 Physics of Medical Imaging	12
HIPI 446 Radiation Protection*	**12**
HIPI 447 Radiotherapy Physics	12

Level 4 Semester 2

Industrial Automation Specialization:

HIPI 438 Dissertation*	24
HIPI 436 Applied Nuclear and Radiation Physics*	**12**
HIPI 449 AI, Robotics and Mechatronics	12
HIPI 443 Electric Drives & Control Systems*	**12**
HIPI439 Lasers and Laser Technology	12

Medical Physics Specialization:

HIPI 436 Applied Nuclear and Radiation Physics*	**12**
HIPI 451 AI and Advanced Biomedical Imaging	12
HIPI 438 Dissertation*	**24**
HIPI 452 Advanced Biomedical Instrumentation*	**12**
HIPI439 Lasers and Laser Technology	12

Module Synopsis

HIPI131 Instrument Measurement

Introduction to instrumentation and their applications, measurement terminologies and principles of different measurement techniques. Types of bridges for measurements of resistance, capacitance and inductance. Universal Instrumentation amplifiers. Types of sensors and their applications, Temperature sensor calibration, working principle and their maximum and minimum threshold. Ultrasonic transducer and their applications. Working principle and calibration. Airflow/Pressure Sensor and their working. Working of accelerometer for vibration and speed, LDR sensor and their working principle. Motion sensor and its applications.

HIPI132 Industrial Automation

Introduction to the Importance of the automation of an industrial system. Basic concepts: plant, control, operator, sensors, drives, open loop control, closed loop control, continuous processes, discrete processes, mixed processes, batch processing. Functional and physical architecture of control systems, automation pyramid and function of each level. Technological elements of each level: sensor networks, fieldbuses, controllers (PLCs), instrumentation, drives, robots, plant buses, RTUs, local area networks and control centres. OSI communications model. Control types: centralized, distributed and real time control. Methodologies for discrete processes control, classic methods: Boole's algebra, wired logic definitions. Sequential and Combinational logic. PLC programming; Execution models and architecture of a PLC. Ladder diagram language, Instruction language, Functional block language and Structure text language. Industrial communications network types; OSI model, Sensor networks, Device networks, Fieldbus networks. Industrial standards. IoT in automation systems; Definitions, Models, Design considerations, Architectures, Risk and

Security. Control Centers; Communication systems, supervision and control systems (SCADA), Functions, Basic models of organization. Integration with other systems. Web Technology; Man/machine interface design and Information model. Case Study: energy systems, industrial plants, intelligent management of buildings, home automation and security systems. Design strategies and implementation.

HCSCI132 Principles of Programming Language

This module examines the concepts and structures governing the design and implementation of programming languages. It presents an introduction to the concepts behind compilers and runtime representations of programming languages; features of programming languages supporting abstraction and polymorphism; and the procedural, functional, object-oriented, and concurrent programming paradigms. Programs are required in languages illustrating each of these paradigms.

HIPI133 Mechanics
Inertia framework reference; motion in two dimensions, three dimensions. Particle dynamics, rotational dynamic systems, gravitation, mechanical oscillations, properties of matter, fluid mechanics. Special Relativity: Space-time frame reference Galilean transformation, simultaneity of event, Einstein special relativity theory and Lorentz transformation, time dilation and length contraction, velocity transformation, fluid mechanics.

HMAT131 Calculus I
The principle of mathematical induction. The real number system: Functions: Limits of functions. Continuity. Sequences: Differentiation: Derivatives of functions of a single variable. Integration: Method of substitution, integration by parts and reduction formulae, fundamental theorem of calculus.

CS131 Communication Skills
Refer to the Communications Skills Department

HIPI134 Modern Physics
The nuclear atom: Thompson and Rutherford; The Bohr Model of hydrogen atom; Quantisation of energy; Line spectra; Rydberg constant. Black Body radiation; radiation and quantization of light; Plank distribution; Compton effects; Continuous and discrete energy spectrum; Frank- Hertz experiment; spontaneous and stimulated emission; Wein and Plank distribution.Wave particle duality; Uncertainty principle; Nuclear Physics.

HIPI135 Waves, Oscillations & Thermodynamics
Oscillations and SHM; Wave motion and interaction, standing waves, sound propagation and effects; EM waves, ray model of light, the wave model of light. Temperature, thermal effects on solids, ideal gases; Kinetic theory of gases; 1st and 2nd law of thermodynamics. The Einstein model of a solid. The ideal gas. Temperature. Paramagnetism Mechanical equilibrium and pressure. Diffusive equilibrium and chemical potential. Engines and refrigerators. Free energy and chemical thermodynamics. Boltzmann-Maxwell statistics. Partition function. Elements of quantum statistics. Bose-Einstein and Fermi-Dirac distributions

HMAT132 Linear Mathematics I
Complex numbers: De Moivre's theorem polynomials and roots of polynomial equations. Matrices and determinants: solutions of simultaneous linear equations, applications to geometry and vectors. Differential equations: separable, homogeneous, exact, integrating factors, linear equation with constant coefficients.

HIPI136 Electricity & Magnetism

Static charges: Electric charge, Coulomb's law, Electric fields, moving point charges in an electric field, lines of force, electric dipoles, Electric flux, Gauss's law, Electric potential, capacitors, energy in a capacitor and dielectrics. Charge distributions, force vector at a point; superposition. Current and Resistance: Conductors, current density, Kirchhoff's laws, DC network theorems, Wheatstone bridge, pd and resistance, capacitance and capacitors. Magnetic Fields: Force between currents, magnetic fields, magnetic flux density, magnetic intensity, magnets in magnetic fields, magnetic dipole moments, Torque on a current loop, motion of charge in a magnetic field, Bio-Savart law, Ampere's law, Induction and Inductance, Faraday's and lentz's law, self and mutual inductance, energy storage in inductors in magnetic fields, Superconductivity, Principle of electric motor and generator, Laurenz's force, Alternating currents: LR, LC and LCR circuits, ac circuits, phasor notation, power in ac circuits and ac networks, frequency filtering and tuning circuits, transformer, capacitor and inductor circuit in a generator, Eddy currents. Time constants and waveforms.

HIPI137 Fundamentals of Instrumentation & Metrology
Basic concepts of instrumentation, generalized instrumentation systems block diagram representation, scope of instrumentation in Industrial organization. Measurement systems, impedance matching and loading, dynamic characteristics. Metrology: Errors in measurement and Traceability: Errors and sources of errors, Uncertainties' types; calibration; Transducers and Sensors - Industrial Flow meters.

HIPI231 Computational Physics
This module introduces their use for modelling and simulation of physical systems. Good programming practice is emphasised throughout the module so you should begin to appreciate the discipline of programming beyond that of developing simulations of simple physical systems. Assessment is based on written laboratory reports/mini projects and computer programming tasks. Hands-on application of Scientific Software for Numerical Analysis; Simulation and Modelling of Dynamic Systems and control processes- Programming, Simulation and Modelling of dynamic systems and control systems through Scientific Computing in Maple and, COMSOL, MULTIPHYSICS/ C++ and other specialist Packages.

HTENG137 Digital Electronics and Systems
Numerical systems and binary arithmetic. Binary Coded Decimal (BCD) and ASCII. Logic gates and truth tables. Boolean algebra theorems. Karno (K) - maps. Combinational logic - minimizing logic expressions using Boolean algebra theorems and K-maps. Universal gates (NAND and NOR) implementation of combinational logic circuits. Combinational logic applications: adders and subtractors, magnitude comparators, multipliers, encoders and decoders, multiplexers and de-multiplexers. Digital integrated circuits. Integrated circuit families: RTL, DTL, TTL, ECL, NMOS, PMOS, CMOS. Bi-Stable systems – flip flops: R-S type, T-type, D-type, J-K-type and Master-Slave type. Sequential logic circuits: counters and registers and applications. Finite State Machines (FSMs). Memories: RAM, ROM, EAROM. PROM, EPROM. Programmable Logic Devices (PLDs).

HMAT231 Ordinary Differential Equations
This module aims at providing a broad introduction to differential equations (ordinary and partial), First order differential equations: Second order differential equations: 3D coordinate systems: Introduction to 2nd order partial differential equations, their solutions written in terms of a Fourier series; the one-dimensional wave equation; the time dependent Schrodinger equation; the Laplace equation in two dimensions; the Diffusion equation. Fourier Series:. Fourier transforms: mathematical definition, relationship with Fourier series, inverse Fourier transform, Fourier transform pair, Fourier integrals; definition of odd and even functions. Dirac delta-function; Fourier transform of a Gaussian; Fourier transforms applied to diffraction, Young's slits Lagrange's Equations including the following: Generalised coordinates and forces, Holonomic constraints, The Lagrangian and Lagrange's equations.

TCNP201 Technopreneurship
Introduction: Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy; Innovation and creativity, Qualities of an entrepreneur.
Small business model and financial issues: Developing a business model, Basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation, How to form and register a small business in Zimbabwe.
New Product development (NDP): Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities, Designing your product/service: design thinking; process thinking, strategic thinking; The NPD process: idea generation, idea screening, concept testing, market strategy development, business financial analysis, prototyping, test marketing, commercialization.
Developing and Protecting Intellectual Property: Concept of intellectual property, Theory behind IP protection, Intellectual Property (IP)-driven vs non-IP driven technopreneurshipTrade secrets, Copyrights, Trademarks, Patent and Trademark protection and its significance, Basics of patenting, legislation governing IP in Zimbabwe; Case studies of successful technopreneurs.

GS211 Gender Studies Refer to Gender studies department

HIPI232 Industrial Communications and Networks
The course covers the standards and techniques needed to design and maintain a communications system in an industrial production environment such as listed in the following: Data Communications, Electronic data communications and automated process control, Data Communication Terminology, Serial Communications Standards, Data Link Layer Basics and Data Encoding, ISOIOSI Reference Model, LAN Technologies to automate process control, Ethernet Technology, Cabling and Configuration Rules, Repeaters, Bridges, Routers, and Gateways, TCP/IP, Error detection and correction schemes, Ethernet Readiness for an industrial environment, Industrial Ethernet Design Techniques.
Industrial Ethernet Design Techniques, PLC software systems, PLC Networks in an automated industrial setting, Industrial Networks and Fieldbuses, Wireless communication systems. The Five Rules for troubleshooting industrial data communication networks.

HIPI233 Environmental Instrumentation
Introduction: Necessity of Instrumentation & Control for the environment, sensor requirement for environment. Instrumentation methodologies: Ultraviolet analyzers, total hydrocarbon analyzers using flame ionization detector, Gas chromatography in environmental analysis, photo ionization, portable & stationary analytical instruments.
Quality of water: Standards of raw & treated water, sources of water & their natural quality, effects of water quality. Water quality parameters: Thermal conductivity, detectors, Opacity monitors, pH analyzers & their application, conductivity analyzers & their application. Water treatment: Requirement of water treatment facilities, process design.

Sedimentation & flotation: General equation for settling or rising of discrete particles, hindered settling, effect of temperature, viscosity, efficiency of an ideal settling basin, reduction in efficiency due to various causes, sludge, storage & removal, design criteria of settling tank, effect of temperature on coagulation. Ground water monitoring: Level measurement in ground water monitoring wells, laboratory analysis of ground water samples, instrumentation in ground water monitoring, instrumentation in assessment of soil & ground water pollution.

Waste water monitoring: Automatic waste water sampling, optimum waste water sampling locations, and waste water measurement techniques. Instrumentation set up for waste water treatment plant. Latest methods of waste water treatment plants.
Air pollution: definitions, energy environment relationship, importance of air pollution, air pollution from thermal power plant, their characteristics & control. Air sampling methods & equipment, analytical methods for air pollution studies. Control of air pollution. Flue gas analysis for pollution control –Measurement of CO, carbon di-oxide, NOX and SOX, dust and smoke measurement. Chromatography –Basic principles of liquid and gas chromatography –Column details –Detectors for chromatography –Thermal conductivity detector –Flame ionization detector –Flame photometric detector –Electron capture detector –Effect of temperature programming – High-pressure liquid chromatography (HPLC).
Air monitoring: measurement of ambient air quality. Flow monitoring: Air flow measurement, gas flow, non-open channel flow measurement, open channel wastewater flow measurement. Rain water harvesting: necessity, methods, rate of NGOs municipal corporation, Govt., limitations. Quality assurance of storage water

HTENG233 Electromagnetic Theory
Maxwell's equations. Laplace and Poisson equations and their solution. Boundary conditions. Plane waves in a perfect dielectric; propagation in imperfect dielectric. Propagation in imperfect conductors, skin effect. Generalized wave equation, field distributions in rectangular waveguide. Radiation field, dipoles, radiation resistance, impedance, mutual impedance, linear arrays.

HTENG234 Telematics and Industrial Telemetry
Instrumentation Standard Protocols and automation networks, bus standards, third party interface, object linking and embedding for Process Control. Wireless instrumentation- Wireless HART, ISA100.11A, Foundation Fieldbus H1: frame structure, programming. Buses, Controlnet, Industrial Ethernet. PLC and SCADA/DCS.

HIPI234 Digital Systems and Interfacing
This module enables the design and construction of instrumentation systems based on digital techniques and safely interface digital systems to the real world. Aspects of the module include: System hardware and software; Bus design and timing, processor and local buses, bridge and bus hierarchy, fault-tolerant; Parallel, serial, and Internet communication, RS232, USB, SATA, GPIB, PCI, SCSI; A/D and D/A; System design process, design entry, signal integrity, PCB testing.

HIPI238 Mini Project
Apply Physics principles in a short demonstration project.

HIPI235 Optics and Optical Instrumentation
Fundamentals of Waves- wavelength, period, displacement, amplitude, speed, phase and phase difference, wave superposition, reflection at a boundaries and discontinuities, coherence, coherence length Interference in thin films, thin film equation. Huygen's Principle Single slit, Young's double slit experiment, diffraction grating, Michelson Interferometer Fresnel and Fraunhofer Diffraction Polarisation, Malus's Law, polarizing filter. Monochromator Spectrometer

HIPI236 Introduction to Condensed Matter Physics
The course covers the structure of solids and other phases of condensed matter. The first part will introduce materials which are not solid, or crystalline, and neither liquids, but are still

condensed matter, such as colloidal fluids or glasses, gels and liquid crystals. The remainder of the course will be focused on providing a basic understanding of the electrical, thermal and mechanical properties of solids, and in particular metals and semiconductors. The aim is to give a firm grounding in the core concepts.

HIPI240 Biophysics
The module covers areas such as hearing and echolocation, ultrasound, quantum nature of vision, biophysics of neural spike, structural biophysics, radiation and radiobiology, bio potentials and bioenergetics. Systemic Biophysics, Biophysics and Fluid Flow, Biophysics and Gas Transport, Physics of Audition, Physics of Vision: Physical Methods in Biology and Medicine, X-Ray diffraction, Isotope labelling, Photodynamic therapy, Dynamic light scattering, Circular Dichroism(CD).Quantum, Dot imaging, Raman Scattering. Fluorescence and UV-vis light spectroscopy, canning probe microscopy

HIPI241 Biomedical Instrumentation
This course covers theory of operation, circuit analysis, troubleshooting techniques, and medical applications for a variety of instruments and devices. Topics include electrodes, transducers, instrumentation amplifiers, electrocardiographs, monitors, recorders, defibrillators, ESU units, and related equipment used in clinical laboratories, intensive care units, and research facilities. Upon completion, students should be able to calibrate, troubleshoot, repair, and certify that instrumentation meets manufacturer's original specifications.

HIPI431 Process Control Technology
Process characteristics: Incentives for process control, Process Variables types and selection criteria, Process degree of freedom, Oscillations and Damping, Characteristics of physical System: Resistance, Capacitive and Combination of both. Elements of Process Dynamics, Types of processes- Dead time, Single/multi-capacity, self-Regulating/non self-regulating, Interacting /non-interacting, Linear/non-linear, and Selection of control action for them. Study of Liquid Processes, Gas Processes, Flow Processes, Thermal Processes with respect to the above concepts.

HIPI442 Quantum Mechanics
Formal treatment of quantum mechanics. Topics include wave packets and free particle motion, the Schrodinger equation, harmonic oscillator, piecewise constant potentials, central forces and angular momentum, and the hydrogen atom.

HIPI433 Advanced Industrial Automation
Distributed Control System basics DCS, PLC, HMI and SCADA functionality; Distributed Control Systems Engineering, configuration and programming, functions. Enhanced functions like Advanced process control, fuzzy logic, ANN. Process safety and Safety Management Systems; Basics of Advance Process Control and Optimization. Batch Process Control Requirements; Batch Process Control Application and Case Study. Analysis of Control Loop: Steady state gain, Process gain, Valve gain, Process time constant, Variable time Constant, Transmitter gain, Variable pressure drop. Analysis of Flow Control, Pressure Control, Liquid level Control, Temperature control, SLPC-features, faceplate, functions, MLPC- features, faceplate, functions, SLPC and MLPC comparison. Scaling: types of scaling, examples of scaling: Feedback Control: Basic principles, Elements of the feedback Loop, Block Diagram, Control Performance Measures for Common Input Changes, Selection of Variables for Control Approach to Process Control. Factors in Controller Tuning, Determining Tuning Constants for Good Control Performance, Correlations for tuning Constants, Fine Tuning of the controller tuning Constants. The performance of feedback

Systems, Practical Application. Industrial robotics; pneumatics; machine vision systems; automated assembly; design for automation; and Industry 4.0 (such as machine-to-machine communications and data analysis).

HTENG434 Project Management
Project proposal writing- types of proposals; Project definition, life cycle, and systems approach; Project scoping, work definition, and work breakdown structure (WBS); Project time estimation and scheduling using GANTT, PERT and CPM. Project costing, budgeting, and financial appraisal; Project control and management, using standard tools of cost and schedule variance analysis; project management use-case through practical, example projects; use of computers in project management, some software tools for PM e.g. MS Project; PM techniques e.g. PRINCE2.

HCSE432 Embedded Systems
This module covers Embedded Systems concepts. Topics include nature of embedded systems, specific problems, special issues; role in computer engineering; embedded microcontrollers, embedded software; real time systems, problems of timing and scheduling; testing and performance issues, reliability; low power computing, energy sources, leakage; design methodologies, software tool support for the development of such systems; problems of maintenance and upgrade; networked embedded systems.

HIPI444 Advanced SCADA and PLCs
Explore major SCADA applications, SCADA system components, and architecture. The course defines and explains Programmable Logic Controller (PLC) and architecture, configuration, installation, maintenance, and programming.

HIPI434 Medical Physics and Biomedical Instrumentation
Introduction to bioelectric potential, bio-amplifiers, components of man Instrument system, types of bio-medical systems, design factors and limitations of biomedical instruments. Cardiac vascular system. Blood pressure measurements: - direct, indirect. Respiratory system: types of volume, types of measurements, Instrumentation of respiratory system. Medical Imaging systems.

HIPI436 Applied Nuclear and Radiation Physics
Radiation and matter: Heavy charged particle interactions; Nuclear reactions & scattering. Detecting Radioactivity & Radiation: Gas counters; Scintillation counters, solid-state spectrometers and neutron detectors. Applications: Solar nuclear Physics; Radiation Dating; Mossnauer Spectroscopy; Nuclear Energy & Nuclear Reactor Design Technology; Applications in industry; Medicine; Material analysis and Forensics. Non-ionizing radiation and how to address safe use of non-ionizing radiation producing devices. This includes tanning beds, ultra-violet light curing of materials, lasers used in entertainment, medicine and industry, and radiofrequency sources such as WiFi networks, cell phones and industrial heat sealers. Use knowledge of mathematics, science, and applied sciences to determine exposure limits to all types of non-ionizing radiation. Radiation Control and Policing.

HIPI445 Physics of medical imaging
Introduction of medical imaging, namely X-ray, nuclear medicine, magnetic resonance and ultrasound. It enables students to develop an understanding of the physics principles underlying these imaging techniques and an awareness of their clinical applications. It also discusses the mathematical principle involved in image formation and processing and provides experience in their use.

HIPI446 Radiation Protection

This module provides instruction on the nature of radiation and radioactivity, the hazard and risk it presents, the necessary control measures, and the application and method of compliance with relevant radiation safety legislation.

HIPI447 Radiotherapy Physics
Radiation therapy involves the therapeutic use of controlled doses of radiation for cancer treatment in hospitals. The major topic areas include Introduction and a Radiobiological Basis for Radiotherapy, Radiation Dosimetry, Dose Calibration Protocols, Equipment in Radiotherapy, Radiotherapy Prescription and Treatment Planning, Dose Calculation, Radiotherapy Treatment Techniques, Advanced Topics in Radiotherapy.

HIPI449 AI, Robotics and Mechatronics
Apply AI concepts to robotic devices to solve real-world challenges and integrate AI technologies with traditional automation systems such as heating, ventilation and air conditioning (HVAC) and progressive assembly systems to create intelligent automation solutions.

HIPI438 Dissertation
Provides students with the opportunity to design, undertake or conduct an independent piece of research of study related to their programme of study under the guidance of a supervisor who is usually a member of the academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

HTENG434 Project Management
Project proposal writing- types of proposals; Project definition, life cycle, and systems approach; Project scoping, work definition, and work breakdown structure (WBS); Project time estimation and scheduling using GANTT, PERT and CPM. Project costing, budgeting, and financial appraisal; Project control and management, using standard tools of cost and schedule variance analysis; project management use-case through practical, example projects; use of computers in project management, some software tools for PM e.g. MS Project; PM techniques e.g. PRINCE2.

HIPI439 Lasers and Laser Technology
Introduction to laser physics: Classical electron oscillator model. Fundamental optical processes: absorption, spontaneous emission, stimulated emission. Basic principles of lasers: Gain processes, optical feedback, spectral linewidth. Laser performance: Frequency and intensity distribution, temporal behavior, pulsed systems. Laser systems: Solid, liquid and gaseous systems. Applications of lasers: CD players, holography, optical fibre communications, gas sensing and LIDAR, medical applications. Pumping processes: optical pumping, radiative and transfer efficiency, Quantum efficiencies for absorption and power, Electrical pumping: electron impact excitation, ionization balance equation, pump rate calculations. Optical systems; Matrix formulation of geometrical optics, Fabray-Parot interferometer, Fox and Lee treatment. Confocal resonator, Gaussion beam propagation and ABCD Law. Stability condition, unstable resonators, hard edge, unstable resonators. Resonators design, and transformation of impedance multilayer optical systems. Applications in optical instruments, metrology & medicine.

HIPI452 Advanced Biomedical Instrumentation
Analysis and design of transducers and signal processors; measurements of physical, chemical, biological, and physiological variables; special purpose medical instruments, systems design, storage and display, grounding, noise, and electrical safety

HIPI441 Renewable Energy Technology of Sustainability
The Renewable and Sustainable Energy Systems course provides an understanding of the conversion principles and technology behind various renewable energy sources. Topics include:
The principles of operation of the broad spectrum of renewable energy technologies; conduct preliminary resource assessments for a variety of renewable energy technologies; analysis of energy technologies from a systems perspective; The technical challenges for each of the renewable sources; the economic, technical, and sustainability issues involved in the integration of renewable energy systems.

HIPI432 Acoustics & Industrial Noise
Basic acoustics, terminology, measurement metrics, physiology. Analysis of sounds, energy equivalent sound level, measurement techniques, FFT, octave & 1/3 octave band frequency analysis. Sound sources such as machines, roads, pipelines. Environmental noise, sound propagation outdoors, assessment criteria, noise modelling software. Architectural acoustics, sound absorption, sound transmission, assessment criteria. Noise control methods such as barriers, enclosures, vibration isolation. HVAC acoustics and noise control methods.

HIPI442 Quantum Mechanics
Origins of Quantum Mechanics. Introductory Wave Mechanics: Schrodinger's equation and its physical interpretation. Energy eigenvalues and energy functions. Potential wells, harmonic oscillator Step potentials and barrier potentials and tunnelling. Operators, particle in a 2-D box. The Hydrogen Atom: Radial solutions, Radial probability densities, energy level. Separation of variables. Spherical harmonics & angular momentum.

HIPI443 Electric Drives & Control System
Fundamentals of advanced electric machines and motor drives. Applications in industry automation, robotics, automotive and traction systems, home appliances, aerospace, etc. Different drive systems for conventional and advanced machines are presented in this module. Fundamentals of electric machines, basic principles of variable speed controls, field orientation theory, direct torque control, vector control of AC drives, induction machines, switched reluctance and synchronous reluctance motors, permanent magnet brushless DC drives, converter topologies of DC and AC drives, and sensorless operation. Selection of motors and drives, calculating the ratings, speed control, position control, starting, and braking are also covered. Drives for DC machines, induction motors, brushless DC, switched reluctance, and stepper motors are explained. Simulation projects using COMSOL Multiphysics, Maple, PSPICE and MATALB are parts of module.

HIPI451 AI and Advanced Biomedical Imaging
Digital image processing has various applications ranging from remote sensing and entertainment to medical applications. This course explores a few major areas of digital image processing at an advanced level, with a primary emphasis on medical applications. Topics covered include image segmentation, image registration, and image processing using Image Processing Tools.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN INDUSTRIAL PHYSICS (MIP)

Duration: 2 years

Actual Credit Load	360
Minimum Credit Load:	360
Maximum Credit Load:	420
Maximum MBKS Credit Load:	288
ZNQF Level:	9

1.0 PURPOSE OF THE PROGRAMME

The purpose of a Master's degree in Industrial Physics is to provide students with advanced knowledge and skills in the field of physics and its application to industrial settings. The program aims to bridge the gap between theoretical physics principles and their practical implementation in industries, equipping graduates with the expertise to address complex technical challenges and contribute to the development of innovative solutions in industrial sectors. The programme also prepares students for further studies and research.

15. ENTRY REQUIREMENTS

Prospective students must hold at least a good first degree in Physics, Applied Physics, Industrial Manufacturing or any other related degree from any university recognized by Midlands State University.

16. PROGRAMME CHARACTERISTICS

This Master's Degree in Industrial Physics adopts an interdisciplinary approach, incorporating concepts and methodologies from various fields such as physics, engineering, and science. This approach enables students to understand and address complex industrial challenges from multiple perspectives. The program also includes a core curriculum that covers advanced topics in physics, such as Electrodynamics, Thermal and Statistical Physics, Semiconductor, with specializations in Medical Physics, Energy and Environmental Physics, and Industrial Automation. This Master's programs include research components that allow students to engage in independent or collaborative research projects, thereby providing an opportunity to explore specific areas of interest within industrial physics and contribute to the advancement of knowledge in the field.

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

A Master's degree in Industrial Physics offers diverse career opportunities in various sectors that require the application of physics principles in industrial settings such as:

- Research and Development (R&D),
- Quality Control and Testing,
- Process Optimization,
- Industrial Automation

5.0 PROGRAMME DELIVERY

The programme is delivered over a two year period during which the students are engaged in a number of lectures, tutorials, practical investigations, field visits, journal article writing, research projects, problem based learning, blended learning, student-led learning and seminar presentations.

6.0 LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Apply physics knowledge and skills in solving theoretical and practical problems in a wide range of industrial settings, devising and using unconventional techniques.
- Solve physics problems using qualitative and quantitative reasoning including sophisticated mathematical techniques.
- Apply knowledge and skills to start new technology-driven businesses and professionally manage projects.
- Communicate physics reasoning in oral and in written form.
- Demonstrate professional and ethical responsibility.
- Conduct independent research or work successfully in a technical position.

7.0 GENERAL PROVISIONS

The degree programme shall normally be studied over a minimum of four semesters on a block-release basis. This programme is a taught Masters programme with modular work in the first year. The research project is to be done in the second year.

7.1 Dissertation

7.1.2 The dissertation shall be done in level two during which time the student is expected to undertake research

7.1.3 Each candidate is expected to submit a dissertation of at least one hundred (100) pages.

7.1.4 The format of the dissertation shall be provided by the Department of Applied Physics and Telecommunications

8.0 PROGRAMME ASSESSMENT

Coursework

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of the final mark for practical courses.

Written Examinations

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses.

Practical Examination

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0 PROVISIONS FOR PROGRESSION

9.1 A student must pass modules worth at least 180 credits in the first level in order to proceed to the second level

9.2 To graduate a student should have accumulated at least 360 credits

10. FAILURE TO SATISFY EXAMINERS

Refer to section 21 of the General Academic Regulations

11.0 GRADING AND DEGREE CLASSIFICATION

Refer to section 12 of the General Academic Regulations

12.0 DEGREE WEIGHTING

For the purpose of degree classification, the parts of the degree programme are weighted as follows:

Leve1 (50%)

Level 2 (50%)

13.0 PROGRAMME STRUCTURE

N.B. ***Denotes core modules which are not Minimum Body of Knowledge and Skills**

****Denotes core modules which are Minimum Body of Knowledge and Skills**

Level 1 Semester 1

Code	Module Description	Credits
**MIP 801	Semiconductor Device Physics	18

**MIP 802	Thermal and Statistical Physics	18
**MIP 803	Computational Physics and Scientific Computing	18
*MIP 804	Electrodynamics	18
*MIP 805	Advanced Optics and Lasers	18

Level 1 Semester 2

Option 1 – Medical Physics

Code	Module Description	Credits
**MIP 806	Microprocessor and Embedded Systems	18
**MIP 807	Radiation Safety and Quality Management	18
*MIP 808	Physics of Non-ionizing Radiation	18
**MIP 809	Biomedical Electronics and Instrumentation	18

Option 2 – Energy and Environmental Physics

Code	Module Description	Credits
**MIP 810	Energy Physics	18
**MIP 811	Renewable Energies Systems Design	18
*MIP 812	Atmospheric Physics	18
**MIP 813	Environmental Physics	18

Option 3 – Industrial Automation

Code	Module Description	Credits
**MIP 814	Instrumentation Electronics and Metrology	18
**MIP 806	Microprocessor and Embedded Systems	18
*MIP 815	SCADA and Distributed Control Systems	18
**MIP 816	Artificial Intelligence, Robotics and Automation	18

Level 2 Semester 1

Code	Module Description	Credits
**MIP 901	Advanced Techniques for Scientific Research and Innovation	18

Level 2 Semester 2

Code	Module Description	Credits
**MIP 902	Dissertation	180

MODULE SYNOPSES

MIP 801 Semiconductor Device Physics

This module provides both fundamental theory and hands-on experience on semiconductor physics, devices, fabrication and analysis techniques. The module will cover first the basic semiconductor physics such as crystal structures, doping and band gaps etc, devices such as diodes and transistors and then various measurement techniques to probe the morphology, compositions, electrical, magnetic and optical properties of materials and devices at nano/atomic scales, along with both conventional nanofabrication techniques (e.g, e-beam lithography) and non-conventional nanofabrication techniques (e.g, self-assembly).

MIP 802 Thermal and Statistical Physics

Identify, critically analyse and describe the statistical nature of concepts and laws in thermodynamics, in particular: entropy, temperature, chemical potential, free energies, partition functions. Expertly use statistical physics methods, such as Boltzmann distribution, Gibbs distribution, Fermi-Dirac and Bose-Einstein distributions to solve complex problems in physical systems. Apply the concepts and principles of black-body radiation to analyze radiation phenomena in thermodynamic systems.

MIP 803 Computational Physics and Scientific Computing

This module will introduce a range of computational and analytic methods that can be used to model the properties and dynamics of physical systems. Computational methods in solving problems in physics, Programming tactics, numerical methods and their implementation, together with methods of linear algebra. These computational methods are applied to problems in physics, including the modelling of classical physical systems to quantum systems,
as well as data analysis such as linear and nonlinear fits to data sets.

MIP 804 Electrodynamics

Geometrical vectors and vector fields, Calculating with the div, grad and curl, Index notation, unit vectors and coordinate systems, Maxwell's equations and the Lorentz Force, Gauge transformations and a particle in an electromagnetic field, The Laplace equation and the method of images, Separation of variables and Legendre polynomials, Electric and magnetic multipole moments, The wave equation, polarisation, phase and group velocities, Energy and momentum in electromagnetic field, Electric and magnetic dipole radiation, Radiation from accelerated charges, Macroscopic media Waves in dielectrics, conductors and plasmas, Frequency-dependent refractivity and anomalous dispersion, Reflection and transmission of waves, Waveguides and coaxial cables, Cavities, Revision of special relativity, Covariant Maxwell's equations and the transformation of fields

MIP 806 Microprocessor and Embedded Systems

Operating Systems (OS) and Real-Time Operating Systems (RTOS). Embedded RTOS. Software and Hardware development methods and tools: Run-time libraries. Writing a library. Porting kernels. Concurrent Programming and Concurrent Programming Constructs. Task Scheduling and Task Interaction. Basic Scheduling methods, scheduling algorithms. Tasks, threads and processes. Context switching. Multitasking. Communication, Synchronisation. Semaphores and critical sections. Example RTOS systems. (E.g. Embedded Linux, Windows CE, Micrium, VxWorks etc.). Programming and debugging Embedded Systems. Practical examples and case studies.

MIP 807 Radiation Safety and Quality Management

Fundamentals of Radiation and Radioactivity, Radiation Biology, Radiation Dose Limits and ALARA, Personnel Monitoring, Access Control and Postings, Emergency and Spill Procedures, Radiation Dose Limits Contamination Control, and Employee Responsibilities. Quality Management Systems, Governance, Risk Management and Compliance

MIP 808 Physics of Non-ionising Radiation
The interaction processes of particles with matter. Interaction of ionising and non-ionising radiation with biological systems. Interaction of ultrasound with tissue, Interaction with red cells, attenuation, absorption, reflection and refraction.

MIP 809 Biomedical Electronics and Instrumentation

Electrical safety for medical devices. The basic building blocks of medical instruments; the instrumentation amplifier, filters, analogue to digital conversion. Recording and display devices. Physiological measurements; the ECG, EEG, Pulse Oximeter and Evoked Potentials. Electrical therapeutics; the defibrillator and electrical diathermy. Computers in medicine: picture archiving and retrieval; the physiological signals. Telemetry and ambulatory monitoring. Biomedical device development process, involving subject knowledge from electronics, control theory, microcontrollers, MATLAB programming, machine design. Biomaterials and Bio-fabrication. Integration of tissue engineering and bioreactors into biomedical engineering.

MIP 810 Energy Physics
Introduction to Energy Physics, Definition of energy and its significance in physics, Energy forms and their interconversion, Conservation of energy principles, Energy Generation, Fossil fuel-based energy generation Renewable energy sources (solar, wind, hydro, geothermal, biomass), Nuclear energy generation (fission and fusion), Energy Conversion and Storage, Thermodynamics and energy conversion processes, Energy storage technologies (batteries, supercapacitors, flywheels, pumped hydro, etc.), Energy efficiency and energy loss considerations. Energy Utilization and Applications, Energy in transportation (electric vehicles, hybrid vehicles, fuel cells), Energy in buildings (heating, cooling, lighting, insulation), Industrial energy systems and processes (cogeneration, waste heat recovery), Energy Policy and Economics, Energy policy frameworks and regulations, Economic analysis of energy systems, Environmental and sustainability considerations.

MIP 811 Renewable Energies Systems Design
Introduction to Renewable Energy Systems, Overview of renewable energy sources (solar, wind, hydro, geothermal, biomass), Environmental and social benefits of renewable energy, Global energy demand and the role of renewables in the energy transition, Resource Assessment and Site Analysis, Techniques for assessing renewable energy resources (solar radiation, wind speed, hydro potential), Site selection criteria and considerations, Data collection and analysis for resource assessment, Design Principles and System Components, Design considerations for solar photovoltaic (PV) systems, Design principles for wind energy systems Design of small-scale hydroelectric systems, Biomass energy system design and optimization, Integration and System Design, Grid integration of renewable energy systems, Hybrid renewable energy systems (solar-wind, solar-hydro, etc.), Energy storage and grid management for intermittent renewable, Designing for off-grid and remote applications, System Performance Analysis and Optimization, Performance metrics for renewable energy systems, Modelling and simulation tools for system analysis, Optimization techniques for system design and operations, Economic analysis and feasibility studies, Project Planning and Implementation, Project development lifecycle and stages, Permitting and regulatory requirements, Financial and risk analysis, Project management and execution.

MIP 812 Atmospheric Physics
Introduction to Atmospheric Physics, Overview of Earth's atmosphere, Composition and structure of the atmosphere, Key atmospheric properties and variables, Atmospheric Thermodynamics, Laws of thermodynamics applied to the atmosphere, Atmospheric stability and lapse rates, Moisture and cloud formation, Atmospheric convection and thunderstorms, Atmospheric Dynamics, Atmospheric circulation patterns, Forces and motion in the

atmosphere, General circulation and global weather systems, Synoptic meteorology and weather forecasting, Atmospheric Radiation, Solar radiation and its interaction with the atmosphere, Absorption, scattering, and emission processes, Radiative transfer in the atmosphere, Greenhouse effect and climate change, Atmospheric Chemistry, Composition and sources of atmospheric gases, Chemical reactions and kinetics in the atmosphere, Ozone depletion and stratospheric chemistry, Air pollution and its impacts on human health and the environment, Data Analysis and Modelling, Statistical analysis of atmospheric data, Time series analysis and spectral methods, Numerical modelling techniques for atmospheric simulations, Introduction to climate models and climate projections, Mesoscale meteorology and weather phenomena, Atmospheric boundary layer dynamics, Aerosols and their impact on climate and air quality, Remote sensing and satellite observations.

MIP 813 Environmental Physics

Introduction to Environmental Physics, Overview of environmental physics and its interdisciplinary nature, Key concepts and principles in environmental physics, Measurement techniques and instrumentation in environmental research, Atmospheric Physics and Climate Change, Radiative transfer and the greenhouse effect, Climate patterns and variability, Global climate models and climate projection, Impacts of climate change on ecosystems and society, Air Pollution and Atmospheric Chemistry, Sources and types of air pollutants, Atmospheric chemistry and chemical reactions, Dispersion and transport of air pollutants, Health and environmental impacts of air pollution, Hydrological Processes and Water Resources, Water cycle and its component, Precipitation and evaporation processes, Surface and groundwater dynamics, Water quality and pollution, Environmental Biophysics and Ecosystem Dynamics, Energy and mass transfer in ecosystems, Photosynthesis and carbon cycle, Soil physics and nutrient cycling, Ecological modelling and ecosystem services Environmental Remote Sensing and GIS, Remote sensing techniques for environmental monitoring, Satellite data analysis for climate and environmental studies, Geographic Information Systems (GIS) applications in environmental research, Integration of remote sensing and GIS for environmental analysis.

MIP 814 Instrumentation Electronics and Metrology

The module combines measurement theory and physics to establish a deep understanding of the leading transducer applications. The focus will be to describe the key features of widely used measurement techniques and will be able to show how transducers are combined with microprocessor devices to create robust and reliable industrial instruments such as pressure transmitters, flow metering systems and temperature transmitters. The module will introduce students to the latest practices in industrial instrument communication networks including wireless technology and field bus.

MIP 815 SCADA and Distributed Control Systems

SCADA systems hardware and software, a review of typical DCS and SCADA systems, DCS controllers and configuration. Structured programming based on the 61131-3 standard. Alarm system management, configuration, reporting, and maintenance. Examination of the implementation of a complete SCADA and DCS system. Case studies of SCADA and DCS projects and operations.

MIP 816 Artificial Intelligence, Robotics and Automation

Supervised learning, clustering, regression and time-series analysis. Data pre-processing and system evaluation. Characteristics of commonly used algorithms such as Decision Trees, K-nearest neighbours, Neural Networks, linear regression, and K-means clustering. Different applications of machine learning to industrial automation. Condition monitoring, system identification, and image processing for robotics. Software tools that can be used to implement machine learning algorithms. Apply machine learning to a particular industrial automation problem. understanding how robots perceive the world, build models and plans, and take

actions to manipulate their environment. Examine corporate Operations Strategy, Process strategies, Tactical issues, aggregate production planning and master production scheduling

MIP901 Advanced Techniques for Scientific Research and Innovation
The scientific method and systems thinking approach, Mathematical and computational techniques, theoretical and abstractions, statistics, measurement uncertainties and experimental techniques. Basics of IPs, Innovation and Engineering Research.

MIP 902 Dissertation

The module provides students with the opportunity to design, undertake or conduct an independent piece ofresearch of study related to their programme of study under the guidance of a supervisor who is usually a member of the academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN MOBILE AND WIRELESS COMMUNICATIONS (MMW)

Duration:	2 years
Minimum Credit load:	288
Maximum Credit load:	360
Maximum MBKs Credit Load:	288
ZNQF Level:	9

1.0 PURPOSE OF THE PROGRAMME

- Understanding the fundamentals of wireless transmission, coding and signal processing, building wireless transceivers and designing intelligent wireless networks.
- Knowledge in wireless communications, both at physical and network layers.
- Capability to design and implement wireless solutions, e.g., for future networks, Internet-of-Things (IoT) devices, and smart efficient wireless sensor applications.
- Relevant skills in the latest radio engineering methods, tools, and technologies,

and the ability to design RF electronics for smartphones and base stations of mobile systems.

2.0. ENTRY REQUIREMENTS

Prospective students must hold at least a good first degree in communication engineering, electronic engineering, electrical engineering, computer engineering or a closely related subject from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

3.1 Areas of Study

Advanced wireless communication systems, Communication networks, Radio channels, Radio engineering, Radio Access and Networks, RF Engineering. Multiple antenna systems, Mobile network protocols, Electromagnetic Field Theory, Stochastical and Digital signal processing, Antennas, Electronics, Information theory, Advanced Electronics, Microelectronic Device Physics

3.2 Specialist Focus

This is a unique programme that brings deep understanding of currently operational mobile and wireless networks taking into account the considerations for advancements towards future, better and more advanced wireless and mobile communication networks.

3.3 Orientation

Research and innovation-oriented. Teaching and learning are academically and professionally oriented and focused on practical aspects.

3.4 Distinctive Feature

The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach.

4 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 Employability

Network architects, wireless technology specialists, wireless design professionals, consultants, Development engineers, Maintenance engineers, Project managers, network systems analysts, planners, Radio network designers, Research engineers, academia and researchers in the industry.

4.2 Further Studies

Doctoral studies in Electronic and communication technologies, information and communication engineering, radio electronics, signal and information processing, computer science and telecommunications, semiconductor physics, nanotechnology or related areas.

5.0 PROGRAMME DELIVERY

5.1 Teaching and Learning Methods

Lectures, tutorials, practical work, group presentations, group work, mini research projects, industrial visits, research project, and individual independent study.

5.2 Assessment Methods

Written and oral examinations, tests, laboratory reports, seminar presentations, final year dissertations, continuous assessments

Continuous assessments:	Test (15%),
Practical work	(20%),
Assignments	(5%)
Written Examinations:	60%

6.0 LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Demonstrate advanced knowledge of the scientific foundations of the sub-fields included in the field of wireless communication.
- Identify and discuss the fundamental operational and design problems of wireless communication systems.
- Discuss basic technical standards related to 2G/3G/4G wireless systems, Wi-Fi and sensor networks
- Apply basic techniques to design radio links and basic mobile communication systems.
- Analyse an entire system as well as the sub-systems in wireless communication.
- Demonstrate an understanding of the nature of research and development in wireless communications.
- Develop abilities to setup experiments and analyse system performance using wireless systems, hardware and software.

7.0. GENERAL PROVISIONS

The programme shall normally be studied over a minimum period of 4 semesters on a block release basis.

8.0 PROGRAMME ASSESMENT

8.1 Coursework 1076

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of final mark for practical courses.

8.2 Written Examinations

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses

8.3 Practical Examination

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1 A student must pass all core modules to proceed to level 2.

9.2. All level 1 and level 2 core modules should be passed before proceeding to the dissertation

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

To be awarded a Master of Science degree a student must pass each core module listed in the programme and must have accrued at least 336 credits.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1	50%
Level 2	50%

13.0. PROGRAMME STRUCTURE

Level 1 Semester 1

Code	Module Description	Credits
MMW731	Mobile Communication Systems*	24
MMW732	RAN – Radio Access networks*	24
MMW733	Data communication and networking*	24
MMW734	Mobile network protocols*	24

Level 1 Semester 2

MMW735	Simulation of Mobile Communications*	24
MMW736	Wireless transceiver design and implementation*	24
MMW737	RF Engineering*	24
MMW738	Advanced Research Methods*	24

Level 2 Semester 1 (choose any THREE modules)

MMW831	5G and Future networks	24
MMW832	Broadband Wireless Communications	24
MMW833	Advanced Digital Signal Processing	24
MMW834	Multiple Antenna Systems	24
MMW835	Modelling and Performance	24
MMW836	Communication technologies for Internet of Things (IoT)	24

Level 2 Semester 2

MMW 840	Dissertation*	96

MODULE SYNOPSIS

MMW731 Mobile Communication Systems

Cellular wireless networks: architecture, frequency reuse, multiplexing, multiple-access, broadcast, power-control, handover, interference, examples of TDMA-, CDMA- and LTE-based cellular networks, etc. Path loss, large-scale fading, small-scale fading; Power budget of mobile links. Doppler spread and coherent time, delay spread and coherent bandwidth; flat fading and frequency selective fading. 1st generation (1G) system, 2G system, 3G system, 4G system, 5G system, and beyond 5G system. Wireless multiple-access techniques, including randomized medium access (ALOHA and CSMA), FDMA, TDMA, CDMA, SDMA.

MMW732 RAN – Radio Access Networks

Radio access networks: (GRAN: GSM radio access network, GERAN: essentially the same as GRAN but specifying the inclusion of EDGE packet radio services, UTRAN: UMTS radio access network, E-UTRAN: The Long-Term Evolution (LTE) high speed and low latency radio access network). Understanding of the underlying architecture of RAN in terms of planes, Software Defined Networking (SDN), network functions virtualization (NFV) techniques such as network slicing and high MIMO that are key elements for RAN flexibility and 5G.

MMW733 Data Communication and Networking

Introduction to data networks: layered architectures, TCP/IP and UDP protocol architecture, OSI model, main functions of different layers, relationship between layers, etc. Architecture and protocols of typical communications networks: flow control, congestion control, error control, etc. in wireless networks. Routing algorithms, delay

modelling, multiple-access principles and basic queuing theory and basic principles of wireless LANs, wireless ad-hoc networks, wireless sensor networks, etc. communication security protocols and techniques; TLS/SSL, HTTP, HTTPS , FTP , SMTP etc.

MMW734 Mobile Network Protocols

A study in depth looking into protocols which govern mobile and wireless communications (MAC, BRAIN, HAMAC, ARCMA and the considerations taken into account when choosing a protocol to implement in a network to suit the services provided. Overview of mobility concepts like Mobile IP, SIP and SCTP as well as quality of service/quality of experience, mobile network deployment, heterogeneous network access and performance evaluation.

MMW735 Simulation of Mobile Communications

Using MATLAB as a simulation tool to implement and characterize various wireless communication schemes. Signal processing, wireless Communication block diagram, Analogue and digital modulation, Constellation diagram, Eye diagram, Spectral analysis, Baseband simulation, Bit Error Ratio computation, Monte Carlo simulation, Equalisation, Carrier simulation

MMW736 Wireless Transceiver Design and Implementation

Noise and Noise figure: Sources of noise, Noise models, Noise figure, Cascaded noise figure, Measurement of noise figure. Link budgets: Sources of loss, Link loss equations, Maximum noise figure, Maximum range. The superheterodyne: Filter selectivity, Adjacent channels, Image frequencies, Multiple stage superheterodyne, The frequency mixer, Intermodulation products. Transceiver Design: Components used in transceiver designs, Typical transceiver design examples, Specifying amplifiers, mixers. Synchronisation: The timing and carrier synchronisation problem, Timing sync methods, delay locked loop and DSP equivalent, Open-loop timing sync, Zero Crossing detection, Carrier sync methods, Carrier regeneration, Costas loop introduction, Decision directed. Review of Passive Filters: Synthesis of doubly terminated filters, Low-Pass to Band-Pass transformation, Filter implementations - ceramic, SAW, Fractional bandwidth, Software-assisted specification and design, Review of Matching, Maximum power transfer, RLC Matching networks. Power amplifiers: Class A, B, AB, C, Power efficiency, Linearization techniques, and Suitability of PA classes to modulation types. Antennas: Fundamental model: Types of omnidirectional antennas.

MMW737 RF Engineering

Components at RF, resonant circuits; Q factor, Fixed Q, High Q and low Q impedance matching; Semiconductors at RF and BJT models at RF; s-parameters. Small-signal RF amplifier design: bias networks, stability and matching. Smith chart matching network design. Amplifier design using Smith charts: stability, gain control and matching network design. Low-noise design: choice of bias point and noise figure. Principles of microwave systems; microwave amplifier design; Transmitter architectures; Amplifier distortion: modelling, intercept point, harmonic and inter-modulation distortion. Large-signal RF amplifier types and classes.

MMW738 Advanced Research Methods

Research ethics, concept of research, defining research problem, problem identification, delimiting the research problem. Choosing a project: project styles and research methodologies; fitting a project to the researcher's interests, ambitions and capabilities; some characteristics of excellent projects. Preparing a project synopsis: aims and

objectives; deliverables; added value. Finding relevant literature: available resources and tools; primary and secondary sources; Writing research papers: planning the structure of your paper; framing the question; writing styles; tools and resources; citing and referencing; avoiding plagiarism; proofreading and corrections. Evaluating project outcomes: critical analysis; scope and limitations; evaluation styles: experimental, analytical, unit-testing, user analysis.

MMW831 5G and Future networks

Fundamentals of 5G mobile telecommunication: Standardisation, Air interface, Waveforms, MIMO and massive MIMO methods, Densification, SON and backhaul technologies, Cooperative communication. Study of structure and architecture of 5g base stations, eNodeB, Applications of 5G in automation, IoT etc. Explore life beyond 5G, Introduction 6G technology; features, characteristics, protocols for realisation.

MMW832 Broadband Wireless Communications

Design of broadband wireless communications systems: Data Link Control (DLC), Physical (PHY) layers: Fading, Interference, Hidden and Exposed Terminals and the possible absence of central control/centralised communication, high spectral efficiency and provision of QoS. Multiple Access and Duplexing strategies. Principles of Link Adaptation, ARQ and HARQ. The 802.11 and 802.11e MACs. CSMA/CA. Hidden and Exposed Nodes and RTS/CTS. Poor QoS provision in 802.11 and the improvements in 802.11e. TSPEC. 802.11a/b/g Physical Layers, Bluetooth MAC and PHY, WiMax MACs and PHYs and DOCSIS support. Single Frequency Networking and Mesh capabilities.

MMW833 Advanced Digital Signal and Image Processing

Introduction to the signal and image processing techniques, including analogue and digital filter design; FIR, IIR etc. Study of transform domain with particular focus on FFT and STFT and their applications. Concepts of statistical signal processing: estimation and detection theories. Development of: Optimal DSP programs, Real-time DSP algorithms for communications applications, Analysis of DSP device architectures and interfacing with DSP microprocessors, General purpose processors, Digital Signal processors, Modern FPGA architectures and their configuration process, Implementation of hardware designs in FPGA devices, C/C++ programming for hardware design, Comparison of FPGA and DSP features. Introduction to data compression and array processing. Introduction to Digital Image, Digital Image Processing System, Sampling and Quantization, Representation of Digital Image, Connectivity, and Image File Formats: BMP, TIFF and JPEG. Gray Level Transformations, Zero Memory Point Operations, Histogram Processing, Histogram equalization. Neighbourhood Processing, Spatial Filtering, Smoothing and Sharpening Filters, and Median Filter.

MMW834 Multiple antenna systems

Key techniques proposed for the 5G wireless systems. Principles of multiple-input multiple-output (MIMO), MIMO techniques for future wireless systems. MIMO transceiver optimization. Theoretical foundations of multi-user MIMO and massive MIMO, with an emphasis on principle characteristics, and implementation challenges. Cooperative Wireless Communications. Principles of relay communications. Multi-cell

cooperative communications: Uplink/downlink cooperative processing. MIMO technology including; Cooperative MIMO, MIMO, Multiuser MIMO and Massive MIMO

MMW835 Modelling and Performance

Background material: probability, conditional probability, Markov models, Queue modelling of OS, e.g. multi-tasking, proof (and uses) of Little's law. Workload modelling: exponential versus Pareto; call centre analysis. Simulation-how to generate random numbers from arbitrary distributions, steady state versus terminating; output analysis; some simple simulation applications. Reliability theory: oriented towards electronic systems, though e.g. passive component failure, and then to microprocessor (embedded software) systems through s/w failures Network Science: introduction to the fundamental ideas in network science: graph theory, network metrics, network models, network robustness. Approach to modelling emergence and topological robustness of supply networks, communication networks and general human-technology interaction.

MMW836 Communication technologies for Internet of Things (IoT)

IoT components and architecture; Technologies in each layer of the architecture; Sensors and Sensing technology; Data transmission and connectivity; IoT platforms, including offerings by Google and Apple; IoT specific data processing and analytics; IoT and open data; IoT user interface issues including data models and semantics; Applications of IoT including smart cities, smart homes; Ongoing and future IoT challenges. LSEPI issues related to setting infrastructure for IoT systems and handling IoT data. Radio Frequency Identification (RFID); Near Field Communication (NFC); Wireless Sensor Networks: covering its major concepts in node sensing, wireless transmission characteristics, medium access protocols, and routing protocols; Wireless Personal Area Networks such as the ones using IEEE802.15.4 standard, Zigbee, Zwave; Low Power Wide Area Networks such as LoRa and Sigfox systems; and Power line communications.

MMW840 Dissertation

The module provides students with an opportunity to design, undertake or conduct an independent piece of research of study related to their programme of study under the guidance of a supervisor who is usually a member of the academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN OPTIC FIBRE AND PHOTONIC ENGINEERING (MOF)

Duration:	2 years
Minimum Credit load:	336
Maximum Credit load:	360
Maximum MBKs Credit Load:	288

ZNQF Level: 9

1.0 PURPOSE OF THE PROGRAMME

- Provide students with advanced knowledge of optical fibre and photonic technologies.
- Give students the opportunity to work in a research-led environment using state-of-the-art facilities.
- Gain specialist knowledge of technologies that harness the power of light, such as lasers and optical fibres.
- Develop research skills applicable to a career in research and development.
- Stimulate interest in the subject using a variety of teaching and learning methods

2.0. ENTRY REQUIREMENTS

Prospective students must hold at least a second class first degree in Physics, Applied Physics or any other material science related degree from any university recognized by Midlands State University.

3.0 PROGRAMME CHARACTERISTICS

3.1 *Areas of Study*

Specialised fibre design and fabrication, optoelectronics, passive fibre devices, photonic signal processing, optical wave-guides, lasers, photonic materials and structures, biophotonics and sensors, microelectronic device physics, nanophysics and advanced electronics

3.2 *Specialist Focus*

This is a unique programme that brings the applications of fibre based, optical and photonic technologies in areas as diverse as optical sensing, manufacturing, medicine, defence and telecommunication.

3.3 *Orientation*

Research and innovation oriented. Teaching and learning are academically and professionally oriented and focused on practical aspects

3.4 *Distinctive Features*

The programme builds the research-technology-innovation continuum and focuses on knowledge development and application using a student-centred approach

4.0 CAREER OPPORTUNITIES AND FURTHER EDUCATION

4.1 ***Employability***

Fiber Optics Engineer, Optical Scientist, Laser Systems Engineer, Electro Optical Engineer, Optomechanical Engineer, Optical Fabricator, System Engineer, system designer, research and design scientists, researchers in industry and academia.

4.2 ***Further Studies***

Doctoral studies in Opto-Mechatronics Engineering, Quantum Optics and Optoelectronics, Laser, Photonics and Vision, Signal and Information processing, Optics and Photonics, Fibre Telecommunications, Physics, material science, semiconductor Physics, nanotechnology or related areas.

5.0 PROGRAMME DELIVERY

5.1 ***Teaching and Learning Methods***

Lectures, tutorials, practicals, group presentations, group work, mini research projects, industrial visits, research project, and individual independent study

5.2 ***Assessment Methods***

Written and oral examinations, tests, laboratory reports, seminar presentations, final year dissertations, continuous assessments

Continuous assessments:

Test	(15%),
Practicals	(20%),
Assignments	(5%)
Written Examinations:	60%

6.0 LEARNING OUTCOMES

On successful completion of this programme the graduate will be able to:

- Work competently with lasers, optical fibres and related technology to gather experimental data.
- Work with a range of optical systems, including lasers and optical fibre-based devices, and their applications in real-world settings.
- Process, analyse, and display data from optical experiments.
- Perform practical tasks such as fusion splicing and fibre characterisation.
- Employ techniques for analysing and characterising optical systems.
- Understand scientific and engineering principles underlying a range of optical fibre and photonics technologies.
- Use tools and techniques used to design, construct, measure, analyse a range of optical systems, including lasers and optical fibre-based devices.

- Understand current research relevant to optical fibres and photonics.

7.0. GENERAL PROVISIONS

7.1 The programme shall be studied over a minimum of period of 4 semester on a block release basis

8.0 PROGRAMME ASSESSMENT

8.1 *Coursework*

Coursework shall comprise written assignments, practical work, projects, reports, presentations and tests and shall account for 25% of the final mark for theoretical courses and 40% of the final mark for practical courses.

8.2 *Written Examinations*

A three-hour final examination, which shall contribute to 75% of the final mark for theoretical courses and 60% for practical courses

8.3 *Practical Examination*

The duration of a practical examination, where it is deemed necessary, and their contribution to the final mark, will depend on the module.

9.0. PROVISION FOR PROGRESSION

9.1 A student must pass all core modules to proceed to level 2.

9.2. All level 1 and level 2 core modules should be passed before proceeding to the dissertation.

10.0. FAILURE TO SATISFY THE EXAMINERS

Refer to Section 11 of the General Regulations.

11.0. GRADING AND DEGREE CLASSIFICATION

To be awarded a Masters of Science degree a student must pass each core module listed in that programme and must have accrued at least 336 credits.

12.0. DEGREE WEIGHTING

For the purpose of degree classification, the weighting of the results of the different levels shall be:

Level 1 50%

Level 2 50%

13.0. PROGRAMME STRUCTURE

Level 1 Semester 1	**Credits**
MOF731 Advanced Quantum Mechanics*	24
MOF732 Optics and Photonics*	24
MOF733 Optical Communication Systems*	24
MOF734 Optical Fibre Sensors*	24
Level 1 Semester 2	
MOF735 Lasers Physics and Applications	24
MOF736 Optical Engineering*	24
MOF737 Ultrafast Physics & Plasmas*	24
MOF738 Advanced Research Methods*	24
Level 2 Semester 1 (choose any three modules)	
MOF831 Laboratory and study skills	24
MOF832 Fabrication and Characterization of Optoelectronic Devices	24
MOF833 Nano-Optics	24
MOF834 Optical Systems in Medicine and Life Science	24
MOF835 Advanced Digital signal processing	24
Level 2 Semester 2	
MO840 Dissertation*	96

Module Synopsis

MOF731 Advanced Quantum Mechanics
A study of quantum mechanics including combination of two or more quantum mechanical systems, addition of angular momentum, time independent perturbation theory, and time dependent perturbation theory.

MOF732 Optics and Photonics
Wave optics, physical optics and introductory laser physics. Wave optics content: solutions of the wave equation, including Hermite-Gaussian laser modes, optical cavities, Fresnel and Fraunhofer diffraction integrals; Laser Amplifiers, Semiconductor photon source, Electro-optics, Semiconductor photon detectors

MOF733 Optical Communication Systems

Introduction to Network types, Network architectures, Basic network types, Long-haul transmission systems, Metropolitan-area networks, Local-area networks. Basic characteristics of Fibres; Linear impairments, Dispersion, Polarization. Passive and Active components, Multiplexing and Modulation, spectrally efficient formats. Direct detection vs. differential detection & Coherent receivers. The module provides knowledge about optical networks and systems with applications ranging from photonic interconnects, to fiber-to-the-home (FTTH), optical metro and long-haul networks, and automotive and industrial automation. The role of various network layers will be discussed in conjunction with relevant standards and protocols. Physical-layer specifications of relevant photonic components and system design trade-offs will be introduced.

Systems considerations in communication systems, Capacity of a transmission system, Capacity limits in fibre systems, Performance evaluation, Noise in transmission, Control of optical nonlinearities, Exploiting fibre nonlinearities and the role of electronic digital signal processing

MOF734 Optical Fibre Sensors

Governing standards for sensing systems. Optical sensing principles (temperature, strain, stress, pressure, refractive index, etc.). Fibre types and materials for optical fibre sensing (silica based, polymer based, etc.).

Point sensors (Fibre Bragg gratings, long period gratings, and microfibres/nanowires). Design, fabrication and characterisation of point sensors. Distributed sensors (Brillouin scattering based, Raman scattering based, Rayleigh scattering based).

Design, fabrication and characterisation of distributed sensors. Fibre gyroscopes. Fibre based gas and chemical sensors. Optical fibre sensors for extreme and harsh environments (high temperature and strain, shock, high radiation).

Principles and application of optical fibre sensors in medicine and life sciences. Principles and application of optical sensors in oil and gas exploration. Principles and application of optical sensors in civil engineering, e.g., structural monitoring and aircraft navigation.

MOF735 Lasers Physics and Applications

The principles of laser operation will be discussed, with reference to commonly used laser systems. The course provides knowledge of the laser as a fundamental tool of contemporary science and technology. The course will give a detailed and mathematical introduction to gain media, laser cavities, Gaussian beams, and their combination into many forms of laser.

- Properties of coherent light: temporal and spatial coherence, beam propagation
- Interaction of light and matter: spontaneous & stimulated emission, cross-sections & rate equations
- Gaussian beam optics, optical resonators, cavity stability
- Generation of coherent light: energy storage, resonator modes, laser threshold
- Laser dynamics: steady state and transient behaviour, relaxation oscillations, Q-switching, mode locking
- Practical laser systems:
- Ultra-fast lasers, properties of ultra-fast pulses, X-ray and attosecond pulses

MOF736 Optical Engineering

The course focuses on the practical aspects of designing optical components and instruments such as lenses, microscopes, optical sensors and measurement systems, and optical disc systems (e.g. CD, DVD, HVD). The course explains the layout of modern optical systems and gives an overview over available technology, materials, costs, design methods, as well as optical design software.

MOF737 Ultrafast Physics & Plasmas

The course provides an introduction to laser-plasma interaction, in particular with very high power and ultrashort pulses, and the resulting applications in radiation sources from the terahertz to the X-ray region, laser fusion and laser-based particle acceleration.

MOF737 Advanced Research Methods

Research ethics, concept of research, defining research problem, problem identification, delimiting the research problem. Choosing a project: project styles and research methodologies; fitting a project to researcher's interests, ambitions and capabilities; some characteristics of excellent projects. Preparing a project synopsis: aims and objectives; deliverables; added value. Finding relevant literature: available resources and tools; primary and secondary sources; Writing research papers: planning the structure of your paper; framing the question; writing styles; tools and resources; citing and referencing; avoiding plagiarism; proofreading and corrections. Evaluating project outcomes: critical analysis; scope and limitations; evaluation styles: experimental, analytical, unit-testing, user analysis.

MOF831 Laboratory and study skills

In the Laboratory part students will carry out a selection of experiments from the list below and make a short presentation on one of the experiments.

Fibre optics and optical waveguiding, Semiconductor pn junctions, Experimental Neodymium YAG Laser, Electro-Optic Effect and Modulation of Laser Light, Optical spectroscopy, Laser modes and speed of light, Fluorescence of laser glasses or Diode Lasers.

MOF832 Fabrication and Characterisation of Optoelectronic Devices

The students build knowledge on process technology for the fabrication of a range of optoelectronic devices, including LEDs, solar cells, laser diodes, photodiodes, etc. They learn to compare the advantages of different technological approaches, including their economic boundary conditions. This is a technological-based course where students will use their prior fundamental knowledge to gain a firm grasp on the fabrication sequences and characterisation (optical, electrical, electronic, materials) steps that are required to realise the above devices.

MOF833 Nano-Optics

This module will focus on general concepts and device developments in the rapidly evolving field of nanoscale optics and photonics. Topics covered include; nano-scale and near-field optics, near-field optical probes, quantum confined materials, plasmonics. Nanophotonic devices to be discussed include: photonic crystals, nanolasers, single-photon sources, nanostructured solar cells and sensors

MOF834 Optical Systems in Medicine and Life Science

The basic optical and optoelectronic principles are applied in the design of modern medical devices and routine diagnostic equipment. Some examples of optical applications that may be covered are imaging, spectroscopy, illumination, adaptive optics, communication, detection and metrology especially in Medicine and Life Science. These systems will be used to teach fundamentals of systems engineering, optical system design, quantifying performance for optical systems, specification of optical components and professional engineering skills.

MOF835 Advanced Digital Signal and Image Processing

Introduction to the signal and image processing techniques, including analogue and digital filter design; FIR, IIR etc. Study of transform domain with particular focus on FFT and STFT and their applications. Concepts of statistical signal processing: estimation and detection theories. Development of: Optimal DSP programs, Real-time DSP algorithms for communications applications, Analysis of DSP device architectures and interfacing with DSP microprocessors, General purpose processors, Digital Signal processors, Modern FPGA architectures and their configuration process, Implementation of hardware designs in FPGA devices, C/C++ programming for hardware design, Comparison of FPGA and DSP features. Introduction to data compression and array processing. Introduction to Digital Image, Digital Image Processing System, Sampling and Quantization, Representation of Digital Image, Connectivity, and Image File Formats: BMP, TIFF and JPEG. Gray Level Transformations, Zero Memory Point Operations, Histogram Processing, Histogram equalization. Neighbourhood Processing, Spatial Filtering, Smoothing and Sharpening Filters, and Median Filter.

MOF840 Dissertation

The module provides students with the opportunity to design, undertake or conduct an independent piece of research of study related to their programme of study under the guidance of a supervisor who is usually a member of the academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

MIP838 Nuclear Metrology

Introduction to radiation metrology, Calibration standards, Liquid scintillation counter
Naturally Occurring Radioactive Materials,
Nuclear data: Why is it important: How to measure it: Where to get data from:
Nuclear forensics, Primary standards, Quality assurance, Radiochemical separation, Radiochemistry, Reference materials, Source preparation, Uncertainties.

MIP839 Radiation Laboratory

A range of experiments designed to provide familiarity with the full set of radiation detector systems, including alpha and beta particle detection, X- and gamma-ray spectroscopy and neutron detection.

MIP840 Nondestructive Testing

Radiography: Principle of radiography, types of radiography, equipment for neutron

radiography, x-ray radiography, equipment for x-ray radiography, advantages and applications of fluoroscopy and photo fluoroscopy
Electromagnetic methods: Principle of electromagnetic testing, mathematical analysis, flaw detection in conductors, various types of instruments used and advantages of various electromagnetic methods for crack detection etc.
Ultrasonic methods: Principle of ultrasonic testing, generation of ultrasonic waves, equipment details for ultrasonic checking, methods of wave propagation, methods of flaw detection, various methods of ultrasonic testing, advantages of ultrasonic methods for flaw detection and crack location
Holography: Principle of holography, method of holographic recording, method of holographic reconstruction, advantages of this technique and applications of holographic methods for non-destructive testing.
Liquid penetrant testing: Principle of liquid penetrates testing, types of dyes and penetrants used in this testing technique and application of liquids for detecting sub-surface defects.
Magnetic particle testing: Principles of magnetic particle testing, details of equipment used and methods of crack detection by magnetic particle testing
Hardness testing: Brinnel hardness testing, Rockwell hardness tests, shore hardness testing, Vicker hardness testing and theory behind various hardness testing methods.

MIP841 Industrial Electronics
Introduction to various electronic components and systems used in modern industry. Operational amplifier principles and applications include comparators (zero and non-zero crossing detectors), voltage followers, and inverting and non-inverting amplifiers. Subtraction, summing (mixer), difference and compound amplifiers and active filters. Operational amplifiers circuits are configured to make up complex analog circuits. Examples of these include the temperature controller and the pulse width modulation technique of DC motor speed control. The importance of digital computers used in modern industrial processes is stressed. Thyristors, photosensitive devices, optically coupled devices, and timer control circuits and various transducers are introduced.

MIP842 Computer Interfacing
Introduction to basic concepts and techniques for interfacing a micro-controller to external devices for data collection and process control and developing the related software required. Transfer and converting analog variables into the digital form needed for processing for data acquisition and real-time control systems, design and construct simple control, data logging system incorporating input/output to and from external devices and design simple control system for stepper and DC motor.

MIP843 Advanced Embedded Systems
Characteristics of embedded computing applications and challenges in advanced embedded computing system design
. The embedded system design process
. Processor architectures 1089
. Instruction sets, processor and System on Chip architectures
. Hardware and Software System design and analysis techniques
. Processes and operating systems
. Embedded communication architectures and multiprocessing
. System design methodologies
. Various embedded systems will be examined as case studies

MIP844 Advanced Digital Signal and Image Processing
Introduction to the signal and image processing techniques, including analogue and

digital filter design; FIR, IIR etc. Study of transform domain with particular focus on FFT and STFT and their applications. Concepts of statistical signal processing: estimation and detection theories. Development of: Optimal DSP programs, Real-time DSP algorithms for communications applications, Analysis of DSP device architectures and interfacing with DSP microprocessors, General purpose processors, Digital Signal processors, Modern FPGA architectures and their configuration process, Implementation of hardware designs in FPGA devices, C/C++ programming for hardware design, Comparison of FPGA and DSP features. Introduction to data compression and array processing. Introduction to Digital Image, Digital Image Processing System, Sampling and Quantization, Representation of Digital Image, Connectivity, and Image File Formats: BMP, TIFF and JPEG. Gray Level Transformations, Zero Memory Point Operations, Histogram Processing, Histogram equalization. Neighbourhood Processing, Spatial Filtering, Smoothing and Sharpening Filters, and Median Filter.

MIP845 Renewable Energies

This course provides an introduction to the principles of renewable energy. The course covers different types of renewable and alternative energy sources, and discuss their basic principles of operation, achievable efficiency, cost, and their effect on the environment. The impact of the new energy technologies on the environment, obstacles to their wide implementation in industrial and consumer applications, and the role of social attitudes and government planning, financial investments and incentives are also introduced.

MIP846 Low Carbon Energy Solutions

This course explores the challenges associated with climate change, energy and environmental policies from multiple perspectives, disciplines and scales. Students will examine the evolving science and policy of climate change, observe the conflicts between energy companies and other interest, discuss technology innovations and its impact on reducing greenhouse gases emission. Parallel to that, local energy and environmental policies will be scrutinized in a context of global politics. Through various case studies, students are encouraged to grasp the nature of national development strategies and the risks and challenges the country is facing in implementing its “green policies”.

MIP847 Solar Cell Technology

Photovoltaic Systems: Technology of photovoltaic systems, approaches for measuring the performance of individual systems under real operating conditions. different approaches used in system level modelling, tools to perform real analyses. Cell, module and system performance. The implications of latitude, shading, temperature, and system geometry. models for diffuse vs direct sunlight. Physical and statistical approaches to system modelling, data from real PV systems. Solar Cell Laboratory: measure and characterize solar cell performance and solar cell materials using standard research laboratory techniques e.g. measuring solar cell efficiency, characterisation of degradation rates, measuring irradiance and light spectrum, and using a cryostat to determine operating characteristics at low temperatures. Fabrication of thin film solar cells and the characterization of photovoltaic materials using photoluminescence and absorption spectroscopy. assembling a silicon photovoltaic module and measure its performance in outdoor conditions Innovation in Solar Energy: innovation process. history of innovation in solar photovoltaics up to modern day markets and systems. Innovation and business idea proposal, reviewed and critiqued by peers and academic supervisors along with external industry experts. Low Carbon

Energy, Science and Technology: The need for various different technologies that can help to meet the world's energy needs without releasing large amounts of CO2 into the atmosphere. Various different technologies will be introduced and a select number will be studied in more detail. make a reasoned comparison between the different low carbon technologies backed by sound scientific understanding of their limitations and advantages.

MIP848 Nanotechnology in Energy Conversion and Storage

This course is designed to provide the student with an understanding of the contributions and potential contributions of advances in nanotechnology to various sustainable energy production/conversion, storage and utilization technologies. Students will be introduced to solar, hydrogen, new generation of batteries and supercapacitors as the most significant contributions of nanotechnology to the energy sector.

MIP849 Interfacing clean energy systems

The unit introduces technology which sits between the energy generation device (solar cell, turbine etc.) and the distribution system and includes interfacing to high-efficiency loads. The unit will give an overview of the structure, function and applications of electrical energy conversion systems, allowing students to select technology and undertake top level design and performance calculations.

Power Electronic Converters

Three main classes of power electronic converter (AC/AC, DC/DC, AC/DC); components, operational principles and steady-state characteristics, limitations, efficiency, control of systems; system modelling.

Electromechanical Energy Conversion

Motor, generator and actuator technologies, characteristics and efficiency; generator, drive and actuator control and optimisation; effect of typical end-user applications, including electric vehicles.

Energy Interfacing

Interfacing to renewable energy sources; interfacing to cleaner technologies; full-system energy flows to/from supply and to/from loads; selection of technologies and configurations; system modelling and control; case studies of energy efficiency enhancement; energy storage, system reliability and condition monitoring.

MIP850 Dissertation

The module provides students with an opportunity to design, undertake or conduct an independent piece of research of study related to their programme of study under the guidance of a supervisor who is usually a member of the academic staff of the department. Runs over two semesters: Regular report backs to the departmental board by the supervisor. The project is continually assessed throughout two semesters. A student undertakes a viva for the project.

DEPARTMENT OF FOOD SCIENCE AND NUTRITION

REGULATIONS FOR THE BACHELOR OF SCIENCE HONOURS DEGREE IN FOOD SCIENCE AND TECHNOLOGY (FST)

Duration	4 years
Minimum Credit Load	480
Maximum Credit Load	540
Programme Credit Load	540
MBKs Credit Load	420
ZNQF Level	8

1 **ENTRY REQUIREMENTS**

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry: At least 2 Advanced Level passes including Chemistry, Biology or Food Science and one subject from, Physics, Mathematics or Agriculture

Special Entry: National Diploma or Higher National Diploma in relevant field

Mature Entry: Refer to General Academic Regulations

2 LEARNING OUTCOMES

1. Ability to integrate chemical, biochemical, microbiological, engineering and sensory analysis principles to develop foods that are safe, nutritious, and flavorful

2. Ability to use the necessary mathematical, computer, statistical, and critical thinking skills to design experiments, collect and interpret data, and predict the impact of raw material and processing changes on food safety and quality.

3. Demonstrate laboratory skills and knowledge to identify the important microorganisms in foods, and the conditions under which they grow. Familiarize with methods, including sanitation procedures, to control spoilage and maintain the safety of foods.

4. Demonstrate ability to evaluate quality of new food products using objective and subjective methodologies.

5. Demonstrate ability to innovate food products and processes as well as industrialize them

3 ASSESSMENT

Coursework:	40%
Test	*15%*
Practicals	*20%*
Assignments	*5%)*
Written Examinations:	60%

PROGRAMME STRUCTURE N.B * Denotes core modules and modules in bold are MBKs CODE MODULE DESCRIPTION CREDITS

Level 1 Semester 1	Credits
FST 111* Introduction to Food Science and Technology	12
FST 112 Cell Biology and Genetics	12
FST 113* Organic chemistry	12
HCSCI 131 Introduction to Computers and Computer Applications	12
CS 131 Basics of Communication Skills	12
FST 114 Foundation Mathematics for Food Scientists	12

Level 1 Semester 2	Credits
FST 121* Food Chemistry	12
FST 122* Food Engineering 1	12
FST 123* Food Microbiology	12
FST 124* Food Processing and Preservation Technology	12
Choose one elective	
FST 125 Food Biochemistry	12
FTS 126 Analytical biochemistry	12
FST 127 Principles of Nutrition	12

Level 2 Semester 1	Credits
GS 211 Introduction to Gender Studies	12
TCNP 201 Technopreneurship	12
FST 211 Fruits and Vegetables Technology	12
FST 212* Dairy Technology	12
FST 213* Meat, Poultry and Fish Technology	12
FST 214 Cereals and Legume Technology	12

Level 2 Semester 2	
FST 221* Food analysis and Sensory evaluation	12
FST 222* Research Methods and Statistics	12
FST 223 Quality Assurance and Food Laws	12
FST 224 Fats and Oils Technology	12
Choose one elective	
FST 225 Beverage Technology	12
FST 226 Sugar and Confectionary Technology	12
FST 227 Bakery Technology	12

Level 3 Semesters 1	
FST 331 Work-Related Learning I	40

Level 3 Semester 2	
FST 332 Work-Related Learning II	80

Level 4 Semester 1	Credits
FST 411 Food Engineering II	12
FST 412* Food Safety and Plant sanitation	12
FST 413 Nutraceuticals and Functional foods	12

FST 414 Food Packaging technology	12
FST 415*Food product development and Marketing	12

Level 4 Semester 2

FST 421 Food Fermentation technology	12
FST 422*Advanced Food Microbiology and Biotechnology	12
FST 423 Food Production Management	12
FST 424 Food Toxicology	12
FST 425* Dissertation	24

MODULE SYNOPSES

CS 131 Basics of Communication skills
Refer to the Department of Communication Skills

HCSCI 131 Introduction to Computers and Computer Applications
Refer to the Department of Computer Science.

TCNP 201 Technopreneurship

Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, and product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

GS231 Introduction to Gender Studies

This module will empower the students with knowledge and skills that enable them to be gender sensitive in the University, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Students gain insight into accounts of gender studies in Science and Technology

FST 111 Introduction to Food Science and Technology

The importance of Food Science and Technology lies in the fact that it has capability to provide food to our population through scientific conservations, eliminating avoidable losses and making available more balanced and nutritious food. This module introduces the basics concepts of food science and food technology. To understand the history and evolution of food processing. The module is designed to enable the students to gain an insight into basic aspect of fruit and vegetable processing, to understand the importance of advance techniques in food analysis, acquire knowledge of availability of raw materials, regarding processing of animal proteins, recent advances in dairy technology, novel foodstuffs, and awareness of importance of cereal, legume and oil seed technology. The students will also gain knowledge about various packaging materials and importance of packaging.

FST 112 Cell Biology and Genetics

Cell biology focuses on the structure and function of the components of living cells (such as the cell membrane, the flagella responsible for cell movements, cell differentiation in developing organisms, abnormal cell division, and metabolic interactions between cell compartments) and how food and the diet is integrated. The module will further introduce the interaction of foods, and individual food constituents, with the biological systems and can be defined on a molecular basis i.e. the cell. Specifically, the module

aims to underline approaches to investigate how genes determine physical traits, including susceptibility to diseases and inherited disorders.

FST 113 Organic Chemistry
The module introduces food science students to basic organic chemistry concepts. The focus is on the study of carbon compounds, their structure, bonding and nomenclature. Furthermore, this will also include the study of alkanes, alkenes, alkynes, alkyl halides and their reactivity with emphasis on reaction mechanisms, addition, substitution, elimination reactions and stereochemistry. The chemistry of aromatic compounds, free radicals, macromolecules, carbohydrates, lipids, amino acids, peptides, proteins, and synthesis of natural and complex synthetic polymers will also be covered.

FST 114 Foundation Mathematics for Food Scientists
Mathematics is the language of science, and the study of any science requires some degree of mathematical reasoning and understanding. Mathematical principles in the food science and technology field address the theoretical foundations. Components include basic linear algebra, Functions, Graphs, Differential and Integral Calculus. Students will be expected to apply techniques from calculus to a variety of applied problems in food science.

FST 121 Food Chemistry
The core competency of this course is for students to differentiate chemical interactions and reactions of food components and their effect on sensory, nutritional, and functional properties of foods, and how processing influences these properties. Food chemistry-definition and importance, Carbohydrates-chemical reactions, functional properties of sugars and polysaccharides in foods. Applications and preparations of sugars and polysaccharides. Protein and amino acids: structure, classifications, sources, denaturation and functional properties of proteins. Maillard browning. Lipids: classification, and use of lipids in foods, physical and chemical properties, effects of processing on functional properties. Vitamins and Minerals, Effect of processing on vitamins and minerals. Enzymatic browning in foods and industrial applications of enzymes. Water in food, water activity and shelf life of food. Natural food flavours, extraction methods and characterization. Pigments in food and their industrial applications. Additives in food processing and preservation. Their functions and safety. Safety and quality evaluation of additives. Various additives such as preservatives, antioxidants, emulsifiers, sequesterants, humectants, stabilizers with respect to chemistry, food uses and functions in formulations

FST 122 Food Engineering Operations
Unit operation in food engineering, Mass and energy balance, Fluid flow, fluid statics,
fluid dynamics, fluid flow applications, Heat transfer-modes of heat transfer, conduction, convection, and radiation, heat exchangers and their designs. Thermal processing-evaporation and concentration, dehydration, drying, blanching, pasteurization, sterilization, distillation and crystallization. Mechanical separation-filtration, membrane concentration, sieving, centrifugation, sedimentation, Mechanical handling-conveying and elevation. Size reduction and classification-mixing, kneading, blending. Application of psychrometric charts in food engineering, Generation of steam and its applications, Measurement of pressure, and flow of fluids. Study of heat exchangers, dryers, elevating and conveying equipment, size reduction equipment, and sieve analysis.

FST 123 Food Microbiology 1095
This module is designed to introduce students to various aspects of food microbiology, organisms associated naturally with foods and those responsible for spoilage. Bacteria, moulds, yeast, viruses, protozoa, algae - their classification and growth characteristics. Conditions favouring the growth, death and survival of microorganisms in foods will also be studied; their immediate and long range effects on foods will be discussed. Issues of foodborne diseases, food spoilage and preservation will be treated. Topics to be covered include; Introduction to food microbiology; factors that affect interactions of microorganisms with foods; microorganisms important in foods: Also included are food spoilage and preservation; food-borne diseases - infections and intoxications; public health and sanitation and

microbiology of water. Students should also learn principles and methods for the microbiological examination of foods; microbiological quality control, and quality schemes.

FST 124 Food Processing and Preservation Technology
Scope and importance of food processing: national and international perspectives,
Principles of Preservation methods, fermentation methods for preservation, and chemical preservations of foods. Food preservation by low-temp: Refrigeration, freezing and freeze-drying. Food preservation by heating: drying, osmotic dehydration, blanching, canning, pasteurization, sterilization, extrusion cooking. Non-thermal preservation: Hydrostatic pressure, dielectric heating, microwave processing, hurdle technology, membrane technology, irradiation.

FST 125 Food Biochemistry
Pathways of metabolism of carbohydrates, proteins, lipids. Enzyme biosynthesis and regulation. Enzymes as biocatalysts – chemistry, classification, mode of action, specificity, assay techniques, isolation and purification, stabilization, enzyme kinetics. Applications of enzymes. Metabolic regulation, hormones, Release of energy and its trapping. Metabolic rate and caloric needs.

FST 126 Analytical Biochemistry
The module aims to provide an introduction to the theories underlying the common analytical techniques applied in the food sciences and technology laboratory. Principles on experimental models, experimental errors and instrument standardization will be covered. The principles, instrumentation and applications of chromatography, electrophoresis, spectroscopy, electrochemical, centrifugation, immunochemical, mass spectrometry and radioisotope techniques will also be covered, biased towards food technology. The development of skills in accuracy and precision in laboratory analysis and logical report writing and representation of analytical data will also be covered.

FSN 127 Principles of Nutrition
Classification, digestion, functions, dietary sources, RDA, clinical manifestations of deficiency and excess and factors affecting absorption of the following : Energy, Carbohydrates, lipids and proteins; Fat soluble vitamins-A, D, E and K; Water soluble vitamins – thiamin, riboflavin, niacin, pyridoxine, folate, vitamin B12 and vitamin C; Minerals – calcium, iron, iodine, fluorine, copper and zinc.

FST 211 Fruit and Vegetable Technology
This module covers the principles and methods of fruit and vegetable preservation. Classification of fruits and vegetables, general composition, enzymatic browning, names and sources of pigments. Principles of storage of fruits and vegetables. Types of storage and preparation of food products post-harvest changes in fruits and vegetables will also be considered.

FST 212 Dairy Technology
This module focuses on the sources, and composition of milk, processing of market milk, standardization, toning of milk, homogenization, pasteurization, sterilization, storage, transportation and distribution of milk. Milk product processing-cream, butter, condensed milk, evaporated milk, whole and skimmed milk powder. Analysis of milk products will be covered in detail.

FST 213 Meat, Poultry and Fish Technology
Meat and poultry are essential components of a balanced diet. Chemical composition and microscopic structure of meat. Meat quality evaluation as well as meat plant sanitation and safety will be covered. Poultry: classification, composition, preservation methods and processing as well as processing of egg products. Students will also learn processing and quality control measures of fish and seafood technology.

FST 214 Cereals and Legume Technology
General introduction to cereals, production trends of cereals, Structure and nutrient distribution in cereals, wheat types, milling of wheat, quality of flour and flour treatment. Students will also learn technology of corn-wet milling and dry milling, corn flakes. Corn starch and its hydrolyzed syrups will also be

considered. Production and utilization of small grains (sorghum, millet, oats etc.). Module will also cover structure and composition of legumes, their importance in diet. Milling and processing of pulses, anti-nutritional factors in legumes will also be covered.

FST 221 Food Analysis and Sensory Science
Food analysis is the discipline dealing with the development, application and study of analytical procedures for characterizing the properties of foods and their constituents. Sensory science is a scientific discipline used to evoke, measure, analyze and interpret those responses to products that are perceived by the senses of sight, smell, touch, taste and hearing hence determining consumer behaviour as well as food descriptive sensory profiles.

FST 222 Research Methods and Statistics
The aim of this module is to provide the student with the essential knowledge of the process of conducting sound scientific research, project writing and develop skills in data handling and statistical analysis. Identify and critically evaluate research data and research sources; Evaluate a range of data collection procedures; Critically analyse, evaluate and utilise research findings; Critically assess the role and value of a range of statistical parameters using appropriate software packages; Evaluate the ethical issues arising from research.

FST 223 Quality Assurance and Food Laws
Standardization; involves standard development processes and roles, the importance of standards in food processing, standards that are relevant to food processing as well as technology (ISO 9001, ISO 22000, ISO 14 000, ISO 31000 etc). Focuses on national, regional, and international food laws; adoption, interpretation and enforcement of laws and regulations governing food processing and food service systems.

FST 224 Fats and Oils Technology
This module will cover the classification of lipids, and types of fatty acids - saturated fatty acids, unsaturated fatty acids, essential fatty acids, trans fatty acids. Rancidity Types - hydrolytic and oxidative rancidity and its prevention. Importance of oil seeds processing. Commercial edible oil sources. Processing of crude oils and refining.

FST 225 Beverage Technology
This module covers the processing and technologies of alcoholic and non-alcoholic beverages; and physical, microbiological, and chemical properties of both raw materials and finished products. Students will use laboratory techniques to analyze and measure important physicochemical parameters of beverages.

FST 226 Sugar and Confectionery Technology
The course will cover Sugar confectionery, Bakers' confectionery, Sweetening agents and Nutrition – confections (micronutrients and calorific value). The confectionery industry will be subdivided into three categories that is sugar confectionery, chocolate confectionery and flour confectionery. Sugar processing and Sugar substitutes regulation are included.

FST 227 Bakery Technology
This module covers the science of bakery product. Bread making: chemistry of dough development, making of bread, types of bread, variants of bread. Products other than bread: pastry, biscuits, wafers, cakes and other chemically leavened products. Dietetics bakery products and Quality control in the bakery will be covered in detail.

FST 411 Food Engineering II
This module covers principles of operation and design of industrial equipment, used in the processing, storage and packaging of foods. Food processing equipment is classified and described according to the basic unit operations, including mechanical transport, mechanical processing and separations, heat transfer operations, evaporation, dehydration, thermal processing, etc. The role of automation in food

industries, benefits of automation and introduction to automation tolls like PAC, PLC and SCADA will be covered.

FST 412 Food Safety and Plant Sanitation
This course is designed to cover the essentials of food safety, food microbiology, sanitary design principles for facilities and equipment, worker hygiene practices, correct procedures for cleaning and sanitizing, food defence and plant security. Food hygiene constitutes a basic necessity of Good Manufacturing/Agricultural Practices and the development of Hazard Analysis Critical Control Point (HACCP), as well being as a component of all benchmarked food safety standards (ISO 22 000).

FST 413 Nutraceuticals and Functional Foods
Functional foods are considered as those foods which are intended to be consumed as part of the normal diet but contain bioactive substances that offer the potential of enhanced health or reduced risk of disease. Consumer awareness and interest in the relationship between diet and health has increased substantially and health is now a major driver for market positioning of foods. The module covers the issues and challenges in the development, evidence testing, marketing and changing regulations controlling functional foods and nutraceuticals. It will prepare students to assist industry in exploiting this growing opportunity.

FST 414 Food Packaging Technology
This module will give a broad and clear overview of the packaging of food products. It will start with the basics of chemical, physiological and microbiological degradation of food products and how packaging can affect these spoilage phenomena. Evaluation of packaging performance, packaging equipment, packaging standards, regulations and labelling will be covered.

FST 415 Food Product Development and Marketing

This module will enable students to understand the new product development process and the strategic features of new product development; develop strategic thinking, planning, and managing abilities throughout the entire new product development process; and understand various techniques in identifying new product opportunities in the food industry. The students will be able to identify new product opportunities and introduce financial, environmental, social, and cultural considerations with regard to design decisions.

FST 421 Fermentation Technology
This module covers the use of microorganisms in fermentation technology and biotechnology. Metabolic activities of microorganisms and their influence on product attributes. Fermenter design, operation, measurement and control in fermentation will also be covered. Production of fermented products and principles of downstream processing and product recovery are essential.

FST 422 Advanced Food Microbiology and Biotechnology
This module is designed to cover the basic principles of biotechnology and applied food microbiology. Introduction to recombinant DNA techniques and how they are applied to genetically modify microorganisms. The use of nucleic acids as tools of rapid detection of microorganisms in foods and regulatory aspects of food biotechnology. This module will enable students to comprehend the theory and concepts of current biotechnology trends as it relates to food safety and security issues. Regulatory and social aspects of biotechnology of foods.

FST 423 Food Production Management

This module is designed to equip students with knowledge of the management skills required to contribute to the industry. This module aims to provide students a comprehensive coverage of the

management of food and agribusiness firms from a managerial perspective by covering the areas of expertise that a manager must master (finance, marketing, operations principles and concepts, business ownership, organizational management and human resources).

FST 424 Food Toxicology
Toxicology concepts covered include dose-response relationships, absorption of toxicants, distribution and storage of toxicants, biotransformation and elimination of toxicants, target organ toxicity, teratogenesis, mutagenesis, carcinogenesis, food allergy, and risk assessment. Examines chemicals of food interest such as food additives, heavy metals, mycotoxins, and pesticides, and how they are tested and regulated.

FST 425 Dissertation
Students will be expected to carry out a research project, writing a complete and coherent dissertation. Understanding and interpreting results, drawing valid conclusions based on objectives and existing and generated information. The module provides students the opportunity to design, undertake or conduct an independent piece of research study related to the programme under the guidance of a supervisor.

REGULATIONS FOR BACHELOR OF SCIENCE HONOURS DEGREE IN NUTRITION SCIENCE (HNS)

Duration:	4 years
Minimum Credit Load:	480
Maximum Credit Load:	540
MBKS Credit Load:	420
Programme Credit Load:	540
ZNQF Level:	8

1. ENTRY REQUIREMENTS

For all entry pathways candidates must have at least five Ordinary Level subjects/ National Foundation Certificates including English Language, Mathematics and a Science subject at grade C or better.

Normal Entry:	At least 2 Advanced Level passes including Chemistry, Biology or Food Science and one subject from, Physics, Mathematics or Agriculture
Special Entry:	National Diploma or Higher National Diploma in relevant field
Mature Entry:	Refer to General Academic Regulations

2 LEARNING OUTCOMES

The outcome of the programme is to equip the graduate with knowledge and skills to;

1. Understand the relationship of diet to health and to apply this knowledge to the provision of nutritional advice to the public, as well as providing safe, wholesome and nutritious food to consumers.
2. Apply knowledge and understanding of the role of diet and nutrition in the cause and prevention of diseases such as obesity, cardiovascular disease, certain cancers and others.
3. Work independently on individual projects and, ultimately, to manage group projects,
4. Develop time management and organisational skills.

3 ASSESSMENT

Coursework: 40%: (Test 15%, Practicals 20%, Assignments 5%

Written Examinations: 60%
Other: a) Research Project is assessed on the basis of a research project (75%), Oral presentation (15%) and student conduct in the laboratory (10%).

4. PROGRAMME STRUCTURE N.B * Denotes core modules and modules in bold are MBKs

CODE	MODULE DESCRIPTION	CREDITS
Level 1 Semester 1		
HNS 111	*Organic Chemistry	12
HNS 112	*Human Anatomy	12
HNS 113	*Principles of Human Physiology	12
HNS 114	*Principles of Nutrition	12
HCSCI 131	Introduction to Computers and Computer Applications	12
CS 131	Basics of Communication skills	12
Level 1 Semester 2		
HNS 121	*Nutrition and Metabolism	12
HNS 122	*Community Nutrition I (Prerequisite for Community Nutrition II)	12
HNS 123	*Nutrition through the life cycle I (Prerequisite for Nutrition through the life cycle II)	12
HNS 124	Food Chemistry	12
HNS 125	Cell biology and Genetics	12
Choose one elective		
HNS 126	Experimental Foods and Food Preparation	12
HNS 127	Introduction to Food, Nutrition and Health	12
HNS 128	Food and Nutrient Analysis	12
Level 2 Semester 1		
HNS 211	*Nutrition Education and Counseling I (Prerequisite for Nutrition Education and Counseling II)	12
HNS 212	*Food Service Management I (Prerequisite for Food Service Management II)	12
HNS 213	*Medical Nutrition Therapy I (Prerequisite for Medical Nutrition Therapy II)	12
HNS 214	*Community Nutrition II	12
GS 211	Introduction to Gender Studies	12
TCNP 201	Technopreneurship	12
Level 2 Semester 2		
HNS 221	*Nutrition Assessment and Surveillance	12
HNS 222	*Nutrition through the life cycle II	12
HNS 223	*Nutrition Education and Counseling II	12
HNS 224	Research Methods and Statistics	12
HNS 225	Food Service Management II	12
Choose one elective		
HNS 226	Food safety and sanitation	12
HNS 227	Drug nutrient interaction	12
HNS 228	Food Toxicology	12

Level 3 Semester 1

HNS 331	Work-Related Learning I	40

Level 3 Semester 2

HNS 332	Work-Related Learning II	80

Level 4 Semester 1

HNS 411	*Professional Practice in Nutrition	12
HNS 412	Food Microbiology	12
HNS 413	*Nutrition Programming, Implementation and Evaluation	12
HNS 414	Food Biochemistry	12
HNS 415	Public Health Nutrition	12
HNS 416	*Medical Nutrition Therapy II	

Level 4 Semesters 2

HNS 421	*Nutritional Epidemiology	12
HNS 422	*Nutrition and Agricultural Systems	12
HNS 425	*Dissertation	

Choose one elective

HNS 423	Advanced Topics in Nutrition	12
HNS 424	Sports Nutrition	12

MODULE SYNOPSES

HCSCI 131 Introduction to Computers and Computer Applications
Basic Computer Concepts, Data Processing Cycle, Number System. Computer Arithmetic, Types of Computer, Evolution of Computers, Concepts of computer languages, Programming construct, Programming data structure, Programming algorithms. Data Communications (Wired, Hybrid and Wireless networks). Variations in Computer Architectures.

CS 131 Basics of Communication skills
This module imparts writing, reading, speaking, and listening skills that make learning more effective. Students are taught skills that they can use even beyond the academic context such as report writing, curriculum vitae writing (CV), use of visual aids. The module equips students with the necessary communication skills essential for their degree studies as well as post university experience. In this module, students are taught and encouraged to incorporate source materials (academic referencing) into their writing, according to the standard academic conventions recommended in their field.

GS 211 Introduction to Gender Studies
This module will empower the student with knowledge and skills that enable them to be gender sensitive in the university, workplace and in all their social interaction. Topics covered include: understanding gender, gender analysis, gender issue in Zimbabwe, redressing gender imbalances, empowerment and strategies for creating gender responsive environment. Every student has to pass this module in order to graduate.

TCNP 201 Technopreneurship
Nature and importance of technopreneurship, Differences between technopreneurship and entrepreneurship; Relationship between technopreneurship and the national economy. Developing a business model and basics of small business management, Risks and stages of funding, Sources of funding, Financial funding for growth, product valuation. Opportunity recognition and creation, Sources of opportunity, Screening technology opportunities. The New Product Development process. Concept of intellectual property and its significance, Basics of patenting, legislation governing IP in Zimbabwe.

HNS 111 *Organic Chemistry
This module comprises of the nomenclature, structural bonding, isomerism, introductory stereochemistry, introduction to chemical reactions and chemical properties of organic compounds and biological compounds, i.e. carbohydrates, amino acids and proteins. Moreover, the module focuses on the practical molecular bonding and structure of organic compounds, aromatic chemistry and organic chemistry aspects

HNS 112 *Human Anatomy
This module focuses on different parts of any organized body to discover anatomical structure. Of emphasis are the, skeletal system, joints, muscular system, central and peripheral nervous system.The module also emphasizes on the special senses, blood, the cardiovascular and lymphatic system, the reproductive system, embryology and the urinary system. It gives the students an understanding of the anatomical structures.

HNS 113 *Principles of Human Physiology
The structure of the human body at macro and cellular level is emphasized in this module. It integrates, synthesizes and describes the role and function of every part of human anatomy and physiology, to enable students to achieve a working knowledge of human physiology in relation to nutritional sciences. The modules also focuses on the structure of organic bodies, organization zootomy, phytonym and anthroponomy will be focused on.

HNS 114 *Principles of Nutrition
This module focuses on the nutritive value of foods and metabolism of essential nutrients. It also evaluates nutritional adequacy of a diet as well as the application of ethical principles of nutrition to the requirements of normal individuals throughout the lifecycle

HNS 121 *Nutrition and Metabolism
The module looks at the physio-biochemical properties of nutrients and their bio-nutrition interrelationships at the cellular and sub cellular levels, carbohydrates, protein, and lipid metabolism and the role of vitamins and minerals in these processes. The module also covers the ppathways of metabolism of carbohydrates, proteins and lipids. Enzyme biosynthesis and regulation is dwelt on in this module. The module also explains enzymes as biocatalysts.

HNS 122 *Community Nutrition I
This module examines the assessment of cross-cultural dietary patterns and food habits, knowledge of national nutrition policies and programmes including integrated nutrition programmes. This module also encompasses components such as: knowledge and understanding of food and nutrition security, nutrition promotion, determining economical lifestyle interrelationships that impact on nutritional problems and educational needs throughout the life cycle of a human being.

HNS 123 *Nutrition throughout the Life Cycle I
The module provides an overview of the nutrient needs of individuals for each of the major stages through the life cycle. It examines the recommended dietary intakes (RDAs) for each stage and explores common nutritional problems associated with each life stage. The module also covers pre-conception

nutrition and associated conditions and recommended interventions. It concludes with infant nutrition and associated conditions and interventions. At each stage, the module emphasizes on nutrition requirements for normal growth and development, and on the consequences of under and over nutrition.

HNS 124 Food Chemistry
The module focuses on the chemistry of major food components: Carbohydrates, Proteins, Lipids and Water. Chemical and nutritional aspects of food processing: implications of different processing techniques on the major food components are also aspects of this module. Functional properties of the major food components as well as modification of functional properties of the major food components in relation to nutrition is covered.

HNS 125 Cell Biology and Genetics
This module looks at genetics, mitosis and meiosis, Mendelian genetics and the chromosome theory of heredity, mutation and variation genetics, introduction to population genetics, the nature of species and specific mate recognition systems. The focus is on the structure of eukaryotic cells, function of different organelles, membrane structures including modification e.g. gap and tight junction as well as transport across membranes.

HNS 126 Experimental Foods and Food Preparation
The focus of this module is on the application of preparation principles and techniques in the preparation of standard food products, principles of management and their application in the planning and preparation of meals. This module focuses on chemical reactions of food constituents, intermolecular interaction, aqueous solutions, chemical equilibrium, acids and bases and formation of precipitates and electron transfer reactions.

HNS 127 Introduction to Food, Nutrition and Health
The module introduces food choice as affected by social factors, religious influences, ethnicity, health, safety, economics, food sensory properties and the introduction to the food supply chain with special emphasis on the nutritional, environmental, ethical and safety issues that are of importance to consumers. Aspects such as hunger - food needs, including food and nutrition security, nature of nutritional problems, approaches to combat over- and under nutrition are emphasized.

HNS 128 Food and Nutrient Analysis
The principles and methods used in nutrient analyses and nutritional assessment are covered in this module. Laboratory introduction to principles and analytical techniques of nutritional research are also covered. The module emphasizes on analytical concepts and skills required to determine nutrient function and nutritional status of individuals. The module covers topics such as methods of nutrient, metabolite, and enzyme analysis in body fluids; methods for assessing individual food intake and nutritional status; and methods for assessing the composition of foods.

HNS 211* Nutrition Education and Counseling I
The aim is to provide knowledge about application and integration of the principles of nutrition and their transmission to groups and individuals. Students learn individual counseling techniques as well as how to present nutrition information to groups.

HNS 212 *Food Service Management I
This module focuses on the introduction to the management of commercial and noncommercial food service systems through the operation of Bevier Café. Students experience managing the procurement production and service of food, as well as the sanitation and maintenance of equipment and facilities. This module also examines, national, regional, and international food laws, adoption, interpretation and enforcement of laws and regulations governing food preparation and food service systems, impact of regulation on food production availability, marketing and safety.

HNS 213 *Medical Nutrition Therapy I
This module examines pre-requisites of nutrition and metabolism, the application of principles of biochemistry and physiology in the study of nutrient metabolism as altered by disease. The physio-biochemical basis of the diet in the treatment of diseases, may include field experiences.

HNS 214 *Community Nutrition II
The module builds on concepts learned in introductory community nutrition that relate to health promotion, food security, policy, program planning and community nutrition throughout the lifecycle. Students develop the skills in community grants application. This module is offered in a seminar format with instructor, guest and student presentations and discussions on specific topics relevant to community nutrition. Field trips to places and events that relate to community nutrition are also part of this module.

HNS 221 *Nutrition Assessment and Surveillance
This module focuses on food and nutrient intake of individuals and populations and environmental and individual factors affecting food intake. Nutritional status of individuals such as anthropometric data collection and interpretation, biochemical parameter interpretation, clinical data collection and interpretation are emphasized. Aspects of program management such as surveillance and monitoring data collection and interpretation and design of nutritional assessment systems are dealt with. Computer programs such as the WHO Anthro, Anthroplus and Stata packages, are utilized.

HNS 222*Nutrition throughout the life cycle II
The module provides an overview of the nutrient needs of individuals for each of the major stages through the life cycle. It examines the recommended dietary intakes (RDAs) for each stage and explores common nutritional problems associated with each life stage. The module covers pre-conception nutrition and associated conditions and recommended interventions. It concludes with infant nutrition and associated conditions and interventions. At each stage, the module emphasizes on nutrition requirements for normal growth and development, and the consequences of under and over nutrition.

HNS 223* Nutrition Education and Counseling II
The aim of this module is to equip students with the necessary knowledge and skills to deal with various problems within the health system. The students acquire knowledge on counseling on HIV/AIDS and post –traumatic counseling, typical counseling techniques and therapeutic process and theories about the nature of emotions, strategies and techniques.

HNS 224* Research Methods and Statistics
This module encompasses research methods, statistical methods involving relationships between populations and sample, collection, organization, and analysis of data and techniques in testing hypothesis with an introduction to regression, correlation and analysis of variance limited to the completely randomized design and the randomized complete-block design. Practical utilization of statistical packages such as the SPSS and Epi- Info is demonstrated to students.

HNS 225 Food Service Management II
This module allows students to gain experience in facility design; equipment selection, use, and care; job analysis and evaluation; human resources planning; management of financial resources; recipe development and volume food production; computer-assisted management; employee training; and applied safety and sanitation standards. Through planning and executing a themed event, students develop other skills required to operate/manage a food service program. Application of quality management in food service operations and facility management is stressed.

HNS 226 Food Safety and Sanitation

The management of food safety includes the role of sampling, monitoring, audits and inspections. Practical application of the principles and practice of HACCP. There is emphasis on microbiological criteria, hygiene control measures and Codex Alimentarius.

HNS 227 Drug Nutrient Interaction
This module focuses on drug classifications relevant to nutrition and their modes of action. Common medication side effects and contraindications relevant to nutrition are encompassed. Drug nutrient interactions are explained.

HNS 228 Food Toxicology
The module covers toxicology concepts such as dose-response relationships, absorption of toxicants, distribution and storage of toxicants, biotransformation and elimination of toxicants, target organ toxicity, teratogenesis, mutagenesis, carcinogenesis, food allergy, and risk assessment. Examines chemicals of food interest such as food additives, heavy metals, mycotoxins, and pesticides, and how they are tested and regulated.

HNS 411* Professional Practice in Nutrition
This is an essential part of the preparation for clinical placement. Nutrition professional issues and regulatory structures, the food service system in health care institutions and clinics, management and communication principles, reflective practice and keeping a professional portfolio are discussed in this module.

HNS 412 Food Microbiology
The module introduces students to the microbial world and made aware of existing tools to explore it. Starting from a historical perspective, the central role that microbes play in health is investigated. Aspects of taxonomic relationships, safe laboratory practice and general techniques in culturing, staining and quantification are covered. The role of microorganisms in the spoilage of food, food poisoning and food borne diseases are also covered.

HNS 413*Nutrition Programming, Implementation and Evaluation
This module focuses on project management, needs assessment, goal setting and outcome assessment. Students should develop team management, conflict resolution, human resource management, quality improvements, appraisals and financial methods, management process, project proposal writing and monitoring and evaluation skills. The module encompasses monitoring and evaluation plans, benchmarking for innovative monitoring and evaluation systems, the stages in results based monitoring and evaluation systems.

HNS 414 Food Biochemistry
The module encompasses the structural and ionic properties of amino acids, peptides, the peptide bond, primary, secondary, tertiary and quaternary structure of proteins. Biochemistry of carbohydrates, thermodynamics and bioenergetics is also covered. Glycolysis, citric acid cycle and electron transport, glycogen metabolism, pentose-phosphate pathway, gluconeogenesis are also important aspects of the module. Understanding of biochemistry of lipids, membrane structure, anabolism and catabolism of lipids, nitrogen metabolism, amino acid biosynthesis and catabolism is essential in the module.

HNS 415 Public Health Nutrition
The module provides an integrated approach covering dietary needs, epidemiological, public health, social and biological aspects of nutritional science. Patterns of food consumption based on biological, psychological, and social needs; and anthropological findings are assessed. Laboratory component illustrates physiological and biochemical principles of nutrition. Critical evaluation of the impact of the promotion of good health through primary prevention of nutrition-related illness in the population.

HNS 416*Medical Nutrition Therapy II
This module involves the application of the principles of normal and therapeutic nutrition, nutrition assessment, nutrition intervention and evaluation as related to the management and treatment of disease states, including causes and treatments of anorexia nervosa, bulimia nervosa lounge eating and overeating from biological, psychological and social perspectives.

HNS 421* Nutrition Epidemiology
This module focuses on the application and integration of nutrition in the etiology of disease. The module also looks at the monitoring of nutritional status of individuals and populations, developing criteria of evaluating interventions to maintain healthy eating in relation to synergy between nutrition and physical activity as far as health and diseases are concerned. The module also looks at determining outbreaks of diseases in relation to geographical location, seasons of the year, living conditions and cultural beliefs.

HNS 422*Nutrition and Agricultural Systems
This module looks at food production, preparation, processing (primary processing of agricultural products), distribution and waste management, global and local food systems and factors affecting the supply of food and sustainable food practices. Marketing of foods, factors affecting accessing food, food consumption patterns and trends and disaster planning is also emphasized.

HNS 423 Advanced Topics in Nutrition
The principles and methods used in nutrient analyses and nutritional assessment are covered in this module. Laboratory introduction to principles and analytical techniques of nutritional research are also aspects of the module. The module also emphasizes analytical concepts and skills required to determine nutrient function and nutritional status of individuals. Topics to be covered by the module also include methods of nutrient, metabolite, and enzyme analysis in body fluids; methods for assessing individual food intake and nutritional status; and methods for assessing the composition of foods as well as Nutrigenetics.

HNS 424 Sports Nutrition
This module focuses on how nutrients affect physical performance, how physical performance can be improved by adopting optimal dietary practice and how exercise and optimal nutrition can prevent disease. Assessment of body composition and physical activity is also be dealt with. Students will acquire practical knowledge on the assessment of body composition and physical activity.

HNS 425 *Dissertation
Students carry out a research project in this module, writing a complete and coherent dissertation. Understanding and interpreting results, drawing valid conclusions based on objectives and existing and generated information. The module provides students the opportunity to design, undertake or conduct an independent piece of research study related to the programme under the guidance of a supervisor.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN FOOD SCIENCE AND NUTRITION (MFSN)

Duration	2 years
Minimum Credit Load	288
Maximum Credit Load	384
MBKs Credit Load	288

ZNQF Level 9

1. ENTRY REQUIREMENTS

Normal Entry
Applicants should be holders of at least an upper second class degree in any of the following degree programmes: Food Science and nutrition, Biosciences, Biochemistry, Chemistry, Agriculture (Animal Science), Nutritional Sciences or any other related degree from any university recognized by Midlands State University.

Special Entry
A holder of lower second or third class any of the above-mentioned programmes or equivalent and relevant post-qualification experience.

2. LEARNING OUTCOMES

1. Possess advanced knowledge and skills required in troubleshooting the current and forecast technical challenges of the food industry

2. Have an understanding of nutrition and health aspects of food production and processing

3. An ethical interface between technically oriented professionals, various stakeholders and the general public

3. PROGRAMME ASSESSMENT

Continuous assessments: 40% (Test 15%, Practicals, 20%, Assignments, 5%)

Written Examinations: 60%

4. PROGRAMME STRUCTURE

N.B * Denotes core modules

CODE	MODULE DESCRIPTION	CREDITS
Level 1 Semester 1		
MFSN 731*	Food Production and Processing systems	24
MFSN 732*	Food Microbiology and Biotechnology	24
MFSN 733*	Nutrition and Public Health	24
MFSN 734*	Professional Ethics	24

Level 1 Semester 2		
MFSN 736*	Food Analysis and Sensory Science	24
MFSN 737*	Communication Skills in Nutritional Health	24
MFSN 738*	Total Quality Management and Food Safety	24
MFSN 739*	Food Policy and Programme Planning	24
MFSN 740*	Nutrition Through the Lifecycle	24

Level 2 Semester 1

MFSN 831*	Research Methods and Statistics	24

Elective Modules: Choose any two

MFSN 735	HIV/AIDS Nutritional aspects of food	24
MFSN 832	Advanced Laboratory technology	24
MFSN 833	Advanced Food Packaging	24
MFSN 834	Clinical Nutrition	24
MFSN 835	Dietetics	24
MFSN 836	Advanced Food Chemistry	24
MFSN 837	Environmental Management in Food Science	24
MFSN 838	Food Toxicology	24
MFSN 840	Food Systems for Sustainable Diets	24
MFSN 841	Global Nutrition and Health	24

Level 2 Semester 2

MFSN 839	Dissertation	48

MODULE SYNOPSES

MFSN 731 Food Production and Processing Systems
An applied module investigating various food processing techniques and their underlying principles. The outline should cover raw materials used in food processing. Sorting and grading of agricultural products. Food processing techniques, to include microbiological, biochemical, nutritional and technological aspects. Principles of processing foods and Techniques to include, freezing, chilling, canning, intermediate moisture foods, and new technologies such as extrusion and aseptic processing. Brewing, fermentation, meat, dairy, cereal technologies are also covered. Visits to local food processing units. Research within the field of food science and technology is both wide ranging and rapidly changing. This gives rise to an increasingly diverse number of interesting and important applications.

MFSN 732 Food Microbiology and Biotechnology
This module informs on the part played by each of the major groups of micro-organisms in the production and spoilage of foodstuffs. In addition the changes, which occur in foods during spoilage, are highlighted as are the risks associated with such changes. Topics include Food and Microorganisms, Principles of Food preparation, Food Spoilage, contamination and Preservation, Food Fermentations, Food borne diseases and Food poisoning, Food sanitation, control and inspection. Food biotechnology focuses on Basic tool in Biotechnology (molecular tools and bioreactors), Applications of Food biotechnology, Biosafety and Bio-ethics.

MFSN 733 Nutrition and Public Health
This module aims to provide comprehensive training in public health nutrition in a global setting with the choice of specializing in a number of topics and in a range of contexts. It provides an integrated programme covering dietary, epidemiological, public health, social and biological aspects of nutritional science. Specialist topics include maternal and child nutrition, nutrition in emergencies, nutrition programme planning, evaluation and monitoring, and nutritional epidemiology. The course attracts graduates from various disciplines who wish to equip themselves for public health nutrition research and teaching, operational work in the field or community nutrition programmes, work in public health

nutrition, including health promotion and nutrition education, and for policy and programme planning in nutrition. Graduates work for local and national governments, NGOs, international agencies, academic institutions, and in fields ranging from food emergencies to research on nutrient-gene interactions. The module has been designed to focus on nutritional problems in developing countries and transitional societies. By the end of this module students should be able to: demonstrate an advanced knowledge of public health nutrition, at biological, social and policy levels; assess critically, select and apply a range of appropriate research skills and techniques, from anthropometry and information on dietary intake to broader analytical skills; interpret and synthesize different types of data used to analyze and assess nutritional problems at population and population sub-group levels; evaluate critically the findings of scientific studies of public health nutrition; disseminate and present findings of research in a range of formats and contexts; identify and formulate appropriate responses and intervention strategies to address nutritional issues, taking into account the public health and social policy contexts, and apply knowledge of effective teamwork and communication skills to solve problems and achieve goals.

MFSN 734 Professional Ethics

As part of ethical development, students are required to complete the Professional Ethics module. This will give exposure to a range of ethical perspectives and includes several self-tests which require to reflect on one's own ethical behavior and values. The Code of Professional Ethics lays down the standards of integrity, professionalism and confidentiality.

MFSN 735 Nutrition and HIV/AIDS

The coexistence of high rates of malnutrition and HIV/AIDS creates a challenge for nutritionists. Providing adequate nutrition at community and individual level remains a problem. The new challenge calls for the acceleration of both short-term and long-term effects to combat malnutrition and its effects on morbidity and mortality associated with HIV/AIDS. The module covers aspects involving: Introduction to HIV/AIDS, Link between nutrition and HIV/AIDS, Food security components in HIV/AIDS nutrition care and support, Nutrition management of HIV/AIDS- related symptoms, Nutritional care and support for vulnerable groups and Management of drug and food interactions in HIV/AIDS therapy.

MFSN 736 Food Analysis and Sensory Science

This module will be of value to students from a broad range of backgrounds who wish to develop their understanding of the scientific principles and practical methods which underlie the analytical techniques and methods used to assess food standards and quality, and the sensory perception of food products, with an emphasis on evaluating development of products. The principles underlying a range of analytical methods [e.g. Kjeldhal, Soxhlet, calorimetry] used to perform nutrient analysis will be discussed and performed in order to evaluate the nutrient content of food samples. The principles of sensory evaluation of food will be discussed, including the relevant physiological aspects of the senses such as taste thresholds. Sensory evaluation methodology such as taste panels and the statistical analysis of results (including profiling) will be studied. The procedures involved in developing new and existing products from product concept to product launch (including packaging, labelling, marketing, pricing and advertising) will be undertaken in the context of recipe/ product development case studies and practical exercises.

MFSN 737 Communication Skills in Nutritional Health

This module is designed to improve communication skills for the nutritionists in information dissemination. It focuses on: Communication skills (i.e., oral and written communication, listening; Critical thinking/problem solving skills (i.e., creativity, common sense, resourcefulness, scientific

reasoning, analytical thinking, interviewing; Professionalism skills (i.e., ethics, integrity, respect for diversity); Life-long learning skills; Interaction skills (i.e., teamwork, mentoring, leadership, networking, interpersonal skills, etc.); Information acquisition skills (i.e., written and electronic searches, databases, Internet; Organizational skills (i.e., time management, project management, etc.); Commit to the highest standards of professional integrity and ethical values. The students should also be able to independently research scientific and nonscientific information as well as work and/or interact with individuals from diverse cultures.

MFSN 738 Total Quality Management and Food Safety

This module is designed to provide the student with an appreciation of the value of quality assurance and quality control procedures in the production of food, satisfying both legal requirements and the customer's need for quality and safety. Awareness of recognised quality and safety systems such as International Standards Organisation (ISO) and Hazard Analysis Critical Control Point (HACCP) should be increased, together with the ability to apply them in setting up of an effective food safety management system. An overview of current food quality and safety systems to include Quality Control, Quality Assurance, Total Quality Management and relevant legislation. Quality programmes, practices and systems such as ISO9001:2000, ISO9004:2000, Good Manufacturing Practice, HACCP. The management of food safety includes the role of sampling, monitoring, audits and inspections. Practical application of the principles and practice of HACCP.

MFSN 739 Food Policy and Programme Planning

This module gives students the opportunity to examine a number of current food policies, both in terms of how they are formulated and also the impact they have on a given population. The student will be encouraged to critically analyse policies and their associated issues in relation to consumers, industry and Government. Lectures and seminars will introduce national and international policy makers and case studies will be used to illustrate the process of policy formulation and the impact that policies have in a wider context. The assessment will allow students to individually assess a given policy and consider the wider implications for ever-evolving polices. Various common themes are likely to emerge including Food and Nutrition policies in Zimbabwe, and around the world, including agencies involved in their evolution and implication. Several examples of current policies will be reviewed as case studies, including consideration of their impact on relevant stakeholders. Issues relating to food processing, food policy and the consumer. Genetically modified foods and their impact on health, policy and the consumer.

MFSN 740 Nutrition Through the Life Cycle

This module is intended to provide an overview of nutrition by looking at it from the perspective of each life stage. The module provides a brief on basic nutrition science principles and looks at the nutrition assessment parameters to give a background on how nutrition problems are identified. It also focuses on selected life-stages to outline nutritional needs and discusses conditions commonly associated with each of the life-stages. In addition, it looks at current nutrition-related issues of concern in Zimbabwe.

MFSN 831 Research Methods and Statistics

This module is for all students on the MSc programme. It includes: Design of experiments, Variables and distributions; summarising data, Sampling variability of a mean, Analysis of quantitative data; comparing two means, Sampling variability of proportions, Analysis of categorical data; comparing two proportions, Regression and correlation (computer session practical), ANOVA and multiple regression (paper critique practical). Emphasis will be on practical and applied skills. Methods will be illustrated using examples of relevance to Food Science and Nutrition students. By the end of this module students should be able to: describe the role of statistical methods in food/ public health research; present results effectively by making appropriate displays, summaries, and tables of data; appreciate the problem of sampling variation, and the role of statistical methods in quantifying this; select an appropriate statistical method for the analysis of simple data sets; correctly interpret the results of statistical analyses reported

in the food/health literature; perform simple statistical analyses using software packages; and interpret their findings from statistical analyses and present these findings in a clear, concise, and logical manner.

MFSN 832 Advanced Laboratory Technology
Application of food preparation principles and techniques in the preparation of standard food products. Analytical techniques involving Spectroscopy, chromatography, Microbiological techniques, Immunoassay and electrophoresis are covered. Other useful topics include Method selection and validation, Good Laboratory practice (GLP), Laboratory Management, Safety and Health.

MFSN 833 Advanced Food Packaging
Emphasis should be given to aspects of food packaging including Materials, Design techniques, Environmental considerations, Food ethics and Economics. There is also an element of improved technologies for the problems unique to food packaging like flexible packaging. It is also a topical issue in Food product development and clean technology concepts.

MFSN 834 Clinical Nutrition
This module examines the metabolic demands on nutrients in acute and chronic illness; a critical evaluation will be made of recent advances in specific nutrition therapies for each disease category, incorporating clinical case studies. As the field of nutritional health sciences continues to develop at a rapid pace the key skills to access, interpret and evaluate critically current literature are integral to this programme. It is designed to facilitate an understanding of the processes that underlie methods of artificial nutrition support; this programme includes digestive physiology, energy and nutrient requirements, nutritional assessment, and formulation of enteral and parenteral formulae. Aspects of the current best feeding practice are reviewed together with the organisation of nutrition support teams in the hospital and community settings. There is also need to focus on Evidence-based approach to the effect of nutritional status on disease outcome, Nutrient metabolism in health and disease, Gastrointestinal physiology and the uptake of nutrients, Nutrition support, encompassing oral nutritional supplements, enteral and parenteral nutrition, in the hospital and community setting, Nutritional Assessment in the hospital and community setting, How different diseases affect nutritional status, Gene-nutrient interactions, Metabolic stress during critical illness, Nutritional modulation of immune response, Energy imbalance; under- and over-nutrition.

MFSN 835 Dietetics
The dietician is a practitioner who applies the principles of nutrition in a variety of settings. The dietician assists clients in managing their diet either as part of disease management (clinical setting) or as part of the wider prevention of nutrition-related disease in communities. The Master's level dietician is able to lead and contribute to ongoing developments that will enhance the professional status of the dietician together with enabling implementation of clinical governance and clinical effectiveness in particular ways: through effective and evidence-based practice, justified clinical decision making and identifying the need for reflective practice. This module focuses on current practices in dietetics in relation to trends in diet issues. It focuses on topics as clinical Dietetics, Research Strategies in Nutrition & Dietetics, Professional Studies, Behavioural Sociology, Awareness of Dietetic Practice, Community Dietetics and Contemporary Nutrition Issues, Psychology of Behaviour, Building on Dietetic Practice.

MFSN 836 Advanced Food Chemistry
Food chemistry, a major aspect of food science, deals with the composition and properties of food and the chemical changes it undergoes during handling processing and storage. Food chemists are typically concerned with identifying the molecular determinants of material properties and chemical reactivity of food matrices, and how this understanding is effectively applied to improve formulation, processing and

storage stability of foods. The module covers a broad range of topics ranging from the chemistry of food components (Water, carbohydrates, lipids, and proteins) to minor food components (Vitamins, Minerals, Colorants, Flavors, Sweeteners, Food Additives, neutraceuticals and toxicants). Aspects of Food systems are also covered including dispersal systems, and physical and chemical interactions of components in food systems.

MFSN 837 Environmental Management in Food Science
This module is of relevance to students interested in the assessment and control of environmental problems associated with food production and public health. This module consolidates knowledge of key methodological tools: risk assessment as a structured method to estimate quantitative health impacts, especially for chemicals; risk management as a framework for assessing and controlling environmental hazards. Teaching considers the roles of scientific, socio-cultural, historical, economic and political factors in influencing the management of environmental risk. The module should cover the principles and legislation governing, as well as techniques in Environmental management.

MFSN 838 Advanced Food Toxicology
The module focuses on general aspects and regulatory aspects of food toxicology There is need to look at pesticides and other industrial chemicals, hormones, mycotoxins, mutagens and carcinogens in heat-processed food; modification of carcinogen response by antioxidants, food irradiation, food intolerance, methylxanthines; use and safety of novel foods, and the future of food toxicology.

MFSN 839 Dissertation
The module is aimed at engendering students with a spirit of enquiry into a research-based dissertation. A broad range of research is encompassed within the expertise based in the Food Science and Nutrition field may include a single focused investigation, or a series of discipline specific investigations, ranging from 'wet'-laboratory based, ICT-based, questionnaire/ survey- based, to 'dry' theoretically-based projects.

MFSN 840 Food Systems for Sustainable Diets
In this module, food systems are explored from a dietary perspective and components related to the diet gaps for different population groups at national, regional and global level are discussed. Possible actions to transform the food system to address the gaps and how these actions can be leveraged in a sustainable way are discussed.

MFSN 841 Global Nutrition and Health
The module dwells on the emerging current issues of concern in Zimbabwe and other low-middle income countries as well the global arena. The module also focuses on the vital nutrition programming concepts as well the Sustainable Development Goals (SDGs) and the global targets in relation to nutrition.

REGULATIONS FOR MASTER OF SCIENCEDEGREE IN FOOD PROCESSING TECHNOLOGY (MFPT)

Duration	2 years
Minimum Credit Load	288
Maximum Credit Load	384
Programme Credit Load	384
MBKs Credit Load	288
ZNQF Level	9

1 .**ENTRY REQUIREMENTS**

NORMAL ENTRY
Applicants should be holders of at least an upper second class in any of the following programmes, Food Science, Food Technology, Food Science and Nutrition, Biosciences, Biochemistry, Chemistry, Animal Science, and Agricultural Sciences, or other relevant qualifications.

SPECIAL ENTRY
A holder of lower second or third classes any of the above-mentioned programmes or equivalent and relevant post qualification experience.

2 LEARNING OUTCOMES

1. Have detailed scientific knowledge and understanding of the (bio)-chemical, physical, nutritional and microbiological changes that occur in raw materials during the transformation into food products.
2. Be able to explain the theory and working principle, give detailed and illustrative examples of conventional and modern processing technologies and their real-life applications in the food industry.
3. Have detailed scientific knowledge in different fields of product technology such as vegetable products, dairy products, meat products, fish products, cereal derived products and fermented products.
4. Be able to acquire practical skills that will assist them to undertake independent research to help solve the food industry problems and offer feasible solutions

3. PROGRAMME ASSESSMENT

Continuous assessments: 40%
Test (15%), Practicals (20%), Assignments (5%)
Written Examinations: 60%

4. **PROGRAMME STRUCTURE**

N.B * Denotes core modules

CODE	MODULE DESCRIPTION	CREDITS
Level 1 Semester 1		
*MFPT 701	Food Processing Systems	24
*MFPT 702	Advanced Food Microbiology and Biotechnology	24
*MFPT 703	Advanced Food Chemistry	24
MFPT 704	Food Analysis and Sensory Science	24
*MFPT 705	Food Quality Assurance, Safety and Plant Sanitation	24
Level 1 Semester 2		
*MFPT 706	Processing of Plant Foods	24
MFPT 707	Food Business Operations Management	24
*MFPT 708	Food Engineering and Processing Technology	24
*MFPT 709	Research Methods and Statistics	24
Level 2 Semester 1		

*MFPT 801	Processing of Animal Foods	24

Electives (Select two)

MFPT 802	Food Product Development	24
MFPT 803	Nutraceuticals and Functional foods	24
MFPT 804	Food Packaging Technology	24
MFPT 805	Food additives, contaminants and toxicology	24
MFPT 806	Advanced Fermentation technology	24

Level 2 Semester 2

*MFPT 807	Dissertation	96

MODULE SYNOPSES

MFPT 701 Food Processing Systems
An applied module investigating various food processing techniques and their underlying principles which provides students with an overview of all elements required in food processing. The outline covers raw materials used in food processing, and the various important stages involved in the transformation of agricultural produce into convenience foods. Food processing techniques, include microbiological, biochemical, nutritional, and technological aspects. The module covers key processing techniques which include, freezing, drying, chilling, canning, intermediate moisture foods. Research within the field of food science and technology is both wide ranging and rapidly changing. This gives rise to an increasingly diverse number of interesting and important applications.

MFPT 702 Advanced Food Microbiology and Biotechnology
This module explores the most important advances in food microbiology and biotechnology, with special emphasis on world trends on the challenges that the food industry faces. The main focus is on recent technological developments in microbiology and biotechnology with a bias towards their application in food production and processing. The main highlights of the module include novel and sustainable methods in food microbiology and food biotechnology that pay attention to impacts on consumer health and food security.

MFPT 703 Advanced Food Chemistry
The module deals with the composition and properties of food and the chemical changes it undergoes during handling processing and storage. Food chemists are typically concerned with identifying the molecular determinants of material properties and chemical reactivity of food matrices, and how this understanding is effectively applied to improve the formulation, processing and storage stability of foods. The module covers a broad range of topics ranging from the chemistry of food components (Water, carbohydrates, lipids, and proteins) to minor food components (Vitamins, Minerals, Colorants, Flavors, Sweeteners, Food Additives, nutraceuticals and toxicants). Aspects of Food systems are also covered including dispersal systems, and physical and chemical interactions of components in food systems.

MFPT 704 Food Analysis and Sensory Science
Food analysis is the discipline dealing with the development, application and study of analytical procedures for characterizing the properties of foods and their constituents. Sensory evaluation is defined as a scientific method used to evoke, measure, analyze and interpret human responses to products as perceived through the senses of sight, smell, touch, taste, and hearing. This module informs on the various methods of determining food constituents, how they affect physicochemical properties and their effects on consumers' perception of food products. The module covers scientific methods

applied in the chemical laboratory as well as the sensory laboratory - where human beings are used as measuring instruments in determining the sensory characteristics of food products.

MFPT 705 Food Quality Assurance, Safety and Plant Sanitation
This module provides advanced training in quality assurance and quality control procedures in the production of food, satisfying both legal requirements and the customer's need for quality and safety. The students acquire knowledge of food safety management systems, Total Quality Management, Quality control, relevant food legislation, and risk analysis. The management of food safety includes the role of sampling, monitoring, audits and inspections. Practical application of the principles and practice of Hazard Analysis Critical Control Point (HACCP), Good Manufacturing Practice and other relevant prerequisite programs will be covered.

MFPT 706 Processing of Plant Foods
Cereals, legumes and pulses are an important part of dietaries and contribute substantially to the nutrient intake of human beings. The module aims to acquaint students with production trends, structure, composition, quality evaluation and processing technologies for product development and value addition of various cereals, legumes and pulses. The module explores the pretreatment processes which are done to improve the nutritional quality of Cereals, legumes and pulses. Furthermore, the processing techniques involved in extracting their respective starches are studied

MFPT 707 Food Business Operations Management
This course introduces participants to the concepts and practices of operations management in the context of food processing and manufacturing businesses. It conveys an understanding of the role of supervisory and management staff in the design, operation, management and improvement of food production operations such that customer requirements are continually met and legal requirements are adhered to.

MFPT 708 Food Engineering and Processing Technology
This module focuses on the concepts of heat transfer in food processing, food rheology, freezing of foods, food dehydration, kinetics of chemical reactions in foods, refrigeration and thermal process calculations. Furthermore, alternative methods of food processing (thermal process innovations and non-thermal technologies) will be explored. The module also covers physical principles, engineering design of novel processing technologies and efficacy. Important processing principles such as combined heat and mass transfer and the consequences for product structure and quality are discussed.

MFPT 709 Research Methods and Statistics
This module is for all students in the MSc programme and covers quantitative and qualitative research methods and statistics specific to Food Processing Technology. By the end of this module students should be able to do a proper evaluation and critical appraisal of research data and reports; citing research and research proposals; understand and perform statistical analyses and interpret results using statistical software packages. Furthermore, the students are able to present research findings in a clear, concise, and logical manner.

MFPT 801 Processing of Animal Foods

The products derived from animal flesh and by-products are an important component of the human diet through the provision of essential nutrients. These commodities are usually minimally or fully processed to ensure preservation, safety or the development of new products. The module covers the physicochemical properties of milk, as are the industrial and traditional processing methods and technologies involved in the conversion of milk to other dairy products such as cheese, yoghurt and butter. Furthermore, the chemistry, structure, composition, properties, uses, and methods and technologies of processing animal food such as eggs, fish meat and meat products are also examined. By the end of the module, the students are able to critically analyse the methods and technologies used in the production of animal-based products.

MFPT 802 Food Product Development
Among methods of industrial research, product development is a combination and application of natural sciences with the social sciences – of food science and processing with marketing and consumer science – into one type of integrated research whose aim is the development of new products. This module aims at a detailed exploration of various concepts and their application in the development and assessment of new food products using traditional and novel food processing methods.

MFPT 803 Nutraceuticals and Functional Foods
Nutraceuticals, functional food ingredients and dietary supplements are important for health promotion and disease risk reduction. This module deals with health promoting nutritional factors and bioactive constituents, their potential health implications and mechanisms of action. Special emphasis is given to linking the molecular and chemical structures of biologically active components in foods to their nutritional and pharmacological effects on human health and wellness.

MFPT 804 Advanced Food Packaging Technology
Emphasis should be given to aspects of food packaging including Materials, Design techniques, Engineering principles of packaging, Environmental considerations, Food ethics and Economics. The module covers recent and improved technologies in food packaging which include principles and applications of active and intelligent packaging, nanotechnology, antimicrobial packaging, edible coatings and films and sustainable packaging.

MFPT 805 Food Additives, Contaminants and Toxicology
While food additives are intentionally added to food in order to exhibit certain desirable characteristics, contaminants are all the other undesirable constituents that are present naturally or unintentionally. Food toxicology is the study of the nature, properties, effects, and detection of poisonous substances in food or food animal feed and their disease manifestation in humans. This module comprehensively deals with food additives, contaminants and their toxicological effects on consumers.

MFPT 806 Advanced Fermentation Technology
This module provides in-depth coverage of food fermentation. Particular emphasis is given to the microbiology and chemistry of fermentation, physiology and metabolism of lactic acid bacteria, yeasts and moulds, using selected food fermentations as examples. Major topics include microbiology and starter cultures of selected fermented foods, chemistry and flavour of selected fermented foods, alcoholic and non-alcoholic fermentation by yeasts, lactic acid bacterial fermentations and fungal fermentations, and health implications of selected fermented foods.

MFPT 807 Dissertation
The module is aimed at providing students with an opportunity for an indepth investigation of a relevant research-based dissertation topic. The application of knowledge acquired from the modules in the program is useful to the students as they undertake independent research which is a requirement for this module. It offers a platform for the application of research methods specific to Food Processing and related disciplines. A broad range of research is encompassed within the expertise in the Food Processing and related fields and the students will have an opportunity to help solve some of the problems within the respective fields. Furthermore, other skills which include approach to the presentation, delivery and appraisal of your written and oral presentations will be fostered through the module.

REGULATIONS FOR MASTER OF SCIENCE DEGREE IN FOOD SAFETY AND QUALITY MANAGEMENT (MFSQ)

Duration	2 years
Minimum Credit Load	288
Maximum Credit Load	384
Programme Credit Load	360
MBKs Credit Load	288
ZNQF Level	10

1.**ENTRY REQUIREMENTS**

NORMAL ENTRY

Applicants should be holders of at least an upper second class in any of the following programmes, Food Science and Nutrition, Food Science and Technology, Biosciences, Biochemistry, Chemistry, Animal Science, Environmental Health, Public Health, Veterinary Sciences, Agricultural Sciences, and Human Medicine, or other relevant qualifications.

SPECIAL ENTRY

A holder of lower second or third class of any of the above-mentioned programmes or equivalent with relevant post-qualification experience.

2. **LEARNING OUTCOMES**

1. Have an understanding of principles of food quality and how they can be applied for the benefit of an ever changing market.
2. Have an understanding of potential hazards in the food chain and prerequisite programmes that can prevent these hazards.
3. Have an understanding of various Food safety and quality management systems.
4. Be able to use available tools and models to calculate risk as well as communicate it to relevant stakeholders for the development of both Food safety objectives (FSO) and Appropriate level of protection.
5. Be able to appreciate and apply both food safety and consumer legislation at local and international levels.
6. Have an understanding of how drugs and pesticides can be hazards in the food chain and the development of antimicrobial resistance.

3. **PROGRAMME ASSESSMENT**

Continuous assessments: 40%

Test (15%), Practicals (20%), Assignments (5%)

Written Examinations: 60%

PROGRAMME STRUCTURE

N.B * Denotes core modules

CODE	MODULE DESCRIPTION	CREDITS
	Level 1 Semester 1	
*MFSQ 701	Total Quality Management	24
*MFSQ 702	Advanced Food Microbiology and Toxicology	24

*MFSQ 703	Food Safety and Risk Analysis	24
*MFSQ 704	Food Borne Disease Surveillance and Control	24

Level 1 Semester 2

*MFSQ 705	Research Methods and Statistics	24
*MFSQ 706	Food Safety, Quality Management Systems and International Standards	24
*MFSQ 707	Food Safety and Quality Governance, Legislation, Policy and Inspection	24
MFSQ 708	Water Quality, Sanitation and Environmental Hygiene	24

Level 2 Semester 1

*MFSQ 801	Statistical Process Control	24

Electives (Choose 2)

MFSQ 802	Sensory Aspects of Food Quality	24
MFSQ 803	Laboratory Quality Assurance	24
MFSQ 804	Agri-Food Safety, Antimicrobial Resistance and one Health	24
MFSQ 805	Port Health, Food Safety, Quality and International Trade	24
MFSQ 806	Food Chemistry	24
MFSQ 807	Food Business Operations Management	24

Level 2 Semester 2

*MFSQ 808	Dissertation	96

MODULE SYNOPSES

MFSQ 701 Total Quality Management

Total quality management (TQM) is a philosophy, methodology and system of tools aimed to create and maintain mechanism of organization's continuous improvement. It involves all departments and employees into improvement of processes and products. This module helps participants understand total quality concept and techniques for managing, controlling, and improving quality. The module exposes participants to contemporary knowledge and techniques of TQM.

MFSQ 702 Advanced Food Microbiology and Toxicology

This module informs on the part played by each of the major groups of microorganisms in the food chain. In addition, the changes, which occur in foods during spoilage, are highlighted, as are the risks associated with such changes. In addition to microbial toxins and pathogens the module further focuses on aspects of food toxicology including effects of pesticides and other industrial chemicals, hormones, mycotoxins, mutagens and carcinogens in food; modification of carcinogen response by antioxidants,

food irradiation, food intolerance, methylxanthines; use and safety of novel foods. The module further addresses LD50 calculations and effects of various toxins on the health.

MFSQ 703 Food Safety and Risk Analysis
This module addresses the meaning of food safety and types of hazards in the food chain. It further provides knowledge on prerequisite programs (PRPs) as ways of preventing occurrence of hazards in the food chain. The module also provides knowledge on principles, methods, and approaches related to food safety and risk analysis. The modules covers risk assessment, risk management risk communication and predominant qualitative and quantitative risk analysis softwares such as @Risk and Risk Ranger among others.

MFSQ 704 Food Borne Disease Surveillance and Control
The purpose of this module is to introduce students to major foodborne pathogens, their epidemiology and associated disease burdens, approaches to outbreak investigation and control of foodborne illness. This module focuses on principles and methods for surveillance of food-borne diseases in humans, surveillance and monitoring of diseases and pathogens in animals that affect public health, investigation of outbreaks, and application of surveillance, monitoring, and outbreak investigation for the assessment of food safety hazards.

MFSQ 705 Research Methods and Statistics
The module develops foundational knowledge of qualitative and quantitative research methods, evaluation of research and statistical applications to relevant research data. The module focuses on application of statistical methods in food science and the use of statistical software in modern data analysis. Good research practice, procedures, ethics and data management will be covered. Students should develop skills in building an argument and communicating it orally and in writing.

MFSQ 706 Food Safety, Quality Management Systems and International Standards
With dynamic and complex food processing environments, management of food safety and quality has proved to be quite challenging. This module provides guidance into various voluntary standards available to manage food safety and quality ranging from ISO 22000, 9001, BRC and Global GAP among others. It further introduces the use of food safety schemes such as FSSC 22000 as opposed to the usual systems. Evaluation of the systems is also integral to the module, this includes auditing, MAS and other diagnostic tools.

MFSQ 707 Food Safety and Quality Governance, Legislation, Policy and Inspection
The module covers the essentials of food and consumer laws in Zimbabwe and on global scale. It introduces the concept of a food control system in food safety governance and provides the role of various ministries and pieces of legislation in the food control system. The module looks at how food and consumer-related offenders are dealt with by the law both locally and globally. The international food law as Codex in known is also addressed in this module. The module also looks at how local legislation fits into the Codex in the dynamic and complex field of food processing.

MFSQ 708 Water Quality, Sanitation and Environmental Hygiene
The module serves to introduce the important concepts and key concepts that are used in environmental health and hygiene. The module seeks to explain the scope of environmental health and describe the links between hygiene, water, sanitation and human health. It describes the steps in environmental health planning and gives an overview of the management of hygiene and environmental health at the community level. The module examines the importance of safe drinking water, the sources and treatment of water, as well as its application in the food industry.

MFSQ 801 Statistical Process Control
The module is designed to provide students with a basic understanding of the importance of statistical quality control concepts in controlling and improving the production process and to give students a

practical knowledge of using statistical methods in analysing the production and service processes. Major topics include tools for descriptive and predictive statistical analysis, design and use of various control charts for quality control, process characterization and capability analysis, design of experiments, acceptance sampling and continuous improvement. The emphasis is on ensuring that the students gain both a broad perspective of quality control as well as the technical skills necessary to implement quality control in any industrial setting.

MFSQ 802 Sensory Aspects of Food Quality
This module provides advanced elements of food quality, safety and sensory science. In particular, it will discuss the concept of quality and its components with a particular focus on the sensory science elements. The sensory science component gives an overview of objective (analytical) sensory testing and concentrates on developing practical skills within analytical sensory testing. The module examines the role of sensory evaluation in marketing of food and beverages, physiological and psychological factors affecting sensory perception and relationships between sensory properties and product acceptability as well as recent advances in sensory methods.

MFSQ 803 Laboratory Quality Assurance
This module provides the necessary knowledge, attitudes and skills needed to efficiently implement and maintain laboratory management systems. Students gain an understanding of the applicability of tests, in relation to the resources available to both chemistry and microbiology laboratories and practical experience of working within a laboratory quality assurance system such as ISO 17025. The competencies of the students should ensure that the food produced is compliant with private and public standard requirements.

MFSQ 804 Agri-Food Safety, Antimicrobial Resistance and one Health
The module provides information on zoonotic diseases and how they impede human health. It also looks at the various drugs, pesticides, herbicides and other chemicals used in the agriculture sector and their control. It further provides mechanisms of antimicrobial resistance and how agro-chemicals contribute to antimicrobial resistance. Further, the module addresses how antimicrobial resistance is a threat to human health and the economy. In addition, the module provides room for students to come up with potential solutions from the farm up to the policymaking level. The module also looks at the one health approach in dealing with zoonosis, antimicrobial resistance and food safety at the national level.

MFSQ 805 Port Health, Food Safety, Quality and International Trade
The food industry has become a global village that has seen purchases of raw materials and final products across the globe. More often in times of foodborne disease outbreaks, many nations restrict entry of certain food products. This module looks at various trade agreements and their implications on food safety. It further addresses various practices at the ports of entry. Additionally, the module addresses the structures of port health departments of various nations (Developing vs. Developed), their strengths and weaknesses and implications on food safety, quality and economics of trade.

15. MFSQ 806 Food Chemistry
The module deals with the composition and properties of food and the chemical changes it undergoes during handling processing and storage. Food chemists are typically concerned with identifying the molecular determinants of material properties and chemical reactivity of food matrices, and how this understanding is effectively applied to improve the formulation, processing and storage stability of foods. The module covers a broad range of topics ranging from the chemistry of food components (Water, carbohydrates, lipids, and proteins) to minor food components (Vitamins, Minerals, Colorants, Flavors, Sweeteners, Food Additives, nutraceuticals and toxicants). Aspects of Food systems are also covered including dispersal systems, physical and chemical interactions of components in food systems.

MFSQ 807 Food Business Operations Management
This course introduces participants to the concepts and practices of operations management in the context of food processing and manufacturing businesses. It conveys an understanding of the role of

supervisory and management staff in the design, operation, management and improvement of food production operations such that customer requirements are continually met and legal requirements are adhered to.

MFSQ 808 Dissertation

The module is aimed at engendering students with a spirit of enquiry into a research-based dissertation. A broad range of research is encompassed within the expertise based in the Food Safety and Quality Management field and may include a single focused investigation, or a series of discipline specific investigations, ranging from 'wet'-laboratory based, ICT-based, questionnaire/survey-based, to 'dry' theoretically-based projects.

LIBRARY DEPARTMENT

INTRODUCTION

The Midlands State University (MSU) Library comprises multiple individual libraries, including the main and nine branch libraries. These libraries provide critical academic support to the university's mission in various areas, such as teaching and learning, research, innovation, community engagement, and industrialisation, in line with national development strategies. The MSU Library is crucial in enabling the Education 5.0 design, which aims to leverage natural resources and intellectual capital to promote and achieve sustainable, inclusive socio-economic development. The Library provides top-quality, innovative, and user-focused services, spaces, and collections in a welcoming study environment that encourages teaching and learning, research, and innovation.

1. SERVICES

The Library offers a variety of information access services to its clients to assist them in their research needs and these include:

a) Book Borrowing
b) Inter-Library Loan
c) Reprographic Services
d) Research Support
e) E-resources access
f) Reading and Study Spaces

2. LIBRARY SPACES

The MSU Library has established spaces to provide comprehensive support for the university's teaching and learning, research, and innovation. The aim of creating these spaces is to enable users to explore their minds while enjoying the atmosphere of these areas.

2.1 Reading and Study

These Sections of the libraries are designed to provide readers with a peaceful and studious environment. Users can fully immerse themselves in their research and independent reading with comfortable reading desks and carrels. The spaces also offer wifi access, enabling users to access online resources and conduct research efficiently.

2.2 Learning Commons

A great feature in MSU libraries is the learning commons, which provide access to networked desktop computers with high-speed internet. These spaces also serve as venues for undergraduates, allowing individuals to enhance their research skills and knowledge.

2.3 Research Commons

This space is primarily designed for postgraduate students who wish to research independently.

2.4 24-Hour Study Centres

These are dedicated spaces opened 24 hours a day and designed to facilitate a focused, quiet, and comfortable study experience. The 24-Hour Study Centres are fitted with plug-top desktops for plugging in laptops and have WIFI hotspots.

2.5 Branch Libraries

The MSU Libraries are available on all campuses and learning centres due to the university's multicampus structure. The libraries are conveniently located and easily accessible for all students and staff. All 10 libraries have plug-top desks and wireless hotspots for internet connectivity, which support the University's Bring Your Device (BYOD) policy. This allows students to have the best experience while using the facilities.

3. MSU LIBRARY COLLECTIONS

3.1 Print Book Collection

Over the years, the MSU Library has established a robust hardcopy book collection of books that are well-selected recent and relevant titles of books necessary for programs on offer.

3.2 E-Books

The Midlands State University Library has extensive eBooks covering all university programs. These eBooks are accessible to students and staff both on and off campus.

3.3 E-Journals

The library also has a significant collection of full-text electronic journal collections (e-journals) that can be accessed on and off campus. Like the e-books, the e-journal collections support all programmes the University offers.

3.4 E-Reserves

The Library provides eReserves in the form of prescribed e-books that support all programmes and are accessible to staff and students through the Library's subject-based electronic guides.

3.5 Special Collections 1123

The Special Collections of the MSU Library include various items such as print journals, past examination papers, novels, African Writers Series, materials from International Organizations, Government publications, maps, multimedia collections, and the MSU History book.

3.6 Reference Collection

The Library offers access to in-house standard references such as encyclopedias, handbooks, and dictionaries to assist researchers with quick reference inquiries. You can approach the Client Services librarian at the Information Desk for assistance.

3.7 Core Collection

The Library Core Collection is an essential set of hard-copy textbooks critical to the University's academic programs. The faculty librarians work closely with academic staff to ensure the collection is curated precisely and in relevance.

3.8 Institutional Repository

The Midlands State University Institutional Repository is an open-access digital archive that collects, maintains, disseminates, and distributes the research output of Midlands State University (MSU) community members. The electronic archive promotes MSU's intellectual output in line with the Education 5.0 design to advance research and innovation. This open-access platform contains research papers, dissertations and theses, books and book chapters, and conference papers presented by MSU researchers.

4. LIBRARY MEMBERSHIP

The following categories of clients are entitled to use the MSU Libraries:

a) Registered students of the Midlands State University.
b) Teaching and Non-teaching Staff.
c) Approved Readers, e.g. students from other Universities and professionals.
d) MSU Alumni

5. PROCEDURES FOR ACCESSING LIBRARY SPACES AND SERVICES

a) Registered students and staff shall have access to all campus libraries upon presenting their valid university identification cards (IDs).
b) Books are issued to clients upon presenting their valid university ID cards.
c) The ID card holder is liable for all transactions for which his or her card is used.
d) Approved Readers (who include Alumni Association members; Students and staff from the Zimbabwe University Libraries Consortium; students and staff from colleges; government officials and general researchers from recognised institutions are required to pay a prescribed fee to have access to Library spaces for studying and research. Approved readers cannot borrow books from the Library but can

only use them in-house.

6. MSU LIBRARIES GENERAL RULES AND REGULATIONS

The primary purpose of these rules is to safeguard the common interest of all library users and to enable the Library to function as efficiently as possible. Here is a brief outline of some MSU Library Rules and Regulations.

- The right to use library facilities is exclusively reserved for bonafide students and staff of MSU.
- Users must present a valid MSU ID Card to the security personnel upon entering the Library.
- A borrower will be held responsible for all material that is taken out under their name and may not lend such material to other people. Borrower privileges are not transferable.
- Lost staff or student cards must be reported immediately to the Libraries.
- Any acts of theft or attempted theft of library items are not tolerated in the library.
- Personal belongings like bags should be deposited in the Baggage area against a token/disc. Loss of the token will attract a fine.
- Smoking and eating in the Library is strictly prohibited.
- Drinking water is only allowed at designated water dispenser points.
- Cellphones and laptops should be put on silent mode or otherwise attract a fine.
- Mutilation of Library materials by marking, underlining, or removing pages or portions of pages is strictly prohibited.
- The Library does not issue out torn or defaced books; should readers lose, damage, or deface library resources they will be requested to replace them.
- All items taken in and out of the Library are subject to security checks at the Library exit point.
- Loss of Library material should be reported immediately.
- Library materials are not to be re-shelved by users as this leads to misplacement of a book. A misplaced book is like a lost book.
- Journals, dissertations, and past exam papers are reference sources for in-house use only.
- Users should not remove/ unplug computer cables/connections, network cables, and other accessories in the library.
- All forms of payments are to be done at the cash office (Bursar) or bank, receipts are to be brought to the Library for verification.
- Clients with overdue fines lost library books or both will be denied clearance and access to results until the balances are settled.
- Library staff on duty shall not be held responsible for the loss of any personal belongings of

users in the Library.

- Changing the settings and display of the computers in the Library is not permitted.

NB: Anyone who violates the rules and regulations of the library would be liable to lose the privilege of library membership and may be barred from using the library facilities. The Librarian reserves the right to suspend the membership of any client/ user found misbehaving in the Library.

7. MSU LIBRARY CONTACT DETAILS

LIBRARY FACULTY LIBRARIANS

NAME	FACULTY	CONTACT DETAILS
R Masimbiti	Agriculture Environment and Natural Resources Management.	masimbitir@staff.msu.ac.zw
P Chigamani	Arts and Humanities	chigamanip@staff.msu.ac.zw
E Poswo	Built Environment Art and Design	poswoe@staff.msu.ac.zw
K Tokwe	Business Sciences	tokwek@staff.msu.ac.zw
E Matsa	Education	matsae@staff.msu.ac.zw
P Chigamani	Geosciences & Engineering	chigamanip@staff.msu.ac.zw
S Mutavayi	Law	mutavayis@staff.msu.ac.zw
S Mazhindu	Medicine & Health Sciences	mazhindus@staff.msu.ac.zw
K Mafa	Postgraduate Studies	mafak@staff.msu.ac.zw
G Bonde	Science & Technology	bondeg@staff.msu.ac.zw
Mr. C. Chuma	School of Social Work	chumac@staff.msu.ac.zw
Mr. F. Masunungure	Social Sciences	masununguref@staff.msu.ac.zw

a) Chat widget – chat with a librarian through the libguides accessible through the library website.
b) WhatsApp platform – +263786755935
c) Email Address – libraryresearchsupportservices@library.msu.ac.zw
d) Facebook – https://www.facebook.com/msulib